



**IEEE Standard for Information Technology—
Telecommunications and Information Exchange between Systems
Local and Metropolitan Area Networks—
Specific Requirements**

**Part 11: Wireless LAN Medium Access Control
(MAC) and Physical Layer (PHY) Specifications**

**Amendment 1:
Enhancements for High-Efficiency WLAN**

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.11ax™-2021
(Amendment to IEEE Std 802.11-2020)

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Approved 9 February 2021
IEEE SA Standards Board

Abstract: Modifications to both the IEEE 802.11 physical layer (PHY) and medium access control (MAC) sublayer for high-efficiency operation in frequency bands between 1 GHz and 7.125 GHz are defined in this amendment to IEEE Std 802.11-2020.

Keywords: dense deployment, high efficiency, IEEE 802.11™, IEEE 802.11ax™, MAC, medium access control, OFDMA, orthogonal frequency division multiple access, PHY, physical layer, wireless local area network, WLAN

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This amendment defines modifications to both the IEEE 802.11 physical layer (PHY) and medium access control (MAC) sublayer for high-efficiency operation in frequency bands between 1 GHz and 7.125 GHz.

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**IEEE Standard for Information Technology—
Telecommunications and Information Exchange between Systems
Local and Metropolitan Area Networks—
Specific Requirements**

**Part 11: Wireless LAN Medium Access Control
(MAC) and Physical Layer (PHY) Specifications**

**Amendment 1:
Enhancements for High-Efficiency WLAN**

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strike through~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in tables, figures, or equations by removing the existing figure or equation and replacing it with a new one. Editorial instructions, change markings and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

1. Overview

1.4 Word usage

Change the second paragraph in 1.4 as follows:

The construction “between x and y ”, “ x to y ” or “ x - y ” represents an inclusive range (i.e., the range includes both values x and y). The construction “B x -B y ” represents bits x to y ; it is not the subtraction of bit y from bit x .

1.5 Terminology for mathematical, logical, and bit operations

Insert the following text into 1.5 after the description of “ $\log_2(x)$ ”:

$\log_{10}(x)$ is the logarithm of x to the base 10. For example, $\log_{10}(100)$ is 2.

2. Normative references

Insert the following reference into Clause 2 in alphanumeric order:

IETF RFC 8110, Opportunistic Wireless Encryption, Harkins, D., and W. Kumari, Mar. 2017.

3. Definitions, acronyms, and abbreviations

3.1 Definitions

Change the following definitions in 3.1 as shown:

aggregate medium access control (MAC) service data unit (A-MSDU): A structure that contains one or more MSDUs and is ~~transported within a single (unfragmented) data medium access control (MAC) MAC protocol data units (MPDUs)~~ transmitted in one or more QoS Data frames with the same sequence number.

multi-user multiple input, multiple output (MU-MIMO): A technique by which multiple stations (STAs), each with one or more antennas, either simultaneously transmit to a single STA or simultaneously receive from a single STA independent data streams over the same ~~radio frequencies~~ subcarriers.

NOTE — IEEE Std 802.11 supports only downlink (DL) MU-MIMO. See ~~downlink multi-user multiple input, multiple output (DL MU-MIMO)~~ (in 3.2).

non-high-throughput (non-HT): A modifier meaning ~~neither not~~ high throughput (HT), ~~not nor~~ very high throughput (VHT), and not high efficiency (HE).

physical layer (PHY) protocol data unit (PPDU): The unit of data exchanged between ~~two peer~~ PHY entities to provide the PHY data service.

Insert the following definitions into 3.1 in alphabetic order:

orthogonal frequency division multiple access (OFDMA): An orthogonal frequency division multiple (OFDM)-based multiple access technique by which multiple stations (STAs) either simultaneously transmit to a single STA or simultaneously receive from a single STA independent data streams over different groups of subcarriers.

single input, single output (SISO): A technique by which a station (STA) transmits to or receives from a single STA a single space-time stream.

3.2 Definitions specific to IEEE 802.11

Change the following definitions in 3.2 as shown:

20 MHz mask physical layer (PHY) protocol data unit (PPDU): One of the following PPDU:

- a) A Clause 17 PPDU transmitted using the 20 MHz transmit spectral mask defined in Clause 17.
- b) A Clause 18 orthogonal frequency division multiplexing (OFDM) PPDU transmitted using the transmit spectral mask defined in Clause 18.
- c) A high-throughput (HT) PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to HT_CBW20 and the parameter CH_OFFSET equal to CH_OFF_20 transmitted using the 20 MHz transmit spectral mask defined in Clause 19.
- d) A very high throughput (VHT) PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to CBW20 transmitted using the 20 MHz transmit spectral mask defined in Clause 21.
- e) A Clause 17 PPDU transmitted by a VHT STA using the 20 MHz transmit spectral mask defined in Clause 21.
- f) An HT PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to HT_CBW20 and the parameter CH_OFFSET equal to CH_OFF_20 transmitted by a VHT STA using the 20 MHz transmit spectral mask defined in Clause 21.
- g) A high-efficiency (HE) PPDU with TXVECTOR parameter CH_BANDWIDTH equal to CBW20 transmitted using the 20 MHz transmit spectral mask defined in Clause 27.
- h) A Clause 17 PPDU transmitted by an HE STA using the 20 MHz transmit spectral mask defined in Clause 27.

20 MHz physical layer (PHY) protocol data unit (PPDU): A Clause 15 PPDU, Clause 17 PPDU (when using 20 MHz channel spacing), Clause 16 PPDU, Clause 18 orthogonal frequency division multiplexing (OFDM) PPDU, Clause 19 20-MHz high-throughput (HT) PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to HT_CBW20, ~~or~~ Clause 21 20-MHz very high throughput (VHT) PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to CBW20, or Clause 27 20-MHz high-efficiency (HE) PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to CBW20.

40 MHz mask physical layer (PHY) protocol data unit (PPDU): One of the following PPDU:

- a) A 40 MHz high-throughput (HT) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to HT_CBW40) transmitted using the 40 MHz transmit spectral mask defined in Clause 19.
- b) A 40 MHz non-HT duplicate PPDU (TXVECTOR parameter CH_BANDWIDTH equal to NON_HT_CBW40) transmitted by a non-very high throughput (non-VHT) STA using the 40 MHz transmit spectral mask defined in Clause 19.
- c) A 40 MHz non-HT duplicate PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW40) transmitted by a very high throughput (VHT) STA using the 40 MHz transmit spectral mask defined in Clause 21.
- d) A 20 MHz HT PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to HT_CBW20 and the CH_OFFSET parameter equal to either CH_OFF_20U or CH_OFF_20L transmitted using the 40 MHz transmit spectral mask defined in Clause 19.
- e) A 20 MHz VHT PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to CBW20 transmitted using the 40 MHz transmit spectral mask defined in Clause 21.
- f) A 40 MHz VHT PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to CBW40 transmitted using the 40 MHz transmit spectral mask defined in Clause 21.
- g) A 40 MHz HT PPDU (TXVECTOR parameter CH_BANDWIDTH equal to HT_CBW40) transmitted by a VHT STA using the 40 MHz transmit spectral mask defined in Clause 21.
- h) A 20 MHz non-HT PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW20) transmitted using the 40 MHz transmit spectral mask defined in Clause 19.

- i) A 20 MHz non-HT PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW20) transmitted by a VHT STA using the 40 MHz transmit spectral mask defined in Clause 21.
- j) A 40 MHz high-efficiency (HE) PPDU with TXVECTOR parameter CH_BANDWIDTH equal to CBW40 transmitted using the 40 MHz transmit spectral mask defined in Clause 27.
- k) A 40 MHz VHT PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW40) transmitted by an HE STA using the 40 MHz transmit spectral mask defined in Clause 21.
- l) A 40 MHz non-HT duplicate PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW40) transmitted by an HE STA using the 40 MHz transmit spectral mask defined in Clause 19.

40 MHz physical layer (PHY) protocol data unit (PPDU): A 40 MHz high-throughput (HT) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to HT_CBW40), or a 40 MHz non-HT duplicate PPDU (TXVECTOR parameter CH_BANDWIDTH equal to NON_HT_CBW40 or TXVECTOR parameter CH_BANDWIDTH equal to CBW40), or a 40 MHz very high throughput (VHT) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW40), or a Clause 27 40-MHz high-efficiency (HE) PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to CBW40.

80 MHz mask physical layer (PHY) protocol data unit (PPDU): ~~A PPDU that is transmitted using the 80 MHz transmit spectral mask defined in Clause 21 and that is one~~ One of the following PPDUs:

- a) An 80 MHz very high throughput (VHT) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80) transmitted using the 80 MHz transmit spectral mask defined in Clause 21.
- b) An 80 MHz non-high-throughput (non-HT) duplicate PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80) transmitted using the 80 MHz transmit spectral mask defined in Clause 21.
- c) A 20 MHz non-HT, high-throughput (HT), or VHT PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW20) transmitted using the 80 MHz transmit spectral mask defined in Clause 21.
- d) A 40 MHz non-HT duplicate, HT, or VHT PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW40) transmitted using the 80 MHz transmit spectral mask defined in Clause 21.
- e) An 80 MHz high-efficiency (HE) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80) transmitted using the 80 MHz transmit spectral mask defined in Clause 27.

80 MHz physical layer (PHY) protocol data unit (PPDU): A Clause 21 80-MHz very high throughput (VHT) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80), or a Clause 21 80-MHz non-high-throughput (non-HT) duplicate PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80), or a Clause 27 80-MHz high-efficiency (HE) PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to CBW80.

80+80 MHz mask physical layer (PHY) protocol data unit (PPDU): ~~A PPDU that is transmitted using the 80+80 MHz transmit spectral mask defined in Clause 21 and that is one~~ One of the following PPDUs:

- a) An 80+80 MHz very high throughput (VHT) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80+80) transmitted using the 80+80 MHz transmit spectral mask defined in Clause 21.
- b) An 80+80 MHz non-high-throughput (non-HT) duplicate PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80+80) transmitted using the 80+80 MHz transmit spectral mask defined in Clause 21.
- c) An 80+80 MHz high-efficiency (HE) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80+80) transmitted using the 80+80 MHz transmit spectral mask defined in Clause 27.

80+80 MHz physical layer (PHY) protocol data unit (PPDU): A Clause 21 80+80 MHz very high throughput (VHT) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80+80), or a Clause 21 80+80 MHz non-high-throughput (non-HT) duplicate PPDU (TXVECTOR parameter

CH_BANDWIDTH equal to CBW80+80), or a Clause 27 80+80 MHz high-efficiency (HE) PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to CBW80+80.

160 MHz mask physical layer (PHY) protocol data unit (PPDU): ~~A PPDU that is transmitted using the 160 MHz transmit spectral mask defined in Clause 21 and that is one~~ One of the following PPDU:

- A 160 MHz very high throughput (VHT) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW160) transmitted using the 160 MHz transmit spectral mask defined in Clause 21.
- A 160 MHz non-high-throughput (non-HT) duplicate PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW160) transmitted using the 160 MHz transmit spectral mask defined in Clause 21.
- A 20 MHz non-HT, high-throughput (HT), or VHT PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW20) transmitted using the 160 MHz transmit spectral mask defined in Clause 21.
- A 40 MHz non-HT duplicate, HT, or VHT PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW40) transmitted using the 160 MHz transmit spectral mask defined in Clause 21.
- An 80 MHz non-HT duplicate or VHT PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW80) transmitted using the 160 MHz transmit spectral mask defined in Clause 21.
- A 160 MHz high-efficiency (HE) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW160) transmitted using the 160 MHz transmit spectral mask defined in Clause 27.

160 MHz physical layer (PHY) protocol data unit (PPDU): A Clause 21 160-MHz very high throughput (VHT) PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW160), or a Clause 21 160-MHz non-high-throughput (non-HT) duplicate PPDU (TXVECTOR parameter CH_BANDWIDTH equal to CBW160), or a Clause 27 160-MHz high-efficiency (HE) PPDU with the TXVECTOR parameter CH_BANDWIDTH equal to CBW160.

bandwidth signaling transmitter address (TA): A TA that is used by a very high throughput (VHT) station (STA) or a high-efficiency (HE) STA to indicate the presence of additional signaling related to the bandwidth to be used in a subsequent transmission in an enhanced distributed channel access (EDCA) transmission opportunity (TXOP). It is represented by the IEEE medium access control (MAC) individual address of the transmitting VHT-STA but with the Individual/Group bit set to 1.

fragmentation: The process of partitioning a medium access control (MAC) service data unit (MSDU), aggregate MAC service data unit (A-MSDU) or MAC management protocol data unit (MMPDU) into a sequence of smaller MAC protocol data units (MPDUs) prior to transmission. The process of recombining a set of fragment MPDUs into an MSDU, A-MSDU, or MMPDU is known as defragmentation. These processes are described in 5.8.1.9 of ISO/IEC 7498-1:1994.

medium access control (MAC) management protocol data unit (MMPDU): The unit of data exchanged between two peer MAC entities, using services of the physical layer (PHY), to implement the MAC management protocol. The MMPDU is transported in one or more Management frames. The MMPDU might include a Mesh Control field or management message integrity code element (Management MIC element), but does not include a MAC header, a frame check sequence (FCS), or any other security encapsulation overhead.

NOTE—The MMPDU occupies a position in the management plane similar to that of the MAC service data unit (MSDU) in the data plane. An MSDU, A-MSDU, or MMPDU is transmitted in one or more MAC protocol data units (MPDUs) (with the Type field set to Data, Data, or Management, respectively). ~~An MSDU can be carried in an aggregate MAC service data unit (A-MSDU).~~ An A-MSDU is transmitted in one MPDU. An MSDU, A-MSDU, or MMPDU can be carried (in an MPDU) in an aggregate MAC protocol data unit (A-MPDU)

multiple basic service set identifier (BSSID) set: A collection of ~~cooperating~~ access points (APs), such that all APs use a common operating class, channel, receive antenna connector, and transmit antenna

connector and advertise information for multiple BSSIDs using Beacon or Probe Response frames sent by the AP corresponding to the transmitted BSSID.

multi-user (MU) physical layer (PHY) protocol data unit (PPDU): A PPDU that carries one or more PHY service data units (PSDUs) for one or more stations (STAs) using the downlink multi-user multiple input, multiple output (DL-MU-MIMO) technique, orthogonal frequency division multiple access (DL OFDMA) technique, or a combination of the two techniques, or that carries a PSDU for an AP and is in high-efficiency (HE) MU PPDU format.

nontransmitted basic service set (BSS) identifier (BSSID): A BSSID corresponding to one of the basic service sets (BSSs) when the multiple BSSID capability is supported that is not transmitted explicitly, but that can be derived from the information encoded in Probe Response, Beacon and directional multi-gigabit (DMG) Beacon frames and Neighbor reports.

individual target wake time (TWT): A specific time or set of times negotiated between two for individual stations (STAs) at which the STAs are expected to be awake in order to exchange frames with each other STAs.

Insert the following definitions into 3.2 in alphanumeric order:

20 MHz-only non-access-point (non-AP) high-efficiency station (HE STA): A non-AP HE STA that, for the frequency band in which it is operating, indicates in the Supported Channel Width Set subfield in the HE PHY Capabilities Information field in the HE Capabilities element that it does not support operating with a channel width greater than 20 MHz.

20 MHz operating non-access-point (non-AP) high-efficiency station (HE STA): A non-AP HE STA that is operating in 20 MHz channel width mode, such as a 20 MHz-only non-AP HE STA or an HE STA that has reduced its operating channel width to 20 MHz using operating mode indication (OMI).

80 MHz operating non-access-point (non-AP) high-efficiency station (HE STA): A non-AP HE STA that is operating in 80 MHz channel width mode, such as a non-AP STA (excluding the 20 MHz-only non-AP HE STA) which is not capable of 160 MHz operation or a non-AP STA that has reduced its operating channel width to 80 MHz using operating mode indication (OMI).

ack-enabled single traffic identifier (TID) aggregate medium access control (MAC) protocol data unit (A-MPDU) (ack-enabled single-TID A-MPDU): An A-MPDU that contains at least two A-MPDU subframes, where the TIDs differ and where only one of the A-MPDU subframes includes a tagged MPDU that solicits the acknowledgment context.

NOTE—A Management frame that solicits an acknowledgment in an ack-enabled single-TID A-MPDU is treated as if it had a TID 15.

ack-enabled multi traffic identifier (TID) aggregate medium access control (MAC) protocol data unit (A-MPDU) (ack-enabled multi-TID A-MPDU): An A-MPDU where at least one tagged MPDU that solicits acknowledgment context is aggregated in the A-MPDU, and MPDUs from more than one TID that solicit Ack acknowledgment or Block Ack acknowledgment context are aggregated in the A-MPDU.

NOTE—A Management frame that solicits an acknowledgment in an ack-enabled multi-TID A-MPDU is treated as if it had a TID 15.

basic service set (BSS) color: An identifier for a BSS or for a set of BSSs belonging to a multiple basic service set identifier (BSSID) set or a co-hosted BSSID set.

broadcast resource unit (RU): A resource unit in a high-efficiency (HE) multi-user (MU) physical layer (PHY) protocol data unit (PPDU) transmitted by an access point (AP) that is intended for either unassociated STAs or more than one associated STA in the basic service set (BSS) or in any of the other BSSs in the multiple BSSID set to which the AP's BSS belongs.

broadcast target wake time (TWT): A specific time or set of times broadcast by an access point (AP) to multiple non-AP stations (STAs) at which the non-AP STAs are expected to be awake in order to exchange frames with the AP.

co-hosted basic service set identifier (BSSID) set: A collection of access points (APs) such that all APs use a common operating class, channel, receive antenna connector, and transmit antenna connector, and each AP advertises information for its BSSID using Beacon or Probe Response frames.

co-located access point (AP) set: A set of two or more APs in the same physical device.

NOTE 1—APs in the co-located set might be operating on the same or different channel.

NOTE 2—The APs that are part of a co-located AP set and that are operating on the same channel might form a co-hosted basic service set identifier (BSSID) set or multiple BSSID set.

detected access point (AP): An AP might be detected by a station (STA) if the STA and the AP are on the same channel and in range.

downlink (DL) high-efficiency (HE) multi-user (MU) physical layer (PHY) protocol data unit (PPDU) (DL HE MU PPDU): An HE MU PPDU transmitted by an access point (AP). This PPDU carries one or more PHY service data units (PSDUs) for one or more users.

enhanced multiple basic service set identifier (BSSID) advertisement (EMA) access point (AP): An AP with `dot11MultiBSSIDImplemented` set to true that supports enhancements related to the discovery of nontransmitted BSSIDs.

extended range (ER) beacon: A Beacon frame transmitted in a high-efficiency (HE) ER single-user (SU) physical layer (PHY) protocol data unit (PPDU) to form an ER basic service set (BSS).

high-efficiency (HE) basic service set (BSS): A BSS in which the transmitted Beacon frame includes an HE Operation element.

high-efficiency (HE) beacon: A Beacon frame transmitted in a HE single-user (SU) physical layer (PHY) protocol data unit (PPDU).

high-efficiency (HE) beamformee: An HE station (STA) that receives an HE physical layer (PHY) protocol data unit (PPDU) that was transmitted using a beamforming steering matrix.

high-efficiency (HE) beamformer: An HE station (STA) that transmits an HE physical layer (PHY) protocol data unit (PPDU) using a beamforming steering matrix.

high-efficiency (HE) extended range (ER) single-user (SU) physical layer (PHY) protocol data unit (PPDU) (HE ER SU PPDU): A PPDU transmitted with HE ER SU PPDU format. This PPDU carries a single PHY service data unit (PSDU).

high-efficiency (HE) masked HE-long training field (HE-LTF) sequence mode: An HE-LTF mode used in an HE TB PPDU. The masked HE-LTF sequence mode does not have any pilot subcarriers in the HE-LTF field and uses a masked HE-LTF sequence generated by multiplying an orthogonal code (distinct to each spatial stream) over groups of subcarriers.

high-efficiency (HE) modulation and coding scheme (HE-MCS): A specification of the HE physical layer (PHY) parameters that consists of modulation order (BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM) and forward error correction (FEC) coding rate (1/2, 2/3, 3/4, 5/6) and that is used in an HE PHY protocol data unit (PPDU).

high-efficiency (HE) multi-user (MU) physical layer (PHY) protocol data unit (PPDU) (HE MU PPDU): A PPDU transmitted with HE MU PPDU format.

high-efficiency (HE) no pilot HE-long training field (HE-LTF) mode: An HE-LTF mode used in an HE triggered-based (TB) PPDU. The no pilot HE-LTF mode does not have any pilot subcarriers in the HE-LTF field and does not use a masked HE-LTF sequence. *See also:* **high-efficiency (HE) masked HE-long training field (HE-LTF) sequence mode.**

high-efficiency (HE) physical layer (PHY) protocol data unit (PPDU): A Clause 27 PPDU that is not a Clause 21 PPDU.

high-efficiency (HE) single stream pilot: The same pilot sequence is applied to all spatial time streams for a given resource allocation.

high-efficiency (HE) single stream pilot HE-long training field (HE-LTF) mode: An HE-LTF mode used in an HE single-user (SU), HE extended range (ER) SU, HE multi-user (MU) and HE trigger-based (TB) PPDU. The HE single stream pilot HE-LTF mode has single stream pilot subcarriers in the HE-LTF field.

high-efficiency (HE) single-user (SU) physical layer (PHY) protocol data unit (PPDU) (HE SU PPDU): A PPDU transmitted with HE SU PPDU format. This PPDU carries a single PHY service data unit (PSDU).

high-efficiency (HE) trigger-based (TB) physical layer (PHY) protocol data unit (PPDU) (HE TB PPDU): A PPDU transmitted with HE TB PPDU format. This PPDU carries a single PHY service data unit (PSDU).

individually addressed resource unit (RU): A resource unit in a high-efficiency (HE) multi-user (MU) physical layer (PHY) protocol data unit (PPDU) transmitted by an access point (AP) or a tunneled direct link setup (TDLS) peer station (STA) that is intended for a single associated non-AP STA or a TDLS peer STA, respectively.

non-ack-enabled single traffic identifier (TID) aggregate medium access control (MAC) protocol data unit (A-MPDU) (non-ack-enabled single-TID A-MPDU): A legacy A-MPDU with the exception that a Trigger frame can be aggregated.

non-ack-enabled multi traffic identifier (TID) aggregate medium access control (MAC) protocol data unit (A-MPDU) (non-ack-enabled multi-TID A-MPDU): An A-MPDU where tagged MPDUs are not aggregated and the aggregated untagged MPDUs from at least two TIDs solicit block acknowledgment.

non-orthogonal frequency division multiple access (non-OFDMA) high-efficiency (HE) physical layer (PHY) protocol data unit (PPDU) (non-OFDMA HE PPDU): A 20 MHz HE PPDU with a 242-tone resource unit (RU), a 40 MHz HE PPDU with a 484-tone RU, an 80 MHz HE PPDU with a 996-tone RU, or a 160 MHz or 80+80 MHz HE PPDU with a 2×996-tone RU.

non-spatial reuse group (non-SRG): An adjective indicating the quality of not being a member of a particular spatial reuse group or the quality of not originating from a station (STA) that is a member of a basic service set (BSS) that is part of a particular spatial reuse group.

opportunistic power save (OPS) access point (AP): A high-efficiency (HE) AP that supports the OPS mechanism.

opportunistic power save (OPS) mechanism: A power save mechanism to allow OPS non-access-point (non-AP) stations (STAs) to opportunistically go to doze state or be unavailable for a defined period.

opportunistic power save (OPS) non-access-point (non-AP) station (STA): A non-AP high-efficiency (HE) STA that supports the OPS mechanism.

opportunistic power save (OPS) period: A period during which an OPS non-access-point (non-AP) station (STA) is allowed to go to doze state or be unavailable if it received an indication that it will not be scheduled by its associated OPS access point (AP).

orthogonal frequency division multiple access (OFDMA) high-efficiency (HE) physical layer (PHY) protocol data unit (PPDU) (OFDMA HE PPDU): A 20 MHz HE PPDU with resource units (RUs) smaller than 242-tone, or a 40 MHz HE PPDU with RUs smaller than 484-tone, or an 80 MHz HE PPDU with RUs smaller than 996-tone, or a 160 MHz or 80+80 MHz HE PPDU with RUs smaller than 2×996-tone.

overlapping basic service set (OBSS) packet detect (PD): A packet detection level used for spatial reuse procedure.

parameterized spatial reuse (PSR) opportunity: A spatial reuse opportunity that is established based on the value of a Spatial Reuse field in the HE-SIG-A field of a high-efficiency (HE) trigger-based (TB) physical layer (PHY) protocol data unit (PPDU) and/or the UL Spatial Reuse subfield in the Common Info field of a Trigger frame.

parameterized spatial reuse reception (PSRR) physical layer (PHY) protocol data unit (PPDU) (PSRR PPDU): A PPDU that contains a Trigger frame that has a value in the UL Spatial Reuse subfield of the Common Info field that is neither PSR_DISALLOW nor PSR_AND_NON_SRG_OBSS_PD_PROHIBITED.

parameterized spatial reuse transmission (PSRT) physical layer (PHY) protocol data unit (PPDU) (PSRT PPDU): A PPDU that is transmitted during a parameterized spatial reuse (PSR) opportunity by an HE STA when PSR conditions for PSR-based spatial reuse operation are satisfied and that has the SR PPDU subfield of the CAS Control field equal to 1.

random access resource unit (RA-RU): A resource unit (RU) allocated in a Trigger frame to support the uplink (UL) orthogonal frequency division multiple access (OFDMA) based random access (UORA) procedure.

reported access point (AP): An AP that is described in an element such as a Neighbor Report element or a Reduced Neighbor Report element.

reporting access point (AP): An AP that is transmitting an element, such as a Neighbor Report element or a Reduced Neighbor Report element, describing a reported AP.

resource unit (RU): A group of 26, 52, 106, 242, 484, 996, or 2×996 subcarriers as an allocation of subcarriers for transmission.

spatial reuse (SR): The transmission of a physical layer (PHY) protocol data unit (PPDU) on the medium under certain conditions when a PPDU has been detected that would otherwise have prevented the transmission.

spatial reuse group (SRG): A group of basic service sets (BSSs) identified by their BSS colors or partial basic service set identifiers (BSSIDs) for overlapping basic service set packet detect (OBSS PD) based spatial reuse operation with SRG OBSS PD level.

station (STA) 6G: A STA that is operating on a channel that belongs to any operating class that has a value of 5.950 for the entry in the “Channel starting frequency” column of Table E-4.

tagged media access control (MAC) protocol data unit (MPDU) (tagged MPDU): An MPDU carried in an aggregate MPDU (A-MPDU) subframe that has the EOF/Tag field in the MPDU delimiter set to 1.

target beacon transmission time (TBTT) scheduled station (STA): A non-access-point (non-AP) STA that has negotiated the TBTT of the first Beacon frame and the wake interval between subsequent Beacon frames that it intends to receive.

target beacon transmission time (TBTT) scheduling access point (AP): An AP that has negotiated with a non-AP station (STA) the TBTT of the first Beacon frame and the wake interval between subsequent Beacon frames that the non-AP STA intends to receive.

target wake time (TWT) scheduled station (STA): A STA that follows the broadcast TWT schedules provided in a broadcast TWT element.

target wake time (TWT) scheduling access point (AP): An AP that schedules broadcast TWTs and provides these broadcast TWT schedules in a broadcast TWT element.

triggered uplink access (TUA): A mechanism by which one or more non-access-point (non-AP) stations (STAs) simultaneously participate in an uplink (UL) transmission to an access point (AP) using resource units (RUs) allocated in the preceding Trigger frame.

triggering frame: A Trigger frame or a frame carrying a TRS Control subfield.

triggering physical layer (PHY) protocol data unit (PPDU): A PPDU carrying a triggering frame.

untagged medium access control (MAC) protocol data unit (MPDU) (untagged MPDU): An MPDU carried in an aggregate MPDU (A-MPDU) subframe that has the EOF/Tag field in the MPDU delimiter set to 0.

uplink (UL) high-efficiency (HE) multi-user (MU) physical layer (PHY) protocol data unit (PPDU) (UL HE MU PPDU): An HE MU PPDU transmitted by a non-access-point (non-AP) station (STA). This PPDU carries a single PHY service data unit (PSDU).

NOTE—The UL HE MU PPDU has an HE-SIG-B field that contains additional information (e.g., the identifier of the transmitter) that can be used by the recipient of the UL HE MU PPDU to determine the transmitter of the PPDU even in cases where the Data field of the PPDU is not received. For example, this allows the originator of persistently failing PPDU to be identified.

uplink (UL) orthogonal frequency division multiple access (OFDMA)-based random access (UORA): A random access mechanism for non-access-point (non-AP) high-efficiency (HE) stations (STAs) to participate in uplink OFDMA transmissions in one or more designated random access resource units (RUs).

3.4 Acronyms and abbreviations

Insert the following abbreviations into 3.4 in alphabetic order:

A-Control	aggregated control
BQR	bandwidth query report
BQRP	bandwidth query report poll
BFRP	beamforming report poll
BSR	buffer status report
BSRP	buffer status report poll
CAS	command and status
CCDF	complementary cumulative distribution function
CQI	channel quality indication
DCM	dual carrier modulation
DL	downlink
DL MU	downlink multi-user
HE	high efficiency
MU-BAR	multi-user block ack request
MUEDCATimer	multi-user EDCA timer
MU-RTS	multi-user request to send
NFRP	NDP feedback report poll
OBO	orthogonal frequency division multiple access (OFDMA) random access backoff
OCW	orthogonal frequency division multiple access (OFDMA) contention window
OFDMA	orthogonal frequency division multiple access
OM	operating mode
OMI	operating mode indication
PD	packet detect
PPE	physical layer (PHY) packet extension
PPET	physical layer (PHY) packet extension threshold
PSR	parameterized spatial reuse
QTP	quiet time period
RA-RU	random access resource unit
RDP	reverse direction protocol
ROM	receive operating mode
RPL	received power level
RU	resource unit
SF	scaling factor
SR	spatial reuse
SRG	spatial reuse group
TB	trigger-based
TOM	transmit operating mode
TRS	triggered response scheduling
TUA	triggered uplink access
UL	uplink
UL MU	uplink multi-user
UORA	uplink orthogonal frequency division multiple access (OFDMA) based random access
UPH	uplink (UL) power headroom

4. General description

4.3 Components of the IEEE Std 802.11 architecture

Insert the following subclause (4.3.15a) after 4.3.15:

4.3.15a High-efficiency (HE) STA

The IEEE 802.11 HE STA operates in frequency bands between 1 GHz and 7.125 GHz.

In the 5 GHz and 6 GHz bands, the following apply:

- An HE STA is also a VHT STA if operating in the 5 GHz band, except that a 20 MHz-only HE STA does not support 40 MHz and 80 MHz channel widths
- An HE STA is also a STA that supports the OFDM PHY defined in Clause 17 if operating in the 6 GHz band
- Support for 20 MHz operating channel width is mandatory in an HE STA
- Support for 40 MHz and 80 MHz operating channel width is mandatory in an HE STA that is not a 20 MHz-only non-AP HE STA
- Support for 160 MHz and 80+80 MHz operating channel width is optional in an HE STA

In the 2.4 GHz band, the following apply:

- An HE STA is also an HT STA
- Support for 20 MHz operating channel width is mandatory in an HE STA
- Support for 40 MHz operating channel width is optional in an HE STA

The main PHY features in an HE STA that are not present in VHT STA or HT STA are the following:

- Mandatory support for DL and UL OFDMA
- Mandatory support for DL MU-MIMO by an HE AP that supports 4 or more spatial streams when MU-MIMO is done on the entire PPDU bandwidth
- Mandatory support for DL MU-MIMO reception for a non-AP HE STA
- Mandatory support for the HE sounding protocol to support beamforming for a non-AP STA beamformee and optional otherwise
- Optional support for HE-MCSs 10 and 11
- Optional support for UL MU-MIMO
- Optional support for preamble puncturing

NOTE—Preamble puncturing is a mechanism whereby OFDMA is used to avoid transmissions in certain subcarriers.

The main MAC features in an HE STA that are not present in VHT STA or HT STA are the following:

- In an AP, mandatory support for the role of operating mode indication (OMI) responder and optional support for the role of OMI initiator
- In an AP, mandatory support for individual target wake time (TWT) operation
- In a non-AP STA, mandatory support for two NAV operation
- In a non-AP STA, mandatory support for multiple BSSID operation
- In an AP, optional support for two NAV operation
- In a non-AP STA, optional support for the roles of OMI initiator and responder
- In a non-AP STA, optional support for individual TWT operation
- Optional support for dynamic fragmentation levels 1, 2, and 3

- Optional support for broadcast TWT
- Optional support for UL OFDMA-based random access (UORA)
- Optional support for spatial reuse operation
- Optional support for multi-TID A-MPDU operation
- Optional support for ER BSS
- Optional support for the NDP feedback report

An HE AP sends a Trigger frame to initiate UL MU operation using UL OFDMA or UL MU-MIMO transmissions or a frame containing a TRS Control subfield to initiate UL OFDMA transmissions. The frame initiating these transmissions in the uplink direction is a triggering frame. The triggering frame identifies non-AP STAs participating in UL MU operation and assigns RUs and/or spatial streams to these STAs. Multi-STA BlockAck frames can be used by the AP to acknowledge the frames transmitted by multiple non-AP STAs. The scheduling of these Trigger frames can be set up between a non-AP STA and the AP using TWT operation to save power and reduce collisions.

These features can reduce protocol overhead and increase aggregate network throughput (e.g., DL and UL OFDMA, DL/UL MU-MIMO), enhance peak link throughput (e.g., HE-MCS 10, 11), enhance dense network efficiency (e.g., spatial reuse), and/or enhance power conservation (e.g., TWT). These features can improve the average throughput per STA in a BSS by a factor of four, compared to VHT.

4.3.19 Wireless network management

4.3.19.8 Event reporting

Change 4.3.19.8 as follows:

Event requests enable a STA to request a non-AP STA to send particular real-time event reports. The types of events include transition, RSNA, WNM log, BSS color collision, BSS color in use, and peer-to-peer link events. A transition event is transmitted after a non-AP STA successfully completes a BSS transition. Transition events are used to diagnose transition performance problems. An RSNA event report describes the type of Authentication used for the RSNA. RSNA events are used to diagnose security and authentication performance problems. A WNM log event report enables a non-AP STA to transmit a set of WNM log event messages to the requesting STA. WNM log event reports are used to access the contents of a STA's WNM log. A BSS color collision event report enables a non-AP HE STA to signal BSS color collision to its associated AP. A BSS color in use event report enables a non-AP HE STA to signal a BSS color in use by the non-AP HE STA to its associated AP. A peer-to-peer link event report enables a non-AP STA to inform the requesting STA that a peer-to-peer link has been established. Peer-to-peer link event reports are used to monitor the use of peer-to-peer links in the network.

6. Layer management

6.3 MLME SAP interface

6.3.3 Scan

6.3.3.2 MLME-SCAN.request

6.3.3.2.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.3.2.2 as follows:

The primitive parameters are as follows:

```
MLME-SCAN.request(  
    ...,  
    Short SSID List,  
    HE Capabilities,  
    HE 6 GHz Band Capabilities,  
    VendorSpecificInfo  
)
```

Insert the following rows into the untitled table in 6.3.3.2.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
Short SSID List	A list of short SSIDs	As defined in 9.4.2.262	One or more Short SSID fields that are optionally present if dot11ShortSSIDList is true.
HE Capabilities	As defined in HE Capabilities element	As defined in 9.4.2.248	Specifies the parameters in the HE Capabilities element that are supported by the STA. The parameter is present if dot11HEOptionImplemented is true; otherwise, it is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters in the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true; otherwise, it is not present.

6.3.3.3 MLME-SCAN.confirm

6.3.3.3.2 Semantics of the service primitive

Insert the following rows at the end of the untitled BSSDescriptionSet parameter table (introduced by "Each BSSDescription consists of....") in 6.3.3.3.2:

Name	Type	Valid range	Description	IBSS adoption
TWT	As defined in frame format	As defined in 9.4.2.199	The value from the TWT element. The parameter is optionally present if dot11HEOptionImplemented is true, dot11TWTOptionActivated is true, and a TWT element was present in the broadcast Probe Response or Beacon frame from which the BSSDescriptionSet was determined. Otherwise, the parameter is not present.	Do not adopt
HE Capabilities	As defined in frame format	As defined in 9.4.2.248	The value from the HE Capabilities element. The parameter is present if dot11HEOptionImplemented is true and an HE Capabilities element was present in the Probe Response or Beacon frame from which the BSSDescription was determined. Otherwise, the parameter is not present.	Do not adopt
HE 6 GHz Band Capabilities	As defined in frame format	As defined in 9.4.2.263	The value from the HE 6 GHz Band Capabilities element. The parameter is present if dot11HE6GOptionImplemented is true and an HE 6 GHz Band Capabilities element was present in the Probe Response or Beacon frame from which the BSSDescription was determined. Otherwise, the parameter is not present.	Do not adopt
HE Operation	As defined in frame format	As defined in 9.4.2.249	The value from the HE Operation element. The parameter is present if dot11HEOptionImplemented is true and an HE Operation element was present in the Probe Response or Beacon frame from which the BSSDescriptionSet was determined. Otherwise, the parameter is not present.	Adopt

Name	Type	Valid range	Description	IBSS adoption
UORA Parameter Set	As defined in frame format	As defined in 9.4.2.250	The value from the UORA Parameter Set element. The parameter is optionally present if dot11OFDMARandomAccessOptionImplemented is true and a UORA Parameter Set element was present in the Probe Response or Beacon frame from which the BSSDescriptionSet was determined. Otherwise, the parameter is not present.	Do not adopt
MU EDCA Parameter Set	As defined in frame format	As defined in 9.4.2.251	The value from the MU EDCA Parameter Set element. The parameter is optionally present if dot11HEOptionImplemented is true and a MU EDCA Parameter Set element was present in the Probe Response or Beacon frame from which the BSSDescriptionSet was determined. Otherwise, the parameter is not present.	Do not adopt
Spatial Reuse Parameter Set	As defined in frame format	As defined in 9.4.2.252	The value from the Spatial Reuse Parameter Set element. The parameter is optionally present if dot11HEOptionImplemented is true and a Spatial Reuse Parameter Set element was present in the Probe Response or Beacon frame from which the BSSDescriptionSet was determined. Otherwise, the parameter is not present.	Do not adopt
NDP Feedback Report Parameter Set	As defined in frame format	As defined in 9.4.2.253	The value from the NDP Feedback Report Parameter Set element. The parameter is optionally present if dot11HEOptionImplemented is true and an NDP Feedback Report Parameter Set element was present in the Probe Response or Beacon frame from which the BSSDescriptionSet was determined. Otherwise, the parameter is not present.	Do not adopt
BSS Color Change Announcement	As defined in frame format	As defined in 9.4.2.254	The value from the BSS Color Change Announcement element. The parameter is optionally present if dot11HEOptionImplemented is true and a BSS Color Change Announcement element was present in the Probe Response or Beacon frame from which the BSSDescriptionSet was determined. Otherwise, the parameter is not present.	Do not adopt

Name	Type	Valid range	Description	IBSS adoption
ESS Report	As defined in frame format	As defined in 9.4.2.256	The value from the ESS Report element. The parameter is optionally present if an ESS Report element was present in the Probe Response or Beacon frame from which the BSSDescriptionSet was determined. Otherwise, the parameter is not present.	Do not adopt
HE BSS Load	As defined in frame format	As defined in 9.4.2.259	The value from the HE BSS Load element. The parameter is optionally present if dot11QBSSLoadImplemented and dot11HEOptionImplemented are true and an HE BSS Load element was present in the Probe Response or Beacon frame from which the BSSDescriptionSet was determined. Otherwise, the parameter is not present.	Do not adopt
Multiple BSSID Configuration	As defined in frame format	As defined in 9.4.2.260	The value from the Multiple BSSID Configuration element. The parameter is present if dot11MultiBSSIDImplemented is true and the Multiple BSSID Configuration element was present in the Probe Response or Beacon frame from which the BSSDescription was determined. Otherwise, the parameter is not present.	Do not adopt

Change the following row in the untitled BSSDescriptionFromFD parameter table (introduced by "Each BSSDescriptionFromFDSet parameter is") in 6.3.3.3.2 as shown:

Name	Type	Valid range	Description
Reduced Neighbor Report	As defined in 9.4.2.170 <u>One or more Reduced Neighbor Report elements</u>	As defined in 9.4.2.170	The information of the Reduced Neighbor Information field of the received FILS Discovery frame related to APs in the neighborhood of the reporting AP or APs co-located with the reporting AP. This parameter is optional.

6.3.4 Synchronization

6.3.4.2 MLME-JOIN.request

6.3.4.2.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.4.2.2 as follows:

The primitive parameters are as follows:

```
MLME-JOIN.request(  
    ...,  
    HE Capabilities,  
    HE 6 GHz Band Capabilities,  
    VendorSpecificInfo  
)
```

Insert the following rows into the untitled table in 6.3.4.2.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element	As defined in 9.4.2.248	Specifies the parameters within the HE Capabilities element that are supported by the STA. The parameter is present if dot11HEOptionImplemented is true; otherwise, this parameter is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters within the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true; otherwise, this parameter is not present.

6.3.4.2.4 Effect of receipt

Insert the following at the end of 6.3.4.2.4:

If the MLME of an HE STA receives an MLME-JOIN.request primitive with a SelectedBSS parameter containing a Basic HE-MCS And NSS Set field in the HE Operation parameter that contains any unsupported <HE-MCS, NSS> tuple, then the MLME response in the resulting MLME-JOIN.confirm primitive shall contain a ResultCode parameter that is not set to the value SUCCESS.

6.3.7 Associate

6.3.7.2 MLME-ASSOCIATE.request

6.3.7.2.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.7.2.2 as follows:

The primitive parameters are as follows:

```
MLME-ASSOCIATE.request(  
    ...,  
    HE Capabilities,  
    HE 6 GHz Band Capabilities,
```

Channel Switch Timing,
UL MU Power Capabilities,
VendorSpecificInfo
)

Insert the following rows into the untitled table in 6.3.7.2.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element	As defined in 9.4.2.248	Specifies the parameters in the HE Capabilities element that are supported by the STA. The parameter is present if dot11HEOptionImplemented is true; otherwise, it is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters in the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true; otherwise, it is not present.
Channel Switch Timing	As defined in the Channel Switch Timing element	As defined in 9.4.2.63	Provides information regarding the channel switch timing. The parameter is optionally present if dot11HESubchannelSelectiveTransmissionImplemented is true; otherwise, it is not present.
UL MU Power Capabilities	As defined in UL MU Power Capabilities element	As defined in 9.4.2.264	Provides information regarding the UL MU power capabilities for the non-AP STA. The UL MU Power Capabilities element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.

6.3.7.3 MLME-ASSOCIATE.confirm

6.3.7.3.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.7.3.2 as follows:

The primitive parameters are as follows:

```
MLME-ASSOCIATE.confirm(
    ...,
    HE Capabilities,
    HE 6 GHz Band Capabilities,
    HE Operation,
    UORA Parameter Set,
    MU EDCA Parameter Set,
    Spatial Reuse Parameter Set,
    NDP Feedback Report Parameter Set,
    BSS Color Change Announcement,
    ESS Report,
    VendorSpecificInfo
)
```

Change the following row in the untitled table in 6.3.7.3.2 as follows:

Name	Type	Valid range	Description
TWT	TWT element	As defined in 9.4.2.199	Specifies the parameters in the TWT element. This parameter is optionally present if dot11TWTOptionActivated is true <u>and the TWT element is present in the Association Request frame that elicited the Association Response frame or the TWT Requester Support field in the HE Capabilities element in the Association Request frame that elicited the Association Response frame is 1</u> ; otherwise, it is not present.

Insert the following rows into the untitled table in 6.3.7.3.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element	As defined in 9.4.2.248	Specifies the parameters within the HE Capabilities element that are supported by the STA. The parameter is present if dot11HEOptionImplemented is true; otherwise, this parameter is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters within the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true; otherwise, this parameter is not present.
HE Operation	As defined in HE Operation element	As defined in 9.4.2.249	Provides additional information for operating the HE BSS. The parameter is present if dot11HEOptionImplemented is true; otherwise, it is not present.
UORA Parameter Set	As defined in UORA Parameter Set element	As defined in 9.4.2.250	Indicates the metrics of the OFDMA-based random access mechanism. The parameter is optionally present if dot11OFDMARandomAccessOptionImplemented is true; otherwise, it is not present.
MU EDCA Parameter Set	As defined in MU EDCA Parameter Set element	As defined in 9.4.2.251	Indicates MU EDCA parameters. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
Spatial Reuse Parameter Set	As defined in Spatial Reuse Parameter Set element	As defined in 9.4.2.252	Indicates parameters needed by STAs when performing OBSS PD-based spatial reuse. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
NDP Feedback Report Parameter Set	As defined in NDP Feedback Report Parameter Set element	As defined in 9.4.2.253	Indicates NDP Feedback Report parameter values. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.

Name	Type	Valid range	Description
BSS Color Change Announcement	As defined in BSS Color Change Announcement element	As defined in 9.4.2.254	Indicates information on BSS Color Change. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
ESS Report	As defined in ESS Report element	As defined in 9.4.2.256	Provides information on ESS to assist BSS transition. The parameter is optionally present.

6.3.7.4 MLME-ASSOCIATE.indication

6.3.7.4.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.7.4.2 as follows:

The primitive parameters are as follows:

```
MLME-ASSOCIATE.indication(
    ...,
    HE Capabilities,
    HE 6 GHz Band Capabilities,
    Channel Switch Timing,
    UL MU Power Capabilities,
    VendorSpecificInfo
)
```

Insert the following rows into the untitled table in 6.3.7.4.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element	As defined in 9.4.2.248	Specifies the parameters within the HE Capabilities element that are supported by the peer STA. The parameter is present if dot11HEOptionImplemented is true and the HE Capabilities element is present in the Association Request frame received from the STA; otherwise, this parameter is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters within the HE 6 GHz Band Capabilities element that are supported by the peer STA. The parameter is present if dot11HE6GOptionImplemented is true and the HE 6 GHz Band Capabilities element is present in the Association Request frame received from the STA; otherwise, this parameter is not present.
Channel Switch Timing	As defined in the Channel Switch Timing element	As defined in 9.4.2.63	Provides information regarding the channel switch timing. The parameter is optionally present if dot11HESubchannelSelectiveTransmissionImplemented is true and the Channel Switch Timing element is present in the Association Request frame received from the STA; otherwise, it is not present.

Name	Type	Valid range	Description
UL MU Power Capabilities	As defined in UL MU Power Capabilities element	As defined in 9.4.2.264	Provides information regarding the UL MU power capabilities for the non-AP STA. The UL MU Power Capabilities element is present if dot11HEOptionImplemented is true and the HE Capabilities element and UL MU Power Capabilities element is present in the Association Request frame received from the STA; otherwise, it is not present.

6.3.7.5 MLME-ASSOCIATE.response

6.3.7.5.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.7.5.2 as follows:

The primitive parameters are as follows:

```
MLME-ASSOCIATE.response(
    ...,
    HE Capabilities,
    HE 6 GHz Band Capabilities,
    HE Operation,
    UORA Parameter Set,
    MU EDCA Parameter Set,
    Spatial Reuse Parameter Set,
    NDP Feedback Report Parameter Set,
    BSS Color Change Announcement,
    ESS Report,
    VendorSpecificInfo
)
```

Change the following row in the untitled table in 6.3.7.5.2 as follows:

Name	Type	Valid range	Description
TWT	TWT element	As defined in 9.4.2.199	Specifies the parameters in the TWT element. This parameter is optionally present if dot11TWTOptionActivated is true <u>and the TWT element is present in the Association Request frame that elicited the Association Response frame or the TWT Requester Support field in the HE Capabilities element in the Association Request frame that elicited the Association Response frame is 1;</u> otherwise, it is not present.

Insert the following rows into the untitled table in 6.3.7.5.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element	As defined in 9.4.2.248	Specifies the parameters within the HE Capabilities element that are supported by the STA. The parameter is present if dot11HEOptionImplemented is true; otherwise, this parameter is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters within the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true; otherwise, this parameter is not present.
HE Operation	As defined in HE Operation element	As defined in 9.4.2.249	Provides additional information for operating the HE BSS. The parameter is present if dot11HEOptionImplemented is true; otherwise, it is not present.
UORA Parameter Set	As defined in UORA Parameter Set element	As defined in 9.4.2.250	Indicates the metrics of the OFDMA-based random access mechanism. The parameter is optionally present if dot11OFDMARandomAccessOptionImplemented is true; otherwise, it is not present.
MU EDCA Parameter Set	As defined in MU EDCA Parameter Set element	As defined in 9.4.2.251	Indicates MU EDCA parameters. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
Spatial Reuse Parameter Set	As defined in Spatial Reuse Parameter Set element	As defined in 9.4.2.252	Indicates parameters needed by STAs when performing OBSS PD-based spatial reuse. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
NDP Feedback Report Parameter Set	As defined in NDP Feedback Report Parameter Set element	As defined in 9.4.2.253	Indicates NDP Feedback Report parameter values. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
BSS Color Change Announcement	As defined in BSS Color Change Announcement element	As defined in 9.4.2.254	Indicates information on BSS Color Change. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
ESS Report	As defined in ESS Report element	As defined in 9.4.2.256	Provides information on ESS to assist BSS transition. The parameter is optionally present.

6.3.8 Reassociate

6.3.8.2 MLME-REASSOCIATE.request

6.3.8.2.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.8.2.2 as follows:

The primitive parameters are as follows:

```
MLME-REASSOCIATE.request(  
    ...,  
    HE Capabilities,  
    HE 6 GHz Band Capabilities,  
    Channel Switch Timing,  
    UL MU Power Capabilities,  
    VendorSpecificInfo  
)
```

Insert the following rows into the untitled table in 6.3.8.2.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element	As defined in 9.4.2.248	Specifies the parameters in the HE Capabilities element that are supported by the STA. The parameter is present if dot11HEOptionImplemented is true; otherwise, it is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters in the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true; otherwise, it is not present.
Channel Switch Timing	As defined in the Channel Switch Timing element	As defined in 9.4.2.63	Provides information regarding the channel switch timing. The parameter is optionally present if dot11HESubchannelSelectiveTransmissionImplemented is true; otherwise, it is not present.
UL MU Power Capabilities	As defined in UL MU Power Capabilities element	As defined in 9.4.2.264	Provides information regarding the UL MU power capabilities for the non-AP STA. The UL MU Power Capabilities element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.

6.3.8.3 MLME-REASSOCIATE.confirm

6.3.8.3.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.8.3.2 as follows:

The primitive parameters are as follows:

```
MLME-REASSOCIATE.confirm(
    ...,
    HE Capabilities,
    HE 6 GHz Band Capabilities,
    HE Operation,
    UORA Parameter Set,
    MU EDCA Parameter Set,
    Spatial Reuse Parameter Set,
    NDP Feedback Report Parameter Set,
    BSS Color Change Announcement,
    ESS Report,
    VendorSpecificInfo
)
```

Change the untitled table in 6.3.8.3.2 as follows:

Name	Type	Valid range	Description
TWT	TWT element	As defined in 9.4.2.199	Specifies the parameters in the TWT element. This parameter is optionally present if dot11TWTOptionActivated is true <u>and the TWT element is present in the Reassociation Request frame that elicited the Reassociation Response frame or the TWT Requester Support field in the HE Capabilities element in the Reassociation Request frame that elicited the Reassociation Response frame is 1</u> ; otherwise, it is not present.

Insert the following rows into the untitled table in 6.3.8.3.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element	As defined in 9.4.2.248	Specifies the parameters within the HE Capabilities element that are supported by the STA. The parameter is present if dot11HEOptionImplemented is true; otherwise, this parameter is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters within the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true; otherwise, this parameter is not present.
HE Operation	As defined in HE Operation element	As defined in 9.4.2.249	Provides additional information for operating the HE BSS. The parameter is present if dot11HEOptionImplemented is true; otherwise, it is not present.

Name	Type	Valid range	Description
UORA Parameter Set	As defined in UORA Parameter Set element	As defined in 9.4.2.250	Indicates the metrics of the OFDMA-based random access mechanism. The parameter is optionally present if dot11OFDMARandomAccessOptionImplemented is true; otherwise, it is not present.
MU EDCA Parameter Set	As defined in MU EDCA Parameter Set element	As defined in 9.4.2.251	Indicates MU EDCA parameters. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
Spatial Reuse Parameter Set	As defined in Spatial Reuse Parameter Set element	As defined in 9.4.2.252	Indicates parameters needed by STAs when performing OBSS PD-based spatial reuse. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
NDP Feedback Report Parameter Set	As defined in NDP Feedback Report Parameter Set element	As defined in 9.4.2.253	Indicates NDP Feedback Report parameter values. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
BSS Color Change Announcement	As defined in BSS Color Change Announcement element	As defined in 9.4.2.254	Indicates information on BSS Color Change. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
ESS Report	As defined in ESS Report element	As defined in 9.4.2.256	Provides information on ESS to assist BSS transition. The parameter is optionally present.

6.3.8.4 MLME-REASSOCIATE.indication

6.3.8.4.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.8.4.2 as follows:

The primitive parameters are as follows:

```
MLME-REASSOCIATE.indication(
    ...,
    HE Capabilities,
    HE 6 GHz Band Capabilities,
    Channel Switch Timing,
    UL MU Power Capabilities,
    VendorSpecificInfo
)
```

Insert the following rows into the untitled table in 6.3.8.4.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element.	As defined in 9.4.2.248	Specifies the parameters within the HE Capabilities element that are supported by the peer STA. The parameter is present if dot11HEOptionImplemented is true and the HE Capabilities element is present in the Reassociation Request frame received from the STA; otherwise, this parameter is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters within the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true and the HE 6 GHz Band Capabilities element is present in the Reassociation Request frame received from the STA; otherwise, this parameter is not present.
Channel Switch Timing	As defined in the Channel Switch Timing element	As defined in 9.4.2.63	Provides information regarding the channel switch timing. The parameter is optionally present if dot11HESubchannelSelectiveTransmissionImplemented is true and the Channel Switch Timing element is present in the Reassociation Request frame received from the STA; otherwise, it is not present.
UL MU Power Capabilities	As defined in UL MU Power Capabilities element	As defined in 9.4.2.264	Provides information regarding the UL MU power capabilities for the non-AP STA. The UL MU Power Capabilities element is present if dot11HEOptionImplemented is true and the HE Capabilities element and UL MU Power Capabilities element is present in the Association Request frame received from the STA; otherwise, it is not present.

6.3.8.5 MLME-REASSOCIATE.response

6.3.8.5.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.8.5.2 as follows:

The primitive parameters are as follows:

```
MLME-REASSOCIATE.response(
    ...,
    HE Capabilities,
    HE 6 GHz Band Capabilities,
    HE Operation,
    UORA Parameter Set,
    MU EDCA Parameter Set,
    Spatial Reuse Parameter Set,
    NDP Feedback Report Parameter Set,
```

BSS Color Change Announcement.
ESS Report.
VendorSpecificInfo
)

Change the following row in the untitled table in 6.3.8.5.2 as shown:

Name	Type	Valid range	Description
TWT	TWT element	As defined in 9.4.2.199	Specifies the parameters in the TWT element. This parameter is optionally present if dot11TWTOptionActivated is true <u>and the TWT element is present in the Reassociation Request frame that elicited the Reassociation Response frame or the TWT Requester Support field in the HE Capabilities element in the Reassociation Request frame that elicited the Reassociation Response frame is 1</u> ; otherwise, it is not present.

Insert the following rows into the untitled table in 6.3.8.5.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element	As defined in 9.4.2.248	Specifies the parameters within the HE Capabilities element that are supported by the STA. The parameter is present if dot11HEOptionImplemented is true; otherwise, this parameter is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters within the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true; otherwise, this parameter is not present.
HE Operation	As defined in HE Operation element	As defined in 9.4.2.249	Provides additional information for operating the HE BSS. The parameter is present if dot11HEOptionImplemented is true; otherwise, it is not present.
UORA Parameter Set	As defined in UORA Parameter Set element	As defined in 9.4.2.250	Indicates the metrics of the OFDMA-based random access mechanism. The parameter is optionally present if dot11OFDMARandomAccessOptionImplemented is true; otherwise, it is not present.
MU EDCA Parameter Set	As defined in MU EDCA Parameter Set element	As defined in 9.4.2.251	Indicates MU EDCA parameters. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
Spatial Reuse Parameter Set	As defined in Spatial Reuse Parameter Set element	As defined in 9.4.2.252	Indicates parameters needed by STAs when performing OBSS PD-based spatial reuse. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.

Name	Type	Valid range	Description
NDP Feedback Report Parameter Set	As defined in NDP Feedback Report Parameter Set element	As defined in 9.4.2.253	Indicates NDP Feedback Report parameter values. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
BSS Color Change Announcement	As defined in BSS Color Change Announcement element	As defined in 9.4.2.254	Indicates information on BSS Color Change. The parameter is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
ESS Report	As defined in ESS Report element	As defined in 9.4.2.256	Provides information on ESS to assist BSS transition. The parameter is optionally present.

6.3.11 Start

6.3.11.2 MLME-START.request

6.3.11.2.2 Semantics of the service primitive

Change the primitive parameter list in 6.3.11.2.2 as follows:

The primitive parameters are as follows:

```
MLME-START.request(
    ...,
    HE Capabilities,
    HE 6 GHz Band Capabilities,
    HE Operation,
    VendorSpecificInfo
)
```

Insert the following rows into the untitled table in 6.3.11.2.2 before the "VendorSpecificInfo" row:

Name	Type	Valid range	Description
HE Capabilities	As defined in HE Capabilities element.	As defined in 9.4.2.248	Specifies the parameters within the HE Capabilities element that are supported by the STA. The parameter is present if dot11HEOptionImplemented is true; otherwise, this parameter is not present.
HE 6 GHz Band Capabilities	As defined in HE 6 GHz Band Capabilities element	As defined in 9.4.2.263	Specifies the parameters within the HE 6 GHz Band Capabilities element that are supported by the STA. The parameter is present if dot11HE6GOptionImplemented is true; otherwise, this parameter is not present.
HE Operation	As defined in HE Operation element.	As defined in 9.4.2.249	Provides additional information for operating the HE BSS. The parameter is present if dot11HEOptionImplemented is true; otherwise, this parameter is not present.

6.3.11.2.4 Effect of receipt

Insert the following paragraph at the end of 6.3.11.2.4:

If the MLME of an HE STA receives an MLME-START.request primitive with a Basic HE-MCS And NSS Set field in the HE Operation parameter that contains any unsupported <HE-MCS, NSS> tuple, then the MLME response in the resulting MLME-START.confirm primitive shall not contain a ResultCode parameter that is set to SUCCESS.

Insert the following subclauses (6.3.118 through 6.3.118.5.4) after 6.3.117.2.4:

6.3.118 Quiet time period

6.3.118.1 Introduction

This mechanism supports quiet time period operation.

6.3.118.2 MLME-QTP.request

6.3.118.2.1 Function

This primitive requests a quiet time period for the quiet time period operation.

6.3.118.2.2 Semantics of the service primitive

The primitive parameters are as follows:

MLME-QTP.request (

Peer MAC Address,
 Dialog Token,
 Quiet Period Offset,
 Quiet Period Duration,
 Quiet Period Interval,
 Repetition Count,
 Service Specific Identifier

)

Name	Type	Valid range	Description
Peer MAC Address	MAC address	Any valid individually addressed MAC address	The address of the peer MAC entity to which the QTP Request frame is to be sent.
Dialog Token	Integer	0–65 535	The dialog token to identify the QTP Request frame.
Quiet Period Offset	Integer	0–255	Indicates the offset of the first QTP from the TBTT, expressed in TUs.
Quiet Period Duration	Integer	1–65 535	Indicates the duration of the QTP in units of 32 μ s.
Quiet Period Interval	Integer	1–255	Indicates the requested interval between the start of two consecutive QTPs, expressed in TUs.

Name	Type	Valid range	Description
Repetition Count	Integer	0–255	Indicates the number of requested QTPs. A repetition count equal to 0 indicates the setup time of the QTP is for a one time operation. Repetition count equal to 255 indicates the setup of the QTP is canceled.
Service Specific Identifier	Integer	0–65 535	Indicates an identifier assigned by a peer-to-peer application to identify frame exchanges using peer-to-peer links.

6.3.118.2.3 When generated

This primitive is generated by the SME to request that a QTP Request frame to be sent to its associated AP.

6.3.118.2.4 Effect of receipt

On receipt of this primitive, the MLME constructs and transmits a QTP Request frame.

6.3.118.3 MLME-QTP.indication

6.3.118.3.1 Function

This primitive indicates that a QTP Request frame has been received for the quiet time period operation.

6.3.118.3.2 Semantics of the service primitive

The primitive parameters are as follows:

MLME-QTP.indication (

Peer MAC Address,
 Dialog Token,
 Status Code,
 Quiet Period Offset,
 Quiet Period Duration,
 Quiet Period Interval,
 Repetition Count,
 Service Specific Identifier

)

Name	Type	Valid range	Description
Peer MAC Address	MAC address	Any valid individually addressed MAC address	The address of the peer MAC entity from which the QTP Request frame is received.
Dialog Token	Integer	0–255	The dialog token to identify the QTP Request frame.
Quiet Period Offset	Integer	0–255	Indicates the offset of the first QTP from the TBTT, expressed in TUs.
Quiet Period Duration	Integer	1–65 535	Indicates the duration of the QTP in units of 32 μ s.

Name	Type	Valid range	Description
Quiet Period Interval	Integer	1–255	Indicates the requested interval between the start of two consecutive QTPs, expressed in TUs.
Repetition Count	Integer	0–255	Indicates the number of requested QTPs. A repetition count equal to 0 indicates the setup time of the QTP is for a one time operation. Repetition count equal to 255 indicates the setup of the QTP is canceled.
Service Specific Identifier	Integer	0–65 535	Indicates an identifier assigned by a peer-to-peer application to identify frame exchanges using peer-to-peer links.

6.3.118.3.3 When generated

This primitive is generated by the MLME when a QTP Request frame is received.

6.3.118.3.4 Effect of receipt

On receipt of this primitive, the SME constructs and transmits a QTP Response frame.

6.3.118.4 MLME-QTP.response

6.3.118.4.1 Function

This primitive requests the transmission of quiet time period information to a peer entity, in response to a QTP Request frame for the quiet time period operation.

6.3.118.4.2 Semantics of the service primitive

The primitive parameters are as follows:

MLME-QTP.response (

Peer MAC Address,
 Dialog Token,
 Status Code,
 Quiet Period Offset,
 Quiet Period Duration,
 Quiet Period Interval,
 Repetition Count,
 Service Specific Identifier

)

Name	Type	Valid range	Description
Peer MAC Address	MAC address	Any valid individually addressed MAC address	The address of the peer MAC entity from which the QTP Request frame is received.
Dialog Token	Integer	0–255	The dialog token to identify the QTP Request frame.
Status Code	Integer	0–255	Indicates the status of a requested operation.

Name	Type	Valid range	Description
Quiet Period Offset	Integer	1–255	Indicates the offset of the first QTP from the TBTT, expressed in TUs.
Quiet Period Duration	Integer	1–65 535	Indicates the duration of the QTP in units of 32 μ s.
Quiet Period Interval	Integer	1–255	Indicates the responded interval between the start of two consecutive QTPs, expressed in TUs.
Repetition Count	Integer	0–255	Indicates the number of responded QTPs. A repetition count equal to 0 indicates the setup time of the QTP is for a one time operation. Repetition count equal to 255 indicates the setup of the QTP is canceled.
Service Specific Identifier	Integer	0–65 535	Indicates an identifier assigned by a peer-to-peer application to identify frame exchanges using peer-to-peer links.

6.3.118.4.3 When generated

This primitive is generated by the SME to request that a QTP Response frame be sent to a peer entity as a response to an MLME-QTP.indication primitive.

6.3.118.4.4 Effect of receipt

On receipt of this primitive, the SME constructs and transmits a QTP Response frame.

6.3.118.5 MLME-QTP.confirm

6.3.118.5.1 Function

This primitive reports the result of a QTP request to send a QTP Response frame for the quiet time period operation.

6.3.118.5.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-QTP.confirm (
    Peer MAC Address,
    Dialog Token,
    Status Code,
    Quiet Period Offset,
    Quiet Period Duration,
    Quiet Period Interval,
    Repetition Count,
    Service Specific Identifier
)
```

Name	Type	Valid range	Description
Peer MAC Address	MAC address	Any valid individually addressed MAC address	The address of the peer MAC entity from which the QTP Response frame is received.
Dialog Token	Integer	0–255	The dialog token to identify the QTP Response frame.
Status Code	Integer	0–255	Indicates the status of a requested operation.
Quiet Period Offset	Integer	1–255	Indicates the offset of the first QTP from the TBTT, expressed in TUs.
Quiet Period Duration	Integer	1–65 535	Indicates the duration of the QTP in units of 32 μ s.
Quiet Period Interval	Integer	1–255	Indicates the responded interval between the start of two consecutive QTPs, expressed in TUs.
Repetition Count	Integer	0–255	Indicates the number of responded QTPs. A repetition count equal to 0 indicates the setup time of the QTP is for a one time operation. Repetition count equal to 255 indicates the setup of the QTP is canceled.
Service Specific Identifier	Integer	0–65 535	Indicates an identifier assigned by a peer-to-peer application to identify frame exchanges using peer-to-peer links.

6.3.118.5.3 When generated

This primitive is generated by the MLME when the STA receives a QTP Response frame from the AP.

6.3.118.5.4 Effect of receipt

The SME is notified of the results of the QTP Request frame.

6.5 PLME SAP interface

6.5.6 PLME-TXTIME.confirm

6.5.6.2 Semantics of the service primitive

Change the second and third paragraphs of 6.5.6.2 as follows:

The TXTIME represents the time, in microseconds, required to transmit the PPDU described in the corresponding PLME-TXTIME.request primitive. If the calculated time includes a fractional microsecond and the TXVECTOR parameter FORMAT in the corresponding PLME-TXTIME.request primitive is not HE_SU, HE_MU, HE_TB, or HE_ER_SU, a non-DMG STA rounds the TXTIME value to the next higher integer. A non-DMG STA does not round the TXTIME value up or down if the TXVECTOR parameter FORMAT in the corresponding PLME-TXTIME.request primitive is HE_SU, HE_MU, HE_TB, or HE_ER_SU. A DMG STA does not round the TXTIME value up or down (see 20.11.3).

The parameter PSDU_LENGTH[] is an array of size TXVECTOR parameter NUM_USERS if the TXVECTOR parameter FORMAT is VHT. If the TXVECTOR parameter FORMAT is HE_SU, HE_ER, or HE_TB, the size of the parameter PSDU_LENGTH[] array is always 1. If the TXVECTOR parameter FORMAT is HE_MU, the size of the parameter PSDU_LENGTH[] array is determined by TXVECTOR parameters RU_ALLOCATION and STA_ID. Each value in the array indicates the number of octets required to fill the PPDU for the user represented by that array index. The parameter is present only when the TXVECTOR parameter FORMAT is VHT, HE_SU, HE_ER_SU, HE_MU, or HE_TB.

8. PHY service specification

8.3 Detailed PHY service specifications

8.3.4 Basic service and options

8.3.4.2 PHY-SAP inter-(sub)layer service primitives

Insert the following row at the end of Table 8-2:

Table 8-2—PHY SAP inter-(sub)layer service primitives

Primitive	Request	Indication	Confirm
PHY-TRIGGER	X		

8.3.4.3 PHY SAP service primitives parameters

Change Table 8-3 as follows (not all rows are shown):

Table 8-3—PHY SAP service primitive parameters

Parameter	Associated primitive	Value
...		
STATE	PH-CCA.indication	(BUSY, [channel-list]) (BUSY[, channel-list[, per20bitmap]]) (IDLE, [per20bitmap])
...		
<u>TRIGVECTOR</u>	<u>PHY-TRIGGER.request</u>	<u>A set of parameters</u>

8.3.4.4 Vector descriptions

Change the second and third paragraphs of 8.3.4.4 as follows:

The ~~Clause 19—HT~~ PHY TXVECTOR and RXVECTOR contain additional parameters related to the ~~operation of the Clause 19—HT~~ PHY modes of operation as described in 19.2. In certain modes of operation, the DATARATE parameter is replaced by MCS, CH_BANDWIDTH, and GI_TYPE values. The mapping from these values to data rate is defined in 19.5.

The ~~Clause 21—VHT~~ PHY TXVECTOR and RXVECTOR contain additional parameters related to the ~~operation of the Clause 21—VHT~~ PHY modes of operation as described in 21.2. In certain modes of operation, the DATARATE parameter is replaced by MCS, CH_BANDWIDTH, NUM_STS, STBC, and GI_TYPE values. The mapping from these values to data rate is defined 21.5, where VHT-MCS is MCS and N_{SS} is $NUM_STS / (STBC + 1)$.

Insert the following paragraphs at the end of 8.3.4.4:

The HE PHY TXVECTOR and RXVECTOR contain additional parameters related to the HE PHY modes of operation as described in 27.2. In certain modes of operation, the DATARATE parameter is replaced by

MCS, CH_BANDWIDTH, RU_ALLOCATION, NUM_STS, STBC, GI_TYPE, and DCM values. The mapping from these values to data rate is defined in 27.5, where HE-MCS is MCS and N_{SS} is $NUM_STS / (STBC + 1)$.

The HE PHY TRIGVECTOR contains parameters for UL MU operation (see Table 27-2).

8.3.5 PHY SAP detailed service specification

8.3.5.2 PHY-DATA.request

8.3.5.2.2 Semantics of the service primitive

Change 8.3.5.2.2 as follows:

The primitive provides the following parameters:

```
PHY-DATA.request(  
    DATA_  
    USER_INDEX_  
    STA_INDEX  
)
```

The DATA parameter is an octet of value X'00' to X'FF'.

The USER_INDEX parameter (typically identified as u for a VHT STA; for MU usage, see the NOTE for MU usage at the end of Table 21-1) is present for a VHT MU PPDU and indicates the index of the user in the TXVECTOR to which the accompanying DATA octet applies; otherwise, this parameter is not present.

The STA_INDEX parameter (identified as the STA_ID parameter; see STA_ID parameter in Table 27-1 and 26.11.1) is present for an HE MU PPDU; otherwise, this parameter is not present. If the TXVECTOR parameter UPLINK_FLAG is 0, this parameter indicates the STA or group of STAs that is the recipient of an RU to which the accompanying DATA octet applies. If the TXVECTOR parameter UPLINK_FLAG is 1, this parameter indicates the STA that is the transmitter of an RU to which the accompanying DATA octet applies.

8.3.5.3 PHY-DATA.indication

8.3.5.3.2 Semantics of the service primitive

Change 8.3.5.3.2 as follows:

The primitive provides the following parameter:

```
PHY-DATA.indication(  
    DATA_  
    STA_INDEX  
)
```

The DATA parameter is an octet of value X'00' to X'FF'.

The STA_INDEX parameter (identified as an AID of the transmitter of the DATA contained in an HE TB PPDU; see STA_ID parameter in Table 27-1) is present for an HE TB PPDU and indicates the STA of an RU from which the accompanying DATA octet applies; otherwise, this parameter is not present.

8.3.5.10 PHY-CCARESET.request

8.3.5.10.3 When generated

Insert the following paragraph at the end of 8.3.5.10.3:

This primitive is also generated by the MAC sublayer for the local PHY entity when the spatial reuse conditions defined in 26.10 are met.

8.3.5.12 PHY-CCA.indication

8.3.5.12.2 Semantics of the service primitive

Change the primitive parameter list in 8.3.5.12.2 as follows:

The primitive provides the following parameters:

```
PHY-CCA.indication(
    STATE,
    IPI-REPORT,
    channel-list,
    per20bitmap
)
```

Change the following rows in Table 8-5 as shown:

Table 8-5—The channel-list parameter entries

Channel-list parameter entry	Meaning
primary	In an HT STA that is not <u>neither a VHT STA nor an HE STA</u> , indicates that the primary 20 MHz channel is busy according to the rules specified in 19.3.19.5.2. In a VHT STA <u>that is not an HE STA</u> , indicates that the primary 20 MHz channel is busy according to the rules specified in 21.3.18.5.3. In a TVHT STA, indicates that the primary channel is busy according to the rules specified in 22.3.18.6.3. <u>In an HE STA, indicates that the primary 20 MHz channel is busy according to the rules specified in 27.3.20.6.3.</u>
secondary	In an HT STA that is not <u>neither a VHT STA nor an HE STA</u> , indicates that the secondary channel is busy according to the rules specified in 19.3.19.5.5. In a VHT STA <u>that is not an HE STA</u> , indicates that the secondary 20 MHz channel is busy according to the rules specified in 21.3.18.5.4. In a TVHT STA, indicates that the secondary channel is busy according to the rules specified in 22.3.18.6.4. <u>In an HE STA, indicates that the secondary 20 MHz channel is busy according to the rules specified in 27.3.20.6.4.</u>
secondary40	<u>In a VHT STA that is not an HE STA, indicates that the secondary 40 MHz channel is busy according to the rules specified in 21.3.18.5.4.</u> In a TVHT STA, indicates that the secondary TVHT_2W channel is busy according to the rules specified in 22.3.18.6.4. <u>In an HE STA, indicates that the secondary 40 MHz channel is busy according to the rules specified in 27.3.20.6.4.</u>

Table 8-5—The channel-list parameter entries (continued)

Channel-list parameter entry	Meaning
secondary80	<u>In a VHT STA that is not an HE STA, indicates that the secondary 80 MHz channel is busy according to the rules specified in 21.3.18.5.4.</u> <u>In an HE STA, indicates that the secondary 80 MHz channel is busy according to the rules specified in 27.3.20.6.4.</u>

Insert the following paragraph at the end of 8.3.5.12.2:

If the STA is an HE STA with an operating channel width greater than 20 MHz, then the per20bitmap parameter is present; otherwise, it is absent. If present, the per20bitmap parameter is a bitmap where each bit represents the busy/idle status of a 20 MHz subchannel in the operating channel width as defined in 27.3.20.6.5.

8.3.5.12.3 When generated

Change the first paragraph and note as follows:

For Clause 15 to Clause 20 PHYs, this primitive is generated within aCCATime of the occurrence of a change in the status of the primary channel from channel idle to channel busy or from channel busy to channel idle or when the entries of the channel-list parameter change. For Clause 21 and Clause 22 PHYs, this primitive is generated when the status of the channel(s) changes from channel idle to channel busy or from channel busy to channel idle or when the entries of the channel-list parameter change. This includes the period of time when the PHY is receiving data. For Clause 27 PHYs, this primitive is generated when the status of the channel(s) changes from channel idle to channel busy or from channel busy to channel idle, when the entries of the channel-list parameter change, or when the per20bitmap parameter changes. The timing of PHY-CCA.indication primitives related to transitions on secondary channel(s) is PHY specific. Refer to specific PHY clauses for details about CCA behavior for a given PHY.

~~NOTE For the VHT PHY, the timing information is omitted here and is defined in 21.3.18.5.~~

Insert the following subclauses (8.3.5.16 through 8.3.5.16.4) after 8.3.5.15.4:

8.3.5.16 PHY-TRIGGER.request

8.3.5.16.1 Function

This primitive is a request by the MAC sublayer to the local PHY entity to set parameters for the receipt of HE TB PPDU.

8.3.5.16.2 Semantics of the service primitive

The primitive provides the following parameter:

```
PHY-TRIGGER.request (
    TRIGVECTOR
)
```

The TRIGVECTOR parameter provides the PHY with the information needed to demodulate of the expected HE TB PPDUs.

8.3.5.16.3 When generated

This primitive is issued by the MAC sublayer to the PHY entity after issuing PHY-TXSTART.request and before receiving HE TB PPDU.

8.3.5.16.4 Effect of receipt

On receipt of this primitive, the PHY entity configures the receiver module for the expected arrival of the HE TB PPDU.

9. Frame formats

9.2 MAC frame formats

9.2.4 Frame fields

9.2.4.1 Frame Control field

9.2.4.1.3 Type and Subtype subfields

Change Table 9-1 as follows (not all rows are shown):

Table 9-1—Valid type and subtype combinations

Type value B3 B2	Type description	Subtype value B7 B6 B5 B4	Subtype description
...			
01	Control	0000- 0010 0001	Reserved
<u>01</u>	<u>Control</u>	<u>0010</u>	<u>Trigger</u>
...			
01	Control	0101	VHT/ <u>HE</u> NDP Announcement
...			

9.2.4.1.8 More Data subfield

Change the third and fourth paragraphs of 9.2.4.1.8 as follows (including splitting each paragraph into two paragraphs):

An AP optionally sets the More Data subfield to 1 in Ack frames sent to a non-DMG and non-SIG non-HE STA and in Ack, BlockAck, and Multi-STA BlockAck frames sent to an HE STA. An HE AP indicates that it supports setting the More Data subfield to 1 in these control response frames by setting the More Data Ack subfield to 1 in the QoS Info field of elements it includes in frames transmitted to the STA.

The AP can set the More Data subfield to 1 to indicate that it has a pending transmission for the STA if it from which it has received a frame that contains a QoS Capability element QoS Info field in which the More Data Ack subfield is equal to 1 from the STA and that has one or more ACs that are delivery enabled and that is in PS mode to indicate that the AP has a pending transmission for the STA. one of the following conditions is true:

- The STA is in PS mode and has one or more ACs that are delivery enabled (see 11.2.3.6).
- The STA is in PS mode and is a TWT requester or a TWT scheduled STA (see 26.8).

A TDLS peer STA optionally sets the More Data subfield to 1 in Ack frames sent to a non-HE STA and in Ack, BlockAck, and Multi-STA BlockAck frames sent to an HE STA. An HE TDLS peer STA indicates that it supports setting the More Data subfield to 1 in these control response frames by setting the More Data Ack subfield to 1 in the QoS Info field of the QoS Capability element it includes in frames transmitted to the STA.

The TDLS peer STA can set the More Data subfield to 1 to indicate that it has a pending transmission for the STA if it has received from the STA a TDLS Setup Request frame or TDLS Setup Response frame that has TDLS peer PSM enabled and that has the More Data Ack subfield equal to 1 in the QoS Info field of the QoS Capability element of its transmitted TDLS Setup Request frame or TDLS Setup Response frame to indicate that it has a pending transmission for the STA, and one of the following conditions is true:

- The STA has TDLS peer PSM enabled (see 11.2.3.6).
- The STA is in PS mode and is a TWT requester or a TWT scheduled STA (see 26.8).

9.2.4.1.10 +HTC subfield

Change 9.2.4.1.10 as follows:

The +HTC subfield is set as follows:

- It is set to 1 in a QoS Data or Management frame transmitted with the TXVECTOR parameter FORMAT set to HT_GF, HT_MF, VHT, or S1G to indicate that the frame contains an HT Control field.
- It is set to 1 in an RTS frame transmitted with the TXVECTOR parameter FORMAT set to S1G to indicate that the intended recipient of the frame has permission to extend the TXOP as described in 10.54.5.4.
- It is set to 1 in a QoS Data or Management frame transmitted by a CMMG STA to indicate that the frame contains a CMMG variant HT Control field.
- It is set to 1 in a QoS Data, QoS Null, or Management frame transmitted by an HE STA to another HE STA to indicate that the frame contains an HT Control field.

Otherwise, the +HTC subfield is set to 0.

NOTE—The +HTC subfield is always set to 0 for frames transmitted by a DMG STA.

9.2.4.3 Address fields

9.2.4.3.5 DA field

Change 9.2.4.3.5 as follows:

The DA field contains an IEEE MAC individual or group address that identifies the MAC entity or entities intended as the final recipient(s) of the MSDU (or fragment thereof) or A-MSDU (or fragment thereof), as defined in 9.3.2.1, contained in the frame body field.

9.2.4.3.6 SA field

Change 9.2.4.3.6 as follows:

The SA field contains an IEEE MAC individual address that identifies the MAC entity from which the transfer of the MSDU (or fragment thereof) or A-MSDU (or fragment thereof), as defined in 9.3.2.1, contained in the frame body field was initiated. The Individual/Group bit is always transmitted as a 0 in the source address.

9.2.4.5 QoS Control field

9.2.4.5.2 TID subfield

Change the first paragraph in 9.2.4.5.2 as follows:

The TID subfield identifies the TC or TS to which the corresponding MSDU (or fragment thereof) or A-MSDU (or fragment thereof) in the Frame Body field belongs. The TID subfield also identifies the TC or TS of traffic for which a TXOP is being requested, through the setting of TXOP duration requested or queue size. The encoding of the TID subfield depends on the access policy (see 9.4.2.29) and is shown in Table 9-12. Additional information on the interpretation of the contents of this field appears in 5.1.1.3.

9.2.4.5.4 Ack Policy Indicator subfield

Change Table 9-13 as shown (not all rows are shown):

Table 9-13—Ack policy

Ack policy	Ack Policy Indicator subfield		Other conditions	Meaning
	Bit 0	Bit 1		
...				
No Explicit Acknowledgment	0	1	Bit 6 of the Frame Control field (see 9.2.4.1.3) is equal to 1 <u>and the frame is not carried in an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU that contains a frame that solicits a response in an HE TB PPDU</u>	There might be a response frame to the frame that is received, but it is neither the Ack frame nor any Data frame of subtype +CF-Ack. This ack policy is used for QoS CF-Poll and QoS CF-Ack +CF-Poll Data frames. NOTE—Bit 6 of the Frame Control field (see 9.2.4.1.3) indicates the absence of a Frame Body field in a QoS Data frame. When If equal to 1, the QoS Data frame contains no Frame Body field, and any response is generated in response to a QoS CF-Poll or QoS CF-Ack +CF-Poll frame, but does not signify an acknowledgment of data.
PSMP Ack	0	1	Bit 6 of the Frame Control field (see 9.2.4.1.3) is equal to 0 <u>and the frame is not carried in an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU that contains a frame that solicits a response in an HE TB PPDU</u>	The acknowledgment for a frame indicating PSMP Ack when it appears in a PSMP downlink transmission time (PSMP-DTT) is to be received in a later PSMP uplink transmission time (PSMP-UTT). The acknowledgment for a frame indicating PSMP Ack when it appears in a PSMP-UTT is to be received in a later PSMP-DTT. See 10.30.2.7.
<u>HETP Ack</u>	<u>0</u>	<u>1</u>	<u>The frame is carried in an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU that contains a frame that solicits a response in an HE TB PPDU</u>	<u>The addressed recipient returns an Ack, Compressed BlockAck, or Multi-STA BlockAck frame carried in an HE TB PPDU a SIFS after the PPDU, subject to reception of a triggering frame in the PPDU, as defined in 10.3.2.13.2 and 26.5.2.</u>
...				

9.2.4.5.6 Queue Size subfield

Change 9.2.4.5.6 as follows:

The Queue Size subfield is an 8-bit field that indicates the amount of buffered traffic for a given TC or TS at the non-AP non-HE STA sending the this frame that contains this subfield and the amount of buffered traffic for a given TC or TS at the non-AP HE STA for transmission to the HE STA identified by the receiver address of the frame that contains this subfield. The Queue Size subfield is present in QoS Data frames with bit 4 of the QoS Control field set to 1 sent by a non-AP STA~~s~~ and in QoS Null frames with bit 4 of the QoS Control field set to 1 sent by a non-AP HE STA. The AP might use information contained in the Queue Size subfield to determine the TXOP duration assigned to the STA or to determine the UL resources assigned to the non-AP HE STA (see 26.5.2).

In a frame sent by a non-HE STA or sent to a non-HE STA, the following apply:

- ~~The Queue Size subfield is set to the~~ The queue size value is the approximate total size, rounded up to the nearest multiple of 256 octets and expressed in units of 256 octets, of all MSDUs and A-MSDUs buffered at the STA (excluding the MSDU or A-MSDU of the present QoS Data frame) in the delivery queue used for MSDUs and A-MSDUs with TID values equal to the value in the TID subfield of this QoS Control field.
- ~~The Queue Size subfield is set to~~ A queue size value of 0 is used solely to indicate the absence of any buffered traffic in the queue used for the specified TID.
- ~~The Queue Size subfield is set to~~ A queue size value of 254 is used for all sizes greater than 64 768 octets.
- ~~The Queue Size subfield is set to~~ A queue size value of 255 is used to indicate an unspecified or unknown size.

Insert the following text, Table 9-13a, and Equation (9-a) at the end of 9.2.4.5.6:

In a frame sent by a non-AP HE STA to an HE STA, the remainder of the subclause applies.

The queue size, QS , is the approximate total size in octets, of all MSDUs and A-MSDUs buffered at the STA (including the MSDUs or A-MSDUs in the same PSDU as the frame containing the Queue Size subfield) in the delivery queue used for MSDUs and A-MSDUs with TID values equal to the value in the TID subfield of this QoS Control field.

NOTE 1—The queue size is based on data received by the STA at the MAC SAP (MA-UNITDATA.request).

NOTE 2—Buffered MSDUs are those that have been received in an MA-UNITDATA.request but that have not been successfully transmitted and have not been discarded.

The Queue Size subfield consists of a Scaling Factor subfield in B14–B15 of the QoS Control subfield and an unscaled value, UV , in B8–B13 of the QoS Control subfield. The Scaling Factor subfield provides the scaling factor, SF , with an encoding that is shown in Table 9-24f (in 9.2.4.6a.4). A non-AP HE STA sets the Queue Size subfield in a QoS frame it transmits as shown in Table 9-13a.

Table 9-13a—Queue Size subfield encoding by a non-AP HE STA

Queue Size subfields		Queue Size, QS
Scaling Factor	UV	
0	0	0
0	$\text{Ceil}(QS, 16) / 16$	$0 < QS \leq 1008$
1	0	$1008 < QS \leq 1024$
1	$\text{Ceil}(QS - 1024, 256) / 256$	$1024 < QS \leq 17\,152$
2	0	$17\,152 < QS \leq 17\,408$
2	$\text{Ceil}(QS - 17\,408, 2048) / 2048$	$17\,408 < QS \leq 146\,432$
3	0	$146\,432 < QS \leq 148\,480$
3	$\text{Ceil}(QS - 148\,480, 32\,768) / 32\,768$	$148\,480 < QS \leq 2\,147\,328$
3	62	$QS > 2\,147\,328$
3	63	Unspecified or unknown

An HE STA obtains the queue size, QS , from a received QoS Control field, which contains a scaling factor and an unscaled value, as follows:

$$QS = \begin{cases} 16 \times UV, & \text{if the Scaling Factor subfield is 0} \\ 1024 + 256 \times UV, & \text{if the Scaling Factor subfield is 1} \\ 17\,408 + 2048 \times UV, & \text{if the Scaling Factor subfield is 2} \\ 148\,480 + 32\,768 \times UV, & \text{if the Scaling Factor subfield is 3 and UV subfield is } < 62 \\ > 2\,147\,328, & \text{if the Scaling Factor subfield is 3 and UV subfield is 62} \\ \text{Unspecified or Unknown}, & \text{if the Scaling Factor subfield is 3 and UV subfield is 63} \end{cases} \quad (9\text{-a})$$

The queue size value of QoS Data frames containing fragments might remain constant in all fragments even if the amount of queued traffic changes as successive fragments are transmitted (see 10.23.3.5.1). If the QoS Data frames containing fragments are carried in an A-MPDU, the queue size values of the MPDU containing the fragments are set according to the rules in 10.12.1.

9.2.4.6 HT Control field

9.2.4.6.1 General

Change the first through fourth paragraphs in 9.2.4.6.1 as follows (including deleting Figure 9-11 and inserting Table 9-13b):

The HT Control field is always present in a Control Wrapper frame and is present in QoS Data, QoS Null, and Management frames as determined by the +HTC subfield of the Frame Control field as defined in 9.2.4.1.10.

NOTE 1—The only control frame subtype for which HT Control field is present is the Control Wrapper frame. A Control frame that is described as +HTC (e.g., an RTS+HTC, CTS+HTC, BlockAck+HTC or BlockAckReq+HTC frame) implies the use of the Control Wrapper frame to carry that Control frame.

The format of the HT Control field transmitted by a non-CMMG STA is shown in Figure 9-11 defined in Table 9-13b.

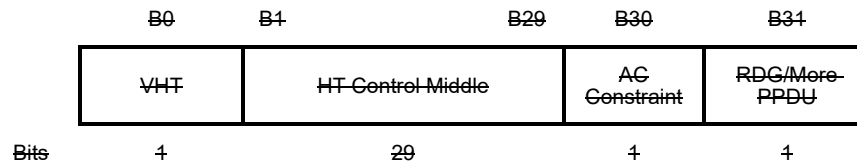


Figure 9-11—Non-CMMG variant HT Control field format

Table 9-13b—HT Control field format

Variant	B0	B1	B2–B29	B30	B31
HT	0		HT Control Middle	AC Constraint	RDG/More PPDU
VHT	1	0	VHT Control Middle	AC Constraint	RDG/More PPDU
HE	1	1	A-Control		

The HT Control field transmitted by a non-CMMG STA has ~~two forms, three variants:~~ the HT variant, ~~and~~ the VHT variant, ~~and the HE variant.~~ The variant formats are differentiated by the values of B0 and B1 as defined in Table 9-13b. ~~The two forms differ in the format of the HT Control Middle subfield, described in 9.2.4.6.2 for the HT variant and in 9.2.4.6.3 for the VHT variant and in the value of the VHT subfield.~~

The HT Control Middle subfield is defined in 9.2.4.6.2, and the VHT Control Middle subfield is defined in 9.2.4.6.3. The A-Control subfield is defined in 9.2.4.6.3a.

~~The VHT subfield of the HT Control field indicates whether the HT Control Middle subfield is the VHT Variant or the HT Variant. The VHT subfield is set to 1 to indicate that the HT Control Middle subfield is the VHT Variant and is set to 0 to indicate that the HT Control Middle subfield is the HT Variant.~~

9.2.4.6.3 VHT variant

Change the first paragraph of 9.2.4.6.3 and Figure 9-16 as follows:

In a non-SIG STA, the format of the VHT Control Middle subfield of the VHT variant HT Control field is shown in Figure 9-16.

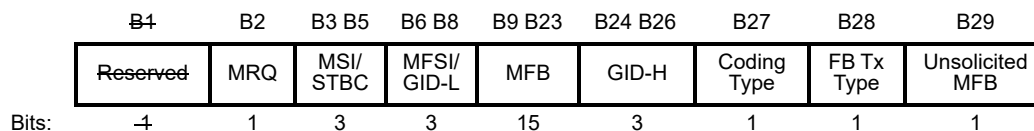


Figure 9-16—VHT Control Middle subfield of the VHT variant HT Control field format

Insert the following subclause (9.2.4.6.3a, including Figure 9-19a, Figure 9-19b, and Table 9-22a) after 9.2.4.6.3:

9.2.4.6.3a HE variant

The format of the A-Control subfield of the HE variant HT Control field is shown in Figure 9-19a.

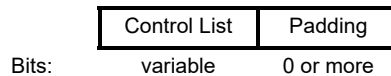


Figure 9-19a—A-Control subfield of the HE variant HT Control field format

The A-Control subfield is 30 bits in length.

The Control List subfield contains one or more Control subfields. The format of each Control subfield is shown in Figure 9-19b.

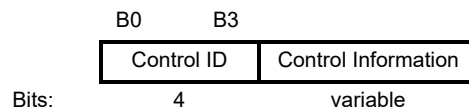


Figure 9-19b—Control subfield format

The Control ID subfield indicates the type of information carried in the Control Information subfield. The length of the Control Information subfield is fixed for each value of the Control ID subfield that is not reserved. The values of the Control ID subfield and the associated length of the Control Information subfield are defined in Table 9-22a.

Table 9-22a—Control ID subfield values

Control ID value	Meaning	Length of the Control Information subfield (bits)	Content of the Control Information subfield
0	Triggered response scheduling (TRS)	26	See 9.2.4.6a.1.
1	Operating mode (OM)	12	See 9.2.4.6a.2.
2	HE link adaptation (HLA)	26	See 9.2.4.6a.3.
3	Buffer status report (BSR)	26	See 9.2.4.6a.4.
4	UL power headroom (UPH)	8	See 9.2.4.6a.5.
5	Bandwidth query report (BQR)	10	See 9.2.4.6a.6.
6	Command and status (CAS)	8	See 9.2.4.6a.7.
7–14	Reserved		
15	Ones need expansion surely (ONES)	26	Set to all 1s.

The Padding subfield, if present, follows the last Control subfield and is set to a sequence of zeros so that the length of the A-Control subfield carried in the HT Control field is 30 bits.

Insert the following subclauses (9.2.4.6a through 9.2.4.6a.7, including Figure 9-22a through Figure 9-22h and Table 9-24a through Table 9-24f) after 9.2.4.6.4:

9.2.4.6a Control subfield variants of an A-Control subfield

9.2.4.6a.1 TRS Control

The Control Information subfield in a TRS Control subfield contains triggered response scheduling (TRS) information for soliciting an HE TB PPDU that follows an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU carrying the Control subfield (see 26.5.2.2). See 26.5.2.4 for details on allowed content in an A-MPDU carried in an HE TB PPDU. The format of the subfield is shown in Figure 9-22a.

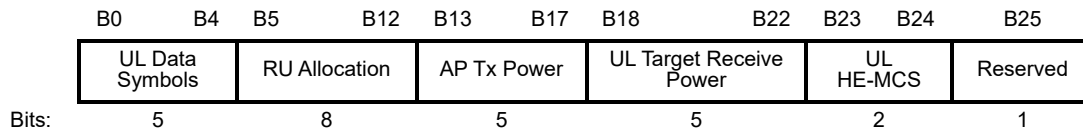


Figure 9-22a—Control Information subfield format in a TRS Control subfield

NOTE 1—A TRS Control subfield is not included in a non-HE PPDU.

The UL Data Symbols subfield indicates the number of OFDM symbols in the Data field of the HE TB PPDU response and is set to the number of OFDM symbols minus 1.

The RU Allocation subfield indicates the resource unit (RU) assigned for transmitting the HE TB PPDU response and the encoding is defined in 9.3.1.22.1.

The AP Tx Power subfield indicates the AP's combined transmit power at the transmit antenna connector of all the antennas used to transmit the triggering PPDU in units of dBm/20 MHz. The transmit power in dBm/20 MHz, P_{TX} , is calculated as $P_{TX} = -20 + 2 \times F_{val}$, where F_{val} is the value of the AP Tx Power subfield, except for the value 31, which is reserved.

The UL Target Receive Power subfield indicates the expected receive signal power, measured at the AP's antenna connector and averaged over the antennas, for the HE portion of the HE TB PPDU transmitted on the assigned RU as defined in Table 9-24a.

Table 9-24a—UL Target Receive Power subfield in TRS Control field

UL Target Receive Power subfield	Description
0–30	The expected receive signal power, in units of dBm, is $Target_{pwr} = -90 + 2 \times F_{val}$, where F_{val} is the subfield value
31	The STA transmits the HE TB PPDU at the STA's maximum transmit power for the assigned HE-MCS. NOTE—The expected receive signal power is then the STA's maximum transmit power for the assigned HE-MCS minus the path loss.

NOTE 2—A STA might transmit the HE TB PPDU at a transmit power that is below the transmit power needed to achieve the expected receive signal power due to hardware or regulatory limits (see 27.3.15.2).

The UL HE-MCS subfield indicates the HE-MCS, in the range HE-MCS 0 to 3, to be used by the receiving STA for the HE TB PPDU is set to the HE-MCS index (see 27.5).

9.2.4.6a.2 OM Control

The Control Information subfield in an OM Control subfield contains information related to the operating mode (OM) change of the STA transmitting the frame containing this information (see 26.9). The format of the subfield is shown in Figure 9-22b.

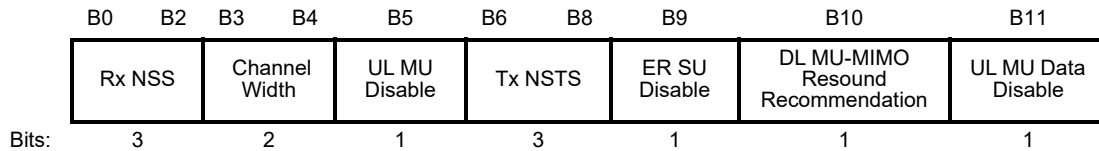


Figure 9-22b—Control Information subfield format in an OM Control subfield

If the operating channel width of the STA is greater than 80 MHz, then the Rx NSS subfield indicates the maximum number of spatial streams, N_{SS} , that the STA supports in reception for PPDU bandwidths less than or equal to 80 MHz and is set to $N_{SS} - 1$. If the operating channel width of the STA is less than or equal to 80 MHz, then the Rx NSS subfield indicates the maximum number of spatial streams, N_{SS} , that the STA supports in reception and is set to $N_{SS} - 1$.

If the operating channel width of the STA is greater than 80 MHz, then the maximum number of spatial streams that the STA supports in reception for PPDU bandwidths greater than 80 MHz is defined in 26.9.

The Channel Width subfield indicates the operating channel width supported by the STA for both reception and transmission. It is set to 0 for 20 MHz, 1 for primary 40 MHz, 2 for primary 80 MHz, and 3 for 160 MHz and 80+80 MHz. The value 0 indicates a primary 20 MHz, unless the STA is an HE SST STA in which case it indicates any of the negotiated 20 MHz subchannels of the SST operation (see 26.8.7).

The allowed UL MU operations and frame types that can be transmitted as a response to a triggering frame are determined by the UL MU Disable subfield, UL MU Data Disable subfield and the recipient's setting of the OM Control UL MU Data Disable RX Support subfield in the HE Capabilities element, as indicated in Table 9-24b.

If the OM Control field is transmitted by an HE AP, then the UL MU Disable and UL MU Data Disable subfields are reserved.

A non-AP STA sets the Tx NSTS subfield to $N_{STS} - 1$, where N_{STS} is the maximum number of space-time streams that the non-AP STA supports in transmission. If the OM Control field is transmitted by an HE AP, then the Tx NSTS subfield is reserved.

A non-AP HE STA sets the ER SU Disable subfield to 1 to indicate that 242-tone HE ER SU PPDU reception is disabled and to 0 to indicate that 242-tone HE ER SU PPDU reception is enabled. If the OM Control field is transmitted by an HE AP, then the ER SU Disable subfield is reserved.

A non-AP HE STA sets the DL MU-MIMO Resound Recommendation subfield to 1 to indicate that the STA suggests that the AP either resound the channel or increase the channel sounding frequency with the STA. The subfield is set to 0 to indicate that the non-AP HE STA has no recommendation on the AP channel sounding frequency. If the OM Control field is transmitted by an HE AP, then the DL MU-MIMO Resound Recommendation subfield is reserved.

Table 9-24b—UL MU Disable and UL MU Data Disable subfields encoding

UL MU Disable subfield	UL MU Data Disable subfield	Interpretation by an AP that transmits a value of 0 in the OM Control UL MU Data Disable RX Support	Interpretation by an AP that transmits a value of 1 in the OM Control UL MU Data Disable RX Support
0	0	All trigger based UL MU transmissions are enabled by the STA as defined in 26.5.2.	All trigger based UL MU transmissions are enabled by the STA as defined in 26.5.2.
0	1	N/A	Trigger based UL MU Data frame transmissions in response to a Basic Trigger frame are suspended by the STA as defined in 26.9.3. Other trigger based UL MU transmissions remain enabled by the STA as defined in 26.9.3.
1	0	All trigger based UL MU transmissions are suspended by the STA. The STA will not respond to a received triggering frame.	All trigger based UL MU transmissions are suspended by the STA. The STA will not respond to a received triggering frame.
1	1	Reserved	Reserved

9.2.4.6a.3 HLA Control

The Control Information subfield in an HLA Control subfield contains information related to the HE link adaptation (HLA) procedure (see 26.13). The format of the subfield is shown in Figure 9-22c.

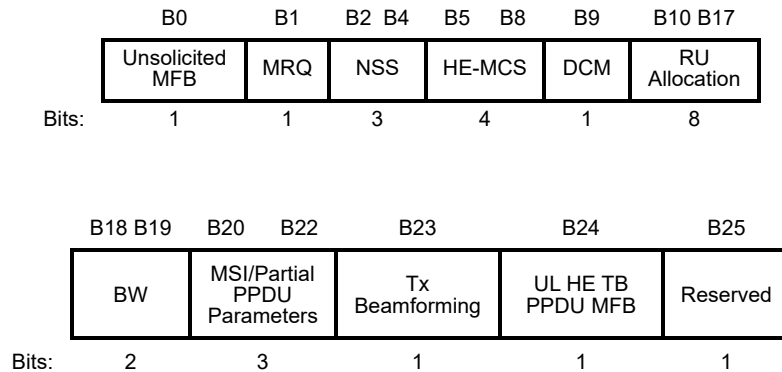


Figure 9-22c—Control Information subfield format in an HLA Control subfield

The HLA Control subfields are defined in Table 9-24c.

Table 9-24c—HLA Control subfields

Subfield	Meaning	Definition
Unsolicited MFB	Unsolicited MFB indicator	Set to 1 if the HLA Control is an unsolicited MFB. Set to 0 if the HLA Control is an MRQ or a solicited MFB.
MRQ	HLA feedback request indicator	Set to 1 and set Unsolicited MFB subfield to 0 to request an HLA feedback. Set to 0 and set Unsolicited MFB subfield to 0 to respond to an HLA request. If the Unsolicited MFB subfield is 1, the MRQ subfield is reserved.
NSS	Recommended number of spatial stream	If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 0 or if the Unsolicited MFB subfield is 0 and the MRQ subfield is 0, the NSS subfield indicates the recommended number of spatial streams to the PPDU sent to the STA, N_{SS} , and is set to $N_{SS} - 1$. If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 1, the NSS subfield indicates the recommended number of spatial streams to the HE TB PPDU sent from the STA, N_{SS} , and is set to $N_{SS} - 1$. Otherwise, this subfield is reserved.
HE-MCS	Recommended HE-MCS	If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 0 or if the Unsolicited MFB subfield is 0 and the MRQ subfield is 0, the HE-MCS subfield indicates the recommended HE-MCS of the PPDU sent to the STA, and is set to the HE-MCS index (see 27.5). If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 1, the HE-MCS subfield indicates the recommended HE-MCS of the HE TB PPDU sent from the STA, and is set to the HE-MCS index (see 27.5). Otherwise, this subfield is reserved.
DCM	Recommended usage of dual carrier modulation (DCM)	If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 0 or if the Unsolicited MFB subfield is 0 and the MRQ subfield is 0, the DCM subfield indicates the recommended usage of DCM. This subfield is set to 1 if DCM is recommended to the PPDU sent to the STA and is set to 0 otherwise. If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 1, the DCM subfield indicates the recommended usage of DCM. This subfield is set to 1 if DCM is recommended to the HE TB PPDU sent from the STA and is set to 0 otherwise. Otherwise, this subfield is reserved.

Table 9-24c—HLA Control subfields (continued)

Subfield	Meaning	Definition
RU Allocation	RU of the recommended HE-MCS/RU specified by MFB requester to get feedback	If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 0, the RU Allocation subfield indicates the RU for which the recommended HE-MCS applies to the PPDU sent to the STA, as defined in 26.13. If the Unsolicited MFB subfield is 0 and the MRQ subfield is 1, the RU subfield indicates the RU requested by the MFB requester to get feedback. The RU Allocation subfield is interpreted with the BW subfield to specify the RU. The RU index encoding is as defined in Table 9-29i. If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 1, the RU Allocation subfield indicates the RU for which the recommended HE-MCS applies to the HE TB PPDU sent from the STA, as defined in 26.13 and that the actual allocation of the RU can be ignored by the recipient. Otherwise, this subfield is reserved.
BW	Bandwidth of the recommended HE-MCS/Bandwidth specified by MFB requester to get feedback	If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 0, the BW subfield indicates the bandwidth for which the recommended HE-MCS applies to the PPDU sent to the STA, as defined in 26.13. If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 1, the BW subfield indicates the bandwidth for which the recommended HE-MCS applies to the HE TB PPDU sent from the STA, as defined in 26.13. If the Unsolicited MFB subfield is 0 and the MRQ subfield is 1, the BW subfield indicates the bandwidth requested by the MFB requester to get feedback. Set to 0 for 20 MHz. Set to 1 for 40 MHz. Set to 2 for 80 MHz. Set to 3 for 160 MHz or 80+80 MHz. Otherwise, this subfield is reserved.
MSI/Partial PPDU Parameters	Partial parameters of the measured PPDU/ MRQ sequence identifier	If the Unsolicited MFB subfield is 0 and the MRQ subfield is 1, the MSI/Partial PPDU Parameters subfield contains a sequence number in the range 0 to 6 that identifies the specific HE-MCS feedback request. If the Unsolicited MFB subfield is 0 and the MRQ subfield is 0, the MSI/Partial PPDU Parameters subfield contains a sequence number in the range 0 to 6 that responds to the specific solicited HE-MCS feedback request. If the Unsolicited MFB subfield is 1, the MSI/Partial PPDU Parameters subfield contains the PPDU Format and Coding Type subfields as shown in Figure 9-22d.

Table 9-24c—HLA Control subfields (continued)

Subfield	Meaning	Definition
Tx Beamforming	Transmission type of the measured PPDU	If the Unsolicited MFB subfield is 1 and the UL HE TB PPDU MFB subfield is 0, then the Tx Beamforming subfield indicates whether the PPDU from which the unsolicited MFB was estimated is beamformed. Set to 0 for non-beamformed PPDU. Set to 1 for beamformed PPDU. Otherwise, this subfield is reserved.
UL HE TB PPDU MFB	UL HE TB PPDU MFB indication	If the Unsolicited MFB subfield is 1, a value of 1 in this subfield indicates that the NSS, HE-MCS, DCM, BW and RU Allocation fields represent the recommended MFB for the HE TB PPDU sent from the STA as defined in 26.13. Otherwise, this subfield is reserved.

The format of the MSI/Partial PPDU Parameters subfield is defined in Figure 9-22d.

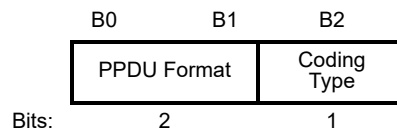


Figure 9-22d—MSI/Partial PPDU Parameters subfield format if the Unsolicited MFB subfield is 1

The PPDU Format subfield indicates the format of the PPDU from which the unsolicited MFB was estimated:

- Set to 0 for an HE SU PPDU.
- Set to 1 for an HE MU PPDU.
- Set to 2 for an HE ER SU PPDU.
- Set to 3 for an HE TB PPDU.

The Coding Type subfield contains the coding information of the PPDU from which the unsolicited MFB was estimated:

- Set to 0 for BCC.
- Set to 1 for LDPC.

9.2.4.6a.4 BSR Control

The Control Information subfield in a BSR Control subfield contains buffer status information used for UL MU operation (see 26.5.2). The format of the subfield is shown in Figure 9-22e.

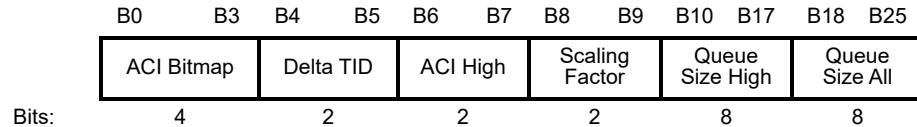


Figure 9-22e—Control Information subfield format in a BSR Control subfield

The ACI Bitmap subfield indicates the access categories for which the buffer status is reported and its encoding is shown in Table 9-24d. Each bit of the ACI Bitmap subfield is set to 1 to indicate that the buffer status of the corresponding AC is included in the Queue Size All subfield, and set to 0 otherwise, except that if the ACI Bitmap subfield is 0 and the Delta TID subfield is 3, then the buffer status of all 8 TIDs is included (see Table 9-24e).

Table 9-24d—ACI Bitmap subfield encoding

B0	B1	B2	B3
AC_BE	AC_BK	AC_VI	AC_VO

The Delta TID subfield, together with the values of the ACI Bitmap subfield, indicate the number of TIDs for which the STA is reporting the buffer status. The encoding of the Delta TID subfield is defined in Table 9-24e.

Table 9-24e—Delta TID subfield encoding

Number of bits in the ACI Bitmap subfield that are set to 1	Mapping of Delta TID subfield value and number of TIDs, N_{TID}
0	Values 0 to 2 are not applicable; Value 3 indicates 8 TIDs (i.e., all ACs have traffic).
1	Value 0 indicates 1 TID; Value 1 indicates 2 TIDs; Values 2 to 3 are not applicable.
2	Value 0 indicates 2 TID; Value 1 indicates 3 TIDs; Value 2 indicates 4 TIDs; Value 3 is not applicable.
3	Value 0 indicates 3 TID; Value 1 indicates 4 TIDs; Value 2 indicates 5 TIDs; Value 3 indicates 6 TIDs.
4	Value 0 indicates 4 TID; Value 1 indicates 5 TIDs; Value 2 indicates 6 TIDs; Value 3 indicates 7 TIDs.

NOTE 1—The number of TIDs can be obtained as $N_{TID} = N_{ones} + D_{Val}$, where N_{ones} is the number of bits set to one in the AC Bitmap subfield, and D_{Val} is the value of the Delta TID subfield, except if N_{ones} is equal to 0 for which there is the $N_{TID} = 8$ case.

NOTE 2—The Delta TID might be used by an AP to determine the setting of the TID Aggregation Limit field in the User Info field addressed to the STA in a subsequent Basic Trigger frame.

The ACI High subfield indicates the ACI of the AC for which the BSR is indicated in the Queue Size High subfield. The ACI to AC mapping is shown in Table 9-154.

NOTE 1—It is up to the non-AP STA that reports the buffer status to determine the queue that deserves higher priority with respect to the other queues. The determination might be based on the time the traffic has been outstanding, QoS delay requirements, amount of buffered traffic, etc., and is beyond the scope of this standard.

The Scaling Factor subfield indicates the unit SF , in octets, of the Queue Size High and Queue Size All subfields. The encoding of the Scaling Factor subfield is shown in Table 9-24f.

Table 9-24f—Scaling Factor subfield encoding

Scaling Factor subfield	Scaling factor, SF (octets)
0	16
1	256
2	2 048
3	32 768

The Queue Size High subfield indicates the amount of buffered traffic, in units of SF octets, for the AC identified by the ACI High subfield that is intended for the STA identified by the receiver address of the frame containing the BSR Control subfield.

The Queue Size All subfield indicates the amount of buffered traffic, in units of SF octets, for all the ACs identified by the ACI Bitmap subfield that is intended for the STA identified by the receiver address of the frame containing the BSR Control subfield.

The queue size values in the Queue Size High and Queue Size All subfields are the total sizes, rounded up to the nearest multiple of SF octets, of all MSDUs and A-MSDUs buffered at the STA (including the MSDUs or A-MSDUs in the same PSDU as the frame containing the BSR Control subfield) in the delivery queues used for MSDUs and A-MSDUs with AC(s) that are specified in the ACI High and ACI Bitmap subfields, respectively.

NOTE 2—The queue size is based on data received by the STA at the MAC SAP (MA-UNITDATA.request). Any data in layers above the MAC is not taken into account.

NOTE 3—Buffered MSDUs are those that have been received in an MA-UNITDATA.request but that have not been successfully transmitted and have not been discarded.

A queue size value of 254 in the Queue Size High and Queue Size All subfields indicates that the amount of buffered traffic is greater than $254 \times SF$ octets. A queue size value of 255 in the Queue Size High and Queue Size All subfields indicates that the amount of buffered traffic is an unspecified or unknown size.

The queue size value of the QoS Data frames containing the fragments might remain constant in all fragments even if the amount of queued traffic changes as successive fragments are transmitted (see 10.23.3.5.1). If the QoS Data frames containing fragments are carried in the A-MPDU, the queue size values of the MPDUs containing the fragments are set according to the rules in 10.8.

9.2.4.6a.5 UPH Control

The Control Information subfield in an UPH Control subfield contains the UL power headroom (UPH) used for power pre-correction (see 26.5.2.4). The format of the subfield is shown in Figure 9-22f.

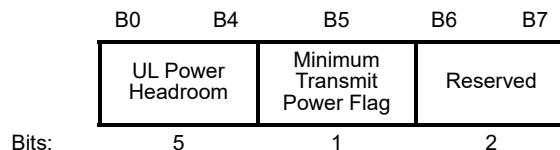


Figure 9-22f—Control Information subfield format in a UPH Control subfield

The UL Power Headroom subfield indicates the available UL power headroom, in units of dB, for the current HE-MCS (see 26.5.2.4). The UL Power Headroom subfield carries a value 0 to 31 that maps to 0 dB to 31 dB.

The Minimum Transmit Power Flag subfield is set to 1 to indicate that the minimum transmit power for the current HE-MCS is reached by the STA and set to 0 otherwise.

9.2.4.6a.6 BQR Control

The Control Information subfield in a BQR Control subfield contains the bandwidth query report (BQR) used for bandwidth query report operation to assist HE MU transmission (see 26.5.2). The format of the subfield is shown in Figure 9-22g.

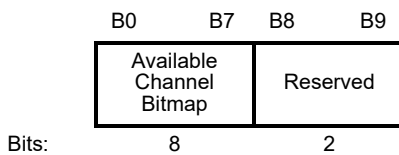


Figure 9-22g—Control Information subfield format in a BQR Control subfield

The Available Channel Bitmap subfield contains a bitmap indicating the subchannels available at the STA transmitting the BQR. Each bit in the bitmap corresponds to a 20 MHz subchannel within the operating channel width of the BSS in which the STA is associated, with the LSB corresponding to the lowest numbered operating subchannel of the BSS. The bit in position X in the bitmap is set to 1 to indicate that the subchannel $X + 1$ is idle; otherwise, it is set to 0 to indicate that the subchannel is busy or unavailable. Availability of each 20 MHz subchannel is based on the ED-based CCA defined in 27.3.20.6.5 and is reported for the 20 MHz subchannels located in the operating channel of the reporting STA when the WM is idle as defined in 10.3.2.1 and in 26.5.2.5.

9.2.4.6a.7 CAS Control

The Control Information subfield in a CAS Control subfield contains the command and status (CAS) control. The format of the subfield frame is shown in Figure 9-22h.

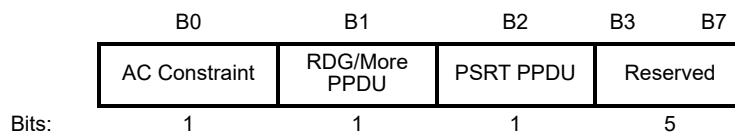


Figure 9-22h—Control Information subfield format in a CAS Control subfield

The AC Constraint subfield is defined in Table 9-14, except that a value of 1 indicates to an HE STA that the response can contain RD Data frames from the same AC or higher priority ACs as defined in 10.29.4.

The RDG/More PPDU subfield is defined in Table 9-15.

The PSRT PPDU subfield indicates whether the PPDU carrying the frame with the CAS Control subfield is an PSRT PPDU. The PSRT PPDU subfield is set to 1 if the PPDU is an PSRT PPDU; otherwise, it is set to 0.

9.2.4.7 Frame Body field

9.2.4.7.1 General

Change as Table 9-25 follows:

**Table 9-25—Maximum data unit sizes (in octets) and durations
(in microseconds)**

	Non-HT non-VHT non-HE non-SIG non-DMG PPDU and non-HT duplicate PPDU	HT PPDU	VHT PPDU	<u>HE PPDU</u>	SIG PPDU	DMG PPDU
MMPDU size	2304	2304	See NOTE 1	<u>See NOTE 1</u>	See NOTE 1	2304
MSDU size	2304	2304	2304	<u>2304</u>	2304	7920
A-MSDU size	3839 or 4065 (see NOTE 2) (HT STA, see also Table 9-184), or N/A (non-HT STA, see also 10.11)	3839 or 7935 (see also Table 9-184)	See NOTE 3	<u>2.4 GHz band: 3839 or 7935 (see also Table 9-184)</u> <u>Otherwise: see NOTE 3</u>	See NOTE 3	7935
MPDU size	See NOTE 4	See NOTE 5	3895 or 7991 or 11 454 (see also Table 9-271)	<u>2.4 GHz band: see NOTE 5</u> <u>Otherwise: 3895 or 7991 or 11 454 (see also Table 9-271)</u> <u>See NOTE 7</u>	3895 or 7991 (see also Table 9-300)	See NOTE 5
PSDU size	$2^{12}-1$ (see Table 15-5, Table 16-4, Table 17-21, Table 18-5)	$2^{16}-1$ (see Table 19-25)	4 692 480 ($\sim 2^{22.16}$) (see Table 21-28)	<u>6 500 631 ($\sim 2^{22.63}$)</u> (see <u>Table 27-54</u>)	797 160 ($\sim 2^{19.60}$) (see Table 23-40)	$2^{18}-1$ (see Table 20-30)

**Table 9-25—Maximum data unit sizes (in octets) and durations
(in microseconds) (continued)**

	Non-HT non-VHT non-HE non-SIG non-DMG PPDU and non-HT duplicate PPDU	HT PPDU	VHT PPDU	<u>HE PPDU</u>	SIG PPDU	DMG PPDU
PPDU duration	See NOTE 6	5484 (HT_MF, see 10.27.4) or 10 000 (HT_GF, see Table 19-25)	5484 (see Table 21-28)	<u>5484</u> (see <u>Table 27-54</u>)	27 840 (see Table 23-40)	2000 (see Table 20-30)
NOTE 1—No direct constraint on the maximum MMPDU size; indirectly constrained by the maximum MPDU size (see 9.3.3.1). NOTE 2—Indirect constraint from the maximum PSDU size: $2^{12}-1$ octets minus the minimum QoS Data frame overhead (26 octets for the MAC header and 4 octets for the FCS). NOTE 3—No direct constraint on the maximum A-MSDU size; indirectly constrained by the maximum MPDU size. NOTE 4—No direct constraint on the maximum MPDU size; indirectly constrained by the maximum MSDU/MMPDU or (for HT STAs only) A-MSDU size. NOTE 5—No direct constraint on the maximum MPDU size; indirectly constrained by the maximum A-MSDU size. NOTE 6—No direct constraint on the maximum duration, but an L_LENGTH value above 2332 might not be supported by some receivers (see NOTE 2 in 10.27.4). <u>NOTE 7—The maximum MPDU size might be greater than the size declared as supported by the recipient if the MPDU is an HE Compressed Beamforming/CQI frame.</u>						

9.2.5 Duration/ID field (QoS STA)

9.2.5.2 Setting for single and multiple protection under enhanced distributed channel access (EDCA)

Change the second paragraph of 9.2.5.2 as follows:

The STA selects between single and multiple protection when it transmits the first frame of a TXOP. All subsequent frames transmitted by the STA in the same TXOP use the same class of duration settings. A STA always uses multiple protection in a TXOP that includes the following:

- Frames that have the RDG/More PPDU subfield equal to 1
- PSMP frames
- VHT/HE NDP Announcement frames, ~~or~~ Beamforming Report Poll frames, or BFRP Trigger frames
- SIG Beacon frames
- Frames transmitted by an SIG STA with the TXVECTOR parameter RESPONSE INDICATION equal to Long Response

Change item a) and item b) in the lettered list of 9.2.5.2 as follows, and renumber the subsequent notes in this subclause accordingly:

- a) Single protection settings.
- 1)
- 1a) In an MU-RTS Trigger frame, the Duration/ID field is set to the estimated time, in microseconds, required to transmit the pending frame(s), plus one CTS frame, plus the time to transmit the solicited HE TB PPDU if required, plus the time to transmit the acknowledgment for the solicited HE TB PPDU if required, plus applicable IFSSs.

NOTE 1—The pending frame(s) include a triggering frame if required.

- ...
- 3)
- 3a) In an MU-BAR Trigger frame, BSRP Trigger frame, GCR MU-BAR Trigger frame, BQRP Trigger frame, and NFRP Trigger frame, the Duration/ID field is set to the time required to transmit the solicited HE TB PPDU plus one SIFS.
- ...
- 6)
- 7) In a Basic Trigger frame, the Duration/ID field is set to the estimated time required to transmit the solicited HE TB PPDU, plus the estimated time required to transmit the acknowledgment for the solicited HE TB PPDU if required, plus applicable SIFSs.
- b) Multiple protection settings. The Duration/ID field is set to a value D as follows:

- ...
- 4) Else $T_{END-NAV} - T_{PPDU} \leq D \leq T_{TXOP-REMAINING} - T_{PPDU}$
where

$T_{SINGLE-MSDU}$	is the estimated time required for the transmission of the allowed frame exchange sequence defined in 10.23.2.9 (for a TXOP limit of 0), including applicable IFS durations
$T_{PENDING}$	is the estimated time required for the transmission of — Pending MPDUs of the same AC — Any associated immediate response frames — Any HT NDP, VHT NDP, <u>HE sounding NDP</u>, or Beamforming Report Poll frame transmissions and explicit feedback response frames — Applicable IFSSs — Any RDG — Any BDT — Any pending QoS Null frame exchanges by paged STAs — Any pending PS-Poll or NDP PS-Poll frame exchanges by paged STAs
T_{TXOP}	is the duration given by dot11EDCATableTXOPLimit (dot11QAPEDCATableTXOPLimit for the AP) for that AC
$T_{TXOP-REMAINING}$	is T_{TXOP} less the time already used time within the TXOP
$T_{END-NAV}$	is the remaining duration of any NAV set by the TXOP holder, or 0 if no NAV has been established
T_{PPDU}	is the time required for transmission of the current PPDU

NOTE 2—The rules allowing or disallowing the transmission of MPDUs with different ACs are described in 10.23.2.7, 10.23.2.8, and 26.6.3.

NOTE 3—The estimated time to transmit an acknowledgment in response to the frames carried in a solicited HE TB PPDU might be inexact. The TXOP holder might use the maximum time required to transmit the acknowledgment as the estimated time.

9.2.5.7 Setting for control response frames

Change the beginning text before the now seventh paragraph ("In an NDP CTS frame transmitted") of 9.2.5.7 as follows:

This subclause describes how to set the Duration/ID field for CTS, Ack, and BlockAck frames transmitted by a QoS STA.

In a CTS frame that is not part of a dual CTS sequence transmitted in response to an RTS frame, the Duration/ID field is set to the value obtained from the Duration/ID field of the RTS frame that elicited the response minus the time, in microseconds, between the end of the PPDU carrying the RTS frame and the end of the PPDU carrying the CTS frame.

In a CTS frame that is transmitted in response to an MU-RTS Trigger frame, the Duration/ID field is set to the value obtained from the Duration/ID field of the MU-RTS Trigger frame that elicited the CTS frame minus the time, in microseconds, between the end of the PPDU carrying the MU-RTS Trigger frame and the end of the PPDU carrying the CTS frame.

In an Ack frame, the Duration/ID field is set to the value obtained from the Duration/ID field of the frame that elicited the response minus the time, in microseconds between the end of the PPDU carrying the frame that elicited the response and the end of the PPDU carrying the Ack frame.

In a BlockAck frame transmitted in response to a BlockAckReq frame, MU-BAR Trigger frame, or transmitted in response to a frame containing an implicit block ack request, or frame carried in HE TB PPDU under single protection settings, the Duration/ID field is set to the value obtained from the Duration/ID field of the frame that elicited the response minus the time, in microseconds between the end of the PPDU carrying the frame that elicited the response and the end of the PPDU carrying the BlockAck frame.

Under multiple protection settings, the Duration/ID field in a BlockAck frame transmitted in response to a frame carried in HE TB PPDU is set according to the multiple protection settings defined in 9.2.5.2.

9.2.5.8 Setting for other response frames

Change 9.2.5.8 as follows:

In any frame transmitted by a STA that is not the TXOP holder and is not specified by 9.2.5.1 to 9.2.5.7, the Duration/ID field is set to the value obtained from the Duration/ID field of the frame that elicited the response minus the time, in microseconds, between the end of the PPDU carrying the frame that elicited the response and the end of the PPDU carrying the frame. If the frame is contained in an HE TB PPDU, the time is equal to SIFS plus the duration of the HE TB PPDU, where the duration of the HE TB PPDU is defined in Equation (27-134).

9.3 Format of individual frame types

9.3.1 Control frames

9.3.1.2 RTS frame format

Change the third paragraph in 9.3.1.2 as follows:

The TA field is the address of the STA transmitting the RTS frame or the bandwidth signaling TA of the STA transmitting the RTS frame. In an RTS frame transmitted by a VHT STA or an HE STA in a non-HT or

non-HT duplicate format to another VHT STA or HE STA, the scrambling sequence carries the TXVECTOR parameters CH_BANDWIDTH_IN_NON_HT and DYN_BANDWIDTH_IN_NON_HT (see 10.3.2.7) and the TA field is a bandwidth signaling TA.

9.3.1.3 CTS frame format

Change the second paragraph of 9.3.1.3 as follows:

~~When~~ If the CTS frame is a response to an RTS frame, the RA field of the CTS frame is set to the address from the TA field of the RTS frame with the Individual/Group bit set to 0. ~~When~~ If the CTS frame is the first frame in a frame exchange, the RA field is set to the MAC address of the transmitter. If the CTS frame is a response to an MU-RTS Trigger frame, the RA field of the CTS frame is set to the address from the TA field of the MU-RTS Trigger frame.

9.3.1.5 PS-Poll frame format

9.3.1.5.1 General

Change the second paragraph of 9.3.1.5.1 as follows:

The BSSID (RA) field is set to the address of the STA contained in the AP. The TA field value is the address of the STA transmitting the frame or a bandwidth signaling TA. In a PS-Poll frame transmitted by a VHT STA or an HE STA in a non-HT or non-HT duplicate format and where the scrambling sequence carries the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT, the TA field value is a bandwidth signaling TA.

9.3.1.6 CF-End frame format

Change the last paragraph of 9.3.1.6 as follows:

~~When~~ If transmitted by a non-DMG STA, the BSSID (TA) field is the address of the STA contained in the AP, except that the Individual/Group bit of the BSSID (TA) field is set to 1 in a CF-End frame transmitted by a VHT STA to a VHT AP or an HE STA to an HE AP in a non-HT or non-HT duplicate format to indicate that the scrambling sequence carries the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT. ~~When~~ If transmitted by a DMG STA, the TA field is the MAC address of the STA transmitting the frame.

9.3.1.7 BlockAckReq frame format

9.3.1.7.1 Overview

Change the fourth paragraph in 9.3.1.7.1 as follows:

The TA field value is the address of the STA transmitting the BlockAckReq frame or a bandwidth signaling TA. In a BlockAckReq frame transmitted by a VHT STA or an HE STA in a non-HT or non-HT duplicate format and where the scrambling sequence carries the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT, the TA field value is a bandwidth signaling TA.

Change Figure 9-36 as follows:

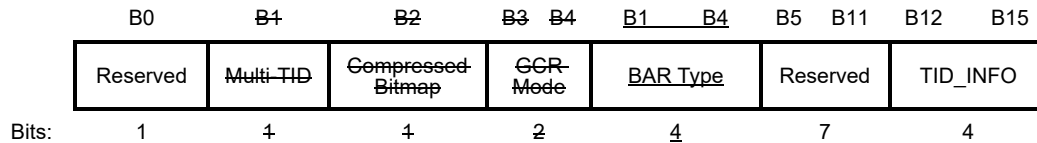


Figure 9-36—BAR Control field format

Change the fifth paragraph in 9.3.1.7.1 as follows:

The values of the Multi-TID, Compressed Bitmap, and GCR Mode subfields BAR Type subfield indicates the BlockAckReq frame variants used, as indicated defined in Table 9-27.

Replace Table 9-27 with the following table:

Table 9-27—BlockAckReq frame variant encoding

BAR Type	BlockAckReq frame variant
0	Reserved
1	Extended Compressed
2	Compressed
3	Multi-TID
4–5	Reserved
6	GCR
7–9	Reserved
10	GLK-GCR
11–15	Reserved

9.3.1.8 BlockAck frame format

9.3.1.8.1 Overview

Change the third paragraph in 9.3.1.8.1 as follows:

The RA field of the BlockAck frame is the address of the recipient STA that requested the a BlockAck frame that is not a Multi-STA BlockAck variant is set to the TA field of the soliciting frame or the address of the recipient STA from which Data frames are acknowledged. The RA field of a Multi-STA BlockAck frame is set as described in 9.3.1.8.7.

Change Figure 9-42 as follows:

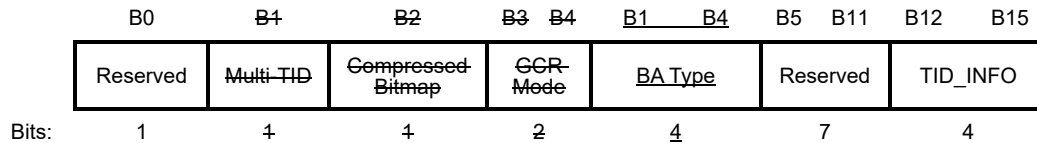


Figure 9-42—BA Control field format

Change the sixth and seventh paragraphs in 9.3.1.8.1 as follows:

The values of the Multi-TID, Compressed Bitmap, and GCR Mode subfields of BA Type subfield in the BA Control field ~~determine~~ indicates the BlockAck frame variants ~~represented~~, as ~~indicated~~ defined in Table 9-28.

NOTE—Reference to “a BlockAck” frame without any other qualification from other subclauses applies to any of the variants, unless specific exclusions are called out.

The GCR Mode subfield ~~indicates whether the GCR BlockAck frame was sent~~ is used in response to a GCR BlockAckReq frame, ~~or~~ and the GLK-GCR BlockAck frame is used in response to a GLK-GCR BlockAckReq frame.

Replace Table 9-28 with the following table:

Table 9-28—BlockAck frame variant encoding

BA Type	BlockAck frame variant
0	Reserved
1	Extended Compressed
2	Compressed
3	Multi-TID
4–5	Reserved
6	GCR
7–9	Reserved
10	GLK-GCR
11	Multi-STA
12–15	Reserved

9.3.1.8.2 Compressed BlockAck variant

Change 9.3.1.8.2 as follows (including changing Figure 9-43 and inserting Table 9-28a):

The TID_INFO subfield of the BA Control field of the Compressed BlockAck frame contains the TID for which this BlockAck frame is sent.

The BA Information field of the Compressed BlockAck frame ~~comprises the Block Ack Starting Sequence Control subfield and the Block Ack Bitmap subfield, as is shown in Figure 9-43. The Block Ack Starting Sequence Control subfield is shown in Figure 9-37. The Starting Sequence Number subfield of the Block Ack Starting Sequence Control subfield contains the sequence number of the first MSDU or A-MSDU for which this BlockAck frame is sent. This subfield is defined in 10.25.6.5. The Fragment Number subfield of the Block Ack Starting Sequence Control subfield is set to 0.~~

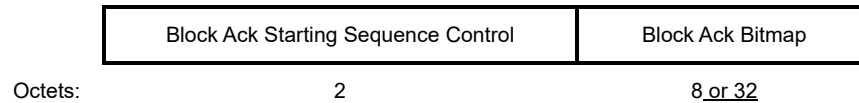


Figure 9-43—BA Information field format (Compressed BlockAck)

The Fragment Number subfield of the Block Ack Starting Sequence Control field is set as defined in Table 9-28a. The Fragment Number subfield of the Block Ack Starting Sequence Control subfield is set to all 0s if the Compressed BlockAck frame is sent to or from a non-HE STA.

Table 9-28a—Fragment Number subfield encoding for the Compressed BlockAck variant

<u>Fragment Number subfield</u>			<u>Fragmentation level 3 (ON/OFF)</u>	<u>Block Ack Bitmap subfield length (octets)</u>	<u>Maximum number of MSDUs/A-MSDUs that can be acknowledged</u>
<u>B3</u>	<u>B2–B1</u>	<u>B0</u>			
<u>0</u>	<u>0</u>	<u>0</u>	<u>OFF</u>	<u>8</u>	<u>64</u>
<u>0</u>	<u>1</u>	<u>0</u>		<u>Reserved</u>	<u>Reserved</u>
<u>0</u>	<u>2</u>	<u>0</u>		<u>32</u>	<u>256</u>
<u>0</u>	<u>3</u>	<u>0</u>		<u>Reserved</u>	<u>Reserved</u>
<u>0</u>	<u>0</u>	<u>1</u>	<u>ON</u>	<u>8</u>	<u>16</u>
<u>0</u>	<u>1</u>	<u>1</u>		<u>Reserved</u>	<u>Reserved</u>
<u>0</u>	<u>2</u>	<u>1</u>		<u>32</u>	<u>64</u>
<u>0</u>	<u>3</u>	<u>1</u>		<u>Reserved</u>	<u>Reserved</u>
<u>1</u>	<u>Any</u>	<u>Any</u>		<u>Reserved</u>	<u>Reserved</u>

NOTE—A Compressed BlockAck frame with B0 of the Fragment Number subfield set to 1 is not sent to an HE STA whose Dynamic Fragmentation Support subfield in the HE Capabilities element it transmits is not set to 3 (see 26.3).

If B0 of the Fragment Number subfield is 0, the Block Ack Bitmap subfield of the BA Information field of the Compressed BlockAck frame indicates the receive status of up to 64 or 256 MSDUs and/or A-MSDUs depending upon the value of B2–B1 in the Fragment Number subfield as shown in Table 9-28a. The Block Ack Bitmap subfield of the BA Information field of the Compressed BlockAck frame is used to indicate the received status of up to 64 entries, where each entry represents an MSDU or an A-MSDU. Each bit that is equal to 1 in the compressed Block Ack Bitmap subfield acknowledges the reception of a single MSDU or A-MSDU in the order of sequence number, with the first bit of the Block Ack Bitmap subfield corresponding to the MSDU (or fragment thereof) or A-MSDU (or fragment thereof) with the sequence number that matches the Starting Sequence Number subfield of the Block Ack Starting Sequence Control subfield.

If B0 of the Fragment Number subfield is 1, the Block Ack Bitmap subfield of the BA Information field of the Compressed BlockAck frame indicates the receive status of up to 16 or 64 MSDUs and/or A-MSDUs depending upon the value B2–B1 in the Fragment Number subfield as shown in Table 9-28a. If bit position n of the Block Ack Bitmap subfield is 1, it acknowledges receipt of an MPDU with sequence number value SN and fragment number value FN with $n = 4 \times (SN - SSN) + FN$, where SSN is the value of the Starting Sequence Number subfield of the Block Ack Starting Sequence Control subfield and the operations on the sequence numbers are performed modulo 4096. If bit position n of the Block Ack Bitmap subfield is 0, it indicates that the MPDU has not been received.

NOTE—If the B0 of the Fragment Number subfield is equal to 1, then the Block Ack Bitmap subfield is split into (Block Ack Bitmap subfield length)/4 subbitmaps, each of which indicates receive status for 4 fragments of each of the MSDUs or A-MSDUs as indicated in Table 9-28a.

9.3.1.8.5 GCR Block Ack variant

Change the third paragraph in 9.3.1.8.5 as follows:

The GCR Group Address subfield is set to the value from the Group Address subfield of the GCR BAR Information field in the BlockAckReq frame to which the BlockAck frame is sent in response. If the BlockAck frame is sent in response to a GCR MU-BAR Trigger frame, the GCR Group Address subfield is set to the value from the RA field in the GCR MU-BAR Trigger frame.

Insert the following subclause (9.3.1.8.7, including Figure 9-47a through Figure 9-47d, Table 9-28b, and Table 9-28c) after 9.3.1.8.6:

9.3.1.8.7 Multi-STA BlockAck variant

The Multi-STA BlockAck frame is supported if either UL MU or multi-TID A-MPDU operation is supported and acknowledges MPDUs carried in an HE TB PPDU or multi-STA multi-TID, multi-STA single-TID, or single-STA multi-TID A-MPDUs.

An HE AP that sends a Multi-STA BlockAck frame where the Per AID TID Info fields are addressed to more than one STA sets the RA field to the broadcast address. An HE AP that sends a Multi-STA BlockAck frame where the Per AID TID Info fields are all addressed to a single recipient STA and that is sent in response to an HE TB PPDU sets the RA field of the Multi-STA BlockAck frame to either the address of the recipient STA or to the broadcast address. An HE AP that sends a Multi-STA BlockAck frame where the Per AID TID Info fields are all addressed to a single recipient STA and that is not sent in response to an HE TB PPDU sets the RA field of the Multi-STA BlockAck frame to the address of the recipient STA.

A non-AP HE STA sets the RA field to the TA field of the soliciting frame or to the address of the recipient STA whose Data or Management frames are acknowledged.

The TID_INFO subfield of the BA Control field of the Multi-STA BlockAck frame is reserved.

The BA Information field of the Multi-STA BlockAck frame comprises one or more Per AID TID Info subfields as defined in Figure 9-47a.

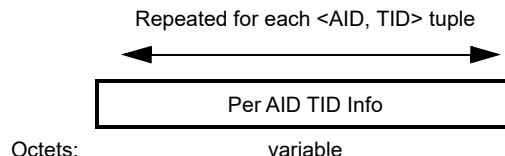


Figure 9-47a—BA Information field format (Multi-STA BlockAck)

The AID TID Info subfield is shown in Figure 9-47b.

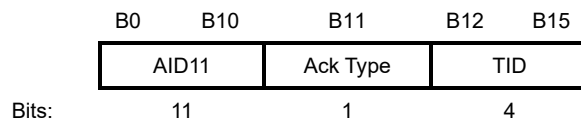


Figure 9-47b—AID TID Info subfield format

The AID11 subfield carries the 11 LSBs of the AID of the non-AP STA for which the Per AID TID Info subfield is intended. The format of the Per AID TID Info subfield depends on the value of the AID11 subfield. If the Multi-STA BlockAck frame is sent to an AP, the AID11 subfield is set to 0. A value of 2045 in the AID11 subfield is used as an identifier for any unassociated STA. If the AID11 subfield is set to 2045, then the Ack Type subfield and TID subfield are set to 0 and 15, respectively.

NOTE 1—More than one Per AID TID Info subfield with the same value in the AID11 subfield but different values in the TID subfield can be present in the Multi-STA BlockAck frame.

If the AID11 subfield of the AID TID Info subfield is not 2045, then the Per AID TID Info subfield has the format shown in Figure 9-47c.

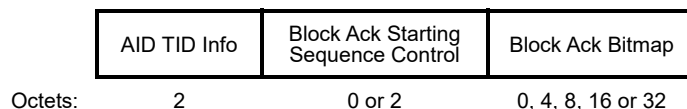


Figure 9-47c—Per AID TID Info subfield format if the AID11 subfield is not 2045

If the AID11 subfield is not 2045, then the context and the presence of each optional subfield in a Per AID TID Info subfield in a Multi-STA BlockAck frame is defined in Table 9-28b.

Table 9-28b—Context of the Per AID TID Info subfield and presence of optional subfields if the AID11 subfield is not 2045

Ack Type subfield values	TID subfield values	Presence of Block Ack Starting Sequence Control subfield and Block Ack Bitmap subfields	Context of a Per AID TID Info subfield in a Multi-STA BlockAck frame
0	0–7	Present	Block acknowledgment context: Sent as an acknowledgment to QoS Data frames that solicit a BlockAck frame response or to a BlockAckReq frame.

Table 9-28b—Context of the Per AID TID Info subfield and presence of optional subfields if the AID11 subfield is not 2045 (continued)

Ack Type subfield values	TID subfield values	Presence of Block Ack Starting Sequence Control subfield and Block Ack Bitmap subfields	Context of a Per AID TID Info subfield in a Multi-STA BlockAck frame
1	0–7	Not present	Acknowledgment context: Sent as an acknowledgment to a QoS Data or QoS Null frame that solicits an Ack frame response.
0 or 1	8–13	N/A	Reserved
0	14	N/A	Reserved
1	14	Not present	All ack context: Sent as an acknowledgment to an A-MPDU that contains an MPDU that solicits an immediate response and all MPDUs contained in the A-MPDU are received successfully.
0	15	N/A	Reserved
1	15	Not present	Management/PS-Poll frame acknowledgment context: Sent as an acknowledgment to a Management or PS-Poll frame.
NOTE 1—Additional rules for acknowledgment, block acknowledgment and the all ack context are defined in 26.4.2 for a multi-TID A-MPDU.			
NOTE 2—As HE STAs do not use HCCA (see 10.23.1), TID values from 8 to 15 are not used in QoS Data frames.			

If the Ack Type subfield is 0, the Fragment Number subfield encoding indicates the length of the BlockAck bitmap subfield as defined in Table 9-28c.

Table 9-28c—Fragment Number subfield encoding for the Multi-STA BlockAck variant

Fragment Number subfield			Fragmentation level 3 (ON/OFF)	Block Ack Bitmap subfield length (octets)	Maximum number of MSDUs/A-MSDUs that can be acknowledged
B3	B2–B1	B0			
0	0	0	OFF	8	64
0	1	0		16	128
0	2	0		32	256
0	3	0		4	32
0	0	1	ON	8	16
0	1	1		16	32
0	2	1		32	64
0	3	1		4	8

Table 9-28c—Fragment Number subfield encoding for the Multi-STA BlockAck variant (*continued*)

Fragment Number subfield			Fragmentation level 3 (ON/OFF)	Block Ack Bitmap subfield length (octets)	Maximum number of MSDUs/A-MSDUs that can be acknowledged
B3	B2–B1	B0			
1	Any	Any		Reserved	Reserved
NOTE—A Multi-STA BlockAck frame with B0 of the Fragment Number subfield set to 1 cannot be sent to an HE STA, unless the HE Capabilities element received from the HE STA has the Dynamic Fragmentation Support subfield equal to 3 (see 26.3).					

If B0 of the Fragment Number subfield of the Block Ack Starting Sequence Control subfield is 0, the BA Information field of the Multi-STA BlockAck frame contains an 8-octet, 16-octet, 32-octet, or 4-octet Block Ack Bitmap subfield depending on B2–B1 of the Fragment Number subfield as defined in Table 9-28c indicating the receive status of up to 64, 128, 256, or 32 MSDUs (or fragments thereof) and/or A-MSDUs (or fragments thereof), respectively. Each bit that is equal to 1 in the Block Ack Bitmap subfield acknowledges the reception of a single MSDU (or fragment thereof) or A-MSDU (or fragment thereof) in the order of sequence number with the first bit of the Block Ack Bitmap subfield corresponding to the MSDU or A-MSDU with the sequence number that matches the value of the Starting Sequence Number subfield of the Block Ack Starting Sequence Control subfield.

If B0 of the Fragment Number subfield of the Block Ack Starting Sequence Control subfield is 1, the Block Ack Bitmap subfield of the BA Information field of the Multi-STA BlockAck frame indicates the receive status of up to 16, 32, 64, or 8 MSDUs and/or A-MSDUs depending on B2–B1 of the Fragment Number subfield as shown in Table 9-28c. If bit position n of the Block Ack Bitmap subfield is 1, it acknowledges receipt of an MPDU with sequence number value SN and fragment number value FN with $n = 4 \times (SN - SSN) + FN$, where SSN is the value of the Starting Sequence Number subfield of the Block Ack Starting Sequence Control subfield and the operations on the sequence numbers are performed modulo 4096. If bit position n of the Block Ack Bitmap subfield is 0, it indicates that the MPDU has not been received.

NOTE 2—If B0 of the Fragment Number subfield is 1, then the Block Ack Bitmap subfield is split into (Block Ack Bitmap subfield length)/4 subbitmaps, each of which indicates receive status for 4 fragments of each of the MSDUs or A-MSDUs as indicated in Table 9-28c. For an A-MSDU, only the first bit of the subbitmap is used if fragmentation is not allowed in an A-MSDU.

If the AID11 subfield of the AID TID Info subfield is 2045, then the Per AID TID Info subfield has the format shown in Figure 9-47d, where the RA subfield indicates the MAC address of an unassociated STA for which the Per AID TID Info subfield is intended.

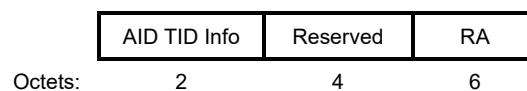


Figure 9-47d—Per AID TID Info subfield format if the AID11 subfield is 2045

NOTE 3—An associated non-AP HE STA that receives a Multi-STA BlockAck frame as a response from its AP and does not support the UORA procedure ignores the 10 octets following the AID TID Info subfield that are the remainder of the Per AID TID Info subfield if the AID11 subfield is 2045 and parses the following Per AID TID Info subfields if any.

Change the title of 9.3.1.19 as follows:

9.3.1.19 VHT/HE NDP Announcement frame format

Change the beginning text (including Figure 9-59) before the now eighth paragraph ("The Sounding Dialog Token Number subfield") in 9.3.1.19 as follows:

The VHT/HE NDP Announcement frame has two variants, the VHT NDP Announcement frame and the HE NDP Announcement frame. The two formats are distinguished by the setting of the HE subfield in the Sounding Dialog Token field.

The frame format of the VHT NDP Announcement frame is shown in Figure 9-58.

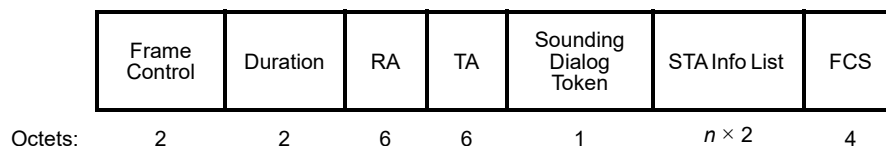


Figure 9-58—VHT NDP Announcement frame format

The Duration field is set as defined in 9.2.5.

The VHT/HE NDP Announcement frame contains at least one STA Info field. If the VHT/HE NDP Announcement frame contains only one STA Info field, then the RA field is set to the address of the STA that can provide feedback (see 10.36.5.2). If the VHT/HE NDP Announcement frame contains more than one STA Info field, then the RA field is set to the broadcast address.

The TA field is set to the address of the STA transmitting the VHT/HE NDP Announcement frame or the bandwidth signaling TA of the STA transmitting the VHT/HE NDP Announcement frame. In a VHT/HE NDP Announcement frame transmitted by a VHT or HE STA in a non-HT or non-HT duplicate format and where the scrambling sequence carries the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT, the TA field is set to a bandwidth signaling TA.

The format of the Sounding Dialog Token field is shown in Figure 9-59.

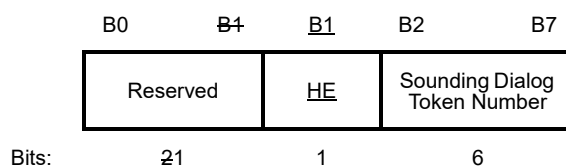


Figure 9-59—Sounding Dialog Token field format

The HE subfield in the Sounding Dialog Token field is set to 0 to identify the frame as a VHT NDP Announcement frame and set to 1 to identify the frame as an HE NDP Announcement frame.

Insert the following text (including Figure 9-61a through Figure 9-61d, Table 9-29a, and Table 9-29b) at the end of 9.3.1.19:

The format of the HE NDP Announcement frame is shown in Figure 9-61a.

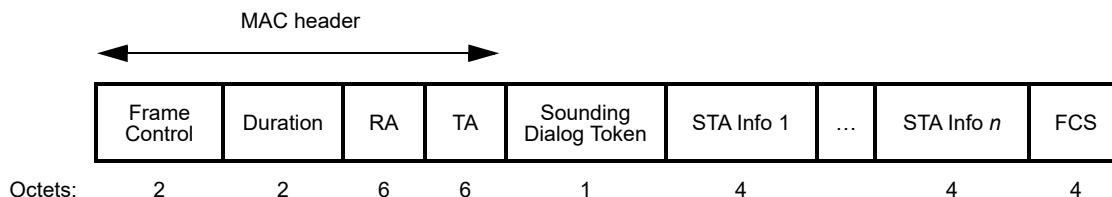


Figure 9-61a—HE NDP Announcement frame format

The Duration, RA, and TA fields are set as in a VHT NDP Announcement frame.

The HE subfield in the Sounding Dialog Token field is set to 1 to identify the frame as an HE NDP Announcement frame.

The Sounding Dialog Token Number field in the Sounding Dialog Token field contains a value selected by the beamformer to identify the HE NDP Announcement frame.

The format of the STA Info field in an HE NDP Announcement Frame if the AID11 subfield is not set to 2047 is defined in Figure 9-61b.

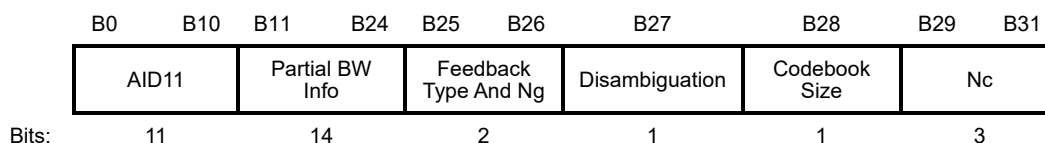


Figure 9-61b—STA Info field format in an HE NDP Announcement frame if the AID11 subfield is not 2047

An HE NDP Announcement frame contains at most 1 STA Info field per STA.

If the AID11 subfield is not 2047, then it contains an identifier of a STA expected to process the following HE sounding NDP and prepare the sounding feedback.

The Partial BW Info subfield is defined in Figure 9-61c.

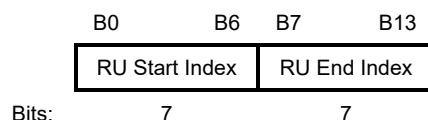


Figure 9-61c—Partial BW Info subfield format

The RU Start Index subfield in the Partial BW Info subfield indicates the first 26-tone RU for which the HE beamformer is requesting feedback. The RU End Index subfield of the Partial BW Info subfield indicates the last 26-tone RU for which the HE beamformer is requesting feedback. The value of the RU Start Index subfield is less than or equal to the value of the RU End Index subfield. The RU Start Index subfield and RU End Index subfield depends on the bandwidth of the HE NDP Announcement frame, which is indicated by the TXVECTOR parameter CH_BANDWIDTH if the frame is carried in an HE, VHT, or HT PPDU and by the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT if the frame is carried in a non-HT duplicate PPDU, and is 20 MHz if the frame is carried in non-HT PPDU. The values of the RU Start Index subfield and RU End Index subfield are each selected from the following:

- Values 0 to 8 if the bandwidth of the HE NDP Announcement frame is 20 MHz, where 0 indicates 26-tone RU 1 and 8 indicates 26-tone RU 9. Values 9–127 are reserved. See Table 27-7.
- Values 0 to 17 if the bandwidth of the HE NDP Announcement frame is 40 MHz, where 0 indicates 26-tone RU 1 and 17 indicates 26-tone RU 18. Values 18–127 are reserved. See Table 27-8.
- Values 0 to 36 if the bandwidth of the HE NDP Announcement frame is 80 MHz, where 0 indicates 26-tone RU 1 and 36 indicates 26-tone RU 37. Values 37–127 are reserved. See Table 27-9.
- Values 0 to 73 if the bandwidth of the HE NDP Announcement frame is 160 MHz, where 0 indicates 26-tone RU 1 and 73 indicates 26-tone RU 74. Values 74–127 are reserved.
- Values 0 to 73 if the bandwidth of the HE NDP Announcement frame is 80+80 MHz, where 0 indicates 26-tone RU 1 in the lower 80 MHz frequency segment, 36 indicates 26-tone RU 37 in the lower 80 MHz frequency segment, 37 indicates 26-tone RU 1 in the upper 80 MHz frequency segment, and 73 indicates 26-tone RU 37 in the upper 80 MHz frequency segment. Values 74–127 are reserved. For 80+80 MHz, feedback is not requested for the gap between the 80 MHz segments. See Table 27-9.

The Feedback Type And Ng and Codebook Size subfields for HE TB sounding are defined in Table 9-29a.

Table 9-29a—Feedback Type And Ng subfield and Codebook Size subfield encoding for HE TB sounding

Feedback Type And Ng		Codebook Size	Description
B25	B26	B28	
0	0	0	SU, $N_g = 4$, quantization resolution $(\phi, \psi) = \{4, 2\}$
0	0	1	SU, $N_g = 4$, quantization resolution $(\phi, \psi) = \{6, 4\}$
0	1	0	SU, $N_g = 16$, quantization resolution $(\phi, \psi) = \{4, 2\}$
0	1	1	SU, $N_g = 16$, quantization resolution $(\phi, \psi) = \{6, 4\}$
1	0	0	MU, $N_g = 4$, quantization resolution $(\phi, \psi) = \{7, 5\}$
1	0	1	MU, $N_g = 4$, quantization resolution $(\phi, \psi) = \{9, 7\}$
1	1	0	CQI
1	1	1	MU, $N_g = 16$, quantization resolution $(\phi, \psi) = \{9, 7\}$

The Feedback Type And Ng and Codebook Size subfields for HE non-TB sounding are defined in Table 9-29b.

Table 9-29b—Feedback Type And Ng subfield and Codebook Size subfield encoding for HE non-TB sounding

Feedback Type And Ng		Codebook Size	Description
B25	B26	B28	
0	Reserved	Reserved	SU
1	1	0	CQI

The Disambiguation subfield is set to 1.

NOTE—Setting the Disambiguation subfield to 1 prevents a non-HE VHT STA from wrongly identifying its AID in the HE NDP Announcement frame. The Disambiguation subfield coincides with the MSB of the AID12 subfield of a VHT NDP Announcement frame if the HE NDP Announcement field is parsed as VHT NDP Announcement frame by a non-HE VHT STA. The MSB of the AID12 subfield is always 0 since the maximum AID is 2007.

In a broadcast HE NDP Announcement frame that has more than one STA Info field with a value other than 2047 in the AID11 field the following applies to each STA Info subfield with a value other than 2047:

- If the Feedback Type subfield indicates SU or MU, the N_c subfield indicates the number of columns, N_c , in the compressed beamforming feedback matrix and is set to $N_c - 1$
- If the Feedback Type subfield indicates CQI, the N_c subfield indicates the number of space-time streams, N_c , in the CQI report and is set to $N_c - 1$

In an individually addressed HE NDP Announcement frame with a single STA Info field, the STA Info field having a value in the AID11 field other than 2047, the N_c subfield is reserved.

The format of the STA Info field in an HE NDP Announcement frame if the AID11 subfield is set to 2047 is defined in Figure 9-61d.



**Figure 9-61d—STA Info subfield format in an HE NDP Announcement frame
if the AID11 subfield is 2047**

The Disallowed Subchannel Bitmap subfield indicates the 20 MHz subchannels and the 242-tone RUs that are present in HE sounding NDPs announced by the HE NDP Announcement frame and the 242-tone RUs that are to be included in requested sounding feedback. A 20 MHz subchannel is as defined in Clause 17 for the portions of the PPDU that use a tone plan as specified in Clause 17 and a 242-tone RU is as defined in 27.3.2). The lowest numbered bit of the Disallowed Subchannel Bitmap subfield corresponds to the 20 MHz subchannel that lies within the BSS bandwidth and that has the lowest frequency of the set of all 20 MHz subchannels within the BSS bandwidth. Each successive bit in the bitmap corresponds to the next higher frequency 20 MHz subchannel. A bit in the bitmap is set to 1 to indicate that for the corresponding 20 MHz subchannel, no energy is present in the HE sounding NDP associated with this HE NDP Announcement frame. For each disallowed 20 MHz subchannel, the 242-tone RU that is most closely aligned in frequency with the 20 MHz subchannel is disallowed for PPDU that use a tone plan as specified in Clause 27. STAs addressed by the HE NDP Announcement frame do not include tones from disallowed 242-tone RUs when determining the average SNR of space time streams 1 to N_c and when generating requested sounding feedback. If a 20 MHz subchannel and its corresponding 242-tone RU is not disallowed, the corresponding bit in the bitmap is set to 0.

Insert the following subclauses [9.3.1.22 through 9.3.1.22.9, including Figure 9-64a through Figure 9-64l, Table 9-29c through Table 9-29k, and Equation (9-b)] after 9.3.1.21:

9.3.1.22 Trigger frame format

9.3.1.22.1 General

A Trigger frame allocates resources for and solicits one or more HE TB PPDU transmissions. The Trigger frame also carries other information required by the responding STA to send an HE TB PPDU.

The format for the Trigger frame is defined in Figure 9-64a.

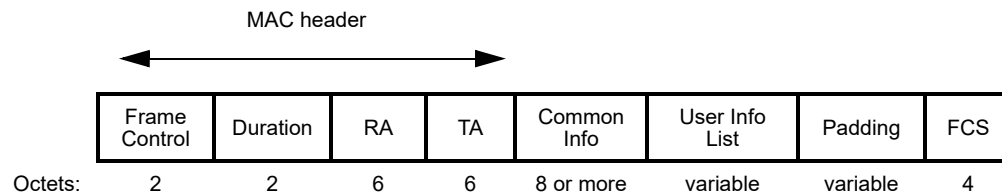


Figure 9-64a—Trigger frame format

The Duration field is set as defined in 9.2.5.

The RA field is set as follows:

- For a Trigger frame that is not a GCR MU-BAR, NFRP or MU-RTS Trigger frame, and that has one User Info field and the AID12 subfield of the User Info field contains the AID of a non-AP STA, the RA field is set to the address of that STA
- For a Trigger frame that has at least one User Info field with the AID12 subfield that allocates an RA-RU, the RA field is set to the broadcast address
- For a Trigger frame that is not a GCR MU-BAR Trigger frame and that has more than one User Info field, the RA field is set to the broadcast address
- For a Trigger frame that is an NFRP Trigger frame or MU-RTS Trigger frame, the RA field is set to the broadcast address
- For a Trigger frame that is a GCR MU-BAR Trigger frame, the RA field is set to the MAC address of the group for which reception status is being requested

The TA field is the address of the STA transmitting the Trigger frame if the Trigger frame is addressed to STAs that belong to a single BSS. The TA field is the transmitted BSSID if the Trigger frame is addressed to STAs from at least two different BSSs of the multiple BSSID set. The rules for setting of the TA field are defined in 26.5.2.2.4.

The Common Info field is defined in Figure 9-64b.

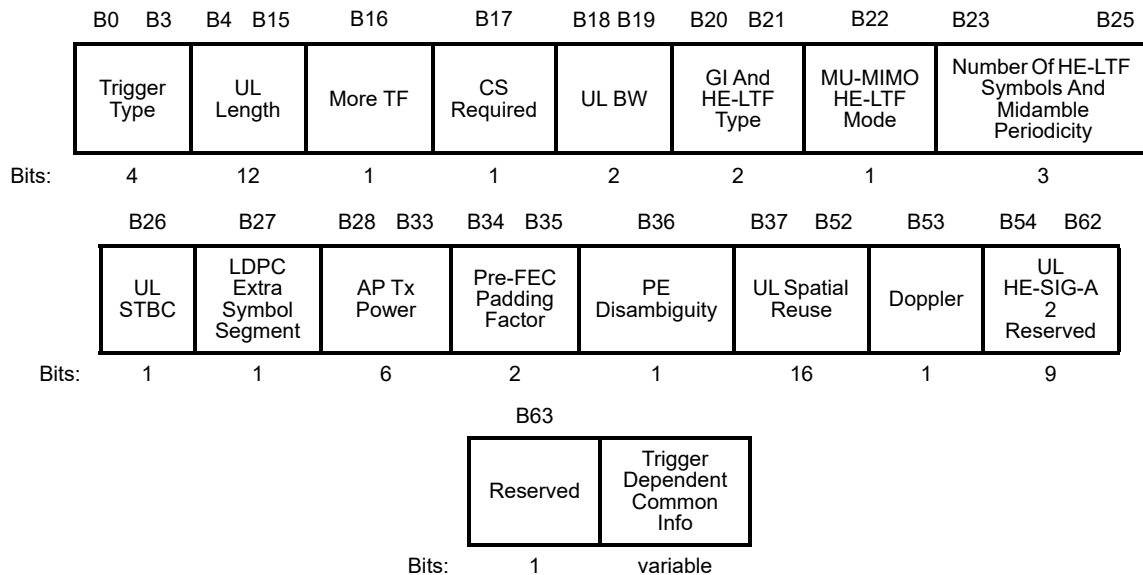


Figure 9-64b—Common Info field format

The Trigger Type subfield identifies the Trigger frame variant and its encoding is defined in Table 9-29c.

Table 9-29c—Trigger Type subfield encoding

Trigger Type subfield value	Trigger frame variant
0	Basic
1	Beamforming Report Poll (BFRP)
2	MU-BAR
3	MU-RTS
4	Buffer Status Report Poll (BSRP)
5	GCR MU-BAR
6	Bandwidth Query Report Poll (BQRP)
7	NDP Feedback Report Poll (NFRP)
8–15	Reserved

The UL Length subfield of the Common Info field indicates the value of the L-SIG LENGTH field of the solicited HE TB PPDU.

The More TF subfield of the Common Info field indicates whether a subsequent Trigger frame is scheduled for transmission. The More TF subfield is set as defined in 26.8.2 and 26.8.3.2.

The CS Required subfield of the Common Info field is set to 1 to indicate that the STAs identified in the User Info fields are required to use ED to sense the medium and to consider the medium state and the NAV in determining whether to respond. The CS Required subfield is set to 0 to indicate that the STAs identified in the User Info fields are not required to consider the medium state or the NAV in determining whether to respond. See 26.5.2.3 and 26.5.2.5 for details.

The UL BW subfield of the Common Info field indicates the bandwidth in the HE-SIG-A field of the HE TB PPDU and is defined in Table 9-29d.

Table 9-29d—UL BW subfield encoding

UL BW subfield value	Description
0	20 MHz
1	40 MHz
2	80 MHz
3	80+80 MHz or 160 MHz

The GI And HE-LTF Type subfield of the Common Info field indicates the GI and HE-LTF type of the HE TB PPDU response. The GI And HE-LTF Type subfield encoding is defined in Table 9-29e.

Table 9-29e—GI And HE-LTF Type subfield encoding

GI And HE-LTF Type subfield value	Description
0	1x HE-LTF + 1.6 μ s GI
1	2x HE-LTF + 1.6 μ s GI
2	4x HE-LTF + 3.2 μ s GI
3	Reserved

The MU-MIMO HE-LTF Mode subfield of the Common Info field indicates the HE-LTF mode for an HE TB PPDU that has an RU that spans the entire bandwidth and that is assigned to more than one non-AP STA (i.e., for UL MU-MIMO) when the GI And HE-LTF Type subfield of the Common Info field indicates either 2x HE-LTF + 1.6 μ s GI or 4x HE-LTF + 3.2 μ s GI, as defined in Table 9-29e. Otherwise, this subfield is set to indicate HE single stream pilot HE-LTF mode. See Table 9-29f.

Table 9-29f—MU-MIMO HE-LTF Mode subfield encoding

MU-MIMO HE-LTF subfield value	Description
0	HE single stream pilot HE-LTF mode
1	HE masked HE-LTF sequence mode

If the Doppler subfield of the Common Info field is 0, then the Number Of HE-LTF Symbols And Midamble Periodicity subfield of the Common Info field indicates the number of HE-LTF symbols present in the HE TB PPDU and is encoded as follows:

- 0 for 1 HE-LTF symbol
- 1 for 2 HE-LTF symbols
- 2 for 4 HE-LTF symbols
- 3 for 6 HE-LTF symbols
- 4 for 8 HE-LTF symbols
- 5–7 are reserved

If the Doppler subfield of the Common Info field is 1, then the Number Of HE-LTF Symbols And Midamble Periodicity subfield indicates the number of HE-LTF symbols and the periodicity of the midamble and is encoded as follows:

- 0 for 1 HE-LTF symbol and 10 symbol midamble periodicity
- 1 for 2 HE-LTF symbols and 10 symbol midamble periodicity
- 2 for 4 HE-LTF symbols and 10 symbol midamble periodicity
- 4 for 1 HE-LTF symbol and 20 symbol midamble periodicity
- 5 for 2 HE-LTF symbols and 20 symbol midamble periodicity

- 6 for 4 HE-LTF symbols and 20 symbol midamble periodicity
- 3 and 7 are reserved

The UL STBC subfield of the Common Info field indicates the status of STBC encoding for the solicited HE TB PPDU. It is set to 1 to indicate STBC encoding and set to 0 otherwise.

The LDPC Extra Symbol Segment subfield of the Common Info field indicates the status of the LDPC extra symbol segment. It is set to 1 if the LDPC extra symbol segment is present in the solicited HE TB PPDU and set to 0 otherwise.

The AP Tx Power subfield of the Common Info field indicates the AP's combined transmit power at the transmit antenna connector of all the antennas used to transmit the triggering PPDU in units of dBm/20 MHz. The transmit power in dBm/20 MHz, P_{TX} , is calculated as $P_{TX} = -20 + F_{Val}$, where F_{Val} is the value of the AP Tx Power subfield, except for the values above 60, which are reserved.

The Pre-FEC Padding Factor and PE Disambiguity subfields are defined in Table 9-29g and have the same encoding as their respective subfields in HE SIG-A (see Table 27-20).

Table 9-29g—Pre-FEC Padding Factor and PE Disambiguity subfields

Subfield	Description	Encoding
Pre-FEC Padding Factor	Indicates the pre-FEC padding factor	Set to 0 to indicate a pre-FEC padding factor of 4 Set to 1 to indicate a pre-FEC padding factor of 1 Set to 2 to indicate a pre-FEC padding factor of 2 Set to 3 to indicate a pre-FEC padding factor of 3
PE Disambiguity	Indicates PE disambiguity	Set to 1 if the condition in Equation (27-118) is met; otherwise, it is set to 0

The UL Spatial Reuse subfield of the Common Info field carries the values to be included in the Spatial Reuse fields in the HE-SIG-A field of the solicited HE TB PPDU. The format of the UL Spatial Reuse subfield is shown in Figure 9-64c, where each Spatial Reuse n subfield, $1 \leq n \leq 4$, is set to the same value as its corresponding subfield in the HE-SIG-A field of the HE TB PPDU, which are defined in Table 27-21.

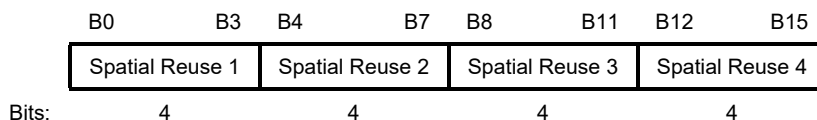


Figure 9-64c—UL Spatial Reuse subfield format

The Doppler subfield of the Common Info field is set to 1 to indicate that a midamble is present in the HE TB PPDU and set to 0 otherwise.

The UL HE-SIG-A2 Reserved subfield of the Common Info field carries the value to be included in the Reserved field in the HE-SIG-A2 subfield of the solicited HE TB PPDU. An HE AP sets the UL HE-SIG-A2 Reserved subfield to all 1s.

The Trigger Dependent Common Info subfield in the Common Info field is optionally present based on the value of the Trigger Type field (see 9.3.1.22.2 to 9.3.1.22.9).

The User Info List field contains zero or more User Info fields.

The User Info field is defined in Figure 9-64d for all Trigger frame variants, except the NFRP Trigger frame, which is defined in 9.3.1.22.9.

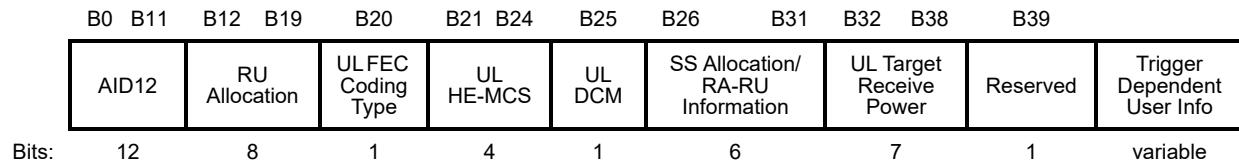


Figure 9-64d—User Info field format

The AID12 subfield in the User Info field is encoded as defined in Table 9-29h:

Table 9-29h—AID12 subfield encoding

AID12 subfield	Description
0	User Info field allocates one or more contiguous RA-RUs for associated STAs
1–2007	User Info field is addressed to an associated STA whose AID is equal to the value in the AID12 subfield
2008–2044	Reserved
2045	User Info field allocates one or more contiguous RA-RUs for unassociated STAs
2046	Unallocated RU
2047–4094	Reserved
4095	Start of Padding field

If the AID12 subfield is 2046, then the remaining subfields in the User Info field are reserved, except for the RU Allocation subfield, which indicates the RU location of the unallocated RU.

If the AID12 subfield is 4095, then the remaining subfields in the User Info field are not present.

The RU Allocation subfield along with the UL BW subfield in the Common Info field identifies the size and the location of the RU. If the UL BW subfield indicates 20 MHz, 40 MHz, or 80 MHz PPDU, then B0 of the RU Allocation subfield is set to 0. If the UL BW subfield indicates 80+80 MHz or 160 MHz, then B0 of the RU Allocation subfield is set to 0 to indicate that the RU allocation applies to the primary 80 MHz channel and set to 1 to indicate that the RU allocation applies to the secondary 80 MHz channel. The mapping of B7–B1 of the RU Allocation subfield for a Trigger frame that is not an MU-RTS Trigger frame is defined in Table 9-29i. See 9.3.1.22.5 for the encoding of the RU Allocation subfield in an MU-RTS Trigger frame.

Table 9-29i—B7–B1 of the RU Allocation subfield

B7–B1 of the RU Allocation subfield	UL BW subfield	RU size	RU Index
0–8	20 MHz, 40 MHz, 80 MHz, 80+80 MHz or 160 MHz	26	RU1 to RU9, respectively
9–17	40 MHz, 80 MHz, 80+80 MHz or 160 MHz		RU10 to RU18, respectively
18–36	80 MHz, 80+80 MHz or 160 MHz		RU19 to RU37, respectively
37–40	20 MHz, 40 MHz, 80 MHz, 80+80 MHz or 160 MHz	52	RU1 to RU4, respectively
41–44	40 MHz, 80 MHz, 80+80 MHz or 160 MHz		RU5 to RU8, respectively
45–52	80 MHz, 80+80 MHz or 160 MHz		RU9 to RU16, respectively
53, 54	20 MHz, 40 MHz, 80 MHz, 80+80 MHz or 160 MHz	106	RU1 and RU2, respectively
55, 56	40 MHz, 80 MHz, 80+80 MHz or 160 MHz		RU3 and RU4, respectively
57–60	80 MHz, 80+80 MHz or 160 MHz		RU5 to RU8, respectively
61	20 MHz, 40 MHz, 80 MHz, 80+80 MHz or 160 MHz	242	RU1
62	40 MHz, 80 MHz, 80+80 MHz or 160 MHz		RU2
63, 64	80 MHz, 80+80 MHz or 160 MHz		RU3 and RU4, respectively
65	40 MHz, 80 MHz, 80+80 MHz or 160 MHz	484	RU1
66	80 MHz, 80+80 MHz or 160 MHz		RU2
67	80 MHz, 80+80 MHz or 160 MHz	996	RU1
68	80+80 MHz or 160 MHz	2×996	RU1
NOTE—If the UL BW subfield indicates 80+80 MHz or 160 MHz, the description indicates the RU index for the primary 80 MHz channel or secondary 80 MHz channel as indicated by B0 of the RU Allocation subfield.			

If the UL BW subfield indicates 20 MHz, the mapping of the RU index to RU is defined in Table 27-7 in increasing order.

If the UL BW subfield indicates 40 MHz, the mapping of the RU index to RU is defined in Table 27-8 in increasing order.

If the UL BW subfield indicates 80 MHz, 160 MHz, or 80+80 MHz, the mapping of the RU index to RU is defined in Table 27-9 in increasing order.

If the UL BW subfield indicates 160 MHz or 80+80 MHz, B7–B1 of the RU Allocation subfield is set to 68 and B0 is set to 1 to indicate a 2×996-tone RU. A non-AP STA ignores B0 for 2×996-tone RU indication.

If the AID12 subfield is in the range 1 to 2007, then the RU Allocation subfield indicates the RU allocated to the STA identified by the AID12 subfield. If the AID12 subfield is 0 or 2045, then the RU Allocation subfield indicates the starting RU of one or more contiguous RA-RUs allocated by the User Info field. If the AID12 subfield is 2046, then the RU Allocation subfield indicates an unallocated RU.

If there is more than one RA-RU (i.e., the Number Of RA-RU subfield of this User Info field has a value greater than 0), then the allocated RUs are contiguous and the RU sizes of all RA-RUs are the same and equal to the size of the first RU. Further, all the remaining subfields of the User Info field apply to all the RA-RUs.

The UL FEC Coding Type subfield of the User Info field indicates the code type of the solicited HE TB PPDU. The UL FEC Coding Type subfield is set to 0 to indicate BCC and set to 1 to indicate LDPC.

The UL HE-MCS subfield of the User Info field indicates the HE-MCS of the solicited HE TB PPDU. The encoding of the UL HE-MCS subfield is defined in 27.3.7.

The UL DCM subfield of the User Info field indicates DCM of the solicited HE TB PPDU. The UL DCM subfield is set to 1 to indicate that DCM is used in the solicited HE TB PPDU as defined in 27.3.12.15. The UL DCM subfield is set to 0 to indicate that DCM is not used. The UL DCM subfield is set to 0 if the UL STBC subfield of the Common Info field is set to 1.

If the AID12 subfield is either 0 or 2045, then B26–B31 of the User Info field is the RA-RU Information subfield; otherwise, B26–B31 of the User Info field is the SS Allocation subfield.

The SS Allocation subfield of the User Info field indicates the spatial streams of the solicited HE TB PPDU and the format is defined in Figure 9-64e.

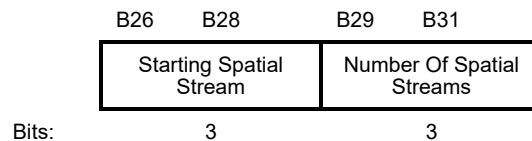


Figure 9-64e—SS Allocation subfield format

The Starting Spatial Stream subfield indicates the starting spatial stream and is set to the starting spatial stream minus 1 (see 26.5.2.3.3).

The Number Of Spatial Streams subfield indicates the number of spatial streams, and is set to the number of spatial streams minus 1.

The RA-RU Information subfield of the User Info field indicates the RA-RU information and the format is defined in Figure 9-64f.

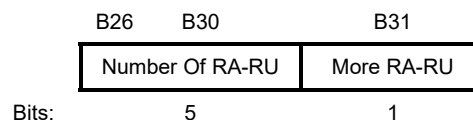


Figure 9-64f—RA-RU Information subfield format

The Number Of RA-RU subfield indicates the number of contiguous RUs allocated for UORA. The value of the Number Of RA-RU subfield is equal to the number of contiguous RA-RUs minus 1.

NOTE—The starting spatial stream and the number of spatial streams of the HE TB PPDU transmitted on each RA-RU are 1.

The More RA-RU subfield is set to 1 to indicate that RA-RUs of the type indicated by the AID12 subfield in this User Info field (see Table 9-29h) are allocated in subsequent Trigger frames that are sent until the end of the TWT SP in which the Trigger frame carrying this field is sent. Otherwise, the subfield is set to 0. The More RA-RU subfield is reserved if the More TF field in the Common Info field is set to 0.

The UL Target Receive Power subfield indicates the expected receive signal power, measured at the AP's antenna connector and averaged over the antennas, for the HE portion of the HE TB PPDU transmitted on the assigned RU and is defined in Table 9-29j.

Table 9-29j—UL Target Receive Power subfield in Trigger frame

UL Target Receive Power subfield	Description
0–90	The expected receive signal power, in units of dBm, is $Target_{pwr} = -110 + F_{val}$, where F_{val} is the subfield value
91–126	Reserved
127	The STA transmits the HE TB PPDU at the STA's maximum transmit power for the assigned HE-MCS. The expected receive signal power is then the STA's maximum transmit power for the assigned HE-MCS minus the path loss.

The Trigger Dependent User Info subfield in the User Info field is optionally present based on the value of the Trigger Type field (see 9.3.1.22.2 to 9.3.1.22.9).

The Padding field is optionally present in a Trigger frame to extend the frame length to give the recipient STAs enough time to prepare a response for transmission a SIFS after the frame is received. The Padding field, if present, is at least two octets in length and is set to all 1s. If the Padding field is present in a Trigger frame, its length is computed as described in 26.5.2.2.3.

9.3.1.22.2 Basic Trigger frame format

The Trigger Dependent Common Info subfield is not present in the Basic Trigger frame. The Trigger Dependent User Info subfield of the Basic Trigger frame is defined in Figure 9-64g.

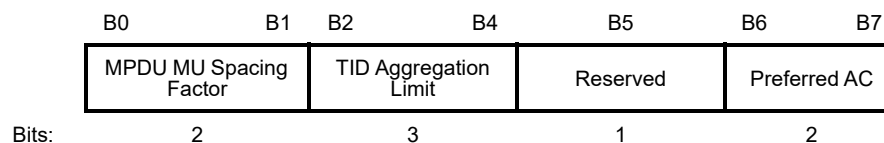


Figure 9-64g—Trigger Dependent User Info subfield format in the Basic Trigger frame

The MPDU MU Spacing Factor subfield is used for calculating the value by which the minimum MPDU start spacing is multiplied (see 10.12.3).

The TID Aggregation Limit subfield indicates the MPDUs allowed in an A-MPDU carried in the HE TB PPDU and the maximum number of TIDs that can be aggregated by the STA in the A-MPDU and is set as defined in 26.5.2.2.4.

The value in the TID Aggregation Limit subfield in Trigger frame is less than or equal to $MT + 1$, where MT is the value indicated in the Multi-TID Aggregation Tx Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element transmitted by the non-AP STA that is the intended receiver of the User Info field.

The Preferred AC subfield indicates the lowest AC that is recommended for aggregation of MPDUs in the A-MPDU contained in the HE TB PPDU sent as a response to the Trigger frame. The encoding of the Preferred AC subfield is as defined in Table 9-154.

9.3.1.22.3 BFRP Trigger frame format

The Trigger Dependent Common Info subfield is not present in the BFRP Trigger frame. The Trigger Dependent User Info subfield of the BFRP Trigger frame is defined in Figure 9-64h.

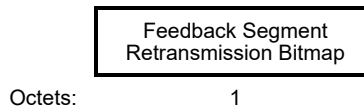


Figure 9-64h—Trigger Dependent User Info subfield format in the BFRP Trigger frame

The Feedback Segment Retransmission Bitmap subfield indicates the requested feedback segments of an HE compressed beamforming report. If the bit in position n ($n = 0$ for LSB and $n = 7$ for MSB) is 1, then the feedback segment with the Remaining Feedback Segments subfield in the HE MIMO Control field equal to n is requested. If the bit in position n is 0, then the feedback segment with the Remaining Feedback Segments subfield in the HE MIMO Control field equal to n is not requested.

9.3.1.22.4 MU-BAR Trigger frame format

The Trigger Dependent Common Info subfield is not present in the MU-BAR Trigger frame. The Trigger Dependent User Info subfield for the MU-BAR Trigger frame is defined in Figure 9-64i.

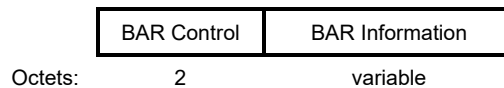


Figure 9-64i—Trigger Dependent User Info subfield format in the MU-BAR Trigger frame

The BAR Control subfield is defined in 9.3.1.7 and indicates either a Compressed BlockAckReq variant or a Multi-TID BlockAckReq variant.

The BAR Information subfield is defined in 9.3.1.7.

9.3.1.22.5 MU-RTS Trigger frame format

The Trigger Dependent Common Info subfield and Trigger Dependent User Info subfield are not present in the MU-RTS Trigger frame.

The UL BW subfield in the Common Info field indicates the bandwidth of the PPDU carrying the MU-RTS Trigger frame and is defined in Table 9-29d.

The UL Length, GI And HE-LTF Type, MU-MIMO HE-LTF Mode, Number Of HE-LTF Symbols And Midamble Periodicity, UL STBC, LDPC Extra Symbol Segment, AP Tx Power, Pre-FEC Padding Factor, PE Disambiguity, UL Spatial Reuse, Doppler and UL HE-SIG-A2 Reserved subfields in the Common Info field are reserved.

The UL HE-MCS, UL FEC Coding Type, UL DCM, SS Allocation/RA-RU Information and UL Target Receive Power fields in the User Info field are reserved.

The RU Allocation subfield in the User Info field addressed to the STA indicates whether the CTS frame is transmitted on the primary 20 MHz channel, primary 40 MHz channel, primary 80 MHz channel, 160 MHz channel, or 80+80 MHz channel.

B0 of the RU Allocation subfield is set to 0 to indicate primary 20 MHz channel, primary 40 MHz channel and primary 80 MHz channel. For 160 MHz and 80+80 MHz indication, B0 of the RU Allocation subfield is set to 1. A non-AP STA ignores B0 for 160 MHz and 80+80 MHz indication.

B7–B1 of the RU Allocation subfield is set to indicate the primary 20 MHz channel as follows:

- 61 if the primary 20 MHz channel is the only 20 MHz channel or the lowest frequency 20 MHz channel in the primary 40 MHz channel or primary 80 MHz channel
- 62 if the primary 20 MHz channel is the second lowest frequency 20 MHz channel in the primary 40 MHz channel or primary 80 MHz channel
- 63 if the primary 20 MHz channel is the third lowest frequency 20 MHz channel in the primary 80 MHz channel
- 64 if the primary 20 MHz channel is the fourth lowest frequency 20 MHz channel in the primary 80 MHz channel

B7–B1 of the RU Allocation subfield is set to indicate the primary 40 MHz channel as follows:

- 65 if the primary 40 MHz channel is the only 40 MHz channel or the lowest frequency 40 MHz channel in the primary 80 MHz channel
- 66 if the primary 40 MHz channel is the second lowest frequency 40 MHz channel in the primary 80 MHz channel

B7–B1 of the RU Allocation subfield is set to 67 to indicate the primary 80 MHz channel.

B7–B1 of the RU Allocation subfield is set to 68 to indicate the primary and secondary 80 MHz channel.

The settings for B7–B1 of the RU Allocation subfield are illustrated in Figure 9-64j.

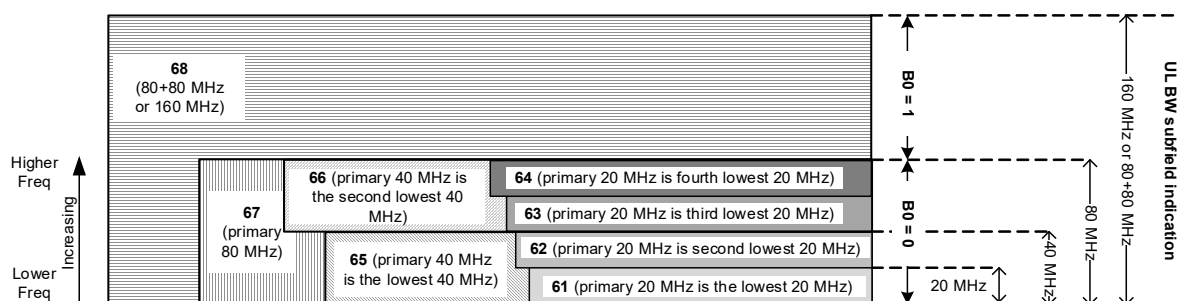


Figure 9-64j—UL BW subfield and B7–B1 of RU Allocation subfield in MU-RTS Trigger frame

9.3.1.22.6 BSRP Trigger frame format

The Trigger Dependent Common Info subfield and Trigger Dependent User Info subfield are not present in the BSRP Trigger frame.

9.3.1.22.7 GCR MU-BAR Trigger frame format

The Trigger Dependent Common Info subfield of the GCR MU-BAR Trigger frame is defined in Figure 9-64k. The Trigger Dependent User Info subfield is not present in the GCR MU-BAR Trigger frame.

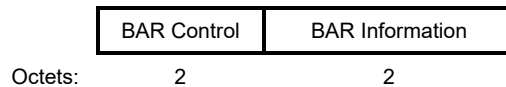


Figure 9-64k—Trigger Dependent Common Info subfield format in the GCR MU-BAR Trigger frame

The BAR Control subfield is defined in 9.3.1.7 and indicates a GCR BlockAckReq variant.

The BAR Information subfield is defined in 9.3.1.7.5, except that the GCR Group Address field is not present.

NOTE—A GCR MU-BAR Trigger frame is not a type of MU BAR Trigger frame.

9.3.1.22.8 BQRP Trigger frame format

The Trigger Dependent Common Info subfield and Trigger Dependent User Info subfield are not present in the BQRP Trigger frame.

9.3.1.22.9 NFRP Trigger frame format

The UL BW subfield in the Common Info field indicates the bandwidth of the NDP feedback report response.

The UL STBC, LDPC Extra Symbol Segment, Pre-FEC Padding Factor, PE Disambiguity, UL Spatial Reuse, and Doppler subfields in the Common Info field are reserved.

The Number Of HE-LTF Symbols And Midamble Periodicity subfield in the Common Info field indicates the number of HE-LTF symbols present in the NDP feedback report response and is set to 1.

The GI And HE-LTF Type subfield in the Common Info field is set to 2.

The Trigger Dependent Common Info subfield is not present.

The User Info field for NFRP Trigger frame is defined in Figure 9-64l.

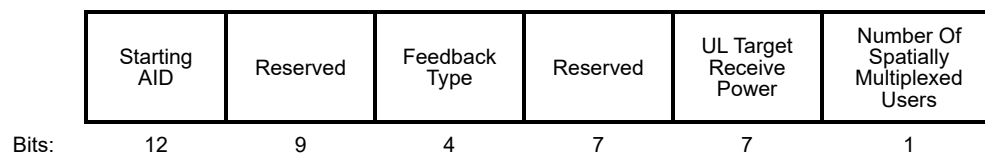


Figure 9-64l—User Info field format in the NFRP Trigger frame

The Feedback Type subfield encoding is defined in Table 9-29k.

Table 9-29k—Feedback Type subfield encoding

Value	Description
0	Resource request
1–15	Reserved

The scheduled non-AP HE STAs are identified by a range of AIDs. The Starting AID field defines the first AID of the range of AIDs that are scheduled to respond to the NFRP Trigger frame.

The UL Target Receive Power subfield indicates the expected receive signal power, measured at the AP's antenna connector and averaged over the antennas, for the HE portion of the HE TB PPDU transmitted on the assigned RU and is defined in Table 9-29j.

The Number Of Spatially Multiplexed Users subfield indicates the number of STAs that are multiplexed on the same set of tones in the same RU, and is encoded as the number of STAs minus 1.

The number of STAs, N_{STA} , that are scheduled to respond to the NFRP Trigger frame is calculated using Equation (9-b).

$$N_{STA} = 18 \times 2^{BW} \times (MultiplexingFlag + 1) \quad (9-b)$$

where BW is the value of the UL BW subfield in the Common Info field of the NFRP Trigger frame, and $MultiplexingFlag$ is the value of the Number Of Spatially Multiplexed Users subfield.

9.3.2 Data frames

9.3.2.1 Format of Data frames

9.3.2.1.2 Address and BSSID fields

Change the first paragraph in 9.3.2.1.2 as follows:

The content of the address fields of Data frames is dependent upon the values of the To DS and From DS subfields in the Frame Control field and whether the Frame Body field contains either an MSDU (or fragment thereof) or an ~~entire~~ A-MSDU (or fragment thereof), as determined by the A-MSDU Present subfield of the QoS Control field (see 9.2.4.5.9). The content of the address fields transmitted by nonmesh STAs is defined in Table 9-30. The content of the address fields transmitted by mesh STAs is defined in 9.3.5, and the content of the fields transmitted by GLK STAs is defined in 10.65. Where the content of a field is shown as not applicable (N/A), the field is omitted. Note that Address 1 always holds the receiver address of the intended receiver (or, in the case of group addressed frames, receivers) and that Address 2 always holds the address of the STA that is transmitting the frame.

Change the third and fourth paragraphs in 9.3.2.1.2 as follows (NOTE 1 remains unchanged):

The DA field contains the destination of the MSDU (or fragment thereof) or A-MSDU (or fragment thereof) in the Frame Body field.

The SA field contains the address of the MAC entity that initiated the MSDU (or fragment thereof) or A-MSDU (or fragment thereof) in the Frame Body field.

Change the sixth paragraph in 9.3.2.1.2 as follows:

When a Data frame carries a basic A-MSDU (or fragment thereof), the DA and SA values related to each MSDU carried by the A-MSDU are carried within the A-MSDU subframe header. Zero, one, or both of these fields are present in the Address 1 and Address 2 fields as indicated in Table 9-30.

9.3.3 (PV0) Management frames

9.3.3.2 Beacon frame format

Change Table 9-32 as follows (not all rows are shown):

Table 9-32—Beacon frame body

Order	Information	Notes
...		
19	EDCA Parameter Set	The EDCA Parameter Set element is present if dot11QosOptionImplemented is true, and dot11MeshActivated is false, and the QoS Capability element is not present; <u>otherwise, it is not present.</u>
20	QoS Capability	The QoS Capability element is present if dot11QosOptionImplemented is true, and dot11MeshActivated is false, and <u>neither the EDCA Parameter Set element nor the MU EDCA Parameter Set element is not present; otherwise, it is not present.</u>
...		
33	HT Capabilities	The HT Capabilities element is present when dot11HighThroughputOptionImplemented is true <u>and the STA is not a STA 6G.</u>
34	HT Operation	The HT Operation element is included by an AP and a mesh STA when dot11HighThroughputOptionImplemented is true <u>and the AP or mesh STA is not a STA 6G.</u>
...		
56	VHT Capabilities	The VHT Capabilities element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G.</u>
57	VHT Operation	The VHT Operation element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G and is optionally present if dot11HEOptionImplemented is true;</u> otherwise, it is not present.

Table 9-32—Beacon frame body (continued)

Order	Information	Notes
58	Transmit Power Envelope element	One Transmit Power Envelope element is present for each distinct combination of values of the Local Maximum Transmit Power Unit Interpretation subfield and Maximum Transmit Power Category subfield that is supported for the BSS if both of the following conditions are met: — Either dot11VHTOptionImplemented or dot11ExtendedSpectrumManagementImplemented is true. — Either dot11SpectrumManagementRequired is true or dot11RadioMeasurementActivated is true. Otherwise, this element is not present. NOTE—In a 6 GHz HE AP, both dot11VHTOptionImplemented (see 26.17.1) and dot11SpectrumManagementRequired (see 26.17.2.1) are true.
...		
63	Reduced Neighbor Report	The One or more Reduced Neighbor Report elements are is optionally present if dot11TVHTOptionImplemented, or dot11FILSActivated, or dot11CollocatedRNRImplemented is true; otherwise, they are not present.
...		
76	<u>Multiple BSSID Configuration</u>	<u>The Multiple BSSID Configuration element is optionally present if dot11MultiBSSIDImplemented is true.</u>
77	<u>HE Capabilities</u>	<u>The HE Capabilities element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
78	<u>HE Operation</u>	<u>The HE Operation element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
79	<u>TWT</u>	<u>The TWT element is optionally present if dot11TWTOptionActivated is true; otherwise, it is not present.</u>
80	<u>UORA Parameter Set</u>	<u>The UORA Parameter Set element is optionally present if dot11OFDMARandomAccessOptionImplemented is true; otherwise, it is not present.</u>
81	<u>BSS Color Change Announcement</u>	<u>The BSS Color Change Announcement element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
82	<u>Spatial Reuse Parameter Set</u>	<u>The Spatial Reuse Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
83	<u>MU EDCA Parameter Set</u>	<u>The MU EDCA Parameter Set element is present if dot11HEOptionImplemented is true, dot11MeshActivated is false, dot11MUEDCAParametersActivated is true, and the QoS Capability element is not present; otherwise, it is not present.</u>
84	<u>ESS Report</u>	<u>The ESS Report element is optionally present.</u>
85	<u>NDP Feedback Report Parameter Set</u>	<u>The NDP Feedback Report Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
86	<u>HE BSS Load</u>	<u>The HE BSS Load element is optionally present if dot11QBSSLoadImplemented and dot11HEOptionImplemented are true.</u>

Table 9-32—Beacon frame body (continued)

Order	Information	Notes
87	<u>HE 6 GHz Band Capabilities</u>	<u>The HE 6 GHz Band Capabilities element is present if dot11HEOptionImplemented and dot11HE6GOptionImplemented are true.</u>
88	<u>TWT Constraint Parameters</u>	<u>The TWT Constraint Parameters element is optionally present if dot11TWTOptionActivated is true; otherwise, it is not present.</u>
...		

9.3.3.5 Association Request frame format

Change Table 9-34 as follows (not all rows are shown):

Table 9-34—Association Request frame body

Order	Information	Notes
...		
13	HT Capabilities	The HT Capabilities element is present when dot11HighThroughputOptionImplemented is true <u>and the STA is not a STA 6G.</u>
...		
22	VHT Capabilities	The VHT Capabilities element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G.</u>
...		
46	<u>HE Capabilities</u>	<u>The HE Capabilities element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
47	<u>Channel Switch Timing</u>	<u>The Channel Switch Timing element is optionally present if dot11HESubchannelSelectiveTransmissionImplemented is true; otherwise, it is not present.</u>
48	<u>HE 6 GHz Band Capabilities</u>	<u>The HE 6 GHz Band Capabilities element is present if dot11HEOptionImplemented and dot11HE6GOptionImplemented are true; otherwise, it is not present.</u>
49	<u>UL MU Power Capabilities</u>	<u>The UL MU Power Capabilities element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
50	<u>TWT Constraint Parameters</u>	<u>The TWT Constraint Parameters element is optionally present if dot11TWTOptionActivated is true; otherwise, it is not present.</u>
...		

9.3.3.6 Association Response frame format

Change Table 9-35 as follows (not all rows are shown):

Table 9-35—Association Response frame body

Order	Information	Notes
...		
15	HT Capabilities	The HT Capabilities element is present when dot11HighThroughputOptionImplemented is true <u>and the STA is not a STA 6G.</u>
16	HT Operation	The HT Operation element is included by an AP and a mesh STA when dot11HighThroughputOptionImplemented is true <u>and the STA is not a STA 6G.</u>
...		
29	VHT Capabilities	The VHT Capabilities element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G.</u>
30	VHT Operation	The VHT Operation element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G and is optionally present if dot11HEOptionImplemented is true;</u> otherwise, it is not present.
...		
40	TWT	The TWT element is present if dot11TWTOptionActivated is true and the TWT element is present in the Association Request frame that elicited the Association Response frame. <u>The TWT element is optionally present if dot11TWTOptionActivated is true, dot11HEOptionImplemented is true, and the TWT Requester Support field in the HE Capabilities element in the Association Request frame that elicited this Association Response frame is 1.</u> <u>Otherwise, the TWT element is not present.</u> <u>If the TWT element is present in the Association Request frame that solicits the Association Response frame but the TWT element is not present in the Association Response frame, then the STA can transmit another TWT request frame after association.</u>
...		
<u>57</u>	<u>HE Capabilities</u>	<u>The HE Capabilities element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>58</u>	<u>HE Operation</u>	<u>The HE Operation element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>59</u>	<u>BSS Color Change Announcement</u>	<u>The BSS Color Change Announcement element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>60</u>	<u>Spatial Reuse Parameter Set</u>	<u>The Spatial Reuse Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>61</u>	<u>MU EDCA Parameter Set</u>	<u>The MU EDCA Parameter Set element is present if dot11HEOptionImplemented and dot11MUEDCAParametersActivated are true; otherwise, it is not present.</u>

Table 9-35—Association Response frame body (continued)

Order	Information	Notes
<u>62</u>	<u>UORA Parameter Set</u>	<u>The UORA Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>63</u>	<u>ESS Report</u>	<u>The ESS Report element is optionally present.</u>
<u>64</u>	<u>NDP Feedback Report Parameter Set</u>	<u>The NDP Feedback Report Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>65</u>	<u>HE 6 GHz Band Capabilities</u>	<u>The HE 6 GHz Band Capabilities element is present if dot11HEOptionImplemented and dot11HE6GOptionImplemented are true.</u>
<u>66</u>	<u>TWT Constraint Parameters</u>	<u>The TWT Constraint Parameters element is optionally present if dot11TWTOptionActivated is true; otherwise, it is not present.</u>
...		

9.3.3.7 Reassociation Request frame format

Change Table 9-36 as follows (not all rows are shown):

Table 9-36—Reassociation Request frame body

Order	Information	Notes
...		
16	HT Capabilities	The HT Capabilities element is present when dot11HighThroughputOptionImplemented is true <u>and the STA is not a STA 6G.</u>
...		
27	VHT Capabilities	The VHT Capabilities element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G.</u>
...		
<u>51</u>	<u>HE Capabilities</u>	<u>The HE Capabilities element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>52</u>	<u>Channel Switch Timing</u>	<u>The Channel Switch Timing element is optionally present if dot11HESubchannelSelectiveTransmissionImplemented is true; otherwise, it is not present.</u>
<u>53</u>	<u>HE 6 GHz Band Capabilities</u>	<u>The HE 6 GHz Band Capabilities element is present if dot11HEOptionImplemented and dot11HE6GOptionImplemented are true.</u>
<u>54</u>	<u>UL MU Power Capabilities</u>	<u>The UL MU Power Capabilities element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
...		

9.3.3.8 Reassociation Response frame format

Change Table 9-37 as follows (not all rows are shown):

Table 9-37—Reassociation Response frame body

Order	Information	Notes
...		
16	HT Capabilities	The HT Capabilities element is present when dot11HighThroughputOptionImplemented is true <u>and the STA is not a STA 6G.</u>
17	HT Operation	The HT Operation element is included by an AP when dot11HighThroughputOptionImplemented is true <u>and the AP is not a STA 6G.</u>
...		
32	VHT Capabilities	The VHT Capabilities element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G.</u>
33	VHT Operation	The VHT Operation element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G</u> and is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
...		
43	TWT	The TWT element is present if dot11TWTOptionActivated is true and the TWT element is present in the Reassociation Request frame that elicited this Reassociation Response frame. <u>The TWT element is optionally present if dot11TWTOptionActivated is true, dot11HEOptionImplemented is true, and the TWT Requester Support field in the HE Capabilities element in the Reassociation Request frame that elicited this Reassociation Response frame is 1.</u> <u>Otherwise, the TWT element is not present.</u> <u>If the TWT element is present in the Reassociation Request frame that solicits the Reassociation Response frame but the TWT element is not present in the Reassociation Response frame, then the STA can transmit another TWT request frame after association.</u>
...		
<u>61</u>	<u>HE Capabilities</u>	<u>The HE Capabilities element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>62</u>	<u>HE Operation</u>	<u>The HE Operation element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>63</u>	<u>BSS Color Change Announcement</u>	<u>The BSS Color Change Announcement element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>64</u>	<u>Spatial Reuse Parameter Set</u>	<u>The Spatial Reuse Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>65</u>	<u>MU EDCA Parameter Set</u>	<u>The MU EDCA Parameter Set element is present if dot11HEOptionImplemented is true and dot11MUEDCAParametersActivated is true; otherwise, it is not present.</u>

Table 9-37—Reassociation Response frame body (continued)

Order	Information	Notes
<u>66</u>	<u>UORA Parameter Set</u>	<u>The UORA Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>67</u>	<u>ESS Report</u>	<u>The ESS Report element is optionally present.</u>
<u>68</u>	<u>NDP Feedback Report Parameter Set</u>	<u>The NDP Feedback Report Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>69</u>	<u>HE 6 GHz Band Capabilities</u>	<u>The HE 6 GHz Band Capabilities element is present if dot11HEOptionImplemented and dot11HE6GOptionImplemented are true.</u>
...		

9.3.3.9 Probe Request frame format

Change Table 9-38 (not all rows are shown):

Table 9-38—Probe Request frame body

Order	Information	Notes
...		
<u>7</u>	HT Capabilities	The HT Capabilities element is present when dot11HighThroughputOptionImplemented is true <u>and the STA is not a STA 6G.</u>
...		
<u>17</u>	VHT Capabilities	The VHT Capabilities element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G.</u>
...		
<u>34</u>	<u>HE Capabilities</u>	<u>The HE Capabilities element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>35</u>	<u>Known BSSID</u>	<u>The Known BSSID element is optionally present if dot11MultiBSSIDImplemented is true.</u>
<u>36</u>	<u>HE 6 GHz Band Capabilities</u>	<u>The HE 6 GHz Band Capabilities element is present if dot11HEOptionImplemented and dot11HE6GOptionImplemented are true.</u>
<u>37</u>	<u>Short SSID List</u>	<u>The Short SSID List element is optionally present if dot11ShortSSIDListImplemented is true; otherwise, it is not present.</u>
<u>38</u>	<u>TWT Constraint Parameters</u>	<u>The TWT Constraint Parameters element is optionally present if dot11TWTOptionActivated is true; otherwise, it is not present.</u>
...		

9.3.3.10 Probe Response frame format

Change Table 9-39 as follows (not all rows are shown):

Table 9-39—Probe Response frame body

Order	Information	Notes
...		
31	HT Capabilities	The HT Capabilities element is present when dot11HighThroughputOptionImplemented is true <u>and the STA is not a STA 6G.</u>
32	HT Operation	The HT Operation element is included by an AP and a mesh STA when dot11HighThroughputOptionImplemented is true <u>and the AP is not a STA 6G.</u>
...		
58	VHT Capabilities	The VHT Capabilities element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G.</u>
59	VHT Operation	The VHT Operation element is present when dot11VHTOptionImplemented is true <u>and the STA is not a STA 6G and is optionally present if dot11HEOptionImplemented is true;</u> otherwise, it is not present.
60	Transmit Power Envelope element	One Transmit Power Envelope element is present for each distinct <u>combination of values of the Local Maximum Transmit Power Unit Interpretation subfield and Maximum Transmit Power Category subfield</u> that is supported for the BSS if both of the following conditions are met: — <u>Either dot11VHTOptionImplemented or dot11ExtendedSpectrumManagementImplemented is true.</u> — <u>Either dot11SpectrumManagementRequired is true or dot11RadioMeasurementActivated is true.</u> Otherwise, this element is not present. NOTE—In a 6 GHz HE AP, both dot11VHTOptionImplemented (see 26.17.1) and dot11SpectrumManagementRequired (see 26.17.2.1) are <u>true.</u>
...		
65	Reduced Neighbor Report	The <u>One or more Reduced Neighbor Report elements are is</u> optionally present if dot11TVHTOptionImplemented, or dot11FILSActivated, or dot11ColocatedRNRImplemented <u>is true;</u> otherwise, they are not present.
...		
92 94	RSN Extension	...
93	<u>Multiple BSSID Configuration</u>	<u>The Multiple BSSID Configuration element is optionally present if dot11MultiBSSIDImplemented is true.</u>
94	<u>HE Capabilities</u>	<u>The HE Capabilities element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
95	<u>HE Operation</u>	<u>The HE Operation element is present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>

Table 9-39—Probe Response frame body (continued)

Order	Information	Notes
<u>96</u>	<u>TWT</u>	<u>The TWT element is optionally present within broadcast Probe Response frames if dot11TWTOptionActivated, dot11HEOptionImplemented and dot11FILSOmitReplicateProbeResponses are true; otherwise, it is not present. If the TWT element is present, then the Negotiation Type subfield of the TWT element is 2.</u>
<u>97</u>	<u>UORA Parameter Set</u>	<u>The UORA Parameter Set element is optionally present if dot11OFDMARandomAccessOptionImplemented is true; otherwise, it is not present.</u>
<u>98</u>	<u>BSS Color Change Announcement</u>	<u>The BSS Color Change Announcement element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>99</u>	<u>Spatial Reuse Parameter Set</u>	<u>The Spatial Reuse Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>100</u>	<u>MU EDCA Parameter Set</u>	<u>The MU EDCA Parameter Set element is present if dot11HEOptionImplemented is true and dot11MUEDCAParametersActivated is true; otherwise, it is not present.</u>
<u>101</u>	<u>ESS Report</u>	<u>The ESS Report element is optionally present.</u>
<u>102</u>	<u>NDP Feedback Report Parameter Set</u>	<u>The NDP Feedback Report Parameter Set element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
<u>103</u>	<u>HE BSS Load</u>	<u>The HE BSS Load element is optionally present if dot11QBSSLoadImplemented and dot11HEOptionImplemented are true.</u>
<u>104</u>	<u>HE 6 GHz Band Capabilities</u>	<u>The HE 6 GHz Band Capabilities element is present if dot11HEOptionImplemented and dot11HE6GOptionImplemented are true.</u>
<u>105</u>	<u>TWT Constraint Parameters</u>	<u>The TWT Constraint Parameters element is optionally present if dot11TWTOptionActivated is true; otherwise, it is not present.</u>
...		

9.3.3.13 Action frame format

Change Table 9-43 as follows (not all rows are shown):

Table 9-43—Action frame body and Action No Ack frame body

Order	Information
...	
Last – 2	One or more vendor-specific elements are optionally present. These elements are absent when the Category subfield of the Action field is Vendor-Specific, Vendor-Specific Protected, or Self-protected; or when the Category subfield of the Action field is VHT and the VHT Action subfield of the Action field is VHT Compressed Beamforming; <u>or when the Category subfield of the Action field is HE and the HE Action subfield of the Action field is HE Compressed Beamforming/CQI.</u>
...	

9.3.4 Extension frames

9.3.4.2 DMG Beacon

Insert the following row into Table 9-45 in numeric order:

Table 9-45—DMG Beacon frame body

Order	Information	Notes
55	Multiple BSSID Configuration	The Multiple BSSID Configuration element is optionally present if dot11MultiBSSIDImplemented is true.

9.3.4.3 S1G Beacon frame format

Insert the following row into Table 9-46 in numeric order:

Table 9-46—Minimum and full set of optional elements

Order	Information	Notes	Allowed in minimum set	Allowed in full set
17	Multiple BSSID Configuration	The Multiple BSSID Configuration element is optionally present if dot11MultiBSSIDImplemented is true.	No	Yes

9.4 Management and Extension frame body components

9.4.1 Fields that are not elements

9.4.1.7 Reason Code field

Insert the following row into Table 9-49 in numeric order, and change the related Reserved row accordingly:

Table 9-49—Reason codes

Reason code	Name	Meaning
71	POOR_RSSI_CONDITIONS	Disassociated due to poor RSSI.

9.4.1.9 Status Code field

Change Table 9-50 as follows (not all rows are shown):

Table 9-50—Status codes

Status code	Name	Meaning
...		
18	REFUSED_BASIC_RATES_MISMATCH	Association denied due to requesting STA not supporting all of the data rates in the BSSBasicRateSet parameter, the Basic HT-MCS Set field of the HT Operation parameter, or the Basic VHT-MCS And NSS Set field in the VHT Operation parameter, <u>or the Basic HE-MCS And NSS Set field in the HE Operation parameter.</u>
...		
124	<u>DENIED_HE_NOT_SUPPORTED</u>	Reserved. <u>Association denied because the requesting STA does not support HE features.</u>
...		

9.4.1.11 Action field

Insert the following rows into Table 9-51 in numeric order, and change the related Reserved row accordingly:

Table 9-51—Category values

Code	Meaning	See subclause	Robust	Group addressed privacy
30	HE	9.6.31	No	No
31	Protected HE	9.6.32	Yes	No

9.4.1.17 QoS Info field

Change Figure 9-100 as follows:

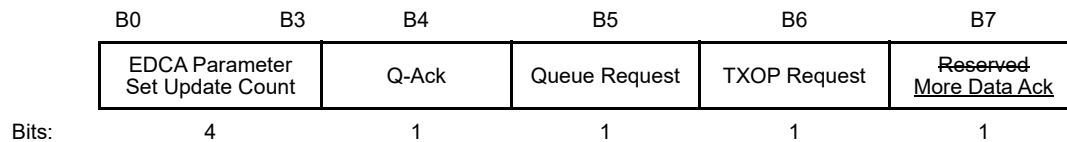


Figure 9-100—QoS Info field format when sent by an AP

Change the third paragraph in 9.4.1.17 as follows:

The EDCA Parameter Set Update Count subfield indicates when the EDCA parameters, and for an HE BSS, the MU EDCA parameters have changed is described in (see 10.2.3.2).

Insert the following paragraph into 9.4.1.17 after the sixth paragraph (“APs set the TXOP Request subfield”):

An HE AP sets the More Data Ack subfield to 1 to indicate that it can generate individually addressed Ack and BlockAck frames with the More Data bit in the Frame Control field equal to 1; otherwise, the AP sets the More Data Ack subfield to 0. For a non-HE AP, the More Data Ack subfield is reserved.

Change the now 12th paragraph of 9.4.1.17 as follows:

Non-AP non-HE STAs set the More Data Ack subfield to 1 to indicate that they can process Ack frames with the More Data bit in the Frame Control field equal to 1 and remain in the awake state. Non-AP HE STAs set the More Data Ack subfield to 1 to indicate that they can process Ack and BlockAck frames with the More Data bit in the Frame Control field equal to 1 and remain in the awake state. Non-AP STAs set the More Data Ack subfield to 0 otherwise. ~~For APs, the More Data Ack subfield is reserved.~~

Insert the following paragraph at the end of 9.4.1.17:

An HE TDLS peer STA uses the More Data Ack subfield to indicate support for both processing and generating Ack and BlockAck frames.

9.4.1.45 Band ID field

Insert the following row into Table 9-67 in numeric order, and change the Reserved row accordingly:

Table 9-67—Band ID field

Band ID value	Meaning
7	6 GHz

9.4.1.53 Operating Mode field

Change the following row in Table 9-80 as shown:

Table 9-80—Subfield values of the Operating Mode field

Subfield	Description
Rx NSS	If the STA that transmits the Operating Mode field (STA1) and the receiver of the Operating Mode field (STA2) are not both HE STAs and if the Rx NSS Type subfield is 0, <u>then</u> this field, combined with other information described in 9.4.2.157.3, indicates the maximum number of spatial streams that the STA STA1 can receive. If the STA that transmits the Operating Mode field (STA1) and the receiver of the Operating Mode field (STA2) are both HE STAs and if the Rx NSS Type subfield is 0, then the following apply: — <u>The value of this field, combined with other information described in 9.4.2.157.3, indicates the maximum number of spatial streams that the HE STA can receive in a VHT PPDU.</u> — <u>The value of this field, combined with other information described in 9.4.2.248.4, indicates the maximum number of spatial streams that STA1 can receive in an HE PPDU.</u> If the Rx NSS Type subfield is 1, this field indicates the maximum number of spatial streams that the STA can receive as a beamformee in an SU PPDU using a beamforming steering matrix derived from a VHT Compressed Beamforming report with Feedback Type subfield indicating MU in the corresponding VHT Compressed Beamforming frame sent by the STA. In a non-SIG STA: Set to 0 for $N_{SS} = 1$ Set to 1 for $N_{SS} = 2$... Set to 7 for $N_{SS} = 8$ In an SIG STA: Set to 0 for $N_{SS} = 1$ Set to 1 for $N_{SS} = 2$ Set to 2 for $N_{SS} = 3$ Set to 3 for $N_{SS} = 4$ NOTE—In a STA with dot11VHTExtendedNSSBWCcapable equal to true, NSS might be further modified for VHT PPDU per Table 9-81. <u>In an HE STA with dot11VHTExtendedNSSBWCcapable equal to true, NSS might be further modified for HE PPDU per Equation (9-2a).</u>

Insert the following text at the end of 9.4.1.53:

The maximum number of spatial streams that the STA supports in reception for a given HE-MCS as a function of the received HE PPDU bandwidth at an HE STA transmitting an Operating Mode field is defined as

$$\text{floor} (Rx\text{-}NSS\text{-}from\text{-}OMF \times (Max\text{-}HE\text{-}NSS\text{-}at\text{-}BW / Max\text{-}HE\text{-}NSS\text{-}at\text{-}80)) \quad (9\text{-}2a)$$

where

Rx-NSS-from-OMF is Rx NSS from the Operating Mode field transmitted by the STA
Max-HE-NSS-at-BW is the maximum NSS among all HE-MCS at *BW* MHz from the Supported HE-MCS And NSS Set field transmitted by the STA
Max-HE-NSS-at-80 is the maximum N_{SS} among all HE-MCS at 80 MHz from the Supported HE-MCS And NSS Set field transmitted by the STA

NOTE—For operating mode between two HE STAs, the Rx NSS subfield indicates the maximum number of spatial streams at channel widths less than or equal to 80 MHz.

9.4.1.60 TWT Information field

Change Figure 9-142 as follows:

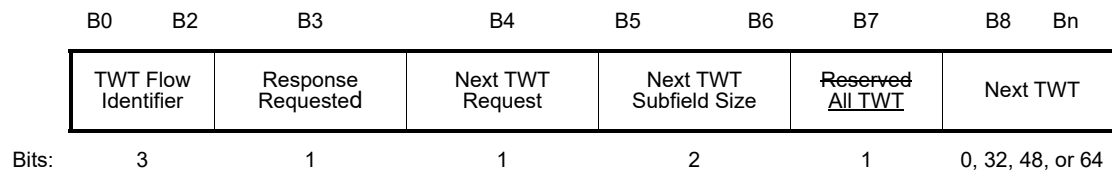


Figure 9-142—TWT Information field format

Change the third paragraph in 9.4.1.60 as follows:

The TWT Flow Identifier subfield contains the TWT flow identifier for which TWT information is requested or being provided. The TWT Flow Identifier subfield is reserved if the All TWT subfield is 1.

Insert the following paragraph into 9.4.1.60 before the last paragraph (“The Next TWT subfield”):

The All TWT subfield is set to 1 by an HE STA to indicate that the TWT Information frame reschedules all TWTs as defined in 26.8.4. Otherwise, it is set to 0.

Insert the following subclauses (9.4.1.64 through 9.4.1.67, including Figure 9-144a and Table 9-91a through Table 9-91h) after 9.4.1.63:

9.4.1.64 HE MIMO Control field

The HE MIMO Control field is defined in Figure 9-144a.

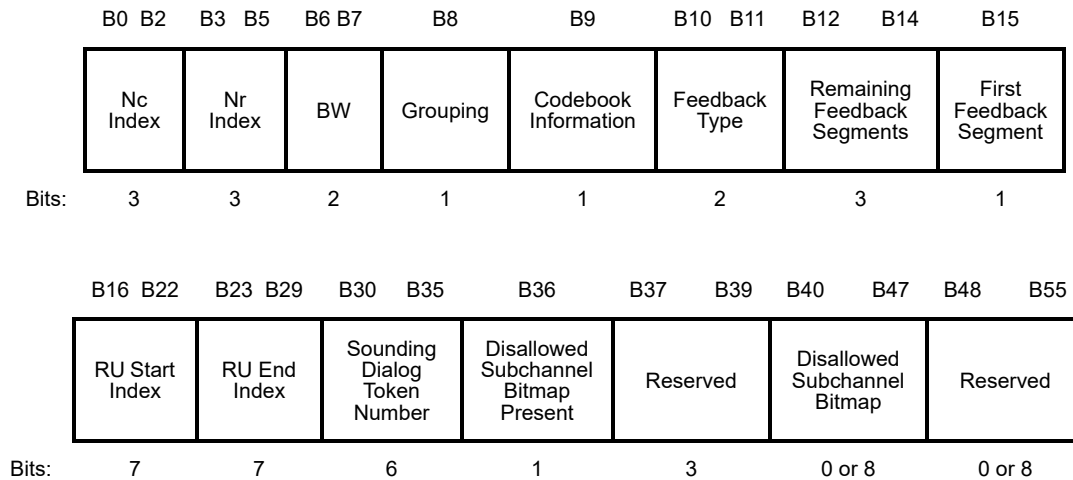


Figure 9-144a—HE MIMO Control field format

The subfields of the HE MIMO Control field are defined in Table 9-91a.

Table 9-91a—HE MIMO Control field encoding

Subfield	Description
Nc Index	If the Feedback Type subfield indicates SU or MU, the Nc Index subfield indicates the number of columns, N_c, in the compressed beamforming feedback matrix and is set to $N_c - 1$. If the Feedback Type subfield indicates CQI, the Nc Index subfield indicates the number of space time streams, N_c, in the CQI Report and is set to $N_c - 1$.
Nr Index	If the Feedback Type subfield indicates SU or MU, then the Nr Index subfield indicates the number of rows, N_r, in the compressed beamforming feedback matrix and is set to $N_r - 1$. The value 0 is reserved. If the Feedback Type subfield indicates CQI, then the Nr Index subfield is reserved.
BW	Indicates the channel width used to determine the starting and ending subcarrier indices when interpreting the RU Start Index and RU End Index subfields. Set to 0 for 20 MHz. Set to 1 for 40 MHz. Set to 2 for 80 MHz. Set to 3 for 160 MHz and 80+80 MHz.

Table 9-91a—HE MIMO Control field encoding (*continued*)

Subfield	Description
Grouping	If the Feedback Type subfield indicates SU or MU, then the Grouping subfield indicates the subcarrier grouping, N_g, used for the compressed beamforming feedback matrix: Set to 0 for $N_g = 4$. Set to 1 for $N_g = 16$. If the Feedback Type subfield indicates CQI, then the Grouping subfield is reserved.
Codebook Information	Indicates the size of codebook entries. If the Feedback Type subfield indicates SU: Set to 0 for 4 bits for ϕ and 2 bits for ψ. Set to 1 for 6 bits for ϕ and 4 bits for ψ. If the Feedback Type subfield indicates MU: Set to 0 for 7 bits for ϕ and 5 bits for ψ. Set to 1 for 9 bits for ϕ and 7 bits for ψ. If the Feedback Type subfield indicates CQI, then the Codebook Information subfield is reserved. NOTE—The codebook size for MU Feedback with $N_g = 16$ is limited to $(\phi, \psi) = \{9, 7\}$.
Feedback Type	Indicates the feedback type: Set to 0 for SU. Set to 1 for MU. Set to 2 for CQI. 3 is reserved.
Remaining Feedback Segments	Indicates the number of remaining feedback segments for the associated HE Compressed Beamforming/CQI frame: Set to 0 for the last feedback segment of a segmented report or the only feedback segment of an unsegmented report. Set to a value between 1 and 7 for a feedback segment that is not the last feedback segment of a segmented report. In a retransmitted feedback segment, the subfield is set to the same value associated with the feedback segment in the original transmission.
First Feedback Segment	Set to 1 for the first feedback segment of a segmented report or the only feedback segment of an unsegmented report. Set to 0 if not the first feedback segment or if the HE Compressed Beamforming Report field and HE MU Exclusive Beamforming Report field are not present in the frame. In a retransmitted feedback segment, the subfield is set to the same value associated with the feedback segment in the original transmission. NOTE—The First Feedback Segment subfield is always set to 0 if the Feedback Type subfield indicates CQI because the HE Compressed Beamforming/CQI Report frame is always less than 11 454 octets in length.
RU Start Index	The starting RU index indicates the first 26-tone RU for which the HE beamformer is requesting feedback.
RU End Index	The ending RU index indicates the last 26-tone RU for which the HE beamformer is requesting feedback.
Sounding Dialog Token Number	Set to the same value as the Sounding Dialog Token Number field in the corresponding HE NDP Announcement frame.

In an HE Compressed Beamforming/CQI frame not carrying all or part of an HE compressed beamforming/CQI report (see 26.7 for a description of such a case), the N_c Index, N_r Index, BW, Grouping, Codebook Information, Feedback Type and Sounding Dialog Token Number subfields are reserved, the First Feedback Segment subfield is set to 0 and the Remaining Feedback Segments subfield is set to 7.

The Disallowed Subchannel Bitmap Present subfield indicates whether a Disallowed Subchannel Bitmap subfield and a reserved field of 8 bits are present in the HE MIMO Control field. These subfields are present if the Disallowed Subchannel Bitmap Present subfield is equal to 1. These subfields are not present if the Disallowed Subchannel Bitmap Present subfield is equal to 0.

The Disallowed Subchannel Bitmap subfield is defined in 9.3.1.19.

9.4.1.65 HE Compressed Beamforming Report field

The HE Compressed Beamforming Report field carries the average SNR of each space-time stream and compressed beamforming feedback matrices V for use by a transmit beamformer to determine steering matrices Q , as described in 10.34.3 and 19.3.12.3.

The size of the HE Compressed Beamforming Report field depends on the values in the HE MIMO Control field. The HE Compressed Beamforming Report field contains HE Compressed Beamforming Report information or successive (possibly zero-length) portions thereof in the case of segmented HE compressed beamforming/CQI report (see 26.7.4). HE Compressed Beamforming Report information is included in the HE compressed beamforming/CQI report if the Feedback Type subfield in the HE MIMO Control field indicates SU or MU. If the HE MIMO Control field contains a Disallowed Subchannel Bitmap subfield, then the HE Compressed Beamforming Report field does not include information for tones that are included within 242-tone RUs that are indicated as disallowed by the bitmap.

The HE Compressed Beamforming Report information contains the channel matrix elements indexed, first, by matrix angles in order shown in Table 9-71 and, second, by data and pilot subcarrier index from lowest frequency to highest frequency. An explanation of how these angles are generated from the beamforming feedback matrix V is given in 19.3.12.3.6, where N_c is the number of columns in a compressed beamforming feedback matrix determined by the N_c Index subfield of the HE MIMO Control field, and N_r is the number of rows in a compressed beamforming feedback matrix determined by the N_r Index subfield of the HE MIMO Control field.

The beamforming feedback matrix V is formed by the beamformee as follows. The beamformer transmits an HE sounding NDP with $N_{STS,NDP}$ space-time streams, where $N_{STS,NDP}$ takes a value between 2 and 8. Based on this HE sounding NDP, the beamformee estimates the $N_{RX,BFEE} \times N_{STS,NDP}$ channel, and based on that channel it determines a $N_r \times N_c$ orthogonal matrix V , where N_r and N_c satisfy Equation (9-1). $N_{RX,BFEE}$ is the number of receiver chains used to receive the HE sounding NDP at the beamformee.

Further restrictions on N_c are described in 27.2. The angles are quantized as defined in Table 9-74 with b_ψ defined by the Codebook Information field of the HE MIMO Control field (see 9.4.1.64).

The HE Compressed Beamforming Report information has the structure and order defined in Table 9-91b, where N_a is the number of angles used for the compressed beamforming feedback matrix (see Table 9-71).

In Table 9-91b, N_s is the number of subcarriers for which a compressed beamforming feedback matrix is sent back to the beamformer. A beamformer or beamformee, depending upon which of the two determines the feedback parameters, reduces N_s by using a method referred to as grouping, in which only a single compressed beamforming feedback matrix is reported for each group of N_g adjacent subcarriers. N_s is a function of the BW, RU Start Index, RU End Index and Grouping subfields in the HE MIMO Control field (see 9.4.1.64).

Table 9-91b—HE Compressed Beamforming Report information

Field	Size (bits)	Meaning
Average SNR of Space-Time Stream 1	8	Signal-to-noise ratio at the beamformee for space-time stream 1 averaged over all data subcarriers. See Table 9-77.
...		...
Average SNR of Space-Time Stream Nc	8	Signal-to-noise ratio at the beamformee for space-time stream Nc averaged over all data subcarriers. See Table 9-77.
Compressed beamforming feedback matrix V for subcarrier $k = scidx(0)$	$Na \times (b_\phi + b_\psi)/2$	Compressed beamforming feedback matrix defined in Table 9-71.
Compressed beamforming feedback matrix V for subcarrier $k = scidx(1)$	$Na \times (b_\phi + b_\psi)/2$	Compressed beamforming feedback matrix defined in Table 9-71.
Compressed beamforming feedback matrix V for subcarrier $k = scidx(2)$	$Na \times (b_\phi + b_\psi)/2$	Compressed beamforming feedback matrix defined in Table 9-71.
...		...
Compressed beamforming feedback matrix V for subcarrier $k = scidx(Ns - 1)$	$Na \times (b_\phi + b_\psi)/2$	Compressed beamforming feedback matrix defined in Table 9-71.

Subcarrier indices $scidx(0)$ and $scidx(Ns - 1)$ are identified by the RU Start Index and RU End Index subfields, respectively, together with the BW and Grouping subfields, as defined in Table 9-91c and Table 9-91d.

Table 9-91c—Subcarrier indices $scidx(0)$ and $scidx(Ns - 1)$ for $Ng = 4$

RU Index	20 MHz		40 MHz		80 MHz		160 MHz		80+80 MHz	
	S	E	S	E	S	E	S	E	S	E
0	−122	−96	−244	−216	−500	−472	−1012	−984	−500(L)	−472(L)
1	−96	−68	−220	−192	−476	−448	−988	−960	−476(L)	−448(L)
2	−68	−40	−192	−164	−448	−420	−960	−932	−448(L)	−420(L)
3	−44	−16	−164	−136	−420	−392	−932	−904	−420(L)	−392(L)
4	−16	16	−136	−108	−392	−364	−904	−876	−392(L)	−364(L)
5	16	44	−112	−84	−368	−340	−880	−852	−368(L)	−340(L)
6	40	68	−84	−56	−340	−312	−852	−824	−340(L)	−312(L)
7	68	96	−56	−28	−312	−284	−824	−796	−312(L)	−284(L)
8	96	122	−32	−4	−288	−260	−800	−772	−288(L)	−260(L)
9			4	32	−260	−232	−772	−744	−260(L)	−232(L)
10			28	56	−232	−204	−744	−716	−232(L)	−204(L)
11			56	84	−204	−176	−716	−688	−204(L)	−176(L)

Table 9-91c—Subcarrier indices $scidx(0)$ and $scidx(N_s - 1)$ for $N_g = 4$ (continued)

RU Index	20 MHz		40 MHz		80 MHz		160 MHz		80+80 MHz	
	S	E	S	E	S	E	S	E	S	E
12			84	112	−180	−152	−692	−664	−180(L)	−152(L)
13			108	136	−152	−124	−664	−636	−152(L)	−124(L)
14			136	164	−124	−96	−636	−608	−124(L)	−96(L)
15			164	192	−100	−72	−612	−584	−100(L)	−72(L)
16			192	220	−72	−44	−584	−556	−72(L)	−44(L)
17			216	244	−44	−16	−556	−528	−44(L)	−16(L)
18					−16	16	−528	−496	−16(L)	16(L)
19					16	44	−496	−468	16(L)	44(L)
20					44	72	−468	−440	44(L)	72(L)
21					72	100	−440	−412	72(L)	100(L)
22					96	124	−416	−388	96(L)	124(L)
23					124	152	−388	−360	124(L)	152(L)
24					152	180	−360	−332	152(L)	180(L)
25					176	204	−336	−308	176(L)	204(L)
26					204	232	−308	−280	204(L)	232(L)
27					232	260	−280	−252	232(L)	260(L)
28					260	288	−252	−224	260(L)	288(L)
29					284	312	−228	−200	284(L)	312(L)
30					312	340	−200	−172	312(L)	340(L)
31					340	368	−172	−144	340(L)	368(L)
32					364	392	−148	−120	364(L)	392(L)
33					392	420	−120	−92	392(L)	420(L)
34					420	448	−92	−64	420(L)	448(L)
35					448	476	−64	−36	448(L)	476(L)
36					472	500	−40	−12	472(L)	500(L)
37							12	40	−500(H)	−472(H)
38							36	64	−476(H)	−448(H)
39							64	92	−448(H)	−420(H)
40							92	120	−420(H)	−392(H)
41							120	148	−392(H)	−364(H)
42							144	172	−368(H)	−340(H)
43							172	200	−340(H)	−312(H)
44							200	228	−312(H)	−284(H)

Table 9-91c—Subcarrier indices $scidx(0)$ and $scidx(N_s - 1)$ for $N_g = 4$ (continued)

RU Index	20 MHz		40 MHz		80 MHz		160 MHz		80+80 MHz	
	S	E	S	E	S	E	S	E	S	E
45							224	252	−288(H)	−260(H)
46							252	280	−260(H)	−232(H)
47							280	308	−232(H)	−204(H)
48							308	336	−204(H)	−176(H)
49							332	360	−180(H)	−152(H)
50							360	388	−152(H)	−124(H)
51							388	416	−124(H)	−96(H)
52							412	440	−100(H)	−72(H)
53							440	468	−72(H)	−44(H)
54							468	496	−44(H)	−16(H)
55							496	528	−16(H)	16(H)
56							528	556	16(H)	44(H)
57							556	584	44(H)	72(H)
58							584	612	72(H)	100(H)
59							608	636	96(H)	124(H)
60							636	664	124(H)	152(H)
61							664	692	152(H)	180(H)
62							688	716	176(H)	204(H)
63							716	744	204(H)	232(H)
64							744	772	232(H)	260(H)
65							772	800	260(H)	288(H)
66							796	824	284(H)	312(H)
67							824	852	312(H)	340(H)
68							852	880	340(H)	368(H)
69							876	904	364(H)	392(H)
70							904	932	392(H)	420(H)
71							932	960	420(H)	448(H)
72							960	988	448(H)	476(H)
73							984	1012	472(H)	500(H)

NOTE 1—S denotes subcarrier index $scidx(0)$, identified by the RU Start Index subfield; E denotes subcarrier index $scidx(N_s - 1)$, identified by the RU End Index subfield.

NOTE 2— $x(L)$ denotes subcarrier index x in the frequency segment lower in frequency, and $x(H)$ denotes subcarrier index x in the frequency segment higher in frequency.

Table 9-91d—Subcarrier indices $scidx(0)$ and $scidx(N_s - 1)$ for $N_g = 16$

RU Index	20 MHz		40 MHz		80 MHz		160 MHz		80+80 MHz	
	S	E	S	E	S	E	S	E	S	E
0	−122	−84	−244	−212	−500	−468	−1012	−980	−500(L)	−468(L)
1	−100	−68	−228	−180	−484	−436	−996	−948	−484(L)	−436(L)
2	−68	−36	−196	−164	−452	−420	−964	−932	−452(L)	−420(L)
3	−52	−4	−164	−132	−420	−388	−932	−900	−420(L)	−388(L)
4	−20	20	−148	−100	−404	−356	−916	−868	−404(L)	−356(L)
5	4	52	−116	−84	−372	−340	−884	−852	−372(L)	−340(L)
6	36	68	−84	−52	−340	−308	−852	−820	−340(L)	−308(L)
7	68	100	−68	−20	−324	−276	−836	−788	−324(L)	−276(L)
8	84	122	−36	−4	−292	−260	−804	−772	−292(L)	−260(L)
9			4	36	−260	−228	−772	−740	−260(L)	−228(L)
10			20	68	−244	−196	−756	−708	−244(L)	−196(L)
11			52	84	−212	−164	−724	−676	−212(L)	−164(L)
12			84	116	−180	−148	−692	−660	−180(L)	−148(L)
13			100	148	−164	−116	−676	−628	−164(L)	−116(L)
14			132	164	−132	−84	−644	−596	−132(L)	−84(L)
15			164	196	−100	−68	−612	−580	−100(L)	−68(L)
16			180	228	−84	−36	−596	−548	−84(L)	−36(L)
17			212	244	−52	−4	−564	−516	−52(L)	−4(L)
18					−20	20	−532	−492	−20(L)	20(L)
19					4	52	−508	−460	4(L)	52(L)
20					36	84	−476	−428	36(L)	84(L)
21					68	100	−444	−412	68(L)	100(L)
22					84	132	−428	−380	84(L)	132(L)
23					116	164	−396	−348	116(L)	164(L)
24					148	180	−364	−332	148(L)	180(L)
25					164	212	−348	−300	164(L)	212(L)
26					196	244	−316	−268	196(L)	244(L)
27					228	260	−284	−252	228(L)	260(L)
28					260	292	−252	−220	260(L)	292(L)
29					276	324	−236	−188	276(L)	324(L)
30					308	340	−204	−172	308(L)	340(L)
31					340	372	−172	−140	340(L)	372(L)

Table 9-91d—Subcarrier indices $scidx(0)$ and $scidx(N_s - 1)$ for $N_g = 16$ (continued)

RU Index	20 MHz		40 MHz		80 MHz		160 MHz		80+80 MHz	
	S	E	S	E	S	E	S	E	S	E
32					356	404	−156	−108	356(L)	404(L)
33					388	420	−124	−92	388(L)	420(L)
34					420	452	−92	−60	420(L)	452(L)
35					436	484	−76	−28	436(L)	484(L)
36					468	500	−44	−12	468(L)	500(L)
37							12	44	−500(H)	−468(H)
38							28	76	−484(H)	−436(H)
39							60	92	−452(H)	−420(H)
40							92	124	−420(H)	−388(H)
41							108	156	−404(H)	−356(H)
42							140	172	−372(H)	−340(H)
43							172	204	−340(H)	−308(H)
44							188	236	−324(H)	−276(H)
45							220	252	−292(H)	−260(H)
46							252	284	−260(H)	−228(H)
47							268	316	−244(H)	−196(H)
48							300	348	−212(H)	−164(H)
49							332	364	−180(H)	−148(H)
50							348	396	−164(H)	−116(H)
51							380	428	−132(H)	−84(H)
52							412	444	−100(H)	−68(H)
53							428	476	−84(H)	−36(H)
54							460	508	−52(H)	−4(H)
55							492	532	−20(H)	20(H)
56							516	564	4(H)	52(H)
57							548	596	36(H)	84(H)
58							580	612	68(H)	100(H)
59							596	644	84(H)	132(H)
60							628	676	116(H)	164(H)
61							660	692	148(H)	180(H)
62							676	724	164(H)	212(H)
63							708	756	196(H)	244(H)

Table 9-91d—Subcarrier indices $scidx(0)$ and $scidx(N_s - 1)$ for $N_g = 16$ (continued)

RU Index	20 MHz		40 MHz		80 MHz		160 MHz		80+80 MHz	
	S	E	S	E	S	E	S	E	S	E
64							740	772	228(H)	260(H)
65							772	804	260(H)	292(H)
66							788	836	276(H)	324(H)
67							820	852	308(H)	340(H)
68							852	884	340(H)	372(H)
69							868	916	356(H)	404(H)
70							900	932	388(H)	420(H)
71							932	964	420(H)	452(H)
72							948	996	436(H)	484(H)
73							980	1012	468(H)	500(H)
NOTE 1—S denotes subcarrier index $scidx(0)$, identified by the RU Start Index subfield; E denotes subcarrier index $scidx(N_s - 1)$, identified by the RU End Index subfield.										
NOTE 2— $x(L)$ denotes subcarrier index x in the frequency segment lower in frequency, and $x(H)$ denotes subcarrier index x in the frequency segment higher in frequency.										

Subcarrier indices $scidx(i)$, $i = 0, \dots, N_s - 1$ are the subset of the subcarrier indices identified by the BW and Grouping subfields, as defined in Table 9-91e, starting with $scidx(0)$ and ending with $scidx(N_s - 1)$, in the order given.

NOTE 1—This implicitly defines N_s .

NOTE 2—For full-bandwidth feedback, subcarrier indices $scidx(i)$, $i = 0, \dots, N_s - 1$ are the entire superset shown in Table 9-91e, in the order given.

Table 9-91e—Subcarrier indices for compressed beamforming feedback matrix

Channel width	N_g	Superset of subcarrier indices ($scidx$)
20 MHz	4	−122, −120, −116, ..., −8, −4, −2, 2, 4, 8, ..., 116, 120, 122
	16	−122, −116, −100, ..., −20, −4, −2, 2, 4, 20, ..., 100, 116, 122
40 MHz	4	−244, −240, ..., −8, −4, 4, 8, ..., 240, 244
	16	−244, −228, ..., −20, −4, 4, 20, ..., 228, 244
80 MHz	4	−500, −496, ..., −8, −4, 4, 8, ..., 496, 500
	16	−500, −484, ..., −20, −4, 4, 20, ..., 484, 500
160 MHz	4	−1012, −1008, ..., −520, −516, −508, −504, ..., −16, −12, 12, 16, ..., 504, 508, 516, 520, ..., 1008, 1012
	16	−1012, −996, ..., −532, −516, −508, −492, ..., −28, −12, 12, 28, ..., 492, 508, 516, 532, ..., 996, 1012

Table 9-91e—Subcarrier indices for compressed beamforming feedback matrix (*continued*)

Channel width	N_g	Superset of subcarrier indices ($scidx$)
80+80 MHz	4	–500(L), –496(L), ..., –8(L), –4(L), 4(L), 8(L), ..., 496(L), 500(L), –500(H), –496(H), ..., –8(H), –4(H), 4(H), 8(H), ..., 496(H), 500(H)
	16	–500(L), –484(L), ..., –20(L), –4(L), 4(L), 20(L), ..., 484(L), 500(L), –500(H), –484(H), ..., –20(H), –4(H), 4(H), 20(H), ..., 484(H), 500(H)
NOTE 1— $x(L)$ denotes subcarrier index x in the frequency segment lower in frequency, and $x(H)$ denotes subcarrier index x in the frequency segment higher in frequency.		
NOTE 2—“...” denotes an arithmetic progression in N_g increments.		
NOTE 3—Pilot subcarriers are not skipped.		

The Average SNR of Space-Time Stream i subfield in Table 9-91b is an 8-bit 2s complement integer defined in Table 9-77.

The $AvgSNR_i$ in Table 9-77 is found by computing the SNR per subcarrier in decibels for the subcarriers identified in Table 9-91c for $N_g = 4$ and Table 9-91d for $N_g = 16$, and then computing the arithmetic mean of those values. Each SNR value per subcarrier in stream i (before being averaged) corresponds to the SNR associated with column i of the beamforming feedback matrix V determined at the beamformee. Each SNR corresponds to the predicted SNR at the beamformee when the beamformer applies all columns of the matrix V .

The computation of the $AvgSNR_i$ values does not include channel information from subcarriers that lie within 242-tone RUs that are indicated as punctured by the Disallowed Subchannel Bitmap subfield, if present, of the HE NDP Announcement frame that solicited the feedback.

Padding is not present between angles in the HE Compressed Beamforming Report information, even if they correspond to different subcarriers. If the size of the HE Compressed Beamforming Report information is not an integer multiple of 8 bits, up to seven 0s are appended to the end of the field to make its size an integer multiple of 8 bits.

9.4.1.66 HE MU Exclusive Beamforming Report field

The HE MU Exclusive Beamforming Report field carries explicit feedback in the form of delta SNRs. The information in the HE Compressed Beamforming Report field and the HE MU Exclusive Beamforming Report field can be used by the transmit MU beamformer to determine the steering matrices Q , as described in 27.3.3.1.

The size of the HE MU Exclusive Beamforming Report field depends on the values in the HE MIMO Control field. The HE MU Exclusive Beamforming Report field contains HE MU Exclusive Beamforming Report information or successive (possibly zero-length) portions thereof in the case of segmented HE compressed beamforming/CQI report (see 26.7.4). HE MU Exclusive Beamforming Report information is included in the HE compressed beamforming/CQI report (in addition to HE Compressed Beamforming Report information) if the Feedback Type subfield in the HE MIMO Control field indicates MU. If the HE MIMO Control field contains a Disallowed Subchannel Bitmap subfield, then the HE MU Exclusive Beamforming Report field does not include information for tones that are included within 242-tone RUs that are indicated as disallowed by the bitmap.

The HE MU Exclusive Beamforming Report information consists of Delta SNR subfields for each of the space-time streams, 1 to N_c , of a subset of subcarriers typically spaced N_g apart, where N_g is signaled in the Grouping subfield of the HE MIMO Control field. The subset of subcarriers starts from the lowest frequency subcarrier and continues to the highest frequency subcarrier. The subcarrier indices of the feedback for each Delta SNR subfield are identical to the subcarrier indices for the compressed beamforming feedback matrix V .

NOTE—The feedback subcarrier spacings are mostly equal to N_g , but there are a few exceptions, generally around the RU edge and the DC tone, where extra feedback subcarriers are added to improve the channel interpolation/extrapolation quality.

No padding is present between $\Delta SNR_{k,i}$ in the HE MU Exclusive Beamforming Report field, even if they correspond to different subcarriers. The subset of subcarriers included is determined by the values of the RU Start Index, RU End Index, and Grouping subfields of the HE MIMO Control field. For each subcarrier included, the deviation in dB of the SNR of that subcarrier for each column of V relative to the average SNR of the corresponding space-time stream is computed using Equation (9-2), except that k is the subcarrier index in the range $scidx(0)$, ..., $scidx(N_s - 1)$ and $\overline{SNR}_i$ is the average SNR of space-time stream i reported in the Average SNR of Space-Time Stream i field of the HE Compressed Beamforming Report Information field. In Equation (9-2), the average SNR value is computed only for subcarriers that are not indicated as disallowed by a Disallowed Subchannel Bitmap subfield, when it is present.

The HE MU Exclusive Beamforming Report information has the structure and order defined in Table 9-91f.

Table 9-91f—HE MU Exclusive Beamforming Report information

Field	Size (bits)	Meaning
Delta SNR for space-time stream 1 for subcarrier $k = scidx(0)$	4	$\Delta SNR_{scidx(0),1}$ as defined in Equation (9-2) as modified above.
...		...
Delta SNR for space-time stream N_c for subcarrier $k = scidx(0)$	4	$\Delta SNR_{scidx(0),N_c}$ as defined in Equation (9-2) as modified above.
Delta SNR for space-time stream 1 for subcarrier $k = scidx(1)$	4	$\Delta SNR_{scidx(1),1}$ as defined in Equation (9-2) as modified above.
...		...
Delta SNR for space-time stream N_c for subcarrier $k = scidx(1)$	4	$\Delta SNR_{scidx(1),N_c}$ as defined in Equation (9-2) as modified above.
...		...
Delta SNR for space-time stream 1 for subcarrier $k = scidx(N_s - 1)$	4	$\Delta SNR_{scidx(N_s - 1),1}$ as defined in Equation (9-2) as modified above.
...		...
Delta SNR for space-time stream N_c for subcarrier $k = scidx(N_s - 1)$	4	$\Delta SNR_{scidx(N_s - 1),N_c}$ as defined in Equation (9-2) as modified above.

In Table 9-91f, N_s and $scidx()$ are defined in 9.4.1.65.

9.4.1.67 HE CQI Report field

The HE CQI Report field carries the per-RU average SNRs of each space-time stream, where each per-RU average SNR is the arithmetic mean of the SNR in decibels over a 26-tone RU for which the feedback is being requested. The HE CQI Report field contains information about the quality of the link.

The size of the HE CQI Report field depends on the values in the HE MIMO Control field. The HE CQI Report field contains HE CQI Report information. HE CQI Report information is included in the HE compressed beamforming/CQI report if the Feedback Type subfield in the HE MIMO Control field indicates CQI feedback. If the HE MIMO Control field contains a Disallowed Subchannel Bitmap subfield, then the HE CQI Report field does not include information for tones that are included within 26-tone RUs that are indicated as disallowed by the bitmap.

The HE CQI Report field has the structure and order defined in Table 9-91g.

Table 9-91g—HE CQI Report information

Field	Size (bits)	Meaning
Average SNR for space-time stream 1 for RU index $k = \text{ruidx}(0)$	6	SNR at the beamformee for space-time stream 1 averaged over 26-tone RU. See Table 9-91h.
...		...
Average SNR for space-time stream N_c for RU index $k = \text{ruidx}(0)$	6	SNR at the beamformee for space-time stream N_c averaged over 26-tone RU. See Table 9-91h.
Average SNR for space-time stream 1 for RU index $k = \text{ruidx}(1)$	6	SNR at the beamformee for space-time stream 1 averaged over 26-tone RU. See Table 9-91h.
...		...
Average SNR for space-time stream N_c for RU index $k = \text{ruidx}(1)$	6	SNR at the beamformee for space-time stream N_c averaged over 26-tone RU. See Table 9-91h.
...		...
Average SNR for space-time stream 1 for RU index $k = \text{ruidx}(N_{cqi} - 1)$	6	SNR at the beamformee for space-time stream 1 averaged over 26-tone RU. See Table 9-91h.
...		...
Average SNR for space-time stream N_c for RU index $k = \text{ruidx}(N_{cqi} - 1)$	6	SNR at the beamformee for space-time stream N_c averaged over 26-tone RU. See Table 9-91h.

N_{cqi} is the number of RU indices for which the CQI report is sent back to the beamformer. $N_{cqi} = (\text{ruidx}(N_{cqi} - 1) - \text{ruidx}(0)) + 1$ – Disallowed # of RU26, where $\text{ruidx}(0)$ and $\text{ruidx}(N_{cqi} - 1)$ are equal to the RU Start Index and RU End Index subfields, respectively. The RU index $\text{ruidx}(i) = \text{ruidx}(i - 1) + 1$, where $1 \leq i \leq N_{cqi} - 2$. The 26-tone RU subcarrier indices for 20 MHz, 40 MHz, and 80 MHz are defined in Table 27-7, Table 27-8, and Table 27-9, respectively.

The Average SNR of space-time stream i for the RU index k subfield in the Table 9-91g is a 6-bit 2s complement integer whose definition is shown in Table 9-91h.

Table 9-91h—Average SNR of RU index k for space-time stream i subfield

Average SNR of RU index k for space-time stream i	$AvgSNR_{k,i}$ (dB)
−32	≤ -10
−31	−9
−30	−8
...	...
30	52
31	≥ 53

The $AvgSNR_{k,i}$ in Table 9-91h is found by computing the arithmetic mean of the SNR per subcarrier in decibels for space-time stream i over the subcarriers in RU index k for which the feedback is being requested. The SNR per subcarrier calculation is defined in 9.4.1.65.

Padding is not present between per-RU average SNRs of each space-time stream information, even if they correspond to different RUs and space-time streams. If the size of the HE CQI Report information is not an integer multiple of 8 bits, up to seven 0s are appended to the end of the field to make its size an integer multiple of 8 bits.

9.4.2 Elements

9.4.2.1 General

Insert the following rows into Table 9-92 in numeric order, and delete or change the related Reserved rows accordingly:

Table 9-92—Element IDs

Element	Element ID	Element ID Extension	Extensible	Fragmentable
HE Capabilities (see 9.4.2.248)	255	35	Yes	No
HE Operation (see 9.4.2.249)	255	36	Yes	No
UORA Parameter Set element (see 9.4.2.250)	255	37	Yes	No
MU EDCA Parameter Set (see 9.4.2.251)	255	38	Yes	No
Spatial Reuse Parameter Set element (see 9.4.2.252)	255	39	Yes	No
NDP Feedback Report Parameter Set element (see 9.4.2.253)	255	41	Yes	No
BSS Color Change Announcement (see 9.4.2.254)	255	42	Yes	No
Quiet Time Period (see 9.4.2.255)	255	43	Yes	No

Table 9-92—Element IDs (*continued*)

Element	Element ID	Element ID Extension	Extensible	Fragmentable
ESS Report (see 9.4.2.256)	255	45	Yes	No
OPS (see 9.4.2.257)	255	46	Yes	No
HE BSS Load (see 9.4.2.259)	255	47	Yes	No
Multiple BSSID Configuration (see 9.4.2.260)	255	55	Yes	No
Known BSSID (see 9.4.2.261)	255	57	No	No
Short SSID List (see 9.4.2.262)	255	58	No	No
HE 6 GHz Band Capabilities (see 9.4.2.263)	255	59	Yes	No
UL MU Power Capabilities (see 9.4.2.264)	255	60	Yes	No

9.4.2.3 Supported Rates and BSS Membership Selectors element

Change Table 9-93 as follows (not all rows are shown):

Table 9-93—BSS membership selector value encoding

Value	Feature	Interpretation
...		
126	VHT PHY	Support for the mandatory features of Clause 21 is required in order to join the BSS that was the source of the Supported Rates and BSS Membership Selectors element or Extended Supported Rates and BSS Membership Selectors element containing this value, <u>unless the STA is a 20 MHz-only non-AP HE STA and the BSS is an HE BSS.</u>
...		
<u>122</u>	<u>HE PHY</u>	<u>Support for the mandatory features of Clause 27 is required in order to join the BSS that was the source of the Supported Rates and BSS Membership Selectors element or Extended Supported Rates and BSS Membership Selectors element containing this value.</u>

9.4.2.5 TIM element

9.4.2.5.1 General

Insert the following text at the end of 9.4.2.5.1:

If included in an OPS frame or a FILS Discovery frame by an OPS AP for aperiodic opportunistic power save (see 26.14.3), the following apply:

- The DTIM Count field is reserved
- The DTIM Period field is reserved

- Bit N in the traffic indication virtual bitmap that corresponds to an OPS non-AP STA with AID N is determined as follows:
 - Bit N in the traffic indication virtual bitmap is set to 0 if the OPS AP does not intend to transmit to the OPS non-AP STA including to trigger the OPS non-AP STA for an UL MU transmission during the OPS period.
 - Otherwise, bit N in the traffic indication virtual bitmap for the OPS non-AP STA is set to 1.
- Bit N in the traffic indication virtual bitmap that corresponds to a non-OPS non-AP STA with AID N is determined as follows:
 - Bit N in the traffic indication virtual bitmap is set to 1 to indicate that AP has buffered frames for the STA and set to 0 otherwise.

If included in a TIM frame or a FILS Discovery frame by an OPS AP for periodic opportunistic power save (see 26.14.3), the following apply:

- The DTIM Count field is reserved
- The DTIM Period field is reserved
- Bit N in the traffic indication virtual bitmap that corresponds to an OPS non-AP STA with AID N is determined as follows:
 - Bit N in the traffic indication virtual bitmap is set to 0 if the OPS AP does not intend to transmit to the OPS non-AP STA including to trigger the OPS non-AP STA for an UL MU transmission during the TWT SP and before the next TWT SP.
 - Otherwise, bit N in the traffic indication virtual bitmap for the OPS non-AP STA is set to 1.
- Bit N in the traffic indication virtual bitmap that corresponds to a non-OPS non-AP STA with AID N is determined as follows:
 - Bit N in the traffic indication virtual bitmap is set to 1 to indicate that AP has buffered frames for the STA and set to 0 otherwise.

9.4.2.8 Country element

Change the fifth paragraph in 9.4.2.8 as follows (Figure 9-165 remains unchanged):

If dot11OperatingClassesRequired is true, then the Triplet field is composed of zero or more Subband Triplet fields followed by one or more Operating/Subband Sequences, as shown in Figure 9-165. If the Country element is included in a frame transmitted in the 6 GHz band, the Triplet field is composed of zero Subband Triplet fields and only has one or more Operating/Subband Sequence fields. Each Operating/Subband Sequence is composed of one Operating Triplet field followed by one Subband Triplet Sequence field, as shown in Figure 9-166. Each Subband Triplet Sequence field is composed of zero or more Subband Triplet fields. If dot11OperatingClassesRequired is true, the number of triplets in the Triplet

field is $Q = N + \sum_{m=1}^M (1 + P(m))$, where N is the total number of Subband Triplet fields and M is the total

number of Operating/Subband Sequences contained in Country element and $P(m)$ is the number of Subband Triplet fields making up Operating/Subband Sequence field m .

Insert the following paragraph and notes into 9.4.2.8 after NOTE 2, and change the subsequent note in this subclause to “NOTE 5”:

An Operating/Subband Sequence field contains zero Subband Triplet fields if all the following conditions are true:

- The operating class table number indicated in the Country String field is Table E-4 (see dot11CountryString).
- The “Channel starting frequency” column in Table E-4 is greater than or equal to 5.925 and less than or equal to 7.125 for the operating class indicated in the Operating Class field.
- The “Channel spacing” column in Table E-4 is greater than or equal to 40 MHz for the operating class indicated in the Operating Class field.

NOTE 3—Any Operating Triplet field for an operating class for which the “Channel starting frequency” column in Table E-4 is greater than or equal to 5.925 and less than or equal to 7.125 can be omitted from the Country element (see 10.22.3).

NOTE 4—The Transmit Power Envelope element is always used for TPC for operating classes in the 6 GHz band instead of Subband Triplet fields (see 26.15.8).

Insert the following paragraph and note into 9.4.2.8 after the now 14th paragraph (“The Maximum Transmit Power Level field”):

The Maximum Transmit Power Level field is reserved if it is within an Operating/Subband Sequence field with the operating class for which the “Channel starting frequency” column in Table E-4 is greater than or equal to 5.925 and less than or equal to 7.125.

NOTE 6—Maximum transmit power information for channels in the 6 GHz band is conveyed using the Transmit Power Envelope element (see 10.22.4).

9.4.2.26 Extended Capabilities element

Change Table 9-153 as follows, and change the related Reserved rows accordingly (not all rows are shown):

Table 9-153—Extended Capabilities element

Bit	Information	Notes
...		
77	<u>TWT Requester Support</u>	<u>A STA sets the TWT Requester Support field to 1 if dot11TWTOptionActivated is true and TWT requester functionality is supported. Otherwise, the STA sets the TWT Requester Support field to 0. See 10.47.</u>
78	<u>TWT Responder Support</u>	<u>A STA sets the TWT Responder Support field to 1 if dot11TWTOptionActivated is true and TWT responder functionality is supported. Otherwise, the STA sets the TWT Responder Support field to 0. See 10.47.</u>
79	<u>OBSS Narrow Bandwidth RU In OFDMA Tolerance Support</u>	<u>An AP STA sets the OBSS Narrow Bandwidth RU In OFDMA Tolerance Support field to 1 if dot11OBSSNarrowBWRUinOFDMATolerated is true, and sets it to 0 otherwise.</u> <u>A non-AP STA sets the OBSS Narrow Bandwidth RU In OFDMA Tolerance Support field to 0.</u>

Table 9-153—Extended Capabilities element (continued)

Bit	Information	Notes
80	Complete List of NonTxBSSID Profiles	This field is reserved for a non-AP STA or when the AP has dot11MultiBSSIDImplemented set to false. When set to 1, Set to 1 to indicates that the frame carrying this element includes a complete list of nontransmitted BSSID profiles. When set to 0, Set to 0 by a non-HE AP if there is no indication about the completeness of the list of nontransmitted BSSID profiles in the frame. Set to 0 by an HE AP to indicate that the frame carrying this element does not include a complete list of nontransmitted BSSID profiles. Also see 11.1.3.8.
...		
83	<u>Enhanced Multi-BSSID Advertisement Support</u> Reserved	<u>This field is reserved for a non-AP STA or when the AP has dot11MultiBSSIDImplemented set to false.</u> <u>Set to 1 to indicate that the AP supports enhancements related to discovery and advertisement of nontransmitted BSSIDs. Set to 0 otherwise.</u> Also see 11.1.3.8.
...		
86	<u>OCT</u> Reserved	<u>The non-AP STA sets the OCT field to 1 when dot11OCTOptionImplemented is true and sets it to 0 otherwise.</u> <u>This field is reserved for an AP.</u>
...		
89	<u>TWTParameters Range Support</u>	<u>Set to 1 to indicate support for reception of a TWT Setup frame that contains two TWT elements (see 10.47.9); otherwise, set to 0.</u>
<u>90–n</u>	<u>Reserved</u>	

9.4.2.28 EDCA Parameter Set element

Change the fourth paragraph in 9.4.2.28 as follows:

The format of the QoS Info field is defined in 9.4.1.17 ~~when sent by an AP. The QoS Info field contains the EDCA Parameter Set Update Count subfield, which is initially set to 0 and is incremented each time any of the announced EDCA parameters changes. This subfield is used by non-AP STAs to determine whether the EDCA parameter set has changed and requires updating the appropriate MIB attributes.~~

Insert NOTE 1 into 9.4.2.28 after the fourth paragraph, and change the subsequent note in this subclause to “NOTE 2”:

NOTE 1—The QoS Info field contains the EDCA Parameter Set Update Count subfield, which indicates when the EDCA parameters and, for an HE BSS, the MU EDCA parameters have changed (see 10.2.3.2).

9.4.2.29 TSPEC element

Change the first paragraph in 9.4.2.29 as follows:

The TSPEC element contains the set of parameters that define the characteristics and QoS expectations of a traffic flow, in the context of a particular STA, for use by the HC or PCP and STA(s) or a mesh STA and its peer mesh STAs in support of QoS traffic transfer using the procedures defined in 11.4 and 11.21.16.3, or for use by HE STAs in support of HE AP scheduling for EDCA or MU operations (see 26.5). The element information format comprises the items as defined in this subclause, and the structure is defined in Figure 9-298.

Change the fourth paragraph in 9.4.2.29, including Table 9-158, Table 9-159, and Table 9-160, as follows (Table 9-161 remains unchanged):

The subfields of the TS Info field are defined as follows:

- The Traffic Type subfield is set to 1 for a periodic traffic pattern (e.g., isochronous TS of MSDUs or A-MSDUs, with constant or variable sizes, that are originated at fixed rate) or set to 0 for an aperiodic, or unspecified, traffic pattern (e.g., asynchronous TS of low-duty cycles).
- The TSID subfield contains a value that is a TSID. Note that the MSB (bit 4 in TS Info field) of the TSID subfield is always set to 1 when the TSPEC element is included within an ADDTS Response frame.
- The Direction subfield specifies the direction of data carried by the TS as defined in Table 9-158.

Table 9-158—Direction subfield encoding

Bit 5	Bit 6	Usage
0	0	Uplink, defined as follows: — Non-DMG BSS: MSDUs or A-MSDUs are sent from the non-AP <u>non-HE</u> STA to HC, <u>or from the non-AP HE STA to the HE AP</u> . — DMG BSS: MSDUs or A-MSDUs are sent by the originator of the ADDTS Request frame.
1	0	Downlink, defined as follows: — Non-DMG BSS: MSDUs or A-MSDUs are sent from the HC to the non-AP <u>non-HE</u> STA <u>or from the HE AP to the non-AP HE STA</u> . — DMG BSS: MSDUs or A-MSDUs are sent by the recipient of the ADDTS Request frame.
0	1	Direct link (MSDUs or A-MSDUs are sent from the non-AP STA to another non-AP STA).
1	1	Bidirectional link (equivalent to a downlink request plus an uplink request, each direction having the same parameters). The fields in the TSPEC element specify resources for a single direction. Double the specified resources are required to support both streams.

- The Access Policy subfield specifies the access method to be used for the TS and is defined in Table 9-159.

Table 9-159—Access Policy subfield

Bit 7	Bit 8	Usage
0	0	Reserved.
1	0	For non-HE STAs: Contention based channel access (EDCA). For HE STAs: EDCA or MU based access (see 26.5).
0	1	Controlled channel access (HCCA for non-DMG STAs and SPCA for DMG STAs).
1	1	Controlled and contention based channel access (HCCA, EDCA mixed mode (HEMM) for non-DMG STAs; SPCA, EDCA mixed mode (SEMM) for DMG STAs).

- The Aggregation subfield is valid only when the access method is HCCA or SPCA or when the access method is EDCA and the Schedule subfield is equal to 1. It is set to 1 by a non-AP STA to indicate that an aggregate schedule is required. It is set to 1 by the AP if an aggregate schedule is being provided to the STA. It is set to 0 otherwise. In all other cases, the Aggregation subfield is reserved.
- The APSD subfield is set to 1 to indicate that automatic PS delivery is to be used for the traffic associated with the TSPEC and set to 0 otherwise.
- The UP subfield indicates the actual value of the UP to be used for the transport of MSDUs or A-MSDUs belonging to this TS ~~when~~ if relative prioritization is required. When the TCLAS element is present in the request, the UP subfield in TS Info field of the TSPEC element is reserved.
- The TS Info Ack Policy subfield indicates whether MAC acknowledgments are required for MPDUs or A-MSDUs belonging to this TSID and the form of those acknowledgments. The encoding of the TS Info Ack Policy subfield is shown in Table 9-160.

Table 9-160—TS Info Ack Policy subfield encoding

Bit 14	Bit 15	Usage
0	0	Normal Acknowledgment Ack. The addressed recipient returns an Ack or QoS +CF-Ack frame after a SIFS, according to the procedures defined in 10.3.2.11 and 10.23.3.5.
1	0	No Ack: The recipient(s) do not acknowledge the transmission.
0	1	Reserved.
1	1	Block Ack: A separate block ack mechanism described in 10.25 is used.

- The Schedule subfield specifies the requested type of schedule. The setting of the subfield when the access policy is EDCA is shown in Table 9-161. When the Access Policy subfield is equal to any value other than EDCA, the Schedule subfield is reserved. When the Schedule and APSD subfields are equal to 1, the AP sets the Aggregation subfield to 1, indicating that an aggregate schedule is being provided to the STA.

Table 9-161—Setting of Schedule subfield

APSD	Schedule	Usage
0	0	No Schedule
1	0	Unscheduled APSD
0	1	Scheduled PSMP or GCR-SP
1	1	Scheduled APSD

Change the 10th and 11th paragraphs in 9.4.2.29 as follows:

The Inactivity Interval field contains an unsigned integer that specifies the minimum amount of time, in microseconds, that can elapse without arrival or transfer of an MPDU belonging to the TS before this TS is deleted by the MAC entity at the HC for a non-HE STA or before the information provided in this TSPEC is considered invalid at HE STAs for HE STAs.

For non-HE STAs, the The Suspension Interval field contains an unsigned integer that specifies the minimum amount of time, in microseconds, that can elapse without arrival or transfer of an MSDU belonging to the TS before the generation of successive QoS(+)CF-Poll is stopped for this TS. A value of 4 294 967 295 ($= 2^{32} - 1$) disables the suspension interval, indicating that polling for the TS is not to be interrupted based on inactivity. The suspension interval is always less than or equal to the inactivity interval. The Suspension Interval field is reserved for HE STAs.

9.4.2.36 Neighbor Report element

Change Figure 9-337 as follows:

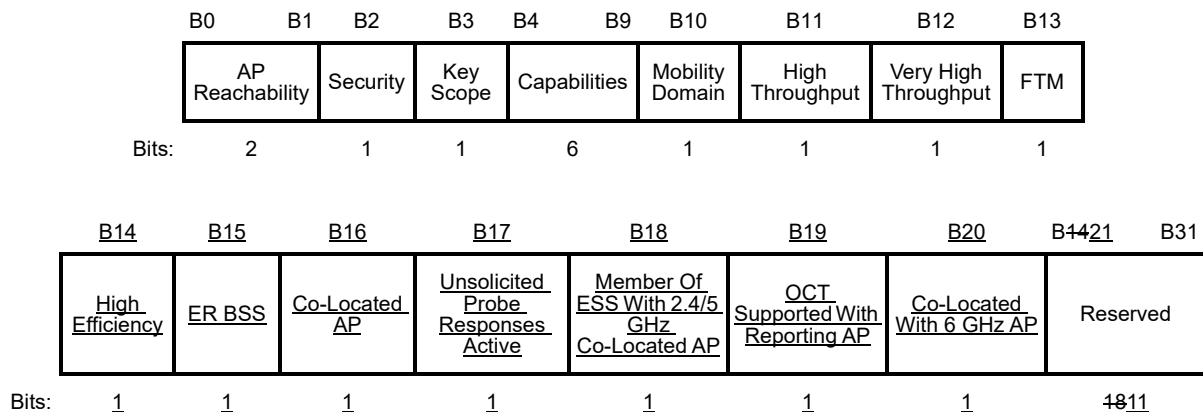


Figure 9-337—BSSID Information field format

Change the 10th and 11th paragraphs in 9.4.2.36 as follows:

The High Throughput bit is set to 1 to indicate that the AP represented by this BSSID is an HT AP ~~including and that the HT Capabilities element (or HT Operation element), if included as a subelement in the report, is in its Beacons, and that the contents of that HT Capabilities element are identical in content to the HT Capabilities element (or HT Operation element) included in the neighboring AP's Beacon advertised by the AP sending the report. Otherwise, the High Throughput subfield is set to 0.~~

The Very High Throughput bit is set to 1 to indicate that the AP represented by this BSSID is a VHT AP and that the VHT Capabilities element ~~(or VHT Operation element), if included as a subelement in the report, is identical in content to the VHT Capabilities element (or VHT Operation element) included in the neighboring AP's Beacon frame. Otherwise, the Very High Throughput subfield is set to 0.~~

Delete the 13th paragraph (“Bits 14-31 are reserved.”) in 9.4.2.36.

Insert the following text into 9.4.2.36 after the 12th paragraph (“The FTM field”):

The High Efficiency subfield is set to 1 to indicate that the AP represented by this BSSID is an HE AP and that the HE Capabilities element (or HE Operation element), if included as a subelement in the report, is identical in content to the HE Capabilities element (or HE Operation element) included in the neighboring AP’s Beacon frame. Otherwise, the High Efficiency subfield is set to 0.

The ER BSS subfield is set to 1 if the BSS corresponding to the HE AP representing this BSSID is an ER BSS transmitting Beacon frames using an HE ER SU PPDU (see 26.17.6). Otherwise, the ER BSS subfield is set to 0.

The Co-Located AP subfield is set to 1 to indicate that the AP reported in this Neighbor Report element is in the same co-located AP set as the AP sending the Neighbor Report element.

The Unsolicited Probe Responses Active subfield is set to 1 if the reported AP is part of an ESS where all the APs that operate in the same channel as the reported AP and that might be detected by a STA receiving this frame [see the definition of “detected access point (AP)” in 3.2] have dot11UnsolicitedProbeResponseOptionActivated equal to true and so are transmitting unsolicited Probe Response frames every 20 TUs or less (see 26.17.2.3). It is set to 0 otherwise or if the reporting AP does not have that information.

The Member Of ESS With 2.4/5 GHz Co-Located AP subfield is set to 1 if the reported AP is part of an ESS where each AP in the ESS and operating in the same band as the reported AP (irrespective of the operating channel in that band) that might be detected by a STA receiving this frame [see the definition of “detected access point (AP)” in 3.2] has dot11MemberOfColocated6GHzESSOptionActivated equal to true and also has a corresponding AP operating in the 2.4 GHz or 5 GHz bands that is in the same co-located AP set as that AP. It is set to 0 otherwise or if the reporting AP does not have that information. It is reserved if the reported AP is operating in the 2.4 GHz or 5 GHz bands.

NOTE—This subfield indicates that the reported AP is part of an ESS that has no 6 GHz-only APs that might be detected by a STA receiving this frame. This means that all APs operating in the 6 GHz band that are part of that ESS that might be detected by a STA receiving this frame can be discovered in the 2.4 GHz and/or 5 GHz bands.

The OCT Supported With Reported AP subfield is set to 1 to indicate that OCT is supported to exchange MMPDUs with the AP reported in the Neighbor Report element (see 11.31.5), through over-the-air transmissions with the AP sending the Neighbor Report element. It is set to 0 otherwise.

The Co-Located With 6 GHz AP subfield is set to 1 to indicate that the AP reported by the Neighbor Report element is in the same co-located AP set as a 6 GHz AP and that the 6 GHz AP can be discovered by receiving Management frames (as described in 26.17.2.3 and 11.53) sent by the reported AP. It is set to 0 otherwise.

Insert the following rows into Table 9-173 in numeric order, and change the related Reserved row accordingly:

Table 9-173—Optional subelement IDs for Neighbor Report

Subelement ID	Name	Extensible
193	HE Capabilities	Yes
194	HE Operation	Yes
195	BSS Load	
196	HE BSS Load	Yes
197	SSID	

Insert the following paragraphs into 9.4.2.36 immediately before the last paragraph (“The Vendor Specific subelement”):

The HE Capabilities subelement has the same format as the HE Capabilities element (see 9.4.2.248).

The HE Operation subelement has the same format as the HE Operation element (see 9.4.2.249).

The BSS Load subelement has the same format as the BSS Load element (see 9.4.2.27).

The HE BSS Load subelement has the same format as the HE BSS Load element (see 9.4.2.259).

The SSID subelement has the same format as the SSID element (see 9.4.2.2).

9.4.2.45 Multiple BSSID element

Change the third and now fifth paragraphs in 9.4.2.45 as follows (including splitting the third paragraph into two paragraphs):

The MaxBSSID Indicator field contains a value assigned to n , where 2^n is the maximum number of BSSIDs in the multiple BSSID set, including the reference BSSID (see 11.10.14). The maximum value of n is 8.

NOTE 1— $1 \leq n \leq 8$ since the BSSID Index field in 9.4.2.73 indicates the number of BSSIDs in a multiple BSSID set.

The actual number of BSSIDs in the multiple BSSID set is not explicitly signaled. BSSID(i) corresponding to the i^{th} BSSID in the multiple BSSID set is derived as follows:

A0-A1-A2-A3-A4-A5 = Reference BSSID

$B = A5 \bmod 2^n$

$A5(i) = A5 - B + ((B + i) \bmod 2^n)$

$\text{BSSID}(i) = \text{A0-A1-A2-A3-A4-A5}(i)$

~~NOTE 1~~ NOTE 2—For example, for $n = 3$ and Reference BSSID = 8c-fd-0f-7f-1e-f5:

A5 = f5

B = 5

A5(5) = f2 and BSSID(5) = 8c-fd-0f-7f-1e-f2

A5(2) = f7 and BSSID(2) = 8c-fd-0f-7f-1e-f7

~~NOTE 2~~ NOTE 3—This definition uses the hexadecimal address representation defined in IEEE Std 802.

~~NOTE 3~~ NOTE 4—The BSSID index as defined in 9.4.2.73 cannot be larger than 255, which effectively limits n to 8.

When the Multiple BSSID element is transmitted in a Beacon, DMG Beacon, or Probe Response frame, the reference BSSID is the BSSID field of the frame. The AP or DMG STA determines the number of Multiple BSSID elements. The AP or DMG STA does not fragment a nontransmitted BSSID profile subelement for a single BSSID across two Multiple BSSID elements, unless the length contents of the nontransmitted BSSID profile subelement exceeds 255 octets or are greater than the number of octets remaining in the Multiple BSSID element after taking into account the mandatory fields (i.e., Element ID, Length, MaxBSSID Indicator) and any preceding Nontransmitted BSSID Profile subelement(s) carried in the element (see 11.1.3.8.2). When the Multiple BSSID element is transmitted as a subelement in a Neighbor Report element, the reference BSSID is the BSSID field in the Neighbor Report element.

Change the now eighth paragraph in 9.4.2.45 as follows:

~~The Nontransmitted BSSID Profile subelement contains a list of elements for one or more APs or DMG STAs that have nontransmitted BSSIDs. A nontransmitted BSSID profile carried in one or more Nontransmitted BSSID Profile subelements across one or more multiple BSSID elements in the same frame contains a list of elements for the AP or the DMG STA that has a nontransmitted BSSID and is defined as follows:~~

- ~~— For each nontransmitted BSSID, t~~ The Nontransmitted BSSID Capability element (see 9.4.2.71) is the first element included, followed by a variable number of elements, in the order defined in Table 9-32 for a non-DMG non-S1G AP, Table 9-45 for a DMG AP, or Table 9-46 for a S1G AP.
- ~~— The SSID element (see 9.4.2.2) and Multiple BSSID-Index element (see 9.4.2.73) are included as the second and third elements, respectively in the Nontransmitted BSSID Profile subelement.~~
- ~~— The FMS Descriptor element (see 9.4.2.74) is included in the Nontransmitted BSSID Profile subelement if dot11FMSActivated is true for the BSS using this nontransmitted BSSID and if the Multiple BSSID element is included in a Beacon frame.~~
- ~~— Any element specific to the BSS or with content that is different from the transmitted BSSID.~~
- ~~— The Timestamp and Beacon Interval fields, TIM, DSSS Parameter Set, IBSS Parameter Set, Country, Channel Switch Announcement, Extended Channel Switch Announcement, Wide Bandwidth Channel Switch, Transmit Power Envelope, Supported Operating Classes, IBSS DFS, ERP Information, HT Capabilities, HT Operation, VHT Capabilities, VHT Operation, S1G Beacon Compatibility, Short Beacon Interval, S1G Capabilities, and S1G Operation, HE Capabilities, HE 6 GHz Band Capabilities, HE Operation, BSS Color Change Announcement, and Spatial Reuse Parameter Set elements are not included in the Nontransmitted BSSID Profile subelement; the values of these elements for each nontransmitted BSSID are always the same as the corresponding transmitted BSSID element values.~~
- ~~— When included in the Nontransmitted BSSID Profile subelement for this nontransmitted BSSID, the Non-Inheritance element (see 9.4.2.240) appears as the last element in the profile and carries a list of elements that are not inherited by this nontransmitted BSSID from the transmitted BSSID.~~

Insert NOTE 5 and the following paragraph into 9.4.2.45 after the now eighth paragraph:

NOTE 5—A Reduced Neighbor Report element is not carried in the Nontransmitted BSSID Profile subelement. When present in the frame, the values of fields in the element, except the Same SSID subfield(s) apply to all the BSSs in the multiple BSSID set.

Each Nontransmitted BSSID Profile subelement contains only elements for a BSS with a nontransmitted BSSID.

9.4.2.63 Channel Switch Timing element

Change the last paragraph in 9.4.2.63 as follows:

The Switch Timeout field is set to a time in units of microseconds. The STA sending the Channel Switch Timing element waits for the first Data frame exchange on the off-channel for Switch Timeout microseconds before switching back to base channel. The time is measured from the end of the last symbol of the Ack frame that is transmitted in response to TDLS Channel Switch Response frame, as seen on the WM. If transmitted in a (Re)Association Request frame, the Switch Timeout field is not present in the Channel Switch Timing element.

9.4.2.66 Event Request element

9.4.2.66.1 Event Request definition

Insert the following rows into Table 9-193 in numeric order, and change the related Reserved row accordingly:

Table 9-193—Event Type field definitions for event requests and reports

Name	Event Type
BSS Color Collision	4
BSS Color In Use	5

9.4.2.67 Event Report element

9.4.2.67.1 Event Report definition

Change the sixth paragraph in 9.4.2.67.1 as follows:

If the Event Report Status field is 0 (Successful) and the Event Type field is neither 4 (BSS Color Collision) nor 5 (BSS Color In Use), then t~~The Event TSF, UTC Offset, Event Time Error, and Event Report fields are present only when.~~If the Event Report Status field is 0 (Successful) and the Event Type field is either 4 (BSS Color Collision) or 5 (BSS Color In Use), then Event TSF and Event Report fields are present.
Otherwise, the Event TSF, UTC Offset, Event Time Error, and Event Report fields are not present.

Change the 10th paragraph in 9.4.2.67.1 as follows:

The Event Report field contains the specification of a single event report, as described in 9.4.2.67.2 to 9.4.2.67.5, 9.4.2.67.7, and 9.4.2.67.8.

Insert the following subclauses (9.4.2.67.7 and 9.4.2.67.8) after 9.4.2.67.6:

9.4.2.67.7 BSS color collision event report

The Event Report field for a BSS color collision event report is 8 octets in length with each bit representing a BSS color value. A value of 1 at a bit position indicates that the BSS color value corresponding to that position is in use by OBSS as detected by the reporting non-AP HE STA.

9.4.2.67.8 BSS color in use event report

The Event Report field for a BSS color in use event report is 1 octet in length. If a reporting non-AP HE STA communicates with a peer STA with a BSS color that is different from the BSS color used by its associated AP, the Event Report field is set to the BSS color used in the communication with the peer STA.

A non-AP HE STA sets the Event Report field to 0 to cancel a previously sent BSS color in use event report. See 11.21.2.8.

9.4.2.73 Multiple BSSID-Index element

Change the third, fourth, and fifth paragraphs in 9.4.2.73 as follows:

The BSSID Index field is a value between 1 and $2^n - 1$ that identifies the nontransmitted BSSID, where n is a nonzero, positive integer value (see MaxBSSID Indicator field in 9.4.2.45).

The DTIM Period field indicates the DTIM period for the BSSID as defined in 9.4.2.5.1. This field is not present when the Multiple BSSID-Index element is included in the Probe Response frame.

The DTIM Count field indicates the DTIM count for the BSSID as defined in 9.4.2.5.1. This field is not present when the Multiple BSSID-Index element is included in the Probe Response frame.

9.4.2.139 ADDBA Extension element

Change Figure 9-580 as follows:

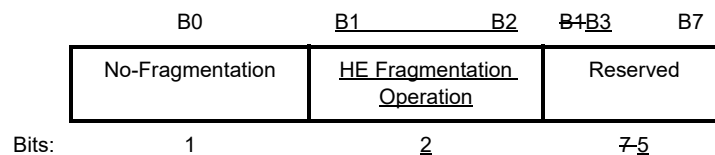


Figure 9-580—ADDBA Capabilities field format

Change the fifth paragraph in 9.4.2.139 as follows:

The No-Fragmentation subfield determines whether a fragmented MSDU can be carried in the MPDU sent under the block ack agreement. ~~When~~ If this subfield set to 1 in the ADDBA Request frame, it indicates that the non-HE originator is not fragmenting sent MSDUs. ~~When~~ If this subfield set to 1 in the ADDBA Response frame, it indicates that the non-HE recipient is not capable of receiving fragmented MSDUs. The No-Fragmentation subfield is reserved when transmitted by an HE STA to another HE STA.

Insert the following paragraph and Table 9-265a at the end of 9.4.2.139:

If transmitted by an HE STA, the HE Fragmentation Operation subfield indicates the level of dynamic fragmentation that is under negotiation for the TID indicated in the ADDBA frame as defined in Table 9-265a.

Table 9-265a—HE Fragmentation Operation subfield

Value	Meaning in ADDBA Request frame	Meaning in ADDBA Response frame
0	The originator does not intend to send fragmented MSDUs or A-MSDUs (if supported by the transmitter) for the TID specified in the Block Ack Parameter Set field of the ADDBA Request frame.	The recipient does not support the reception fragmented MSDUs or A-MSDUs for the TID specified in the Block Ack Parameter Set field of the ADDBA Response frame.
1	The originator intends to send fragmented MSDUs or A-MSDU (if supported by the transmitter) under fragmentation level 1 (see 26.3.2.2) for the TID specified in the Block Ack Parameter Set field of the ADDBA Request frame.	The recipient supports the reception of fragmented MSDUs or A-MSDU (if supported) under fragmentation level 1 only for the TID specified in the Block Ack Parameter Set field of the ADDBA Response frame.

Table 9-265a—HE Fragmentation Operation subfield (*continued*)

Value	Meaning in ADDBA Request frame	Meaning in ADDBA Response frame
2	The originator intends to send fragmented MSDUs or A-MSDUs (if supported by the transmitter) under fragmentation level 2 (see 26.3.2.3) for the TID specified in the Block Ack Parameter Set field of the ADDBA Request frame.	The recipient supports the reception of receiving fragmented MSDUs or A-MSDU (if supported) under fragmentation levels 1 and 2 for the TID specified in the Block Ack Parameter Set field of the ADDBA Response frame.
3	The originator intends to send fragmented MSDUs or A-MSDUs (if supported by the transmitter) under fragmentation level 3 (see 26.3.2.4) for the TID specified in the Block Ack Parameter Set field of the ADDBA Request frame.	The recipient supports the reception of fragmented MSDUs or A-MSDU (if supported) under fragmentation levels 1, 2 and 3 for the TID specified in the Block Ack Parameter Set field of the ADDBA Response frame.

9.4.2.157 VHT Capabilities element

9.4.2.157.3 Supported VHT-MCS and NSS Set field

Insert the following text at the end of 9.4.2.157.3, and change the previous note in this subclause to “NOTE 1”:

The value of Max VHT NSS for a given MCS is equal to the smaller of

- The maximum value of n for which the Max VHT-MCS for n SS has a value that indicates support for that MCS or
- The maximum supported N_{SS} as indicated in by the value of the Rx NSS field of the OM Control subfield (and further defined in the Table 26-9).

NOTE 2—A VHT-MCS indicated as supported in the VHT-MCS Map fields for a particular number of spatial streams might not be valid at all bandwidths (see 21.5); might be limited by the declaration of Tx Highest Supported Long GI Data Rates and Rx Highest Supported Long GI Data Rates; and might be affected by additional rate selection constraints in 10.6.13.3, the value of the Extended NSS BW Support field of the VHT Capabilities Information field (see 9.4.2.157.2), and the 160/80+80 BW subfield of the Operating Mode field (see 9.4.1.53).

9.4.2.159 Extended BSS Load element

Change the description of “ $T_{busy,W1}$ ” in the variable list for Equation (9-3) as follows:

$T_{busy,W1}$ is computed as the sum of the times from PHY-CCA.indication(BUSY, {W2}, per20bitmap) to the next issue of a PHY-CCA.indication primitive and that overlap the measurement interval, for $W1 = 20, 40$, or 80 , ~~and where W2 equals secondary, secondary40, or secondary80 for $W1 = 20, 40$, or 80 , respectively~~.
For a VHT AP, for $W1 = 20, 40, 80$, W2 equals secondary, secondary40, or secondary80, respectively.
For an HE AP, for $W1 = 20$, W2 equals secondary or per20bitmap has the bit corresponding to the primary 20 MHz channel equal to 0 and the bit corresponding to the secondary 20 MHz channel equal to 1; for $W1 = 40$, W2 equals secondary40 or per20bitmap has the bits corresponding to the primary 20 MHz and secondary 20 MHz channels equal to 0, and at least one bit corresponding to any 20 MHz subchannel in the secondary 40 MHz channel equal to 1; for $W1 = 80$, W2 equals secondary80 or per20bitmap has the bits corresponding to the primary 20 MHz, secondary 20 MHz, and secondary 40 MHz channels equal to 0, and at least one bit corresponding to any 20 MHz subchannel in the secondary 80 MHz channel is equal to 1.

9.4.2.161 Transmit Power Envelope element

Change 9.4.2.161 as follows (including changing Figure 9-616, Figure 9-617, and Table 9-276; inserting Table 9-275a and Figure 9-617a; and deleting Table 9-277):

The Transmit Power Envelope element conveys the local or regulatory maximum transmit powers for various transmission bandwidths or channels within the bandwidth of the BSS. The format of the Transmit Power Envelope element is shown in Figure 9-616.

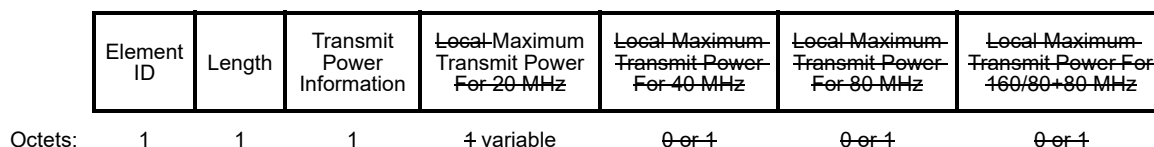


Figure 9-616—Transmit Power Envelope element format

The Element ID and Length fields are defined in 9.4.2.1.

The format of the Transmit Power Information field is defined in Figure 9-617.

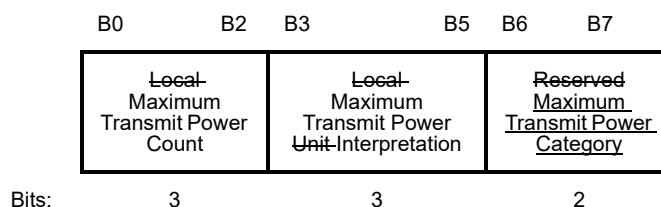


Figure 9-617—Transmit Power Information field format

The Maximum Transmit Power Interpretation subfield indicates the contents of the Maximum Transmit Power field and interpretation of the Maximum Transmit Power Count field and is defined in Table 9-275a.

Table 9-275a—Maximum Transmit Power Interpretation subfield encoding

Value	Interpretation of the Maximum Transmit Power field
0	Local EIRP
1	Local EIRP PSD (power spectral density)
2	Regulatory client EIRP
3	Regulatory client EIRP PSD
4–7	Reserved

NOTE—This table is expected to be updated only if regulatory domains mandate the use of transmit power control with limits that cannot be converted into one of the currently defined interpretations.

The Maximum Transmit Power Category subfield indicates a category for which the maximum transmit powers apply. A value of 0 indicates the default category; the interpretation of other values depends on the country; see E.2.7 for 6 GHz operation for specific countries. In bands other than the 6 GHz band, this subfield is reserved.

~~The~~ If the Maximum Transmit Power Interpretation subfield is 0 or 2 (EIRP), the ~~Local~~ Maximum Transmit Power Count subfield indicates the number of ~~Local~~ Maximum Transmit Power For X MHz subfields (where $X = 20, 40, 80$, or $160/80+80$) minus 1 in the Maximum Transmit Power field of the Transmit Power Envelope element, as shown in Table 9-276.

Table 9-276—Meaning of ~~Local~~ Maximum Transmit Power Count subfield if the Maximum Transmit Power Interpretation subfield is 0 or 2

Value	Field(s) present
0	Local Maximum Transmit Power For 20 MHz.
1	Local Maximum Transmit Power For 20 MHz and Local Maximum Transmit Power For 40 MHz.
2	Local Maximum Transmit Power For 20 MHz, Local Maximum Transmit Power For 40 MHz, and Local Maximum Transmit Power For 80 MHz.
3	Local Maximum Transmit Power For 20 MHz, Local Maximum Transmit Power For 40 MHz, Local Maximum Transmit Power For 80 MHz, and Local Maximum Transmit Power For 160/80+80 MHz. For TVHT STAs, reserved.
4–7	Reserved

The ~~Local Maximum Transmit Power Unit Interpretation~~ subfield provides additional interpretation for the units of the ~~Local Maximum Transmit Power For X MHz~~ fields (where $X = 20, 40, 80$, or $160/80+80$) and is defined in Table 9-277. Allowed values are further constrained as defined in Annex E.

Table 9-277—Definition of ~~Local Maximum Transmit Power Unit Interpretation~~ subfield

Value	Unit interpretation of the Local Maximum Transmit Power For X MHz fields
0	EIRP
1–7	Reserved
NOTE—This table is expected to be updated only if regulatory domains mandate the use of transmit power control with limits that cannot be converted into an EIRP value per transmission bandwidth.	

If the Maximum Transmit Power Interpretation subfield is 0 or 2 (EIRP), the format of the Maximum Transmit Power field is defined in Figure 9-617a.

Maximum Transmit Power For 20 MHz	Maximum Transmit Power For 40 MHz	Maximum Transmit Power For 80 MHz	Maximum Transmit Power For 160/80+80 MHz
Octets: 1	0 or 1	0 or 1	0 or 1

**Figure 9-617a—Maximum Transmit Power field format
if the Maximum Transmit Power Interpretation subfield is 0 or 2**

~~Local~~ Maximum Transmit Power For X MHz subfields (where $X = 20, 40, 80$, or $160/80+80$) define the local maximum transmit power limit of X MHz PPDUs, except for an HE TB PPDU where X MHz is the bandwidth of the pre-HE modulated fields of the HE TB PPDU transmitted by a STA. Each ~~Local~~ Maximum Transmit Power For X MHz subfield is encoded as an 8-bit 2s complement signed integer in the range -64 dBm to 63 dBm with a 0.5 dB step. Setting this field to 63.5 dBm indicates 63.5 dBm or higher (i.e., no local maximum transmit power constraint).

In frames transmitted by a TVHT STA, the ~~Local~~ Maximum Transmit Power For 20 MHz subfield indicates the local maximum transmit power for TVHT_W bandwidth; the ~~Local~~ Maximum Transmit Power For 40 MHz subfield indicates the local maximum transmit power for TVHT_2W or TVHT_W+W bandwidth; the ~~Local~~ Maximum Transmit Power For 80 MHz subfield indicates the local maximum transmit power for TVHT_4W or TVHT_2W+2W bandwidth; and the ~~Local~~ Maximum Transmit Power For 160/80+80 MHz subfield is not included in the Transmit Power Envelope element.

Change “Local Maximum Transmit Power Unit Interpretation subfield” to “Maximum Transmit Power Interpretation subfield” throughout IEEE Std 802.11-2020 (i.e., in 9.4.2.162, 9.4.2.174, 9.6.2.6, 9.6.7.7, Table 9-419, 11.7.4, 11.38.4, 11.48.3, and Table D-2).

Insert the following text, Figure 9-617b, and a new Table 9-277 at the end of 9.4.2.161:

If the Maximum Transmit Power Interpretation subfield is 1 or 3 (EIRP PSD), the format of the Maximum Transmit Power field is shown in Table 9-617b.

Maximum Transmit PSD 1	Maximum Transmit PSD 2	...	Maximum Transmit PSD N
Octets: 1	0 or 1		0 or 1

**Figure 9-617b—Maximum Transmit Power field format
if Maximum Transmit Power Interpretation subfield is 1 or 3**

The Maximum Transmit Power Count subfield determines the value of an integer N as defined in Table 9-277 which specifies the format and interpretation of the Maximum Transmit Power field as described below.

**Table 9-277—Meaning of Maximum Transmit Power Count subfield
if Maximum Transmit Power Interpretation subfield is 1 or 3**

Value	N
0	0
1	1
2	2
3	4
4	8
5–7	Reserved to indicate values of N greater than 8

If N is 0, then the Maximum Transmit Power field contains one Maximum Transmit PSD subfield that represents the maximum transmit PSD for a PPDU of any bandwidth within the BSS bandwidth.

If N is greater than 0, then the Maximum Transmit Power field has N octets, with N representing the number of 20 MHz channels for which a maximum transmit PSD is indicated. The X^{th} octet (X = integer ranging from 1 to N) of the Maximum Transmit Power field is the Maximum Transmit PSD X subfield, which indicates the maximum transmit PSD for the X^{th} 20 MHz channel.

If the BSS bandwidth is 20, 40, 80, or 160 MHz, then the Maximum Transmit PSD 1- N subfields correspond to 20 MHz channels from lowest to highest frequency, respectively, within the indicated bandwidth. If N is equal to 1, 2, 4, or 8 for 20, 40, 80, or 160 MHz BSS bandwidth, respectively, the indicated bandwidth is the BSS bandwidth. If N is greater than 0 and less than 2, 4, or 8 for 40, 80, or 160 MHz BSS bandwidth, respectively, then the indicated bandwidth is the primary 20 MHz, primary 40 MHz, or primary 80 MHz channel for N equal to 1, 2, or 4, respectively. If N is greater than 1, 2, or 4 for 20, 40, or 80 MHz BSS bandwidth, respectively, then the indicated bandwidth is wider than the BSS bandwidth. In this case, the Maximum Transmit PSD 1- M subfields correspond to the 20 MHz channels from lowest to highest frequency, respectively, within the BSS bandwidth where M is 1, 2, or 4 for 20, 40, or 80 MHz BSS bandwidth, respectively. The Maximum Transmit PSD ($M+1$)- N subfields are reserved for future use (see 10.22.4).

If the BSS bandwidth is 80+80 MHz, N is less than or equal to 8. If N is equal to 8 and the BSS bandwidth is 80+80 MHz, the Maximum Transmit PSD 1-4 subfields correspond to the 20 MHz channels from lowest to highest frequency, respectively, within the 80 MHz segment lower in frequency; the Maximum Transmit PSD 5-8 subfields correspond to the 20 MHz channels from lowest to highest frequency, respectively, within the 80 MHz segment higher in frequency. If N is greater than 0 and less than 8 for 80+80 MHz BSS bandwidth, then the bandwidth indicated by the Maximum Transmit PSD 1- N subfields is the primary 20 MHz, primary 40 MHz, or primary 80 MHz channel for N equal to 1, 2, or 4, respectively. In this case, the Maximum Transmit PSD 1- N subfields correspond to 20 MHz channels from lowest to highest frequency, respectively, within the indicated bandwidth.

Values of the Maximum Transmit Power Count field between 5 and 7 are reserved for future use to indicate values of N greater than 8. If N is greater than 8, the Maximum Transmit PSD 1-8 subfields correspond to the 20 MHz channels from lowest to highest frequency, respectively, within the 160 MHz channel containing the primary 20 MHz channel. See 10.22.4.

The Maximum Transmit PSD X subfield is encoded as an 8-bit 2s complement signed integer. The value of -128 indicates that the corresponding 20 MHz channel cannot be used for transmission. The value of $+127$ indicates that no maximum PSD limit is specified for the corresponding 20 MHz channel. For all other values Y of the subfield (i.e., -127 to $+126$, inclusive), the maximum transmit PSD in the corresponding 20 MHz channel is $Y/2$ dBm/MHz (i.e., ranging from -63.5 to $+63$ dBm/MHz).

9.4.2.170 Reduced Neighbor Report element

9.4.2.170.2 Neighbor AP Information field

Change the sixth paragraph in 9.4.2.170.2 as follows:

The TBTT Information Length subfield indicates the length of each TBTT Information field included in the TBTT Information Set field of the Neighbor AP Information field. ~~When~~If the TBTT Information Field Type subfield is ~~set to 0~~, the TBTT Information Length subfield

- Contains the length in octets of each TBTT Information field that is included in the TBTT Information Set field of the Neighbor AP Information field.
- Is set to 1, 2, 5, 6, 7, 8, 9, ~~or 11, 12, or 13~~; other values are reserved.
- Indicates the TBTT Information field contents as shown in Table 9-281.

Change Table 9-281 as follows:

Table 9-281—TBTT Information field contents

TBTT Information Length subfield value	TBTT Information field contents
1	The Neighbor AP TBTT Offset subfield
<u>2</u>	<u>The Neighbor AP TBTT Offset subfield and the BSS Parameters subfield</u>
5	The Neighbor AP TBTT Offset subfield and the Short SSID subfield
<u>6</u>	<u>The Neighbor AP TBTT Offset subfield, the Short SSID subfield, and the BSS Parameters subfield</u>
7	The Neighbor AP TBTT Offset subfield and the BSSID subfield
<u>8</u>	<u>The Neighbor AP TBTT Offset subfield, the BSSID subfield, and the BSS Parameters subfield</u>
<u>9</u>	<u>The Neighbor AP TBTT Offset subfield, the BSSID subfield, the BSS Parameters subfield, and the 20 MHz PSD subfield</u>
11	The Neighbor AP TBTT Offset subfield, the BSSID subfield, and the Short SSID subfield

Table 9-281—TBTT Information field contents (*continued*)

TBTT Information Length subfield value	TBTT Information field contents
<u>12</u>	<u>The Neighbor AP TBTT Offset subfield, the BSSID subfield, the Short SSID subfield, and the BSS Parameters subfield</u>
0, 2, 4, 6, 8, 10, 12, 255 <u>0, 3, 4, 10</u>	Reserved
<u>13</u>	<u>The Neighbor AP TBTT Offset subfield, the BSSID subfield, the Short SSID subfield, the BSS Parameters subfield, and the 20 MHz PSD subfield</u>
<u>14–255</u>	<u>The first 13 octets of the field contain the Neighbor AP TBTT Offset subfield, the BSSID subfield, the Short SSID subfield, the BSS Parameters subfield, and the 20 MHz PSD subfield (i.e., same contents as when the length of the TBTT Information field is 13). The remaining octets are reserved.</u>

Change Figure 9-632 as follows:

	Neighbor AP TBTT Offset	BSSID (optional)	Short SSID (optional)	<u>BSS parameters</u>	<u>20 MHz PSD</u>
Octets:	1	0 or 6	0 or 4	<u>0 or 1</u>	<u>0 or 1</u>

Figure 9-632—TBTT Information field format

Change the 13th paragraph in 9.4.2.170.2 as follows (including splitting the paragraph into two paragraphs):

The Neighbor AP TBTT Offset subfield indicates the offset in TUs, rounded down to nearest TU, to the following:

- The next TBTT of the reported AP ~~an AP's BSS~~ from the immediately prior TBTT of the AP that transmits this element ~~if the reported AP is not part of a multiple BSSID set or is the transmitted BSSID of a multiple BSSID set.~~
- The next TBTT of the transmitted BSSID of the multiple BSSID set of the reported AP from the immediately prior TBTT of the AP that transmits this element if the reported AP is part of a multiple BSSID set and is a nontransmitted BSSID.

The value 254 indicates an offset of 254 TUs or higher. The value 255 indicates an unknown offset value.

Insert the following text and Figure 9-632a at the end of 9.4.2.170.2, and change the previous note in this subclause to “NOTE 1”:

The format of the BSS Parameters subfield is defined in Figure 9-632a.

B0	B1	B2	B3	B4	B5	B6	B7
OCT Recommended	Same SSID	Multiple BSSID	Transmitted BSSID	Member Of ESS With 2.4/5 GHz Co-Located AP	Unsolicited Probe Responses Active	Co-Located AP	Reserved
Bits: 1	1	1	1	1	1	1	1

Figure 9-632a—BSS Parameters subfield format

The OCT Recommended subfield is set to 1 to indicate that OCT is recommended to exchange MMPDUs with the AP identified in the TBTT Information field (see 11.31.5), through over-the-air transmissions with the AP sending the Reduced Neighbor Report element. It is set to 0 otherwise.

The Same SSID subfield is set to 1 to indicate that the reported AP has the same SSID as the reporting AP. It is set to 0 otherwise.

The Multiple BSSID subfield is set to 1 to indicate that the reported AP is part of a multiple BSSID set. It is set to 0 otherwise.

The Transmitted BSSID subfield is set to 1 to indicate that the reported AP is a transmitted BSSID. It is set to 0 if the reported AP is a nontransmitted BSSID. It is reserved if the Multiple BSSID subfield is set to 0.

The Member Of ESS With 2.4/5 GHz Co-Located AP subfield is set to 1 if the reported AP is part of an ESS where each AP in the ESS and operating in the same band as the reported AP (irrespective of the operating channel in that band) that might be detected by a STA receiving this frame [see the definition of “detected access point (AP)” in 3.2] has dot11MemberOfCollocated6GHzESSOptionActivated equal to true and also has a corresponding AP operating in the 2.4 GHz or 5 GHz bands that is in the same co-located AP set as that AP. It is set to 0 otherwise or if the reporting AP does not have that information. It is reserved if the reported AP is operating in the 2.4 GHz or 5 GHz bands.

NOTE 2—This subfield indicates that the reported AP is part of an ESS that has no 6 GHz-only APs that might be detected by a STA receiving this frame. This means that all APs operating in the 6 GHz band that are part of that ESS that might be detected by a STA receiving this frame can be discovered in the 2.4 GHz and/or 5 GHz bands.

The Unsolicited Probe Responses Active subfield is set to 1 if the reported AP is part of an ESS where all the APs that operate in the same channel as the reported AP and that might be detected by a STA receiving this frame [see the definition of “detected access point (AP)” in 3.2] have dot11UnsolicitedProbeResponseOptionActivated equal to true and are transmitting unsolicited Probe Response frames every 20 TUs or less (see 26.17.2.3). It is set to 0 otherwise or if the reporting AP does not have that information.

The Co-Located AP subfield is set to 1 if every AP in this Neighbor AP Information field is in the same co-located AP set as the transmitting AP. It is set to 0 otherwise.

The 20 MHz PSD subfield, if present, indicates a maximum transmit power for the Default category, with unit interpretation of PSD EIRP in dBm/MHz (see 9.4.2.161 and 11.7.5), corresponding to the primary 20 MHz channel of the reported AP. The maximum transmit power is encoded as a 2s complement signed integer. The value –128 is reserved. The value +127 indicates that no maximum transmit power is specified for the corresponding 20 MHz channel. For all other values *Y* of the subfield (i.e., –127 to +126),

the maximum transmit power in the 20 MHz channel is $Y/2$ dBm/MHz (i.e., ranging from -63.5 to $+63$ dBm/MHz).

NOTE 3—For example, suppose the reported AP transmits, in Beacon and Probe Response frames, one Transmit Power Envelope element with the Maximum Transmit Power For 20 MHz subfield indicating 20 dBm (regulatory client EIRP), then the 20 MHz PSD subfield indicates the equivalent PSD limit of 7 dBm/MHz with the value 14.

9.4.2.177 FILS Request Parameters element

Insert the following row into Table 9-288, and change the Reserved row accordingly:

Table 9-288—PHY Support Criterion subfield

Value	Explanation
3	Indicates that a responding FILS STA is HE capable.

9.4.2.199 TWT element

Replace Figure 9-686 with the following figure:

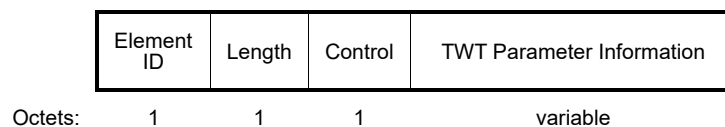


Figure 9-686—TWT element format

Change Figure 9-687 as follows:

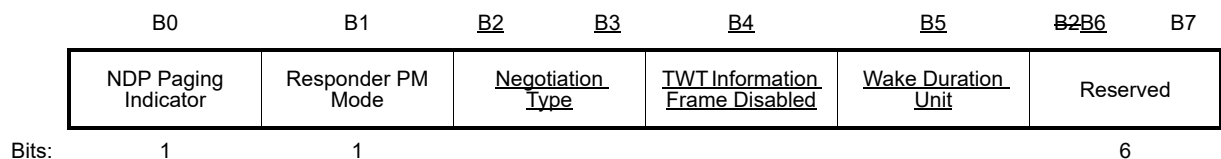


Figure 9-687—Control field format

Insert the following paragraphs, Table 9-296a, Figure 9-687a, and Figure 9-687b into 9.4.2.199 after the fifth paragraph (“The Responder PM Mode subfield...”):

The Negotiation Type subfield indicates whether the information included in the TWT element is for the negotiation of parameters of broadcast or individual TWT(s) or a Wake TBTT interval. The MSB of the Negotiation Type subfield is the Broadcast field.

The TWT Information Frame Disabled subfield is set to 1 to indicate that the reception of TWT Information frames is disabled by the STA; otherwise, it is set to 0.

The Wake Duration Unit subfield indicates the unit of the Nominal Minimum TWT Wake Duration field. The Wake Duration Unit subfield is set to 0 if the unit is 256 us and is set to 1 if the unit is a TU. A non-HE STA sets the Wake Duration Unit subfield to 0.

If the Broadcast field of the Negotiation Type subfield is 1, then one or more broadcast TWT parameter sets are contained in the TWT element (see Figure 9-687b). If the Broadcast field of the Negotiation Type subfield is 0, then only one Individual TWT parameter set is contained in the TWT element (see Figure 9-687a). An S1G STA sets the Negotiation Type subfield to 0.

A TWT element that has the Broadcast field in the Control field set to 1 is referred to as broadcast TWT element.

The Negotiation Type subfield determines the interpretation of the Target Wake Time, TWT Wake Interval Mantissa, and TWT Wake Interval Exponent subfields of the TWT element as defined in Table 9-296a.

Table 9-296a—Interpretation of Negotiation Type subfield and of Target Wake Time, TWT Wake Interval Mantissa, and TWT Wake Interval Exponent subfields

Negotiation Type subfield	Target Wake Time subfield	TWT Wake Interval Mantissa and TWT Wake Interval Exponent subfields	Description
0	A future Individual TWT SP start time	Interval between individual TWT SPs	Individual TWT negotiation between TWT requesting STA and TWT responding STA or individual TWT announcement by TWT responder. See 10.47 and 26.8.2. The TWT element contains one individual TWT parameter set.
1	Next Wake TBTT time	Interval between wake TBTTs	Wake TBTT and wake interval negotiation between TWT scheduled STA and TWT scheduling AP. See 26.8.6. The TWT element contains one individual TWT parameter set.
2	A future Broadcast TWT SP start time	Interval between broadcast TWT SPs	Provide broadcast TWT schedules to TWT scheduled STAs by including the TWT element in broadcast Management frames sent by TWT scheduling AP. See 26.8.3.2. The TWT element contains one or more broadcast TWT parameter sets.
3	A future Broadcast TWT SP start time	Interval between broadcast TWT SPs	Manage memberships in broadcast TWT schedules by including the TWT element in individually addressed Management frames sent by either a TWT scheduled STA or a TWT scheduling AP. See 26.8.3. The TWT element contains one or more broadcast TWT parameter sets.

The TWT Parameter Information field contains a single Individual TWT Parameter Set field with format defined in Figure 9-687a if the Broadcast subfield in the Control field is 0 and contains one or more Broadcast TWT Parameter Set fields with format defined in Figure 9-687b if the Broadcast subfield of the Control field is 1. The number of Broadcast TWT Parameter Set fields present is determined by the values of the Last Broadcast Parameter Set subfields of the Request Type fields.

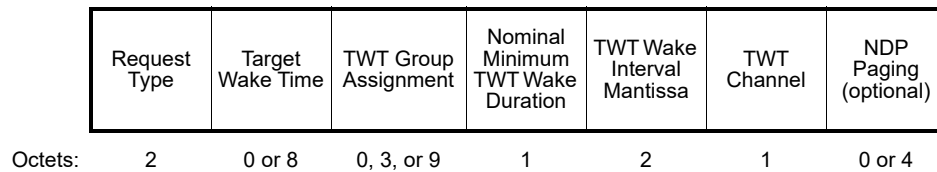


Figure 9-687a—Individual TWT Parameter Set field format

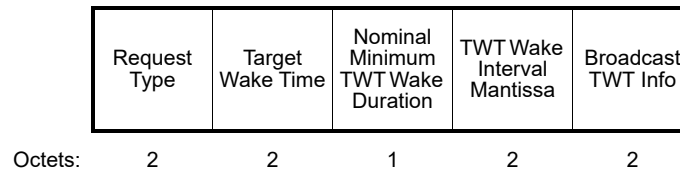


Figure 9-687b—Broadcast TWT Parameter Set field format

Change the following block of text (i.e., now 13th through 22nd paragraphs) in 9.4.2.199 as shown (including changing Figure 9-688 and Table 9-297 and inserting Figure 9-688a and Table 9-297a):

The format of the Request Type field of the Individual TWT Parameter Set field is shown in Figure 9-688 and of a Broadcast TWT Parameter Set field is shown in Figure 9-688a.

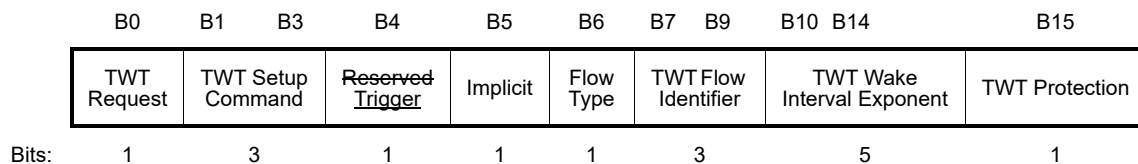


Figure 9-688—Request Type field format in Individual TWT Parameter Set field

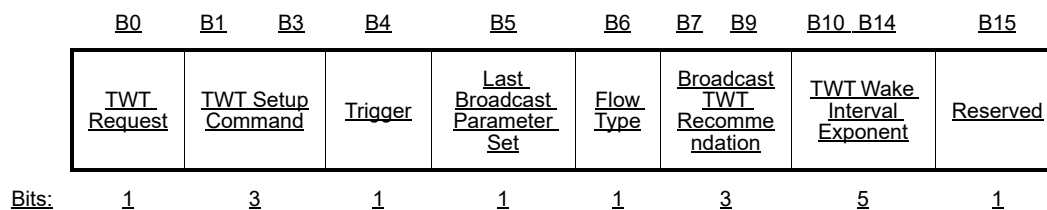


Figure 9-688a—Request Type field format in Broadcast TWT Parameter Set field

A STA that transmits a TWT element with the TWT Request subfield equal to 1 is a TWT requesting STA or TWT scheduled STA. Otherwise, it is a TWT responding STA or TWT scheduling STA.

The TWT Setup Command subfield values indicate the type of TWT command. The use of the TWT Setup Command field for the negotiation of individual and broadcast TWT is described, as shown in Table 9-297. The entries in the table apply to cases where the Negotiation Type subfield is not 1. For TWT Setup Command field use when the Negotiation Type subfield is 1, see 26.8.6.

Table 9-297—TWT Setup Command field values

TWT Setup Command field value	Command name	Description when transmitted by a TWT requesting STA	Description when transmitted by a TWT responding STA
0	Request TWT	The Target Wake Time field of the TWT element contains 0s as the TWT responding STA specifies the target wake time value for this case, other TWT parameters are suggested by the TWT requesting STA in the TWT request (see NOTE). <u>A TWT requesting or TWT scheduled STA requests to join a TWT without specifying a target wake time.</u> <u>This command is valid if the TWT Request field is equal to 1; otherwise, the command is not applicable.</u>	N/A
1	Suggest TWT	TWT requesting STA includes a set of TWT parameters such that if the requested target wake time value and/or other TWT parameters cannot be accommodated, then the TWT setup might still be accepted. <u>A TWT requesting or TWT scheduled STA requests to join a TWT and specifies a suggested set of TWT parameters with the possibility that if the requested target wake time and/or other TWT parameters cannot be accommodated, then the TWT setup might still be accepted by the TWT requesting or TWT scheduled STA.</u> <u>This command is valid if the TWT Request field is equal to 1; otherwise, it is not applicable.</u>	N/A
2	Demand TWT	TWT requesting STA includes a set of TWT parameters such that if the requested target wake time value and/or other TWT parameters cannot be accommodated, then the TWT setup will be rejected. <u>A TWT requesting or TWT scheduled STA requests to join a TWT and specifies a demanded set of TWT parameters. If the demanded set of TWT parameters is not accommodated by the responding STA or TWT scheduling AP, then the TWT requesting STA or TWT scheduled STA will reject the TWT setup.</u> <u>This command is valid if the TWT Request field is equal to 1; otherwise, it is not applicable.</u>	N/A

Table 9-297—TWT Setup Command field values (continued)

TWT Setup Command field value	Command name	Description when transmitted by a TWT requesting STA	Description when transmitted by a TWT responding STA
3	TWT Grouping	N/A <u>The TWT responding STA suggests TWT group parameters that are different from the suggested or demanded TWT parameters of the TWT requesting STA.</u> <u>This command is valid if the TWT Request field is 0 and the Negotiation Type subfield is 0 and sent by an SIG STA; otherwise, it is not applicable.</u>	TWT responding STA suggests TWT group parameters that are different from the suggested or demanded TWT parameters of the TWT requesting STA
4	Accept TWT	N/A <u>A TWT responding STA or TWT scheduling AP accepts the TWT request with the TWT parameters (see NOTE) indicated in the TWT element transmitted by the TWT requesting STA or TWT scheduled STA. This value is also used in unsolicited TWT responses.</u> <u>This command is valid if the TWT Request field is 0; otherwise, it is not applicable.</u>	TWT responding STA accepts the TWT request with the TWT parameters (See NOTE) indicated in the TWT element transmitted by the responding STA
5	Alternate TWT	N/A <u>A TWT responding STA or TWT scheduling AP suggests TWT parameters that are different from those suggested by the TWT requesting STA or TWT scheduled STA.</u> <u>This command is valid if the TWT Request field is 0; otherwise, it is not applicable.</u>	TWT responding STA suggests TWT parameters that are different from TWT requesting STA suggested or demanded TWT parameters
6	Dictate TWT	N/A <u>A TWT responding STA or TWT scheduling AP indicates TWT parameters that are different from those suggested by the TWT requesting STA or TWT scheduled STA.</u> <u>This command is valid if the TWT Request field is 0; otherwise, it is not applicable.</u>	TWT responding STA demands TWT parameters that are different from TWT requesting STA TWT suggested or demanded parameters
7	Reject TWT	N/A <u>A TWT responding STA or TWT scheduling AP rejects setup, or a TWT scheduling AP terminates an existing broadcast TWT, or a TWT scheduled STA terminates its membership in a broadcast TWT.</u>	TWT responding STA rejects TWT setup
NOTE—TWT Parameters are TWT, Nominal Minimum TWT Wake Duration, TWT Wake Interval, and TWT Channel subfield values indicated in the TWT element. <u>The Trigger subfield value indicated in the TWT element is also a TWT parameter for an HE STA.</u>			

The Trigger field indicates whether or not the TWT SP indicated by the TWT element includes triggering frames as defined in 26.8. The Trigger field is set to 1 to indicate that at least one triggering frame is transmitted during the TWT SP. The Trigger field is set to 0 otherwise.

~~When transmitted by a TWT requesting STA, the Implicit subfield is set to 1 to request indicate an implicit TWT and is set to 0 to indicate an explicit TWT.~~

~~When transmitted by a TWT requesting STA, the Implicit subfield is set to 0 to request an explicit TWT.~~

The Last Broadcast Parameter Set subfield is set to 0 to indicate that another broadcast TWT Parameter set follows this set. The Last Broadcast Parameter Set subfield is set to 1 to indicate that this is the last broadcast TWT Parameter set in the broadcast TWT element.

The Flow Type subfield indicates the type of interaction between the TWT requesting STA or TWT scheduled STA and the TWT responding STA or TWT scheduling AP at a TWT. Setting the Flow Type subfield to 0 indicates an announced TWT in which the TWT requesting STA or TWT scheduled STA will send a PS-Poll or an APSD trigger frame (see 11.2.3.5) to signal its awake state to the TWT responding STA or TWT scheduling AP before a frame that is not a Trigger frame is sent from the TWT responding STA or TWT scheduling AP to the TWT requesting STA or TWT scheduled STA. A TWT SP that is set up under an announced TWT agreement is an announced TWT SP. Setting the Flow Type subfield to 1 indicates an unannounced TWT in which the TWT responding STA or TWT scheduling AP will send a frame to the TWT requesting STA or TWT scheduled STA at TWT without waiting to receive a PS-Poll or an APSD trigger frame from the TWT requesting STA or TWT scheduled STA. A TWT SP that is set up under an unannounced TWT agreement is an unannounced TWT SP.

NOTE—The TWT requesting STA is expected to send the PS-Poll or APSD trigger frame if the TWT is a trigger-enabled TWT.

The TWT Flow Identifier subfield contains a 3-bit value ~~which that~~ identifies the specific information for this TWT request uniquely from other requests made between the same TWT requesting STA and TWT responding STA pair. The Broadcast TWT Recommendation subfield contains a value that indicates recommendations on the types of frames that are transmitted by TWT scheduled STAs and scheduling AP during the broadcast TWT SP, encoded according to the Broadcast TWT Recommendation field for a broadcast TWT element as defined in Table 9-297a. The Broadcast TWT Recommendation is reserved if transmitted by a TWT scheduled STA.

**Table 9-297a—Broadcast TWT Recommendation field
for a broadcast TWT element**

<u>Broadcast TWT Recommendation field value</u>	<u>Description when transmitted in a broadcast TWT element</u>
0	<u>No constraints on the frames transmitted during a broadcast TWT SP.</u>
1	<u>Frames transmitted during a broadcast TWT SP by a TWT scheduled STA are recommended to be limited to solicited status and solicited feedback:</u> — <u>PS-Poll and QoS Null frames</u> — <u>Feedback can be contained in the QoS Control field or in the HE variant HT Control field of the frame, if either is present (see 26.5.2, 26.9, 26.13, etc.)</u> — <u>Feedback in an HE TB feedback NDP, if solicited by the AP (see 26.5.7)</u> — <u>BQRs (see 26.5.6)</u> — <u>BSRs (see 26.5.5)</u> — <u>Frames that are sent as part of a sounding feedback exchange (see 26.7)</u> — <u>Management frames: Action or Action No Ack frames</u> — <u>Control response frames</u> <u>Trigger frames transmitted by the TWT scheduling AP during the broadcast TWT SP do not contain RUs for random access (see 26.8.3.2 and 26.5.4), otherwise, there are no other restrictions on the frames transmitted by the TWT scheduling AP.</u>

**Table 9-297a—Broadcast TWT Recommendation field
for a broadcast TWT element *(continued)***

Broadcast TWT Recommendation field value	Description when transmitted in a broadcast TWT element
<u>2</u>	<u>Frames transmitted during a broadcast TWT SP by a TWT scheduled STA are recommended to be limited to solicited status and solicited feedback:</u> — PS-Poll and QoS Null frames — Feedback can be contained in the QoS Control field or in the HE variant HT Control field of the frame, if either is present (see 26.5.2, 26.9, 26.13, etc.) — BQRs (see 26.5.6) — BSRs (see 26.5.5) — Frames that are sent as part of a sounding feedback exchange (see 26.7) — Management frames: Action, Action No Ack frames or (Re)Association Request — Control response frames <u>Trigger frames transmitted by the TWT scheduling AP during the broadcast TWT SP contain at least one RU for random access (see 26.8.3.2 and 26.5.4); otherwise, there are no restrictions on the frames transmitted by the TWT scheduling AP.</u>
<u>3</u>	<u>No constraints on the frames transmitted during a broadcast TWT SP, except that the AP transmits a TIM frame or a FILS Discovery frame including a TIM element at the beginning of each TWT SP (see 26.14.3.2).</u>
<u>4–7</u>	<u>Reserved</u>

In a TWT element transmitted by a TWT requesting or TWT scheduled STA, the TWT wake interval is equal to the average time that the ~~TWT requesting~~ STA expects to elapse between successive TWT SPs start times (see Table 9-296a). In a TWT element transmitted by a TWT responding STA or TWT scheduling AP, the TWT wake interval is equal to the average time that the ~~TWT responding~~ STA expects to elapse between successive TWT SPs start times. In a TWT element contained in a TWT request that is sent by the TWT scheduled STA to negotiate its wake intervals, the TWT wake interval indicates the value of the wake interval (see 26.8.6). The TWT Wake Interval Exponent subfield is set to the value of the exponent of the TWT wake interval value in microseconds, base 2. The TWT wake interval of the requesting STA is equal to $(\text{TWT Wake Interval Mantissa}) \times 2^{(\text{TWT Wake Interval Exponent})}$.

~~When~~ If transmitted by a TWT requesting STA or a TWT scheduled STA and the TWT Setup Command subfield contains a value corresponding to the command "Suggest TWT" or "Demand TWT", the Target Wake Time field contains a ~~positive~~ an unsigned integer corresponding to a TSF time at which the STA requests to wake, ~~or 0 when the TWT Setup Command subfield contains the value corresponding to the command "Request TWT".~~ If transmitted by a TWT requesting STA or a TWT scheduled STA and the TWT Setup Command subfield contains the value corresponding to the command "Request TWT", the Target Wake Time field contains the value 0. The Target Wake Time field is 8 octets if the Broadcast field is 0; otherwise, it is 2 octets with the lowest bit of the 2 octets corresponding to bit 10 of the relevant TSF value. ~~When~~ If a TWT responding STA with dot11TWTGroupingSupport equal to 0 transmits a TWT element to the TWT requesting STA, the TWT element contains a value in the Target Wake Time field corresponding to a TSF time at which the TWT responding STA requests the TWT requesting STA to wake for the corresponding TWT SP and it does not contain the TWT Group Assignment field.

The block of text from the now 23rd paragraph ("When a TWT responding STA") through the now 30th paragraph ("A non-AP STA uses") (including Figure 9-689 and Table 9-298) remains unchanged.

Change the following block of text (i.e., now 31st through 40th paragraphs) in 9.4.2.199 as shown (including inserting Figure 9-689a):

The Nominal Minimum TWT Wake Duration field indicates the minimum amount of time, in the units of 256 μ s indicated by the Wake Duration Unit subfield, that the TWT requesting STA or TWT scheduled STA is expected expects that it needs to be awake in order to complete the frame exchanges associated with the TWT flow identifier for the period of TWT wake interval, where TWT wake interval is the average time that the TWT requesting STA or TWT scheduled STA expects to elapse between successive TWT SPs.

The TWT Wake Interval Mantissa subfield is set to the value of the mantissa of the TWT wake interval value in microseconds, base 2.

The Broadcast TWT Info subfield is defined in Figure 9-689a.

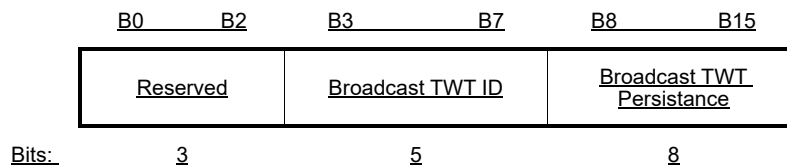


Figure 9-689a—Broadcast TWT Info subfield format

Within a TWT element that includes a TWT setup command value of Request TWT, Suggest TWT, or Demand TWT, the Broadcast TWT ID, if present, indicates a specific Broadcast TWT in which the transmitting STA is requesting to participate. Within a TWT element that includes a TWT setup command value of Accept TWT, Alternate TWT, Dictate TWT, or Reject TWT, the Broadcast TWT ID, if present, indicates a specific Broadcast TWT for which the transmitting STA is providing TWT parameters. Within a TWT element that includes a TWT setup command value of TWT Grouping, the Broadcast subfield is 0 and the Broadcast TWT ID, is not present. The value 0 in the Broadcast TWT ID subfield indicates the broadcast TWT whose membership corresponds to all STAs that are members of the BSS corresponding to the BSSID of the Management frame carrying the TWT element and that is permitted to contain Trigger frames with RA-RUs for unassociated STAs.

The Broadcast TWT Persistence subfield indicates the number of TBTTs during which the Broadcast TWT SPs corresponding to this broadcast TWT Parameter set are present. The number of beacon intervals during which the Broadcast TWT SPs are present is equal to the value in the Broadcast TWT Persistence subfield plus 1, except that the value 255 indicates that the Broadcast TWT SPs are present until explicitly terminated.

When transmitted by a TWT requesting STA that is negotiating SST operation, the TWT Channel field contains a bitmap indicating the channel the STA requests to use as a temporary primary channel during a TWT SP. When transmitted by a TWT responding STA that is negotiating SST operation, the TWT Channel field contains a bitmap indicating which channel the TWT requesting STA is allowed to use as a temporary channel during the TWT SP. When transmitted by a STA that is not negotiating SST operation, the TWT Channel field is reserved.

The TWT Channel field includes a bitmap that provides the channel that is being negotiated by a STA as a temporary channel during a TWT SP. Each bit in the bitmap corresponds to one minimum width channel for the band in which the TWT responding STA's associated BSS is currently operating, with the least significant bit corresponding to the lowest numbered channel of the operating channels of the BSS. In an SIG BSS, the minimum width channel is equal to the SST Channel Unit field of the SST Operation element if such an element has been previously received or is equal to 1 MHz for a BSS with a BSS primary channel width of 1 MHz and 2 MHz for a BSS with a BSS primary channel width of 2 MHz if no such element has been previously received from the AP to which the SST STA is associated. In an HE BSS, the

minimum width channel is equal to 20 MHz. Setting a position in the bitmap transmitted to 1 by a TWT requesting STA means that operation with that channel as the temporary primary channel is requested during a TWT SP. Setting a position in the bitmap transmitted to 1 by a TWT responding STA means that operation with that channel as the primary channel is allowed during the TWT SP. The TWT Channel field is used by an S1G STA as defined in 10.53 and is used by an HE STA as defined in 26.8.7. If the TWT channel field is 0 then the STA operates as define in 10.47 or 26.8.2.

A TWT requesting STA sets the TWT Protection subfield to 1 to request the TWT responding STA to provide protection of the set of TWT SPs corresponding to the requested TWT flow identifier by

- Allocating RAW(s) that restrict access to the medium during the TWT SP(s) for ~~the~~ ~~that~~ ~~(those)~~ TWTs that are set up within an S1G BSS.
- Enabling NAV protection during the TWT SP(s) for the TWTs that are set up within an HE BSS.

A TWT requesting STA sets the TWT Protection subfield to 0 if TWT protection ~~by RAW allocation~~ is not requested for the corresponding TWT(s).

~~When transmitted by a TWT responding STA that is an AP, the TWT Protection subfield indicates whether the TWT SP(s) identified in the TWT element will be protected. A TWT responding STA or TWT scheduling AP sets the TWT Protection subfield to 1 to indicate that the TWT SP(s) corresponding to the TWT flow identifier(s) of the TWT element will be protected by~~

- Allocating RAW(s) that restrict access to the medium during the TWT SP(s) for ~~the~~ ~~that~~ ~~(those)~~ TWT(s) where the TWT responding STA is an S1G STA.
- Enabling NAV protection during the TWT SP(s) for the TWTs where the TWT responding STE is an HE STA.

A TWT responding STA sets the TWT Protection subfield to 0 to indicate that the TWT SP(s) identified in the TWT element might not be protected ~~from S1G STAs in TIM mode by allocating RAW(s).~~

Insert the following subclauses [9.4.2.248 through 9.4.2.264, including Figure 9-788a through Figure 9-788ak, Table 9-322a through Table 9-322h, and Equation (9-3a) through Equation (9-3c)] after 9.4.2.247:

9.4.2.248 HE Capabilities element

9.4.2.248.1 General

An HE STA declares that it is an HE STA by transmitting the HE Capabilities element.

The HE Capabilities element contains a number of fields that are used to advertise the HE capabilities of an HE STA. The HE Capabilities element is defined in Figure 9-788a.

Element ID	Length	Element ID Extension	HE MAC Capabilities Information	HE PHY Capabilities Information	Supported HE-MCS And NSS Set	PPE Thresholds (optional)
Octets:	1	1	6	11	4, 8 or 12	variable

Figure 9-788a—HE Capabilities element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The HE MAC Capabilities Information, HE PHY Capabilities Information, Supported HE-MCS And NSS Set, and PPE Thresholds fields are defined in the subclauses below.

9.4.2.248.2 HE MAC Capabilities Information field

The format of the HE MAC Capabilities Information field is defined in Figure 9-788b.

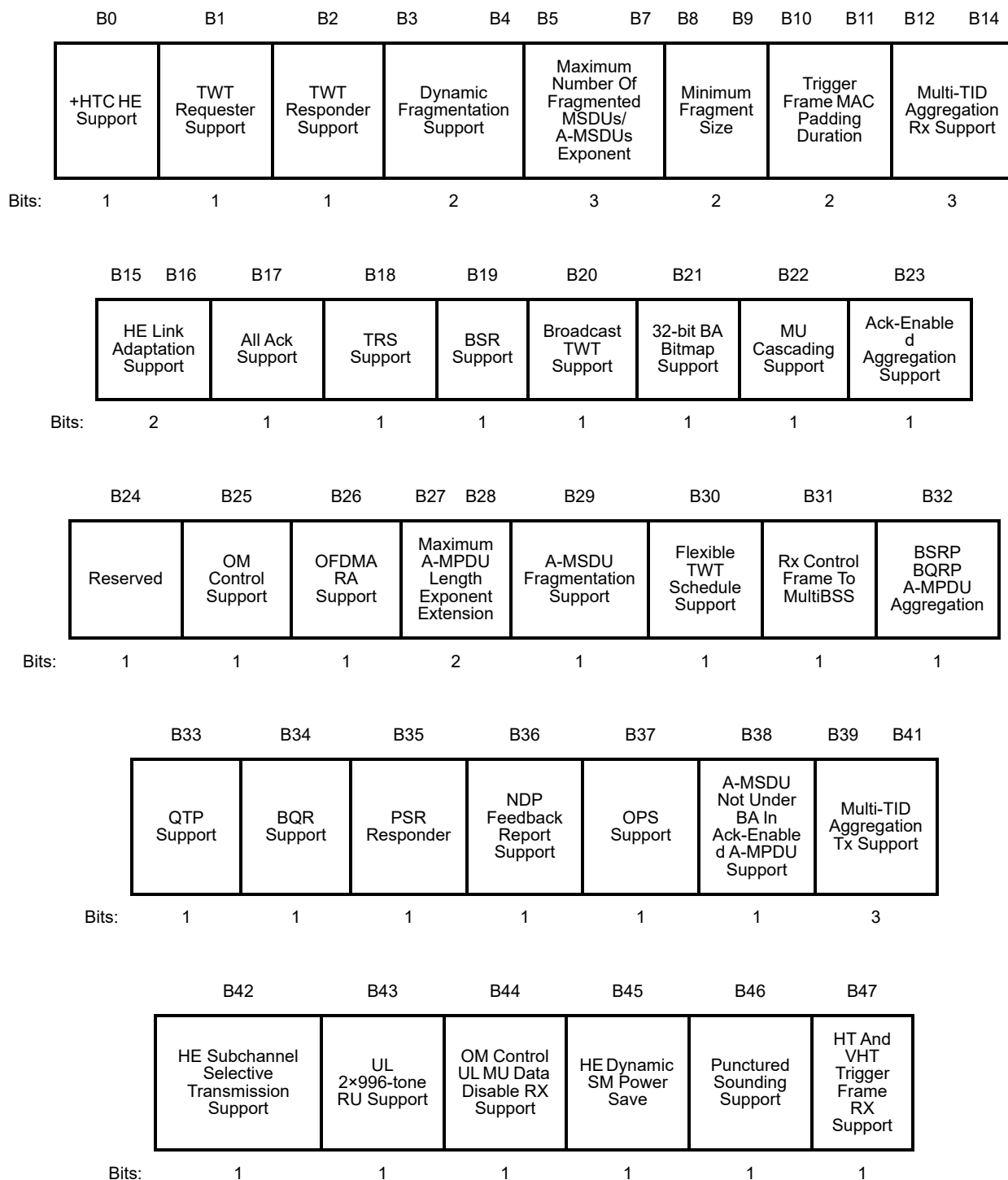


Figure 9-788b—HE MAC Capabilities Information field format

The subfields of the HE MAC Capabilities Information field are defined in Table 9-322a.

Table 9-322a—Subfields of the HE MAC Capabilities Information field

Subfield	Definition	Encoding
+HTC-HE Support	Indicates support for the reception of a frame that carries an HE variant HT Control field.	For a non-AP STA: Set to 1 if the STA supports reception of an HE variant HT Control field based on the description in 10.8. Set to 0 otherwise. An AP sets the +HTC-HE Support subfield to 1.
TWT Requester Support	Indicates support for the role of TWT requesting STA as described in 26.8.	Set to 1 if dot11TWTOptionActivated is true and the STA supports TWT requesting STA functionality (see 26.8). Set to 0 otherwise.
TWT Responder Support	Indicates support for the role of TWT responder STA as described in 26.8.	Set to 1 if dot11TWTOptionActivated is true and the STA supports TWT responder STA functionality (see 26.8). Set to 0 otherwise. An AP sets the TWT Responder Support subfield to 1.
Dynamic Fragmentation Support	Indicates the level of dynamic fragmentation that is supported by a STA as a recipient.	Set to 0 if the STA does not support dynamic fragmentation. Set to 1 if the STA supports level 1 dynamic fragmentation (see 26.3.1). Set to 2 if the STA supports level 2 dynamic fragmentation (see 26.3.1). Set to 3 if the STA supports level 3 dynamic fragmentation (see 26.3.1).
Maximum Number Of Fragmented MSDUs/A-MSDUs Exponent	Indicates the maximum number of fragmented MSDUs and/or A-MSDUs (if supported by the recipient) that the STA is capable of receiving concurrently.	If the Dynamic Fragmentation Support subfield is greater than 0: The maximum number of fragmented MSDUs and/or A-MSDUs, N_{max} , defined by this field is $N_{max} = 2^{\text{Maximum Number Of Fragmented MSDUs/A-MSDUs Exponent}}$, except that a value 7 in the Maximum Number Of Fragmented MSDUs/A-MSDUs Exponent subfield indicates that there is no restriction. Reserved if the Dynamic Fragmentation Support subfield is 0.

Table 9-322a—Subfields of the HE MAC Capabilities Information field (continued)

Subfield	Definition	Encoding
Minimum Fragment Size	Indicates the minimum frame body size in octets of the first fragment of an MSDU, A-MSDU (if supported), or MMPDU that is supported by the recipient STA.	If the Dynamic Fragmentation Support subfield is greater than 0: Set to 0 to indicate no minimum frame body size. Set to 1 to indicate a minimum frame body size of 128 octets. Set to 2 to indicate a minimum frame body size of 256 octets. Set to 3 to indicate a minimum frame body size of 512 octets. Reserved if the Dynamic Fragmentation Support subfield is 0.
Trigger Frame MAC Padding Duration	Indicates <i>MinTrigProcTime</i> , which is used in 26.5.2.2.3.	For a non-AP STA: Set to 0 to indicate 0. Set to 1 to indicate 8 μs. Set to 2 to indicate 16 μs. The value 3 is reserved. Reserved for an AP.
Multi-TID Aggregation Rx Support	Indicates the number of TIDs of QoS Data frames that an HE STA can receive in a multi-TID A-MPDU as described in 26.6.3.	Set to the number of TIDs minus 1 of QoS Data frames that an HE STA can receive in a multi-TID A-MPDU.
Multi-TID Aggregation Tx Support	Indicates the number of TIDs of QoS Data frames that an HE STA can transmit in a multi-TID A-MPDU as described in 26.6.3.	Set to the number of TIDs minus 1 of QoS Data frames that an HE STA can transmit in a multi-TID A-MPDU.
HE Link Adaptation Support	Indicates support for link adaptation using the HLA Control subfield.	If the +HTC-HE Support subfield is 1: Set to 0 (No Feedback) if the STA does not provide HE MFB. Set to 2 (Unsolicited) if the STA can receive and provide only unsolicited HE MFB. Set to 3 (Solicited and unsolicited) if the STA is capable of receiving and providing HE MFB in response to HE MRQ and if the STA can receive and provide unsolicited HE MFB. The value 1 is reserved. HE MFB and HE MRQ are MFB and MRQ using HLA Control subfield, respectively. Reserved if the +HTC-HE Support subfield is 0.
All Ack Support	Indicates support for the reception of a Multi-STA BlockAck frame under the all ack context (see 26.4.2)	Set to 1 if supported. Set to 0 otherwise.

Table 9-322a—Subfields of the HE MAC Capabilities Information field (continued)

Subfield	Definition	Encoding
TRS Support	For a non-AP STA, indicates support for receiving a frame with a TRS Control subfield.	For a non-AP STA that has set the +HTC-HE Support subfield to 1: Set to 1 if the STA supports reception of the TRS Control subfield. Set to 0 otherwise. Reserved for an AP or if the +HTC-HE Support subfield is 0.
BSR Support	For an AP, indicates support for receiving a frame with a BSR Control subfield. For a non-AP STA, indicates support for generating a frame with a BSR Control subfield.	If the +HTC-HE Support subfield is 1: Set to 1 if the STA supports the BSR Control subfield functionality. Set to 0 otherwise. Reserved if the +HTC-HE Support subfield is 0.
Broadcast TWT Support	For a non-AP STA, indicates support for the role of TWT scheduled STA. For an AP indicates support for the role of TWT scheduling AP as described in 26.8.3.	Set to 1 if the STA supports broadcast TWT functionality. Set to 0 otherwise.
32-bit BA Bitmap Support	Indicates support for the reception of a Multi-STA BlockAck frame that has a Per AID Info subfield addressed to it with a 32-bit Block Ack Bitmap subfield.	Set to 1 if the STA supports reception of a Multi-STA BlockAck frame that has a Per AID Info subfield addressed to it with a 32-bit Block Ack Bitmap subfield. Set to 0 otherwise.
MU Cascading Support	Indicates support for participating in an MU cascading sequence (see 26.5.3).	For an HE AP: Set to 1 to indicate that the AP is capable of transmitting an A-MPDU that is constructed following the MU cascade sequence rules (see 26.5.3) under MU cascade operation. Set to 0 otherwise. For a non-AP HE STA: Set to 1 to indicate that the non-AP STA is capable of receiving an A-MPDU that is constructed following the MU cascade sequence rules (see 26.5.3). Set to 0 otherwise.
Ack-Enabled Aggregation Support	Indicates support by a STA to receive an A-MPDU that contains two or more frames at least one of which solicits an Ack frame or acknowledgment context in a Multi-STA BlockAck frame as described in 26.6.3 and 26.5.1.1.	Set to 1 if the STA supports reception of this A-MPDU format. Set to 0 otherwise.

Table 9-322a—Subfields of the HE MAC Capabilities Information field (continued)

Subfield	Definition	Encoding
OM Control Support	Indicates support for receiving a frame with an OM Control subfield.	If the +HTC-HE Support subfield is 1 in a non-AP STA: Set to 1 if the non-AP STA supports reception of the OM Control subfield. Set to 0 otherwise. Reserved if the +HTC-HE Support subfield is 0 in a non-AP STA. An AP sets the OM Control Support subfield to 1.
OFDMA RA Support	For a non-AP STA, indicates support for the OFDMA random access procedure. For an AP, indicates support for sending Trigger frames that allocate RA-RUs. See 26.5.4.	Set to 1 if supported. Set to 0 otherwise.
Maximum A-MPDU Length Exponent Extension	Indicates the exponent extension for the maximum A-MPDU length supported in reception (see 26.6).	Set to the value of the maximum A-MPDU exponent extension value.
A-MSDU Fragmentation Support	Indicates support for the reception of fragmented A-MSDUs.	If the Dynamic Fragmentation Support subfield is not 0: Set to 1 to indicate support for the receipt of fragmented A-MSDUs. Set to 0 to indicate that reception of fragmented A-MSDUs is not supported. Reserved if the Dynamic Fragmentation Support subfield is 0.
Flexible TWT Schedule Support	Indicates support for the reception of TWT Information frames with flexible TWT schedules as defined in 26.8.4.4.	Set to 1 if the STA supports reception of a TWT Information frame with flexible TWT schedules. Set to 0 otherwise.
Rx Control Frame To MultiBSS	For a non-AP STA associated with a BSS corresponding to a nontransmitted BSSID, indicates support for the reception of a Control frame with TA equal to the transmitted BSSID.	For a non-AP STA: Set to 1 if supported. Set to 0 otherwise. Reserved for an AP.
BSRP BQRP A-MPDU Aggregation	For a non-AP STA, indicates whether the STA accepts a BSRP Trigger frame or BQRP Trigger frame that is aggregated with other Control, Data, and Management frames in an A-MPDU destined to the STA.	For a non-AP STA: Set to 1 if supported. Set to 0 otherwise. Reserved for an AP.
QTP Support	Indicates support for quiet time period (QTP) operation as described in 26.17.5.	Set to 1 if supported. Set to 0 otherwise.

Table 9-322a—Subfields of the HE MAC Capabilities Information field (continued)

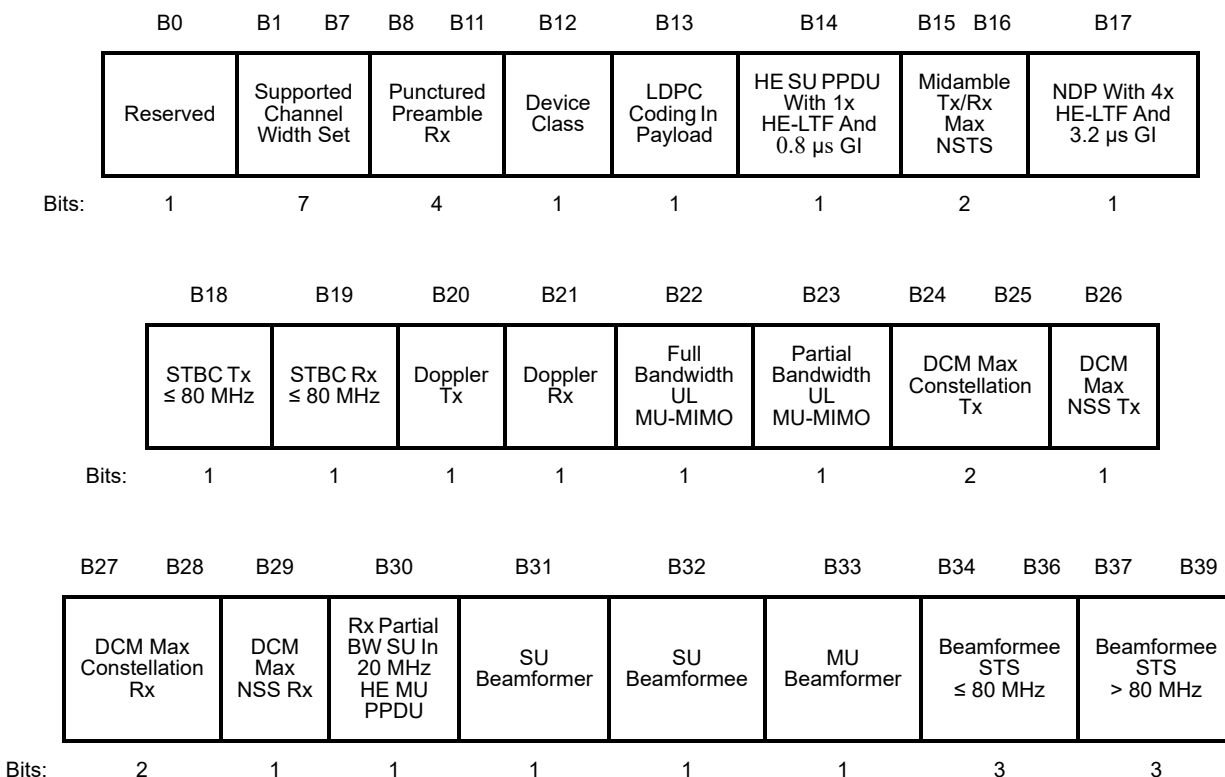
Subfield	Definition	Encoding
BQR Support	For an AP, indicates support for receiving a frame with a BQR Control subfield. For a non-AP STA, indicates support for generating a frame with a BQR Control subfield.	If the +HTC-HE Support subfield is 1: Set to 1 if the STA supports the BQR Control subfield functionality. Set to 0 otherwise. Reserved if the +HTC-HE Support subfield is 0.
PSR Responder	Indicates support for the role of PSR responder.	Set to 1 if the STA supports the role of PSR responder. Set to 0 otherwise.
NDP Feedback Report Support	For an AP, indicates support for the NDP feedback report procedure. For a non-AP STA, indicates support for responding to an NFRP Trigger frame.	Set to 1 if supported. Set to 0 otherwise.
OPS Support	For an AP, indicates support for encoding OPS information in the TIM element of FILS Discovery frames, TIM frames or OPS frames as described in 26.14.3.2. For a non-AP STA, indicates support for receiving the opportunistic power save encoded TIM elements.	Set to 1 if supported. Set to 0 otherwise.
A-MSDU Not Under BA In Ack-Enabled A-MPDU Support	Indicates support by a STA to receive an ack-enabled single-TID A-MPDU that carries an A-MSDU that is not under a block ack agreement.	Set to 1 if supported. Set to 0 otherwise.
HE Subchannel Selective Transmission Support	Indicates whether an HE STA supports an HE subchannel selective transmission operation described in 26.8.7.	Set to 1 if supported. Set to 0 otherwise.
UL 2×996-tone RU Support	Indicates support by a non-AP STA to receive a TRS Control subfield or a Trigger frame with a User Info field addressed to the STA with the RU Allocation subfield of the TRS Control subfield or the User Info field indicating 2×996-tone RU.	For a non-AP STA: Set to 1 if the STA supports reception of a TRS Control subfield with the RU Allocation subfield indicating a 2×996-tone RU or a Trigger frame with a User Info field addressed to the STA with the RU Allocation subfield indicating 2×996-tone RU. Set to 0 otherwise. Reserved for an AP.
OM Control UL MU Data Disable RX Support	Indicates whether an AP supports interpretation of the UL MU Data Disable subfield of the OM Control subfield as described in 26.5.2.	For an AP: Set to 1 if supported. Set to 0 otherwise. Reserved for a non-AP STA.

Table 9-322a—Subfields of the HE MAC Capabilities Information field (continued)

Subfield	Definition	Encoding
HE Dynamic SM Power Save	Indicates support for HE dynamic SM power save. See 26.14.4.	For a non-AP STA: Set to 1 if supported. Set to 0 if not supported. Reserved for an AP.
Punctured Sounding Support	Indicates support for punctured sounding as described in 26.7.	Set to 1 if dot11HEPuncturedSoundingOptionImplemented is true (see 26.7). Set to 0 otherwise.
HT And VHT Trigger Frame Rx Support	Indicates support for receiving a Trigger frame in an HT PPDU and receiving a Trigger frame in a VHT PPDU.	For a non-AP STA: Set to 1 if supported. Set to 0 otherwise. Reserved for an AP.

9.4.2.248.3 HE PHY Capabilities Information field

The format of the HE PHY Capabilities Information field is defined in Figure 9-788c.



(continued)

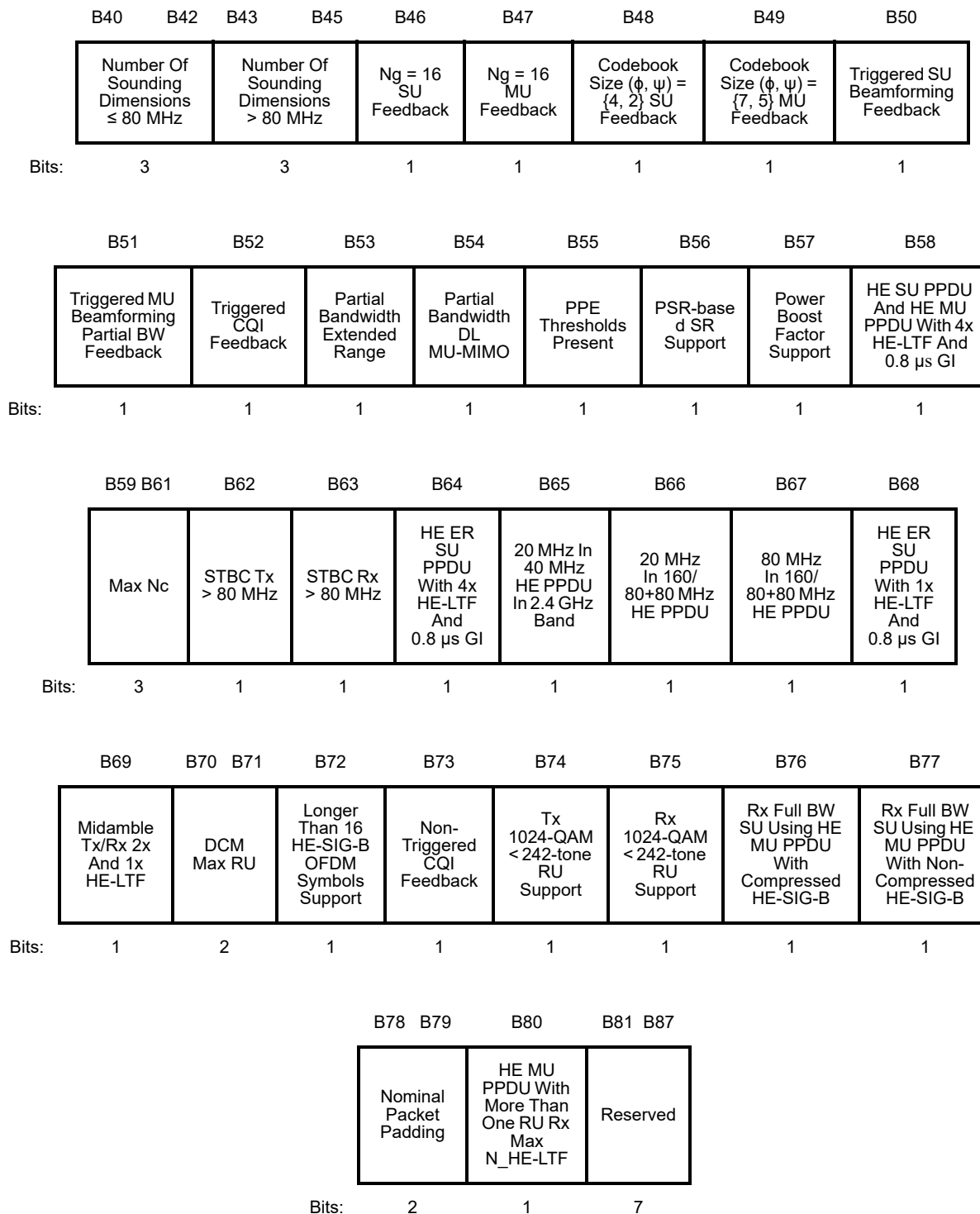


Figure 9-788c—HE PHY Capabilities Information field format

The subfields of the HE PHY Capabilities Information field are defined in Table 9-322b.

Table 9-322b—Subfields of the HE PHY Capabilities Information field

Subfield	Definition	Encoding
Supported Channel Width Set	In the 2.4 GHz band: — B0 indicates support for a 40 MHz channel width. — B1, B2, and B3 are reserved. — B4 indicates support of 242-tone RUs in a 40 MHz HE MU PPDU if a non-AP STA operates with a 20 MHz channel width and the 20 MHz In 40 MHz HE PPDU In 2.4 GHz subfield is 1; otherwise, B4 is reserved. — B5 and B6 are reserved. In the 5 GHz and 6 GHz bands: — B0 is reserved. — B1 indicates support for a 40 MHz and 80 MHz channel width. — B2 indicates support for a 160 MHz channel width. — B3 indicates support for a 160/80+80 MHz channel width. — B4 is reserved. — B5 indicates support of 242-tone RUs in a — 40 MHz and 80 MHz HE MU PPDU if a non-AP STA operates with 20 MHz channel width and the 20 MHz In 160/80+80 MHz HE PPDU subfield is set to 0, or — 40 MHz, 80 MHz, 160 MHz, 80+80 MHz HE MU PPDU if a non-AP STA operates with 20 MHz channel width and the 20 MHz In 160/80+80 MHz HE PPDU subfield is set to 1; — Otherwise, B5 is reserved. — B6 is reserved.	B0 is set to 0 if not supported. B0 is set to 1 if supported. B1 is set to 0 if not supported. B1 is set to 1 if supported. B2 is set to 0 if not supported. B2 is set to 1 if supported. If B2 is 1, then B1 is set to 1. B3 is set to 0 if not supported. B3 is set to 1 if supported. If B3 is 1, then B2 is set to 1. B4 is set to 0 if not supported. B4 is set to 1 if supported. B5 is set to 0 if not supported. B5 is set to 1 if supported. NOTE 1—If a non-AP STA operates with 20 MHz channel width and the 20 MHz In 40 MHz HE PPDU In 2.4 GHz subfield is 0, then B4 is set to 0. NOTE 2—If a non-AP STA operates with 20 MHz channel width and the 20 MHz In 160/80+80 MHz HE PPDU subfield is 0, then the 242-tone RU in a 160 MHz and 80+80 MHz HE MU PPDU in the 5 GHz band or 6 GHz band is not supported.

Table 9-322b—Subfields of the HE PHY Capabilities Information field (*continued*)

Subfield	Definition	Encoding
Punctured Preamble Rx	B0 indicates support for the reception of an 80 MHz preamble where the only punctured subchannel is the secondary 20 MHz channel. B1 indicates support for the reception of an 80 MHz preamble where the only punctured subchannel is one of the two 20 MHz subchannels in the secondary 40 MHz channel. B2 indicates support for the reception of a 160 MHz or 80+80 MHz preamble where the only punctured subchannels are the secondary 20 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel. See Table 27-20 for allowed 20 MHz punctured subchannel combinations. B3 indicates support for the reception of a 160 MHz or 80+80 MHz preamble where the only punctured subchannels are zero, one or both of the 20 MHz subchannels in the secondary 40 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel; at least one 20 MHz subchannel is punctured. See Table 27-20 for allowed 20 MHz punctured subchannel combinations.	B0 is set to 0 if not supported. B0 is set to 1 if supported. B1 is set to 0 if not supported. B1 is set to 1 if supported. B2 is set to 0 if not supported. B2 is set to 1 if supported. B3 is set to 0 if not supported. B3 is set to 1 if supported.
Device Class	For a non-AP STA, indicates whether the STA is a Class A or a Class B device.	For a non-AP STA: Set to 1 for a Class A device. Set to 0 for a Class B device. Reserved for an AP.
LDPC Coding In Payload	Indicates support for the transmission and reception of LDPC encoded PPDU.	Set to 0 if not supported. Set to 1 if supported. NOTE—Set to 1 by a STA that supports more than 4 spatial streams, an HE PPDU bandwidth greater than 20 MHz, HE-MCS 10, or HE-MCS 11.
HE SU PPDU With 1x HE-LTF And 0.8 μ s GI	Indicates support of the reception of an HE SU PPDU with 1x HE-LTF and 0.8 μ s guard interval duration.	Set to 0 if not supported. Set to 1 if supported.
Midamble Tx/Rx Max NSTS	If the Doppler Rx subfield is 1, indicates the maximum number of space-time streams supported for reception when a midamble is present in the Data field. If the Doppler Tx subfield is 1, indicates the maximum number of space-time streams supported for transmission when a midamble is present in the Data field. If both Doppler Rx and Doppler Tx subfields are 1, indicates the maximum number of space-time streams supported for transmission and reception when a midamble is present in the Data field.	Set to 0 for 1 space-time stream. Set to 1 for 2 space-time streams. Set to 2 for 3 space-time streams. Set to 3 for 4 space-time streams.

Table 9-322b—Subfields of the HE PHY Capabilities Information field (*continued*)

Subfield	Definition	Encoding
NDP With 4x HE-LTF And 3.2 μ s GI	For a beamformee, indicates support for receiving an HE sounding NDP with 4x HE-LTF and 3.2 μ s guard interval duration.	If the SU Beamformee field is 1: Set to 0 if not supported. Set to 1 if supported.
STBC Tx $\leq$ 80 MHz	Indicates support for the transmission of an HE TB PPDU using STBC that has a bandwidth less than or equal to 80 MHz.	For a non-AP STA: Set to 0 if not supported. Set to 1 if supported. Reserved for an AP.
STBC Rx $\leq$ 80 MHz	Indicates support for the reception of an HE PPDU using STBC that has a bandwidth less than or equal to 80 MHz.	Set to 0 if not supported. Set to 1 if supported.
Doppler Tx	Indicates support for transmitting HE PPDU with midamble.	Set to 0 if not supported. Set to 1 if supported.
Doppler Rx	Indicates support for receiving HE PPDU with midamble.	Set to 0 if not supported. Set to 1 if supported.
Full Bandwidth UL MU-MIMO	For an AP, indicates support for MU-MIMO reception of an HE TB PPDU on an RU that spans the entire PPDU bandwidth (UL MU-MIMO). For a non-AP STA, indicates support for the transmission of an HE TB PPDU on an RU that spans the entire PPDU bandwidth (UL MU-MIMO).	Set to 0 if not supported. Set to 1 if supported.
Partial Bandwidth UL MU-MIMO	For an AP, indicates support for receiving an HE TB PPDU on an RU where MU-MIMO is employed and where the RU does not span the entire PPDU bandwidth (UL MU-MIMO in OFDMA). For a non-AP STA, indicates support for transmitting an HE TB PPDU on an RU where MU-MIMO is employed and where the RU does not span the entire PPDU bandwidth (UL MU-MIMO in OFDMA). NOTE—The RU is a 106-tone or larger RU.	Set to 0 if not supported. Set to 1 if supported.
DCM Max Constellation Tx	Indicates the maximum supported constellation for DCM in the Data field of an HE TB PPDU that the STA is capable of transmitting.	For a non-AP STA: Set to 0 if DCM is not supported. Set to 1 for BPSK. Set to 2 for QPSK. Set to 3 for 16-QAM. Reserved for an AP.
DCM Max NSS Tx	Indicates the maximum number of spatial streams supported for transmission when DCM is used in the Data field of an HE TB PPDU.	For a non-AP STA, if the DCM Max Constellation Tx subfield is not 0: Set to 0 for 1 spatial stream. Set to 1 for 2 spatial streams. Reserved for an AP or if the DCM Max Constellation Tx subfield is 0.

Table 9-322b—Subfields of the HE PHY Capabilities Information field (*continued*)

Subfield	Definition	Encoding
DCM Max Constellation Rx	Indicates the maximum supported constellation for DCM in both the Data field and HE-SIG-B field that the STA is capable of receiving.	Set to 0 if DCM is not supported. Set to 1 for BPSK. Set to 2 for QPSK. Set to 3 for 16-QAM.
DCM Max NSS Rx	Indicates the maximum number of spatial streams supported for reception when DCM is used in the Data field.	Set to 0 for 1 spatial stream. Set to 1 for 2 spatial streams.
Rx Partial BW SU In 20 MHz HE MU PPDU	For an AP, TDLS STA or IBSS STA, indicates support for the reception of a 20 MHz HE MU PPDU with just a 106-tone RU. NOTE—It is mandatory for a non-AP HE STA to support reception of a 20 MHz HE MU PPDU with just a 106-tone RU.	For an AP, TDLS STA or IBSS STA: Set to 0 if not supported. Set to 1 if supported. Reserved for a non-AP STA.
SU Beamformer	Indicates support for operation as an SU beamformer.	Set to 0 if not supported. Set to 1 if supported. NOTE—Set to 1 by an AP with support for 4 or more spatial streams.
SU Beamformee	Indicates support for operation as an SU beamformee.	For an AP: Set to 0 if not supported. Set to 1 if supported. Set to 1 for a non-AP STA.
MU Beamformer	Indicates support for operation as an MU beamformer.	For an AP: Set to 0 if not supported. Set to 1 if the SU Beamformer field is 1 and operation as an MU beamformer is supported. Set to 0 for a non-AP STA. NOTE—Set to 1 by an AP with support for 4 or more spatial streams.
Beamformee STS ≤ 80 MHz	For a PPDU bandwidth less than or equal to 80 MHz, indicates the maximum number of space-time streams that the STA can receive in an HE sounding NDP, which is also the maximum total number of space-time streams over all the users that can be sent in a DL MU-MIMO transmission on an RU that includes that STA, where the RU might or might not span the entire PPDU bandwidth.	If the SU Beamformee subfield is 1: Set to the maximum number of space-time streams that the STA is capable of receiving in an HE sounding NDP minus 1. The minimum value of this field is 3. Reserved if the SU Beamformee field is 0.

Table 9-322b—Subfields of the HE PHY Capabilities Information field (*continued*)

Subfield	Definition	Encoding
Beamformee STS > 80 MHz	For a PPDU bandwidth greater than 80 MHz, indicates the maximum number of space-time streams that the STA can receive in an HE sounding NDP, which is also the maximum total number of space-time streams over all the users that can be sent in a DL MU-MIMO transmission on an RU that includes that STA, where the RU might or might not span the entire PPDU bandwidth.	If the SU Beamformee subfield is 1: Set to the maximum number of space-time streams that the STA is capable of receiving in an HE sounding NDP minus 1. The minimum value of this field is 3. Reserved if the SU Beamformee subfield is 0 or the Supported Channel Width Set field does not indicate support for bandwidths greater than 80 MHz.
Number Of Sounding Dimensions ≤ 80 MHz	For bandwidth less than or equal to 80 MHz, it indicates the beamformer's capability indicating the maximum value of the TXVECTOR parameter NUM_STS for an HE sounding NDP.	If the SU Beamformer subfield is 1: Set to the supported maximum TXVECTOR parameter NUM_STS value minus 1. Reserved if the SU Beamformer subfield is 0.
Number Of Sounding Dimensions > 80 MHz	For bandwidth greater than 80 MHz, indicates the beamformer's capability indicating the maximum value of the TXVECTOR parameter NUM_STS for an HE sounding NDP.	If the SU Beamformer subfield is 1: Set to the supported maximum TXVECTOR parameter NUM_STS value minus 1. Reserved if the SU Beamformer subfield is 0 or the Supported Channel Width Set field does not indicate support for bandwidths greater than 80 MHz.
Ng = 16 SU Feedback	Indicates HE beamformee support for a subcarrier grouping of 16 in the HE Compressed Beamforming Report field for SU feedback.	Set to 0 if not supported. Set to 1 if supported.
Ng = 16 MU Feedback	Indicates HE beamformee support for a subcarrier grouping of 16 in the HE Compressed Beamforming Report field for MU feedback.	Set to 0 if not supported. Set to 1 if supported.
Codebook Size (ϕ , ψ) = {4, 2} SU Feedback	Indicates HE beamformee support for a codebook size (ϕ , ψ) = {4, 2} in the HE Compressed Beamforming Report field for SU feedback.	Set to 0 if not supported. Set to 1 if supported.
Codebook Size (ϕ , ψ) = {7, 5} MU Feedback	Indicates HE beamformee support for a codebook size (ϕ , ψ) = {7, 5} in the HE Compressed Beamforming Report field for MU feedback.	Set to 0 if not supported. Set to 1 if supported.
Triggered SU Beamforming Feedback	For an AP, indicates support for the reception of partial- and full-bandwidth SU feedback in an HE TB sounding sequence. For a non-AP STA, indicates support for the transmission of partial- and full-bandwidth SU feedback in an HE TB sounding sequence.	Set to 0 if not supported. Set to 1 if supported.

Table 9-322b—Subfields of the HE PHY Capabilities Information field (*continued*)

Subfield	Definition	Encoding
Triggered MU Beamforming Partial BW Feedback	For an AP, indicates support for the reception of partial-bandwidth MU feedback in an HE TB sounding sequence. For a non-AP STA, indicates support for the transmission of partial-bandwidth MU feedback in an HE TB sounding sequence.	Set to 0 if not supported. Set to 1 if supported.
Triggered CQI Feedback	For an AP, indicates support for the reception of partial- and full-bandwidth CQI feedback in an HE TB sounding sequence. For a non-AP STA, indicates support for the transmission of partial- and full-bandwidth CQI feedback in an HE TB sounding sequence.	Set to 0 if not supported. Set to 1 if supported.
Partial Bandwidth Extended Range	Indicates support for the transmission and reception of an HE ER SU PPDU in which the HE modulated fields are transmitted over the higher frequency 106-tone RU within primary 20 MHz channel.	Set to 0 if not supported. Set to 1 if supported.
Partial Bandwidth DL MU-MIMO	For a non-AP STA, indicates support for the reception of a DL MU-MIMO transmission on an RU in an HE MU PPDU where the RU does not span the entire PPDU bandwidth (DL MU-MIMO within OFDMA).	For a non-AP STA: Set to 0 if not supported. Set to 1 if supported. NOTE—A non-AP STA that sets this field to 0 supports receiving a partial-bandwidth RU allocated to a single user within an HE MU PPDU where some other RU is employing DL MU-MIMO. Reserved for an AP.
PPE Thresholds Present	Indicates whether the PPE Thresholds field is present.	Set to 1 if PPE Thresholds field is present. Set to 0 otherwise.
PSR-based SR Support	Indicates support for PSR-based SR operation.	Set to 0 if not supported. Set to 1 if supported.
Power Boost Factor Support	Indicates that the STA supports a power boost factor for the RUs in an HE MU PPDU in the range [0.5, 2].	Set to 0 if not supported. Set to 1 if supported.
HE SU PPDU And HE MU PPDU With 4x HE-LTF And 0.8 μ s GI	Indicates support for the reception of an HE SU PPDU and HE MU PPDU with 4x HE-LTF and 0.8 μ s guard interval duration.	Set to 0 if not supported. Set to 1 if supported. This subfield is set to 1 if the HE ER SU PPDU With 4x HE-LTF And 0.8 μ s GI subfield is 1.
Max Nc	Indicates the maximum supported Nc for an HE compressed beamforming/CQI report.	If the SU Beamformee subfield is 1: Set to the maximum supported Nc for an HE compressed beamforming/CQI report minus 1. Reserved if the SU Beamformee subfield is 0.

Table 9-322b—Subfields of the HE PHY Capabilities Information field (*continued*)

Subfield	Definition	Encoding
STBC Tx > 80 MHz	Indicates support for the transmission of an HE TB PPDU using STBC that has a bandwidth greater than 80 MHz.	For a non-AP STA: Set to 0 if not supported. Set to 1 if supported. Reserved for an AP or if the Supported Channel Width Set field does not indicate support for bandwidths greater than 80 MHz.
STBC Rx > 80 MHz	Indicates support for the reception of an HE PPDU using STBC that has a bandwidth greater than 80 MHz.	Set to 0 if not supported. Set to 1 if supported. Reserved if the Supported Channel Width Set field does not indicate support for bandwidths greater than 80 MHz.
HE ER SU PPDU With 4x HE-LTF And 0.8 μ s GI	Indicates support for the reception of an HE ER SU PPDU with 4x HE-LTF and 0.8 μ s guard interval duration.	Set to 0 if not supported. Set to 1 if supported.
20 MHz In 40 MHz HE PPDU In 2.4 GHz Band	Indicates support for 26-, 52-, and 106-tone RU mapping for a 20 MHz operating non-AP HE STA that is the receiver of a 40 MHz HE MU PPDU in 2.4 GHz band, or the transmitter of a 40 MHz HE TB PPDU in 2.4 GHz band.	Set to 0 if not supported. Set to 1 if supported. NOTE—Set to 1 if B0 of the Supported Channel Width Set subfield is 1. Reserved for an AP.
20 MHz In 160/80+80 MHz HE PPDU	Indicates support for 26-, 52-, and 106-tone RU mapping for a 20 MHz operating non-AP HE STA that is the receiver of an 80+80 MHz or a 160 MHz HE MU PPDU, or the transmitter of an 80+80 MHz or 160 MHz HE TB PPDU.	Set to 0 if not supported. Set to 1 if supported. NOTE—Set to 1 if B2 of the Supported Channel Width Set subfield is 1. Reserved for an AP.
80 MHz In 160/80+80 MHz HE PPDU	Indicates supports of 160/80+80 MHz OFDMA for a non-AP HE STA that sets B1 of Supported Channel Width Set to 1, and sets B2 and B3 of Supported Channel Width Set each to 0, when operating with 80 MHz channel width. The capability bit is applicable while receiving an 80+80 MHz or a 160 MHz HE MU PPDU, or transmitting an 80+80 MHz or a 160 MHz HE TB PPDU.	Set to 0 if not supported. Set to 1 if supported. NOTE—Set to 1 if B2 of the Supported Channel Width Set subfield is 1. Reserved for an AP.
HE ER SU PPDU With 1x HE-LTF And 0.8 μ s GI	Indicates support of the reception of an HE ER SU PPDU with 1x HE-LTF and 0.8 μ s guard interval duration.	Set to 0 if not supported. Set to 1 if supported.

Table 9-322b—Subfields of the HE PHY Capabilities Information field (*continued*)

Subfield	Definition	Encoding
Midamble Tx/Rx 2x And 1x HE-LTF	If the Doppler Rx subfield is 1, indicates support for receiving midambles with 2x HE-LTF, 1x HE-LTF in HE SU PPDU if the HE SU PPDU With 1x HE-LTF And 0.8 μs GI subfield is 1, and 1x HE-LTF in HE ER SU PPDU if the HE ER SU PPDU With 1x HE-LTF And 0.8 μs GI subfield is 1. If the Doppler Tx subfield is 1, indicates support for transmitting midambles with 2x HE-LTF, 1x HE-LTF in HE TB PPDU when allowed. If both the Doppler Rx and Doppler Tx subfields are 1, indicates support for receiving midambles with 2x HE-LTF, 1x HE-LTF in HE SU PPDU if the HE SU PPDU With 1x HE-LTF And 0.8 μs GI subfield is 1, and 1x HE-LTF in HE ER SU PPDU if the HE ER SU PPDU With 1x HE-LTF And 0.8 μs GI subfield is 1; and also support for transmitting midambles with 2x HE-LTF, 1x HE-LTF in HE TB PPDU when allowed.	Set to 0 if not supported. Set to 1 if supported.
DCM Max RU	If the DCM Max Constellation Tx subfield is greater than 0, then the DCM Max RU subfield indicates the maximum RU size that the STA might transmit with DCM applied. If the DCM Max Constellation Rx subfield is greater than 0, then the DCM Max RU subfield indicates the maximum RU size with DCM applied that the STA can receive. If both the DCM Max Constellation Tx subfield and DCM Max Constellation Rx subfield are 0, then this subfield is reserved.	Set to 0 for 242-tone RU Set to 1 for 484-tone RU Set to 2 for 996-tone RU Set to 3 for 2×996-tone RU
Longer Than 16 HE-SIG-B OFDM Symbols Support	For a non-AP STA, indicates support for receiving a DL HE MU PPDU where the number of OFDM symbols in the HE-SIG-B field is greater than 16.	Set to 0 if not supported. Set to 1 if supported. A 20 MHz-only non-AP HE STA sets this to 0.
Non-Triggered CQI Feedback	For an AP, indicates support for the reception of full-bandwidth non-triggered CQI feedback. For a non-AP STA, indicates support for the transmission of full-bandwidth non-triggered CQI feedback.	Set to 0 if not supported. Set to 1 if supported.
Tx 1024-QAM < 242-tone RU Support	For a non-AP STA, indicates support for the transmission of 1024-QAM on a 26-, 52-, and 106-tone RU.	For a non-AP STA: Set to 0 if not supported. Set to 1 if supported. Reserved for an AP.
Rx 1024-QAM < 242-tone RU Support	Indicates support for the reception of 1024-QAM on a 26-, 52-, and 106-tone RU.	Set to 0 if not supported. Set to 1 if supported.

Table 9-322b—Subfields of the HE PHY Capabilities Information field (*continued*)

Subfield	Definition	Encoding
Rx Full BW SU Using HE MU PPDU With Compressed HE-SIG-B	Indicates support for reception of an HE MU PPDU with a single user allocated an RU spanning the entire PPDU bandwidth and a compressed HE-SIG-B field format.	Set to 0 if not supported. Set to 1 if supported.
Rx Full BW SU Using HE MU PPDU With Non-Compressed HE-SIG-B	Indicates support for reception of an HE MU PPDU with a bandwidth less than or equal to 80 MHz, a single user allocated an RU spanning the entire PPDU bandwidth and a noncompressed HE-SIG-B field format.	Set to 0 if not supported. Set to 1 if supported.
Nominal Packet Padding	Indicates the nominal packet padding to be used for all constellations, N_{SS} and RU allocations the STA supports if the PPE Thresholds Present subfield is set to 0.	Set to 0 if the nominal packet padding is 0 μ s for all constellations, N_{SS} and RU allocations the STA supports. Set to 1 if the nominal packet padding is 8 μ s for all constellations, N_{SS} and RU allocations the STA supports. Set to 2 if the nominal packet padding is 16 μ s for all constellations, N_{SS} and RU allocations the STA supports. The value 3 is reserved. Reserved if the PPE Thresholds Present subfield is 1.
HE MU PPDU With More Than One RU Rx Max N_{HE-LTF}	Indicates the maximum number of HE-LTF symbols, N_{HE-LTF} , that the STA is capable of receiving in an HE MU PPDU with more than one RU.	For a non-AP STA: Set to 0 if the number of HE-LTF symbols, N_{HE-LTF} , that the STA is capable of receiving in an HE MU PPDU equals $N_{HE-LTF}(\text{Beamformee STS} + 1)$, where the $N_{HE-LTF}(\text{Beamformee STS} + 1)$ is the lookup function shown in Table 21-13 with $N_{STS, total}$ taken as Beamformee STS + 1. NOTE—For a PPDU bandwidth less than or equal to 80 MHz, use the value defined in the Beamformee STS $\leq$ 80 MHz subfield. For a PPDU bandwidth greater than 80 MHz, use the value defined in Beamformee STS $>$ 80 MHz subfield. Set to 1 if the maximum number of HE-LTF symbols, N_{HE-LTF} , that the STA is capable of receiving in an HE MU PPDU is 8. Reserved for an AP.

9.4.2.248.4 Supported HE-MCS And NSS Set field

The Supported HE-MCS And NSS Set field indicates the combinations of HE-MCSs and spatial streams that a STA supports for reception and the combinations that it supports for transmission. The format of the field is shown in Figure 9-788d.

Rx HE-MCS Map ≤ 80 MHz	Tx HE-MCS Map ≤ 80 MHz	Rx HE-MCS Map 160 MHz	Tx HE-MCS Map 160 MHz	Rx HE-MCS Map 80+80 MHz	Tx HE-MCS Map 80+80 MHz
Octets: 2	2	0 or 2	0 or 2	0 or 2	0 or 2

Figure 9-788d—Supported HE-MCS And NSS Set field format

The subfields of the Supported HE-MCS And NSS Set field, and their presence, are defined in Table 9-322c.

Table 9-322c—Subfields of the Supported HE-MCS And NSS Set field

Subfield	Definition	Encoding
Rx HE-MCS Map ≤ 80 MHz	If the operating channel width of the STA is greater than 80 MHz, indicates the maximum value of the RXVECTOR parameter MCS of a PPDU that can be received by the STA for a PPDU with bandwidth less than or equal to 80 MHz for each number of spatial streams. If the operating channel width of this STA is less than or equal to 80 MHz, indicates the maximum value of the RXVECTOR parameter MCS for a PPDU that can be received by the STA for each number of spatial streams.	The format and encoding of this subfield are defined in Figure 9-788d and the associated description.
Tx HE-MCS Map ≤ 80 MHz	If the operating channel width of this STA is greater than 80 MHz, indicates the maximum value of the TXVECTOR parameter MCS of a PPDU that can be transmitted by the STA for a PPDU with bandwidth less than or equal to 80 MHz for each number of spatial streams. If the operating channel width of this STA is less than or equal to 80 MHz, indicates the maximum value of the TXVECTOR parameter MCS for a PPDU that can be transmitted by the STA for each number of spatial streams.	The format and encoding of this subfield are defined in Figure 9-788d and the associated description.
Rx HE-MCS Map 160 MHz	If the operating channel width of the STA is 160 MHz, indicates the maximum value of the RXVECTOR parameter MCS for a 160 MHz PPDU that can be received by this STA for each number of spatial streams.	The format and encoding of this subfield are defined in Figure 9-788d and the associated description. The Rx HE-MCS Map 160 MHz subfield is present if B2 of the Supported Channel Width Set subfield of the HE PHY Capabilities Information field is 1; otherwise, it is not present.

Table 9-322c—Subfields of the Supported HE-MCS And NSS Set field (continued)

Subfield	Definition	Encoding
Tx HE-MCS Map 160 MHz	If the operating channel width of the STA is 160 MHz, indicates the maximum value of the TXVECTOR parameter MCS for a 160 MHz PPDU that can be transmitted by this STA for each number of spatial streams.	The format and encoding of this subfield are defined in Figure 9-788d and the associated description. The Tx HE-MCS Map 160 MHz subfield is present if B2 of the Supported Channel Width Set subfield of the HE PHY Capabilities Information field is 1; otherwise, it is not present.
Rx HE-MCS Map 80+80 MHz	If the operating channel width of the STA is 160 MHz, indicates the maximum value of the RXVECTOR parameter MCS for an 80+80 MHz PPDU that can be received by this STA for each number of spatial streams.	The format and encoding of this subfield are defined in Figure 9-788d and the associated description. The Rx HE-MCS Map 80+80 MHz subfield is present if B3 of the Supported Channel Width Set subfield of the HE PHY Capabilities Information field is 1; otherwise, it is not present.
Tx HE-MCS Map 80+80 MHz	If the operating channel width of the STA is 160 MHz, indicates the maximum value of the TXVECTOR parameter MCS for an 80+80 MHz PPDU that can be transmitted by this STA for each number of spatial streams.	The format and encoding of this subfield are defined in Figure 9-788d and the associated description. The Tx HE-MCS Map 80+80 MHz subfield is present if B3 of the Supported Channel Width Set subfield of the HE PHY Capabilities Information field is 1; otherwise, it is not present.

The Rx HE-MCS Map and Tx HE-MCS Map subfields have the format shown in Figure 9-788e.

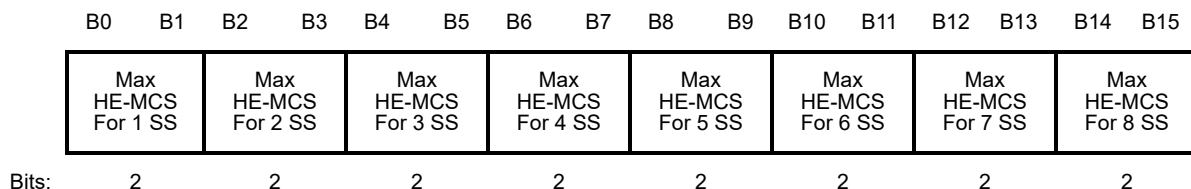


Figure 9-788e—Rx HE-MCS Map subfield, Tx HE-MCS Map subfield, and Basic HE-MCS And NSS Set field format

The Max HE-MCS For n SS subfield (where $n = 1, \dots, 8$) is encoded as follows:

- 0 indicates support for HE-MCS 0–7 for n spatial streams.
- 1 indicates support for HE-MCS 0–9 for n spatial streams.
- 2 indicates support for HE-MCS 0–11 for n spatial streams.
- 3 indicates that n spatial streams is not supported for HE PPDU.

The maximum receive N_{SS} for a given HE-MCS is equal to the smaller of

- The maximum value of n for which the Max HE-MCS For n SS has a value that indicates support for that HE-MCS or
- The maximum supported N_{SS} as indicated by the value of the Rx NSS field of the Operating Mode Notification frame if the value of Rx NSS Type is 0 or of the OM Control subfield.

NOTE 1—An HE-MCS indicated as supported in the Rx HE-MCS Map fields for a particular number of spatial streams might not be valid at all bandwidths (see 27.5) and might be affected by 26.15.4.3.

The maximum transmit N_{SS} for a given HE-MCS is equal to the smaller of

- The maximum value of n for which the Max HE-MCS For n SS has a value that indicates support for that HE-MCS (0, 1, or 2 for HE-MCS 0-7, 1 or 2 for HE-MCS 8-9, 2 for HE-MCS 10-11) or
- The maximum supported NSTS as indicated by the value of the Tx NSTS field of the OM Control subfield sent by a non-AP STA.

NOTE 2—An HE-MCS indicated as supported in the Tx HE-MCS Map fields for a particular number of space-time streams might not be valid at all bandwidths (see 27.5) and might be affected by 26.15.4.3.

9.4.2.248.5 PPE Thresholds field

The PPE Thresholds field determines the nominal packet padding value (see 26.12) for an HE PPDU of a particular RU allocation size and NSTS value. The format of the PPE Thresholds field is defined in Figure 9-788f.

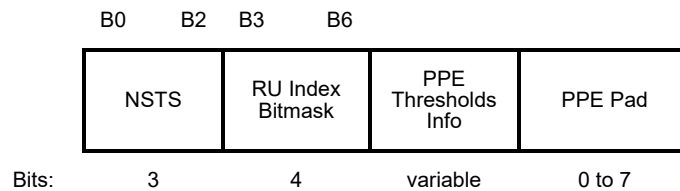


Figure 9-788f—PPE Thresholds field format

The NSTS subfield contains an unsigned integer that is the number of NSTS values minus 1 for which PPE threshold values are included in the PPE Thresholds List subfield.

The RU Index Bitmask subfield contains a bitmask that indicates whether the PPE Thresholds Info field contains PPET16 and PPET8 values for the four possible RU sizes indicated in Table 9-322e. The PPET16 and PPET8 values for RU allocation index k is present in the PPE Thresholds Info field only if bit k of the RU Index Bitmask subfield (bit $3 + k$ of the PPE Thresholds field) is 1. For example, if B0 of the RU Index Bitmask subfield (B3 of the PPE Thresholds field) is 1, PPET16 and PPET8 values are present in the PPE Thresholds Info field for the RU allocation size corresponding to RU allocation index 0 (242-tone RU). If B0 of the RU Index Bitmask subfield is 0, PPET16 and PPET8 values are not present in the PPE Thresholds Info field for the RU allocation size corresponding to RU allocation index 0.

The PPE Thresholds List field contains $6 \times (NSTS + 1)$ bits, where $NSTS$ is the value in the NSTS field, for every bit in the RU Index Bitmask subfield that is nonzero. The format of the PPE Thresholds Info field is defined in Figure 9-788g.

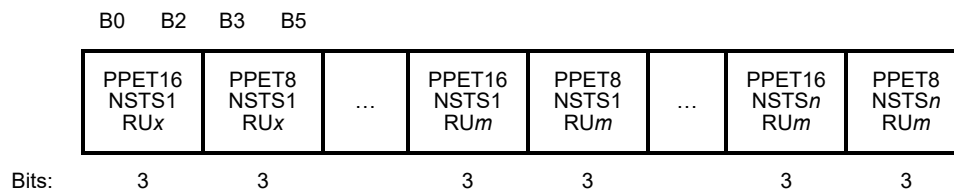


Figure 9-788g—PPE Thresholds Info field format

The PPET16 and PPET8 subfields for various NSTS and RU values appear in increasing NSTS value and increasing RU index value order. Lower numbered PPE Thresholds Info field bits contain PPET16 and PPET8 subfields corresponding to lower numbered NSTS values. Within a set of PPET16 and PPET8

subfields corresponding to a single value of $NSTS$, lower numbered PPE Thresholds Info field bits contain PPET16 and PPET8 subfields corresponding to lower numbered RU index values. The PPET16 $NSTS_n$ RUB and PPET8 $NSTS_n$ RUB subfields are present for all values of n and b where $1 \leq n \leq (NSTS + 1)$ and where $b = [x, \dots, m]$ is the set of integers equal to the ordered list of bit positions of all bits that are set to 1 in the RU Index Bitmask subfield, with x being the lowest value.

Each PPET8 $NSTS_n$ RUB and PPET16 $NSTS_n$ RUB subfield contains an integer as defined in Table 9-322d, which is used to compute the nominal packet padding value.

Table 9-322d—Constellation index

Constellation Index	Corresponding Transmission Constellation
0	BPSK
1	QPSK
2	16-QAM
3	64-QAM
4	256-QAM
5	1024-QAM
6	Reserved
7	None

The value of the PPET8 $NSTS_n$ RUB subfield is always less than the value of the PPET16 $NSTS_n$ RUB subfield, except if the PPET8 subfield is 7.

The RU allocation index for each RU allocation size is defined in Table 9-322e.

Table 9-322e—RU allocation index

RU allocation index	RU allocation size
0	242
1	484
2	996
3	2×996

The PPE Pad field contains all 0s. The number of bits in the PPE Pad field is the least number of bits required to round the length of the PPE Thresholds Info field to an integer number of octets.

9.4.2.249 HE Operation element

The operation of HE STAs in an HE BSS is controlled by the following:

- The HT Operation element and the HE Operation element if operating in the 2.4 GHz band.
- The HT Operation element, VHT Operation element (if present) and the HE Operation element if operating in the 5 GHz band.
- The HE Operation element if operating in the 6 GHz band.

The format of the HE Operation element is defined in Figure 9-788h.

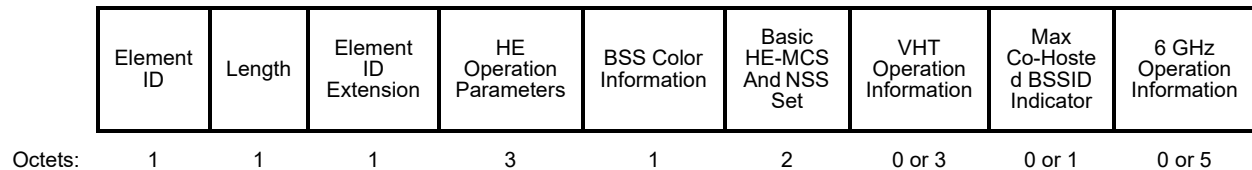


Figure 9-788h—HE Operation element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The format of the HE Operation Parameters field is defined in Figure 9-788i.

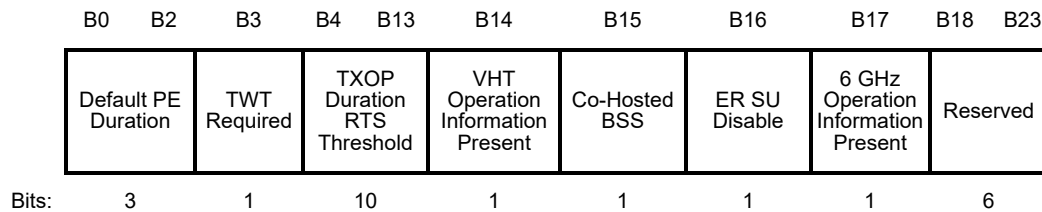


Figure 9-788i—HE Operation Parameters field format

The Default PE Duration subfield indicates the PE field duration in units of 4 μ s for an HE TB PPDU that is solicited with a TRS Control subfield and its use is defined in 26.5.2.3. Values 5–7 of the Default PE Duration subfield are reserved.

The TWT Required subfield is set to 1 to indicate that the AP requires its associated non-AP HE STAs that have declared support for TWT by setting any one of TWT Requester Support or TWT Responder Support or Broadcast TWT Support subfield in HE Capabilities element that it transmits to 1 to operate in the role of either TWT requesting STA by following the rules in 26.8.2, or TWT scheduled STA by following the rules in 26.8.3 and set to 0 otherwise.

The TXOP Duration RTS Threshold subfield enables an HE AP to manage RTS/CTS usage by non-AP HE STAs that are associated with it (see 26.2.1). The TXOP Duration RTS Threshold subfield contains the TXOP duration RTS threshold in units of 32 μ s, which enables the use of RTS/CTS, except for the value 1023. The value 1023 indicates that TXOP duration-based RTS is disabled. The value of 0 is allowed in Beacon and Probe Response frames and indicates that the previously announced TXOP duration RTS threshold remains in effect. In all other frames, the value of 0 is reserved.

The VHT Operation Information Present subfield is set to 1 to indicate that the VHT Operation Information field is present in the HE Operation element and set to 0 otherwise. The VHT Operation Information Present subfield is set as defined in 26.17.

The Co-Hosted BSS subfield is set to 1 to indicate that the AP transmitting this element shares the same operating class, channel, receive antenna connector, and transmit antenna connector with at least one other AP that is providing its BSS information by transmitting Beacon and Probe Response frames. Otherwise, the subfield is set to 0. An AP operating in the 6 GHz band, a TDLS STA, an IBSS STA, a mesh STA, or an AP with dot11MultiBSSImplemented equal to true sets the subfield to 0.

The ER SU Disable subfield indicates whether 242-tone HE ER SU PPDU reception by the AP is disabled or enabled. The ER SU Disable subfield is set to 1 to indicate that it is disabled and set to 0 to indicate that it is enabled.

The 6 GHz Operation Information Present field is set to 1 to indicate that the 6 GHz Operation Information field is present and set to 0 otherwise. The 6 GHz Operation Information Present field is set to 1 by an AP operating in the 6 GHz band.

The BSS Color Information field is defined in Figure 9-788j.

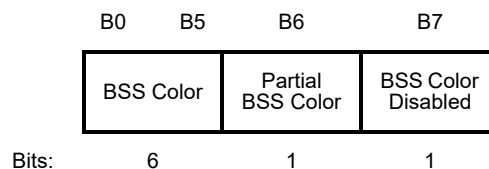


Figure 9-788j—BSS Color Information field format

The BSS Color subfield is an unsigned integer whose value is the BSS Color of the BSS corresponding to the AP, IBSS STA, mesh STA or TDLS STA that transmitted this element and is set as defined in 26.17.3.

The Partial BSS Color subfield is set to 1 to indicate that an AID assignment rule based on the BSS color as defined in 26.17.4 is applied for the BSS. Otherwise, the Partial BSS Color subfield is set to 0.

The BSS Color Disabled subfield is set to 1 to disable the use of color for the BSS as described in 26.17.3.3; otherwise, it is set to 0.

The Basic HE-MCS And NSS Set field indicates the HE-MCSs for each number of spatial streams in HE PPDUs that are supported by all HE STAs in the BSS (including IBSS and MBSS) in transmit and receive. The Basic HE-MCS And NSS Set field is defined in Figure 9-788e.

The format of the VHT Operation Information field is defined in Figure 9-613 and its subfields are defined in Table 9-175. The VHT Operation Information field is present if the VHT Operation Info Present field is 1; otherwise, it is not present.

The Max Co-Hosted BSSID Indicator field contains a value assigned to n , where 2^n is the maximum number of BSSIDs in the co-hosted BSSID set as defined in 26.17.7. This field is present if the Co-Hosted BSS subfield in HE Operation Parameters field is 1 and is not present otherwise.

NOTE—The Max Co-Hosted BSSID Indicator field does not provide the exact number or the identity of each co-hosted BSSID.

The 6 GHz Operation Information field provides channel and bandwidth information related to 6 GHz operation (see 27.3.23.2). The format of the 6 GHz Operation Information field is defined in Figure 9-788k.

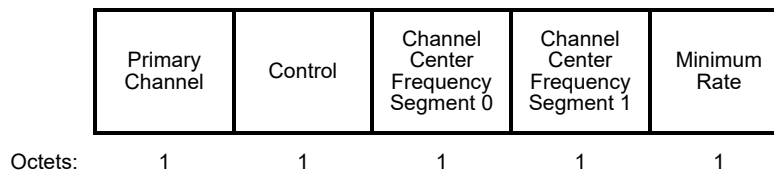


Figure 9-788k—6 GHz Operation Information field format

The Primary Channel field indicates the channel number of the primary channel in the 6 GHz band.

The Control field is defined in Figure 9-788l.

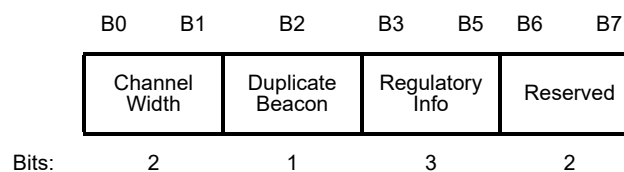


Figure 9-788l—Control field format

The Channel Width field indicates the BSS channel width and is set to 0 for 20 MHz, 1 for 40 MHz, 2 for 80 MHz, and 3 for 80+80 or 160 MHz.

The Duplicate Beacon subfield is set to 1 if the AP transmits Beacon frames in non-HT duplicate PPDU with a TXVECTOR parameter CH_BANDWIDTH value that is up to the BSS bandwidth and is set to 0 otherwise.

The Regulatory Info subfield carries information related to regulatory rules specific to the country in which the BSS is operating in, which is identified by the Country String field in the Country element. The interpretation of the Regulatory Info subfield is in E.2.7. If E.2.7 does not list information for the country in which the BSS is operating, then the Regulatory Info subfield is reserved.

The Channel Center Frequency Segment 0 field indicates the channel center frequency index for the 20 MHz, 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz channel on which the BSS operates in the 6 GHz band. If the BSS channel width is 160 MHz or 80+80 MHz, then the Channel Center Frequency Segment 0 field indicates the channel center frequency index of the primary 80 MHz.

The Channel Center Frequency Segment 1 field indicates the channel center frequency index of the 160 MHz channel on which the BSS operates in the 6 GHz band. If the channel width is 80+80 MHz, then it indicates the channel center frequency index of the secondary 80 MHz.

The Minimum Rate field indicates the minimum rate, in units of 1 Mb/s, that the non-AP STA is allowed to use for sending PPDU (see 26.15.4.3), where the rate is obtained with an N_{SS} that is less than or equal to 3 and an MCS that is less than or equal to 3.

9.4.2.250 UORA Parameter Set element

The metrics of the OFDMA-based random access mechanism (see 26.5.4) are signaled in the UORA Parameter Set element. The format of the UORA Parameter Set element is defined in Figure 9-788m.

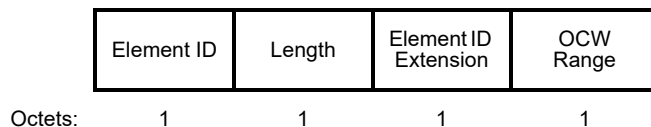


Figure 9-788m—UORA Parameter Set element format

The Element ID, Length, Element ID Extension fields are defined in 9.4.2.1.

The OCW Range field indicates the minimum and maximum values of the OCW (OFDMA contention window) derived from the fields defined in Figure 9-788n.

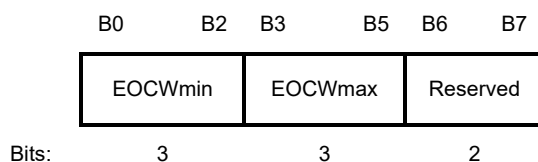


Figure 9-788n—OCW Range field format

The EOCWmin subfield indicates the minimum value of OCW for the initial HE TB PPDU transmission using UORA. The *OCWmin* parameter is used by a STA either for an initial transmission or following a successful HE TB PPDU transmission and is derived as follows:

$$OCWmin = 2^{EOCWmin} - 1$$

where

EOCWmin is the value in the EOCWmin subfield

The EOCWmax subfield indicates the maximum value of OCW for UORA. The *OCWmax* parameter used by a STA for its retransmission attempts of UORA and is derived as follows:

$$OCWmax = 2^{EOCWmax} - 1$$

where

EOCWmax is the value in the EOCWmax subfield

9.4.2.251 MU EDCA Parameter Set element

In an infrastructure BSS, the MU EDCA Parameter Set element is used by the AP to control the use of EDCA by non-AP HE STAs following particular UL MU HE TB PPDU transmissions, as defined in 26.2.7. The most recent MU EDCA Parameter Set element received by a non-AP HE STA is used to update the appropriate MIB values.

The format of the MU EDCA Parameter Set element is defined in Figure 9-788o.

Element ID	Length	Element ID Extension	QoS Info	MUAC_BE Parameter Record	MUAC_BK Parameter Record	MUAC_VI Parameter Record	MUAC_VO Parameter Record
Octets: 1	1	1	1	3	3	3	3

Figure 9-788o—MU EDCA Parameter Set element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The format of the QoS Info field is defined in 9.4.1.17 when sent by the AP.

NOTE—The QoS Info field contains the EDCA Parameter Set Update Count subfield, which indicates when the EDCA parameters and, for an HE BSS, the MU EDCA parameters have changed (see 10.2.3.2).

The format of the MU AC_BE, MU AC_BK, MU AC_VI, and MU AC_VO Parameter Record fields are identical and defined in Figure 9-788p.

ACI/AIFSN	ECWmin/ECWmax	MU EDCA Timer
Octets: 1	1	1

Figure 9-788p—MU AC Parameter Record field format

The format of the ACI/AIFSN field is defined in Figure 9-262 and the encoding of its subfields is defined in 9.4.2.28, except that the value 0 in the AIFSN field indicates that EDCA is disabled for the duration specified by the MUEDCATimer for the corresponding AC.

The format of the ECWmin/ECWmax field is defined in Figure 9-263, and the encoding of its subfields is defined in 9.4.2.28.

The MU EDCA Timer field indicates the duration of time, in units of 8 TUs, during which the HE STA uses the MU EDCA parameters for the corresponding AC, as defined in 26.2.7, except that the value 0 is reserved.

9.4.2.252 Spatial Reuse Parameter Set element

The Spatial Reuse Parameter Set element provides information needed by STAs when performing OBSS PD-based spatial reuse as defined in 26.10.2 and PSR-based spatial reuse as defined in 26.10.3. The format of the Spatial Reuse Parameter Set element is defined in Figure 9-788q.

Element ID	Length	Element ID Extension	SR Control	Non-SRG OBSS PD Max Offset	SRG OBSS PD Min Offset	SRG OBSS PD Max Offset	SRG BSS Color Bitmap	SRG Partial BSSID Bitmap
Octets: 1	1	1	1	0 or 1	0 or 1	0 or 1	0 or 8	0 or 8

Figure 9-788q—Spatial Reuse Parameter Set element format

The Element ID, Element ID extension and Length fields are defined in 9.4.2.1.

The SR Control field is defined in Figure 9-788r.

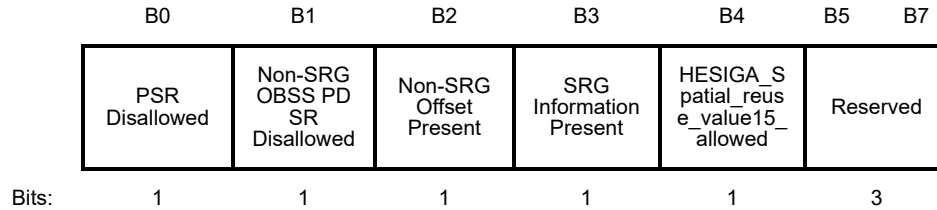


Figure 9-788r—SR Control field format

The PSR Disallowed subfield in the SR Control field indicates whether PSR-based SR transmissions are allowed at non-AP STAs that are associated with the AP that transmitted this element. PSR-based SR transmissions are disallowed when the PSR Disallowed subfield is 1. PSR-based SR transmissions are allowed when the PSR Disallowed subfield is 0. The PSR Disallowed subfield also affects the value of the SPATIAL_REUSE parameter of the TXVECTOR as described in 26.11.6.

The Non-SRG OBSS PD SR Disallowed subfield in the SR Control field indicates whether non-SRG OBSS PD SR transmissions are allowed or not at non-AP STAs that are associated with the AP that transmitted this element. Non-SRG OBSS PD SR transmissions are disallowed when the Non-SRG OBSS PD SR Disallowed subfield is 1. Non-SRG OBSS PD SR transmissions are allowed when the Non-SRG OBSS PD SR Disallowed subfield is 0.

The Non-SRG Offset Present subfield in the SR Control field indicates whether the Non-SRG OBSS PD Max Offset field is present in the element. If the Non-SRG Offset Present subfield is 1, then the Non-SRG OBSS PD Max Offset field is present; otherwise, the Non-SRG OBSS PD Max Offset field is not present.

The SRG Information Present subfield in the SR Control field indicates whether the SRG OBSS PD Min Offset, SRG OBSS PD Max Offset, SRG BSS Color Bitmap, and SRG Partial BSSID Bitmap fields are present in the element. If the SRG Information Present subfield is 1, then the subfields are present; otherwise, the fields are not present.

The HESIGA_Spatial_reuse_value15_allowed subfield in the SR Control field indicates whether non-AP STAs that are associated with the AP that transmitted this element can set the TXVECTOR parameter SPATIAL_REUSE to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED. The subfield is set as described in 26.11.6.

The Non-SRG OBSS PD Max Offset field contains an unsigned integer that is added to –82 dBm to generate the value of the Non-SRG OBSS PD Max parameter.

The SRG OBSS PD Min Offset field contains an unsigned integer that is added to –82 dBm to generate the value of the SRG OBSS PD Min parameter.

The SRG OBSS PD Max Offset field contains an unsigned integer that is added to –82 dBm to generate the value of the SRG OBSS PD Max parameter.

The SRG BSS Color Bitmap field is a bitmap that indicates the BSS color values used by members of the SRG of which the transmitting STA is a member. Each bit of the bitmap corresponds to one of the 64 BSS colors, where the lowest numbered bit corresponds to BSS color value 0 and the highest numbered bit corresponds to BSS color value 63. A BSS color value is used by at least one BSS that is a member of the same SRG of the transmitting STA if the corresponding bit of the bitmap is 1. If a bit in the bitmap is 0, then no BSS in the same SRG of the transmitting STA uses the corresponding BSS color value. The bit in the bitmap that corresponds to the BSS color value 0 is reserved.

The SRG Partial BSSID Bitmap field is a bitmap that indicates the Partial BSSID values used by members of the SRG of which the transmitting STA is a member. Each bit of the bitmap corresponds to one of the 64 possible values of BSSID[39:44], where the lowest numbered bit corresponds to Partial BSSID value 0 and the highest numbered bit corresponds to Partial BSSID value 63. A Partial BSSID value is used by at least one BSS that is a member of the same SRG of the transmitting STA if the corresponding bit of the bitmap is 1. If a bit in the bitmap is 0, then no BSS in the same SRG of the transmitting STA uses the corresponding Partial BSSID value.

9.4.2.253 NDP Feedback Report Parameter Set element

The format of the NDP feedback report Parameter Set element is defined in Figure 9-788s.

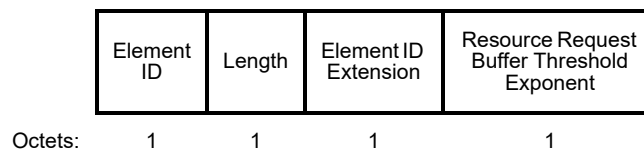


Figure 9-788s—NDP Feedback Report Parameter Set element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The Resource Request Buffer Threshold Exponent field is used to calculate the buffer threshold between two different resource requests as defined in 26.5.7.4. The resource request buffer threshold value is equal to $2^{(\text{Resource Request Buffer Threshold Exponent})}$ octets.

The resource request buffer threshold is equal to 256 octets if no NDP Feedback Report Parameter Set element is sent by the AP.

9.4.2.254 BSS Color Change Announcement element

The BSS Color Change Announcement element is used by an AP to advertise an upcoming BSS color change and the new BSS color that will take effect after the BSS color change (see 26.17.3.4). The format of the BSS Color Change Announcement element is shown in Figure 9-788t.

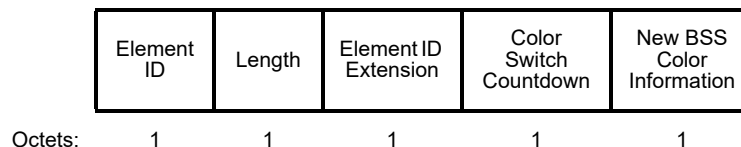


Figure 9-788t—BSS Color Change Announcement element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The Color Switch Countdown field is set to the number of TBTTs until the HE AP sending the BSS Color Change Announcement element switches to the new BSS color. The value 1 indicates that the switch occurs at the next TBTT (the ensuing Beacon frame advertises in the BSS Color subfield of the BSS Color Information field the new BSS color). The value 0 is reserved.

The format of the New BSS Color Information field is defined in Figure 9-788u. The New BSS Color subfield is set to the new BSS color value that the HE AP intends to use starting from the TBTT at which the color switch countdown reaches 0.



Figure 9-788u—New BSS Color Information field format

9.4.2.255 Quiet Time Period element

9.4.2.255.1 General

The format of the Quiet Time Period element is shown in Figure 9-788v.

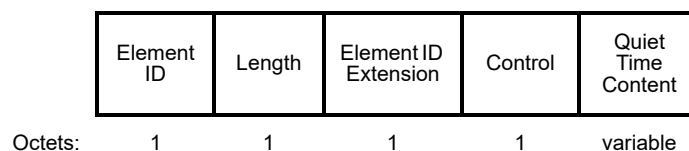


Figure 9-788v—Quiet Time Period element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

A one octet Control field specifies the subtype of the Quiet Time Period element. The two LSBs define the subtype and are referred to as the Quiet Time Period Subtype field. The remaining 6 bits are reserved. Table 9-322f shows the encoding of the Quiet Time Period Subtype field.

Table 9-322f—Quiet Time Period Subtype field encoding

Quiet Time Period Subtype field value	Meaning
0	Quiet Time Period Setup
1	Quiet Time Period Request
2	Quiet Time Period Response
3	Reserved

The Quiet Time Content field is a variable length field and carries information of quiet time operation indicated by the value in the Control field.

9.4.2.255.2 Quiet Time Period Setup subtype

The Quiet Time Period Setup subtype defines a period of time for QTP (see 26.17.5). The QTP can be used by an AP to mitigate interference by reducing the contention from HE STAs in a period that gives preference to HE STAs participating in the exchange of specific frames using peer-to-peer link.

The content of the Quiet Time Content subfield in the Quiet Time Period Setup subtype is shown Figure 9-788w.

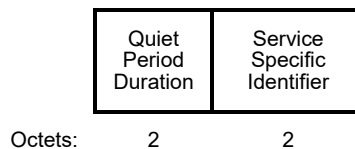


Figure 9-788w—Quiet Time Content subfield format in Quiet Time Period Setup subtype

The Quiet Period Duration field is set to the duration of the QTP, in units of 32 μ s, that is no larger than the value indicated in the Quiet Period Interval subtype field of the Quiet Time Period Request subtype sent by the QTP requesting STA.

The Service Specific Identifier field contains an identifier assigned by a peer-to-peer application to identify specific frame exchanges using peer-to-peer links during which HE STAs that have requested participation of the specified frame exchanges might transmit frames during the quiet time period.

9.4.2.255.3 Quiet Time Period Request subtype

The Quiet Time Period Request subtype defines a periodic sequence of QTPs that the QTP requesting STA requests the QTP AP to schedule.

The content of the Quiet Time Content subfield in the Quiet Time Period Request subtype is shown Figure 9-788x.

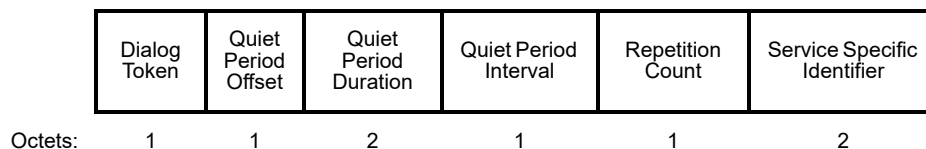


Figure 9-788x—Quiet Time Content subfield format in Quiet Time Period Request subtype

The Dialog Token field identifies the Quiet Time Period Response subtype to which the Quiet Time Period Request subtype corresponds.

The Quiet Period Offset field is set to the offset of the first QTP from the TBTT expressed in TUs.

The Quiet Period Interval field is set to the requested interval between the start of two consecutive QTPs, expressed in TUs.

The Quiet Period Duration field is set to the duration of the QTP in units of 32 μ s.

The Repetition Count field is set to the number of requested QTPs. A repetition count equal to 0 indicates the setup time of the QTP is for an one time operation. Repetition count equals to 0xFF indicates the setup of the QTP is canceled.

The Service Specific Identifier field contains an identifier assigned by a peer-to-peer application to identify specific frame exchanges using peer-to-peer links during which HE STAs that have requested the participation of the specified frame exchanges might transmit frames during the quiet time period.

9.4.2.255.4 Quiet Time Period Response subtype

The Quiet Time Period Response subtype defines the feedback information from the AP that received the Quiet Time Period Request element. If an AP decides not to accept the value requested by the QTP requesting STA, the AP can set different values carried in the Quiet Period Response frame.

The content of the Quiet Time Content subfield in the Quiet Time Period Response subtype is shown in Figure 9-788y.

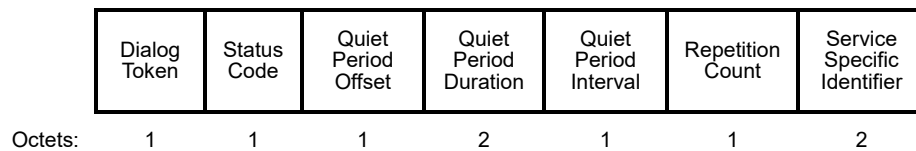


Figure 9-788y—Quiet Time Content subfield format in Quiet Time Period Response subtype

The Dialog Token field identifies the Quiet Time Period Request subtype to which this Quiet Time Period Response subtype corresponds.

The Status Code field indicates the status of a requested operation. The value of the status code is shown in Table 9-322g.

Table 9-322g—Status codes

Value	Meaning
0	Success
1	Reject
2	Counter
3–255	Reserved

The Quiet Period Offset field is set to the offset of the first QTP from the TBTT expressed in TUs.

The Quiet Period Interval field is set to the interval between the start of two consecutive quiet time periods, expressed in TUs.

The Quiet Period Duration field is a one octet field with resolution of 32 μ s.

The Repetition Count field is set to the number of requested QTPs.

The Service Specific Identifier field contains an identifier assigned by a peer-to-peer application to identify specific frame exchanges using peer-to-peer links which HE STAs that have requested the participation of the specific frame exchanges might transmit frames during the quiet time period.

9.4.2.256 ESS Report element

The format of the ESS Report element is shown in Figure 9-788z.

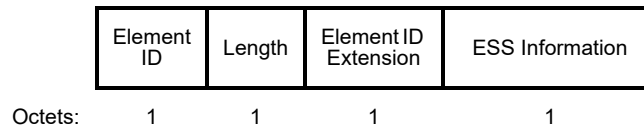


Figure 9-788z—ESS Report element format

The Element ID, Length and Element ID Extension fields are defined in 9.4.2.1.

The format of the ESS Information field is defined in Figure 9-788aa.

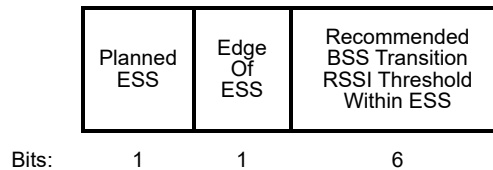


Figure 9-788aa—ESS Information field format

The Planned ESS subfield indicates whether the BSS is part of an ESS that is planned with several BSSs in an overlapping configuration. This subfield is set to 1 to indicate that the ESS is deployed to ensure blanket coverage over the Extended Service Area (ESA). Otherwise, this subfield is set to 0 and the Edge Of ESS and Recommended BSS Transition RSSI Threshold Within ESS subfields are reserved.

The Edge Of ESS subfield indicates whether the BSS is at the edge of the ESS. This subfield is set to 1 to indicate the BSS is at the edge of the ESS. Otherwise, this subfield is set to 0.

The Recommended BSS Transition RSSI Threshold Within ESS subfield indicates the RSSI below which an associated STA is recommended to initiate BSS transition to a neighbor BSS belonging to the ESS.

The resolution for the Recommended BSS Transition RSSI Threshold Within ESS subfield is 1 dB. The encoding is defined in Table 9-322h.

Table 9-322h—Recommended BSS Transition RSSI Threshold Within ESS subfield encoding

Value	Description
0–62	–100 dBm to –38 dBm
63	No recommendation

The use of the ESS Report element is described in 11.21.7.5.

9.4.2.257 OPS element

The OPS element provides information needed by STAs when operating with OPS as defined in 27.14.3. The format of the OPS element is shown in Figure 9-788ab.

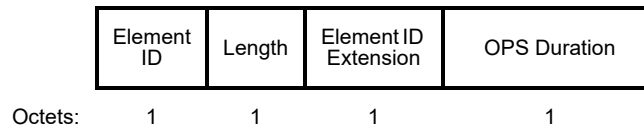


Figure 9-788ab—OPS element format

The Element ID, Element ID extension and Length fields are defined in 9.4.2.1.

If the OPS element is included in an OPS frame or a FILS Discovery frame, the OPS Duration field indicates the OPS period duration, during which a STA can go to doze state if it is explicitly not scheduled during that period, as defined in 26.14.3. The OPS Duration field is encoded in units of milliseconds.

9.4.2.258 TWT Constraint Parameters element

The TWT Constraint Parameters element provides TWT constraint parameters that can be used during the establishment of individual TWT agreements and/or broadcast TWT schedules. The format of the TWT Constraint Parameters element is shown in Figure 9-788ac.

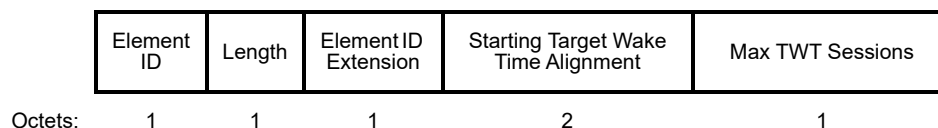


Figure 9-788ac—TWT Constraint Parameters element format

The Starting Target Wake Time Alignment field contains a positive integer n that indicates a recommended time for the start of the first TWT SP of a TWT agreement. A value of n indicates that the first start time is recommended to be an integer multiple of $n + 1$ TUs [i.e., $(\text{Target Wake Time}) \bmod (n + 1) = 0$].

The Max TWT Sessions field contains the maximum number of TWT sessions that a STA is capable of establishing with a peer STA.

9.4.2.259 HE BSS Load element

The HE BSS Load element reported by the AP contains information on utilization, frequency underutilization and spatial stream underutilization. The element format is defined in Figure 9-788ad. A STA receiving the element might use the information it conveys in an implementation-specific AP selection algorithm.

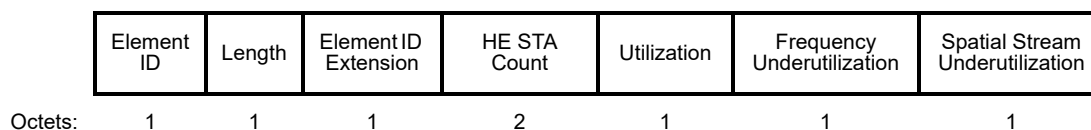


Figure 9-788ad—HE BSS Load element format

The Element ID, Length and Element ID extension fields are defined in 9.4.2.1.

The HE STA Count field indicates the total number of STAs currently associated with this BSS that declare that they are HE STAs by transmitting their HE Capabilities elements.

The Utilization field is defined as the fraction of time, linearly scaled with 255 representing 100%, that the AP sensed the medium was busy due to a transmission between the AP and HE STAs, as indicated by the physical carrier sense (CS) mechanism. When more than one channels are in use for the BSS, the Utilization field value is calculated only for the primary channel. This field value is computed using Equation (9-3a).

$$\text{Utilization} = \left\lfloor \frac{T_{\text{busy}}}{\text{dot11ChannelUtilizationBeaconInterval} \times \text{dot11BeaconPeriod} \times 1024} \times 255 \right\rfloor \quad (9-3a)$$

The Frequency Underutilization field is defined as the fraction of time, linearly scaled with 255 representing 100%, that the AP has underutilized frequency domain resources for given busy time of the medium. This field value is computed using Equation (9-3b).

$$\text{Frequency Underutilization} = \left\lfloor \frac{T_{\text{busy}} - \sum_{i=1}^N \left\{ \left(\sum_{j=1}^{N_{RU}} B_{j,i} \times RU_j \right) \times T_i \right\}}{T_{\text{busy}}} \times 255 \right\rfloor \quad (9-3b)$$

The Spatial Stream Underutilization field is defined as the fraction of time, linearly scaled with 255 representing 100%, that the AP has underutilized spatial domain resources for given busy time of the medium. This percentage is computed using Equation (9-3c).

$$\text{Spatial Stream Underutilization} = \left\lfloor \frac{N_{\text{maxSS}} \times T_{\text{busy}} - \sum_{i=1}^N \left\{ \left(\sum_{j=1}^{N_{RUM}} N_{SS,j,i} \times RUM_j \right) \times T_i \right\}}{N_{\text{maxSS}} \times T_{\text{busy}}} \times 255 \right\rfloor \quad (9-3c)$$

where

dot11ChannelUtilizationBeaconIntervals

represents the number of consecutive beacon intervals during which the channel busy time is measured (see 9.4.2.27).

T_{busy} is the number of microseconds during which CCA indicated the channel was busy due to a transmission between the AP and HE STAs during the measurement duration. The resolution of the CCA busy measurement is in microseconds

T_i is the time interval, in units of microseconds, during which the primary 20 MHz channel is busy due to a transmission between the AP and HE STAs

N is the number of busy events that occurred during the total measurement time that is less than or equal to dot11ChannelUtilizationBeaconIntervals consecutive beacon intervals

N_{RU} is the number of RUs that are allocated within the BSS bandwidth during time interval T_i

RU_j is a normalizing factor depending on the RU size and equals the ratio of the j^{th} RU size to the maximum RU size within the BSS bandwidth, i.e., if the j^{th} RU is a 26-tone RU and the BSS bandwidth is 20 MHz, then $RU_j = 26/242$

$B_{j,i}$ is 1 if the j^{th} RU is occupied or interfered in the busy time T_i ; otherwise, it is 0. Any 20 MHz subchannels that are not occupied by a PPDU are regarded as interfered RUs if the bandwidth of PPDU is less than the BSS bandwidth.

- N_{maxSS} is the maximum number of spatial streams supported by the AP
- N_{RUM} is the number of RUs with a size of at least 106 tones and that are allocated within the BSS bandwidth during time interval T_i
- RUM_j is a normalizing factor depending on the RU size. RUM is applied to RUs with a size of at least 106 tones and equals the ratio of the j^{th} RU size to the maximum RUM size within the BSS bandwidth, i.e., if the j^{th} RUM is a 106-tone RU and the BSS bandwidth is 20 MHz, then $RUM_j = 106/242$.
- $N_{SS,j,i}$ is the number of streams over the j^{th} RUM in the busy time T_i .

If T_{busy} is 0, the Utilization field, Frequency Underutilization field and Spatial Stream Underutilization field are reserved.

9.4.2.260 Multiple BSSID Configuration element

The Multiple BSSID Configuration element is used to provide configuration information for a multiple BSSID set.

The format of the Multiple BSSID Configuration element is shown in Figure 9-788ae.

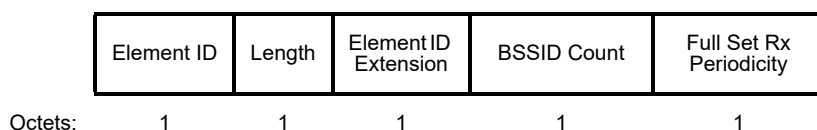


Figure 9-788ae—Multiple BSSID Configuration element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The BSSID Count field carries the total number of active BSSIDs in the multiple BSSID set.

Full Set Rx Periodicity field indicates the least number of Beacon frames or DMG Beacon frames a STA needs to receive in order to discover all the active nontransmitted BSSIDs in the set.

9.4.2.261 Known BSSID element

The Known BSSID element identifies the nontransmitted BSSIDs that a non-AP STA has discovered so far. A non-AP STA can include this element in a directed Probe Request frame to discover other nontransmitted BSSIDs not known to the requesting STA.

The format of the Known BSSID element is defined in Figure 9-788af.

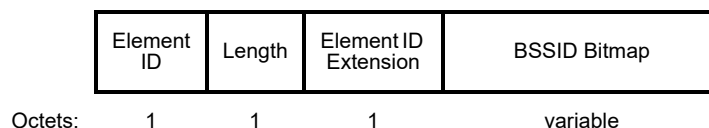


Figure 9-788af—Known BSSID element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The format of the BSSID Bitmap field is as defined in Figure 9-788ag.

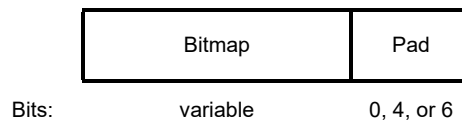


Figure 9-788ag—BSSID Bitmap field format

The Bitmap subfield has a length of 2^n bits where n is the value carried in the MaxBSSID Indicator field of the Multiple BSSID element advertised by the AP to which the Probe Request frame is being sent. Bit position 0 is reserved. The remainder of the bits represent one of $2^n - 1$ possible BSSID Index values (see 9.4.2.73) in the multiple BSSID set. A value of 1 at bit position k indicates that the non-AP STA has knowledge of nontransmitted BSSID with BSSID index k . Otherwise, the bit is set to 0.

The Pad subfield contains additional bits set to 0 to make the total number of bits in the BSSID Bitmap field equal to an integer number of octets.

9.4.2.262 Short SSID List element

The format of the Short SSID List Element is shown in Figure 9-788ah.

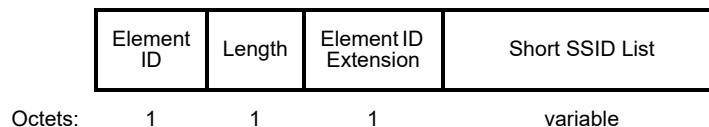


Figure 9-788ah—Short SSID List element format

The Element ID, Length and Element ID Extension fields are defined in 9.4.2.1.

The Short SSID List field contains the one or more four octet Short SSID fields for which the STA is requesting information. The use of the Short SSID List element and frames is described in 11.1.4.3.2. The Short SSID field is defined in 9.4.2.170.3.

9.4.2.263 HE 6 GHz Band Capabilities element

An HE STA operating in the 6 GHz band declares its extended capabilities by transmitting the HE 6 GHz Band Capabilities element. The HE 6 GHz Band Capabilities element is defined in Figure 9-788ai.

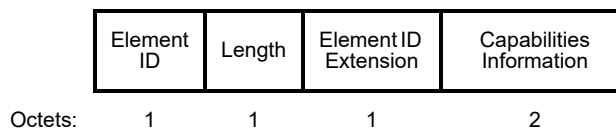


Figure 9-788ai—HE 6 GHz Band Capabilities element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The Capabilities Information field is defined in Figure 9-788aj.

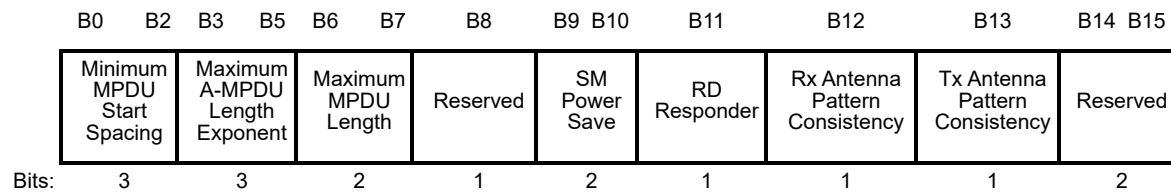


Figure 9-788aj—Capabilities Information field format

The Minimum MPDU Start Spacing subfield is defined in 9.4.2.55.3.

The Maximum A-MPDU Length Exponent subfield and Maximum MPDU Length subfield are defined in Table 9-271.

The SM Power Save subfield is defined in defined in Table 9-184.

The RD Responder subfield is defined in defined in Table 9-187.

The Rx Antenna Pattern Consistency subfield is defined in defined in Table 9-271.

The Tx Antenna Pattern Consistency subfield is defined in defined in Table 9-271.

9.4.2.264 UL MU Power Capabilities element

The UL MU Power Capabilities element indicates the relative maximum transmit power that a STA is capable of transmitting an HE TB PPDU for each HE-MCS when using RU size greater than or equal to 242 subcarriers. The format of the UL MU Power Capabilities element is shown in Figure 9-788ak.

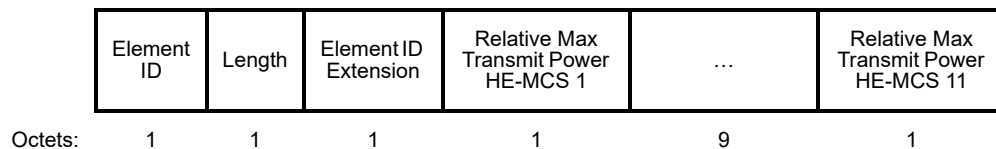


Figure 9-788ak—UL MU Power Capabilities element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The UL MU Power Capabilities element contains 11 Relative Max Transmit Power HE-MCS n fields in ascending order of HE-MCS from 1 to 11.

The Relative Max Transmit Power HE-MCS n field (where $n = 1, \dots, 11$) is an unsigned integer in dB and contains the difference between the nominal maximum transmit power in dBm of an HE TB PPDU using an RU size greater than or equal to 242 subcarriers for HE-MCS 0 and that for HE-MCS n . The Relative Max Transmit Power HE-MCS n fields corresponding to HE-MCSs that are not supported by the HE STA that transmits this element are reserved.

9.6 Action frame format details

Change the title of 9.6.1 as follows:

9.6.1 Introduction-General

Insert the following paragraph at the end of 9.6.1:

In the Action field formats defined in 9.6, the fields and elements listed are always present unless stated otherwise.

9.6.7 Public Action details

9.6.7.16 TDLS Discovery Response frame format

Insert the following row into Table 9-374 in numeric order:

Table 9-374—TDLS Discovery Response frame Action field format

Order	Information	Notes
19	HE Capabilities	The HE Capabilities element is present if dot11HEOptionImplemented is true; otherwise, it is not present. The HE Capabilities element is defined in 9.4.2.248

9.6.7.36 FILS Discovery frame format

Change Table 9-383 as follows (not all rows are shown):

Table 9-383—FILS Discovery frame format

Order	Information	Notes
...		
4	Reduced Neighbor Report elements	The One or more Reduced Neighbor Report elements are is optionally present if dot11FILSActivated or dot11ColocatedRNRImplemented is true; otherwise, they are not present.
5	FILS Indication element	The FILS Indication element is optionally present if dot11FILSActivated is true; otherwise, it is not present.
6	Roaming Consortium element	The Roaming Consortium element is optionally present if dot11FILSActivated is true; otherwise, it is not present.
7	<u>TIM element</u>	The TIM element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.
8	<u>TWT element</u>	The TWT element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present. If present, the Broadcast field of the TWT element is 1.

Table 9-383—FILS Discovery frame format (continued)

Order	Information	Notes
9	<u>OPS element</u>	<u>The OPS element is optionally present if dot11HEOptionImplemented is true; otherwise, it is not present.</u>
10	<u>Transmit Power Envelope element</u>	One Transmit Power Envelope element is optionally present for each distinct combination of values of the Maximum Transmit Power Interpretation subfield and Maximum Transmit Power Category subfield that is supported for the BSS if both of the following conditions are met: — <u>Either dot11VHTOptionImplemented or dot11ExtendedSpectrumManagementImplemented is true.</u> — <u>Either dot11SpectrumManagementRequired or dot11RadioMeasurementActivated is true.</u> <u>NOTE—In a 6 GHz HE AP, both dot11VHTOptionImplemented (see 26.17.1) and dot11SpectrumManagementRequired (see 26.17.2.1) are true.</u>

Change the header row of Table 9-384 as follows:

Table 9-384—BSS Operating Channel Width

BSS Operating Channel Width field	HR/DSSS, OFDM, ERP, HT, or VHT, <u>or HE</u> BSS operating channel width	TVHT BSS operating channel width
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Insert the following row into Table 9-386 in numeric order, and change the Reserved row accordingly:

Table 9-386—PHY Index subfield

PHY Index subfield	PHY
4	HE (see Clause 27)

Change Table 9-387 as follows:

Table 9-387—FILS Minimum Rate

FILS Minimum Rate subfield	PHY Index subfield is 0 (HR/DSSS)	PHY Index subfield is 1 (ERP-OFDM)	PHY Index subfield is 2 (HT)	PHY Index subfield is 3 (VHT or TVHT)	<u>PHY Index subfield is 4 (HE)</u>
0	1 Mb/s	6 Mb/s	MCS 0	MCS 0	<u>HE-MCS 0</u>
1	2 Mb/s	9 Mb/s	MCS 1	MCS 1	<u>HE-MCS 1</u>
2	5.5 Mb/s	12 Mb/s	MCS 2	MCS 2	<u>HE-MCS 2</u>
3	11 Mb/s	18 Mb/s	MCS 3	MCS 3	<u>HE-MCS 3</u>
4	Reserved	24 Mb/s	MCS 4	MCS 4	<u>HE-MCS 4</u>
5–7	Reserved	Reserved	Reserved	Reserved	<u>Reserved</u>

Change the 36th paragraph (right after Table 9-389) in 9.6.7.36 as follows:

The Channel Center Frequency Segment 1 subfield is set to the index of the channel center frequency of the frequency segment 1 for an 80+80 MHz VHT or HE BSS; if the FILS Discovery frame is transmitted as a non-HT duplicate PPDU at an 80+80 MHz channel bandwidth; otherwise, the subfield is not present.

Insert the following text at the end of 9.6.7.36:

The TIM element is defined in 9.4.2.5 and is included for operation as defined in 26.14.3.

The TWT element is defined in 9.4.2.199 and is included with the Broadcast field set to 1 to aid an unassociated STA determine the target transmission time of Trigger frames that contain at least one User Info field that allocates RA-RUs for unassociated STAs (see 26.5.4.5 and 26.8.3.1).

The OPS element is defined in 9.4.2.257.

The Transmit Power Envelope element is defined in 9.4.2.161.

9.6.12 TDLS Action field formats

9.6.12.2 TDLS Setup Request Action field format

Change Table 9-414 as follows (not all rows are shown):

Table 9-414—Information for TDLS Setup Request Action field

Order	Information	Notes
...		
19	AID	The AID element containing the AID of the STA sending the frame is present if dot11VHTOptionImplemented, <u>dot11HEOptionImplemented</u> , or dot11SISOOptionImplemented is true.
...		
<u>23</u>	<u>HE Capabilities</u>	<u>The HE Capabilities element is present if dot11HEOptionImplemented is true; otherwise, it is not present. The HE Capabilities element is defined in 9.4.2.248.</u>
<u>24</u>	<u>TWT</u>	<u>The TWT element is optionally present if dot11TWTOptionActivated is true; otherwise, it is not present.</u> <u>The Trigger subfield and the Negotiation Type subfield of the TWT element are set to 0.</u>
<u>25</u>	<u>HE 6 GHz Band Capabilities</u>	<u>The HE 6 GHz Band Capabilities element is present if dot11HEOptionImplemented and dot11HE6GOptionImplemented are true; otherwise, it is not present.</u>

9.6.12.3 TDLS Setup Response Action field format

Change Table 9-415 as follows (not all rows are shown):

Table 9-415—Information for TDLS Setup Response Action field

Order	Information	Notes
...		
20	AID	The AID element containing the AID of the STA sending the frame is present if dot11VHTOptionImplemented, dot11HEOptionImplemented, or dot11SISOOptionImplemented is true.
...		
25	<u>HE Capabilities</u>	<u>The HE Capabilities element is present if dot11HEOptionImplemented is true and the Status Code is SUCCESS; otherwise, it is not present. The HE Capabilities element is defined in 9.4.2.248.</u>
26	<u>TWT</u>	<u>The TWT element is present if dot11TWTOptionActivated is true and the TWT element is present in the TDLS Setup Request frame that elicited this TDLS Setup Response frame. The TWT element is optionally present if dot11TWTOptionActivated is true and the TWT Requester Support field or the TWT Responder Support field is equal to 1 in the HE Capabilities in the TDLS Setup Request frame that elicited this TDLS Setup Response frame. Otherwise, the TWT element is not present.</u> <u>The Trigger subfield and the Negotiation Type subfield of the TWT element are set to 0.</u>
27	<u>HE 6 GHz Band Capabilities</u>	<u>The HE 6 GHz Band Capabilities element is present if dot11HEOptionImplemented and dot11HE6GOptionImplemented are true; otherwise, it is not present.</u>

9.6.12.4 TDLS Setup Confirm Action field format

Insert the following row into Table 9-416 in numeric order:

Table 9-416—Information for TDLS Setup Confirm Action field

Order	Information	Notes
14	HE Operation	The HE Operation element is present when dot11HEOptionImplemented is true, the TDLS Setup Response frame contains an HE Capabilities element, and the Status Code is SUCCESS; otherwise, it is not present. The HE Operation element is defined in 9.4.2.249.

9.6.15 Self-protected Action frame details

9.6.15.2 Mesh Peering Open frame format

9.6.15.2.2 Mesh Peering Open frame details

Insert the following rows into Table 9-436 in numeric order:

Table 9-436—Mesh Peering Open frame Action field format

Order	Information	Notes
22	HE Capabilities	The HE Capabilities element is present when dot11HEOptionImplemented is true; otherwise, it is not present.
23	HE Operation	The HE Operation element is present when dot11HEOptionImplemented is true; otherwise, it is not present.

9.6.15.3 Mesh Peering Confirm frame format

9.6.15.3.2 Mesh Peering Confirm frame details

Insert the following rows into Table 9-437 in numeric order:

Table 9-437—Mesh Peering Confirm frame Action field format

Order	Information	Notes
18	HE Capabilities	The HE Capabilities element is present when dot11HEOptionImplemented is true; otherwise, it is not present.
19	HE Operation	The HE Operation element is present when dot11HEOptionImplemented is true; otherwise, it is not present.

9.6.24 Unprotected S1G Action frame details

9.6.24.1 Unprotected S1G Action field

Change the first paragraph in 9.6.24.1 as follows:

Several Action frame formats are defined to support S1G functionality. An Unprotected S1G Action field, in the octet immediately after the Category field, differentiates the Unprotected S1G Action frame formats. The Unprotected S1G Action field values associated with each frame format within the Unprotected S1G category are defined in Table 9-494.

9.6.24.8 TWT Setup frame format

Change Table 9-501 as follows:

Table 9-501—TWT Setup frame Action field format

Order	Information
1	Category
2	Unprotected SIG Action
3	Dialog Token
4	<u>One or two</u> TWT elements (9.4.2.199)

Change the end of 9.6.24.8 as follows:

In a TWT Setup frame with a TWT Request field that is ~~equal-set~~ to 1, the Dialog Token field is set to a nonzero value chosen by the transmitting STA to identify the request/response transaction. In a TWT Setup frame with a TWT Request field ~~equal-set~~ to 0 that is sent in response to a TWT Setup frame with a TWT Request field that is equal to 1, the Dialog Token field is set to the value copied from the corresponding received TWT Setup frame with a TWT Request field equal to 1. In a TWT Setup frame with a TWT Request field set to 0 that is not sent in response to a TWT Setup frame with a TWT Request field equal to 1, the Dialog Token field is set to 0.

A TWT Setup frame contains only one TWT element, except if used for the establishment of a TWT agreement with a range of TWT parameter values (see 10.47.9). In this case, an additional TWT element is present.

9.6.24.9 TWT Teardown frame format

Change the end of 9.6.24.9 (starting with the fourth paragraph) as follows (including changing Figure 9-965 and inserting Figure 9-965a and Figure 9-965b):

~~The TWT Flow field contains the TWT Flow Identifier field and 5 reserved bits as shown in Figure 9-965. The format of the TWT Flow field if the Negotiation Type subfield is 0 or 1 is defined in Figure 9-965. The format of the TWT Flow field if the Negotiation Type field is 2 is defined in Figure 9-965a. The format of the TWT Flow field if the Negotiation Type subfield is 3 is defined in Figure 9-965b.~~

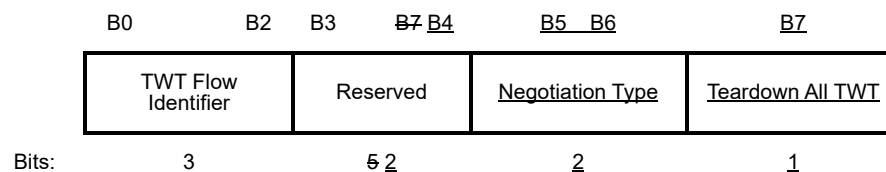


Figure 9-965—TWT Flow field format if the Negotiation Type subfield is 0 or 1

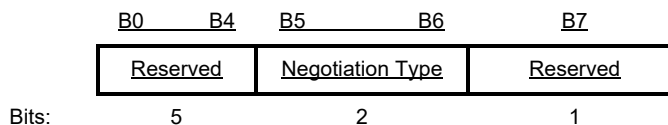


Figure 9-965a—TWT Flow field format if the Negotiation Type subfield is 2

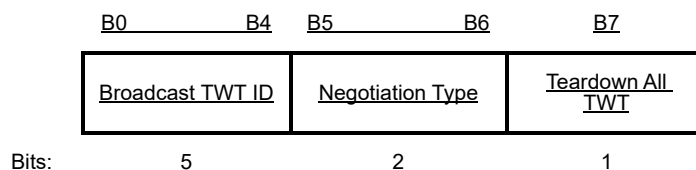


Figure 9-965b—TWT Flow field format if the Negotiation Type subfield is 3

The TWT Flow Identifier/Broadcast TWT ID field contains the TWT Flow Identifier when the Negotiation Type subfield is 0 or 1 and contains the Broadcast TWT ID field when the Negotiation Type subfield is 3. The TWT Flow Identifier field and the Broadcast TWT ID field are is defined in 9.4.2.199 and are reserved if the Teardown All TWT field is 1.

In a TWT Teardown frame, the The TWT Flow Identifier field in a TWT Teardown frame is set to the value of the TWT Flow Identifier field of the TWT element in the frame that successfully concluded the setup of the TWT that is the subject of the teardown request. The Broadcast TWT ID field of a TWT Teardown frame is set to the value of the Broadcast TWT identifier of the broadcast TWT schedule that is subject of the teardown request.

The Negotiation Type subfield indicates the type of negotiation that is subject to the teardown request and is set as defined in Table 9-296a. An S1G STA sets the Negotiation Type subfield to 0. The Negotiation Type subfield is reserved if the Teardown All TWT field is 1.

The Teardown All TWT field is set to 1 by an HE STA to indicate that the TWT Teardown frame tears down all TWTs as defined in 26.8. Otherwise, it is set to 0.

9.6.24.12 TWT Information frame format

Change the first paragraph in 9.6.24.12 as follows:

The TWT Information frame is sent by a STA to request or deliver information about a TWT ~~agreement~~ and is transmitted by either STA of an existing TWT agreement or is transmitted by a STA to a peer STA that has indicated support of its reception. The Action field of the TWT Information frame contains the information shown in Table 9-505.

Insert the following subclauses (9.6.31 through 9.6.32.3, including Table 9-526a through Table 9-526g and Figure 9-970a) after 9.6.30.2:

9.6.31 HE Action frame details

9.6.31.1 HE Action field

An HE Action field, in the octet immediately after the Category field, differentiates the HE Action frame formats. The HE Action field values associated with each frame format within the HE category are defined in Table 9-526a.

Table 9-526a—HE Action field values

Value	Meaning
0	HE Compressed Beamforming/CQI
1	Quiet Time Period
2	OPS
3–255	Reserved

9.6.31.2 HE Compressed Beamforming/CQI frame format

The HE Compressed Beamforming/CQI frame is an Action No Ack frame of category HE. The Action field of an HE Compressed Beamforming/CQI frame contains the information shown in Table 9-526b.

Table 9-526b—HE Compressed Beamforming/CQI frame Action field format

Order	Information
1	Category
2	HE Action
3	HE MIMO Control (see 9.4.1.64)
4	HE Compressed Beamforming Report (see 9.4.1.65)
5	HE MU Exclusive Beamforming Report (see 9.4.1.66)
6	HE CQI Report (see 9.4.1.67)

The Category field is defined in Table 9-51.

The HE Action field is defined in Table 9-526a.

The presence and contents of the HE Compressed Beamforming Report field, HE MU Exclusive Beamforming Report field and HE CQI Report field are dependent on the values of the Feedback Type subfield of the HE MIMO Control field (see 9.4.1.65, 9.4.1.66, and 9.4.1.67).

A Vendor Specific element is not present in the HE Compressed Beamforming/CQI frame.

9.6.31.3 Quiet Time Period action frame format

The Quiet Time Period action frame is an Action No Ack frame of category HE. The Action field of a Quiet Time Period contains the information shown in Table 9-526c.

Table 9-526c—Quiet Time Period frame body

Order	Information
1	Category
2	HE Action
3	Quiet Time Period element (see 9.4.2.255)

The Category field is defined in Table 9-51.

The HE Action field is defined in Table 9-526a.

The Quiet Time Period element is defined in 9.4.2.255.

9.6.31.4 OPS frame format

The OPS frame is an Action No Ack frame of category HE. The Action field of an OPS frame contains the information shown in Table 9-526d.

Table 9-526d—OPS frame Action field format

Order	Information
1	Category
2	HE Action
3	TIM element (see 9.4.2.5)
4	OPS element (see 9.4.2.257)

The Category field is defined in Table 9-51.

The HE Action field is defined in Table 9-526a.

9.6.32 Protected HE Action frame details

9.6.32.1 Protected HE Action field

A Protected HE Action field, in the octet immediately after the Category field, differentiates the Protected HE Action frame formats. The Protected HE Action field values associated with each frame format within the HE category are defined in Table 9-526e.

Table 9-526e—Protected HE Action field values

Value	Meaning
0	HE BSS Color Change Announcement
1	MU EDCA Reset
1–255	Reserved

9.6.32.2 HE BSS Color Change Announcement frame format

The HE BSS Color Change Announcement frame is an Action or Action No Ack frame of category Protected HE. The Action field of an HE BSS Color Change Announcement frame contains the information shown in Table 9-526f.

Table 9-526f—HE BSS Color Change Announcement frame Action field format

Order	Information
1	Category
2	Protected HE Action
3	BSS Color Change Announcement element (see 9.4.2.254)

NOTE—An HE AP might send an HE BSS Color Change Announcement frame as an Action frame to a STA that appears to have missed a color change announcement and is transmitting with the old color.

The Category field is defined in Table 9-51.

The Protected HE Action field is defined in Table 9-526e.

9.6.32.3 MU EDCA Reset frame format

The MU EDCA Reset frame is an Action frame of category Protected HE. The Action field of an MU EDCA Reset frame contains the information shown in Table 9-526g.

Table 9-526g—MU EDCA Reset frame Action field format

Order	Information
1	Category
2	Protected HE Action
3	MU EDCA Control

The Category field is defined in Table 9-51.

The Protected HE Action field is defined in Table 9-526e.

The MU EDCA Control field is defined in Figure 9-970a.

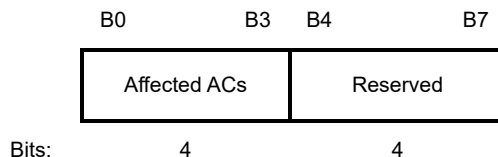


Figure 9-970a—MU EDCA Control field format

The Affected ACs subfield contains a bitmap that indicates the ACs for which the MUEDCATimer[AC] as defined in 26.2.7 are reset to 0 upon receipt of the MU EDCA Reset frame for affected STAs. Each bit in the bitmap corresponds to one AC with B0 mapped to AC_BK, B1 mapped to AC_BE, B2 mapped to AC_VI and B3 mapped to AC_VO. A value of 1 in the bit position corresponding to a given AC indicates that the MUEDCATimer[AC] for that AC is reset to 0; otherwise, the value of the bit is 0.

9.7 Aggregate MPDU (A-MPDU)

9.7.1 A-MPDU format

Change the fourth paragraph in 9.7.1 as follows:

The EOF Padding field is shown in Figure 9-972. This is present only in a VHT₁, ~~or~~ S1G₁, or HE PPDU.

Change the sixth and eighth paragraphs and NOTE 1 in 9.7.1 as follows:

In a VHT₁, ~~or~~ S1G₁, or HE PPDU, the following padding is present, as determined by the rules in 10.12.6:

- 0–3 octets in the Padding subfield of the final A-MPDU subframe (see Figure 9-973) before any EOF padding subframes. The content of these octets is unspecified.
- Zero or more EOF padding subframes in the EOF Padding Subframes subfield.
- 0–3 octets in the EOF Padding Octets subfield. The content of these octets is unspecified.

An A-MPDU pre-EOF padding refers to the contents of the A-MPDU up to, but not including, the EOF Padding field.

NOTE 1—A-MPDU pre-EOF padding includes any A-MPDU subframes with 0 in the MPDU Length field and 0 in the EOF/Tag field inserted in order to meet the minimum MPDU start spacing requirement.

The maximum length of an A-MPDU in an HT PPDU is 65 535 octets. The maximum length of an A-MPDU in a DMG PPDU is 262 143 octets. The maximum length of an A-MPDU pre-EOF padding in a VHT PPDU is 1 048 575 octets. The maximum length of an A-MPDU pre-EOF padding in an HE PPDU is 6 500 631 octets. The length of an A-MPDU addressed to a particular STA can be further constrained as described in 10.12.2.

Change Table 9-527 as follows:

Table 9-527— MPDU delimiter fields (non-DMG)

Field	Size (bits)	Description
EOF/Tag	1	End of frame indication <u>if the MPDU Length field is 0</u> . Set to 1 in an A-MPDU subframe that has 0 in the MPDU Length field and that is used to pad the A-MPDU in a VHT <u>or HE</u> PPDU as described in 10.12.6. Set to 1 in the MPDU delimiter of an S-MPDU as described in 10.12.7. <u>Tagged/untagged indication if the MPDU Length field is nonzero. Set to 1 in an MPDU delimiter preceding a QoS Data frame or Management frame soliciting an Ack frame or Per AID TID Info field with the Ack Type field set to 1 in a Multi-STA BlockAck frame in a response that is contained in an ack-enabled multi-TID A-MPDU as described in 26.6.3.4 and ack-enabled single-TID A-MPDU as described in 26.6.3.2. Set to 0 otherwise.</u>
Reserved	1	
MPDU Length	14	Length of the MPDU in octets. Set to 0 if no MPDU is present. An A-MPDU subframe with 0 in the MPDU Length field is used as defined in 10.12.3 to meet the minimum MPDU start spacing requirement and also to pad the A-MPDU to fill the available octets in a VHT <u>or HE</u> PPDU as defined in 10.12.6.
CRC	8	8-bit CRC of the preceding 16 bits (see 9.7.2).
Delimiter Signature	8	Pattern that may be used to detect an MPDU delimiter when scanning for an MPDU delimiter. The unique pattern is 0x4E, which is the ASCII value of the character 'N' (see NOTE).
NOTE—The ASCII value of the character 'N' was chosen as the unique pattern for the value in the Delimiter Signature field.		

Change the 11th paragraph in 9.7.1 as follows:

The format of the MPDU Length field when transmitted by a non-DMG STA is shown in Figure 9-976. The MPDU Length Low subfield contains the 12 low order bits of the MPDU length. In a VHT or HE PPDU, the MPDU Length High subfield contains the two high order bits of the MPDU length. In an HT PPDU, the MPDU Length High subfield is reserved.

Replace Equation (9-5) with the following equation (variable list remains unchanged):

$$L_{MPDU} = \begin{cases} L_{low} + L_{high} \times 4096, & \text{for a VHT and HE PPDU} \\ L_{low}, & \text{for an HT PPDU} \\ L, & \text{for a DMG PPDU} \end{cases} \quad (9-5)$$

Change NOTE 2 in 9.7.2 as follows:

NOTE 2—The format of the MPDU Length field maintains a common encoding structure for ~~both HE, VHT, and HT~~ PPDU_s. For HT PPDU_s, only the MPDU Length Low subfield is used, while for VHT and HE PPDU_s, both subfields are used.

9.7.3 A-MPDU contents

Change the text of 9.7.3, and change Table 9-529, Table 9-530 (title only), Table 9-531, and Table 9-533 as follows (Table 9-532 and Table 9-534 remain unchanged):

In a non-DMG PPDU, an A-MPDU is a sequence of A-MPDU subframes carried in a single PPDU with one of the following combinations of RXVECTOR or TXVECTOR parameter values:

- The FORMAT parameter set to VHT.
- The FORMAT parameter set to HT_MF or HT_GF and the AGGREGATION parameter set to 1.
- The FORMAT parameter set to S1G, S1G_DUP_1M, or S1G_DUP_2M and the AGGREGATION parameter set to 1.
- The FORMAT parameter set to HE_SU, HE_MU, HE_TB, or HE_ER_SU.

An A-MPDU carried in an HE SU PPDU, HE ER SU PPDU, HE TB PPDU, or HE MU PPDU can include MPDUs with different values of the TID field as described in 26.6.3.

In a DMG PPDU, an A-MPDU is a sequence of MPDUs carried in a single PPDU with the TXVECTOR/RXVECTOR AGGREGATION parameter set to 1.

All of the MPDUs within an A-MPDU are addressed to the same RA. All of the MPDUs within an A-MPDU have the same TA. All QoS Data frames within an A-MPDU that have a TID for which an HT-immediate block ack agreement exists have the same value for the Ack Policy Indicator subfield of the QoS Control field.

All protected MPDUs within an A-MPDU have the same Key ID.

The Duration/ID fields in the MAC headers of all MPDUs in an A-MPDU carry the same value. The Duration/ID fields in the MAC headers of the MPDUs in the A-MPDUs carried in the same a VHT MU PPDU and an HE MU PPDU all carry the same value.

NOTE 1—The reference point for the Duration/ID field is the end of the PPDU carrying the MPDU. Setting the Duration/ID field to the same value in the case of A-MPDU aggregation means that each MPDU consistently specifies the same NAV setting.

An A-MPDU is transmitted in one of the contexts specified in Table 9-529 as defined by the description in the “Definition of context” column, ~~independently of whether the A-MPDU is contained in a VHT MU PPDU or an SU PPDU.~~ The content of an A-MPDU depends on the context in which it is transmitted as defined in the tables below. Ordering of MPDUs within an A-MPDU is not constrained, except where noted in these tables. See 10.12.1.

Table 9-529—A-MPDU contexts

Name of context	Definition of context	Table defining permitted contents
Non-HE Data Enabled Immediate Response	The A-MPDU is transmitted outside a PSMP sequence by a TXOP holder or an RD responder including potential immediate responses.	Table 9-530
Data Enabled No Immediate Response	The A-MPDU is transmitted outside a PSMP sequence by a TXOP holder, <u>TXOP responder when transmitted by an HE STA to another HE STA, and the A-MPDU that</u> does not include or solicit an immediate response. See NOTE.	Table 9-531

Table 9-529—A-MPDU contexts (continued)

Name of context	Definition of context	Table defining permitted contents
PSMP	The A-MPDU is transmitted within a PSMP sequence.	Table 9-532
Control Response	The A-MPDU is transmitted by a STA that is neither a TXOP holder nor an RD responder, <u>or the A-MPDU is transmitted by an HE AP in response to an HE TB PPDU, and the transmitter that also needs to transmit one of the following immediate response frames:</u> — Ack frame — BlockAck frame with a TID for which an HT-immediate block ack agreement exists — <u>Multi-STA BlockAck frame for acknowledging multi-TID A-MPDU</u>	Table 9-533
S-MPDU context	The A-MPDU is transmitted within a VHT PPDU <u>or an HE PPDU</u> and contains an S-MPDU.	Table 9-534
<u>HE Non-Ack-Enabled Single-TID Immediate Response</u>	<u>The A-MPDU is transmitted by a TXOP holder or TXOP responder in an HE PPDU and solicits block acknowledgment for a single TID.</u>	<u>Table 9-534a</u>
<u>HE Ack-Enabled Single-TID Immediate Response</u>	<u>The A-MPDU is transmitted by a TXOP holder or TXOP responder in an HE PPDU and solicits single acknowledgment.</u>	<u>Table 9-534b</u>
<u>HE Non-Ack Enabled Multi-TID Immediate Response</u>	<u>The A-MPDU is transmitted by a TXOP holder or TXOP responder in an HE PPDU, and solicits block acknowledgments for multiple TIDs.</u>	<u>Table 9-534c</u>
<u>HE Ack-Enabled Multi-TID Immediate Response</u>	<u>The A-MPDU is transmitted by a TXOP holder or TXOP responder in an HE PPDU, and solicits at least one acknowledgment and zero or more block acknowledgments.</u>	<u>Table 9-534d</u>
NOTE—This context includes cases when no response is generated.		

NOTE 2—If the number qualifier for a frame subtype does not exist, zero or more frames of the frame subtype can be aggregated in the A-MPDU.

A VHT MU PPDU, or S1G MU PPDU, and HE MU PPDU does not carry more than one A-MPDU that contains one or more MPDUs soliciting an immediate response if the immediate response is carried in a PPDU that is not an HE TB PPDU. An HE MU PPDU can carry more than one A-MPDU each of which contains one or more MPDUs soliciting an immediate response if the immediate response is carried in an HE TB PPDU.

NOTE 2.3—The TIDs present in a data enabled A-MPDU context are also constrained by the channel access rules (for a TXOP holder; see 10.23.2 and 10.23.3), the TXOP responder rules (see 26.6, and 26.5.2) and the RD response rules (for an RD responder, see 10.29.4). This is not shown in these tables.

NOTE 3.4—If a STA supports A-MSDUs of 7935 octets (indicated by the Maximum A-MSDU Length field in the HT Capabilities element), A-MSDUs transmitted by that STA within an A-MPDU carried in a PPDU with FORMAT HT_MF or HT_GF are constrained so that the length of the QoS Data frame carrying the A-MSDU is no more

than 4095 octets. The 4095-octet MPDU length limit does not apply to A-MPDUs carried in VHT, HE or DMG PPDU. The use of A-MSDU within A-MPDU might be further constrained as described in 9.4.1.13 through the operation of the A-MSDU Supported field.

Table 9-530—A-MPDU contents in the non-HE data enabled immediate response context

The body of Table 9-530 remains unchanged.

Table 9-531—A-MPDU contents in the data enabled no immediate response context

MPDU description	Conditions
Data without a block ack agreement	QoS Data frames with a TID that does not correspond to a block ack agreement. These have No Ack ack policy and the A-MSDU Present subfield equal to 0.
Action No Ack	Action No Ack frames.
<u>Trigger</u>	<u>If sent by an HE AP, Trigger frames where the Trigger Type field indicates Basic Trigger frame, BSRP Trigger frame, or BQRP Trigger frame.</u> <u>The Trigger frames are the first MPDUs of the A-MPDU, unless the A-MPDU also carries an Ack or BlockAck frame in which case the Trigger frames are included immediately after the Ack or BlockAck frame.</u> <u>See NOTE 1 and NOTE 2.</u>
<u>QoS Null frame with No Ack ack policy</u>	<u>If sent to an HE STA, QoS Null frames with No Ack ack policy.</u>
<u>NOTE 1—Only an HE AP is allowed to include a Trigger frame in the A-MPDU. The presence of more than one copy of a Trigger frame in an A-MPDU might increase the probability of the successful reception of the Trigger frame. The content of all Trigger frames in the A-MPDU is the same.</u> <u>NOTE 2—The BSRP and BQRP Trigger frames can be aggregated with other MPDUs in the A-MPDU if the receiver has indicated the support of receiving these trigger types in the BSRP BQRP A-MPDU Aggregation field in the HE Capabilities element.</u>	

Table 9-532 remains unchanged.

Table 9-533—A-MPDU contents in the control response context

MPDU	Conditions
Ack	Ack frame transmitted in response to an MPDU that requires an Ack frame.
BlockAck	<u>Compressed BlockAck frame with a TID that corresponds to an HT-immediate block ack agreement.</u> <u>Multi-STA BlockAck frame if the preceding PPDU is either an HE TB PPDU that solicits an immediate response (see 26.4.4.5) or an HE PPDU that carries a multi-TID A-MPDU or ack-enabled multi-TID A-MPDU (see 26.6.3).</u>
<u>One of Ack and compressed BlockAck frame is present at the start of the A-MPDU between two STAs that are not both HE STAs; these are not present other than at the start of the A-MPDU.</u> <u>One of these—One of Ack, Compressed BlockAck, and Multi-STA BlockAck frame is present at the start of the A-MPDU between two HE STAs; these are not present other than at the start of the A-MPDU.</u>	

Table 9-533—A-MPDU contents in the control response context *(continued)*

MPDU	Conditions
Action No Ack	<u>In an A-MPDU between two STAs that are not both HE STAs:</u> BRP +HTC frames. Action No Ack +HTC frames containing an explicit feedback response. Action No Ack frames that are Flow Suspension frames or Flow Resumption frames. <u>In an A-MPDU between two HE STAs: Action No Ack frames.</u>
<u>QoS Null frame with No Ack ack policy</u>	<u>If sent to an HE STA. QoS Null frames with No Ack ack policy.</u>

Table 9-534 remains unchanged.

Insert Table 9-534a through Table 9-534d into 9.7.1 after Table 9-534:

**Table 9-534a—A-MPDU contents in the
HE non-ack-enabled single-TID immediate response context**

MPDU	Conditions
Ack	If the preceding PPDU contains an MPDU that requires an Ack frame response, a single Ack frame at the start of the A-MPDU. At most one untagged MPDU that is Ack, Compressed BlockAck, or Multi-STA BlockAck frame is present
HT-immediate BlockAck	If the preceding PPDU contains an implicit or explicit block ack request for a TID for which an HT-immediate block ack agreement exists, at most one Compressed BlockAck frame for this TID at the start of the A-MPDU. If the preceding PPDU contains explicit block ack requests for multiple TIDs or a multi-TID A-MPDU, at most one Multi-STA BlockAck frame at the start of the A-MPDU.
Action No Ack	Untagged MPDUs that are Action No Ack frames.
QoS Null frame with No Ack ack policy	Untagged MPDUs that are QoS Null frames with No Ack ack policy.

**Table 9-534a—A-MPDU contents in the
HE non-ack-enabled single-TID immediate response context (continued)**

MPDU	Conditions	
Data frames sent under an HT-immediate block ack agreement	One or more QoS Data frames with the same TID, which corresponds to an HT-immediate block ack agreement See NOTE 1.	One of the following is present: — One or more untagged MPDUs, each of which is a QoS Data frame with Implicit BAR, HETP Ack, or Block Ack ack policy and belonging to a block ack agreement, zero or more untagged MPDUs, each of which is a Basic Trigger, BSRP Trigger, or BQRP Trigger frame. — One untagged MPDU that is BlockAckReq frame, zero or more untagged MPDUs each of which is a QoS Null frame with No Ack ack policy. — One or more untagged MPDUs each of which is a Basic Trigger, MU-BAR Trigger, BQRP Trigger, or BSRP Trigger frame. The MU-BAR Trigger frame solicits block acknowledgment for one TID.
Compressed BlockAckReq	At most one BlockAckReq frame with a TID that corresponds to an HT-immediate block ack agreement. This frame is the last MPDU in the A-MPDU. BlockAckReq is not present if any QoS Data frames are present.	
Trigger	Trigger frames where the Trigger Type field is Basic Trigger, MU-BAR Trigger, BQRP Trigger or BSRP Trigger. MU BAR Trigger frame is not present if any QoS Data frames are present. The Trigger frames are the first MPDUs of the A-MPDU, unless the A-MPDU also carries an Ack or BlockAck frame in which case the Trigger frames are included immediately after the Ack or BlockAck frame. See NOTE 2 and NOTE 3.	
NOTE 1—MPDUs from the same TID all have the same ack policy, which is Implicit BAR, HETP Ack or Block Ack.		
NOTE 2—Only an HE AP is allowed to include a Trigger frame in the A-MPDU. The presence of more than one copy of a Trigger frame in an A-MPDU might increase the probability of the reception of the Trigger frame. The content of all Trigger frames in the A-MPDU is the same.		
NOTE 3—The BSRP and BQRP Trigger frames can be aggregated with other MPDUs in the A-MPDU if the receiver has indicated the support of receiving these trigger types in the BSRP BQRP A-MPDU Aggregation field in the HE Capabilities element.		

**Table 9-534b—A-MPDU contents in the
HE ack-enabled single-TID immediate response context**

MPDU	Conditions	
Ack	If the preceding PPDU contains an MPDU that requires an Ack frame response, a single Ack frame at the start of the A MPDU.	At most one untagged MPDU that is an Ack, Compressed BlockAck, or Multi-STA BlockAck frame is present
HT-immediate BlockAck	If the preceding PPDU contains an implicit or explicit block ack request for a TID for which an HT-immediate block ack agreement exists, at most one Compressed BlockAck frame for this TID at the start of the A-MPDU. If the preceding PPDU contains explicit block ack requests for multiple TIDs or a multi-TID A-MPDU, at most one Multi-STA BlockAck frame at the start of the A-MPDU.	
Action No Ack	Untagged MPDUs that are Action No Ack frames.	
Data frames not sent under an HT-immediate block ack agreement	At most one Data frame with a TID that does not correspond to an HT-immediate block ack agreement and Normal Ack or HETP Ack ack policy.	One of the following: — A tagged MPDU that is a QoS Data frame with Normal Ack or HETP Ack ack policy — A tagged MPDU that is a Management frame that solicits an immediate response And at least one untagged MPDU that is a QoS Null frame with No Ack ack policy, Basic Trigger frame, BSRP Trigger frame, or BQRP Trigger frame.
Data frames sent under an HT-immediate block ack agreement	At most one QoS Data frame with a TID that corresponds to an HT-immediate block ack agreement	
QoS Null frame with No Ack ack policy	QoS Null frames with No Ack ack policy.	
Management frame	At most one Management frame that solicits an acknowledgment.	
Trigger	Basic Trigger, BQRP Trigger or BSRP Trigger frames. The Trigger frames are the first MPDUs of the A-MPDU, unless the A-MPDU also carries an Ack or BlockAck frame in which case the Trigger frames are included immediately after the Ack or BlockAck frame. See NOTE 1 and NOTE 2.	
NOTE 1—Only an HE AP is allowed to include a Trigger frame in the A-MPDU. The presence of more than one copy of a Trigger frame in an A-MPDU might increase the probability of the reception of the Trigger frame. The content of all Trigger frames in the A-MPDU is the same. NOTE 2—The BSRP Trigger and BQRP Trigger frames can be aggregated with other MPDUs in the A-MPDU if the receiver has indicated the support of receiving these Trigger frame types in the BSRP BQRP A-MPDU Aggregation field in the HE Capabilities element. NOTE 3—The single Management frame that solicits the acknowledgment in ack-enabled single-TID A-MPDU is treated as single-TID frame, e.g., soliciting Ack of TID 15 in multi-STA BlockAck frame.		

**Table 9-534c—A-MPDU contents in the
HE non-ack-enabled multi-TID immediate response context**

MPDU	Conditions	
Ack	If the preceding PPDU contains an MPDU that requires an Ack frame response, a single Ack frame at the start of the A MPDU.	At most one tagged MPDU that is an Ack, Compressed BlockAck, or Multi-STA BlockAck frame is present
HT-immediate BlockAck	If the preceding PPDU contains an implicit or explicit block ack request for a TID for which an HT-immediate block ack agreement exists, at most one Compressed BlockAck frame for this TID at the start of the A-MPDU. If the preceding PPDU contains explicit block ack requests for multiple TIDs or a multi-TID A-MPDU, at most one Multi-STA BlockAck frame at the start of the A-MPDU.	
Action No Ack	Untagged MPDUs that are Action No Ack frames.	
QoS Null frame with No Ack ack policy	Untagged MPDUs that are QoS Null frames with No Ack ack policy.	
Data frames sent under an HT-immediate block ack agreement	QoS Data frames with different TIDs each of which corresponds to an HT-immediate block ack agreement. See NOTE 1.	One of the following is present: - Two or more untagged MPDUs that are QoS Data frames that belong to two or more block ack agreements and with Implicit BAR, HETP Ack, or Block Ack ack policy, zero or more untagged MPDUs each of which is a Trigger frame. The Trigger frame is a Basic Trigger, BSRP Trigger, or BQRP Trigger frame - One untagged MPDU that is a Multi-TID BlockAckReq frame. - One or more untagged MPDUs each of which is an MU-BAR Trigger frame that solicits block acknowledgment for more than one TID.
Immediate BlockAckReq	At most one multi-TID BlockAckReq frame with TIDs that correspond to HT-immediate block ack agreements This frame is the last MPDU in the A-MPDU. Multi-TID BlockAckReq frame is not present if any QoS Data frames are present.	
Trigger	Basic Trigger, MU-BAR Trigger, BQRP Trigger or BSRP Trigger frames. MU-BAR Trigger frame is not present if any QoS Data frames are present. The Trigger frames are the first MPDUs of the A-MPDU, unless the A-MPDU also carries an Ack or BlockAck frame in which case the Trigger frames are included immediately after the Ack or BlockAck frame. See NOTE 2 and NOTE 3.	
NOTE 1—MPDUs from the same TID all have the same ack policy, which is Implicit BAR, HETP Ack or Block Ack. NOTE 2—Only an HE AP is allowed to include a Trigger frame in the A-MPDU. Multiple Trigger frames in one A-MPDU increases the robustness. The content of all Trigger frames in the A-MPDU is the same. NOTE 3—The BSRP Trigger and BQRP Trigger frames can be aggregated with other MPDUs in the A-MPDU if the receiver has indicated the support of receiving these trigger types in the BSRP BQRP A-MPDU Aggregation field in the HE Capabilities element.		

**Table 9-534d—A-MPDU contents in the
HE ack-enabled multi-TID immediate response context**

MPDU	Conditions		
Ack	If the preceding PPDU contains an MPDU that requires an Ack frame response, a single Ack frame at the start of the A MPDU.	At most one untagged MPDU that is an Ack, Compressed BlockAck, or Multi-STA BlockAck frame is present	
HT-immediate BlockAck	If the preceding PPDU contains an implicit or explicit block ack request for a TID for which an HT-immediate block ack agreement exists, at most one Compressed BlockAck frame for this TID at the start of the A-MPDU. If the preceding PPDU contains explicit block ack requests for multiple TIDs or a multi-TID A-MPDU, at most one Multi-STA BlockAck frame at the start of the A-MPDU.		
Action No Ack	Untagged MPDUs that are Action No Ack frames.		
QoS Null frame with No Ack ack policy	Untagged MPDUs that are QoS Null frames with No Ack ack policy.		
Data frames without an HT-immediate block ack agreement	One or more QoS Data frames with each with different TIDs where none of the TID have HT-immediate block ack agreement See NOTE 1.	One of the following is present: — Tagged MPDUs each of which is a QoS Data frame with Normal Ack or HETP Ack ack policy and where the TIDs of the QoS Data frames differ if there is more than one, zero or more untagged MPDUs each of which is a Basic Trigger, BSRP Trigger, or BQRP Trigger frame. — Zero or more untagged MPDUs each of which is a QoS Data frame with Implicit BAR, HETP Ack, or Block Ack ack policy and belonging to a block ack agreement, one or more tagged MPDUs each of which is a QoS Data frame with Normal Ack or HETP Ack ack policy and where the TIDs of the QoS Data frames differ if there is more than one, a tagged MPDU that is a Management frame, zero or more untagged MPDUs each of which is a Basic Trigger, BSRP Trigger, or BQRP Trigger frame. — Zero or more untagged MPDUs each of which is a QoS Data frame with Implicit BAR, HETP Ack, or Block Ack ack policy and belonging to a block ack agreement, and two or more tagged MPDUs each of which is a QoS Data frame with Normal Ack or HETP Ack ack policy and where the TIDs of the QoS Data frames differ if there is more than one, zero or more untagged MPDUs each of which is a Basic Trigger, BSRP Trigger, or BQRP Trigger frame.	
Data frames sent under an HT-immediate block ack agreement	One of the following: — One or more QoS Data frames with a TID that corresponding to an HT-immediate block ack agreement. — QoS Data frames with TIDs that correspond to two or more HT-immediate block ack agreements. See NOTE 1.		
Management	At most one Management frame that solicits an acknowledgment		
Trigger	One or more Basic Trigger, BQRP Trigger or BSRP Trigger frames. The Trigger frames are the first MPDUs of the A-MPDU, unless the A-MPDU also carries an Ack or BlockAck frame in which case the Trigger frames are included immediately after the Ack or BlockAck frame. See NOTE 2 and NOTE 3.		

**Table 9-534d—A-MPDU contents in the
HE ack-enabled multi-TID immediate response context *(continued)***

MPDU	Conditions
	NOTE 1—MPDUs with the same TID all have the same ack policy, which is Implicit BAR, HETP Ack or Block Ack. NOTE 2—Only an HE AP is allowed to include a Trigger frame in the A-MPDU. Multiple Trigger frames in one A-MPDU increases the robustness. The content of all Trigger frames in the A-MPDU is the same. NOTE 3—The BSRP Trigger and BQRP Trigger frames can be aggregated with other MPDUs in the A-MPDU if the receiver has indicated the support of receiving these trigger types in the BSRP BQRP A-MPDU Aggregation field in the HE Capabilities element.

10. MAC sublayer functional description

10.1 Introduction

Change 10.1 as follows:

The MAC functional description is presented in this clause. The architecture of the MAC sublayer, including the distributed coordination function (DCF), the hybrid coordination function (HCF), the mesh coordination function (MCF), the triggered UL access (TUA), and their coexistence in an IEEE 802.11 LAN are introduced in 10.2. These functions are expanded on in 10.3, 10.23, and 10.24, and 26.2. Fragmentation and defragmentation are defined in 10.4 and 10.5. Multirate support is addressed in 10.6. A number of additional restrictions to limit the cases in which MSDUs are reordered or discarded are described in 10.7. Operation across regulatory domains is defined in 10.22. The block ack mechanism is described in 10.25. The No Ack mechanism is described in 10.26. The protection mechanism is described in 10.27. Rules for processing MAC frames are described in 10.28.

10.2 MAC architecture

10.2.1 General

Replace Figure 10-1 with the following figure:

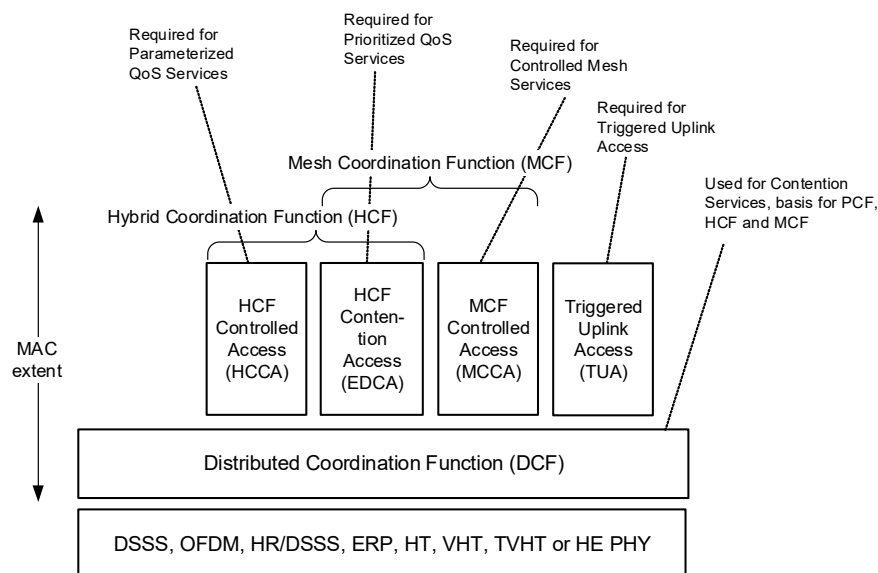


Figure 10-1—Non-DMG non-CMMG non-S1G STA MAC architecture

Change the second paragraph in 10.2.1 as follows:

In a non-DMG non-CMMG non-S1G STA:

- The MAC provides the HCF, ~~and MCF, and TUA~~ services using the services of the DCF. The MAC also provides direct access to the DCF service.
- The HCF is present in QoS STAs and absent otherwise.
- The MCF is present in mesh STAs and absent otherwise.
- The TUA is present in non-AP HE STAs and absent otherwise.

10.2.3 Hybrid coordination function (HCF)

10.2.3.2 HCF contention based channel access (EDCA)

Insert the following paragraph into 10.2.3.2 after the fourth paragraph (“The QoS AP”):

An HE AP can additionally provide MU EDCA parameters for non-AP HE STAs as defined in 26.2.7.

Change the now sixth paragraph in 10.2.3.2 as follows:

A QoS STA, on receiving an updated EDCA Parameter Set element, should update the dot11EDCATable as soon as practical in the implementation and shall update the dot11EDCATable within an interval of time equal to one beacon interval. QoS STAs update the dot11EDCATable and store the EDCA Parameter Set update count value in the QoS Info field. An SIG STA shall update its dot11EDCATable if its STA type is indicated by the STA Type subfield contained in the received EDCA Parameter Set element (see 10.61). An AP may change the EDCA access parameters by changing the EDCA Parameter Set element in the Beacon frame, Probe Response frame, and (Re)Association Response frame. However, the AP should change them only rarely. A QoS STA shall use the EDCA Parameter Set Update Count ~~Value~~ subfield in the QoS Capability element of all Beacon frames to determine whether the STA is using the current EDCA Parameter Values. If the EDCA Parameter Set update count value in the QoS Capability element is different from the value that has been stored, the QoS STA shall query the updated EDCA parameter values by sending a Probe Request frame to the AP.

Change the now 14th paragraph in 10.2.3.2 as follows:

BlockAckReq and BlockAck frames shall be sent using the same access category for medium access as the corresponding QoS Data frames. When a BlockAckReq, Multi-TID BlockAckReq, or Multi-STA BlockAck frame is transmitted by an HE STA, its operation is defined in 26.4.

Insert the following subclause (10.2.4a) after 10.2.4:

10.2.4a Triggered uplink access (TUA)

A non-AP HE STA supports trigger-based UL access methods. Triggered UL access (TUA) is used when an HE AP triggers one or more non-AP HE STAs to transmit HE TB PPDU_s simultaneously. The optional UL OFDMA-based random access (UORA) additionally allows a non-AP HE STA to access one of a number of resource units (RUs) designated for random access by the HE AP. See 26.5.2 and 26.5.4.

Change the title and first paragraph of 10.2.5 as follows:

10.2.5 Combined use of DCF, ~~and HCF,~~ and TUA

The DCF, ~~and the hybrid coordination function HCF,~~ and TUA are defined so they may operate within the same BSS. The HCF access methods (controlled and contention based) operate sequentially. Sequential operation allows the polled and contention based access methods to alternate, within intervals as short as the time to transmit a frame exchange sequence, under rules defined in 10.23.

10.3 DCF

10.3.1 General

Change the sixth paragraph in 10.3.1 as follows:

The virtual CS mechanism is achieved by distributing reservation information announcing the impending use of the medium. The exchange of RTS and CTS frames prior to the actual Data frame is one means of distribution of this medium reservation information. The RTS and CTS frames contain a Duration field that defines the period of time that the medium is to be reserved to transmit the actual Data frame and the returning Ack frame. A STA receiving either the RTS frame (sent by the originating STA) or the CTS frame (sent by the destination STA) shall process the medium reservation. Thus, a STA might be unable to receive from the originating STA and yet still know about the impending use of the medium to transmit a Data frame. The exchange of an MU-RTS Trigger frame and simultaneous CTS frame responses by HE STAs prior to the actual Data frames is another means of distribution of this medium reservation information.

Change the 11th paragraph in 10.3.1 as follows:

The use of the RTS/CTS mechanism under control of dot11RTSThreshold___if dot11TXOPDurationRTSThreshold is 1023 or is not present is described in 10.3.5.

Insert the following paragraph into 10.3.1 after the 11th paragraph:

The use of the RTS/CTS mechanism is under the control of dot11TXOPDurationRTSThreshold if dot11TXOPDurationRTSThreshold is present and is not 1023. If this mechanism is enabled, a non-AP HE STA shall initiate a TXOP that is used for individually addressed frames with an RTS/CTS exchange as defined in 26.2.1.

10.3.2 Procedures common to the DCF and EDCAF

10.3.2.1 CS mechanism

Change the fourth paragraph in 10.3.2.1 as follows:

The NAV maintains a prediction of future traffic on the medium based on duration information that is announced in RTS/CTS frames by non-DMG STAs, in MU-RTS Trigger/CTS frames by HE STAs as defined in 26.2.6, and in RTS/DMG CTS frames by DMG STAs prior to the actual exchange of data. The duration information is also available in the MAC headers of all frames other than PV1 MAC frames and PS-Poll frames, and during the BTI, the A-BFT, the ATI, the CBAP, and the SP. The duration information in a frame transmitted by an SIG STA is also available in PS-Poll+BDT frames, in NDP CTS frames, in NDP Ack frames whose Idle Indication field value is 0, and in NDP_2M PS-Poll-Ack frames whose Idle Indication field is 0. The duration information might also be available in the RXVECTOR parameter TXOP_DURATION when an HE PPDU is received (see 26.11.5).

Change the seventh paragraph in 10.3.2.1 as follows:

The CS mechanism combines the NAV state, and in SIG STAs also the RID state, and the STA's transmitter status with physical CS to determine the busy/idle state of the medium. The NAV and RID may be thought of as counters that count down to 0 at a uniform rate. In non-SIG STAs, when the NAV counter is 0, the virtual CS indication is that the medium is idle; when the counter is nonzero, the indication is busy. In SIG STAs, when both NAV and RID counters are 0, the virtual CS indication is that the medium is idle; when either the

NAV counter or the RID counter is nonzero, the indication is that the medium is busy. The virtual CS indication of medium for HE STAs with two NAVs is described in 26.2.4. If a DMG STA supports multiple NAVs as defined in 10.39.10 and all counters are 0, the virtual CS indication is that the medium is idle; when at least one of the counters is nonzero, the indication is busy. The medium shall be determined to be busy when the STA is transmitting.

10.3.2.3 IFS

10.3.2.3.3 SIFS

Insert the following paragraph at the beginning of 10.3.2.3.3 (the now second paragraph starts with “The SIFS is”):

An HE STA that transmits an HE TB PPDU a SIFS after the end of a received PPDU follows the procedure in 26.5.2.3.

10.3.2.3.7 EIFS

Change 10.3.2.3.7 as follows:

A DCF shall use EIFS before transmission, when it determines that the medium is idle immediately following reception of a frame for which the PHY-RXEND.indication primitive contained an error or a frame for which the FCS value was not correct. Similarly, a STA's EDCA mechanism under HCF shall use the EIFS-DIFS+AIFS[AC] interval. The duration of an EIFS is defined in 10.3.7. The EIFS or EIFS-DIFS+AIFS[AC] interval shall begin following indication by the PHY that the medium is idle after detection of the erroneous frame, without regard to the virtual CS mechanism. The STA shall not begin a transmission until the expiration of the later of the NAV and EIFS or EIFS-DIFS+AIFS[AC]. The EIFS and EIFS-DIFS+AIFS[AC] are defined to provide enough time for another STA to acknowledge what was, to this STA, an incorrectly received frame before this STA commences transmission. Reception of an error-free frame or reception of an HE PPDU with the RXVECTOR parameter TXOP_DURATION that is not set to UNSPECIFIED during the EIFS or EIFS-DIFS+AIFS[AC] resynchronizes the STA to the actual busy/idle state of the medium, so the EIFS or EIFS-DIFS+AIFS[AC] is terminated and medium access (using DIFS or AIFS as appropriate and, if necessary, backoff) continues following reception of ~~that the error-free frame~~ or following an expected end of the received HE PPDU. At the expiration or termination of the EIFS or EIFS-DIFS+AIFS[AC], the STA reverts to the NAV and physical CS to control access to the medium.

EIFS shall not be invoked for an A-MPDU if one or more of its frames are received correctly. EIFS shall not be invoked if the RXVECTOR parameter TXOP_DURATION of a received HE PPDU is not set to UNSPECIFIED.

10.3.2.4 Setting and resetting the NAV

Change the first and second paragraphs in 10.3.2.4 as follows:

This subclause describes the setting and resetting of the NAV for non-DMG STAs and DMG STAs that support a single NAV. DMG STAs that support multiple NAVs shall update their NAVs according to the procedures described in 10.39.10. HE STAs with two NAV timers shall update their NAV timers according to the procedures described in 26.2.4.

A STA that receives at least one valid frame in a PSDU can update its NAV with the information from any valid Duration field in the PSDU. When the received frame's RA is equal to the STA's own MAC address,

the STA shall not update its NAV. Further, when the received frame is a DMG CTS frame and its TA is equal to the STA's own MAC address, the STA shall not update its NAV. For all other received frames the STA shall update its NAV when the received Duration is greater than the STA's current NAV value. Upon receipt of a PS-Poll frame, a STA, except for an S1G STA for which the RXVECTOR parameter RESPONSE_INDICATION of the received PS-Poll frame is NDP Response, shall update its NAV settings as appropriate under the data rate selection rules using a duration value equal to the time, in microseconds, required to transmit one Ack frame plus one SIFS, but only when the new NAV value is greater than the current NAV value. If the calculated duration includes a fractional microsecond, that value is rounded up to the next higher integer. When the NAV is reset, a PHY-CCARESET.request primitive shall be issued. This NAV update operation is performed when the PHY-RXEND.indication primitive is received, except when the PHY-RXEND.indication primitive is received before the end of the PPDU. In this exception, the NAV update is performed at the expected end of the PPDU.

Insert the following paragraphs and notes into 10.3.2.4 after the fourth paragraph (“In addition to the NAV....”), and change the subsequent note in this subclause to “NOTE 4”:

An HE AP that is not a TXOP holder shall update the NAV with the duration information indicated by the RXVECTOR parameter TXOP_DURATION for an HE PPDU if all of the following conditions are met and shall not update the NAV with the duration information indicated by the RXVECTOR parameter TXOP_DURATION if one or more of the following conditions are not met:

- The RXVECTOR parameter TXOP_DURATION is not UNSPECIFIED.
- The HE AP does not receive a frame with a Duration field in the PPDU.
- The duration indicated by the RXVECTOR parameter TXOP_DURATION is greater than the current NAV value of the HE AP.

An HE AP that is a TXOP holder shall update the NAV with the duration information indicated by the RXVECTOR parameter TXOP_DURATION for an HE PPDU if all of the following conditions are met and shall not update the NAV with the duration information indicated by the RXVECTOR parameter TXOP_DURATION if one or more of the following conditions are not met:

- The RXVECTOR parameter TXOP_DURATION is not UNSPECIFIED.
- The HE AP does not receive a frame with a Duration field in the PPDU.
- The duration indicated by the RXVECTOR parameter TXOP_DURATION is greater than the current NAV value of the HE AP.
- The RXVECTOR parameter BSS_COLOR is not equal to the BSS color of the HE AP.

NOTE 2—A non-AP HE STA maintains two NAVs, but an HE AP might maintain only one NAV (see 26.2.4).

NOTE 3—If a STA receives an HE PPDU with the duration information indicated by both a frame with a Duration field and the RXVECTOR parameter TXOP_DURATION, then the duration information indicated by the RXVECTOR parameter TXOP_DURATION is ignored.

Change the now eighth and ninth paragraphs in 10.3.2.4 as follows:

A STA that used information from an RTS frame or MU-RTS Trigger frame as the most recent basis to update its NAV setting is permitted to reset its NAV if no PHY-RXSTART.indication primitive is received from the PHY during a NAVTimeout period starting when the MAC receives a PHY-RXEND.indication primitive corresponding to the detection of the RTS frame or MU-RTS Trigger frame.

In non-DMG BSS, NAVTimeout period is equal to $(2 \times \text{aSIFSTime}) + (\text{CTS_Time}) + \text{aRxPHYStartDelay} + (2 \times \text{aSlotTime})$. In a non-S1G STA, if an RTS frame is used for the most recent NAV update, the CTS_Time shall be calculated using the length of the CTS frame and the data rate at which the RTS frame used for the most recent NAV update was received. If an MU-RTS Trigger frame was used for the most recent NAV

update, CTS_Time shall be calculated using the length of the CTS frame and the 6 Mb/s data rate (see 26.2.6).
In an S/G STA, ~~the~~ CTS_Time shall be calculated using the time required to transmit an NDP CTS frame that is equal to NDPTxTime, which is specified in 10.3.2.5.2.

10.3.2.6 RTS/CTS with fragmentation

Change the second paragraph in 10.3.2.6 as follows:

Each frame contains information that defines the duration of the next transmission. The duration information from RTS frames shall be used to update the NAV to indicate busy until the end of Ack frame 0. The duration information from the CTS frame shall also be used to update the NAV to indicate busy until the end of Ack frame 0. Both Fragment 0 and Ack frame 0 shall contain duration information to update the NAV to indicate busy until the end of Ack frame 1. This shall be done by using the Duration/ID field in the Data and Ack frames. This shall continue until the last fragment, which shall have a duration of one Ack time plus one SIFS, and its Ack frame, which shall have its Duration/ID field set to 0. Each fragment and Ack frame acts as a virtual RTS frame and CTS frame; therefore, no further RTS/CTS frames need to be generated after the RTS/CTS that began the frame exchange sequence—even though the PSDUs carrying subsequent fragments may be larger than ~~dot11RTSThreshold~~.

10.3.2.13 MU acknowledgment procedure

Insert the following subclause heading (for 10.3.2.13.1) immediately after the heading for 10.3.2.13:

10.3.2.13.1 Acknowledgment procedure for DL MU PPDU in SU PPDU

Change the text in 10.3.2.13.1 as follows (Figure 10-13 and Figure 10-14 remain unchanged):

The acknowledgment procedure performed by a STA that receives MPDUs that were transmitted within a VHT MU PPDU or an HE MU PPDU is the same as the acknowledgment procedure for MPDUs that were not transmitted within a VHT MU PPDU or an HE MU PPDU sent by an AP, except if the STA is an HE STA that follows the rules defined in 26.3 and in 26.4.

NOTE 1—All MPDUs transmitted within a VHT MU PPDU or an HE MU PPDU are contained within A-MPDUs, and the rules specified in 9.7.3 prevent an immediate response carried in an SU PPDU to more than one of the A-MPDUs.

Responses to A-MPDUs within a VHT MU PPDU or an HE MU PPDU for DL transmission that are not immediate responses to the VHT MU PPDU or the HE MU PPDU are transmitted in response to explicit BlockAckReq frames by the AP. Examples of VHT MU PPDU frame exchange sequences are shown in Figure 10-13 and Figure 10-14.

Recovery within the TXOP that contains a VHT MU PPDU or an HE MU PPDU can be performed according to the rules of 10.23.2.8. BlockAckReq~~uest~~ frames related to A-MPDUs within a VHT MU PPDU or an HE MU PPDU can be transmitted in a TXOP separate from the one that contained the VHT MU PPDU or the HE MU PPDU.

NOTE 2—A BlockAck frame is sent in immediate response to the BlockAckReq frame for HT-immediate block ack. An Ack frame might be sent in immediate response carried in an SU PPDU to an S-MPDU in the VHT MU PPDU or the HE MU PPDU. Responses to S-MPDUs for more than one STA contained in an HE MU PPDU are transmitted as specified in 10.3.2.13.2. A Multi-STA BlockAck frame is sent in immediate response to a Multi-TID BlockAckReq frame.

NOTE 3—A BlockAckReq~~uest~~ frame would typically not be sent to a STA in the case where the A-MPDU to the STA contained no MPDUs requiring immediate acknowledgment. It could be sent if MPDUs in a previous A-MPDU remain unacknowledged.

Insert the following subclauses (10.3.2.13.2 and 10.3.2.13.3, including Figure 10-14a through Figure 10-14d) after 10.3.2.13.1:

10.3.2.13.2 Acknowledgment procedure for DL MU PPDU in MU format

A non-AP STA shall not set the ack policy to HETP Ack.

A non-AP STA that is the recipient, within an HE MU PPDU, of a QoS Data frame or QoS Null frame with HETP Ack ack policy, of an MU-BAR Trigger frame or a GCR MU-BAR Trigger frame, or of a Management frame that solicits acknowledgment, shall send the immediate response according to the scheduling information that is carried either in the Trigger frame(s) or TRS Control subfield. If a Basic Trigger frame (see 9.3.1.22) or frame carrying a TRS Control subfield (see 9.2.4.6a.1) is not received, then the STA shall not respond.

An example of UL OFDMA acknowledgment to an HE MU PPDU is shown in Figure 10-14a.

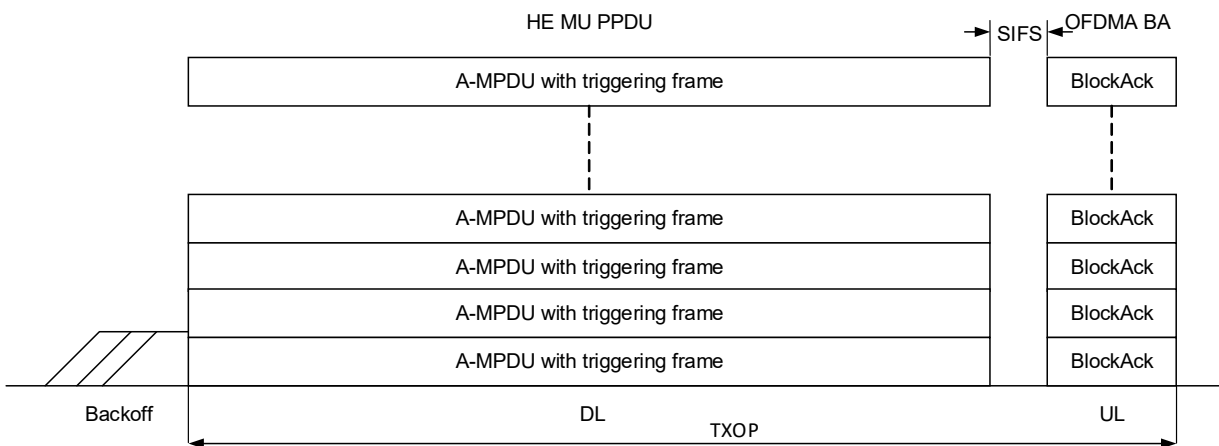


Figure 10-14a—Example of HE MU PPDU transmission with immediate UL OFDMA acknowledgment

An AP may use an MU-BAR Trigger frame or a GCR MU-BAR Trigger frame to solicit acknowledgment frames from multiple HE STAs to which the AP has sent QoS Data frames with Block Ack ack policy or from which the AP has not received immediate acknowledgment frames after sending QoS Data frames with HETP Ack ack policy in an HE MU PPDU.

10.3.2.13.3 Acknowledgment procedure for UL MU transmission

An AP that receives frames from more than one STA that are part of an UL MU transmission (see 9.42.2) and that require an immediate acknowledgment (i.e., a QoS Data frame with Normal Ack or Implicit BAR ack policy or a Management frame other than an Action No Ack frame), shall send an immediate acknowledgment in either an SU PPDU (see 26.4.4.5) or an HE MU PPDU (see 26.4.4.6). The Multi-STA BlockAck frame may be transmitted in a non-HT PPDU, non-HT duplicate PPDU, HT PPDU, VHT PPDU, HE SU PPDU, HE ER SU PPDU, or HE MU PPDU. After the reception of an UL frame requiring acknowledgment, transmission of the DL acknowledgment shall commence after a SIFS, without regard to the busy/idle state of the medium. When an AP transmits an immediate acknowledgment in an HE MU PPDU in response to an A-MPDU sent in an HE TB PPDU, the AP should send it within the 20 MHz channel(s) where the pre-HE modulated fields of the HE TB PPDU sent by the STA are located. The immediate acknowledgment is an Ack frame, Compressed BlockAck frame, or Multi-STA BlockAck frame.

An example of multiple BlockAck frames sent in DL MU is shown in Figure 10-14b.

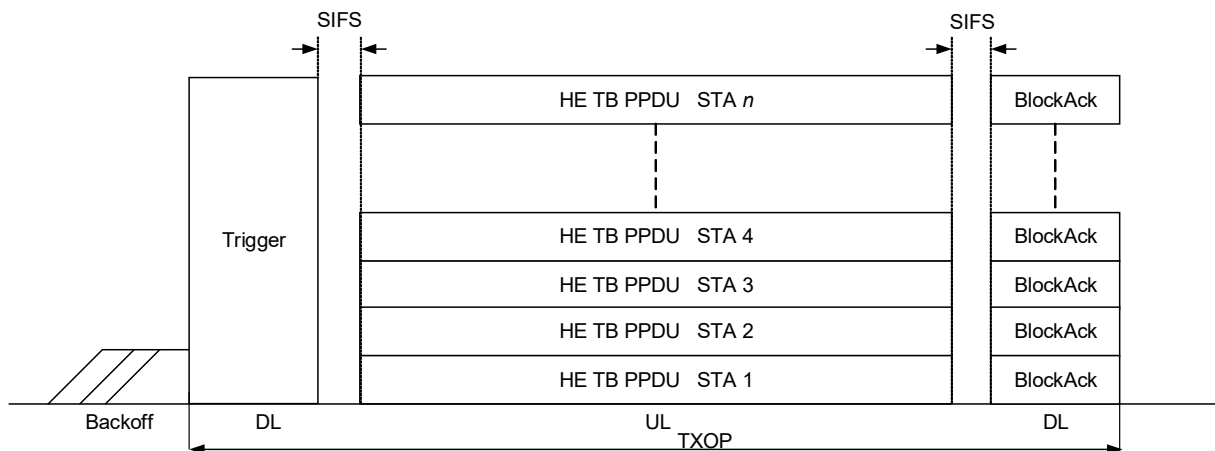


Figure 10-14b—Example of UL MU transmission with immediate DL MU transmission containing individually addressed BlockAck frames acknowledging frames received from respective STAs

An example of a Multi-STA BlockAck frame acknowledgment in a non-HT PPDU, HT PPDU, VHT PPDU, HE SU PPDU, or HE ER SU PPDU is given in Figure 10-14c.

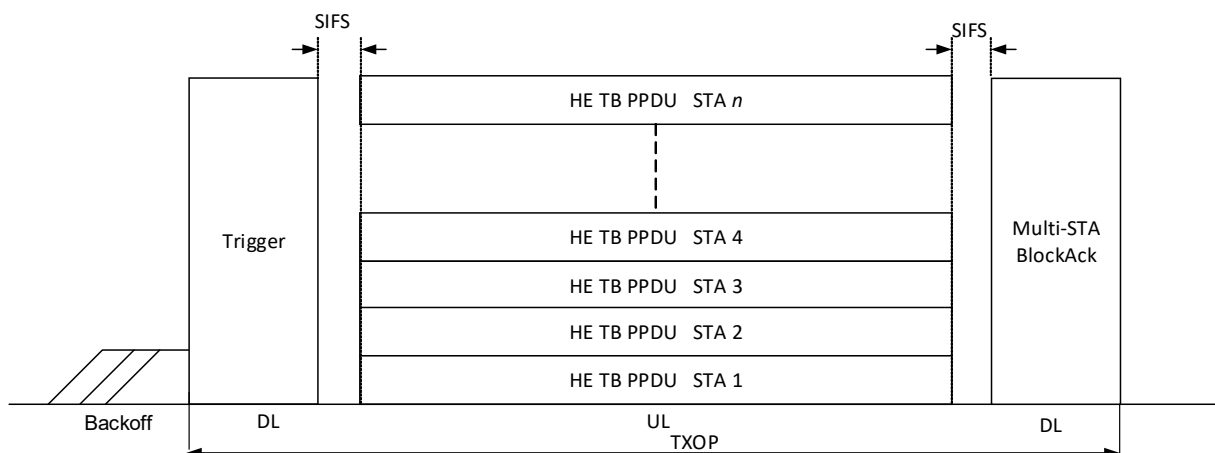


Figure 10-14c—Example of UL MU transmissions with immediate Multi-STA BlockAck frame acknowledging MPDUs

An example of a Multi-STA BlockAck frame acknowledgment in a non-HT duplicate PPDU is given in Figure 10-14d.

The Ack Policy Indicator subfield of a QoS Data frame sent in an HE TB PPDU shall not be set to Block Ack.

A STA may send a BlockAckReq frame or Multi-TID BlockAckReq frame to solicit the acknowledgment frame(s) from an AP.

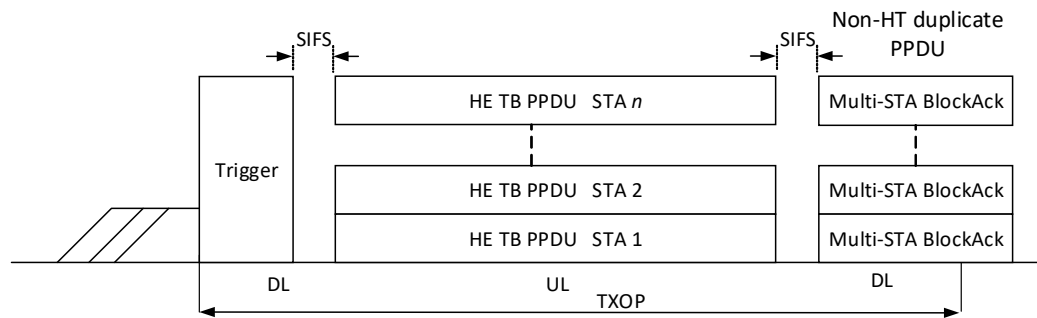


Figure 10-14d—Example of UL MU transmissions with immediate DL non-HT duplicate PPDU containing Multi-STA BlockAck frame

10.3.3 Random backoff time

Insert NOTE 2 into 10.3.3 after the second-to-last paragraph (“The SLRC shall ... when an Ack frame”), and change the note earlier in this subclause to “NOTE 1”:

NOTE 2—For non-HE STAs that use DCF for channel access, dot11TXOPDurationRTSThreshold is not present and the use of the RTS/CTS mechanism is under the control of dot11RTSThreshold.

10.3.4 DCF access procedure

10.3.4.4 Recovery procedures and retransmit limits

Insert the following note into 10.3.4.4 after the fourth paragraph (“After transmitting a frame”):

NOTE—For non-HE STAs that use DCF for channel access, dot11TXOPDurationRTSThreshold is not present, and the use of the RTS/CTS mechanism is under the control of dot11RTSThreshold.

10.3.5 Individually addressed MPDU transfer procedure

Change 10.3.5 as follows:

~~A~~—If dot11TXOPDurationRTSThreshold is 1023 or not present, a STA shall use an RTS/CTS preceding a frame exchange including an individually addressed Data or Management frame when the length of the PSDU is greater than the length threshold indicated by dot11RTSThreshold. If dot11TXOPDurationRTSThreshold is present and is not 1023, a non-AP HE STA using EDCA shall use an RTS/CTS exchange as defined in 26.2.1. A STA may also use an RTS/CTS exchange to protect the transmission of for individually addressed frames when it is necessary to distribute the NAV, or when it is necessary to establish protection (see 10.27), or for other purposes.

NOTE 1—If dot11RTSThreshold is 0, an RTS/CTS exchange precedes all frame exchanges including an individually addressed Data or Management frame, except for a non-AP STA with dot11TXOPDurationRTSThreshold present and not equal to 1023. In this exception, use of an RTS/CTS exchange is controlled by dot11TXOPDurationRTSThreshold (see 26.2.1).

NOTE 2—A non-AP STA that transmits an HE TB PPDU outside the context of an RDG is exempt from requirements related to dot11RTSThreshold and dot11TXOPDurationRTSThreshold because the STA is not the TXOP holder.

~~When~~~~If~~ an RTS/CTS exchange is used, the PPDU containing the PSDU shall be transmitted starting one SIFS after the end of the CTS frame.

NOTE ~~2~~~~3~~—No regard is given to the busy or idle status of the medium when transmitting this PSDU.

~~When~~~~If~~ an RTS/CTS exchange is not used, the PSDU shall be transmitted following the success of the basic access procedure. With or without the use of the RTS/CTS exchange procedure, the STA that is the destination of a Data frame shall follow the acknowledgment procedure.

10.3.7 DCF timing relations

Insert the following note after Equation (10-8) and its variable description, and change the note at the end of this subclause to “NOTE 3”:

NOTE 2—This also applies to an HE STA with dot11DynamicEIFSActivated set to true that receives an HE PPDU with the RXVECTOR parameter TXOP_DURATION set to UNSPECIFIED that causes the EIFS.

10.3.8 Signal extension

Change the first paragraph in 10.3.8 as follows:

Transmissions of PSDUs with the TXVECTOR parameter FORMAT of type NON_HT with NON_HT_MODULATION values of ERP-OFDM and NON_HT_DUP_OFDM and transmissions of frames with the TXVECTOR parameter FORMAT with values of HT_MF, ~~and HT_GF, HE_SU, HE_MU, HE_ER_SU, or HE_TB~~ include a period of no transmission of duration aSignalExtension, except for RIFS transmissions. The purpose of this signal extension is to enable the NAV value of Clause 16 STAs to be set correctly.

Change the title of 10.4 as follows:

10.4 MSDU, A-MSDU, and MMPDU fragmentation

Change the third and fourth paragraphs in 10.4 as follows:

A fragment is an MPDU, the Frame Body field of which carries only a portion of an MSDU, A-MSDU, or MMPDU. When data are to be transmitted, the number of octets in the fragment (before processing by the security mechanism) shall be limited by dot11FragmentationThreshold and the number of octets in the MPDU that have yet to be assigned to a fragment at the instant the fragment is constructed for the first time. Once a fragment is transmitted for the first time, its frame body content and length shall be fixed until it is successfully delivered to the immediate receiving STA.

A STA shall be capable of receiving MPDUs, containing all or part of an MSDU, of arbitrary length that is less than or equal to the maximum MSDU size as specified in Table 9-25 ~~defined in 9.2.4.7~~, plus any security encapsulation overhead, plus MAC header and FCS.

Insert the following paragraph at the end of 10.4:

An HE STA may also use dynamic fragmentation as defined in 26.3.2 if the conditions in 26.3.1 are met. Dynamic fragmentation allows A-MSDUs to be fragmented.

Change the title of 10.5 as follows:

10.5 MSDU, A-MSDU, and MMPDU defragmentation

Insert the following paragraph at the end of 10.5:

An HE STA may also use dynamic defragmentation as defined in 26.3.3 if the conditions in 26.3.1 are met. Dynamic fragmentation allows A-MSDUs to be fragmented.

10.6 Multirate support

10.6.1 Overview

Change the fifth and sixth paragraphs in 10.6.1 as follows:

For specific PHYs, the value of the Duration/ID field is determined using the PLME-TXTIME.request primitive and the PLME-TXTIME.confirm primitive. These specific PHYs are defined in:

- Clause 16 for HR/DSSS
- Clause 17 for OFDM
- Clause 18 for ERP
- Clause 19 for HT
- Clause 20 for DMG
- Clause 21 for VHT
- Clause 22 for TVHT
- Clause 24 for CDMG
- Clause 25 for CMMG
- Clause 27 for HE

The two PLME-TXTIME primitives are defined in the respective PHY specifications:

- 16.3.4 for HR TXTIME calculation
- 17.4.3 for OFDM TXTIME calculation
- 18.5.3.2 for ERP-OFDM TXTIME calculation
- 19.4.3 for HT TXTIME calculation
- 20.11.3 for DMG PLME TXTIME calculation
- 21.4.3 for VHT PLME TXTIME calculation
- 22.4.3 for TVHT PLME TXTIME calculation
- 25.14.3 for CMMG PLME TXTIME calculation
- 27.4.3 for HE PLME TXTIME calculation

10.6.5 Rate selection for Data and Management frames

10.6.5.1 Rate selection for non-STBC Beacon and non-STBC PSMP frames

Change the second paragraph in 10.6.5.1 as follows:

If the BSSBasicRateSet parameter is not empty, a non-STBC PSMP frame or a non-STBC Beacon frame that is not an ER beacon or HE beacon shall be transmitted in a non-HT PPDU using one of the rates included in the BSSBasicRateSet parameter. An ER beacon is transmitted as defined in 26.15.5, and an HE beacon is transmitted as defined in 26.15.6.

10.6.5.3 Rate selection for other group addressed Data and Management frames

Change the first paragraph in 10.6.5.3 as follows:

This subclause describes the rate selection rules for group addressed Data and Management frames, excluding the following:

- Non-STBC Beacon and non-STBC PSMP frames
- ER beacon and HE beacon
- STBC group addressed Data and Management frames
- Data frames located in an FMS stream (see 11.21.8)
- Group addressed frames transmitted to the GCR concealment address (see 11.21.16.3.5)
- Group addressed Data and Management frames transmitted in an HE ER SU PPDU (see 26.15.5)
- Group addressed Data and Management frames transmitted in an HE SU PPDU (see 26.15.6)
- Group addressed Data and Management frames transmitted in an HE MU PPDU (see 26.15.7)

10.6.6 Rate selection for Control frames

10.6.6.1 General rules for rate selection for Control frames

Insert item f) at the end of the lettered list in 10.6.6.1:

- f) A control response frame may be carried in a VHT PPDU or HT PPDU if the eliciting frame was a Fine Timing Measurement frame carried in a VHT PPDU or HT PPDU, respectively.

Insert the following paragraph at the end of 10.6.6.1:

An HE STA that transmits a Trigger frame, Multi-STA BlockAck frame, or HE/VHT NDP Announcement frame addressed to more than one STA shall use a rate, HT-MCS, <VHT-MCS, NSS> tuple, or <HE-MCS, NSS> tuple that is supported by all recipient STAs.

10.6.6.6 Channel Width selection for Control frames

Change the second paragraph in 10.6.6.6 as follows:

If a VHT or HE STA transmits to another VHT or HE STA a Control frame that is not an RTS frame or a CF-End frame, if that Control frame is an HE NDP Announcement frame or elicits a control response frame, ~~or a VHT Compressed Beamforming frame, or an HE Compressed Beamforming/CQI frame,~~ and

- If the Control frame is transmitted in a non-HT duplicate PPDU (channel width 40 MHz or wider), the transmitting VHT or HE STA shall set the TA field to a bandwidth signaling TA.
- If the Control frame is transmitted in a non-HT PPDU (channel width 20 MHz), the transmitting VHT or HE STA may set the TA field to a bandwidth signaling TA.

10.6.10 Modulation classes

Change Table 10-9 as follows:

Table 10-9—Modulation classes

Description of modulation	Condition that selects this modulation class			
	Clause 15 to Clause 18 PHYs, or Clause 20 PHY, or Clause 24 PHY, or Clause 25 PHY	Clause 19 PHY	Clause 21 PHY	<u>Clause 27 PHY</u>
DSSS and HR/DSSS	Clause 15 or Clause 16 transmission	FORMAT is NON_HT. NON_HT_MODULATION is ERP-DSSS or ERP-CCK.	N/A	<u>FORMAT is NON_HT. NON_HT_MODULATION is ERP-DSSS or ERP-CCK.</u>
ERP-OFDM	18.4 transmission	FORMAT is NON_HT. NON_HT_MODULATION is ERP-OFDM.	N/A	<u>FORMAT is NON_HT. NON_HT_MODULATION is ERP-OFDM.</u>
OFDM	Clause 17 transmission	FORMAT is NON_HT. NON_HT_MODULATION is OFDM or NON_HT_DUP_OFDM.	FORMAT is NON_HT. NON_HT_MODULATION is OFDM or NON_HT_DUP_OFDM.	<u>FORMAT is NON_HT. NON_HT_MODULATION is OFDM or NON_HT_DUP_OFDM.</u>
HT	N/A	FORMAT is HT_MF or HT_GF.	FORMAT is HT_MF or HT_GF.	<u>FORMAT is HT_MF or HT_GF.</u>
DMG Control	Clause 20 transmission and MCS is 0	NA	NA	<u>N/A</u>
DMG SC	Clause 20 transmission and $1 \leq \text{MCS} \leq 12.6$	NA	NA	<u>N/A</u>
DMG Low-power SC	Clause 20 transmission and $25 \leq \text{MCS} \leq 31$	NA	NA	<u>N/A</u>
VHT	N/A	N/A	FORMAT is VHT.	<u>FORMAT is VHT.</u>
CDMG Control	Clause 24 transmission and MCS is 0	N/A	N/A	<u>N/A</u>
CDMG SC	Clause 24 transmission and $1 \leq \text{MCS} \leq 16$	N/A	N/A	<u>N/A</u>
CDMG Low-power SC	Clause 24 transmission and $17 \leq \text{MCS} \leq 23$	N/A	N/A	<u>N/A</u>

Table 10-9—Modulation classes (continued)

Description of modulation	Condition that selects this modulation class			
	Clause 15 to Clause 18 PHYs, or Clause 20 PHY, or Clause 24 PHY, or Clause 25 PHY	Clause 19 PHY	Clause 21 PHY	<u>Clause 27 PHY</u>
CMMG Control	Clause 25 transmission and MCS is 0	N/A	N/A	<u>N/A</u>
CMMG SC	Clause 25 transmission and $1 \leq \text{MCS} \leq 8$	N/A	N/A	<u>N/A</u>
CMMG OFDM	Clause 25 transmission and $9 \leq \text{MCS} \leq 16$	N/A	N/A	<u>N/A</u>
<u>HE</u>	<u>N/A</u>	<u>N/A</u>	<u>N/A</u>	<u>FORMAT is HE_SU, HE_ER_SU, HE_MU, or HE_TB.</u>

10.6.11 Non-HT basic rate calculation

Change 10.6.11 as follows (including changing Table 10-10) (footnote 29 remains unchanged):

This subclause defines how to convert an HT-MCS, ~~or a VHT-MCS, or an HE-MCS~~ to a non-HT basic rate for the purpose of determining the rate of the response frame. It consists of two steps as follows:

- Use the modulation and coding rate determined from the HT-MCS (defined in 19.5), ~~or VHT-MCS (defined in 21.5), or HE-MCS (defined in 27.5)~~ to locate a non-HT reference rate by lookup into Table 10-10.²⁹ In the case of an MCS with UEQ, the modulation of stream 1 is used.
- The non-HT basic rate is the highest rate in the BSSBasicRateSet that is less than or equal to this non-HT reference rate.

NOTE 1—The selection of a non-HT basic rate for the frame sent in response to an HE PPDU is not influenced by DCM encoding in the HE PPDU.

NOTE 2—In a TVWS band, the non-HT reference rate is scaled as described in 22.2.4.

Table 10-10—Non-HT reference rate

Modulation	Coding rate (R)	Non-HT reference rate (Mb/s)
BPSK	1/2	6
BPSK	3/4	9
QPSK	1/2	12
QPSK	3/4	18
16-QAM	1/2	24
16-QAM	3/4	36

Table 10-10—Non-HT reference rate (*continued*)

Modulation	Coding rate (R)	Non-HT reference rate (Mb/s)
64-QAM	1/2	48
64-QAM	2/3	48
64-QAM	3/4	54
64-QAM	5/6	54
256-QAM	3/4	54
256-QAM	5/6	54
<u>1024-QAM</u>	<u>3/4</u>	<u>54</u>
<u>1024-QAM</u>	<u>5/6</u>	<u>54</u>

10.6.12 Channel Width in non-HT and non-HT duplicate PPDUs

Change the first paragraph in 10.6.12 as follows:

A non-VHT STA shall include neither the CH_BANDWIDTH_IN_NON_HT parameter nor the DYN_BANDWIDTH_IN_NON_HT parameter in either of the TXVECTOR or RXVECTOR for NON_HT PPDUs. A non-VHT STA shall not set the TA field to a bandwidth signaling TA. A VHT STA shall include neither the CH_BANDWIDTH_IN_NON_HT parameter nor the DYN_BANDWIDTH_IN_NON_HT parameter in the TXVECTOR of a non-HT PPDUs containing frames addressed to a non-VHT STA. A VHT STA shall not set the TA field to a bandwidth signaling TA in a frame addressed to a non-VHT STA. The TA field shall not be set to a bandwidth signaling TA in a frame carried in a DSSS/CCK PPDU. A VHT STA that includes the DYN_BANDWIDTH_IN_NON_HT parameter in the TXVECTOR shall also include the CH_BANDWIDTH_IN_NON_HT parameter in the TXVECTOR. A VHT STA shall not include the DYN_BANDWIDTH_IN_NON_HT parameter in the TXVECTOR for transmitted frames other than RTS frames with bandwidth signaling TA and that are sent in a non-HT PPDU. A STA that transmits an RTS frame with a bandwidth signaling TA shall include the DYN_BANDWIDTH_IN_NON_HT parameter in the TXVECTOR. A VHT STA shall include both the CH_BANDWIDTH_IN_NON_HT and DYN_BANDWIDTH_IN_NON_HT parameters in the RXVECTOR if the PPDU format is NON_HT.

10.6.13 Rate selection constraints for VHT STAs

10.6.13.3 Additional rate selection constraints for VHT PPDUs

Change the text of 10.6.13.3 as follows (Table 10-11 remains unchanged):

~~The following apply for a STA that transmits a VHT PPDU with a number of spatial streams (NSS) less than or equal to 4:~~

- ~~— If the channel width of the PPDU is equal to CBW20 or CBW40, then the STA should not use a <VHT-MCS, NSS> tuple if the VHT-MCS is equal to 0, 1, 2, or 3 and the HT-MCS with value VHT-MCS + 8×(NSS – 1) is marked as unsupported in the Rx-MCS bitmask of the HT Capabilities element of the receiver STA.~~
- ~~— If the channel width of the PPDU is equal to CBW80, CBW160, or CBW80+80, then the STA should not use a <VHT-MCS, NSS> tuple if the VHT-MCS is equal to 0 or 1 and both the HT-MCS~~

~~values $2 \times \text{VHT-MCS} + 8 \times (\text{NSS} - 1)$ and $2 \times \text{VHT-MCS} + 1 + 8 \times (\text{NSS} - 1)$ are marked as unsupported in the Rx MCS bitmask of the HT Capabilities element of the receiver STA.~~

A STA should not transmit a 20 MHz or 40 MHz VHT PPDU with a <VHT-MCS, NSS> tuple that has VHT-MCS 0, 1, 2, or 3 and NSS less than or equal to 4 to a receiver STA that has marked as unsupported the HT-MCS with value $\text{VHT-MCS} + 8 \times (\text{NSS} - 1)$ in the Rx MCS Bitmask subfield in the Supported MCS Set field in the HT Capabilities element it transmits. The transmission of a 20 MHz or 40 MHz VHT PPDU with VHT-MCS greater than 3 is not subject to this constraint.

A STA should not transmit an 80 MHz, 160 MHz, or 80+80 MHz VHT PPDU with a <VHT-MCS, NSS> tuple that has VHT-MCS 0 or 1 and NSS less than or equal to 4 to a receiver STA that has marked as unsupported the HT-MCS values of both $2 \times \text{VHT-MCS} + 8 \times (\text{NSS} - 1)$ and $2 \times \text{VHT-MCS} + 1 + 8 \times (\text{NSS} - 1)$ in the Rx MCS Bitmask subfield in the Supported MCS Set field in the HT Capabilities element it transmits. The transmission of an 80 MHz, 160 MHz, or 80+80 MHz VHT PPDU with VHT-MCS greater than 1 is not subject to this constraint.

An example tabulation of this behavior is given in Table 10-11.

10.8 HT Control field operation

Change 10.8 as follows (including inserting Table 10-11a):

If dot11HTControlFieldSupported is true, a STA shall set the +HTC Support subfield of the HT Extended Capabilities field of the HT Capabilities element to 1 in HT Capabilities elements that it transmits. If dot11VHTControlFieldOptionImplemented is true, a STA shall set the +HTC-VHT Support subfield of the VHT Capabilities Information field of the VHT Capabilities element to 1 in VHT Capabilities elements that it transmits. If dot11HEControlFieldOptionImplemented is true, a STA shall set the +HTC-HE Support subfield in the HE MAC Capabilities Information field to 1 in the HE Capabilities elements that it transmits.

A STA in which at least one of dot11RDResponderOptionImplemented, dot11MCSFeedbackOptionImplemented, and dot11AlternateEDCAActivated is equal to true shall set has dot11HTControlFieldSupported or dot11VHTControlFieldOptionImplemented or both equal to true. A STA that has at least one of dot11TRSOOptionImplemented, dot11OMIOptionImplemented, dot11HEBSRControlImplemented, dot11HEBQRControlImplemented, dot11RDResponderOptionImplemented, and dot11SRResponderOptionImplemented equal to true or has dot11HEMCSFeedbackOptionImplemented greater than zero shall set dot11HEControlFieldOptionImplemented to true. An HE AP shall set dot11HEControlFieldOptionImplemented to true.

An HT variant HT Control field shall not be present in a frame addressed to a STA, unless that STA declares support for +HTC-HT in the HT Extended Capabilities field of its HT Capabilities element (see 9.2.4.55).

A VHT variant HT Control field shall not be present in a frame addressed to a STA, unless that STA declares support for +HTC-VHT in the VHT Capabilities Information field of its VHT Capabilities element or in the SIG Capabilities Information field of SIG Capabilities elements that it transmits.

NOTE—An HT STA that does not support +HTC (HT, ~~or~~ VHT, or HE variant) that receives a +HTC frame addressed to another STA still performs the CRC on the actual length of the MPDU and uses the Duration/ID field to update the NAV, as described in 10.3.2.4.

An HE variant HT Control field shall not be present in a frame addressed to a STA, unless that STA declares support for +HTC-HE in the HE MAC Capabilities Information field in the HE Capabilities element. The HE variant HT Control field carried in the frame may contain one or more Control subfields under the conditions listed in Table 10-11a.

Table 10-11a—Conditions for including Control subfield variants

<u>Control subfield variant</u>	<u>Condition</u>
<u>TRS</u>	<u>The transmitting AP expects an HE TB PPDU that follows the TRS information as described in 26.5.2.2 and the recipient non-AP STA has set the TRS Support subfield in the HE MAC Capabilities Information field in the HE Capabilities elements it transmits to 1.</u>
<u>OM</u>	<u>The transmitting STA changes its operating mode, as described in 26.9 and the recipient STA has set the OM Control Support subfield in the HE MAC Capabilities Information field in the HE Capabilities elements it transmits to 1.</u>
<u>HLA</u>	<u>The transmitting STA follows the HE link adaptation procedure, as described in 26.13 and the recipient STA has set the HE Link Adaptation Support subfield in the HE MAC Capabilities Information field in the HE Capabilities elements it transmits to a nonzero value.</u>
<u>BSR</u>	<u>The transmitting non-AP STA follows the corresponding buffer status report procedure, as described in 26.5.5 and the recipient AP has set the BSR Support subfield in the HE MAC Capabilities Information field in the HE Capabilities elements it transmits to 1.</u>
<u>UPH</u>	<u>The transmitting non-AP STA follows the UL MU operation procedure, as described in 26.5.2.3.</u>
<u>BQR</u>	<u>The transmitting non-AP STA follows the bandwidth query report procedure, as described in 26.5.2 and the recipient AP has set the BQR Support subfield in the HE MAC Capabilities Information field in the HE Capabilities elements it transmits to 1.</u>
<u>CAS</u>	<u>The transmitting STA follows either:</u> — <u>The reverse direction protocol procedure described in 10.29 and the recipient STA has set the RD Responder in the HT Extended Capabilities field in the HT Capabilities elements it transmits to 1, or</u> — <u>The PSR procedure described in 26.10.3 and the recipient STA has set the SR Responder subfield of the HE MAC Capabilities Information field in the HE Capabilities elements it transmits to 1.</u>
<u>ONES</u>	<u>The transmitting STA may include a ONES Control subfield in an MPDU that is not carried in an HE TB PPDU (see 26.5.2.4).</u>

If the HT Control field is present in an MPDU aggregated in an A-MPDU, then all MPDUs of the same frame type (i.e., having the same value for the Type subfield of the Frame Control field) aggregated in the same A-MPDU shall contain the same HT Control field. If MSDU(s) received in an MPDU are delivered via an MA-UNITDATA.indication, the drop eligible parameter of the MA-UNITDATA.indication is set to true if the MPDU contains an HT variant HT Control field and the DEI subfield is equal to 1 and set to false otherwise.

An HE STA that transmits a frame containing an A-Control subfield shall include at least one Control subfield in the A-Control subfield, and the included Control subfields shall be those that are supported by the receiving STAs, unless the Control ID subfield is 15.

An HE STA that receives a Control subfield in an A-Control subfield with a Control ID subfield value that is not recognized or not supported by the HE STA shall ignore the Control subfield and the remainder of the A-Control subfield. If more than one Control subfield is present in an A-Control subfield, the Control subfields shall not have the same Control ID value.

An HE STA that receives a ONES Control subfield shall ignore the remainder of the A-Control subfield.

If the HT variant HT Control field is present in an MPDU, the DEI subfield provides information on the drop eligibility of the contents of the received MPDU. When there are insufficient resources in a STA, the STA arbitrarily discards frames in order to recover from the lack of resources. With the information from the DEI subfield, a STA may selectively drop frames with the DEI subfield set to 1 in preference to frames with the DEI subfield set to 0, if resources are insufficient. Note that this might not help in the recovery in all conditions, and the STA might still have to fall back to the arbitrary discard strategy. The mechanisms for determining whether resources are insufficient or when to discard MSDUs or A-MSDUs are beyond the scope of this standard.

If the value of dot11S1GControlFieldOptionImplemented is true, an S1G STA shall set the +HTC-VHT Capable subfield of the S1G Capabilities Information field of the S1G Capabilities element that it transmits to 1.

An S1G shall not use an HT Control field other than a VHT variant HT Control field. An S1G STA shall not use a VHT variant HT Control field for any purpose other than link adaptation (see 10.32.3).

10.9 Control Wrapper operation

Change 10.9 as follows:

A STA supporting the HT Control field that receives a Control Wrapper frame shall process it as though it received a frame of the subtype of the wrapped frame. An HE STA shall not send a Control Wrapper frame to another HE STA.

10.11 A-MSDU operation

Change the third paragraph in 10.11 as follows:

If the A-MSDU Fragmentation Support subfield in the MAC Capabilities Information field in the HE Capabilities element transmitted by the recipient STA is 0, then an A-MSDU shall be carried, without fragmentation, within a single QoS Data frame. If the A-MSDU Fragmentation Support subfield in the HE Capabilities element transmitted by the recipient STA is 1, then an A-MSDU may be fragmented, and each fragment is sent to the recipient in a QoS Data frame.

Change the fifth paragraph in 10.11 as follows:

The channel access rules for a QoS Data frame carrying an A-MSDU (or fragment thereof) are the same as for a Data frame carrying an MSDU (or fragment thereof) of the same TID.

Change the 16th paragraph in 10.11 as follows:

A STA shall support the reception of an A-MSDU, where the A-MSDU is carried in a QoS Data frame with Normal Ack ack policy in the following cases:

- By an HT STA ~~when if~~ the A-MSDU is not aggregated within an A-MPDU
- By a VHT ~~or HE STA when if~~ the A-MSDU is sent as an S-MPDU
- For a CMMG STA if the A-MSDU is not aggregated within an A-MPDU

Insert the following paragraph into 10.11 after the 17th paragraph (“The use of an A-MSDU”):

An HE STA shall not transmit an A-MSDU that is carried in a QoS Data frame for which no block ack agreement exists and that is part of an ack-enabled single-TID A-MPDU, unless the recipient has set the A-MSDU Not Under BA In Ack-Enabled A-MPDU Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element to 1.

Change the now 21st paragraph and NOTE 3 in 10.11 as follows:

The length of an A-MSDU transmitted in a VHT PPDU or HE PPDU is limited by the maximum MPDU size supported by the recipient STA (see 10.12.5).

NOTE 3—An A-MSDU that meets the A-MSDU length limit for transmission in a VHT PPDU or HE PPDU might exceed the A-MSDU length limit for an HT PPDU, in which case it cannot be retransmitted in an HT PPDU.

10.12 A-MPDU operation

10.12.1 A-MPDU contents

Change the third paragraph in 10.12.1 as follows:

~~When~~ If an A-MPDU contains multiple QoS Control fields, then bits 4 of all the QoS Control fields shall have the same value, and bits 8–15 of these QoS Control fields that have the same value in bits 0–3 shall be identical have the same value.

10.12.2 A-MPDU length limit rules

Change 10.12.2 as follows:

A STA indicates in the Maximum A-MPDU Length Exponent field in its HT Capabilities element the maximum A-MPDU length that it can receive in an HT PPDU. A STA indicates in the Maximum A-MPDU Length Exponent field in its VHT Capabilities element the maximum length of the A-MPDU pre-EOF padding that it can receive in a VHT PPDU. A STA indicates in the Maximum A-MPDU Length Exponent field in its S1G Capabilities element the maximum length of the A-MPDU pre-EOF padding that it can receive in an S1G PPDU. A STA indicates in the Maximum A-MPDU Length Exponent field in its DMG Capabilities element the maximum A-MPDU length that it can receive in a DMG PPDU. A STA indicates the maximum length of the A-MPDU pre-EOF padding that it can receive in an HE PPDU in the Maximum A-MPDU Length Exponent field in its HT Capabilities, VHT Capabilities, and HE 6 GHz Band Capabilities elements (if present) and in the Maximum A-MPDU Length Exponent Extension field in its HE Capabilities element.

A VHT STA that sets the Maximum A-MPDU Length Exponent field in its VHT Capabilities element to a value in the range 0 to 3 shall set the Maximum A-MPDU Length Exponent in its HT Capabilities to the same value. A VHT STA that sets the Maximum A-MPDU Length Exponent field in the VHT Capabilities element to a value larger than 3 shall set the Maximum A-MPDU Length Exponent in its HT Capabilities element to 3.

Using the Maximum A-MPDU Length Exponent fields in the HT Capabilities, ~~and VHT Capabilities, HE Capabilities, and HE 6 GHz Band Capabilities elements (if present),~~ the STA establishes at association the maximum length of an A-MPDU pre-EOF padding that can be sent to it. An HT STA shall be capable of receiving A-MPDUs of length up to the value indicated by the Maximum A-MPDU Length Exponent field in its HT Capabilities element. A VHT STA shall be capable of receiving A-MPDUs where the A-MPDU pre-EOF padding length is up to the value indicated by the Maximum A-MPDU Length Exponent field in its VHT Capabilities element. An S1G STA that sets the A-MPDU Supported subfield in the S1G Capabilities element to 1 shall be capable of receiving A-MPDUs where the A-MPDU pre-EOF padding length is up to the value indicated by the Maximum A-MPDU Length Exponent field in its S1G Capabilities element.

An HE STA shall be capable of receiving A-MPDUs where the A-MPDU pre-EOF padding length is up to the value indicated by the Maximum A-MPDU Length Exponent field in its HT Capabilities and VHT Capabilities elements and the Maximum A-MPDU Length Exponent Extension field in its HE Capabilities element in the 2.4 GHz or 5 GHz bands. An HE STA shall be capable of receiving A-MPDUs where the A-MPDU pre-EOF padding length is up to the value indicated by the Maximum A-MPDU Length Exponent Extension field in the HE Capabilities element and the Maximum A-MPDU Length Exponent field in HE 6 GHz Band Capabilities element in the 6 GHz band.

A STA shall not transmit an A-MPDU in an HT PPDU that is longer than the value indicated by the Maximum A-MPDU Length Exponent field in the HT Capabilities element received from the intended receiver. MPDUs in an A-MPDU carried in an HT PPDU shall be limited to a maximum length of 4095 octets. A STA shall not transmit an A-MPDU in a VHT PPDU where the A-MPDU pre-EOF padding length is longer than the value indicated by the Maximum A-MPDU Length Exponent field in the VHT Capabilities element received from the intended receiver. An S1G STA shall not transmit an A-MPDU in an S1G PPDU where the A-MPDU pre-EOF padding length field is longer than the value indicated by the Maximum A-MPDU Length Exponent field in the S1G Capabilities element received from the intended receiver. A STA shall not transmit an A-MPDU in a DMG PPDU that is longer than the value indicated by the Maximum A-MPDU Length Exponent field in the DMG Capabilities element received from the intended receiver.

A STA shall not transmit an A-MPDU in an HE PPDU where the A-MPDU pre-EOF padding length is greater than the value indicated by the Maximum A-MPDU Length Exponent field in the HT Capabilities and VHT Capabilities elements and the Maximum A-MPDU Length Exponent Extension field in its HE Capabilities elements received from the intended receiver in the 2.4 GHz or 5 GHz bands. A STA shall not transmit an A-MPDU in an HE PPDU where the A-MPDU pre-EOF padding length is greater than the value indicated by the Maximum A-MPDU Length Exponent Extension field in the HE Capabilities element and the Maximum A-MPDU Length Exponent field in the HE 6 GHz Band Capabilities element received from the intended receiver in the 6 GHz band.

An S1G STA shall not transmit an A-MPDU, except for an S-MPDU, to an S1G STA from which it received a frame containing an S1G Capability element with the A-MPDU Supported subfield equal to 0.

Change the title and content of 10.12.3 as follows [including inserting Equation (10-11a)]:

10.12.3 Minimum MPDU Start Spacing field rules

If the intended receiver is a non-HE STA, a STA shall not start the transmission of more than one MPDU within the time limit described in the Minimum MPDU Start Spacing field declared by the intended receiver. If the intended receiver is an HE STA, an HE STA shall not start the transmission of more than one QoS Data frame, QoS Null frame, or Management frame within the time limit described in the Minimum MPDU Start Spacing field declared by the intended receiver. To satisfy this requirement, the number of octets between the start of two consecutive MPDUs in an A-MPDU, N , measured at the PHY SAP, shall be equal to or greater than meet the condition defined by Equation (10-11a).

$$N \geq \begin{cases} t_{MMSS} \times r/8, & \text{if the A-MPDU is not carried in an HE TB PPDU} \\ t_{MMSS} \times 2^{MMSF} \times r/8, & \text{if the A-MPDU is carried in an HE TB PPDU} \end{cases} \quad (10-11a)$$

where

t_{MMSS}	is the time (in microseconds) defined in the “Encoding” column of Table 9-185 for an HT STA, of Table 9-300 for an S1G STA for the value of the Minimum MPDU Start Spacing field, and of Table 9-251 for a DMG STA for the value of the Minimum MPDU Start Spacing field
$MMSF$	is the value of the MPDU MU Spacing Factor subfield of the User Info field addressed to the HE STA in the Trigger frame soliciting the HE TB PPDU (see 9.3.1.22)
r	is the value of the PHY Data Rate (in megabits per second) defined in 19.5 for HT PPDUs, in 21.5 for VHT PPDUs, in 23.5 for S1G PPDUs, and in Clause 20 for a DMG STA

If necessary, in order to satisfy this requirement, a STA shall add padding between MPDUs in an A-MPDU. Any such padding shall be in the form of one or more MPDU delimiters with the MPDU Length field set to 0.

QoS Null frames transmitted by DMG STAs are not subject to this spacing, i.e., no MPDU delimiters with zero length need to be inserted after the MPDU immediately preceding the QoS Null frame in an A-MPDU.

10.12.4 A-MPDU aggregation of group addressed Data frames

Change NOTE 2 in 10.12.4 as follows:

NOTE 2—As a VHT STA and an HE STA are is an HT STA, NOTE 1 also applies to VHT APs, and VHT mesh STAs, HE APs, and HE mesh STAs.

Change the fourth paragraph in 10.12.4 as follows:

When a STA transmits a PPDU containing at least one A-MPDU that contains MPDUs with an RA that is a group address, the following shall apply:

- If the PPDU is an HT PPDU ~~transmitted by an AP~~, the maximum A-MPDU length exponent value is the minimum value in the Maximum A-MPDU Length Exponent subfield of the A-MPDU Parameters field of the HT Capabilities elements across ~~of~~ all HT STAs associated with the transmitting AP or across all peer HT mesh STAs.

- If the PPDU is an HT PPDU transmitted by a mesh STA, the maximum A-MPDU length exponent value is the minimum value in the Maximum A-MPDU Length Exponent subfield of the A-MPDU Parameters field of the HT Capabilities element of all peer HT mesh STAs.
- If the PPDU is a VHT PPDU, the ~~value of~~ maximum A-MPDU length exponent ~~value that applies~~ is the minimum value in the Maximum A-MPDU Length Exponent subfields of the ~~A-MPDU Parameters fields of the~~ VHT Capabilities elements across all VHT STAs associated with the transmitting AP or across all peer VHT mesh STAs.
- If the PPDU is an HE PPDU sent in the 2.4 GHz or 5 GHz band, the maximum A-MPDU length exponent value is the minimum value in the Maximum A-MPDU Length Exponent subfield of the VHT Capabilities elements across all HE STAs associated with the transmitting AP or across all peer HE mesh STAs.
- If the PPDU is an HE PPDU sent in the 6 GHz band, the maximum A-MPDU length exponent value is the minimum value in the Maximum A-MPDU Length Exponent subfield of the HE 6 GHz Band Capabilities elements across all HE STAs associated with the transmitting AP or across all peer HE mesh STAs.
- If the PPDU is a VHT PPDU, the ~~value of~~ minimum MPDU start spacing ~~value that applies~~ is the maximum value in the Minimum MPDU Start Spacing subfields of the A-MPDU Parameters fields of the HT Capabilities elements across all VHT STAs associated with the transmitting AP or across all peer VHT mesh STAs ~~of the transmitting mesh STA~~.
- If the PPDU is an HT PPDU ~~transmitted by an AP~~, the minimum MPDU start spacing value ~~that applies~~ is the maximum value in the Minimum MPDU Start Spacing subfield of the A-MPDU Parameters field of the HT Capabilities elements across ~~of~~ all HT STAs associated with the transmitting AP or across all peer HT mesh STAs.
- ~~If the PPDU is an HT PPDU transmitted by a mesh STA, the minimum MPDU start spacing value that applies is the maximum value in the Minimum MPDU Start Spacing subfield of the A-MPDU Parameters field of the HT Capabilities element of all peer HT mesh STAs.~~
- If the PPDU is an HE PPDU sent in the 2.4 GHz or 5 GHz band, the minimum MPDU start spacing value is the maximum value in the Minimum MPDU Start Spacing subfield of the A-MPDU Parameters field of the HT Capabilities elements across all HE STAs associated with the transmitting AP or across all peer HE mesh STAs.
- If the PPDU is an HE PPDU sent in the 6 GHz band, the minimum MPDU start spacing value is the maximum value in the Minimum MPDU Start Spacing subfield of the HE 6 GHz Band Capabilities elements across all HE STAs associated with the transmitting AP or across all peer HE mesh STAs.
- If the PPDU is a DMG PPDU, the maximum A-MPDU length exponent value ~~that applies~~ is the minimum value in the Maximum A-MPDU Length Exponent subfield of the A-MPDU Parameters field of the DMG Capabilities element of all DMG STAs associated with the AP or PCP.
- If the PPDU is a DMG PPDU, the minimum MPDU start spacing value ~~that applies~~ is the maximum value in the Minimum MPDU Start Spacing subfield of the A-MPDU Parameters field of the DMG Capabilities element of all DMG STAs associated with the AP or PCP.
- If the PPDU is an S1G PPDU, the ~~value of~~ maximum A-MPDU length exponent ~~value that applies~~ is the minimum value in the Maximum A-MPDU Length Exponent subfields of the S1G Capabilities Information field of the S1G Capabilities elements across all S1G STAs associated with the transmitting AP.
- If the PPDU is an S1G PPDU, the ~~value of~~ minimum MPDU start spacing ~~value that applies~~ is the maximum value in the Minimum MPDU Start Spacing subfields of the S1G Capabilities Information field of the S1G Capabilities elements across all S1G STAs associated with the transmitting AP.

Change the title of 10.12.6 as follows:

10.12.6 A-MPDU padding for VHT PPDU, HE PPDU, or S1G PPDU

Change the second-to-last paragraph in 10.12.6 as follows:

An A-MPDU subframe with EOF set to 0 shall not be added after any A-MPDU subframe with EOF set to 1, except if the A-MPDU subframe is in an ack-enabled single-TID A-MPDU or ack-enabled multi-TID A-MPDU (see 9.7.3).

Change the title and text of 10.12.7 as follows:

10.12.7 Setting the EOF/Tag field of the MPDU delimiter

The EOF/Tag field may be set to 1 in an A-MPDU subframe carried in a VHT PPDU, HE PPDU, or S1G PPDU if the subframe's MPDU Length field is nonzero and the subframe is the only subframe that has a nonzero MPDU Length field. The EOF/Tag field of each A-MPDU subframe with an MPDU Length field with a nonzero value that is not the only A-MPDU subframe with MPDU Length field with a nonzero value in the A-MPDU carried in a VHT PPDU or S1G PPDU shall be set to 0. The EOF/Tag field shall be set to 0 in all A-MPDU subframes that are carried in an HT PPDU.

An MPDU that is the only MPDU in an A-MPDU and that is carried in an A-MPDU subframe with 1 in the EOF/Tag field is called an *S-MPDU*.

An MPDU that is carried in an A-MPDU subframe with the EOF/Tag field in the MPDU delimiter set to 1 is called a *tagged MPDU*. An S-MPDU is a tagged MPDU, but a tagged MPDU is not necessarily an S-MPDU.

An MPDU that is carried in an A-MPDU subframe with the EOF/Tag field in the MPDU delimiter set to 0 is called an *untagged MPDU*.

10.13 PPDU duration constraint

Insert the following paragraph at the end of 10.13:

An HE STA shall not transmit an HE PPDU that has a duration (as determined by the PHY-TXTIME.confirm primitive defined in 6.5.6) that is greater than *aPPDUMaxTime* defined in Table 27-54.

10.15 Low-density parity check code (LDPC) operation

Insert the following paragraph at the end of 10.15:

An HE STA shall not transmit a frame in an HE PPDU with the TXVECTOR parameter FEC_CODING set to LDPC_CODING, unless the frame is addressed to an HE STA for which the LDPC Coding In Payload subfield in the HE Capabilities element received from that STA contained a value of 1 and dot11HELDPCCodingInPayloadImplemented is true.

10.22 Operation across regulatory domains

10.22.3 Operation with operating classes

Change 10.22.3 as follows:

The following, and only the following, are extended spectrum management capable: a VHT STA, an HE STA, and a STA that has dot11ExtendedSpectrumManagementImplemented true. A non-VHT STA that has dot11ExtendedSpectrumManagementImplemented true shall indicate that it is extended spectrum management capable using the Extended Spectrum Management Capable field of the Extended Capabilities element.

When communicating with a STA that supports global operating classes, all requests and Action and Action No Ack frames that convey elements containing operating classes shall use global operating class values.

When dot11OperatingClassesImplemented is true, the following statements apply:

- When dot11OperatingClassesRequired is false, or where operating classes domain information is not present in a STA, that STA is not required to change its operation in response to an element that contains an operating class.
- When dot11OperatingClassesRequired is true, or where operating classes domain information is present in a STA, the STA shall indicate current operating class information in the Country element and Supported Operating Classes element, except for the following cases:
 - ~~that a VHT STA may omit, from the Country element, any Operating Triplet field for an Operating Class for which the “Channel spacing” column indicates 80 MHz or wider and for which the “Behavior limits set” column in the applicable table in Annex E contains only a blank entry or either or both of “80+” and “UseEirpForVHTTxPowEnv.”~~
 - An HE STA may omit from the Country element any Operating Triplet field for an Operating Class for which the “Channel starting frequency” column in Table E-4 is greater than or equal to 5.925 and less than or equal to 7.125.
- When dot11OperatingClassesRequired and dot11ExtendedChannelSwitchActivated are true and a STA is capable of operating as specified in more than one operating class, the STA shall include the Supported Operating Classes element in (Re)Association Request and Response frames.
- When dot11OperatingClassesRequired is true, or where operating classes domain information is present and the STA parsing a Country element finds an invalid First Channel Number field or Operating Class field with a value that is reserved, the STA shall ignore the remainder of the Country element and shall parse any remaining management frame body for additional elements.
- When dot11OperatingClassesRequired is true and the STA supports one or more global operating classes, or where global operating classes domain information is present in a STA, the STA shall indicate current operating class information in the Country element and Supported Operating Classes element using the country string for the global operating classes, except for the following cases:
 - ~~that a VHT STA may omit from the Country element any Operating Triplet field for an Operating Class for which the “Channel spacing” column indicates 80 MHz or wider and for which the “Behavior limits set” column in the applicable table in Annex E contains only a blank entry or either or both of “80+” and “UseEirpForVHTTxPowEnv.”~~
 - An HE STA may omit from the Country element any Operating Triplet field for an Operating Class for which the “Channel starting frequency” column in Table E-4 is greater than or equal to 5.925 and less than or equal to 7.125.

10.22.4 Operation with the Transmit Power Envelope element

Change 10.22.4 as follows:

A STA that is not operating in the 6 GHz band, is extended spectrum management capable, and ~~that~~ has ~~dot11SpectrumManagementRequired~~ or ~~dot11RadioMeasurementActivated~~ equal to true shall determine a local maximum transmit power from a Transmit Power Envelope element for which the ~~Local-Maximum Transmit Power~~ ~~Unit-Interpretation~~ subfield indicates EIRP.

A STA that is operating in the 6 GHz band shall determine local and regulatory client maximum transmit powers from Transmit Power Envelope element(s) according to local regulations known at the STA (see E.2.7). A STA shall ignore Transmit Power Envelope element(s) indicating transmit power category values that the STA is unable to interpret for the current country.

NOTE 1—The Default category value (0) is applicable to, and so can be interpreted for, all countries (see 11.7.5). An AP in the 6 GHz band has dot11SpectrumManagementRequired equal to true and therefore transmits a Country element in Beacon and Probe Response frames.

A STA that sends two or more Transmit Power Envelope elements in a frame shall order the elements by increasing values of their ~~Local-Maximum Transmit Power~~ ~~Unit-Interpretation~~ subfields.

A STA that is operating in the 6 GHz band that sends two or more Transmit Power Envelope elements in a frame with the same value in the Maximum Transmit Power Interpretation subfield shall order the elements by increasing values of their Maximum Transmit Power Category subfields.

NOTE 2—The Maximum Transmit Power Category subfield is reserved, except in the 6 GHz band.

If a STA that is extended spectrum management capable finds an unknown value in the ~~Local-Maximum Transmit Power~~ ~~Unit-Interpretation~~ subfield in a Transmit Power Envelope element, then the STA shall ignore that and subsequent Transmit Power Envelope elements.

A STA that receives two or more Transmit Power Envelope elements in the same frame with known values in their ~~Local-Maximum Transmit Power~~ ~~Unit-Interpretation~~ subfields shall process all of the elements according to the local regulations known at the STA.

NOTE 3—If a STA receives two Transmit Power Envelope elements, each with a known value in the ~~Local-Maximum Transmit Power~~ ~~Unit-Interpretation~~ subfield, then the expected possibilities are as follows:

- a) The STA complies with either element (shared spectrum),
- b) The STA complies with both elements (tightened regulations), or
- c) The STA complies with the second element (changed regulations).

If a STA receives a Transmit Power Envelope element with the Maximum Transmit Power Interpretation subfield equal to 1 or 3 (EIRP PSD) and the Maximum Transmit Power Count subfield indicating a value of N greater than 1, 2, 4, or 8 from an AP that the STA identifies to have BSS bandwidth of 20, 40, 80, or 160/80+80 MHz, respectively, then the STA shall use the Maximum Transmit PSD 1– M subfields (M equal to 1, 2, 4, or 8 for BSS bandwidth of 20, 40, 80, or 160/80+80 MHz, respectively) to determine the maximum transmit PSD for each 20 MHz channel within the BSS bandwidth. The STA shall ignore the Maximum Transmit PSD X subfields with $X > M$.

NOTE 4—This might occur when the AP supports PHY mode(s) unknown to the STA and the actual BSS bandwidth is wider than the BSS bandwidth recognized by the STA.

10.23 HCF

10.23.1 General

Change the third paragraph in 10.23.1 as follows:

HCCA is not used by ~~either DMG₂ or S1G₂~~ and HE STAs.

10.23.2 HCF contention based channel access (EDCA)

10.23.2.2 EDCA backoff procedure

Change the second paragraph in 10.23.2.2 as follows:

For the purposes of this subclause, transmission failure of an MPDU is defined as follows:

- After transmitting an MPDU (even if it is carried in an A-MPDU, ~~or as part of a VHT or S1G MU PPDU,~~ or as part of an HE MU PPDU that is sent using TXVECTOR parameter NUM_USERS > 1) that requires an immediate response:
 - The STA shall wait for a timeout interval of duration aSIFSTime + aSlotTime + aRxPHYStartDelay, starting when the MAC receives a PHY-TXEND.confirm primitive. If a PHY-RXSTART.indication primitive does not occur during the timeout interval, the transmission of the MPDU has failed.
 - If a PHY-RXSTART.indication primitive does occur during the timeout interval, the STA shall wait for the corresponding PHY-RXEND.indication primitive to recognize a valid response MPDU (see Annex G) that either does not have a TA field or is sent by the recipient of the MPDU requiring a response. If anything else, including any other valid frame, is recognized, the transmission of the MPDU has failed.
- The nonfinal (re)transmission of an MPDU that is delivered using the GCR unsolicited retry retransmission policy (10.23.2.12.2) is defined to be a failure.
- In all other cases, the transmission of the MPDU has not failed.

Change the fourth and fifth paragraphs in 10.23.2.2 as follows:

The backoff procedure shall be invoked by an EDCAF ~~when~~ if any of the following events occurs:

- a) An MA-UNITDATA.request primitive is received that causes an MPDU corresponding to the EDCAF's AC to be queued for transmission such that all of the following are true:
 - 1) One of the transmit queues associated with that AC has now become non-empty.
 - 2) Any other transmit queues associated with that AC are empty.
 - 3) The backoff counter has a value of 0 for that AC.
 - 4) The medium is busy on the primary channel as indicated by any of the following:
 - Physical CS
 - Virtual CS
 - A nonzero TXNAV timer value
 - For a mesh STA that has dot11MCCAActivated true, a nonzero RAV timer value
- b) For the EDCAF that is the TXOP holder, the transmission of the final PPDU transmitted by the TXOP holder during the TXOP has completed, the final PPDU does not solicit an HE TB PPDU, and the TXNAV timer has expired.
- c) For the EDCAF that is the TXOP holder, the transmission of an MPDU in the initial PPDU of a TXOP fails, as defined in this subclause, and the initial PPDU does not solicit an HE TB PPDU.

- d) A transmission attempt by the EDCAF collides internally with another EDCAF of an AC that has higher priority, that is, two or more EDCAFs in the same STA are granted a TXOP at the same time.
- e) The transmission of at least one MPDU in the final PPDU transmitted by the TXOP holder during the TXOP for that AC has completed, the PPDU contains an MPDU that solicits an HE TB PPDU and the TXNAV timer has expired.
- f) The transmission of all MPDUs in the initial PPDU of a TXOP fails, as defined in this subclause, and the PPDU contains an MPDU that solicits an HE TB PPDU.
- g) If explicitly indicated, such as in 26.17.2.3.3.

In addition, the backoff procedure may be invoked by an EDCAF ~~when~~ if

- h) ~~e)~~ For the EDCAF that is the TXOP holder, the transmission by the TXOP holder of an MPDU in a non-initial PPDU of a TXOP fails, as defined in this subclause, and an MPDU in the non-initial PPDU does not solicit an HE TB PPDU.
- i) For the EDCAF that is the TXOP holder, the transmission by the TXOP holder of all MPDUs in a non-initial PPDU of a TXOP fails, as defined in this subclause, and the PPDU contains an MPDU that solicits an HE TB PPDU.

Change the seventh, eighth, and ninth paragraphs in 10.23.2.2 as follows (including combining the seventh and eighth paragraphs into one paragraph):

If the backoff procedure is invoked for reason a) above, CW[AC] and QSRC[AC] shall be left unchanged. If the backoff procedure is invoked for reason b) or f) above, CW[AC] shall be reset to CWmin[AC], and QSRC[AC] shall be set to 0.

If the backoff procedure is invoked for reason c), d), ~~or e)~~, g), h), or i) above, CW[AC] and QSRC[AC] shall be updated as follows:

- If QSRC[AC] is less than dot11ShortRetryLimit,
 - QSRC[AC] shall be incremented by 1.
 - CW[AC] shall be set to the lesser of CWmax[AC] and $2^{QSRC[AC]} \times (CWmin[AC] + 1) - 1$.
- Else
 - QSRC[AC] shall be set to 0.
 - CW[AC] shall be set to CWmin[AC].

NOTE 3—An HE STA updates its local MIB variables related to CWmin and CWmax as defined in 26.2.7.

Insert the following paragraph at the end of 10.23.2.2:

After transmission of an MPDU in an HE TB PPDU, an HE STA resumes the EDCA backoff procedure without modifying CW or the backoff counter for the associated EDCAF regardless of whether the STA has received the corresponding acknowledgment frame in response to the MPDU sent in the HE TB PPDU.

10.23.2.4 Obtaining an EDCA TXOP

Change the 11th paragraph of 10.23.2.4 as follows:

A STA shall save the TXOP holder address for the BSS in which it is associated, ~~which~~. The TXOP holder address is the MAC address from the Address 2 field of the frame that initiated a frame exchange sequence, except ~~when~~ if this is a CTS frame, in which case the TXOP holder address is the Address 1 field. If the TXOP holder address is obtained from a Control frame, a VHT STA or HE STA shall save the

nonbandwidth signaling TA value obtained from the Address 2 field. If a non-VHT non-HE STA receives an RTS frame with the RA matching the MAC address of the STA and the MAC address in the TA field in the RTS frame matches the saved TXOP holder address, then the STA shall send the CTS frame after SIFS without regard for, and without resetting, its NAV. If a VHT STA or HE STA receives an RTS frame with the RA matching the MAC address of the STA and the nonbandwidth signaling TA value obtained from the Address 2 field in the RTS frame matches the saved TXOP holder address, then the STA shall send the CTS frame after SIFS without regard for, and without resetting, its NAV. If a CMMG STA receives an RTS frame with the RA matching the MAC address of the STA and the TA value obtained from the Address 2 field in the RTS frame matches the saved TXOP holder address, then the STA shall send the CTS frame after SIFS without regard for, and without resetting, its NAV. When a STA receives a frame addressed to it that requires an immediate response, except for RTS and Trigger frames (see 26.5.2.5), it shall transmit the response independent of its NAV. The saved TXOP holder address shall be cleared when the NAV is reset or when the NAV counts down to 0.

Insert the following paragraph at the end of 10.23.2.4:

During an EDCA TXOP, the Address 2 field excluding the Individual/Group bit of all Control frames sent by an HE STA that is a TXOP holder and carried in a PPDU that is not an HE MU PPDU shall be set to the same address value.

Change the title of 10.23.2.5 as follows:

10.23.2.5 EDCA channel access in a VHT, HE, or TVHT BSS

Insert the following paragraph into 10.23.2.5 after the first paragraph (“If the MAC receives”):

If the MAC receives a PHY-CCA.indication primitive with the per20bitmap parameter present, the parameter indicates the busy/idle status of each of the 20 MHz subchannels that comprise the operating channel width.

Change the now fifth paragraph and the notes in 10.23.2.5 as follows:

If a STA is permitted to begin a TXOP (as defined in 10.23.2.4) and the STA has at least one MSDU pending for transmission for the AC of the permitted TXOP, the STA shall perform exactly one of the following actions:

- a) Transmit a 160 MHz or 80+80 MHz mask PPDU if the secondary channel, the secondary 40 MHz channel, and the secondary 80 MHz channel were idle during an interval of PIFS immediately preceding the start of the TXOP.
- b) Transmit an 80 MHz mask PPDU on the primary 80 MHz channel if both the secondary channel and the secondary 40 MHz channel were idle during an interval of PIFS immediately preceding the start of the TXOP.
- c) Transmit a 40 MHz mask PPDU on the primary 40 MHz channel if the secondary channel was idle during an interval of duration 1) DIFS if the PPDU is transmitted in the 2.4 GHz band or 2) PIFS otherwise, immediately preceding the start of the TXOP.
- d) Transmit a 20 MHz mask PPDU on the primary 20 MHz channel.
- e) Restart the channel access attempt by invoking the backoff procedure as specified in 10.23.2 as though the medium is busy on the primary channel as indicated by either physical or virtual CS and the backoff counter has a value of 0.

- f) Transmit a TVHT_4W or TVHT_2W+2W mask PPDU if the secondary TVHT_W channel and the secondary TVHT_2W channel were idle during an interval of PIFS immediately preceding the start of the TXOP.
- g) Transmit a TVHT_2W or TVHT_W+W mask PPDU if the secondary TVHT_W channel was idle during an interval of PIFS immediately preceding the start of the TXOP.
- h) Transmit a TVHT_W mask PPDU on the primary TVHT_W channel.
- i) Transmit an 80 MHz HE MU PPDU where in the preamble the only punctured subchannel is the secondary 20 MHz channel, if all of the 20 MHz subchannels (other than the primary 20 MHz channel) that are not punctured were idle during an interval of PIFS immediately preceding the start of the TXOP.
- j) Transmit an 80 MHz HE MU PPDU where in the preamble the only punctured subchannel is one of the two 20 MHz subchannels in the secondary 40 MHz channel, if all of the 20 MHz subchannels that are not punctured were idle during an interval of PIFS immediately preceding the start of the TXOP.
- k) Transmit a 160 MHz or 80+80 MHz HE MU PPDU where in the preamble the only punctured subchannels are the secondary 20 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel, if all of the 20 MHz subchannels that are not punctured were idle during an interval of PIFS immediately preceding the start of the TXOP. If two of the 20 MHz subchannels in the secondary 80 MHz channel are punctured, these are either the lower two or the higher two. No more than two adjacent 20 MHz subchannels are punctured across the preamble, for a 160 MHz preamble.
- l) Transmit a 160 MHz or 80+80 MHz HE MU PPDU where in the preamble the only punctured subchannels are zero, one, or both of the 20 MHz subchannels in the secondary 40 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel, if all of the 20 MHz subchannels that are not punctured were idle during an interval of PIFS immediately preceding the start of the TXOP. At least one 20 MHz subchannel is punctured. If two of the 20 MHz subchannels in the secondary 80 MHz channel are punctured, these are either the lower two or the higher two. No more than two adjacent 20 MHz subchannels are punctured across the preamble, for a 160 MHz preamble.

NOTE 1—In the case of rule e), the STA selects a new random number using the current value of CW[AC], and the retry counts are not updated [as described in 10.23.2.8; backoff procedure invoked for event a)].

NOTE 2—~~For both an HT and a VHT STA, an EDCA TXOP is obtained.~~ An HT, VHT, or HE STA obtains an EDCA TXOP based on activity on the primary channel (see 10.23.2.4). The width of transmission is determined by the CCA status of the nonprimary channels during ~~the an interval of duration 1) DIFS if transmitting in the 2.4 GHz band or 2) PIFS otherwise,~~ before transmission (see ~~VHT description in 10.3.2).~~

NOTE 3—In the case of rule j), there is only one idle 20 MHz subchannel in the secondary 40 MHz channel, and the other 20 MHz subchannel in the secondary 40 MHz channel is preamble punctured.

10.23.2.7 Sharing an EDCA TXOP

Change the text of 10.23.2.7 as follows (Figure 10-26 and Figure 10-27 remain unchanged):

The AC associated with the EDCAF that gains an EDCA TXOP is referred to as the *primary AC*. Frames from ACs other than the primary AC shall not be included in the TXOP, with the following exceptions (TXOP sharing):

- For a multi-TID A-MPDU in an HE MU PPDU transmitted by an AP, frames from any AC may be included as defined in 26.6.3. Otherwise, fFrames from a higher priority AC may be included when at least one frame from the primary AC has been transmitted and all frames from the primary AC have been transmitted.

NOTE 1—The frames from a higher priority AC might be included in successive PPDU in the TXOP and/or in one or more MU PPDU.

- When an VHT AP supports DL-MU-MIMO, frames from a lower priority AC may be included in an VHT MU PPDU with the TXVECTOR parameter NUM_USERS > 1 when these frames do not increase the duration of the VHT MU PPDU beyond that required for the transmissions of the frames of the primary AC and any frames from a high priority AC. For a given user, any frames from the primary AC shall be transmitted first, and then any frames from a higher priority AC immediately next.
- When an HE AP transmits an HE MU PPDU, frames from higher or lower priority AC may be included in the HE MU PPDU when these frames do not increase the duration of the HE MU PPDU beyond that required for the transmissions of the frames of the primary AC and any frames from a high priority AC. For a given user, any frames from the primary AC shall be transmitted first, and then any frames from a higher priority AC immediately next.

The EDCAF remains bound by the TXOP limit for its AC (i.e., the primary AC), irrespective of the AC(s) of the frames transmitted during the TXOP.

When sharing, the TXOP limit that applies is the TXOP limit of the primary AC.

With respect to admission control (see 10.23.4.2), all frames transmitted under TXOP sharing shall be treated as if they were from the primary AC.

NOTE 2—An AP can protect an immediate response by preceding the DL-MU-MIMO PPDU (which might have TXVECTOR parameter NUM_USERS > 1) with an RTS/CTS, MU-RTS Trigger/CTS frame exchange, or a CTS-to-self transmission.

An illustration of TXOP sharing is shown in Figure 10-26 and Figure 10-27. The AP has frames in queues of three of its ACs. The TXOP was obtained by AC_VI or AC_BE, as shown, and is shared by the other two ACs. The frames target three STAs, STA-1 to STA-3.

10.23.2.8 Multiple frame transmission in an EDCA TXOP

Change the first paragraph in 10.23.2.8 as follows:

A frame exchange, in the context of multiple frame transmission in an EDCA TXOP, may be one of the following:

- A frame not requiring immediate acknowledgment (such as a group addressed frame or a frame transmitted with an ack policy that does not require immediate acknowledgment) or an A-MPDU containing only such frames.
- A frame requiring immediate acknowledgment (such as an individually addressed frame transmitted with an ack policy that requires immediate acknowledgment) or an A-MPDU containing at least one such frame, followed after SIFS by a corresponding acknowledgment frame.
- A triggering frame or an A-MPDU containing at least one such frame, followed after SIFS by an HE TB PPDU where the HE TB PPDU is optionally followed after SIFS by an acknowledgment.
- One of the following: ~~Either~~
 - A VHT NDP Announcement frame followed after SIFS by a VHT NDP followed after SIFS by an A-MPDU containing one or more VHT Compressed Beamforming frames, ~~or~~
 - A Beamforming Report Poll frame followed after SIFS by an A-MPDU containing one or more VHT Compressed Beamforming frames.
 - An HE NDP Announcement frame followed after SIFS by an HE sounding NDP followed after SIFS by a PPDU containing one or more HE Compressed Beamforming/CQI frames.

- A broadcast HE NDP Announcement frame followed after SIFS by an HE sounding NDP followed after SIFS by a BFRP Trigger frame followed by HE TB PPDU.
- A BFRP Trigger frame followed after SIFS by an HE TB PPDU containing one or more HE Compressed Beamforming/CQI frames.

Change the sixth paragraph in 10.23.2.8 as follows:

If a TXOP is protected by an RTS or CTS frame carried in a non-HT or a non-HT duplicate PPDU, the TXOP holder shall set the TXVECTOR parameter CH_BANDWIDTH of a PPDU as follows:

- To be the same or narrower than the RXVECTOR parameter CH_BANDWIDTH_IN_NON_HT of the last received CTS frame in the same TXOP, if the RTS frame with a bandwidth signaling TA and TXVECTOR parameter DYN_BANDWIDTH_IN_NON_HT set to Dynamic has been sent by the TXOP holder in the last RTS/CTS exchange.
- Otherwise, to be the same or narrower than the TXVECTOR parameter CH_BANDWIDTH of the RTS frame that has been sent by the TXOP holder in the last RTS/CTS exchange in the same TXOP.

Insert the following paragraph into 10.23.2.8 after the sixth paragraph:

If a TXOP is protected by an MU-RTS Trigger frame or CTS frame carried in a non-HT or a non-HT duplicate PPDU, the TXOP holder shall set the TXVECTOR parameter CH_BANDWIDTH of a PPDU as follows:

- To be the same or narrower than the TXVECTOR parameter CH_BANDWIDTH of the MU-RTS Trigger frame that has been sent by the TXOP holder in the last MU-RTS Trigger/CTS frame exchange in the same TXOP, if the RU Allocation subfields of the MU-RTS Trigger frame for all intended receivers are equal to a value that corresponds to the channel bandwidth that is indicated in the UL BW subfield in the Common Info field of the MU-RTS Trigger frame.
- Otherwise, to be the same or narrower than the TXVECTOR parameter CH_BANDWIDTH of the preceding PPDU that it has transmitted in the same TXOP.

Change the now eighth and ninth paragraphs in 10.23.2.8 as follows:

If there is no RTS/CTS or MU-RTS Trigger/CTS frame exchange in non-HT duplicate format in a TXOP, and the TXOP includes at least one non-HT duplicate frame exchange that does not include a PS-Poll, then the TXOP holder shall set the TXVECTOR parameter CH_BANDWIDTH of a PPDU sent after the first non-HT duplicate frame that is not a PS-Poll to be the same or narrower than the TXVECTOR parameter CH_BANDWIDTH of the initial frame in the first non-HT duplicate frame exchange in the same TXOP.

If there is no non-HT duplicate frame in a TXOP, the TXOP holder shall set the TXVECTOR parameter CH_BANDWIDTH of a non-initial PPDU to be the same or narrower than the TXVECTOR parameter CH_BANDWIDTH of the preceding PPDU that it has transmitted in the same TXOP, subject to the following constraints:

- If the preceding PPDU is a DL HE MU PPDU with preamble puncture, the TXOP holder shall set the TXVECTOR parameter CH_BANDWIDTH of the non-initial PPDU to a value whose corresponding 20 MHz channels are within a set of 20 MHz channels where pre-HE modulated fields of the preceding PPDU are located.
- If the non-initial PPDU is a DL HE MU PPDU with preamble puncture, the TXOP holder shall set the TXVECTOR parameter RU_ALLOCATION of the non-initial PPDU to a value whose corresponding RU is within a set of 20 MHz channels where pre-HE modulated fields of the preceding PPDU are located.

10.23.2.9 TXOP limits

Change 10.23.2.9 as follows:

The duration of a TXOP is the time a STA obtaining a TXOP (the TXOP holder) maintains uninterrupted control of the medium, and it includes the time required to transmit frames sent as an immediate response to TXOP holder transmissions. The TXOP holder shall, subject to the exceptions below, ensure that the duration of a TXOP does not exceed the TXOP limit, when nonzero.

The TXOP limits are advertised by the AP in the EDCA Parameter Set element in Beacon and Probe Response frames transmitted by the AP.

A TXOP limit of 0 indicates that the TXOP holder may transmit or cause to be transmitted (as responses) the following within the current TXOP:

- a) One of the following at any rate, subject to the rules in 10.6:
 - 1) One or more SU PPDUs carrying fragments of a single MSDU or MMPDU
 - 2) An SU PPDU, ~~or a VHT MU PPDU, an HE MU PPDU, or an HE TB PPDU~~ carrying a single MSDU, a single MMPDU, a single A-MSDU, or a single A-MPDU
 - 3) A VHT MU PPDU or an HE MU PPDU carrying A-MPDUs to different users (a single A-MPDU to each user)
 - 4) A QoS Null frame or PS-Poll frame that is not an PS-Poll+BDT frame
 - 5) A Basic Trigger frame, BSRP Trigger frame, or BQRP Trigger frame
 - 6) An HE TB PPDU carrying A-MPDUs from different users (a single A-MPDU from each user)
- b) Any required acknowledgments
- c) Any frames required for protection, including one of the following:
 - 1) An RTS/CTS or MU-RTS Trigger/CTS frame exchange
 - 2) CTS-to-self
 - 3) Dual CTS as specified in 10.3.2.10
- d) Any frames required for beamforming as specified in 10.31, 10.36.5, 26.7, and 10.42
- e) Any frames required for link adaptation as specified in 10.32 and 26.13
- f) Any number of BlockAckReq, MU-BAR Trigger, Multi-TID BlockAckReq, or GCR MU-BAR Trigger frames

NOTE 1—This is a rule for the TXOP holder. A TXOP responder need not be aware of the TXOP limit nor of when the TXOP was started.

NOTE 2—This rule prevents the use of RD, BDT, and TXOP sharing when the TXOP limit is 0.

When dot11OCBAActivated is true, TXOP limits shall be 0 for each AC.

The TXOP holder may exceed the TXOP limit only if it does not transmit more than one Data or Management frame in the TXOP, only if it does not transmit a DL-MU-MIMO PPDU in the TXOP, and only for the following situations:

- Retransmission of an MPDU, not in an A-MPDU consisting of more than one MPDU, where the size of the retransmitted MPDU is the same as the initially transmitted MPDU.
- Transmission of an MSDU or MMPDU less than 600 octets by an S1G non-sensor STA.
- Transmission of a fragment of an MSDU or MMPDU, the fragment being less than 256 octets, by an S1G non-sensor STA.
- Initial transmission of an MSDU under a block ack agreement, where the MSDU is not in an A-MPDU consisting of more than one MPDU and the MSDU is not in an A-MSDU.

- Transmission of a Control frame or a QoS Null frame, not in an A-MPDU consisting of more than one MPDU.
- Initial transmission of a non-dynamic fragment of an MSDU or MMPDU (see 10.4), if a previous fragment of that MSDU or MMPDU was retransmitted.
- Transmission of a non-dynamic fragment of an MSDU or MMPDU fragmented into 16 fragments.
- Transmission of the 16th dynamic fragment of an MSDU or MMPDU.
- Initial transmission of the first dynamic fragment of an MSDU or MMPDU, where the size of the first fragment is equal to the minimum fragment size specified by the receiver STA and the MSDU or MMPDU is not in an A-MPDU consisting of more than one MPDU.
- Transmission of an A-MPDU consisting of the initial transmission of a single MPDU not containing an MSDU and that is not an individually addressed Management frame.
- Transmission of a group addressed MPDU, not in an A-MPDU consisting of more than one MPDU.
- Transmission of a null data PPDU (NDP).
- Transmission of a VHT NDP Announcement frame and NDP or transmission of a Beamforming Report Poll frame, where these fit within the TXOP limit and ~~it is only~~ the response and the immediately preceding SIFS ~~that~~ cause the TXOP limit to be exceeded.
- Transmission of one of the following sequences, provided that the sequence fits within the TXOP limit and that only the response and the immediately preceding SIFS cause the TXOP limit to be exceeded:
 - An HE NDP Announcement frame and HE sounding NDP
 - An HE NDP Announcement frame and HE sounding NDP and BFRP Trigger frame
 - A BFRP Trigger frame

Except as described above, a STA shall fragment an individually addressed MSDU or MMPDU so that the initial transmission of the first fragment does not cause the TXOP limit to be exceeded.

NOTE 3—The TXOP limit is not exceeded for the following situations:

- a) Initial transmission of an MPDU containing an unfragmented though fragmentable (see 10.2.6 and 26.3) MSDU/MMPDU.
- b) Initial transmission of the first fragment of a fragmented MSDU/MMPDU, ~~except for an if the~~ MSDU/MMPDU is fragmented into 16 fragments.
- c) Initial transmission of an A-MSDU.
- d) Initial transmission of a non-dynamic fragment of a fragmented MSDU/MMPDU, if no previous fragment of that MSDU/MMPDU was retransmitted, ~~except for an if the~~ MSDU/MMPDU is fragmented into 16 fragments.
- e) Initial transmission of a dynamic fragment of a fragmented MSDU/MMPDU, except for either the first dynamic fragment of a fragmented MSDU/MMPDU using the minimum fragment size specified by the receiver STA, or the 16th dynamic fragment of a fragmented MSDU/MMPDU.
- f) ~~⇒~~ Transmission of an A-MPDU consisting of a single MPDU containing an A MSDU or individually addressed Management frame, unless this is a retransmission of that MPDU.
- g) ~~⇒~~ Transmission of an A-MPDU consisting of more than one MPDU, even if some or all of the MPDUs are retransmissions.
- h) Transmission of a Trigger frame, other than a BFRP Trigger frame, where either the Trigger frame or its response does not fit within the TXOP limit.

If the TXOP holder exceeds the TXOP limit, it should use as high a PHY rate as possible to minimize the duration of the TXOP.

The duration of a TXOP for a mesh STA that has dot11MCCAActivated true shall not exceed the time between the start of the TXOP and the end of the current MCCAOP reservation.

NOTE 4—The rules in this subclause also apply to priority-downgraded MSDUs and A-MSDUs (see 10.23.4.2).

If the Duration field in a frame carried in an HE TB PPDU is set to 0, the HE TB PPDU shall not include any frames that solicit a control response frame from the AP.

10.23.2.10 Truncation of TXOP

Change the sixth paragraph in 10.23.2.10 as follows:

In a non-DMG BSS, a non-S1G non-HE STA shall interpret the reception of a CF-End frame as a NAV reset, i.e., it resets its NAV to 0 at the end of the PPDU containing this frame. After receiving a CF-End frame with a matching BSSID(TA) without comparing Individual/Group bit, an AP may respond by transmitting a CF-End frame after SIFS. An HE STA interprets the reception of a CF-End frame as defined in 26.2.5.

Change the ninth paragraph in 10.23.2.10 as follows:

Non-DMG non-S1G non-HE STAs ~~that are not S1G STAs~~ that receive a CF-End frame reset their NAV and can start contending for the medium without further delay. An HE STA that receives a CF-End frame may start contending for the medium without further delay as defined in 26.2.5. A DMG STA that receives a CF-End frame can start contending for the medium at the end of the time interval equal to the value in the Duration/ID field of the frame if none of its NAVs has a nonzero value (10.39.10).

10.23.2.11 Termination of TXOP

Insert the following row at the end of Table 10-19:

Table 10-19—Modulation classes eligible for TXOP termination

Modulation classes eligible for TXOP termination (see Table 10-9)
HE

Change Table 10-20 as follows:

Table 10-20—Rate and modulation class of a final transmission in a TXOP

Modulation class and data rate of immediately preceding frame in TXOP	Rate and modulation class of final transmission
DSSS or HR/DSSS with long preamble, data rate > 1 Mb/s	1 Mb/s DSSS
HR/DSSS with short preamble, data rate > 2 Mb/s	2 Mb/s HR/DSSS short preamble
Other eligible modulation classes, data rate > 6 Mb/s <u>except 6 Mb/s OFDM</u>	6 Mb/s OFDM

10.23.2.12 Retransmit procedures

10.23.2.12.1 General

Insert the following paragraph at the end of 10.23.2.12.1:

If an HE STA transmits an HE TB PPDU, the QSRC[AC] for the associated EDCAF is not changed.

10.23.3 HCF controlled channel access (HCCA)

10.23.3.5 HCCA transfer rules

10.23.3.5.3 Use of RTS/CTS

Change the first paragraph in 10.23.3.5.3 as follows:

In order to provide improved NAV protection, a STA may send an RTS frame as the first frame of any frame exchange sequence without regard for dot11RTSThreshold or dot11TXOPDurationRTSThreshold.

10.25 Block acknowledgment (block ack)

10.25.2 Setup and modification of the block ack parameters

Change the 11th paragraph in 10.25.2 as follows (including splitting the paragraph into two paragraphs):

When a block ack agreement is established between two HT STAs, two DMG STAs, or two SIG STAs, the originator may change the size of its transmission window if the value in the Buffer Size field of the ADDBA Response frame is larger than the value in the ADDBA Request frame, subject to the following conditions:

- The transmission window is not greater than the value in the Buffer Size field of the ADDBA Response frame.
- The transmission window is not greater than 64 if the sender or receiver of the ADDBA Response frame is a non-HE STA.
- The transmission window is not greater than 256 if the sender and receiver of the ADDBA Response frame are HE STAs.

If the value in the Buffer Size field of the ADDBA Response frame is smaller than the value in the ADDBA Request frame, the originator shall change the size of its transmission window (*WinSize_O*) so that it meets the following conditions:

- Is not greater than the value in the Buffer Size field of the ADDBA Response frame. ~~and~~
- Is not greater than ~~the value~~ 64 if the sender or receiver of the ADDBA Response frame is a non-HE STA.
- Is not greater than 256 if the sender and receiver of the ADDBA Response frame are HE STAs.

10.25.5 Selection of BlockAck and BlockAckReq variants

Change the beginning of 10.25.5 up to the now sixth paragraph (“The SIG recipient ... NDP ADDBA”) as follows:

~~The Compressed Bitmap subfield of the BA Control field or BAR Control field shall be set to 1 in all BlockAck and BlockAckReq frames sent from one HT STA to another HT STA and shall be set to 0 otherwise.~~

The Multi-TID BlockAck variant shall be used for all BlockAck frames related to an HT-immediate agreement transmitted inside a PSMP sequence and shall not be used otherwise. For non-HE STAs, the Multi-TID BlockAckReq variant shall be used for all BlockAckReq frames related to an HT-immediate agreement transmitted inside a PSMP sequence and shall not be used otherwise. The Multi-TID BlockAckReq variant can be used between HE STAs to solicit a Multi-STA BlockAck frame for Multi-TID A-MPDUs. ~~subfield of the BA Control field shall be set to 1 in all BlockAck frames related to an HT-~~

~~immediate agreement transmitted inside a PSMP sequence and shall be set to 0 otherwise. The Multi-TID subfield of the BAR Control field shall be set to 1 in all BlockAckReq frames related to an HT-immediate agreement transmitted inside a PSMP sequence and shall be set to 0 otherwise.~~

~~In a DMG BSS, if the Compressed BlockAckReq variant is used related to an HT-immediate agreement, then all of the following BlockAck and BlockAckReq frames transmitted as part of the HT-immediate agreement shall use the Compressed BlockAck and Compressed BlockAckReq variants. Bitmap subfield of the BAR Control field within a BlockAckReq frame related to an HT-immediate agreement is equal to 1, then all of the following BlockAck and BlockAckReq frames transmitted as part of the HT-immediate agreement shall have the Compressed Bitmap subfield of the BA Control and BAR Control fields set to 1. In this case, the Multi-TID subfield of the BA Control field and BAR Control field shall be set to 0 in all BlockAck and BlockAckReq frames transmitted as part of the HT-immediate agreement.~~

~~In a DMG BSS, if the Extended Compressed BlockAckReq variant is used related to an HT-immediate agreement, then all of the following BlockAck and BlockAckReq frames transmitted as part of the HT-immediate agreement shall use the Extended Compressed BlockAck and Extended Compressed BlockAckReq variants. Compressed Bitmap subfield of the BAR Control field within a BlockAckReq frame related to an HT-immediate agreement is equal to 0, then all of the following BlockAck and BlockAckReq frames transmitted as part of the HT-immediate agreement shall have the Compressed Bitmap subfield of the BA Control and BAR Control fields set to 0. In this case, the Multi-TID subfield of the BA Control field and BAR Control field shall be set to 1 in all BlockAck and BlockAckReq frames transmitted as part of the HT-immediate agreement.~~

Where the terms *BlockAck* and *BlockAckReq* are used within 10.25.6, the appropriate variant according to this subclause (e.g., Compressed, Multi-TID) is referenced by the generic term. The term *BlockAck* as used in 10.25.6 includes the additional NDP_1M BlockAck, NDP_2M BlockAck, BAT, and BlockAck frame variants.

GCR BlockAck and GCR BlockAckReq variants shall be used in a GCR block ack agreement. GLK-GCR BlockAck and GLK-GCR BlockAckReq variants shall be used in a GLK-GCR block ack agreement.

The GCR Mode subfield of the BAR Control field (see Table 9-27) shall be set to

- GCR BlockAck: in all BlockAckReq frames within a GCR block ack agreement, or
- GLK-GCR BlockAck: in all BlockAckReq frames within a GLK-GCR block ack agreement.

The GCR Mode subfield in the corresponding BA Control field (see Table 9-28) shall be set to

- GCR BlockAck: in all BlockAck frames within a GCR block ack agreement, or
- GLK-GCR BlockAck: in all BlockAck frames within a GLK-GCR block ack agreement.

10.25.6 HT-immediate block ack extensions

10.25.6.1 Introduction to HT-immediate block ack extensions

Insert the following paragraph at the end of 10.25.6.1:

If the B0 of Fragment Number subfield of a Compressed BlockAck frame or Multi-STA BlockAck frame is equal to 0, then *BitmapLength* represents the maximum length, in bits, of the Block Ack Bitmap subfield in the Compressed BlockAck frame and Multi-STA BlockAck frame for a particular TID; otherwise, *BitmapLength* is derived by dividing by 4 the length represented in the Block Ack Bitmap subfield in the Compressed BlockAck frame and Multi-STA BlockAck frame for a particular TID. For a non-HE STA, *BitmapLength* is 64. For an HE STA, *BitmapLength* is negotiated when the block ack agreement is established as defined in 26.4.3.

10.25.6.3 Scoreboard context control during full-state operation

Change the first paragraph in 10.25.6.3 as follows (the list of rules remains unchanged):

For each HT-immediate block ack agreement that uses full-state operation, a recipient shall maintain a block acknowledgment record. This record includes a bitmap, indexed by sequence number; a 12-bit unsigned integer starting sequence number, $WinStart_R$, representing the lowest sequence number position in the bitmap; $WinEnd_R$, representing the highest sequence number in the current transmission window; and the maximum transmission window size, $WinSize_R$, which is set to the smaller of ~~64~~ $BitmapLength$ and the value of the Buffer Size field of the associated ADDBA Response frame that established the block ack agreement. A STA implementing full-state operation for an HT-immediate block ack agreement shall maintain the block acknowledgment record for that agreement according to the following rules:

10.25.6.5 Generation and transmission of BlockAck frames by an HT STA, DMG STA, or S1G STA

Change the second paragraph in 10.25.6.5 as follows:

When responding with a BlockAck frame to either a received BlockAckReq frame or a received QoS Data frame with Implicit BAR ack policy during either full-state operation or partial-state operation, any adjustment to the value of $WinStart_R$ according to the procedures defined within 10.25.6.3 and 10.25.6.4 shall be performed before the generation and transmission of the response BlockAck frame. The Starting Sequence Number subfield of the Block Ack Starting Sequence Control subfield of the BlockAck frame shall be set to any value in the range $(WinEnd_R - \del{63} \text{ $BitmapLength + 1$ })$ to $WinStart_R$. The Starting Sequence Number subfield stored in the Starting Sequence Control field of NDP BlockAck and BAT frames shall be set to $WinStart_R$. The values in the recipient's record of status of MPDUs beginning with the MPDU for which the Sequence Number subfield value is equal to $WinStart_R$ and ending with the MPDU for which the Sequence Number subfield value is equal to $WinEnd_R$ shall be included in the bitmap of the BlockAck frame. The bitmap of the NDP BlockAck frame is protected using the encoding procedure described in 10.57.

Change the fourth paragraph in 10.25.6.5 as follows:

When responding with a BlockAck frame to either a received BlockAckReq frame or a received QoS Data frame with Implicit BAR ack policy during either full-state or partial-state operation, if the adjusted value of $WinEnd_R$ is less than the value of the starting sequence number of the BlockAck frame plus ~~63~~ $BitmapLength - 1$, within the bitmap of the BlockAck frame, the status of MPDUs with sequence numbers that are greater than the adjusted value of $WinEnd_R$ shall be reported as not received (i.e., the corresponding bit in the bitmap shall be set to 0).

10.25.6.6 Receive reordering buffer control operation

10.25.6.6.1 General

Change the fourth paragraph in 10.25.6.6.1 as follows:

$WinEnd_B$ is initialized to $WinStart_B + WinSize_B - 1$, where $WinSize_B$ is set to the smaller of ~~64~~ $BitmapLength$ and the value of the Buffer Size field of the ADDBA Response frame that established the block ack agreement.

10.25.8 GCR and GLK-GCR block ack

Change the title of 10.25.8.1 as follows:

10.25.8.1 **Introduction General**

Insert the following paragraph at the end of 10.25.8.1:

An HE AP shall not send a GCR MU-BAR Trigger frame to a non-AP HE STA if the most recently received Extended Capabilities element from the STA does not indicate support for Robust AV Streaming or Advanced GCR.

10.25.8.4 GCR block ack BlockAckReq and BlockAck frame exchanges

Change the second and third paragraphs in 10.25.8.4 as follows (NOTE 1 remains unchanged):

~~When~~ If the retransmission policy for a group address is GCR Block Ack, an originator shall not transmit more than the GCR buffer size number of A-MSDUs with the RA field set to the GCR concealment address and the DA field of the A-MSDU subframe set to the GCR group address before sending a BlockAckReq frame to one of the STAs that has a GCR block ack agreement for this group address. The RA field of the BlockAckReq frame shall be set to the MAC address of the destination STA. Upon reception of the BlockAck frame, an originator may send a BlockAckReq frame to another STA that has a block ack agreement for this group address, and this process may be repeated multiple times. If the originator has a GCR block ack agreement with one or more of the HE STAs for this group address, the originator may send a GCR MU-BAR Trigger frame to one or more of the HE STAs that are in the awake state. Upon reception of the BlockAck frame from one or more HE STAs, the originator may send a GCR MU-BAR Trigger frame to one or more other HE STAs that have a GCR block ack agreement, and this process may be repeated multiple times.

~~When~~ If a recipient receives a BlockAckReq frame with the GCR Group Address subfield equal to a GCR group address, the recipient shall transmit a BlockAck frame at a delay of SIFS after the BlockAckReq frame. The BlockAck frame acknowledges the STA's reception status of the block of group addressed frames requested by the BlockAckReq frame. If an HE STA receives a GCR MU-BAR Trigger frame with the AID12 subfield set to the 12 LSBs of the AID of the HE STA, the HE STA shall include the BlockAck frame in the HE TB PPDU sent in response to the Trigger frame. The BlockAck frame acknowledges the HE STA's reception status of the block of group addressed frames requested by the GCR MU-BAR Trigger frame.

Insert the following paragraph and Figure 10-37a into 10.25.8.4 after the fourth paragraph ("Figure 10-34 shows"):

Figure 10-37a shows another example of a frame exchange when the GCR block ack retransmission policy is used. The HE AP sends several A-MSDUs using the GCR block ack retransmission policy. The HE AP then sends a GCR MU-BAR Trigger frame to group members 1 and 2 of the GCR group, waits for the BlockAck frames, then sends a GCR MU-BAR Trigger frame to group members 3 and 4, and waits for the BlockAck frames. The HE AP then sends a BlockAckReq frame to group member 5, which is a non-HE STA, and waits for the BlockAck frame. After receiving the BlockAck frames, the HE AP determines whether any A-MSDUs need to be retransmitted and sends additional A-MSDUs (some of which might be retransmissions of previous A-MSDUs) using the GCR block ack retransmission policy.

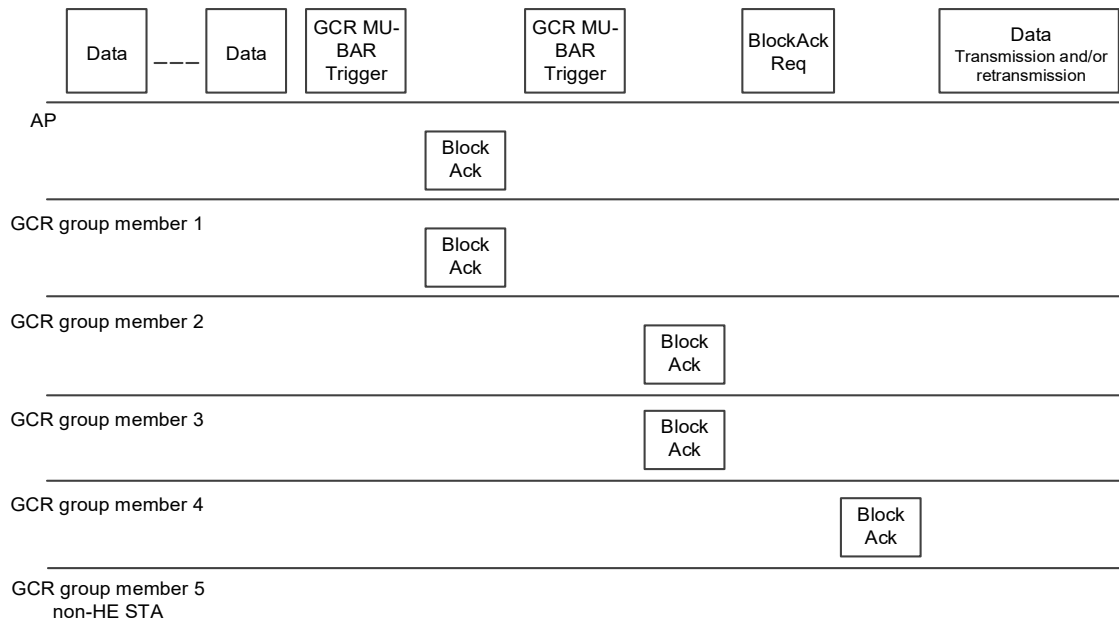


Figure 10-37a—Example of a frame exchange with GCR MU-BAR Trigger frames

Change the now sixth, seventh, and eighth paragraphs in 10.25.8.4 as follows (NOTE 2 remains unchanged):

After completing the BlockAckReq or GCR MU-BAR Trigger frame and BlockAck frame exchanges, the originator determines, from the information provided in the BlockAck bitmap and from the missing BlockAck frame, ~~which, if any, the A-MSDUs, if any, that need to be retransmitted.~~

An originator adopting the GCR block ack retransmission policy for a GCR group address chooses a lifetime limit for the group address. The originator may vary the lifetime limit for the group address at any time and may use different lifetime limits for different GCR group addresses. The originator transmits and retries each A-MSDU until the appropriate lifetime limit is reached or until each one has been received by all group members to which a BlockAckReq frame or a GCR MU-BAR Trigger frame has been sent, whichever occurs first.

For GCR streams with retransmission policy equal to GCR Block Ack, an originator may regularly send a BlockAckReq frame with the GCR Group Address subfield in the BAR Information field set to the GCR group address and the Block Ack Starting Sequence Control subfield set to the Sequence Number field of the earliest A-MSDU of the GCR stream that has not been acknowledged by all group members and has not expired due to lifetime limits, in order to minimize buffering latency at receivers in the GCR group. An originator may also send a GCR MU-BAR Trigger frame with the AID12 fields set to the 12 LSBs of the AIDs of HE STAs that transmit the BlockAck frames and the Block Ack Starting Sequence Control subfield set to the Sequence Number field of the earliest A-MSDU of the GCR stream that has not been acknowledged by all group members and has not expired due to lifetime limits, in order to minimize buffering latency at receivers in the GCR group.

Change the now 12th paragraph in 10.25.8.4 as follows (NOTE 3 remains unchanged):

If the beginning of such reception does not occur during the first slot time following a SIFS, then the originator may perform error recovery by retransmitting a BlockAckReq frame or a GCR MU-BAR Trigger frame PIFS after the previous BlockAckReq frame or GCR MU-BAR Trigger frame if ~~when~~ both of the following conditions are met:

- The carrier sense mechanism (see 10.3.2.1) indicates that the medium is idle at the TxPIFS slot boundary (see Figure 10-25) after the expected start of a BlockAck frame, and
- The remaining duration of the GCR TXOP is longer than the total time required to retransmit the GCR BlockAckReq frame or GCR MU-BAR Trigger frame plus one slot time.

10.27 Protection mechanisms

Insert the following subclause (10.27.6) after 10.27.5:

10.27.6 Protection rules for HE STAs

An HE STA operating in the 2.4 GHz band is subject to all of the rules for HT STAs that apply to that band, except that a PPDU with the TXVECTOR parameter FORMAT set to HE_SU, HE_ER_SU, HE_MU, or HE_TB may be substituted for a PPDU with the TXVECTOR parameter FORMAT set to HT_MF.

An HE STA operating in the 5 GHz band is subject to all of the rules for VHT STAs that apply to that band, except that a PPDU with the TXVECTOR parameter FORMAT set to HE_SU, HE_ER_SU, HE_MU, or HE_TB may be substituted for a PPDU with the TXVECTOR parameter FORMAT set to VHT.

Additionally, an HE STA can use the MU-RTS/CTS frame exchange procedure.

10.29 Reverse direction protocol

10.29.1 General

Change the second paragraph in 10.29.1 as follows:

An HT STA indicates support of the RD feature as an RD responder using the RD Responder subfield of the HT Extended Capabilities field of the HT Capabilities element or the RD Responder subfield of the HE 6 GHz Band Capabilities element. A STA shall set the RD Responder subfield to 1 in frames that it transmits containing the HT Capabilities element in the 2.4 GHz or 5 GHz band and sets the RD Responder subfield to 1 in frames that it transmits containing the HE 6 GHz Band Capabilities element in the 6 GHz band if dot11RDRResponderOptionImplemented is true. Otherwise, the STA shall set the RD Responder subfield to 0. In an HE STA, the RDG/More PPDU subfield and the AC Constraint subfield are present in the CAS Control subfield.

10.29.2 Reverse direction (RD) exchange sequence

Change NOTE 2 in 10.29.2 as follows:

NOTE 2—If the RD responder is a VHT AP, the RD response burst can contain VHT MU PPDU that might have TXVECTOR parameter NUM_USERS > 1. If the RD responder is an HE AP, the RD response burst can contain HE MU PPDU that might have TXVECTOR parameter NUM_USERS > 1.

Insert the following paragraph into 10.29.2 after NOTE 3:

If the RD initiator is an HE STA and the RD responder is an HE AP, the RD response burst may contain one or more Basic Trigger frames. The Basic Trigger frames shall trigger the RD initiator and at least one other STA in a full-bandwidth UL MU-MIMO transmission.

10.29.3 Rules for RD initiator

Change the third through sixth paragraphs in 10.29.3 as follows:

Transmission of a +HTC or DMG frame by an RD initiator with the RDG/More PPDU subfield equal to 1 (either transmitted as a non-A-MPDU frame or within an A-MPDU) indicates that the duration indicated by the Duration/ID field is available for the RD response burst and RD initiator final PPDU (if present). Transmission of an MPDU by an HE RD initiator that contains a CAS Control subfield with the RDG/More PPDU subfield equal to 1 indicates that the duration indicated by the Duration/ID field is available for the RD response burst and RD initiator final PPDU (if present).

An RD initiator that sets the RDG/More PPDU field to 1 in a +HTC or DMG frame transmitted during a TXOP shall set the AC Constraint subfield to 1 in that frame if the TXOP was gained through the EDCA channel access mechanism and shall otherwise set it to 0. An RD initiator that sets the RDG/More PPDU field to 1 in a DMG frame transmitted during an SP can set the AC Constraint subfield to 1 to limit the Data frames transmitted by the RD responder. An HE STA RD initiator that sets the RDG/More PPDU field to 1 in a CAS Control subfield in a frame transmitted during a TXOP may set the AC Constraint subfield in the CAS Control subfield to 1.

An RD initiator shall not transmit a +HTC or DMG frame with the RDG/More PPDU subfield set to 1 that requires a response MPDU that is not one of the following frames:

- Ack
- Compressed BlockAck
- Extended Compressed BlockAck
- Multi-STA BlockAck

Subject to TXOP or SP constraints, after transmitting an RDG PPDU, an RD initiator may transmit its next PPDU as follows:

- a) *Normal continuation:* The RD initiator may transmit its next PPDU a minimum of a SIFS after receiving a response PPDU that meets one of the following conditions:
 - 1) Contains one or more received +HTC or DMG frames with the RDG/More PPDU subfield equal to 0.
 - 2) In an HT STA or an SIG STA, contains one or more received frames that are capable of carrying the HT Control field but did not contain an HT Control field.
 - 3) Contains a received frame that requires an immediate response.
 - 4) In a DMG STA, none of the correctly received frames in the PPDU carry the QoS Control field.
- b) *Error recovery:* For TXOPs and non-DMG SPs, the RD initiator may transmit its next PPDU ~~when~~ if the CS mechanism (see 10.3.2.1) indicates that the medium is idle at the TxPIFS slot boundary (see Figure 10-25). For DMG SPs, the RD initiator shall not transmit its next PPDU earlier than PIFS following its last PPDU transmission. Transmission is a continuation of the current TXOP or SP.

Insert NOTE 5 into 10.29.3 after NOTE 4, and renumber the subsequent notes in this subclause accordingly:

NOTE 5—Control response frames generated by HE STAs do not carry the HT Control field.

10.29.4 Rules for RD responder

Change the second paragraph in 10.29.4 as follows:

The recipient of an RDG may decline the RDG by

- Not transmitting any frames following the RDG PPDU ~~when if~~ no response is otherwise required, or
- Transmitting a control response frame with the RDG/More PPDU subfield set to 0, or
- Transmitting a control response frame that contains no HT Control field, ~~or~~
- Transmitting a control response frame aggregated with other MPDUs with the RDG/More PPDU subfield set to 0.

Change fifth and sixth paragraphs in 10.29.4 as follows (including splitting the sixth paragraph into two paragraphs):

An RD responder shall not transmit an MPDU (either individually or aggregated within an A-MPDU) that is not one of the following frames:

- Ack
- Compressed BlockAck
- Compressed BlockAckReq
- Extended Compressed BlockAck
- Extended Compressed BlockAckReq
- Multi-STA BlockAck
- QoS ~~D~~data
- QoS Null
- Management
- Basic Trigger

~~If the AC Constraint subfield is equal to 1, the RD responder shall transmit Data frames of only the same AC as the last frame received from the RD initiator. If the AC Constraint subfield is equal to 1 in the last frame received from an RD initiator.~~

- A non-HE RD responder shall transmit Data frames of only the same AC as the last frame received from the RD initiator.
- An HE RD responder may transmit an A-MPDU or multi-TID A-MPDU with MPDUs from one or more ACs that have a priority that is equal to or higher than the lowest priority AC of the MPDU(s) carried in the last PPDU received from the RD initiator (see 10.12) and if the RD initiator is an HE STA subject to the additional rules defined in 26.6.3.

For a BlockAckReq or BlockAck frame, the AC is determined by examining the TID field. For a Management frame, the AC is AC_VO. The RD initiator shall not transmit a +HTC or DMG MPDU with the RDG/More PPDU subfield set to 1 from which the AC cannot be determined. If the AC Constraint subfield is equal to 0, the RD responder may transmit Data frames of any TID.

Change the now eighth paragraph in 10.29.4 as follows:

During an RD response burst any PPDU transmitted by an RD responder shall contain at least one MPDU with an Address 1 field that matches the MAC address of the RD initiator or at least one Trigger frame that addresses the RD initiator, and the inclusion of traffic to STAs other than the RD initiator in a VHT MU PPDU, ~~or an SIG MU PPDU, or HE MU PPDU~~ shall not increase the duration of the ~~VHT MU PPDU~~ beyond that required to transport the traffic to the RD initiator. The RD responder shall not transmit ~~any frame causing a~~

frame that is not a Basic Trigger frame and that causes a response after SIFS with an Address 1 field that does not match the MAC address of the RD initiator. The RD responder shall not transmit any PPDU with a CH_BANDWIDTH that is wider than the CH_BANDWIDTH of the PPDU containing the frame(s) that delivered the RD grant.

Insert the following paragraphs into 10.29.4 after the now eighth paragraph:

An RD responder that transmits a Basic Trigger frame shall set the CS Required subfield to 1 and shall allocate a number of streams for the RD initiator that is not smaller than the number of streams of the RD initiator's last PPDU.

If an RD initiator sets the RDG/More PPDU field to 1 in a +HTC frame transmitted during a TXOP and sets the AC Constraint subfield to 1 in that frame, the RD responder shall set the Preferred AC subfield of the Trigger Dependent User Info field in the Trigger frame to the same AC as the last frame received from the RD initiator.

10.36 Null data PPDU (NDP) sounding

10.36.6 Transmission of a VHT NDP

Insert the following paragraph into 10.36.6 after the second paragraph (“The number of space-time streams”):

If an HE STA transmits a VHT NDP where at least one of the intended recipients of the VHT NDP is an HE STA, then the following conditions apply:

- If the bandwidth of an VHT NDP is less than or equal to 80 MHz, the number of space-time streams sounded as indicated by the TXVECTOR parameter NUM_STS shall not exceed the value indicated in the Beamformee STS $\leq$ 80 MHz subfield in the HE Capabilities element of any intended HE STA recipient of the VHT NDP.
- If the bandwidth of an VHT NDP is greater than 80 MHz, the number of space-time streams sounded as indicated by the TXVECTOR parameter NUM_STS shall not exceed the value indicated in the Beamformee STS $>$ 80 MHz subfield in the HE Capabilities element of any intended HE STA recipient of the VHT NDP.

10.47 Target wake time (TWT)

10.47.1 TWT overview

Change the first paragraph in 10.47.1 as follows:

Target wake times (TWTs) allow STAs to manage activity in the BSS by scheduling STAs to operate at different times in order to minimize contention and to reduce the required amount of time that a STA utilizing a power management mode needs to be awake. TWTs can be individual TWTs, which are described in 10.47 and 26.8.2, or broadcast TWTs, which are described in 26.8.3.

Change the fifth paragraph in 10.47.1 as follows:

STAs that request a TWT agreement are called TWT requesting STAs and the STAs that respond to their requests are TWT responding STAs. A TWT requesting STA is assigned specific times to wake and exchange frames with the TWT responding STA. A TWT requesting STA communicates wake scheduling

information to its TWT responding STA and the TWT responding STA devises a schedule and delivers TWT values to the TWT requesting STA when a TWT agreement has been established between them. When explicit TWT is employed, a TWT requesting STA wakes and performs a frame exchange and receives the next TWT information in a response from the TWT responding STA as described in 10.47.3. When implicit TWT is used, the TWT requesting STA calculates the Next TWT by adding a fixed value to the current TWT value as described in 10.47.4. STAs need not be made aware of the TWT values of other STAs. A TWT requesting STA and a TWT responding STA shall set the Negotiation Type subfield to 0 in the TWT element of transmitted frames containing the TWT element, except when the STAs are HE STAs. Additional TWT setup exchanges between HE STAs for individual TWT operation are defined in 26.8.

Change the following block of text (i.e, seventh through now tenth paragraphs, NOTE 1, and NOTE 2) in 10.47.1, and renumber the subsequent notes in this subclause accordingly:

An S1G STA with dot11TWTOptionActivated equal to true and that operates in the role of TWT requesting STA shall set the TWT Requester Support subfield to 1 in all S1G Capabilities elements that it transmits. An S1G STA with dot11TWTOptionActivated equal to true and that operates in the role of TWT responding STA shall set the TWT Responder Support subfield to 1 in all S1G Capabilities elements that it transmits.

A STA that is not an S1G STA and is not an HE STA and with dot11TWTOptionActivated equal to true and that operates in the role of TWT requesting STA shall set the TWT Requester Support subfield to 1 in all Extended Capabilities elements that it transmits. A STA that is not an S1G STA and is not an HE STA and with dot11TWTOptionActivated equal to true and that operates in the role of TWT responding STA shall set the TWT Responder Support subfield to 1 in all Extended Capabilities elements that it transmits.

If the TWT Responder Support subfield of the S1G Capabilities element, HE Capabilities element, or Extended Capabilities element received from its associated AP is equal to 1, a non-AP STA with dot11TWTOptionActivated equal to true may transmit a TWT element to the AP with a value of Request TWT, Suggest TWT, or Demand TWT in the TWT Setup Command field and with the TWT Request field equal to 1.

An AP with dot11TWTOptionActivated equal to true shall transmit a TWT element to a STA that is associated to the AP and from which it received a frame containing a TWT element that contained a value of Request TWT, Suggest TWT, or Demand TWT in the TWT Setup Command field and with the TWT Request field equal to 1. The transmitted TWT element shall be included in the frame that is the appropriate response frame to the received frame. The AP shall include a value of Accept TWT, Alternate TWT, Dictate TWT, or Reject TWT in the TWT Setup Command field of the response and shall set the TWT Request field to 0. If the AP response's TWT Setup Command field includes anything other than Accept TWT or Reject TWT, the STA should send a new request for a TWT value by sending another frame that contains a TWT element, modifying the parameters of the request to indicate, for example, an acceptance of a proposed alternate TWT or dictated TWT value. If the STA receives a TWT response to a TWT request with the TWT Setup Command field value of Accept TWT, then the STA has successfully completed a TWT setup with that STA for the TWT Flow Identifier indicated in the TWT response and the STA becomes a TWT requesting STA and the STA may enter the doze state until the TSF matches the next TWT value of the STA, provided that the STA has indicated that it is in a power save mode and no other condition requires the STA to remain awake. The AP becomes a TWT responding STA of the TWT requesting STA.

NOTE 1—A TWT responding STA might choose a TWT Flow Identifier for the TWT response that is different from the TWT Flow Identifier of a received TWT request.

NOTE 2—A TWT requesting STA might renegotiate the TWT parameters of an existing TWT agreement by sending to the TWT responding STA a TWT request with a Flow Identifier that corresponds to that TWT agreement. The TWT response sent by the TWT responding STA containing the TWT Setup Command field of Accept TWT will indicate whether the newly requested TWT parameters are accepted or whether the previously negotiated TWT parameters are still in place.

Insert the following paragraphs and Table 10-32a into 10.47.1 after the new NOTE 2:

The result of an exchange of TWT Setup frames between a TWT requesting STA and a TWT responding STA is defined in Table 10-32a.

- “Request TWT” indicates that the requesting STA does not provide a set of TWT parameters for a TWT agreement, leaving the choice of parameters to the responding STA.
- “Suggest TWT” indicates that the requesting STA offers a set of preferred TWT parameters for a TWT agreement but might accept alternative TWT parameters that the responding STA indicates.
- “Demand TWT” indicates that the requesting STA will currently accept only the indicated TWT parameters for a TWT agreement.

For transmissions by a responding STA,

- “Accept TWT” indicates that the responding STA has initiated a TWT agreement with the given parameters.
- “Alternate TWT” indicates a counter-offer of TWT parameters (although alternative TWT parameters might be accepted as well) without the creation of a TWT agreement.
- “Dictate TWT” indicates that no TWT agreement is created, but one is likely to be accepted only if the requesting STA transmits a new TWT setup request with the indicated TWT parameters (i.e., no other TWT parameters will be accepted).

For transmissions by a responding STA as part of a negotiation for a new TWT agreement, “Reject TWT” indicates that the negotiation has ended in failure to create a new TWT agreement.

Table 10-32a—TWT setup exchange command interpretation

TWT Setup Command field in an initiating frame	TWT Setup Command field value in a response frame	TWT condition after the completion of the exchange
Request TWT, Suggest TWT, or Demand TWT	No frame transmitted	No new individual TWT agreement exists with the TWT flow identifier corresponding to the TWT flow identifier in the initiating frame. No new individual TWT agreement exists.
Demand TWT	Accept TWT	An individual TWT agreement exists that uses the TWT parameters identified in the initiating frame. The TWT parameters in the response frame match the TWT parameters of the initiating frame.
Suggest TWT or Request TWT	Accept TWT	An individual TWT agreement exists and uses the TWT parameters identified in the response frame.
Demand TWT or Suggest TWT	Alternate TWT	No individual TWT agreement exists with the associated TWT flow identifier. The responder is offering an alternative set of parameters to those indicated in the initiating frame. The requesting STA can send a new request with any set of TWT parameters, and the responder might create an individual TWT agreement using those parameters.

Table 10-32a—TWT setup exchange command interpretation (continued)

TWT Setup Command field in an initiating frame	TWT Setup Command field value in a response frame	TWT condition after the completion of the exchange
Demand TWT or Suggest TWT	Dictate TWT	No individual TWT agreement exists with the associated TWT flow identifier. The responder STA offers an alternative set of parameters to those indicated in the initiating frame. The responder STA indicates that it is not willing to accept any other TWT parameters from the requesting STA at this time. The requesting STA can send a new request, but will receive an Accept TWT only if it uses the dictated TWT parameters.
Request TWT, Suggest TWT, or Demand TWT	Reject TWT	No individual TWT agreement exists with the associated TWT flow identifier. The responding STA will not create any new individual TWT agreement with the requester at this time.
NOTE 1—Negotiation Type subfield of the TWT element contained in these frames is equal to 0.		
NOTE 2—The initiating frame is a TWT request, and the response frame is a TWT response.		

Change the 26th paragraph in 10.47.1 as follows:

A TWT requesting STA indicates which single channel it desires to use as a temporary primary channel during a TWT SP by setting a single bit to 1 within the TWT Channel field of the TWT element, according to the mapping described for that field. A TWT responding STA indicates which single channel the TWT requesting STA is permitted to use as a temporary primary channel during a TWT SP by setting a single bit to 1 within the TWT Channel field of the TWT element, according to the mapping described for that field. In an S1G BSS, during ~~During~~ a TWT SP, access to a channel that is not the primary channel of the BSS shall be performed according to the procedure described in 10.52.

Insert the following paragraphs at the end of 10.47.1:

In an HE BSS, during a trigger-enabled TWT SP, access to a channel that is not the primary channel of the BSS shall be performed according to the procedure described in 26.8.7.

If a STA has received a TWT Constraint Parameters element from another STA, it should not initiate a new TWT agreement or join a broadcast TWT schedule with that STA if the total number of TWT sessions/schedules between the two STAs would exceed the value of the Max TWT Sessions field in the TWT Constraint Parameters element.

If a STA has received a TWT Constraint Parameters element from another STA and includes the Target Wake Time field in a TWT element transmitted to initiate a new TWT agreement with that STA, then it should set the Target Wake Time field to an integer multiple of $n + 1$ TUs [i.e., $(\text{Target Wake Time}) \bmod (n + 1) = 0$] where n is the value from the Starting Target Wake Time Alignment field of the most recent TWT Constraint Parameters element received from that STA.

Insert the following subclause (10.47.9) after 10.47.8:

10.47.9 TWT parameter ranges

A STA that supports receiving two TWT elements in a TWT Setup frame sets the TWT Parameters Range Support field to 1 in the Extended Capabilities elements that it transmits.

A TWT requesting STA may send a TWT request that contains two TWT elements to a TWT responding STA if it has received an Extended Capabilities element from the TWT responding STA with the TWT Parameters Range Support field equal to 1. The TWT requesting STA shall either set the TWT Setup Command fields of the two TWT elements to Suggest TWT or set the TWT Setup Command fields of the two TWT elements to Demand TWT.

A TWT responding STA may send a TWT response that contains two TWT elements to a TWT requesting STA if it has received an Extended Capabilities element from the TWT requesting STA with the TWT Parameters Range Support field equal to 1. The TWT responding STA shall either set the TWT Setup Command fields of the two TWT elements to Alternate TWT or set the TWT Setup Command fields of the two TWT elements to Dictate TWT.

If a STA sends a TWT request or a TWT response with two TWT elements:

- It shall set the TWT Flow Identifier field or the Broadcast TWT ID field of the two TWT elements to the same value.
- For a TWT parameter of the proposed TWT agreement that is carried in the Target Wake Time field or Nominal Minimum TWT Wake Duration field, and for the TWT wake interval parameter, the STA may indicate either a single value or a range of values. For other TWT parameters, the STA shall indicate a single value. To indicate a single value for a parameter, the STA sets the value of that parameter to be the same in each TWT element. To indicate a range of values for a parameter, the STA sets the value of that parameter differently in each TWT element. The range indicated for a parameter is the range that starts with the lower value and includes all values up to the higher value, irrespective of their order in the TWT elements, except for the TWT Wake Interval Mantissa and the TWT Wake Interval Exponent fields. The range indicated for the TWT wake interval begins with the lower value of the two TWT wake interval values and ends with the higher value of the two TWT wake interval values, where the TWT wake interval is determined from the TWT Wake Interval Mantissa and TWT Wake Interval Exponent fields as specified in 9.4.2.199.

If a TWT requesting STA sent a TWT request that contains two TWT elements to a TWT responding STA, the TWT responding STA shall set each TWT parameter of the TWT response to a value selected from the range of values provided in the TWT request, if the TWT Setup Command field of the TWT response is Accept TWT and the parameter included a range in the corresponding TWT request.

If a TWT responding STA sent a TWT response that contains two TWT elements to a TWT requesting STA with a TWT Setup Command field value of Dictate TWT, that response implies that a TWT agreement is likely to be accepted only if the requesting STA transmits a new TWT request with the TWT parameters set to a value selected from the range of values provided in the TWT response. If a TWT responding STA sent a TWT response that contains two TWT elements to a TWT requesting STA with a TWT Setup Command field value of Alternate TWT, that response implies a counter-offer of TWT parameters by the TWT responding STA, which can be indicated in the form of a range of values (although alternative TWT parameters might be accepted as well).

11. MLME

11.1 Synchronization

11.1.3 Maintaining synchronization

11.1.3.8 Multiple BSSID procedure

Insert the following subclause heading (for 11.1.3.8.1) immediately after the heading for 11.1.3.8:

11.1.3.8.1 General

Change 11.1.3.8.1 as follows, using the original first three paragraphs of 11.1.3.8 as published in IEEE Std 802.11-2020 (see the new 11.1.3.8.4 and 11.1.3.8.5 for the remaining original text from 11.1.3.8):

A STA that supports the Multiple BSSID capability has `dot11MultiBSSIDImplemented` equal to true and shall set to 1 the Multiple BSSID field of the Extended Capabilities elements that it transmits. Support for the Multiple BSSID capability is mandatory for a FILS STA and non-AP HE STA. An AP that supports enhancements related to the discovery and advertisement of a nontransmitted BSSID shall set the Enhanced Multi-BSSID Advertisement Support bit in the Extended Capabilities element to 1 and is referred to as an EMA AP. A 6 GHz AP with `dot11MultiBSSIDImplemented` equal to true and advertising a partial list of nontransmitted BSSID profiles shall operate as an EMA AP. If an HE AP operating in the 2.4 GHz or 5 GHz bands has `dot11MultiBSSIDImplemented` equal to true, advertises a partial list of nontransmitted BSSID profiles, and intends a non-AP STA to discover the complete list of nontransmitted BSSID profiles, where a complete list of nontransmitted BSSID profile comprises only BSSIDs that are discoverable, then the HE AP shall operate as an EMA AP.

A BSSID is discoverable if the AP includes information of that BSSID in its Beacon and Probe Response frames (though not necessarily every frame).

~~The nontransmitted BSSID profile shall include the SSID element (see 9.4.2.2) and Multiple BSSID Index element (see 9.4.2.73) for each of the supported BSSIDs. The AP or PCP may include all other elements allowed per 9.4.2.45 in the nontransmitted BSSID profile. The AP or PCP may include two or more Multiple BSSID elements containing elements for a given BSSID index in one Beacon frame or DMG Beacon frame. If two or more are given, the profile is considered to be the complete set of all elements given in all such Multiple BSSID elements sharing the same BSSID index. Since the Multiple BSSID element is also present in Probe Response frames, an AP or PCP may choose to advertise the complete or a partial profile of a BSS corresponding to a nontransmitted BSSID only in the Probe Response frames. In addition, the AP or PCP may choose to include only a partial list of nontransmitted BSSID profiles in the Beacon frame or DMG Beacon frame or to include different sets of nontransmitted BSSID profiles in different Beacon frames or DMG Beacon frames. An AP advertising a complete list of nontransmitted BSSID profiles shall set the Complete List Of NonTx BSSID Profiles field of Extended Capabilities element to 1.~~

~~NOTE—A non-AP STA can send a Probe Request frame to an AP to gather information about all BSSIDs in the multiple BSSID set when the AP advertises partial list of nontransmitted BSSID profiles.~~

An AP with `dot11MultiBSSIDImplemented` equal to true shall set the Co-Hosted BSS subfield in HE Operation element that it transmits to 0.

The BSSID of the AP belonging to a multiple BSSID set is referred to as the transmitted BSSID if the AP includes the Multiple BSSID element in the Beacon frame that it transmits. In a multiple BSSID set, there shall not be more than one AP corresponding to the transmitted BSSID. The BSSID of an AP belonging to a

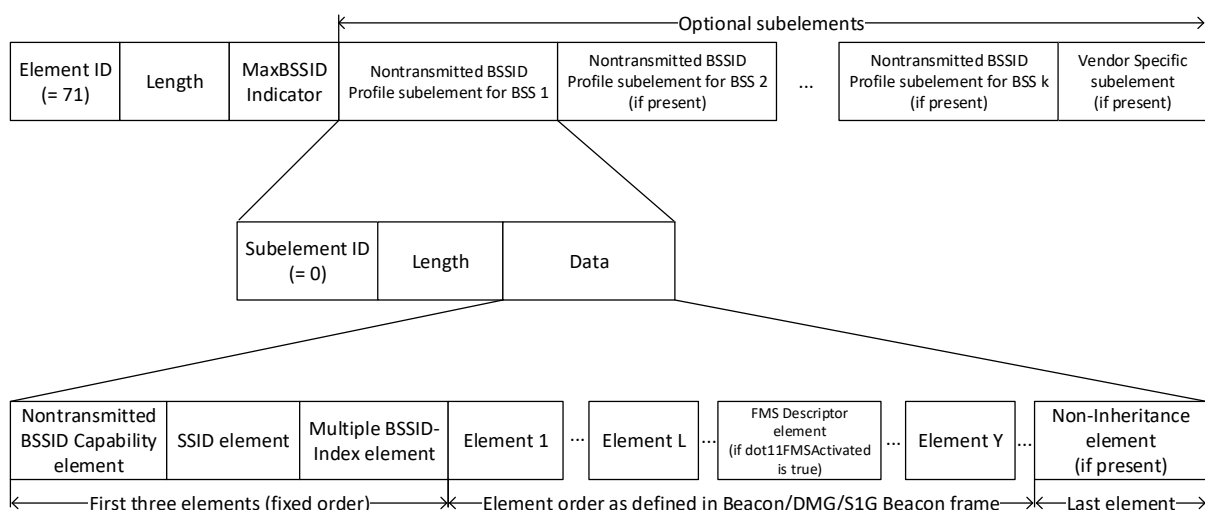
multiple BSSID set is a nontransmitted BSSID if the AP's BSSID is derived according to 9.4.2.45 and 9.4.2.73. Among all AP STAs in a multiple BSSID set, only the AP corresponding to the transmitted BSSID shall transmit a Beacon frame.

See 11.1.4.3.4 for the rules governing transmission of Probe Response frames in a multiple BSSID set.

Insert the following subclauses (11.1.3.8.2 and 11.1.3.8.3, including Figure 11-3a and Figure 11-3b) after 11.1.3.8.1:

11.1.3.8.2 Nontransmitted BSSID profile

A nontransmitted BSSID profile represents information about a particular nontransmitted BSSID. It consists of a set of elements that are carried in a single Nontransmitted BSSID Profile subelement, unless the subelement cannot fit in a single Multiple BSSID element. In this case, the nontransmitted BSSID profile is fragmented and is carried in more than one Nontransmitted BSSID Profile subelements across more than one Multiple BSSID element in the same frame. Each nontransmitted BSSID profile, at a minimum, shall include the elements that are mandatory for that BSS (i.e., Nontransmitted BSSID Capability element, SSID element, Multiple BSSID-Index element as described in 9.4.2.45). An example of Multiple BSSID element carrying one or more Nontransmitted BSSID Profile subelements is shown in Figure 11-3a. The figure also shows the order in which the elements are present within each Nontransmitted BSSID Profile subelement.



**Figure 11-3a—Example of Multiple BSSID element carrying
Nontransmitted BSSID Profile subelements**

An AP shall not carry a nontransmitted BSSID profile across multiple Multiple BSSID elements in a frame, unless the nontransmitted BSSID profile cannot fit in one multiple BSSID element due to the size limit of the multiple BSSID element.

If there is a need to fragment a nontransmitted BSSID profile across more than one Multiple BSSID element in a frame, an AP shall not fragment an element in the profile across multiple Multiple BSSID elements, and it shall place the next element in that profile as the first element in the first Nontransmitted BSSID Profile subelement of the immediately following Multiple BSSID element. An AP shall not fragment a nontransmitted BSSID profile across two frames. If a frame carries multiple Multiple BSSID elements, the MaxBSSID Indicator field in all the Multiple BSSID elements shall carry the same value.

An example of a nontransmitted BSSID profile fragmented across two Multiple BSSID elements in a frame is shown in Figure 11-3b.

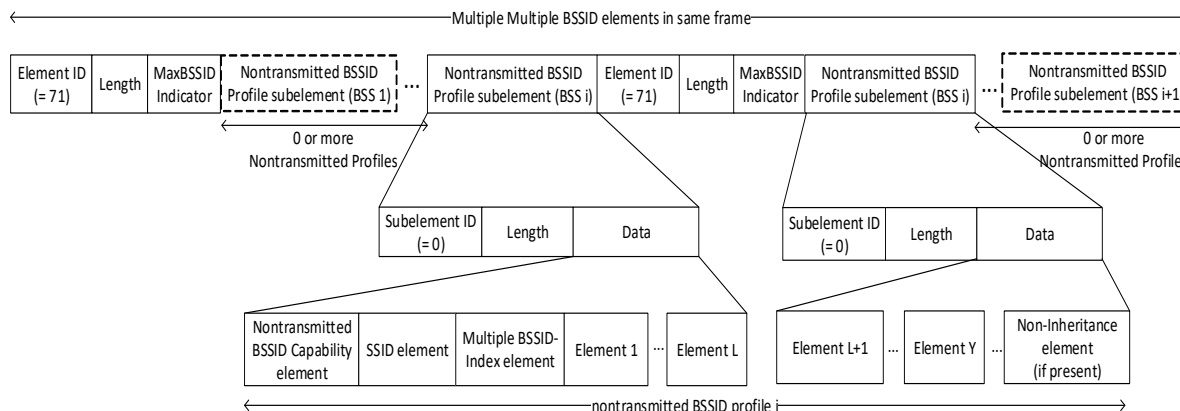


Figure 11-3b—Example of nontransmitted BSSID profile fragmented across multiple Multiple BSSID elements

NOTE—As described in 9.4.3, the Length field of the Nontransmitted BSSID Profile subelement indicates the number of octets only in the Data field of the subelement.

11.1.3.8.3 Discovery of a nontransmitted BSSID profile

An AP or PCP may choose to include only a partial list of nontransmitted BSSID profiles in the Beacon frame, S1G Beacon frame, or DMG Beacon frame or to include different sets of nontransmitted BSSID profiles in different Beacon frames, S1G Beacon frames, or DMG Beacon frames. An AP corresponding to the transmitted BSSID may choose to include only a partial list of nontransmitted BSSID profiles in an unsolicited broadcast Probe Response frame or a Probe Response frame sent in response to a Probe Request frame with Address 3 field set to wildcard BSSID and SSID set to wildcard. An AP advertising a complete list of nontransmitted BSSID profiles shall set the Complete List Of NonTxBSSID Profiles field of Extended Capabilities element to 1.

An EMA AP operating in the 2.4 GHz or 5 GHz band that transmits a Beacon or Probe Response frame carrying a partial list of nontransmitted BSSID profiles should include in the frame a Reduced Neighbor Report element carrying information for at least the nontransmitted BSSIDs that are not present in the Multiple BSSID element carried in that frame.

A 6 GHz-only EMA AP that transmits a Beacon or broadcast Probe Response frame carrying a partial list of nontransmitted BSSID profiles shall include a Reduced Neighbor Report element carrying information for at least the nontransmitted BSSIDs that are discoverable and not present in the Multiple BSSID element carried in that frame.

A 6 GHz-only EMA AP that transmits a FILS Discovery frame shall include a Reduced Neighbor Report element carrying information on all nontransmitted BSSIDs in the multiple BSSID set that are discoverable.

The EMA AP does this to aid the fast discovery of all nontransmitted BSSIDs in the multiple BSSID set that are discoverable.

NOTE 1—A FILS Discovery frame received from a 6 GHz AP with the Multiple BSSIDs Presence Indicator subfield equal to 1 and not carrying a Reduced Neighbor Report element implies that the AP's Beacon frame carries a complete list of nontransmitted BSSID profiles or that the information of the nontransmitted BSSID(s) is advertised in the 2.4 GHz or 5 GHz band by a co-located AP (see 11.53).

An EMA AP advertising a partial list of BSSID profiles, shall include the Multiple BSSID Configuration element (see 9.4.2.260) in its Beacon frame, S1G Beacon frame, or DMG Beacon frame and in any Probe Response frame it sends to indicate the configuration of the multiple BSSID set.

An AP shall set the BSSID Count field of the Multiple BSSID Configuration element to indicate the number of active BSSIDs in the multiple BSSID set, and shall set the Full Set Rx Periodicity field to indicate the number of beacons a scanning STA is required to receive in order to discover all the active nontransmitted BSSIDs in the set. An AP corresponding to the transmitted BSSID shall respond with a Probe Response frame carrying one or more Multiple BSSID elements that include, at a minimum, the profiles for the nontransmitted BSSIDs requested by the soliciting Probe Request frame.

NOTE 2—A nontransmitted BSSID profile is requested if, in the soliciting Probe Request frame, the SSID or BSSID matches the SSID or the BSSID, respectively, of the BSS corresponding to the nontransmitted BSSID.

An unassociated non-AP STA may transmit a Probe Request frame containing a Known BSSID element (see 9.4.2.261) and addressed to an EMA AP corresponding to the transmitted BSSID to gather information on nontransmitted BSSIDs that it has not discovered.

An EMA AP that transmits a Probe Response frame in response to a Probe Request frame containing a Known BSSID element shall, at a minimum, include the nontransmitted BSSID profiles not known to the requesting STA, unless the AP is unable to fit all of them in the response frame.

An EMA AP that includes a partial list of nontransmitted BSSID profiles in its Beacon frame, S1G Beacon frame, or DMG Beacon frame, shall advertise a particular nontransmitted BSSID profile in a repeating pattern such that the profile is present in at least one beacon in a sequence of beacons indicated by the Full Set Rx Periodicity field of the Multiple BSSID Configuration element, unless the membership of the multiple BSSID set changes. An EMA AP shall include a nontransmitted BSSID profile in the DTIM beacon of that BSS so that STAs associated to that AP can receive the profile (and any updates to the BSS configuration) without having to wake up for additional beacons. An EMA AP shall select the DTIM interval for a nontransmitted BSSID as a multiple of the value carried in the Full Set Rx Periodicity field of the Multiple BSSID Configuration element. Annex AA provides several example configurations.

NOTE 3—An AP corresponding to a nontransmitted BSSID advertises any changes to its BSS operational parameters during the beacon interval that follows the profile's DTIM beacon. For example, an AP corresponding to the nontransmitted BSSID can send a broadcast Disassociation frame to disassociate all STAs associated with it.

NOTE 4—In order to aid fast discovery of nontransmitted BSSIDs via passive scanning, it is recommended that an AP select a small value for the Full Set Rx Periodicity field.

Insert the following subclause heading for 11.1.3.8.4 after 11.1.3.8.3:

11.1.3.8.4 Inheritance of element values

Change 11.1.3.8.4 as follows, using the original fourth and fifth paragraphs of 11.1.3.8 as published in IEEE Std 802.11-2020:

~~When a station receives a Beacon frame or DMG Beacon frame with a Multiple BSSID element that consists of a nontransmitted BSSID profile with only the mandatory elements, it may inherit the complete profile from a previously received Beacon frame, DMG Beacon frame, or Probe Response frame, or it may send a Probe Request frame to obtain the complete BSSID profiles. Each Beacon element not transmitted in a nontransmitted BSSID subelement is inherited from previous Beacon, DMG Beacon, or Probe Response~~

~~frame in which the element is present, except for the Quiet element, which shall take effect only in the Beacon frame or DMG Beacon frame that contains it and not carry forward as a part of the inheritance. An AP or PCP is not required to include all supported nontransmitted BSSID profiles in a Probe Response frame, and may choose to only include a subset based on any criteria. When a nontransmitted BSSID profile is present in the one or more Multiple BSSID elements of the a Probe Response frame or a Beacon frame, the AP or PCP shall include all elements that are specific to this BSS. An element is considered to be specific to a BSS if its value is different from the corresponding element advertised by the transmitted BSSID or if the nontransmitted BSSID satisfies the condition as specified in the Table 9-32 for a non-DMG non-S1G AP, Table 9-45 for a DMG AP, or Table 9-46 for a S1G AP for that element to be present while the transmitted BSSID does not satisfy the corresponding condition. If any of the optional elements carried in the Probe Response frame, Beacon frame, DMG Beacon frame, or S1G Beacon frame of the transmitted BSSID are not present in a nontransmitted BSSID profile, the corresponding values are the element values to use for the nontransmitted BSSID are the values of the corresponding element of the transmitted BSSID unless the element is listed in the Non-Inheritance element (if included) in the nontransmitted BSSID profile for that BSS.~~

A non-AP and non-PCP STA derives its nontransmitted BSSID value according to 9.4.2.45 and 9.4.2.73.

Insert the following subclause heading for 11.1.3.8.5 after 11.1.3.8.4:

11.1.3.8.5 Traffic advertisement in a multiple BSSID set

Change 11.1.3.8.5 as follows, using the original sixth and seventh paragraphs of 11.1.3.8 as published in IEEE Std 802.11-2020:

The Partial Virtual Bitmap field of the TIM element carried in the Beacon, S1G Beacon, or TIM frame shall indicate the presence or absence of traffic to be delivered to all stations associated with an AP corresponding to a transmitted or nontransmitted BSSID. The first 2^n bits of the bitmap are reserved for the indication of group addressed frame for the transmitted and all nontransmitted BSSIDs (see 9.4.2.5.1). The AID space is shared by all BSSs, and the lowest AID value that shall be assigned to a non-S1G STA is 2^n (see 9.4.2.5). The value of the 11 LSBs of the AID assigned to an S1G STA shall be greater than 2^n . The Encoded Blocks that contain these first 2^n AIDs (if any) shall precede the Encoded Blocks that contain AIDs for the S1G STAs in the S1G Partial Virtual Bitmap field of each page. Each BSS of the Multiple BSSID set may have a different DTIM interval, which is signaled in the DTIM Period and DTIM Count fields that are present in the Multiple BSSID-Index element carried in the nontransmitted BSSID profile for that BSS.

Based upon its knowledge of the capability of associated non-AP STAs to support the multiple BSSID capability, as indicated by the corresponding field in the Extended Capabilities element and the content of the traffic indication virtual bitmap, an AP shall encode the Partial Virtual Bitmap and the Bitmap Control fields of the TIM element using one of the three methods (Method A, Method B, or Method C) defined in 9.4.2.5.1. Specifically, a non-S1G AP shall use Method B if it determines that the bit for each associated non-AP STA in the traffic indication virtual bitmap that is reconstructed by each non-AP STA from the received TIM element encoded using Method B is set correctly. Otherwise, a non-S1G AP shall use Method A, and an S1G AP shall use Method C.

NOTE—If all the recipients of the TIM element are STAs that support the multiple BSSID capability (e.g., when the TIM element is carried in HE beacon, ER beacon, FILS Discovery frame, or OPS frame where all the addressees are non-AP HE STAs), the transmitting AP uses Method B to encode the Partial Virtual Bitmap and the Bitmap Control fields of the TIM element.

Multiple BSSID rate selection is defined in 10.6.9.

11.1.4 Acquiring synchronization, scanning

11.1.4.3 Active scanning and probing procedures

11.1.4.3.2 Active scanning procedure for a non-DMG STA

Change items b), c), and d) in the lettered list of 11.1.4.3.2 as follows:

- b) If the STA is a FILS STA or a 6 GHz HE STA, set the FILSProbeTimer to 0 and starts the FILSProbeTimer. While the FILSProbeTimer is less than dot11FILSProbeDelay, the STA may skip a probe request transmission and proceed to step g) after setting the ActiveScanningTimer to 0 and starting the ActiveScanningTimer, if one of the following conditions matches:
 - 1) The STA receives a broadcast Probe Request frame that the SME considers to be suitable to discover a candidate AP for association.
 - 2) The STA receives one or more of Probe Response, Beacon, Measurement Pilot, or FILS Discovery frames that identify an AP that the SME considers a suitable candidate for association.
 - 3) The STA successfully sent a Probe Request frame by following the UORA procedure as defined in 26.5.4.

NOTE—How an SME considers a probe request or AP suitable is beyond the scope of this standard.

- c) Perform the basic access procedure as defined in 10.3.4.2. While waiting for access to WM, the STA may send one or more Probe Request frames by following the UORA procedure and proceed to step g). Send a probe request to the broadcast destination address. The probe request is sent with the SSID and BSSID from the received MLME-SCAN.request primitive. If dot11SSIDListActivated is true and the SSID List parameter is present in the MLME-SCAN.request primitive, then one or more SSID List elements should be present in the probe request, indicating all SSIDs in the SSID List parameter if possible.
- d) When dot11SSIDListActivated is true and either or both of the SSID List and Short SSID List parameters are the SSID List parameter is present in the MLME-SCAN.request primitive, send zero or more probe requests to the broadcast destination address, each with ~~an one or more SSIDs~~ indicated in the SSID List and/or Short SSID List parameters and the BSSID from the MLME-SCAN.request primitive. A Probe Request frame that contains a Short SSID List element shall have the SSID field of the SSID element set to the SSID of a known AP or set to the one-octet value 128 if the STA does not know any SSID. These additional probe requests [following step c)] should only carry SSIDs not indicated in the step c) probe request. The basic access procedure (10.3.4.2) is performed prior to each probe request transmission.

11.1.4.3.4 Criteria for sending a response

Change item g) in the lettered list of 11.1.4.3.4 as follows:

- g) The STA is not a mesh STA and none of the following criteria are met:
 - 1) The SSID in the Probe Request frame is the wildcard SSID.
 - 2) The SSID in the Probe Request frame matches the SSID of the STA's BSS.
 - 3) The STA is an AP that is in the same co-located AP set as a 6 GHz AP, the SSID in the Probe Request frame matches the SSID of the 6 GHz AP, and the STA reports the co-located 6 GHz AP in Beacon and Probe Response frames (see 11.53).
 - 4) ~~3)~~ The STA is a member of a multiple BSSID set, and the SSID in the Probe Request frame matches any of the SSIDs of the members of that multiple BSSID set.

- 5) ~~4) The dot11SSIDListImplemented is true, and the SSID List element is present in the Probe Request frame and includes the SSID of the STA's BSS.~~
- 6) dot11SSIDListImplemented is true, the STA is an AP that is in the same co-located AP set as a 6 GHz AP, the SSID List element is present in the Probe Request frame and includes the SSID corresponding to the co-located 6 GHz AP, and the AP reports the co-located 6 GHz AP in Beacon and Probe Response frames (see 11.53).
- 7) dot11ShortSSIDListImplemented is true, and the Short SSID List element is present in the Probe Request frame and includes the Short SSID field corresponding to the SSID of the STA's BSS.
- 8) dot11ShortSSIDListImplemented is true, the STA is an AP that is in the same co-located AP set as a 6 GHz AP, the Short SSID List element is present in the Probe Request frame and includes the Short SSID field corresponding to the SSID of the 6 GHz AP, and the AP reports this 6 GHz AP in its Beacon and Probe Response frames (see 11.53).

Insert the following item into the numbered list after the third paragraph (“A FILS STA shall not response”) in 11.4.3.2.3, and renumber the subsequent list items accordingly:

- 4) If the FILS Criteria field is present in the FILS Requests Parameters element and the PHY Support Criterion of the FILS Criteria field of the FILS Request Parameters element is 3 and the responding STA is not HE capable.

11.1.4.3.11 Enhanced FILS active scanning to preferred AP

Insert the following elements into the dashed list after the first paragraph (“A FILS non-AP STA may”) in 11.1.4.3.11 immediately before Vendor Specific element:

- TWT element
- MU EDCA Parameter Set element
- NDP Feedback Report Parameter set element
- UORA Parameter Set element
- Spatial Reuse Parameter Set element
- HE BSS Load element
- Quiet Time Parameter element

11.2 Power management

11.2.3 Power management in a non-DMG infrastructure network

11.2.3.2 Non-AP STA power management modes

Change the first paragraph in 11.2.3.2 as follows:

A non-AP STA can be in one of two power management modes:

- Active mode: The STA receives and transmits frames at any time if the STA is in awake state. The A non-HE STA remains in the awake state. An HE STA remains in the awake state, unless the STA is unavailable. A STA that is unavailable is not capable of receiving PPDUs. A STA is permitted to be unavailable as described in 26.14.3, 26.14.1, and 26.8.4.4.
- Power save (PS) mode: The STA enters the awake state to receive or transmit frames. The STA remains in the doze state otherwise.

11.2.3.6 AP operation

Change the first paragraph in 11.2.3.6 as follows:

An AP shall maintain for each currently associated STA a Power Management status that indicates in which power management mode the STA is currently operating. APs that implement and signal their support of APSD shall maintain for each currently associated STA an APSD and an access policy status that indicates whether the STA is presently using APSD and shall maintain the schedule (if any) for the STA. An AP shall, depending on the power management mode of the STA, temporarily buffer BUs destined to the STA. An AP implementing APSD shall, if a STA is using APSD and is in PS mode, temporarily buffer BUs destined to that STA. No BUs addressed directly to STAs operating in the active mode shall be buffered for power management reasons. An HE AP should not transmit to an HE STA if the STA might be unavailable, as defined in 26.8.4.4 and 26.14.3, unless the transmission is solicited by the STA.

Change items f) and g) in the lettered list of 11.2.3.6 as follows:

- f) When dot11FMSActivated is false, the AP shall transmit all buffered non-GCR-SP non-SYNRA group addressed BUs immediately after every DTIM or during broadcast TWT SPs within that beacon interval as defined in 26.8.3.2.

When dot11FMSActivated is true and the AP has established an FMS delivery interval for a multicast stream, the AP shall transmit all non-GCR-SP non-SYNRA group addressed BUs belonging to that particular FMS stream immediately after the DTIM that has the Current Count field of the FMS Counter field set to 0 for that particular FMS stream or during broadcast TWT SPs within that beacon interval as defined in 26.8.3.2.

The More Data subfield of each group addressed frame shall be set to indicate the presence of further buffered non-GCR-SP group addressed BUs that will be delivered using MPDUs with an RA other than a SYNRA. If the AP is unable, before the primary or secondary TBTT following the DTIM, to transmit all of the buffered non-GCR-SP group addressed BUs that will be delivered using MPDUs with an RA other than a SYNRA, then the AP shall set the bit for AID 0 (zero) in the TIM element to 1 for a single BSSID or set the corresponding group address bit to 1 for multiple BSSIDs, as defined in 9.4.2.5, and when dot11FMSActivated is true, shall set the appropriate bits in the FMS Descriptor element as described in 9.4.2.74 to indicate for which non-GCR-SP non-SYNRA group addresses there are still buffered BUs, until all buffered non-GCR-SP group addressed BUs that will be delivered using MPDUs with an RA other than a SYNRA have been transmitted.

When the AP transmits an STBC DTIM or TIM Beacon frame, the AP shall retransmit all non-GCR-SP group addressed BUs that will be delivered using MPDUs with an RA other than a SYNRA and that were transmitted following the non-STBC DTIM or TIM Beacon frame, except that they are transmitted using the basic STBC MCS. It may be the case that a complete set of buffered non-GCR-SP non-SYNRA group addressed BUs is sent over a period of time during which non-STBC and STBC transmissions are interleaved, but the transition from non-STBC group addressed transmissions to STBC group addressed transmissions shall be preceded by the transmission of an STBC Beacon frame and the transition from STBC group addressed transmissions to non-STBC group addressed transmissions shall be preceded by the transmission of a non-STBC Beacon frame.

- g) When the AP receives a PS-Poll frame from a STA that is in PS mode, it shall forward to the STA a buffered BU. The AP shall respond after a SIFS either with a Data or Management frame, or with an Ack frame, in which case the corresponding Data or Management frame is delayed. Until the transmission of this BU either has succeeded or is presumed failed (when maximum retries are exceeded), the AP shall acknowledge but ignore all PS-Poll frames from the same STA. This prevents a retried PS-Poll frame from being treated as a new request to deliver a buffered BU.

For a STA using U-APSD, the AP transmits one BU destined for the STA from any AC that is not delivery-enabled in response to a PS-Poll frame from the STA. The AP should transmit the BU from

the highest priority AC that is not delivery-enabled and that has a buffered BU. When all ACs associated with the STA are delivery-enabled, the AP transmits one BU from the highest priority AC that has a BU. Upon receiving a PS-Poll frame, the S1G AP that intends to respond with immediate Data frame may use the RTS/CTS scheme to protect the transmission of the frame.

An S1G AP that sends an acknowledgment frame of type (NDP) Ack or NDP PS-Poll-Ack in response to an (NDP)PS-Poll/trigger frame that is received from an S1G STA shall set the More Data subfield of the acknowledgment frame to 0 when no BU is buffered for the STA; otherwise, it shall set it to 1. The reception of the acknowledgment frame provides the following indications to the S1G STA:

- 1) If the More Data subfield is equal to 0, it indicates that no service period starts for the STA and that it may enter the doze state,
- 2) If the More Data subfield is equal to 1, it indicates that a service period starts for the STA after a time T, starting from the end of the acknowledgment frame, after which the S1G STA shall remain in the awake state until a frame is received from the S1G AP that has the EOSP subfield equal to 1. The time T is equal to one of the following:
 - i) 0 if the acknowledgment frame is an Ack frame or is an NDP (PS-Poll-)Ack frame with the Idle Indication subfield equal to 0.
 - ii) The value indicated in the Duration field of the frame if the frame is an NDP (PS-Poll-)Ack frame with the Idle Indication subfield equal to 1.

For a STA in PS mode and not using U-APSD, the AP shall set the More Data subfield of the response Data or Management frame to 1 to indicate the presence of further buffered BUs (not including the BU currently being transmitted) for the polling STA.

For a STA using U-APSD, the AP shall set the More Data subfield to 1 to indicate the presence of further buffered BUs (not including the BU currently being transmitted) that do not use delivery-enabled ACs. When all ACs associated with the STA are delivery-enabled, the AP shall set the More Data subfield to 1 to indicate the presence of further buffered BUs (not including the BU currently being transmitted) using delivery-enabled ACs.

If there are buffered BUs to transmit to the STA, the AP may set the More Data bit in a QoS +CF-Ack frame to 1 in response to a QoS Data frame to indicate that it has one or more pending BUs buffered for the PS STA identified by the RA in the QoS +CF-Ack frame. An AP may also set the More Data bit in an Ack frame to 1 in response to a QoS Data frame to indicate that it has one or more pending BUs buffered for the PS STA identified by the RA in the Ack frame, if that PS STA has set the More Data Ack subfield in the QoS Info field QoS Capability element to 1. An HE AP may also set the More Data bit in a BlockAck or Multi-STA BlockAck frame to 1 to indicate that it has one or more pending BUs buffered for the HE PS STA identified by the RA in the BlockAck or Multi-STA BlockAck frame, if that HE PS STA has set the More Data Ack subfield in the QoS Info field to 1. An HE AP indicates support for sending Ack, BlockAck, or Multi-STA BlockAck frames with a nonzero More Data subfield by setting the More Data Ack subfield to 1 in the QoS Info field of frames it transmits.

Unless indicated above, the AP shall set the More Data bit to 0.

11.2.3.7 Receive operation for STAs in PS mode

Change item e) in the lettered list of 11.2.3.7 as follows:

- e) When dot11FMSActivated is false and ReceiveDTIMs is true, the STA shall wake up early enough to be able to receive either every non-STBC DTIM or every STBC DTIM sent by the AP of the BSS. When dot11FMSActivated is true and ReceiveDTIMs is true and the STA has been granted by the AP an alternate delivery interval for a multicast stream, the STA shall wake up before the non-STBC

DTIM or STBC DTIM having Current Count of FMS Counter field set to 0 for that particular FMS stream.

A STA that stays awake to receive group addressed BUs shall elect to receive all group addressed non-STBC transmissions or all group addressed STBC transmissions and remain awake until the More Data subfield of the appropriate type (non-STBC or STBC) of group addressed BUs indicates there are no further buffered group addressed BUs of that type, or until a TIM is received indicating there are no more buffered group addressed BUs of that type, or until an FMS Descriptor element is received indicating that there are no further buffered group addressed BUs for which the STA has previously received an FMS Response element in a frame that has a value in Address 1 that matches its MAC address or that has an Address 1 value that is a group address corresponding to a group of which it is a member and that was transmitted by the AP with which it is associated and which had an Element Status value in the FMS Status subelement of “Accept”. If a STA receives a QoS +CF-Ack frame from its AP with the More Data bit equal to 1, then the STA shall operate ~~exactly as~~ if it received a TIM with its AID bit equal to 1. If a STA has set the More Data Ack subfield in the QoS Info field QoS Capability element to 1, then if it receives an Ack frame from its AP with the More Data bit equal to 1, the STA shall operate ~~exactly as~~ if it received a TIM with its AID bit equal to 1. If an HE STA has set the More Data Ack subfield in the QoS Info field to 1, then if it receives a BlockAck or Multi-STA BlockAck frame from its AP with the More Data bit equal to 1, the STA shall operate as if it received a TIM with its AID bit equal to 1. For example, a STA that is using the PS-Poll delivery method shall issue a PS-Poll frame to retrieve a buffered BU. See also 10.3.6.

11.2.3.9 STAs operating in active mode

Change 11.2.3.9 as follows:

A STA operating in this mode shall have its receiver activated continuously, unless the STA is allowed to be temporarily unavailable through the opportunistic power save mechanism defined in 26.14.3 or through the intra-PPDU power save mechanism defined in 26.14.1 or 26.8.4.4; such STAs do not need to interpret the TIM elements in Beacon frames.

11.2.3.15 TIM Broadcast

Change the third paragraph in 11.2.3.15 as follows (including splitting the paragraph into two paragraphs and inserting NOTE 1), and renumber the subsequent notes in this subclause accordingly:

A non-AP STA may activate the TIM broadcast service by including a TIM Broadcast Request element in a TIM Broadcast Request frame, Association Request frame, or Reassociation Request frame that is transmitted to the AP, which specifies the requested interval between TIM frame transmissions (the TIM broadcast interval). On receipt of a properly formatted TIM Broadcast Request element in a TIM Broadcast Request frame, Association Request frame, or Reassociation Request frame, the AP shall include a TIM Broadcast Response element in the corresponding TIM Broadcast Response frame, Association Response frame, or Reassociation Response frame, when `dot11TIMBroadcastActivated` is true. A non-AP STA shall transmit a TIM Broadcast Request only if the associated AP has indicated support for TIM Broadcast by setting the TIM Broadcast field of the Extended Capabilities elements that it transmits to 1.

NOTE 1—An OPS AP that transmits TIM frames as described in 26.14.3 is expected to encode the TIM bits such that an associated non-AP STA that does not support OPS operation can use the information received in the TIM frame as it would do when receiving a TIM frame transmitted following the TIM Broadcast procedure.

When the requested TIM broadcast interval is acceptable, the AP shall include a TIM Broadcast Response element specifying the requested TIM broadcast interval and a Status field indicating “Accept” when no TSF timestamp is present in the TIM frames, or “Accept, timestamp present in TIM frames” when a TSF

timestamp is present in the TIM frames. When the AP overrides the requested TIM broadcast interval, it shall include a TIM Broadcast Response element specifying a different TIM broadcast interval and a Status field indicating “Overridden” when no TSF timestamp is present in the TIM frames, or “Overridden, timestamp present in TIM frames” when a TSF timestamp is present in the TIM frames, and include in the TIM Broadcast Response element the smallest TIM broadcast interval that is currently active. Otherwise, the AP shall include a TIM Broadcast Response element with a Status field indicating “Denied.” The Status field in a TIM Broadcast Response element that is included in an Association Response frame or Reassociation Response frame has implications only for the TIM Broadcast negotiation.

Change the now 11th paragraph in 11.2.3.15 as follows:

The AP shall increase the value (modulo 256) of the Check Beacon field in the next transmitted TIM frame(s) when a critical update occurs to any of the elements inside the Beacon frame. The following events shall classify as a critical update:

- a) Inclusion of a Channel Switch Announcement element
- b) Inclusion of an Extended Channel Switch Announcement element
- c) Modification of the EDCA parameters element
- d) Inclusion of a Quiet element
- e) Modification of the DSSS Parameter Set
- f) Modification of the HT Operation element
- g) Inclusion of a Wide Bandwidth Channel Switch element
- h) Inclusion of a Channel Switch Wrapper element
- i) Inclusion of an Operating Mode Notification element
- j) Inclusion of a Quiet Channel element
- k) Modification of the VHT Operation element
- l) Modification of the HE Operation element
- m) Insertion of a Broadcast TWT element
- n) Inclusion of the BSS Color Change Announcement element
- o) Modification of the MU EDCA Parameter Set element
- p) Modification of the Spatial Reuse Parameter Set element
- q) Modification of the UORA Parameter Set element

Insert NOTE 4 into 11.2.3.15 after the now 11th paragraph as follows:

NOTE 4—Modification of an element means that at least one value of a field in the element is changed. Inclusion of an element means that the element is included in a Beacon frame. The insertion of an element means that the element was not present in the previous Beacon frame, is present in the current Beacon frame, and will be carried in the next Beacon frame.

11.2.6 SM power save

Change 11.2.6 as follows:

A STA consumes power on all active receive chains, even though they are not necessarily required for the actual frame exchange. The SM power save feature allows a non-AP HT STA in an infrastructure BSS to operate with only one active receive chain for a significant portion of time.

The STA controls which receive chains are active through the PHY-RXCONFIG.request primitive specifying a PHYCONFIG_VECTOR parameter ACTIVE_RXCHAIN_SET that indicates which of its receive chains should be active.

The basic rules for a STA are defined below. Additional rules for an HE STA in dynamic SM power save mode that sets the HE Dynamic SM Power Save subfield to 1 in the HE MAC Capabilities Information field in the HE Capabilities element it transmits are defined in 26.14.4.

In dynamic SM power save mode, the STA enables its multiple receive chains when it receives the start of a frame exchange sequence addressed to it. Such a frame exchange sequence shall start with a single-spatial stream individually addressed frame that is not a Trigger frame, that requires an immediate response, and that is addressed to the STA in dynamic SM power save mode. An RTS/CTS sequence may be used for this purpose. The STA shall, subject to its spatial stream capabilities (see 9.4.2.55.4 and 9.4.2.157.3) and operating mode (see 11.40), be capable of receiving a PPDU that is sent using more than one spatial stream a SIFS after the end of its response frame transmission. ~~The STA switches to the multiple receive chain mode when it receives the frame addressed to it and switches back~~ may switch back to the single receive chain mode immediately when after the end of the frame exchange sequence ends.

NOTE—A STA in dynamic SM power save mode cannot distinguish between an RTS/CTS sequence that precedes a MIMO transmission and any other RTS/CTS and, therefore, always enables its multiple receive chains ~~when it receives~~ if it responds to the RTS addressed to it.

The STA can determine the end of the frame exchange sequence through any of the following:

- It receives an individually addressed frame addressed to another STA.
- It receives a frame with a TA that differs from the TA of the frame that started the TXOP.
- It receives a PPDU and classifies the PPDU as inter-BSS PPDU (see 26.2.2).
- It receives an HE MU PPDU where the RXVECTOR parameter BSS_COLOR is the BSS color of the BSS in which the STA is associated, the RXVECTOR parameter does not have any STA_ID of an RU that identifies the STA as the recipient or one of the recipients of the RU (see 26.11.1), and the BSS Color Disabled subfield in the most recently received HE Operation element from the AP with which the STA is associated is 0.
- The CS mechanism (see 10.3.2.1) indicates that the medium is idle at the TxPIFS slot boundary (defined in 10.3.7).

In static SM power save mode, the STA maintains only a single receive chain active.

The STA may use the SM Power Save frame to communicate its SM power save state. The STA may also use the SM Power Save subfield in the HT Capabilities element of its (Re)Association Request frame or the SM Power Save subfield in the HE 6 GHz Band Capabilities element of its (Re)Association Request frame to achieve the same purpose. Using the (Re)Association Request frame ~~The latter~~ allows the STA to use only a single receive chain immediately after (re)association.

A STA that has one or more TDLS links shall not operate in SM power save mode.

Changes to the number of active receive chains are made only after the SM power save mode indication has been successfully delivered (i.e., by acknowledgment of a frame carrying the HT Capabilities element, ~~or~~ by acknowledgment of a frame carrying the HE 6 GHz Band Capabilities element, or by acknowledgment of an SM Power Save frame). The SM power save mode indication shall be transmitted using an individually addressed frame.

11.3 STA authentication and association

11.3.3 Frame filtering based on STA state

Change item a) in the lettered list of 11.3.3 as follows:

- a) Class 1 frames
 - 1) Control frames
 - i) RTS
 - ...
 - xiii) In an HE BSS Basic Trigger frame and Multi-STA BlockAck frame

11.3.5 Association, reassociation, and disassociation

11.3.5.3 AP or PCP association receipt procedures

Insert item h1) into the lettered list of 11.3.5.3 after item h):

- h1) The SME shall refuse an association request from an HE STA that does not support all of the <HE-MCS, NSS> tuples indicated by the Basic HE-MCS And NSS Set field of the HE Operation parameter in the MLME-START.request primitive.

11.3.5.5 AP or PCP reassociation receipt procedures

Insert item h1) into the lettered list of 11.3.5.5 after item h):

- h1) The SME shall refuse a reassociation request from an HE STA that does not support all of the <HE-MCS, NSS> tuples indicated by the Basic HE-MCS And NSS Set field of the HE Operation parameter in the MLME-START.request primitive.

11.3.8 Neighbor report information upon rejection with suggested BSS transition

Insert the following paragraph at the end of 11.3.8:

An HE STA that has received from an HE AP an Authentication or (Re)Association Response frame that has the Status Code field set to REJECTED_WITH_SUGGESTED_BSS_TRANSITION and that includes one or more Neighbor Report elements for BSSs that are part of the ESS of the HE AP shall, if it re-attempts to associate with the ESS, select an AP from (one of) the Neighbor Report element(s).

11.4 TS operation

11.4.4 TS setup

11.4.4.4 TS setup procedures for both AP and non-AP STA initiation

Change the first paragraph in 11.4.4.4 as follows:

The non-AP STA's SME decides that a TS needs to be created or non-AP HE STA traffic characteristics and QoS requirements need to be provided. The mechanism to provide traffic characteristics and QoS requirements is described in 26.5.8. How it does this, and how it selects the TSPEC or DMG TSPEC parameters, is beyond the scope of this standard. The SME generates an MLME-ADDTTS.request primitive containing a TSPEC or DMG TSPEC. A TSPEC or DMG TSPEC may also be generated autonomously by the MAC without any initiation by the SME.

11.7 TPC procedures

11.7.1 General

Change the first paragraph in 11.7.1 as follows:

Regulations ~~that apply to the 5 GHz band~~ in most regulatory domains require WLANs operating in the 5 GHz band or 6 GHz band to use transmitter power control, involving specification of a regulatory maximum transmit power and a mitigation requirement for each allowed channel. This standard describes such a mechanism, referred to as *transmit power control (TPC)*.

11.7.2 Association based on transmit power capability

Change the fourth paragraph and note in 11.7.2 as follows:

If a STA sends a Country element, a Power Constraint element, and a Transmit Power Envelope element, where the interpretation of the Maximum Transmit Power Level field in the Country element for a 20 MHz or 40 MHz Subband Triplet field is the same as the ~~Local~~-Maximum Transmit Power ~~Unit~~-Interpretation subfield, then at least one of local power constraints indicated by the ~~Local~~-Maximum Transmit Power For 20 MHz and ~~Local~~-Maximum Transmit Power For 40 MHz subfields in the Transmit Power Envelope element shall be the same as the indicated local power constraint expressed by the combination of Country element and Power Constraint element.

NOTE—An example of the interpretation of the Maximum Transmit Power Level field in the Country element for a 20 MHz or 40 MHz Subband Triplet field being the same as the ~~Local~~-Maximum Transmit Power ~~Unit~~-Interpretation subfield is when both are EIRP.

11.7.3 Peering based on transmit power capability

Change the fourth paragraph in 11.7.3 as follows:

If a STA sends a Country element, a Power Constraint element, and a Transmit Power Envelope element, where the interpretation of the Maximum Transmit Power Level field in the Country element for a 20 MHz or 40 MHz Subband Triplet field is the same as the ~~Local~~-Maximum Transmit Power ~~Unit~~-Interpretation subfield, then at least one of the local power constraints indicated by ~~a) the~~ ~~Local~~-Maximum Transmit Power For 20 MHz subfield or ~~b) the~~ ~~Local~~-Maximum Transmit Power For 40 MHz subfield in the Transmit Power Envelope element shall be the same as the indicated local power constraint expressed by the combination of Country element and Power Constraint element.

11.7.5 Specification of regulatory and local maximum transmit power levels

Change 11.7.5 as follows:

A STA shall determine a regulatory maximum transmit power for the current channel by selecting the minimum of the following:

- Any regulatory maximum transmit power received in a Country element from the AP in its BSS, a PCP in its PBSS, another STA in its IBSS, or a neighbor peer mesh STA in its MBSS.
- If the STA is extended spectrum management capable, any regulatory client maximum transmit power received in a Transmit Power Envelope element from the AP in its BSS, another STA in its IBSS, or a neighbor peer mesh STA in its MBSS.
- Any regulatory maximum transmit power for the channel in the current regulatory domain known by the STA from other sources.

A STA shall determine a local maximum transmit power for the current channel by selecting the minimum of the following:

- Unless the STA is extended spectrum management capable and has received a Transmit Power Envelope element for a channel width of 20 MHz and 40 MHz, any local maximum transmit power received in the combination of a Country element and a Power Constraint element from the AP in its BSS, a PCP in its PBSS, another STA in its IBSS, or a neighbor peer mesh STA in its MBSS.
- If the STA is extended spectrum management capable, any local maximum transmit power received in a Transmit Power Envelope element from the AP in its BSS, another STA in its IBSS, or a neighbor peer mesh STA in its MBSS.
- Any local maximum transmit power for the channel in the current regulatory domain known by the STA from other sources.

NOTE 1—A STA might receive a maximum transmit power in a Transmit Power Envelope element from the AP in its BSS, another STA in its IBSS, or a neighbor peer mesh STA in its MBSS in various management frames, including Beacon frames, Probe Response frames, FILS Discovery frames, and (prior to a channel switch) New Transmit Power Envelope elements (in Channel Switch Wrapper elements, Future Channel Guidance elements, Channel Switch Announcement elements/frames, or Extended Channel Switch Announcement elements/frames). Other sources from which a STA might receive a maximum transmit power for a channel include Reduced Neighbor Report elements (20 MHz PSD subfield) sent by a (co-located) AP. If this information is received by a STA, any requirements on its usage depend on local regulations known at the STA (see E.2).

NOTE 2—The determination of a maximum transmit power from Transmit Power Envelope element(s) is specified in 10.22.4.

The Local Power Constraint field of any transmitted Power Constraint element and each ~~Local~~ Maximum Transmit Power For X MHz subfield (where $X = 20, 40, 80$, or $160/80+80$) in the Transmit Power Envelope element shall each be set to its respective value that allows the mitigation requirements to be satisfied in the current channel.

~~NOTE 4.3~~—The local maximum transmit power for the channel needs to meet the mitigation requirements for the channel in the current regulatory domain. The conservative approach is to set the local maximum transmit power level equal to the regulatory maximum transmit power level minus the mitigation requirement. However, it might be possible to satisfy the mitigation requirement using a higher local maximum transmit power level. A lower local maximum transmit power level might be used for other purposes (e.g., range control, reduction of interference).

The regulatory and local maximum transmit powers may change in a STA during the life of an infrastructure BSS and an MBSS. However, network stability needs to be considered when deciding how often or by how much these maximums are changed. The regulatory and local maximum transmit powers shall not change during the life of an IBSS.

An AP, IBSS STA, or mesh STA that is not operating in the 6 GHz band shall advertise the regulatory maximum transmit power for that STA's operating channel in Beacon frames and Probe Response frames

using a Country element. An AP, IBSS STA, or mesh STA shall advertise the local maximum transmit power for that STA's operating channel in Beacon frames and Probe Response frames using the combination of a Country element and a Power Constraint element.

If an AP, IBSS STA, or mesh STA that is not operating in the 6 GHz band is extended spectrum management capable, it shall advertise the local maximum transmit power for that STA's operating channel in Beacon frames and Probe Response frames using one Transmit Power Envelope element for each distinct value of the ~~Local~~ Maximum Transmit Power ~~Unit~~ Interpretation subfield that is supported by the BSS, IBSS, or MBSS, respectively. Each Transmit Power Envelope element shall include a local power constraint for all channel widths supported by the BSS.

If an AP, IBSS STA, or mesh STA is operating in the 6 GHz band, it shall include Transmit Power Envelope element(s) in Beacon and Probe Response frames as follows:

- a) For each distinct combination of values of the Maximum Transmit Power Interpretation subfield and Maximum Transmit Power Category subfield that are supported by the BSS, IBSS, or MBSS, respectively, one Transmit Power Envelope element is included that indicates the regulatory client maximum transmit power for the STA's operating channel.
- b) At a minimum, a regulatory client maximum transmit power is indicated where the Maximum Transmit Power Category subfield is set to 0 (Default). If different regulatory limits apply to different categories, the value of the limit advertised for Default category shall be the minimum of those limits.
- c) If a local maximum transmit power is lower than a regulatory client maximum transmit power for the same combination of values of the Maximum Transmit Power Interpretation subfield and Maximum Transmit Power Category subfield, a Transmit Power Envelope element is included that indicates that local maximum transmit power.
- d) Each Transmit Power Envelope element that is included indicates a power constraint for all channel widths supported by the BSS (if an EIRP constraint is advertised) or for all 20 MHz channels within the bandwidth of the BSS (if an EIRP PSD constraint is advertised).

NOTE 4—The requirement in item b) above ensures that a STA, even if it does not know how to interpret the Regulatory Info subfield (in the HE Operation element) and/or nondefault values of the Maximum Transmit Power Category subfield (in Transmit Power Envelope elements) for the current country, can determine a regulatory client maximum transmit power for use with the AP. This default value might be lower than the actual regulatory client maximum transmit power for specific categories.

NOTE 5—The regulatory client maximum transmit power is the regulatory limit for client devices, such as non-AP STAs, which might be different from the regulatory limit for APs.

NOTE 6—STAs operating in the 6 GHz band are extended spectrum management capable (see 10.22.3).

NOTE 7—If a Transmit Power Envelope element in a Beacon or Probe Response frame in the 6 GHz band indicates EIRP PSD limits, the value of N (see Table 9-277) is either equal to 0 or equal to the BSS bandwidth, in megahertz, divided by 20 (i.e., 1, 2, 4, or 8 for 20, 40, 80, or 160/80+80 MHz BSS bandwidth, respectively).

NOTE 8—An AP, IBSS STA, or mesh STA does not need to send Transmit Power Envelope elements indicating maximum transmit EIRP if it sends Transmit Power Envelope elements indicating maximum transmit PSD and those PSD values are sufficient to ensure regulatory limits on total EIRP are always met for all transmission bandwidths of STAs in the BSS.

STAs that are extended spectrum management capable and have `dot11RadioMeasurementActivated` equal to true should be able to reduce their EIRP to 0 dBm.

NOTE 2.9—When the local maximum transmit power is set by an AP for radio resource management, a typical low value for the local power constraint is 0 dBm. A STA that cannot reduce its transmit power to this level or below will not be able to associate to the AP.

The PCP in a PBSS shall advertise both the regulatory maximum transmit power and the local maximum transmit power for its operating channel in Announce and Probe Response frames (using a Country element for the regulatory maximum and a combination of a Country element and a Power Constraint element for the

local maximum). In addition, it should advertise both regulatory and local maximum transmit power in DMG Beacon frames.

When `dot11SpectrumManagementRequired` is false and `dot11RadioMeasurementActivated` is true, an AP or a PCP may include a Power Constraint element and a Transmit Power Envelope element in Beacon, DMG Beacon, Announce, and Probe Response frames.

11.7.6 Transmit power selection

Change 11.7.6 as follows:

A STA may select any transmit power for transmissions in a channel within the following constraints:

- A STA shall determine a regulatory maximum transmit power and a local maximum transmit power for a channel in the current regulatory domain before transmitting in the channel.
- An AP shall use a transmit power less than or equal to the regulatory maximum transmit power level for the channel. The AP shall also meet any regulatory mitigation requirement.
- A non-AP STA shall use a transmit power less than or equal to the minimum or the local maximum transmit power level and regulatory client maximum transmit power for the channel.

11.10 Radio measurement procedures

11.10.14 Multiple BSSID set

Change the first paragraph and note in 11.10.14 as follows:

A multiple BSSID set is characterized as follows:

- All members of the set use a common operating class, channel, channel access functions, receive antenna connector, and transmit antenna connector.
- The size of the set is 2^n , for a selected value of n ; $1 \leq n \leq 8$.
 - $1 \leq n \leq 8$ if `dot11MultiBSSIDImplemented` is true.
 - $1 \leq n \leq 46$ if `dot11MultiBSSIDImplemented` (if present) is false and `dot11RMMeasurementPilotActivated` is nonzero.
- All BSSIDs within the multiple BSSID set are assigned in a way that they are not available as MAC addresses for STAs using a different operating class, channel, receive antenna connector, or transmit antenna connector.

NOTE—For example, if the APs within BSSs with BSSIDs 16, 17, and 27 share the same operating class, channel, receive antenna connector, and transmit antenna connector, and the range of MAC addresses from 16 to 31 inclusive is ~~are~~ not assigned to other STAs using a different receive antenna connector or transmit antenna connector, then the BSSIDs 16, 17, and 27 are members of a multiple BSSID set. The set is described by $n = 4$ ($2^n = 16$) with BSSIDs in the range 16 to 31. The set cannot be described by $n = 8$, for instance, since at least one of the BSSIDs in the range 0 to 255 might be used as a BSSID by an AP that does not share the same operating class, channel, receive antenna connector, and transmit antenna connector.

11.20 Tunnelled direct-link setup

11.20.1 General

Change the seventh paragraph in 11.20.1 as follows:

Features ~~that are not supported by the BSS but~~ that are supported by both TDLS peer STAs may be used on a TDLS direct link between those STAs regardless of whether they are supported by the BSS. An example is the use of an HT-MCS on a TDLS direct link between HT STAs when these STAs are associated with a non-HT BSS. Features that are supported by the BSS shall follow the BSS rules when they are used on a TDLS direct link on the base channel. The channel width of a TDLS direct link on the base channel shall not exceed the channel width of the BSS of which the TDLS peer STAs are members, except when the TDLS Wider Bandwidth subfield in the Extended Capabilities element of the TDLS Setup Request frame or the TDLS Setup Response frame is 1 for both TDLS peer STAs.

Insert the following paragraph into 11.20.1 before the last paragraph (“The SIG Operation element”):

The HE Operation element shall be present in a TDLS Setup Confirm frame when both STAs are HE capable.

11.21 Wireless network management procedures

11.21.2 Event request and report procedures

11.21.2.1 Event request and event report

Change the first paragraph in 11.21.2.1 as follows:

The Event Request and Event Report frames enable network real-time diagnostics. A STA with dot11EventsActivated equal to true shall support event requests and reports and shall set to 1 the Event field of the Extended Capabilities elements that it transmits. If dot11EventsActivated is true and the Event Type field is neither BSS Color Collision nor BSS Color In Use, a STA shall log all Transition, RSNA, peer-to-peer, and WNM log events, including the corresponding TSF, UTC Offset, and Event Time Error. An HE STA that has dot11EventsActivated equal to true and reports BSS color collisions shall log all BSS color collision events, including the TSF value when the STA finished logging the events that are reported (see 11.21.2.7).

Insert the following paragraph into 11.21.2.1 after the first paragraph:

A STA with either dot11AutonomousBSSColorCollisionReportingImplemented equal to true or dot11AutonomousBSSColorInUseReportingImplemented equal to true shall set the Event field of the Extended Capabilities elements that it transmits to 1.

Insert the following subclauses (11.21.2.7 and 11.21.2.8) after 11.21.2.6:

11.21.2.7 BSS color collision event

The BSS color collision event report enables a non-AP HE STA to inform its associated AP whether a BSS color collision has occurred. The report carries information about the BSS color used by OBSSs that the reporting STA is able to detect (see 26.17.3.5.2).

11.21.2.8 BSS color in use event

If a non-AP HE STA communicates with a peer STA with a BSS color that is different from the BSS color used by its associated AP, the non-AP HE STA may send a BSS color in use event report to its associated AP in which the Event Report field has the BSS color used in the communication with the peer STA.

A non-AP HE STA shall send a BSS color in use event report to its associated AP with the Event Report field set to 0 to cancel a previously sent BSS color in use event report.

An AP shall not transmit frames to a non-AP HE STA during a TXOP, if it has received a BSS color in use event report from that non-AP HE STA with a nonzero BSS color in the Event Report field and the AP ignores an inter-BSS PPDU with the same BSS color value as the one carried in the Event Report field to obtain a TXOP by following the procedure in 26.10.2.2 and 26.10.2.3.

11.21.6 Fine timing measurement (FTM) procedure

11.21.6.4 Measurement exchange

Change the 10th paragraph in 11.21.6.4 as follows:

For the Fine Timing Measurement frames transmitted during the FTM session:

- The responding STA shall not use a bandwidth wider than it indicated in the initial Fine Timing Measurement frame.
- The responding STA shall not use an HE format if it indicated VHT, HT-mixed, or non-HT format in the initial Fine Timing Measurement frame.
- The responding STA shall not use a VHT format if it indicated HT-mixed or non-HT format in the initial Fine Timing Measurement frame.
- The responding STA shall not use an HT format if it indicated non-HT format in the initial Fine Timing Measurement frame.

11.21.7 BSS transition management for network load balancing

11.21.7.1 BSS transition capability

Change the third paragraph in 11.21.7.1 as follows:

Implementation of BSS transition management is optional for a WNM STA that is not a non-AP HE STA. ~~When~~ If dot11BSSTransitionActivated is true, dot11WirelessManagementImplemented shall be true. A STA with dot11BSSTransitionActivated equal to true shall support BSS transition management and shall set to 1 the BSS Transition field of the Extended Capabilities elements that it transmits. A non-AP HE STA shall have dot11BSSTransitionImplemented and dot11BSSTransitionActivated equal to true.

11.21.7.4 BSS transition management response

Change the second paragraph in 11.21.7.4 as follows:

The STA's SME may include the result of its BSS transition decision in the Target BSSID field and BTM Status Code field in the MLME-BTM.response primitive. A BTM Status Code field set to a value of 0 (i.e., Accept) indicates the STA will transition from the current BSS. If a non-HE STA's SME receives an

MLME-BTM.indication primitive indicating a BSS transition management request ~~with which that~~ it is unable to comply ~~with~~, it may issue an MLME-BTM.response primitive with a status code indicating rejection.

Insert the following subclause (11.21.7.5) after 11.21.7.4:

11.21.7.5 Planned ESS

The AP may transmit an ESS Report element (see 9.4.2.256) to indicate whether it is in a planned ESS, to assist associated STAs' roaming. If it indicates that it is in a planned ESS, it indicates whether it is at the physical edge of the ESS and can provide a recommendation on the RSSI level to consider for BSS transition.

If the AP transmits an ESS Report element, it shall set the Planned ESS subfield in the ESS Information field to 1 if it is part of an ESS that is planned with several BSSs in overlapping configuration, whereby an associated STA may adjust its BSS transition algorithms accordingly. Otherwise, it shall set the Planned ESS subfield to 0.

If the Planned ESS subfield is 1, then the AP shall set the Edge Of ESS subfield in the ESS Information field of the ESS Report element to 1 if the AP's BSS is at the edge of an ESS (e.g., exit of a building). Otherwise, it shall set the Edge Of ESS subfield to 0.

NOTE 1—A non-AP STA could then prepare for more aggressive roaming, or the device in which the non-AP STA is located could then prepare for switching to a different system. However, how the non-AP STA uses the edge of ESS information is implementation specific and beyond the scope of this standard.

If the Planned ESS subfield is 1, then the AP shall set the Recommended BSS Transition RSSI Threshold Within ESS subfield in the ESS Information field of the ESS Report element to indicate the beacon RSSI below which an associated STA should initiate a BSS transition. Otherwise, it shall set the Recommended BSS Transition RSSI Threshold Within ESS subfield to 0.

NOTE 2—A non-AP STA could then use the Recommended BSS Transition RSSI Threshold Within ESS subfield to modify when it starts scanning for a new BSS. However, how the non-AP STA adjusts its BSS transition algorithms is implementation specific and beyond the scope of this standard.

The value of the Edge Of ESS subfield and the Recommended BSS Transition RSSI Threshold Within ESS subfield may be changed by the AP if conditions in the ESS change. An AP shall not change the value of the Planned ESS subfield over the lifetime of the BSS.

11.31 Multi-band operation

11.31.5 On-channel Tunneling (OCT) operation

Change the first paragraph in 11.31.5 as follows (including splitting the paragraph into two paragraphs):

~~OCT allows a STA of a multi-band capable device to transmit an MMPDU that was constructed by a different STA of the same device. OCT allows the following:~~

- A STA of a multi-band capable device or a STA that is in the same device as another STA to transmit or forward an MMPDU that was constructed by, addressed by, or addressed to a different STA in the same device.
- An AP to transmit or forward an MMPDU that was constructed by, addressed by, or addressed to another AP if either one of the APs sends a Reduced Neighbor Report element with a TBTT

Information field describing the other AP or if either one of the APs sends a Neighbor Report element describing the other AP, and where both APs support OCT.

An MMPDU transmitted this way is referred to as an *OCT MMPDU*. The MLME of the nontransmitting STA that constructs or is the destination of an OCT MMPDU is referred to as an *NT-MLME*. The MLME of the STA that transmits or receives an OCT MMPDU over the air is referred to as a *TR-MLME*. An NT-MLME that constructs an OCT MMPDU destined to a peer NT-MLME does so according to the capabilities of the STA that contains the peer NT-MLME.

Change the now third paragraph in 11.31.5 as follows (including spitting the paragraph into two paragraphs):

Either of the following conditions indicates that a STA supports OCT and has dot11OCTOptionImplemented equal to true:

- A STA supports OCT if the OCT Not Supported subfield within the STA's Multi-band element is 0.
- A non-AP STA supports OCT if the OCT field is equal to true in the Extended Capabilities elements it transmits.
- If a reporting AP sends a frame with a Reduced Neighbor Report element with a TBTT Information field describing a reported AP that has the OCT Recommended subfield equal to 1, then both the reporting AP and the reported AP support OCT.
- If a reporting AP sends a frame with a Neighbor Report element describing a reported AP that has the OCT Supported With Reporting AP subfield equal to 1, then both the reporting AP and the reported AP support OCT.

A STA should not perform OCT with a peer STA that does not support OCT. A STA that does not support OCT shall ignore a received OCT MMPDU.

Change the now 11th paragraph in 11.31.5 as follows:

To transmit a tunneled MMPDU, the SME of a ~~multi-band capable device STA~~ STA that supports OCT generates an OCT MLME request primitive that includes the peer Multi-band element and the local Multi-band element. If the OCT MLME request primitive is the MLME-SCAN.request primitive with the ScanType parameter set to ACTIVE, the BSSID field within the peer Multi-band element shall be set to the value of the BSSID parameter in the MLME-SCAN.request primitive, and the BSSID field within the local Multi-band element shall be set to an individual MAC address.

11.45 Fast Initial Link Setup (FILS) procedures

11.45.2 FILS Discovery frame generation and usage

11.45.2.1 FILS Discovery frame transmission

Change the third and fourth paragraph in 11.45.2.1 as follows:

If an AP transmits a FILS Discovery frame as a non-HT duplicate PPDU in an 80+80 MHz channel bandwidth, the Channel Center Frequency Segment 1 field shall be present in the FILS Discovery frame and shall be set to the channel center frequency of the frequency segment 1 for an 80+80 MHz VHT or HE operating channel.

A FILS AP should transmit FILS Discovery frame(s) in every beacon interval. The interval between the transmission of a Beacon frame and a subsequent FILS Discovery frame shall be no less than the interval

indicated in `dot11FILSFDFrameBeaconMinimumInterval`. The transmission interval between subsequent FILS Discovery frames by an AP in a beacon interval shall be no less than the interval indicated in `dot11FILSFDFrameBeaconMinimumInterval`. If `dot11FILSFDFrameBeaconMaximumInterval` is not equal to 0, and if a Beacon frame, broadcast Probe Response frame, or FD frame has not been transmitted by an AP for a period that is equal to `dot11FILSFDFrameBeaconMaximumInterval`, that AP shall queue for transmission a FD frame, broadcast Probe Response frame, or a Beacon frame, unless the next TBTT is within a duration indicated by the value of `dot11FILSFDFrameBeaconMinimumInterval`.

Insert the following text at the end of 11.45.2.1:

The Address 1 field of the FILS Discovery frame shall be set to the broadcast address.

For the APs in a multiple BSSID set, only the AP corresponding to the transmitted BSSID may transmit a FILS Discovery frame; other APs corresponding to nontransmitted BSSIDs shall not transmit a FILS Discovery frame. If `dot11MultiBSSIDImplemented` is true, then the following applies to the fields in the FILS Discovery frame:

- The SSID or Short SSID field shall be set to the SSID or short SSID, respectively, of the transmitted BSSID.
- The FILS Capability field shall be present, and the Multiple BSSIDs Presence Indicator subfield shall be set to 1.

A 6 GHz AP that is sending FILS Discovery frames shall include at least one Transmit Power Envelope element, where the Maximum Transmit Power Category subfield is set to 0 (Default), including a regulatory client transmit power limit for at least 20 MHz bandwidth (for EIRP) or at least the AP's primary 20 MHz channel (for EIRP PSD).

NOTE 1—The AP is not required to include power constraints for bandwidths greater than 20 MHz in FILS Discovery frames, even if they are supported by the BSS. Therefore, when a Transmit Power Envelope element in a FILS Discovery frame indicates EIRP PSD limits, the value of N (see Table 9-277) can take any valid value equal to or less than the BSS bandwidth, in megahertz, divided by 20.

NOTE 2—A Transmit Power Envelope element sent in a FILS Discovery frame by a 6 GHz AP can be used by a STA to determine a transmit power limit for 20 MHz PPDU's corresponding to the 6 GHz AP prior to having received a Beacon or Probe Response frame from that AP. A STA might, for example, determine a transmit power limit based on this information when sending a Probe Request frame with 20 MHz PPDU bandwidth during active scanning on the 6 GHz AP's channel.

11.49 Reduced neighbor report

Change the first paragraph in 11.49 as follows:

~~In Beacon and Probe Response frames, a Reduced Neighbor Report element may be transmitted by an AP with `dot11TVHTOptionImplemented` or `dot11FILSActivated` true. In FILS Discovery frames, a Reduced Neighbor Report element is optionally sent by a FILS AP. A Reduced Neighbor Report element contains information on neighbor APs, co-located APs, or a combination of both. An AP that operates in the 2.4 GHz or 5 GHz band and that is in the same co-located AP set as one or more 6 GHz APs shall follow the rules in 11.53 for including a Reduced Neighbor Report element in Beacon and Probe Response frames. A Reduced Neighbor Report element might not be exhaustive either by choice or by the fact that there may be neighbor APs not known to the reporting AP. An AP that intends to report neighboring or co-located APs may include more than one Reduced Neighbor Report element in a Beacon, Probe Response, or FILS Discovery frame if the reported APs do not all fit in a single Reduced Neighbor Report element; otherwise, it shall not include more than one Reduced Neighbor Report element.~~

Insert the following paragraphs and NOTE 1 into 11.49 after the first paragraph, and renumber the subsequent notes in this subclause accordingly:

An AP with dot11MultiBSSIDImplemented equal to true shall not include a Reduced Neighbor Report element in the Nontransmitted BSSID Profile subelement of the Multiple BSSID element.

NOTE 1—The Beacon, Probe Response, or FILS Discovery frame of an AP with dot11MultiBSSIDImplemented equal to true can carry a Reduced Neighbor Report element. In this case, the values of the fields in the Reduced Neighbor Report element apply to all the BSSs in the multiple BSSID set, except for the Same SSID subfield(s), whose value applies only to the BSS corresponding to the transmitted BSSID.

If an AP reported in a TBTT Information field in a Reduced Neighbor Report element is not part of a multiple BSSID set, then the BSS Parameters subfield, if included, shall have the Multiple BSSID subfield set to 0. If an AP reported in a TBTT Information field in a Reduced Neighbor Report element is a transmitted BSSID, then the BSS Parameters subfield, if included, shall have the Multiple BSSID subfield set to 1 and the Transmitted BSSID subfield set to 1. If an AP reported in a TBTT Information field in a Reduced Neighbor Report element is a nontransmitted BSSID, then the BSS Parameters subfield, if included, shall have the Multiple BSSID subfield set to 1 and the Transmitted BSSID subfield set to 0.

An HE AP with dot11MultiBSSIDImplemented equal to true may advertise one or more nontransmitted BSSIDs in the multiple BSSID set by including the Reduced Neighbor Report element in its Beacon, Probe Response, or FILS Discovery frames with the BSS Parameters subfield of the TBTT Information field containing the Co-Located subfield set to 1, the Multiple BSSID subfield set to 1, and the Transmitted BSSID subfield set to 0 and with the Operating Class and Channel Number subfields of the Neighbor AP Information field set to the operating class and channel number, respectively, of the transmitting AP (i.e., the transmitted BSSID).

A reporting AP should set the OCT Recommended subfield to 1 in the BSS Parameters subfield of a TBTT Information field in a Reduced Neighbor Report element if both the reporting AP and the reported AP have dot11OCTOptionImplemented equal to true and the Co-Located AP subfield is 1 in the BSS Parameters subfield corresponding to the reported AP. A reporting AP may set the OCT Recommended subfield to 1 in the BSS Parameters subfield of a TBTT Information field in a Reduced Neighbor Report element if both the reporting AP and the reported AP have the same SSID and have dot11OCTOptionImplemented equal to true and the Co-Located AP subfield is 0 in the BSS Parameters subfield corresponding to the reported AP. If the OCT Recommended subfield is 1 and the Co-Located AP subfield is 1 in the Neighbor AP Information field describing a reported HE AP in the Reduced Neighbor Report element, then a non-AP STA that has dot11OCTOptionImplemented equal to true should use the OCT procedure described in 11.31.5 to perform active scanning, authentication, and/or association with the reported AP through over-the-air transmissions with the AP that sent the Reduced Neighbor Report element. If the OCT Recommended subfield is 1 and the Co-Located AP subfield is 0 in the Neighbor AP Information field describing a reported HE AP in the Reduced Neighbor Report element, then a non-AP STA that has dot11OCTOptionImplemented equal to true may use the OCT procedure described in 11.31.5 to perform active scanning, authentication, and/or association with the reported AP through over-the-air transmissions with the AP that sent the Reduced Neighbor Report element.

Change the now 13th paragraph in 11.49 as follows:

A STA that receives a Neighbor AP Information field with a recognized TBTT Information Field Type subfield but an unrecognized TBTT Information Length subfield ~~shall ignore that Neighbor AP Information field and continue to process remaining Neighbor AP Information fields~~ has two possible ways of processing the received information:

- a) Ignore that Neighbor AP Information field, and continue to process the subsequent Neighbor AP Information fields, or

- b) Process the first 13 octets of each TBTT Information field of the Neighbor AP Information field as if the TBTT Information Length subfield had value 13, ignore the remaining TBTT Information Length minus 13 octets of each TBTT Information field of the Neighbor AP Information field, and continue to process the subsequent Neighbor AP Information fields.

If the unrecognized TBTT Information Length value is less than or equal to 13, the STA shall follow alternative a). If the unrecognized TBTT Information Length value is greater than 13, an HE STA shall follow alternative b), and a non-HE STA shall follow either alternative a) or b).

Insert the following text at the end of 11.49:

If an AP that operates in the 2.4 GHz or 5 GHz band advertises in Reduced Neighbor Report elements a 6 GHz AP that is in the same co-located AP set as itself, the AP shall include the 20 MHz PSD subfield in the TBTT Information field corresponding to that 6 GHz AP.

When a 20 MHz PSD subfield is present in a TBTT Information field that reports a 6 GHz AP, its value shall be set such that the resulting allowed maximum transmit power for the primary 20 MHz channel is equal to the minimum of the regulatory client maximum transmit powers indicated by the Transmit Power Envelope element(s) transmitted by the reported AP in Beacon and Probe Response frames.

NOTE 5—Country-specific operating requirements that relate to use of the Reduced Neighbor Report element are defined in E.2.7.

NOTE 6—A 20 MHz PSD subfield in a Reduced Neighbor Report element sent in Beacon and Probe Response frames by an AP that is in the same co-located AP set as a 6 GHz AP can be used by a STA to determine a local transmit power limit for 20 MHz PPDU's corresponding to a 6 GHz AP, prior to having received a Beacon or Probe Response frame from that AP. The value in the 20 MHz PSD subfield can be used by any STA, although for some categories it may result in determination of a lower transmit power limit than would be determined by (subsequent) reception of a Transmit Power Envelope element. A STA might, for example, determine a transmit power limit based on this information when sending a Probe Request frame with 20 MHz PPDU bandwidth during active scanning on the 6 GHz AP's channel.

Insert the following subclause (11.53) after 11.52:

11.53 Out-of-band discovery of a 6 GHz BSS

An AP that operates in the 2.4 GHz or 5 GHz band and that is in the same co-located AP set as one or more 6 GHz APs shall include in Beacon and Probe Response frames that it transmits a Reduced Neighbor Report element with the Co-Located AP subfield in the BSS Parameters subfield in the TBTT Information field set to 1 to provide at least the operating channels and operating classes of those 6 GHz APs.

NOTE 1—The Reduced Neighbor Report element might contain information on 6 GHz APs that are not in the same co-located AP set as the transmitting AP. In this case the Co-Located AP subfield is set to 0.

An AP responds to a probe request by following the rules defined in 11.1.4.3.4.

If neither of the following conditions is met:

- The AP transmits an individually addressed Probe Response frame to a STA that has signaled that it does not support operating in the 6 GHz band (see 9.4.2.53), nor
- The AP operating in the 6 GHz band does not intend to be discovered by STAs,

then the following applies:

- If an AP operating in the 2.4 GHz or 5 GHz band is in the same co-located AP set as one or more 6 GHz APs and has the same SSID as those 6 GHz APs, then the Beacon and Probe Response frames transmitted by the AP or by the transmitted BSSID of the same Multiple BSSID set as the AP shall include, for each of these 6 GHz APs, a TBTT Information field in a Reduced Neighbor Report

element with the BSSID field set to the BSSID of the 6 GHz AP, and with either the Short SSID field set to the short SSID of the 6 GHz AP or the Same SSID subfield in the BSS Parameters subfield set to 1.

- If an AP operating in the 2.4 GHz or 5 GHz band is in the same co-located AP set as a 6 GHz AP and has a different SSID, and if no other AP in the same co-located AP set and operating in the 2.4 GHz or 5 GHz band is indicating the 6 GHz AP in a Reduced Neighbor Report element of the Beacon and Probe Response frames it transmits, then Beacon and Probe Response frames transmitted by the AP (or by the transmitted BSSID of the same Multiple BSSID set as the AP) shall include a TBTT Information field in a Reduced Neighbor Report element with the BSSID field and the Short SSID field set to the BSSID and short SSID of the 6 GHz AP, respectively.

If the AP reported in the TBTT Information field in the Reduced Neighbor Report element is a 6 GHz AP, the reporting AP shall include the BSS Parameters subfield in the TBTT Information field and shall follow the rules in 11.49 to set the Multiple BSSID subfield, the Transmitted BSSID subfield, the Co-Located AP subfield, and the OCT Recommended subfield.

A STA receiving a frame containing a Reduced Neighbor Report element describing a reported 6 GHz AP with the OCT Recommended subfield set to 1 in the BSS Parameters subfield shall follow the rules in 11.49 to perform active scanning, authentication, and/or association with the reported AP.

An AP that operates in the 2.4 GHz or 5 GHz band and that is in the same co-located AP set as one or more 6 GHz APs shall include the Advertisement Protocol element in Beacon and Probe Response frames that it transmits and shall support responding with a Neighbor Report ANQP-element (9.4.5.19) carrying one or more Neighbor Report elements (see 9.4.2.36) that include at least the SSID information of all the 6 GHz APs in the same co-located AP set, except the 6 GHz APs that do not intend to be discovered.

NOTE 2—The Neighbor Report ANQP-element can also carry Neighbor Report elements containing information on 6 GHz APs that are not in the same co-located AP set.

NOTE 3—The AP might respond with a GAS comeback delay of zero.

NOTE 4—If the Same SSID subfield is set to 0 in the BSS Parameters of a reported 6 GHz AP, a non-AP STA that does not know the short SSID of the reported 6 GHz AP and that intends to discover the SSID of the reported 6 GHz AP might:

- a) Use the OCT procedure described in 11.31.5 to send a Probe Request frame to the reported AP through over-the-air transmissions with the reporting AP, if the OCT Recommended subfield is 1 in the Neighbor AP Information field describing the reported AP.
- b) Use the ANQP procedure described in 11.22.3.3 to send an ANQP request with a Query ID corresponding to Neighbor Report to the reporting AP to retrieve the SSID of the 6 GHz APs, including the reported AP.
- c) Send a Probe Request frame to the reported AP including the BSSID of the reported AP.
- d) Send a Probe Request frame to the reported AP including the short SSID of the reported AP.
- e) Perform passive scanning in the operating channel of the reported AP.

An AP may set the Unsolicited Probe Responses Active subfield to 1 for a reported AP in a Reduced Neighbor Report element or Neighbor Report element in a frame it transmits if all 6 GHz APs of the same ESS as the reported AP that operate in the same channel as the reported AP and that might be detected by a STA receiving this frame [see the definition of “detected access point (AP)” in 3.2] have dot11UnsolicitedProbeResponseOptionActivated equal to true and so are transmitting unsolicited Probe Response frames every 20 TUs (see 26.17.2.3.2). Otherwise, the AP shall set the Unsolicited Probe Responses Active subfield to 0.

An AP may set the Member Of ESS With 2.4/5 GHz Co-Located AP subfield to 1 in a Reduced Neighbor Report element in a frame it transmits if the reported AP is a 6 GHz AP and is part of an ESS where each AP in the ESS that is operating in the same band as the reported AP and that might be detected by a STA receiving this frame (irrespective of the operating channel) has

dot11MemberOfCollocated6GHzESSOptionActivated equal to true and also has a corresponding AP operating in the 2.4 GHz or 5 GHz band that is in the same co-located AP set as that AP. See the definition of “detected access point (AP)” in 3.2.

NOTE 5—This subfield indicates that the reported AP is part of an ESS that has no 6 GHz-only APs that might be detected by a STA receiving this frame. In other words, all 6 GHz APs that are part of that ESS that might be detected by a STA receiving this frame can be discovered in the 2.4 GHz and/or 5 GHz bands.

12. Security

12.5 RSNA confidentiality and integrity protocols

12.5.3 CTR with CBC-MAC protocol (CCMP)

12.5.3.3 CCMP cryptographic encapsulation

12.5.3.3.2 PN processing

Change the first paragraph in 12.5.3.3.2 as follows:

The PN is incremented by a positive number for each MPDU. The PN shall be incremented in steps of 1 for constituent MPDUs of fragmented MSDUs, A-MSDUs, and MMPDUs. For PV0 MPDUs, the PN shall never repeat for a series of encrypted MPDUs using the same temporal key. For PV1 MPDUs, the PN shall never repeat for a series of encrypted MPDUs using the same temporal key and TID/ACI.

12.5.5 GCM protocol (GCMP)

12.5.5.3 GCMP cryptographic encapsulation

12.5.5.3.2 PN processing

Change the first paragraph in 12.5.5.3.2 as follows:

The PN is incremented by a positive number for each MPDU. The PN shall be incremented in steps of 1 for constituent MPDUs of fragmented MSDUs, A-MSDUs, and MMPDUs. The PN shall never repeat for a series of encrypted MPDUs using the same temporal key.

Insert the following subclauses (12.12, 12.12.1, and 12.2.2) after 12.11.2.7:

12.12 Constraints on allowed security parameters

12.12.1 Introduction

This standard defines various security mechanisms of which some are deprecated or obsolete and are not considered to meet current requirements for security. Such mechanisms enable interoperability with legacy devices, but are not suitable for new uses where legacy support is not needed. Subclause 12.12 describes constraints on security parameter selection for some use cases.

12.12.2 Security constraints in the 6 GHz band

The following apply to a STA operating in the 6 GHz band:

- The STA shall not use the following pre-RSNA security methods:
 - WEP
 - Open System authentication without encryption
 - Shared Key authentication
- The STA shall not use the following cipher suite selectors:
 - 00-0F-AC:0 (Use group cipher suite)

- 00-0F-AC:1 (WEP-40)
- 00-0F-AC:2 (TKIP)
- 00-0F-AC:5 (WEP-104)
- The STA should use Opportunistic Wireless Encryption, as specified in IETF RFC 8110, when connecting in an infrastructure BSS without authentication (as a replacement for Open System authentication without encryption).
- The STA shall not use the following AKM suite selectors:
 - 00-0F-AC:2 (PSK)
 - 00-0F-AC:4 (FT authentication using PSK)
 - 00-0F-AC:6 (PSK with SHA-256)
 - 00-0F-AC:19 (FT authentication using PSK with SHA-384)
 - 00-0F-AC:20 (PSK with SHA-384)
- The STA should use SAE (AKM suite selectors 00-0F-AC:8 and/or 00-0F-AC:9) if authenticating using a password where IEEE Std 802.1X is not used (as a replacement for PSK).
- The STA shall use management frame protection (MFPR=1) when using RSN.

14. MLME mesh procedures

14.2 Mesh discovery

14.2.4 Mesh STA configuration

Change 14.2.4 as follows:

The mesh STA configuration consists of the mesh profile (see 14.2.3), the Supported Rates and BSS Membership Selectors element, the Extended Supported Rates and BSS Membership Selectors element, the HT Operations element (if present), ~~and the VHT Operations element (if present), and the HE Operation element (if present).~~

Mesh STA configurations are identical if the following conditions hold:

- The mesh profiles are identical.
- The BSSBasicRateSet parameter of the MLME-START.request primitive is identical to the basic rate set indicated by the Supported Rates and BSS Membership Selectors element and Extended Supported Rates and BSS Membership Selectors element, if present, received in the MLME-MESHEERINGMANAGEMENT.indication primitive.
- For HT mesh STAs, the Basic HT-MCS Set field of the HT Operation parameter of the MLME-START.request primitive is identical to the HT Operation element received in the MLME-MESHEERINGMANAGEMENT.indication primitive.
- For VHT mesh STAs, the Basic VHT-MCS And NSS Set field in the VHT Operation element of the MLME-START.request primitive is identical to the Basic VHT-MCS And NSS Set field in the VHT Operation element received in the MLME-MESHPEERINGMANAGEMENT.indication primitive.
- For HE mesh STAs, the Basic HE-MCS And NSS Set field in the HE Operation element of the MLME-START.request primitive is identical to the Basic HE-MCS And NSS Set field in the HE Operation element received in the MLME-MESHPEERINGMANAGEMENT.indication primitive.

14.2.7 Candidate peer mesh STA

Insert item f) into the lettered list of 14.2.7 after item e), and reletter the subsequent list item accordingly:

- f) If both the scanning mesh STA and the discovered neighbor STA are HE STAs, the mesh STA uses the same value for the Basic HE-MCS And NSS Set field in its HE Operation element as received in the Beacon or Probe Response frame from the neighbor mesh STA.

18. Extended Rate PHY (ERP) specification

18.2 PHY-specific service parameter list

Insert the following rows at the end of Table 18-1:

Table 18-1—TXVECTOR parameters

Parameter	Value
CH_BANDWIDTH_IN_NON_HT	If present, CBW20 or CBW40.
DYN_BANDWIDTH_IN_NON_HT	If present, Static or Dynamic.

Insert the following rows at the end of Table 18-3:

Table 18-3—RXVECTOR parameters

Parameter	Value
CH_BANDWIDTH_IN_NON_HT	If present, CBW20 or CBW40.
DYN_BANDWIDTH_IN_NON_HT	If present, Static or Dynamic.

Insert the following text (Clause 26 and Clause 27) after Clause 25:

26. High-efficiency (HE) MAC specification

26.1 Introduction

An HE STA supports the MAC and MLME functions defined in Clause 26 in addition to the MAC functions defined in Clause 10, the MLME functions defined in Clause 11, and the security functions defined in Clause 12, except when the functions in Clause 26 supersede the functions in Clause 10 or Clause 11. A frame successfully transmitted by a non-AP STA in response to a Basic Trigger frame is a successful frame exchange initiated by the STA as referred to in Clause 11 and Clause 12.

26.2 HE channel access

26.2.1 TXOP duration-based RTS/CTS

In an HE BSS, the use of RTS/CTS can be TXOP duration based or PSDU length based. An HE AP may configure a non-AP HE STA to use the TXOP duration-based RTS/CTS exchanges to help mitigate interference in dense environments.

An HE AP may set the TXOP Duration RTS Threshold subfield in the HE Operation Parameters field in the HE Operation element it transmits to a value between 1 and 1022 to enable TXOP duration-based RTS/CTS exchanges of its associated STAs. The AP may set the TXOP Duration RTS Threshold field to 1023 to disable TXOP duration-based RTS/CTS exchanges of its associated STAs. In Beacon and Probe Response frames, the AP may set the TXOP Duration RTS Threshold field to 0 to make no changes to TXOP duration-based RTS/CTS exchanges of its associated STAs.

If the TXOP Duration RTS Threshold subfield in the HE Operation Parameters field in the most recently received HE Operation element sent by the AP with which a non-AP HE STA is associated is equal to a nonzero value, then the non-AP HE STA shall set dot11TXOPDurationRTSThreshold to the value of the TXOP Duration RTS Threshold subfield. Otherwise, the non-AP HE STA shall not update dot11TXOPDurationRTSThreshold.

The TXOP duration-based RTS/CTS exchange is disabled at a non-AP HE STA if dot11TXOPDurationRTSThreshold is 1023. The TXOP duration-based RTS/CTS exchange is enabled at a non-AP HE STA when dot11TXOPDurationRTSThreshold is less than 1023.

A non-AP HE STA shall use an RTS/CTS exchange to initiate a TXOP if TXOP duration-based RTS/CTS exchange is enabled at a non-AP HE STA and the following conditions are met:

- The STA intends to transmit individually addressed frames to the HE AP or to a TDLS peer STA
- The TXOP duration is greater than or equal to $32 \mu\text{s} \times \text{dot11TXOPDurationRTSThreshold}$

Otherwise, the non-AP HE STA follows the rules defined in 10.3.1.

26.2.2 Intra-BSS and inter-BSS PPDU classification

A STA shall classify a received PPDU as an inter-BSS PPDU if at least one of the following conditions is true:

- The RXVECTOR parameter BSS_COLOR is not 0 and is not the BSS color of the BSS of which the STA is a member.
- The PPDU is a VHT PPDU with RXVECTOR parameter PARTIAL_AID not equal to the BSSID[39:47] of the BSS in which the STA is associated or any of the other BSSs in the same multiple BSSID set or co-hosted BSSID set to which its BSS belongs and the RXVECTOR parameter GROUP_ID is 0.
- The PPDU is a VHT PPDU with RXVECTOR parameter PARTIAL_AID[5:8] not equal to the 4 LSBs of the BSS color announced by the BSS of which the STA whose dot11PartialBSSColorImplemented is equal to true is a member and RXVECTOR parameter GROUP_ID equal to 63 when the Partial BSS Color field in the most recent HE Operation element is 1.
- The PPDU is either a VHT MU PPDU or an HE MU PPDU with the RXVECTOR parameter UPLINK_FLAG equal to 0, and the STA is an AP.
- The PPDU carries a frame that has a BSSID field, the value of which is not the BSSID of the BSS in which the STA is associated or any of the other BSSs in the same multiple BSSID set or co-hosted BSSID set to which its BSS belongs or the wildcard BSSID.
- The PPDU carries a frame that does not have a BSSID field but has both an RA field and TA field, neither value of which is equal to the BSSID of the BSS in which the STA is associated or any of the other BSSs in the same multiple BSSID set or co-hosted BSSID set to which its BSS belongs. The Individual/Group bit in the TA field value is forced to 0 prior to comparison.

A STA shall classify the received PPDU as an intra-BSS PPDU if at least one of the following conditions is true:

- The RXVECTOR parameter BSS_COLOR of the PPDU carrying the frame is the BSS color of the BSS of which the STA is a member or the BSS color of any TDLS links to which the STA belongs if the STA is an HE STA associated with a non-HE AP.
- The PPDU is a VHT PPDU with RXVECTOR parameter PARTIAL_AID equal to the BSSID[39:47] of the BSS in which the STA is associated or any of the other BSSs in the same multiple BSSID set or co-hosted BSSID set to which its BSS belongs and the RXVECTOR parameter GROUP_ID equal to 0.
- The PPDU is a VHT PPDU with RXVECTOR parameter PARTIAL_AID[5:8] equal to the 4 LSBs of the BSS color announced by of the BSS of which the STA whose dot11PartialBSSColorImplemented is equal to true is a member, the RXVECTOR parameter GROUP_ID is equal to 63, and the Partial BSS Color field in the most recent HE Operation element is 1.
- The PPDU carries a frame that has an RA, TA, or BSSID field value that is equal to the BSSID of the BSS or the BSSID of any BSS in which the STA is associated or any of the other BSSs in the same multiple BSSID set or co-hosted BSSID set to which its BSS belongs. The Individual/Group bit in the TA field value is forced to the value 0 prior to the comparison.
- The PPDU carries a Control frame that does not have a TA field and that has an RA field value that matches the saved TXOP holder address of the BSS or any BSS in which the STA is associated or any of the other BSSs in the same multiple BSSID set or co-hosted BSSID set to which its BSS belongs.

NOTE—See 10.19 for the definition of PARTIAL_AID[5:8] and BSSID[39:47].

Otherwise, the PPDU cannot be determined as an intra-BSS or inter-BSS PPDU.

If, based on the MAC address information of a frame carried in a received PPDU, the received PPDU satisfies both intra-BSS and inter-BSS conditions, then the received PPDU is classified as an intra-BSS PPDU.

If the received PPDU satisfies the intra-BSS conditions using the RXVECTOR parameter BSS_COLOR and also satisfies the inter-BSS conditions using MAC address information of a frame carried in the PPDU, then the classification made using the MAC address information takes precedence.

If a STA determines that the BSS color is disabled (see 26.17.3.3), then the RXVECTOR parameter BSS_COLOR of a PPDU shall not be used to classify the PPDU.

26.2.3 SRG PPDU identification

Identification of SRG and non-SRG PDUs is used during SRG OBSS PD spatial reuse operation as described in 26.10.

A non-AP HE STA that has received a Spatial Reuse Parameter Set element from its associated AP with a value of 1 in the SRG Information Present subfield shall use information provided in the Spatial Reuse Parameter Set element to identify BSSs that are members of the STA's SRG to determine if a received inter-BSS PPDU is an SRG PPDU. An HE AP may use an SRG that is different from the SRG that it transmits to other STAs in Spatial Reuse Parameter Set elements to determine whether a received inter-BSS PPDU is an SRG PPDU.

A received HE PPDU that is an inter-BSS PPDU is an SRG PPDU if the bit in the SRG BSS Color Bitmap field indexed by the value of the RXVECTOR parameter BSS_COLOR is 1 (see 9.4.2.252). A received VHT PPDU that is an inter-BSS PPDU is an SRG PPDU if the GROUP_ID parameter of the RXVECTOR has a value of 0 and the bit in the SRG Partial BSSID Bitmap field that corresponds to the numerical value of PARTIAL_AID[0:5] of the RXVECTOR is 1 (see 9.4.2.252).

A received PPDU that is an inter-BSS PPDU is an SRG PPDU if BSSID information from a frame carried in the PPDU is correctly received and the bit in the SRG Partial BSSID Bitmap field that corresponds to the numerical value of BSSID[39:44] is 1.

A VHT PPDU that is an inter-BSS PPDU and that is received with RXVECTOR parameter GROUP_ID equal to 0 is an SRG PPDU if the bit in the SRG Partial BSSID Bitmap field that corresponds to the numerical value of bits [39:44] of the RA field of any correctly received frame from the PPDU is 1.

A VHT PPDU that is an inter-BSS PPDU and that is received with RXVECTOR parameter GROUP_ID equal to 63 is an SRG PPDU if the bit in the SRG Partial BSSID Bitmap field that corresponds to the numerical value of bits [39:44] of the TA field of any correctly received frame from the PPDU is 1.

An HE SU PPDU, HE ER SU PPDU, or HE MU PPDU that is an inter-BSS PPDU and that is received with the RXVECTOR parameter UPLINK_FLAG equal to 1 is an SRG PPDU if the bit in the SRG Partial BSSID Bitmap field that corresponds to the numerical value of bits [39:44] of the RA field of any correctly received frame from the PPDU is 1.

Otherwise, the PPDU is not determined to be an SRG PPDU. A non-AP HE STA that has not received a Spatial Reuse Parameter Set element from its associated AP with a value of 1 in the SRG Information Present subfield shall not classify any received PDUs as an SRG PPDU. An HE AP that has not transmitted a Spatial Reuse Parameter Set element with a value of 1 in the SRG Information Present subfield may classify received PDUs as SRG PDUs using information that it has not transmitted.

26.2.4 Updating two NAVs

A non-AP HE STA shall maintain two NAVs, and an HE AP may maintain two NAVs: an intra-BSS NAV and a basic NAV. The intra-BSS NAV is updated by an intra-BSS PPDU. The basic NAV is updated by an inter-BSS PPDU or a PPDU that cannot be classified as intra-BSS or inter-BSS. The mechanism by which a PPDU is classified intra-BSS or inter-BSS is defined in 26.2.2.

Maintaining two NAVs is beneficial in dense deployment scenarios in which a STA requires protection from frames transmitted by STAs within its BSS, i.e., intra-BSS, and wants to avoid interference from frames transmitted by STAs in a neighboring BSS, i.e., inter-BSS. For example, in a TXOP initiated by the AP with which a STA is associated for an HE TB PPDU transmission, the intra-BSS NAV of the STA is set by the AP to prevent the STA from contending for the channel. The basic NAV of the STA is not updated by transmissions from the AP during the TXOP so that if the basic NAV of the STA is nonzero and the STA receives, from the AP, a Trigger frame with the CS Required subfield equal to 1, the STA will not respond (see 26.5.2.5).

The requirements in 10.3.2.1 apply to an HE STA maintaining two NAVs, with the exception of the virtual CS indication of medium. For an HE STA maintaining two NAVs, if both the NAV timers are 0, the virtual CS indication is that the medium is idle; if at least one of the two NAV timers is nonzero, the virtual CS indication is that the medium is busy.

The procedure in 10.3.2.9 applies to an HE STA maintaining two NAVs, and the NAV referred by the description in 10.3.2.4 is the basic NAV.

The duration information is indicated by a frame as follows:

- If there is a Duration field in the frame, then the duration information is indicated by the Duration field.
- If the frame is a PS-Poll, then the duration information is equal to the time, in microseconds, required to transmit one Ack frame plus one SIFS under the data rate selection rules. If the calculated duration information includes a fractional microsecond, that duration information is rounded up to the next higher integer.

A STA shall update the intra-BSS NAV with the duration information indicated by the received frame in a PSDU if and only if all the following conditions are met:

- The frame is identified as intra-BSS according to the rule described in 26.2.2.
- The indicated duration is greater than the current intra-BSS NAV value.
- The RA of the received frame is not the STA's MAC address; or the STA is not a TXOP holder, and the PPDU carrying the frame does not contain a frame that solicits an immediate response from the STA; or the STA is not a TXOP holder, and the received frame is a Trigger frame.

A STA shall update the basic NAV with the duration information indicated by the received frame in a PSDU if and only if all the following conditions are met:

- The frame is identified as inter-BSS or cannot be identified as intra-BSS or inter-BSS according to the rule described in 26.2.2.
- The indicated duration is greater than the current basic NAV value.
- The RA of the received frame is not the STA's MAC address.

A STA that is a TXOP holder shall not update the intra-BSS NAV with the duration information indicated by the RXVECTOR parameter TXOP_DURATION.

A STA that is not a TXOP holder shall update the intra-BSS NAV with the duration information indicated by the RXVECTOR parameter TXOP_DURATION for an HE PPDU if and only if all the following conditions are met:

- The RXVECTOR parameter TXOP_DURATION is not UNSPECIFIED.
- The PPDU that carried information of the RXVECTOR parameter is identified as intra-BSS according to the rule described in 26.2.2.
- The STA does not receive a frame with a Duration field in the PPDU.
- The duration information indicated by the RXVECTOR parameter TXOP_DURATION is greater than the current intra-BSS NAV of the STA.

A STA shall update the basic NAV with the duration information indicated by the RXVECTOR parameter TXOP_DURATION for an HE PPDU if and only if all the following conditions are met:

- The RXVECTOR parameter TXOP_DURATION is not UNSPECIFIED.
- The PPDU that carried information for the RXVECTOR parameter is identified as inter-BSS or cannot be identified as intra-BSS or inter-BSS according to the rule described in 26.2.2.
- The STA does not receive a frame with a Duration field in the PPDU.
- The duration information indicated by the RXVECTOR parameter TXOP_DURATION is greater than the current basic NAV of the STA.

NOTE 1—If a PS-Poll frame is received in an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU, then the RXVECTOR parameter TXOP_DURATION does not indicate duration information (see 26.11.5).

NOTE 2—If a STA receives an HE PPDU with the duration information indicated by both a frame with a Duration field and the RXVECTOR parameter TXOP_DURATION, then the duration information indicated by the RXVECTOR parameter TXOP_DURATION is ignored.

NOTE 3—The additional rules of NAV consideration for a STA that is solicited for an immediate response are described in 10.3.2.9, 10.3.2.11, and 26.5.2.5.

A PHY-CCARESET.request primitive shall be issued if one of the following condition is met:

- One NAV is reset, and the other NAV timer is 0.
- Both NAVs are reset simultaneously.

The NAVs (if any) are updated at the expected end of the PPDU as defined in 10.3.2.4.

An HE STA that used information from an RTS or MU-RTS Trigger frame as the most recent basis to update its NAV may reset the NAV that is updated by the RTS or MU-RTS Trigger frame if no PHY-RXSTART.indication primitive is received from the PHY during a period with a duration of $2 \times \text{aSIFSTime} + \text{CTS_Time} + \text{aRxPHYStartDelay} + 2 \times \text{aSlotTime}$ starting when the MAC receives a PHY-RXEND.indication primitive corresponding to the detection of the RTS or MU-RTS Trigger frame (see 10.3.2.4 for the definition of CTS_Time).

26.2.5 Truncation of TXOP

A non-AP HE STA that is not associated with an HE AP shall interpret the reception of a CF-End frame as a NAV reset, i.e., it resets its maintained NAV to 0 at the end of the PPDU containing this frame.

An HE AP that maintains one NAV (see 10.3.2.1) and receives a CF-End frame should reset the NAV, unless either of following conditions is met:

- The received CF-End frame is carried in an inter-BSS PPDU, and the most recently updated NAV was due to a frame carried in an intra-BSS PPDU (see 26.2.2).
- The received CF-End frame is carried in an intra-BSS PPDU, and the most recently updated NAV was due to a frame carried in an inter-BSS PPDU (see 26.2.2).

An HE STA that maintains two NAVs (see 26.2.4) and receives a CF-End frame should reset the basic NAV if the received CF-End frame is carried in an inter-BSS PPDU and reset the intra-BSS NAV if the received CF-End frame is carried in an intra-BSS PPDU.

An HE STA that maintains two NAVs may reset both NAVs if the received CF-End frame is carried in an intra-BSS PPDU and the basic NAV was updated due to a frame carried in a PPDU that cannot be identified as either an inter-BSS PPDU or intra-BSS PPDU.

An HE STA that receives a CF-End frame and resets all its maintained NAV(s) can start contending for the medium without further delay.

26.2.6 MU-RTS Trigger/CTS frame exchange procedure

26.2.6.1 General

The MU-RTS Trigger/CTS frame exchange procedure allows an AP to initiate a TXOP and protect the TXOP frame exchanges. An AP may transmit an MU-RTS Trigger frame to solicit simultaneous CTS frame responses from one or more non-AP STAs.

Figure 26-1 shows an example of the exchange of MU-RTS and simultaneous CTS responses to protect the DL MU PPDU and the acknowledgment responses.

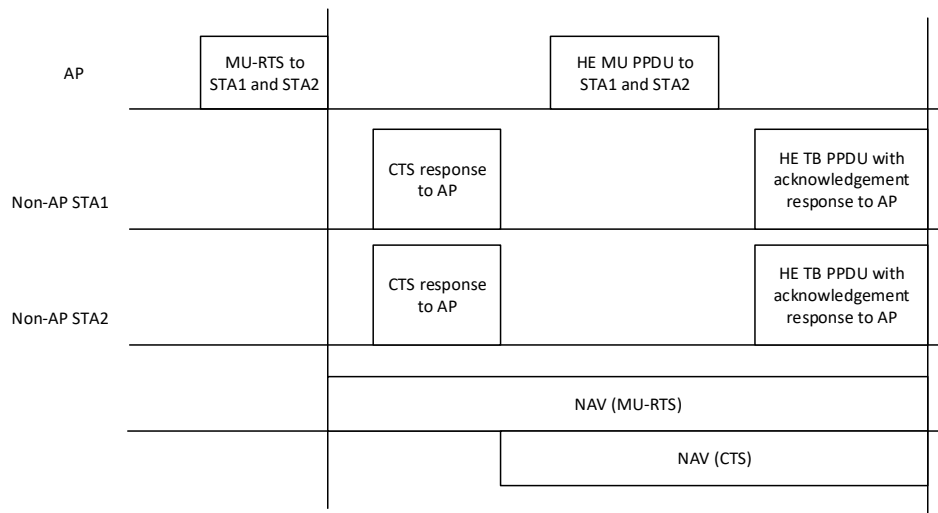


Figure 26-1—Example of MU-RTS/CTS/DL MU PPDU/Acknowledgment Response and NAV setting

Figure 26-2 shows an example of the exchange of MU-RTS and simultaneous CTS responses to protect the HE TB PPDU and Multi-STA BlockAck frame.

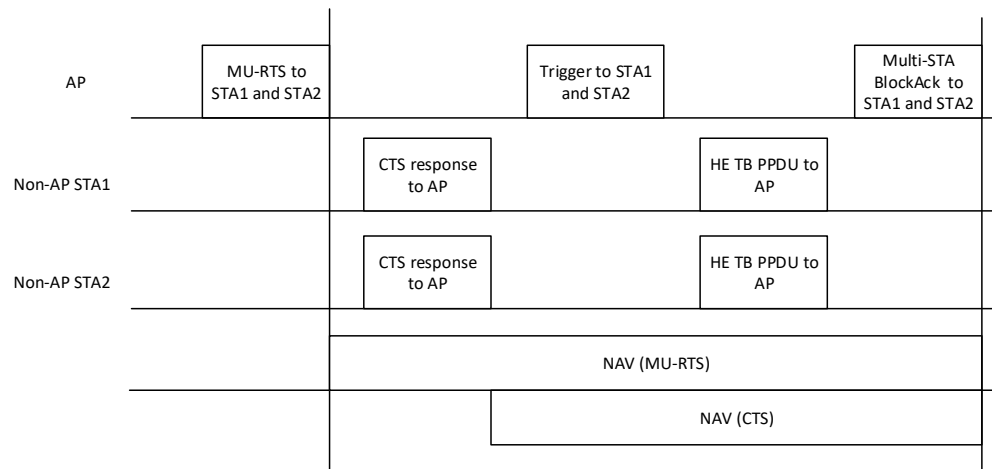


Figure 26-2—Example of MU-RTS/CTS/Trigger/HE TB PPDU/Multi-STA BlockAck and NAV setting

26.2.6.2 MU-RTS Trigger frame transmission

In each 20 MHz channel occupied by the PPDU that contains an MU-RTS Trigger frame, the transmitter of the MU-RTS Trigger frame shall request at least one non-AP STA to send a CTS frame response that occupies the 20 MHz channel. The transmitter of an MU-RTS Trigger frame shall not request a non-AP STA to send a CTS frame response in a 20 MHz channel that is not occupied by the PPDU that contains the MU-RTS Trigger frame.

After transmitting an MU-RTS Trigger frame, the AP shall wait for a CTSTimeout interval of $aSIFSTime + aSlotTime + aRxPHYStartDelay$ that begins when the MAC receives the PHY-TXEND.confirm primitive for the transmitted MU-RTS Trigger frame. If the MAC does not receive a PHY-RXSTART.indication primitive during the CTSTimeout interval, the AP shall conclude that the transmission of the MU-RTS Trigger frame has failed, and, if the MU-RTS Trigger frame initiated a TXOP, the AP shall invoke its backoff procedure. If the MAC receives a PHY-RXSTART.indication primitive during the CTSTimeout interval, then the MAC shall wait for the corresponding PHY-RXEND.indication primitive to determine whether the MU-RTS Trigger frame transmission was successful. The receipt of a CTS frame from any non-AP STA addressed by the MU-RTS Trigger frame before the PHY-RXEND.indication primitive shall be interpreted as the successful transmission of the MU-RTS Trigger frame, permitting the frame exchange sequence to continue. The receipt of any other type of frame shall be interpreted as a failure of the MU-RTS Trigger frame transmission. In this instance, the AP may process the received frame and, if the MU-RTS Trigger frame initiated a TXOP, shall invoke its backoff procedure at the PHY-RXEND.indication primitive.

NOTE—If an AP transmits an MU-RTS Trigger frame in an already initiated TXOP and the MU-RTS Trigger frame transmission fails, then the AP might perform a PIFS recovery as described in 10.23.2.8 or invoke the backoff procedure described in 10.23.2.2.

An MU-RTS Trigger frame shall not be carried in a VHT MU PPDU or an HE MU PPDU.

26.2.6.3 CTS frame response to an MU-RTS Trigger frame

If a non-AP STA receives an MU-RTS Trigger frame, the non-AP STA shall commence the transmission of a CTS frame response at the SIFS time boundary after the end of a received PPDU when all the following conditions are met:

- The MU-RTS Trigger frame has one of the User Info fields addressed to the non-AP STA. The User Info field is addressed to a non-AP STA if the AID12 subfield is equal to the 12 LSBs of the AID of the STA and the MU-RTS Trigger frame is sent by the AP with which the non-AP STA is associated or by the AP corresponding to the transmitted BSSID if the non-AP STA is associated with an AP corresponding to a nontransmitted BSSID and has indicated support for receiving Control frames with TA field set to the transmitted BSSID by setting the Rx Control Frame To MultiBSS subfield to 1 in the HE Capabilities element that the non-AP STA transmits.
- The UL MU CS condition indicates that the medium is idle (see 26.5.2.5).

Otherwise, the non-AP STA shall not send a CTS frame response.

NOTE 1—The RU Allocation subfield in the User Info field addressed to the non-AP STA indicates whether the CTS frame response is to be sent on the primary 20 MHz channel, primary 40 MHz channel, primary 80 MHz channel, 160 MHz channel, or 80+80 MHz channel as described in 9.3.1.22.5.

NOTE 2—A combination of virtual CS and ED-based CCA during the SIFS after the PPDU containing the MU-RTS Trigger frame is used to determine the state of the medium (see 26.5.2.5).

The CTS frame sent in response to an MU-RTS Trigger frame shall be carried in a non-HT or non-HT duplicate PPDU (see Clause 17) with a 6 Mb/s rate and with the TXVECTOR parameter SCRAMBLER_INITIAL_VALUE set to the same value as the RXVECTOR parameter SCRAMBLER_INITIAL_VALUE of the PPDU carrying the MU-RTS Trigger frame. The PPDU carrying the CTS frame shall be transmitted on the 20 MHz channels indicated in the RU Allocation subfield of the User Info field of the MU-RTS Trigger frame.

NOTE 3—A bandwidth signaling TA is not used in an MU-RTS Trigger frame or a CTS frame response to an MU-RTS Trigger frame (see 9.3.1.22 and 9.3.1.3). As a result, the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT is not present when transmitting an MU-RTS Trigger frame or CTS frame response to an MU-RTS Trigger frame. In Figure 17-7 the first 7 bits of scrambling sequence of an MU-RTS Trigger frame or CTS frame response to an MU-RTS Trigger frame are not defined by Table 17-7.

The Power Management and More Data subfields in a CTS frame sent in response to an MU-RTS Trigger frame shall be set to 0.

NOTE 4—Other subfields of the Frame Control field of the CTS frames sent in response to an MU-RTS Trigger frame are set as described in Figure 9-26.

Figure 26-3 shows an example of the exchange of an MU-RTS Trigger frame and simultaneous CTS frame responses on the primary 40 MHz channel. In this example, MU-RTS Trigger frame is transmitted in a 40 MHz non-HT duplicate PPDU on the primary 40 MHz channel. Further, the MU-RTS Trigger frame requests non-AP STA1 to transmit a CTS frame response in a non-HT PPDU on the primary 20 MHz channel and non-AP STA2 to transmit a CTS frame response in a 40 MHz non-HT duplicate PPDU on the primary 40 MHz channel.

A non-AP STA that transmits a CTS frame in response to an MU-RTS Trigger frame shall follow the pre-correction requirements defined in 27.3.15.3.

NOTE 5—The AP Tx Power and UL Target Receive Power fields in an MU-RTS Trigger frame are reserved, and there is no power pre-correction requirement for the CTS frame sent in response to an MU-RTS Trigger frame (i.e., the AP that sends the MU-RTS frame does not specify the transmission power of the solicited CTS frame).

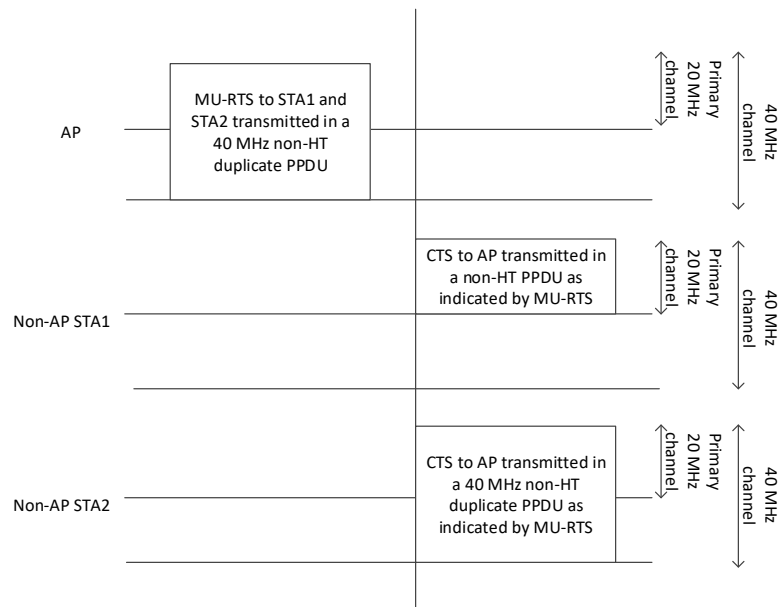


Figure 26-3—Example of MU-RTS Trigger frame soliciting CTS frame responses on primary 40 MHz channel

26.2.7 EDCA operation using MU EDCA parameters

An HE AP shall set the QoS Info field of an MU EDCA Parameter Set element (if present) to the same value as the QoS Info field of an EDCA Parameter Set element (if present). An HE AP may change the MU EDCA parameters by including the MU EDCA Parameter Set element with updated MU EDCA parameters in the Beacon frames and Probe Response frames it transmits. The EDCA Parameter Set Update Count subfield in the QoS Info field of the EDCA Parameter Set element and MU EDCA Parameter Set element is incremented every time any EDCA parameters or MU EDCA parameters change.

A non-AP HE STA shall update the dot11EDCATable and dot11MUEDCATable that correspond to fields in an EDCA Parameter Set element or an MU EDCA Parameter Set element within an interval of time equal to one beacon interval after receiving an updated EDCA or MU EDCA parameter set from its associated AP. When updating its MIB attributes, an HE STA stores the value of the EDCA Parameter Set Update Count subfield in the QoS Info field of the received EDCA Parameter Set element or MU EDCA Parameter Set element.

A non-AP HE STA shall use the EDCA Parameter Set Update Count subfield in the QoS Capability element of Beacon frames, where present, to determine whether the STA is using the current EDCA (and optionally MU EDCA) parameter values. If the EDCA Parameter Set Update Count subfield value is different from the value that has been stored, the STA shall query the updated EDCA (and any MU EDCA) parameter values by sending a Probe Request frame to the AP.

NOTE 1—If the QoS Capability element is present in a Beacon frame, the EDCA Parameter Set element and the MU EDCA Parameter Set element are not present. In this case, the only way for an HE STA to obtain the updated parameters is to send a Probe Request frame to the AP.

A non-AP HE STA that receives a Basic Trigger frame that contains a User Info field addressed to the STA shall update its CWmin[AC], CWmax[AC], AIFSN[AC], and MUEDCATimer[AC] state variables to the values contained in the dot11MUEDCATable, for all the ACs from which at least one QoS Data frame was transmitted successfully in an HE TB PPDU in response to the Trigger frame. A QoS Data frame is

transmitted successfully by the STA in an HE TB PPDU for an AC if it requires immediate acknowledgment and the STA receives an immediate acknowledgment for that frame, or if the QoS Data frame does not require immediate acknowledgment.

The MUEDCATimer[AC] state variable is updated with the value contained in the MU EDCA Timer subfield of the MU EDCA Parameter Set element. The backoff counter maintenance corresponding to the updated state variables shall follow the rules in 10.23.2.2, except that if AIFSN[AC] is 0, then the EDCAF corresponding to that AC shall be suspended until the MUEDCATimer[AC] reaches 0 or is reset to 0. The updated MUEDCATimer[AC] shall start at the end of the immediate response if the transmitted HE TB PPDU contains at least one QoS Data frame for that AC that requires immediate acknowledgment, and shall start at the end of the HE TB PPDU if the transmitted HE TB PPDU does not contain any QoS Data frames for that AC that require immediate acknowledgment.

In a non-AP HE STA, each MUEDCATimer[AC] shall uniformly count down without suspension to 0 when its value is nonzero.

NOTE 2—A non-AP STA that sends a frame to the AP with an OM Control subfield containing a value of 1 in the UL MU Disable subfield or a value of 0 in the UL MU Disable subfield and a value of 1 in the UL MU Data Disable subfield does not participate in UL MU operation. As such it is exempt from updating its EDCA access parameters to the values contained in the MU EDCA Parameter Set element as defined in this subclause.

NOTE 3—A non-AP STA does not update its state variables to the values contained in the MU EDCA Parameter Set element if any of the following applies:

- a) The Trigger frame addressed to the STA is not a Basic Trigger frame.
- b) The STA does not include QoS Data frames in the HE TB PPDU response sent in response to the Basic Trigger frame.
- c) The STA transmits the HE TB PPDU in response to a Basic Trigger frame following the rules defined in 26.5.4.

NOTE 4—The TXOP limits are not updated by the procedure defined in this subclause, but by the procedure in 10.23.2.9.

A non-AP STA that sends frames that are not addressed to its associated AP may use the EDCA parameters values that are contained in the most recently received EDCA Parameter Set element sent by the AP with which the STA is associated or the default EDCA parameter values (see Table 9-155), following the rules in 10.2.3.2.

If the MUEDCATimer[AC] of a non-AP HE STA reaches 0, either by counting down or due to a reset following the reception of an MU EDCA Reset frame, the STA shall update CWmin[AC], CWmax[AC], and AIFSN[AC] to the values that are contained in the most recently received EDCA Parameter Set element sent by the AP with which the STA is associated.

A non-AP HE STA that sends a frame with an OM Control subfield with the UL MU Disable subfield set to 1 or with the UL MU Disable subfield set to 0 and the UL MU Data Disable subfield set to 1 as defined in 26.9.3 may set the MUEDCATimer[AC] for all ACs to 0 on receiving an immediate acknowledgment from the OMI responder. The STA continues the current EDCA backoff procedure without modifying the QSRC[AC], QLRC[AC] or the backoff counter for the associated EDCAF, regardless of whether the MUEDCATimer[AC] has reached zero, until the STA invokes a new EDCA backoff procedure. The STA follows the rules defined in 10.23.2.2 for updating CWmin, CWmax, and AIFSN for that AC.

A non-AP HE STA that receives an individually addressed MU EDCA Reset frame from its associated AP may reset the MUEDCATimer[AC] to 0 for an AC if the bit corresponding to that AC in the Affected ACs subfield is equal to 1 when the MUEDCATimer[AC] of the STA is not equal to 0. The STA may invoke a new EDCA backoff procedure after the MUEDCATimer[AC] is reset for that AC and after CWmin[AC], CWmax[AC], and AIFSN[AC] are updated for that AC, as per this subclause, in response to the MUEDCATimer[AC] reset.

26.2.8 Multiple frame transmission in an EDCA TXOP in the 6 GHz band

A STA that operates in the 6 GHz band and transmits multiple frames shall follow the rules defined in 10.23.2.7 with the exceptions described in this subclause.

NOTE 1—In the 6 GHz band, the TXOP field in the HE-SIG-A field can be understood by all STAs. The TXOP field provides TXOP protection equivalent to non-HT duplicate RTS/CTS.

In a TXOP without non-HT duplicate PPDU, if the TXOP is protected by the TXOP field in the HE-SIG-A field of the first HE PPDU in the TXOP, i.e., the TXOP field is not set to UNSPECIFIED, the TXOP holder shall set the TXVECTOR parameter CH_BANDWIDTH of the subsequent PPDU in the TXOP to indicate a bandwidth that is the same or narrower than the bandwidth indicated by the CH_BANDWIDTH parameter of the first HE PPDU in the TXOP.

NOTE 2—Frame exchanges in a TXOP that is protected by RTS/CTS follow the rules in 10.23.2.8.

Additionally, if the first HE PPDU whose TXOP field in the HE-SIG-A field is not set to UNSPECIFIED is a DL HE MU PPDU with preamble puncture, then the TXOP holder shall use the 20 MHz channels for the non-initial PPDU that are within the set of 20 MHz channels where pre-HE modulated fields of the first HE PPDU whose TXOP field in the HE-SIG-A field is not set to UNSPECIFIED are located.

Within an obtained TXOP that does not include HE PPDU whose TXOP field in the HE-SIG-A field is not UNSPECIFIED nor non-HT duplicate PPDU, the TXOP holder shall set the TXVECTOR parameter CH_BANDWIDTH of a non-initial PPDU to be equal to or less than the TXVECTOR parameter CH_BANDWIDTH of the preceding PPDU that was transmitted in the same TXOP, subject to the following constraints:

- If the preceding PPDU is a DL HE MU PPDU with preamble puncture, the TXOP holder shall set the TXVECTOR parameter CH_BANDWIDTH of the non-initial PPDU to a value whose corresponding 20 MHz channels are within a set of 20 MHz channels where pre-HE modulated fields of the preceding PPDU are located.
- If the non-initial PPDU is a DL HE MU PPDU with preamble puncture, the TXOP holder shall set the TXVECTOR parameter RU_ALLOCATION of the non-initial PPDU to a value whose corresponding RU is within a set of 20 MHz channels where pre-HE modulated fields of the preceding PPDU are located.

26.3 Fragmentation and defragmentation

26.3.1 General

An HE STA shall follow the fragmentation procedures defined in 10.2.6, 10.4, and 10.5, except that some of the rules are relaxed as defined in the subclauses of 26.3. Dynamic fragmentation provides further flexibility in aggregating data to fit in a constrained PPDU duration (see 26.5).

Subclauses 10.2.6 and 10.4 define the procedure to generate uniformly fragmented MSDUs or MMPDUs, where the length of each fragment is the same, except for the last fragment. These fragments are not included in an A-MPDU under HT-immediate block ack agreements.

An HE STA can negotiate the use of different levels of dynamic fragmentation:

- Level 1: support for one dynamic fragment that is a non-A-MPDU; no support for dynamic fragments in an A-MPDU that does not contain an S-MPDU.

- Level 2: support for dynamic fragments in an A-MPDU that does not contain an S-MPDU, subject to the following conditions:
 - No more than one dynamic fragment of any given MSDU or A-MSDU in the A-MPDU; this dynamic fragment is sent under a block ack agreement.
 - No more than one MPDU that is a dynamic fragment of an MMPDU in the A-MPDU (see 26.6.2 and 26.6.3).
- Level 3: support for dynamic fragments in an A-MPDU that does not contain an S-MPDU, subject to the following conditions:
 - No more than four dynamic fragments of any given MSDU or A-MSDU in the A-MPDU; these dynamic fragments are sent under a block ack agreement.
 - No more than one MPDU that is a dynamic fragment of an MMPDU in the A-MPDU (see 26.6.2 and 26.6.3).

NOTE 1—An HE STA that is operating in level 2 or level 3 also supports transmitting and receiving level 1 dynamic fragments.

NOTE 2—Under level 2, there might be multiple Data frames that are dynamic fragments, but these have different sequence numbers and/or UPs.

This subclause defines the procedure for generating nonuniformly fragmented MSDUs, A-MSDUs, or MMPDUs, where the length of each fragment is not necessarily the same. The length of the first fragment shall be greater than or equal to the minimum fragment size indicated in the Minimum Fragment Size subfield in the HE Capabilities Information field in the HE Capabilities element sent by the recipient STA. If the length of the MSDU, A-MSDU, or MMPDU is less than the minimum fragment size, then the MSDU, A-MSDU, or MMPDU shall not be fragmented. The fragments generated with dynamic fragmentation are referred to as dynamic fragments.

These procedures allow the inclusion of one dynamic fragment that is a non-A-MPDU in level 1, level 2, and level 3 and allow the inclusion of one or more dynamic fragments within an A-MPDU that contains one or more MPDUs in level 2 and level 3.

An HE STA follows the rules defined in 26.3.2 for generating these fragments and the rules defined in 26.3.3 for defragmenting of the received dynamic fragments. In 26.3.2 and 26.3.3, the HE STA follows the fragmentation level that is indicated in the Dynamic Fragmentation Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element it transmits, unless an ADDBA Extension element is included in the ADDBA Request and ADDBA Response frames exchanged during the block ack setup procedure. In this case, the HE STA follows the fragmentation level that is indicated in an ADDBA Extension element in the ADDBA Response frame it receives (see 26.3.3.1) during the block ack setup procedure.

An HE STA shall set the HE Fragmentation Operation subfield in the ADDBA Extension element, if present, in the ADDBA Request or ADDBA Response frame to a value that is less than or equal to the value of the Dynamic Fragmentation Support subfield in the HE Capabilities element it transmits.

An HE STA shall set the HE Fragmentation Operation subfield in the ADDBA Extension element, if present, in the ADDBA Response frame to a value that is less than or equal to the value of HE Fragmentation Operation subfield in the ADDBA Extension element, if present, in the received ADDBA Request frame.

26.3.2 Dynamic fragmentation

26.3.2.1 General

With level 1 and level 2 dynamic fragmentation, the following apply:

- An originator STA transmitting one or more dynamic fragments shall solicit an immediate response from the recipient STA for each of the fragments.
- The originator STA shall transmit the fragments in order as defined in 10.4.

With level 3 dynamic fragmentation, not all dynamic fragments require an immediate response, and dynamic fragments are not required to be sent in order.

An HE STA may transmit dynamic fragments of an A-MSDU, provided the A-MSDU Fragmentation Support subfield in the HE Capabilities element transmitted by the recipient is 1.

The originator STA shall not transmit concurrently the dynamic fragments of a number of outstanding MSDUs and A-MSDUs (if supported) to the same recipient STA that is greater than N_{max} , where N_{max} for MSDUs and A-MSDUs (if supported) is calculated based on the Maximum Number of Fragmented MSDUs/A-MSDUs subfield in the HE Capabilities element transmitted by the recipient STA. The term *outstanding* refers to an MPDU containing all or part of an MSDU, A-MSDU, or MMPDU for which transmission has been started and for which delivery of the MSDU, A-MSDU, or MMPDU has not yet been completed (i.e., an acknowledgment of the final fragment has not been received and the MSDU, A-MSDU, or MMPDU has not been discarded due to retries, lifetime, or some other reason).

NOTE—Dynamic fragments of outstanding MMPDUs are not counted in the check against the N_{max} limit.

An originator STA may retransmit the full MSDU, A-MSDU, or MMPDU if all the previously transmitted dynamic fragments of that MSDU, A-MSDU, or MMPDU have explicitly failed at the receiving STA. An originator STA may retransmit a failed fragment if one or more of the previously transmitted fragments of that MSDU, A-MSDU, or MMPDU have explicitly failed at the receiving STA (see the next paragraph for a definition of the term *explicitly failed*). The frame body length and contents of the retransmitted fragment shall be the same as the initially transmitted fragment and shall remain fixed for the lifetime of the MSDU, A-MSDU, or MMPDU at that STA, except when the frame is not encrypted, all the fragments preceding the initial transmitted fragment were received, and all the fragments following the initial transmitted fragment either have explicitly failed or have not been transmitted. In such cases, the frame body length and contents of the retransmitted fragment may be different from the initially transmitted fragment.

A fragment has explicitly failed at the receiving STA if the originator STA receives an immediate response that contains the following:

- A valid first MPDU that is not an Ack frame, Compressed BlockAck frame, or Multi-STA BlockAck frame, of which all the preceding pre-EOF MPDU delimiters are received.
- A Multi-STA BlockAck frame that does not contain a BA Information field with TID equal to that of the fragment.
- A Compressed BlockAck frame or Multi-STA BlockAck frame that contains a BA Information field with TID equal to that of the fragment, but does not acknowledge receipt of the fragment.

An originator STA shall not transmit to a recipient STA an MPDU that is not carried in an A-MPDU or an A-MPDU that carries dynamic fragments that do not satisfy the conditions in 26.3.2.2, 26.3.2.3, and 26.3.2.4.

26.3.2.2 Level 1 dynamic fragmentation

An originator STA may transmit one dynamic fragment of an MSDU, A-MSDU (if supported by the recipient), or MMPDU that is a non-A-MPDU and that is not sent under a block ack agreement to a recipient STA using level 1 dynamic fragmentation if the Dynamic Fragmentation Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element received from the recipient STA is 1, 2, or 3.

An originator STA may transmit a fragmented MSDU or A-MSDU (if supported by the recipient) under a block ack agreement to a recipient STA using level 1 dynamic fragmentation, provided the following condition is met: the Dynamic Fragmentation Support subfield in the HE Capabilities element received from the recipient STA is 1, 2, or 3; and for the block ack agreement associated with the TID of the MSDU or A-MSDU, the HE Fragmentation Operation subfield is 1 if the ADDBA Extension element is present in the ADDBA Response frame received from the recipient STA.

26.3.2.3 Level 2 dynamic fragmentation

An originator STA may transmit fragmented MSDUs or A-MSDU (if supported by the recipient) under a block ack agreement to a recipient STA using level 2 dynamic fragmentation, provided one of the following conditions is met:

- The Dynamic Fragmentation Support subfield in the HE Capabilities element received from the recipient STA is 2; and for the block ack agreement associated with the TID of the MSDU or A-MSDU, the HE Fragmentation Operation subfield is 2 if the ADDBA Extension element is present in the ADDBA Response frame received from the recipient STA.
- The Dynamic Fragmentation Support subfield in the HE Capabilities element received from the recipient STA is 3; and for the block ack agreement associated with the TID of the MSDU or A-MSDU, the ADDBA Extension element is present, and the HE Fragmentation Operation subfield is 2 in the ADDBA Response frame received from the recipient STA.

An originator STA may transmit to a recipient STA the following:

- One dynamic fragment of an MSDU, A-MSDU (if supported by the recipient), or MMPDU that is a non-A-MPDU using level 1 dynamic fragmentation.
- Up to one dynamic fragment of an MSDU or A-MSDU (if supported by the recipient) for each MSDU or A-MSDU and up to one dynamic fragment of an MMPDU in an A-MPDU where the A-MPDU contains at least one dynamic fragment using level 2 dynamic fragmentation; the dynamic fragment of an MSDU or A-MSDU shall be sent under a block ack agreement.

NOTE—The originator STA follows the rules in 10.12.8 for generating the S-MPDU, the rules in 10.25.6.7 for generating the A-MPDU, and the rules in 26.6.3 for generating a multi-TID A-MPDU that can contain the fragment of the MMPDU.

26.3.2.4 Level 3 dynamic fragmentation

An originator STA may transmit fragmented MSDUs or A-MSDU (if supported by the recipient) under a block ack agreement or fragmented MMPDU to a recipient STA using level 3 dynamic fragmentation, provided the following conditions are met:

- The Dynamic Fragmentation Support subfield in the HE Capabilities element received from the recipient STA is 3.
- For the block ack agreement associated with the TID of the MSDU or A-MSDU, the HE Fragmentation Operation subfield is 3 if the ADDBA Extension element is present in the ADDBA Response frame received from the recipient STA.

Level 3 fragmentation allows multiple fragments of an MSDU or A-MSDU included in the same A-MPDU, reducing the fragments' transmission delay.

An originator STA may transmit to a recipient STA the following:

- One dynamic fragment of an MSDU, A-MSDU (if supported by the recipient), or MMPDU that is a non-A-MPDU using level 1 dynamic fragmentation.
 - The originator STA shall follow the rules defined in 10.12.8 for generating the S-MPDU.
- Up to four dynamic fragments of an MSDU or A-MSDU (if supported by the recipient) for each MSDU or A-MSDU and up to one dynamic fragment of an MMPDU in an A-MPDU, where the A-MPDU contains at least one dynamic fragment using level 3 dynamic fragmentation; the dynamic fragment of an MSDU or A-MSDU shall be sent under block ack agreement.
 - The originator STA shall set the Fragment Number subfield of each MPDU to a value less than 4.
 - The originator STA shall follow the rules defined in 10.25.6.7 for generating the A-MPDU with the exception that the A-MPDU shall contain MPDUs whose range of the Sequence Number subfields does not exceed $B_L / 4$, where B_L is the length of the Block Ack Bitmap field of the Compressed BlockAck or Multi-STA BlockAck frame that corresponds to a TID of a transmitted fragment (see 10.25.6 and 26.4).
 - The originator STA shall ensure that $WinSize_O$ does not exceed $B_L / 4$, where B_L is the length of the Block Ack Bitmap field of the Compressed BlockAck or Multi-STA BlockAck frame that corresponds to a TID of a transmitted fragment (see 10.25.6 and 26.4).

26.3.3 Dynamic defragmentation

26.3.3.1 General

An HE STA shall set the Dynamic Fragmentation Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element it transmits to 0 if it does not support dynamic fragments. Otherwise, the HE STA shall set the Dynamic Fragmentation Support subfield as follows:

- Set to 1, 2, or 3 if the STA supports reception of dynamic fragments following the procedure defined in 26.3.2.2.
- Set to 2 or 3 if the STA supports reception of dynamic fragments following the procedure defined in 26.3.2.3.
- Set to 3 if the STA supports reception of dynamic fragments following the procedure defined in 26.3.2.4.

An HE STA shall set `dot11HEDynamicFragmentationLevel` to the value of Dynamic Fragmentation Support subfield in the HE Capabilities element it transmits if it supports reception of dynamic fragments.

If a fragment is sent under level 1, level 2, or level 3 fragmentation, it belongs to dynamic fragment.

Defragmentation of dynamic fragments shall follow the rules defined in 10.5 with the following exceptions:

- The recipient STA shall support the concurrent reception of dynamic fragments of a number of *outstanding* MSDUs and A-MSDUs when supported from the same transmitting STA that is equal to N_{max} , where N_{max} for MSDUs and A-MSDUs (if supported) is obtained from the Maximum Number Of Fragmented MSDUs/A-MSDUs Exponent subfield in the HE Capabilities element transmitted by the STA. The term *outstanding* refers to an MPDU containing all or part of an MSDU or A-MSDU for which transmission has been started and for which delivery of the MSDU or A-MSDU has not yet been completed (i.e., an acknowledgment of the final fragment has not been received and the MSDU, A-MSDU, or MMPDU has not been discarded due to retries, lifetime, or some other reason). The recipient STA shall support the concurrent reception of dynamic fragments of one outstanding MMPDU from a transmitting STA.
- The recipient STA may be subject to the receive timer rules for each of the MSDUs, A-MSDUs, and MMPDUs defined in 10.5.

A STA that has `dot11AMSDUFragmentationOptionImplemented` true shall set the A-MSDU Fragmentation Support subfield in the HE Capabilities element to 1. Otherwise, the STA shall set the A-MSDU Fragmentation Support subfield in the HE Capabilities element to 0.

26.3.3.2 Level 1 dynamic defragmentation

Upon reception of a non-A-MPDU or A-MPDU that carries a dynamic fragment, the recipient STA responds with an Ack frame or a Multi-STA BlockAck frame if the received fragment is contained in an MPDU that solicits an immediate response. The receiver STA shall follow the rules defined in 10.3.2.11 for generating the Ack frame for the soliciting dynamic fragment that is an MPDU or carried in an S-MPDU or ack-enabled single-TID A-MPDU and the rules defined in 26.4 for generating the Multi-STA BlockAck frame that contains the acknowledgment for the soliciting S-MPDU that carries one dynamic fragment and carried in an HE TB PPDU, ack-enabled single-TID A-MPDU or ack-enabled multi-TID A-MPDU.

26.3.3.3 Level 2 dynamic defragmentation

Upon reception of a non-A-MPDU that is a dynamic fragment or A-MPDU that carries one or more dynamic fragments, the recipient STA responds with one of the following frames:

- An Ack frame or Multi-STA BlockAck frame if the received fragment is a non-A-MPDU or contained in an A-MPDU and solicits the immediate response. The recipient STA shall follow the rules defined in 10.3.2.11 for generating the Ack frame for the soliciting dynamic fragment that is an MPDU or carried in an S-MPDU or ack-enabled single-TID A-MPDU and the rules defined in 26.4 for generating the Multi-STA BlockAck frame that contains the acknowledgment for the soliciting S-MPDU carried in an HE TB PPDU, ack-enabled single-TID A-MPDU, or ack-enabled multi-TID A-MPDU.
- A Compressed BlockAck frame or Multi-STA BlockAck frame if the received fragments of an MSDU or A-MSDU (if any) (up to one fragment for each MSDU or A-MSDU) are contained in an A-MPDU that contains an MPDU that solicits an immediate response. The recipient STA shall follow the rules defined in 10.25.6.5 for generating the BlockAck frame and the rules in 26.4 for generating the Multi-STA BlockAck frame, except that the STA shall:
 - Set to 0 the LSB of the Fragment Number subfield in the Block Ack Starting Sequence Control subfield of the Compressed BlockAck frame or Multi-STA BlockAck frame that corresponds to a TID of a received fragment that is sent under a block ack agreement.
 - Set to 1 each bit of the Block Ack Bitmap field that corresponds to a Sequence Number subfield and TID subfield of a received fragment that is sent under a block ack agreement contained in the soliciting A-MPDU that is not an S-MPDU.
- The STA shall update the corresponding block acknowledgment record for an MSDU or A-MSDU for which fragments are received only if that MSDU or A-MSDU is successfully reconstructed (see 10.5). Otherwise, the STA shall not update the block ack record for that MSDU or A-MSDU.

NOTE—The recipient STA sets the bits of the Block Ack Bitmap field that correspond to a Sequence Number and TID subfield of an MPDU that is not a fragment as defined in 10.25.6.5.

A recipient STA shall discard any fragments of incomplete MSDUs or A-MSDUs that have been received during an HT-immediate block ack session for a TID if it receives a BlockAckReq frame from the originator STA for that TID if the fragments have a Sequence Number field value that is less than the value of the Starting Sequence Number field of the BlockAckReq frame or have a sequence number that is less than *WinStartB* (see 10.25.6.6). The comparison of the two values is performed circular modulo 4096 as described in 10.25.1.

26.3.3.4 Level 3 dynamic defragmentation

Upon reception of a non-A-MPDU that is a dynamic fragment or A-MPDU that carries one or more dynamic fragments, the recipient STA responds with one of the following frames:

- An Ack frame or a Multi-STA BlockAck frame if the received fragment is a non-A-MPDU or contained in an A-MPDU that solicits the immediate response. The recipient STA shall follow the rules defined in 10.3.2.11 for generating the Ack frame for the soliciting dynamic fragment that is a MPDU or carried in an S-MPDU or ack-enabled single-TID A-MPDU and the rules defined in 26.4 for generating the Multi-STA BlockAck frame that contains the acknowledgment for the soliciting S-MPDU carried in an HE TB PPDU, ack-enabled single-TID A-MPDU, or ack-enabled multi-TID A-MPDU.
- A Compressed BlockAck frame or Multi-STA BlockAck frame if the received fragments of an MSDU or A-MSDU (if any) corresponding to a TID (one or more fragments for each MSDU or A-MSDU) are contained in an A-MPDU where at least one received MPDU's Fragment Number field corresponding to the same TID as the received fragments is of nonzero value that solicits the immediate response. The recipient STA shall follow the rules in 10.25.6.5 for generating the BlockAck frame and the rules in 26.4 for generating the Multi-STA BlockAck frame, except that the STA shall
 - Set to 1 the LSB of the Fragment Number subfield in the Block Ack Starting Sequence Control subfield of the Compressed BlockAck frame or Multi-STA BlockAck frame that corresponds to a TID of a received fragment that is sent under a block ack agreement.
 - Set to 1 each bit in position B of the Block Ack Bitmap field that corresponds to a received fragment that is sent under a block ack agreement and shall set it to 0 otherwise, with B calculated as $B = 4 \times (SN - SSN) + FN$, where the operations on the sequence numbers are performed modulo 4096.
 SN is the value of the Sequence Number subfield of an MPDU containing the fragment for which the receive status is indicated.
 SSN is the value of the Starting Sequence Number subfield of the Block Ack Starting Sequence Control subfield of the BlockAck frame.
 FN is the value in the Fragment Number subfield.
- A Compressed BlockAck frame or Multi-STA BlockAck frame if the received fragments of an MSDU or A-MSDU (if any) corresponding to a TID (one or more fragments for each MSDU or A-MSDU) are contained in an A-MPDU where all the received MPDUs' Fragment Number fields corresponding to the same TID as the received fragments are of zero value that solicits the immediate response. The recipient STA shall follow the rules in 10.25.6.5 for generating the BlockAck frame and the rules in 26.4 for generating the Multi-STA BlockAck frame, except that the STA shall
 - Set to 0 the LSB of the Fragment Number subfield in the Block Ack Starting Sequence Control subfield of the Compressed BlockAck frame or Multi-STA BlockAck frame that corresponds to a TID of a received fragment that is sent under a block ack agreement.
 - Set to 1 each bit of the Block Ack Bitmap field that corresponds to a Sequence Number subfield and TID subfield of a received fragment that is sent under a block ack agreement contained in the soliciting A-MPDU that is not an S-MPDU.
- The STA shall update the corresponding block acknowledgment record for an MSDU or A-MSDU for which fragments are received only if that MSDU or A-MSDU is successfully reconstructed (see 10.5). Otherwise, the STA shall not update the block ack record for that MSDU or A-MSDU.

NOTE—The recipient STA sets the bits of the Block Ack Bitmap field that correspond to a Sequence Number and TID subfield of an MPDU that is not a fragment as defined in 10.25.6.5.

The recipient STA shall discard any fragments of incomplete MSDUs or A-MSDUs that have been received during an HT-immediate block ack session for a TID if it receives a BlockAckReq frame from the originator STA for that TID and the fragments have a Sequence Number field value that is less than the value of the Starting Sequence Number field of the BlockAckReq frame or have a sequence number that is less than *WinStartB* (see 10.25.6.6). The comparison of the two values is performed circular modulo 4096 as described in 10.25.1.

26.4 HE acknowledgment procedure

26.4.1 Overview

The HE acknowledgment procedure builds on the features defined for HT-immediate block ack (see 10.25.6), with the following extensions:

- Support for a Multi-STA BlockAck frame
- Support for a MU-BAR Trigger frame
- Support for a Multi-TID BlockAckReq frame
- Support for BlockAck Bitmap field lengths of 32, 64, 128, and 256
- Acknowledging QoS Data frames with two or more TIDs using a Multi-STA BlockAck frame
- Acknowledging QoS Data frames with one or more TIDs and a Management frame using a Multi-STA BlockAck frame
- Acknowledging all MPDUs in a PPDU using a variant of the Multi-STA BlockAck frame
- Acknowledging MPDUs from multiple associated STAs using a single Multi-STA BlockAck frame
- Acknowledging MPDUs from multiple unassociated STAs with a single Multi-STA BlockAck frame

An HE STA shall be able to respond with Compressed BlockAck frames if HT-immediate block ack is supported in the role of recipient (see 10.25.6.1). An HE STA shall be able to respond with a Multi-STA BlockAck frame if multi-TID A-MPDU operation (26.6.3) is supported in the role of recipient.

A non-AP HE STA that is associated with an AP and that sends a Multi-STA BlockAck frame shall set the AID11 subfield in the Per AID TID Info field of the Multi-STA BlockAck frame to 0 and the RA field to the MAC address of the intended recipient. A non-AP HE STA that is not associated with an AP shall not send a Multi-STA BlockAck frame.

An HE AP that sends a Multi-STA BlockAck frame where the Per AID TID Info fields are addressed to more than one STA shall set the RA field to the broadcast address. An HE AP that sends a Multi-STA BlockAck frame where the Per AID TID Info fields are all addressed to a single recipient STA and that is sent in response to an HE TB PPDU may set the RA field of the Multi-STA BlockAck frame either to the address of the recipient STA or to the broadcast address. An HE AP that sends a Multi-STA BlockAck frame where the Per AID TID Info fields are all addressed to a single recipient STA and that is not sent in response to an HE TB PPDU shall set the RA field of the Multi-STA BlockAck frame to the address of the recipient STA.

An HE AP that sends a Multi-STA BlockAck frame to an associated STA shall set the AID11 subfield in the Per AID TID Info field of the Multi-STA BlockAck frame to the 11 LSBs of the AID of the intended STA. An HE AP that sends a Multi-STA BlockAck frame to an unassociated STA shall set the AID11 subfield in the Per AID TID Info field of the Multi-STA BlockAck frame to 2045.

An HE STA that transmits a Multi-STA BlockAck frame shall use a rate, HT-MCS, <VHT-MCS, NSS> tuple, or <HE-MCS, NSS> tuple that is supported by all recipient STAs.

An HE STA that receives a Multi-STA BlockAck frame that is a response to frames requiring acknowledgment shall examine the Per AID TID Info field received in the Multi-STA BlockAck frame and shall process each Per AID TID Info field using the procedure defined in 26.4.2.

A non-AP HE STA that receives a Multi-STA BlockAck frame that is a response to frames requiring acknowledgment but that do not belong to an established block ack agreement shall examine each Per AID TID Info field received in the Multi-STA BlockAck frame as follows:

- If the Ack Type field is 1 and the TID field is less than 8, then the Per AID TID Info field indicates the acknowledgment of a tagged MPDU that is a QoS Data frame with the indicated TID. The BA Information field is addressed to the STA if the AID of the BA Information field contains the STA's AID, and is processed according to the procedure defined in 26.4.2.
- If the Ack Type field is 1 and the TID field is 15, then the Per AID TID Info field indicates the acknowledgment of a tagged MPDU that is a Management frame that solicits acknowledgment or a PS-Poll frame. The BA Information field is addressed to the STA if the AID of the BA Information field contains the STA's AID, and is processed according to the procedure defined in 26.4.2.
- If the Ack Type field is 0, the AID field is 2045, and the TID field is 15, then the Per AID TID Info field indicates the acknowledgment of a tagged MPDU that is a Management frame soliciting immediate acknowledgment. The RA field in the Per AID TID Info field is the MAC address of an unassociated STA for which the Per AID TID Info subfield is intended. The BA Information field is addressed to the STA if the RA field of the BA Information field contains the STA's MAC address, and is processed according to the procedure defined in 26.4.2.

An HE AP with dot11MultiBSSIDImplemented equal to true shall not send to a non-AP STA that is associated with an AP corresponding to a nontransmitted BSSID in the multiple BSSID set a Multi-STA BlockAck frame with the TA field set to the transmitted BSSID, unless the HE AP has received from the non-AP STA an HE Capabilities element with the Rx Control Frame To MultiBSS subfield in HE MAC Capabilities Information field equal to 1.

An AP that transmits a Multi-STA BlockAck frame addressed to HE STAs shall set the TA field of the frame to the MAC address of the AP, unless dot11MultiBSSIDImplemented is true and the Multi-STA BlockAck frame is directed to STAs from at least two different BSSs of the multiple BSSID set. In this case, the AP shall set the TA field of the frame to the transmitted BSSID.

NOTE—An AP sets the TA field of the Multi-STA BlockAck frame that is not carried in HE MU PPDU to the transmitted BSSID when the TXOP is obtained from the transmitted BSSID (see 10.23.2.4).

An HE STA that transmits a Multi-TID BlockAckReq frame in a PPDU that is not an HE TB PPDU shall set the TID Value subfields in the Per TID Info subfields of the BAR Information field of the Multi-TID BlockAckReq frame to TIDs that correspond to ACs that have the same or higher priority as the primary AC. An HE STA that transmits a Multi-TID BlockAckReq frame in an HE TB PPDU may set each of the TID Value subfields in the Per TID Info subfields of the BAR Information field of the Multi-TID BlockAckReq frame to a TID that corresponds to any AC.

An HE STA that transmits a BlockAckReq frame in an HE TB PPDU may set the TID subfield in the AID TID Info field in the BAR Information field of the BlockAckReq frame to a TID that corresponds to any AC.

26.4.2 Acknowledgment context in a Multi-STA BlockAck frame

A recipient of an A-MPDU shall set the Ack Type subfield and TID subfield in the Per AID TID Info field of the Multi-STA BlockAck frame sent as a response depending on the acknowledgment context as follows:

- An HE AP that receives, prior to association, an A-MPDU that includes one tagged MPDU that is a Management frame soliciting an acknowledgment but does not include any other MPDUs may generate a Multi-STA BlockAck frame using the procedure defined in the *pre-association ack context* [described in item b) below in this subclause].
- An HE STA that receives an A-MPDU that does not include a tagged MPDU but does include one or more untagged MPDUs that are QoS Data frames with Normal Ack or Implicit BAR ack policy belonging to the same block ack agreement may generate a Multi-STA BlockAck frame as follows:
 - If all MPDUs in the A-MPDU are received successfully, then the recipient may follow the procedure defined in the *all ack context* [described in item a) below in this subclause].
 - Otherwise, the recipient shall follow the procedure defined in the *block ack context* [described in item d) below in this subclause].
- If an HE STA supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1 and if the A-MPDU includes a tagged MPDU that is a Management frame that solicits acknowledgment and includes one or more MPDUs (either tagged MPDUs or untagged MPDUs) that are QoS Data frames with Normal Ack or Implicit BAR ack policy, then the recipient shall generate Multi-STA BlockAck frame as follows:
 - If all the MPDUs in the A-MPDU are received successfully, then the recipient may follow the procedure defined in the *all ack context*.
 - Otherwise:
 - For the MPDU that is a Management frame, the recipient shall create a Per AID TID info field using the procedure defined in the *ack context* [described in item c) below in this subclause] with the TID value set to 15.
 - For the tagged MPDUs that are QoS Data frames, the recipient shall create a Per AID TID info field using the procedure defined in the *ack context* with the TID set to the TID of the QoS Data frame.
 - For the untagged MPDUs that are QoS Data frames, the recipient shall create a Per AID TID info field using the procedure defined in *block ack context* with the TID set to the TID of the QoS Data frame.
- If an HE STA supports multi-TID aggregation and the A-MPDU does not include a tagged MPDU but does include untagged MPDUs that are QoS Data frames with Implicit BAR ack policy and belonging to more than one block ack agreement, then the recipient shall generate a Multi-STA BlockAck frame as follows:
 - If all MPDUs in the A-MPDU are received successfully, then the recipient may follow the procedure defined in the *all ack context*.
 - Otherwise, for each TID included the received A-MPDU, the recipient shall create a per AID TID info field using the procedure defined in the *block ack context* with the TID set to the TID of the QoS Data frame.

NOTE—A STA indicates the maximum number of Per AID TID Info fields with the same AID excluding the one for a Management frame that it can include in the Multi-STA BlockAck frame in the Multi-TID Aggregation Rx Support field in the HE Capabilities element it transmits.

The procedures for the different acknowledgment contexts for generating a Multi-STA BlockAck frame are defined as follows:

- a) *All ack context*: If the originator had set the All Ack Support subfield in the HE Capabilities element to 1, then the recipient may set the Ack Type field to 1 and the TID subfield to 14 to indicate the reception of all the MPDUs carried in the eliciting A-MPDU or multi-TID A-MPDU. Otherwise, the recipient shall not set the Ack Type field to 1 and the TID subfield to 14. The Multi-STA BlockAck frame shall contain only one Per AID TID Info field addressed to an originator in the Multi-STA BlockAck frame. The recipient determines that all the MPDUs carried in the eliciting A-MPDU were received if there were no MPDU delimiter CRC errors and no MPDU FCS errors in that A-MPDU.
- b) *Pre-association ack context*: A recipient receiving a Management frame from an unassociated STA that requires an acknowledgment shall set the Ack Type field to 0, AID subfield to 2045, and TID field to 15 in the Per AID TID Info field and set the RA field of the Per AID TID Info field to the intended recipient's MAC address to indicate the reception of that Management frame.
- c) *Ack context*: A recipient that sets the Ack-Enabled Aggregation Support subfield in the HE Capabilities element to 1 and that receives a tagged MPDU soliciting acknowledgment shall set the Ack Type field to 1 and, if the tagged MPDU is a QoS Data frame, set the TID field to the TID of the QoS Data frame or, if the tagged MPDU is a Management frame or PS-Poll frame, set the TID field to 15.

If a received A-MPDU contains more than one tagged MPDU that solicits an immediate acknowledgment, then the Multi-STA BlockAck frame shall contain multiple Per AID TID Info fields, with Ack Type field equal to 1, one for each such received tagged MPDU requesting an acknowledgment.

The TID field is set to the TID of the QoS Data or QoS Null frame that is being acknowledged and set to 15 for a PS Poll frame or Management frame that is being acknowledged.

- d) *Block ack context*: The recipient shall set the Ack Type field to 0 and the TID field of a Per AID TID Info field to the TID value of MPDUs requesting block acknowledgment that are carried in the eliciting A-MPDU or multi-TID A-MPDU.

The Multi-STA BlockAck frame may contain multiple occurrences of these Per AID TID Info fields addressed to an originator, one for each MPDU that is requesting block acknowledgment. In such cases, the Block Ack Starting Sequence Control and Block Ack Bitmap fields shall be set according to 10.25.6 for each block ack session and according to 26.3 for each block ack session with dynamic fragmentation.

The allowed values for the TID field in this context are 0 to 7 (for indicating block acknowledgment of QoS Data frames).

Variable bitmap lengths may be included in the Per AID TID Info field when the originator and recipient negotiate their use as defined in 26.4.3.

The Ack Type subfield(s) in a Multi-STA BlockAck frame shall be set to 0 if the Multi-STA BlockAck frame is sent in response to an MU-BAR Trigger frame.

Upon receipt of a Multi-STA BlockAck frame, the originator shall examine each Per AID TID Info field and shall perform the following operations:

- If the AID subfield is 0 for an AP originator or the non-AP STA's AID for a non-AP STA originator, the Ack Type field is 0, and the TID field is less than 8, then the BlockAck Starting Sequence Control, TID, and Block Ack Bitmap fields of the Per AID TID Info field are processed according to 10.25.6, 26.3, and 26.4.4.
- If the AID subfield is 2045, the Ack Type field is 0, and the TID field is 15, then the Per AID TID Info field indicates the acknowledgment of a single Management frame sent by the unassociated STA as defined by the acknowledgment context.

- If the AID subfield is 0 for an AP originator or the non-AP STA's AID for a non-AP STA originator, the Ack Type field is 1, and the TID is less than or equal to 7 or is equal to 15, then the Per AID TID Info field indicates the acknowledgment of an tagged MPDU that is a QoS Data frame identified by the value of the TID, a Management frame, or a PS-Poll frame.
- If the AID subfield is 0 for an AP originator or the non-AP STA's AID for a non-AP STA originator, the Ack Type field is 1, and the TID subfield of AID TID Info field is 14, then the Per AID TID Info field indicates the acknowledgment of all MPDUs carried in the eliciting A-MPDU as defined by the acknowledgment context.

If an associated non-AP STA that does not support the UORA procedure receives a Multi-STA BlockAck frame in response to an eliciting frame, and if the Multi-STA BlockAck frame contains a Per AID TID Info subfield with the AID11 subfield set to 2045, then the STA shall ignore this Per AID TID Info subfield, and shall continue to parse the following Per AID TID Info subfields (if any).

26.4.3 Negotiation of block ack bitmap lengths

Both the Compressed BlockAck frame and Multi-STA BlockAck frame allow different Block Ack Bitmap subfield lengths. The length of the Block Ack Bitmap subfield is indicated in the Fragment Number subfield of the Block Ack Starting Sequence Control field as defined in 9.3.1.8. The allowed Block Ack Bitmap lengths for each of the negotiated buffer sizes are defined in Table 26-1.

Table 26-1—Negotiated buffer size and Block Ack Bitmap subfield length

Negotiated buffer size	Block Ack Bitmap subfield length in a Compressed BlockAck frame (bits)	Block Ack Bitmap subfield length in a Multi-STA BlockAck frame (bits)
1–64	64	32 or 64
65–128	64 or 256	32, 64, or 128
129–256	64 or 256	32, 64, 128, or 256
NOTE—A 32-bit Block Ack Bitmap subfield length is not allowed, unless the originator has set the 32-bit BA Bitmap Support field in the HE MAC Capabilities Information field in the HE Capabilities element to 1.		

An HE STA that transmits a Compressed BlockAck frame or a Multi-STA BlockAck frame shall use a Block Ack Bitmap subfield length identified in Table 26-1 for the negotiated buffer size of the block ack agreement to which the BA Information field corresponds.

The recipient is allowed to respond with a Block Ack Bitmap subfield in the BA Information field that is less than the maximum allowed Block Ack Bitmap for the negotiated buffer size. The length of the Block Ack Bitmap subfield in a Compressed BlockAck frame or a Multi-STA BlockAck frame may be less than the maximum allowed Block Ack Bitmap but shall be sufficient to include the recipient's scoreboard state for MPDUs beginning with the MPDU for which the Sequence Number subfield value is $WinStart_R$ and ending with a successfully received MPDU for which the Sequence Number subfield is less than or equal to $WinEnd_R$.

The recipient shall not include in the Buffer Size field of an ADDBA Response frame a value that would cause the BlockAck Bitmap length of its block ack responses to exceed the BlockAck Bitmap length that is derived by the Buffer Size field of the ADDBA Request frame sent by the originator. When the Buffer Size field in the ADDBA Request frame is 0, the Buffer Size field of an ADDBA Response frame is in the range 1 to 64.

NOTE 1—Refer to Block Ack Bitmap subfield length identified in Table 26-1 for the negotiated buffer size of the block ack agreement.

A recipient shall not include in a Multi-STA BlockAck frame a Per AID TID Info field with a 32-bit BlockAck Bitmap field addressed to an originator if the 32-bit BA Bitmap Support field in the HE MAC Capabilities Information field in the HE Capabilities element received from that originator is 0.

NOTE 2—A Multi-STA BlockAck frame might include Per AID TID Info fields with a 32-bit BlockAck Bitmap field addressed to other originators, and the nonsupporting originator needs to be able to parse these fields to locate a possible Per AID TID Info field addressed to it.

The originator of a BlockAckReq frame, MU-BAR Trigger frame, GCR MU-BAR Trigger frame, or a A-MPDU that includes QoS Data frames that solicits an immediate BlockAck frame response or Management frame that solicits acknowledgment shall set the Duration field accounting for the largest BlockAck Bitmap length based on negotiated buffer size.

A recipient shall not transmit a Compressed BlockAck frame or a Multi-STA BlockAck frame with the LSB of the Fragment Number subfield set to 1, unless the recipient has received from the originator an HE Capabilities element with the Dynamic Fragmentation Support subfield equal to 3. If the LSB of the Fragment Number subfield of the BlockAck frame is 1, then the Block Ack Bitmap fields are set as defined in 26.3.2.4.

26.4.4 Per-PPDU acknowledgment selection rules

26.4.4.1 General

A STA that transmits a PPDU can solicit different immediate responses for frames contained in the PPDU by using the Ack Policy Indicator subfield of QoS Data or QoS Null frames, the type of the frame, PPDU format, number of TIDs in the A-MPDU, and the EOF/Tag field setting of the A-MPDU delimiter.

26.4.4.2 Responding to an HE SU PPDU or HE ER SU PPDU with an SU PPDU

An HE STA that receives an HE SU PPDU or HE ER SU PPDU carrying an A-MPDU that includes MPDUs, solicits acknowledgment, and does not include a triggering frame shall respond using an SU PPDU as follows:

- a) If the A-MPDU includes only one MPDU and the MPDU is a tagged MPDU that either is a QoS Data frame or QoS Null frame with Normal Ack ack policy or is a Management frame that solicits acknowledgment, then the STA shall respond with an Ack frame.
- b) If the A-MPDU includes only one MPDU and the MPDU is an EOF MPD that is a PS-Poll frame, the STA shall respond with an Ack frame or a QoS Data frame.
- c) If the HE STA supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1, and if the A-MPDU includes more than one MPDU, only one of which solicits acknowledgment and the MPDU that solicits acknowledgment is a tagged MPDU that either is a QoS Data frame or a QoS Null frame with Normal Ack ack policy or is a Management frame that solicits acknowledgment, then the HE STA shall respond with an Ack frame.
- d) If the A-MPDU does not include a tagged MPDU but does include one or more untagged MPDUs that are QoS Data frames belonging to the same block ack agreement and with the Ack Policy Indicator subfield equal to Implicit BAR for at least one MPDU, then the STA shall respond with a Compressed BlockAck frame as defined in 10.25.6.5. Alternatively, if all the MPDUs carried in the eliciting A-MPDU were received, the STA may respond with a Multi-STA BlockAck frame with Ack Type field set to 1 and the TID field set to 14 as defined in 26.4.2 if the originator of the A-MPDU has indicated support for the all ack context by setting the All Ack Support subfield in the HE MAC Capabilities Information field to 1.

- e) If the HE STA supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1, and if the A-MPDU includes a Management frame that solicits an acknowledgment and includes one or more QoS Data frames with Normal Ack or Implicit BAR ack policy, then the STA shall respond with a Multi-STA BlockAck frame as defined in 26.4.2.
- f) If the HE STA supports multi-TID aggregation and if the A-MPDU includes two or more QoS Data frames with Implicit BAR ack policy and belonging to more than one block ack agreement, then the STA shall respond with a Multi-STA BlockAck frame as defined in 26.4.2.

26.4.4.3 Responding to an HE MU PPDU with an SU PPDU

If an AP intends to solicit an immediate response in an SU PPDU, the following apply:

- An AP shall set the Ack Policy Indicator subfield in the QoS Data and QoS Null frames to Normal Ack or Implicit BAR in at most one A-MPDU in the HE MU PPDU (see 10.3.2.13.1 for an example of this sequence).
- The A-MPDUs in the HE MU PPDU shall not contain a Management frame that solicits acknowledgment.

An HE STA that receives an HE MU PPDU with an A-MPDU that contains MPDUs that solicit acknowledgment and that does not include a triggering frame shall respond using an SU PPDU as follows:

- a) If the A-MPDU carries only one MPDU and the MPDU is a tagged MPDU that is either a QoS Data frame or QoS Null frame with Normal Ack ack policy, then the STA shall respond with an Ack frame.
- b) If the HE STA supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1, and if the A-MPDU includes more than one MPDU, only one of which solicits acknowledgment and the MPDU that solicits acknowledgment is a tagged MPDU that is a QoS Data frame or a QoS Null frame with Normal Ack ack policy, then the HE STA shall respond with an Ack frame.
- c) If the A-MPDU does not include a tagged MPDU but does include one or more untagged MPDUs that are QoS Data frames belonging to the same block ack agreement and with the Ack Policy Indicator subfield equal to Implicit BAR for at least one MPDU, then the STA shall respond with a Compressed BlockAck frame as defined in 10.25.6.5. Alternatively, if all the MPDUs carried in the eliciting A-MPDU were received, the STA may respond with a Multi-STA BlockAck frame with the Ack Type set to 1 and the TID field set to 14 as defined in 26.4.2 if the originator of the A-MPDU has indicated support for the all ack context by setting the All Ack Support subfield in the HE MAC Capabilities Information field to 1.
- d) If the HE STA supports multi-TID aggregation and if the A-MPDU includes two or more QoS Data frames with Implicit BAR ack policy addressed to it and belonging to more than one block ack agreement, then the STA shall respond with a Multi-STA BlockAck frame as defined in 26.4.2.

NOTE—A control response frame carried in an SU PPDU that is an immediate response to an HE MU PPDU follows the rules defined in 10.6.6.5.

An AP that sends an HE MU PPDU shall not set the Ack Policy Indicator subfield to Normal Ack or Implicit BAR for any of the MPDUs carried in the HE MU PPDU if the PPDU containing a control response would occupy one or more 20 MHz channels where pre-HE modulated fields of the HE MU PPDU are not located.

26.4.4.4 Responding to an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU with an HE TB PPDU

An AP that sends an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU that contains an MPDU that solicits an immediate response carried in an HE TB PPDU shall set the Ack Policy Indicator subfield to

HETP Ack for each of the QoS Data frames for which it intends to solicit an immediate response (see 10.3.2.13.2). If a Management frame that solicits acknowledgment is carried in an HE MU PPDU, then the response is carried in an HE TB PPDU. A non-AP STA that receives an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU with an A-MPDU that contains a QoS Data frame with HETP Ack ack policy addressed to it or contains a Management frame that solicits an immediate acknowledgment shall not respond, unless it has also received a triggering frame in the A-MPDU.

A non-AP STA that receives an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU with an A-MPDU that contains one or more MPDUs soliciting acknowledgment and includes a triggering frame shall respond with an HE TB PPDU as follows:

- a) If the A-MPDU includes only one MPDU and the MPDU is a tagged MPDU that either is a QoS Data or QoS Null frame with HETP Ack ack policy or is a Management frame that solicits acknowledgment, then the STA shall respond with an Ack frame.
- b) If the HE STA supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1, and if the A-MPDU includes more than one MPDU, only one of which solicits acknowledgment and the MPDU that solicits acknowledgment is a tagged MPDU that either is a QoS Data or QoS Null frame with HETP Ack ack policy or is a Management frame that solicits acknowledgment, then the HE STA shall respond with an Ack frame.
- c) If the A-MPDU does not include a tagged MPDU but does include one or more untagged MPDUs that are QoS Data frames belonging to the same block ack agreement and with the Ack Policy Indicator subfield equal to HETP Ack for at least one MPDU, then the STA shall respond with a Compressed BlockAck frame as defined in 10.25.6.5. Alternatively, if all the MPDUs carried in the eliciting A-MPDU were received, the STA may respond with a Multi-STA BlockAck frame with the Ack Type set to 1 and the TID field set to 14 as defined in 26.4.2 if the originator of the A-MPDU has indicated support for the all ack context by setting the All Ack Support subfield in the HE MAC Capabilities Information field to 1.
- d) If the HE STA supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1, and if the A-MPDU includes a Management frame that solicits an acknowledgment and contains one or more QoS Data frames with HETP Ack or Implicit BAR ack policy, then the STA shall respond with a Multi-STA BlockAck frame as defined in 26.4.2.
- e) If the HE STA supports multi-TID aggregation and if the A-MPDU includes two or more QoS Data frames with HETP Ack ack policy and belonging to more than one block ack agreement, then the STA shall respond with a Multi-STA BlockAck frame.

26.4.4.5 Responding to an HE TB PPDU with an SU PPDU

A non-AP STA that sends an HE TB PPDU as a response to a Basic Trigger frame shall set the Ack Policy Indicator subfield of the QoS Data frames or QoS Null frames to Normal Ack or Implicit BAR (see 10.3.2.13.3 for an example of this sequence).

If the HE TB PPDU carries MPDUs only from one STA and if the HE AP intends to send the response in an SU PPDU, then the HE AP shall respond using an SU PPDU as follows:

- a) If the A-MPDU includes only one MPDU, and the MPDU is a tagged MPDU that either is a QoS Data frame or QoS Null frame with Normal Ack ack policy or is a Management frame that solicits acknowledgment, then the HE AP shall respond with either an Ack frame or a Multi-STA BlockAck frame with the Ack Type field set to 1.
- b) If the HE AP supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1, and if the A-MPDU includes more than one MPDU, only one of which solicits acknowledgment and the MPDU that solicits acknowledgment is a tagged MPDU that either is a QoS Data or QoS Null frame with Normal Ack

ack policy or is a Management frame that solicits acknowledgment, then the HE AP shall respond with an Ack frame or a Multi-STA BlockAck frame with the Ack Type field set to 1.

- c) If the A-MPDU does not include a tagged MPDU but does include one or more untagged MPDUs that are QoS Data frames belonging to the same block ack agreement and with Ack Policy Indicator subfield equal to Implicit BAR for at least one MPDU, then the HE AP shall respond with a Compressed BlockAck frame as defined in 10.25.6.5 or a Multi-STA BlockAck frame with the Ack Type field set to 0. Alternatively, if all the MPDUs carried in the eliciting A-MPDU were received, the HE AP may respond with a Multi-STA BlockAck frame with the Ack Type field set to 1 and the TID field set to 14 if the originator of the A-MPDU has indicated support for the all ack context by setting the All Ack Support subfield in the HE MAC Capabilities Information field to 1, as defined in 26.4.2.
- d) If the HE AP supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1, and if the A-MPDU carries a Management frame that solicits acknowledgment and carries one or more QoS Data frames with Implicit BAR ack policy, then the HE AP shall respond with a Multi-STA BlockAck frame as defined in 26.4.2.
- e) If the HE AP supports multi-TID aggregation and if the A-MPDU includes two or more QoS Data frames with Normal Ack or Implicit BAR ack policy and belonging to more than one block ack agreement, then the HE AP shall respond with a Multi-STA BlockAck frame as defined in 26.4.2.

If the AP receives HE TB PPDU from more than one STA, and if the AP intends to send the response in an SU PDU, then the AP shall respond with a Multi-STA BlockAck frame carried in an SU PDU that contains the appropriate settings in each Per AID TID Info field addressed to each STA as defined in 26.4.2.

26.4.4.6 Responding to an HE TB PDU with an HE MU PDU

A non-AP STA that sends an HE TB PDU as a response to a Basic Trigger frame that solicits an immediate response shall set the Ack Policy Indicator subfield to Normal Ack or Implicit BAR for each of the QoS Data frames carried in the A-MPDU (see 10.3.2.13.3 for an example of this sequence).

If an HE AP sends a response to an HE TB PDU that it received using an HE MU PDU, then the AP shall respond to each A-MPDU that it received using the following procedure:

- a) If the A-MPDU received from a STA includes only one MPDU and the MPDU is a tagged MPDU that either is a QoS Data frame or QoS Null frame with Normal Ack ack policy or is a Management frame that solicits acknowledgment, then the STA shall respond with an Ack frame or a Multi-STA BlockAck frame with the Ack Type field set to 1 carried in the HE MU PDU.
- b) If the HE AP supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1, and if the A-MPDU includes more than one MPDU, only one of which solicits acknowledgment and the MPDU that solicits acknowledgment is a tagged MPDU that either is a QoS Data frame with Normal Ack ack policy or is a Management frame that solicits acknowledgment, then the HE AP shall respond with an Ack frame or a Multi-STA BlockAck frame with the Ack Type field set to 1 carried in the HE MU PDU.
- c) If the A-MPDU does not include a tagged MPDU but does include one or more untagged MPDUs that are QoS Data frames belonging to the same block ack agreement and with the Ack Policy Indicator subfield equal to Implicit BAR for at least one MPDU, then the HE AP shall respond with a Compressed BlockAck frame as defined in 10.25.6.5 or a Multi-STA BlockAck frame with the Ack Type field set to 0. Alternatively, if all the MPDUs carried in the eliciting A-MPDU were received, the HE AP may respond with a Multi-STA BlockAck frame with the Ack Type field set to 1 and the TID field set to 14 if the originator of the A-MPDU has indicated support for the all ack context by setting the All Ack Support subfield in the HE MAC Capabilities Information field to 1 as defined in 26.4.2 carried in the HE MU PDU.

- d) If the HE AP supports ack-enabled aggregation by setting the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field to 1, and if the A-MPDU carries a Management frame that solicits acknowledgment and carries one or more QoS Data frames with Implicit BAR ack policy, then the HE AP shall respond with a Multi-STA BlockAck frame as defined in 26.4.2 carried in the HE MU PPDU.
- e) If the HE AP supports multi-TID aggregation and if the A-MPDU includes two or more QoS Data frames with Implicit BAR ack policy and belonging to more than one block ack agreement, then the HE AP shall respond with a Multi-STA BlockAck frame as defined in 26.4.2.

In addition, an AP with `dot11MultiBSSIDImplemented` equal to true may do one of the following:

- For each BSS belonging to the multiple BSSID set for which the AP has received an HE TB PPDU, the AP responds with a Multi-STA BlockAck frame with RA field set to the broadcast address and carried in a DL HE MU PPDU (see 9.3.1.8.7). The AP shall set the TXVECTOR parameter STA_ID for the RU carrying the Multi-STA BlockAck frame to the value of the BSSID Index field as defined in 26.11.1.
- The AP may respond with a Multi-STA BlockAck frame with the TA field set to the transmitted BSSID and carried in a DL HE MU PPDU to acknowledge the STA's transmission, if the recipient non-AP STA is associated with an AP corresponding to a nontransmitted BSSID of the multiple BSSID set and the AP has received an HE Capabilities element from the STA with the Rx Control Frame To MultiBSS subfield equal to 1 (see 9.3.1.8.7). The AP shall set the TXVECTOR parameter STA_ID for the RU carrying the Multi-STA BlockAck frame to 2047.

NOTE—An AP includes at most one Ack or BlockAck frame (group addressed Multi-STA BlockAck frame included) in an A-MPDU as specified in Table 9.7.3.

26.4.5 HE block acknowledgment request and response rules

An HE AP may solicit BlockAck frame responses from multiple HE STAs using an MU-BAR Trigger frame or GCR MU-BAR Trigger frame. An HE AP shall not send a Multi-TID BlockAckReq (as part of a User Info field addressed to the STA in an MU-BAR Trigger frame or as a BlockAckReq frame) to a STA that has not indicated support for multi-TID A-MPDU. The Block Ack Bitmap length of the block ack sent in response to an eliciting Multi-TID BlockAckReq frame, BlockAckReq frame, GCR MU-BAR Trigger frame, or MU-BAR Trigger frame is determined as defined in 26.4.3.

An HE STA that receives a BlockAckReq frame or an MU-BAR Trigger frame that contains a Compressed BlockAckReq variant in the User Info field addressed to the STA or receives a GCR MU-BAR Trigger frame that contains a Compressed BlockAckReq variant in the Common Info field shall respond with a Compressed BlockAck frame as defined in 10.25.6 or a Multi-STA BlockAck frame as defined in 26.4 with the Starting Sequence Number subfield set to the Starting Sequence Number subfield of the Block Ack Request Starting Sequence Control subfield and the length of the Block Ack Bitmap subfield calculated as defined in 26.4.3.

An HE STA that receives a Multi-TID BlockAckReq frame or an MU-BAR Trigger frame that contains a Multi-TID BlockAckReq variant in the User Info field addressed to the STA or receives a GCR MU-BAR Trigger frame that contains a Multi-TID BlockAckReq variant in the Common Info field shall respond with a Multi-STA BlockAck frame that contains a Per AID TID Info field with a Block Ack Bitmap subfield for each of the TIDs (with values less than 8) contained in the BlockAckReq frame, with the Starting Sequence Number subfield set to the Starting Sequence Number subfield of the Block Ack Request Starting Sequence Control subfield, and with the length of the Block Ack Bitmap subfield calculated as defined in 26.4.3.

A non-AP HE STA that responds to a Compressed BlockAckReq frame, Multi-TID BlockAckReq frame, MU-BAR Trigger frame, or GCR MU-BAR Trigger frame with a Multi-STA BlockAck frame shall set the Ack Type subfield of the Multi-STA BlockAck frame to 0.

26.5 MU operation

26.5.1 HE DL MU operation

26.5.1.1 General

HE DL MU operation allows an AP to transmit simultaneously to one or more non-AP STAs in DL OFDMA, DL MU-MIMO, or both.

The PSDU on each RU in an HE MU PPDU shall be padded to end at the same time, indicated by the L-SIG field as described in 27.3.11.5.

The padding procedure for each A-MPDU in an HE MU PPDU is defined in 26.6.2.2.

An AP shall not transmit an HE MU PPDU with an RU that is narrower than the PPDU bandwidth and that is allocated to more than one STA (DL MU-MIMO), unless the AP has received from each STA an HE Capabilities element with the Partial Bandwidth DL MU-MIMO subfield in the HE PHY Capabilities Information field equal to 1.

An AP shall not send to a STA an A-MPDU contained in an HE MU PPDU that contains one of the following combinations of frames, unless the AP has received from the STA an HE Capabilities element with the Ack-Enabled Aggregation Support subfield in the HE MAC Capabilities Information field equal to 1:

- A QoS Data frame with HETP Ack ack policy carried in an A-MPDU subframe with the EOF/Tag field set to 1 and a Trigger frame.
- A Management frame carried in an A-MPDU subframe with the EOF/Tag field set to 1, where the Management frame is a Disassociation frame, (Re)Association Response frame, Authentication frame, or Action frame, and a Trigger frame.

The AP shall follow the EDCA procedure defined in 10.23 and the following additional rules:

- If at least one of the frame exchanges in HE DL MU operation requires an immediate response (i.e., the AP includes at least one triggering frame) and if the AP receives an immediate response with at least one correct MPDU from at least one of the solicited STAs, the frame exchange is successful.
- If at least one of the frame exchanges in HE DL MU operation requires an immediate response (i.e., the AP includes at least one triggering frame) and if the AP receives no immediate response, the frame exchange is not successful.
- The AP follows the MPDU aggregation rules in 26.6 that supersede the rules in 10.23.2.7.

An AP shall not transmit an HE MU PPDU where the number of OFDM symbols in the HE-SIG-B field is greater than 16 to a non-AP STA with a 20 MHz operating channel width.

26.5.1.2 RU addressing in an HE MU PPDU

The Type and Subtype subfields in the Frame Control field and address type (individually addressed or group addressed) of MPDUs may be different across A-MPDUs in different PSDUs within the same HE MU PPDU.

An AP includes in the TXVECTOR for an HE MU PPDU at least one parameter STA_ID for each RU in the HE MU PPDU as defined in 26.11.1. The AP shall not include in the TXVECTOR more than one parameter STA_ID with the same value, unless the value is 2046 (indicating an unallocated RU).



An HE AP with `dot11MultiBSSIDImplemented` equal to false shall not include in an HE MU PPDU anything other than one or more of the following:

- One or more individually addressed RUs, corresponding to the parameter `STA_ID` equal to the AID(s) of STA(s) associated with the AP, to carry information intended for the STA(s).
- A broadcast RU corresponding to parameter `STA_ID` equal to 0 to carry information intended for the STAs associated with the AP that are not the recipient of a individually addressed RU.
- A broadcast RU corresponding to parameter `STA_ID` equal to 2045 to carry information intended for STAs not associated with the AP.
- One or more RUs corresponding to parameter `STA_ID` equal to 2046 for unassigned RUs.

An HE AP with `dot11MultiBSSIDImplemented` equal to true shall not include in an HE MU PPDU anything other than one or more of the following:

- One or more individually addressed RUs, corresponding to the parameter `STA_ID` equal to the AID(s) of STA(s) associated with any AP in the multiple BSSID set, to carry information intended for that associated STA.
- A broadcast RU corresponding to parameter `STA_ID` equal to 0 to carry information intended for STAs associated with the AP corresponding to the transmitted BSSID and not the recipient of an individually addressed RU.
- A broadcast RU corresponding to parameter `STA_ID` equal to the BSSID Index of a BSSID in a multiple BSSID set to carry information intended for STAs associated with the AP corresponding to that BSSID and not the recipient of an individually addressed RU.
- A broadcast RU corresponding to parameter `STA_ID` equal to 2047 to carry information intended for STAs associated with the APs in the multiple BSSID set and not the recipient of an individually addressed RU or another broadcast RU corresponding to parameter `STA_ID` equal 0 or equal to the BSSID Index of a BSSID in a multiple BSSID set.
- A broadcast RU corresponding to parameter `STA_ID` equal to 2045 to carry information intended for STAs not associated with any AP in the multiple BSSID set.
- One or more RUs corresponding to parameter `STA_ID` equal to 2046 for unassigned RUs.

A non-AP STA that receives an HE MU PPDU where the `RXVECTOR` includes a parameter `STA_ID` that matches the 11 LSBs of the non-AP STA's AID shall disregard any broadcast RU in the HE MU PPDU. A non-AP STA that receives an HE MU PPDU where the `RXVECTOR` includes a parameter `STA_ID` that is equal to the BSSID Index of the BSSID of the AP with which the STA is associated (see 9.4.2.73) shall disregard a broadcast RU corresponding to parameter `STA_ID` equal to 2047.

For an AP with `dot11MultiBSSIDImplemented` equal to false, the AP shall set the TA and BSSID fields (when present) to the BSSID of the AP's BSS for all the frames carried in a broadcast RU that corresponds to parameter `STA_ID` equal to 0 or 2045.

For an AP with `dot11MultiBSSIDImplemented` equal to true, the AP shall set the TA and BSSID fields (when present) to the transmitted BSSID for all the frames carried in a broadcast RU corresponding to parameter `STA_ID` equal to 0 or 2045 or 2047.

For an AP with `dot11MultiBSSIDImplemented` equal to true, the AP shall set the TA and BSSID fields (when present) to the BSSID of the corresponding nontransmitted BSSID for the all frames carried in a broadcast RU corresponding to parameter `STA_ID` equal to the BSSID Index of a nontransmitted BSSID in a multiple BSSID set.

26.5.1.3 RU allocation in an HE MU PPDU

An AP shall not transmit a 40 MHz HE MU PPDU in the 2.4 GHz band with an RU allocated to a 20 MHz operating non-AP HE STA, unless the AP has received from the 20 MHz operating non-AP HE STA an HE Capabilities element with the 20 MHz In 40 MHz HE PPDU In 2.4 GHz Band subfield in the HE PHY Capabilities Information field equal to 1.

An AP shall not transmit a 160 MHz or 80+80 MHz HE MU PPDU with an RU allocated to a 20 MHz operating non-AP HE STA, unless the AP has received from the 20 MHz operating non-AP HE STA an HE Capabilities element with the 20 MHz In 160/80+80 MHz HE PPDU subfield in the HE PHY Capabilities Information field equal to 1.

An AP shall not transmit a 160 MHz or 80+80 MHz HE MU PPDU with an RU allocated to an 80 MHz operating non-AP HE STA, unless the AP has received from the 80 MHz operating non-AP HE STA an HE Capabilities element with the 80 MHz In 160/80+80 MHz HE PPDU subfield in the HE PHY Capabilities Information field equal to 1.

An AP shall follow the RU restriction rules defined in 27.3.2.8 when assigning an RU to a 20 MHz operating non-AP STA in a 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz HE MU PPDU. An AP shall follow the rules in 27.3.2.7 and 27.3.2.9 or the SST subchannel (if applicable) in which the STA is operating (see 26.8.7) if allocating RUs to a non-AP STA.

An AP shall not transmit a 40 MHz HE MU PPDU in the 2.4 GHz band with a 242-tone RU allocated to a 20 MHz operating non-AP HE STA, unless the AP has received from the 20 MHz operating non-AP HE STA an HE Capabilities element with B4 of the Supported Channel Width Set subfield of the HE PHY Capabilities Information field equal to 1.

An AP shall not transmit a 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz HE MU PPDU in the 5 GHz band with a 242-tone RU allocated to a 20 MHz operating non-AP HE STA, unless the AP has received from the 20 MHz operating non-AP HE STA an HE Capabilities element with B5 of the Supported Channel Width Set subfield of the HE PHY Capabilities Information field equal to 1.

An HE MU PPDU shall have a sufficient number of RUs allocated to users such that all of the following conditions are satisfied:

- a) At least $N \times 4 \times 26$ subcarriers are modulated by the allocated RUs within the entire PPDU, where N is the number of 20 MHz subchannels that are not preamble punctured in the PPDU.
- b) For each 20 MHz subchannel S within the bandwidth of the HE MU PPDU, at least 2×26 subcarriers are modulated by the allocated RUs in the 20 MHz subchannel S if all of the following are true:
 - 1) At least one RU is allocated in the 20 MHz subchannel S .
 - 2) Transmitter is an AP.
 - 3) The AP is operating in an operating class for which the behavior limits set listed in Annex E includes the DFS_50_100_Behavior.
 - 4) The AP has received at least one Beacon frame from OBSS B within the past $\text{dot11ObssNbRuToleranceTime}$ in the current operating channel in which any of the following are true:
 - i) The Extended Capabilities element is not present.
 - ii) The OBSS Narrow Bandwidth RU In OFDMA Tolerance Support field in the Extended Capabilities element is not present.
 - iii) The OBSS Narrow Bandwidth RU In OFDMA Tolerance Support field in the Extended Capabilities element is 0.
 - 5) The 20 MHz subchannel S overlaps with the operating bandwidth of the OBSS B .
- c) At least one RU is allocated in the primary 20 MHz.

26.5.2 UL MU operation

26.5.2.1 General

UL MU operation allows an AP to solicit simultaneous immediate response frames from one or more non-AP HE STAs. A non-AP HE STA shall follow the rules in this subclause for the transmission of response frames in an HE TB PPDU, unless the Trigger frame is an MU-RTS Trigger frame, in which case the response is a CTS frame sent in a non-HT PPDU (see 26.2.6).

A non-AP STA shall not send a triggering frame.

An HE STA that is a mesh STA shall not transmit and shall not receive HE TB PPDU.

A non-AP HE STA shall set the TRS Support subfield in the HE Capabilities element it transmits to 1 if its `dot11TRSOOptionImplemented` is true; otherwise, the STA shall set it to 0.

A non-AP HE STA shall set the UL 2×996-tone RU Support subfield in HE Capabilities element to 1 if it supports receiving a frame that carries a TRS Control subfield that allocates a 2×996-tone RU or a Trigger frame with User Info field addressed to the STA with RU Allocation subfield indicating a 2×996-tone RU.

If a non-AP HE STA supports transmitting an HE TB PPDU that uses UL MU-MIMO within an RU that spans the entire PPDU bandwidth, then the STA shall set `dot11HEFullBWULMUMIMOImplemented` to true and the Full Bandwidth UL MU-MIMO subfield of the HE PHY Capabilities Information field in the HE Capabilities element it transmits to 1. Otherwise, the HE STA shall set `dot11HEFullBWULMUMIMOImplemented` to false and the Full Bandwidth UL MU-MIMO subfield to 0.

If a non-AP HE STA with `dot11HEFullBWULMUMIMOImplemented` equal to true also supports transmitting an HE TB PPDU that uses UL MU-MIMO within an RU that does not span the entire PPDU bandwidth, then the STA shall set `dot11HEPartialBWULMUMIMOImplemented` to true and the Partial Bandwidth UL MU-MIMO subfield in the HE PHY Capabilities Information field in the HE Capabilities element it transmits to 1. Otherwise, the non-AP HE STA shall set `dot11HEPartialBWULMUMIMOImplemented` to false and the Partial Bandwidth UL MU-MIMO subfield to 0.

If a non-AP STA has `dot11HEDeviceClass` equal to ClassA (1), then it indicates Class A in the Device Class subfield in the HE PHY Capabilities Information field in the HE Capabilities element, and the HE TB PPDU it transmits shall meet the Class A requirements in 27.3.15. Otherwise, it indicates Class B in the Device Class subfield in the HE PHY Capabilities Information field in the HE Capabilities element, and the HE TB PPDU it transmits shall meet the Class B requirements in 27.3.15.

A non-AP HE STA with `dot11HTVHTTriggerOptionImplemented` equal to true shall set the HT And VHT Trigger Frame Rx Support field to 1 in the HE MAC Capabilities Information field in HE Capabilities elements that it transmits. A non-AP HE STA with `dot11HTVHTTriggerOptionImplemented` equal to false shall set the HT And VHT Trigger Frame Rx Support field to 0.

26.5.2.2 Rules for soliciting UL MU frames

26.5.2.2.1 General

An HE AP shall not allocate an RU for a 40 MHz HE TB PPDU to a 20 MHz operating non-AP HE STA in the 2.4 GHz band, unless the AP has received from the 20 MHz operating non-AP HE STA an HE Capabilities element with the 20 MHz In 40 MHz HE PPDU In 2.4 GHz Band subfield in the HE PHY Capabilities Information field in its HE Capabilities element equal to 1.

An HE AP shall not allocate an RU for an 160 MHz or 80+80 MHz HE TB PPDU to a 20 MHz operating non-AP HE STA, unless the AP has received from the 20 MHz operating non-AP HE STA an HE Capabilities element with the 20 MHz In 160/80+80 MHz HE PPDU in the HE PHY Capabilities Information field equal to 1.

An AP shall not allocate to a 20 MHz operating non-AP HE STA a 242-tone RU for a 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz HE TB PPDU transmission.

An AP shall not transmit a Trigger frame soliciting an HE TB PPDU that uses UL MU-MIMO within an RU that does not span the entire PPDU bandwidth to a non-AP STA from which it has not received an HE Capabilities element with the Partial Bandwidth UL MU-MIMO subfield of the HE PHY Capabilities Information field equal to 1.

An AP shall not transmit a Trigger frame soliciting an HE TB PPDU that uses UL MU-MIMO within an RU that spans the full bandwidth to a non-AP STA from which it has not received an HE Capabilities element with the Full Bandwidth UL MU-MIMO subfield of the HE PHY Capabilities Information field equal to 1.

An AP shall not send a frame that carries a TRS Control subfield to a non-AP STA that has not set the TRS Support subfield to 1 in the HE MAC Capabilities Information field in the HE Capabilities element it transmits.

An AP shall not send a frame that carries a TRS Control subfield that allocates a 2×996-tone RU to a non-AP STA or a Trigger frame with a User Info field that allocates a 2×996-tone RU to a non-AP STA, unless the AP has received from the non-AP STA an HE MAC Capabilities element with the UL 2×996-tone RU Support subfield in the MAC Capabilities Information field equal to 1.

NOTE 1—An AP does not send a Trigger frame containing a User Info field with AID12 subfield carrying the 12 LSBs of the AID of a non-AP STA or a frame addressed to a non-AP STA that carries a TRS Control subfield if the AP has received from the non-AP STA an OM Control subfield with UL MU Disabled subfield set to 1 and UL MU Data Disable subfield set to 0 (see 26.9.3).

An AP that transmits a PPDU may solicit an HE TB PPDU from one or more non-AP STAs through one of the following mechanisms:

- Including in the PPDU one or more Trigger frames that include one or more User Info fields with one of the following AID12 subfield settings:
 - The AID12 subfield is set to the 12 LSBs of the AID of the non-AP STA if the User Info field is addressed to a STA that is associated with the AP.
 - The AID12 subfield is set to the 12 LSBs of the AID of the non-AP STA if the User Info field is addressed to a STA that is associated with an AP corresponding to a nontransmitted BSSID in a multiple BSSID set to which the AP belongs, the TA field of the Trigger frame is set to the transmitted BSSID, and the non-AP STA has set the Rx Control Frame To MultiBSS subfield in the HE Capabilities element it transmits to 1.
 - The AID12 subfield indicates that one or more contiguous RA-RUs are allocated (see 26.5.4).
- Including in the PPDU one or more individually addressed frames that include a TRS Control subfield and that are carried in one of the following:
 - An S-MPDU that solicits an immediate Ack frame (see 10.12.8).
 - An A-MPDU that solicits an immediate BlockAck frame (see 10.25.6.7).
 - A multi-TID A-MPDU that solicits an immediate Multi-STA BlockAck frame (see 26.6.3).

The AP shall follow the rules in 26.3.2 if any MPDUs are fragments.

More than one Trigger frame may be aggregated in an A-MPDU. If more than one Trigger frame is aggregated in an A-MPDU, all of them shall have the same content. An AP may include a frame carrying a TRS Control subfield in an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU.

NOTE 2—The TRS Control subfields within MPDUs carried in an A-MPDU have the same value (see 10.8).

An A-MPDU shall not include both a Trigger frame and a frame carrying a TRS Control subfield.

An AP shall not transmit a triggering frame if all of the following conditions are satisfied:

- The AP is operating in an operating class for which the behavior limits set listed in Annex E includes the DFS_50_100_Behavior (see Table E-1).
- The AP has received at least one Beacon frame from OBSS *B* within the past $\text{dot11ObssNbRuToleranceTime}$ in the current operating channel in which any of the following is true:
 - The Extended Capabilities element is not present.
 - The OBSS Narrow Bandwidth RU In OFDMA Tolerance Support field in the Extended Capabilities element is not present.
 - The OBSS Narrow Bandwidth RU In OFDMA Tolerance Support field in the Extended Capabilities element is 0.
- The triggering frame allocates at least one 26-tone RU whose location in frequency overlaps with the operating bandwidth of the OBSS *B*.

If a non-AP HE STA operating in an operating class for which the behavior limits set listed in Annex E includes the DFS_50_100_Behavior does not respond to a triggering frame in which the non-AP HE STA is allocated a 26-tone RU, then the AP should not allocate a 26-tone RU to the non-AP HE STA in the next triggering frame.

An AP shall not transmit a Trigger frame with a User Info field addressed to a non-AP STA in an HT or VHT PPDU, unless the AP has received from the non-AP STA an HE Capabilities element with the HT And VHT Trigger Frame Rx Support subfield in HE MAC Capabilities Information field equal to 1.

An AP shall not use the short guard interval for an HT or VHT PPDU that carries a Trigger frame. A Trigger frame shall not be carried in a DSSS or HR/DSSS PPDU. An AP shall not use STBC encoding for a PPDU that carries a triggering frame.

26.5.2.2.2 Requirements for allocating resources

An AP that sends a Basic Trigger frame, a BQRP Trigger frame, or a BSRP Trigger frame that is not aggregated with a QoS Data frame with HETP Ack ack policy or with a Management frame that solicits acknowledgment shall allocate sufficient resources for the responding STA indicated by each User Info field for the solicited associated non-AP STA or the UORA response from an associated non-AP STA to send at least one QoS Null frame that carries an HT Control field.

NOTE—An AP that sends a Basic Trigger frame uses an implementation-specific method to decide the amount of resources indicated for UORA allocations.

An AP that solicits an Ack or BlockAck frame carried in an HE TB PPDU from a non-AP STA shall allocate sufficient resources for the non-AP STA to send the expected acknowledgment. If the AP solicits a BlockAck frame, the allocated resource shall be sufficient for the non-AP STA to send the BlockAck frame with Block Ack Bitmap subfields that are the negotiated block ack bitmap length (see 26.4.3).

An AP that sends a BFRP Trigger frame shall allocate sufficient resources for the HE TB PPDU response from each HE beamformee to include all the solicited feedback, including feedback that is segmented and including an HT Control field in each frame.

26.5.2.2.3 Padding for a triggering frame

An AP transmitting a PPDU that contains a BCC-encoded triggering frame soliciting a response from a non-AP STA shall ensure that the number of bits in the PSDU following the last bit of *SCH* is at least $L_{PAD,MAC}$ as defined in Equation (26-1), which is based on the *MinTrigProcTime* indicated by the non-AP STA (see Table 9-322a), where *SCH* is either

- The User Info field addressed to the STA of the last or only Trigger frame, or
- The last TRS Control subfield in the PSDU.

$$L_{PAD,MAC} = N_{DBPS} m_{PAD} \quad (26-1)$$

where

N_{DBPS} is defined in Table 17-4 for a non-HT PPDU, Table 19-7 for an HT PPDU, Table 21-6 for a VHT PPDU, and Table 27-15 for an HE PPDU. If the triggering frame is carried in an HE MU PPDU, N_{DBPS} is replaced by $N_{DBPS,u}$ of the target user in Equation (26-1).

m_{PAD} is as follows:

- For a non-HT PPDU, HT PPDU, and VHT PPDU:

$$m_{PAD} = \begin{cases} 0, & \text{if } MinTrigProcTime \text{ is } 0 \\ 2, & \text{if } MinTrigProcTime \text{ is } 8 \mu s \\ 4, & \text{if } MinTrigProcTime \text{ is } 16 \mu s \end{cases}$$

- For an HE PPDU:

$$m_{PAD} = \begin{cases} 0, & \text{if } MinTrigProcTime \text{ is } 0 \\ 1, & \text{if } MinTrigProcTime \text{ is } 8 \text{ or } 16 \mu s \end{cases}$$

An AP transmitting a Trigger frame that contains at least one User Info field with AID12 subfield indicating allocation of one or more contiguous RA-RUs for associated STAs shall ensure that the number of bits following the last bit of *SCH* is at least $L_{PAD,MAC}$ as defined in Equation (26-1), which is based on the largest *MinTrigProcTime* of all associated non-AP STAs, where *SCH* is the last User Info field with AID12 subfield equal to either 0 or 2046.

An AP transmitting a Trigger frame that contains at least one User Info field with AID12 subfield indicating allocation of one or more contiguous RA-RUs for unassociated non-AP STAs should ensure that the number of bits following the last bit of *SCH* is at least $4 \times N_{DBPS}$ for a non-HT PPDU, HT PPDU, or VHT PPDU or N_{DBPS} for an HE PPDU, where *SCH* is the last User Info field with AID12 subfield equal to either 2045 or 2046.

An AP transmitting an NFRP Trigger frame shall ensure that the number of bits following the last User Info field with an AID12 subfield not equal to 4095 is at least $4 \times N_{DBPS}$ for a non-HT PPDU, HT PPDU, or VHT PPDU or N_{DBPS} for an HE PPDU.

An AP may use any type of padding to satisfy the *MinTrigProcTime* requirement of a non-AP STA, such as using the Padding field in a Trigger frame, using post-EOF A-MPDU padding, or aggregating other MPDUs in the A-MPDU.

If a triggering frame is LDPC encoded, then the transmitting AP ensures that $T_{TrigProc}$ meets the following requirements:

- $T_{TrigProc}$ shall be greater than or equal to the $MinTrigProcTime$ specified by the non-AP STAs that are the recipients of the Trigger frame.
- For a Trigger frame that contains at least one User Info field with AID12 subfield indicating allocation of one or more contiguous RA-RUs for associated STAs, $T_{TrigProc}$ shall be greater than or equal to the largest $MinTrigProcTime$ of all associated non-AP STAs.
- For a Trigger frame that contains at least one User Info field with AID12 subfield indicating allocation of one or more contiguous RA-RUs for unassociated STAs, $T_{TrigProc}$ should be at least 16 μ s.
- For an NFRP Trigger frame, $T_{TrigProc}$ shall be at least 16 μ s.

$T_{TrigProc}$ is defined as the duration of PPDU that is after the OFDM symbol containing the last coded bit of the LDPC codeword that encodes the last bit of SCH minus $T_{PE,nominal}$ defined in 27.3.13.

26.5.2.2.4 Allowed settings of the Trigger frame fields and TRS Control subfield

An AP with `dot11MultiBSSIDImplemented` equal to true shall not send a Trigger frame (other than an NFRP Trigger frame) with the TA field set to the transmitted BSSID to a non-AP STA that is associated with an AP corresponding to a nontransmitted BSSID in the multiple BSSID set, unless the AP has received an HE Capabilities element from non-AP STA with the Rx Control Frame To MultiBSS subfield in the HE MAC Capabilities Information field equal to 1. An AP with `dot11MultiBSSIDImplemented` equal to true may send an NFRP Trigger frame with the TA field set to the transmitted BSSID to a non-AP STA that is associated with an AP corresponding to a nontransmitted BSSID in a multiple BSSID set. An AP with `dot11MultiBSSIDImplemented` equal to true shall not send an individually addressed Trigger frame with the TA field set to the transmitted BSSID to a non-AP STA associated with an AP corresponding to a nontransmitted BSSID.

An AP that transmits a Trigger frame shall set the TA field of the frame to the MAC address of the AP, except if `dot11MultiBSSIDImplemented` is true and the Trigger frame is directed to non-AP STAs from at least two different BSSs of a multiple BSSID set. In this case, the AP shall set the TA field of the frame to the transmitted BSSID.

The AP shall set the UL Length subfield of a Trigger frame to the value given by Equation (27-11) with $m = 2$.

If an AP transmits a Trigger frame that allocates an RU that spans the entire HE TB PPDU bandwidth and assigns the RU to more than one non-AP STA (i.e., for UL MU-MIMO) and with the GI And HE-LTF Type subfield of the Common Info field set to indicate either 2x HE-LTF + 1.6 μ s GI or 4x HE-LTF + 3.2 μ s GI, the AP may set the MU-MIMO HE-LTF Mode subfield in the Common Info field of the Trigger frame to indicate either HE single stream pilot HE-LTF mode or HE masked HE-LTF sequence mode. Otherwise, the AP shall set the MU-MIMO HE-LTF Mode subfield in the Common Info field to indicate HE single stream pilot HE-LTF mode.

An AP that transmits Trigger frames in more than one A-MPDU in an HE MU PPDU shall set each subfield, except the Trigger Type, More TF, CS Required, and Trigger Dependent Common Info subfields, in the Common Info field of the Trigger frame in one A-MPDU to the same value as the corresponding subfield in the Common Info field of the Trigger frames in the other A-MPDUs.

An AP that transmits frames carrying a TRS Control subfield in more than one A-MPDU in an HE MU PPDU shall set the UL Data Symbols and AP Tx Power subfields of the TRS Control subfield in the frames of one A-MPDU to the same value as the corresponding subfields of the TRS Control subfield in the frames of the other A-MPDUs.

An AP that transmits one or more Trigger frames in one or more A-MPDUs and frames carrying a TRS Control subfield in one or more other A-MPDUs in an HE MU PPDU shall set the Common Info field of the Trigger frames and the TRS Control subfields in each A-MPDU as follows:

- The UL Length subfield in the Common Info field of the Trigger frames and the UL Data Symbols subfield in the TRS Control subfields indicate the same HE TB PPDU duration.
- The AP Tx Power subfield in the Common Info field of the Trigger frames and the AP Tx Power subfield in the TRS Control subfields indicate the same transmit power.
- In the Common Info field of the Trigger frames:
 - The MU-MIMO HE-LTF Mode and UL STBC subfields are set to 0.
 - The Number Of HE-LTF Symbols And Midamble Periodicity subfield is set to 0.
 - The Doppler subfield is set to 0.
 - The Pre-FEC Padding Factor subfield is set to 0 (i.e., to indicate a pre-FEC padding factor of 4).
 - The UL Length subfield and the PE Disambiguity subfield are set to indicate the PE duration calculated with Equation (27-114) is the default PE duration value, which is indicated by the AP in the Default PE Duration subfield of the HE Operation element it transmits.
 - The UL Spatial Reuse subfield is set to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED.
 - If the TXVECTOR parameters HE_LTF_TYPE and GI_TYPE of the HE PPDU carrying the Trigger frame are either 4xHE-LTF and 3u2s_GI, respectively, or 2xHE-LTF and 1u6s_GI, respectively, then the GI And HE-LTF Type subfield is set to 2. Otherwise, the GI And HE-LTF Type subfield is set to 1.

NOTE 1—A non-AP STA obtains the information required to prepare an HE TB PPDU explicitly and implicitly. Explicit information is obtained in the triggering frame contained in the triggering PPDU. Implicit information is obtained in previously exchanged frames with the AP, e.g., in the BSS Color and the Default PE Duration subfields of the HE Operation element, or from default values specified in 26.5.2.3.

An AP shall not set any subfields of the Common Info field of a Trigger frame to a value that is not supported by all the recipient non-AP STAs of the Trigger frame and the AP. An AP shall not set any subfields of the User Info field of a Trigger frame to a value that is not supported by the recipient non-AP STA of the User Info field and the AP. An AP shall set the values of the subfields of the Common Info field and User Info field of a Trigger frame such that the combination together would cause the solicited non-AP STA to construct a valid HE TB PPDU. An AP shall not set any subfields of a TRS Control subfield to a value that is not supported by the recipient non-AP STA of the TRS Control subfield and the AP. An AP shall set the values of the subfields of a TRS Control subfield such that the combination together would cause the solicited non-AP STA to construct a valid HE TB PPDU. If an RU is allocated to only one non-AP STA, the Starting Spatial Stream subfield for that non-AP STA shall be set to 0.

NOTE 2—Subclauses 27.3.4 and 27.3.6.10 specify the requirements on each of the other remaining subfields, such that the solicited non-AP STAs each construct a valid HE TB PPDU in response to the Trigger frame.

An AP shall follow the RU restriction rules defined in 27.3.2.8 when assigning an RU to a 20 MHz operating non-AP STA for a 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz HE TB PPDU. An AP shall not set the RU Allocation subfield of the User Info field of a triggering frame that is addressed to a non-AP STA to a value such that the RU allocated to the STA lies outside the channel in which the STA is operating (see 27.3.2.6 and 27.3.2.9) or outside the SST subchannel (if applicable) in which the STA is operating (see 26.8.7).

If a Trigger frame is transmitted in the broadcast RU of an HE MU PPDU, then the Trigger frame shall not include any User Info fields addressed to a non-AP STA that is identified as a recipient of another RU or spatial stream of the same HE MU PPDU.

A TRS Control subfield shall not be included in a group addressed frame.

If an AP transmits one or more Trigger frames or frames carrying a TRS Control subfield, then the frames shall collectively elicit HE TB PPDU responses such that at least one scheduled RU is allocated for each 20 MHz channel occupied by the eliciting PPDU. An AP shall not allocate an RU in any 20 MHz channel that is not occupied by the immediately preceding DL PPDU.

An AP may indicate an unallocated RU in a Trigger frame by including a User Info field with the AID12 subfield set to 2046. The AP shall place any User Info fields with the AID12 subfield set to 2046 after User Info fields with the AID12 subfield set to a value less than 2046.

An AP shall not transmit a Trigger frame that contains more than one User Info field with the same value in the AID12 subfield, unless the value in the AID12 subfield is 0 or greater than 2007. The AP shall place User Info fields with the same value in the AID12 subfield together as a contiguous block in the Trigger frame. The AP shall place User Info fields with the AID12 subfield set to 0 or a value greater than 2007 after User Info fields with the AID12 subfield set to a value in the range 1 to 2007 (if any present).

An AP that transmits an individually addressed Trigger frame shall include only one User Info field in the Trigger frame with the AID12 subfield set to the 12 LSBs of the AID of the non-AP STA addressed by the RA field.

An example of User Info field ordering in a Trigger frame and the relationship to RU locations is shown in Figure 26-4.

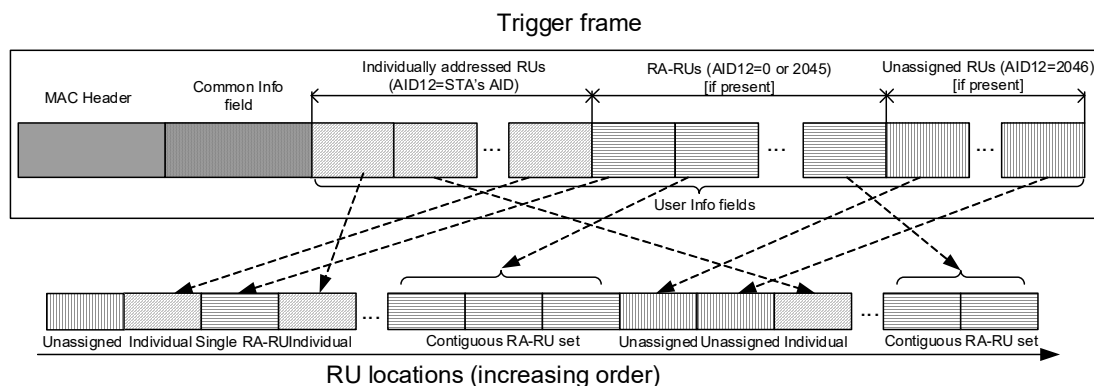


Figure 26-4—Example of User Info field ordering and RU location mapping

An AP that sends a Basic Trigger frame shall set the TID Aggregation Limit subfield of a User Info field that is addressed to a non-AP STA to a value between 0 and 1 plus the value of the Multi-TID Aggregation Tx Support subfield in the HE Capabilities element most recently received from the non-AP STA. A value 0 in the TID Aggregation Limit subfield indicates to the non-AP STA that it shall not solicit any immediate response for the MPDUs that the non-AP STA aggregates in the HE TB PPDU. A value of 7 in the TID Aggregation Limit subfield indicates to the non-AP STA that it may aggregate QoS Data frames from any number of different TID values in the multi-TID A-MPDU. Table 26-2 provides a summary of the possible combinations.

Table 26-2—Relation between TID Aggregation Limit field, Multi-TID Aggregation Tx Support field, and solicited immediate response

AP: TID Aggregation Limit field	Non-AP STA: Multi-TID Aggregation Tx Support field	Non-AP STA: Solicited immediate response?
0	Any value	No
1	Any value	Yes for at most one TID and/or one Management frame
<i>tal</i> (0 to <i>mta</i> + 1)	<i>mta</i>	Yes at most <i>tal</i> TIDs and/or at most one Management frame
Legend: <i>tal</i> is the value in the TID Aggregation Limit field. <i>mta</i> is the value in the Multi-TID Aggregation Tx Support field.		

The AP may assign any ACI value defined in Table 9-154 in the Preferred AC subfield in the Trigger Dependent User Info field for an HE STA and identified by the AID12 subfield of the User Info field of a Basic Trigger frame. If the AP does not have a recommendation, then it shall set the Preferred AC subfield to the value 0 (AC_BE).

When transmitting a PPDU carrying a triggering frame, an AP should not apply transmit beamforming to the following:

- Non-HT PPDUs
- Fields prior to the HT-STF field of an HT mixed format PPDU
- Pre-VHT modulated fields of an VHT PPDU
- Pre-HE modulated fields of an HE PPDU

An AP shall transmit an HE PPDU that carries a triggering frame with the TXVECTOR parameter BEAM_CHANGE set to 1.

An AP should set the TID Aggregation Limit subfield in the User Info fields of a Basic Trigger frame to 0 if the CS Required subfield in the Common Info field of the Basic Trigger frame is 0.

An AP shall set the TID Aggregation Limit subfield in the User info fields of a Basic Trigger frame to 0 if the solicited HE TB PPDU is the last PPDU of the TXOP.

NOTE 3—An HE TB PPDU is the last PPDU if the Duration/ID field is equal to 0 in the MPDU(s) contained in the HE TB PPDU.

26.5.2.2.5 AP access procedures for UL MU operation

The AP shall follow the EDCA procedure defined in 10.23 and the additional rules in this subclause.

If an AP receives an immediate response with at least one frame from at least one non-AP STA solicited by a triggering frame, the frame exchange is successful.

If an AP does not receive an immediate response with at least one frame from at least one non-AP STA solicited by a PPDU that contains at least one Trigger frame, then the frame exchange is not successful, and the AP shall follow the backoff procedure in 10.23.2.2.

An AP may use any AC for sending a PPDU that contains only Trigger frames.

If the PPDU contains frames that are not Trigger frames in addition to a Trigger frame, then the AP follows the MPDU aggregation rules in 26.6, which supersede the rules in 10.23.2.7.

26.5.2.3 Non-AP STA behavior for UL MU operation

26.5.2.3.1 General

A non-AP STA shall not send an HE TB PPDU, unless it is explicitly triggered by an AP in one of the operation modes described in this subclause.

The interframe space between a PPDU that contains a triggering frame and the HE TB PPDU is a SIFS.

A non-AP STA shall not transmit an HE TB PPDU if all of the conditions in 26.5.2.3.2 are satisfied. Otherwise, a non-AP STA shall transmit an HE TB PPDU a SIFS after a received PPDU if all of the following conditions are met:

- The received PPDU contains either a Trigger frame (that is not an MU-RTS variant) with a User Info field addressed to the non-AP STA or a frame addressed to the non-AP STA that contains an TRS Control subfield. A User Info field in the Trigger frame is addressed to a non-AP STA if one of the following conditions are met:
 - The AID12 subfield is equal to the 12 LSBs of the AID of the non-AP STA, and the Trigger frame is sent by the AP with which the non-AP STA is associated or by the AP corresponding to the transmitted BSSID if the non-AP STA is associated with an AP corresponding to a nontransmitted BSSID and has indicated support for receiving Control frames with TA field set to the transmitted BSSID by setting the Rx Control Frame To MultiBSS subfield to 1 in the HE Capabilities element that the STA transmits.
 - The AID12 subfield indicates allocation of one or more contiguous RA-RUs for associated STAs, the non-AP STA is associated with the AP that sent the Trigger frame, the non-AP STA supports the UORA procedure, and the conditions in 26.5.4 are satisfied.
 - The AID12 subfield indicates allocation of one or more contiguous RA-RUs for unassociated STAs, the non-AP STA is not associated with the AP that sent the Trigger frame, the non-AP STA supports the UORA procedure, the conditions in 26.5.4 are satisfied, and the resource to which the non-AP STA gains access is sufficient for the non-AP STA to include the pending frame.
- The CS Required subfield in the Trigger frame is 1, and the UL MU CS condition described in 26.5.2.5 indicates the medium is idle; or the CS Required subfield in a Trigger frame is 0; or the response was solicited by a frame containing a TRS Control subfield.
- The UL MU Disable subfield is 0, and the UL MU Data Disable subfield is 0 in the most recent OM Control subfield (if any) sent by the non-AP STA to the AP; or the UL MU Disable subfield is 0, the UL MU Data Disable subfield is 1 in the most recent OM Control subfield (if any) sent by the non-AP STA to the AP, and the frame that is being triggered is an acknowledgment (see 26.9.3).

A non-AP STA addressed by a User Info field in a Trigger frame (i.e., the AID12 subfield is equal to the 12 LSBs of the AID of the non-AP STA) may ignore the remainder of User Info fields in the Trigger frame.

A non-AP STA generates the A-MPDU carried in the HE TB PPDU as defined in 26.5.2.4.

26.5.2.3.2 Conditions for not responding with an HE TB PPDU

A non-AP STA shall not transmit an HE TB PPDU that is not an HE TB feedback NDP if all the following conditions are satisfied:

- The non-AP STA is operating in an operating class for which the behavior limits set listed in Annex E includes the DFS_50_100_Behavior (see Table E-1).
- The HE TB PPDU would be in response to one of the following:
 - A Trigger frame containing a User Info field with AID12 subfield carrying the 12 LSBs of the AID of the STA.
 - A frame addressed to the non-AP STA that includes a TRS Control subfield.
 - A Trigger frame that allocates at least one RA-RU.
- The RU is a 26-tone RU.
- The non-AP STA has received at least one Beacon frame within the past $\text{dot11ObssNbRuToleranceTime}$ from an AP with which the non-AP STA is not associated, and the Beacon frame meets any of the following conditions:
 - The Extended Capabilities element is not present.
 - The OBSS Narrow Bandwidth RU In OFDMA Tolerance Support field in the Extended Capabilities element is not present.
 - The OBSS Narrow Bandwidth RU In OFDMA Tolerance Support field in the Extended Capabilities element is 0.

A non-AP STA may choose to not respond to a Trigger frame that contains one or more subfields in the Common Info field or in the User Info field addressed to or selected by the non-AP STA with values that are not recognized, are not supported, or cannot be satisfied by the non-AP STA.

NOTE—The User Info field in this context corresponds to the one directed to the non-AP STA (i.e., value in the AID12 subfield matches the STA's AID) or the one allocating an RA-RU (single or within a contiguous set) that is selected by the non-AP STA.

A non-AP STA may choose to not respond a TRS Control subfield in a frame addressed to the non-AP STA if the TRS Control subfield contains one or more subfields with values that are not recognized, are not supported, or cannot be satisfied by the non-AP STA.

A non-AP STA shall update the intra-BSS NAV (see 26.2.4) based on the duration information of the triggering frame even if it decides to not respond to the frame.

26.5.2.3.3 TXVECTOR parameters for HE TB PPDU response to Trigger frame

A non-AP STA transmitting an HE TB PPDU in response to a Trigger frame that is not an MU-RTS Trigger frame shall set the TXVECTOR parameters as follows (subject to, in the case of an NFRP Trigger frame, the exceptions given in 26.5.7.2):

- The FORMAT parameter is set to HE_TB.
- The TRIGGER_METHOD parameter is set to TRIGGER_FRAME.
- The TXOP_DURATION parameter is set as defined in 26.11.5.
- The BSS_COLOR parameter is set as follows:
 - If the Trigger frame was received in an HE PPDU, then set to the value of the RXVECTOR parameter BSS_COLOR of the HE PPDU.
 - If the Trigger frame was received in a non-HE PPDU, then set to the value of the active BSS color as defined in 26.11.4.
- The L_LENGTH parameter is set to the value indicated by the UL Length subfield in the Common Info field of the Trigger frame.

- The GI_TYPE and HE_LTF_TYPE parameters are set to the value indicated by the GI and HE-LTF Type subfield of the Common Info field of the Trigger frame.
- The NUM_STS parameter is set to the number of space-time streams indicated by the Number Of Spatial Streams subfield of the SS Allocation field of the User Info field and UL STBC subfield in the Common Info field of the Trigger frame. The NUM_STS parameter is set to 1 if the HE TB PPDU is sent on an RA-RU following the UORA procedure.
- The CH_BANDWIDTH parameter is set to the value of the UL BW subfield in the Common Info field of the Trigger frame.
- The HE_LTF_MODE parameter is set to the value indicated by the MU-MIMO HE-LTF Mode subfield of the Common Info field of the Trigger frame if the HE_LTF_TYPE parameter does not indicate 1x HE-LTF and the Trigger frame indicated full-bandwidth MU-MIMO. Otherwise, the parameter is not present.
- The NUM_HE_LTF parameter is set to the value indicated by the Number Of HE-LTF Symbols And Midamble Periodicity subfield of the Common Info field of the Trigger frame.
- The STBC parameter is set to the value indicated by the UL STBC subfield of the Common Info field of the Trigger frame.
- The LDPC_EXTRA_SYMBOL parameter is set to the value indicated by the LDPC Extra Symbol Segment subfield of the Common Info field of the Trigger frame.
- The SPATIAL_REUSE parameter is set to the value of the UL Spatial Reuse subfield in the Common Info field of the eliciting Trigger frame.
- The DOPPLER parameter is set to the value of the Doppler subfield in the Common Info field of the Trigger frame.
- The MIDAMBLE_PERIODICITY parameter is present if the Doppler subfield in the Common Info field of the Trigger frame is 1. If present, it is set to the value indicated by the Number Of HE-LTF Symbols And Midamble Periodicity subfield in the Common Info field of the Trigger frame.
- The HE_SIG_A2_RESERVED parameter is set to the value of the UL HE-SIG-A2 Reserved subfield in the Common Info field of the Trigger frame.
- The MCS parameter is set to the value of the UL HE-MCS subfield in the User Info field of the Trigger frame.
- The DCM parameter is set to the value indicated by the UL DCM subfield of the User Info field of the Trigger frame.
- The STARTING_STS_NUM parameter is set to the value of the Starting Spatial Stream subfield in the SS Allocation field in the User Info field of the Trigger frame. The STARTING_STS_NUM parameter is set to 0 if the HE TB PPDU is sent on an RA-RU following the UORA procedure.
- The FEC_CODING parameter is set to the value indicated by the UL FEC Coding Type subfield of the User Info field of the Trigger frame.
- The RU_ALLOCATION parameter is set as follows:
 - If the RU is not an RA-RU or an RA-RU with the Number Of RA-RU subfield of the User Info subfield of the Trigger frame set to 0, it is set to the value indicated by the RU Allocation subfield of the User Info subfield of the Trigger frame.
 - If the RU is the k^{th} RU of a set of contiguous RA-RUs starting with an RA-RU with Number Of RA-RU subfield of the User Info subfield of the Trigger frame set to a nonzero value, it is set to the value indicated by the RU Allocation subfield of the corresponding User Info subfield of the Trigger frame plus k minus 1.
- The TXPWR_LEVEL_INDEX parameter is set to a value based on the computed transmission power (see 27.3.15.2) for an HE TB PPDU and the value of the AP Tx Power subfield in the Common Info field and the UL Target Receive Power subfield in the User Info field of the Trigger frame.

- The HE_TB_PE_DISAMBIGUITY parameter is set to the value indicated by the PE Disambiguity subfield in the Common Info field in the Trigger frame.
- The PRE_FEC_PADDING_FACTOR parameter is set to the value indicated by the Pre-FEC Padding Factor subfield in the Common Info field in the Trigger frame.

26.5.2.3.4 TXVECTOR parameters for HE TB PPDU response to TRS Control subfield

A non-AP STA transmitting an HE TB PPDU in response to a frame containing a TRS Control subfield shall set the TXVECTOR parameters as follows:

- The FORMAT parameter is set to HE_TB.
- The TRIGGER_METHOD parameter is set to TRS.
- The L_LENGTH parameter is computed as described in Equation (27-11) using the TXTIME value. The TXTIME is defined by Equation (27-136) where N_{SYM} is set to $F_{VAL} + 1$, where F_{VAL} is the value of the UL Data Symbols subfield of the TRS Control subfield.
- The RU_ALLOCATION and MCS parameters are set to the values of the RU Allocation and UL HE-MCS subfields of the TRS Control subfield, respectively.
- The CH_BANDWIDTH parameter is set to the value of the RXVECTOR parameter CH_BANDWIDTH of the soliciting DL HE PPDU (see Table 27-1).
- The BSS_COLOR and DCM parameters are set to the values of the RXVECTOR parameters BSS_COLOR and DCM of the soliciting DL HE PPDU, respectively.
- The DOPPLER parameter is set to 0, and the MIDAMBLE_PERIODICITY parameter is absent.
- The NUM_HE_LTF parameter is set to 1.
- The STARTING_STS_NUM parameter is set to 0.
- The STBC parameter is set to 0, and the NUM_STS parameter is set to 1 (the HE_LTF_MODE parameter is not present).
- The FEC_CODING parameter is set to BCC_CODING if the RU Allocation subfield indicates an RU that is smaller than a 484-tone RU; otherwise, it is set to LDPC_CODING.
- The LDPC_EXTRA_SYMBOL parameter is set to 0 if the RU Allocation subfield indicates less than a 484-tone RU; otherwise, it is set to 1.
- The SPATIAL_REUSE parameter is set to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED.
- The DEFAULT_PE_DURATION parameter is set to the default PE duration value for UL MU response scheduling, which is indicated by the AP in the Default PE Duration subfield of the HE Operation element it transmits.
- The TXOP_DURATION parameter is set as defined in 26.11.5.
- The HE_SIG_A2_RESERVED parameter is set to 511 (all 1s).
- If the RXVECTOR parameters HE_LTF_TYPE and GI_TYPE of HE MU PPDU carrying the frame with the TRS Control subfield are either 4xHE-LTF and 3u2s_GI, respectively, or 2xHE-LTF and 1u6s_GI, respectively, then the HE_LTF_TYPE and GI_TYPE parameters are set to 4xHE-LTF and 3u2s_GI, respectively. Otherwise, the HE_LTF_TYPE and GI_TYPE parameters are set to 2xHE-LTF and 1u6s_GI, respectively.
- The TXPWR_LEVEL_INDEX parameter is set to a value based on the computed transmission power (see 27.3.15.2) for an HE TB PPDU and the value of the AP Tx Power subfield of the TRS Control subfield and the UL Target Receive Power subfield of the TRS Control subfield.

NOTE 1—A non-AP STA transmitting an HE TB PPDU in response to a frame carrying a TRS Control subfield considers both physical CS and virtual CS to be 0 (see 26.5.2.5).

NOTE 2—The only permissible values for CH_BANDWIDTH are CBW20, CBW40, CBW80, CBW80+80, and CBW160 if the triggering PPDU is an HE SU PPDU or HE MU PPDU. The only permissible value for CH_BANDWIDTH is CBW20 if the triggering PPDU is an HE ER SU PPDU.

26.5.2.3.5 RA field for frames carried in an HE TB PPDU

The RA field of the frames sent in response to a MU-RTS Trigger frame is set as defined in 9.3.1.3. The RA field of the frames sent in response of a GCR MU-BAR Trigger frame or MU-BAR Trigger frame is set as defined in 9.3.1.8. The RA field of the QoS Null frames, QoS Data frames, and Management frames sent in response to a Trigger frame shall be set to the MAC address of the destination AP (see 9.3.2.1 and 9.3.3.1). The RA field of a QoS Null frame or Action No Ack frame sent in response to a frame carrying TRS Control subfield shall be the MAC address of the destination AP (see 9.3.2.1 and 9.3.3.1).

NOTE 1—If `dot11MultiBSSIDImplemented` is true and the TA field of the soliciting Trigger frame contains the transmitted BSSID, the destination AP is the BSSID to which the non-AP STA intends to send the frame.

NOTE 2—All MPDUs within an A-MPDU carried in an HE TB PPDU have the same RA (see 9.7.3). The settings of the address fields of MPDUs within the A-MPDU depend on the type and subtype of the MPDU as defined in 9.3.

26.5.2.4 A-MPDU contents in an HE TB PPDU

A non-AP STA that receives a triggering frame other than an MU-RTS Trigger frame or an NFRP Trigger frame and that transmits an HE TB PPDU response shall follow the A-MPDU padding procedure described in 26.6.2.3 and construct the A-MPDU carried in the HE TB PPDU as described below in this subclause, provided the AP allocates sufficient resources for the non-AP STA to include MPDU(s) in the A-MPDU. Otherwise, the non-AP STA is not required to include MPDUs in the A-MPDU and includes only padding in the A-MPDU.

NOTE 1—The responses to a MU-RTS Trigger frame and a NFRP Trigger frame are exempt from these construction rules since the MU-RTS Trigger frame does not solicit an HE TB PPDU and the NFRP Trigger frame solicits an HE TB PPDU that does not carry an A-MPDU.

A non-AP STA shall follow the rules in 26.5.4.5 to construct an HE TB PPDU in response to a Trigger frame that is not an MU-RTS Trigger frame from an AP with which it is not associated and that allocates RA-RUs for unassociated STAs.

A non-AP STA that responds to a DL MU PPDU containing one or more frames addressed to it that include a TRS Control subfield follows the rules defined in 10.3.2.11 for generating the Ack frame, the rules defined in 10.25.6.5 for generating the BlockAck frame, and the rules defined in 26.4 for generating the Multi-STA BlockAck frame if at least one of the received MPDUs solicits an immediate acknowledgment. The contents of the A-MPDU carried in the HE TB PPDU shall be as defined in the following:

- Table 9-533 if at least one of the received MPDUs solicits an immediate acknowledgment.
- Table 9-531 with the exception that the A-MPDU does not contain QoS Data frames, if none of the received MPDUs solicit an immediate acknowledgment.

NOTE 2—The non-AP STA additionally follows the rules in 26.3.2 if fragments are present in the soliciting A-MPDU.

NOTE 3—An AP might transmit an HE MU PPDU with an RU allocated to STA-ID 2045 with an A-MPDU that includes a Management frame that is addressed to an unassociated non-AP STA, solicits an acknowledgment, and carries a TRS Control subfield. The TRS Control subfield allocates resources for the unassociated non-AP STA to respond with an HE TB PPDU that carries the acknowledgment.

An associated non-AP STA that responds to a Basic Trigger frame addressed to it shall construct the A-MPDU carried in the HE TB PPDU as defined in the following:

- Table 9-533 if the Trigger frame is contained in an A-MPDU and the non-AP STA received in the same A-MPDU at least one MPDU that solicits an immediate acknowledgment. The TID Aggregation Limit field of the User Info field addressed to the non-AP STA may have any value.

- Table 9-531 with the exception that the A-MPDU does not contain QoS Data frames, if the Trigger frame either is not carried in an A-MPDU or is carried in an A-MPDU but the non-AP STA receives no other MPDUs that solicit an immediate acknowledgment. The TID Aggregation Limit field of the User Info field addressed to the non-AP STA may have any value.
- Table 9-534 with the following restrictions:
 - The S-MPDU shall be a control response frame if the non-AP STA received in the same A-MPDU at least one MPDU that solicits an immediate acknowledgment. The TID Aggregation Limit field of the User Info field addressed to the non-AP STA may have any value.
 - The S-MPDU may be a Multi-TID BlockAckReq frame, provided that the number of TIDs present in the Multi-TID BlockAckReq frame does not exceed the TID aggregation limit indicated by the TID Aggregation Limit field of the User Info field addressed to the non-AP STA.
 - The S-MPDU may be a QoS Data, QoS Null, Management, BlockAckReq, or PS-Poll frame if the TID Aggregation Limit field of the User Info field addressed to the non-AP STA is greater than 0.
 - Otherwise, the S-MPDU shall be a QoS Data, QoS Null, or Management frame that does not solicit an immediate acknowledgment.
- Table 9-534a or Table 9-534c if the TID Aggregation Limit field of the User Info field addressed to the non-AP STA in the Trigger frame is greater than 0 and the non-AP STA intends to carry one or more untagged MPDUs (see 10.12 and 26.6.3.3). The A-MPDU is subject to the following restrictions:
 - It shall contain a control response frame if the non-AP STA received in the same A-MPDU at least one MPDU that solicits an immediate acknowledgment.
 - The number of TIDs present in the A-MPDU shall count towards reaching the TID aggregation limit indicated by the TID Aggregation Limit field of the User Info field addressed to the non-AP STA in the Trigger frame.
- Table 9-534b or Table 9-534d if the TID Aggregation Limit field of the User Info field addressed to the non-AP STA in the Trigger frame is greater than 0 and the non-AP STA intends to carry an ack-enabled single-TID A-MPDU or ack-enabled multi-TID A-MPDU (see 26.6.3.1 and 26.6.3.4). The A-MPDU is subject to the following restrictions:
 - It shall contain a control response frame if the non-AP STA receives at least another MPDU that solicits an immediate acknowledgment.
 - The number of TIDs present in the A-MPDU, in either QoS Data or BlockAckReq frames, shall count towards reaching the TID aggregation limit that is obtained from the TID Aggregation Limit field of the User Info field addressed to the non-AP STA in the Trigger frame.

If the associated non-AP STA has no frames pending or is unable to include pending frames in response to a Basic Trigger frame because the allocated resource is insufficient, then the associated non-AP STA shall include in the A-MPDU at least one QoS Null frame if the allocated resource is sufficient for containing a QoS Null frame.

NOTE 4—The non-AP STA can transmit one QoS Null frame as defined in Table 9-534 or one or more QoS Null frames as defined in Table 9-531.

A non-AP STA that responds to a BFRP Trigger frame addressed to it shall construct an A-MPDU carried in the HE TB PPDU with one or more HE Compressed Beamforming/CQI frames (see 26.7); other frames shall not be allowed in the A-MPDU.

NOTE 5—It is not always possible to fragment an HE compressed beamforming/CQI report (see 26.7.4). If the length is insufficient to contain the HE compressed beamforming/CQI report requested by a BFRP Trigger frame, no feedback is sent.

A non-AP STA that responds to an MU-BAR Trigger frame addressed to it shall construct the A-MPDU carried in the HE TB PPDU as defined in Table 9-533. The non-AP STA includes either a BlockAck frame or a Multi-STA BlockAck frame in the A-MPDU as defined in 26.4.

A non-AP STA that responds to a GCR MU-BAR Trigger frame addressed to it shall construct the A-MPDU carried in the HE TB PPDU as defined in Table 9-533. The non-AP STA includes a GCR BlockAck frame in the A-MPDU as defined in 10.25.8.

A non-AP STA that responds to a BSRP or BQRP Trigger frame addressed to it and that is not aggregated with any MPDUs that solicit an immediate acknowledgment shall construct the A-MPDU carried in the HE TB PPDU as defined in Table 9-531 with the exception that the A-MPDU does not contain QoS Data frames. The non-AP STA shall include in the A-MPDU at least one QoS Null frame. A non-AP STA that responds to a BSRP or BQRP Trigger frame addressed to it and that is aggregated with at least one MPDU that solicits an immediate acknowledgment, shall construct the A-MPDU carried in the HE TB PPDU as defined in Table 9-533.

NOTE 6—The frame type of MPDUs may be different across A-MPDUs within the same HE TB PPDU subject to A-MPDU context.

NOTE 7—A non-AP STA follows the rules in 26.6.3 for aggregating the QoS Data frames with multiple TIDs in HE TB PDUs.

A non-AP STA may set `dot11HEUPHControlActivated` to false if the most recent OM Control field sent (if any) to the AP had the UL MU Disable field equal to 1; otherwise, the non-AP STA shall set `dot11HEUPHControlActivated` to true.

A STA that transmits an HE TB PPDU transmits the decibel value of its UL power headroom, HR_{STA} , in the UPH Control subfield of frames (that can carry an HE-variant HT Control field) carried in the HE TB PPDU, to assist in the AP's HE-MCS selection. The UL power headroom for the assigned HE-MCS is defined in Equation (26-2).

$$HR_{STA} = Tx_{pwr}^{Max} - Tx_{pwr}^{STA} \quad (26-2)$$

where

Tx_{pwr}^{Max}	represents the maximum UL transmit power of an HE TB PPDU with the assigned HE-MCS after considering hardware capability, regulatory requirements, and local maximum transmit power levels (see 11.7.5) as well as non-IEEE-802.11 in-device coexistence requirements
Tx_{pwr}^{STA}	represents the current UL transmit power of the HE TB PPDU for the assigned HE-MCS, which is determined by power control and subject to the non-AP STA's capabilities and other requirements as defined in 27.3.15.2
HR_{STA}	is the UL power headroom, in dB, of the HE TB PPDU, the encoding of which is specified in 9.2.4.6a.5

NOTE 8—If the Minimum Transmit Power Flag subfield in the UPH Control subfield is 1, then the non-AP STA is transmitting the HE TB PPDU at its minimum Tx_{pwr}^{STA} for the assigned HE-MCS. The UL power headroom is calculated for the assigned HE-MCS and is independent of TOM parameters provided in any OM Control field contained in the same A-Control field (see 26.9.3).

NOTE 9—The uplink power headroom is not normalized to 20 MHz bandwidth, unlike the value in the AP Tx Power subfield.

A non-AP STA shall include an HE variant HT Control field containing the UPH Control subfield in the frames carried in the A-MPDU of the HE TB PPDU with the following exceptions:

- A UPH Control subfield is not included in any frame if the remaining space in the A-MPDU, after inclusion of solicited frames that are required to be included in the A-MPDU but cannot contain an HE variant HT Control field, is not sufficient to contain a frame that can be included in the A-MPDU and can contain an HE variant HT Control field.
- A UPH Control subfield is not included in a frame if the frame contains Control subfields other than UPH Control and ONES Control subfields and the remaining space in the HE variant HT Control field of the frame is not sufficient to contain a UPH Control subfield as well.
- A UPH Control subfield is not included in a frame that is a Control frame.

A non-AP STA shall not include a ONES Control subfield in the HE variant HT Control field of the frames carried in an HE TB PPDU.

26.5.2.5 UL MU CS mechanism

The ED-based CCA and virtual CS functions are used to determine the state of the medium if CS is required before responding to a received Trigger frame. ED-based CCA for the UL MU CS mechanism is defined in 27.3.20.6.4, and virtual CS is defined in 10.3.2.1.

If the CS Required subfield in a received Trigger frame is 0 or a frame that includes a TRS Control subfield and solicits a response is received, then the non-AP STA may respond without regard to the busy/idle state of the medium.

NOTE 1—Responding without regard to the busy/idle state of the medium means that a non-AP STA can respond without the need to check the medium indication from physical CS and virtual CS (i.e., basic NAV and intra-BSS NAV).

A non-AP STA does not consider the intra-BSS NAV in determining whether to respond to a Trigger frame sent by the AP with which the non-AP STA is associated.

A non-AP STA considers the basic NAV in determining whether to respond to a Trigger frame sent by the AP with which the non-AP STA is associated.

A non-AP STA considers the NAV in determining whether to respond to a Trigger frame sent by an AP with which the non-AP STA is not associated, through the UORA procedure (see 26.5.4), unless the NAV was set by a frame originating from the AP sending the Trigger frame.

NOTE 2—A non-AP STA considers the intra-BSS NAV associated with an AP in determining whether to respond to a Trigger frame with RU allocations for unassociated STAs sent by another AP.

NOTE 3—The details of how a non-AP STA is solicited by the Trigger frame for transmission are described in 26.5.2.2.4.

NOTE 4—If a non-AP STA responds to a Trigger frame from an AP with which it is not associated through the UORA procedure, the method to identify that a NAV was set by a frame originating from the AP sending the Trigger frame is implementation specific. For example, the non-AP STA can save the TXOP holder address and match the saved TXOP holder address with the TA field of the Trigger frame.

For a non-AP STA that is solicited by a Trigger frame for transmission, the indication of the virtual CS is described as follows. If the non-AP STA does not consider any NAV, then the virtual CS indicates idle. If all NAVs that the non-AP STA considers have their NAV counter equal to 0, then the virtual CS indicates idle. Otherwise, the virtual CS indicates busy.

If the CS Required subfield in a Trigger frame is 1, then the non-AP STA shall consider the status of the CCA [using energy detect defined in 27.3.20.6.2 and the virtual carrier sense (NAV)] during the SIFS between the Trigger frame and the PPDU sent in response to the Trigger frame. In this case, the non-AP STA shall sense the medium using energy detect after receiving the PPDU that contains the Trigger frame (i.e., during the SIFS), and it shall perform the energy detect at least in the subchannel that contains the non-AP STA's UL allocation, where the sensed subchannel consists of one or more 20 MHz channels. The non-AP STA may transmit the solicited PPDU if the 20 MHz channels containing the RUs allocated in the Trigger frame are considered idle. If the non-AP STA detects that the 20 MHz channels containing the allocated RUs are not all idle, then the non-AP STA shall not transmit.

NOTE 5—The solicited PPDU is a non-HT or non-HT duplicate PPDU if the Trigger frame is an MU-RTS Trigger frame (see 26.2.6); otherwise, the solicited PPDU is an HE TB PPDU (see 26.5.2.3).

The CS Required subfield in the MU-RTS Trigger frame shall be set to 1.

An AP that transmits a BFRP Trigger frame with the UL Length subfield in the Common Info field set to a value greater than 76 shall set the CS Required subfield in the Common Info field to 1.

An AP that transmits a Basic, BSRP, MU-BAR, BQRP, or GCR MU-BAR Trigger frame shall set the CS Required subfield to 1, unless one of the following conditions is met:

- The RA of the Trigger frame is an individually addressed non-AP STA's MAC address, a QoS Data frame with HETP Ack ack policy and/or a Management frame that solicits an acknowledgment is aggregated with the Trigger frame in an A-MPDU, and the UL Length subfield in the Common Info field of the Trigger frame is less than or equal to 418.
- The Trigger frame is either an MU-BAR or a GCR MU-BAR Trigger frame, and the UL Length subfield in the Common Info field of the Trigger frame is less than or equal to 418.
- The UL Length subfield in the Common Info field of the Trigger frame is less than or equal to 76.

NOTE 6—The threshold value 418 of the UL Length subfield in the Common Info field of the Trigger frame is obtained from the maximum HE TB PPDU duration, 584 μ s, that can be solicited by the TRS Control subfield based on Equation (27-16). This duration is the sum of 20 μ s for the L-STF, L-LTF, and L-SIG fields; 20 μ s for the RL-SIG, HE-SIG-A, and HE-STF fields; 16 μ s for the 4x HE-LTF field with 3.2 μ s GI; 512 μ s for 32 OFDM symbols in the Data field with 3.2 μ s GI; and 16 μ s for the PE field (see 9.2.4.6a.1, 26.5.2.3, and 27.3.4).

NOTE 7—The UL Length subfield value 76 is derived from the duration 128 μ s, which is the duration of an HE TB PPDU with 4 HE-LTF symbols and a PE field.

An AP may transmit an NFRP Trigger frame with the CS Required subfield set to 0 or 1.

26.5.3 MU cascading sequence

An MU cascading sequence is a frame exchange sequence between an AP and one or more non-AP STAs in which the AP, within a single PPDU, acknowledges one or more frames from a STA and triggers the STA for a further UL transmission. An example of an MU cascading sequence is shown in Figure 26-5. In this figure the second HE MU PPDU contains an Ack or BlockAck frame and a triggering frame.

An AP shall not transmit an A-MPDU to a non-AP STA that includes an Ack or BlockAck frame together with a triggering frame, unless both the AP and the non-AP STA have indicated support by setting the MU Cascading Support subfield to 1 in the MAC Capabilities Information field in the HE Capabilities element they transmit.

NOTE—An A-MPDU sent by an AP in an MU cascading sequence typically includes, in addition to the acknowledgement and triggering frames, one or more QoS Data frames with HETP Ack ack policy and/or a Management frame soliciting acknowledgement, subject to the rules in 26.6.

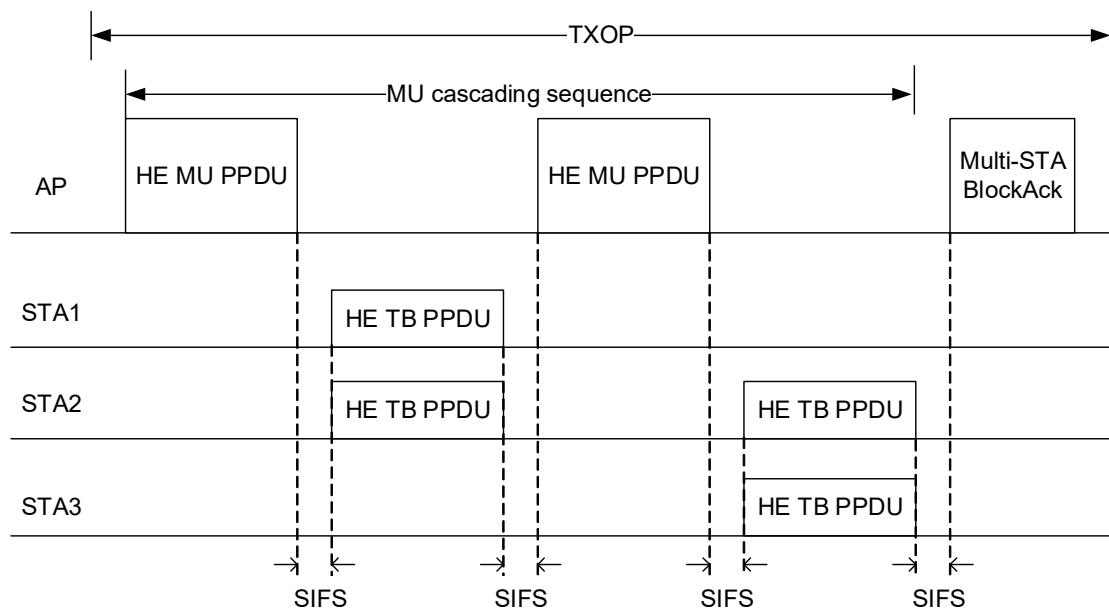


Figure 26-5—Example of MU cascading sequence

In an MU cascading sequence, the receivers of an HE MU PPDU may differ from the transmitters of preceding HE TB PPDUs. An MU cascading sequence is sent within a single TXOP.

An AP follows the procedure in 26.5.2.2.5.

26.5.4 UL OFDMA-based random access (UORA)

26.5.4.1 General

An HE STA with `dot11OFDMARandomAccessOptionImplemented` equal to true shall set the OFDMA RA Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element to 1. Otherwise, it shall set the OFDMA RA Support subfield to 0.

NOTE 1—A non-AP STA that does not support UORA can contend for the WM using EDCA for sending UL frames to the AP with which it intends to communicate.

A non-AP STA with `dot11OFDMARandomAccessOptionImplemented` set to true shall follow the procedure defined in 26.5.4.3 and 26.5.4.5 to contend for an eligible RA-RU.

An HE AP may transmit a Basic Trigger frame, BQRP Trigger frame, or BSRP Trigger frame that contains one or more RUs for random access. An AP that transmits a Trigger frame that is not a Basic Trigger frame shall not set the AID12 subfield of any User Info field of the frame to indicate allocation of one or more RA-RUs for unassociated STAs. An AP that transmits a Trigger frame that is not a Basic Trigger frame, BQRP Trigger frame, or BSRP Trigger frame shall not set the AID12 subfield of any User Info field of the frame to indicate allocation of one or more RA-RUs for associated STAs.

NOTE 2—An non-AP HE STA that changes the maximum number of receiving spatial streams to 1 and sets the OFDMA RA Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element to 1 cannot receive a Trigger frame sent with more than one spatial stream. As a result, if the Trigger frame indicates eligible RA-RUs for associated non-AP STAs, the non-AP HE STA cannot perform UORA defined in 26.5.4). An AP might take this behavior into consideration when sending a Trigger frame indicating eligible RA-RUs for associated non-AP STAs.

An HE AP that transmits a Basic Trigger frame should set the TID Aggregation Limit subfield in the User Info field indicating an RA-RU to 0 or 1.

An AP shall set the More RA-RU subfield in the User Info field to 1 if it intends to transmit additional Trigger frames in the current broadcast TWT SP that allocate RA-RUs matching the AID12 subfield value in the User Info field (see Table 9-29h).

A non-AP HE STA shall ignore the More RA-RU subfield if the More TF field in the Trigger frame is equal to 0.

The HE AP may include the UORA Parameter Set element (see 9.4.2.250) in Management frames that it transmits. The AP shall indicate the range of OFDMA contention window (OCW) in the UORA Parameter Set element for non-AP STAs to initiate random access following the Trigger frame transmission. An AP corresponding to a nontransmitted BSSID in a multiple BSSID set shall follow the rules in 11.1.3.8.4.

NOTE 3—An AP with `dot11MultiBSSIDImplemented` equal to true can allocate RA-RUs for non-AP STAs associated with different BSSIDs in a multiple BSSID set by transmitting a DL MU PPDU carrying broadcast RUs, one per BSS in the set (see 26.5.1.2), with an A-MPDU in each broadcast RU carrying a Trigger frame with at least one User Info field with the AID12 subfield set to 0.

A non-AP HE STA shall maintain an internal OCW and an internal OBO counter. OCW is an integer in the range OCW_{min} to OCW_{max} . The OCW_{min} and OCW_{max} parameters are defined by `dot11OCWmin` and `dot11OCWmax`. A non-AP HE STA shall update `dot11OCWmin` and `dot11OCWmax` from an UORA Parameter Set element within an interval of time equal to one beacon interval after receiving an updated UORA Parameter Set element carried in a Beacon, Probe Response, or (Re)Association Response frame transmitted by its associated AP, unless the non-AP HE STA is associated with an AP corresponding to a nontransmitted BSSID of a multiple BSSID set. In this case, it shall update `dot11OCWmin` and `dot11OCWmax` by following the rules in 11.1.3.8.4.

Each time a non-AP HE STA associates with a different AP and prior to the initial attempt of RA-RU transmission towards it, the non-AP STA shall set the value of OCW to the OCW_{min} value and shall initialize its OBO counter in the range 0 to OCW as defined in 26.5.4.3.

NOTE 4—For a non-AP STA with `dot11MultiBSSIDImplemented` equal to true, associating with a different AP includes associating with an AP corresponding to a different BSSID in the same multiple BSSID set.

26.5.4.2 Eligible RA-RUs

A non-AP STA that is the intended receiver of a User Info field in a Trigger frame (i.e., the AID12 subfield equal to the 12 LSBs of the AID of the non-AP STA) shall not contend for an RA-RU that is indicated by a Trigger frame contained in the same PPDU and shall not decrement its OBO counter.

A non-AP STA shall consider an RU as an eligible RA-RU if it is capable of transmitting an HE TB PPDU in that RU according to the parameters indicated in the Common Info field and in the User Info field that allocates the RU (as described in 26.5.2.3), the non-AP STA is associated with the BSS whose BSSID is the value in the TA field of the Trigger frame, and the RA-RU is allocated for associated STAs.

NOTE 1—If the STA is a 20 MHz operating non-AP HE STA, RUs that are excluded as specified in 27.3.2.8 are not present in the list of eligible RA-RUs.

A non-AP STA may consider an RU as an eligible RA-RU if it supports all the transmit parameters indicated in the Common Info field and in the User Info field that allocates that RU (as described in 26.5.2.3), the non-AP STA is not associated with the BSS, and the RA-RU is allocated for unassociated STAs.

An HE AP may allocate a contiguous set of RUs for random access by setting the Number Of RA-RU subfield in the User Info field of the Trigger frame to a value greater than 1. The RA-RU indicated by the RU Allocation subfield in the User Info field shall represent the starting RU of the set. The size of all RA-RUs in the set shall be the same and equal to the size of the RA-RU indicated by the RU Allocation subfield in the User Info field. The remaining subfields of the User Info field apply to each RA-RU in the set. An AP allocating a contiguous set of RA-RUs in a Trigger frame with an UL BW subfield that indicates 80+80 MHz or 160 MHz shall set the Number Of RA-RUs subfield so that all the RA-RUs in the set lie in one 80 MHz frequency segment.

NOTE 2—An AP can transmit a Trigger frame carrying more than one User Info field, each allocating a single or a contiguous set of RA-RUs, to ensure that an RA-RU set does not overlap with other RUs allocated by the frame.

A non-AP HE STA shall determine the total number of eligible RA-RUs in a contiguous set from the Number Of RA-RU subfields (see Table 9-64f) in the User Info field corresponding to an eligible RA-RU, excluding RA-RUs that are not within its operating bandwidth.

A non-AP HE STA may consider as eligible RA-RUs all or a subset of the RA-RUs indicated by the User Info fields in a Trigger frame that carries more than one User Info field that allocates RA-RUs. The number of eligible RA-RUs that the non-AP STA considers for OBO countdown and transmission (see 26.5.4.3) shall be the total number of eligible RA-RUs indicated by the selected subset of User Info fields.

NOTE 3—A STA that considers only a subset of User Info fields can randomly select User Info fields from the available set of User Info fields that allocate RA-RUs so that the UORA contention is not concentrated at the RA-RU set indicated by the first User Info field.

26.5.4.3 Transmission procedure for UORA

In this subclause, the transmit procedure using RA-RUs is described with respect to UORA parameters. The procedure is also illustrated in Figure 26-6.

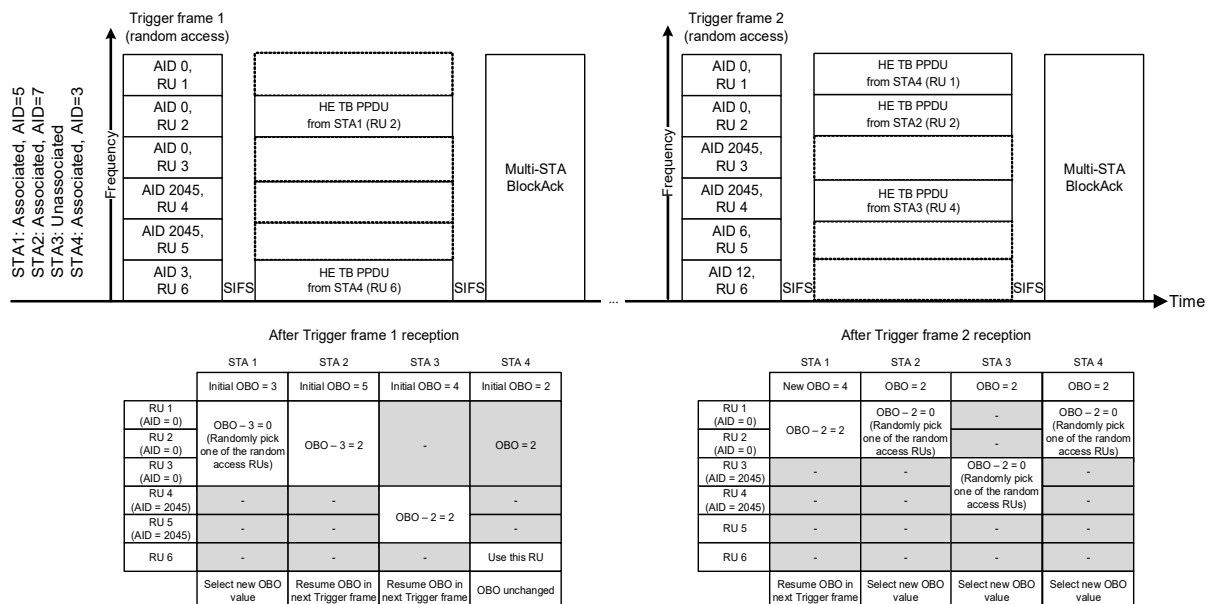


Figure 26-6—Illustration of UORA procedure

A non-AP STA shall not contend for an eligible RA-RU or decrement its OBO counter if it does not have pending frames for the AP.

If an HE STA has a pending frame for the AP upon the reception of a Trigger frame containing at least one eligible RA-RU and if the OBO counter of an HE STA is not greater than the number of eligible RA-RUs in a Trigger frame from that AP, then the HE STA shall set its OBO counter to zero and randomly select one of the eligible RA-RUs to be considered for transmission. Otherwise, the HE STA decrements its OBO counter by the number of eligible RA-RUs in the Trigger frame.

In the example in Figure 26-6:

- Before Trigger frame 1 was sent by the AP, HE STA 1, STA 2, STA 3, and STA 4 had initial OBO values of 3, 5, 4, and 2, respectively.
- Upon receiving Trigger frame 1:
 - STA 4, which is associated with the AP and has pending frames for the AP, is allocated a dedicated RU (RU6). The STA does not contend for RA-RUs and instead transmits its pending frames on RU6.
 - STA 1 and STA 2, both associated with the AP and having pending frames for the AP, decrement their respective OBO counters by the number of eligible RA-RUs indicated in the Trigger frame (i.e., three RA-RUs for associated STAs). Since STA 1's OBO counter decrements to 0, it transmits its pending frames on RU2 that it randomly selects from the eligible set of RUs (i.e., RU1, RU2, and RU3). Since STA 2's OBO counter decrements to a nonzero value, it maintains the new OBO value (2) until it receives a later Trigger frame carrying RA-RUs for associated STAs.
 - STA 3, which is not associated with the AP but has a pending frame for the AP, decrements its OBO counter by the number of eligible RA-RUs indicated in the Trigger frame (i.e., two RA-RUs for unassociated STAs). Since STA 3's OBO counter decrements to a nonzero value, it maintains the new OBO value (2) until it receives a later Trigger frame carrying RA-RUs for unassociated STAs.
- After transmission of HE TB PPDU in response to Trigger frame 1:
 - STA 4 has additional frames pending for the AP. Therefore, it maintains its initial OBO value (2) until it receives a later Trigger frame carrying RA-RUs for associated STAs.
 - STA 1 has additional frames pending for the AP and randomly selects a new OBO value (4).
- Upon receiving Trigger frame 2:
 - STA 1, STA 2, and STA 4 decrement their respective OBO counters by the number of eligible RA-RUs (two in this case). Since STA 2 and STA 4's OBO counters decrements to 0, they both transmit their pending frames on a randomly selected RU (RU2 in the case of STA 2 and RU1 in the case of STA 4). If either STAs have additional frames pending for the AP, each would randomly select a new OBO value. Since STA 1's OBO decrements to a nonzero value, it maintains the new OBO value (2) until it receives a later Trigger frame carrying RA-RUs for associated STAs.
 - STA 3 decrements its OBO counter by the number of eligible RA-RUs (two in this case). Since the STA's OBO counter decrements to 0, it transmits its pending frame on a randomly selected RU (RU4 in this case).

A non-AP STA shall follow the rules in 26.5.2.3 to construct an HE TB PPDU and shall follow the rules as defined in 26.5.2.5 to determine the state of the medium before transmitting the HE TB PPDU. If CS is required and the selected RU is considered busy, then the non-AP STA shall not transmit the HE TB PPDU and the non-AP STA shall set its OBO counter to a random value drawn from a uniform distribution in the range 0 to OCW.

The MU acknowledgment procedure for UORA follows the procedure as defined in 10.3.2.13.3.

If a non-AP STA transmits an HE TB PPDU that contains a frame that solicits an immediate response in an RA-RU and the expected response is not received, the transmission is considered unsuccessful. Otherwise, the transmission is considered successful. After each successful HE TB PPDU transmission in an RA-RU, a

non-AP HE STA shall set the value of OCW to the OCW_{min} obtained from the most recent OCW_{min} indicated in the UORA Parameter Set element from the HE AP or the default (if UORA Parameter Set element was not received) and shall initialize its OBO counter to an integer value randomly selected from a uniform distribution in the range 0 to OCW. The non-AP STA shall follow the retransmission procedure defined in 26.5.4.4 if the transmission is not successful.

NOTE—A non-AP STA that transmits an HE TB PPDU in response to a Trigger frame that is not an MU-RTS Trigger frame that allocates RA-RU(s) by following the UORA procedure does not update its state variables to the values contained in the MU EDCA Parameter Set element (see 26.2.7).

26.5.4.4 Retransmission procedure for UORA

A non-AP STA whose HE TB PPDU transmission sent in an RA-RU of a Trigger frame is unsuccessful, may attempt to retransmit the failed PPDU using EDCA or as a response to a Trigger frame.

If the HE TB PPDU is not successfully transmitted in the selected RA-RU, then the non-AP STA shall update its OCW to $2 \times OCW + 1$ when the OCW is less than the value of OCW_{max} , and shall randomly select its OBO counter in the range of 0 and OCW. Once the OCW reaches OCW_{max} for successive retransmission attempts, the OCW shall remain at the value of OCW_{max} until the OCW is reset as described in 26.5.4.3.

A non-AP STA shall update its OCW value under the condition that the updated OCW remains in the range OCW_{min} to OCW_{max} obtained from the most recently received UORA Parameter Set element (see 9.4.2.250). If the updated OCW becomes greater than OCW_{max} as a consequence of receiving a modified UORA Parameter Set element, then the non-AP STA shall set the value of OCW to the new OCW_{max} value.

26.5.4.5 Additional considerations for unassociated STAs

An AP transmitting a Trigger frame that allocates one or more RA-RUs for unassociated STAs shall transmit the Trigger frame in an HE PPDU so that an unassociated non-AP STA can determine the AP's BSS color.

An HE AP that supports the UORA procedure and broadcast TWT operation and that intends to transmit Trigger frames that allocate one or more RA-RUs for unassociated STAs shall schedule the transmission of at least one such Trigger frame within each TWT SP corresponding to a Broadcast TWT Parameter Set field in a TWT element with a Broadcast TWT ID subfield equal to 0, Flow Type subfield equal to 0, Trigger subfield equal to 1, and Broadcast TWT Recommendation subfield equal to 2.

An AP that receives an Authentication frame within an RA-RU should schedule for transmission, at a time no less than 3 TUs and no greater than 5 TUs subsequent to the transmission of the Authentication frame that is a response to that reception, a Trigger frame that allocates one or more RA-RUs for unassociated STAs.

NOTE—This Trigger frame provides a potential opportunity for an unassociated STA that transmitted the Authentication frame to transmit another Authentication frame or an Association Request frame.

An AP that supports the UORA procedure, broadcasts TWT operation, operates a BSS with a width of 80 MHz or greater, and transmits a Trigger frame that allocates one or more RA-RUs for unassociated STAs shall include at least 2 RA-RUs for unassociated STAs for at least one transmission of such a Trigger frame within a Broadcast TWT SP that meets the conditions described above.

An AP that intends to transmit Trigger frames that allocates one or more RA-RUs for unassociated STAs should transmit FILS Discovery frames as described in 11.45.2.1 at regular intervals within a beacon period to assist unassociated STA discovery of the BSS and its operating parameters.

The TWT scheduling AP (see 26.8.3.1) that supports the UORA procedure may include a broadcast TWT element in FILS Discovery frames and in broadcast Probe Response frames to indicate the TWT SPs during which the AP intends to schedule for transmission at least one Trigger frame allocating one or more RA-RUs for STAs not associated with the AP. The broadcast TWT element shall carry only a broadcast TWT parameter set with the Broadcast TWT Recommendation subfield set to 2, the Trigger subfield set to 1, the Flow Type subfield set to 0, and the Broadcast TWT ID subfield set to 0. The AP transmits broadcast Probe Response frames if it has `dot11FILSOmitReplicateProbeResponses` equal to true.

A non-AP STA that receives a FILS Discovery frame from an AP with which it is not associated may use the values carried in the frame to determine the operating parameters for that AP and may use the information when responding to a Trigger frame from the AP containing RA-RUs for unassociated non-AP STAs.

An unassociated non-AP STA that has not received an UORA Parameter Set element from the AP with which it intends to communicate shall use the default OCW values as defined in 26.5.4.1. Each time an unassociated non-AP STA communicates with a different AP using random access, it shall initialize its OCW using the default values or the parameters from the UORA Parameter Set element received from that AP and shall initialize its OBO counter as defined in 26.5.4.3.

An unassociated non-AP STA that supports the UORA and TWT procedure may begin listening for Trigger frames at the start of a particular broadcast TWT SP after receiving a Beacon frame, a broadcast Probe Response frame, or a FILS Discovery frame containing a TWT element with a Broadcast TWT Parameter Set field that has the Broadcast TWT ID subfield equal to 0, the Flow Type subfield equal to 0, the Trigger subfield equal to 1, and the Broadcast TWT Recommendation subfield equal to 2.

A non-AP STA shall include at most one Management frame that is an S-MPDU when it transmits an HE TB PPDU in response to a Trigger frame that is not an MU-RTS Trigger frame sent by an AP with which the non-AP STA is not associated.

An AP that receives Management frames from one or more unassociated non-AP STAs carried in HE TB PPDU transmitted on RA-RUs shall respond with a Multi-STA BlockAck frame carried either in an SU PPDU or in a DL HE MU PPDU on a broadcast RU corresponding to parameter STA-ID equal to 2045.

An AP with `dot11FILSOmitReplicateProbeResponses` equal to true shall follow the procedure defined in 11.1.4.3.4 to respond with a broadcast Probe Response frame or the next Beacon frame if it receives one or more Probe Request frames via the UORA procedure.

26.5.5 Buffer status report operation

A non-AP STA delivers buffer status reports (BSRs) to assist its AP in allocating UL MU resources. The non-AP STA can either implicitly deliver BSRs in the QoS Control field or BSR Control subfield of any frame transmitted to the AP (unsolicited BSR) or explicitly deliver BSRs in any frame sent to the AP in response to a BSRP Trigger frame (solicited BSR). The buffer status reported in the QoS Control field consists of a queue size value for a given TID (see 9.2.4.5.6). The buffer status reported in the BSR Control field consists of an ACI bitmap, delta TID, a high priority AC, and two queue sizes (see 9.2.4.6a.4).

An HE STA shall set the BSR Support subfield in the HE Capabilities element it transmits to 1 if `dot11HEBSRControlImplemented` is true; otherwise, the HE STA shall set the BSR Support subfield to 0.

A non-AP STA reports its buffer status (unsolicited BSR) to the AP with which it is associated in the QoS Control field in QoS Null and QoS Data frames and in the BSR Control subfield (if present) in QoS Null, QoS Data, and Management frames as defined below:

- The HE STA shall report the queue size for a given TID in the Queue Size subfield of the QoS Control field in QoS Data or QoS Null frames it transmits; the STA may set the Queue Size subfield to 255 to indicate an unknown/unspecified queue size for that TID.
 - The HE STA may aggregate multiple QoS Data frames or QoS Null frames in an A-MPDU to report the queue size for different TIDs. The HE STA shall follow the A-MPDU aggregation rules defined in 26.6.3 for aggregating QoS Data frames with multiple TIDs. The HE STA does not follow the rules defined in 26.6.3 for QoS Null frames with No Ack ack policy.
- The HE STA may report the buffer status in the BSR Control subfield of frames it transmits if the AP has indicated its support in the BSR Support subfield of its HE Capabilities element; otherwise, the STA shall not report the buffer status in the BSR Control subfield.
 - The HE STA shall report the queue size for its preferred AC, indicated by the ACI High subfield, in the Queue Size High subfield of the BSR Control subfield; the STA may set the Queue Size High subfield to 255 to indicate an unknown/unspecified queue size for that AC.
 - The HE STA shall report the queue size for the ACs, indicated by the ACI Bitmap subfield, in the Queue Size All subfield of the BSR Control subfield; the STA may set the Queue Size All subfield to 255 to indicate an unknown/unspecified BSR for those ACs.
 - The HE STA shall set the Delta TID subfield according to Table 9-24e and the Scaling Factor subfield as defined in 9.2.4.6a.4.

NOTE 1—The STA can send an unsolicited BSR in response to certain Trigger frames except MU-RTS and BSRP (with or without RA-RUs, as defined in 26.5.2.3 and 26.5.4), or it can send the unsolicited BSR after accessing the WM using EDCA.

NOTE 2—The STA might include a BSR Control subfield in a QoS Data or QoS Null frame. In this case the Queue Size subfield in the QoS Control field and the Queue Size High and Queue Size All subfields in the BSR Control subfield might differ, and any of the three subfields might be set to 255 to indicate unspecified/unknown queue size. The STA might include only the BSR Control subfield in a Management frame.

An AP can also solicit one or more associated non-AP STAs for their BSR(s) by sending a BSRP Trigger frame (see 9.3.1.22.6). The non-AP STA responds (solicited BSR) as defined below:

- The non-AP STA that receives a BSRP Trigger frame shall follow the rules defined in 26.5.2.3 to generate the HE TB PPDU if the Trigger frame contains the 12 LSBs of the non-AP STA's AID in any of the User Info fields; otherwise, if the non-AP STA's buffers are not empty and the non-AP STA supports the UORA procedure, it may follow the rules defined in 26.5.4 to gain access to an RA-RU and generate the HE TB PPDU when the Trigger frame contains one or more RA-RUs.
- The non-AP STA shall include in the HE TB PPDU one or more QoS Null frames containing one or more of the following:
 - The QoS Control field(s) with Queue Size subfields for each of the TIDs for which the non-AP STA has queue size to report to the AP.
 - The BSR Control subfield with the Queue Size All subfield indicating the queue size for the ACs, indicated by the ACI Bitmap subfield, for which the non-AP STA has queue size to report to the AP if the AP has indicated its support in the BSR Support subfield of its HE Capabilities element. The non-AP STA shall set Delta TID, Scaling Factor, ACI High, and Queue Size High subfields of the BSR Control subfield as defined in 9.2.4.6a.4.
- The non-AP STA shall not solicit an immediate response for the frames carried in the HE TB PPDU (e.g., the Ack Policy Indicator subfield of a QoS Null frame shall not be set to Normal Ack or Implicit BAR).

NOTE 3—As with unsolicited BSR, the STA might include a BSR Control subfield in a QoS Null frame that is sent in response to the BSRP Trigger frame. In this case, the Queue Size subfield in the QoS Control field and the Queue Size High and Queue Size All subfields in the BSR Control subfield might differ, and any of the three subfields might be set to 255 to indicate an unspecified/unknown queue size.

NOTE 4—An AP does not send a BSRP Trigger frame containing the 12 LSBs of the AID of the non-AP STA that sets the UL MU Disable field to 1.

An AP may include a BSRP Trigger frame together with other Control, Data, and Management frames in one A-MPDU to a non-AP STA if the HE Capabilities element received from the non-AP STA has the BSRP BQRP A-MPDU Aggregation field equal to 1. A non-AP STA constructs the A-MPDU contained in the HE TB PPDU sent in response to a BSRP Trigger frame as described in 26.5.2.4.

The NDP feedback report procedure described in 26.5.7 can be used for buffer status feedback operation. An AP that sent an NFRP Trigger frame to one or more non-AP STAs may send a BSRP Trigger frame to those non-AP STAs to get more precise buffer status information.

26.5.6 Bandwidth query report operation

A non-AP STA may send bandwidth query reports (BQRs) to assist its AP in allocating DL MU and UL MU resources. The non-AP STA may either implicitly deliver BQRs in the BQR Control subfield of a frame transmitted to the AP (unsolicited BQR) or explicitly deliver BQRs in a frame sent to the AP in response to a BQRP Trigger frame (solicited BQR).

An HE STA shall set the BQR Support subfield in the HE Capabilities element it transmits to 1 if dot11HEBQRControlImplemented is true; otherwise, the HE STA shall set the BQR Support subfield to 0.

The HE STA may report the channel availability information as specified in 27.3.20.6.5 in the BQR Control subfield of frames it transmits if the AP has indicated its support in the BQR Support subfield of its HE Capabilities element; otherwise, the STA shall not report the channel availability information in the BQR Control subfield.

The availability of each 20 MHz subchannel reported in the BQR Control field shall be reported for the 20 MHz subchannels located in the operating channel of the reporting STA (see 9.2.4.6a.6).

NOTE 1—The STA can send an unsolicited BQR in response to certain Trigger frames except NFRP, MU-RTS, and BQRP (with or without RA-RUs, as defined in 26.5.2.3 and 26.5.4), or it can send the unsolicited BQR after accessing the WM using EDCA.

An AP may solicit BQRs from a non-AP STA only if the non-AP STA has indicated support by setting the BQR Support field in the HE Capabilities element it transmits to 1; otherwise, the AP shall not solicit BQRs from the non-AP STA. An AP may solicit BQRs from one or more non-AP HE STAs that support generating BQRs, by sending a BQRP Trigger frame (see 9.3.1.22.8).

An AP shall not transmit a BQR Trigger frame with the User Info field addressed to a non-AP STA, unless it has received from the non-AP STA an HE Capabilities element with the BQR Support subfield equal to 1.

If a non-AP STA supports BQR operation and receives a BQRP Trigger frame with a User Info field addressed to the non-AP STA, then the non-AP STA shall respond by following the procedure in 26.5.2.3.1.

If a non-AP STA supports both BQR operation and the UORA procedure and receives a BQRP Trigger frame from the AP with which it is associated and that allocates RA-RUs for associated STAs but that does contain a User Info field addressed to the non-AP STA, then the non-AP STA shall respond by following the procedure in 26.5.4.

If the non-AP STA responds with an HE TB PPDU using one of the above procedures, then the A-MPDU carried in the HE TB PPDU shall include one or more QoS Null frames containing a BQR Control subfield with the channel availability information of the STA. The non-AP STA shall not solicit an immediate response for the frames carried in the HE TB PPDU. The Ack Policy Indication subfield of the frame shall be set to No Ack.

NOTE 2—An AP does not send a BQRP Trigger frame addressed to a STA that has set the UL MU Disable field to 1.

NOTE 3—The channel availability information of the STA is limited to the STA's operating channel width. See 9.2.4.6a.6.

An AP may include a BQRP Trigger frame together with other Control, Data, and Management frames in one A-MPDU to a STA if the HE Capabilities element received from the STA has the BSRP BQRP A-MPDU Aggregation field equal to 1. A non-AP STA constructs the A-MPDU contained in the HE TB PPDU sent in response to a BQRP Trigger frame as described in 26.5.2.4.

26.5.7 NDP feedback report procedure

26.5.7.1 General

The NDP feedback report procedure allows an HE AP to collect feedback that is not channel sounding from multiple non-AP HE STAs.

An HE AP sends an NFRP Trigger frame to solicit NDP feedback report response from many non-AP STAs that are identified by a range of scheduled AIDs in the Trigger frame. The NDP feedback report response from a non-AP STA is an HE TB feedback NDP (see 27.3.18). A non-AP STA uses the information carried in the NFRP Trigger frame to know if it is scheduled and, in this case, to derive the parameters for the transmission of the response.

The NDP feedback report procedure is described in 26.5.7.2 through 26.5.7.5.

26.5.7.2 STA behavior

A non-AP STA shall set the NDP Feedback Report Support subfield in the HE Capabilities element to 1 if it supports NDP feedback report and set it 0, otherwise.

A non-AP STA shall not transmit an NDP feedback report response, unless it is explicitly enabled by an AP in one of the operation modes described in this subclause. The interframe space between a PPDU that contains an NFRP Trigger frame and the NDP feedback report poll response is SIFS. A non-AP STA shall commence the transmission of an NDP feedback report response at the SIFS time boundary after the end of a received PPDU if all the following conditions are met:

- The received PPDU contains an NFRP Trigger frame.
- The non-AP STA is scheduled by the NFRP Trigger frame.
- The NDP feedback report support subfield in HE MAC Capabilities Information field is set to 1.
- The non-AP STA intends to provide a response to the type of the NDP feedback contained in the NFRP Trigger frame, as described in 26.5.7.4.

A non-AP STA that does not satisfy all of the above conditions shall not respond to the NFRP Trigger frame.

A non-AP STA is scheduled to respond to the NFRP Trigger frame if all the following conditions are met:

- The non-AP STA is associated with an AP corresponding to the BSSID indicated in the TA field of the NFRP Trigger frame; or the non-AP STA is associated with an AP corresponding to a nontransmitted BSSID of a multiple BSSID set, and the TA field of the NFRP Trigger frame is set to the transmitted BSSID of that multiple BSSID set.
- The non-AP STA's AID is greater than or equal to the starting AID and less than the starting AID + N_{STA} , using the Starting AID subfield in the eliciting Trigger frame and N_{STA} as the total number of non-AP STAs that are scheduled to respond to the NFRP Trigger frame. N_{STA} is calculated by the following equation, using UL BW subfield and Number Of Spatially Multiplexed Users subfield from the eliciting Trigger frame:

$$N_{STA} = 18 \times 2^{BW} \times (\text{value of the Number Of Spatially Multiplexed Users subfield} + 1)$$

A non-AP STA shall obtain NDP feedback report parameter values from the most recently received NDP Feedback Report Parameter Set element carried in a Beacon, Probe Response, or (Re)Association Response frame from its associated AP, unless the non-AP STA is associated with an AP corresponding to a nontransmitted BSSID of a multiple BSSID set. In this case, it shall follow the rules in 11.1.3.8.4 to determine the NDP feedback parameter values. If the NDP Feedback Report Parameter Set element is not received in a Management frame with a TA equal to the BSSID of the associated AP or to the transmitted BSSID of the multiple BSSID set, the non-AP STA shall use default values for the NDP Feedback Report parameters.

An NDP feedback report response is an HE TB feedback NDP as defined in 27.3.4.

A non-AP STA transmitting an NDP feedback report shall set the TXVECTOR parameters as for transmitting an HE TB PPDU in response to a Trigger frame that is not an MU-RTS Trigger frame as described in 26.5.2.3, except for the following parameters:

- The FORMAT parameter shall be set to HE_TB.
- The APEP_LENGTH parameter shall be set to 0.
- The RU_ALLOCATION parameter shall be set to be maximum RU size for the BW.
- The RU_TONE_SET_INDEX parameter shall be set using the following equation, with the value of the Starting AID subfield in the User Info field of the eliciting Trigger frame:

$$RU_TONE_SET_INDEX = 1 + ((AID - \text{Starting AID}) \bmod (18 \times 2^{BW}))$$

- The NUM_STS parameter shall be set to 1.
- The SPATIAL_REUSE parameter shall be set to PSR_DISALLOW.
- The STARTING_STS_NUM parameter shall be set with the following equation, using the values of the Starting AID subfield in the User Info field of the eliciting Trigger frame:

$$STARTING_STS_NUM = \lfloor (AID - \text{Starting AID}) / 18 / 2^{BW} \rfloor$$

- The MCS parameter shall be set to 0.
- The DCM parameter shall be set to 0.
- The FEC_CODING parameter shall be set to BCC_CODING.
- The TXPWR_LEVEL_INDEX parameter shall be set to the value based on the Transmit Power Control for an HE TB PPDU and based on the value of the AP Tx Power subfield and the UL Target Receive Power subfield in the User Info field of the eliciting Trigger frame (see 27.3.15.2).

NOTE—The subcarriers for each RU_TONE_SET index are contained in a 20 MHz channel and can be transmitted by a 20 MHz operating STA.

26.5.7.3 AP behavior

An AP shall set the NDP Feedback Report Support subfield in the HE Capabilities element to 1 if it supports NDP feedback report and set it 0 otherwise.

An AP may include the NDP Feedback Report Parameter Set element in Beacon frames, Probe Response frames, and (Re)Association Response frames in order to modify parameters for NDP Feedback Report operation. The procedure of NDP Feedback report described in this subclause allows operation even if the NDP Feedback Report Parameter Set element is not sent by the AP.

The NFRP Trigger frame shall be transmitted in a non-HT PPDU or HT PPDU, or as a tagged MPDU in a VHT, HE ER SU PPDU, or HE SU PPDU.

An AP that transmits an NFRP Trigger frame shall set the TA field of the frame to the MAC address of the AP, unless `dot11MultiBSSIDImplemented` is true and the Trigger frame is directed to STAs from at least two different BSSs of a multiple BSSID set. In this case, the AP shall set the TA field of the frame to the transmitted BSSID.

Following the transmission from an AP of an NFRP Trigger frame, multiple STAs may simultaneously send NDP feedback report responses to the AP. Based on the RXVECTOR parameter `NDP_REPORT`, which provides the detected status array for the resources of each spatial stream and tone set assigned by the Trigger frame, the AP can derive the list of AIDs from the resources to which an NDP feedback report response was sent and their responses.

The AP shall not send any acknowledgment in response to the reception of NDP feedback report responses.

26.5.7.4 NDP feedback report for a resource request

If the Feedback Type subfield in the User Info field of the NFRP Trigger frame is 0, a STA that is scheduled may send an NDP feedback report response in order to signal to the AP that it is in the awake state and that it might have frames in its queues for UL MU. If the STA does not satisfy either of the conditions in Table 26-3, then the STA shall not respond to the NFRP Trigger frame.

Each STA that is scheduled is assigned a `STARTING_STS_NUM` and an `RU_TONE_SET_INDEX` to transmit a `FEEDBACK_STATUS` bit.

The meaning of the `FEEDBACK_STATUS` bit is defined in Table 26-3:

Table 26-3—FEEDBACK_STATUS description

FEEDBACK_STATUS	Condition
0	The STA is in the awake state and reports buffered octets for transmission not exceeding the resource request buffer threshold. NOTE—The STA can use this value to indicate that it is in the awake state even if it does not have any buffered octets for transmission, e.g., to solicit delivery of BUs known to be buffered at the AP.
1	The STA is in the awake state and reports buffered octets for transmission exceeding the resource request buffer threshold.

The resource request buffer threshold is equal to $2^{(\text{Resource request buffer threshold exponent})}$ octets, using the Resource Request Buffer Threshold Exponent subfield in the most recently received NDP Feedback Report Parameter Set element sent by the AP with which the STA is associated. The resource request buffer threshold is equal to 256 octets if the STA did not receive an NDP Feedback Report Parameter Set element from the AP with which the STA is associated.

26.5.7.5 Power save operation with NDP feedback report procedure

An HE AP that sends an NFRP Trigger frame to a non-AP STA in PS mode and receives an NDP feedback report response from the STA shall assume the STA to be in, or to have transitioned to, the awake state and shall follow the rules defined in

- 11.2.3.5 to deliver BUs to the STA as if the STA had sent a U-APSD trigger frame if the STA is using U-APSD.
- 11.2.3.1 to deliver BUs to the STA as if the STA had sent a PS-Poll frame, if the STA is not using U-APSD.
- 26.8 for delivery of BUs during the TWT SP if the STA operates with TWT.

NOTE—As the AP does not send an acknowledgment in response to the NDP Feedback Report response, a STA that sends an NDP Feedback Report response and indicates that it is in the awake state might still schedule for transmission a PS-Poll or U-APSD trigger frame in order to indicate that it is in the awake state.

26.5.8 Use of TSPEC by HE STAs

In addition to the TS Setup operations as described in 11.4.4, a non-AP HE STA may use a TSPEC contained in a Basic ADDTS Request frame to provide its traffic characteristics and QoS requirements to an HE AP that supports the reception of Basic ADDTS Request frame in order to facilitate efficient scheduling for HE AP's UL and DL MU operations. A TSPEC provided by a non-AP HE STA is used by a receiving HE AP to facilitate the creation of a schedule for contention based channel access (EDCA) or MU operation. How the HE AP uses the information provided by the non-AP HE STA is beyond the scope of this standard.

An HE AP transmits an ADDTS Response frame as a response to ADDTS Request frame as described in 11.4.4.

A non-AP HE STA should send a DELTS frame with the corresponding TSID if the traffic associated with the TSID has been terminated. When receiving a DELTS from a non-AP HE STA, the HE AP shall consider the information provided in the TSPEC as no longer valid.

26.5.9 UL MU transmit power capabilities

A non-AP HE STA may use the UL MU Power Capabilities element in a (Re)Association Request frame in order to inform an HE AP of the relative maximum transmit power at which the non-AP HE STA is capable of transmitting an HE TB PPDU for each HE-MCS when using an RU size greater than or equal to 242 subcarriers.

An HE AP might use this information as an input into the algorithm used to schedule the non-AP HE STA for UL MU transmission. The specification of the algorithm is beyond the scope of this standard.

26.6 A-MPDU operation in an HE PPDU

26.6.1 General

A-MPDU operation for an HE PPDU follows the procedures defined in 10.12 and the additional rules in this subclause.

An HE STA that sends a Class 1 frame or a Class 2 frame in an HE PPDU shall send the frame as an S-MPDU (see Table 9-534).

An HE STA that sends a VHT Capabilities element, an HT Capabilities element, or an HE 6 GHz Band Capabilities element and an HE Capabilities element with Maximum A-MPDU Length Exponent Extension field of 0 shall support in reception an A-MPDU pre-EOF padding with maximum length defined in 10.12.2.

An HE STA that sends a VHT Capabilities element and an HE Capabilities element with Maximum A-MPDU Length Exponent Extension subfield greater than 0 shall support reception of an HE PPDU with an A-MPDU pre-EOF padding as defined in 10.12.2, except that the maximum length for the A-MPDU pre-EOF padding shall be equal to $\min(2^{(20 + \text{Maximum A-MPDU Length Exponent Extension})} - 1, 6\,500\,631)$. An HE STA that sets the Maximum A-MPDU Length Exponent Extension subfield in the HE Capabilities element to a value greater than 0 shall set the Maximum A-MPDU Length Exponent subfield of the VHT Capabilities element to 7.

NOTE 1—The value 6 500 631 is defined in Table 9-25 as the upper bound of A-MPDU size.

An HE STA that does not send a VHT Capabilities element but sends an HT Capabilities element and an HE Capabilities element with Maximum A-MPDU Length Exponent Extension subfield greater than 0 shall support in reception an A-MPDU pre-EOF padding in an HE PPDU as defined in 10.13.2, except that the maximum length for the A-MPDU pre-EOF padding shall be equal to $2^{(16 + \text{Maximum A-MPDU Length Exponent Extension})} - 1$. An HE STA that sets the Maximum A-MPDU Length Exponent Extension subfield in the HE Capabilities element to a value greater than 0 shall set the Maximum A-MPDU Length Exponent subfield of the HT Capabilities element to 3.

NOTE 2—An HE STA that is a VHT STA sends a VHT Capabilities element. An HE STA that is not a VHT STA does not send a VHT Capabilities element.

An HE STA that sends an HE 6 GHz Band Capabilities element and an HE Capabilities element with Maximum A-MPDU Length Exponent Extension subfield greater than 0 shall support in reception an A-MPDU pre-EOF padding in an HE PPDU as defined in 10.12.2, except that the maximum length for the A-MPDU pre-EOF padding shall be equal to $\min(2^{(20 + \text{Maximum A-MPDU Length Exponent Extension})} - 1, 6\,500\,631)$. An HE STA that sets the Maximum A-MPDU Length Exponent Extension subfield in the HE Capabilities element to a value greater than 0 shall set the Maximum A-MPDU Length Exponent subfield of the HE 6 GHz Band Capabilities element to 7.

An HE STA shall not transmit an A-MPDU in an HE PPDU to a STA that exceeds the maximum A-MPDU length capability indicated in the HE Capabilities, VHT Capabilities, and HT Capabilities elements received from the recipient STA. If a VHT Capabilities element is received from the recipient STA, then the maximum A-MPDU length capability is derived from the Maximum A-MPDU Length Exponent Extension subfield in the HE Capabilities and the Maximum A-MPDU Length Exponent subfield in the VHT Capabilities element. Otherwise, the maximum A-MPDU length capability is derived from the Maximum A-MPDU Length Exponent subfields in the HE Capabilities element and the Maximum A-MPDU Length Exponent subfield in the HT Capabilities element or in the HE 6 GHz Band Capabilities element.

An HE STA may transmit an HE SU PPDU or HE MU PPDU that carries an A-MPDU with contents defined in Table 9-531 or Table 9-534a.

An A-MPDU with any number of QoS Null frames with any TID and with No Ack ack policy and aggregated with or without other frames may be transmitted to a recipient STA in an HE PPDU that is not an HE TB PPDU regardless of the value of the Multi-TID Aggregation Rx/Tx Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element received from the recipient STA.

An A-MPDU with any number of QoS Null frames with any TID and with No Ack ack policy and aggregated with or without other frames in an A-MPDU may be transmitted in the HE TB PPDU regardless of the value of the TID Aggregation Limit subfield and the value of the Preferred AC subfield in the Basic Trigger frame and the value of the Multi-TID Aggregation Rx Support of the AP that solicits the A-MPDU.

NOTE 3—A QoS Null frame with Normal Ack or Implicit BAR ack policy is not allowed to be sent in an A-MPDU, except for a QoS Null frame with Normal Ack ack policy in an S-MPDU (as defined in Table 9-531, Table 9-533, Table 9-534a, Table 9-534b, Table 9-534c, and Table 9-534d).

26.6.2 A-MPDU padding in an HE PPDU

26.6.2.1 General

In an HE PPDU, a STA shall not add an A-MPDU subframe with the EOF/Tag field set to 1 and with the MPDU Length field set to 0 before an A-MPDU subframe with a nonzero MPDU Length field. In an HE PPDU, a STA should add an A-MPDU subframe with the EOF/Tag field set to 1 and with the MPDU Length field set to 0 (i.e., EOF padding) immediately after the last A-MPDU subframe with a nonzero MPDU Length field.

26.6.2.2 A-MPDU padding in an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU

An HE STA that transmits a DL HE MU PPDU that contains one or more PSDUs, each of which carries an A-MPDU, shall construct the A-MPDUs as described in 10.12.6, except that one or more A-MPDU subframes with a nonzero MPDU Length field and with an EOF/Tag field equal to 1 may be added if the A-MPDU is an ack-enabled single-TID A-MPDU (see 26.6.3).

An HE STA that transmits an HE SU PPDU, HE ER SU PPDU, or UL HE MU PPDU that contains one A-MPDU shall construct the A-MPDU as described in 10.12.6, except that one or more A-MPDU subframes with a nonzero MPDU Length field and with an EOF/Tag field equal to 1 may be added if the A-MPDU is an ack-enabled single-TID A-MPDU (see 26.6.3).

26.6.2.3 A-MPDU padding in an HE TB PPDU

A non-AP STA that transmits an HE TB PPDU shall construct the PSDU carried in the HE TB PPDU as described in this subclause.

The STA calculates the PSDU_LENGTH as defined in 6.5 and initializes A-MPDU_Length to 0.

The STA may add A-MPDU subframes to the A-MPDU contained in the PSDU, provided that the following constraints are fulfilled:

- The A-MPDU content constraints (see 10.12.1 and 26.6.3) for the intended recipient
- The Length limit constraints (see 9.7.1 and 10.12.2) for the intended recipient
- The MPDU start spacing constraints (see 10.12.3) for the intended recipient

and provided that the following conditions are met:

- The A-MPDU subframes have a value greater than 0 in the MPDU Length field, or have a value 0 in the MPDU Length field and a value 0 in the EOF/Tag field.
- After incrementing the A-MPDU_Length with the length of each such added A-MPDU subframe, the relationship $A-MPDU_Length \leq PSDU_LENGTH$ is true.

Padding is then added so that the resulting A-MPDU contains exactly PSDU_LENGTH octets as follows:

- First, while A-MPDU_Length < PSDU_LENGTH and A-MPDU_Length mod 4 ≠ 0, add an octet to the final A-MPDU subframe's Padding subfield, and increment A-MPDU_Length by 1.
- Then, while A-MPDU_Length + 4 ≤ PSDU_LENGTH, add an EOF padding subframe to the EOF Padding Subframes field, and increment A-MPDU_Length by 4.
- Finally, while A-MPDU_Length < PSDU_LENGTH, add an octet to the EOF Padding Octets subfield, and increment A-MPDU_Length by 1.

An A-MPDU pre-EOF padding is constructed from each user from any of the following:

- A-MPDU subframes constructed from the MPDUs available for transmission from any AC that is selected by the STA.
- A-MPDU subframes with 0 in the MPDU Length field and 0 in the EOF/Tag field.

26.6.3 Multi-TID A-MPDU and ack-enabled single-TID A-MPDU

26.6.3.1 General

A non-ack-enabled multi-TID A-MPDU is an A-MPDU that does not contain a tagged MPDU but does contain untagged MPDUs from at least two TIDs (see Table 9-534c).

An ack-enabled multi-TID A-MPDU is an A-MPDU that contains at least one tagged MPDU that solicits acknowledgment and one or more MPDUs from at least one TID (see Table 9-534d).

An ack-enabled single-TID A-MPDU is an A-MPDU that contains one tagged MPDU that solicits acknowledgment and one or more untagged MPDUs that do not solicit an immediate response (see Table 9-534b).

An HE STA with dot11HEAMPDUwithMultipleTIDOptionImplemented equal to true shall set the Multi-TID Aggregation Rx Support subfield to a nonzero value in the HE MAC Capabilities Information field in the HE Capabilities element it transmits. An HE STA with dot11HEAMPDUwithMultipleTIDOptionImplemented equal to false shall set the Multi-TID Aggregation Rx Support subfield to 0.

An HE STA with dot11AckEnabledAMPDUOptionImplemented equal to true shall set the Ack-Enabled Aggregation Support subfield to 1 in the HE MAC Capabilities Information field in the HE Capabilities element it transmits. An HE STA with dot11AckEnabledAMPDUOptionImplemented equal to false shall set the Ack-Enabled Aggregation Support subfield to 0.

A multi-TID A-MPDU is either a non-ack-enabled multi-TID A-MPDU or an ack-enabled multi-TID A-MPDU. A first HE STA may transmit a non-ack-enabled multi-TID A-MPDU to a second HE STA if the first HE STA has received from the second STA an HE Capabilities element where the Multi-TID Aggregation Rx Support subfield is nonzero. A first HE STA may transmit an ack-enabled multi-TID A-MPDU or a non-ack-enabled multi-TID A-MPDU to a second HE STA if the first HE STA has received from the second HE STA an HE Capabilities element where the Multi-TID Aggregation Rx Support subfield is nonzero and the Ack-Enabled Aggregation Support subfield is 1. Otherwise, the first HE STA shall not transmit a multi-TID A-MPDU to the second HE STA.

An HE STA shall not transmit a multi-TID A-MPDU or ack-enabled single-TID A-MPDU in a VHT PPDU or a HT PPDU.

A non-AP STA shall not send a non-ack-enabled multi-TID A-MPDU in an HE TB PPDU, unless it is in response to a Basic Trigger frame where the TID Aggregation Limit field of the User Info field addressed to the STA is greater than 1. A non-AP STA shall not send an ack-enabled multi-TID A-MPDU in an HE TB PPDU, unless it is in response to a Basic Trigger frame where the TID Aggregation Limit field of the User Info field addressed to the STA is greater than 0.

NOTE 1—An ack-enabled multi-TID A-MPDU solicited by a Basic Trigger frame with TID Aggregation Limit field equal to 1 can contain one Management frame that solicits acknowledgment and one or more QoS Data frames from the same TID if the AP supports reception of ack-enabled multi-TID A-MPDUs.

A multi-TID A-MPDU shall not be transmitted in an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU, unless the TXOP limit is greater than 0 for the AC that is used to gain access to the medium. The AC used to gain access to the medium is the primary AC (see 10.23.2.9). If the TXOP limit of the primary AC is greater than 0, then the STA may aggregate QoS Data frames from one or more TIDs in the A-MPDU under the following conditions:

- The A-MPDU shall be carried in either an HE SU PPDU or an HE ER SU PPDU transmitted by the STA within the obtained TXOP or in an HE MU PPDU transmitted by a non-AP STA within the obtained TXOP.
- The A-MPDU shall contain one or more MPDUs with any of the TIDs that correspond to the primary AC.
- If no more MPDUs can be aggregated in the A-MPDU from any of the TIDs that correspond to the primary AC, then the A-MPDU may additionally contain one or more MPDUs with TIDs that do not correspond to the primary AC if the TIDs correspond to any AC that has a higher priority with respect to the primary AC and the addition of these MPDUs does not cause the STA to exceed the TXOP limit of the primary AC.

An HE AP may aggregate MPDUs from any TIDs in a multi-TID A-MPDU for DL HE MU PPDU transmission, and the number of TIDs in the multi-TID A-MPDU shall not be more than the Multi-TID Aggregation Rx Support announced by the recipient.

The Multi-STA BlockAck frame is used to acknowledge the MPDUs in a multi-TID A-MPDU as defined in 26.4.

In a multi-TID A-MPDU, MPDUs with the same TID are not necessarily contiguous.

A non-AP HE STA that transmits a multi-TID A-MPDU in an HE TB PPDU should aggregate QoS Data frames in the following order:

- First, any and all MPDUs that correspond to the Preferred AC subfield of the Trigger Dependent User Info field addressed to the STA in the Trigger frame.
- Then, any and all MPDUs that correspond to any AC that has a higher priority.
- Then, any MPDUs that correspond to any AC that has a lower priority.

The aggregation of the QoS Data frames is subject to the following:

- The limit indicated in the TID Aggregation Limit subfield in the Trigger Dependent User Info field addressed to the STA in the Trigger frame.
- The restrictions in Table 26-2.
- The duration indicated in the UL Length subfield in the Common Info field of the Trigger frame.

NOTE 2—These rules permit a STA to not aggregate any MPDUs from the preferred AC or a higher priority AC, and permit a STA to aggregate MPDUs in any order.

NOTE 3—A multi-TID A-MPDU allows the aggregation of a Management frame regardless of the value indicated in the Multi-TID Aggregation Rx Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element as long as the indicated value of the TID Aggregation Limit subfield in the Trigger Dependent User Info field of a the Basic Trigger frame is nonzero.

An non-AP HE STA that transmits a single-TID A-MPDU in an HE TB PPDU should select the TID in the following order:

- A TID that corresponds to the Preferred AC subfield of the Trigger Dependent User Info field addressed to the STA in the Trigger frame if there is a corresponding MPDU for that TID.
- Otherwise, a TID that corresponds to any AC that has a higher priority, if there is a corresponding MPDU for that TID.
- Otherwise, a TID that corresponds to any AC that has a lower priority.

The TID selection is subject to the following:

- The limit indicated in the TID Aggregation Limit subfield in the Trigger Dependent User Info field addressed to the STA in the Trigger frame.
- The duration indicated in the UL Length subfield in the Common Info field of the Trigger frame.

NOTE 4—These rules permit a STA to not select MPDUs from the preferred AC or a higher priority AC.

26.6.3.2 Ack-enabled single-TID A-MPDU operation

An ack-enabled single-TID A-MPDU is an A-MPDU with contents defined in Table 9-534b.

NOTE—An ack-enabled single-TID A-MPDU does not contain more than one of the following frames: (a) a QoS Data frame or (b) a Management frame that solicits acknowledgment. In this case, the Management frame that solicits acknowledgment is treated as if it were a QoS Data frame with TID equal to 15.

An HE STA shall not transmit an ack-enabled single-TID A-MPDU to a recipient STA, unless it has received from the recipient STA an HE Capabilities element with the Ack-Enabled Aggregation Support subfield equal to 1.

26.6.3.3 Non-ack-enabled multi-TID A-MPDU operation

A non-ack-enabled multi-TID A-MPDU is an A-MPDU with contents defined in Table 9-534c.

NOTE—A non-ack-enabled multi-TID A-MPDU might include other frames such as a Trigger frame, BlockAck frame, or QoS Null frame (see Table 9-530).

An HE STA shall not transmit a non-ack-enabled multi-TID A-MPDU to a recipient STA, unless it has received from the recipient STA an HE Capabilities element where the Multi-TID Aggregation Rx Support subfield is nonzero.

A STA that receives a non-ack-enabled multi-TID A-MPDU responds as defined in 26.4.4.

26.6.3.4 Ack-enabled multi-TID A-MPDU operation

An ack-enabled multi-TID A-MPDU is an A-MPDU with contents defined in Table 9-534d.

QoS Data frames with the same TID shall have the same ack policy.

QoS Data frames with the same TID shall be carried in A-MPDU subframes with the same value in the EOF/Tag field setting.

In an ack-enabled multi-TID A-MPDU, the EOF/Tag field in the MPDU delimiter of each A-MPDU subframe carrying a frame that solicits an Ack frame as defined in Table 9-534d shall be set to 1. The EOF/Tag field in all other A-MPDU subframes in the A-MPDU that carry frames shall be set to 0.

An HE STA shall not transmit an ack-enabled multi-TID A-MPDU to a recipient STA, unless it has received from the recipient STA an HE Capabilities element where the Multi-TID Aggregation Rx Support subfield is nonzero and the Ack-Enabled Aggregation Support subfield is 1.

A STA that receives an ack-enabled multi-TID A-MPDU responds as defined in 26.4.4.

A STA that transmits an ack-enabled multi-TID A-MPDU that contains at least two tagged MPDUs shall ignore the immediate response if it is an Ack frame.

26.7 HE sounding protocol

26.7.1 General

Transmit beamforming and DL MU-MIMO require knowledge of the channel state to compute a steering matrix that is applied to the transmit signal to optimize reception at one or more receivers. HE STAs use the HE sounding protocol to determine the channel state information. The HE sounding protocol provides explicit feedback mechanisms, defined as HE non-trigger-based (non-TB) sounding and HE trigger-based (TB) sounding, where the HE beamformee measures the channel using a training signal (i.e., an HE sounding NDP) transmitted by the HE beamformer and sends back a transformed estimate of the channel state. The HE beamformer uses this estimate to derive the steering matrix.

The HE beamformee returns an estimate of the channel state in an HE compressed beamforming/CQI report carried in one or more HE Compressed Beamforming/CQI frames. There are three types of HE compressed beamforming/CQI report:

- SU feedback: The HE compressed beamforming/CQI report consists of an HE Compressed Beamforming Report field.
- MU feedback: The HE compressed beamforming/CQI report consists of an HE Compressed Beamforming Report field and HE MU Exclusive Beamforming Report field.
- CQI feedback: The HE compressed beamforming/CQI report consists of an HE CQI Report field.

NOTE—Use of HE TB sounding does not necessarily imply MU feedback. HE TB sounding is also used to obtain SU feedback and CQI feedback.

The HE compressed beamforming/CQI report is carried in a single HE Compressed Beamforming/CQI frame if the resulting frame is less than or equal to 11 454 octets in length (see 26.7.3). Otherwise, the HE beamforming feedback is segmented, and each segment is carried in an HE Compressed Beamforming/CQI frame.

An HE beamformer shall support a maximum MPDU length for the HE compressed beamforming/CQI report that is the minimum of 11 454 octets and the maximum length of the HE compressed beamforming/CQI report that the HE beamformer intends to solicit from its HE beamformees.

26.7.2 HE sounding protocol

An SU beamformer is an HE STA that sets the SU Beamformer subfield in the HE PHY Capabilities Information field in the HE Capabilities element it transmits to 1.

An SU beamformee is an HE STA that sets the SU Beamformee subfield in the HE PHY Capabilities Information field in the HE Capabilities element it transmits to 1. A non-AP HE STA shall set the SU Beamformee subfield to 1. An HE AP may set the SU Beamformee subfield to 1.

An MU beamformer is an HE AP that sets the MU Beamformer subfield in the HE PHY Capabilities Information field in the HE Capabilities element it transmits to 1. An HE AP that indicates support for 4 or more space-time streams in the Tx HE-MCS Map ≤ 80 MHz subfield in the Supported HE-MCS And NSS field in the HE Capabilities element shall set the MU Beamformer subfield to 1. A non-AP HE STA shall set the MU Beamformer subfield to 0. An MU beamformer is also an SU beamformer and shall set the SU Beamformer subfield to 1.

NOTE—A non-AP STA might use the setting of the MU Beamformer subfield to determine the AP with which it will associate.

A non-AP HE STA shall support operation as an MU beamformee. An HE AP does not support operation as an MU beamformee.

The term *HE beamformer* refers to both the SU beamformer and MU beamformer. The term *HE beamformee* refers to both the SU beamformee and MU beamformee.

The type of feedback (SU, MU, or CQI) solicited by an HE beamformer from an HE beamformee is indicated in the Feedback Type And Ng and Codebook subfields in the STA Info field identifying the HE beamformee in the HE NDP Announcement frame as defined in Table 9-29a and Table 9-29b.

The bandwidth (partial or full) of the feedback solicited by an HE beamformer from an HE beamformee depends on the Partial BW Info subfield in the STA Info field identifying the HE beamformee in the HE NDP Announcement frame, the bandwidth of the HE NDP Announcement frame, and the value of the Disallowed Subchannel Bitmap subfield, if present. Full bandwidth feedback is solicited if the RU Start Index subfield in the Partial BW Info subfield is 0, the Disallowed Subchannel Bitmap subfield is absent or contains all zeroes, and the RU End Index subfield in the Partial BW subfield is the value shown in Table 26-4 where partial bandwidth is not supported by the HE beamformer, for the bandwidth of the HE NDP Announcement frame.

Other settings of the Partial BW Info subfield solicit partial-bandwidth feedback. Punctured sounding is indicated by the inclusion of a nonzero Disallowed Subchannel Bitmap subfield in the HE NDP Announcement frame. In such a case, the disallowed subchannels are applied to the tone information to be included in the feedback after selecting tones for feedback based on the RU Start Index and RU End Index subfield values and HE NDP Announcement frame bandwidth as described above. See Table 26-4.

The RU Start Index and RU End Index subfields shall cover at least one allowed subchannel.

An SU beamformer may solicit full-bandwidth SU feedback from an SU beamformee in an HE non-TB sounding sequence. An SU beamformer shall not solicit partial-bandwidth SU feedback in an HE non-TB sounding sequence. An SU beamformer may solicit partial-bandwidth or full-bandwidth SU feedback from an SU beamformee in an HE TB sounding sequence if the SU beamformee indicates support by setting the Triggered SU Beamforming Feedback subfield in the HE PHY Capabilities Information field in the HE Capabilities element it transmits to 1.

An MU beamformer may solicit full-bandwidth MU feedback from an MU beamformee in an HE TB sounding sequence. An MU beamformer may solicit partial-bandwidth MU feedback from an MU beamformee in an HE TB sounding sequence if the MU beamformee indicates support by setting the Triggered MU Beamforming Partial BW Feedback subfield to 1. An MU beamformer shall not solicit MU feedback in an HE non-TB sounding sequence.

An MU beamformer may solicit full-bandwidth or partial-bandwidth CQI feedback from an MU beamformee in an HE TB sounding sequence if the MU beamformee indicates support by setting the Triggered CQI Beamforming Feedback subfield to 1.

An MU beamformer may solicit full-bandwidth CQI feedback from an MU beamformee in an HE non-TB sounding sequence if the MU beamformee indicates support by setting the Non-Triggered CQI Beamforming Feedback subfield to 1.

An HE beamformer may solicit punctured SU feedback from an HE beamformee in an HE TB sounding sequence if the HE beamformee indicates support for punctured sounding by setting the Punctured Sounding Support subfield in the HE Capabilities elements it transmits to 1. An HE beamformer shall indicate punctured subchannels in the HE sounding NDP of an HE TB sounding sequence by setting the appropriate bits of the Disallowed Subchannel Bitmap subfield of the STA Info field with the AID11 subfield set to 2047 in the preceding HE NDP Announcement frame. An SU beamformer that indicates punctured subchannels in an HE NDP Announcement frame in an HE TB sounding sequence shall set the TXVECTOR parameter INACTIVE_SUBCHANNELS of the non-HT duplicate PPDU carrying the HE NDP Announcement frame and the HE sounding NDP as described in 26.11.7.

An SU beamformee that supports punctured sounding shall generate feedback corresponding to the subchannels indicated in the STA Info field addressed to it in an HE NDP Announcement frame, but excluding subcarriers that are disallowed according to the value of the Disallowed Subchannel Bitmap subfield in that frame.

An HE beamformer shall not send an HE NDP Announcement frame that initiates an HE TB sounding sequence with a STA Info field identifying an HE beamformee if the STA Info field and the PHY Capabilities Information field in the HE Capabilities element most recently received from the HE beamformee meet any of the following conditions (see Table 9-29a):

- The Feedback Type And Ng subfield in the STA Info field indicates SU and $N_g = 16$, and the $N_g = 16$ SU Feedback subfield in the HE PHY Capabilities Information field is 0.
- The Feedback Type And Ng subfield in the STA Info field indicates MU and $N_g = 16$, and the $N_g = 16$ MU Feedback subfield in the HE PHY Capabilities Information field is 0.
- The Feedback Type And Ng subfield in the STA Info field indicates SU, the Codebook Size subfield indicates codebook resolution $(\phi, \psi) = \{4, 2\}$, and the Codebook Size $(\phi, \psi) = \{4, 2\}$ SU Feedback subfield in the HE PHY Capabilities Information field is 0.
- The Feedback Type And Ng subfield in the STA Info field indicates MU, the Codebook Size subfield in the STA Info field indicates codebook resolution $(\phi, \psi) = \{7, 5\}$, and the Codebook Size $(\phi, \psi) = \{7, 5\}$ MU Feedback subfield in the HE PHY Capabilities Information field is 0.
- The Feedback Type And Ng and Codebook Size subfields in the STA Info field indicate CQI, and the Triggered CQI Beamforming Feedback subfield in the HE PHY Capabilities Information field is 0.
- The Feedback Type And Ng subfield in the STA Info field indicates MU, the Partial BW Info subfield in the STA Info field indicates partial bandwidth, and the Triggered MU Beamforming Partial BW Feedback subfield in the HE PHY Capabilities Information field is 0.
- The Feedback Type And Ng subfield indicates SU, and the Triggered SU Beamforming Feedback subfield in the HE PHY Capabilities Information field is 0.

An HE beamformee indicates the maximum number of HE-LTF symbols it can receive in a 20 MHz, 40 MHz, or 80 MHz HE sounding NDP in the Beamformee STS ≤ 80 MHz subfield in the PHY Capabilities Information field in the HE Capabilities element it transmits.

An HE beamformee shall set the Beamformee STS ≤ 80 MHz subfield to indicate a maximum number of HE-LTF symbols of 4 or greater.

An HE beamformee that supports 160 MHz or 80+80 MHz channel widths indicates the maximum number of HE-LTF symbols it can receive in a 160 MHz or 80+80 MHz HE sounding NDP in the Beamformee STS > 80 MHz subfield in the PHY Capabilities Information field in the HE Capabilities element it transmits.

An HE beamformer shall not transmit a 20 MHz, 40 MHz, or 80 MHz HE sounding NDP with a TXVECTOR parameter NUM_STS that is greater than the maximum number of HE-LTF symbols indicated in the Beamformee STS $\leq$ 80 MHz subfield of any STA identified by a STA Info field in the preceding HE NDP Announcement frame.

An HE beamformer shall not transmit a 160 MHz or 80+80 MHz HE sounding NDP with a TXVECTOR parameter NUM_STS that is greater than the maximum number of HE-LTF symbols indicated in the Beamformee STS > 80 MHz subfield of any STA identified by a STA Info field in the preceding HE NDP Announcement frame.

An HE beamformer indicates the maximum number of HE-LTF symbols it might transmit in a 20 MHz, 40 MHz, or 80 MHz HE sounding NDP in the Number Of Sounding Dimensions $\leq$ 80 MHz subfield.

An HE beamformer indicates the maximum number of HE-LTF symbols it might transmit in an 80+80 MHz or 160 MHz HE sounding NDP in the Number Of Sounding Dimensions > 80 MHz subfield.

An HE beamformer shall not transmit a 20 MHz, 40 MHz, or 80 MHz HE sounding NDP where the number of HE-LTF symbols exceeds the value indicated in the Number Of Sounding Dimensions $\leq$ 80 MHz subfield.

An HE beamformer shall not transmit an 80+80 MHz or 160 MHz HE sounding NDP where the number of HE-LTF symbols exceeds the value indicated in the Number Of Sounding Dimensions > 80 MHz subfield.

26.7.3 Rules for HE sounding protocol sequences

An HE non-TB sounding sequence is initiated by an HE beamformer with an individually addressed HE NDP Announcement frame comprising exactly one STA Info field, followed after SIFS by an HE sounding NDP. The HE beamformee responds after SIFS with an HE Compressed Beamforming/CQI frame.

The AID11 subfield of the STA Info field shall be set to the AID of the STA identified by the RA field of the HE NDP Announcement frame, or to 0 if the STA identified by the RA field is a mesh STA, AP, or IBSS STA.

An example of an HE non-TB sounding sequence with a single HE beamformee is shown in Figure 26-7.

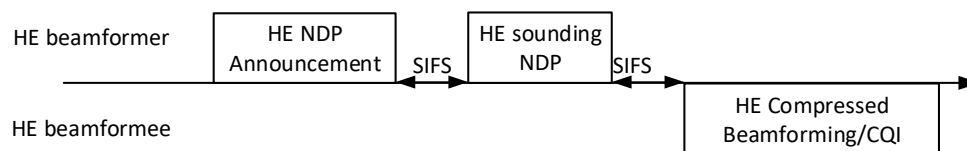


Figure 26-7—Example of HE non-TB sounding

An HE beamformer that initiates the HE non-TB sounding sequence shall transmit the HE NDP Announcement frame with a single STA Info field, and the value in the AID11 field of that STA Info field shall be other than 2047 and set to the AID of the STA identified by the RA field or to 0 if the STA identified by the RA field is a mesh STA, AP, or IBSS member STA.

An HE beamformer may initiate an HE non-TB sounding sequence with an HE beamformee to solicit SU feedback over full bandwidth.

An HE beamformer shall not initiate an HE non-TB sounding sequence with an HE NDP Announcement frame that has a Partial BW Info subfield that indicates less than full bandwidth (see Table 26-4).

An HE TB sounding sequence is initiated by an HE beamformer with a broadcast HE NDP Announcement frame with two or more STA Info fields, followed after a SIFS by an HE sounding NDP, followed after a SIFS by a BFRP Trigger frame. Each HE beamformee responds after a SIFS with an HE Compressed Beamforming/CQI frame.

An example of an HE TB sounding sequence with more than one HE beamformee is shown in Figure 26-8.

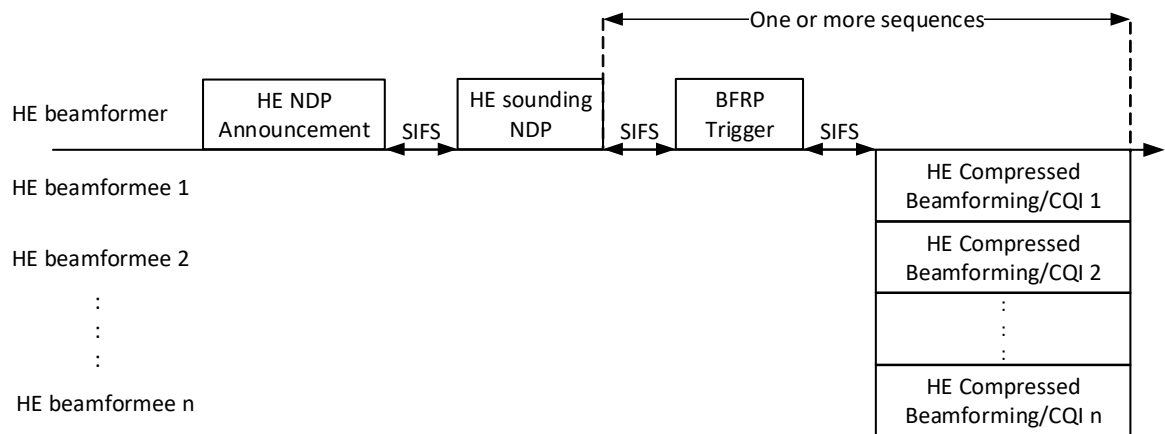


Figure 26-8—Example of HE TB sounding

An HE beamformer that initiates an HE TB sounding sequence shall transmit the HE NDP Announcement frame with two or more STA Info fields and the RA field set to the broadcast address.

An HE beamformer may initiate an HE TB sounding sequence to solicit MU feedback over full bandwidth.

An HE beamformer may initiate an HE TB sounding sequence to solicit a feedback variant only if the feedback variant is computed based on parameters supported by the HE beamformee; otherwise, the HE beamformer shall not solicit a feedback variant computed based on parameters not supported by the HE beamformee (see 26.7.2).

An HE AP with dot11MultiBSSIDImplemented equal to true shall not send an HE NDP Announcement frame with the TA field set to the transmitted BSSID to a non-AP STA that is associated with an AP corresponding to a nontransmitted BSSID in the multiple BSSID set, unless the AP has received from the non-AP STA an HE Capabilities element with the Rx Control Frame To MultiBSS subfield in the HE MAC Capabilities Information field equal to 1. An AP with dot11MultiBSSIDImplemented equal to true shall not send an individually addressed HE NDP Announcement frame with the TA field set to the transmitted BSSID to a non-AP STA associated with an AP corresponding to a nontransmitted BSSID.

An AP that transmits an HE NDP Announcement frame identifying HE STAs shall set the TA field of the frame to the MAC address of the AP, unless `dot11MultiBSSIDImplemented` is true and the HE NDP Announcement frame identifies STAs from at least two different BSSs of the multiple BSSID set. In this case, the AP shall set the TA field of the frame to the transmitted BSSID. If the HE NDP Announcement frame is transmitted in a non-HT duplicate PPDU, then the TA field of the HE NDP Announcement frame is a bandwidth signaling TA (see 10.6.6.6).

An HE beamformer that transmits an HE NDP Announcement frame to an HE beamformee that is an AP, TDLS peer STA, mesh STA, or IBSS STA shall include one STA Info field in the HE NDP Announcement frame and shall set the AID11 field in the STA Info field of the frame to 0.

An HE beamformer that is an AP and that transmits an HE NDP Announcement frame to one or more HE beamformees shall set the AID11 field in the STA Info field identifying a non-AP STA to the 11 LSBs of the AID of the non-AP STA.

An HE NDP Announcement frame shall not include multiple STA Info fields that have the same value in the AID11 subfield.

An HE beamformer that transmits an HE NDP Announcement frame initiating an HE TB sounding sequence may include a STA Info field with an AID11 subfield value of 2047 to indicate disallowed subchannels during punctured channel operation. When present, the STA Info field with AID11 value of 2047 shall be the first STA Info field in the frame. An HE beamformer that transmits an HE NDP Announcement frame shall not include more than one STA Info field with an AID11 subfield value of 2047. An HE beamformer that transmits an HE NDP Announcement frame initiating a HE non-TB sounding sequence shall not include a STA Info field with an AID11 subfield value of 2047.

In an HE TB sounding sequence, a STA Info field in the HE NDP Announcement frame that solicits SU or MU feedback indicates the subcarrier grouping (N_g), codebook size, and number of columns (N_c) to be used by the HE beamformee identified by the STA Info field for the generation of the SU or MU feedback.

In an HE non-TB sounding sequence where the STA Info field in the HE NDP Announcement frame solicits SU feedback, the N_g , codebook size, and the N_c used for the generation of the SU feedback are determined by the HE beamformee.

In an HE TB sounding sequence, a STA Info field in the HE NDP Announcement frame that solicits CQI feedback indicates the N_c to be used by the HE beamformee identified by the STA Info field for the generation of the CQI feedback.

In an HE non-TB sounding sequence where the STA Info field in the HE NDP Announcement frame solicits CQI feedback, the N_c used for the generation of the CQI feedback is determined by the HE beamformee.

An HE beamformer that has initiated an HE TB sounding sequence may send another BFRP Trigger frame in the same TXOP as shown in Figure 26-8. The HE beamformer uses the additional BFRP Trigger frames to solicit HE compressed beamforming/CQI reports from HE beamformees not addressed in a previous BFRP Trigger frame or to solicit retransmission of an HE compressed beamforming/CQI report. An HE beamformer shall not transmit a BFRP Trigger frame that identifies a STA identified in the HE NDP Announcement frame of an HE TB sounding sequence, unless it is in the same TXOP as the HE TB sounding sequence.

An HE beamformer that transmits an HE NDP Announcement frame as part of an HE TB sounding sequence shall set the N_c subfield of the STA Info field to indicate a value less than or equal to the minimum of the following:

- The maximum number of supported spatial streams according to the corresponding HE beamformee's Rx HE-MCS Map ≤ 80 MHz, Rx HE-MCS Map 160 MHz, and Rx HE-MCS Map 80+80 MHz subfields in the Supported HE-MCS And NSS Set field in the HE Capabilities element sent by the HE beamformee.
- The maximum number of supported spatial streams according to the Rx NSS subfield value in the most recently received Operating Mode Notification frame, Operating Mode Notification element with the Rx NSS Type subfield equal to 0, or OM Control subfield sent by the corresponding HE beamformee (see 26.9).
- The maximum N_c indicated by the Max N_c subfield in the HE PHY Capabilities Information field in the HE Capabilities element sent by the HE beamformee.

An HE beamformer that transmits an HE NDP Announcement frame shall set the RU Start Index and RU End Index subfields in a STA Info field to indicate the starting 26-tone RU and the ending 26-tone RU, respectively, of the solicited HE compressed beamforming/CQI report (see 9.3.1.19). For preamble punctured sounding, the RU Start Index and RU End Index correspond to the bandwidth before puncturing, and the Disallowed Subchannel Bitmap subfield is used to indicate the subcarriers that are punctured in the HE sounding NDP and in the solicited feedback.

NOTE 1—In order to maximize the channel estimate quality after interpolation/extrapolation resulting from grouping, the subcarriers for which feedback is requested using the RU Start Index and RU End Index subfields do not necessarily lie within the actual RU as defined in Table 27-7, Table 27-8, and Table 27-9. Rather, they are defined in Table 9-91c and Table 9-91d.

The bandwidth of the HE NDP Announcement frame is obtained from the RXVECTOR parameter CH_BANDWIDTH of the HE NDP Announcement frame if received in an HE, VHT, or HT PPDU; is obtained from the RXVECTOR parameter CH_BANDWIDTH_IN_NON_HT if the HE NDP Announcement frame is received in a non-HT duplicate PPDU; and is 20 MHz if the HE NDP Announcement frame is received in a non-HT PPDU.

In an HE non-TB sounding sequence, an HE beamformer shall solicit full-bandwidth feedback. In an HE TB sounding sequence, an HE beamformer shall solicit full-bandwidth feedback in a STA Info field identifying an HE beamformee that has not indicated support for partial-bandwidth feedback. In an HE TB sounding sequence, an HE beamformer may solicit full-bandwidth or partial-bandwidth feedback in a STA Info field identifying an HE beamformee that has indicated support for partial-bandwidth feedback (see 26.7.2).

For 80+80 MHz, feedback is not requested for the gap between the 80 MHz segments.

The HE beamformer shall set the TXVECTOR parameter CH_BANDWIDTH or CH_BANDWIDTH_IN_NON_HT, the RU Start Index field, and the RU End Index field of the HE NDP Announcement frame, depending on the operating channel width and partial BW support of the HE beamformee, as defined in Table 26-4. The bandwidth of the HE NDP Announcement frame is determined before applying puncturing based on disallowed subchannels.

Table 26-4—Settings for BW, RU Start Index, and RU End Index fields in HE NDP Announcement frame

Partial bandwidth supported by HE beamformer	Operating channel width of the HE beamformee (MHz)	Bandwidth of HE NDP Announcement frame	RU Start Index field	RU End Index field
No	20, 40, 80, 160/80+80	20 MHz	0	8
Yes	20, 40, 80, 160/80+80		0–8	0–8
No	40, 80, 160/80+80	40 MHz	0	17
Yes	40, 80, 160/80+80		0–17	0–17
No	80, 160/80+80	80 MHz	0	36
Yes	80, 160/80+80		0–36	0–36
No	160/80+80	160 MHz or 80+80 MHz	0	73
Yes	160/80+80		0–73	0–73

NOTE 1—The value of the RU Start Index field is always less than or equal to the value of the RU End Index field.

NOTE 2—Partial BW feedback can be solicited only with an HE TB sounding sequence and cannot be solicited with an HE non-TB sounding sequence.

The HE beamformer shall use the lowest 26-tone RU, which is the lower bound of the starting 26-tone in the RU Start Index subfield of a STA Info field that is equal to the maximum of the following:

- The minimum 26-tone RU located within the channel width indicated in the VHT Operation Information field of either the HE Operation element or the VHT Operation element, whichever is present, and within the channel width indicated in the HT Operation element.
- The minimum 26-tone RU located within the channel width indicated in the most recently received Operating Mode Notification frame, Operating Mode Notification element with the Rx NSS Type subfield equal to 0, or OM Control subfield sent by the corresponding HE beamformee (see 26.9).

The HE beamformer shall use the highest 26-tone RU, which is the upper bound of the ending 26-tone RU in the RU End Index subfield of a STA Info field that is equal to the minimum of the following:

- The maximum 26-tone RU located within the channel width indicated in the VHT Operation Information field of either the HE Operation element or the VHT Operation element, whichever is present, and within the channel width indicated in the HT Operation element.
- The maximum 26-tone RU located within the channel width indicated in the most recently received Operating Mode Notification frame, Operating Mode Notification element with the Rx NSS Type subfield equal to 0, or OM Control field sent by the corresponding HE beamformee (see 26.9).

In an HE non-TB sounding sequence soliciting SU feedback, B26 (in the Feedback Type And Ng subfield), the Codebook Size subfield, and the Nc subfield in the STA Info field of the HE NDP Announcement frame are reserved.

In an HE non-TB sounding sequence soliciting CQI feedback, the Nc subfield in an HE NDP Announcement frame is reserved.

An HE beamformee that receives an HE NDP Announcement frame soliciting SU feedback as part of an HE non-TB sounding sequence shall generate an HE compressed beamforming/CQI report for SU feedback with N_c in the range 1 to 8, $N_g = 4$ or $N_g = 16$, and codebook size $(\phi, \psi) = \{4, 2\}$ or $(\phi, \psi) = \{6, 4\}$. The HE beamformee shall transmit the HE compressed beamforming/CQI report a SIFS after the HE sounding NDP.

An HE beamformee that receives an HE NDP Announcement frame soliciting CQI feedback as part of an HE non-TB sounding sequence shall generate an HE compressed beamforming/CQI report for CQI feedback with N_c determined by the HE beamformee.

An HE beamformee that receives an HE NDP Announcement frame soliciting CQI feedback as part of an HE TB sounding sequence shall generate an HE compressed beamforming/CQI report for CQI feedback with N_c determined by the HE beamformer.

An HE beamformee that receives an HE NDP Announcement frame from an HE beamformer with which it is associated and that contains the HE beamformee's MAC address in the RA field (indicating a non-TB sounding sequence) and also receives an HE sounding NDP a SIFS after the HE NDP Announcement frame shall transmit its HE compressed beamforming/CQI report a SIFS after the HE sounding NDP. The TXVECTOR parameter CH_BANDWIDTH for the PPDU containing the HE compressed beamforming/CQI report shall be set to indicate a bandwidth not wider than that indicated by the RXVECTOR parameter CH_BANDWIDTH of the HE sounding NDP.

An HE beamformee that receives an HE NDP Announcement frame as part of an HE TB sounding sequence with a STA Info field identifying it soliciting SU or MU feedback shall generate an HE compressed beamforming/CQI report using the feedback type, N_g , codebook size, and N_c indicated in the STA Info field. If the HE beamformee then receives a BFRP Trigger frame with a matching STA Info field, the HE beamformee transmits an HE TB PPDU containing the HE compressed beamforming/CQI report following the rules defined in 26.5.2.3. If the HE NDP Announcement frame has the TA field set to the transmitted BSSID and the HE beamformee is a non-AP STA associated with an AP corresponding to a nontransmitted BSSID that supports receiving Control frames with TA field set to the transmitted BSSID, then the HE compressed beamforming/CQI report sent in response shall have the RA field set to as defined in 26.5.2.3.5.

NOTE 2—A non-AP HE beamformee that transmits an OM Control subfield with the UL MU Disable field set to 1 does not respond to BFRP Trigger frames (see 26.9).

An HE beamformee that is a non-AP STA that transmits an HE Compressed Beamforming/CQI Report shall set the RU Start Index and RU End Index subfields of the HE MIMO Control field to indicate the range of tones for which compressed beamforming/CQI information is provided. If the HE NDP Announcement frame that solicited the feedback includes a Disallowed Subchannel Bitmap field with a nonzero value, then a beamformee that indicates support for punctured sounding by setting the Punctured Sounding Support subfield in the HE Capabilities elements that it transmits to 1 shall include a Disallowed Subchannel Bitmap subfield in the solicited feedback with the same value as the Disallowed Subchannel Bitmap subfield of the HE NDP Announcement frame that solicited the feedback to indicate subcarriers for which feedback information is not provided from within the range of subcarriers indicated by the RU Start Index and RU End Index subfields.

An HE beamformee that transmits HE compressed beamforming feedback shall include neither the HE Compressed Beamforming Report information nor the HE MU Exclusive Beamforming Report information if the transmission duration of the PPDU carrying the HE Compressed Beamforming Report information and any HE MU Exclusive Beamforming Report information would exceed the maximum PPDU duration.

The Sounding Dialog Token Number field in the HE MIMO Control field shall be set to the same value as the Sounding Dialog Token Number field in the corresponding HE NDP Announcement frame.

An HE beamformer that sends a BFRP Trigger frame shall set the Feedback Segment Retransmission Bitmap fields of the BFRP Trigger frame to all 1s, except when the HE beamformer intends to solicit the retransmission of segmented feedback as defined in 26.7.4.

NOTE 3—The BFRP Trigger frame contains one or more User Info fields, each of which identifies an HE beamformee.

The SNR per subcarrier computation is recommended to be done on at least 4 subcarriers in a 26-tone RU.

26.7.4 Rules for generating segmented feedback

If the HE compressed beamforming/CQI report solicited by the HE beamformer would result in an HE Compressed Beamforming/CQI frame that exceeds 11 454 octets in length, then the HE compressed beamforming/CQI report shall be split into up to 8 feedback segments. Each feedback segment shall be included in a separate HE Compressed Beamforming/CQI frame and shall contain successive portions of the HE compressed beamforming/CQI report. Each feedback segment shall be of equal length, except the last feedback segment that may be smaller. Each HE Compressed Beamforming/CQI frame that includes a feedback segment that is not the last feedback segment shall have a length of 11 454 octets. Each feedback segment is identified by the value of the Remaining Feedback Segments subfield and the First Feedback Segment subfield in the HE MIMO Control field as defined in 9.4.1.64; the other nonreserved subfields of the HE MIMO Control field shall be the same for all feedback segments. All feedback segments shall be sent in a single A-MPDU contained in a PPDU and shall be included in the A-MPDU in the descending order of the Remaining Feedback Segments subfield values.

An HE beamformer that sends a BFRP Trigger frame, in its first attempt to retrieve an HE compressed beamforming/CQI report from an HE beamformee, shall solicit all possible feedback segments by setting all of the bits in the Feedback Segment Retransmission Bitmap subfield to 1 in the User Info field identifying the HE beamformee.

An HE beamformer that fails to receive some or all of the feedback segments of the HE compressed beamforming/CQI report from the HE beamformee may solicit the selective retransmission of missing feedback segments by sending a BFRP Trigger frame that indicates in the Feedback Segment Retransmission Bitmap subfield of the User Info field identifying the HE beamformee the list of feedback segments solicited for retransmission (see 9.3.1.22.3).

NOTE 1—In an HE non-TB sounding sequence, if the HE beamformer does not receive all feedback segments from the HE beamformee, the HE beamformer cannot use a BFRP Trigger frame to request retransmission of the feedback segments. In this case the HE beamformee can only repeat the entire non-TB sounding sequence.

An HE beamformer that fails to receive the first feedback segment (identified by the First Feedback Segment field set to 1), may solicit the selective retransmission of the missing feedback segments assuming the HE compressed beamforming/CQI report is split into 8 feedback segments. The HE beamformer may also solicit the retransmission of all feedback segments by setting all of the bits in the Feedback Segment Retransmission Bitmap subfield to 1 in the User Info field identifying the HE beamformee.

An HE beamformee that transmits an HE compressed beamforming/CQI report including the HE Compressed Beamforming Report information and any HE MU Exclusive Beamforming Report information in response to a BFRP Trigger frame shall either transmit only the feedback segments indicated in the Feedback Segment Retransmission Bitmap field in the User Info field of the BFRP Trigger frame identifying the HE beamformee or transmit all the feedback segments available at the HE beamformee, excluding the feedback segments that do not exist at the HE beamformee.

NOTE 2—If an HE beamformer solicits the missing feedback segments from a beamformee and does not receive a response from the beamformee, the HE beamformer might either initiate an HE TB sounding sequence or transmit an additional BFRP Trigger frame to the HE beamformee.

26.7.5 HE sounding NDP transmission

The TXVECTOR parameters for an HE sounding NDP shall be set as follows:

- FORMAT is set to HE_SU.
- APEP_LENGTH is set to 0.
- HE_LTF_TYPE is set to either 2xHE-LTF or 4xHE-LTF.
- If HE_LTF_TYPE is 2xHE-LTF, then GI_TYPE is set to either 0u8s_GI or 1u6s_GI.
- If HE_LTF_TYPE is 4xHE-LTF, then GI_TYPE is set to 3u2s_GI.
- NUM_STS indicates two or more space-time streams if the Feedback Type field in the HE MIMO Control field of the preceding HE NDP Announcement frame indicates either SU or MU, or one or more space-time streams if the Feedback Type field in the HE MIMO Control field of the preceding HE NDP Announcement frame indicates CQI.
- CH_BANDWIDTH is set to the same value as the TXVECTOR parameter CH_BANDWIDTH in the preceding HE NDP Announcement frame.
- INACTIVE_SUBCHANNELS is set to the value of the Disallowed Subchannel Bitmap subfield of the STA Info field with the AID11 subfield set to 2047 in the preceding HE NDP Announcement frame.
- SPATIAL_REUSE is set to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED (see 26.11.6).
- BSS_COLOR is set to the value indicated in the BSS Color subfield of the HE Operation element received or transmitted by the HE AP.
- TXOP_DURATION set to either 127 or a value defined in Equation (26-3).

$$\max(\min(8 \left\lfloor \frac{D_{\text{HE_NDPA}} - \text{SIFS} - \text{TXTIME}}{8} \right\rfloor, 504), \min(128 \left\lfloor \frac{D_{\text{HE_NDPA}} - \text{SIFS} - \text{TXTIME}}{128} \right\rfloor, 8448)) \quad (26-3)$$

where

$D_{\text{HE_NDPA}}$ is the value of the Duration/ID field in the MAC header of the preceding HE NDP Announcement frame

TXTIME is the transmission time of an HE sounding NDP defined in Equation (27-136)

The Supported HE-MCS and NSS Set field in the HE Capabilities element transmitted by the transmitter and the receiver of the HE sounding NDP do not affect the values used for the NUM_STS parameter for the TXVECTOR of an HE sounding NDP.

The destination of an HE sounding NDP is equal to the RA of the immediately preceding HE NDP Announcement frame.

The source of an HE sounding NDP is equal to the TA of the immediately preceding HE NDP Announcement frame.

26.8 TWT operation

26.8.1 General

Target wake time (TWT) allows an AP to manage activity in the BSS in order to minimize contention between STAs and to reduce the required amount of time that a STA utilizing a power management mode needs to be awake. This is achieved by allocating STAs to operate at nonoverlapping times and/or frequencies, and concentrate the frame exchanges in predefined service periods.

An HE STA negotiates individual TWT agreements, as defined in 10.47, subject to the additional rules and restrictions that are defined in 26.8.2 and 26.8.7. A non-AP HE STA establishes membership in broadcast TWT schedules, as defined in 26.8.3, which are used as defined in 26.8.3.3, 26.14.2, and 26.14.3. An HE AP delivers broadcast TWT parameter sets to non-AP HE STAs as described in 26.8.3.2, 26.14.2, and 26.14.3.

A STA does not need to be aware of the values of TWT parameters of the TWT agreements of other STAs in the BSS of the STA or of TWT agreements of STAs in other BSSs. A STA does not need to be aware that a TWT service period (SP) is used to exchange frames with other STAs. Frames transmitted during a TWT SP are carried in any PPDU format supported by the pair of STAs that have established the TWT agreement corresponding to that TWT SP, including HE MU PPDU, HE TB PPDU, etc.

An HE STA with `dot11TWTOptionImplemented` equal to true shall set the following:

- The TWT Requester Support subfield to 1 in the HE Capabilities element that it transmits if it supports operating in the role of a TWT requesting STA; otherwise, to 0.
- The TWT Responder Support subfield to 1 in the HE Capabilities elements that it transmits if it supports operating in the role of a TWT responding STA; otherwise, to 0.
- The Broadcast TWT Support subfield to 1 in the HE Capabilities element that it transmits if it supports operating in the role of a TWT scheduled STA or in the role of a TWT scheduling AP; otherwise, to 0.

An HE AP shall set the TWT Responder Support subfield of the Extended Capabilities element and HE Capabilities element to 1.

An HE AP may request TWT participation by all associated STAs that have declared support for TWT. A non-AP STA declares support for the role of TWT requesting STA by setting the TWT Requester Support subfield in the Extended Capabilities element or in the HE Capabilities element to 1 and declares support for the role of TWT scheduled STA by setting the Broadcast TWT Support subfield in the HE Capabilities element to 1. The HE AP makes the request for TWT participation by setting the TWT Required subfield to 1 in HE Operation elements it transmits. A STA that supports TWT and that has received an HE Operation element with the TWT Required subfield equal to 1 from the HE AP with which it is associated shall either negotiate individual TWT agreements, as defined in 26.8.2, or participate in broadcast TWT operation, as defined in 26.8.3.

NOTE—The AP sets the TWT Required subfield to 1 if it is not available outside TWT SPs (see 26.8.2 and 26.8.3). The AP might not be available outside TWT SPs if it sets the Responder PM Mode subfield to 1 (see 10.47.7).

26.8.2 Individual TWT agreements

An HE STA may negotiate individual TWT agreements with another HE STA as defined in 10.47.1, except that the STA

- May set the Responder PM Mode subfield to 1 if it is a TWT responding STA that intends to go to doze state outside of TWT SPs.
 - If the TWT responding STA is an AP then it may set the Responder PM Mode subfield to 1 only if all non-AP STAs that are associated to it indicate support of TWT and the AP has set the TWT Required subfield to 1 in the HE Operation element it transmits; otherwise, it shall set the Responder PM Mode subfield to 0.
 - An AP that sets the Responder PM Mode subfield to 1 follows the rules defined in 10.47.7.
- Shall set the Implicit subfield to 1 and the NDP Paging Indicator subfield to 0 in all TWT elements that it transmits during the TWT setup.

- May set the Trigger subfield to 1 in the TWT element it transmits during the TWT setup to negotiate a trigger-enabled TWT.
 - A successful TWT agreement whose Trigger subfield in the TWT response sent by the AP is 1 is a trigger-enabled TWT; otherwise, it is not a trigger-enabled TWT.
- Shall set the TWT Channel subfield in the TWT element it transmits to 0, unless the HE STA sets up a subchannel selective transmission operation as defined in 26.8.7.
- May set the TWT Protection field to 1 to indicate that TXOPs within the TWT SPs shall be initiated with a NAV protection mechanism, such as (MU) RTS/CTS or CTS-to-self frame; otherwise, it shall set it to 0.
- Shall not use the RAW mechanism for protection of TWT SPs.

An HE STA that successfully sets up a TWT agreement with another HE STA shall follow the rules defined in 10.47.1 and 10.47.4, except that all the additional rules defined in 26.8 supersede all the respective rules defined in 10.47.1 and 10.47.4. A TWT or TWT SP that is set up under an implicit TWT agreement is an implicit TWT or implicit TWT SP, respectively (see 10.47.1). A TWT or TWT SP that is set up under a trigger-enabled TWT agreement is a trigger-enabled TWT or trigger-enabled TWT SP, respectively. The TWT responding STA of a trigger-enabled TWT agreement is an HE AP.

An example of individual TWT operation is shown in Figure 26-9, where the AP is the TWT responding STA and STA 1 and STA 2 are the TWT requesting STAs.

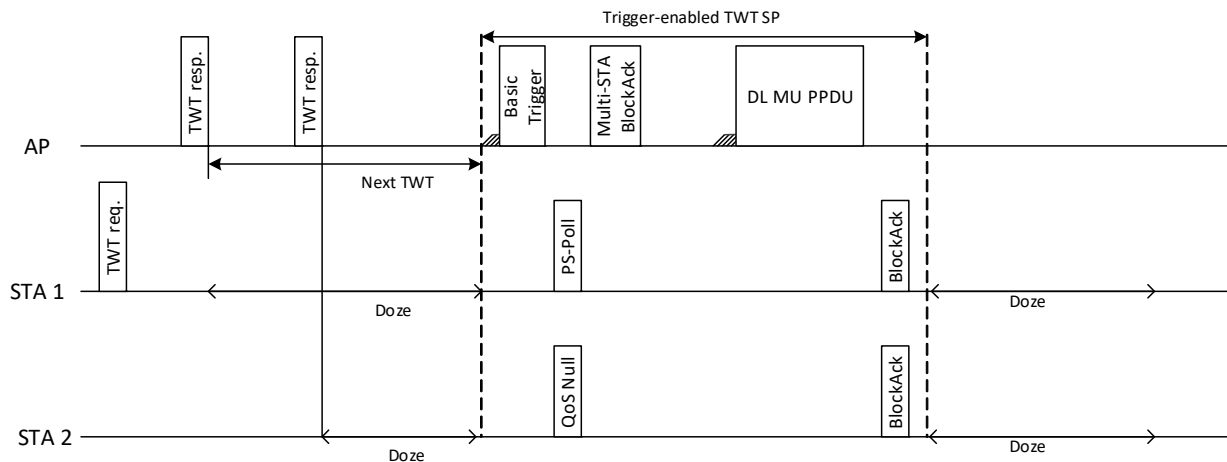


Figure 26-9—Example of individual TWT operation

In this example, STA 1 sends a TWT request to the TWT responding STA to setup a trigger-enabled TWT agreement. The TWT responding STA accepts the TWT agreement with STA 1 and confirms the acceptance in the TWT response sent to STA 1. Subsequently, the TWT responding STA sends an unsolicited TWT response to STA 2 to set up a trigger-enabled TWT agreement with STA 2. Both these TWT agreements are set up as announced TWTs. During the trigger-enabled TWT SP, the TWT responding STA sends a Basic Trigger frame to which the TWT requesting STAs indicate that they are awake during the TWT SP. STA 1 indicates that it is awake by sending a PS-Poll frame, and STA 2 indicates that it is awake by sending a QoS Null frame in response to the Basic Trigger frame. STA 1 and STA 2 receive their DL BUs in a subsequent exchange with the TWT responding STA and go to doze state outside of this TWT SP.

An HE STA may execute the individual TWT setup exchanges defined in Table 26-5 in addition to the exchanges defined in 10.47. An HE STA that intends to set up an individual TWT shall set the Negotiation Type subfield to 0 as defined in 10.47 or as defined in Table 26-5. The HE STA may respond to the TWT request with a TWT response that has the Negotiation Type subfield equal to 3 as indicated in Table 26-5 to provide recommended broadcast TWT schedules for the requesting STA.

Table 26-5—TWT setup exchange for unsolicited TWT and recommended broadcast TWT switch

TWT Setup Command field in an initiating frame	TWT Setup Command field in a response frame	TWT condition after the completion of the exchange
Request TWT, Suggest TWT, or Demand TWT with Broadcast subfield = 0	Accept TWT with Broadcast subfield = 1	This response is not allowed.
Request TWT, Suggest TWT, or Demand TWT with Broadcast subfield = 0	Dictate TWT with Broadcast subfield = 1	No individual TWT agreement exists with the associated TWT Flow identifier. One or more broadcast TWT schedules exist that use the TWT parameters identified in the response frame including a Broadcast TWT ID. The broadcast TWT schedules are not necessarily newly created. The responding STA will not create any new individual TWT agreement with the requester at this time. The STA transmitting the initiating frame is not a member of the broadcast TWT; however, the STA is recommended to join any of the broadcast TWT schedules.
Accept TWT with Broadcast subfield set to 0	No frame transmitted	The STA receiving this frame now has an individual TWT agreement with the transmitter of the frame where the parameters of the individual TWT agreement are identified by the initiating frame.
Alternate TWT or Dictate TWT with Broadcast subfield = 0	No frame transmitted	The STA receiving this frame is not, through the receipt of this frame, a member of the TWT identified by the initiating frame but can use the information provided to create a request to set up a TWT in a subsequent initiating frame that it transmits.
NOTE 1—The Negotiation Type subfield in the TWT element contained in these frames is 0 if the Broadcast subfield is 0 and is 3 if the Broadcast subfield is 1. NOTE 2—The initiating frame and response frame settings not listed in the tables in 10.47 or 26.8 are not allowed. The initiating frame is a TWT request if the TWT element carried in the frame has the TWT Request field set to 1; otherwise, it is a TWT response (see Table 9-297). The response frame is a TWT response if the TWT element contained in the frame has the TWT Request field equal to 0.		

An HE STA that successfully sets up an individual TWT agreement and operates in PS mode may listen to Beacon frames, but is exempt from the requirements for receiving Beacon frames as defined in 11.2.3.1. The HE STA follows the rules in 11.2.3 to receive group addressed frames.

NOTE 1—An HE AP sets the bit in the TIM element of the Beacon frame that corresponds to the AID of the TWT requesting STA to 1 to indicate the presence of available buffered BUs for the STA (see 11.2.3.7).

NOTE 2—The TWT responding STA might inform the TWT requesting STA, if it supports TIM Broadcast, of any critical update (as defined in 11.2.3.15) by sending a Management frame to the TWT requesting STA when the STA is in the awake state.

An HE STA may tear down an individual TWT agreement by sending a TWT Teardown frame with the Negotiation Type subfield set to 0. An HE STA may tear down all individual TWT agreements by sending a TWT Teardown frame with the Teardown All TWT field set to 1.

An HE AP may send an unsolicited TWT response with the Trigger subfield equal to 1 to a non-AP HE STA that has set the TWT Requester Support subfield to 1 in the HE Capabilities elements that it transmits to the AP. The TWT response shall have one of these values in the TWT Setup Command field: Accept TWT, Alternate TWT, or Dictate TWT. An unsolicited TWT response with TWT Setup Command field of Alternate TWT or Dictate TWT contains an advisory notification to the recipient of TWT parameters that are likely to be accepted by the AP if the recipient transmits a subsequent TWT request to the AP that includes those TWT parameters. An unsolicited TWT response with the TWT Setup Command field of Accept TWT creates a TWT agreement between the two STAs. A STA that receives an unsolicited TWT response with the TWT Setup Command field of Accept TWT may transmit a TWT Teardown frame to delete the unsolicited individual TWT agreement.

NOTE 3—The HE AP might send an unsolicited TWT response to a non-AP HE STA with a TWT Flow Identifier that corresponds to an existing TWT agreement. The unsolicited TWT response with TWT Setup Command field of Accept TWT will indicate new TWT parameters that are different from the previously negotiated TWT parameters for that TWT agreement.

An HE STA shall not transmit BAT, TACK, or STACK frames, which are allowed in 10.47.2.

A TWT requesting STA should not transmit frames to the TWT responding STA outside of negotiated TWT SPs and should not transmit frames that are not contained within HE TB PPDU to the TWT responding STA within trigger-enabled TWT SPs.

NOTE 4—The TWT requesting STA decides which frames to transmit within or outside a TWT SP; and while it is recommended that the TWT requesting STA not transmit using EDCA within or outside TWT SPs, the TWT requesting STA might still do so. If the STA decides to transmit, then the STA might contend for access to the medium as defined in 10.23.2 and in 26.2.7.

The TWT responding STA of a trigger-enabled TWT agreement shall schedule for transmission a Trigger frame for the TWT requesting STA, as described in 26.5.2, within each TWT SP for that TWT agreement, except that the Trigger frame may be replaced by a frame carrying a TRS Control subfield, provided that the frame is carried in a DL MU PPDU and the AP allocates enough resources in the HE TB PPDU for the STA to at least deliver its BSRs in response to the soliciting DL MU PPDU. The TWT responding STA should solicit buffer status reports from the TWT requesting STA at the start of the TWT SP following the procedure described in 26.5.5 or as described in 26.5.7. The TWT responding STA that intends to schedule for transmission additional Trigger frames during a trigger-enabled TWT SP shall set the More TF subfield in the Common Info field of the Trigger frame to 1 to indicate that it will schedule for transmission another Trigger frame within the same TWT SP. The TWT responding STA shall set the More TF subfield to 0 when the Trigger frame is the last scheduled Trigger frame of the TWT SP or when the Trigger frame is scheduled for transmission outside of a TWT SP.

NOTE 5—The TWT responding STA can cancel the transmission of a scheduled Trigger frame if the STA gains access to the wireless medium outside of the TWT SP. The TWT responding STA does not schedule for transmission a Trigger frame for the TWT requesting STA when the TWT agreement is not a trigger-enabled TWT agreement or when the TWT requesting STA has sent an OM Control subfield that has the UL MU Disable subfield equal to 1 (see 26.9).

NOTE 6—If the AP replaces the Trigger frame with a frame carrying a TRS Control field, then it is recommended that the AP allocate enough resources in subsequent Trigger frames sent during the TWT SP so that the STA can send as much as possible of the data reported in the BSR.

A TWT requesting STA transmits an HE TB PPDU as a response to a Trigger frame that identifies it and is sent during a trigger-enabled TWT SP (see 26.5.2). A TWT requesting STA that is in PS mode and is awake shall include a PS-Poll frame or a U-APSD trigger frame in the HE TB PPDU if the TWT is an announced TWT, unless the STA has already transmitted a PS-Poll or U-APSD trigger frame or transmitted any other indication that the STA is in the awake state within that TWT SP or has, previous to the start of the TWT SP but after the end of the most recent TWT SP, indicated to the AP that it is currently in the awake state. The STA may include other frames in the HE TB PPDU when other rules do not prohibit their inclusion (see 26.5.2 and 9.7.3).

NOTE 7—A Trigger frame identifies a TWT requesting STA if it is sent by the AP with which the STA is associated and the frame contains the 12 LSBs of the STA's AID in any of its User Info fields. The Trigger frame can have multiple recipients, each of which is identified by the presence of the 12 LSBs of the recipient's AID in any of its User Info fields (see 26.5.2), and can have in the TA field the MAC address of the AP or the transmitted BSSID under the conditions defined in 26.5.2.2.4.

NOTE 8—Other indications that the STA is in the awake state are the transmission of an HE TB feedback NDP in response to an NFRP Trigger frame (see 26.5.7) or the transmission of a frame that indicates that the STA is in active mode (see 11.2.3.2).

A TWT responding STA that receives a PS-Poll frame or a U-APSD trigger frame or any other indication from a TWT requesting STA in PS mode, during or before an announced TWT SP but after the end of the most recent TWT SP, that the TWT requesting STA is in the awake state during the TWT SP shall follow the rules defined in 11.2.3.6, except that the TWT responding STA should deliver to the TWT requesting STA as many buffered BUs as are available at the TWT responding STA, provided that the BU delivery does not exceed the duration of the TWT SP, the TWT requesting STA has indicated that it is in the awake state for that TWT SP, and the TWT requesting STA has not entered the doze state (see 26.8.4.2 and 26.8.5).

NOTE 9—The indication that the TWT requesting STA is in the awake state for that TWT SP might be a PS-Poll, U-APSD trigger frame, or any frame for which an immediate response is solicited and that follows the rules in 11.2.3.2 but the corresponding immediate response frame is not received by the TWT requesting STA. Other indications that the STA is in the awake state are the transmission of an HE TB feedback NDP in response to an NFRP Trigger frame (see 26.5.7) or the transmission of a frame that indicates that the STA is in active mode (see 11.2.3.2).

A TWT responding STA that sends frames to a TWT requesting STA that is in PS mode during an unannounced TWT SP shall follow the rules defined in 11.2.3.6, except that the TWT responding STA should deliver to the TWT requesting STA as many buffered BUs as are available at the TWT responding STA, provided that the BU delivery does not exceed the duration of the TWT SP and the TWT requesting STA has not entered the doze state (see 26.8.4.2 and 26.8.5).

NOTE 10—The TWT responding STA can deliver the buffered BUs in A-MPDUs sent under a block ack agreement if the TWT is an announced TWT and the TWT requesting STA is awake for that TWT SP, or if the TWT is an unannounced TWT (at the start of which the TWT requesting STA is assumed to already be awake). The buffered BUs can be delivered in multiple PPDUs transmitted within the TWT SP. The TWT responding STA can transmit frames to TWT requesting STA after the end of the TWT SP if the STA is in active mode.

A TWT responding STA may transmit to a TWT requesting STA that is in active mode at any time (see 11.2.3.2). A TWT responding STA may transmit to a TWT requesting STA that is in PS mode and awake outside of a TWT SP following the rules in 11.2.3.6.

26.8.3 Broadcast TWT operation

26.8.3.1 General

A TWT scheduling AP is an HE AP with `dot11TWTOptionActivated` equal to true that sets the Broadcast TWT Support field in the HE Capabilities element it transmits to 1 and that follows the rules for TWT scheduling APs in 26.8.3.2, for power save with UORA and TWT in 26.14.2, and for periodic opportunistic power save (OPS) defined in 26.14.3.

A TWT scheduling AP includes a broadcast TWT element in the Beacon frame as described in 26.8.3.2. An AP corresponding to a nontransmitted BSSID in a multiple BSSID set shall follow the rules in 11.1.3.8.4.

A TWT scheduling AP may include a TWT element with the Negotiation Type subfield equal to 3 in a (Re)Association Response frame or in a TWT setup frame to assign the recipient STA to a broadcast TWT schedule without having received a request from the STA to become a member of the broadcast TWT schedule if that STA has set the Broadcast TWT Support field of HE Capabilities element it transmits to 1.

A non-AP HE STA shall obtain TWT parameter values from the most recently received TWT element carried in a Beacon, Probe Response, or (Re)Association Response frame from its associated AP, unless the non-AP HE STA is associated with an AP corresponding to a nontransmitted BSSID of a multiple BSSID set. In this case, it shall follow the rules in 11.1.3.8.4 to determine the TWT parameter values.

A TWT scheduled STA is a non-AP HE STA that sets the Broadcast TWT Support field in the HE Capabilities element it transmits to 1 and receives a broadcast TWT element transmitted by an HE AP that is a TWT scheduling AP.

A TWT scheduled STA follows the schedule provided by the TWT scheduling AP as described in 26.8.3.3 and, in addition, the rules in 26.14.2 if the STA supports the UORA procedure and the rules in 26.14.3 if the STA supports OPS operation. A TWT scheduled STA can negotiate the wake TBTT and wake interval for Beacon frames it intends to receive as described in 26.8.6 or can join a particular broadcast TWT as described below in this subclause.

An example of broadcast TWT operation is shown in Figure 26-10, where the AP is the TWT scheduling AP and STA 1 and STA 2 are the TWT scheduled STAs.

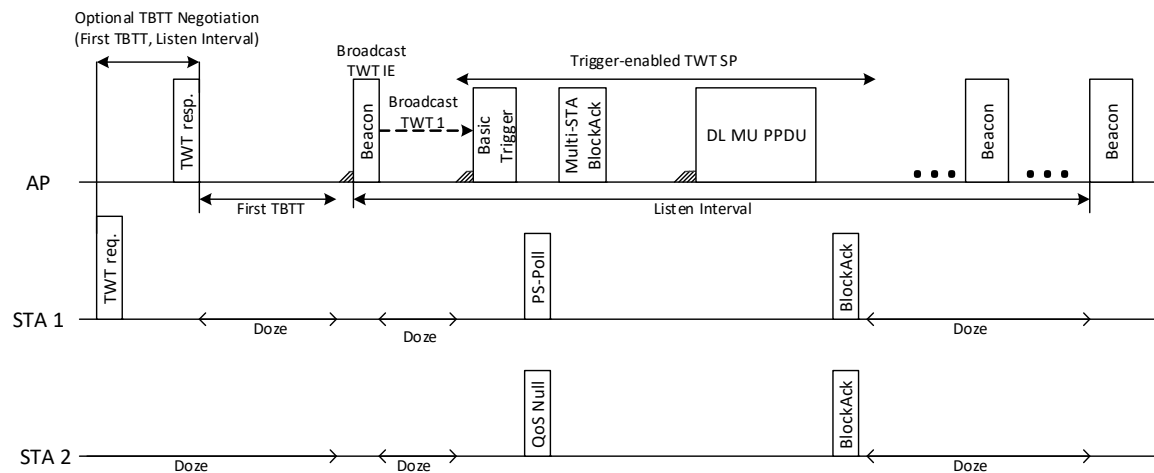


Figure 26-10—Example of broadcast TWT operation with optional TBTT negotiation

The TWT scheduling AP includes a broadcast TWT element in the Beacon frame that indicates a broadcast TWT SP during which the AP intends to send Trigger frames, or DL BUs to the TWT scheduled STAs. STA 1 and STA 2 wake to receive the Beacon frame to determine the broadcast TWT. During the trigger-enabled TWT SP, the AP sends a Basic Trigger frame to which STA 1 and STA 2 indicate that they are awake during the TWT SP. STA 1 indicates that it is awake by sending a PS-Poll, and STA 2 indicates that it is awake by sending a QoS Null frame in response to the Basic Trigger frame. STA 1 and STA 2 receive their DL BUs in a subsequent exchange with the AP and go to doze state outside of this TWT SP.

Each broadcast TWT is uniquely identified by the <broadcast TWT ID, MAC address> tuple, where the broadcast TWT ID is the value of the Broadcast TWT ID subfield and is greater than 0 and the MAC address is the address of the TWT scheduling AP.

Broadcast TWT schedules are advertised by the TWT scheduling AP in frames that carry TWT elements with the Negotiation Type subfield set to 2 as described in 26.8.3.2. Broadcast TWT schedules that are intended for member TWT scheduled STAs are identified by a Broadcast TWT ID subfield that is greater than 0, and broadcast TWT schedules that are intended for all TWT scheduled STAs are identified by a Broadcast TWT ID subfield equal to 0.

Negotiations to become a member of or terminate membership in a broadcast TWT, identified by a Broadcast TWT ID subfield greater than 0, are performed with an exchange of frames that carry TWT elements with the Negotiation Type subfield set to 3 as described in 26.8.3.3.

The TWT scheduling AP may send an unsolicited TWT response with the Trigger subfield set to 1 to a non-AP HE STA that has set the Broadcast TWT Support subfield to 1 in the HE Capabilities elements that it transmits to the AP. The TWT response shall indicate one of the following values in the TWT Setup Command field: Accept TWT, Alternate TWT, or Dictate TWT. An unsolicited TWT response with TWT Setup Command field indicating Alternate TWT or Dictate TWT contains an advisory notification to the recipient of TWT parameters that are likely to be accepted by the AP if the recipient transmits a subsequent TWT request to the AP that includes those TWT parameters. An unsolicited TWT response with a TWT Setup Command field that indicates Accept TWT allocates a broadcast TWT schedule to the receiving STA. A STA that receives an unsolicited TWT response with a TWT Setup Command field that indicates Accept TWT may transmit a TWT Teardown frame or a TWT response with TWT Setup Command field indicating Reject TWT to withdraw from the unsolicited broadcast TWT schedule.

26.8.3.2 Rules for TWT scheduling AP

A TWT scheduling AP may include a broadcast TWT element in a Beacon frame that is scheduled at a TBTT (see 11.1.3.2). The TWT scheduling AP shall include one or more TWT parameter sets in the TWT element, and each TWT parameter set may indicate a periodic occurrence of TWTs. The TWT scheduling AP shall set the Last Broadcast Parameter Set subfield to 0 in each TWT parameter set, except for the last (or only) TWT parameter set of the TWT element that shall have the Last Broadcast Parameter Set subfield set to 1. The TWT scheduling AP shall set the NDP Paging Indicator subfield to 0 and the Negotiation Type subfield to 2 and may set the Responder PM Mode subfield to 0 in the TWT element (see 10.47.7). Each TWT parameter set specifies the TWT parameters of a specific broadcast TWT that are valid within a broadcast TWT SP. Each specific broadcast TWT is identified as indicated in 26.8.3.1. Individual STAs may have membership in broadcast TWTs as the result of negotiation with a TWT scheduling AP as described in 26.8.3.1.

The TWT scheduling AP sets the TWT parameters of each TWT parameter set as described below in this subclause.

The TWT scheduling AP shall set the TWT Request subfield to 0 and the TWT Setup Command subfield as defined in Table 26-6 and shall include the broadcast TWT element in the Beacon frames for as long as there is at least one active broadcast TWT schedule. Broadcast TWT announcements are broadcast TWT schedules advertised in broadcast TWT elements contained in broadcast Management frames.

Table 26-6—Broadcast TWT announcements

TWT Setup Command field in an initiating frame	TWT Setup Command field in a response frame	Condition after the completion of the exchange
Accept TWT	No frame transmitted	Only an HE AP is permitted to transmit this sequence. TWT scheduled STAs that receive this frame use the provided TWT parameters to determine the broadcast TWT schedule. The broadcast TWT schedule is identified by the broadcast TWT ID and the TA of the initiating frame.
Alternate TWT	No frame transmitted	When transmitted by a TWT scheduling AP, some of the parameters of the broadcast TWT schedule identified by the broadcast TWT ID and the TA of the initiating frame will change at the TBTT that occurs after the Broadcast TWT Persistence field of that broadcast TWT parameter set reaches 0. The new parameters will be present in the first Beacon frame transmitted by the TWT scheduling AP at the TBTT that has a broadcast TWT parameter set with the same broadcast TWT ID and same TA, but with the TWT command value set to Accept TWT.
Reject TWT	No frame transmitted	When transmitted by a TWT scheduling AP, the broadcast TWT schedule identified by the broadcast TWT ID and the TA of the initiating frame will be terminated at the TBTT that occurs after the Broadcast TWT Persistence field of that broadcast TWT parameter reaches 0. The termination occurs at the TBTT at which a Beacon frame is transmitted by the TWT scheduling AP that does not include a broadcast TWT parameter set with the same broadcast TWT ID and same TA as the initiating frame.
NOTE 1—The Negotiation Type subfield of the TWT element contained in these frames is 2. NOTE 2—The initiating frame and response frame settings not listed in the tables in 10.47 or 26.8 are not allowed. The initiating frame is a TWT response. NOTE 3—MMPDUs that contain a broadcast TWT element generated by a TWT scheduling AP can be broadcast Probe Response, FILS Discovery, and Beacon frames. The TWT element has the TWT Request field equal to 0 and the Negotiation Type subfield equal to 2. The TWT scheduling AP can include a TWT parameter set with Broadcast TWT ID value 0 to indicate a TWT allocated for all STAs, and Broadcast TWT ID greater than 0 to indicate a TWT intended to TWT scheduled STAs that are members of that broadcast TWT.		

The TWT scheduling AP shall set the Broadcast TWT Persistence subfield for each broadcast TWT to the number of TBTTs for which the Broadcast TWT schedule will be in existence, counting forward from the current TBTT. The AP may change the value of the Broadcast TWT Persistence subfield for any Broadcast TWT within any transmitted TWT element. If the AP reduces the value of the subfield, it shall not reduce the value by more than one as compared to the value transmitted during the immediately preceding beacon interval. If the AP increases the value of the Broadcast TWT Persistence subfield, it may increase the value by any amount as compared to the value transmitted during the immediately preceding TBTT.

A TWT scheduling AP that sets the TWT Setup Command subfield to Reject TWT shall indicate the TBTT at which the periodic broadcast TWT will be terminated by setting the value of the Broadcast TWT Persistence subfield to indicate the number of TBTTs that remain until the broadcast TWT schedule is terminated. The broadcast TWT schedule terminates at the next TBTT that follows the TBTT at which the TWT scheduling AP transmits the broadcast TWT element with Broadcast TWT Persistence subfield for that broadcast TWT schedule equal to 0. A TWT scheduling AP may terminate the membership of a TWT

scheduled STA in all broadcast TWTs by transmitting a TWT Teardown frame with the Teardown All TWT field set to 1.

A TWT scheduling AP that sets the TWT Setup Command subfield to Alternate TWT shall indicate the TBTT at which the periodic broadcast TWT parameter set will be modified by setting the Broadcast TWT Persistence subfield to indicate the number of TBTTs that remain until the broadcast TWT schedule is modified. The broadcast TWT schedule will be modified at the next TBTT that follows the TBTT at which the TWT scheduling AP transmits the broadcast TWT element with Broadcast TWT Persistence subfield for that broadcast TWT schedule equal to 0. The AP shall include in the broadcast TWT element the future broadcast TWT parameter set that will take effect at that TBTT. The future broadcast TWT parameter set shall have the same values in the TWT Setup Command and Broadcast TWT ID subfields as the current broadcast TWT parameter set that is being modified and switch the TWT Setup Command subfield from Alternate TWT to Accept TWT at that TBTT. The future broadcast TWT parameter set shall be in a Broadcast TWT Parameter Set field that is located after the Broadcast TWT Parameter Set field that contains the current broadcast TWT parameter set.

NOTE 1—TWT scheduled STAs follow the broadcast TWT parameters that are included in the current broadcast TWT parameter set and switch to following the broadcast TWT parameters in the future broadcast TWT parameter set only if the TWT Setup Command field is equal to Accept TWT in the Broadcast TWT Parameter Set field that contains the future broadcast TWT parameter set.

A TWT scheduling AP should indicate Alternate TWT or Reject TWT in the TWT Setup Command field of the broadcast TWT element for as many TBTTs as needed to exceed the longest interval any STA is expected to not receive Beacon frames either when the TWT parameters of a periodic TWT will change, or when the periodic TWT specified by that TWT parameter set will be terminated.

The TWT scheduling AP shall set the Trigger field to 1 to indicate a trigger-enabled TWT. Otherwise, it shall set the Trigger field to 0 (i.e., the TWT is not a trigger-enabled TWT). The AP is not expected to schedule for transmission Trigger frames during a non-trigger-enabled TWT SP and is expected to schedule Trigger frames during a trigger-enabled TWT SP as described below in this subclause.

The TWT scheduling AP shall schedule for transmission of a Trigger frame addressed to one or more TWT scheduled STAs during a trigger-enabled TWT SP, except that the Trigger frame may be replaced by a frame carrying a TRS Control subfield, provided that the frame is carried in a DL MU PPDU and the AP allocates enough resources in the HE TB PPDU for the STA to at least deliver its BSRs in response to the soliciting DL MU PPDU. A TWT scheduling AP should not include the 12 LSBs of the STA's AID in a User Info field of a Trigger frame transmitted within a broadcast TWT SP, unless the STA is in the awake state, has established membership in the broadcast TWT with that Broadcast TWT ID, or has indicated to receive the Beacon frame preceding the beacon interval that contains this TWT SP (see 26.8.6).

The TWT scheduling AP that schedules for transmission additional Trigger frames during a trigger-enabled TWT SP shall set the More TF subfield in the Common Info field of the Trigger frame to 1 to indicate that it will schedule for transmission another Trigger frame within the same TWT SP. The TWT scheduling AP shall set the More TF subfield to 0 if the Trigger frame is the last scheduled Trigger frame of the TWT SP or if the Trigger frame is scheduled for transmission outside of a trigger-enabled TWT SP. The TWT scheduling AP should poll as many STAs as possible among TWT scheduled STAs that are members of that nonzero Broadcast TWT ID so that the STAs can perform a frame exchange with the TWT scheduling AP during that TWT SP.

NOTE 2—The TWT scheduling AP does not intend to schedule for transmission of a Trigger frame for the TWT scheduled STA when the broadcast TWT is not a trigger-enabled TWT or when the TWT scheduled STA has sent an OM Control subfield that has the UL MU disable bit equal to 1 (see 26.9).

NOTE 3—The TWT scheduling AP can cancel the transmission of a scheduled Trigger frame if the AP gains access to the wireless medium outside of the TWT SP.

NOTE 4—If the AP replaces the Trigger frame with a frame carrying a TRS Control field, then it is recommended that the AP allocate enough resources in subsequent Trigger frames sent during the TWT SP so that the STA can send as much as possible of the data reported in the BSR.

The TWT scheduling AP shall set the Flow Type field to 1 to indicate an unannounced TWT. Otherwise, it shall set the Flow Type field to 0 to indicate an announced TWT.

The TWT scheduling AP should schedule delivery of individually addressed DL BUs during unannounced TWT SPs with nonzero Broadcast TWT ID subfield.

The TWT scheduling AP shall set the Broadcast TWT Recommendation subfield according to Table 9-297a. The TWT scheduling AP shall set the Trigger field to 1 if the Broadcast TWT Recommendation subfield is 1 or 2 and may set the Trigger field to any value if the Broadcast TWT Recommendation subfield is 0 or 3.

A TWT scheduling AP that has advertised a broadcast TWT with a Broadcast TWT ID equal to 0 shall schedule the following:

- The delivery of group addressed DL BUs during the broadcast TWT SPs located within the beacon interval that follows the DTIM Beacon frame if the TWT parameter set indicated non-trigger enabled unannounced TWT SP and had the Broadcast TWT Recommendation subfield equal to 0.
- The transmission of a Trigger frame that does not contain an RA-RU during the broadcast TWT SPs if the TWT parameter set indicated trigger-enabled announced TWT SP and had the Broadcast TWT Recommendation subfield equal to 1. The Trigger frame shall contain at least one User Info field addressed to a TWT scheduled STA whose TIM bit in the Beacon frame is 1 and that is not a member of any nonzero broadcast TWT during this beacon interval.
- The transmission of a Trigger frame that contains at least one RA-RU (see 26.5.4) during the broadcast TWT SPs if the TWT parameter set indicated a trigger enabled announced TWT SP and had the Broadcast TWT Recommendation subfield set to 2 (see 26.14.2).
- The transmission of a TIM frame or FILS Discovery frame at the start of a broadcast TWT SP if the TWT parameter set indicated a non-trigger enabled unannounced TWT SP and had the Broadcast TWT Recommendation subfield set to 3 (see 26.14.3.2).

A Trigger frame transmitted during a broadcast TWT SP whose TWT parameter set has the Broadcast TWT Recommendation subfield equal to 0 or 3 may contain zero or more RA-RUs (see 26.5.4). A Trigger frame transmitted during a broadcast TWT SP whose TWT parameter set has the Broadcast TWT Recommendation subfield equal to 1 shall contain no RA-RU.

The TWT scheduling AP shall set the Target Wake Time field to the TSF timer [10: 25] that corresponds to the next TWT that is scheduled for this TWT parameter set when it queues for transmission the frame that contains the TWT element. The TSF timer at which the next TWT is scheduled has bits 0 to 9 equal to 0 and bits 26 to 63 equal to the same value as the respective bits in the current TSF timer.

The TWT scheduling AP shall include a nonzero value for the TWT wake interval in the TWT Wake Interval Exponent and TWT Wake Interval Mantissa fields for a periodic TWT and a zero value for an aperiodic TWT.

The TWT parameters are valid for each successive TWT of a periodic TWT and for the only TWT of an aperiodic TWT.

The TWT scheduling AP shall include a unique value in the Broadcast TWT ID subfield for each Broadcast TWT to allow identification of each Broadcast TWT, unless the TWT Setup Command field is Alternate TWT or the Broadcast TWT ID subfield is zero.

NOTE 5—The broadcast TWT element contains two Broadcast TWT Parameter Set fields with the same Broadcast TWT ID subfield value if the TWT Setup Command field indicates Alternate TWT in one of the Broadcast TWT Parameter Set fields. The broadcast TWT element might contain multiple Broadcast TWT Parameter Set fields with the Broadcast TWT ID subfield equal to 0.

A TWT scheduling AP that receives a PS-Poll or a U-APSD trigger frame or any other indication from a TWT scheduled STA in PS mode, during or before a specific announced TWT SP but after the end of the most recent TWT SP preceding the specific TWT SP (if any), that the TWT scheduled STA is in the awake state during the TWT SP shall follow the rules defined in 11.2.3.6, except that the AP should deliver to the TWT scheduled STA as many buffered BUs as are available at the AP, provided that the BU delivery does not exceed the duration of the TWT SP, the TWT scheduled STA has indicated that it is in the awake state for that TWT SP, and the TWT scheduled STA has not entered the doze state (see 26.8.4.3 and 26.8.5).

NOTE 6—The indication that the TWT scheduled STA is in the awake state for that TWT SP might be a PS-Poll, U-APSD trigger frame, or any frame for which an immediate response is solicited and that follows the rules in 11.2.3.2 but the corresponding immediate response frame is not received by the TWT scheduled STA. Other indications that the STA is in the awake state are the transmission of an HE TB feedback NDP in response to an NFRP Trigger frame (see 26.5.7) or the transmission of a frame that indicates that the STA is in active mode (see 11.2.3.2).

A TWT scheduling AP that sends frames to a TWT scheduled STA that is in PS mode during an unannounced TWT SP shall follow the rules defined in 11.2.3.6, except that the AP should deliver to the TWT scheduled STA as many buffered BUs as available at the AP, provided that the BU delivery does not exceed the duration of the TWT SP and the TWT scheduled STA has not entered the doze state (see 26.8.4.3 and 26.8.5).

NOTE 7—The TWT scheduling AP can deliver the buffered BUs in A-MPDUs sent under a BlockAck agreement if the TWT is an announced TWT and the TWT scheduled STA is awake for that TWT SP or if the TWT is an unannounced TWT (at the start of which the TWT scheduled STA is assumed to already be awake). The buffered BUs can be delivered in multiple PPDU's transmitted within the TWT SP. The TWT scheduling AP can exceed the duration of the TWT SP if the TWT scheduled STA is in active mode.

A TWT scheduling AP may transmit to a TWT scheduled STA that is in active mode at any time (see 11.2.3.2). A TWT scheduling AP may transmit to a TWT scheduled STA that is in PS mode and awake outside of a TWT SP following the rules in 11.2.3.6.

A TWT scheduling AP that receives a TWT element with the TWT Request field equal to 1, the Negotiation Type subfield equal to 3, and the TWT Setup Command field set to Suggest or Demand may respond with a frame containing a TWT element as shown in Table 26-7 (in 26.8.3.3).

A TWT scheduling AP that receives a TWT element with the TWT Request field equal to 1, the Negotiation Type subfield equal to 3, and the TWT Setup Command field set to Reject shall delete the membership of the STA corresponding to the TA of the MMPDU that contained the TWT element from the broadcast TWT schedule that has the Broadcast TWT ID value that is equal to the value of the Broadcast TWT ID field of the TWT element.

A TWT scheduling AP may transmit a broadcast TWT announcement at any time. Valid broadcast TWT announcements are described in Table 26-6.

26.8.3.3 Rules for TWT scheduled STA

A TWT scheduled STA that receives a broadcast TWT element in a Beacon frame shall follow the rules defined in this subclause to interact with the TWT scheduling AP.

A TWT scheduled STA should not transmit frames to the TWT scheduling AP outside of broadcast TWT SPs and should not transmit frames that are not contained within HE TB PPDU to the TWT scheduling AP within trigger-enabled broadcast TWT SPs, except that the STA can transmit frames within negotiated individual TWT SPs as defined in 26.8.2.

NOTE 1—The TWT scheduled STA decides which frames to transmit within or outside a TWT SP; and while it is recommended that the TWT scheduled STA not transmit using EDCA within or outside TWT SPs, the TWT scheduled STA might still do so. If the STA decides to transmit, then the STA might contend for accessing the medium as defined in 10.23.2 and in 26.2.7.

A TWT scheduled STA may request to become a member of a broadcast TWT by transmitting a frame to its associated AP that contains a TWT element with the Negotiation Type subfield set to 3 and the TWT Setup Command field set to Request TWT, Suggest TWT, or Demand TWT. The TWT Parameter set indicates the Broadcast TWT ID of the broadcast TWT that the STA is requesting to join. See Table 26-7.

A TWT scheduled STA may terminate membership in a broadcast TWT by transmitting a frame to its associated AP that contains a TWT element with the Negotiation Type subfield set to 3 and the TWT Setup Command field set to Reject TWT or by transmitting a TWT Teardown frame that has the Negotiation Type subfield set to 3. A TWT scheduled STA may terminate membership in all broadcast TWTs by transmitting a TWT Teardown frame with the Teardown All TWT field set to 1.

A TWT scheduled STA that receives a TWT element with the TWT Request field equal to 0, the Negotiation Type subfield equal to 3, and the TWT Setup Command field indicating Accept TWT is a member of the broadcast TWT identified by the <broadcast TWT ID, MAC address> tuple, where the broadcast TWT ID is the value of the Broadcast TWT ID subfield in the TWT element and the MAC address that is the TA of the MMPDU that contained the TWT element is equal to the MAC address of the AP with which the STA is associated, regardless of whether the TWT scheduled STA had previously transmitted a corresponding TWT element to the AP with the TWT Setup Command field indicating Request TWT, Suggest TWT, or Demand TWT.

Valid broadcast TWT membership exchanges are described in Table 26-7.

Table 26-7—Broadcast TWT membership exchanges

TWT Setup Command field in an initiating frame	TWT Setup Command field in a response frame	Condition after the completion of the exchange
Demand TWT	Accept TWT	A broadcast TWT schedule exists or has been created with the TWT parameters indicated in the initiating frame and repeated in the response frame. The TWT scheduled STA transmitting the initiating frame is a member of the Broadcast TWT schedule identified by the Broadcast TWT ID and the TA of the response frame.
Request TWT or Suggest TWT	Accept TWT	A broadcast TWT schedule exists or has been created with the TWT parameters indicated in the response frame. The TWT scheduled STA transmitting the initiating frame is a member of the broadcast TWT schedule identified by the broadcast TWT ID and the TA of the response frame.

Table 26-7—Broadcast TWT membership exchanges (*continued*)

TWT Setup Command field in an initiating frame	TWT Setup Command field in a response frame	Condition after the completion of the exchange
Suggest TWT or Demand TWT	Alternate TWT	No new broadcast TWT schedule has been created with the TWT parameters indicated in the initiating frame. The TWT scheduling AP is offering an alternative set of parameters to those indicated in the initiating frame as a means of negotiating TWT parameters with the TWT scheduled STA. The TWT scheduled STA can send a new request with any set of TWT parameters, and the TWT scheduling AP might create a new broadcast TWT schedule using the parameters indicated in the response frame.
Suggest TWT or Demand TWT	Dictate TWT	A broadcast TWT schedule either is created or already exists and is using the TWT parameters identified in the response frame, including a broadcast TWT ID. The TWT scheduling AP will not create any new broadcast TWT schedule with the TWT scheduled STA at this time. The TWT scheduled STA transmitting the initiating frame is not a member of the broadcast TWT schedule identified by the broadcast TWT ID and the TA of the response frame. The TWT scheduled STA can send a new request, but will receive an Accept TWT only if it uses the dictated TWT parameters.
Request TWT, Suggest TWT, or Demand TWT	Reject TWT	The TWT scheduled STA transmitting the initiating frame is not a member of a broadcast TWT identified by the broadcast TWT ID and the TA of the response frame, if such a broadcast TWT exists. The TWT scheduling AP will not accept any new request from the TWT scheduled STA to join or create a broadcast TWT at this time.
Accept TWT	No frame transmitted	Not permitted to be transmitted by a TWT scheduled STA. When transmitted by a TWT scheduling AP, the recipient STA's membership in the broadcast TWT schedule identified by the broadcast TWT ID and the TA of the initiating frame is established.
Alternate TWT or Dictate TWT	No frame transmitted	Not permitted to be transmitted by a TWT scheduled STA. When transmitted by a TWT scheduling AP, the TWT scheduled STA receiving this frame is not, through the receipt of this frame, a member of the broadcast TWT identified by the initiating frame. The TWT scheduled STA can use the information provided to create a request to join a TWT in a subsequent initiating frame that it transmits.

Table 26-7—Broadcast TWT membership exchanges (*continued*)

TWT Setup Command field in an initiating frame	TWT Setup Command field in a response frame	Condition after the completion of the exchange
Reject TWT	No frame transmitted	When transmitted by a TWT scheduled STA, the transmitting STA's membership in the broadcast TWT schedule identified by the broadcast TWT ID and the RA of the initiating frame is terminated. When transmitted by a TWT scheduling AP, the receiving STA's membership in the broadcast TWT schedule identified by the broadcast TWT ID and the TA of the initiating frame is terminated.
NOTE 1—The Negotiation Type subfield of the TWT element contained in these frames is 3. NOTE 2—The initiating frame and response frame settings not listed in the tables in 10.47 or 26.8 are not allowed. The initiating frame is a TWT request if the TWT element contained in the frame has the TWT Request field equal to 1 (see Table 9-297); otherwise, it is a TWT response. The response frame is a TWT response. NOTE 3—In addition to these exchanges, the TWT scheduling AP might respond to an initiating frame that solicits membership in a broadcast TWT schedule with an indication or solicitation of the establishment of an individual TWT agreement. NOTE 4—MMPDUs that contain a broadcast TWT element generated by a TWT scheduled STA can be (Re)Association Request frames and can be TWT Setup frames with TWT Request field equal to 1. The TWT element has the Negotiation Type subfield equal to 3 and the Broadcast TWT ID(s) that the STA intends to join or withdraw. MMPDUs that contain a broadcast TWT element generated by a TWT scheduled AP can be (Re)Association Response frames and can be TWT Setup frames with TWT Request field equal to 0. The TWT element has the Negotiation Type subfield equal to 3 and the Broadcast TWT ID(s) to which the STA is assigned or from which the STA is withdrawn.		

A TWT scheduled STA that is in PS mode may enter the doze state after receiving a Beacon frame with a TWT element indicating the existence of a broadcast TWT and shall be in the awake state at the broadcast TWT start times for which the STA has indicated it will be awake by any of the following means:

- Establishing a membership for the unannounced broadcast TWT with the applicable broadcast TWT IDs.
- Negotiating a wake TBTT and wake interval between Beacon frames that the STA receives, as defined in 26.8.6.
- Having sent a PS-Poll or U-APSD trigger frame during the beacon interval.
- Having sent another indication that it is in the awake state during that beacon interval.

NOTE 2—Other indications that the STA is in the awake state are the transmission of an HE TB feedback NDP in response to an NFRP Trigger frame (see 26.5.7) or the transmission of a frame that indicates that the STA is in active mode (see 11.2.3.2).

NOTE 3—The STA might indicate that it will not be awake at certain broadcast TWT start times by sending a TWT Information frame. The AP might indicate to a STA that it need not be awake at certain broadcast TWT start times by sending a TWT information frame (see 26.8.4).

A TWT scheduled STA is not required to be in the awake state at broadcast TWT SP start times corresponding to the broadcast TWT that has the broadcast TWT ID value of 0.

A TWT scheduled STA that did not receive a Beacon frame at a TBTT shall act as if it had received the expected Beacon frame containing a TWT element for a broadcast TWT if the missed beacon corresponds to a TBTT that is within the next n TBTTs beyond the most recently received Beacon frame that included

a TWT element for that broadcast TWT, where n is equal to 1 plus the value obtained from the Broadcast TWT Persistence subfield of the corresponding Broadcast TWT, except that n is infinite if the Broadcast TWT Persistence subfield is 255.

A TWT scheduled STA transmits an HE TB PPDU as a response to a Trigger frame that is addressed to it and is sent during a trigger-enabled TWT SP (see 26.5.2). A TWT scheduled STA that is in PS mode and is awake during an announced TWT SP shall include a PS-Poll frame or a U-APSD trigger frame in the HE TB PPDU if it intends to solicit buffered BUs from the TWT scheduling AP (see 11.2.3.7), unless the STA has already transmitted within that TWT SP a PS-Poll or U-APSD trigger frame, or has transmitted any other indication that the STA is in the awake state within that TWT SP, or has, previous to the start of the TWT SP but after the end of the most recent TWT SP, indicated to the AP that it is currently in the awake state. A TWT scheduled STA that is in PS mode shall transition to the awake state at the start of an unannounced TWT SP of which it is a member. The STA may include other frames in the HE TB PPDU when other rules do not prohibit their inclusion (see 26.5.2 and 9.7.3).

NOTE 4—A TWT scheduling AP sets the bit in the TIM element of the Beacon frame that corresponds to the AID of the TWT scheduled STA to 1 to indicate that it expects the TWT scheduled STA to solicit available buffered BUs (see 11.2.3.7).

NOTE 5—Other indications that the STA is in the awake state are the transmission of an HE TB feedback NDP in response to an NFRP Trigger frame (see 26.5.7) or the transmission of a frame that indicates that the STA is in active mode (see 11.2.3.2).

A TWT scheduled STA should not send frames that do not satisfy the Broadcast TWT Recommendation subfield recommendations in Table 9-297a during the corresponding TWT SP(s). Frames sent as a response to a Trigger frame are subject to further restrictions as defined in 26.5.2 and 9.7.3.

26.8.4 Use of TWT Information frames

26.8.4.1 General

An HE STA may transmit a TWT Information frame to its peer STA during an individual TWT agreement, during broadcast TWT schedule, or at any time as defined in 26.8.4.2, 26.8.4.3, and 26.8.4.4, respectively.

NOTE 1—An HE AP might include multiple TWT Information frames, each addressed to a different peer STA, in an HE MU PPDU (see 26.5.1).

The TWT Information frame shall have the Response Requested subfield equal to 0, the Next TWT Request subfield equal to 0, and one of the following:

- The Next TWT subfield that is equal to a nonzero value if the frame is transmitted by a TWT **responding STA**, by a TWT scheduling AP, or by any HE STA to a peer STA that has set the Flexible TWT Schedule Support field to 1 in the HE Capabilities element it transmits.
 - The value of the Next TWT subfield shall be selected from existing TWT values for an individual TWT agreement if the Flexible TWT Schedule Support field in the HE Capabilities element received from the peer STA is 0 and shall be selected from existing TWT values for a broadcast TWT schedule regardless of the value of the Flexible TWT Schedule Support field received from the peer STA.
 - The Next TWT subfield may contain any nonzero value if the Flexible TWT Schedule Support field in the HE Capabilities element received from the peer STA is 1.
 - The All TWT subfield is 1 if the resumption applies to all broadcast TWT schedules followed by the TWT scheduled STA and/or to all individual TWT agreements followed by the TWT responding STA.

- A Next TWT subfield that is present if the frame is transmitted by a TWT **requesting STA**, by a TWT scheduled STA, or by any HE STA to a peer STA that has set the Flexible TWT Schedule Support field to 1 in the HE Capabilities element it transmits.
 - The Next TWT subfield indicates the earliest TWT at which the individual TWT agreement or broadcast TWT schedule is resumed and shall be selected from existing TWT values for that TWT agreement or broadcast TWT schedule if the Flexible TWT Schedule Support field in the HE Capabilities element received from the peer STA is 0.
 - The All TWT subfield is 1 if the resumption applies to all broadcast TWT schedules followed by the TWT scheduled STA and/or to all individual TWT agreements followed by the TWT requesting STA.
 - The Next TWT subfield may contain any nonzero value if Flexible TWT Schedule Support field in the HE Capabilities element received from the peer STA is 1.
- A Next TWT subfield that is not present if the frame is transmitted by a **TWT requesting STA** or a TWT scheduled STA to indicate suspension of the individual TWT agreement or broadcast TWT schedule.
 - The All TWT subfield is 1 if the suspension applies to all broadcast TWT schedules followed by the TWT scheduled STA and/or to all individual TWT agreements followed by the TWT requesting STA.

NOTE 2—Information exchanged with TWT Information frames does not modify the TWT parameters of any existing TWT session, except if the TWT Information frame is sent under flexible TWT (see 26.8.4.4).

The use of TWT Information frames for suspending and/or resuming existing individual TWT agreements is described in 26.8.4.2. The use of TWT Information frames for suspending and/or resuming existing broadcast TWT schedules is described in 26.8.4.3. The use of TWT Information frames for providing a flexible TWT that is independent of any existing TWT agreements or TWT schedules is described in 26.8.4.4.

26.8.4.2 TWT Information frame exchange for individual TWT

An HE STA that has an individual TWT agreement may transmit a TWT Information frame to a peer STA with which it has the agreement if the peer STA has set the TWT Information Frame Disabled field to 0 in the TWT element sent during TWT setup; otherwise, the HE STA **shall not transmit** a TWT Information frame to the peer STA. The HE STA sets the fields of the TWT Information frame as defined in 26.8.4.1.

A TWT requesting STA that receives a TWT Information frame containing a Next TWT subfield follows the rules in 10.47.4 and the rules below in this subclause.

A TWT requesting STA that receives an acknowledgment in response to a TWT Information frame transmitted by the STA that does not contain a Next TWT subfield shall consider the corresponding TWT agreement suspended until the TWT session is resumed. A TWT requesting STA that receives an acknowledgment in response to a TWT Information frame transmitted by the STA that does contain a Next TWT subfield shall consider the corresponding TWT agreement suspended and shall resume the TWT agreement starting from the value indicated in the Next TWT subfield of the transmitted TWT Information frame. A TWT requesting STA that receives an acknowledgment in response to a TWT Information frame transmitted by the STA assumes that any other individual TWT agreements and broadcast TWT schedules (see 26.8.3) are not affected by the transmission of this TWT Information frame, except when the All TWT subfield of the TWT Information frame is equal to 1. Other default PS procedures are not affected by the transmission of this TWT Information frame (see 11.2).

NOTE—The TWT Flow Identifier, together with the MAC addresses of the TWT requesting STA and TWT responding STA, identifies the TWT agreement for which the TWT Information frame is sent (see 10.47.1).

If the TWT Information frame contains an All TWT subfield equal to 1, then the above rules apply to all individual TWT agreements, except that the resumptions of the respective TWTs shall occur at the first TWT of the respective TWT agreement that occurs not earlier than the Next TWT value contained in the TWT Information frame, regardless of the value of the Flexible TWT Schedule Support field in the HE Capabilities element exchanged between the two STAs.

A TWT requesting STA that is in PS mode and that transmits a TWT Information frame to a TWT responding STA shall suspend the corresponding TWT agreement and may transition to doze state after receiving the acknowledgment even if it has previously transmitted a PS-Poll or U-APSD trigger frame and has not yet received the expected frames from the AP in response. The STA shall resume TWT operation for the corresponding TWT agreement at the specified TWT indicated (if any) in the TWT Information frame. A TWT requesting STA that is in PS mode and that receives a TWT Information frame from a TWT responding STA shall suspend the TWT agreement and may go to doze state after transmitting the acknowledgment even if it has previously transmitted a PS-Poll or U-APSD trigger frame and has not yet received the expected frames from the TWT responding STA in response. The STA shall resume TWT operation for the corresponding TWT agreement at the specified TWT indicated (if any) in the TWT Information frame.

26.8.4.3 TWT Information frame exchange for broadcast TWT

An HE STA that is a TWT scheduling AP may transmit a TWT Information frame to any of the members of a broadcast TWT schedule if the member has set the TWT Information Frame Disabled field to 0 in the TWT element sent when joining the broadcast TWT schedule. An HE STA that is a TWT scheduled STA may transmit a TWT Information frame to the TWT scheduling AP corresponding to a broadcast TWT schedule established by that STA if the AP has set the TWT Information Frame Disabled field to 0 in the broadcast TWT element it transmits. The HE STA sets the fields of the TWT Information frame as defined in 26.8.4.1.

A TWT scheduled STA that receives a TWT Information frame that contains an All TWT subfield equal to 1 follows the rules defined in 26.8.3.3, except that the TWT scheduled STA shall consider all the broadcast TWT schedules as suspended and shall resume each broadcast TWT schedule at the first TWT that occurs not earlier than the Next TWT subfield value contained in the received TWT Information frame.

A TWT scheduled STA that receives an acknowledgment in response to a TWT Information frame transmitted by the STA that contains an All TWT subfield equal to 1 and that does not contain a Next TWT subfield shall consider all broadcast TWT schedules suspended and can follow the default PS procedure defined in 11.2 until the broadcast TWT schedules are resumed.

A TWT scheduled STA that receives an acknowledgment in response to a TWT Information frame transmitted by the STA that contains an All TWT subfield equal to 1 and contains a Next TWT subfield shall suspend all broadcast TWT schedules and shall resume the broadcast TWT schedules at the first scheduled TWT for each respective broadcast TWT schedule that occurs not earlier than the value indicated in the Next TWT subfield contained in the transmitted TWT Information frame, regardless of the values of the Flexible TWT Schedule Support field in the HE Capabilities element exchanged between the two STAs.

NOTE—TWT suspension and resumption as indicated by a TWT Information frame with the All TWT subfield equal to 1 applies to all broadcast TWT schedules of the TWT scheduling AP.

A TWT scheduled STA that is in PS mode and that transmits a TWT Information frame to a TWT scheduling AP shall suspend the corresponding broadcast TWT schedule and may transition to doze state after receiving the acknowledgment, even if it has previously transmitted a PS-Poll or U-APSD trigger frame and has not yet received the expected frames from the TWT scheduling AP in response. The STA shall resume TWT operation for the corresponding broadcast TWT schedule at the specified TWT indicated (if any) in the TWT Information frame. A TWT scheduled STA that is in PS mode and that receives a TWT

Information frame from a TWT scheduling AP shall suspend the corresponding broadcast TWT schedule and may transition to doze state after transmitting the acknowledgment, even if it has previously transmitted a PS-Poll or U-APSD trigger frame and has not yet received the expected frames from the TWT scheduling AP in response. The STA shall resume TWT operation for the corresponding broadcast TWT schedule at the specified TWT indicated (if any) in the TWT Information frame.

26.8.4.4 TWT Information frame exchange for flexible wake time

An HE STA may transmit a TWT Information frame that contains a flexible TWT to a peer STA if the peer STA has set the Flexible TWT Schedule Support field in the HE Capabilities it transmits to 1; otherwise, the HE STA shall not transmit a TWT Information frame that contains a flexible TWT to the peer STA.

A flexible TWT is a nonzero value indicated in the Next TWT subfield of a TWT Information frame with All TWT subfield equal to 0, and the value is independent from any existing TWT values of TWT agreements that the HE STA might be following (if any). The HE STA sets the fields of the transmitted TWT Information frame as defined in 26.8.4.1.

NOTE 1—Flexible TWT support does not depend on the STA's TWT capabilities, i.e., the STA can use flexible TWT without being required to set up an individual TWT agreement or broadcast TWT schedule.

An HE STA that receives an acknowledgment for a TWT Information frame with flexible TWT that contains a TWT Flow Identifier that identifies an existing individual TWT agreement shall replace the next TWT SP start time for that TWT agreement with the value contained in the Next TWT subfield of the TWT Information frame.

A non-AP HE STA that receives an acknowledgment for a TWT Information frame with flexible TWT that contains a TWT Flow Identifier that does not identify any existing individual TWT agreement preserves the PM mode from the time the TWT Information frame was sent to the time indicated in the Next TWT subfield of the TWT Information frame as described below in this subclause.

NOTE 2—If the TWT Information frame has the All TWT field equal to 1, then the TWTs are suspended and resumed as described in 26.8.4.2 and 26.8.4.3.

A non-AP HE STA that transmits a TWT Information frame with flexible TWT to a peer STA

- May go to doze state after receiving the acknowledgment sent in response to the TWT Information frame if it is in PS mode (i.e., the PM subfield of the Frame Control field of the TWT Information frame is 1).
- May be unavailable if it is in active mode (i.e., the PM subfield of the Frame Control field of the TWT Information frame is 0).
- Shall be in the awake state at the time it indicated in the Next TWT subfield of the TWT Information frame.
- Shall be in the PS mode if the PM subfield of the TWT Information frame was 1 and in active mode if the PM subfield of the TWT Information frame was 0.

The STA, once in the awake state, shall follow the rules that correspond to the power management mode of the STA, which are defined in 11.2.3 for the active and PS modes and in 26.8 when the STA operates within TWT SPs.

NOTE 3—An HE AP delivers DL BUs to the STA at or after the flexible TWT indicated in the flexible TWT by following the rules in 11.2.3.6 if the STA does not follow TWT and by following the rules in 26.8 if the STA follows TWT and the delivery falls within a TWT SP. The STA is not required to send a frame at or after the flexible TWT to indicate its awake state to the AP. If the STA is following U-APSD, then the operation is resumed at a time that occurs at the flexible TWT; and if the STA is following an APSD schedule, then the operation is resumed at a time that occurs at or after the flexible TWT.

A non-AP HE STA that receives a TWT Information frame with flexible TWT from a peer STA

- May go to doze state after transmitting the acknowledgment if it is in PS mode.
- May be unavailable if it is in active mode.
- Shall be in the awake state at the time it indicated in the Next TWT subfield of the TWT Information frame.
- Shall be in the PS mode if the STA was in PS mode when it received the TWT Information frame and in active mode if the STA was in active mode when it received the TWT Information frame.

The STA, once in the awake state, shall follow the rules that correspond to the power management mode of the STA, which are defined in 11.2.3 for the active and PS modes and in 26.8 when the STA operates within TWT SPs.

26.8.5 Power save operation during TWT SPs

The following rules apply to TWT SPs for both broadcast TWT schedules and individual TWT agreements where the TWT SP of a broadcast TWT is uniquely identified by the <broadcast TWT ID, MAC address of TWT scheduling AP> tuple and the TWT SP of an individual TWT is uniquely identified by the <TWT flow identifier, MAC address of TWT requesting STA, MAC address of TWT responding STA> triple.

A TWT requesting STA or a TWT scheduled STA that is not in PS mode and that transmits a frame with the Power Management subfield set to 1 during a TWT SP shall remain in the awake state until the AdjustedMinimumTWTWakeDuration time has elapsed from the TWT SP start time or until a TWT SP termination event is detected, whichever occurs first for that particular TWT SP.

A TWT requesting STA or a TWT scheduled STA in PS mode that is in the awake state for a TWT SP may transition to the doze state after AdjustedMinimumTWTWakeDuration time has elapsed from the TWT SP start time even if it has previously transmitted a PS-Poll frame or U-APSD trigger frame and has not yet received the expected frames from the AP in response. For a trigger-enabled TWT SP, if the AdjustedMinimumTWTWakeDuration time has elapsed from the scheduled TWT SP start time and no Trigger frames are received by the STA, the HE STA may enter doze state if no other condition requires the STA to remain awake.

When a **TWT SP termination event** is detected within a TWT SP by a STA in PS mode that is participating in the TWT SP, the STA may transition to the doze state without waiting for the expiration of the AdjustedMinimumTWTWakeDuration time as described in 10.47.1, even if it has previously transmitted a PS-Poll frame or U-APSD trigger frame and has not yet received the expected frames from the AP in response.

A TWT requesting STA or a TWT scheduled STA shall classify any of the following events as a TWT SP termination event:

- a) The transmission by the TWT requesting STA of an acknowledgment in response to an individually addressed QoS Data or QoS Null frame sent by the TWT responding STA that had the EOSP subfield equal to 1.
- b) The transmission by the TWT scheduled STA of an acknowledgment in response to an individually addressed QoS Data or QoS Null frame sent by the TWT scheduling AP that had the EOSP subfield equal to 1.
- c) The transmission by the TWT requesting STA of an acknowledgment in response to an individually addressed frame that is neither a QoS Data frame nor a QoS Null frame, but that was sent by the TWT responding STA with the More Data field equal to 0.

- d) The transmission by the TWT scheduled STA of an acknowledgment in response to an individually addressed frame that is neither a QoS Data frame nor a QoS Null frame, but that was sent by the TWT scheduling AP with the More Data field equal to 0.
- e) The reception of an individually addressed or broadcast QoS Data or QoS Null frame sent by the TWT responding STA or TWT scheduling AP that does not solicit an immediate response and had the EOSP subfield equal to 1.
- f) The reception of an individually addressed frame that is neither a QoS Data frame nor a QoS Null frame, but that was sent by the TWT responding STA or TWT scheduling AP, does not solicit an immediate response, and had the More Data field equal to 0.
- g) The reception of a Trigger frame sent by the TWT responding STA or TWT scheduling AP that has the More TF field equal to 0 and is not addressed to the TWT requesting STA or TWT scheduled STA, provided that the TWT requesting STA or TWT scheduled STA either is awake for an announced trigger-enabled TWT SP but did not transmit an indication that it is in the awake state to the TWT responding STA or TWT scheduling AP or is awake for an unannounced trigger-enabled TWT SP.
- h) The transmission or reception by the TWT requesting STA of the acknowledgment for a TWT Information frame that satisfies specific conditions in 26.8.4.2 and 26.8.4.4.
- i) The transmission or reception by the TWT scheduled STA of the acknowledgment for a TWT Information frame that satisfies specific conditions in 26.8.4.3 and 26.8.4.4.

The classification of a More Data field equal to 0 in an Ack, BlockAck, or individually addressed Multi-STA BlockAck frame as an event that terminates a TWT SP is possible only when both STAs have indicated support of transmitting or receiving the frame with a nonzero More Data subfield, which is indicated in the More Data Ack subfield of the QoS Info field of frames they transmit (see 11.2.3).

NOTE 1—A STA participating in multiple TWT SPs that overlap in time stays in the awake state until the latest AdjustedMinimumTWTWakeDuration time of all of the TWT SPs expires, except that a TWT SP termination event causes all of the overlapping TWT SPs to terminate.

NOTE 2—A Trigger frame is addressed to the STA if the Trigger frame contains the AID of the STA in one of its User Info fields (see 26.5.2) and has in its TA field either the MAC address of its associated AP or the transmitted BSSID (see 26.5.2.2.4). Otherwise, the Trigger frame is not addressed to the STA. If the Trigger frame contains one or more RA-RUs for which the STA can gain access according to 26.5.4, then the STA can follow the rules defined in 26.14.2 to determine an early TWT SP termination event.

26.8.6 Negotiation of wake TBTT and wake interval

A TBTT scheduled STA that intends to operate in power save mode (see 11.2.3.2) may transmit a TWT request to the TBTT scheduling AP that identifies the wake TBTT of the first Beacon frame and the wake interval between subsequent Beacon frames it intends to receive. The TWT request shall contain the following:

- The Negotiation Type subfield equal to 1 and the TWT Setup Command field equal to Suggest TWT or Demand TWT.
- The requested first wake TBTT in the Target Wake Time field.
- The requested wake interval between consecutive TBTTs in the TWT Wake Interval Mantissa and TWT Wake Interval Exponent fields.
- The requested TBTT wake duration in the Nominal Minimum TWT Wake Duration field.
- All other fields in the TWT element reserved.

A TBTT scheduling AP that receives a TWT request from a STA whose value of the Negotiation Type subfield is 1 shall respond with a TWT response that contains Accept TWT, Alternate TWT, or Reject TWT in the TWT Setup Command field. For an Accept TWT, it shall also contain the following:

- The Negotiation Type subfield equal to 1.
- The allocated first wake TBTT in the Target Wake Time field.
- The allocated wake interval between consecutive TBTTs in the TWT Wake Interval Mantissa and TWT Wake Interval Exponent fields.
- The allocated TBTT wake duration in the Nominal Minimum TWT Wake Duration field.
- All other fields in the TWT element reserved.

After successfully completing the negotiation, the TBTT scheduled STA may go to doze state until its TSF matches the next negotiated wake TBTT, provided that the STA is in power save mode and no other condition requires the STA to remain awake. The TBTT scheduled STA shall be in the awake state to listen to Beacon frames transmitted at negotiated wake TBTTs and shall operate as described in 26.8.3.3.

If the TBTT scheduled STA receives a Beacon frame from the TBTT scheduling AP at or after TBTT, the TBTT scheduled STA may go to doze state until the next wake TBTT if no other condition requires the STA to remain awake. The TBTT scheduled STA may go to doze state after a nominal minimum TBTT wake duration time has elapsed from the TBTT start time if no other condition requires the STA to remain awake.

Either STA that is a party to an established wake TBTT agreement can tear down the wake TBTT agreement by following the teardown procedure described in 10.47.8 and by setting the Negotiation Type subfield to 1 in the TWT Teardown frame.

Table 26-8 summarizes the interactions between devices that negotiate a Wake TBTT agreement.

Table 26-8—Wake TBTT negotiation exchanges

TWT Setup Command field in an initiating frame	TWT Setup Command field in a response frame	Condition after the completion of the exchange
Request TWT	Accept TWT, Alternate TWT, Dictate TWT, Reject TWT, or no response	This exchange is not allowed.
Demand TWT or Suggest TWT	Accept TWT	A Wake TBTT agreement has been created with the Wake TBTT parameters indicated in the initiating frame.
Demand TWT or Suggest TWT	Reject TWT	No Wake TBTT agreement has been created.
Demand TWT or Suggest TWT	Alternate TWT	No Wake TBTT agreement has been created. The TBTT scheduling AP is offering an alternative set of parameters to those indicated in the initiating frame. The TBTT scheduled STA can send a new request with any set of Wake TBTT parameters, and the responder might create a Wake TBTT agreement using those parameters. The TBTT scheduled STA is unlikely to send a new request if the TWT Setup Command is Demand TWT and is very likely to send a new request if the TWT Setup Command is Suggest TWT.
NOTE 1—The Negotiation Type subfield of the TWT element contained in these frames is equal to 1.		
NOTE 2—The initiating frame and response frame settings not listed in the tables in 10.47 or 26.8 are not allowed. The initiating frame is a TWT request, and the response frame is a TWT response.		

26.8.7 HE subchannel selective transmission

26.8.7.1 General

An HE STA that supports HE subchannel selective transmission (SST) operation shall set `dot11HESubchannelSelectiveTransmissionImplemented` to true and shall set the HE Subchannel Selective Transmission Support field in the HE Capabilities element it transmits to 1. An HE STA that does not support HE SST operation shall set the HE Subchannel Selective Transmission Support field in the HE Capabilities element it transmits to 0.

A non-AP HE STA with `dot11HESubchannelSelectiveTransmissionImplemented` equal to true is a HE SST non-AP STA.

An HE AP with `dot11HESubchannelSelectiveTransmissionImplemented` equal to true is an HE SST AP.

An HE SST non-AP STA and an HE SST AP may set up SST operation by negotiating a trigger-enabled TWT as defined in 26.8.2 with the following exceptions:

- The TWT request may have a TWT Channel field with up to one bit set to 1 to indicate the secondary channel requested to contain the RU allocations addressed to the HE SST non-AP STA that is a 20 MHz operating STA.
- The TWT request may have a TWT Channel field with all 4 LSBs or all 4 MSBs set to 1 to indicate whether the primary 80 MHz channel or the secondary 80 MHz channel is requested to contain the RU allocations addressed to the HE SST non-AP STA that is an 80 MHz operating STA.
- The TWT response shall have a TWT Channel field with up to one bit set to 1 to indicate the secondary channel that will contain the RU allocations addressed to the HE SST non-AP STA that is a 20 MHz operating STA.
- The TWT response shall have a TWT Channel field with all 4 LSBs or all 4 MSBs set to 1 to indicate whether the primary 80 MHz channel or the secondary 80 MHz channel will contain the RU allocations addressed to the HE SST non-AP STA that is a 80 MHz operating STA.

26.8.7.2 SST operation

An HE SST non-AP STA and HE SST AP that successfully set up SST operation shall follow the rules defined in this subclause.

If an HE SST AP causes its operating channel or channel width to change and if any secondary channel of a trigger-enabled TWT is not within the new operating channel or channel width, then the HE SST AP and the HE SST non-AP STA implicitly terminate the trigger-enabled TWT.

The HE SST AP follows the rules in 26.8.2 to exchange frames with the HE SST non-AP STA during trigger-enabled TWT SPs, except that the AP shall ensure the following:

- The individually addressed RUs allocated in DL MU PPDU and in Trigger frames addressed to the HE SST non-AP STA are within the subchannel indicated in the TWT Channel field of the TWT response and follow the RU restriction rules defined in 27.3.2.8 if the HE SST non-AP STA is a 20 MHz operating STA and in 27.3.2.9 if the HE SST non-AP STA is an 80 MHz operating STA.
- The TXVECTOR parameter `CH_BANDWIDTH` of a DL MU PPDU is not set to HE-CBW-PUNC160-PRI20, HE-CBW-PUNC80+80-PRI20, HE-CBW-PUNC160-SEC40, or HE-CBW-PUNC80+80-SEC40 if the DL MU PPDU is addressed to at least one HE SST non-AP STA that is an 80 MHz operating STA operating in a secondary channel.
- The trigger-enabled TWT SPs do not overlap with TBTTs at which DTIM Beacon frames are sent.
- The same subchannel is used for all trigger-enabled TWT SPs with the same HE SST non-AP STA that overlap in time.

An HE SST non-AP STA operating on the secondary channel shall not conduct OMI operation as defined in 26.9 or OMN operation as defined in 11.40 to change the operating bandwidth.

The HE SST non-AP STA follows the rules in 26.8.2 to exchange frames with the HE SST AP during trigger-enabled TWT SPs, except that the STA

- Shall be available in the subchannel indicated in the TWT Channel field of the TWT response at TWT start times.
- Shall not access the medium in the subchannel using DCF or EDCAF.
- Shall not respond to Trigger frames addressed to it (see 26.5 and 26.8.2), unless it has performed CCA until a frame is detected by which it can set its NAV, or until a period equal to NAVSyncDelay has transpired, whichever is earlier.
- Shall update its NAV according to 26.2.4 if it receives a PPDU in the subchannel.

An HE SST non-AP STA may include a Channel Switch Timing element in (Re-)Association Request frames it transmits to an HE SST AP to indicate the time required by the STA to switch between different subchannels. The received channel switch time informs the HE SST AP of the duration of time that the HE SST non-AP STA might not be available to receive frames before the TWT start time and after the end of the trigger-enabled TWT SP.

NOTE—An HE SST STA in PS mode is not required to move to the primary channel after the end of the trigger-enabled TWT SP.

26.9 Operating mode indication

26.9.1 General

OMI is a procedure used between an OMI initiator and an OMI responder. An HE STA that transmits a frame including an OM Control subfield is defined as an OMI initiator. An HE STA with `dot11OMIOptionImplemented` equal to true that receives a frame including an OM Control subfield is defined as an OMI responder.

An HE STA with `dot11OMIOptionImplemented` equal to true shall set the OM Control Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element it transmits to 1; otherwise, the HE STA shall set the OM Control Support subfield to 0. An HE AP shall set `dot11OMIOptionImplemented` to true, and the HE AP shall implement the reception of the OM Control subfield.

An OMI initiator may send to an OMI responder an individually addressed QoS Data, QoS Null, or Class 3 Management frame after association that contains the OM Control subfield and that solicits an immediate acknowledgment to indicate a change in its receive operating mode (ROM) as defined in 26.9.2 and/or transmit operating parameters (TOM) as defined in 26.9.3. An OMI responder implements the reception of an individually addressed QoS Data, QoS Null, or Class 3 Management frame that contains the OM Control subfield that indicates a change in ROM and/or TOM parameters.

The OMI initiator shall indicate a change in its ROM parameters by including the OM Control subfield in a QoS Data, QoS Null, or Class 3 Management frame that solicits acknowledgment and is addressed to the OMI responder as defined in 26.9.2.

NOTE 1—Frames that solicit an immediate acknowledgment are, for example, QoS Null frames and QoS Data frames with Normal Ack or Implicit BAR ack policy and Action frames.

An HE STA can change its operating mode setting using either the operating mode notification as described in 11.40 or the operating mode indication (OMI) procedure described in this subclause. An HE STA should not transmit an OM Control subfield and an Operating Mode field in the same PPDU. If a STA transmits

both an OM Control subfield and Operating Mode field in the same PPDU, then the OMI responder shall use the channel width and the maximum number of spatial streams indicated by the most recently received OM Control subfield or Operating Mode field from the OMI initiator.

NOTE 2—An OM Control field is transmitted before an Operating Mode field in the same frame.

The OMI initiator supports receiving PPDU with a bandwidth up to the value indicated by the Channel Width subfield and with a number of spatial streams, N_{ss} , as indicated in the Rx NSS subfield of the OM Control subfield and calculated in the Equation (26-4).

If the operating channel width of the STA is greater than 80 MHz, then the maximum number of spatial streams that the STA supports in reception for a given HE-MCS as a function of the received HE PPDU bandwidth BW at an HE STA transmitting an OM Control subfield is defined in Equation (26-4).

$$\text{floor} (Rx\text{-}NSS\text{-}from\text{-}OMI \times (Max\text{-}HE\text{-}NSS\text{-}at\text{-}BW / Max\text{-}HE\text{-}NSS\text{-}at\text{-}80)) \quad (26-4)$$

where

Rx-NSS-from-OMI is Rx NSS from the OM Control subfield transmitted by the STA
Max-HE-NSS-at-BW is the maximum NSS among all HE-MCS at BW MHz from the Supported HE-MCS And NSS Set field transmitted by the STA as described in 26.15.4
Max-HE-NSS-at-80 is the maximum NSS among all HE-MCS at 80 MHz from the Supported HE-MCS And NSS Set field transmitted by the STA

NOTE 3—If the operating channel width of the STA is greater than 80 MHz, then the Rx NSS subfield indicates the maximum number of spatial streams for PPDU bandwidths that are equal to or less than 80 MHz. If the operating channel width of the STA is less than or equal to 80 MHz, then the Rx NSS subfield indicates the maximum number of spatial streams, N_{SS} , that the STA supports in reception.

The VHT channel width and the VHT NSS allowed at an HE STA transmitting an OM Control subfield are defined in Table 26-9 to determine the allowed NSS when operating as HE STA using channel bandwidth of 160 MHz or 80+80 MHz.

Table 26-9—Setting of VHT Channel Width and VHT NSS at an HE STA transmitting the OM Control subfield

OM Control subfield	VHT capabilities of STA transmitting OM Control subfield		VHT NSS support of STA transmitting the OM Control subfield as a function of the PPDU bandwidth ($\times$ Max VHT NSS) (see requirements R1 and R2 at end of this table)					Location of 160 MHz center frequency if BSS bandwidth is 160 MHz	Location of secondary 80 MHz center frequency if BSS bandwidth is 80+80 MHz
	Channel Width	Supported Channel Width	Extended NSS BW Support	20 MHz	40 MHz	80 MHz	160 MHz	80+80 MHz	
0	0–2	0–3		1					
1	0–2	0–3		1	1				
2	0–2	0–3		1	1	1			
3	0	1		1	1	1	1/2		CCFS2
3	0	2		1	1	1	1/2	1/2	CCFS2
3	0	3		1	1	1	3/4	3/4	CCFS2
3	1	0		1	1	1	1		CCFS1

Table 26-9—Setting of VHT Channel Width and VHT NSS at an HE STA transmitting the OM Control subfield (*continued*)

OM Control subfield	VHT capabilities of STA transmitting OM Control subfield		VHT NSS support of STA transmitting the OM Control subfield as a function of the PPDU bandwidth (× Max VHT NSS) (see requirements R1 and R2 at end of this table)					Location of 160 MHz center frequency if BSS bandwidth is 160 MHz	Location of secondary 80 MHz center frequency if BSS bandwidth is 80+80 MHz
Channel Width	Supported Channel Width	Extended NSS BW Support	20 MHz	40 MHz	80 MHz	160 MHz	80+80 MHz		
3	1	1	1	1	1	1	1/2	CCFS1	CCFS2
3	1	2	1	1	1	1	3/4	CCFS1	CCFS2
3	1	3	2	2	2	2	1	CCFS1	CCFS1
3	2	0	1	1	1	1	1	CCFS1	CCFS1
3	2	3	2	2	2	1	1	CCFS1	CCFS1

R1: NSS support shall be rounded down to the nearest integer.

R2: The maximum NSS support shall be 8.

NOTE 1—Max VHT NSS as indicated by the value of the Rx NSS field. The Rx NSS field indicates the same Max HE NSS and Max VHT NSS. Max VHT NSS is at the bandwidth indicated in the VHT Capabilities element. For all allowed MCS values, the Max VHT NSS values are same, but the supported NSS can be different.

NOTE 2— $1/2 \times$ or $3/4 \times$ Max VHT NSS support might end up being 0, indicating no support.

NOTE 3—Any other combination than the ones listed in this table is reserved.

NOTE 4—CCFS1 refers to the value of the Channel Center Frequency Segment 1 field of the most recently transmitted VHT Operation element (if any) or HE Operation element.

NOTE 5—CCFS2 refers to the value of the Channel Center Frequency Segment 2 field of the most recently transmitted HT Operation element.

NOTE 6—CCFS1 is nonzero when the current BSS bandwidth is 160 MHz or 80+80 MHz and the NSS support is at least Max VHT NSS. CCFS2 is zero in this case.

NOTE 7—CCFS2 is nonzero when the current BSS bandwidth is 160 MHz or 80+80 MHz and the NSS support is less than Max VHT NSS. CCFS1 is zero in this case.

NOTE 8—At most one of CCFS1 and CCFS2 is nonzero.

NOTE 9—A supported multiple of Max VHT NSS applies to both transmit and receive. A supported multiple of Max HE NSS applies to receive.

NOTE 10—Some combinations of Supported Channel Width Set and Extended NSS BW support might not occur in practice.

The OMI initiator shall indicate a change in its TOM parameters by including the OM Control subfield in a QoS Data, QoS Null, or Class 3 Management frame that solicits an immediate acknowledgment frame and is addressed to the OMI responder as defined in 26.9.3.

A non-AP STA OMI initiator that sends an OM Control subfield with UL MU Disable subfield equal to 0, supports transmitting an HE TB PPDU with an RU allocation that is within the operating channel width indicated in the Channel Width subfield and with a number of space-time streams, N_{STS} , that is up to the value indicated by the Tx NSTS subfield of the OM Control subfield as defined in 26.9.3.

NOTE 4—To avoid possible frame loss, an OMI initiator can continue with its current operating channel width, active receive chains, and active transmit chains in HE TB PPDU and not suspend HE TB PPDU or Data frames in HE TB PPDU until it infers that the OMI responder has processed an OM Control subfield from the OMI initiator indicating any of the following changes:

- Reduced operating channel width
- Reduction in the number of active receive chains
- Reduction in the number of active transmit chains in HE TB PPDU
- Suspension of UL MU operation

The OMI initiator might make this inference from any combination of the following:

- By receiving a frame addressed to itself from the second HE STA in a PPDU with a bandwidth and NSS that are less than or equal to the channel width and NSS, respectively, indicated in the OM Control subfield.
- Based on the passage of time in some implementation-dependent way, which is beyond the scope of this standard.

NOTE 5—It might take a long time for a STA to change its operating mode following the transmission of the OM Control subfield. During that time the STA might not be able to receive frames, and this situation may result in frame loss. If a non-AP STA cannot tolerate frame loss during that period, it can set the Power Management subfield of the Frame Control field of the frame that carries the OM Control subfield to 1 to indicate that the STA has entered power save. When the non-AP STA has completed its operating mode change, it can send another frame (such as a QoS Null) with the Frame Control field Power Management subfield set to 0 to indicate that the STA has exited power save.

26.9.2 Receive operating mode (ROM) indication

ROM indication allows the OMI initiator to adapt the maximum operating channel width and/or the maximum number of spatial streams, N_{ss} , it can receive from the OMI responder.

An OMI initiator that sends a frame that includes an OM Control subfield should change its OMI parameters, Rx NSS and Channel Width as follows:

- When the OMI initiator changes a ROM parameter from higher to lower, it should make the change for that parameter only after the TXOP in which it received the immediate acknowledgment from the OMI responder.
- When the OMI initiator changes a ROM parameter from lower to higher, it should make the change for that parameter only after the TXOP in which it expects to receive acknowledgment from the OMI responder.

An OMI initiator that is an HE AP should be capable of receiving within an operating channel width and with N_{ss} that are up to the values of the most recently transmitted Channel Width subfield and Rx NSS subfield that the OMI initiator has successfully indicated in the OM Control subfield or in the Operating Mode field sent to any associated STA.

NOTE 1—In the event of transmission failure of the frame containing the OM Control subfield, the OMI initiator attempts the recovery procedure defined in 10.23.2.8.

The OMI responder shall update the operating channel width and the maximum N_{ss} values as obtained from the Channel Width and Rx NSS subfields, respectively, of the most recently received OM Control subfield sent by the OMI initiator to send SU PPDU and to assign an RU allocation in sent MU PPDU, subject to restrictions defined in 27.3.1.2, addressed to the OMI initiator in subsequent TXOPs. The OMI responder shall update the maximum N_{ss} value to a value determined from the Rx NSS subfield and Table 26-9 if the Channel Width subfield of the OM Control field indicates 160 MHz or 80+80 MHz.

After transmitting the acknowledgment for the frame containing the OM Control subfield, the OMI responder may transmit subsequent SU PPDU or MU PPDU that are addressed to the OMI initiator.

NOTE 2—The acknowledgment is transmitted a SIFS after the frame. A subsequent PPDU is a PPDU that is intended for the OMI initiator and need not be the PPDU immediately following the acknowledgment.

An OMI initiator that is a non-AP STA may set the DL MU-MIMO Resound Recommendation subfield to 1 in the OM Control field in frames addressed to an OMI responder that is an AP to indicate that the non-AP STA suggests that the AP resound the channel with the non-AP STA. An OMI initiator that is a non-AP STA and that has no recommendation on the AP's DL MU-MIMO operation shall set DL MU-MIMO Resound Recommendation subfield to 0.

An OMI responder that receives a frame that carries an OM Control field with the DL MU-MIMO Resound Recommendation field equal to 1 from an OMI initiator may resound the channel or increase the frequency of the channel sounding with the OMI initiator if the OMI responder sends a DL MU-MIMO PPDU addressed to the OMI initiator.

26.9.3 Transmit operating mode (TOM) indication

TOM indication allows the OMI initiator to suspend and resume responding to variants of the Trigger frame and TRS Control subfields per the UL MU Disable and UL MU Data Disable subfields settings as indicated in Table 9-24b or to adapt the maximum operating channel width and/or the maximum number of space-time streams, N_{STS} , that it can transmit in response to a triggering frame sent by the OMI responder

NOTE 1—TOM indication does not relate to transmissions in PPDU's other than HE TB PPDU's. An AP does not perform TOM indication as an OMI initiator.

An OMI initiator that is a non-AP STA may indicate changes in its transmit parameters by sending a frame that contains the OM Control subfield to the OMI responder. The OMI initiator shall set

- The UL MU Disable subfield to 1 to indicate that responding to a triggering frame is suspended (see 26.5.2).
 - An AP that is an OMI initiator shall set the UL MU Disable subfield to 0.
- The Tx NSTS subfield to the maximum N_{STS} that the STA will use for an HE TB PPDU sent in response to a triggering frame.
- The Channel Width subfield to the maximum operating channel width that the STA will use for an HE TB PPDU sent in response to a triggering frame.

NOTE 2—Responding to all Trigger frame variants, including MU-RTS Trigger frames and NFRP Trigger frames, is suspended.

If a non-AP HE STA has received the OM Control UL MU Data Disable RX Support field in the HE Capabilities element set to 1, then the non-AP HE STA, acting as an OMI initiator, may set the UL MU Disable subfield to 0 and the UL MU Data Disable subfield to 1 to indicate that only UL MU Data frame transmission is suspended. In other words, UL MU control response frame transmissions in response to a Basic Trigger frame are not suspended (see 26.5.2), responses to other Trigger frame variants are not suspended, and management frame transmissions are not suspended.

An OMI initiator shall set the UL MU Disable subfield to 0 and the UL MU Data Disable subfield to 0 to indicate resumption or continuation of participation in all triggered UL MU operations.

If an HE AP has set the OM Control UL MU Data Disable RX Support field in the HE Capabilities element it transmits to 0, an associated STA shall not set the UL MU Data Disable subfield in the OM Control field to 1.

An OMI initiator that sent a frame including the OM Control subfield should change its TOM parameters, Tx NSTS, UL MU Disable, UL MU Data Disable, and Channel Width as follows:

- If the OMI initiator changes a TOM parameter from higher to lower, it should make the change for that parameter only after the TXOP in which it received the immediate acknowledgment from the OMI responder.
- If the OMI initiator changes a TOM parameter from lower to higher, it should make the change for that parameter only after the TXOP in which it expects to receive acknowledgment from the OMI responder.

The TOM parameters UL MU Disable and UL MU Data Disable change from higher to lower if their values change from 0 to 1. The change of UL MU Disable from 1 to 0 and UL MU Data Disable from 0 to 1 is a change from lower to higher.

An OMI responder that receives a frame containing an OM Control subfield from an OMI initiator performs the operations described below in this subclause.

An AP OMI responder shall not send any triggering frames to a non-AP STA OMI initiator for subsequent TXOPs (see 26.5.2) if the UL MU Disable subfield is 1 in the most recently received OM Control subfield sent by the STA.

NOTE 3—A device might have multiple radios that can create difficult in-device coexistence challenges. The device might set UL MU Disable subfield to 1 and the UL MU Data Disable subfield to 0 if it has trouble responding to a triggering frame because the timing or high transmit power would cause interference with another radio in the device.

An OMI responder shall consider the OMI initiator as participating in UL MU operation for subsequent TXOPs if the UL MU Disable subfield is 0 in the most recently received OM Control subfield with the following restrictions:

- The maximum N_{STS} that the OMI initiator can transmit in an HE TB PPDU is indicated in the Tx NSTS subfield of the OM Control subfield.
- The maximum operating channel width over which the OMI initiator can transmit in an HE TB PPDU is indicated in the Channel Width subfield of the OM Control subfield.

An OMI responder that has transmitted the OM Control UL MU Data Disable RX Support subfield set to 1 shall regard an OMI initiator as capable of participating in UL MU operation only for transmitting acknowledgments if the UL MU Disable subfield is equal to 0 and the UL MU Data Disable subfield is equal to 1 in the most recently received OM Control subfield from that OMI initiator.

The OMI responder shall indicate a number of spatial streams, N_{SS} , in the User Info field of a Trigger frame, which contains the AID of the OMI initiator, that is less than or equal to the N_{STS} that is calculated from the Tx NSTS subfield of the OM Control subfield received from the OMI initiator.

The OMI responder shall indicate, in the RU Allocation subfield of the User Info field of a triggering frame addressed to the OMI initiator, an RU allocation that is within the operating channel width specified in the Channel Width subfield of the OM Control subfield received from the OMI initiator and subject to the restrictions defined in 27.3.1.2.

26.10 Spatial reuse operation

26.10.1 General

The objective of HE spatial reuse operation is to allow the medium to be reused more often between OBSSs in dense deployment scenarios by the early identification of signals from overlapping basic service sets (OBSSs) and interference management.

There are two independent spatial reuse modes: OBSS PD-based spatial reuse and PSR-based spatial reuse.

An HE AP participating in spatial reuse may request an associated non-AP HE STA to gather information regarding the neighborhood by sending a Beacon request (see 9.4.2.21.7) following the procedure described in 11.11. An HE AP shall not set a measurement mode in a Beacon request to an associated STA to a mode for which the STA has not explicitly indicated support via the RM Enabled Capabilities element (see 9.4.2.44). An HE AP that sends a Beacon request for this purpose

- May request that the non-AP HE STA gather information of BSSs matching a particular BSSID and/or SSID.
- May request that the non-AP HE STA generate a report that is only for the channel on which the requesting AP is operating or to which the requesting AP is considering switching.
- Shall request that the non-AP HE STA include the HE Operation element of neighboring HE APs in order to help determine the BSS color information of the neighboring APs.

A non-AP HE STA that performs spatial reuse operation shall have dot11RadioMeasurementActivated set to true and shall respond to a Beacon request from its associated AP with a Beacon report as described in 11.11.

A Class B device as defined in 27.3.15.3 shall not operate with the procedures defined in this subclause.

26.10.2 OBSS PD-based spatial reuse operation

26.10.2.1 General

OBSS PD-based spatial reuse operation comprises two types of operation. The first type is defined in 26.10.2.2 and allows a STA, under specific conditions, to ignore an inter-BSS PPDU using a non-SRG OBSS PD level. The second type is defined in 26.10.2.3 and allows a STA, under specific conditions, to ignore inter-BSS PPDU that are identified as being SRG PPDU, using an SRG OBSS PD level. In addition to these differences between the two types, Non-SRG OBSS PD Min offset is fixed and defined in the specification while the SRG OBSS PD Min offset can be defined by the AP. A STA may operate using one of the two types, neither type, or both types simultaneously.

26.10.2.2 General operation with non-SRG OBSS PD level

If the PHY of a STA issues a PHY-CCA.indication(BUSY) followed by a PHY-RXSTART.indication due to a PPDU reception, then the STA's MAC sublayer

- a) May issue a PHY-CCARESET.request primitive before the end of the PPDU and not update its basic NAV timer based on the PPDU, or
- b) May not update its basic NAV timer based on the PPDU if all the following conditions are met:
 - 1) The STA has not set the TXVECTOR parameter SPATIAL_REUSE to the value PSR_AND_NON_SRG_OBSS_PD_PROHIBITED in any HE PPDU it has transmitted in the current beacon period and in the previous beacon period.
 - 2) The most recently received Spatial Reuse Parameter Set element from its associated AP had the Non-SRG OBSS PD SR Disallowed subfield equal to 0, or the non-AP STA has not received a Spatial Reuse Parameter Set element from its associated AP, or the STA is an AP and its most recently transmitted Spatial Reuse Parameter Set element had the Non-SRG OBSS PD SR Disallowed subfield equal to 0, or the STA is an AP and has not transmitted a Spatial Reuse Parameter Set element.
 - 3) The received PPDU is an inter-BSS PPDU (see 26.2.2), and the received PPDU is not a non-HT PPDU carrying a response frame (Ack, BlockAck, or CTS frame); or the received PPDU contains a CTS, a PHY-CCA.indication transition from BUSY to IDLE occurred within the

- PIFS time immediately preceding the received CTS, and that transition corresponded to the end of an inter-BSS PPDU that contained an RTS that was ignored following this procedure.
- 4) The STA is operating with an SRG OBSS PD level as described in 26.10.2.3, and the received PPDU is not an SRG PPDU; or the STA is not operating with an SRG OBSS PD level.
 - 5) The SPATIAL_REUSE subfield in the HE-SIG-A field (if present) of the received PPDU is not set to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED.
 - 6) The received signal strength level, which is measured from the L-STF or L-LTF fields of the PPDU or the PHY SYNC field, shortSYNC field, or Long PHY SYNC field, whichever exists and is used to determine PHY-CCA.indication, is below the non-SRG OBSS PD level. The non-SRG OBSS PD level is defined in 26.10.2.4. If the STA has dot11HEPSROptionImplemented set to true, it also follows the rules defined in 26.10.4 to determine non-SRG OBSS PD level.
 - 7) The PPDU is not one of the following:
 - i) A non-HE PPDU that carries a frame where the RA field is equal to the STA MAC address
 - ii) A non-HE PPDU that carries a Public Action frame
 - iii) A non-HE PPDU that carries a VHT/HE NDP Announcement frame or Fine Timing Measurement frame
 - iv) A non-HE NDP

NOTE 1—A STA cannot perform SR over an HE sounding NDP or HE TB feedback NDP (see 26.11.6).

If the frame is carried in an HE ER SU PPDU that is identified as an inter-BSS PPDU (where power of the L-STF/L-LTF symbols is boosted 3 dB), 3 dB shall be subtracted from the received signal strength measured from the L-STF or L-LTF fields of the PPDU, prior to using it to determine PHY-CCA.indication and to comparing it to the non-SRG OBSS PD level as described in this subclause, to compensate for the power difference.

NOTE 2—If a received CF-End frame satisfies the conditions above, neither the issuance of a PHY-CCARESET.request nor the choice to not update the basic NAV timer results in the cancellation of the NAV, which would normally occur following the reception of a CF-End frame.

The PHY-CCARESET.request primitive shall be issued at the end of the PPDU if the PPDU is an HE SU PPDU or an HE ER SU PPDU and the RXVECTOR parameter SPATIAL_REUSE indicates SR_DELAYED.

NOTE 3—A STA sets the TXVECTOR parameter SPATIAL_REUSE to SR_DELAYED in a PPDU if it allows OBSS PD-based spatial reuse operation, but only after the end of the PPDU.

NOTE 4—An AP can get protection equivalent to SR_DELAYED by transmitting the Trigger frame in a non-HT PPDU or HT PPDU with the TXVECTOR parameter AGGREGATION set to 0 instead of in a VHT PPDU.

If the PHY-CCARESET.request primitive is issued before the end of the received PPDU, and a TXOP is initiated within the duration of the received PPDU, then the TXOP and the duration of the transmitted PPDU within that TXOP shall be limited to the duration of the received PPDU if the received PPDU is an HE MU PPDU and the RXVECTOR parameter SPATIAL_REUSE indicates SR_RESTRICTED.

NOTE 5—A STA sets the TXVECTOR parameter SPATIAL_REUSE to SR_RESTRICTED in a PPDU if it allows OBSS PD-based spatial reuse operation, but only before the end of the PPDU.

A STA that ignores a PPDU following the procedure described in this subclause is deemed to perform non-SRG OBSS PD-based spatial reuse.

26.10.2.3 General operation with SRG OBSS PD level

If the PHY of a STA issues a PHY-CCA.indication(BUSY) followed by a PHY-RXSTART.indication due to a PPDU reception, then the STA's MAC sublayer

- a) May issue a PHY-CCARESET.request primitive before the end of the PPDU and not update its basic NAV timer based on the PPDU, or
- b) May not update its basic NAV timer based on the PPDU if all the following conditions are met:
 - 1) The received PPDU is an SRG PPDU (see 26.2.3)
 - 2) The received signal strength level, which is measured from the L-STF or L-LTF fields of the PPDU or the PHY SYNC field, shortSYNC field, or Long PHY SYNC field, whichever exists and is used to determine PHY-CCA.indication, is below the SRG OBSS PD level. The SRG OBSS PD level is defined in 26.10.2.4. If the STA has dot11HEPSROptionImplemented set to true, it also follows the rules defined in 26.10.4 to determine SRG OBSS PD level.
 - 3) The PPDU is not one of the following:
 - i) A non-HE PPDU that carries a frame where the RA field is equal to the STA MAC address
 - ii) A non-HE PPDU that carries a Public Action frame
 - iii) A non-HE PPDU that carries a VHT/HE NDP Announcement frame or Fine Timing Measurement frame
 - iv) A non-HE NDP

NOTE 1—A STA cannot perform SR over an HE sounding NDP or HE TB feedback NDP (see 26.11.6).

If the frame is carried in an HE ER SU PPDU that is identified as an inter-BSS PPDU (where power of the L-STF/L-LTF symbols is boosted 3 dB), 3 dB shall be subtracted from the received signal strength measured from the L-STF or L-LTF fields of the PPDU, prior to using it to determine PHY-CCA.indication and to comparing it to the SRG OBSS PD level as described in this subclause, to compensate for the power difference.

NOTE 2—If a received CF-End frame satisfies the conditions above, neither the issuance of a PHY-CCARESET.request nor the choice to not update the basic NAV timer results in the cancellation of the NAV, which would normally occur following the reception of a CF-End frame.

The PHY-CCARESET.request primitive shall be issued at the end of the PPDU if the PPDU is an HE SU PPDU or an HE ER SU PPDU and the RXVECTOR parameter SPATIAL_REUSE indicates SR_DELAYED.

NOTE 3—An AP can get protection equivalent to SR_DELAYED by transmitting the Trigger frame in a non-HT PPDU or HT PPDU with the TXVECTOR parameter AGGREGATION set to 0 instead of in a VHT PPDU.

If the PHY-CCARESET.request primitive is issued before the end of the received PPDU, and a TXOP is initiated within the duration of the received PPDU, then the TXOP and the duration of the transmitted PPDU within that TXOP shall be limited to the duration of the received PPDU if the received PPDU is an HE MU PPDU and the RXVECTOR parameter SPATIAL_REUSE indicates SR_RESTRICTED.

NOTE 4—The restriction, in addition to the TXOP limit, of the PPDU duration within the TXOP is included in the above paragraph related to SR_RESTRICTED as there are conditions where the TXOP limit can be exceeded (see 10.23.2.9).

An AP that sends a Spatial Reuse Parameter Set element with the SRG Information Present subfield in the SR Control field set to 1 shall set the SRG BSS Color Bitmap and SRG Partial BSSID Bitmap fields as follows:

- If the transmitting AP is in the same ESS as another AP (i.e., with the same SSID and connected by a DS) or is controlled by the same external management entity as another AP (irrespective of SSID), then the transmitting AP may set the SRG BSS Color Bitmap and/or SRG Partial BSSID Bitmap fields that correspond to that other AP to 1.
- Otherwise, the AP shall set the bits in the SRG BSS Color Bitmap and/or SRG Partial BSSID Bitmap to 0.

If an HE AP determines values for dot11SRGAPBSSColorBitmap and dot11SRGAPBSSIDBitmap (i.e., the SRG for the AP's own transmissions), then the values shall be determined according to the rules above in this subclause.

26.10.2.4 Adjustment of OBSS PD and transmit power

If using OBSS PD-based spatial reuse, an HE STA shall maintain an OBSS PD level and may adjust this OBSS PD level in conjunction with its transmit power and a value, $PPDU_BW$, derived from the received PPDU. The adjustment shall be made in accordance with Equation (26-5).

$$OBSS_PD_{level} \leq \max(OBSS_PD_{min}, \min(OBSS_PD_{max}, OBSS_PD_{min} + (TX_PWR_{ref} - TX_PWR))) + \log_{10}(PPDU_BW / 20 \text{ MHz}) \quad (26-5)$$

where $PPDU_BW$ is determined by Table 26-10 using the following RXVECTOR parameter of the received PPDU:

- CH_BANDWIDTH if present
- CH_BANDWIDTH_IN_NON_HT if present and if CH_BANDWIDTH is not present
- DATARATE if neither CH_BANDWIDTH nor CH_BANDWIDTH_IN_NON_HT is present

Table 26-10— $PPDU_BW$ determination

DATARATE	CH_BANDWIDTH or CH_BANDWIDTH_IN_NON_HT	$PPDU_BW$
N/A	CBW20, HT_CBW20, NON_HT_CBW20	20 MHz
N/A	CBW40, HT_CBW40, NON_HT_CBW40	40 MHz
N/A	CBW80, HE-CBW-PUNC80-PRI, HE-CBW-PUNC80-SEC	80 MHz
N/A	CBW160, CBW80+80, HE-CBW-PUNC160-PRI20, HE-CBW-PUNC80+80-PRI20, HE-CBW-PUNC160-SEC40, HE-CBW-PUNC80+80-SEC40	160 MHz
1, 2, 5.5, or 11	N/A	20 MHz

The value of the $OBSS_PD_{level}$ is applicable to the start of a 20 MHz PPDU received on the primary 20 MHz channel. If the bandwidth of the received PPDU differs from 20 MHz, then the value of the $OBSS_PD_{level}$ is increased by $10 \log(\text{bandwidth}/20 \text{ MHz})$, using the bandwidth in MHz indicated by the value of RXVECTOR parameter CH_BANDWIDTH or CH_BANDWIDTH_IN_NON_HT if present.

$TX_PWR_{ref} = 21 \text{ dBm}$ for non-AP STAs.

$TX_PWR_{ref} = 21 \text{ dBm}$ for an AP with the Max HE-MCS For 3 SS subfield in the Tx HE-MCS Map $\leq 80 \text{ MHz}$ subfield in the Supported HE-MCS and NSS Set field of its HE Capabilities element field set to 3.

$TX_PWR_{ref} = 25$ dBm for an AP with the Max HE-MCS For 3 SS subfield in the Tx HE-MCS Map ≤ 80 MHz subfield in the Supported HE-MCS and NSS Set field of its HE Capabilities element field set to a value other than 3.

TX_PWR is the combined transmit power at the transmit antenna connector of all antennas in dBm and is set following the rules in 11.7.6 and, for transmission of HE TB PPDU, also following the rules in 27.3.15.2.

NOTE 1—The TX_PWR_{ref} is 4 dB higher for APs with more than 2 spatial streams because those APs typically have higher transmit power than other devices and the OBSS PD procedure is based on a relative reduction of power.

NOTE 2—A STA might adjust its OBSS PD level, provided that Equation (26-5) is satisfied. As an example, a non-AP HE STA might monitor the beacons transmitted by the AP to which it is associated, measure their RSSI, subtract a value of 25 dB (as an approximation to the required RSNI) to provide a candidate OBSS PD level, and if Equation (26-5) is satisfied, adjust its OBSS PD level to the candidate level.

The adjustment rule is illustrated in Figure 26-11.

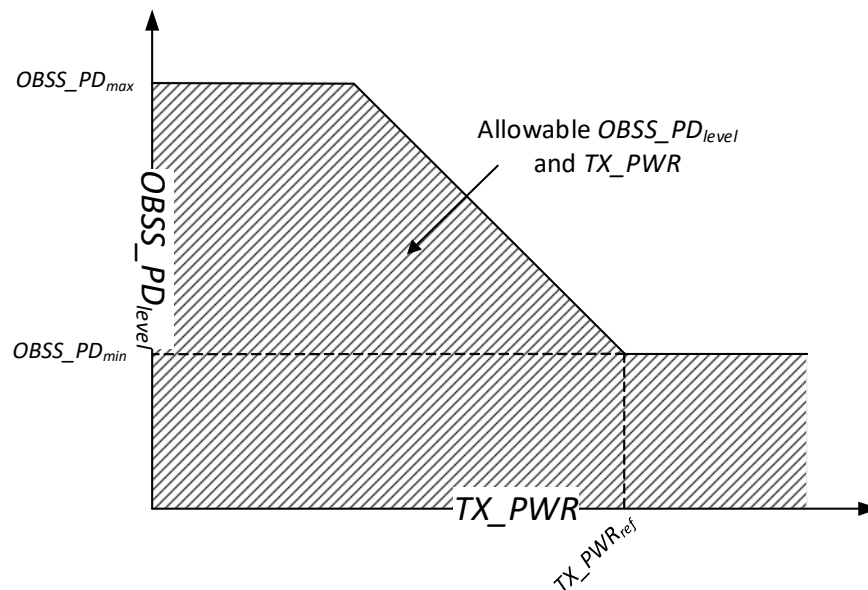


Figure 26-11—Illustration of the adjustment rule for OBSS PD and TX_PWR

An AP may define SRG OBSS PD Min Offset and SRG OBSS PD Max Offset values that are used by its associated STAs to derive an SRG OBSS PD level for determining reception behavior for inter-BSS PPDU that are determined to be SRG PPDU. An AP may define a non-SRG OBSS PD Max Offset value that is used by its associated STAs to derive a non-SRG OBSS PD level for determining reception behavior for inter-BSS PPDU that are not determined to be SRG PPDU. The values of SRG OBSS PD Min Offset, SRG OBSS PD Max Offset, and Non-SRG OBSS PD Max Offset are transmitted to associated STAs within the Spatial Reuse Parameter Set element.

An AP transmitting a Spatial Reuse Parameter Set element shall respect the following constraints:

- $-82 \leq -82 + \text{SRG OBSS PD Min Offset} \leq -62$
- $\text{SRG OBSS PD Min Offset} \leq \text{SRG OBSS PD Max Offset}$
- $-82 + \text{SRG OBSS PD Max Offset} \leq -62$
- $-82 + \text{Non-SRG OBSS PD Max Offset} \leq -62$

An HE STA shall maintain a non-SRG OBSS PD level, with its value selected by respecting the OBSS PD level condition in Equation (26-5) but with Non-SRG OBSS PD Min and Non-SRG OBSS PD Max in place of $OBSS_PD_{min}$ and $OBSS_PD_{max}$, respectively. A non-AP STA shall determine Non-SRG OBSS PD Min and Non-SRG OBSS PD Max according to Table 26-11. An HE AP shall set Non-SRG OBSS PD Min to -82 dBm and Non-SRG OBSS PD Max to -82 dBm + dot11NonSRGAPOBSSPDMaxOffset.

Table 26-11—Non-SRG OBSS PD Min and Non-SRG OBSS PD Max values for non-AP STAs

Non-SRG OBSS PD SR Disallowed field in Spatial Reuse Parameter Set element	Non-SRG Offset Present field in Spatial Reuse Parameter Set element	Value of Non-SRG OBSS PD Min (dBm)	Value of Non-SRG OBSS PD Max (dBm)
Not applicable if the Spatial Reuse Parameter Set element is not received	Not applicable if the Spatial Reuse Parameter Set element is not received	-82	-62
0	0	-82	-62
0	1	-82	$-82 + \text{Non-SRG OBSS PD Max Offset}$
1	N/A	-82	-82

An HE STA shall maintain an SRG OBSS PD level, with its value selected by respecting the OBSS PD level condition in Equation (26-5) but with SRG OBSS PD Min and SRG OBSS PD Max in place of $OBSS_PD_{min}$ and $OBSS_PD_{max}$, respectively. A non-AP STA shall determine SRG OBSS PD Min and SRG OBSS PD Max using Table 26-12. An HE AP shall set SRG OBSS PD Min to $-82 + \text{dot11SRGAPOBSSPDMinOffset}$ dBm and SRG OBSS PD Max to $-82 + \text{dot11SRGAPOBSSPDMaxOffset}$ dBm. An HE AP may transmit SRG OBSS PD Min and SRG OBSS PD Max offset values that are different from the ones that it uses.

Table 26-12—SRG OBSS PD Min and SRG OBSS PD Max values for non-AP STAs

SRG Information Present field in Spatial Reuse Parameter Set element	Value of SRG OBSS PD Min (dBm)	Value of SRG OBSS PD Max (dBm)
Not applicable if the Spatial Reuse Parameter Set element is not received	N/A see NOTE	N/A see NOTE
0	N/A see NOTE	N/A see NOTE
1	$-82 + \text{SRG OBSS PD Min Offset}$	$-82 + \text{SRG OBSS PD Max Offset}$
NOTE—If SRG Information is not present, a STA cannot determine a PPDU to be SRG and therefore will not use SRG OBSS PD Min or SRG OBSS PD Max values.		

The Spatial Reuse Parameter Set element is optionally present in Beacon frames, Probe Response frames, and (Re)Association Response frames.

26.10.2.5 OBSS PD SR transmit power restriction period

If a STA ignores an inter-BSS PPDU following the procedure in 26.10.2.3 using a chosen SRG OBSS PD level, or following the procedure in 26.10.2.2 using a chosen non-SRG OBSS PD level, then the STA shall start an OBSS PD SR transmit power restriction period. This OBSS PD SR transmit power restriction period shall be terminated at the end of the TXOP that the STA gains once its backoff reaches zero.

If a STA starts an OBSS PD SR transmit power restriction period with a chosen non-SRG OBSS PD level, the STA's transmit power as measured at the transmit antenna connector shall be equal to or lower than the TX_PWR_{max} , calculated with the chosen non-SRG OBSS PD level using Equation (26-6) and the appropriate non-SRG parameters according to Table 26-11, for the transmissions of any PPDU that is not carrying a frame that is allowed to be sent without regard to the busy/idle state of the medium until the end of the OBSS PD SR transmit power restriction period.

If a STA starts an OBSS PD SR transmit power restriction period with a chosen SRG OBSS PD level, the STA's transmit power as measured at the transmit antenna connector shall be equal to or lower than the TX_PWR_{max} , calculated with the chosen SRG OBSS PD level using Equation (26-6) and the appropriate SRG parameters according to Table 26-12, for the transmissions of any PPDU that is not carrying a response frame that is allowed to be sent without regard to the busy/idle state of the medium until the end of the OBSS PD SR transmit power restriction period.

NOTE 1—Examples of frames that are transmitted without regard to the busy/idle state of the medium include but are not limited to a frame contained in an HE TB PPDU that is a response to a Trigger frame with the CS Required subfield set to 0 and an Ack or BlockAck frame sent as an immediate response.

A STA may have multiple ongoing OBSS PD SR transmit power restriction periods that overlap in time.

NOTE 2—The STA's transmit power is always equal to or lower than the minimum TX_PWR_{max} among all TX_PWR_{max} from ongoing OBSS PD SR transmit power restriction periods.

NOTE 3—Equation (26-6) is equivalent to the condition defined in Equation (26-5). The STA can derive $OBSS_PD_{level}$ from its transmit power or can derive TX_PWR_{max} from $OBSS_PD_{level}$.

$$TX_PWR_{max} = \begin{cases} \text{unconstrained, if } OBSS_PD_{level} \leq OBSS_PD_{min} \\ TXPWR_{ref} - (OBSS_PD_{level} - OBSS_PD_{min}), \text{ if } OBSS_PD_{max} \geq OBSS_PD_{level} > OBSS_PD_{min} \end{cases} \quad (26-6)$$

NOTE 4—Anytime, even if TX_PWR_{max} is unconstrained, the STA has to respect the transmit power restrictions defined in 11.7.6.

An example of OBSS PD SR operation is shown in Figure 26-12. In this example, STA SR S2

- Receives the PPDU from S1 and, if it classifies the PPDU as inter-BSS PPDU, ignores the PPDU using OBSS PD-based spatial reuse with non-SRG OBSS PD, starts the OBSS PD SR transmit power restriction period 1 with $TX_PWR_{max} 1$, and decrements its backoff counter until the reception of the PPDU from D1.
- Receives the PPDU from D1 and, if it classifies the PPDU as inter-BSS PPDU, ignores the PPDU (if it chooses to do so) using OBSS PD-based spatial reuse with non-SRG OBSS PD, starts the OBSS PD SR transmit power restriction period 2 with $TX_PWR_{max} 2$, and decrements its backoff counter until the reception of the PPDU from S1'.
- Defers during the TXOP S1" set by the intra-BSS PPDU from S1" that belongs to its own BSS and, at the end of the TXOP S1", resumes the decrement of its backoff until the reception of the PPDU from S1'.
- Receives the PPDU from S1' and, if it classifies the PPDU as SRG PPDU, ignores the PPDU (if it chooses to do so) using OBSS PD-based spatial reuse with SRG OBSS PD, starts the OBSS PD SR

transmit power restriction period 3 with TX_PWRmax 3, and decrements its backoff counter until the counter reaches zero because it does not receive the PPDU from D1'.

- Starts transmitting a PPDU with a TX_PWRmax equal to $\min(\text{TX_PWRmax } 1, \text{TX_PWRmax } 2, \text{TX_PWRmax } 3)$ and respects this transmit power restriction until the end of the SR TXOP.

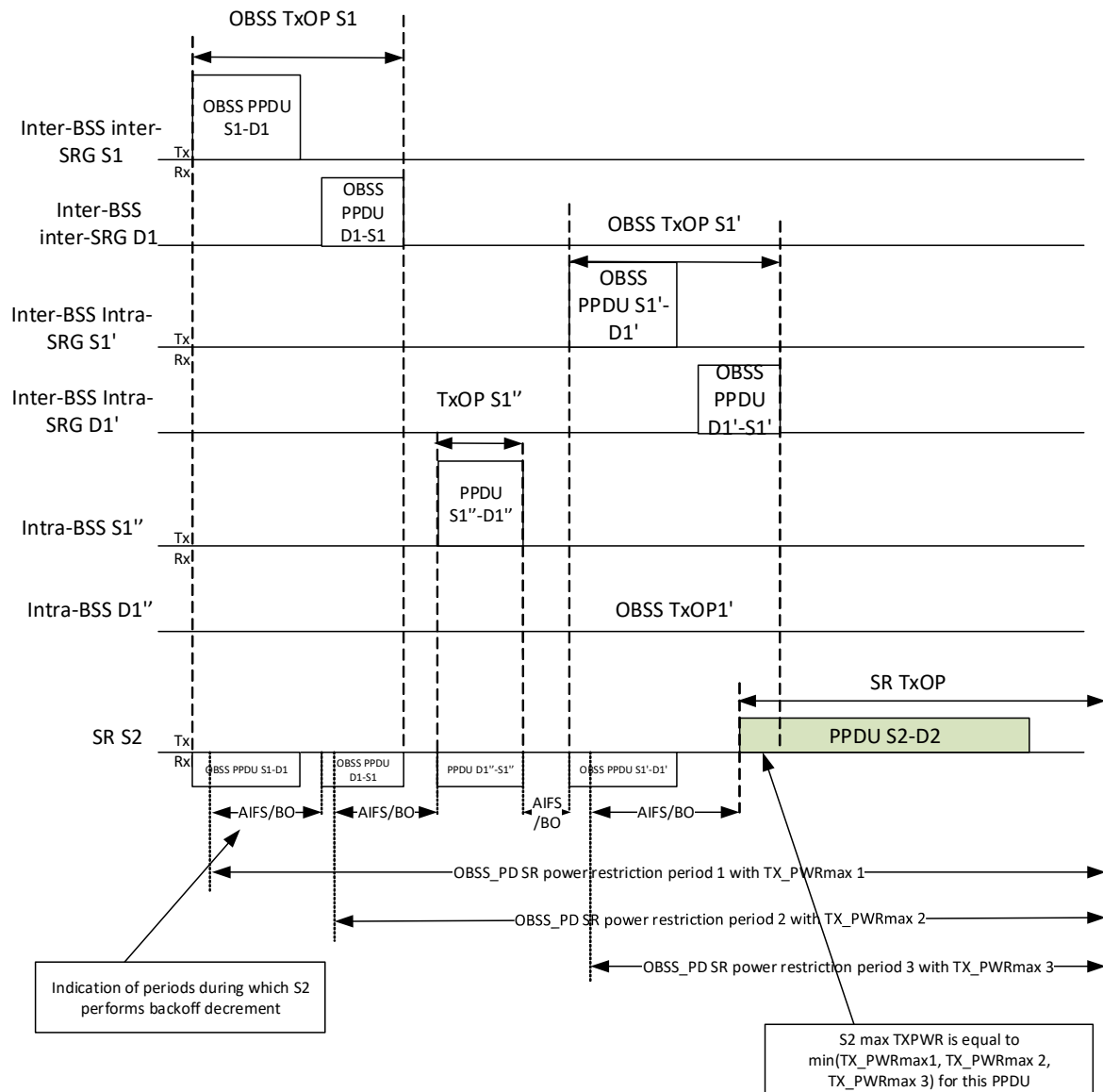


Figure 26-12—Example of OBSS PD SR operation

26.10.2.6 OBSS PD-based spatial reuse backoff procedure

If an HE STA ignores an inter-BSS PPDU following the procedure in 26.10.2.2, the HE STA may resume EDCAF procedures after the PHY-CCARESET.request primitive is sent, provided that the medium condition is not otherwise indicated as BUSY.

26.10.3 PSR-based spatial reuse operation

26.10.3.1 General

An HE STA supporting PSR-based PSRT PPDU transmission indicates this support by setting the PSR-based SR Support subfield to 1 in the HE PHY Capabilities Information field in the HE Capabilities element (see Table 9-322b). An HE-STA supporting PSR-based PSRT PPDU reception indicates this support by setting the PSR Responder subfield to 1 in the HE MAC Capabilities Information field in the HE Capabilities element (see Table 9-322a).

An HE STA shall set the PSR-based SR Support field to 1 in the HE Capabilities element it transmits if it supports transmitting an PSRT PPDU under the conditions specified in this subclause; otherwise, the STA shall set the PSR-based SR Support field to 0.

STAs that receive a Spatial Reuse Parameter Set element from their associated AP that has a value of 1 in the PSR Disallowed subfield shall not perform PSR-based SR transmissions.

An PSR opportunity is identified from the value of the RXVECTOR parameter SPATIAL_REUSE of an HE TB PPDU and/or the contents of a Trigger frame. An HE STA may initiate an SR transmission during an PSR opportunity for the duration of an ongoing PPDU when certain conditions, designed to avoid interfering with the reception of the ongoing PPDU at the recipient, are met. If the RXVECTOR parameter SPATIAL_REUSE of the ongoing PPDU has the value PSR_DISALLOW or PSR_AND_NON_SRG_OBSS_PD_PROHIBITED, no PSR-based SR transmission is allowed for the duration of that PPDU.

An AP sending a Trigger frame may set the Spatial Reuse n field(s) in the UL Spatial Reuse subfield in the Common Info field of the Trigger frame to PSR_DISALLOW or, if permitted by the rules in 26.11.6, to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED to disallow OBSS STAs from performing PSR-based SR transmission during the ensuing uplink PPDU duration. An AP sending a Trigger frame shall not set the SR field in the Common Info field of the Trigger frame to SR_DELAYED or SR_RESTRICTED.

An HE STA shall set the SR Responder subfield in the HE Capabilities element it transmits to 1 if dot11SRResponderOptionImplemented is true; otherwise, the HE STA shall set the SR Responder subfield to 0.

26.10.3.2 PSR-based spatial reuse initiation

An HE STA identifies an PSR opportunity if the following two conditions are met:

- a) The STA receives a PHY-RXSTART.indication corresponding to the reception of a PSRR PPDU that is identified as an inter-BSS PPDU (see 26.2.2).
- b) An PSRT PPDU is queued for transmission, and the intended transmit power of the PSRT PPDU in dBm minus $\log_{10}(PPDU_BW / 20 \text{ MHz})$ dB is below the value of $PSR - RPL$, where $PPDU_BW$ is determined from Table 26-10 using the TXVECTOR parameter CH_BANDWIDTH or CH_BANDWIDTH_IN_NON_HT of the PSRT PPDU, whichever is present, and PSR is the value obtained from Table 27-23 based on at least one of the following:
 - 1) The value of the UL Spatial Reuse field in the Common Info field of the Trigger frame of the PSRR PPDU.
 - 2) The value of the RXVECTOR parameter Spatial Reuse of the HE TB PPDU that follows the PSRR PPDU.

RPL is the combined transmit power at the receive antenna connector, over the PSRR PPDU bandwidth, during the non-HE portion of the HE PPDU preamble of the triggering PPDU, averaged over all antennas used to receive the PPDU.

An HE STA that identifies an PSR opportunity may choose to not perform NAV update operations normally executed based on the receipt of the RXVECTOR parameter TXOP_DURATION and the Trigger frame Duration field. See Figure 26-13. A STA that identifies an PSR opportunity may issue a PHY-CCARESET.request to ignore the associated HE TB PPDU(s) that are triggered by the Trigger frame of the PSRR PPDUs and that occur within $aSIFSTime + aRxPHYStartDelay + 2 \times aSlotTime$ of the end of the last symbol on the air of the PPDU that contained the Trigger frame, provided that the value of the RXVECTOR parameter BSS_COLOR of the HE TB PPDU matches the BSS color of the PSRR PPDUs. A STA that identifies an PSR opportunity shall not transmit an PSRT PPDUs that terminates beyond the PPDU duration of the HE TB PPDU that is triggered by the Trigger frame of the PSRR PPDUs.

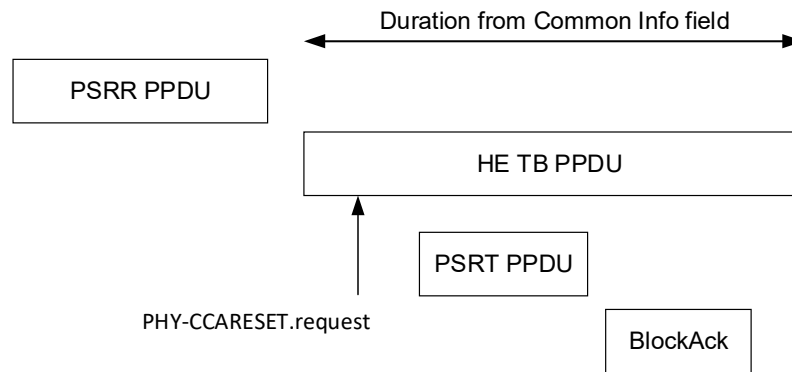


Figure 26-13—PSRR PPDUs spatial reuse

26.10.3.3 PSR-based spatial reuse backoff procedure

If an HE STA identifies an PSR opportunity as allowed in 26.10.3.2, the HE STA may continue the countdown of an existing backoff procedure, provided that the medium condition is not otherwise indicated as BUSY. If the HE STA receives another PPDU during the backoff procedure, it shall suspend its backoff; subsequently, if an PSR opportunity is identified based on the identification of the new PPDU as an PSRR PPDUs, then the STA may resume its backoff procedure. The TXOP that the HE STA gains once its backoff reaches zero shall not extend beyond the PSR opportunity endpoint that is the earliest ending of all durations of all PSRR PPDUs that were used to confirm the PSR opportunity and all durations indicated in the Common Info fields of Trigger frames within all PSRR PPDUs that were used to confirm the PSR opportunity.

If the HE STA is employing $OBSS_PD_{level}$ as a threshold for determination of an IDLE medium condition prior to the reception of an PSRR PPDUs per the rules specified in 26.10.2, the intended transmit power of the next PSRT PPDUs in the transmission queue as measured at the transmit antenna connector shall be equal to or lower than the TX_PWR_{max} , calculated with this specific $OBSS_PD_{level}$ using Equation (26-6).

After a STA has identified the start of an PSR opportunity, and until the PSR opportunity endpoint is reached, the transmission of any PPDU by the STA shall be limited by the transmit power restrictions identified in 26.10.3.

26.10.3.4 UL Spatial Reuse subfield of Trigger frame

An AP with dot11HEPSROptionImplemented set to true that transmits a Trigger frame may determine the value of the UL Spatial Reuse subfield of the Common Info field of the Trigger frame for each 20 MHz subchannel for a 20 MHz, 40 MHz, or 80 MHz PPDU or for each 40 MHz subchannel for an 80+80 MHz or 160 MHz PPDU by selecting the row in Table 27-23 that has a numerical value in the “Meaning” column

that is the highest value that is less than or equal to the value of the computed MAC parameter PSR_INPUT as follows:

$$\text{PSR_INPUT} = \text{TX_PWR}_{AP} + \text{Acceptable Receiver Interference Level}_{AP} \quad (26-7)$$

where

TX_PWR_{AP} is the total power at the antenna connector, in dBm, for that 20 MHz subchannel for a 20 MHz, 40 MHz, or 80 MHz PPDU or for that 40 MHz subchannel for an 80+80 MHz or 160 MHz PPDU, over all antennas used to transmit the PSRR PPDU containing the Trigger frame

Acceptable Receiver Interference Level_{AP} is a value in dBm for that 20 MHz subchannel for a 20 MHz, 40 MHz, or 80 MHz PPDU or for that 40 MHz subchannel for an 80+80 MHz or 160 MHz PPDU and should be set to the expected receive signal power indicated by the UL Target Receive Power subfield in the Trigger frame for the highest HE-MCS of the ensuing HE TB PDUs minus the minimum SNR value that yields $\leq 10\%$ PER for that MCS minus a safety margin value not to exceed 5 dB as determined by the AP

An AP with dot11HEPSROptionImplemented set to true that transmits a Trigger frame may set the value of the UL Spatial Reuse subfield of the Common Info field of the Trigger frame in each 20 MHz bandwidth for a 20 MHz, 40 MHz, or 80 MHz PPDU or in each 40 MHz bandwidth for an 80+80 or 160 MHz PPDU to PSR_DISALLOW.

An AP with dot11HEPSROptionImplemented set to false that transmits a Trigger frame shall set the value of the UL Spatial Reuse subfield of the Common Info field of the Trigger frame in each 20 MHz bandwidth for a 20 MHz, 40 MHz, or 80 MHz PPDU or in each 40 MHz bandwidth for an 80+80 or 160 MHz PPDU to PSR_DISALLOW.

26.10.3.5 PSRT PPDU transmission requirements

An HE STA that identifies an PSR opportunity shall not transmit a frame during the PSR opportunity that elicits a response transmission from a STA from which it has not received an HE Capabilities element with the PSR Responder subfield equal to 1. An HE STA that identifies an PSR opportunity shall not transmit a frame that does not include a CAS Control subfield with the PSRT PPDU subfield set to 1 and that solicits a response transmission during that PSR opportunity.

26.10.3.6 PSRT PPDU reception and response transmission requirements

An HE STA that receives a PPDU that contains at least one frame with a CAS Control subfield with an PSRT PPDU subfield equal to 1 shall not transmit a response PPDU elicited by the received PPDU if all outstanding PSR and OBSS PD transmit power requirements are not met by the response transmission.

26.10.4 Interaction of OBSS PD and PSR-based spatial reuse

An HE STA with dot11HEPSROptionImplemented set to true that receives a PPDU that is identified as an inter-BSS PPDU with a value other than PSR_DISALLOW or PSR_AND_NON_SRG_OBSS_PD_PROHIBITED for the RXVECTOR parameter SPATIAL_REUSE and fails to identify an PSR opportunity based on the receipt of the PPDU shall disable OBSS PD SR operation on this PPDU.

An HE STA with dot11HEPSROptionImplemented set to true that receives a PPDU that is identified as an inter-BSS PPDU with a value other than PSR_DISALLOW, SR_DELAYED, or PSR_AND_NON_SRG_OBSS_PD_PROHIBITED for the RXVECTOR parameter SPATIAL_REUSE and

identifies an PSR opportunity based on the receipt of the PPDU may disable OBSS PD SR operation on this PPDU.

An HE STA with `dot11HEPSROptionImplemented` set to true that receives a PPDU that is identified as an inter-BSS PPDU with a value other than `PSR_DISALLOW` or `PSR_AND_NON_SRG_OBSS_PD_PROHIBITED` in the UL Spatial Reuse subfield of the Common Info field of a Trigger frame and fails to identify an PSR opportunity based on the receipt of the PPDU shall disable OBSS PD SR operation on the HE TB PPDU that is elicited by the Trigger frame.

An HE STA with `dot11HEPSROptionImplemented` set to true that receives a PPDU that is identified as an inter-BSS PPDU with a value other than `PSR_DISALLOW` in the UL Spatial Reuse subfield of the Common Info field of a Trigger frame and identifies an PSR opportunity based on the receipt of the PPDU may disable OBSS PD SR operation on the Data field of the HE TB PPDU that is elicited by the Trigger frame.

26.11 Rules for setting some TXVECTOR parameters for PPDUs transmitted by an HE STA

26.11.1 STA_ID

Each parameter `STA_ID` in the TXVECTOR identifies the STA or group of STAs that is the recipient of an RU in the HE MU PPDU transmitted with the TXVECTOR parameter `UPLINK_FLAG` set to 0. For an individually addressed RU the parameter `STA_ID` is set to the 11 LSBs of the AID of the STA receiving the PSDU contained in that RU. If an RU is intended for one or more unassociated non-AP STAs, then the parameter `STA_ID` for that RU is set to 2045. If an RU is intended for no user, then the parameter `STA_ID` for that RU is set to 2046.

If an RU is intended for an AP (i.e., the TXVECTOR parameter `UPLINK_FLAG` is 1), then the parameter `STA_ID` contains only one element that is set to the 11 LSBs of the AID of the non-AP STA transmitting the PPDU.

NOTE—A non-AP STA can transmit an UL HE MU PPDU to help the AP identify the transmitter of a failed PPDU so that the AP can allocate resources for that non-AP STA in a later TXOP. All unassociated STAs share the same parameter `STA_ID` value (i.e., 2045), which does not uniquely identify the transmitter. Therefore an unassociated STA is not allowed to transmit an UL HE MU PPDU.

If an RU is intended for multiple STAs for MU-MIMO, then multiple STAs identified by STA-IDs in the parameter `STA_ID` will use the same resource unit (see 26.5.2). If an RU is intended for multiple associated STAs and carries a single A-MPDU, then the parameter `STA_ID` is set as follows:

- For an AP with `dot11MultiBSSIDImplemented` equal to false, if the RU is intended for more than one associated STA in the BSS that is not a recipient of an individually addressed RU, the parameter `STA_ID` is set to 0.
- For an AP with `dot11MultiBSSIDImplemented` equal to true, if the RU is intended for more than one associated STA in any of its BSSs that is not a recipient of an individually addressed RU, the parameter `STA_ID` is set to 0 for a transmitted BSSID or to the value of the BSSID Index field corresponding to that BSS (see 9.4.2.73) for a nontransmitted BSSID. The number of such elements shall not exceed the maximum number of BSSs of the multiple BSSID set.
- For an AP with `dot11MultiBSSIDImplemented` equal to true, if the RU is intended for more than one associated STA on any of its BSSs that is not a recipient of an individually addressed RU or another broadcast RU corresponding to parameter `STA_ID` equal 0 or equal to the BSSID Index of a BSSID in a multiple BSSID set, the parameter `STA_ID` is set to 2047.

The parameter STA_ID values between 2008 and 2044 are reserved.

A non-AP STA shall not transmit an HE MU PPDU where the TXVECTOR parameter STA_ID includes more than one entry in the range 1 to 2007.

26.11.2 UPLINK_FLAG

An HE STA transmitting an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU sets the TXVECTOR parameter UPLINK_FLAG as follows:

- A STA transmitting an HE PPDU containing frames that are addressed to an AP shall set the TXVECTOR parameter UPLINK_FLAG to 1, unless the HE PPDU is an HE ER SU PPDU with the TXVECTOR parameter TXOP_DURATION set to UNSPECIFIED and contains an RTS or CTS frame. In this case, the STA may set the TXVECTOR parameter UPLINK_FLAG to 0.
- Otherwise, the HE STA shall set the TXVECTOR parameter UPLINK_FLAG to 0.

26.11.3 BEAM_CHANGE

An HE STA uses the TXVECTOR parameter BEAM_CHANGE to indicate a change in the spatial mapping of the pre-HE-STF portion of the PPDU and the first symbol of HE-LTF (see Table 27-1).

An HE STA that transmits an HE SU PPDU or an HE ER SU PPDU shall set the TXVECTOR parameter BEAM_CHANGE to 1 if one or more of the following conditions are met:

- The number of spatial streams is greater than 2.
- The PPDU is the first PPDU in a TXOP.
- The PPDU carries a Trigger frame.

26.11.4 BSS_COLOR

An HE STA that transmits an HE Operation element shall select and advertise a BSS color as described in 26.17.3.

An HE STA that transmits an HE Operation element shall set the TXVECTOR parameter BSS_COLOR as follows:

- For an HE SU PPDU, HE ER SU PPDU, or DL HE MU PPDU, the parameter BSS_COLOR is set to the value indicated in the BSS Color subfield of the HE Operation element if all the recipient STAs are members of the HE STA's BSS or the PPDU carries at least one triggering frame.
- For an HE SU PPDU, HE ER SU PPDU, or DL HE MU PPDU, the parameter BSS_COLOR is set to 0 if the HE STA expects that at least one intended recipient STA is not a member of the HE STA's BSS and the PPDU does not carry a triggering frame.

A non-AP HE STA that transmits an HE SU PPDU or HE ER SU PPDU to a STA that is not a member of the transmitting STA's HE BSS shall set the TXVECTOR parameter BSS_COLOR to 0.

The active BSS color is one of the following:

- The value of the BSS Color field in the most recently received HE Operation element if an HE STA receives an HE Operation element from a peer HE STA.
- The value of the New BSS Color field in the most recently received BSS Color Change Announcement element if an HE STA receives a BSS Color Change Announcement element from a peer HE STA and the BSS color change TBTT has passed (see 26.17.3.4).

An HE STA shall set the TXVECTOR parameter BSS_COLOR for an HE SU PPDU, HE ER SU PPDU, or UL HE MU PPDU that is addressed to a peer STA to the active BSS color value, if the HE STA has established any of the following:

- An association with the peer STA
- A TDLS link with the peer STA
- An IBSS membership with the peer STA

NOTE 1—A non-AP HE STA sets the TXVECTOR parameter BSS_COLOR for an HE TB PPDU to the active color (see 26.5.2.3).

NOTE 2—An HE mesh STA sets the TXVECTOR parameter BSS_COLOR for an HE PPDU that it transmits to a peer HE mesh STA to the value in the BSS Color subfield of its transmitted HE Operation element.

An HE STA that receives an HE PPDU with RXVECTOR parameter BSS_COLOR with a value between 1 and 63 follows the spatial reuse rule described in 26.10.

NOTE 3—An HE STA that receives an HE PPDU with the RXVECTOR parameter BSS_COLOR equal to 0 might not follow the spatial reuse rules described in 26.10.

An HE STA that receives an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU with the RXVECTOR parameter BSS_COLOR equal to 0 shall not discard the HE PPDU on that basis.

While the BSS Color Disabled subfield is 1, an HE STA shall continue to advertise a nonzero value (same as before the color was disabled) in the BSS Color subfield of HE Operation element and in the TXVECTOR parameter BSS_COLOR of an HE PPDU that it transmits.

NOTE 4—A non-AP HE STA sets the TXVECTOR parameter BSS_COLOR of an HE PPDU that it transmits to the value advertised by the AP with which it intends to communicate even if the AP has disabled BSS color.

If the value of TXVECTOR parameter PARTIAL_AID[5:8] for VHT PDUs with the TXVECTOR parameter GROUP_ID equal to 63 transmitted by an HE AP to any associated STA would not be consistent with the partial BSS color (i.e., BCB(0:3) described in 26.17.4) announced by the HE AP, then the HE AP shall set the Partial BSS Color field in the HE Operation element to 0. Otherwise, the HE AP may set the Partial BSS Color subfield in the HE Operation element to 1 (see 26.17.4).

26.11.5 TXOP_DURATION

The TXVECTOR parameter TXOP_DURATION of an HE PPDU indicates duration information for NAV setting and protection of the TXOP, except that the TXVECTOR parameter TXOP_DURATION is set to UNSPECIFIED to indicate no duration information.

NOTE 1—The value of TXVECTOR parameter TXOP_DURATION is converted to an indication in the TXOP field of the HE-SIG-A field as described in Table 27-18, Table 27-20, and Table 27-21. The indication in the TXOP field of the HE-SIG-A field is converted to the RXVECTOR parameter TXOP_DURATION as described in Table 27-1.

A STA that is not a TXOP responder and that transmits an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU may set the TXVECTOR parameter TXOP_DURATION to UNSPECIFIED.

A STA that transmits an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU that carries a PS-Poll frame shall set the TXVECTOR parameter TXOP_DURATION to UNSPECIFIED.

A STA that is a TXOP responder that transmits an HE SU PPDU, HE ER SU PPDU, or HE TB PPDU shall set the TXVECTOR parameter TXOP_DURATION to UNSPECIFIED if the RXVECTOR parameter TXOP_DURATION of the HE PPDU that solicits a response from the STA is UNSPECIFIED.

An HE AP that has set the BSS Color Disabled field in the HE Operation element to 1 shall set the TXVECTOR parameter TXOP_DURATION to UNSPECIFIED for an HE PPDU that it transmits to non-AP STAs associated to it.

If the BSS Color Disabled field is 1 in the HE Operation element most recently received from an AP by a non-AP STA, then the non-AP STA should set the TXVECTOR parameter TXOP_DURATION to UNSPECIFIED for an HE PPDU that is not an HE TB PPDU sent to that AP.

A STA that transmits an HE TB PPDU shall not set the TXVECTOR parameter TXOP_DURATION to UNSPECIFIED if any one of the following conditions is met:

- The RXVECTOR parameter TXOP_DURATION is present for the PPDU that solicits a response from the STA and is not UNSPECIFIED.
- The PPDU that solicits a response from the STA is not an HE PPDU.

A STA that transmits a frame with a Duration field in an HE PPDU with the TXVECTOR parameter TXOP_DURATION not set to UNSPECIFIED shall set the TXVECTOR parameter TXOP_DURATION to the smaller of the duration information indicated by the Duration field and 8448.

NOTE 2—For a TXOP responder, the Duration field in a frame carried in a response PPDU is set based on the Duration field in a frame carried in the PPDU that solicits a response from the TXOP responder as described in 9.2.5.7 or 9.2.5.8.

If a STA transmits either an HE TB feedback NDP or an HE TB PPDU carrying a PS-Poll frame with the TXVECTOR parameter TXOP_DURATION not set to UNSPECIFIED, it shall calculate the duration information and set the TXVECTOR parameter TXOP_DURATION for the HE TB feedback NDP or HE TB PPDU to the value of the computed duration information. The TXOP responder shall calculate duration information equal to the duration information indicated by the Duration field of the frame that solicits the response minus the time, in microseconds, between the end of the PPDU carrying the frame that is soliciting the HE TB PPDU and the end of the HE TB PPDU. If the calculated duration information includes a fractional microsecond, the duration information is rounded up to the next higher integer. If the calculated duration information is smaller than 8448 μ s, the TXVECTOR parameter TXOP_DURATION shall be set to the calculated duration information. Otherwise, the TXVECTOR parameter TXOP_DURATION shall be set to 8448.

NOTE 3—The time is equal to SIFS plus the duration of the HE TB PPDU, where the duration of the HE TB PPDU is defined in Equation (27-134).

In the 6 GHz band, a TXOP holder shall not set the TXVECTOR parameter TXOP_DURATION for a transmitted HE PPDU to UNSPECIFIED, unless at least one of the following conditions is true:

- The BSS Color Disabled field is 1 in the HE Operation element transmitted within the BSS of which the TXOP holder is a member.
- The HE PPDU carries a PS-Poll frame.

26.11.6 SPATIAL_REUSE

The contents of the Spatial Reuse fields are carried in the TXVECTOR parameter SPATIAL_REUSE for an HE PPDU indicating spatial reuse information. The behavior of STAs upon reception of an HE PPDU with different SPATIAL_REUSE values is described in 26.10.2 and 26.10.3. The different values that may be indicated in the SPATIAL_REUSE parameter of the TXVECTOR are listed in Table 27-22 and Table 27-23. The value PSR_DISALLOW is used to prohibit PSR-based spatial reuse during the transmission of the corresponding PPDU. The value PSR_AND_NON_SRG_OBSS_PD_PROHIBITED is used to prohibit both PSR-based spatial reuse and non-SRG OBSS PD-based spatial reuse during the transmission of the corresponding PPDU. The interpretations of other values are described in this subclause

and in 26.10. The conditions for a STA to set the SPATIAL_REUSE parameter to its different values are described in this subclause.

For a PPDU with a value of HE_TB for the TXVECTOR parameter FORMAT, the SPATIAL_REUSE parameter contains an array of one to four values, depending on the TXVECTOR parameter CH_BANDWIDTH. If the TXVECTOR parameter CH_BANDWIDTH is CBW20, CBW40, or CBW80, the first value in the array is the SPATIAL_REUSE parameter that applies to the lowest frequency 20 MHz subband, the second value, if present, applies to the second lowest frequency 20 MHz subband, the third value, if present, applies to the third lowest frequency 20 MHz subband, and the fourth value, if present, applies to the highest frequency 20 MHz subband. If the TXVECTOR parameter CH_BANDWIDTH is CBW160 or CBW80+80, the first value in the array is the SPATIAL_REUSE parameter that applies to the lowest frequency 40 MHz subband, the second value applies to the second lowest frequency 40 MHz subband, the third value applies to the third lowest frequency 40 MHz subband, and the fourth value applies to the highest frequency 40 MHz subband. Each value in the SPATIAL_REUSE parameter array shall individually conform to the rules in this subclause.

An HE STA that transmits an HE TB PPDU sets the TXVECTOR parameter SPATIAL_REUSE as defined in 26.5.2.3.

An AP with dot11HEPSROptionImplemented set to true that transmits an HE ER SU PPDU that does not contain a Trigger frame should set the TXVECTOR parameter SPATIAL_REUSE to PSR_DISALLOW.

A non-AP STA with dot11HEPSROptionImplemented set to true that transmits an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU may set the TXVECTOR parameter SPATIAL_REUSE, when permitted by other conditions, to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED if the HESIGA_Spatial_reuse_value15_allowed subfield of the SR Control field of the most recently received Spatial Reuse Parameter Set element from its associated AP is equal to 1. Otherwise, the non-AP STA shall set it to PSR_DISALLOW.

An HE STA that transmits an HE TB PPDU determines the value of the TXVECTOR parameter SPATIAL_REUSE according to 26.5.2.3.

An HE AP with dot11HEPSROptionImplemented set to true that transmits an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU may set the TXVECTOR parameter SPATIAL_REUSE to PSR_DISALLOW to disallow OBSS STAs from performing PSR-based SR transmission during the duration of the corresponding PPDU.

An HE STA with dot11HEPSROptionImplemented set to false may set the TXVECTOR parameter SPATIAL_REUSE to PSR_DISALLOW for any PPDU that is not an HE TB PPDU, an HE NDP PPDU, a PPDU containing an HE NDP Announcement frame, or a PPDU containing a response to an HE NDP Announcement frame.

A STA shall set the TXVECTOR parameter SPATIAL_REUSE of an HE PPDU to PSR_DISALLOW or, if permitted by the other rules in this subclause, to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED, if the STA is a non-AP HE STA and the PSR Disallowed subfield of the SR Control field of the most recently received Spatial Reuse Parameter Set element from its associated AP is equal to 1.

An HE STA shall set the TXVECTOR parameter SPATIAL_REUSE to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED for an HE NDP PPDU.

An HE STA shall set the TXVECTOR parameter SPATIAL_REUSE to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED for a PPDU containing an NDP Announcement frame and in any frame that is transmitted as a response to an NDP Announcement frame.

A non-AP HE STA may set the TXVECTOR parameter SPATIAL_REUSE of an HE PPDU to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED if the HESIGA_Spatial_reuse_value15_allowed subfield of the SR Control field of the most recently received Spatial Reuse Parameter Set element from its associated AP is equal to 1. If the HESIGA_Spatial_reuse_value15_allowed subfield of the SR Control field of the most recently received Spatial Reuse Parameter Set element from its associated AP is equal to 0 or if STA has not received a Spatial Reuse Parameter Set element from its associated AP, the STA shall not set the TXVECTOR parameter SPATIAL_REUSE of any HE PPDU to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED, unless the HE PPDU contains an NDP, contains an NDP Announcement frame, or is a frame that is transmitted as a response to an NDP Announcement frame.

An AP HE STA may set the TXVECTOR parameter SPATIAL_REUSE of an HE PPDU to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED if the HESIGA_Spatial_reuse_value15_allowed subfield of the SR Control field of the most recently transmitted Spatial Reuse Parameter Set element is equal to 1. If the HESIGA_Spatial_reuse_value15_allowed subfield of the SR Control field of the most recently transmitted Spatial Reuse Parameter Set element is equal to 0, or if the AP has not transmitted a Spatial Reuse Parameter Set element, the AP shall not set the TXVECTOR parameter SPATIAL_REUSE of any HE PPDU to PSR_AND_NON_SRG_OBSS_PD_PROHIBITED.

An HE AP that transmits an HE SU PPDU or an HE ER SU PPDU that contains a Trigger frame should set the TXVECTOR parameter SPATIAL_REUSE to SR_DELAYED. An HE STA that transmits an HE MU PPDU shall not set the TXVECTOR parameter SPATIAL_REUSE to SR_DELAYED.

An HE STA that transmits an HE SU PPDU or HE ER SU PPDU shall not set the TXVECTOR parameter SPATIAL_REUSE to SR_RESTRICTED.

An HE AP that transmits an HE MU PPDU that contains a Trigger frame should set the TXVECTOR parameter SPATIAL_REUSE to SR_RESTRICTED.

An HE STA that transmits a PPDU that does not contain a Trigger frame shall not set the TXVECTOR parameter SPATIAL_REUSE to SR_DELAYED or SR_RESTRICTED.

26.11.7 INACTIVE_SUBCHANNELS and RU_ALLOCATION

The indication of which subchannels are punctured in an HE sounding NDP or in an HE NDP Announcement frame that is carried in a non-HT duplicate PPDU is conveyed from the MAC to the PHY through the TXVECTOR parameters INACTIVE_SUBCHANNELS and RU_ALLOCATION. The parameter INACTIVE_SUBCHANNELS may be present in the TXVECTOR of a non-HT duplicate PPDU that carries an HE NDP Announcement frame or of an HE sounding PPDU. The parameter INACTIVE_SUBCHANNELS shall not be present otherwise. The setting of the RU_ALLOCATION parameter for other PDUs is specified in 26.5.2.3.3, 26.5.2.3.4, and 26.5.7.2.

INACTIVE_SUBCHANNELS is an 8-bit bitmap with an encoding that is the same as the encoding for the Disallowed Subchannel Bitmap subfield defined in 9.3.1.19. A bit in the INACTIVE_SUBCHANNELS bitmap is set to 1 to indicate that no energy is transmitted on the corresponding subchannel for the corresponding PPDU. The RU_ALLOCATION parameter is set to a value that corresponds to the puncturing signaled by the INACTIVE_SUBCHANNELS bitmap. If a bit in the INACTIVE_SUBCHANNELS bitmap corresponds to a 20 MHz subchannel outside the PPDU bandwidth, then the bit is set to 1.

A STA transmitting an HE sounding NDP may set the TXVECTOR parameter INACTIVE_SUBCHANNELS to any value, provided that the bit representing the primary 20 MHz channel is set to 0.

If an HE AP transmits an HE NDP Announcement frame in a PPDU with punctured channels, then the TXVECTOR parameters FORMAT, NON_HT_MODULATION, INACTIVE_SUBCHANNELS, and CH_BANDWIDTH shall be set as follows:

- The TXVECTOR parameter FORMAT shall be set to NON_HT.
- The TXVECTOR parameter NON_HT_MODULATION shall be set to NON_HT_DUP_OFDM.
- The INACTIVE_SUBCHANNELS parameter may have any value, except that the bit in the bitmap representing the primary 20 MHz subchannel shall be set to 0.
- The CH_BANDWIDTH parameter value shall be set to CBW80 if there are no bits set to 0 in the INACTIVE_SUBCHANNELS bitmap that correspond to any 20 MHz subchannel of the secondary 80 MHz and at least one bit set to 0 that corresponds to any 20 MHz subchannel of the secondary 40 MHz.
- The CH_BANDWIDTH parameter value shall be set to CBW160 if there is at least one bit set to 0 in the INACTIVE_SUBCHANNELS bitmap that corresponds to any 20 MHz subchannel of the secondary 80 MHz.

Each 8 bits of the RU_ALLOCATION are set to 113 (01110001 in binary representation) for the 242-tone RU that is most closely aligned in frequency with the 20 MHz subchannel that is indicated as disallowed by the value 1 in the INACTIVE_SUBCHANNELS parameter. Each 8 bits of the RU_ALLOCATION are set to 192 (11000000 in binary representation) for the 242-tone RU that is most closely aligned in frequency with the 20 MHz subchannel that is indicated as not disallowed by the value 0 in the INACTIVE_SUBCHANNELS parameter.

26.11.8 TRIGGER_RESPONDING

An non-AP HE STA that transmits a non-HT or non-HT duplicate PPDU shall set the TXVECTOR parameter TRIGGER_RESPONDING to true if the PPDU is sent in response to an MU-RTS Trigger frame. Otherwise, the non-AP HE STA shall set the TXVECTOR parameter TRIGGER_RESPONDING to false.

26.11.9 STBC and DCM

An HE STA shall not transmit an HE SU PPDU or HE ER SU PPDU with the TXVECTOR parameter STBC set to 1 if any of the following conditions is met:

- TXVECTOR parameter DCM is set to 1.
- TXVECTOR parameter NUM_STS is not equal to 2.
- TXVECTOR parameters HE_LTF_TYPE and GI_TYPE are set to 4x HE-LTF and 0u8s_GI, respectively.

An HE STA shall not transmit an HE TB PPDU with the TXVECTOR parameter STBC set to 1 if any of the following conditions is met:

- TXVECTOR parameter DCM is set to 1.
- TXVECTOR parameter NUM_STS is not equal to 2.

An HE STA shall not transmit an HE MU PPDU with the TXVECTOR parameter STBC set to 1 if any of the following conditions is met:

- At least one element in the TXVECTOR parameter DCM is set to 1.
- At least one element in the TXVECTOR parameter NUM_STS is not equal to 2.
- Two or more STA_ID parameters in the TXVECTOR use the same RU.

NOTE—Two or more STA_ID parameters in the TXVECTOR using the same RU means that DL MU-MIMO is used.

An HE STA shall not transmit an HE SU PPDU or HE ER SU PPDU with the TXVECTOR parameter STBC or DCM set to 1 if the TXVECTOR parameters HE_LTF_TYPE and GI_TYPE are set to 4x HE-LTF and 0u8s_GI, respectively.

26.12 HE PPDU post-FEC padding and packet extension

An HE STA with dot11PPEThresholdsRequired set to false may set the PPE Thresholds Present subfield in HE Capabilities elements that it transmits to 0.

An HE STA with dot11PPEThresholdsRequired set to true shall set the PPE Thresholds Present subfield in HE Capabilities elements that it transmits to 1.

A STA that sets the PPE Thresholds Present subfield to 0 and the Nominal Packet Padding subfield to 0 in HE Capabilities elements that it transmits has as a nominal packet padding of 0 μ s for all constellations, NSS, and RU allocations that it supports.

A STA that sets the PPE Thresholds Present subfield to 0 and the Nominal Packet Padding subfield to 1 in the HE Capabilities elements that it transmits has a nominal packet padding of 8 μ s for all constellations, NSS, and RU allocations that it supports.

A STA that sets the PPE Thresholds Present subfield to 0 and the Nominal Packet Padding subfield to 2 in the HE Capabilities elements that it transmits has nominal packet padding of 16 μ s for all constellations, NSS, and RU allocations that it supports.

A STA that sets the PPE Thresholds Present subfield to 1 in the HE Capabilities elements that it transmits shall indicate its nominal packet padding per constellation, NSS and RU allocation by setting the subfields of the PPE Thresholds field according to 9.4.2.248 and using the corresponding values from dot11PPEThresholdsMappingTable.

After receiving the PPE Thresholds field from a second STA, the first STA uses the combination of the PPET8 NSTS n RUB subfield and PPET16 NSTS n RUB subfield values to determine the nominal packet padding for HE PDUs that are transmitted to the second STA, using NSTS = n and an RU allocation corresponding to RU Allocation Index b , for each value of NSTS and RU specified by the field. The nominal packet padding is used in computing the PE field duration (see 27.3.13).

NOTE—If the pre-FEC padding factor is 4, then the value of nominal T_{PE} is equal to the nominal packet padding (see Table 27-46).

For all values of n and b for which PPET8 and PPET16 are not present, the nominal packet padding is 0 for HE PDUs that are transmitted to the STA using NSTS = n and an RU allocation corresponding to RU allocation index b . The nominal packet padding as a function of the PPE thresholds, the number of spatial streams, and the RU allocation index is described in Table 26-13.

In Table 26-13, “RU Allocation index = (b + DCM)” means the following. With the exception of a 2×996-tone RU, if DCM is applied in a given RU, the nominal packet padding value is based on the next larger RU size (RU index + 1). For example, if DCM is applied to a 242-tone RU, then the nominal packet padding value for a 484-tone RU is used. If DCM is applied to 106-tone RU, then the nominal packet padding value for a 242-tone RU is used. If DCM is applied to a 2×996-tone RU, then the nominal packet padding value for a 2×996-tone RU is used.

The nominal packet padding value shall be 0 for all RU less than 242, unless the RU size is 106 and DCM is enabled.

Table 26-13—PPE thresholds per PPET8 and PPET16

Result of comparison of constellation index x of an HE PPDU with NSTS value n and RU allocation size that corresponds to RU Allocation index = $(b + \text{DCM})$ to the value in . . .		Nominal packet padding for an HE PPDU transmitted to this STA using constellation index = x , NSTS = n , and RU allocation size that corresponds to RU Allocation index = $(b + \text{DCM})$
PPET8 NSTS n RU $(b + \text{DCM})$ subfield	PPET16 NSTS n RU $(b + \text{DCM})$ subfield	
x greater than or equal to PPET8	x less than PPET16 or PPET16 equal to None	8 μs
x greater than PPET8 or PPET8 equal to None	x greater than or equal to PPET16	16 μs
All other combinations not otherwise listed in this table		0
NOTE—DCM = 1 if the HE PPDU uses DCM; DCM = 0 otherwise.		

A STA transmitting an HE PPDU provides the nominal packet padding in the TXVECTOR parameter NOMINAL_PACKET_PADDING for the minimal PE calculation (see 27.3.13).

A STA transmitting an HE PPDU that carries a broadcast frame shall set the value of the TXVECTOR parameter NOMINAL_PACKET_PADDING to 16 μs . A STA transmitting an HE PPDU that carries a group addressed, but not broadcast, frame shall not set the value of the TXVECTOR parameter NOMINAL_PACKET_PADDING to a value that is less than that required for any of the recipients in the group.

A STA transmitting an HE PPDU to a receiving STA shall include post-FEC padding determined by the pre-FEC padding factor (see 27.3.12); and after including the post-FEC padding, the transmitting STA shall include a packet extension with a duration indicated by the TXVECTOR parameter NOMINAL_PACKET_PADDING (see 27.3.13).

26.13 Link adaptation using the HLA Control subfield

This subclause applies to frame exchange sequences that include PPDUs containing an HE variant HT Control field.

An HE STA shall set the HE Link Adaptation Support subfield in the HE Capabilities Information field in the HE Capabilities element it transmits to the value of dot11HEMCSFeedbackOptionImplemented.

A STA that supports HE link adaptation using the HLA Control subfield shall set the HE Link Adaptation Support subfield in the HE Capabilities Information field in the HE Capabilities element to 2 or 3, depending on its own link adaptation feedback capability. A STA shall not send an MRQ to a STA that has not set the HE Link Adaptation Support subfield to 3 in the HE Capabilities Information field in the HE Capabilities element. A STA shall not send an unsolicited MFB in any frame that contains an HLA Control subfield to a STA that has not set the HE Link Adaptation Support subfield to either 2 or 3 in the HE Capabilities Information field in the HE Capabilities element.

The MFB requester may set the MRQ subfield to 1 and Unsolicited MFB subfield to 0 in the HLA Control subfield of a frame to request a STA to provide link adaptation feedback. In each request, the MFB requester shall set the MSI field to a value ranging from 0 to 6. For the MFB requester, how to choose the MSI value is implementation dependent.

The appearance of more than one instance of an HLA Control subfield with the MRQ field equal to 1 within a single PPDU shall be interpreted by the receiver as a single request for link adaptation feedback.

The MFB requester shall specify the RU index and BW requesting the link adaptation feedback.

On receipt of an HLA Control subfield with the MRQ subfield equal to 1, an MFB responder computes the HE-MCS, NSS, and DCM of the RU and BW specified in the MRQ, and these estimates are based on the same RU of the PPDU carrying the MRQ. The PPDU carrying MRQ shall include the RU requested for MFB. The MFB responder labels the result of this computation with the MSI value from the HLA Control subfield in the received frame carrying the MRQ. The MFB responder may include the received MSI value in the MSI field of the corresponding response frame. In the case of a delayed response, this allows the MFB requester to associate the MFB with the soliciting MRQ.

An MFB responder that sends a solicited MFB shall set the Unsolicited MFB subfield to 0 and MRQ subfield to 0 in the HLA Control subfield.

The STA receiving MFB may use the received MFB to compute the appropriate HE-MCS, DCM, and NSS.

The MFB responder may send a solicited response frame with any of the following combinations of HE-MCS, NSS, and MSI:

- HE-MCS = 15, NSS = 7, MSI = 0–6: the responder will not provide feedback for the request that had the MSI value.
- HE-MCS = valid value, NSS = valid value, MSI = 0–6: the responder is providing feedback for the request that had the MSI value. The MSI value in the response frame matches the MSI value of the MRQ request.

A STA sending unsolicited MFB using the HLA Control subfield shall set the Unsolicited MFB subfield to 1.

Unsolicited HE-MCS, NSS, DCM, BW, and RU estimates reported in an HLA Control subfield sent by a STA are computed based on the most recent PPDU received by the STA that matches the description indicated by the PPDU format, Tx Beamforming, and Coding Type subfields in the same HLA Control subfield.

In an unsolicited MFB response the PPDU Formats, Coding Type, and Tx Beamforming subfields are set according to the RXVECTOR parameters of the received PPDU from which the HE-MCS, RU, BW, and NSS are estimated, as follows:

- The PPDU format subfield is set and encoded as follows:
 - 0 if the parameter FORMAT is equal to HE_SU
 - 1 if the parameter FORMAT is equal to HE_MU
 - 2 if the parameter FORMAT is equal to HE_ER_SU
 - 3 if the parameter FORMAT is equal to HE_TB
- The Coding Type subfield is set to 0 if the parameter FEC_CODING is equal to BCC_CODING and set to 1 if that parameter is equal to LDPC_CODING.
- The Tx Beamforming subfield is set to 1 if the parameter BEAMFORMED is equal to 1 and set to 0 if that parameter is equal to 0.
- The BW subfield shall indicate a bandwidth less than or equal to the bandwidth indicated by the parameter CH_BANDWIDTH.
- The RU subfield indicates the RU at which the recommended HE-MCS is applied. The recommended RU shall be within an RU or a bandwidth in which the received HE PPDU is located.

For either a solicited or an unsolicited response, the recommended HE-MCS and NSS subfields of HLA Control subfield shall be selected from the HE-MCS and NSS set supported by the recipient STA.

The HE-MCS subfield of HLA Control subfield is the recommended data rate, for given transmission properties carried in the RXVECTOR of the PPDU used for MFB estimation, which results in an estimated frame error rate of 10% or lower for an MPDU length of 3895 octets.

NOTE—Some HE PPDU might not be able to carry 3895 octets due to PPDU duration limitations.

If the MFB requester sets the MRQ subfield to 1 and sets the MSI subfield to a value that matches the MSI subfield value of a previous request for which the responder has not yet provided feedback, the responder shall discard or abandon the computation for the MRQ that corresponds to the previous use of that MSI subfield value and start a new computation based on the new request.

A STA may respond immediately to a current request for MFB with a frame containing an MSI field value and NSS, HE-MCS, and DCM subfields that correspond to a request that precedes the current request.

A non-AP HE STA may set the UL HE TB PPDU MFB to 1 in the HLA Control field it transmits to the AP to indicate that the NSS, HE-MCS, DCM, BW, and RU Allocation in the HLA Control field represent the recommended MFB for the HE TB PPDU sent from the non-AP HE STA. The AP should not exceed the recommended RU size indicated in the most recently received RU Allocation field of the HLA Control field when it sends a triggering frame addressed to the STA.

26.14 Power management

26.14.1 Intra-PPDU power save for non-AP HE STAs

Intra-PPDU power save is the power save mechanism for an HE STA to enter the doze state or become unavailable until the end of a received PPDU that is identified as an intra-BSS PPDU. The STA can enter the doze state if it is in PS mode and can become unavailable if it is in active mode (see 11.2.3.2).

A non-AP HE STA that has `dot11IntraPPDUPowerSaveOptionActivated` equal to true operates in intra-PPDU power save mode.

A non-AP HE STA that is in intra-PPDU power save mode may enter the doze state or become unavailable until the end of a PPDU currently being received if one of the following conditions is met:

- The PPDU is an HE MU PPDU where the RXVECTOR parameter `BSS_COLOR` is the BSS color of the BSS in which the STA is associated, the RXVECTOR parameter `UPLINK_FLAG` is 0, the RXVECTOR parameters `STA_ID` do not include the identifier of the STA or the broadcast identifier(s) intended for the STA, and the BSS Color Disabled subfield is 0 in the most recently received HE Operation element from the AP with which it is associated.
- The PPDU is an HE MU PPDU, HE SU PPDU, or HE ER SU PPDU and one of the following conditions is true:
 - The RXVECTOR parameter `BSS_COLOR` is the BSS color of the BSS in which the STA is associated, the RXVECTOR parameter `UPLINK_FLAG` is 1, and the BSS Color Disabled subfield is 0 in the most recently received HE Operation element from the AP with which it is associated.
 - The RXVECTOR parameter `BSS_COLOR` is the BSS color of the BSS in which the STA is associated, the RXVECTOR parameter `UPLINK_FLAG` is 0, a `PHY-RXEND.indication(UnsupportedRate)` primitive was received, and the BSS Color Disabled subfield is 0 in the most recently received HE Operation element from the AP with which it is associated.

- The PPDU is an HE TB PPDU where the RXVECTOR parameter BSS_COLOR is the BSS color of the BSS in which the STA is associated and the BSS Color Disabled subfield is 0 in the most recently received HE Operation element from the AP with which it is associated.
- The PPDU is a VHT PPDU where the RXVECTOR parameter PARTIAL_AID is the BSSID[39:47] of the BSS in which the STA is associated or any of the other BSSs in the same multiple BSSID set or co-hosted BSSID set to which its BSS belongs and the RXVECTOR parameter GROUP_ID is 0.
- The PPDU is a PPDU with
 - An A-MPDU including TA or RA equal to either the BSSID of the BSS in which the STA is associated or any of the other BSSs in the same multiple BSSID set or co-hosted BSSID set to which its BSS belongs and
 - The RA is not the individual MAC address of the STA or the group address(es) of the STA.
- The PPDU is either an HE MU PPDU with the RXVECTOR parameter UPLINK_FLAG set to 0 or a VHT MU PPDU containing an A-MPDU with
 - The RA(s) in the A-MPDU are equal to the STA's individual address and
 - The STA has received in the A-MPDU at least one MPDU delimiter with EOF equal to 1 and with MPDU length field equal to 0.

A non-AP HE STA that is in intra-PPDU power save mode and has entered doze state or has become unavailable shall continue to operate its NAV timers and to consider the medium busy and shall transition to the awake state at the end of the PPDU.

A non-AP HE STA that is in intra-PPDU power save mode may discard a PPDU identified as an inter-BSS PPDU as defined in 26.2.2 until the end of the PPDU.

NOTE—The STA can contend for access to the medium immediately on the expiry of the NAV timers.

26.14.2 Power save with UORA and TWT

This subclause defines the power save mechanisms for a non-AP HE STA that is operating in PS mode and is UORA and TWT capable.

An HE AP may indicate start times for one or more broadcast TWT SPs containing Trigger frames with random access allocations in the broadcast TWT element that is included in a Management frame as described in 26.8.3.2. An example of power save operation is shown in Figure 26-14.

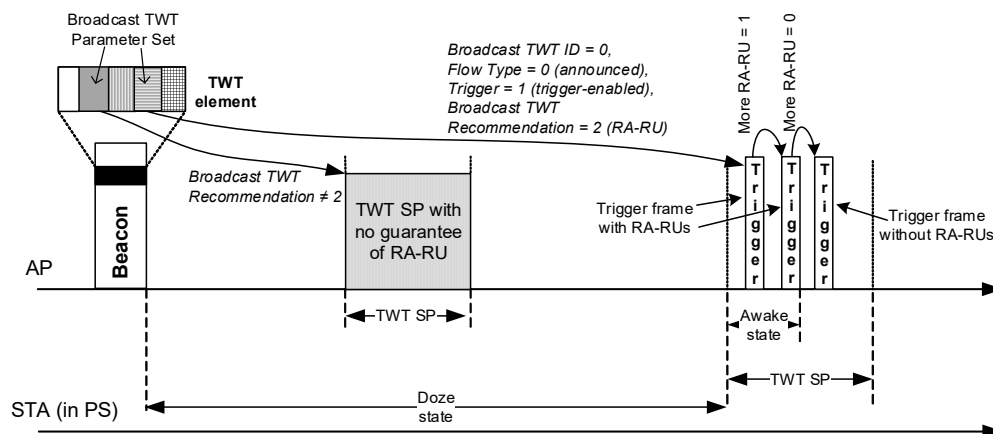


Figure 26-14—Example of power save operation with UORA and TWT

A TWT SP with RA-RU is a TWT SP corresponding to a Broadcast TWT Parameter Set field in a TWT element that has a Broadcast TWT ID subfield equal to 0, Flow Type subfield equal to 0, Trigger subfield equal to 1, and a Broadcast TWT Recommendation subfield equal to 2. An associated HE STA that supports the TWT and UORA procedure when operating in PS mode, upon receiving a Beacon frame from its associated AP carrying a TWT element indicating a schedule for TWT SP(s) with RA-RU, may enter the doze state if no other condition requires it to be awake. The STA may transition to awake state at the start of a TWT SP with RA-RU and follow the procedure in 26.5.4) to send an HE TB PPDU to its associated AP.

An HE STA shall follow the procedure described in 26.8.5 to determine if a TWT SP termination event has occurred and may enter doze state if no other condition requires the STA to remain awake.

A non-AP HE STA shall decrement its OBO counter by following the procedure in 26.5.4.3. If the OBO counter decrements to a nonzero value, then the STA may enter the doze state until the end of the current TWT SP if the STA has not declared to the AP that it is in awake state (as described in 26.8.3.3), no other condition requires it to remain awake, and one of the following conditions is met:

- The More TF subfield in the Common Info field of the Trigger frame is equal to 0.
- The More TF subfield in the Common Info field of the Trigger frame is equal to 1, and the More RA-RU subfield in the User Info field is equal to 0 indicating that subsequent Trigger frames in the current broadcast TWT SP will not include RA-RUs matching the value in AID12 subfield (see Table 9-29h).

26.14.3 Opportunistic power save

26.14.3.1 General

An OPS non-AP STA is a non-AP HE STA that sets the OPS Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element to 1.

An OPS AP is an AP HE STA that sets the OPS Support subfield in the HE MAC Capabilities Information field in HE Capabilities element to 1.

The objective of the opportunistic power save mechanism is to allow OPS non-AP STAs to be unavailable or to be in doze state so that they can save power for a defined period.

The opportunistic power save mechanism has two modes: aperiodic and periodic.

In the aperiodic mode, an OPS AP sends an OPS frame or a FILS Discovery frame at any time to provide the scheduling information for all OPS non-AP STAs for the OPS period that follows the transmission of the OPS frame or FILS Discovery frame. Based on this information, the OPS non-AP STAs that are in active mode may be unavailable during the OPS period, and the OPS non-AP STAs that are in PS mode may be in doze state during the OPS period.

In the periodic mode, an OPS AP splits a beacon interval into several periodic broadcast TWT SPs and provides, at the beginning of each SP, the scheduling information for all OPS non-AP STAs. Based on this information, the OPS non-AP STAs that are in active mode may be unavailable until the next TWT SP, and the OPS non-AP STAs that are in PS mode may be in doze state until the next TWT SP.

26.14.3.2 AP operation for opportunistic power save

To enable aperiodic opportunistic power save, an OPS AP shall schedule for transmission an OPS frame or a FILS Discovery frame with the RA field set to the broadcast address that includes a TIM element (see 9.4.2.5) and an OPS element (see 9.4.2.257). The AP should transmit a FILS Discovery frame instead of an OPS frame if the target transmission time closely aligns with the transmission time of a FILS

Discovery frame. The OPS Duration field in the OPS element shall be set to the duration of the OPS period that immediately follows the transmission of the OPS frame or FILS Discovery frame. The TIM element is encoded specifically as defined in 9.4.2.5 in order to identify which STAs are not scheduled during the OPS period. If the OPS AP sets the bit corresponding to an OPS non-AP STA in the traffic indication virtual bitmap field carried by the Partial Virtual Bitmap of the TIM element of the OPS frame or FILS Discovery frame to 0, the AP should send neither individually addressed frames to the STA nor Trigger frames with a User Info field that addresses the STA during the OPS period.

To enable periodic opportunistic power save, an OPS AP shall include a TWT element in beacons to set a periodic Broadcast TWT SP with the following information:

- The Broadcast TWT Recommendation field set to 3
- The Broadcast TWT ID subfield set to 0

At the beginning of these periodic TWT SPs with the Broadcast TWT Recommendation field set to 3, the AP shall schedule for transmission a TIM frame or a FILS Discovery frame with the RA field set to the broadcast address that includes a TIM element (see 9.4.2.5). The FILS Discovery frame may include an OPS element. The AP should transmit a FILS Discovery frame instead of a TIM frame if the TWT SP start time closely aligns with the transmission time of a FILS Discovery frame. If the OPS AP also operates with TIM Broadcast and uses TIM frames for opportunistic power save, the OPS AP should align the transmission time of a TIM frame for TIM Broadcast with the start time of the broadcast TWT SP with the Broadcast TWT Recommendation field set to 3. If the OPS AP sets the bit corresponding to an OPS non-AP STA in the traffic indication virtual bitmap carried in the Partial Virtual Bitmap field of the TIM element of the TIM frame or FILS Discovery frame to 0, the AP should send neither individually addressed frames to the STA nor Trigger frames with a User Info field that addresses the STA during the TWT SP, until the next TWT SP with the Broadcast TWT Recommendation field set to 3.

26.14.3.3 STA operation for opportunistic power save

With aperiodic opportunistic power save, if an OPS non-AP STA with AID N that is in the awake state receives a TIM element and an OPS element in an OPS frame or a FILS Discovery frame from the associated OPS AP, then the STA may be unavailable if the STA is in active mode or may be in doze state if the STA is in PS mode until the end of the OPS period indicated in the OPS element, if the bit N in the traffic indication virtual bitmap carried in the Partial Virtual Bitmap field of the current TIM element is 0, unless other conditions not related to operation with the OPS AP require the STA to be in the awake state. At the end of the OPS period, the STA shall be in the awake state, unless determined otherwise by other power save protocols.

With periodic opportunistic power save, if an OPS non-AP STA with AID N that is in the awake state receives from the associated OPS AP a TIM element with bit N of the traffic indication virtual bitmap field equal to 0 in a TIM frame or FILS Discovery frame within a broadcast TWT SP with the Broadcast TWT Recommendation field set to 3, then the STA may be unavailable if the STA is in active mode or may be in doze state if the STA is in PS mode during the TWT SP and until the next TWT SP with the Broadcast TWT Recommendation field set to 3, unless other conditions not related to operation with the OPS AP require the STA to be in the awake state.

An OPS non-AP STA shall not operate with TIM broadcast procedure if its associated AP is an OPS AP.

26.14.4 HE dynamic SM power save

A STA that supports HE dynamic SM power save has `dot11HEDynamicSMPowerSaveOptionImplemented` set to true and shall set the HE Dynamic SM Power Save subfield in the HE MAC Capabilities Information field in the HE Capabilities element it transmits to 1.

A non-AP HE STA in dynamic SM power save mode (see 11.2.6) that sets the HE Dynamic SM Power Save subfield in the HE MAC Capabilities Information field in the HE Capabilities element it transmits to 1 shall follow the dynamic SM power save procedures defined in 11.2.6 and shall also enable its multiple receive chains if it responds to a Trigger frame that starts a frame exchange sequence that satisfies the following conditions:

- The Trigger frame is transmitted with a single spatial stream.
- The Trigger frame is from the associated AP or from the AP corresponding to the transmitted BSSID if the non-AP HE STA is associated with an AP corresponding to a nontransmitted BSSID and has indicated support for receiving Control frames with TA set to the transmitted BSSID by setting the Rx Control Frame To MultiBSS subfield to 1 in the HE Capabilities element that the non-AP HE STA transmits.
- The Trigger frame is an MU-RTS Trigger frame, BSRP Trigger frame, or BQRP Trigger frame that includes a User Info field with the AID12 subfield equal to the 12 LSBs of the AID of the non-AP HE STA (see 26.5.2.2.1).

The non-AP HE STA shall, subject to its spatial stream capabilities (see 9.4.2.55.4, 9.4.2.157.3, and 9.4.2.248) and operating mode (see 11.40 and 26.9), be capable of receiving a PPDU that is sent using more than one spatial stream a SIFS after the end of the PPDU that it sends in response. The STA may switch back to single receive chain mode immediately after the end of the frame exchange sequence.

NOTE 1—A Trigger frame always solicits an immediate response.

NOTE 2—A non-AP HE STA that is in dynamic SM power save mode and that sets the HE Dynamic SM Power Save subfield in the HE MAC Capabilities Information field in the HE Capabilities element it transmits to 1 cannot distinguish between a Trigger frame that precedes a MIMO transmission and a Trigger frame that does not precede a MIMO transmission and, therefore, always enables its multiple receive chains if it responds to an MU-RTS Trigger frame, BSRP Trigger frame, or BQRP Trigger frame that has a User Info field addressed to it.

NOTE 3—The STA determines the end of the frame exchange sequence as described in 11.2.6.

26.15 PPDU format, BW, MCS, NSS, and DCM selection rules

26.15.1 General

An HE STA can transmit different PPDU formats, with different transmit parameters, such as channel width, MCS, NSS, DCM. Subclause 26.15 defines the rules followed by an HE STA for selecting these parameters depending on the capabilities of the intended receiver(s) and other considerations.

26.15.2 PPDU format selection

An HE STA that transmits non-HT, HT, or VHT PPDU shall follow the rules in 10.6. An HE STA may transmit an HE SU PPDU to a peer HE STA subject to the restrictions defined in this subclause.

An HE AP may transmit an HE MU PPDU as defined in 26.5.1. A non-AP HE STA transmits HE TB PPDU as defined in 26.5.2.

A STA shall not transmit a 242-tone HE ER SU PPDU to a peer non-AP STA if the most recently received OM Control field from that peer non-AP STA, if any, has the ER SU Disable subfield equal to 1.

A STA shall not transmit a 242-tone HE ER SU PPDU to an AP if the most recently received HE Operation element from that AP has the ER SU Disable subfield equal to 1.

A STA shall not transmit a 106-tone HE ER SU PPDU to a peer STA if the HE Capabilities element received from that peer STA has the Partial Bandwidth Extended Range field equal to 0.

A non-AP STA, TDLS STA, or IBSS STA shall not transmit a 20 MHz HE MU PPDU with only a 106-tone RU to a peer STA, unless it has received from the peer STA an HE Capabilities element with the Rx Partial BW SU In 20 MHz HE MU PPDU subfield in the HE PHY Capabilities Information field equal to 1.

NOTE 1—A non-AP STA transmitting an HE MU PPDU sets the TXVECTOR parameter UPLINK_FLAG to 1 if the PPDU is sent to the AP and to 0 if the PPDU is sent to a TDLS STA (see 26.11.2). The HE MU PPDU format enables the non-AP STA to include its AID (i.e., transmitter's AID if the UPLINK_FLAG is 1 and the receiver's AID if the UPLINK_FLAG is 0) in the PHY header of the PPDU, and its use is beyond the scope of this standard.

An HE STA shall not transmit an HE MU PPDU with a single user being allocated an RU occupying the entire PPDU bandwidth and a compressed HE-SIG-B field to a peer STA, unless the HE STA has received from the peer STA an HE Capabilities element with the Rx Full BW SU Using HE MU PPDU With Compressed HE-SIG-B subfield in the HE PHY Capabilities Information field equal to 1.

An HE STA shall not transmit an HE MU PPDU with a single user being allocated an RU occupying the entire PPDU bandwidth and a noncompressed HE-SIG-B field to a peer STA, unless the PPDU bandwidth is less than or equal to 80 MHz and the HE STA has received from the peer STA an HE Capabilities element with the Rx Full BW SU Using HE MU PPDU With Non-Compressed HE-SIG-B subfield in the HE PHY Capabilities Information field equal to 1.

An HE STA shall send Control frames following the rules defined in 10.6.6 with the following exceptions:

- A Control frame shall be carried in an HT PPDU, VHT PPDU, HE ER SU PPDU, or HE SU PPDU when the Control frame is sent using an STBC frame.
- A Control frame sent by the AP as a response to an HE TB PPDU may be carried in any PPDU format that is supported by the intended receivers.
- A Trigger frame that is not an MU-RTS Trigger frame may be carried in any PPDU format that is supported by the intended receivers subject to the restrictions in 26.5.2.
- A Control frame is carried in an HE TB PPDU if it is sent as a response to a PPDU that contains a Trigger frame that is not an MU-RTS Trigger frame or if it is sent as a response to a PPDU that contains a frame containing a TRS Control subfield (see 26.5.2).
- A Control frame sent by an HE STA as a response to an HE ER SU PPDU that does not contain a triggering frame should be carried in an HE ER SU PPDU, unless the most recent PPDU sent by the HE STA to the recipient of the Control frame, after association, was not an HE ER SU PPDU. In this case, the Control frame should be carried in a non-HT PPDU.
- A Control frame sent by an HE STA as a response to an HE SU PPDU or a non-HT PPDU that does not contain a triggering frame should be carried in a non-HT PPDU, unless the most recent PPDU sent by the HE STA to the recipient of the Control frame, after association, was an HE ER SU PPDU. In this case, the Control frame should be carried in an HE ER SU PPDU.
- A control response frame shall not be sent in an HE ER SU PPDU if the channel bandwidth of the PPDU containing the frame that elicited the response is greater than 20 MHz.
- A Control frame that is not solicited by another frame and is not a Trigger frame may be carried in an HE ER SU PPDU.
- A Control frame that is sent in the 6 GHz band as a response to an HE SU PPDU or HE MU PPDU but that is not carried in an HE TB PPDU may be carried in an HE SU PPDU if the transmit time of HE SU PPDU is less than or equal to the PPDU duration of a non-HT PPDU containing the Control frame sent at the primary rate (see 10.6.6.5.2).

NOTE 2—A change in the format of the PPDU containing the control response frame (between non-HT and HE ER SU PPDU) occurs in subsequent TXOPs. A STA that solicits a control response frame from a responding STA accounts for the PPDU format of the control response frame to calculate the expected duration of the TXOP. The responding STA determines that the most recent PPDU sent to the soliciting STA is received if it receives an immediate acknowledgment by the soliciting STA in response to the PPDU.

NOTE 3—A STA does not transmit a Control frame in an HE ER SU PPDU to a receiving STA, unless the receiving STA indicates that HE ER SU PPDU reception is enabled.

An HE STA should send an Ack frame that is the response to a Fine Timing Measurement frame carried in a VHT PPDU or HT PPDU in the same PPDU format as the PPDU carrying the Fine Timing Measurement frame.

26.15.3 MCS, NSS, BW, and DCM selection

An HE STA shall follow the rules defined in 10.6 and 26.15.4 for selecting the rate, MCS, and NSS and the rules defined in 10.3.2.7, 10.3.2.9, 10.6.6.6, and 10.6.12 for selecting the channel width (BW) of transmitted PPDUs with the following exceptions:

- HE-MCS, NSS, and BW selection for an HE TB PPDU is defined in 26.5.2.3.
- Rate and BW selection for a CTS sent in response to an MU-RTS Trigger frame is defined in 26.2.6.
- A STA that transmits a Control frame carried in a non-HT PPDU that is a response to a frame received in an HE ER SU PPDU shall set the rate of the non-HT PPDU to 6 Mb/s.
- A STA that transmits a Control frame that is an S-MPDU carried in an HE ER SU PPDU and that is a response to a frame received in an HE ER SU PPDU shall use the <HE-MCS, NSS> tuple <HE-MCS 0, 1>.
- NSS and BW selection is further constrained as defined in 11.40, 26.9, 26.15.2, 26.15.4 through 26.15.8, and 26.17.

An HE STA that transmits an HE PPDU to a receiving STA shall use an <HE-MCS, NSS> tuple that is supported by the receiving STA as indicated by the Supported HE-MCS And NSS Set field in the HE Capabilities element that the receiving STA transmits. If the Supported HE-MCS and NSS set of a receiving STA is not known, the transmitting STA shall transmit using a <HE-MCS, NSS> tuple in the basic HE-MCS and NSS set if the basic HE-MCS and NSS set is not empty. Otherwise, the transmitting STA shall transmit using a <HE-MCS, NSS> tuple in the mandatory HE-MCS and NSS set. An HE STA is subject to all of the rules for HT STAs and VHT STAs that apply to its operating band (see 10.27).

An HE STA may transmit an HE PPDU with 1024-QAM on a 26-, 52-, and 106-tone RU to a recipient STA if it has received from the recipient STA an HE Capabilities element with the Rx 1024-QAM < 242-tone RU Support subfield in the HE PHY Capabilities Information field equal to 1; otherwise, the HE STA shall not transmit an HE PPDU with 1024-QAM on a 26-, 52-, and 106-tone RU.

An HE AP shall not set the UL HE-MCS subfield of a User Info field in a Trigger frame to 10 or 11 for a 26-, 52-, or 106-tone RU allocation, unless the User Info field is addressed to a non-AP HE STA from which the HE AP has received an HE Capabilities element with the Tx 1024-QAM < 242-tone RU Support subfield in the HE PHY Capabilities Information field equal to 1.

An HE STA that sends a Control frame in response to a frame carried in an HE SU PPDU, or HE ER SU PPDU, or HE MU PPDU that carries a frame with the Normal Ack or Implicit BAR ack policy shall set the TXVECTOR parameter CH_BANDWIDTH to indicate a channel width that is the same as the channel width indicated by the RXVECTOR parameter CH_BANDWIDTH of the frame eliciting the response. If the most recently received PPDU sent by the responding STA to the soliciting STA after association was an HE ER SU PPDU, the soliciting STA shall set the TXVECTOR parameter CH_BANDWIDTH to CBW20 for an HE SU PPDU and to ER-RU-242 or ER-RU-H-106 for an HE ER SU PPDU.

NOTE 1—A preamble punctured HE MU PPDU cannot carry a frame with Normal Ack or Implicit BAR ack policy whose acknowledgment would be transmitted in one or more punctured 20 MHz channels of the preamble punctured HE MU PPDU (see 26.4.4.3).

If a control response frame is transmitted in an HE SU PPDU or HE MU PPDU, the channel width (CH_BANDWIDTH parameter of the TXVECTOR) shall be selected first according to 10.6.6.6, and then the <HE-MCS, NSS> tuple shall be selected from a set of <HE-MCS, NSS> tuples called the *CandidateMCSSet*. The *CandidateMCSSet* is defined in 10.6.6.5.3, except that the set additionally contains the <HE-MCS, NSS> tuples for an HE STA.

An HE STA may transmit an HE SU PPDU or HE ER SU PPDU with DCM applied to the Data field to a recipient STA if it has received from the recipient STA an HE Capabilities element with the DCM Max Constellation Rx subfield in the HE PHY Capabilities Information field greater than 0; otherwise, the HE STA shall not transmit an HE SU PPDU or HE ER SU PPDU with DCM applied to the Data field to the recipient STA.

An HE STA may transmit to a recipient STA an HE MU PPDU with DCM applied to the HE-SIG-B field and/or an RU in the Data field addressed to the STA if it has received from the recipient STA an HE Capabilities element with the DCM Max Constellation Rx subfield in the HE PHY Capabilities Information field greater than 0; otherwise, the HE STA shall not transmit to the recipient STA an HE MU PPDU with DCM applied to the HE-SIG-B field and/or an RU in the Data field addressed to the STA.

An HE STA transmits an HE TB PPDU with DCM as defined in 26.5.2.3. An HE AP shall not set the DCM subfield of a User Info field in a Trigger frame to 1 if it has not received from the recipient STA an HE Capabilities element with the DCM Max Constellation Tx subfield in the HE PHY Capabilities Information field greater than 0.

An HE STA that transmits an HE PPDU with DCM applied to the Data field to a recipient STA shall use an RU size that is less than or equal to the maximum RU size indicated in the DCM Max RU subfield in the HE PHY Capabilities Information field in the HE Capabilities element received from the recipient STA.

An HE AP that transmits a Trigger frame addressed to a recipient STA that solicits an HE TB PPDU with DCM shall set the RU Allocation subfield in the Trigger frame to indicate an RU size that is less than or equal to the maximum RU size indicated in the DCM Max RU subfield in HE PHY Capabilities Information field in the HE Capabilities element received from the recipient STA.

An HE STA that transmits an HE PPDU with DCM applied to the Data field to a recipient STA shall use an NSS that is less than or equal to the value indicated in the DCM Max NSS Rx subfield in the HE PHY Capabilities Information field in the HE Capabilities element received from the recipient STA.

An HE AP that transmits a Trigger frame with a User Info field addressed to a recipient STA and with the UL DCM subfield in the User Info field set to 1 shall set the Number Of Spatial Streams subfield in the SS Allocation subfield in the User Info field to less than or equal to the DCM Max NSS Tx subfield in HE PHY Capabilities Information field in the HE Capabilities element received from the recipient STA.

An HE AP shall not transmit a Trigger frame with the UL STBC subfield set to 1 and the UL BW subfield set to indicate a bandwidth less than or equal to 80 MHz if at least one User Info field is addressed to a non-AP HE STA from which the HE AP has received an HE Capabilities element with the STBC Tx $\leq$ 80 MHz subfield in HE PHY Capabilities Information field equal to 0.

An HE AP shall not transmit a Trigger frame with the UL STBC subfield set to 1 and the UL BW subfield set to indicate 80+80 MHz or 160 MHz if at least one User Info field is addressed to a non-AP HE STA from which the HE AP has received an HE Capabilities element with the STBC Tx $>$ 80 MHz subfield in HE PHY Capabilities Information field equal to 0.

An HE STA that sends a Control frame in an HE ER SU PPDU format should use

- DCM encoding if the most recently received PPDU sent by the HE STA, after association, to the STA soliciting the Control frame used DCM; otherwise, the STA should not use DCM for the Control frame.
- 106-tone HE ER SU PPDU if the most recently received PPDU sent by the HE STA, after association, to the STA soliciting the Control frame was a 106-tone HE ER SU PPDU; otherwise, the STA should not use a 106-tone HE ER SU PPDU for the Control frame.

NOTE 2—Transmit parameter switching occurs in subsequent TXOPs. A STA that solicits a Control frame from a peer STA accounts for the transmit parameter of the Control frame to calculate the expected duration of the TXOP. The responding STA determines that the most recent PPDU sent to the soliciting STA is received if it receives an immediate acknowledgment by the soliciting STA in response to the PPDU.

26.15.4 Rate selection constraints for HE STAs

26.15.4.1 Receive HE-MCS and NSS Set

The receive HE-MCS and NSS set is the set of <HE-MCS, NSS> tuples for PPDU bandwidths less than or equal to 80 MHz, 160 MHz, or 80+80 MHz PPDU that an HE STA is capable of receiving. The receive HE-MCS and NSS set for a first HE STA is determined by a second HE STA for each <HE-MCS, NSS> tuple, NSS = 1, ..., 8, and PPDU bandwidth (less than or equal to 80 MHz, 160 MHz, or 80+80 MHz) from the Supported HE-MCS And NSS Set field in the HE Capabilities element received from the first HE STA as follows:

- If support for the <HE-MCS, NSS> tuple at that PPDU bandwidth is mandatory (see 27.1.1), then the <HE-MCS, NSS> tuple at that bandwidth is supported by the first HE STA on receive.
- Otherwise, if the Max HE-MCS For n SS subfield ($n = \text{NSS}$) in each Rx HE-MCS Map b subfield for $b \in \{\leq 80 \text{ MHz}, 160 \text{ MHz}, 80+80 \text{ MHz}\}$ indicates support and neither the Operating Mode field nor the OM Control subfield is received from the first HE STA, then the <HE-MCS, NSS> tuple at PPDU bandwidth b for a given operating channel width is supported by the first HE STA on receive as defined in 9.4.2.248.4.
- Otherwise,
 - If the Operating Mode field is received from the first HE STA, the <HE-MCS, NSS> tuple at that PPDU bandwidth for a given operating channel width is supported by the first HE STA on receive as defined in 9.4.2.248.4 and by Equation (9-2a).
 - If the OM Control subfield is received from the first HE STA, the <HE-MCS, NSS> tuple at that PPDU bandwidth for a given operating channel width is supported by the first HE STA on receive as defined in 9.4.2.248.4 and by Equation (26-4).
- Otherwise, the <HE-MCS, NSS> tuple at that PPDU bandwidth is not supported by the first HE STA on receive.

The <HE-MCS, NSS> tuples excluded by 26.15.4.3 can also be eliminated from the receive HE-MCS and NSS set.

An HE STA shall not, unless explicitly stated otherwise, transmit an HE PPDU, unless the <HE-MCS, NSS> tuple and bandwidth used are in the receive HE-MCS and NSS set of the receiving HE STA(s).

26.15.4.2 Transmit HE-MCS and NSS Set

The transmit HE-MCS and NSS set is the set of <HE-MCS, NSS> tuples for PPDU bandwidth less than or equal to 80 MHz, 106 MHz, or 80+80 MHz PPDU that a HE STA is capable of transmitting. The transmit HE-MCS and NSS set of a first HE STA is determined by a second HE STA for each <HE-MCS, NSS> tuple, $NSS = 1, \dots, 8$, and PPDU bandwidth (less than or equal to 80 MHz, 160 MHz, or 80+80 MHz) from the Supported HE-MCS And NSS Set field received from the first HE STA as follows:

- If support for the <HE-MCS, NSS> tuple at that PPDU bandwidth is mandatory (see 27.1.1), then the <HE-MCS, NSS> tuple at that bandwidth is supported by the first HE STA on transmit.
- Otherwise, if the Max HE-MCS For n SS subfield ($n = NSS$) in each Tx HE-MCS Map b subfield for $b \in \{\leq 80 \text{ MHz}, 160 \text{ MHz}, 80+80 \text{ MHz}\}$ indicates support, then the <HE-MCS, NSS> tuple at PPDU bandwidth b for a given operating channel width is supported by the first HE STA on transmit as defined in 9.4.2.248.4.
- Otherwise, the <HE-MCS, NSS> tuple at that PPDU bandwidth is not supported by the first HE STA on transmit.

A non-AP STA may exclude certain numbers of space-time streams, N_{STS} , as defined in 26.9.3 from its transmit HE-MCS and NSS set.

26.15.4.3 Additional rate selection constraints for HE PPDU

A HE STA shall not transmit a 20 MHz or 40 MHz HE PPDU with an <HE-MCS, NSS> tuple that has HE-MCS 0, 1, 2, or 3 and NSS less than or equal to 4 to a receiver HE STA that has marked as unsupported the HT-MCS with value $HE-MCS + 8 \times (NSS - 1)$ in the Rx MCS Bitmask subfield in the Supported MCS Set field in the HT Capabilities element it transmits. The transmission of a 20 MHz or 40 MHz HE PPDU with HE-MCS greater than 3 is not subject to this constraint.

A HE STA shall not transmit an 80 MHz, 160 MHz, or 80+80 MHz HE PPDU with an <HE-MCS, NSS> tuple that has HE-MCS 0 or 1 and NSS less than or equal to 4 to a receiver HE STA that has marked as unsupported the HT-MCS values of both $2 \times HE-MCS + 8 \times (NSS - 1)$ and $2 \times HE-MCS + 1 + 8 \times (NSS - 1)$ in the Rx MCS Bitmask subfield in the Supported MCS Set field in the HT Capabilities element it transmits. The transmission of an 80 MHz, 160 MHz, or 80+80 MHz HE PPDU with HE-MCS greater than 1 is not subject to this constraint.

An example tabulation of this behavior is given in Table 26-14.

Table 26-14—Example of rate selection for HE PPDU

HT-MCSs that are marked as unsupported	<HE-MCS, NSS> tuples that are not used for CBW20 and CBW40	<HE-MCS, NSS> tuples that are not used for CBW80, CBW160, and CBW80+80
0, 8, 16	<0, 1>, <0, 2>, <0, 3>	—
1, 9	<1, 1>, <1, 2>	—
10	<2, 2>	—
3	<3, 1>	—
0, 1	<0, 1>, <1, 1>	<0, 1>
2, 3	<2, 1>, <3, 1>	<1, 1>
0, 1, 8, 9	<0, 1>, <1, 1>, <0, 2>, <1, 2>	<0, 1>, <0, 2>

26.15.4.4 Rx Supported VHT-MCS and NSS Set

For each <VHT-MCS, NSS> tuple, NSS = 1, ..., 8, and bandwidth (20 MHz, 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz) from the Supported VHT-MCS and NSS Set field received from a first HE STA, a second HE STA shall follow the rules in 10.6.13.1 to determine the Rx Supported VHT-MCS and NSS Set of the first HE STA with the following exception:

- If the second HE STA receives OM Control subfield from the first HE STA, the receive HE-MCS and NSS Set of a first HE STA is determined by a second HE STA according to 9.4.2.157.3 and Table 26-9.

NOTE—If the second HE STA receives both an Operating Mode field and an OM Control subfield from the first HE STA, the rules in 26.9.1 apply.

26.15.5 Additional rules for ER beacons and group addressed frames

An AP that transmits group addressed frames in an HE ER SU PPDU shall transmit the HE ER SU PPDU with an <HE-MCS, NSS> tuple where the HE-MCS is a mandatory HE-MCS and NSS = 1.

A group addressed frame (including a Beacon frame) transmitted in an HE ER SU PPDU shall be sent as an S-MPDU (see Table 9-534), except for group addressed Data frames, which are not required to be sent as an S-MPDU but are required to follow the rules in 10.12.4.

The HE AP transmitting the HE ER SU PPDU shall set the TXVECTOR parameters as follows:

- CH_BANDWIDTH to ER-RU-242
- HE_LTF_TYPE to 2xHE-LTF and GI_TYPE to 0u8s_GI or 1u6s_GI, or HE_LTF_TYPE to 4xHE-LTF and GI_TYPE to 3u2s_GI
- FEC_CODING to BCC_CODING
- STBC to 0
- DCM to 0
- DOPPLER to 0
- BEAMFORMED to 0
- NUM_STS to 1
- NOMINAL_PACKET_PADDING to 16 μ s
- NO_SIG_EXTN to false in the 2.4 GHz band and true otherwise
- BEAM_CHANGE as defined in 26.11.3

26.15.6 Additional rules for HE beacons and group addressed frames

An AP that transmits group addressed frames in an HE SU PPDU shall transmit the HE SU PPDU with an <HE-MCS, NSS> tuple where the HE-MCS is a mandatory HE-MCS and NSS = 1.

NOTE—An AP does not send a Beacon frame in an HE SU PPDU (an HE beacon), unless it is operating in the 6 GHz band (see 26.17.2.2).

A group addressed frame (including a Beacon frame) transmitted in an HE SU PPDU shall be sent as an S-MPDU (see Table 9-534), except for group addressed Data frames, which are not required to be sent as an S-MPDU but are required to follow the rules in 10.12.4.

If the HE SU PPDU contains a group addressed frame intended for at least one STA that is not associated to the AP, then the HE AP shall set the TXVECTOR parameters for the HE PPDU as follows:

- CH_BANDWIDTH to CBW20
- HE_LTF_TYPE to 2xHE-LTF and GI_TYPE to 0u8s_GI or 1u6s_GI, or HE_LTF_TYPE to 4xHE-LTF and GI_TYPE to 3u2s_GI
- FEC_CODING to BCC_CODING
- STBC to 0
- DCM to 0
- DOPPLER to 0
- BEAMFORMED to 0
- NOMINAL_PACKET_PADDING to 16 μ s
- NO_SIG_EXTN to false in the 2.4 GHz band and true otherwise
- BEAM_CHANGE as defined in 26.11.3

Otherwise, if the HE SU PPDU contains group addressed frames intended only for associated STAs then the AP shall set the TXVECTOR parameters listed above to values that are indicated as supported by all the intended STAs, except that the CH_BANDWIDTH shall be set to CBW20 if at least one of the intended STAs is currently not in the awake state.

26.15.7 Additional rules for group addressed frames in an HE MU PPDU

An HE AP may include group addressed frames in an HE MU PPDU subject to the rules defined in this subclause.

An HE AP shall not include a Beacon frame in an HE MU PPDU.

An HE AP that includes a group addressed frame in an HE MU PPDU shall ensure that the frame is included in a broadcast RU in the HE MU PPDU. The HE AP shall additionally ensure that the following conditions are satisfied for the broadcast RU:

- The RU allocation shall comply with the rules in 26.5.1.3 and 27.3.2.8.
- The <HE-MCS, NSS> tuple shall have a mandatory HE-MCS and NSS = 1.
- The broadcast RU shall be located within
 - The primary 20 MHz channel if the group addressed frame is a FILS Discovery or a Probe Response frame, except when the primary 20 MHz channel does not coincide with a PSC and the AP is a 6 GHz-only AP. For this exception, the broadcast RU may be in a PSC that is within the BSS operating channel width (see 26.17.2.3). The broadcast RU size shall not exceed 106 subcarriers if the MU PPDU has a bandwidth that is greater than 20 MHz.
 - The primary 20 MHz channel if the group addressed frame is addressed to at least one associated non-AP STA that has not declared to be in the awake state. The broadcast RU size shall not exceed 106 subcarriers if the MU PPDU has a bandwidth that is greater than 20 MHz.
 - A bandwidth that is indicated as supported in reception by one or more associated non-AP STAs, if the group addressed frame is addressed only to those non-AP STAs and the STAs have declared to be in the awake state. The broadcast RU size shall not exceed the minimum common bandwidth that is supported in reception by all STAs in the HE Capabilities element they transmit or in the most recently sent OM Control or OMN frames.
 - The SST subchannel if the group addressed frame is addressed to one or more HE SST STAs, the primary 20 MHz channel does not coincide with the subchannel assigned to the HE SST STAs and the frame is not addressed to any STAs other than the HE SST STAs in that subchannel (see 26.8.7.2). The broadcast RU size shall not exceed 106 subcarriers if the SST subchannel is 20 MHz.

- The TXVECTOR parameters listed below shall be set as follows:
 - HE_LTF_TYPE to 2xHE-LTF and GI_TYPE to 0u8s_GI or 1u6s_GI, or HE_LTF_TYPE to 4xHE-LTF and GI_TYPE to 3u2s_GI
 - FEC_CODING to BCC_CODING
 - STBC to 0
 - DCM to 0
 - DOPPLER to 0
 - BEAMFORMED to 0
 - NOMINAL_PACKET_PADDING to 16 μ s
 - NO_SIG_EXTN to false in the 2.4 GHz band and true otherwise
 - BEAM_CHANGE as defined in 26.11.3
 - STA_ID as defined in 26.11.1

Group addressed frames transmitted in an HE MU PPDU shall be sent as an S-MPDU (see Table 9-535), except for group addressed Data frames, which are not required to be sent as an S-MPDU but are required to follow the rules in 10.12.4.

26.15.8 Additional rules for PPDUs sent in the 6 GHz band

An HE STA that transmits a PPDU that is not sent in response to a Trigger frame in the 6 GHz band and that contains a frame that is not a control response frame with the Address 1 field set to the MAC address of an HE AP with which it is not associated and from which it has received a FILS Discovery frame or an HE Operation element shall ensure that the PPDU meets the following conditions:

- The bandwidth of the PPDU is less than or equal to the operating bandwidth of the HE BSS as indicated in the BSS Operating Channel Width subfield of the FILS Discovery frame or in the Channel Width subfield of the HE Operation element sent by the AP.
- The PPDU is transmitted with a number of spatial streams that is less than or equal to the maximum number of spatial streams of the HE BSS as indicated in the Maximum Number of Spatial Stream subfield of the FILS Discovery frame or in the Basic HE-MCS And NSS Set field of the HE Operation element sent by the AP.
- If the PPDU is an HE PPDU, then the PPDU is transmitted with an <HE-MCS, NSS> tuple providing a data rate that is greater than or equal to the minimum rate indicated in the FILS Minimum Rate field (if present) of the FILS Discovery frame or in the Minimum Rate field of the HE Operation element sent by the AP.
- If the PPDU is a non-HT PPDU, then the PPDU is transmitted with a data rate that is greater than or equal to the minimum of <R, 54 Mb/s>, where R is the minimum rate indicated in the FILS Minimum Rate field (if present) of the FILS Discovery frame or in the Minimum Rate field of the HE Operation element sent by the AP.

An HE STA that transmits a PPDU that is not sent in response to a Trigger frame in the 6 GHz band and that contains a frame that is not a control response frame with Address 1 field set to the MAC address of the AP with which it is associated shall ensure that the PPDU meets the following conditions:

- If the PPDU is a non-HT (duplicate) PPDU, then the PPDU is transmitted with a data rate that is greater than or equal to the minimum of <R, 54 Mb/s>, where R is the minimum rate indicated in the Minimum Rate field of the HE Operation element sent by the AP.
- If the PPDU is an HE PPDU, then the PPDU is transmitted with an <HE-MCS, NSS> tuple providing a data rate that is not less than the data rate indicated in the Minimum Rate field of the HE Operation element sent by the AP.

An HE STA that transmits a PPDU that is not an HE TB PPDU in the 6 GHz band and that contains a frame that is not a control response frame with Address 1 field set to the MAC address of an AP with which it is not associated shall determine a local maximum transmit power for that transmission following the rules in 11.7.5, if the local maximum transmit power is received in Transmit Power Envelope elements and combinations of Country elements and Power Constraint elements in the most recent Beacon or Probe Response frame, on the channel from that AP.

26.16 Midamble parameter setting rules

A STA shall not set the TXVECTOR parameter DOPPLER to 1 for an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU, unless the STA has received from each of the intended recipient STAs an HE Capabilities element with the Doppler Rx subfield in the HE PHY Capabilities Information field equal to 1.

A STA shall not send a Trigger frame with the Doppler subfield in the Common Info field set to 1, unless the STA has received from each of the intended recipient STAs of the Trigger frame an HE Capabilities element with the Doppler Tx subfield in the HE PHY Capabilities Information field equal to 1.

A STA shall not set the TXVECTOR parameter HE_LTF_TYPE to 1xHE-LTF and the parameter DOPPLER to 1 for an HE SU PPDU, unless the STA has received an HE Capabilities element from each of the intended recipient STAs where the HE PHY Capabilities Information field meets the following conditions:

- The Midamble Tx/Rx 2x And 1x HE-LTF subfield is 1.
- The HE SU PPDU With 1x HE-LTF And 0.8 μ s GI subfield is 1.

A STA shall not set the TXVECTOR parameter HE_LTF_TYPE to 1xHE-LTF and the parameter DOPPLER to 1 for an HE ER SU PPDU, unless the HE STA has received from each of the intended recipient STAs an HE Capabilities element in which the HE PHY Capabilities Information field meets the following conditions:

- The Midamble Tx/Rx 2x And 1x HE-LTF subfield is 1.
- The HE ER SU PPDU With 1x HE-LTF And 0.8 μ s GI subfield is 1.

A STA shall not set the TXVECTOR parameter HE_LTF_TYPE to 2xHE-LTF and the parameter DOPPLER to 1 for an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU, unless the HE STA has received an HE Capabilities element with the Midamble Tx/Rx 2x And 1x HE-LTF subfield in the HE PHY Capabilities Information field equal to 1.

A non-AP STA shall not transmit an HE Capabilities element with the Doppler Tx subfield in the HE PHY Capabilities Information field set to 1, unless the non-AP STA meets all of the following conditions:

- The non-AP STA is capable of transmitting an HE TB PPDU with midambles and a 4x HE-LTF.
- If the non-AP STA has set the Midamble Tx/Rx 2x And 1x HE-LTF subfield in the HE PHY Capabilities Information field to 1, the non-AP STA is capable of transmitting an HE TB PPDU with midambles and a 2x HE-LTF.
- If the non-AP STA has set both the Midamble Tx/Rx 2x And 1x HE-LTF and the Full Bandwidth UL MU-MIMO subfields in the HE PHY Capabilities Information field to 1, the non-AP STA is capable of transmitting an HE TB PPDU with midambles, using full-bandwidth UL MU-MIMO, and a 1x HE-LTF.

A STA that transmits an HE SU PPDU or HE ER SU PPDU with the TXVECTOR parameter DOPPLER set to 1 shall not set the TXVECTOR parameter NUM_STS to indicate a number of space-time streams that is greater than indicated in the Midamble Tx/Rx Max NSTS subfield in the HE PHY Capabilities Information subfield in the HE Capabilities element received from any of the intended recipient STAs.

A STA transmitting an HE MU PPDU with the TXVECTOR parameter DOPPLER set to 1 shall not set the TXVECTOR parameter NUM_STS[*u*] to indicate a number of space-time streams that is greater than indicated in the Midamble Tx/Rx Max NSTS subfield in the HE PHY Capabilities Information subfield in the HE Capabilities element received from the intended recipient STA *u*.

An AP transmitting a Trigger frame with the Doppler subfield in the Common Info field set to 1 shall not set the Number Of Spatial Streams subfield in a User Info field in the Trigger frame to indicate a number of space-time streams that is greater than indicated in the Midamble Tx/Rx Max NSTS subfield in the HE PHY Capabilities Information subfield in the HE Capabilities element received from the STA addressed by the User Info field.

26.17 HE BSS operation

26.17.1 Basic HE BSS operation

The Beacon frames generated within an HE BSS contain an HE Operation element.

An HE STA has dot11HEOptionImplemented equal to true.

A STA that is operating in an HE BSS shall be able to receive and transmit at each of the <HE-MCS, NSS> tuple values indicated by the Basic HE-MCS And NSS Set field of the HE Operation parameter of the MLME-START.request primitive and shall be able to receive at each of the <HE-MCS, NSS> tuple values indicated by the Supported HE-MCS and NSS Set field in the HE Capabilities parameter of the MLME-START.request primitive.

The basic HE-MCS and NSS set is the set of <HE-MCS, NSS> tuples that are supported by all HE STAs that are members of an HE BSS. It is established by the STA that starts the HE BSS and indicated by the Basic HE-MCS And NSS Set field of the HE Operation parameter in the MLME-START.request primitive. Other HE STAs determine the basic HE-MCS and NSS set from the Basic HE-MCS And NSS Set field of the HE Operation element in the BSSDescription derived through the scan mechanism (see 11.1.4.1).

An HE STA shall not attempt to join (MLME-JOIN.request primitive) a BSS, unless it supports (i.e., is able to both transmit and receive using) all of the <HE-MCS, NSS> tuples in the basic HE-MCS and NSS set.

NOTE 1—An HE STA does not attempt to (re)associate with an HE AP, unless the STA supports (i.e., is able to both transmit and receive using) all of the <HE-MCS, NSS> tuples in the Basic HE-MCS And NSS Set field in the HE Operation element transmitted by the AP because the MLME-JOIN.request primitive is a necessary precursor to (re)association.

A STA operating in the 2.4 GHz band that sets dot11HEOptionImplemented to true shall set dot11HighThroughputOptionImplemented to true. A STA operating in the 5 GHz or 6 GHz band that sets dot11HEOptionImplemented to true shall set both dot11VHTOptionImplemented and dot11HighThroughputOptionImplemented to true. A non-AP STA that sets dot11HEOptionImplemented to true shall set dot11MultiBSSIDImplemented to true.

An HE STA operating in the 6 GHz band is a VHT STA, except that it is exempt from following VHT and HT functionalities and/or requirements that are not applicable or that are superseded by equivalent HE functionalities and/or requirements (see Clause 26 and Clause 27) and except that it shall use the HE format instead of the VHT, HT_GF, or HT_MF format for PPDU's transmitted in the 6 GHz band. Additional HE functionalities and/or requirements for the 6 GHz band are defined in 26.17.2.

A STA that is an HE AP or an HE mesh STA declares the channel widths at which it is capable of operating in the PHY Capabilities Information field in the HE Capabilities element that it transmits (see Table 9-322b).

An HE AP operating in the 5 GHz or 6 GHz bands shall set B1 in the Supported Channel Width Set field in the PHY Capabilities Information field in the HE Capabilities element to indicate support for 40 MHz and 80 MHz channel width.

A STA transmitting an HT Capabilities element and HE Capabilities element shall set the Supported Channel Width Set subfield of the HT Capabilities element to 1 if either B0 or B1 of the Supported Channel Width Set subfield in the HE Capabilities element is 1, unless the STA is a 20 MHz-only non-AP HE STA. In this case, the Supported Channel Width Set subfield of the HT Capabilities element shall be set to 0.

A STA transmitting a VHT Capabilities element and HE Capabilities element shall set the Supported Channel Width Set subfield of the VHT Capabilities element to indicate the same channel width as indicated in the HE Capabilities element, unless the STA is a 20 MHz-only non-AP HE STA. In this case, the Supported Channel Width Set subfield of the VHT Capabilities element is reserved.

At a minimum, an HE STA sets the Rx MCS Bitmask subfield of the Supported MCS Set field of its HT Capabilities element according to the setting of each Rx HE-MCS Map b subfield for $b \in \{\leq 80 \text{ MHz}, 160 \text{ MHz}, 80+80 \text{ MHz}\}$ of the Supported HE-MCS And NSS Set field of its HE Capabilities element as follows: For each Max HE-MCS For n SS subfield (where $1 \leq n \leq 4$) of each Rx HE-MCS Map b subfield for $b \in \{\leq 80 \text{ MHz}, 160 \text{ MHz}, 80+80 \text{ MHz}\}$ with a value other than 3 (i.e., no support for that number of spatial streams), the STA shall indicate support for HT MCSs $8 \times (n - 1)$ to $8 \times (n - 1) + 7$ in the Rx MCS Bitmask subfield, where n is the number of spatial streams, except for HT-MCSs marked as unsupported as described in 26.15.4.3.

An HE AP or an HE mesh STA shall set the VHT Operation Information Present field in the HE Operation element to 0 if a VHT Operation element is present in the frame that carries the HE Operation element or if the frame that carries the HE Operation element is sent in the 2.4 GHz band. An HE AP or HE mesh STA shall set the VHT Operation Information Present field in the HE Operation element to 1 if a VHT Operation element is not present in the frame that carries the HE Operation element and the frame is sent in the 5 GHz band.

An HE AP or an HE mesh STA that transmits an HE Operation element that has the VHT Operation Information Present field set to 1 shall do one of the following to set the BSS operating channel:

- Set the STA Channel Width subfield and Channel Center Frequency Segment 2 subfield in the HT Operation Information field in the HT Operation element, the Channel Width subfield in the VHT Operation Information field in the HE Operation element, and the Channel Center Frequency Segment 0 and Channel Center Frequency Segment 1 subfields in the VHT Operation Information field in the HE Operation element to indicate the BSS bandwidth as defined in Table 11-23 and Table 11-25 based on the Extended NSS BW Support and Supported Channel Width Set fields.
- Set the STA Channel Width subfield and Channel Center Frequency Segment 2 subfield in the HT Operation Information field in the HT Operation element, the Channel Width subfield in the VHT Operation Information field in the HE Operation element, and the Channel Center Frequency Segment 0 and Channel Center Frequency Segment 1 subfields in the VHT Operation Information field in the HE Operation element to indicate the BSS bandwidth as defined in Table 11-23 and Table 11-25 based on the Rx HE-MCS Map $\leq 80 \text{ MHz}$, Rx HE-MCS Map 160 MHz, and Rx HE-MCS Map 80+80 MHz fields.

NOTE 2—The Channel Center Frequency Segment 2 is 0 if Table 11-24 is applied.

NOTE 3—These two methods give the same result.

The setting of the Channel Center Frequency Segment 0, Channel Center Frequency Segment 1, and Channel Center Frequency Segment 2 subfields is defined in Table 11-24, except that the Max NSS support is provided by the HE STA in frames that contain an HE Capabilities element (see 9.4.2.248) and an Operating Mode field (see 9.4.1.53), where in the table the maximum NSS support refers to the HE maximum NSS support instead of the VHT maximum NSS support for an HE STA.

An HE STA shall determine the channelization using the information in the Primary Channel field of the HT Operation element when operating in 2.4 GHz and using the combination of the information in the Primary Channel field in the HT Operation element and the Channel Center Frequency Segment 0 and Channel Center Frequency Segment 1 subfields in the VHT Operation Information field in the VHT Operation element if operating in the 5 GHz band (see 21.3.14). An HE STA determines the channelization as defined in 26.17.2 if operating in the 6 GHz band.

An HE AP or an HE mesh STA shall set the Secondary Channel Offset subfield in the HT Operation Information field in the HT Operation element to indicate the secondary 20 MHz channel as defined in Table 9-190, if the BSS bandwidth is more than 20 MHz.

An HE STA that is a member of an HE BSS shall follow the rules in 11.38.1 when transmitting a 20 MHz, 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz HE PPDU with the following exceptions:

- An HE TB PPDU sent in response to a triggering frame follows the rules defined in 26.5.2.3.
- An 80 MHz, 160 MHz, or 80+80 MHz DL HE MU PPDU with preamble puncturing may be transmitted if the primary 20 MHz or the primary 40 MHz channel is occupied by the transmission and certain 20 MHz subchannels of the secondary channel are idle (see Table 27-20 and 10.23.2.5).

An HE STA shall not transmit to a recipient HE STA using a channel width that is not indicated as supported in the Supported Channel Width Set subfield in the HE Capabilities element received from that HE STA.

An HE STA shall not transmit to a recipient HE STA using a channel width that exceeds the BSS channel width in the Channel Width field that is contained in the following:

- The HE Operation element most recently exchanged with the recipient STA, if any, and if the Channel Width field is present.
- Otherwise, the VHT Operation element most recently exchanged with the recipient STA, if any.
- Otherwise, the HT Operation element most recently exchanged with the recipient STA, if any.

A STA shall not transmit an HE PPDU to a recipient STA that carries a frame that is not an HE Compressed Beamforming/CQI frame (see 26.7.3) and that exceeds the maximum MPDU length capability indicated in the VHT Capabilities element last received from the recipient STA in the 2.4 GHz or 5 GHz band or, if a VHT Capabilities element has not been received from the recipient STA, that exceeds the maximum A-MSDU length indicated in the HT Capabilities element last received from the recipient STA in the 2.4 GHz or 5 GHz band.

A STA shall not transmit an HE PPDU to a recipient STA that carries a frame that is not an HE Compressed Beamforming/CQI frame (see 26.7.3) and that exceeds the maximum MPDU length capability indicated in the HE 6 GHz Band Capabilities element last received from the recipient STA in the 6 GHz band.

An HE AP shall set the RIFS Mode field in the HT Operation element to 0.

An HE STA follows the rules in 11.38 for channel selection, scanning requirements, channel switching, NAV assertion, and antenna indication when operating in the 5 GHz or 6 GHz band unless explicitly stated otherwise in Clause 26. An HE STA shall additionally follow the rules in 26.17.2 for scanning and operation in the 6 GHz band.

An HE STA shall follow the rules in 11.15 for channel selection, scanning requirements, channel switching, and NAV assertion when operating in 2.4 GHz unless explicitly stated otherwise in Clause 26.

The AP of an ER BSS shall not transmit a Probe Response or (Re)Association Response frame in response to a Probe Request or (Re)Association Request frame, respectively, sent by a non-HE STA. An HE AP that is not operating an ER BSS may set the ER SU Disable subfield in the HE Operation element it transmits to 1.

A STA shall have the same value of maximum VHT NSS defined by its Rx HE-MCS Map ≤ 80 MHz subfield in the HE Capabilities element as the maximum NSS value indicated by its Rx VHT-MCS Map field in the VHT Capabilities element. If a STA supports 160 MHz, the Maximum NSS defined by its Rx VHT-MCS Map field and Extended NSS BW Support field in the VHT Capabilities element at 160 MHz shall not be more than the maximum NSS defined by its Rx HE-MCS Map 160 MHz subfield in the HE Capabilities element at 160 MHz. If a STA supports 80+80 MHz, the maximum NSS defined by its Rx VHT-MCS Map field and Extended NSS BW Support field in the VHT Capabilities element at 80+80 MHz shall not be more than the maximum NSS defined by its Rx HE-MCS Map 80+80 MHz subfield in the HE Capabilities element at 80+80 MHz. For every NSS in VHT Capabilities elements and HE Capabilities elements transmitted by a STA, if the maximum HE-MCS is 9 or more, the maximal VHT-MCS shall be 9. Otherwise, the maximal VHT-MCS shall be the same as the HE-MCS. An HE STA shall not transmit a VHT Capabilities element with the Supported Channel Width Set field equal to 1 and the Extended NSS BW Support field equal to 3 or with the Supported Channel Width Set field equal to 2 and the Extended NSS BW Support field equal to 3.

If an HE STA supports 160 MHz, the maximum NSS defined by its Rx HE-MCS Map 160 MHz subfield for an HE-MCS in the HE Capabilities element at 160 MHz shall not be more than the maximum NSS defined by its Rx HE-MCS Map ≤ 80 MHz subfield for the HE-MCS in the HE Capabilities element at 80 MHz.

If an HE STA supports 80+80 MHz, the maximum NSS defined by its Rx HE-MCS Map 80+80 MHz subfield for an HE-MCS in the HE Capabilities element at 80+80 MHz shall not be more than the maximum NSS defined by its Rx HE-MCS Map ≤ 80 MHz subfield for the HE-MCS in the HE Capabilities element at 80 MHz.

26.17.2 HE BSS operation in the 6 GHz band

26.17.2.1 General

A STA that supports operation in the 6 GHz band sets dot11HE6GOptionImplemented to true.

A STA with dot11HE6GOptionImplemented equal to true and operating in the 6 GHz band is a STA 6G.

An AP with dot11HE6GOptionImplemented equal to true and operating in the 6 GHz band is a 6 GHz AP.

A mesh STA with dot11HE6GOptionImplemented equal to true and operating in the 6 GHz band is a 6 GHz mesh STA.

A non-AP STA with dot11HE6GOptionImplemented equal to true and operating in the 6 GHz band is a 6 GHz non-AP STA.

NOTE—A STA 6G sets dot11HighThroughputOptionImplemented and dot11VHTOptionImplemented to true, but certain VHT and HT functionalities do not apply when operating in the 6 GHz band. See 26.17.1.

A STA 6G shall have dot11ExtendedChannelSwitchActivated, dot11MultiDomainCapabilityActivated, and dot11OperatingClassesRequired equal to true and shall set the Extended Channel Switching field to 1 in the Extended Capabilities elements it transmits.

A STA 6G shall meet the Class A requirements in 27.3.15.

A 6 GHz AP shall indicate support for at least 80 MHz channel width.

A 6 GHz AP shall set the Co-Hosted BSS subfield in HE Operation element to 0.

A STA 6G shall not transmit an HT Capabilities element, VHT Capabilities element, HT Operation element, VHT Operation element, or an HE Operation element that contains a VHT Operation Information field.

A STA 6G shall not transmit in an HE PPDU a frame that is not an HE Compressed Beamforming/CQI frame (see 26.7.3) and that exceeds the maximum MPDU length capability indicated in the HE 6 GHz Band Capabilities element received from the recipient STA.

A 6 GHz AP or 6 GHz mesh STA shall include the 6 GHz Operation Information field in the HE Operation elements it transmits. The AP or mesh STA shall set the Channel Width subfield, the Channel Center Frequency Segment 0, and the Channel Center Frequency Segment 1 subfields of the 6 GHz Operation Information field as defined in Table 26-15, based on the Rx HE-MCS Map ≤ 80 MHz, Rx HE-MCS Map 160 MHz, and Rx HE-MCS Map 80+80 MHz subfields of the Supported HE MCS And NSS Set field in the HE Capabilities element transmitted by the AP.

Table 26-15—6 GHz BSS channel width

Channel Width field	Center Frequency Segment 1 field	BSS channel width (MHz)
0	0	20
1	0	40
2	0	80
3	$CCFS1 > 0$ and $ CCFS1 - CCFS0 = 8$	160
3	$CCFS1 > 0$ and $ CCFS1 - CCFS0 > 16$	80+80
NOTE—CCFS0 represents the value of the Channel Center Frequency Segment 0 field, and CCFS1 represents the value of the Channel Center Frequency Segment 1 field.		

A STA 6G shall determine the BSS channelization using the Primary Channel, Channel Center Frequency Segment 0, and Channel Center Frequency Segment 1 subfields in the 6 GHz Operation Information field in the HE Operation element when operating in 6 GHz band (see 21.3.14 for the channelization and 27.3.23.2 for the equation defining the channel center frequencies in the 6 GHz band).

A STA 6G shall not transmit an HT PPDU.

A STA 6G shall not transmit a VHT PPDU.

A STA 6G shall not transmit a DSSS PPDU.

A STA 6G shall not transmit an HR/DSSS PPDU.

A STA 6G shall not transmit an ERP-OFDM PPDU.

A STA 6G shall set `dot11SpectrumManagementRequired` to true and operate as defined in 11.7.

A 6 GHz AP shall set `dot11FILSOmitReplicateProbeResponses` to true.

A 6 GHz AP may respond with a (Re)Association Response frame with the Status Code field indicating `DENIED_POOR_CHANNEL_CONDITIONS` if it receives a (Re)Association Request frame from a non-AP STA below a minimum RSSI threshold value. A 6 GHz AP may send a Disassociation frame with the Reason Code field indicating `POOR_RSSI_CONDITIONS` to an associated non-AP STA if it receives frames from the STA below a minimum RSSI threshold value for a sufficiently long period of time. How an AP selects a minimum RSSI threshold value or sufficient interval of time is beyond the scope of this standard.

A 6 GHz non-AP STA that receives a (Re)Association Response frame with the Status Code field indicating `DENIED_POOR_CHANNEL_CONDITIONS` or a Disassociation frame with the Reason Code field indicating `POOR_RSSI_CONDITIONS` from a 6 GHz AP should not transmit a (Re)Association Request frame or a Probe Request frame to the AP until one of the following conditions is met:

- Sufficient time has passed since it received the (Re)Association Response frame or Disassociation frame from the AP.
- The STA has determined that a (Re)Association Request frame or Probe Request frame that it transmits will be received by the AP at a sufficiently high RSSI level and in sufficiently good conditions compared with its previous transmission to the AP.

How a 6 GHz non-AP STA determines sufficient time has passed or a suitable RSSI threshold is beyond the scope of this standard.

26.17.2.2 Beacons in the 6 GHz band

A 6 GHz AP transmits Beacon frames as defined in 11.1, and they may be contained in a non-HT PPDU, non-HT duplicate PPDU, or HE SU PPDU.

A 6 GHz AP that transmits a Beacon frame in a non-HT PPDU follows the rules in 10.6.5.1.

A 6 GHz AP that transmits a Beacon frame in a non-HT duplicate PPDU shall follow the rules in 10.6.5.1 and shall set the TXVECTOR parameter `CH_BANDWIDTH` of the PPDU to a value that is up to the operating channel width of the BSS.

If a 6 GHz AP schedules a Beacon frame for transmission in a non-HT duplicate PPDU, then it shall set the Duplicate Beacon subfield to 1 in the 6 GHz Operation Information field of the HE Operation element it transmits; otherwise, the AP shall set the Duplicate Beacon subfield to 0.

A 6 GHz AP that transmits a Beacon frame in an HE SU PPDU shall follow the rules defined in 26.15.6.

NOTE—An AP does not transmit a Beacon frame in an HE SU PPDU or non-HT duplicate PPDU in the 2.4 GHz or 5 GHz bands.

26.17.2.3 Scanning in the 6 GHz band

26.17.2.3.1 General

A STA 6G shall perform the subset of the operations defined for a FILS STA that is described in 26.17.2.3 and may perform all the other operations defined for a FILS STA.

A 6 GHz AP may set `dot11CollocatedRNRImplemented` to true and shall set `dot11ShortSSIDListImplemented` to true. An AP that is in the same co-located AP set as a 6 GHz AP shall set `dot11CollocatedRNRImplemented` to true and `dot11ShortSSIDListImplemented` to true.

A STA 6G shall not transmit a Probe Request frame to the broadcast destination address that includes a Short SSID List element with more than one Short SSID field.

NOTE—In bands other than the 6 GHz band, there might be more than one Short SSID field in a Short SSID List element in a Probe Request frame to the broadcast destination address. A Probe Request frame does not contain more than one Short SSID List element (see Table 9-38). A Probe Request frame transmitted to the broadcast destination address that contains a Short SSID List element has the SSID field of the SSID element set to the SSID of a known AP or set to the one-octet value 128 if the STA does not know any SSID (see 11.1.4.3.2).

26.17.2.3.2 AP behavior for fast passive scanning

A 6 GHz AP that does not share the same co-located AP set as an AP operating in the 2.4 GHz band or 5 GHz band is referred to as a 6 GHz-only AP.

A 6 GHz-only AP that intends to be efficiently discovered by STAs using scanning in the 6 GHz band shall have `dot11FILSFDFrameBeaconMaximumInterval` set to a nonzero value that is less than or equal to 20 TUs.

NOTE 1—A 6 GHz-only AP intends to be efficiently discovered by STAs using scanning in the 6 GHz band if it schedules for transmission FILS Discovery, Beacon, or unsolicited Probe Response frames every 20 TUs or less.

If a 6 GHz AP has `dot11UnsolicitedProbeResponseOptionActivated` equal to true, then the AP shall have `dot11FILSFDFrameBeaconMaximumInterval` set to a nonzero value that is less than or equal to 20 TUs; otherwise, the AP may have `dot11FILSFDFrameBeaconMaximumInterval` set to any value.

A 6 GHz AP that has `dot11FILSFDFrameBeaconMaximumInterval` equal to a nonzero value shall schedule for transmission FILS Discovery frames as described in 11.45.2.1, except that the following apply:

- If the FILS Discovery frame is contained in a DL HE MU PPDU, then it shall be included in a broadcast RU of the DL HE MU PPDU as defined in 26.15.7.
- If `dot11UnsolicitedProbeResponseOptionActivated` is true, all FILS Discovery frames shall be omitted, and an unsolicited broadcast Probe Response frame, instead of each omitted FILS Discovery frame, shall be scheduled for transmission at the target transmit time.
- If `dot11UnsolicitedProbeResponseOptionActivated` is false, then a FILS Discovery frame may be omitted, and an unsolicited broadcast Probe Response frame, instead of the omitted FILS Discovery frame, shall be scheduled for transmission at the target transmit time.

A 6 GHz AP may send an unsolicited broadcast Probe Response frame. The Probe Response frame may be included in a broadcast RU of a DL HE MU PPDU as defined in 26.15.7. The Probe Response may be carried in a non-HT duplicate PPDU in which case the PPDU shall have the TXVECTOR parameter `CH_BANDWIDTH` set to a value that is up to the operating channel width of the BSS.

A 6 GHz AP that transmits a FILS Discovery frame carrying an FD Capability field shall set the PHY Index subfield to 4.

An AP that corresponds to a nontransmitted BSSID does not schedule for transmission FILS Discovery frames (see 11.45.2.1) or unsolicited broadcast Probe Response frames (see 11.1.4.3.4).

See 11.1.3.8 for procedures on advertisement of nontransmitted BSSIDs in a multiple BSSID set.

A 6 GHz-only AP should set up the BSS with a primary 20 MHz channel that coincides with a preferred scanning channel (PSC) (see 26.17.2.3.3).

NOTE 2—An AP might initiate a BSS with a primary channel that coincides with a PSC in order to assist STAs that are scanning the 6 GHz band to discover the BSS. The AP might subsequently switch its operating channel to a non-PSC (e.g., using a CSA mechanism) if it does not expect additional (not yet associated) STAs will need to discover the BSS.

A 6 GHz AP shall not respond to a Probe Request frame if the frame carries a FILS Request Parameters element and the AP is unable to satisfy the response time constraint specified in the Max Channel Time field in the element (see 11.1.4.3.4). If a 6 GHz AP receives a Probe Request frame and responds with a Probe Response frame (per 11.1.4.3.4), the Address 1 field of the Probe Response frame shall be set to the broadcast address, unless the AP is not indicating its actual SSID in the SSID element of its Beacon frames.

26.17.2.3.3 Non-AP STA scanning behavior

The 20 MHz channels in the 6 GHz band, with channel center frequency ($ch_a = \text{Channel starting frequency} - 55 + 80 \times n$, in megahertz) are referred to as *preferred scanning channels* (PSCs). Channel starting frequency is defined in 27.3.23.2, and $n = 1, \dots, 15$.

NOTE 1—PSCs might not all be available in a specific location due to regulatory restrictions. A STA scanning the 6 GHz band knows where these PSCs are located since their position is fixed.

A non-AP STA that is actively scanning a channel in the 6 GHz band shall operate as defined in 11.1.4.3.2, unless a given rule is superseded by the rules defined in 26.17.2.3.

The non-AP STA shall not transmit a Probe Request frame to the broadcast destination address with the Address 3 field set to the wildcard BSSID and the SSID set to the wildcard SSID.

The non-AP STA shall not send a Probe Request frame to the broadcast destination address with the Address 3 field (BSSID) set to the BSSID of an AP from which it has already received a Probe Response or a Beacon frame since the start of its scanning on that channel.

The non-AP STA shall not send a Probe Request frame to the broadcast destination address with the Address 3 field (BSSID) set to the BSSID of a nontransmitted BSSID if it has already received the nontransmitted BSSID profile for that BSSID via a Beacon frame or Probe Response frame sent by the transmitted BSSID since the start of its scanning on that channel.

Until the FILSProbeTimer reaches dot11FILSProbeDelay, the non-AP STA shall not send a Probe Request frame to the broadcast destination address with the SSID field set to the SSID of an AP for which it has received a Reduced Neighbor Report or Neighbor Report element with the Unsolicited Probe Responses Active subfield corresponding to that AP set to 1 and that indicates that the AP is operating in that channel. Until the FILSProbeTimer reaches dot11FILSProbeDelay, the non-AP STA shall not send a Probe Request frame to the broadcast destination address with the Address 3 field set to the BSSID of an AP for which it has received a Reduced Neighbor Report or Neighbor Report element with the Unsolicited Probe Responses Active subfield corresponding to that AP set to 1 and that indicates that the AP is operating in that channel. Until the FILSProbeTimer reaches dot11FILSProbeDelay, the non-AP STA shall not send a Probe Request frame to the broadcast destination address with the Short SSID field set to the short SSID that corresponds to the SSID of an AP for which it has received a Reduced Neighbor Report or Neighbor Report element with the Unsolicited Probe Responses Active subfield corresponding to that AP set to 1 and that indicates that the AP is operating in that channel.

The non-AP STA shall not transmit more than one Probe Request frame to the broadcast destination address with the Address 3 field set to the wildcard BSSID and the SSID field not set to the wildcard SSID during each 20 TU period scanning the channel. The non-AP STA shall not transmit more than three Probe Request frames to the broadcast destination address with Address 3 field set to a non-wildcard BSSID during each 20 TU period scanning the channel.

The non-AP STA shall set `dot11FILSProbeDelay` to a value equal to or greater than 20 TU.

NOTE 2—A non-AP STA waits for at least 20 TU so that it maximizes the probability of receiving FILS Discovery or broadcast Probe Response frames, if any, sent by an AP in that channel (see 26.17.2.3.2).

If the non-AP STA is scanning a channel, then the following apply:

- If the STA has received a FILS Discovery frame indicating that an AP is operating in that channel, or if the STA has received a Reduced Neighbor Report or Neighbor Report element indicating that an AP is operating in that channel, then the STA may, subject to the other rules in this clause, send a Probe Request frame to the broadcast destination address in that channel, with the SSID field set to the SSID that corresponds to that AP or with the Short SSID field of the Short SSID List element set to the short SSID that corresponds to that AP and/or with the Address 3 field set to the BSSID of that AP, starting from step c) of 11.1.4.3.2.
- Otherwise, if the channel is a PSC and the STA has determined the medium to be idle for a continuous period of at least `dot11MinPSCProbeDelay` from the start of the scan on the channel, then the STA may, subject to other rules in this subclause, send a Probe Request frame to the broadcast destination address in that channel, with the SSID field set to the SSID that corresponds to an AP or with the Short SSID field of the Short SSID List element set to the short SSID that corresponds to an AP, and/or with the Address 3 field set to the BSSID of an AP, after invoking the backoff procedure, described in 10.23.2.2, starting from step c) of 11.1.4.3.2.
- Otherwise, if the STA has discovered the presence of an AP in that channel through means that are beyond the scope of this standard and the AP might be detected by the STA [see the definition of “detected access point (AP)” in 3.2], then the STA may send a Probe Request frame to the broadcast destination address in that channel, with the Address 3 field set to the BSSID of that AP, starting from step c) of 11.1.4.3.2.
- Otherwise, if the `FILSProbeTimer` reaches `dot11FILSProbeDelay` and the channel is a PSC, then the STA may, subject to the other rules in this subclause, send a Probe Request to the broadcast destination address in that channel, starting from step c) of 11.1.4.3.2.
- Otherwise, the STA shall not send a Probe Request frame to the broadcast destination address in that channel.

NOTE 3—The STA might send an individually addressed Probe Request frame to an AP for reasons other than active scan (e.g., to obtain an updated EDCA parameter set) even if it has already received a FILS Discovery, Probe Response, or Beacon frame from that AP.

If a non-AP STA sends a Probe Request frame in the 6 GHz band that includes a FILS Request Parameters element, then the non-AP STA shall set the value of PHY Support Criterion subfield in the element to either 0 or 3.

See 11.1.3.8 for procedures on discovery of nontransmitted BSSIDs in a multiple BSSID set.

26.17.2.4 HE STA antenna indication

A 6 GHz HE STA that does not change its receive antenna pattern after association shall set the Rx Antenna Pattern Consistency subfield in the HE 6 GHz Band Capabilities element to 1; otherwise, the STA shall set the Rx Antenna Pattern Consistency subfield in the HE 6 GHz Band Capabilities element to 0.

A 6 GHz HE STA that does not change its transmit antenna pattern after association shall set the Tx Antenna Pattern Consistency subfield in the HE 6 GHz Band Capabilities element to 1; otherwise, the STA shall set the Tx Antenna Pattern Consistency subfield in the HE 6 GHz Band Capabilities element to 0.

26.17.3 BSS color

26.17.3.1 General

BSS color identifies a BSS and assists a STA receiving a PPDU that carries BSS color in identifying the BSS from which the PPDU originates so that the STA can use the channel access rules in 26.10, reduce power consumption as described in 26.14.1, or update its NAV as described in 26.2.4.

All APs that are members of a multiple BSSID set or co-hosted BSSID set shall use the same BSS color.

An HE STA that transmits an HE Operation element shall select an initial BSS color by following the procedure in 26.17.3.2.

An HE STA that transmits an HE Operation element may disable BSS color by following the procedure in 26.17.3.3.

An HE AP may follow the procedure in 26.17.3.4 to select and advertise a new BSS color.

An HE STA may follow the procedure in 26.17.3.5 to determine and report a BSS color collision.

26.17.3.2 Initial BSS color

An HE AP starting an infrastructure BSS or an HE STA starting an IBSS or MBSS shall set the BSS Color subfield of the first HE Operation element it transmits to a value in the range 1 to 63 and shall maintain that value in subsequent HE Operation elements it transmits for the lifetime of the BSS or until the BSS color is changed as described in 26.17.3.4. When selecting a BSS color, the HE STA should consider the BSS colors in use by neighboring OBSSs.

A non-AP HE STA that is associated with an HE AP and that is the initiating STA of a TDLS link shall set the BSS Color subfield of the HE Operation element it transmits to the TDLS peer HE STA to the value indicated in the BSS Color subfield of the HE Operation element received from its associated HE AP. A non-AP HE STA that is associated with a non-HE AP and that is the initiating STA of the TDLS link shall use the same BSS color for all its TDLS links by setting the BSS Color subfield of the HE Operation element it transmits to the TDLS peer HE STA to the value of $\text{dec}(\text{BSSID}[39:44])$ of the non-HE AP or the $\text{dec}(\text{transmitted BSSID}[39:44])$ of the non-HE AP if the AP indicates the support of multiple BSSID in its Extended Capabilities element.

26.17.3.3 Disabling BSS color

An HE STA that transmits an HE Operation element and that decides to disable the use of BSS color in the BSS to which it belongs, for example, after detecting a BSS color collision with an OBSS (see 26.17.3.5), shall set the value of BSS Color Disabled subfield in the HE Operation element to 1 to inform other STAs in the BSS that the BSS color is disabled; otherwise, the HE STA shall set the BSS Color Disabled subfield to 0.

NOTE—It is recommended that an HE STA that transmits an HE Operation element does not disable the BSS color for an extended period of time, so that other HE STAs can benefit from features relying on BSS color.

If the most recently received HE Operation element from the AP with which it is associated contained a value of 1 in the BSS Color Disabled subfield, then the following apply:

- A non-AP HE STA should use the Address 1, Address 2, and Duration/ID fields of the frames contained in the received HE PPDU, instead of the RXVECTOR parameters BSS_COLOR and TXOP_DURATION, to determine whether the STA should update the intra-BSS NAV.
- A non-AP HE STA should use the Address 1 and Address 2 fields of the frames contained in the received HE PPDU, instead of the RXVECTOR parameters BSS_COLOR and STA_ID, to determine whether the STA may go to doze state for the duration of that PPDU (see 26.14.1).

A non-AP HE STA may use the RXVECTOR parameter BSS_COLOR of an HE PPDU to determine whether it should update the intra-BSS NAV (see 26.2.4), and/or the STA may go to doze state for the duration of the PPDU (see 26.14.1) if the most recently received HE Operation element from the AP with which it is associated contained a value of 0 in the BSS Color Disabled subfield.

26.17.3.4 Selecting and advertising a new BSS color

An HE STA that transmits an HE Operation element shall select a BSS color as defined in 26.17.3.2 for its BSS. An HE AP may change the BSS color under certain conditions, for example, if it detects that an OBSS is using the same color. An HE AP shall announce an upcoming BSS color change using the BSS Color Change Announcement element. A non-AP HE STA shall not transmit a BSS Color Change Announcement element.

NOTE—A non-AP HE STA includes a STA participating in an IBSS or MBSS. The color change mechanism described in this subclause does not apply to an IBSS or a mesh BSS since these BSSs do not have a single coordinator.

When changing the BSS color, the HE AP should consider the BSS colors of OBSSs that the HE AP has identified by itself or via the autonomous collision reports received from associated non-AP HE STAs (see 26.17.3.5).

The BSS Color Change Announcement element may be carried in the Beacon, Probe Response, (Re)Association Response, and HE BSS Color Change Announcement frames transmitted by the AP. An HE AP should announce an upcoming BSS color change for a period of time that is sufficiently long for all STAs in the BSS, including STAs in PS mode, to have an opportunity to receive at least one frame carrying a BSS Color Change Announcement element before the BSS color change.

If the Color Switch Countdown field in the BSS Color Change Announcement element has a value greater than 1, then at the next TBTT the AP shall decrement the Color Switch Countdown field value by 1. The BSS color change TBTT is the TBTT following a frame containing a BSS Color Change Announcement element with the Color Switch Countdown field containing the value 1. An HE AP shall not alter the BSS color change TBTT after it has announced a pending BSS color change. An AP belonging to a co-hosted BSSID set (see 26.17.7) should select the value of Color Switch Countdown field such that the BSS color change TBTT interval between the BSSs in the set shall not be greater than one beacon interval of the BSS with largest beacon interval in the set.

During the time leading up to the BSS color change TBTT,

- An HE AP shall set the BSS Color Disabled subfield to 1 and shall continue to advertise the existing BSS color via the BSS Color subfield in the HE Operation element.
- An HE AP shall not change the value it advertises in the New BSS Color subfield of the BSS Color Change Announcement element.
- An HE AP shall set the TXVECTOR parameter BSS_COLOR of an HE PPDU to the existing BSS color.

At the BSS color change TBTT, an HE AP shall

- Set the BSS Color Disabled subfield in the HE Operation element that it transmits to 0, unless the HE AP belongs to a co-hosted BSSID set. In this case, it shall continue to set the BSS Color Disabled subfield to 1 until all the BSSs in the co-hosted BSSID set have passed their respective BSS color change TBTT.
- Start advertising the new BSS color in the BSS Color subfield in the HE Operation element.
- Start using the new BSS color for all frames that it transmits after the TBTT.

A co-hosted AP should not transmit an HE PPDU during the transition period until all the BSSs in the co-hosted set have completed their switch to the new color.

A non-AP HE STA that receives a BSS Color Change Announcement element from an AP shall use the value specified in the New BSS Color field of the element as the BSS color when communicating with that AP following the BSS color change TBTT.

26.17.3.5 Detecting and reporting BSS color collision

26.17.3.5.1 General

An HE AP may determine that a BSS color collision has occurred if it receives HE PDUs on its primary channel from an OBSS STA containing the same BSS color as the one it has selected for its BSS or if it receives autonomous BSS color collision event reports from its associated STAs. The HE AP shall set the BSS Color Disabled subfield to 1 in the HE Operation element that it transmits if the BSS color collision persists for a duration of at least `dot11BSSColorCollisionAPPeriod` and if `dot11BSSColorCollisionAPPeriod` is not `-1`. The value `-1` means that the BSS color is not disabled in the case of a collision.

26.17.3.5.2 Autonomous reporting of BSS color collision

A STA that supports autonomous reporting of BSS color collision shall set `dot11AutonomousBSSColorCollisionReportingImplemented` to true.

A non-AP HE STA that supports autonomous reporting of BSS color collision may send a color collision report to its associated AP when it detects that color collision has occurred. The STA shall declare that a color collision has occurred if it receives, on its associated AP's primary channel, an HE PDU with at least three Address fields in the MAC header and with the same color as its associated BSS in which none of the Address fields match the BSSID of the BSS with which the STA is associated or any of the other BSSs in the same multiple BSSID set or co-hosted BSSID set to which its BSS belongs.

The HE STA's autonomous report shall include BSS color information of all OBSSs from which the STA is able to detect frames in order to help its associated AP select a new nonoverlapping BSS color when the AP decides to switch to a different BSS color.

A non-AP HE STA that is autonomously reporting a BSS color collision shall transmit an Event Report frame (see 9.6.13.3) containing a single Event Report element (see 9.4.2.67). The Event Report element shall carry Event Token field value set to 0 (autonomous report) and Event Type field value set to 4 (BSS Color Collision). The Event Report Status field shall be set to 0 (Successful), and the Event Report field shall carry information about the BSS color used by OBSSs that the reporting STA is able to detect.

A non-AP HE STA that has `dot11AutonomousBSSColorCollisionReportingImplemented` equal to true shall transmit a BSS Color Collision Event Report frame to its associated HE AP when it determines that a color collision has occurred. Subsequently, the STA shall schedule for transmission a BSS color collision event report every `dot11BSSColorCollisionSTAPeriod`, unless the BSS color collision no longer exists or the AP has set the BSS Color Disabled field to 1 in the HE Operation element that it transmits.

26.17.4 AID assignment

An HE AP that transmits an HE Operation element with the Partial BSS Color subfield in the BSS Color Information field set to 1 shall allocate AIDs that meet the constraint in Equation (26-8).

$$AID(5:8) = \text{bin}[(BCB(0:3) - (BSSID[44:47] \oplus BSSID[40:43])) \bmod 2^4, 4] \quad (26-8)$$

where $BCB(0:3)$ represents bits 0 to 3 inclusive of the BSS color in the transmitted HE Operation element with bit 0 being the first transmitted and $AID(5:8)$ represents bits 5 to 8 inclusive of the allocated AID with bit 5 being the first transmitted.

NOTE—See 1.5 for the behavior of the mod operator with a negative first operand.

26.17.5 Quiet HE STAs in an HE BSS

26.17.5.1 General

Quiet time period (QTP) is an optional feature that defines a period of time that is intended to be used primarily for the exchange of specific frames between a STA requesting a QTP and its peers using peer-to-peer links. The particular frames to be exchanged using peer-to-peer links during the QTP are identified by a service-specific identifier. The determination of which frames are associated with the service-specific identifier is beyond the scope of this standard. The method for selection of the service-specific identifier by the peer-to-peer operation is beyond the scope of this standard.

An AP with `dot11QTPOptionImplemented` equal to true is a QTP AP and shall set the QTP Support field to 1 in HE Capabilities elements that it transmits; otherwise, an AP shall set the QTP Support field to 0. A non-AP HE STA with `dot11QTPOptionImplemented` equal to true is a QTP non-AP STA and shall set the QTP Support field to 1 in HE Capabilities elements that it transmits; otherwise, a non-AP HE AP shall set the QTP Support field to 0.

A QTP non-AP STA may request its QTP AP to set up a QTP, and if successful, the QTP AP informs other associated QTP non-AP STAs of the QTP and of the service-specific identifier associated with that QTP.

The QTP mechanism informs other HE STAs of the period and the intended peer-to-peer operation and requests that, during the QTP, the QTP non-AP STAs should not exchange frames that are not associated with the service-specific identifier.

A QTP non-AP STA may ignore the QTP. A QTP non-AP STA that decides to stay quiet during a QTP suspends the decrementing of its backoff counters at the start time of the QTP and resumes them when the QTP ends.

NOTE—Otherwise, a STA that does not stay quiet does not suspend the decrementing of its backoff counters.

26.17.5.2 QTP Requesting STA procedure

A QTP requesting STA is a QTP non-AP STA that requests a quiet time period (QTP). Figure 26-15 illustrates a QTP operation.

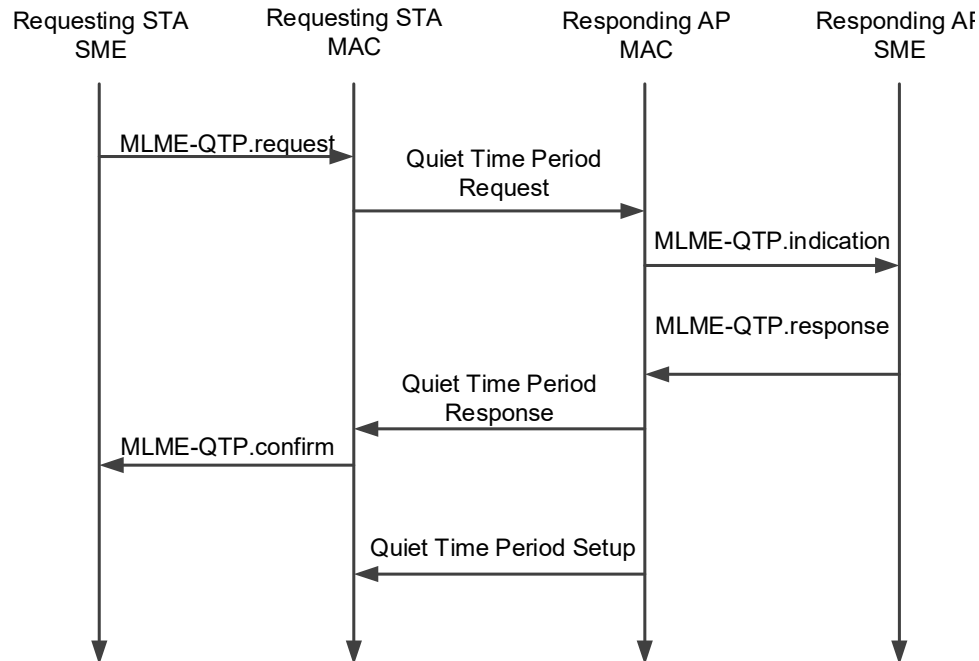


Figure 26-15—Quiet time period operation

Upon the reception of an MLME-QTP.request primitive, a QTP requesting STA shall perform the following procedure:

- If a QTP requesting STA is associated with a QTP AP, the QTP requesting STA sends a QTP Request frame that is a Quiet Time Period Action frame (9.6.31.3) with the Control field of the Quiet Time Period element indicating the Quiet Time Period Request subtype. The QTP Request frame indicates the duration, interval, and type of operation (indicated by the Service Specific Identifier parameter). The QTP requesting STA may include multiple Quiet Time Period elements with Request subtype in one QTP Request frame for multiple types of frames associated with different service-specific identifiers.
- A QTP Response frame is a Quiet Time Period Action frame (9.6.31.3) with the Control field of the Quiet Time Period element indicating Quiet Time Period Response subtype. If a QTP Response frame is received with the dialog token matching the request token with a status code set to a value of SUCCESS, the QTP AP has confirmed the reception of the QTP Request frame, and the MLME shall issue an MLME-QTP.confirm primitive indicating the success of the procedure.
- A QTP Setup frame is a Quiet Time Period Action frame (9.6.31.3) with the Control field of the Quiet Time Period element indicating Quiet Time Period Setup subtype. If a QTP Setup frame is received, at the start time for a QTP, the QTP requesting STA may schedule frames for transmission that are associated with the service-specific identifier indicated in the QTP Setup frame and should not transmit frames that are not associated with the service-specific identifier.

26.17.5.3 Responding AP procedure

Upon receipt of a QTP request, a QTP AP shall operate as follows (Figure 26-15):

- a) The MLME of the AP shall issue an MLME-QTP.response primitive.
- b) Upon receipt of the MLME-QTP.response primitive, the AP may respond by sending a broadcast QTP Response frame.
 - 1) If the status code in the broadcast QTP Response frame is SUCCESS, the AP accepts the request. The AP shall schedule the quiet period(s) according to the accepted request. Contained in the transmitted QTP Response frame is a copy of the dialog token from the QTP requesting STA. The QTP procedure shall be terminated if the number of quiet periods exceeds the value of the Repetition Count field specified.
 - 2) If the status code in the broadcast QTP Response frame is REJECTED, the AP indicates that the request cannot be fulfilled.
 - 3) If the status code in the broadcast QTP Response frame is COUNTERED, the AP counters the request with recommended values, and the current request is rejected. Upon receiving the counter-proposal, a QTP STA can send a QTP Request frame to set up another QTP.
- c) At the start time for a quiet time period, the AP may schedule for transmission a QTP Setup frame. The AP shall set the Quiet Period Duration field of the QTP Setup frame to a value no larger than indicated in the Quiet Period Duration field of the QTP Request frame sent by the QTP requesting STA.

NOTE—The AP is not required to transmit a QTP Setup frame at a scheduled QTP. The interference mitigation protocol is to provide an AP a tool to manage and avoid interference. How or whether the AP will transmit QTP Setup frame at a scheduled QTP is beyond the scope of this standard.

26.17.6 ER beacon generation in an ER BSS

An ER beacon is a Beacon frame carried in an HE ER SU PPDU using a 242-tone RU and transmitted in the primary 20 MHz channel. An ER beacon provides additional link budget for downlink transmissions to compensate for the link budget imbalance between downlink and uplink due to the introduction of UL OFDMA transmission. An HE AP shall operate an ER BSS in addition to a non-ER BSS operated by another AP that is in the same co-located AP set. An ER BSS, if present, shall operate independently of the co-located non-ER BSS that is in the same co-located AP set, and the AP operating the ER BSS shall have a BSSID different from the AP operating the non-ER BSS.

NOTE—An AP that uses ER beacons can balance the link budget by allocating narrow RUs to STAs. An ER BSS is expected to have a larger coverage area than a non-ER BSS.

The HE AP of an ER BSS shall not set the ER SU Disable subfield to 1 in HE Operation elements it transmits.

The HE AP of an ER BSS shall transmit Beacon frames and group addressed frames in HE ER SU PPDU following the rules in 26.15.5.

26.17.7 Co-hosted BSSID set

HE BSSs that are not part of a multiple BSSID set (i.e., `dot11MultiBSSIDImplemented` is false) but share the same operating class, channel, receive antenna connector, and transmit antenna connector belong to a co-hosted BSSID set.

A STA that supports co-hosted BSSID capability shall have `dot11CoHostedBSSIDImplemented` equal to true.

An AP that belongs to a co-hosted BSSID set shall perform the following operations:

- Set the Co-Hosted BSS subfield in the HE Operation element that it transmits to 1.
- Set the Max Co-Hosted BSSID Indicator field in the HE Operation element that it transmits to a nonzero value n , where $1 \leq n \leq 8$, such that 2^n indicates the maximum number of BSSIDs in the co-hosted set.

Members of the co-hosted BSSID set have the same $48 - n$ bits (`BSSID[0:(47 - n)]`) in their BSSIDs.

If its associated AP has set the Co-Hosted BSS subfield in the HE Operation Parameters field to 1, a non-AP STA shall identify a BSS as a co-hosted BSS, if the $48 - n$ bits (`BSSID[0:(47 - n)]`) of the BSSID of the BSS are the same as the $48 - n$ bits (`BSSID[0:(47 - n)]`) of the BSSID of its associated AP, where n is the value carried in the Max Co-Hosted BSSID Indicator field of the HE Operation element transmitted by the associated AP.

27. High-efficiency (HE) PHY specification

27.1 Introduction

27.1.1 Introduction to the HE PHY

Clause 27 specifies the PHY entity for a high-efficiency (HE) orthogonal frequency division multiplexing (OFDM) system. In addition to the requirements in Clause 27, an HE STA shall be capable of transmitting and receiving PPDU that are compliant with the mandatory requirements of the following PHY specifications:

- Clause 19 and Clause 21 if the HE STA supports an operating channel width greater than or equal to 80 MHz and is operating in the 5 GHz band.
- Clause 19 and Clause 21 transmission and reception on 20 MHz channel width (see 26.17.1) if the HE STA is a 20 MHz-only non-AP HE STA and is operating in the 5 GHz band.
- Clause 19 if the HE STA is operating in the 2.4 GHz band.
- Clause 17 if the HE STA is operating in the 6 GHz band.

For 2.4 GHz band operation, the HE PHY is based on HT PHY defined in Clause 19, which in turn is based on the OFDM PHY defined in Clause 17.

For 5 GHz band operation, the HE PHY is based on the VHT PHY defined in Clause 21, which in turn is based on the HT PHY defined in Clause 19, which in turn is further based on the OFDM PHY defined in Clause 17.

For 6 GHz band operation, the HE PHY is based on the OFDM PHY defined in Clause 17.

The HE PHY extends the maximum number of users supported for DL MU-MIMO transmissions to 8 users per resource unit (RU) and provides support for DL and UL orthogonal frequency division multiple access (OFDMA) as well as for UL MU-MIMO. Both DL and UL MU-MIMO transmissions are supported on portions of the PPDU bandwidth (on resource units greater than or equal to 106 tones). In an MU-MIMO resource unit, there is support for up to 8 users with up to 4 space-time streams per user with the total across all users not exceeding 8 space-time streams.

The HE PHY provides support for 20 MHz, 40 MHz, 80 MHz, and 160 MHz contiguous channel widths and support for 80+80 MHz noncontiguous channel width, depending on the frequency band and capability. For PPDU bandwidths greater than or equal to 80 MHz, the HE PHY supports preamble punctured HE MU PPDU transmissions where pre-HE modulated fields (see Figure 27-23 in 27.3.10) are not transmitted in one or more of the nonprimary 20 MHz channels, and RUs associated with those punctured 20 MHz channels as defined in 27.3.11.8.3 are not allocated.

The HE PHY provides support for 0.8 μ s, 1.6 μ s, and 3.2 μ s guard interval durations.

The HE PHY provides support for 3.2 μ s (1x), 6.4 μ s (2x), and 12.8 μ s (4x) HE-LTF symbol durations, excluding the GI duration. The HE PHY supports a DFT period of 3.2 μ s and 12.8 μ s for the pre-HE modulated fields and the HE modulated fields in an HE PPDU, respectively. The HE PHY data subcarrier frequency spacing is a quarter of VHT PHY and HT PHY subcarrier frequency spacing defined in Clause 21 and Clause 19, respectively.

The HE PHY data subcarriers are modulated using BPSK, BPSK DCM, QPSK, QPSK DCM, 16-QAM, 16-QAM DCM, 64-QAM, 256-QAM, and 1024-QAM. Forward error correction (FEC) coding (convolutional or LDPC coding) is used with coding rates of 1/2, 2/3, 3/4, and 5/6.

The HE PHY provides support for midambles. Midambles facilitate updating of the channel estimate during HE PPDU reception and might be of use in high mobility scenarios that often result in significant variations of the wireless channel over the duration of a PPDU.

An HE STA shall support the following features:

- Transmission and reception of an HE SU PPDU that consists of a single RU spanning the entire PPDU bandwidth.
- Transmission and reception of an HE ER SU PPDU that consists of a 242-tone RU spanning the entire primary 20 MHz PPDU bandwidth.
- BCC coding (transmit and receive). BCC coding is not used in the following cases:
 - An HE SU PPDU with a bandwidth greater than 20 MHz
 - An RU of size greater than 242 subcarriers in an HE MU PPDU or an HE TB PPDU
 - An HE SU PPDU with number of spatial streams greater than 4
 - An RU allocated to a single user in an HE MU PPDU or for an HE TB PPDU with a number of spatial streams greater than 4
 - An HE SU PPDU using HE-MCSs 10 or 11
 - An RU in an HE MU PPDU or an HE TB PPDU using HE-MCSs 10 or 11
- LDPC coding (transmit and receive) in all supported HE PPDU types, RU sizes, and number of spatial streams if the STA supports transmitting and receiving in channel bandwidths greater than 20 MHz.
- LDPC coding (transmit and receive) in all supported HE PPDU types, RU sizes, and number of spatial streams if the STA declares support for transmitting or receiving more than 4 spatial streams.
- LDPC coding (transmit and receive) in all supported HE PPDU types, RU sizes, and number of spatial streams if the STA declares support for HE-MCSs 10 and 11 (transmit and receive).
- Single spatial stream HE-MCSs 0 to 7 (transmit and receive) in all supported channel widths for HE SU PDUs.
- HE SU PDUs and HE ER SU PDUs with a 2x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols (transmit and receive).
- HE SU PDUs and HE ER SU PDUs with a 2x HE-LTF and 1.6 μ s GI duration on the HE-LTF and Data field OFDM symbols (transmit and receive).
- HE SU PDUs and HE ER SU PDUs with a 4x HE-LTF and 3.2 μ s GI duration on the HE-LTF and Data field OFDM symbols (transmit and receive).
- Full bandwidth UL MU-MIMO with a 1x HE-LTF and 1.6 μ s GI duration on the HE-LTF and Data field OFDM symbols if the STA supports UL MU-MIMO.
- HE SU PDUs with a 4x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols if the STA supports HE ER SU PDUs with a 4x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols (transmit and receive).
- Single spatial stream HE-MCSs 0 to 2 in primary 20 MHz channel for HE ER SU PDUs.
- HE ER SU PPDU is not used in the following cases:
 - Number of spatial streams greater than 1
 - HE-MCS greater than 2 when 242 subcarriers are used in the Data field OFDM symbols
 - HE-MCS greater than 0 when 106 subcarriers are used in the Data field OFDM symbols
 - Bandwidth greater than 20 MHz
- 20 MHz channel width and all RU sizes and locations applicable to the 20 MHz channel width in 2.4 GHz, 5 GHz, and 6 GHz bands (transmit and receive).

An HE STA may support the following features:

- HE-MCSs 8 to 11 (transmit and receive).
- Two or more spatial streams (transmit and receive).

- DCM (transmit and receive).
- HE SU PPDU and HE ER SU PPDU with a 1x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols (transmit and receive).
- HE SU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols if the STA does not support HE ER SU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on both the HE-LTF and Data field OFDM symbols (transmit and receive).
- HE ER SU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on both the HE-LTF and Data field OFDM symbols (transmit and receive).
- LDPC coding (transmit and receive) if the maximum number of spatial streams the STA is capable of transmitting or receiving in an HE SU PPDU is less than or equal to 4.
- Single spatial stream HE-MCS 0 in the higher frequency 106-tone RU of the primary 20 MHz channel for an HE ER SU PPDU.
- STBC (transmit and receive).

An HE AP shall support the following features:

- Transmission of an HE MU PPDU where none of the RUs utilize MU-MIMO (DL OFDMA).
- Reception of an HE TB PPDU where none of the RUs utilize MU-MIMO (UL OFDMA).
- Transmission of an HE MU PPDU consisting of a single RU spanning the entire PPDU bandwidth and utilizing MU-MIMO (DL MU-MIMO), provided the AP is capable of transmitting 4 or more spatial streams.
- Transmission of the HE-SIG-B field in an HE MU PPDU at HE-MCSs 0 to 5.
- Single spatial stream HE-MCSs 0 to 7 in all supported channel widths and RU sizes for HE MU PPDU (transmit) or HE TB PPDU (receive).
- 40 MHz and 80 MHz channel widths and all RU sizes and locations applicable to the 40 MHz and 80 MHz channel width in 5 GHz and 6 GHz bands (transmit and receive).
- HE MU PPDU with a 2x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols (transmit).
- HE MU PPDU with a 2x HE-LTF and 1.6 μ s GI duration on the HE-LTF and Data field OFDM symbols (transmit).
- Reception of an HE TB PPDU with a 2x HE-LTF and 1.6 μ s GI duration on the HE-LTF and Data field OFDM symbols.
- Reception of an HE TB PPDU with a 4x HE-LTF and 3.2 μ s GI duration on the HE-LTF and Data field OFDM symbols.
- Transmission of an HE MU PPDU with a 4x HE-LTF and 3.2 μ s GI duration on the HE-LTF and Data field OFDM symbols.
- Transmission of an HE MU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols if the HE AP supports HE ER SU PPDU with the same HE-LTF and GI combinations.
- All RU sizes and locations applicable to 40 MHz channel width in the 2.4 GHz band if 40 MHz channel width is supported in the 2.4 GHz band (transmit and receive).

An HE AP may support the following features:

- MU-MIMO transmission on an RU in an HE MU PPDU where the RU does not span the entire PPDU bandwidth (DL MU-MIMO within OFDMA).
- MU-MIMO reception on an RU in an HE TB PPDU where the RU spans the entire PPDU bandwidth (UL MU-MIMO).
- MU-MIMO reception on an RU in an HE TB PPDU where the RU does not span the entire PPDU bandwidth (UL MU-MIMO within OFDMA).

- Reception of the payload on an RU in an HE MU PPDU where RU spans the entire PPDU bandwidth or a 106-tone RU within 20 MHz PPDU bandwidth.
- 40 MHz channel width in the 2.4 GHz band (transmit and receive).
- 160 MHz and 80+80 MHz channel widths and 2×996-tone RU size applicable to the 160/80+80 MHz channel width in the 5 GHz and 6 GHz bands (transmit and receive).
- Transmission of an HE MU PPDU with preamble puncturing.
- HE MU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols if the STA does not support HE ER SU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols (transmit).
- Punctured sounding operation.

A non-AP HE STA shall support the following features:

- Reception of an HE MU PPDU where the RU allocated to the non-AP STA is not utilizing MU-MIMO (DL OFDMA).
- Transmission of an HE TB PPDU where the RU allocated to the non-AP STA is not utilizing MU-MIMO (UL OFDMA).
- Reception of an HE MU PPDU consisting of a single RU spanning the entire PPDU bandwidth and utilizing MU-MIMO (DL MU-MIMO). The maximum number of spatial streams per user the non-AP STA can receive in the DL MU-MIMO transmission shall be equal to the minimum of 4 and the maximum number of spatial streams supported for reception of HE SU PPDU. The non-AP STA shall be able to receive its intended spatial streams in a DL MU-MIMO transmission with a total number of spatial streams across all users of at least 4.
- Reception of an HE MU PPDU with up to 4 HE-LTF OFDM symbols, where the RU allocated to the non-AP STA does not span the entire PPDU bandwidth.
- Responding with the requested beamforming feedback in an HE sounding procedure with the maximum number of space-time streams in the HE sounding NDP that the non-AP STA can respond to being at least 4.
- Reception of the HE-SIG-B field in an HE MU PPDU at HE-MCSs 0 to 5.
- Single spatial stream HE-MCSs 0 to 7 in all supported channel widths and RU sizes for HE MU PPDU (receive) or HE TB PPDU (transmit).
- 40 MHz and 80 MHz channel widths and all RU sizes and locations applicable to the 40 MHz and 80 MHz channel widths in the 5 GHz band (transmit and receive), except for a 20 MHz-only non-AP HE STA in which case the 40 MHz and 80 MHz channel widths, 996-tone RU, and 484-tone RU sizes in the 5 GHz and 6 GHz bands are not applicable.
- Reception of an HE MU PPDU with a 2x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols.
- Reception of an HE MU PPDU with a 2x HE-LTF and 1.6 μ s GI duration on the HE-LTF and Data field OFDM symbols.
- Transmission of an HE TB PPDU with a 2x HE-LTF and 1.6 μ s GI duration on the HE-LTF and Data field OFDM symbols.
- Reception of an HE MU PPDU with a 4x HE-LTF and 3.2 μ s GI duration on the HE-LTF and Data field OFDM symbols.
- Transmission of an HE TB PPDU with a 4x HE-LTF and 3.2 μ s GI duration on the HE-LTF and Data field OFDM symbols.
- Reception of an HE MU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on the HE-LTF and the Data field OFDM symbols if the non-AP HE STA supports HE ER SU PPDU with the same HE-LTF and GI combinations.
- 26-, 52-, and 106-tone RU sizes on locations allowed in 27.3.2.8 in the primary 20 MHz channel within 40 MHz channel width if the non-AP HE STA is a 20 MHz operating non-AP HE STA and

that is capable of supporting 40 MHz channel width (B0 of the Supported Channel Width Set subfield in the HE PHY Capabilities Information field in the HE Capabilities element is 1) (transmit and receive).

- 26-, 52-, and 106-tone RU sizes on locations allowed in 27.3.2.8 in the primary 20 MHz channel within 80 MHz channel width if the non-AP HE STA is a 20 MHz operating non-AP HE STA and is operating in the 5 GHz or 6 GHz bands (transmit and receive).

A non-AP HE STA may support the following:

- Transmission of an HE MU PPDU with a single RU spanning the entire PPDU bandwidth or a 20 MHz HE MU PPDU with a single 106-tone RU in the primary 20 MHz channel.
- 40 MHz channel width in the 2.4 GHz band (transmit and receive). If 40 MHz channel width in the 2.4 GHz band is supported, then all RU sizes and locations applicable to 40 MHz channel width are supported, except for a 20 MHz-only non-AP HE STA in which case the 40 MHz channel width and all RU sizes and locations of 40 MHz channel width in 2.4 GHz band are not applicable.
- 160 MHz and 80+80 MHz channel width and 2×996-tone RU size applicable to the 160 MHz and 80+80 MHz channel width in the 5 GHz and 6 GHz bands (transmit and receive), except for a 20 MHz-only non-AP HE STA in which case the 160 MHz and 80+80 MHz channel width and 2×996-tone RU size in the 5 GHz and 6 GHz bands are not applicable.
- MU-MIMO reception on an RU in an HE MU PPDU where the RU does not span the entire PPDU bandwidth (DL MU-MIMO within OFDMA). The maximum number of spatial streams per user in the DL MU-MIMO within OFDMA transmission that the non-AP STA can receive shall be a minimum of 4 and the maximum number of spatial streams supported for reception of HE SU PPDU. The total number of spatial streams (across all users) in the DL MU-MIMO within OFDMA transmission that the non-AP STA can receive shall be at least 4.
- Reception of an HE MU PPDU with up to 8 HE-LTF OFDM symbols, where the RU allocated to the non-AP STA does not span the entire PPDU bandwidth.
- MU-MIMO transmission on an RU in an HE TB PPDU where the RU spans the entire PPDU bandwidth (UL MU-MIMO). If supported, then the non-AP HE STA shall support transmitting UL MU-MIMO where the total space-time streams summed across all users is less than or equal to 8.
- MU-MIMO transmission on an RU in an HE TB PPDU where the RU does not span the entire PPDU bandwidth (UL MU-MIMO within OFDMA). If supported, then the non-AP HE STA shall support transmitting UL MU-MIMO where the total space-time streams summed across all users is less than or equal to 8.
- The reception of a 160 MHz or 80+80 MHz HE MU PPDU, or the transmission of a 160 MHz or 80+80 MHz HE TB PPDU where the assigned RU is in the primary 80 MHz channel if the non-AP HE STA is capable of up to 80 MHz channel width and operating with 80 MHz channel width.
- HE MU PPDU with a 4x HE-LTF and 0.8 μs GI duration on the HE-LTF and Data field OFDM symbols if the non-AP HE STA does not support HE ER SU PPDU with a 4x HE-LTF and 0.8 μs GI duration on the HE-LTF and Data field OFDM symbols (receive).
- Punctured sounding operation.

A 20 MHz-only non-AP HE STA may support the following:

- 26-, 52-, 106-, and 242-tone RU sizes on locations allowed in 27.3.2.8 in the primary 20 MHz channel within 40 MHz channel width in the 2.4 GHz band if the 20 MHz-only non-AP HE STA does not support the HE subchannel selective transmission operation described in 26.8.7.
- 26-, 52-, 106-, and 242-tone RU sizes on locations allowed in 27.3.2.8 in any 20 MHz channel within 40 MHz channel width in the 2.4 GHz band if the 20 MHz-only non-AP HE STA supports the HE subchannel selective transmission operation as described in 26.8.7.
- 26-, 52-, 106-, and 242-tone RU sizes on locations allowed in 27.3.2.8 in any 20 MHz channel within 40 MHz, 80 MHz, 160 MHz, and 80+80 MHz channel widths in the 5 GHz and 6 GHz bands

if the 20 MHz-only non-AP HE STA supports the HE subchannel selective transmission operation as described in 26.8.7.

A 20 MHz operating non-AP HE STA may support 26-, 52-, 106-, and 242-tone RU sizes on locations allowed in 27.3.2.8 in the primary 20 MHz channel within 160 MHz and 80+80 MHz channel widths in the 5 GHz and 6 GHz bands.

27.1.2 Scope

The services provided to the MAC by the HE PHY consist of the following protocol functions:

- a) A function that maps the PSDU received from the MAC into a PPDU for transmission to one or more receiving STAs.
- b) A function that defines the characteristics and method of transmitting and receiving data through a wireless medium between two or more STAs. Depending on the PPDU format, these STAs support a mixture of HE: Clause 21, Clause 19, Clause 18, Clause 17, Clause 16, and Clause 15 PHYs. A 20 MHz-only non-AP HE STA supports a mixture of HE, Clause 19 and Clause 17 PHYs, and also supports transmission and reception of VHT PPDU of 20 MHz PPDU bandwidth.

27.1.3 HE PHY functions

27.1.3.1 General

The HE PHY contains two functional entities: the PHY function and the physical layer management function (i.e., the PLME). These functions are described in detail in 27.3 and 27.4, respectively. The HE PHY service is provided to the MAC through the PHY service primitives defined in Clause 8. The HE PHY service interface is described in 27.2.

27.1.3.2 PHY management entity (PLME)

The PLME performs management of the local PHY functions in conjunction with the MLME.

27.1.3.3 Service specification method

The models represented by figures and state diagrams are intended to be illustrations of the functions provided. It is important to distinguish between a model and a real implementation. The models are optimized for simplicity and clarity of presentation.

The service of a layer is the set of capabilities that it offers to a user in the next higher layer. Abstract services are specified here by describing the service primitives and parameters that characterize each service. This definition is independent of any particular implementation.

27.1.4 PPDU formats

The structure of the PPDU transmitted by an HE STA is determined by the TXVECTOR parameters as defined in Table 27-1.

The FORMAT parameter determines the overall structure of the PPDU and can take on one of the following values:

- Non-HT format (NON_HT), based on Clause 17 or Clause 18 and including non-HT duplicate format.
- HT-mixed format (HT_MF) as specified in Clause 19.
- HT-greenfield format (HT_GF) as specified in Clause 19.

- VHT format (VHT) as defined in Clause 21.
- HE SU PPDU format (HE_SU) that carries a single PSDU. With this format the HE-SIG-A field is not repeated.
- HE ER SU PPDU format (HE_ER_SU) that carries a single PSDU. It is similar to the HE SU PPDU format, except that the HE-SIG-A field is repeated.
- HE MU PPDU format (HE_MU) that carries one or more PSDUs to one or more users.
- HE TB PPDU format (HE_TB) that carries a single PSDU and is sent in response to a PPDU that carries a triggering frame. The preamble format prior to the HE-STF field is identical to the HE SU PPDU.

27.2 HE PHY service interface

27.2.1 Introduction

The PHY provides an interface to the MAC through an extension of the generic PHY service interface defined in 8.3.4. The interface includes TXVECTOR, RXVECTOR, PHYCONFIG_VECTOR, and TRIG_VECTOR.

The MAC uses the TXVECTOR to supply the PHY with per-PPDU transmit parameters. The PHY uses the RXVECTOR to inform the MAC of the received PPDU parameters. The MAC uses the PHYCONFIG_VECTOR to configure the PHY for operation that is independent of frame transmission or reception. The MAC uses the TRIG_VECTOR to configure the PHY to receive HE TB PPDU over each assigned RU.

27.2.2 TXVECTOR and RXVECTOR parameters

The parameters in Table 27-1 are defined as part of the TXVECTOR parameter list in the PHY-TXSTART.request primitive and/or as part of the RXVECTOR parameter list in the PHY-RXSTART.indication and PHY-RXEND.indication primitives.

Table 27-1—TXVECTOR and RXVECTOR parameters

Parameter	Condition	Value	TXVECTOR	RXVECTOR
FORMAT		Determines the format of the PPDU. Enumerated type: NON_HT indicates Clause 15, Clause 16, Clause 17, Clause 18, or non-HT duplicate PPDU format. In this case, the modulation is determined by the NON_HT_MODULATION parameter. HT_MF indicates HT-mixed format. HT_GF indicates HT-greenfield format. VHT indicates VHT format. HE_SU indicates HE SU PPDU format. HE_MU indicates HE MU PPDU format. HE_ER_SU indicates HE ER SU PPDU format. HE_TB indicates HE TB PPDU format.	Y	Y

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
NON_HT_MODULATION	See corresponding entry in Table 19-1.			
L_LENGTH	FORMAT is HE_SU, HE_MU, or HE_ER_SU	Not present NOTE—The LENGTH field of the L-SIG field for HE PPDU is defined in Equation (27-11) using the TXTIME value defined in 27.4.3, which in turn depends on other parameters including the TXVECTOR parameter APEP_LENGTH.	N	N
	FORMAT is HE_TB	Indicates the value in the LENGTH field of the L-SIG field in the range of 1 to 4095. The value is obtained from the triggering frame to which the HE TB PPDU is a response.	Y	N
	Otherwise	See corresponding entry in Table 19-1 or Table 21-1.		
N_TX	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates the number of transmit chains.	Y	N
	Otherwise	See corresponding entry in Table 21-1.		
EXPANSION_MAT	FORMAT is HE_SU, HE_ER_SU, or HE_TB	Contains a vector in the number of selected subcarriers containing feedback matrices as defined in 27.3.16.2 based on the channel measured during the training symbols of previous HE sounding NDPs or VHT NDPs.	Y	N
	FORMAT is HE_MU	For each user, contains a vector in the number of all the subcarriers in the RU that is assigned to this user. The vector for each subcarrier contains feedback matrices as defined in 27.3.16.2 based on the channel measured during the training symbols of previous HE sounding NDPs or VHT NDPs.	M U	N
	Otherwise	See corresponding entry in Table 21-1.		
CHAN_MAT	FORMAT is HE_SU and PSDU_LENGTH is 0	Contains a vector in the number of selected subcarriers containing feedback matrices as defined in 27.3.16.2 based on the channel measured during the training symbols of previous HE sounding NDP.	N	Y
	FORMAT is HE_MU, HE_ER_SU, or HE_TB. FORMAT is HE_SU and PSDU_LENGTH is greater than 0.	Not present	N	N
	Otherwise	See corresponding entry in Table 21-1.		

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
DELTA_SNR	FORMAT is HE_MU	Contains an array of delta SNR values as defined in 9.4.1.66 based on the channel measured during the training symbols of the received HE sounding NDP.	M U	N
	FORMAT is HE_SU and PSDU_LENGTH greater than 0, or FORMAT is HE_ER_SU or HE_TB		O	N
	FORMAT is HE_SU and PSDU_LENGTH is 0		N	Y
	Otherwise	See corresponding entry in Table 19-1 or Table 21-1.		
RCPI	See corresponding entry in Table 19-1 or Table 21-1.			
SNR	FORMAT is HE_SU and PSDU_LENGTH is 0	Contains an array of average values of received SNR measurements for each spatial stream. SNR indications of 8 bits are supported. Average value of SNR shall be the sum of the decibel values of SNR per subcarrier divided by the number of subcarriers represented in each stream as described in 9.4.1.65.	N	Y
	FORMAT is HE_MU, HE_ER_SU, or HE_TB, or FORMAT is HE_SU and PSDU_LENGTH is greater than 0	Not present	N	N
	Otherwise	See corresponding entry in Table 19-1 or Table 21-1.		
CQI	FORMAT is HE_SU and PSDU_LENGTH is 0	Contains an array of received per-RU average SNRs for each space-time stream, where each per-RU average SNR is the arithmetic mean of the SNR in decibels over a 26-tone RU as described in 9.4.1.67.	N	Y
	Otherwise	Not present	N	N
NO_SIG_EXTN	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates whether signal extension needs to be applied at the end of transmission. Boolean: true indicates that no signal extension is present. false indicates that a signal extension is present.	Y	N
	Otherwise	See corresponding entry in Table 19-1 or Table 21-1.		
FEC_CODING	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates the FEC encoding used. Enumerated type: BCC_CODING indicates BCC coding. LDPC_CODING indicates LDPC coding.	M U	M U
	Otherwise	See corresponding entry in Table 21-1.		

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
LDPC_EXTRA_SYMBOL	FORMAT is HE_TB	Indicates the presence of the LDPC extra symbol segment in an HE TB PPDU. Integer: 1 indicates that an LDPC extra symbol segment is present. 0 indicates that an LDPC extra symbol segment is not present.	Y	N
	Otherwise	Not present	N	N
STBC	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates if STBC is used. For an HE MU PPDU and HE TB PPDU where each RU includes no more than 1 user: Set to 1 to indicate that for all RUs the Data field is STBC encoded. Set to 0 to indicate that in no RU is the Data field STBC encoded. For an HE SU PPDU or HE ER SU PPDU: Set to 1 to indicate that the Data field is STBC encoded. Set to 0 to indicate that the Data field is not STBC encoded.	Y	Y
	Otherwise	See corresponding entry in Table 19-1 or Table 21-1.		
GI_TYPE	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates the length of the GI for the HE-LTF and Data fields. Enumerated type: 0u8s_GI indicates 0.8 μ s. 1u6s_GI indicates 1.6 μ s. 3u2s_GI indicates 3.2 μ s. NOTE—the length of GI for pre-HE modulated fields is 0.8 μ s.	Y	Y
	Otherwise	See corresponding entry in Table 21-1.		
TXPWR_LEVEL_INDEX	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	The allowed values for the TXPWR_LEVEL_INDEX parameter are in the range from 1 to numberOfOctets (dot11TxPowerLevelExtended)/2. This parameter is used to indicate which of the available transmit output power levels defined in dot11TxPowerLevelExtended shall be used for the current transmission.	Y	N
	Otherwise	See corresponding entry in Table 21-1.		

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
RSSI	FORMAT is HE_SU, HE_ER_SU, HE_MU, or HE_TB	The allowed values for the RSSI parameter are in the range 0 to 255 inclusive. This parameter is a measure by the PHY of the power observed at the antennas used to receive the current PPDU measured during the reception of the HE-LTF field. RSSI is intended to be used in a relative manner, and it is a monotonically increasing function of the received power.	N	Y
	Otherwise	See corresponding entry in Table 21-1.		
RSSI_LEGACY	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	The allowed values for the RSSI_LEGACY parameter are in the range 0 to 255 inclusive. This parameter is a measure by the PHY of the power observed at the antennas used to receive the current PPDU measured during the reception of non-HE portion of the HE PPDU preamble. RSSI_LEGACY is intended to be used in a relative manner, and it is a monotonically increasing function of the received power.	N	Y
	Otherwise	Not present	N	N
MCS	FORMAT is HE_SU, HE_MU or HE_TB	Indicates the modulation and coding schemes used in the transmission of the PPDU. Integer: range 0 to 11.	M U	M U
	FORMAT is HE_ER_SU	Indicates the modulation and coding schemes used in the transmission of the PPDU. Integer: range 0 to 2.	Y	Y
	Otherwise	See corresponding entry in Table 21-1.		
DCM	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Set to 1 to indicate that DCM is used for the Data field. Set to 0 to indicate that DCM is not used for the Data field. NOTE—DCM is applicable to only HE-MCSs 0, 1, 3, and 4. DCM is applicable to only 1 and 2 spatial streams.	M U	M U
	Otherwise	Not present		
MCS_SIG_B	FORMAT is HE_MU	Indicates the modulation and coding scheme used for the HE-SIG-B field. Integer: 0 indicates HE-MCS 0. 1 indicates HE-MCS 1. 2 indicates HE-MCS 2. 3 indicates HE-MCS 3. 4 indicates HE-MCS 4. 5 indicates HE-MCS 5.	Y	Y
	Otherwise	Not present	N	N

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
DCM_SIG_B	FORMAT is HE_MU	Set to 1 to indicate that DCM is used for the HE-SIG-B field. Set to 0 to indicate that DCM is not used for the HE-SIG-B field.	Y	Y
	Otherwise	Not present	N	N
SIG_B_COMPRESSION_MODE	FORMAT is HE_MU	Indicates whether the Common field is present in the HE-SIG-B field. Integer: 0 indicates that the Common field is present. 1 indicates that the Common field is not present.	Y	N
	Otherwise	Not present	N	N
REC_MCS	FORMAT is HE_SU, HE_MU, or HE_ER_SU	Indicates the HE-MCS that the receiver recommends.	N	O
	FORMAT is HE_TB	Not present	N	N
	Otherwise	See corresponding entry in Table 21-1.		
CH_BANDWIDTH	FORMAT is HE_SU	Indicates the channel width of the PPDU. Enumerated type: CBW20 for 20 MHz. CBW40 for 40 MHz. CBW80 for 80 MHz. CBW160 for 160 MHz.	Y	Y
	FORMAT is HE_ER_SU	Indicates the channel width of the PPDU. Enumerated type: ER-RU-242 for the 242-tone RU in the primary 20 MHz channel. ER-RU-H-106 for the higher frequency 106-tone RU in the primary 20 MHz channel.	Y	Y

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
CH_BANDWIDTH (continued)	FORMAT is HE_MU	Indicates the channel width of the PPDU. Enumerated type: CBW20 for full 20 MHz. CBW40 for full 40 MHz. CBW80 for full 80 MHz. CBW160 for full 160 MHz. CBW80+80 for 80+80 MHz. HE-CBW-PUNC80-PRI for preamble puncturing in 80 MHz, where in the preamble the only punctured subchannel is the secondary 20 MHz channel. HE-CBW-PUNC80-SEC for preamble puncturing in 80 MHz, where in the preamble the only punctured subchannel is one of the two 20 MHz subchannels in the secondary 40 MHz channel. HE-CBW-PUNC160-PRI20 for preamble puncturing in 160 MHz, where in the preamble the only punctured subchannels are the secondary 20 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel. See Table 27-20 for allowed 20 MHz punctured subchannel combinations. HE-CBW-PUNC80+80-PRI20 for preamble puncturing in 80+80 MHz, where in the preamble the only punctured subchannels are the secondary 20 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel. See Table 27-20 for allowed 20 MHz punctured subchannel combinations. HE-CBW-PUNC160-SEC40 for preamble puncturing in 160 MHz, where in the preamble the only punctured subchannels are zero, one or both of the 20 MHz subchannels in the secondary 40 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel. At least one 20 MHz subchannel is punctured. See Table 27-20 for allowed 20 MHz punctured subchannel combinations. HE-CBW-PUNC80+80-SEC40 for preamble puncturing in 80+80 MHz, where in the preamble the only punctured subchannels are zero, one or both of the 20 MHz subchannels in the secondary 40 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel. At least one 20 MHz subchannel is punctured. See Table 27-20 for allowed 20 MHz punctured subchannel combinations.	Y	Y
	FORMAT is HE_TB	Indicates the Bandwidth field of the HE-SIG-A field in the transmitted or received PPDU. Enumerated type: CBW20 for 20 MHz. CBW40 for 40 MHz. CBW80 for 80 MHz. CBW160 for 160 MHz. CBW80+80 for 80+80 MHz.	Y	Y
	Otherwise	See corresponding entry in Table 19-1 or Table 21-1.		

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
INACTIVE_SUBCHANNELS	FORMAT is HE_MU and CH_BANDWIDTH is any value for preamble puncturing, or FORMAT is NON_HT and NON_HT_MODULATION is NON_HT_DUP_OFDM, or FORMAT is HE_SU and PSDU_LENGTH is 0	Indicates the 20 MHz subchannels that are punctured. A bitmap indexed by the 20 MHz subchannels in ascending order with the LSB indicating the lowest frequency 20 MHz subchannel. A bit is set to 1 to indicate that the corresponding 20 MHz subchannel is punctured and set to 0 to indicate the corresponding 20 MHz subchannel is not punctured.	Y	N
	Otherwise	Not present	N	N
DYN_BANDWIDTH_IN_NON_HT	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Not present	N	N
	Otherwise	See corresponding entry in Table 21-1.		
CH_BANDWIDTH_IN_NON_HT	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Not present	N	N
	Otherwise	See corresponding entry in Table 21-1.		
LENGTH	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Not present	N	N
	Otherwise	See corresponding entry in Table 21-1.		
APEP_LENGTH	FORMAT is HE_SU or HE_ER_SU	Integer.	Y	O
	FORMAT is HE_MU or HE_TB	If 0 and FORMAT is HE_SU, indicates an HE sounding NDP. Otherwise, indicates the number of octets in the range 1 to <i>aPSDUMaxLength</i> in the A-MPDU pre-EOF padding (see Table 27-54) that is carried in the PSDU.	M U	O
	Otherwise	See corresponding entry in Table 19-1 or Table 21-1.		

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
PSDU_LENGTH	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates the number of octets in the PSDU in the range of 0 to <i>aPSDUMaxLength</i> octets (see Table 27-54). A value of 0 indicates an HE sounding NDP.	N	Y
	Otherwise	See corresponding entry in Table 21-1.		
USER_POSITION	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Not present	N	N
	Otherwise	See corresponding entry in Table 21-1.		
NUM_STS	FORMAT is HE_SU	Indicates the number of space-time streams. Integer: range 1 to 8.	Y	Y
	FORMAT is HE_ER_SU	Indicates the number of space-time streams. Integer: range 1 to 2. NOTE—NUM_STS set to 1 is valid only if STBC is not used. NUM_STS set to 2 is valid only if STBC is used.	Y	Y
	FORMAT is HE_MU	Indicates the number of space-time streams. Integer in the range: 1–4 per user per MU-MIMO RU in the TXVECTOR. 1–4 per MU-MIMO RU in the RXVECTOR. 1–8 per RU assigned to no more than 1 user in the TXVECTOR and RXVECTOR. NUM_STS summed over all users per RU is not greater than 8.	M U	Y
	FORMAT is HE_TB	Indicates the number of space-time streams. Integer in the range: 1–4 for a MU-MIMO RU in the TXVECTOR 1–4 per user per MU-MIMO RU in the RXVECTOR 1–8 for an RU assigned to no more than 1 user in the TXVECTOR and RXVECTOR NUM_STS summed over all users per RU is not greater than 8.	Y	N
	Otherwise	See corresponding entry in Table 21-1.		
GROUP_ID	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Not present	N	N
	Otherwise	See corresponding entry in Table 21-1.		

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
PARTIAL_AID	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Not present	N	N
	Otherwise	See corresponding entry in Table 21-1.		
TXOP_DURATION	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates the TXOP duration. Enumerated type or integer: UNSPECIFIED indicates no NAV value specified. 0–8448 indicates a value in units of 1 μs that is used to update the NAV for this TXOP (see 26.2.4). TXVECTOR parameter TXOP_DURATION is converted to a value in the TXOP subfield of the HE-SIG-A field (see Table 27-18, Table 27-20, and Table 27-21) as follows: TXOP_DURATION = UNSPECIFIED: B0–B6 = 127. TXOP_DURATION < 512: B0 = 0, B1–B6 = $\lfloor \text{TXOP_DURATION} / 8 \rfloor$. Otherwise: B0 = 1, B1–B6 = $\lfloor \text{TXOP_DURATION} - 512 / 8 \rfloor$. RXVECTOR parameter TXOP_DURATION is determined from the value in the TXOP subfield of the HE-SIG-A field (see Table 27-18, Table 27-20, and Table 27-21) as follows: B0–B6 = 127: TXOP_DURATION = UNSPECIFIED. B0 = 0: TXOP_DURATION = $8 \times \text{B1–B6}$. Otherwise: TXOP_DURATION = $512 + 128 \times \text{B1–B6}$. See 26.11.5 for more details.	Y	Y
	Otherwise	Not present	N	N
SPATIAL_REUSE	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates the spatial reuse parameter value. There is one value of the parameter for each of an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU. There are one to four values of the parameter present for an HE TB PPDU, with the number of values present dependent on the bandwidth of the PPDU. See the Spatial Reuse field definition in 27.3.11.7.2. See 26.5.2.3 and 26.11.6.	Y	Y
	Otherwise	Not present	N	N
DOPPLER	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Set to 1 to indicate that either the midamble is present for the PPDU or that the channel is with high Doppler. Set to 0 otherwise.	Y	Y
	Otherwise	Not present	N	N

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
NUM_USERS	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Not present NOTE—The number of users for an HE SU PPDU, HE ER SU PPDU, or HE TB PPDU is always 1. The number of users in an RU in an HE MU PPDU is determined by RU_ALLOCATION and STA_ID parameters for that RU.	N	N
	Otherwise	See corresponding entry in Table 21-1.		
RU_ALLOCATION	FORMAT is HE_MU and SIG_B_COMPRESSION_MODE is 0	For the TXVECTOR, indicates the RU Allocation subfield of Common field in the HE-SIG-B field of the transmitted PPDU. 8 bits for a 20 MHz PPDU. 16 bits for a 40 MHz PPDU. 32 bits for an 80 MHz PPDU. 64 bits for a 160 MHz or 80+80 MHz PPDU. See 27.3.11.8.3 for details. For the RXVECTOR, 8 bits are used to indicate the RU allocated in the whole bandwidth. See 9.3.1.22 for details.	Y	Y
	FORMAT is HE_TB	8 bits are used to indicate the RU allocated in the whole bandwidth per user. See 9.3.1.22 for details.	Y	N
	FORMAT is HE_SU, APEP_LENGTH is 0, and CH_BANDWIDTH is not CBW20 or CBW40	For the TXVECTOR, indicates the active RUs. 32 bits for 80 MHz PPDU. 64 bits for 160 MHz and 80+80 MHz PPDU.	Y	N
	FORMAT is NON_HT, NON_HT_MODULATION is NON_HT_DUP_OFDM, and CH_BANDWIDTH is not CBW20 or CBW40	For each 8 bits, only the following values are allowed: 113 (01110001 in binary representation). 192 (11000000 in binary representation). See 27.3.11.8.3 for details.		
	Otherwise	Not present	N	N
BEAMFORMED	FORMAT is HE_SU or HE_ER_SU	Set to 1 if a beamforming steering matrix is applied to the HE modulated fields. Set to 0 otherwise.	Y	Y
	FORMAT is HE_MU or HE_TB	For an RU with no more than 1 user allocated, set to 1 if a beamforming steering matrix is applied to this non-MU-MIMO allocation, and set to 0 otherwise. For each user in an RU assigned to more than 1 user, always set to 0.	M U	O
	Otherwise	See corresponding entry in Table 21-1.		

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
CENTER_26_TONE_RU	FORMAT is HE_MU and CH_BANDWIDTH is CBW80, CBW160, CBW80+80, HE-CBW-PUNC80-PRI, HE-CBW-PUNC80-SEC, HE-CBW-PUNC160-PRI20, HE-CBW-PUNC80+80-PRI20, HE-CBW-PUNC160-SEC40, or HE-CBW-PUNC80+80-SEC40.	Indicate the presence of the center 26-tone RU with regard to bandwidth. If the CH_BANDWIDTH parameter is CBW80, HE-CBW-PUNC80-PRI or HE-CBW-PUNC80-SEC: Set to 1 to indicate that a user is allocated to the center 26-tone RU (see Figure 27-7). Set to 0 to indicate that no user is allocated to the center 26-tone RU. If the CH_BANDWIDTH parameter is CBW160, CBW80+80, HE-CBW-PUNC160-PRI20, HE-CBW-PUNC80+80-PRI20, HE-CBW-PUNC160-SEC40, or HE-CBW-PUNC80+80-SEC40: Set to 0 to indicate that no user is allocated to either the center 26-tone RU of the lower frequency 80 MHz or that of the higher frequency 80 MHz. Set to 1 to indicate that a user is allocated to the center 26-tone RU of the lower frequency 80 MHz. Set to 2 to indicate that a user is allocated to the center 26-tone RU of the higher frequency 80 MHz. Set to 3 to indicate that a user is allocated to each of the center 26-tone RU of the lower frequency 80 MHz and that of the higher frequency 80 MHz individually. See 27.3.11.8.3.	Y	N
	Otherwise	Not present	N	N
HE_LTF_TYPE	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates the type of HE-LTF. Enumerated type: 1xHE-LTF indicates a 1x HE-LTF. 2xHE-LTF indicates a 2x HE-LTF. 4xHE-LTF indicates a 4x HE-LTF. See 27.3.11.10.	Y	Y
	Otherwise	Not present	N	N
HE_LTF_MODE	FORMAT is HE_TB	Integer: Set to 0 to indicate that the transmitted PPDU uses HE single stream pilot HE-LTF mode. Set to 1 to indicate the transmitted PPDU uses HE masked HE-LTF sequence mode. Present for full-bandwidth MU-MIMO not using 1x HE-LTF and not present otherwise. See 27.3.11.10.	O	N
	Otherwise	Not present	N	N

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
NUM_HE_LTF	FORMAT is HE_MU or HE_TB	Indicates the number of OFDM symbols in the HE-LTF field. See 26.5.2.3 and 27.3.11.7.2.	Y	N
	Otherwise	Not present	N	N
HE_SIG_A2_RESERVED	FORMAT is HE_TB	Indicates the Reserved field setting for the HE-SIG-A2 subfield of HE TB PPDU. See 26.5.2.3 and Table 27-21 for details.	Y	N
	Otherwise	Not present	N	N
STARTING_STS_NUM	FORMAT is HE_TB	The sum of the number of space-time streams assigned to other users with lower space-time stream indices than this user.	Y	N
	Otherwise	Not present	N	N
TXOP_PS_NOT_ALLOWED	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Not present	N	N
	Otherwise	See corresponding entry in Table 21-1.		
RX_START_OF_FRAME_OFFSET	See corresponding entry in Table 21-1.			

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
PREAMBLE_TYPE	FORMAT is NON_HT and NON_HT_MODULATION is one of ERP-DSSS ERP-CCK	Enumerated type: SHORTPREAMBLE. LONGPREAMBLE.	Y	Y
	Otherwise	Not present	N	N
NOMINAL_PACKET_PADDING	FORMAT is HE_SU, HE_MU, or HE_ER_SU	The nominal packet padding as defined in 9.4.2.248.5. Possible values are 0 μ s, 8 μ s, and 16 μ s.	M U	N
	Otherwise	Not present	N	N
TRIGGER_METHOD	FORMAT is HE_TB	Indicates the method used to trigger this HE TB PPDU transmission. Enumerated type: TRIGGER_FRAME for Trigger frame. TRS for TRS Control subfield.	Y	N
	Otherwise	Not present	N	N
PRE_FEC_PADDING_FACTOR	FORMAT is HE_TB and TRIGGER_METHOD is TRIGGER_FRAME	Indicates the pre-FEC padding factor used by the HE TB PPDU transmission.	Y	N
	Otherwise	Not present	N	N

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
DEFAULT_PE_DURATION	FORMAT is HE_TB and TRIGGER_METHOD is TRS	Duration of the PE field to be transmitted (see 26.5.2.3). A value 0, 4, 8, 12, or 16 indicating the PE field duration in μ s.	Y	N
	Otherwise	Not present	N	N
BEAM_CHANGE	FORMAT is HE_SU or HE_ER_SU	Integer: Set to 0 to indicate that the spatial mapping of the pre-HE modulated fields are the same as the first symbol of HE-LTF field. Set to 1 to indicate that the spatial mapping of the pre-HE modulated fields are different from the first symbol of HE-LTF field.	Y	Y
	Otherwise	Not present	N	N
BSS_COLOR	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Set to a value in the range 0 to 63 (see 26.11).	Y	Y
	Otherwise	Not present	N	N
UPLINK_FLAG	FORMAT is HE_SU or HE_MU	Set to 1 if the PPDU is addressed to an AP. Set to 0 otherwise.	Y	Y
	FORMAT is HE_ER_SU	Set to 0 if the PPDU is not addressed to an AP, or if the PPDU is addressed to an AP and meets the exception in 26.11.2. Set to 1 otherwise.	Y	Y
	Otherwise	Not present	N	N
STA_ID	FORMAT is HE_MU	Indicates the list of STA-IDs for an HE MU PPDU (see 26.11).	M U	M U
	Otherwise	Not present	N	N

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
SCRAMBLER_INITIAL_VALUE	FORMAT is NON_HT	In TXVECTOR, if present, indicates the value of the Scrambler Initialization field in the SERVICE field, after scrambling. In RXVECTOR, indicates the value of the Scrambler Initialization field in the SERVICE field, prior to descrambling.	O	Y
	FORMAT is HE_MU or HE_TB	Not present	N	N
	FORMAT is VHT and GROUP_ID is neither 0 nor 63	Not present	N	N
	Otherwise	Indicates the value in the Scrambler Initialization field in the SERVICE field prior to descrambling.	N	Y
TRIGGER_RESPONDING	FORMAT is NON_HT	Boolean value: true indicates that the MAC entity requests that the PHY entity follow the pre-correction requirements defined in 27.3.15.3. false indicates that the MAC entity does not request that the PHY entity follow the pre-correction requirements defined in 27.3.15.3.	Y	N
	Otherwise	Not present	N	N
NDP_REPORT	FORMAT is HE_TB and PSDU_LENGTH = 0	Provides the detected status array on the resources assigned by the Trigger frame. The array has N_{STA} entries (N_{STA} is defined in 26.5.7.2) where each entry takes on the value: 1 if transmission is detected on the first group of the tone set. 0 if transmission is detected on the second group of the tone set. NONE if transmission is not detected on either group of the tone set.	N	Y
	Otherwise	Not present	N	N
FEEDBACK_STATUS	FORMAT is HE_TB and APEP_LENGTH = 0	Indicates the value of the one bit used to modulate the tones in each tone set. See 27.3.4 and 26.5.7.	Y	N
	Otherwise	Not present	N	N

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
RU_TONE_SET_INDEX	FORMAT is HE_TB and APEP_LENGTH = 0	Indicates the RU tone set used for an HE TB feedback NDP. See 27.3.18.	Y	N
	Otherwise	Not present	N	N
MIDAMBLE_PERIODICITY	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB, and DOPPLER is 1	Indicates the midamble periodicity in number of OFDM symbols in the Data field. Set to 10 or 20.	Y	N
	Otherwise	Not present	N	N
HE_TB_PE_DISAMBIGUITY	FORMAT is HE_TB and TRIGGER_METHOD is TRIGGER_FRAME	Indicates PE disambiguity for the HE TB PPDU transmission. Set to 0 to indicate no PE disambiguity. Set to 1 to indicate PE disambiguity.	Y	N
	Otherwise	Not present	N	N
NOTE 1—In the “TXVECTOR” and “RXVECTOR” columns, the following apply: Y = Present. N = Not present. O = Optional. MU indicates that the parameter is present once for an HE SU PPDU and HE ER SU PPDU and present per user for an HE MU PPDU. For an HE TB PPDU, MU in the “TXVECTOR” column indicates that the parameter is present once and MU in the “RXVECTOR” column indicates the parameter is not present (the receiver knows the values since they were specified in the triggering PPDU). Parameters specified to be present per user are conceptually supplied as an array of values indexed by u, where u takes values 0 to NUM_USERS – 1. NOTE 2—Refer to Clause 15, Clause 16, Clause 17, Clause 18, Clause 19, and Clause 21 for the TXVECTOR and RXVECTOR parameters that are not present in this table when FORMAT is not HE_SU, HE_MU, HE_ER_SU, or HE_TB.				

27.2.3 TRIGVECTOR parameters

The TRIGVECTOR is carried in a PHY-TRIGGER.request primitive and provides the PHY of the AP with the parameters needed to receive an HE TB PPDU over each assigned RU. The parameters in Table 27-2 are defined as part of the TRIGVECTOR parameter list in the PHY-TRIGGER.request primitive.

Table 27-2—TRIGVECTOR parameters

Parameter	Value
CH_BANDWIDTH	Indicates the bandwidth in the HE-SIG-A field of the expected HE TB PPDU(s). Enumerated type: CBW20 for 20 MHz. CBW40 for 40 MHz. CBW80 for 80 MHz. CBW160 for 160 MHz. CBW80+80 for 80+80 MHz.
UL_LENGTH	Indicates the value of the LENGTH field in the L-SIG field of the expected HE TB PPDU(s).
GI_AND_HE_LTF_TYPE	Indicates the GI and HE-LTF type of the expected HE TB PPDU(s). Enumerated type: 1x HE-LTF + 1.6 μ s GI. 2x HE-LTF + 1.6 μ s GI. 4x HE-LTF + 3.2 μ s GI.
MU_MIMO_HE_LTF_MODE	Indicates the HE-LTF mode of the expected UL MU-MIMO HE TB PPDU(s), if it uses full-bandwidth MU-MIMO and does not use 1x HE-LTF. Set to 0 to indicate that HE single stream pilot HE-LTF mode is used. Set to 1 to indicate that HE masked HE-LTF sequence mode is used.
NUMBER_OF_HE_LTF_SYMBOLS	Indicates the number of HE-LTF symbols present in the expected HE TB PPDU(s). If the parameter DOPPLER is 0: Set to 0 for 1 HE-LTF symbol. Set to 1 for 2 HE-LTF symbols. Set to 2 for 4 HE-LTF symbols. Set to 3 for 6 HE-LTF symbols. Set to 4 for 8 HE-LTF symbols. If the parameter DOPPLER is 1: Set to 0 for 1 HE-LTF symbol. Set to 1 for 2 HE-LTF symbols. Set to 2 for 4 HE-LTF symbols.
MIDAMBLE_PREIODICITY	Indicates the midamble periodicity. Present only if the parameter DOPPLER is 1. Integer value: Set to 10 to indicate a 10 symbol midamble periodicity. Set to 20 to indicate a 20 symbol midamble periodicity.
STBC	Indicates the status of STBC encoding in the expected HE TB PPDU(s). Set to 1 if STBC encoding is used. Set to 0 otherwise.

Table 27-2—TRIGVECTOR parameters (continued)

Parameter	Value
LDPC_EXTRA_SYMBOL	Indicates the status of the LDPC extra symbol segment in the expected HE TB PPDU(s). Set to 1 if LDPC extra symbol segment is present. Set to 0 otherwise.
PRE_FEC_FACTOR	Indicates the pre-FEC padding factor for the expected HE TB PPDU(s). Set to 0 to indicate a pre-FEC padding factor of 4. Set to 1 to indicate a pre-FEC padding factor of 1. Set to 2 to indicate a pre-FEC padding factor of 2. Set to 3 to indicate a pre-FEC padding factor of 3.
PE_DISAMBIGUITY	Indicates the PE disambiguity for the expected HE TB PPDU(s). Set to 0 to indicate no PE disambiguity. Set to 1 to indicate PE disambiguity.
DOPPLER	Indicates a high Doppler mode for the expected HE TB PPDU(s). Set to 1 to indicate that midambles are present. Set to 0 otherwise.
AID12_LIST	Each entry of AID12_LIST is (12-bit) AID of the corresponding HE TB PPDU. See the AID12 subfield in 9.3.1.22. NOTE—The MSB of the 12-bit AID is always 0.
RU_ALLOCATION_LIST	8 bits are used per STA to indicate the RU allocated in the whole bandwidth. See 9.3.1.22.
FEC_CODING_LIST	Each entry indicates the coding type of the corresponding HE TB PPDU. See the UL FEC Coding Type subfield description in 9.3.1.22.
HE_MCS_LIST	Each entry of HE_MCS_LIST indicates the HE-MCS of the corresponding HE TB PPDU. See the UL HE-MCS subfield in 9.3.1.22 for more information of each entry.
UL_DCM_LIST	Each entry indicates whether DCM is applied to the corresponding HE TB PPDU. See the UL DCM subfield description in 9.3.1.22 for details.
SS_ALLOCATION_LIST	Each entry indicates the spatial streams of the corresponding HE TB PPDU. See the SS Allocation subfield description in 9.3.1.22.

27.2.4 PHYCONFIG_VECTOR parameters

The PHYCONFIG_VECTOR carried in a PHY-CONFIG.request primitive for an HE PHY contains an OPERATING_CHANNEL parameter, which identifies the operating or primary channel. The PHY shall set dot11CurrentPrimaryChannel to the value of this parameter.

The PHYCONFIG_VECTOR carried in a PHY-CONFIG.request primitive for an HE PHY contains a CHANNEL_WIDTH parameter, which identifies the operating channel width and takes one of the values 20 MHz, 40 MHz, 80 MHz, 160 MHz, and 80+80 MHz. The PHY shall set dot11CurrentChannelWidth to the value of this parameter. The PHY shall set dot11HECurrentChannelWidthSet to a value that is obtained from the Supported Channel Width Set subfield in the HE Capabilities element transmitted by the STA.

The PHYCONFIG_VECTOR carried in a PHY-CONFIG.request primitive for an HE PHY contains a CENTER_FREQUENCY_SEGMENT_0 parameter, which identifies the center frequency of the channel (or of segment 0 if the CHANNEL_WIDTH parameter indicates 80+80 MHz) and takes a value between 1 and 255. The PHY shall set dot11CurrentChannelCenterFrequencyIndex0 to the value of this parameter.

The PHYCONFIG_VECTOR carried in a PHY-CONFIG.request primitive for an HE PHY contains a CENTER_FREQUENCY_SEGMENT_1 parameter, which takes the value 0 if the CHANNEL_WIDTH parameter does not indicate 80+80 MHz, and otherwise identifies the center frequency of segment 1 and takes a value between 1 and 200. The PHY shall set dot11CurrentChannelCenterFrequencyIndex1 to the value of this parameter.

27.2.5 Effects of CH_BANDWIDTH parameter on PPDU format

Table 27-3 shows the valid combinations of the FORMAT, NON_HT_MODULATION, and CH_BANDWIDTH parameters and the corresponding PPDU format and value of CH_OFFSET (if applicable). Other combinations are reserved.

Table 27-3—Interpretation of FORMAT, NON_HT_MODULATION, CH_BANDWIDTH, and CH_OFFSET parameters

FORMAT	NON_HT_MODULATION	CH_BANDWIDTH	CH_OFFSET	PPDU format
HE	N/A	CBW20	N/A	The STA transmits an HE PPDU of 20 MHz bandwidth. If the BSS bandwidth is wider than 20 MHz, then the transmission shall use the primary 20 MHz channel.
HE	N/A	CBW40	N/A	The STA transmits an HE PPDU of 40 MHz bandwidth. If the BSS bandwidth is wider than 40 MHz, then the transmission shall use the primary 40 MHz channel.
HE	N/A	CBW80	N/A	The STA transmits an HE PPDU of 80 MHz bandwidth. If the BSS bandwidth is wider than 80 MHz, then the transmission shall use the primary 80 MHz channel.
HE	N/A	CBW160	N/A	The STA transmits an HE PPDU of 160 MHz bandwidth.
HE	N/A	CBW80+80	N/A	The STA transmits an HE PPDU of 80+80 MHz bandwidth.
HE	N/A	HE-CBW-PUNC80-PRI	N/A	The STA transmits an 80 MHz HE PPDU where the only punctured subchannel is the secondary 20 MHz channel.
HE	N/A	HE-CBW-PUNC80-SEC	N/A	The STA transmits an 80 MHz HE PPDU where the only punctured subchannel is one of the two 20 MHz subchannels in the secondary 40 MHz channel.

Table 27-3—Interpretation of FORMAT, NON_HT_MODULATION, CH_BANDWIDTH, and CH_OFFSET parameters (continued)

FORMAT	NON_HT_MODULATION	CH_BANDWIDTH	CH_OFFSET	PPDU format
HE	N/A	HE-CBW-PUNC160-PRI20	N/A	The STA transmits a 160 MHz HE PPDU where the only punctured subchannels are the secondary 20 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel. See Table 27-20 for allowed 20 MHz punctured subchannel combinations.
HE	N/A	HE-CBW-PUNC80+80-PRI20	N/A	The STA transmits an 80+80 MHz HE PPDU where the only punctured subchannels are the secondary 20 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel. See Table 27-20 for allowed 20 MHz punctured subchannel combinations.
HE	N/A	HE-CBW-PUNC160-SEC40	N/A	The STA transmits a 160 MHz HE PPDU where the only punctured subchannels are zero, one or both of the 20 MHz subchannels in the secondary 40 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel. At least one 20 MHz subchannel is punctured. See Table 27-20 for allowed 20 MHz punctured subchannel combinations.
HE	N/A	HE-CBW-PUNC80+80-SEC40	N/A	The STA transmits an 80+80 MHz HE PPDU where the only punctured subchannels are zero, one or both of the 20 MHz subchannels in the secondary 40 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel; at least one 20 MHz subchannel is punctured. See Table 27-20 for allowed 20 MHz punctured subchannel combinations.
HT_MF, HT_GF, VHT	See Table 21-2 and Table 19-2.			
NON_HT	If INACTIVE_SUBCHANNELS is not present, see Table 21-2 and Table 19-2.			
NON_HT	If INACTIVE_SUBCHANNELS is present, see Table 27-4.			

Valid combinations of the CH_BANDWIDTH and INACTIVE_SUBCHANNELS parameters when FORMAT is NON_HT and the corresponding PPDU and CH_OFFSET (if applicable) are shown in Table 27-4. Other combinations are reserved.

Table 27-4— Interpretation of CH_BANDWIDTH and INACTIVE_SUBCHANNELS parameters when FORMAT is equal to NON_HT and NON_HT_MODULATION is equal to NON_HT_DUP_OFDM

CH_BANDWIDTH	INACTIVE_SUBCHANNELS	CH_OFFSET	PPDU format
CBW80	All bits set to 1, except for the four bits corresponding to the primary 80 MHz channel, which are set to 0	N/A	The STA transmits a non-HT PPDU of 80 MHz bandwidth. If the BSS bandwidth is wider than 80 MHz, then the transmission shall use the primary 80 MHz channel.
CBW80	The bit corresponding to the primary 20 MHz channel set to 0 and two other bits set to 0 that correspond to any other subchannels in the primary 80 MHz, all other bits set to 1	N/A	The STA transmits a punctured non-HT PPDU of 80 MHz bandwidth. If the BSS bandwidth is wider than 80 MHz, then the transmission shall use the primary 80 MHz channel.
CBW160	All bits set to 0	N/A	The STA transmits a non-HT PPDU of 160 MHz bandwidth.
CBW160	Either the bit corresponding to the secondary 20 MHz channel or zero, one or both bits corresponding to the secondary 40 MHz channel set to 1. Zero to two bits corresponding to 20 MHz subchannels in the secondary 80 MHz channel set to 1. All other bits set to 0. Not all bits set to 0. If two bits corresponding to 20 MHz subchannels in the secondary 80 MHz channel are set to 1 these correspond to the lower two or higher two 20 MHz subchannels. No more than two bits corresponding to adjacent 20 MHz subchannels set to 1.	N/A	The STA transmits a punctured non-HT PPDU of 160 MHz bandwidth.
CBW80+80	All bits set to 0	N/A	The STA transmits a non-HT PPDU of 80+80 MHz bandwidth.

Table 27-4— Interpretation of CH_BANDWIDTH and INACTIVE_SUBCHANNELS parameters when FORMAT is equal to NON_HT and NON_HT_MODULATION is equal to NON_HT_DUP_OFDM (continued)

CH_BANDWIDTH	INACTIVE_SUBCHANNELS	CH_OFFSET	PPDU format
CBW80+80	Either the bit corresponding to the secondary 20 MHz channel or zero, one or both bits corresponding to the secondary 40 MHz channel set to 1. Zero to two bits corresponding to 20 MHz subchannels in the secondary 80 MHz channel set to 1. All other bits set to 0. Not all bits set to 0. If two bits corresponding to 20 MHz subchannels in the secondary 80 MHz channel are set to 1 these correspond to the lower two or higher two 20 MHz subchannels.	N/A	The STA transmits a punctured non-HT PPDU of 80+80 MHz bandwidth.

27.2.6 Support for non-HT, HT, and VHT formats

27.2.6.1 General

An HE STA logically contains Clause 15, Clause 16, Clause 17, Clause 18, Clause 19, Clause 21, and Clause 27 PHYs. The MAC interacts with the PHYs via the Clause 27 PHY service interface, which in turn interacts with the Clause 15, Clause 16, Clause 17, Clause 18, and Clause 19, and Clause 21 PHY service interfaces as shown in Figure 27-1, Figure 27-2, and Figure 27-3.

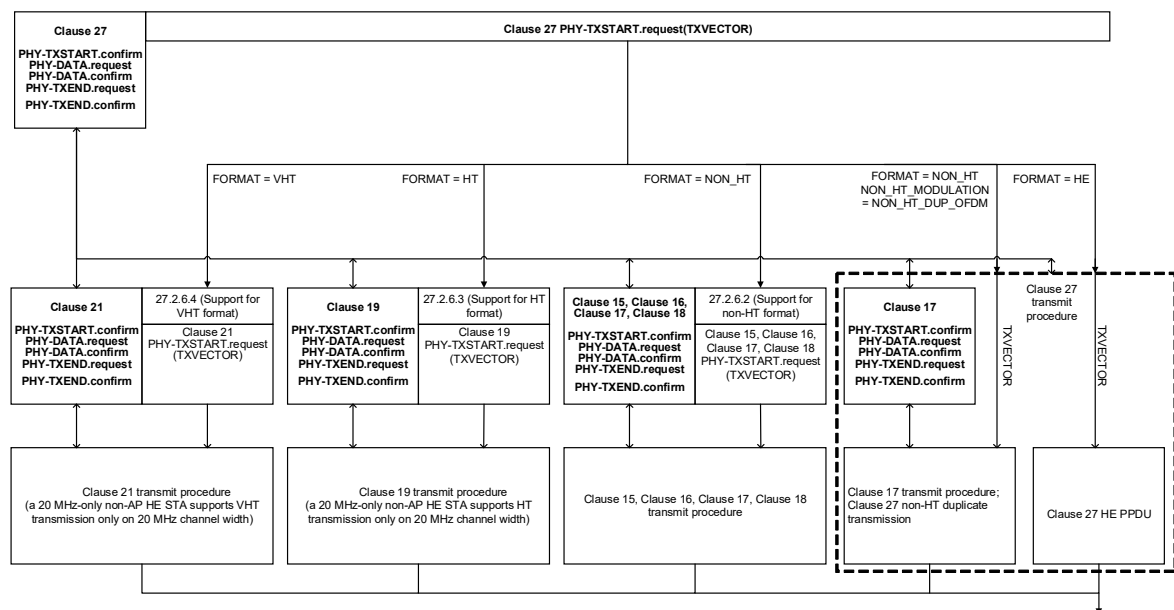


Figure 27-1—PHY interaction on transmit for various PPDU formats

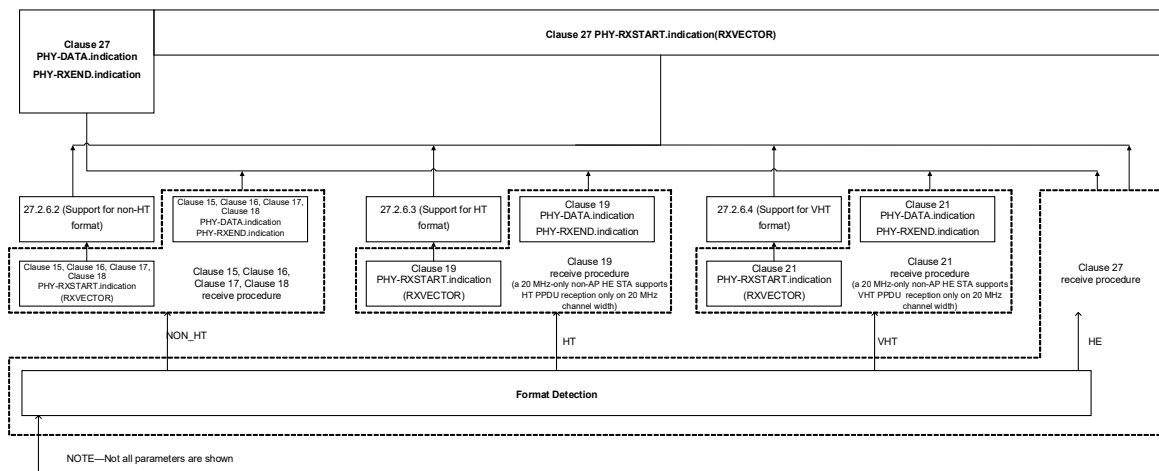


Figure 27-2—PHY interaction on receive for various PPDU formats

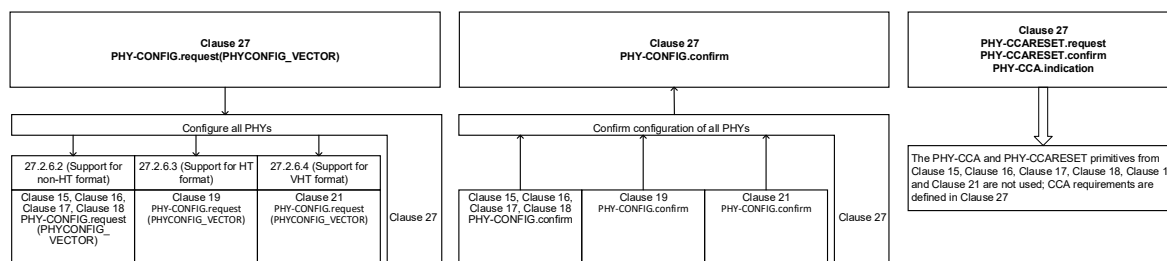


Figure 27-3—PHY-CONFIG and CCA interaction with various PPDU formats

27.2.6.2 Support for non-HT format

The behavior of the HE PHY on receipt of a PHY-TXSTART.request(TXVECTOR) primitive with the FORMAT parameter equal to NON_HT and the NON_HT_MODULATION parameter not equal to NON_HT_DUP_OFDM is defined in Clause 15, Clause 16, Clause 17, and Clause 18 and depends on the parameter NON_HT_MODULATION. If the parameter NON_HT_MODULATION is OFDM or NON_HT_DUP_OFDM, then the following additional requirements apply:

- The requirements in 21.3.9.1
- The requirements in 21.3.17.1 instead of the requirements in 17.3.9.3
- The requirements in 27.3.19.3 instead of the requirements in 17.3.9.7.2

The modulation equation for non-HT duplicate transmission is defined in 27.3.14.

The HE PHY TXVECTOR parameters in Table 27-1 are mapped to Clause 15, Clause 16, Clause 17, and Clause 18 TXVECTOR parameters according to Table 27-5. The HE PHY parameters not listed in the table are not present.

Table 27-5—Mapping of the HE PHY parameters for non-HT operation

HE PHY parameter	2.4 GHz operation defined by Clause 15	2.4 GHz operation defined by Clause 16	2.4 GHz operation defined by Clause 18	5 GHz and 6 GHz operation defined by Clause 17	Parameter list
L_LENGTH	LENGTH	LENGTH	LENGTH	LENGTH	TXVECTOR/ RXVECTOR
L_DATARATE	DATARATE	DATARATE	DATARATE	DATARATE	TXVECTOR/ RXVECTOR
TXPWR_LEVEL_INDEX	TXPWR_LEVEL_INDEX	TXPWR_LEVEL_INDEX	TXPWR_LEVEL_INDEX	TXPWR_LEVEL_INDEX	TXVECTOR
RSSI	RSSI	RSSI	RSSI	RSSI	RXVECTOR
SERVICE	SERVICE	SERVICE	SERVICE	SERVICE	TXVECTOR/ RXVECTOR
RCPI	RCPI	RCPI	RCPI	RCPI	RXVECTOR
CH_BANDWIDTH_IN_NON_HT	discarded	discarded	CH_BANDWIDTH_IN_NON_HT	CH_BANDWIDTH_IN_NON_HT	TXVECTOR/ RXVECTOR
DYN_BANDWIDTH_IN_NON_HT	discarded	discarded	DYN_BANDWIDTH_IN_NON_HT	DYN_BANDWIDTH_IN_NON_HT	TXVECTOR/ RXVECTOR
OPERATING_CHANNEL	discarded	discarded	discarded	OPERATING_CHANNEL	PHYCONFIG_VECTOR

The behavior of the HE PHY on receipt of a PHY-TXSTART.request(TXVECTOR) primitive with the FORMAT parameter equal to NON_HT and the NON_HT_MODULATION parameter equal to NON_HT_DUP_OFDM is defined in 27.3.14.

To support the non-HT format, the HE PHY, on receipt of a PHY-CONFIG.request(PHYCONFIG_VECTOR) primitive, behaves as if it were a Clause 15, Clause 16, Clause 17, or Clause 18 PHY that had received a PHY-CONFIG.request(PHYCONFIG_VECTOR) primitive but without the PHYCONFIG_VECTOR parameters CHANNEL_WIDTH, CENTER_FREQUENCY_SEGMENT_0, and CENTER_FREQUENCY_SEGMENT_1.

As defined in 27.3.22, once a PPDU is received and detected as a non-HT PPDU, the behavior of the HE PHY is defined in Clause 15, Clause 16, Clause 17, or Clause 18 depending on the PPDU format. The RXVECTOR parameters from Clause 15, Clause 16, Clause 17, and Clause 18 are mapped to the HE PHY RXVECTOR parameters as defined in Table 27-5. The HE PHY parameters not listed in the table are not present.

27.2.6.3 Support for HT format

The behavior of an HE PHY on receipt of a PHY-TXSTART.request(TXVECTOR) primitive with the TXVECTOR parameter FORMAT equal to HT_MF or HT_GF is defined in Clause 19 with the following additional requirements:

- The requirements in 21.3.9.2
- The requirements in 27.3.19.3 instead of the requirements in 19.3.18.4

The HE PHY TXVECTOR parameters in Table 27-1 are mapped directly to Clause 19 TXVECTOR parameters in Table 19-1. The HE PHY parameters not listed in Table 19-1 are not present. The PHY shall use a value of CH_OFFSET in the Clause 19 TXVECTOR that is consistent with Table 27-3. A 20 MHz-only non-AP HE STA supports HT transmission only on 20 MHz channel width.

On receipt of a PHY-CONFIG.request(PHYCONFIG_VECTOR) primitive, the HE PHY behaves, for the purposes of HT PPDU transmission and reception, as if it were a Clause 19 PHY that had received PHY-CONFIG.request(PHYCONFIG_VECTOR) primitive but without the PHYCONFIG_VECTOR parameters CHANNEL_WIDTH, CENTER_FREQUENCY_SEGMENT_0, and CENTER_FREQUENCY_SEGMENT_1 and with the PHYCONFIG_VECTOR parameter SECONDARY_CHANNEL_OFFSET set to SECONDARY_CHANNEL_NONE if $\text{dot11CurrentChannelWidth}$ indicates 20 MHz, to SECONDARY_CHANNEL_ABOVE if $f_{P20,idx} < f_{S20,idx}$, or to SECONDARY_CHANNEL_BELOW otherwise.

As defined in 27.3.22, once a PPDU is received and detected as an HT PPDU, the behavior of the HE PHY is defined in Clause 19. The RXVECTOR parameters in Table 19-1 are mapped directly to the RXVECTOR parameters in Table 27-1. The HE PHY parameters not listed in Table 19-1 are not present. A 20 MHz-only non-AP HE STA supports HT reception only on 20 MHz channel width.

27.2.6.4 Support for VHT format

The behavior of an HE PHY on receipt of a PHY-TXSTART.request(TXVECTOR) primitive with the TXVECTOR parameter FORMAT equal to VHT is defined in Clause 21, except that the requirements in 27.3.19.3 apply instead of the requirements in 21.3.17.4.2.

The HE PHY TXVECTOR parameters in Table 27-1 are mapped directly to the Clause 21 TXVECTOR parameters in Table 21-1. The HE PHY parameters not listed in Table 21-1 are not present. The 20 MHz-only non-AP HE STA supports VHT transmission only on 20 MHz channel width.

On receipt of a PHY-CONFIG.request(PHYCONFIG_VECTOR) primitive, the HE PHY behaves, for the purposes of VHT PPDU transmission and reception, as if it were a Clause 21 PHY that received the PHY-CONFIG.request(PHYCONFIG_VECTOR) primitive.

As defined in 27.3.22, once a PPDU is received and detected as an VHT PPDU, the behavior of the HE PHY is defined in Clause 21. The RXVECTOR parameters in Table 21-1 are mapped directly to the RXVECTOR parameters in Table 27-1. The HE PHY parameters not listed in Table 21-1 are not present. A 20 MHz-only non-AP HE STA supports VHT reception only on 20 MHz channel width.

27.3 HE PHY

27.3.1 Introduction

Subclause 27.3 provides the procedure by which PSDUs are converted to and from transmissions on the wireless medium.

During transmission, a PSDU (in the SU case) or one or more PSDUs (in the MU case) are processed (i.e., scrambled and coded) and appended to the PHY preamble to create the PPDU. At the receiver, the PHY preamble is processed to aid in the detection, demodulation, and delivery of the PSDU.

27.3.1.1 MU transmission

The MU transmissions include DL MU transmissions and UL MU transmissions.

DL MU transmission allows an AP to simultaneously transmit information to more than one non-AP STA. For a DL MU transmission, the AP uses the HE MU PPDU format and employs either DL OFDMA or DL MU-MIMO, or a mixture of both. UL MU transmission allows an AP to simultaneously receive information from more than one non-AP STA. UL MU transmissions are preceded by a triggering frame from the AP. The non-AP STAs transmit using the HE TB PPDU format and employ either UL OFDMA or UL MU-MIMO, or a mixture of both.

The HE PHY supports OFDMA transmissions, in both the DL and the UL where different users can occupy different RUs in a PPDU (see 27.3.10). The transmission within an RU in a PPDU may be a single stream to one user (SISO), multiple streams spatially multiplexed to one user (SU-MIMO), or multiple streams spatially multiplexed to multiple users (MU-MIMO).

NOTE—The VHT PHY supports only full-bandwidth DL MU-MIMO as described in 21.3.11.

The HE PHY supports DL MU-MIMO and UL MU-MIMO, for both the full-bandwidth case and the partial-bandwidth case where MU-MIMO is used only on certain RUs in the PPDU. The combination of non-MU-MIMO allocations for some RUs and MU-MIMO allocations on different RUs in one PPDU is also supported.

27.3.1.2 OFDMA

OFDMA is an OFDM-based multiple access scheme where different subsets of subcarriers are allocated to different users, and this scheme allows simultaneous data transmission to or from one or more users. In OFDMA, users are allocated different subsets of subcarriers that can change from one PPDU to the next. The difference between OFDM and OFDMA is illustrated in Figure 27-4. Similar to OFDM, OFDMA employs multiple subcarriers, but the subcarriers are divided into several groups where each group is referred to as an RU. With OFDMA, different transmit powers may be applied to different RUs.

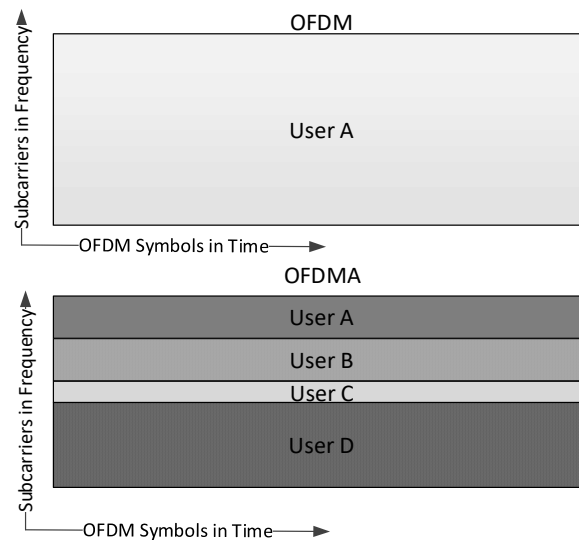


Figure 27-4—Illustration of OFDM and OFDMA concepts

27.3.2 Subcarrier and resource allocation

27.3.2.1 General

An OFDM symbol is constructed of subcarriers, the number of which is a function of the PPDU bandwidth. There are several subcarrier types:

- a) Data subcarriers, which are used for data transmission (see 27.3.2.2).
- b) Pilot subcarriers, which are used for phase information and parameter tracking (see 27.3.2.4).
- c) Unused subcarriers, which are not used for either data or pilot transmission. The unused subcarriers are the DC subcarriers (see 27.3.2.2), the Guard band subcarriers at the band edges (see 27.3.2.2), and the Null subcarriers (see 27.3.2.3).

The following notations are used to describe the indices for a set of subcarriers:

- $[x1:y1]$ represents the set of subcarriers with index k that satisfies $x1 \leq k \leq y1$.
- $[x1:y1, x2:y2]$ represents the set of subcarriers with index k that satisfies either $x1 \leq k \leq y1$ or $x2 \leq k \leq y2$.

27.3.2.2 Resource unit, guard, and DC subcarriers

The RUs defined for DL and UL transmission are as follows: 26-tone RU, 52-tone RU, 106-tone RU, 242-tone RU, 484-tone RU, 996-tone RU, and 2×996-tone RU.

The 26-tone RU, 52-tone RU, 106-tone RU, and 242-tone RU are used in the 20 MHz, 40 MHz, 80 MHz, 160 MHz, and 80+80 MHz HE MU PPDU format. The 52-tone RU, 106-tone RU and 242-tone RU are used in the 20 MHz, 40 MHz, 80 MHz, 160 MHz, and 80+80 MHz HE TB PPDU format. The 26-tone RU is used in the 20 MHz, 40 MHz, 80 MHz, 160 MHz, and 80+80 MHz HE TB PPDU format unless a STA is operating in an operating class for which the behavior limits set listed in Annex E includes the DFS_50_100_Behavior (see 26.5.2.2.1 and 26.5.2.3). The 106-tone RU is used in the HE ER SU PPDU format.

The 484-tone RU is used in the 40 MHz, 80 MHz, 160 MHz, and 80+80 MHz HE MU PPDU and HE TB PPDU formats. The 996-tone RU is used in the 80 MHz, 160 MHz, and 80+80 MHz HE MU PPDU and HE TB PPDU formats. The 2×996-tone RU is used in the 160 MHz and 80+80 MHz HE MU PPDU format and 160 MHz and 80+80 MHz HE TB PPDU format.

The 242-tone and larger RUs are used in the HE SU PPDU format. The 242-tone RU is used in the 20 MHz HE SU PPDU and HE ER SU PPDU formats. The 484-tone RU is used in the 40 MHz HE SU PPDU format. The 996-tone RU is used in the 80 MHz HE SU PPDU format. The 2×996-tone RU is used in the 160 MHz and 80+80 MHz HE SU PPDU formats.

The maximum number of RUs in the 20 MHz, 40 MHz, 80 MHz, 160 MHz, and 80+80 MHz HE PPDU formats is defined in Table 27-6.

An HE MU PPDU using OFDMA transmission can carry a mixture of 26-, 52-, 106-, 242-, 484-, and 996-tone RUs.

Table 27-6—Maximum number of RUs for each channel width

RU type	CBW20	CBW40	CBW80	CBW80+80 and CBW160
26-tone RU	9	18	37	74
52-tone RU	4	8	16	32
106-tone RU	2	4	8	16
242-tone RU	1	2	4	8
484-tone RU	N/A	1	2	4
996-tone RU	N/A	N/A	1	2
2×996 tone RU	N/A	N/A	N/A	1

A 26-tone RU consists of 24 data subcarriers and 2 pilot subcarriers. The positions of the pilots for the 26-tone RU are defined in 27.3.2.4. The locations of the 26-tone RUs are fixed as defined in Table 27-7, Table 27-8, and Table 27-9.

Table 27-7—Data and pilot subcarrier indices for RUs in a 20 MHz HE PPDU and in a non-OFDMA 20 MHz HE PPDU

RU type	RU index and subcarrier range				
26-tone RU	RU 1 [−121: −96]	RU 2 [−95: −70]	RU 3 [−68: −43]	RU 4 [−42: −17]	RU 5 [−16: −4, 4: 16]
	RU 6 [17: 42]	RU 7 [43: 68]	RU 8 [70: 95]	RU 9 [96: 121]	
52-tone RU	RU 1 [−121: −70]	RU 2 [−68: −17]	RU 3 [17: 68]	RU 4 [70: 121]	
106-tone RU	RU 1 [−122: −17]		RU 2 [17: 122]		
242-tone RU	RU 1 [−122: −2, 2:122]				
NOTE 1—The subcarrier index of 0 corresponds to the DC tone. Negative subcarrier indices correspond to subcarriers with a frequency lower than the DC tone, and positive subcarrier indices correspond to subcarriers with a frequency higher than the DC tone.					
NOTE 2—RU 5 is the middle 26-tone RU.					

Table 27-8—Data and pilot subcarrier indices for RUs in a 40 MHz HE PPDU and in a non-OFDMA 40 MHz HE PPDU

RU type	RU index and subcarrier range				
26-tone RU	RU 1 [−243: −218]	RU 2 [−217: −192]	RU 3 [−189: −164]	RU 4 [−163: −138]	RU 5 [−136: −111]
	RU 6 [−109: −84]	RU 7 [−83: −58]	RU 8 [−55: −30]	RU 9 [−29: −4]	
	RU 10 [4: 29]	RU 11 [30: 55]	RU 12 [58: 83]	RU 13 [84: 109]	RU 14 [111: 136]
	RU 15 [138: 163]	RU 16 [164: 189]	RU 17 [192: 217]	RU 18 [218: 243]	
52-tone RU	RU 1 [−243: −192]	RU 2 [−189: −138]	RU 3 [−109: −58]	RU 4 [−55: −4]	
	RU 5 [4: 55]	RU 6 [58: 109]	RU 7 [138: 189]	RU 8 [192: 243]	
106-tone RU	RU 1 [−243: −138]	RU 2 [−109: −4]	RU 3 [4: 109]	RU 4 [138: 243]	
242-tone RU	RU 1 [−244: −3]		RU 2 [3: 244]		
484-tone RU	RU 1 [−244: −3, 3: 244]				
NOTE—The subcarrier index of 0 corresponds to the DC tone. Negative subcarrier indices correspond to subcarriers with a frequency lower than the DC tone, and positive subcarrier indices correspond to subcarriers with a frequency higher than the DC tone.					

Table 27-9—Data and pilot subcarrier indices for RUs in an 80 MHz HE PPDU and in a non-OFDMA 80 MHz HE PPDU

RU type	RU index and subcarrier range				
26-tone RU	RU 1 [−499: −474]	RU 2 [−473: −448]	RU 3 [−445: −420]	RU 4 [−419: −394]	RU 5 [−392: −367]
	RU 6 [−365: −340]	RU 7 [−339: −314]	RU 8 [−311: −286]	RU 9 [−285: −260]	
	RU 10 [−257: −232]	RU 11 [−231: −206]	RU 12 [−203: −178]	RU 13 [−177: −152]	RU 14 [−150: −125]
	RU 15 [−123: −98]	RU 16 [−97: −72]	RU 17 [−69: −44]	RU 18 [−43: −18]	RU 19 [−16: −4, 4: 16]
	RU 20 [18: 43]	RU 21 [44: 69]	RU 22 [72: 97]	RU 23 [98: 123]	RU 24 [125: 150]
	RU 25 [152: 177]	RU 26 [178: 203]	RU 27 [206: 231]	RU 28 [232: 257]	
	RU 29 [260: 285]	RU 30 [286: 311]	RU 31 [314: 339]	RU 32 [340: 365]	RU 33 [367: 392]
	RU 34 [394: 419]	RU 35 [420: 445]	RU 36 [448: 473]	RU 37 [474: 499]	

Table 27-9—Data and pilot subcarrier indices for RUs in an 80 MHz HE PPDU and in a non-OFDMA 80 MHz HE PPDU (*continued*)

RU type	RU index and subcarrier range				
52-tone RU	RU 1 [−499: −448]	RU 2 [−445: −394]	RU 3 [−365: −314]	RU 4 [−311: −260]	
	RU 5 [−257: −206]	RU 6 [−203: −152]	RU 7 [−123: −72]	RU 8 [−69: −18]	
	RU 9 [18: 69]	RU 10 [72: 123]	RU 11 [152: 203]	RU 12 [206: 257]	
	RU 13 [260: 311]	RU 14 [314: 365]	RU 15 [394: 445]	RU 16 [448: 499]	
106-tone RU	RU 1 [−499: −394]	RU 2 [−365: −260]	RU 3 [−257: −152]	RU 4 [−123: −18]	
	RU 5 [18: 123]	RU 6 [152: 257]	RU 7 [260: 365]	RU 8 [394: 499]	
242-tone RU	RU 1 [−500: −259]	RU 2 [−258: −17]	RU 3 [17: 258]	RU 4 [259: 500]	
484-tone RU	RU 1 [−500: −17]		RU 2 [17: 500]		
996-tone RU	RU 1 [−500: −3, 3: 500]				
NOTE 1—The subcarrier index of 0 corresponds to the DC tone. Negative subcarrier indices correspond to subcarriers with a frequency lower than the DC tone, and positive subcarrier indices correspond to subcarriers with a frequency higher than the DC tone.					
NOTE 2—RU 19 is the center 26-tone RU.					

The data and pilot subcarrier indices for a non-OFDMA 80 MHz HE PPDU are the same as those for a 996-tone RU.

The locations of the 26-tone RUs are shown in Figure 27-5, Figure 27-6, and Figure 27-7 for the 20 MHz, 40 MHz, and 80 MHz HE MU PPDU formats or HE TB PPDU formats using OFDMA transmission, respectively. The same structure as used for the 80 MHz HE MU PPDU formats or HE TB PPDU formats using OFDMA transmission is used for both the primary 80 MHz and secondary 80 MHz channels in the 160 MHz and 80+80 MHz HE MU PPDU or HE TB PPDU formats using OFDMA transmission.

A 52-tone RU consists of 48 data subcarriers and 4 pilot subcarriers. The positions of the pilots for the 52-tone RU are defined in 27.3.2.4. The locations of the 52-tone RUs are fixed as defined in Table 27-7, Table 27-8, and Table 27-9 and illustrated in Figure 27-5, Figure 27-6, and Figure 27-7 for the 20 MHz, 40 MHz, and 80 MHz HE MU PPDU formats or HE TB PPDU formats using OFDMA transmission, respectively. The same structure as used in the 80 MHz HE MU PPDU format or HE TB PPDU formats using OFDMA transmission is used for both the Primary 80 MHz and Secondary 80 MHz channels in the 160 MHz and 80+80 MHz HE MU PPDU formats or HE TB PPDU formats using OFDMA transmission.

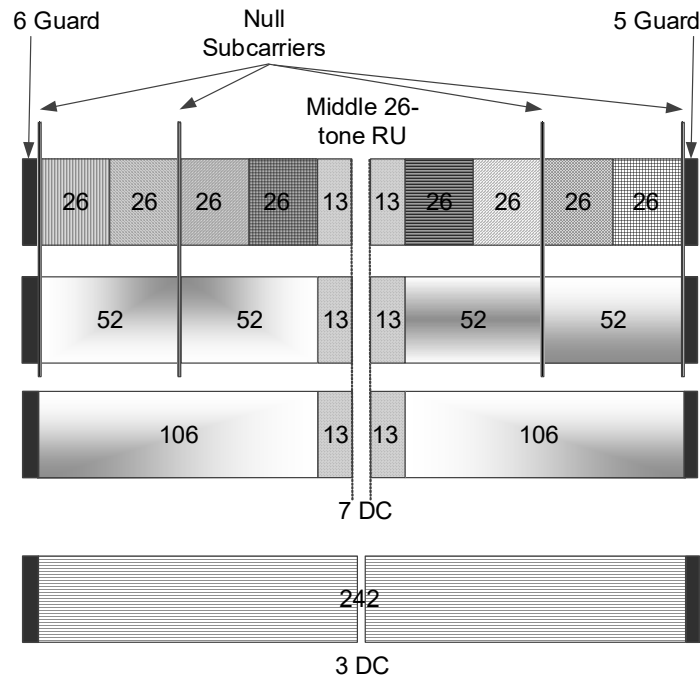


Figure 27-5—RU locations in a 20 MHz HE PDU

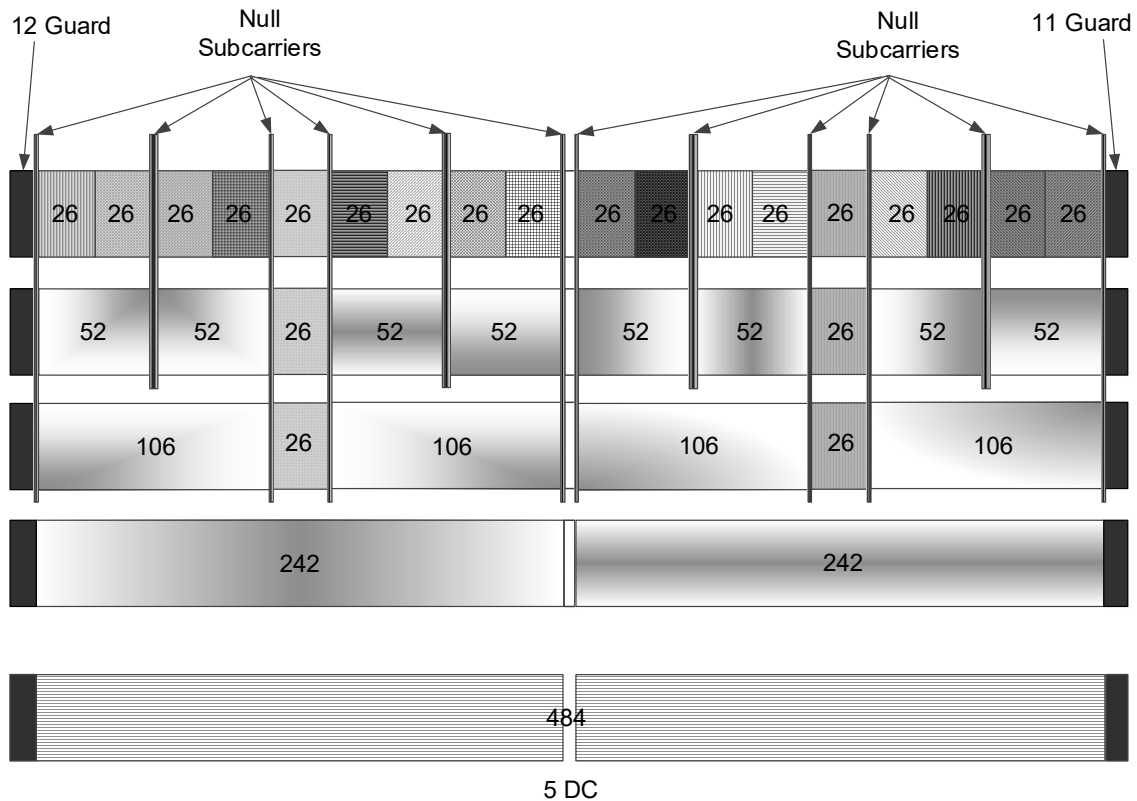


Figure 27-6—RU locations in a 40 MHz HE PDU

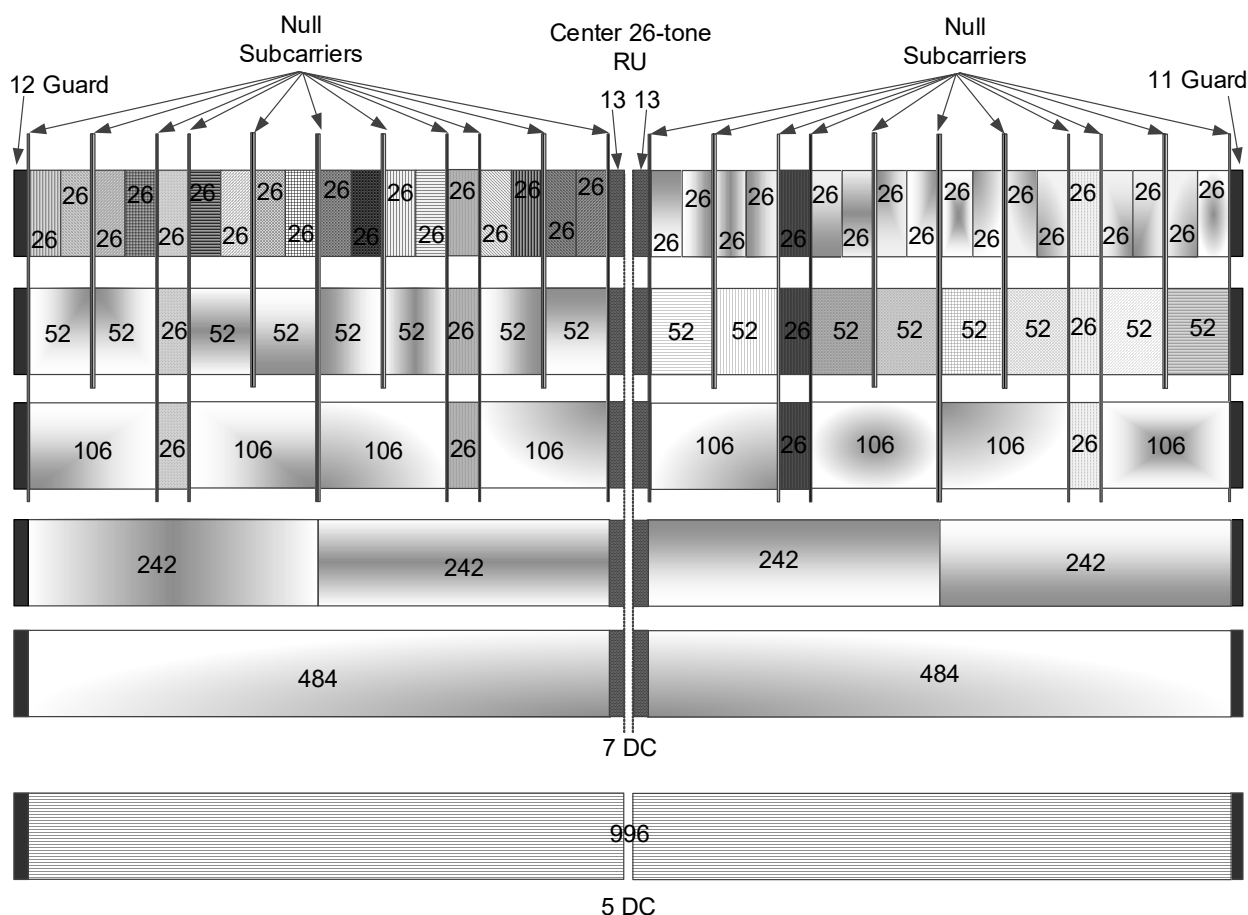


Figure 27-7—RU locations in an 80 MHz HE PPDU

A 106-tone RU consists of 102 data subcarriers and 4 pilot subcarriers. The positions of the pilots for the 106-tone RU are defined in 27.3.2.4. The locations of the 106-tone RUs are fixed as defined in Table 27-7, Table 27-8, and Table 27-9 and illustrated in Figure 27-5, Figure 27-6, and Figure 27-7 for the 20 MHz, 40 MHz, and 80 MHz HE MU PPDU formats or HE TB PPDU formats using OFDMA transmission, respectively. The same structure as used in the 80 MHz HE MU PPDU formats or HE TB PPDU formats using OFDMA transmission is used for both the Primary 80 MHz and Secondary 80 MHz channels in the 160 MHz and 80+80 MHz HE MU PPDU formats or HE TB PPDU formats using OFDMA transmission.

A 242-tone RU consists of 234 data subcarriers and 8 pilot subcarriers. The positions of pilots for the 242-tone RU are defined in 27.3.2.4. The locations of the 242-tone RUs are fixed as defined in Table 27-7, Table 27-8, and Table 27-9 and illustrated in Figure 27-5, Figure 27-6, and Figure 27-7 for the 20 MHz, 40 MHz, and 80 MHz HE PPDU formats, respectively. The same structure as used in the 80 MHz HE PPDU formats is used for both the Primary 80 MHz and Secondary 80 MHz channels in the 160 MHz and 80+80 MHz HE PPDU formats.

A 484-tone RU consists of 468 data subcarriers and 16 pilot subcarriers. The positions of the pilots for the 484-tone RU are defined in 27.3.2.4. The locations of the 484-tone RUs are fixed as defined in Table 27-8 and Table 27-9 and illustrated in Figure 27-6 and Figure 27-7 for the 40 MHz and 80 MHz HE PPDU formats. The same structure as used for the 80 MHz HE PPDU formats is used for both the Primary 80 MHz and Secondary 80 MHz channels in the 160 MHz and 80+80 MHz HE PPDU formats.

A 996-tone RU consists of 980 data subcarriers and 16 pilot subcarriers. The positions of the pilots for the 996-tone RU are defined in 27.3.2.4. The locations of the 996-tone RUs are fixed and located on subcarriers [−1012: −515, −509: −12] and [12: 509, 515: 1012] for each half of the bandwidth, respectively, for the 160 MHz and 80+80 MHz HE PPDU formats.

A 2×996-tone RU consists of two 996-tone RUs, one RU in each of the 80 MHz channels for the 160 MHz and 80+80 MHz HE PPDU formats.

The 20 MHz HE MU PPDU and HE TB PPDU with one or more RUs with less than 242 tones have 7 DC subcarriers located at [−3: 3]. The 20 MHz HE SU PPDU, HE MU PPDU, and HE TB PPDU with a 242-tone RU have 3 DC subcarriers located at [−1: 1]. The 40 MHz HE PPDU has 5 DC subcarriers located at [−2: 2]. The 80 MHz HE MU PPDU and HE TB PPDU with one or more RUs with less than 996 tones have 7 DC subcarriers located at [−3: 3]. The 80 MHz HE SU PPDU, HE MU PPDU, and HE TB PPDU with a 996-tone RU have 5 DC subcarriers located at [−2: 2]. The structure used in the 80 MHz HE PPDU is used for both the Primary 80 MHz and Secondary 80 MHz channels in the 160 MHz and 80+80 MHz HE PPDU. The DC subcarriers are located on subcarriers [−11: 11].

The 20 MHz HE PPDU has 11 guard subcarriers: the 6 lowest frequency subcarriers [−128: −123] and the 5 highest frequency subcarriers [123: 127] as shown in Figure 27-5. The 40 MHz HE PPDU has 23 guard subcarriers: the 12 lowest frequency subcarriers [−256: −245] and the 11 highest frequency subcarriers [245: 255] as shown in Figure 27-6. The 80 MHz HE PPDU has 23 guard subcarriers: the 12 lowest frequency subcarriers [−512: −501] and the 11 highest frequency subcarriers [501: 511] as shown in Figure 27-7. For the 160 MHz and 80+80 MHz HE PPDU, the same number of lowest frequency and highest frequency guard subcarriers as 80 MHz are defined at each edge of the 160 MHz and 80+80 MHz.

27.3.2.3 Null subcarriers

There are null subcarriers between the 26-, 52-, and 106-tone RU locations as illustrated in Figure 27-5, Figure 27-6, and Figure 27-7. The null subcarriers are located near the DC or edge tones to provide protection from transmit center frequency leakage, receiver DC offset, and interference from neighboring RUs. The null subcarriers have zero energy. The indices of the null subcarrier are enumerated in Table 27-10.

Table 27-10— Null subcarrier indices

Channel width	RU size	Null subcarrier indices
20 MHz	26, 52	$\pm 69, \pm 122$
	106	none
	242	none
40 MHz	26, 52	$\pm 3, \pm 56, \pm 57, \pm 110, \pm 137, \pm 190, \pm 191, \pm 244$
	106	$\pm 3, \pm 110, \pm 137, \pm 244$
	242, 484	none

Table 27-10— Null subcarrier indices (continued)

Channel width	RU size	Null subcarrier indices
80 MHz	26, 52	$\pm 17, \pm 70, \pm 71, \pm 124, \pm 151, \pm 204, \pm 205, \pm 258, \pm 259, \pm 312, \pm 313, \pm 366, \pm 393, \pm 446, \pm 447, \pm 500$
	106	$\pm 17, \pm 124, \pm 151, \pm 258, \pm 259, \pm 366, \pm 393, \pm 500$
	242, 484	none
	996	none
160 MHz	26, 52, 106	{null subcarrier indices in 80 MHz – 512, null subcarrier indices in 80 MHz + 512}
	242, 484, 996, 2×996	none

The null subcarriers locations for each 80 MHz frequency segment of an 80+80 MHz HE PPDU shall follow the locations of an 80 MHz HE PPDU.

27.3.2.4 Pilot subcarriers

Within the HE modulated fields (see 27.3.10) of an HE PPDU, pilot subcarriers are present in the Data field and may be present in the HE-LTF field (see 27.3.11.10). The pilot subcarrier indices for the Data field OFDM symbols are defined in 27.3.12.13.

One of three HE-LTF types is used in the HE-LTF field of an HE PPDU: 1x HE-LTF, 2x HE-LTF, or 4x HE-LTF (see 27.3.11.10). If pilot subcarriers are present in the HE-LTF field of an HE PPDU, then, for a 4x HE-LTF and 2x HE-LTF, the pilot subcarrier locations in the HE-LTF field are the same as the pilot subcarrier locations in the Data field. For a 1x HE-LTF, the pilot subcarrier locations in the HE-LTF field are the pilot subcarriers locations in the Data field that are multiples of 4.

27.3.2.5 Resource indication and User identification in an HE MU PPDU

An AP that transmits an HE MU PPDU shall set the UL/DL field in the HE-SIG-A field to 0.

The HE-SIG-B field consists of one or two HE-SIG-B content channels, with each HE-SIG-B content channel conveying user allocation for one or more 20 MHz subchannels. A 20 MHz HE MU PPDU has one HE-SIG-B content channel, while an HE MU PPDU with greater than 20 MHz PPDU bandwidth has two HE-SIG-B content channels.

A full-bandwidth MU-MIMO transmission using the HE MU PPDU format shall have a value of 1 for the HE-SIG-B Compression field in the HE-SIG-A field, where the Common field in the HE-SIG-B field is not present, the HE modulated fields of the PPDU consist of one RU that spans the entire PPDU bandwidth, and the preamble is not punctured. The number of users in the MU-MIMO group is indicated in the Number Of HE-SIG-B Symbols Or MU-MIMO Users field in the HE-SIG-A field. The allocated spatial streams for each user and the total number of spatial streams are indicated in the Spatial Configuration field of the User field in the HE-SIG-B field containing the STA-ID of the designated MU-MIMO STA as defined in Table 27-30.

If the value of the HE-SIG-B Compression field in the HE-SIG-A field is 0, then the RU Allocation subfield in the Common field in each HE-SIG-B content channel indicates the combination of RUs in the current PPDU and the number of User fields included in the corresponding HE-SIG-B content channel for each RU. See 27.3.11.8.2 for a description of the HE-SIG-B content channels.

If there is only one User field (see Table 27-28) for an RU in the HE-SIG-B content channel, then the number of spatial streams for the user in the RU is indicated by the NSTS field in the User field.

If there is more than one User field (see Table 27-29) for an RU in the HE-SIG-B content channel, then the number of allocated spatial streams for each user in the RU is indicated by the Spatial Configuration field of the User field in the HE-SIG-B field. Note that, if the value of the HE-SIG-B Compression field in the HE-SIG-A field is 0, for an RU with 484 or more subcarriers and having two or more intended users, the User fields corresponding to the RU may be split between two HE-SIG-B content channels. In this case, the total number of users and the total number of spatial streams in the RU are the sum of the number of users and number of spatial streams per user, respectively, indicated in both HE-SIG-B content channels. For PPDU bandwidths greater than 20 MHz, the split is an equitable split for full-bandwidth DL MU-MIMO (see 27.3.11.8.4) or a dynamic split when the HE-SIG-B Compression field in the HE-SIG-A field is equal to 0 (see 27.3.11.8.3).

In each HE-SIG-B content channel, the User fields are ordered so that, together with the HE-SIG-B Compression field in the HE-SIG-A field and the Common field in the HE-SIG-B field (if present), the RU location and spatial streams allocated to each user can be identified (see 27.3.11.8.4). If the UL/DL field in the HE-SIG-A field is set to 0, the STA-ID field in each User field indicates the intended recipient user of the corresponding spatial streams and the RU. Otherwise, the STA-ID field indicates the STA that transmits the PPDU in the corresponding spatial streams and the RU.

HE-LTF symbols in the DL HE MU PPDU are used to measure the channel for the space-time streams intended for the STA and can also be used to measure the channel for the interfering space-time streams. To successfully demodulate the space-time streams intended for the STA, it is recommended that the STA use the channel knowledge for all space-time streams to reduce the effect of space-time streams allocated to other users.

If a STA is included as a member of the MU-MIMO group in RU r , its corresponding $N_{STS,r,u}$ contained in the User field in the HE-SIG-B field for user u shall not be zero. If a STA determines that it is not a member of the MU-MIMO group in RU r , then the STA may elect not to process RU r in the remainder of the PPDU.

27.3.2.6 Resource allocation for an HE TB PPDU

UL MU transmissions are preceded by a triggering frame from the AP. The triggering frame indicates the parameters, such as the duration of the HE TB PPDU, RU allocation, target RSSI, and HE-MCS (see 9.3.1.22, 9.2.4.6a.1, and 26.5.2.3), required to transmit an HE TB PPDU.

The Trigger frame indicates whether the UL MU transmission following it uses HE single stream pilot HE-LTF mode or HE masked HE-LTF sequence mode if the HE-LTF type of the HE TB PPDU is 2x HE-LTF or 4x HE-LTF. HE no pilot HE-LTF mode is used if the HE-LTF type of the HE TB PPDU is 1x HE-LTF. If HE single stream pilot HE-LTF mode is used, no masking is applied to the HE-LTF. HE single stream pilot HE-LTF mode is used for any UL OFDMA transmission, including UL OFDMA with MU-MIMO transmissions.

27.3.2.7 20 MHz operating non-AP HE STAs

A 20 MHz-only non-AP HE STA is a non-AP HE STA that indicates support for only 20 MHz channel width for the frequency band in which it is operating using the Supported Channel Width Set subfield in the HE PHY Capabilities Information field in the HE Capabilities element that it transmits (see 9.4.2.248.3). A 20 MHz operating non-AP HE STA is a non-AP HE STA that is operating in 20 MHz channel width mode, such as a 20 MHz-only non-AP HE STA or an HE STA that reduced its operating channel width to 20 MHz using operating mode indication (OMI).

A 20 MHz operating non-AP HE STA shall operate in the primary 20 MHz channel, except when the 20 MHz operating non-AP HE STA sets `dot11HESubchannelSelectiveTransmissionImplemented` equal to true. In this case, the 20 MHz operating non-AP HE STA may operate in any 20 MHz channel within the BSS bandwidth by following the procedure in 26.8.7.

An HE AP operating in the 5 GHz or 6 GHz band shall be able to interoperate with non-AP HE STAs, regardless of the indicated value of B1 in the Supported Channel Width Set subfield in the HE PHY Capabilities Information field in the HE Capabilities element (see 9.4.2.248.3).

A 20 MHz operating non-AP HE STA shall support tone mapping of 26-tone RU, 52-tone RU, 106-tone RU, and 242-tone RU for a 20 MHz HE PPDU (see Table 27-7) in the 2.4 GHz, 5 GHz, and 6 GHz frequency bands.

A 20 MHz operating non-AP HE STA indicates support for tone mapping of 26-tone RU, 52-tone RU, and 106-tone RU for a 40 MHz HE PPDU (see Table 27-8) in the 2.4 GHz band using the 20 MHz In 40 MHz HE PPDU In 2.4 GHz Band subfield in the HE PHY Capabilities Information field in the HE Capabilities element (see 9.4.2.248.3), with the exception of RUs that are restricted from operation as specified in 27.3.2.8.

A 20 MHz operating non-AP HE STA shall support tone mapping of 26-tone RU, 52-tone RU, and 106-tone RU for a 40 MHz HE PPDU (see Table 27-8) in the 5 GHz frequency band and for an 80 MHz HE PPDU (see Table 27-9) in the 5 GHz and 6 GHz bands, with the exception of RUs that are restricted from operation as specified in 27.3.2.8.

A 20 MHz operating non-AP HE STA indicates support for tone mapping of 26-tone RU, 52-tone RU, and 106-tone RU for 80+80 MHz PPDU and 160 MHz HE PDUs using the 20 MHz In 160/80+80 MHz HE PPDU subfield in the HE PHY Capabilities Information field in the HE Capabilities element (see 9.4.2.248.3), with the exception of RUs that are restricted from operation as specified in 27.3.2.8.

A 20 MHz operating non-AP HE STA may support tone mapping of 242-tone RU for the reception of a 40 MHz HE MU PPDU (see Table 27-8) in the 2.4 GHz, 5 GHz, and 6 GHz bands and 80 MHz, 80+80 MHz, and 160 MHz HE MU PDUs (see Table 27-9) in the 5 GHz and 6 GHz bands. This support is indicated in the Supported Channel Width Set subfield in the HE PHY Capabilities Information field in the HE Capabilities element (see 9.4.2.248.3).

27.3.2.8 RU restrictions for 20 MHz operation

If a 20 MHz operating non-AP HE STA is the receiver of a 40 MHz, 80 MHz, 80+80 MHz, or 160 MHz HE MU PPDU or the transmitter of a 40 MHz, 80 MHz, 80+80 MHz, or 160 MHz HE TB PPDU, then the RU tone mapping in 20 MHz is not aligned with the 40 MHz, 80 MHz, 80+80 MHz, or 160 MHz RU tone mapping (see 27.3.2.2).

An AP shall not assign the following RUs to a 20 MHz operating non-AP HE STA where the RU index is defined in Table 27-8:

- 26-tone RU 5 and 14 of a 40 MHz HE MU PPDU and HE TB PPDU

An AP shall not assign the following RUs to a 20 MHz operating non-AP HE STA where the RU index is defined in Table 27-9:

- 26-tone RU 5, 10, 14, 19, 24, 28, and 33 of an 80 MHz HE MU PPDU and HE TB PPDU
- 26-tone RU 5, 10, 14, 19, 24, 28, and 33 of the lower 80 MHz of an 80+80 MHz and 160 MHz HE MU PPDU and HE TB PPDU
- 26-tone RU 5, 10, 14, 19, 24, 28, and 33 of the upper 80 MHz of an 80+80 MHz HE MU PPDU and HE TB PPDU

- 52-tone RU 5 and 12 of an 80 MHz HE MU PPDU and HE TB PPDU
- 52-tone RU 5 and 12 of the lower 80 MHz of an 80+80 MHz and 160 MHz HE MU PPDU and HE TB PPDU
- 52-tone RU 5 and 12 of the upper 80 MHz of an 80+80 MHz HE MU PPDU and HE TB PPDU
- 106-tone RU 3 and 6 of an 80 MHz HE MU PPDU and HE TB PPDU
- 106-tone RU 3 and 6 of the lower 80 MHz of an 80+80 MHz and 160 MHz HE MU PPDU and HE TB PPDU
- 106-tone RU 3 and 6 of the upper 80 MHz of an 80+80 MHz HE MU PPDU and HE TB PPDU

27.3.2.9 80 MHz operating non-AP HE STAs

An HE AP shall not allocate an RU in a 160 MHz or 80+80 MHz HE MU or HE TB PPDU to an 80 MHz operating non-AP HE STA if the non-AP STA has set the 80 MHz In 160/80+80 MHz HE PPDU subfield in the HE PHY Capabilities Information field in the HE Capabilities element to 0.

An HE AP shall not allocate an RU outside of the primary 80 MHz in a 160 MHz or 80+80 MHz HE MU or HE TB PPDU to an 80 MHz operating non-AP HE STA if the non-AP STA has set the HE Subchannel Selective Transmission Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element to 0.

An HE AP shall not allocate an RU outside of the primary 80 MHz in a 160 MHz or 80+80 MHz HE MU or HE TB PPDU to an 80 MHz operating non-AP HE STA if the non-AP STA has set the HE Subchannel Selective Transmission Support subfield in the HE MAC Capabilities Information field in the HE Capabilities element to 1 but has not set up SST operation on the secondary 80 MHz channel with the HE AP.

27.3.3 MU-MIMO

27.3.3.1 DL MU-MIMO

27.3.3.1.1 Supported RU sizes in DL MU-MIMO

A STA that sets the Partial Bandwidth DL MU-MIMO subfield of the HE PHY Capabilities Information field in the HE Capabilities element that it transmits to 1 shall support receiving an RU in an HE MU PPDU where MU-MIMO is employed in the RU, the RU size is greater than or equal to 106 tones, and the RU does not span the entire PPDU bandwidth.

27.3.3.1.2 Maximum number of spatial streams in an HE MU PPDU

An HE STA shall support the reception of DL MU-MIMO transmissions on full bandwidth with a maximum number of space-time streams (per user) that is the minimum of 4 and the maximum number of space-time streams supported for reception of HE SU PPDU. The maximum number of space-time streams supported for reception of HE SU PPDU is indicated for various bandwidths in the Supported HE-MCS and NSS Set field in the HE Capabilities element.

For transmissions using bandwidth less than or equal to 80 MHz, an HE STA shall support reception of DL MU-MIMO transmissions on full bandwidth with the total number of space-time streams (across all users) up to the value indicated by the Beamformee STS $\leq$ 80 MHz subfield in the HE PHY Capabilities Information field in the HE Capabilities element. For transmissions using bandwidth greater than 80 MHz, an HE STA shall support reception of DL MU-MIMO transmissions on full bandwidth with the total number of space-time streams (across all users) up to the value indicated by the Beamformee STS $>$ 80 MHz subfield in the HE PHY Capabilities Information field in the HE Capabilities element.

An HE AP that is capable of transmitting 4 or more space-time streams shall support DL MU-MIMO transmissions on full bandwidth.

All of the aforementioned requirements in this subclause on the per user and total number of space-time-streams are also applicable to an MU-MIMO transmission on an RU in an HE MU PPDU where the RU does not span the entire PPDU bandwidth.

27.3.3.2 UL MU-MIMO

27.3.3.2.1 Introduction

UL MU-MIMO is a technique to allow multiple STAs to transmit simultaneously over the same frequency resource to the receiver. The concept is very similar to SU-MIMO where multiple space-time streams are transmitted simultaneously over the same frequency resource utilizing spatial multiplexing through multiple antennas at the transmitter and receiver. The key difference from SU-MIMO is that in UL MU-MIMO, the transmitted streams originate from multiple STAs.

27.3.3.2.2 Supported RU sizes in UL MU-MIMO

An AP that sets the Partial Bandwidth UL MU-MIMO subfield of the HE PHY Capabilities Information field in the HE Capabilities element that it transmits to 1 shall support receiving an RU in an HE TB PPDU where MU-MIMO is employed in the RU, the RU size is greater than or equal to 106 tones, and the RU does not span the entire PPDU bandwidth.

A non-AP STA that sets the Partial Bandwidth UL MU-MIMO subfield of the HE PHY Capabilities Information field in the HE Capabilities element that it transmits to 1 shall support transmitting an RU in an HE TB PPDU where UL MU-MIMO is employed in the RU, the RU size is greater than or equal to 106 tones, and the RU does not span the entire PPDU bandwidth.

A STA that sets the Partial Bandwidth UL MU-MIMO subfield to 1 shall set the Full Bandwidth UL MU-MIMO subfield in the HE PHY Capabilities Information field to 1.

27.3.3.2.3 MU-MIMO HE-LTF mode

A non-AP STA that sets the Full Bandwidth UL MU-MIMO subfield of the HE PHY Capabilities Information field in the HE Capabilities element it transmits to 1 shall support both the HE single stream pilot HE-LTF mode and the HE masked HE-LTF sequence mode as defined in Equation (27-59) for the transmission of an HE TB PPDU with one RU spanning the entire PPDU bandwidth and the RU using MU-MIMO.

An AP that sets the Partial Bandwidth UL MU-MIMO subfield of the HE PHY Capabilities Information field in the HE Capabilities element it transmits to 1 shall support receiving HE TB PPDU using the HE single stream pilot HE-LTF mode if the HE TB PPDU uses MU-MIMO in an RU that does not span the entire PPDU bandwidth.

A non-AP STA that sets the Partial Bandwidth UL MU-MIMO subfield of the HE PHY Capabilities Information field in the HE Capabilities element it transmits to 1 shall support transmitting an HE TB PPDU using the HE single stream pilot HE-LTF mode if the HE TB PPDU uses MU-MIMO in an RU that does not span the entire PPDU bandwidth.

27.3.3.2.4 Maximum number of spatial streams in UL MU-MIMO

A non-AP STA that supports UL MU-MIMO shall support transmitting an HE TB PPDU using MU-MIMO where

- The number of spatial streams allocated to the non-AP STA ranges from 1 to N , where N is the smaller of 4 and the maximum number of spatial streams supported by the non-AP STA for transmitting HE SU PPDU.
- The number of total spatial streams (summed over all users) is less than or equal to 8.

The maximum number of spatial streams supported by a STA for the transmission of an HE SU PPDU is indicated in the Supported HE-MCS And NSS Set field in the HE Capabilities element.

All the requirements in this subclause on the per user and total number of spatial streams are applicable to both full-bandwidth and partial-bandwidth MU-MIMO.

27.3.4 HE PPDU formats

Four HE PPDU formats are defined: HE SU PPDU, HE MU PPDU, HE ER SU PPDU, and HE TB PPDU. The HE sounding NDP is a variant of the HE SU PPDU and defined in 27.3.17. The HE TB feedback NDP is a variant of the HE TB PPDU and defined in 27.3.18.

NOTE—The HE ER SU PPDU is not a variant of the HE SU PPDU. Requirements related to HE SU PPDU and HE ER SU PPDU are specified separately.

The format of the HE SU PPDU is defined as in Figure 27-8. This PPDU format is used for SU transmission, and in this format, the HE-SIG-A field is not repeated.

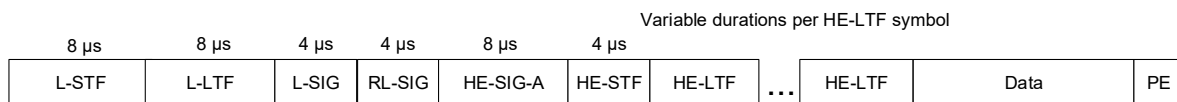


Figure 27-8—HE SU PPDU format

The format of the HE MU PPDU is defined as in Figure 27-9. This format is used for transmission to one or more users if the PPDU is not a response of a Trigger frame. In the HE MU PPDU, the HE-SIG-A field is not repeated. The HE-SIG-B field is present in this format.

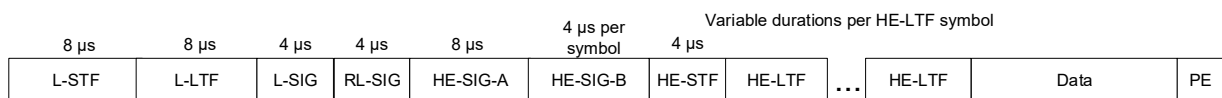


Figure 27-9—HE MU PPDU format

The format of the HE ER SU PPDU is defined as in Figure 27-10. This format is used for SU transmission, and in this format, the HE-SIG-A field is twice as long as the HE-SIG-A field in other HE PPDU formats.

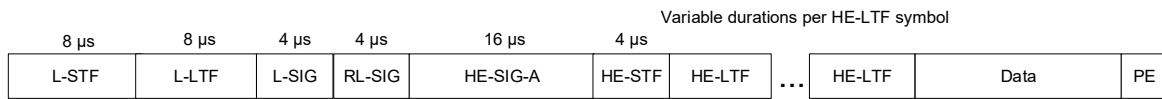


Figure 27-10—HE ER SU PPDU format

The format of the HE TB PPDU is defined as in Figure 27-11. This format is used for a transmission that is a response to a triggering frame from an AP.

The format of the HE TB PPDU is the same as the HE SU PPDU, except that the duration of the HE-STF field in the HE TB PPDU is twice the duration of the HE-STF field in the HE SU PPDU.

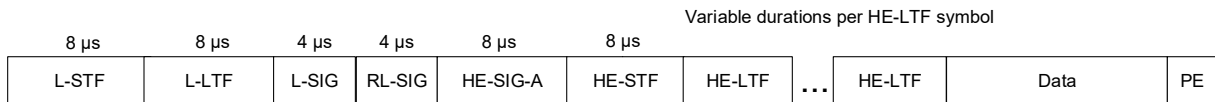


Figure 27-11—HE TB PPDU format

The fields of the HE PPDU formats are summarized in Table 27-11.

Table 27-11—HE PPDU fields

Field	Description
L-STF	Non-HT Short Training field
L-LTF	Non-HT Long Training field
L-SIG	Non-HT SIGNAL field
RL-SIG	Repeated Non-HT SIGNAL field
HE-SIG-A	HE SIGNAL A field
HE-SIG-B	HE SIGNAL B field
HE-STF	HE Short Training field
HE-LTF	HE Long Training field
Data	The Data field carrying the PSDU(s)
PE	Packet extension field

The RL-SIG, HE-SIG-A, HE-STF, HE-LTF, and PE fields are present in all HE PPDU formats. The HE-SIG-B field is present only in the HE MU PPDU. The PE field is defined in 27.3.13.

The L-STF, L-LTF, L-SIG, RL-SIG, HE-SIG-A, and HE-SIG-B fields are referred to as *pre-HE modulated fields*, while the HE-STF, HE-LTF, Data, and PE fields are referred to as *HE modulated fields*.

In the HE TB PPDU, the pre-HE modulated fields are sent only on the 20 MHz channels where the STA's HE modulated fields are located. If the HE modulated fields are located in more than one 20 MHz channel, the pre-HE modulated fields are duplicated over the multiple 20 MHz channels. Figure 27-12 shows how many 20 MHz channels need to be modulated for the pre-HE modulated fields for each RU size and location in an HE TB PPDU.

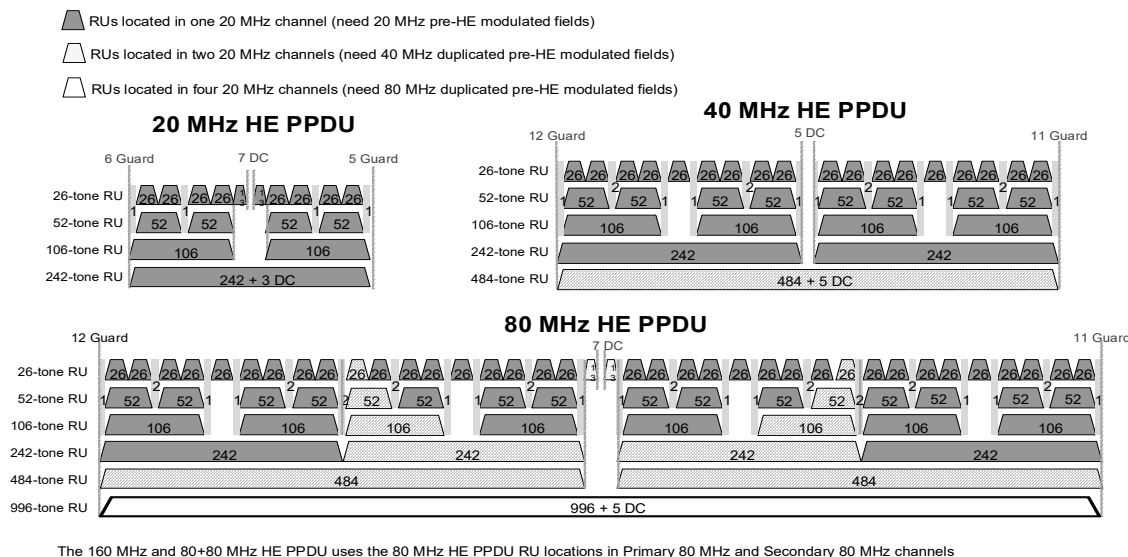


Figure 27-12—Number of 20 MHz channels that need to be modulated for the pre-HE modulated fields for each RU size and location in an HE TB PPDU

A PPDU transmitted with the TXVECTOR parameter NO_SIG_EXTN set to false is followed by a period of duration aSignalExtension without transmission. See 10.3.8.

A signal extension shall be present in a transmitted PPDU if the TXVECTOR parameter NO_SIG_EXTN is false and one of the following conditions apply:

- The TXVECTOR parameter FORMAT is HE, HT_MF or HT_GF
- The TXVECTOR parameter FORMAT is NON_HT and the TXVECTOR parameter NON_HT_MODULATION is ERP-OFDM or NON_HT_DUP_OFDM

A signal extension shall be assumed to be present (for the purpose of timing PHY-RXEND.indication and PHY-CCA.indication primitives, as described below in this subclause and in 27.3.22) in a received PPDU if one of the following conditions apply:

- The RXVECTOR parameter FORMAT is HE, HT_MF or HT_GF
- The RXVECTOR parameter FORMAT is NON_HT and the RXVECTOR parameter NON_HT_MODULATION is ERP-OFDM or NON_HT_DUP_OFDM

A PPDU containing a signal extension is called a signal extended PPDU. When transmitting a signal extended PPDU, the PHY-TXEND.indication primitive shall be emitted a period of aSignalExtension after the end of the actual ending time of the PPDU. When receiving a signal extended PPDU, the PHY-RXEND.indication primitive shall be emitted a period of aSignalExtension after the end of the actual ending time of the PPDU.

27.3.5 Transmitter block diagram

The generation of each field in an HE PPDU uses many of the following blocks:

- a) pre-FEC PHY padding
- b) Scrambler
- c) FEC (BCC or LDPC) encoders
- d) post-FEC PHY padding
- e) Stream parser
- f) Segment parser (for contiguous 160 MHz and noncontiguous 80+80 MHz transmissions)
- g) BCC interleaver
- h) Constellation mapper
- i) DCM tone mapper
- j) Pilot insertion
- k) Replication over multiple 20 MHz (for BW > 20 MHz)
- l) Multiplication by 1st column of P_{HE-LTF}
- m) LDPC tone mapper
- n) Segment deparser
- o) Space time block code (STBC) encoder for one spatial stream
- p) Cyclic shift diversity (CSD) per STS insertion
- q) Spatial mapper
- r) Frequency mapping
- s) Inverse discrete Fourier transform (IDFT)
- t) Cyclic shift diversity (CSD) per chain insertion
- u) Guard interval (GI) insertion
- v) Windowing

Example transmitter block diagrams are shown in Figure 27-13 to Figure 27-22. The actual structure of the transmitter is implementation dependent.

In particular, Figure 27-13 shows the transmit process for the L-SIG, RL-SIG, and HE-SIG-A fields of an HE PPDU using one frequency segment when the Beam Change subfield in the HE-SIG-A field is 1. These transmit blocks are also used to generate the L-STF and L-LTF fields of the HE PPDU when the Beam Change subfield in the HE-SIG-A field is 1, with the following exceptions:

- The BCC encoder and interleaver and the constellation mapper are not used when generating the L-STF and L-LTF fields.
- The BCC interleaver is not applied in the 2nd and the 4th OFDM symbols of the HE-SIG-A field in the HE ER SU PPDU.

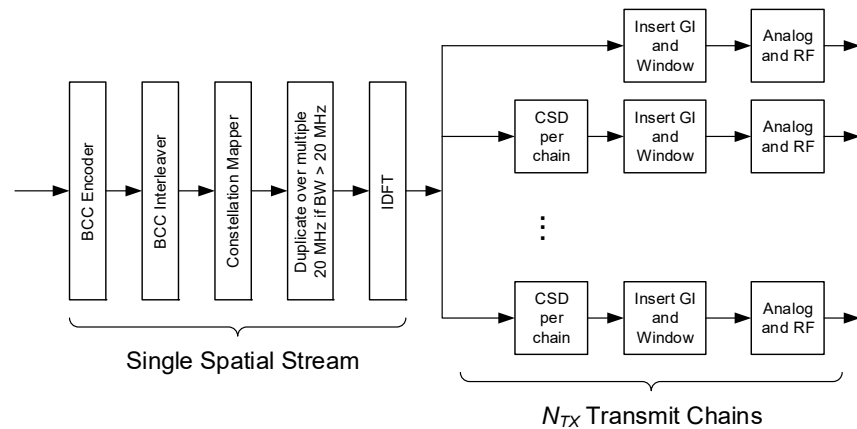


Figure 27-13—Transmitter block diagram for L-SIG, RL-SIG, and HE-SIG-A fields for an HE SU PPDU and HE ER SU PPDU if Beam Change subfield is 1

Figure 27-14 shows the transmit process for the L-SIG, RL-SIG, and HE-SIG-A fields of an HE PPDU using one frequency segment when the Beam Change subfield in the HE-SIG-A field is 0. These transmit blocks are also used to generate the L-STF and L-LTF fields of the HE PPDU if the Beam Change subfield in the HE-SIG-A field is 0, with the following exceptions:

- The BCC encoder and interleaver and the constellation mapper are not used when generating the L-STF and L-LTF fields.
- The BCC interleaver is not applied in the 2nd and the 4th OFDM symbols of the HE-SIG-A field in the HE ER SU PPDU.

NOTE—For an HE MU PPDU, the duplication on 20 MHz channels is subject to the availability of 20 MHz channels in the case of preamble puncturing.

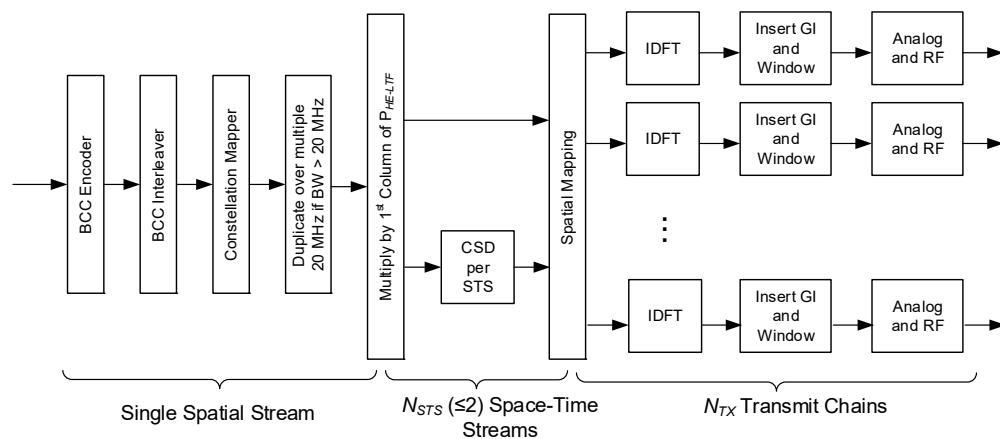


Figure 27-14—Transmitter block diagram for L-SIG, RL-SIG, and HE-SIG-A fields for an HE SU PPDU and HE ER SU PPDU if Beam Change subfield is 0

Figure 27-15 shows the transmit process for the L-SIG, RL-SIG, and HE-SIG-A fields of an HE TB PPDU using one frequency segment. In the HE TB PPDU, the transmission of the pre-HE modulated fields refers to the description in 27.3.4 and is also shown in Figure 27-15. The BCC encoder and interleaver are not used when generating the L-STF and L-LTF fields.

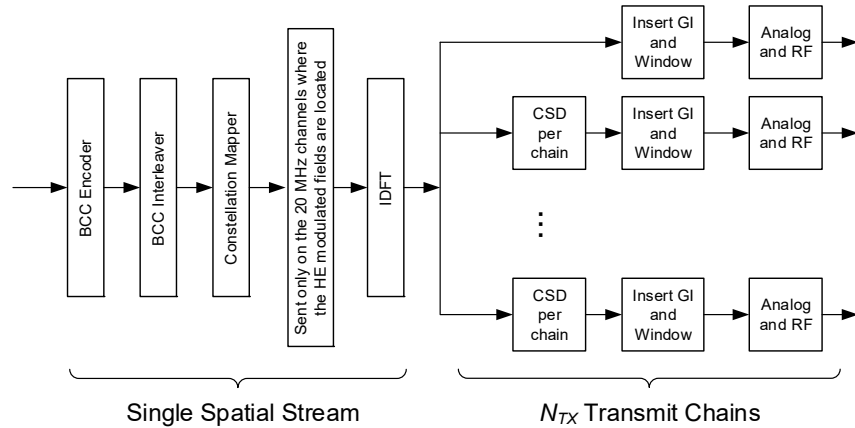


Figure 27-15—Transmitter block diagram for L-SIG, RL-SIG, and HE-SIG-A fields of an HE TB PPDU

Figure 27-16 shows the transmit process for the HE-SIG-B field of an HE MU PPDU using one frequency segment. This block diagram is for transmitting an HE-SIG-B field in one 20 MHz subchannel. Refer to 27.3.11.8.2 for the methods of transmitting an HE-SIG-B field in 40 MHz, 80 MHz, and 160 MHz. The DCM tone mapper, which is defined in 27.3.12.9, is applied only if the HE-SIG-B DCM field in the HE-SIG-A field is 1.

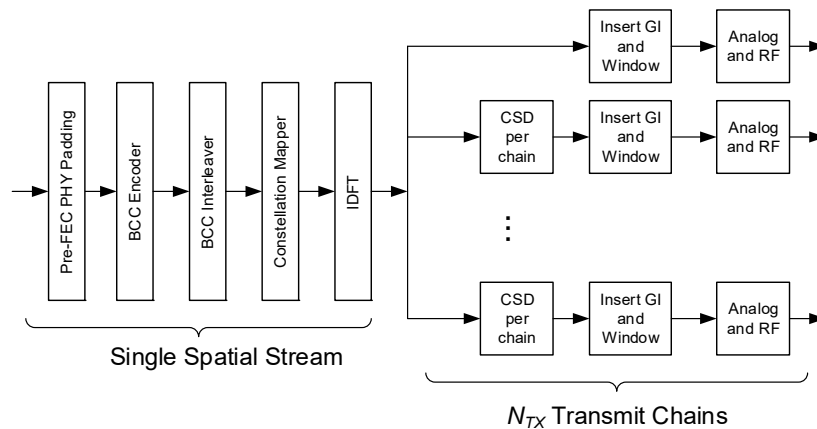


Figure 27-16—Transmitter block diagram for HE-SIG-B field

Figure 27-17 shows the transmitter blocks for the UL transmission or DL non-MU-MIMO transmission of a Data field with BCC encoding on a 26-, 52-, 106-, or 242-tone RU for a single frequency segment if the number of spatial streams is less than or equal to 4. Figure 27-17 applies to the Data field of an HE MU PPDU that is transmitted on an RU allocated to a single user, the Data field of an HE SU PPDU, and the Data field of an HE TB PPDU (regardless of whether it is spatially multiplexed with other users).

The DCM tone mapper, which is part of the constellation mapper, is applied only if DCM is indicated for the RU. A subset of these transmitter blocks consisting of the constellation mapper and CSD blocks and the blocks to the right of, and including, the spatial mapping block is also used to generate the HE-LTF fields. This is illustrated in Figure 27-32 (in 27.3.11.10). A subset of these transmitter blocks consisting of the constellation mapper and CSD blocks and the blocks to the right of, and including, the spatial and frequency mapping block of Figure 27-17 is also used to generate the HE-STF field. This figure also applies to the Data field with BCC encoding in an HE TB PPDU.

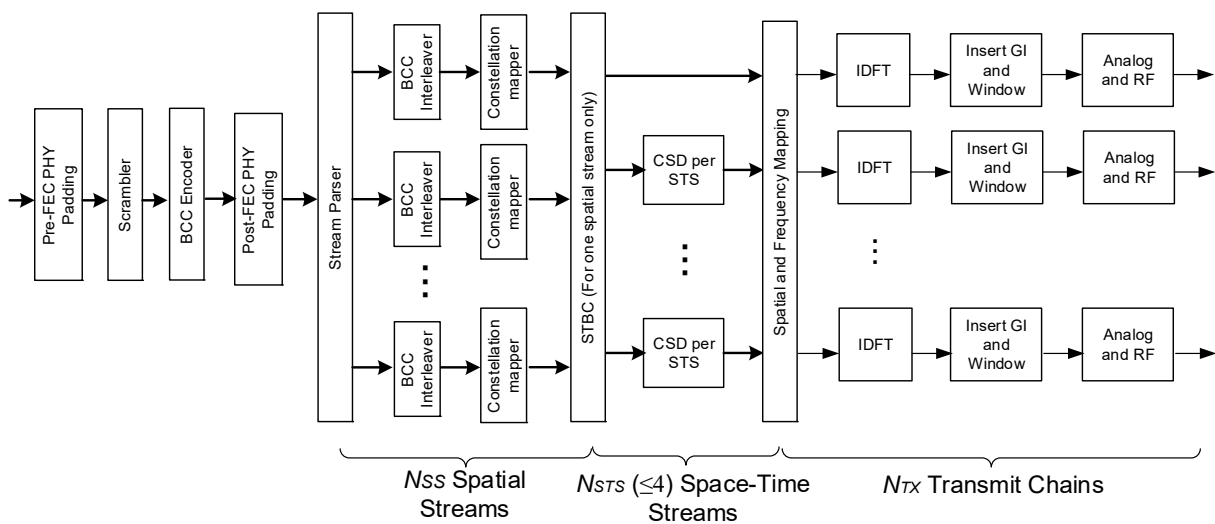


Figure 27-17—Transmitter block diagram for UL transmission or DL non-MU-MIMO transmission of a Data field with BCC encoding on a 26-, 52-, 106-, or 242-tone RU

Figure 27-18 shows the transmitter blocks for the UL transmission or DL non-MU-MIMO transmission of a Data field with LDPC encoding on a 26-, 52-, 106-, 242-, 484-, or 996-tone RU for a single frequency segment. Figure 27-18 applies to the Data field of an HE MU PPDU that is transmitted on an RU allocated to a single user, the Data field of an HE SU PPDU, and the Data field of an HE TB PPDU (regardless of whether it is spatially multiplexed with other users). This figure also applies to the Data field with LDPC encoding in an HE TB PPDU.

The DCM tone mapper, which is part of the constellation mapper, is applied only if DCM is indicated for the RU. This figure also applies to the Data field with LDPC encoding in an HE TB PPDU.

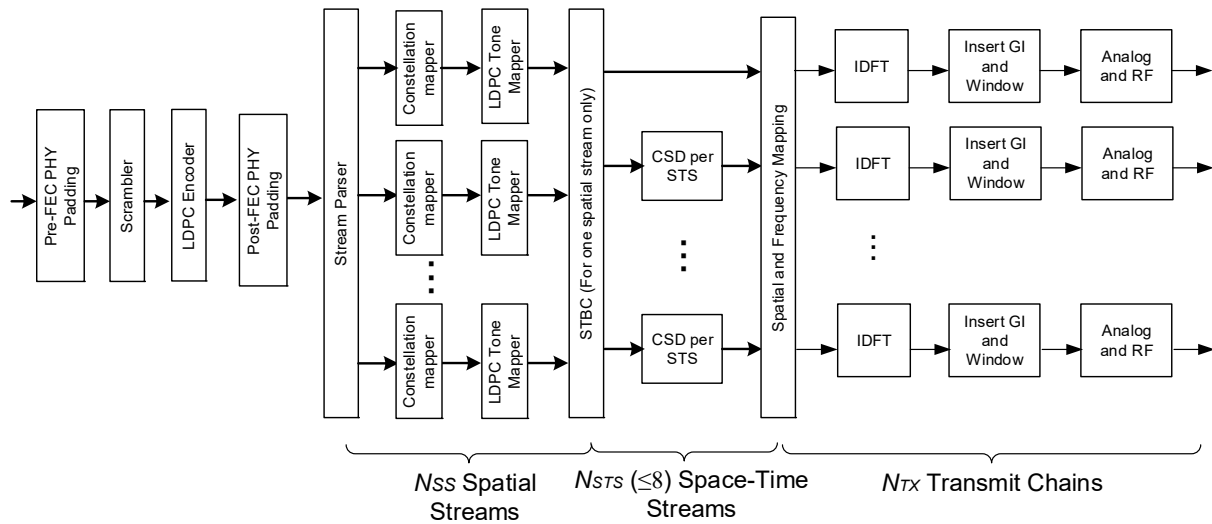


Figure 27-18—Transmitter block diagram for UL transmission or DL non-MU-MIMO transmission of a Data field with LDPC encoding on a 26-, 52-, 106-, 242-, 484-, or 996-tone RU

Figure 27-19 shows the transmitter blocks for the transmission, in an HE MU PPDU, of the Data field with BCC encoding on a 106- or 242-tone RU allocated to more than one user.

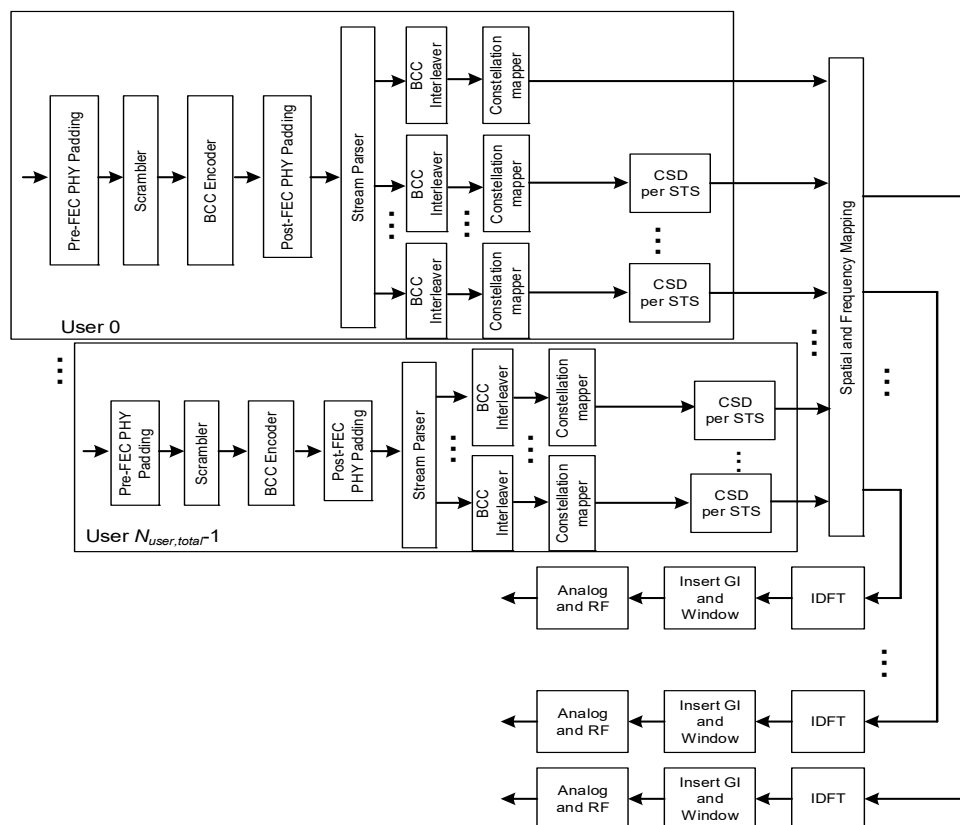


Figure 27-19—Transmitter block diagram for DL MU-MIMO transmission of a Data field with BCC encoding on a 106- or 242-tone RU

Figure 27-20 shows the transmitter blocks for the transmission, in an HE MU PPDU, of the Data field with LDPC encoding on a 106-, 242-, 484-, or 996-tone RU allocated to more than one user.

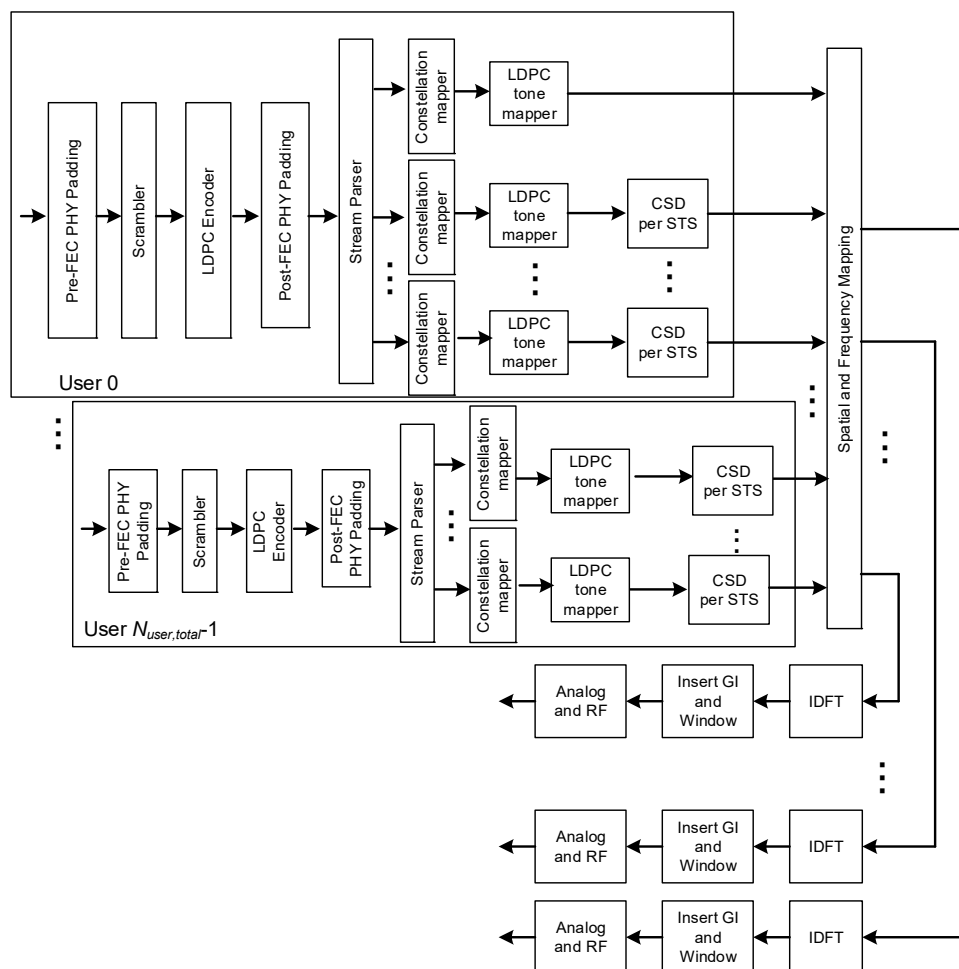


Figure 27-20—Transmitter block diagram for DL MU-MIMO transmission of a Data field with LDPC encoding on a 106-, 242-, 484-, or 996-tone RU

Figure 27-21 shows the transmitter blocks used to generate the Data field of a single-user HE transmission in 160 MHz with LDPC encoding.

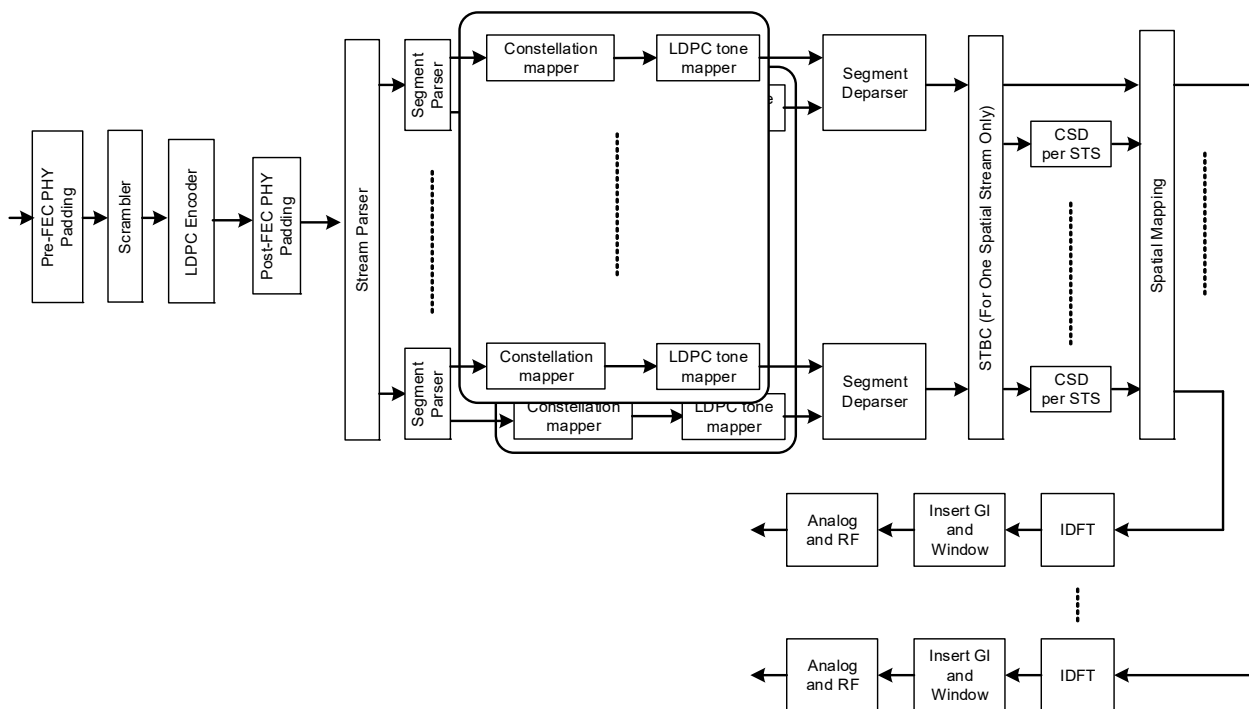


Figure 27-21—Transmitter block diagram for the Data field of an HE SU PPDU in 160 MHz with LDPC encoding

Figure 27-22 shows the transmitter blocks used to generate the Data field of a single-user HE transmission in 80+80 MHz with LDPC encoding.

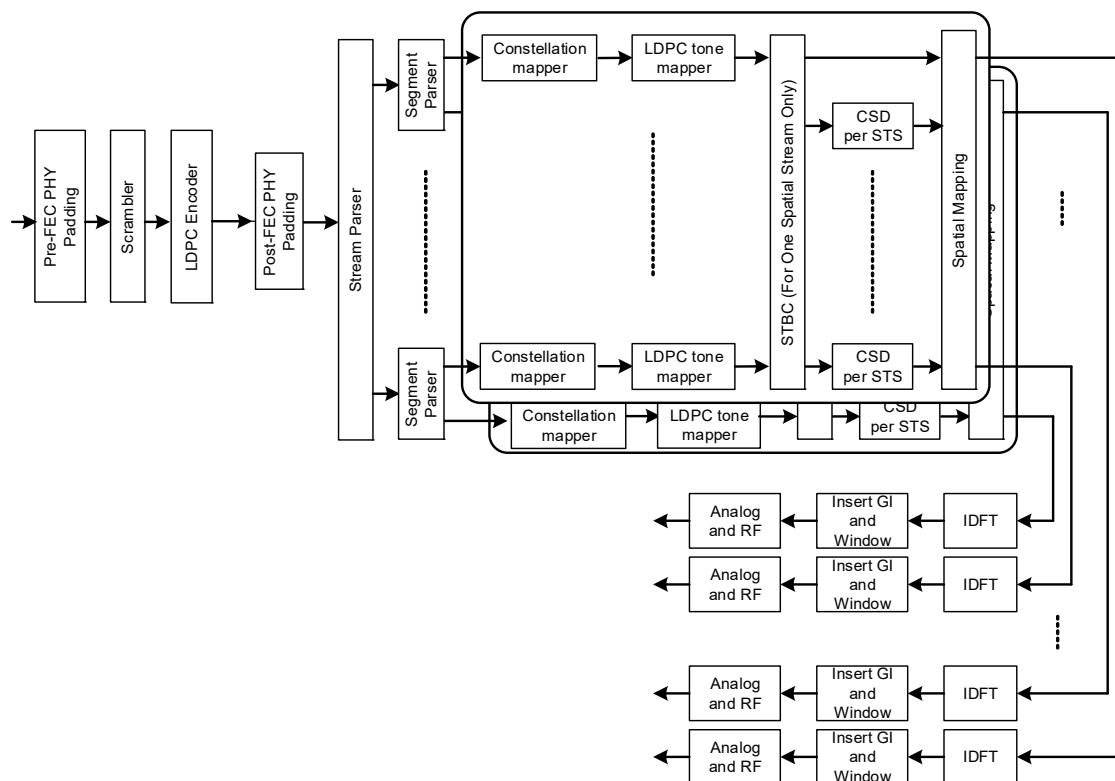


Figure 27-22—Transmitter block diagram for the Data field of an HE SU PPDU in 80+80 MHz with LDPC encoding

27.3.6 Overview of the PPDU encoding process

27.3.6.1 General

Subclause 27.3.6 provides an overview of the HE PPDU encoding process.

27.3.6.2 Construction of L-STF field

Construct the L-STF field as defined in 27.3.11.3 with the following highlights:

- Determine the channel bandwidth from the TXVECTOR parameter CH_BANDWIDTH.
- Sequence generation: Generate the L-STF sequence over the channel bandwidth as described in 27.3.11.3. Apply a 3 dB power boost if transmitting an HE ER SU PPDU as described in 27.3.11.3.
- Phase rotation: Apply appropriate phase rotation for each 20 MHz subchannel as described in 27.3.10 and 21.3.7.5.
- CSD per STS: If the TXVECTOR parameter BEAM_CHANGE is 0, apply CSD per STS for each space-time stream and frequency segment as described in 27.3.11.2.2.

- e) Spatial mapping: If the TXVECTOR parameter BEAM_CHANGE is 0, apply the A matrix and the Q matrix as described in 27.3.11.3.
- f) IDFT: Compute the inverse discrete Fourier transform.
- g) CSD per chain: If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, apply CSD per chain for each transmit chain and frequency segment as described in 27.3.11.2.1.
- h) GI and windowing: Prepend a GI ($T_{GI,Pre-HE}$), and apply windowing as described in 27.3.10.
- i) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.3 Construction of L-LTF field

Construct the L-LTF field as defined in 27.3.11.4 with the following highlights:

- a) Determine the channel bandwidth from the TXVECTOR parameter CH_BANDWIDTH.
- b) Sequence generation: Generate the L-LTF sequence over the channel bandwidth as described in 27.3.11.4. Apply a 3 dB power boost if transmitting an HE ER SU PPDU as described in 27.3.11.4.
- c) Phase rotation: Apply appropriate phase rotation for each 20 MHz subchannel as described in 27.3.10 and 21.3.7.5.
- d) CSD per STS: If the TXVECTOR parameter BEAM_CHANGE is 0, apply CSD per STS for each space-time stream and frequency segment as described in 27.3.11.2.2 before spatial mapping.
- e) Spatial mapping: If the TXVECTOR parameter BEAM_CHANGE is 0, apply the A matrix and the Q matrix as described in 27.3.11.4.
- f) IDFT: Compute the inverse discrete Fourier transform.
- g) CSD per chain: If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, apply CSD per chain for each transmit chain and frequency segment as described in 27.3.11.2.1.
- h) GI and windowing: Prepend a GI ($T_{GI,L-LTF}$), and apply windowing as described in 27.3.10.
- i) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the carrier frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.4 Construction of L-SIG field

Construct the L-SIG field as the SIGNAL field defined in 27.3.11.5 with the following highlights:

- a) Set the RATE subfield in the SIGNAL field to 6 Mb/s. Set the LENGTH, Parity, and Tail fields in the SIGNAL field as described in 27.3.11.5.
- b) BCC encoder: Encode the SIGNAL field by a convolutional encoder at the rate $R = 1/2$ as described in 27.3.12.5.1.
- c) BCC interleaver: Interleave as described in 17.3.5.7.
- d) Constellation mapper: BPSK modulate as described in 27.3.12.9.
- e) Pilot insertion: Insert pilots as described in 27.3.11.5.
- f) Extra subcarrier insertion: Four extra subcarriers are inserted at $k \in \{-28, -27, 27, 28\}$ for channel estimation purpose, and the values on these four extra subcarriers are $\{-1, -1, -1, 1\}$, respectively. Apply a 3 dB power boost to the four extra subcarriers if transmitting an HE ER SU PPDU as described in 27.3.11.5.
- g) Duplication and phase rotation: Duplicate the L-SIG field over each occupied 20 MHz subchannel of the channel bandwidth. Apply appropriate phase rotation for each occupied 20 MHz subchannel as described in 27.3.10 and 21.3.7.5.

- h) CSD per STS: If the TXVECTOR parameter BEAM_CHANGE is 0, apply CSD per STS for each space-time stream and frequency segment as described in 27.3.11.2.2 before spatial mapping.
- i) Spatial mapping: If the TXVECTOR parameter BEAM_CHANGE is 0, apply the A matrix and Q matrix as described in 27.3.11.5.
- j) IDFT: Compute the inverse discrete Fourier transform.
- k) CSD per chain: If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, apply CSD per chain for each transmit chain and frequency segment as described in 27.3.11.2.1.
- l) GI and windowing: Prepend a GI ($T_{GI,Pre-HE}$), and apply windowing as described in 27.3.10.
- m) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.5 Construction of RL-SIG field

Construct the RL-SIG field as the repeat SIGNAL field defined in 27.3.11.6 with the following highlights:

- a) Set the RATE subfield in the repeat SIGNAL field to 6 Mb/s. Set the LENGTH, Parity, and Tail fields in the repeat SIGNAL field as described in 27.3.11.6.
- b) BCC encoder: Encode the repeat SIGNAL field by a convolutional encoder at the rate $R = 1/2$ as described in 27.3.12.5.1.
- c) BCC interleaver: Interleave as described in 17.3.5.7.
- d) Constellation mapper: BPSK modulate as described in 27.3.12.9.
- e) Pilot insertion: Insert pilots as described in 27.3.11.6.
- f) Extra subcarrier insertion: Four extra subcarriers are inserted at $k \in \{-28, -27, 27, 28\}$ for channel estimation purpose, and the values on these four extra subcarriers are $\{-1, -1, -1, 1\}$, respectively. Apply a 3 dB power boost to the four extra subcarriers if transmitting an HE ER SU PPDU as described in 27.3.11.6.
- g) Duplication and phase rotation: Duplicate the RL-SIG field over each occupied 20 MHz subchannel of the channel bandwidth. Apply appropriate phase rotation for each occupied 20 MHz subchannel as described in 27.3.10 and 21.3.7.5.
- h) CSD per STS: If the TXVECTOR parameter BEAM_CHANGE is 0, apply CSD per STS for each space-time stream and frequency segment as described in 27.3.11.2.2 before spatial mapping.
- i) Spatial mapping: If the TXVECTOR parameter BEAM_CHANGE is 0, apply the A matrix and the Q matrix as described in 27.3.11.6.
- j) IDFT: Compute the inverse discrete Fourier transform.
- k) CSD per chain: If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, apply CSD per chain for each transmit chain and frequency segment as described in 27.3.11.2.1.
- l) GI and windowing: Prepend a GI ($T_{GI,Pre-HE}$), and apply windowing as described in 27.3.10.
- m) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.6 Construction of HE-SIG-A field

- a) For an HE SU PPDU, HE MU PPDU, and HE TB PPDU, the HE-SIG-A field consists of two subfields, HE-SIG-A1 and HE-SIG-A2 as defined in 27.3.11.7, and is constructed as follows:
 - 1) Obtain the HE-SIG-A field values from the TXVECTOR. Add the reserved bits, append the calculated CRC, and then append the N_{tail} tail bits as shown in 27.3.11.7. This results in 52 uncoded bits.
 - 2) BCC encoder: Encode the data by a convolutional encoder at the rate $R = 1/2$ as described in 17.3.5.6.

- 3) BCC interleaver: Interleave as described in 27.3.12.8.
 - 4) Constellation mapper: BPSK modulate the first 52 interleaved bits as described in 17.3.5.8 to form the first OFDM symbol of the HE-SIG-A field. BPSK modulate the second 52 interleaved bits to form the second OFDM symbol of the HE-SIG-A field.
 - 5) Pilot insertion: Insert pilots as described in 17.3.5.9.
 - 6) Duplicate and phase rotation: Duplicate the HE-SIG-A OFDM symbols over each occupied 20 MHz subchannel of the channel width. Apply the appropriate phase rotation for each occupied 20 MHz subchannel as described in 27.3.10 and 21.3.7.5.
 - 7) CSD per STS: If the TXVECTOR parameter BEAM_CHANGE is 0, apply CSD per STS for each space-time stream and frequency segment as described in 27.3.11.2.2 before spatial mapping.
 - 8) Spatial mapping: If the TXVECTOR parameter BEAM_CHANGE is 0, apply the A matrix and the Q matrix as described in 27.3.11.7.4.
 - 9) IDFT: Compute the inverse discrete Fourier transform.
 - 10) CSD per chain: If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, apply CSD per chain for each transmit chain and frequency segment as described in 27.3.11.2.1.
 - 11) GI and windowing: Prepend a GI ($T_{GI,Pre-HE}$), and apply windowing as described in 27.3.10.
 - 12) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.
- b) For an HE ER SU PPDU, the HE-SIG-A field consists of four subfields: HE-SIG-A1, HE-SIG-A1-R, HE-SIG-A2, and HE-SIG-A2-R. The HE-SIG-A1 and HE-SIG-A1-R subfields have the same data bits while the HE-SIG-A2 and HE-SIG-A2-R subfields have the same data bits as defined in 27.3.11.7. The HE-SIG-A field is constructed as follows:
- 1) Obtain the HE-SIG-A fields from the TXVECTOR. Add the reserved bits, append the calculated CRC, and then append the N_{tail} tail bits as shown in 27.3.11.7. This results in 52 uncoded bits.
 - 2) BCC encoder: Encode the data by a convolutional encoder at the rate $R = 1/2$ as described in 17.3.5.6.
 - 3) BCC interleaver: Interleave the data bits of the HE-SIG-A1 and HE-SIG-A2 subfields as described in 27.3.12.8. The data bits of the HE-SIG-A1-R and HE-SIG-A2-R subfields are not interleaved.
 - 4) Constellation mapper: BPSK modulate the HE-SIG-A1, HE-SIG-A2, and HE-SIG-A2-R data bits as described in 17.3.5.8 to form the first, third, and fourth OFDM symbol of the HE-SIG-A field, respectively. QPSK modulate the HE-SIG-A1-R encoded data bits to form the second OFDM symbol of the HE-SIG-A field.
 - 5) Pilot insertion: Insert pilots as described in 17.3.5.9.
 - 6) Duplication and phase rotation: Duplicate the HE-SIG-A field over each occupied 20 MHz subchannel of the channel bandwidth. Apply appropriate phase rotation for each occupied 20 MHz subchannel as described in 27.3.10 and 21.3.7.5.
 - 7) CSD per STS: If the TXVECTOR parameter BEAM_CHANGE is 0, apply CSD per STS for each space-time stream and frequency segment as described in 27.3.11.2.2 before spatial mapping.
 - 8) Spatial mapping: If the TXVECTOR parameter BEAM_CHANGE is 0, apply the A matrix and the Q matrix as described in 27.3.11.7.4.
 - 9) IDFT: Compute the inverse Fourier transform.
 - 10) CSD per chain: If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, apply CSD per chain for each transmit chain and frequency segment as described in 27.3.11.2.1.

- 11) GI and windowing: Prepend a GI ($T_{GI,Pre-HE}$), and apply windowing as described in 27.3.10.
- 12) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.7 Construction of HE-SIG-B field

For an HE MU PPDU, the HE-SIG-B field consists of a Common field followed by a User Specific field as defined in 27.3.11.8 and is constructed as follows:

- a) Obtain the HE-SIG-B field values from the TXVECTOR. Add the reserved bits, append the calculated CRC, and then append the N_{tail} tail bits as shown in 27.3.11.8.
- b) BCC encoder: Encode the Common field data and each User Block field data individually by a convolutional encoder as described in 27.3.12.5.1.
- c) BCC interleaver: Interleave as described in 27.3.12.8.
- d) Constellation mapper: Obtain MCS_SIG_B from the TXVECTOR and use it to modulate the interleaved bits as described in 27.3.12.9 to form the HE-SIG-B OFDM symbols.
- e) Pilot insertion: Insert pilots as described in 17.3.5.9.
- f) Duplicate and phase rotation: Duplicate HE-SIG-B OFDM symbols as described in 27.3.11.8.5. Apply the appropriate phase rotation for each 20 MHz subchannel as described in 27.3.10 and 21.3.7.5.
- g) IDFT: Compute the inverse Fourier transform.
- h) CSD per chain: Apply CSD per chain for each transmit chain and frequency segment as described in 27.3.11.2.1.
- i) GI and windowing: Prepend a GI ($T_{GI,Pre-HE}$), and apply windowing as described in 27.3.10.
- j) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.8 Construction of HE-STF field

The HE-STF field is defined in 27.3.11.9 and is constructed as follows:

- a) Sequence generation: Generate the HE-STF in the frequency domain over the bandwidth indicated by the TXVECTOR parameter CH_BANDWIDTH as described in 27.3.11.9. Apply a 3 dB power boost if transmitting an HE ER SU PPDU as described in 27.3.11.9.
- b) CSD: Apply CSD for each space-time stream and frequency segment as described in 27.3.11.2.2.
- c) Spatial mapping: Apply the Q matrix as described in 27.3.11.9.
- d) IDFT: Compute the inverse discrete Fourier transform.
- e) GI and windowing: Prepend a GI, i.e., 0.8 μ s GI for an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU and 1.6 μ s GI for an HE TB PPDU. Apply windowing as described in 27.3.10.
- f) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.9 Construction of HE-LTF field

The HE-LTF field is defined in 27.3.11.10 and is constructed as follows:

- a) Sequence generation: Generate the HE-LTF sequence in frequency domain over the bandwidth indicated by CH_BANDWIDTH as described in 27.3.11.10. Apply a 3 dB power boost if transmitting an HE ER SU PPDU as described in 27.3.11.10.

- b) $A_{\text{HE-LTF}}$ matrix mapping: Apply the $P_{\text{HE-LTF}}$ matrix to the data tones of the HE-LTF sequence, and apply the $R_{\text{HE-LTF}}$ matrix to pilot subcarriers of the HE-LTF sequence, except the UL MU-MIMO transmission not using HE single stream pilot HE-LTF mode as described in 27.3.11.10. There is no pilot mapping for an HE TB feedback NDP.
- c) CSD: Apply CSD for each space-time stream and frequency segment as described in 27.3.11.2.2.
- d) Spatial mapping: Apply the Q matrix as described in 27.3.11.10.
- e) IDFT: Compute the inverse discrete Fourier transform.
- f) GI and windowing: Prepend a GI indicated by the TXVECTOR parameter GI_TYPE, and apply windowing as described in 27.3.10.
- g) Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.10 Construction of the Data field in an HE SU PPDU, HE ER SU PPDU, and HE TB PPDU

27.3.6.10.1 Using BCC

The construction of the Data field in an HE SU PPDU, HE ER SU PPDU, and HE TB PPDU with BCC encoding proceeds as follows:

- a) Construct the SERVICE field as described in 27.3.12.3, and append the PSDU to the SERVICE field.
- b) Pre-FEC padding: Append the pre-FEC pad bits and tail bits as described in 27.3.12.
- c) Scrambler: Scramble the pre-FEC padded data.
- d) BCC encoder: BCC encode as described in 27.3.12.5.1.
- e) Post-FEC padding: Append the post-FEC pad bits and PE field as described in 27.3.12.
- f) Stream parser: Rearrange the output of BCC encoder into blocks as described in 27.3.12.6.
- g) BCC interleaver: Interleave as described in 27.3.12.8.
- h) Constellation mapper: Map to BPSK, BPSK DCM, QPSK, QPSK DCM, 16-QAM, 16-QAM DCM, 64-QAM, or 256-QAM constellation points as described in 27.3.12.9.
- i) STBC: Apply STBC as described in 27.3.12.12.
- j) Pilot insertion: Insert pilots following the steps described in 27.3.12.13.
- k) CSD: Apply CSD for each space-time stream and frequency segment as described in 27.3.11.2.2.
- l) Spatial mapping: Apply the Q matrix as described in 27.3.12.14.
- m) IDFT: In an 80+80 MHz transmission, map each frequency subblock to a separate IDFT. Compute the inverse discrete Fourier transform.
- n) GI and windowing: Prepend a GI determined by the TXVECTOR parameter GI_TYPE, and apply windowing as described in 27.3.10.
- o) Analog and RF: Upconvert the resulting complex baseband waveform with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.10.2 Using LDPC

The construction of the Data field in an HE SU PPDU, HE ER SU PPDU, and HE TB PPDU with LDPC encoding proceeds as follows:

- a) Construct the SERVICE field as described in 27.3.12.3, and append the PSDU to the SERVICE field.

- b) Pre-FEC padding: Append the pre-FEC padding bits as described in 27.3.12. There are no tail bits.
- c) Scrambler: Scramble the pre-FEC padded data.
- d) LDPC encoder: LDPC encode as described in 27.3.12.5.2.
- e) Post-FEC padding: Append the post-FEC pad bits and PE field as described in 27.3.12.
- f) Stream parser: Rearrange the output of LDPC encoder into blocks as described in 27.3.12.6.
- g) Segment parser (if needed): In a 160 MHz or 80+80 MHz transmission with a 2×996-tone RU, divide the output of each stream parser into two frequency subblocks as described in 27.3.12.6. This block is bypassed for 20 MHz, 40 MHz, and 80 MHz transmissions.
- h) Constellation mapper: Map to BPSK, BPSK DCM, QPSK, QPSK DCM, 16-QAM, 16-QAM DCM, 64-QAM, 256-QAM, or 1024-QAM constellation points as described in 27.3.12.9.
- i) LDPC tone mapper: the LDPC tone mapping shall be performed on all LDPC encoded streams as described in 27.3.12.10.
- j) Segment deparser (if needed): In 160 MHz transmission, merge the two frequency subblocks into one frequency segment as described in 27.3.12.11. This block is bypassed for 20 MHz, 40 MHz, 80 MHz, and 80+80 MHz transmissions.
- k) STBC: Apply STBC as described in 27.3.12.12.
- l) Pilot insertion: Insert pilots following the steps described in 27.3.12.13.
- m) CSD: Apply CSD for each space-time stream and frequency segment as described in 27.3.11.2.2.
- n) Spatial mapping: Apply the Q matrix as described in 27.3.12.14.
- o) IDFT: In an 80+80 MHz transmission, map each frequency subblock to a separate IDFT. Compute the inverse discrete Fourier transform.
- p) GI and windowing: Prepend a GI determined by the TXVECTOR parameter GI_TYPE, and apply windowing as described in 27.3.10.
- q) Analog and RF: Upconvert the resulting complex baseband waveform with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.

27.3.6.11 Construction of the Data field in an HE MU PPDU

27.3.6.11.1 General

In an HE MU transmission, the PPDU encoding process is performed independently in an RU on a per user basis up to the input of the spatial mapping block, except that CSD is performed with knowledge of the space-time streams starting index for that user. All user data in an RU is combined and mapped to the transmit chains in the spatial mapping block.

27.3.6.11.2 Using BCC

A Data field with BCC encoding is constructed using steps a) to l) in 27.3.6.10.1 and then applying CSD for an HE MU PPDU as described in 27.3.11.2.2.

27.3.6.11.3 Using LDPC

A Data field with LDPC encoding is constructed using steps a) to l) in 27.3.6.10.2 and then applying CSD for an HE MU PPDU as described in 27.3.11.2.2.

27.3.6.11.4 Combining to form an HE MU PPDU

The per user data is combined as follows:

- a) Spatial mapping: The Q matrix is applied as described in 27.3.12.14. The combining of all user data of an RU is done in this block.
- b) IDFT: Compute the inverse discrete Fourier transform.
- c) GI and windowing: Prepend a GI determined by the TXVECTOR parameter GI_TYPE, and apply windowing as described in 27.3.10.
- d) Analog and RF: Upconvert the resulting complex baseband waveform with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10 and 27.3.11 for details.

27.3.7 HE modulation and coding schemes (HE-MCSs)

The HE-MCS is a compact representation of the modulation and coding used in the Data field of the PPDU. For an HE SU PPDU and an HE ER SU PPDU, it is carried in the HE-SIG-A field. For an HE MU PPDU, it is carried per user in the User Specific field of the HE-SIG-B field. For an HE TB PPDU, it is carried in the User Info field of the Trigger frame soliciting the HE TB PPDU.

Rate-dependent parameters for the full set of HE-MCSs are shown in Table 27-55 to Table 27-110 (in 27.5). These tables give rate-dependent parameters for HE-MCSs with indices 0 to 11; number of spatial streams from 1 to 8; RU options of 26-tone RU, 52-tone RU, 106-tone RU, 242-tone RU, 484-tone RU, and 996-tone RU; and bandwidth options of 20 MHz, 40 MHz, 80 MHz, 160 MHz, and 80+80 MHz.

The HE ER SU PPDU supports only a single 242-tone or 106-tone RU. An HE ER SU PPDU with a 242-tone RU shall be transmitted with only the <HE-MCS, NSS> tuples <HE-MCS 0, 1>, <HE-MCS 1, 1>, and <HE-MCS 2, 1>. An HE ER SU PPDU with a 106-tone RU shall be transmitted with only the <HE-MCS, NSS> tuple <HE-MCS 0, 1>. The 106-tone RU location within the 20 MHz tone plan is fixed as the one that is higher in frequency.

DCM is an optional modulation scheme used for the HE-SIG-B field and the Data field in an HE PPDU. The use of DCM for the HE-SIG-B field in an HE MU PPDU is indicated in the HE-SIG-A field. For the HE-SIG-B field, DCM is applicable to only the HE-SIG-B-MCSs 0, 1, 3, and 4. The use of DCM on the Data field of an HE SU PPDU and HE ER SU PPDU is indicated in the HE-SIG-A field. The use of DCM in the Data field of an HE MU PPDU is indicated in the HE-SIG-B field. For the Data field, DCM is applicable to only the HE-MCSs 0, 1, 3, and 4.

The pre-HE modulated fields (see Figure 27-23) are not transmitted in 20 MHz subchannels in which the preamble is punctured as described in 27.3.11.3.

27.3.8 HE-SIG-B modulation and coding schemes (HE-SIG-B-MCSs)

The HE-SIG-B-MCS is a compact representation of the modulation and coding used in the HE-SIG-B field of the HE MU PPDU. The HE-SIG-B modulation and coding scheme is carried in the HE-SIG-B-MCS subfield of the HE-SIG-A field in the HE MU PPDU and indicates an HE-SIG-B-MCS in the range 0 to 5.

27.3.9 Timing-related parameters

Refer to Table 19-6 and Table 21-5 for timing-related parameters for non-HE PPDU formats.

Table 27-12 defines the timing-related parameters for HE PPDU formats.

Table 27-12—Timing-related constants

Parameter	Values	Description
$\Delta_{F, \text{Pre-HE}}$	312.5 kHz	Subcarrier frequency spacing for the pre-HE modulated fields.
$\Delta_{F, \text{HE}}$	78.125 kHz	Subcarrier frequency spacing for the HE modulated fields.
$T_{DFT, \text{Pre-HE}}$	3.2 μs	IDFT/DFT period for the pre-HE modulated fields.
$T_{DFT, \text{HE}}$	12.8 μs	IDFT/DFT period for the Data field.
$T_{GI, \text{Pre-HE}}$	0.8 μs	Guard interval duration for the pre-HE modulated fields.
$T_{GI, \text{L-LTF}}$	1.6 μs	Guard interval duration for the L-LTF field.
$T_{GI1, \text{Data}}$	0.8 μs	Base guard interval duration for the Data field.
$T_{GI2, \text{Data}}$	1.6 μs	Double guard interval duration for the Data field.
$T_{GI4, \text{Data}}$	3.2 μs	Quadruple guard interval duration for the Data field.
$T_{GI, \text{HE-LTF}}$	$T_{GI1, \text{Data}}$, $T_{GI2, \text{Data}}$ or $T_{GI4, \text{Data}}$ depending on the GI used for data	Guard interval duration for the HE-LTF field, same as $T_{GI, \text{Data}}$.
$T_{GI, \text{Data}}$	$T_{GI1, \text{Data}}$, $T_{GI2, \text{Data}}$ or $T_{GI4, \text{Data}}$ depending on the GI used for data	Guard interval duration for the Data field.
T_{SYM1}	$13.6 \mu\text{s} = T_{DFT, \text{HE}} + T_{GI1, \text{Data}} = 1.0625 \times T_{DFT, \text{HE}}$	OFDM symbol duration with base GI.
T_{SYM2}	$14.4 \mu\text{s} = T_{DFT, \text{HE}} + T_{GI2, \text{Data}} = 1.125 \times T_{DFT, \text{HE}}$	OFDM symbol duration with double GI.
T_{SYM4}	$16 \mu\text{s} = T_{DFT, \text{HE}} + T_{GI4, \text{Data}} = 1.25 \times T_{DFT, \text{HE}}$	OFDM symbol duration with quadruple GI.
T_{SYM}	T_{SYM1} , T_{SYM2} , or T_{SYM4} depending on the GI used for data	OFDM symbol interval for HE PPDU fields. See Table 27-16.
$T_{L\text{-STF}}$	$8 \mu\text{s} = 10 \times T_{DFT, \text{Pre-HE}} / 4$	Non-HT Short Training field duration.
$T_{L\text{-LTF}}$	$8 \mu\text{s} = 2 \times T_{DFT, \text{Pre-HE}} + T_{GI, \text{L-LTF}}$	Non-HT Long Training field duration.
$T_{L\text{-SIG}}$	4 μs	Non-HT SIGNAL field duration.
$T_{RL\text{-SIG}}$	4 μs	Repeated non-HT SIGNAL field duration.
$T_{\text{HE-SIG-A}}$	$8 \mu\text{s} = 2 \times 4 \mu\text{s}$	HE-SIG-A field duration in an HE SU PPDU, HE MU PPDU, and HE TB PPDU.
$T_{\text{HE-SIG-A-R}}$	$16 \mu\text{s} = 4 \times 4 \mu\text{s}$	HE-SIG-A field duration in an HE ER SU PPDU.
$T_{\text{HE-SIG-B}}$	$4 \mu\text{s} = T_{DFT, \text{Pre-HE}} + T_{GI, \text{Pre-HE}}$	Duration of each OFDM symbol in the HE-SIG-B field.
$T_{\text{HE-STF-T}}$	$8 \mu\text{s} = 5 \times 1.6 \mu\text{s}$	HE-STF field duration for an HE TB PPDU.
$T_{\text{HE-STF-NT}}$	$4 \mu\text{s} = 5 \times 0.8 \mu\text{s}$	HE-STF field duration for an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU.
$T_{\text{HE-LTF-1X}}$	3.2 μs	Duration of each 1x HE-LTF OFDM symbol without GI.

Table 27-12—Timing-related constants (continued)

Parameter	Values	Description
$T_{\text{HE-LTF-2X}}$	6.4 μs	Duration of each 2x HE-LTF OFDM symbol without GI.
$T_{\text{HE-LTF-4X}}$	12.8 μs	Duration of each 4x HE-LTF OFDM symbol without GI.
$T_{\text{HE-LTF}}$	$T_{\text{HE-LTF-1X}}$, $T_{\text{HE-LTF-2X}}$ or $T_{\text{HE-LTF-4X}}$ depending upon the HE-LTF duration used	Duration of each OFDM symbol without GI in the HE-LTF field.
$T_{\text{HE-LTF-SYM}}$	$T_{\text{HE-LTF}} + T_{\text{GI,HE-LTF}}$	Duration of each OFDM symbol including GI in the HE-LTF field.
N_{service}	16	Number of bits in the SERVICE field.
$N_{\text{tail}}, N_{\text{tail},u}$	6 for BCC encoder, 0 for LDPC encoder	Number of tail bits per encoder (for user u).
T_{SYML}	4 μs	OFDM symbol duration including GI prior to the HE-STF field.
T_{PE}	0, 4 μs , 8 μs , 12 μs or 16 μs depending on the actual extension duration used	Duration of the PE field.

Table 27-13 defines tone allocation related parameters for a non-OFDMA HE PPDU.

Table 27-13—Subcarrier allocation related constants for the HE-modulated fields in a non-OFDMA HE PPDU

Parameter	CBW20	CBW40	CBW80	CBW80+80	CBW160	Description
N_{SD}	See 27.5					Number of data subcarriers per frequency segment
N_{SP}	8	16	16	16	32	Number of pilot subcarriers per frequency segment
N_{ST}	242	484	996	996	1992	Total number of subcarriers per frequency segment
N_{SR}	122	244	500	500	1012	Highest data subcarrier index per frequency segment
N_{Seg}	1	1	1	2	1	Number of frequency segments
N_{DC}	3	5	5	5	23	Number of null subcarriers at DC per segment

Table 27-14 defines tone allocation related parameters for an OFDMA HE PPDU.

Table 27-14—Subcarrier allocation related constants for RUs in an OFDMA HE PPDU

Parameter	RU Size (subcarriers)							Description
	26	52	106	242	484	996	2×996	
N_{SD}	See 27.5							Number of data subcarriers per RU
N_{SP}	2	4	4	8	16	16	32	Number of pilot subcarriers per RU
N_{ST}	26	52	106	242	484	996	1992	Total number of subcarriers per RU

Table 27-15 defines parameters used frequently in Clause 27.

Table 27-15—Frequently used parameters

Symbol	Explanation
N_{RU}	For pre-HE modulated fields, $N_{RU} = 1$. For HE modulated fields, N_{RU} represents the number of occupied RUs in the transmission.
$N_{user,r}$	For pre-HE modulated fields, $N_{user,r} = 1$. For HE modulated fields, $N_{user,r}$ represents the total number of users in the r^{th} occupied RU of the transmission.
$N_{user,total}$	Total number of users in all occupied RUs of an HE transmission, i.e., $N_{user,total} = \sum_{r=0}^{N_{RU}-1} N_{user,r}$
$N_{CBPS}, N_{CBPS,u}$	Number of coded bits per OFDM symbol for user u , $u = 0, \dots, N_{user,total} - 1$. For an HE SU PPDU and HE ER SU PPDU, $N_{CBPS} = N_{CBPS,0}$. For an HE MU PPDU, N_{CBPS} is undefined.
$N_{CBPSS}, N_{CBPSS,u}$	Number of coded bits per OFDM symbol per spatial stream for user u , $u = 0, \dots, N_{user,total} - 1$. For the Data field of an HE SU PPDU and HE ER SU PPDU, $N_{CBPSS} = N_{CBPSS,0}$. For the Data field of an HE MU PPDU, N_{CBPSS} is undefined.
$N_{DBPS}, N_{DBPS,u}$	Number of data bits per OFDM symbol for user u , $u = 0, \dots, N_{user,total} - 1$. For an HE SU PPDU and HE ER SU PPDU, $N_{DBPS} = N_{DBPS,0}$. For an HE MU PPDU, N_{DBPS} is undefined.
$N_{BPSCS}, N_{BPSCS,u}$	Number of coded bits per subcarrier per spatial stream for user u , $u = 0, \dots, N_{user,total} - 1$. For an HE SU PPDU and HE ER SU PPDU, $N_{BPSCS} = N_{BPSCS,0}$. For an HE MU PPDU, N_{BPSCS} is undefined.
N_{RX}	Number of receive chains.

Table 27-15—Frequently used parameters (continued)

Symbol	Explanation
$N_{STS}, N_{STS,r,u}$	For HE modulated fields, $N_{STS,r,u}$ represents the number of space-time streams in the r^{th} occupied RU for user u, $u = 0, \dots, N_{user,r} - 1$. For STBC, $N_{STS,r,u} = 2$. For an HE SU PPDU and HE ER SU PPDU, $N_{STS} = N_{STS,0,0}$. For an HE MU PPDU, N_{STS} is undefined if any one of the RUs is assigned to more than one user, and $N_{STS} = 2$ if all RUs are assigned to no more than one user and the STBC field is set to 1.
$N_{STS,r,total}$	For HE modulated fields, $N_{STS,r,total}$ is the total number of space-time streams over all the users in the r^{th} occupied RU. $N_{STS,r,total} = \sum_{u=0}^{N_{user,r}-1} N_{STS,r,u}$ For pre-HE modulated fields, $N_{STS,r,total}$ is undefined if the TXVECTOR parameter BEAM_CHANGE is 1 or not present and $N_{STS,r,total} = N_{STS}$ if BEAM_CHANGE is 0. NOTE—$N_{STS,r,total} = N_{STS}$ for an HE SU PPDU and HE ER SU PPDU.
$N_{SS}, N_{SS,r,u}, N_{SS,u}$	Number of spatial streams. For the Data field, $N_{SS,r,u}$ is the number of spatial streams at r^{th} RU for user u, $u = 0, \dots, N_{user,r} - 1$ and $N_{SS,u}$ is the number of spatial streams for user u, $u = 0, \dots, N_{user,total} - 1$. For the Data field of an HE SU PPDU and HE ER SU PPDU, $N_{SS} = N_{SS,0,0}$. For the Data field of an HE MU PPDU, $N_{SS} = \max_{r=0}^{N_{RU}-1} N_{SS,r,total}$.
$N_{SS,r,total}$	For HE modulated fields, $N_{SS,r,total}$ is the total number of spatial streams at r^{th} RU in a PPDU. $N_{SS,r,total} = \sum_{u=0}^{N_{user,r}-1} N_{SS,r,u}$ For pre-HE modulated fields, $N_{SS,r,total}$ is undefined. NOTE—$N_{SS,r,total} = N_{SS}$ for an HE SU PPDU and HE ER SU PPDU.
N_{TX}	Number of transmit chains.
N_{HE-LTF}	The number of OFDM symbols in the HE-LTF field (see 27.3.11.10).
$N_{HE-SIG-B}$	The number of OFDM symbols in the HE-SIG-B field (see 27.3.11.8).
K_r	Set of used subcarrier indices in the r^{th} occupied RU.
R, R_u	R_u is the coding rate for user u, $u = 0, \dots, N_{user,total} - 1$. For an HE SU PPDU, $R = R_0$. For an HE MU PPDU, R is undefined.
$M_{r,u}$	The sum of the number of space-time streams of users prior to user u in RU r. For pre-HE modulated fields, $M_{r,u} = 0$. For HE modulated fields, $M_{r,0} = 0$ for $u = 0$ and $M_{r,u} = \sum_{u'=0}^{u-1} N_{STS,r,u'} \text{ for } u = 1, \dots, N_{user,r} - 1.$

27.3.10 Mathematical description of signals

For a description of the conventions used for the mathematical description of the signals, see 17.3.2.5. In addition, the following notational conventions are used in Clause 27:

- $[Q]_{m,n}$ indicates the element in row m and column n of matrix, where $1 \leq m \leq N_{row}$ and $1 \leq n \leq N_{col}$.
- N_{row} and N_{col} are the number of rows and columns, respectively, of the matrix Q .
- $[Q]_{m:n}$ indicates a matrix consisting of columns m to n of matrix Q .

For a description on subcarrier indices over which the signal is transmitted for non-HT, HT, and VHT PPDU, see 21.3.7.

For a 20 MHz non-OFDMA HE PPDU transmission, the 20 MHz is divided into 256 subcarriers. The signal is transmitted on all or a subset of subcarriers -122 to -2 and 2 to 122 , with 0 being the center subcarrier.

For a 20 MHz OFDMA HE PPDU transmission, the 20 MHz is divided into 256 subcarriers. The signal is transmitted on all or a subset of the subcarriers -122 to -4 and 4 to 122 , with 0 being the center subcarrier.

For a 40 MHz non-OFDMA HE PPDU transmission, the 40 MHz is divided into 512 subcarriers. The signal is transmitted on subcarriers -244 to -3 and 3 to 244 , with 0 being the center subcarrier.

For a 40 MHz OFDMA HE PPDU transmission, the 40 MHz is divided into 512 subcarriers. The signal is transmitted on all or a subset of subcarriers -244 to -3 and 3 to 244 , with 0 being the center subcarrier.

For an 80 MHz non-OFDMA HE PPDU transmission, the 80 MHz is divided into 1024 subcarriers. The signal is transmitted on subcarriers -500 to -3 and 3 to 500 , with 0 being the center subcarrier.

For an 80 MHz OFDMA HE PPDU transmission, the 80 MHz is divided into 1024 subcarriers. The signal is transmitted on all or a subset of the subcarriers -500 to -4 and 4 to 500 , with 0 being the center subcarrier.

For a 160 MHz HE PPDU transmission or a noncontiguous 80+80 MHz transmission, each half 80 MHz bandwidth is divided into 1024 subcarriers, and the subcarriers on which the signal is transmitted in each 80 MHz bandwidth is identical to an 80 MHz HE PPDU transmission, depending on non-OFDMA or OFDMA transmission within the corresponding 80 MHz.

The transmitted signal is described in complex baseband signal notation. The actual transmitted signal on transmit chain i_{TX} and frequency segment i_{Seg} is related to the complex baseband signal by the relation shown in Equation (27-1).

$$r_{RF}^{(i_{Seg}, i_{TX})}(t) = \text{Re} \left\{ \frac{1}{\sqrt{N_{Seg}}} r_{PPDU}^{(i_{Seg}, i_{TX})}(t) \exp(j2\pi f_c^{(i_{Seg})} t) \right\}, \quad i_{Seg} = 0, \dots, N_{Seg} - 1; \quad i_{TX} = 1, \dots, N_{TX} \quad (27-1)$$

where

- N_{Seg} represents the number of frequency segments in the transmit signal as defined in Table 27-13
- $r_{PPDU}^{(i_{Seg}, i_{TX})}$ represents the complex baseband signal of frequency segment i_{Seg} and transmit chain i_{TX}
- $f_c^{(i_{Seg})}$ represents the center frequency of the portion of the PPDU transmitted in frequency segment i_{Seg} . Table 21-7 shows $f_c^{(i_{Seg})}$ as a function of the channel starting frequency, $\text{dot11CurrentChannelWidth}$ (see Table 21-22) and CH_BANDWIDTH where $f_{CH, start}$, $f_{P20, idx}$, $f_{P40, idx}$, and $f_{P80, idx}$ are given in Equation (21-4), Equation (21-5), Equation (21-7), and Equation (21-9), respectively.
 For an HE STA operating in the 6 GHz band, see 27.3.23.2 for the value of channel starting frequency and the valid range of $\text{dot11CurrentChannelCenterFrequencyIndex0}$ and $\text{dot11CurrentChannelCenterFrequencyIndex1}$.

The transmitted RF signal is derived by upconverting the complex baseband signal, which consists of several fields. The timing boundaries for the various fields when the midamble is not present are shown in Figure 27-23, where $N_{\text{HE-LTF}}$ is the number of HE-LTF symbols and is defined in Table 27-15, $N_{\text{HE-SIG-B}}$ is the number of OFDM symbols in the HE-SIG-B field present in an HE MU PPDU, and N_{SYM} is the number of data OFDM symbols.

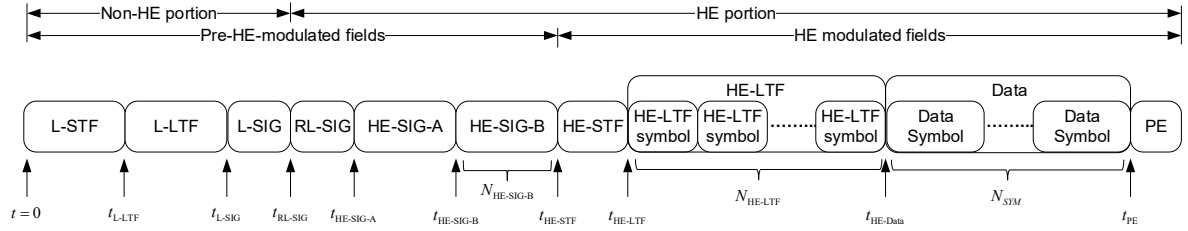


Figure 27-23—Timing boundaries for HE PPDU fields if midamble is not present

NOTE—Data OFDM symbols are OFDM symbols in the Data field of an HE PPDU that are not midamble symbols.

The time offset, t_{Field} , determines the starting time of the corresponding field relative to the start of L-STF ($t = 0$).

The signal transmitted on frequency segment i_{Seg} and transmit chain i_{TX} shall be as shown in Equation (27-2) if midamble is not present.

$$\begin{aligned}
 r_{\text{PPDU}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = & r_{\text{L-STF}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) + r_{\text{L-LTF}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{L-LTF}}) \\
 & + r_{\text{L-SIG}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{L-SIG}}) + r_{\text{RL-SIG}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{RL-SIG}}) + r_{\text{HE-SIG-A}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{HE-SIG-A}}) + r_{\text{HE-SIG-B}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{HE-SIG-B}}) \\
 & + r_{\text{HE-STF}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{HE-STF}}) + r_{\text{HE-LTF}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{HE-LTF}}) + r_{\text{HE-Data}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{HE-Data}}) + r_{\text{HE-PE}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{HE-PE}})
 \end{aligned} \quad (27-2)$$

where

$r_{\text{HE-SIG-B}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t - t_{\text{HE-SIG-B}})$ is only applicable to an HE MU PPDU

$$0 \leq i_{\text{Seg}} \leq N_{\text{Seg}} - 1$$

$$1 \leq i_{\text{TX}} \leq N_{\text{TX}}$$

$$t_{\text{L-LTF}} = T_{\text{L-STF}}$$

$$t_{\text{L-SIG}} = t_{\text{L-LTF}} + T_{\text{L-LTF}}$$

$$t_{\text{RL-SIG}} = t_{\text{L-SIG}} + T_{\text{L-SIG}}$$

$$t_{\text{HE-SIG-A}} = t_{\text{RL-SIG}} + T_{\text{RL-SIG}}$$

$$t_{\text{HE-SIG-B}} = \begin{cases} t_{\text{HE-SIG-A}} + T_{\text{HE-SIG-A}}, & \text{for an HE MU PPDU} \\ \text{undefined}, & \text{otherwise} \end{cases}$$

$$t_{\text{HE-STF}} = \begin{cases} t_{\text{HE-SIG-A}} + T_{\text{HE-SIG-A}}, & \text{for an HE SU PPDU and HE TB PPDU} \\ t_{\text{HE-SIG-A}} + T_{\text{HE-SIG-A-R}}, & \text{for an HE ER SU PPDU} \\ t_{\text{HE-SIG-B}} + N_{\text{HE-SIG-B}} T_{\text{HE-SIG-B}}, & \text{for an HE MU PPDU} \end{cases}$$

$$t_{\text{HE-LTF}} = \begin{cases} t_{\text{HE-STF}} + T_{\text{HE-STF-T}}, & \text{for an HE TB PPDU} \\ t_{\text{HE-STF}} + T_{\text{HE-STF-NT}}, & \text{otherwise} \end{cases}$$

$$t_{\text{HE-Data}} = t_{\text{HE-LTF}} + N_{\text{HE-LTF}} T_{\text{HE-LTF-SYM}}$$

$$t_{\text{HE-PE}} = t_{\text{HE-Data}} + N_{\text{SYM}} T_{\text{SYM}}$$

In an HE SU PPDU, HE MU PPDU, and HE ER SU PPDU, for each field excluding the PE field, $r_{\text{Field}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t)$ is defined as the summation of one or more subfields. Each subfield, $r_{\text{Subfield}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t)$, is defined to be an inverse discrete Fourier transform in Equation (27-3).

$$r_{\text{Subfield}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = w_{T_{\text{Subfield}}}(t) \sum_{r=0}^{N_{\text{RU}}-1} \frac{\alpha_r \beta_r^{\text{Field}}}{\sqrt{N_{\text{Norm}, r}}} \sum_{k \in K_r} \eta_{\text{Field}, k} \sum_{u=0}^{N_{\text{user}, r}-1} \sum_{m=1}^{N_{\text{STS}, r, u}} \left[Q_k^{(i_{\text{Seg}})} \right]_{i_{\text{TX}}, (M_{r, u} + m)} Y_{k, BW} X_{k, r, u}^{(i_{\text{Seg}}, m)} \exp(j2\pi k \Delta_{F, \text{Field}}(t - T_{\text{GI}, \text{Field}} - T_{\text{CS}, \text{HE}}(M_{r, u} + m))) \quad (27-3)$$

In an HE TB PPDU, transmitted by user u in the r^{th} occupied RU, each subfield, $r_{\text{Subfield}, r, u}^{(i_{\text{Seg}}, i_{\text{TX}})}(t)$, is defined in Equation (27-4).

$$r_{\text{Subfield}, r, u}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = \frac{\beta_r^{\text{Field}}}{\sqrt{N_{\text{Norm}, r}}} w_{T_{\text{Subfield}}}(t) \sum_{k \in K_r} \eta_{\text{Field}, k} \sum_{m=1}^{N_{\text{STS}, r, u}} \left[Q_{k, u}^{(i_{\text{Seg}})} \right]_{i_{\text{TX}}, m} Y_{k, BW} X_{k, r, u}^{(i_{\text{Seg}}, m)} \exp(j2\pi k \Delta_{F, \text{Field}}(t - T_{\text{GI}, \text{Field}} - T_{\text{CS}, \text{HE}}(M_{r, u} + m))) \quad (27-4)$$

For an HE SU PPDU, HE MU PPDU, and HE ER SU PPDU, the total power of the time domain HE modulated field signals summed over all transmit chains should not exceed the total power of the time domain pre-HE modulated field signals summed over all transmit chains if the TXVECTOR parameter BEAM_CHANGE is 1 or not present and power boost in HE modulated fields is not present.

For an HE TB PPDU, the total power of the time domain HE modulated field signals summed over all transmit chains may exceed the total power of the time domain pre-HE modulated field signals summed over all transmit chains by up to 3 dB.

For notational simplicity, the parameter BW is omitted from some bandwidth-dependent terms.

In Equation (27-3) and Equation (27-4), the following notations are used:

- $N_{\text{Norm}, r}$ If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, then for pre-HE modulated fields, $N_{\text{Norm}, r} = N_{\text{TX}}$. If the TXVECTOR parameter BEAM_CHANGE is 0, then for pre-HE modulated fields $N_{\text{Norm}, r} = N_{\text{STS}, r, \text{total}}$, where $N_{\text{STS}, r, \text{total}}$ is given in Table 27-15. For HE modulated fields $N_{\text{Norm}, r} = N_{\text{STS}, r, \text{total}}$.
- $w_{\text{Subfield}}(t)$ is a windowing function. An example function, $w_{\text{Subfield}}(t)$, is given in 17.3.2.5.
- K_r For pre-HE modulated fields, K_r is the set of subcarriers indices in the allocated 20 MHz channels. For HE modulated fields in a non-OFDMA HE PPDU, K_r is the set of subcarriers indices from $-N_{\text{SR}}$ to N_{SR} as defined in Table 27-13 excluding DC subcarriers. For HE modulated fields in an OFDMA HE PPDU, K_r is the set of subcarrier indices for the tones in the r^{th} RU as defined in Table 27-7, Table 27-8, and Table 27-9.

$\eta_{Field,k}$ is the power scale factor of the k^{th} subcarrier of a given field within an OFDM symbol, which is $\sqrt{2}$ for all the subcarriers of the L-STF, L-LTF, HE-STF, and HE-LTF fields in the HE ER SU PPDU. For the L-SIG and RL-SIG fields of an HE ER SU PPDU,

$$\eta_{Field,k} = \begin{cases} \sqrt{2}, & k = -28, -27, 27, 28 \\ 1, & \text{otherwise} \end{cases}$$

For the HE-SIG-A and Data fields in an HE ER SU PPDU, $\eta_{Field,k} = 1$. For the pre-HE portion of the HE TB PPDU, $\eta_{Field,k} = \left[\frac{1}{\sqrt{2}}, 1 \right]$, meaning in the range $\frac{1}{\sqrt{2}}$ to 1. For all other fields in other HE PDUs $\eta_{Field,k} = 1$.

α_r is the power boost factor in the range [0.5, 2] of the r^{th} occupied RU in an HE PPDU. For a DL HE MU PPDU, an AP shall limit the ratio between the maximum value of α_r and the minimum value of α_r to 2, unless the Power Boost Factor Support subfield of the HE PHY Capabilities Information field in the HE Capabilities element from all recipient STAs is 1; in this case, the AP can use a ratio of up to 4. For an HE SU PPDU and HE ER SU PPDU, α_r is always set to 1.

β_r^{Field} is the power normalization factor and is defined in Equation (27-5).

$$\beta_r^{Field} = \begin{cases} \frac{\varepsilon_{Field}}{\sqrt{N_{Field}^{\text{Tone}} \cdot \frac{|\Omega_{20\text{MHz}}|}{N_{20\text{MHz}}}}}, & \text{for pre-HE modulated fields} \\ \frac{1}{\sqrt{|K_r^{Field}|}}, & \text{for HE modulated fields in an HE TB PPDU} \\ \sqrt{\frac{|K_r|}{|K_r^{Field}|}} / \sqrt{\sum_{r=0}^{N_{RU}-1} \alpha_r^2 |K_r|}, & \text{otherwise} \end{cases} \quad (27-5)$$

$|K_r|$ is the cardinality of the set of subcarriers K_r .

$|K_r^{Field}|$ is the cardinality of the set of modulated subcarriers within K_r for the HE-STF and Data fields. For the HE-LTF field,

$$|K_r^{\text{HE-LTF}}| = \begin{cases} |K_r|, & \text{for a 4x HE-LTF} \\ |K_r|/2, & \text{for a 2x HE-LTF} \\ |K_r|/4, & \text{for a 1x HE-LTF} \end{cases}$$

N_{RU} is defined in Table 27-15.

$$\varepsilon_{Field} = \begin{cases} \sqrt{\frac{N_{L-LTF}^{\text{Tone}}}{N_{L-SIG}^{\text{Tone}}}}, & \text{for the L-STF and L-LTF fields} \\ 1, & \text{otherwise} \end{cases}$$

N_{Field}^{Tone} is given in Table 27-16, which summarizes the various values of N_{Field}^{Tone} as a function of bandwidth per frequency segment.

Table 27-16—Number of modulated subcarriers and guard interval duration values for HE PPDU fields

Field	N_{Field}^{Tone} as a function of PPDU bandwidth				Guard interval duration
	20 MHz	40 MHz	80 MHz	160 MHz	
L-STF	12	24	48	96	—
L-LTF	52	104	208	416	$T_{GI,L-LTF}$
L-SIG in an HE PPDU	56	112	224	448	$T_{GI,Pre-HE}$
L-SIG in a non-HT duplicate PPDU	—	104	208	416	
RL-SIG	56	112	224	448	$T_{GI,Pre-HE}$
HE-SIG-A	56	112	224	448	$T_{GI,Pre-HE}$
HE-SIG-B	56	112	224	448	$T_{GI,Pre-HE}$
NON_HT_DUP_OFDM-Data (see NOTE)	—	104	208	416	$T_{GI,Pre-HE}$
NOTE—For notational convenience, NON_HT_DUP_OFDM-Data is used as a label for the Data field of a NON_HT PPDU with format type NON_HT_DUP_OFDM.					

$Q_k^{(i_{Seg})}$ is the spatial mapping matrix for the subcarrier k in frequency segment i_{Seg} . For HE modulated fields, $Q_k^{(i_{Seg})}$ is a matrix with N_{TX} rows and $N_{STS,r,total}$ columns. When beamforming or DL MU-MIMO is applied, $Q_k^{(i_{Seg})}$ is a beamforming or DL MU-MIMO steering matrix that is derived from the TXVECTOR parameter EXPANSION_MAT. The beamforming steering matrices and DL MU-MIMO steering matrices are implementation specific. If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, then for pre-HE modulated fields $Q_k^{(i_{Seg})}$ is a column vector with N_{TX} elements with element i_{TX} being $\exp(-j2\pi k \Delta_{F,Pre-HE} T_{CS}^{i_{TX}})$, where $T_{CS}^{i_{TX}}$ represents the cyclic shift for the transmitter chain whose values are defined in 27.3.11.2.1; otherwise it is identical to the spatial mapping $Q_k^{(i_{Seg})}$ for HE modulated fields.

$Q_{k,u}^{(i_{Seg})}$ is the spatial mapping matrix for user u on subcarrier k in frequency segment i_{Seg} . For HE modulated fields, $Q_{k,u}^{(i_{Seg})}$ is a matrix with N_{TX} rows and $N_{STS,r,u}$ columns. For pre-HE modulated fields, $Q_{k,u}^{(i_{Seg})}$ is a column vector with N_{TX} elements with element i_{TX} being $\exp(-j2\pi k \Delta_{F,Pre-HE} T_{CS}^{i_{TX}})$, where $T_{CS}^{i_{TX}}$ represents the cyclic shift for the transmitter chain whose values are defined in 27.3.11.2.1.

- $\Delta_{F, Field}$ is the subcarrier frequency spacing. For pre-HE modulated fields, $\Delta_{F, Field} = \Delta_{F, \text{Pre-HE}}$ given in Table 27-12. For HE modulated fields, $\Delta_{F, Field} = \Delta_{F, \text{HE}}$ given in Table 27-12.
- $M_{r,u}$ is given in Table 27-15.
- $X_{k,r,u}^{(i_{Seg}, m)}$ is the frequency-domain symbol in subcarrier k of user u in the r^{th} RU for frequency segment i_{Seg} of space-time stream m . Some of the $X_{k,r,u}^{(i_{Seg}, m)}$ within $-N_{SR} \leq k \leq N_{SR}$ have a value of zero. Examples of such cases include the DC tones, guard tones on each side of the transmit spectrum, the null subcarriers in an HE OFDMA PPDU, and the unmodulated tones of L-STF, HE-STF, and HE-LTF fields.
- $T_{GI, Field}$ is the guard interval duration used for each OFDM symbol in the field. The value for each field is defined in Table 27-12.
- $T_{CS, \text{HE}}(l)$ For pre-HE modulated fields, if the TXVECTOR parameter BEAM_CHANGE is 1 or not present, $T_{CS, \text{HE}}(l) = 0$. For HE modulated fields and pre-HE modulated fields if the TXVECTOR parameter BEAM_CHANGE is 0, $T_{CS, \text{HE}}(l)$ represents the cyclic shift per space-time stream, whose value is defined in 27.3.11.2.2.
- $\Upsilon_{k, BW}$ is used to represent tone rotation. BW in $\Upsilon_{k, BW}$ is determined by the TXVECTOR parameter CH_BANDWIDTH as defined in Table 27-17. In HE modulated fields, $\Upsilon_{k, BW} = 1$ for all the subcarriers. In pre-HE modulated fields, $\Upsilon_{k, BW}$ is defined as in 21.3.7.5 if TXVECTOR parameter BEAM_CHANGE is 1 and $\Upsilon_{k, BW} = 1$ for all the subcarriers if TXVECTOR parameter BEAM_CHANGE is 0. If the TXVECTOR parameter BEAM_CHANGE is not present (such as in an HE MU PPDU and HE TB PPDU), BEAM_CHANGE is assumed to be 1.

Table 27-17—CH_BANDWIDTH and $\Upsilon_{k, BW}$ for pre-HE modulated fields

CH_BANDWIDTH	$\Upsilon_{k, BW}$
CBW20	$\Upsilon_{k, 20}$
CBW40	$\Upsilon_{k, 40}$
CBW80	$\Upsilon_{k, 80}$
CBW160	$\Upsilon_{k, 160}$
CBW80+80	$\Upsilon_{k, 80}$ per frequency segment
HE-CBW-PUNC80-PRI	$\Upsilon_{k, 80}$
HE-CBW-PUNC80-SEC	$\Upsilon_{k, 80}$
HE-CBW-PUNC160-PRI20	$\Upsilon_{k, 160}$
HE-CBW-PUNC80+80-PRI20	$\Upsilon_{k, 80}$ per frequency segment
HE-CBW-PUNC160-SEC40	$\Upsilon_{k, 160}$
HE-CBW-PUNC80+80-SEC40	$\Upsilon_{k, 80}$ per frequency segment

- $\Omega_{20\text{MHz}}$ is a set of 20 MHz channels where pre-HE modulated fields are located. The set of 20 MHz channels contains one or more values in the range 0 to $N_{20\text{MHz}} - 1$ for an HE TB PPDU, HE sounding NDP, or HE MU PPDU with preamble puncturing, and it contains all values in the range 0 to $N_{20\text{MHz}} - 1$ for other HE PPDU formats.

$|\Omega_{20\text{MHz}}|$ is the cardinality of the set of 20 MHz channels where pre-HE modulated fields are located.

$$N_{20\text{MHz}} = \begin{cases} 1, & \text{if CH_BANDWIDTH is CBW20} \\ 2, & \text{if CH_BANDWIDTH is CBW40} \\ 4, & \text{if CH_BANDWIDTH is CBW80, HE-CBW-PUNC80-PRI,} \\ & \text{HE-CBW-PUNC80-SEC, CBW80+80, HE-CBW-PUNC80+80-PRI20,} \\ & \text{or HE-CBW-PUNC80+80-SEC40} \\ 8, & \text{if CH_BANDWIDTH is CBW160, HE-CBW-PUNC160-PRI20,} \\ & \text{or HE-CBW-PUNC160-SEC40} \end{cases}$$

27.3.11 HE preamble

27.3.11.1 Introduction

The HE preamble consists of pre-HE modulated fields and HE modulated fields. The pre-HE modulated fields for the various HE PPDU formats are the following:

- L-STF, L-LTF, L-SIG, RL-SIG, and HE-SIG-A fields of an HE SU PPDU, HE ER SU PPDU, and HE TB PPDU
- L-STF, L-LTF, L-SIG, RL-SIG, HE-SIG-A, and HE-SIG-B fields of an HE MU PPDU

The HE modulated fields in the preamble for all HE PPDU formats are the HE-STF and HE-LTF fields.

27.3.11.2 Cyclic shift

27.3.11.2.1 Cyclic shift for pre-HE modulated fields

If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, then the cyclic shift value $T_{CS}^{i_{TX}}$ for the L-STF, L-LTF, L-SIG, RL-SIG, HE-SIG-A, and HE-SIG-B fields of the PPDU for transmit chain i_{TX} out of a total of N_{TX} are defined in Table 21-10. In UL MU transmission, the cyclic shift value $T_{CS}^{i_{TX}}$ is based on the local transmit chain indices at each STA.

If the TXVECTOR parameter BEAM_CHANGE is 0, then the cyclic shift value $T_{CS, \text{HE}}(n)$ for the L-STF, L-LTF, L-SIG, RL-SIG, and HE-SIG-A fields is specified in 27.3.11.2.2.

27.3.11.2.2 Cyclic shift for HE modulated fields

The cyclic shift values defined in this subclause apply to the HE-STF, HE-LTF, and Data fields of the HE PPDU if the TXVECTOR parameter BEAM_CHANGE is 1 or not present and apply to the entire PPDU if the TXVECTOR parameter BEAM_CHANGE is 0.

Throughout the HE modulated fields of the preamble, cyclic shifts are applied to prevent unintended beamforming when correlated signals are transmitted in multiple space-time streams. The same cyclic shifts are also applied to these streams during the transmission of the Data field of the HE PPDU. For the r^{th} RU, the cyclic shift value $T_{CS, \text{HE}}(n)$ for the HE modulated fields for space-time stream n out of $N_{STS, r, \text{total}}$ total space-time streams is shown in Table 21-11.

27.3.11.3 L-STF field

If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, the time domain representation of the L-STF field, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , shall be as specified in Equation (27-6). The equation applies to signals up to a contiguous 160 MHz PPDU bandwidth and noncontiguous 80+80 MHz PPDU bandwidth.

$$r_{L-STF}^{(i_{Seg}, i_{TX})}(t) = \frac{\varepsilon}{\sqrt{N_{TX} \cdot N_{L-STF}^{Tone} \cdot \frac{|\Omega_{20MHz}|}{N_{20MHz}}}} w_{T_{L-STF}}(t) \quad (27-6)$$

$$\sum_{i_{BW} \in \Omega_{20MHz}} \sum_{k=-26}^{26} \eta_{L-STF, k} \left(\gamma_{(k - K_{Shift}(i_{BW})), BW} S_{k, 20} \cdot \exp(j2\pi(k - K_{Shift}(i_{BW}))\Delta_{F, Pre-HE}(t - T_{CS}^{i_{TX}})) \right)$$

where

ε is a power scaling factor with the value $\varepsilon = \sqrt{\frac{N_{L-LTF}^{Tone}}{N_{L-SIG}^{Tone}}}$

$\eta_{L-STF, k}$ is a PPDU format-dependent scaling factor for the L-STF field on subcarrier index k with the following value:

$$\eta_{L-STF, k} = \begin{cases} \sqrt{2}, & \text{for an HE ER SU PPDU} \\ \left[\frac{1}{\sqrt{2}}, 1 \right], & \text{for an HE TB PPDU} \\ 1, & \text{otherwise} \end{cases} \quad (27-7)$$

$$K_{Shift}(i) = (N_{20MHz} - 1 - 2i) \cdot 32$$

$T_{CS}^{i_{TX}}$ represents the cyclic shift for transmit chain i_{TX} with a value given in 27.3.11.2.1

N_{L-STF}^{Tone} has the value given in Table 27-16

$S_{k, 20}$ is defined as $S_{-26, 26}$ in Equation (19-8)

i_{BW} is the index of 20 MHz channels, $0 \leq i_{BW} \leq N_{20MHz} - 1$

If the TXVECTOR parameter BEAM_CHANGE is 0, the time domain representation of the L-STF field shall be as specified in Equation (27-8). The equation applies to signals up to a contiguous 160 MHz PPDU bandwidth and noncontiguous 80+80 MHz PPDU bandwidth.

$$r_{L-STF}^{(i_{Seg}, i_{TX})}(t) = \frac{\varepsilon}{\sqrt{N_{STS} \cdot N_{L-STF}^{Tone} \cdot \frac{|\Omega_{20MHz}|}{N_{20MHz}}}} w_{T_{L-STF}}(t) \quad (27-8)$$

$$\sum_{i_{BW}=0}^{N_{20MHz}-1} \sum_{k=-26}^{26} \sum_{m=1}^{N_{STS}} \eta_{L-STF, k} \left(\left[Q_{4(k - K_{Shift}(i_{BW}))}^{(i_{Seg})} \right]_{i_{TX}, m} \left[A_{HE-LTF}^{4(k - K_{Shift}(i_{BW}))} \right]_{m, 1} S_{k, 20} \cdot \exp(j2\pi(k - K_{Shift}(i_{BW}))\Delta_{F, Pre-HE}(t - T_{CS, HE}(m))) \right)$$

where

- $T_{CS, HE}(m)$ represents the cyclic shift for space-time stream m as defined in 27.3.11.2.2
- $Q_k^{(i_{Seg})}$ is the spatial mapping/steering matrix for subcarrier k , in frequency segment i_{Seg} on the data OFDM symbols over subcarrier spacing $\Delta_{F, HE}$ as defined in Table 27-12. Refer to the descriptions in 21.3.10.11.1 for examples of $Q_k^{(i_{Seg})}$.
- A_{HE-LTF}^k is defined in Equation (27-55)
- N_{STS} is the number of space-time streams of the HE-modulated fields in an HE SU PPDU or HE ER SU PPDU as defined in Table 27-15

27.3.11.4 L-LTF field

If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, the time domain representation of the L-LTF field, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , shall be as specified in Equation (27-9). The equation applies to signals up to a contiguous 160 MHz PPDU bandwidth and noncontiguous 80+80 MHz PPDU bandwidth.

$$r_{L-LTF}^{(i_{Seg}, i_{TX})}(t) = \frac{\varepsilon}{\sqrt{N_{TX} \cdot N_{L-LTF}^{Tone} \cdot \frac{|\Omega_{20MHz}|}{N_{20MHz}}}} w_{T_{L-LTF}}(t) \quad (27-9)$$

$$\sum_{i_{BW} \in \Omega_{20MHz}} \sum_{k=-26}^{26} \eta_{L-LTF, k} \left(\begin{matrix} \Upsilon_{(k - K_{Shift}(i_{BW})), BW} L_{k, 20} \\ \cdot \exp(j2\pi(k - K_{Shift}(i_{BW}))\Delta_{F, Pre-HE}(t - T_{GI, L-LTF} - T_{CS}^{i_{TX}})) \end{matrix} \right)$$

where

- $\eta_{L-LTF, k}$ is a PPDU format-dependent scaling factor for the L-LTF field on subcarrier index k with the same value as $\eta_{L-STF, k}$

ε is a power scaling factor with the value $\varepsilon = \sqrt{\frac{N_{L-LTF}^{Tone}}{N_{L-SIG}^{Tone}}}$

$T_{GI, L-LTF}$ is given in Table 27-12

$$K_{Shift}(i) = (N_{20MHz} - 1 - 2i) \cdot 32$$

- $T_{CS}^{i_{TX}}$ represents the cyclic shift for transmit chain i_{TX} with a value given in 27.3.11.2.1

N_{L-LTF}^{Tone} has the value given in Table 27-16

$L_{k, 20}$ is defined as $L_{-26, 26}$ in Equation (17-8)

If the TXVECTOR parameter BEAM_CHANGE is 0, the time domain representation of the L-LTF field shall be as specified in Equation (27-10). The equation applies to signals up to a contiguous 160 MHz PPDU bandwidth and noncontiguous 80+80 MHz PPDU bandwidth.

$$r_{\text{L-LTF}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = \frac{\varepsilon}{\sqrt{N_{\text{STS}} \cdot N_{\text{L-LTF}}^{\text{Tone}} \cdot \frac{|\Omega_{20\text{MHz}}|}{N_{20\text{MHz}}}}} w_{T_{\text{L-LTF}}}(t) \quad (27-10)$$

$$\sum_{i_{\text{BW}}=0}^{N_{20\text{MHz}}-1} \sum_{k=-26}^{26} \sum_{m=1}^{N_{\text{STS}}} \eta_{\text{L-LTF}, k} \left(\left[Q_{4(k-K_{\text{Shift}}(i_{\text{BW}}))}^{(i_{\text{Seg}})} \right]_{i_{\text{TX}}, m} \left[A_{\text{HE-LTF}}^{4(k-K_{\text{Shift}}(i_{\text{BW}}))} \right]_{m, 1} L_{k, 20} \right. \\ \left. \cdot \exp(j2\pi(k-K_{\text{Shift}}(i_{\text{BW}}))\Delta_{F, \text{Pre-HE}}(t-T_{\text{GI, L-LTF}}-T_{\text{CS, HE}}(m))) \right)$$

where

$T_{\text{CS, HE}}(m)$ represents the cyclic shift for space-time stream m as defined in 27.3.11.2.2

$Q_k^{(i_{\text{Seg}})}$ is defined in 27.3.10

$A_{\text{HE-LTF}}^k$ is defined in Equation (27-55)

27.3.11.5 L-SIG field

The L-SIG field is used to communicate rate and length information. The structure of the L-SIG field is defined in Figure 17-5.

In an HE PPDU, the RATE field shall be set to the value representing 6 Mb/s in the “Rate (20 MHz channel spacing)” column of Table 17-6. In a non-HT duplicate PPDU, the RATE field is defined in 17.3.4.2 using the L_DATARATE parameter in the TXVECTOR.

For an HE TB PPDU, the LENGTH field is set to the TXVECTOR parameter L_LENGTH. For an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU, the LENGTH field is set to the value given by the Equation (27-11).

$$\text{Length} = \left\lceil \frac{\text{TXTIME} - \text{SignalExtension} - 20}{4} \right\rceil \times 3 - 3 - m \quad (27-11)$$

where

TXTIME is defined in 27.4.3 (in μs)

m is 1 for an HE MU PPDU and HE ER SU PPDU and 2 otherwise

SignalExtension is 0 μs if the TXVECTOR parameter NO_SIG_EXTN is true and is aSignalExtension as defined in Table 27-54 if the TXVECTOR parameter NO_SIG_EXTN is false

In a non-HT duplicate PPDU, the LENGTH field is defined in 17.3.4.3 using the L_LENGTH parameter in the TXVECTOR.

The Reserved (R) field shall be set to 0.

The Parity (P) field has the even parity of bits 0-16.

The SIGNAL TAIL field shall be set to 0.

The L-SIG field shall be encoded, interleaved, and mapped following the steps described in 17.3.5.6, 17.3.5.7, and 17.3.5.8. The stream of 48 complex numbers generated by these steps is denoted by

$d_k, k = 0, \dots, 47$ and is mapped to subcarriers $[-26, 26]$. In addition, values $[-1, -1, -1, 1]$ are mapped to the extra subcarriers $[-28, -27, 27, 28]$ of the L-SIG field of a 20 MHz HE PPDU. Subcarriers $[-28, -27, 27, 28]$ are also BPSK modulated. Pilots shall be inserted as described in 17.3.5.9.

If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, the time domain waveform of the L-SIG field, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , shall be as given by Equation (27-12).

$$r_{L-SIG}^{(i_{Seg}, i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \cdot N_{L-SIG}^{Tone} \cdot \frac{|\Omega_{20MHz}|}{N_{20MHz}}}} w_{T_{L-SIG}}(t) \quad (27-12)$$

$$\sum_{i_{BW} \in \Omega_{20MHz}} \sum_{k=-28}^{28} \left(\Upsilon_{(k-K_{Shift}(i_{BW})), BW} \eta_{L-SIG, k}(D_{k, 20} + p_0 P_k) \cdot \exp(j2\pi(k - K_{Shift}(i_{BW}))\Delta_{F, Pre-HE}(t - T_{GI, Pre-HE} - T_{CS}^{i_{TX}})) \right)$$

where

$\eta_{L-SIG, k}$ is a PPDU-dependent scaling factor for the L-SIG field on the subcarrier k defined as follows:

$$\eta_{L-SIG, k} = \begin{cases} \sqrt{2}, & k = \pm 28, \pm 27 \text{ for an HE ER SU PPDU} \\ \left[\frac{1}{\sqrt{2}}, 1 \right], & \text{for an HE TB PPDU} \\ 1, & \text{otherwise} \end{cases}$$

$T_{GI, Pre-HE}$ is given in Table 27-12

$$K_{Shift}(i) = (N_{20MHz} - 1 - 2i) \cdot 32$$

$$D_{k, 20} = \begin{cases} 0, & k = 0, \pm 7, \pm 21 \\ -1, & k = -28, -27, 27 \\ 1, & k = 28 \\ d_{M'_{20}(k)}, & \text{otherwise} \end{cases}$$

$$M'_{20}(k) = \begin{cases} k + 26, & -26 \leq k \leq -22 \\ k + 25, & -20 \leq k \leq -8 \\ k + 24, & -6 \leq k \leq -1 \\ k + 23, & 1 \leq k \leq 6 \\ k + 22, & 8 \leq k \leq 20 \\ k + 21, & 22 \leq k \leq 26 \end{cases}$$

P_k is defined in 17.3.5.10

p_0 is the first pilot value in the sequence defined in 17.3.5.10

N_{L-SIG}^{Tone} is defined in Table 27-16

$T_{CS}^{i_{TX}}$ represents the cyclic shift for transmit chain i_{TX} with a value given in 27.3.11.2.1

NOTE— $M'_{20}(k)$ is a “reverse” function of the function $M(k)$ defined in 17.3.5.10.

If the TXVECTOR parameter BEAM_CHANGE is 0, the time domain waveform of the L-SIG field, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , shall be as given by Equation (27-13).

$$r_{L-SIG}^{(i_{Seg}, i_{TX})}(t) = \frac{1}{\sqrt{N_{STS} \cdot N_{L-SIG}^{Tone} \cdot \frac{|\Omega_{20MHz}|}{N_{20MHz}}}} w_{T_{L-SIG}}(t) \quad (27-13)$$

$$\sum_{i_{BW}=0}^{N_{20MHz}-1} \sum_{k=-28}^{28} \sum_{m=1}^{N_{STS}} \left(\left[Q_4^{(i_{Seg})}(k - K_{Shift}(i_{BW})) \right]_{i_{TX}, m} \left[A_{HE-LTF}^{4(k - K_{Shift}(i_{BW}))} \right]_{m, 1} \eta_{L-SIG, k}(D_{k, 20} + p_0 P_k) \right) \cdot \exp(j2\pi(k - K_{Shift}(i_{BW}))\Delta_{F, Pre-HE}(t - T_{GI, Pre-HE} - T_{CS, HE}(m)))$$

where

- $T_{CS, HE}(m)$ represents the cyclic shift for space-time stream m as defined in 27.3.11.2.2
- $Q_k^{(i_{Seg})}$ is defined in 27.3.10
- A_{HE-LTF}^k is defined in Equation (27-55)

27.3.11.6 RL-SIG field

The RL-SIG field is a repeat of the L-SIG field and is used to differentiate an HE PPDU from a non-HT PPDU, HT PPDU, and VHT PPDU.

If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, the time domain waveform of the RL-SIG field, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , shall be as given by Equation (27-14).

$$r_{RL-SIG}^{(i_{Seg}, i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \cdot N_{RL-SIG}^{Tone} \cdot \frac{|\Omega_{20MHz}|}{N_{20MHz}}}} w_{T_{RL-SIG}}(t) \quad (27-14)$$

$$\sum_{i_{BW} \in \Omega_{20MHz}} \sum_{k=-28}^{28} \left(\Upsilon_{(k - K_{Shift}(i_{BW})), BW} \eta_{L-SIG, k}(D_{k, 20} + p_1 P_k) \right) \cdot \exp(j2\pi(k - K_{Shift}(i_{BW}))\Delta_{F, Pre-HE}(t - T_{GI, Pre-HE} - T_{CS}^{i_{TX}}))$$

If the TXVECTOR parameter BEAM_CHANGE is 0, the time domain waveform of the RL-SIG field, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , shall be as given by Equation (27-15).

$$r_{RL-SIG}^{(i_{Seg}, i_{TX})}(t) = \frac{1}{\sqrt{N_{STS} \cdot N_{RL-SIG}^{Tone} \cdot \frac{|\Omega_{20MHz}|}{N_{20MHz}}}} w_{T_{RL-SIG}}(t) \quad (27-15)$$

$$\sum_{i_{BW}=0}^{N_{20MHz}-1} \sum_{k=-28}^{28} \sum_{m=1}^{N_{STS}} \left(\left[Q_4^{(i_{Seg})}(k - K_{Shift}(i_{BW})) \right]_{i_{TX}, m} \left[A_{HE-LTF}^{4(k - K_{Shift}(i_{BW}))} \right]_{m, 1} \eta_{L-SIG, k}(D_{k, 20} + p_1 P_k) \right) \cdot \exp(j2\pi(k - K_{Shift}(i_{BW}))\Delta_{F, Pre-HE}(t - T_{GI, Pre-HE} - T_{CS, HE}(m)))$$

where

- p_1 is the second pilot value in the sequence defined in 17.3.5.10
- other variables are defined after Equation (27-1), Equation (27-3), Equation (27-4), Equation (27-6), Equation (27-8), and Equation (27-12)

27.3.11.7 HE-SIG-A field

27.3.11.7.1 General

The HE-SIG-A field carries information necessary to interpret HE PPDU. The integer fields of the HE-SIG-A field are transmitted in unsigned binary format, LSB first, where the LSB is in the lowest numbered bit position.

27.3.11.7.2 Content

The HE-SIG-A field for an HE SU PPDU or an HE ER SU PPDU contains the fields listed in Table 27-18.

Table 27-18—HE-SIG-A field of an HE SU PPDU and HE ER SU PPDU

Bit	Field	Number of bits	Description
HE-SIG-A1 subfield of HE-SIG-A field			
B0	Format	1	Differentiate an HE SU PPDU and HE ER SU PPDU from an HE TB PPDU: Set to 1 for an HE SU PPDU and HE ER SU PPDU.
B1	Beam Change	1	Set to 1 to indicate that the pre-HE modulated fields of the PPDU may be spatially mapped differently from the first symbol of the HE-LTF. Equation (27-6), Equation (27-9), Equation (27-12), Equation (27-14), Equation (27-16), and Equation (27-18) apply if the Beam Change field is 1. Set to 0 to indicate that the pre-HE modulated fields of the PPDU are spatially mapped the same way as the first symbol of the HE-LTF on each subcarrier. Equation (27-8), Equation (27-10), Equation (27-13), Equation (27-15), Equation (27-17), and Equation (27-19) apply if the Beam Change field is 0.
B2	UL/DL	1	Indicates whether the PPDU is sent UL or DL. Set to 1 if the PPDU is addressed to an AP. Set to 0 otherwise. See TXVECTOR parameter UPLINK_FLAG.
B3–B6	HE-MCS	4	For an HE SU PPDU: Set to n for HE-MCS n , where $n = 0, 1, 2, \dots, 11$. Values 12–15 are reserved. For an HE ER SU PPDU with Bandwidth field set to 0 (242-tone RU): Set to n for HE-MCS n , where $n = 0, 1, 2$. Values 3–15 are reserved. For an HE ER SU PPDU with Bandwidth field set to 1 (upper frequency 106-tone RU): Set to 0 for HE-MCS 0. Values 1–15 are reserved.
B7	DCM	1	Indicates whether DCM is applied to the Data field for the HE-MCS indicated. See Table 27-19 for the interpretation of this field. NOTE—DCM is applicable to only HE-MCSs 0, 1, 3, and 4. DCM is applicable to only 1 and 2 spatial streams.
B8–B13	BSS Color	6	An identifier of the BSS. See TXVECTOR parameter BSS_COLOR.

Table 27-18—HE-SIG-A field of an HE SU PPDU and HE ER SU PPDU (continued)

Bit	Field	Number of bits	Description
B14	Reserved	1	Reserved and set to 1.
B15–B18	Spatial Reuse	4	Indicates whether spatial reuse modes are allowed during the transmission of this PPDU: Set to a value from Table 27-22 (see 26.11.6 and 26.10).
B19–B20	Bandwidth	2	For an HE SU PPDU: Set to 0 for 20 MHz. Set to 1 for 40 MHz. Set to 2 for 80 MHz. Set to 3 for 160 MHz and 80+80 MHz. For an HE ER SU PPDU: Set to 0 for 242-tone RU. Set to 1 for upper frequency 106-tone RU within the primary 20 MHz. Values 2 and 3 are reserved.
B21–B22	GI+HE-LTF Size	2	Indicates the GI duration and HE-LTF size. See Table 27-19 for the interpretation of this field.
B23–B25	NSTS And Midamble Periodicity	3	If the Doppler field is 0, indicates the number of space-time streams. Set to the number of space-time streams minus 1. For an HE ER SU PPDU, values 2–7 are reserved. If the Doppler field is 1, then B23-B24 indicates the number of space time streams, up to 4, and B25 indicates the midamble periodicity. B23-B24 is set to the number of space time streams minus 1. For an HE ER SU PPDU, values 2 and 3 are reserved. B25 is set to 0 for 10 symbol midamble periodicity, to 1 for 20 symbol midamble periodicity. See TXVECTOR parameter MIDAMBLE_PERIODICITY.
HE-SIG-A2 subfield of HE-SIG-A field			
B0–B6	TXOP	7	Set to 127 to indicate no duration information. Set to a value less than 127 to indicate the closest minimum bound on the duration information for NAV setting and protection of the TXOP as follows: If B0 is 0, the TXOP duration indicated is B1-B6, in units of 8 μ s. If B0 is 1, the TXOP duration indicated is B1-B6, in units of 128 μ s, plus 512 μ s. See TXVECTOR parameter TXOP_DURATION.
B7	Coding	1	Indicates whether BCC or LDPC is used: Set to 0 to indicate BCC Set to 1 to indicate LDPC
B8	LDPC Extra Symbol Segment	1	Indicates the presence of the LDPC extra symbol segment: Set to 1 if an LDPC extra symbol segment is present. Set to 0 if an LDPC extra symbol segment is not present. Reserved and set to 1 if the Coding field is 0.
B9	STBC	1	Indicates whether STBC is applied to the Data field. See Table 27-19 for the interpretation of this field.

Table 27-18—HE-SIG-A field of an HE SU PPDU and HE ER SU PPDU (continued)

Bit	Field	Number of bits	Description
B10	Beamformed	1	Set to 1 if a beamforming steering matrix is applied to the HE modulated fields. Set to 0 otherwise.
B11–B12	Pre-FEC Padding Factor	2	Indicates the pre-FEC padding factor: Set to 0 to indicate a pre-FEC padding factor of 4. Set to 1 to indicate a pre-FEC padding factor of 1. Set to 2 to indicate a pre-FEC padding factor of 2. Set to 3 to indicate a pre-FEC padding factor of 3.
B13	PE Disambiguity	1	Indicates PE disambiguity as defined in 27.3.13.
B14	Reserved	1	Reserved and set to 1.
B15	Doppler	1	Set to 1 if one of the following applies: — The number of OFDM symbols in the Data field is larger than the signaled midamble periodicity plus 1 and the midamble is present. — The number of OFDM symbols in the Data field is less than or equal to the signaled midamble periodicity plus 1 (see 27.3.12.16, the midamble is not present, but the channel is fast varying. It recommends that midamble may be used for the PPDUs of the reverse link. Set to 0 otherwise.
B16–B19	CRC	4	CRC for bits 0–41 of the HE-SIG-A field (see 27.3.11.7.3). Bits 0–41 of the HE-SIG-A field correspond to bits 0–25 of the HE-SIG-A1 subfield followed by bits 0–15 of the HE-SIG-A2 subfield.
B20–B25	Tail	6	Used to terminate the trellis of the convolutional decoder. Set to 0.

For an HE SU PPDU and HE ER SU PPDU, the DCM, STBC, and GI+HE-LTF Size subfields in the HE-SIG-A field are interpreted as defined in Table 27-19.

Table 27-19—Interpretation of DCM, STBC, and GI+HE-LTF Size subfields

Subfields in the HE-SIG-A field of an HE SU PPDU or HE ER SU PPDU (see Table 27-18)			Interpretation		
DCM	STBC	GI+HE-LTF Size	DCM Applied	STBC Applied	GI+HE-LTF size
0	0	0	No	No	1x HE-LTF and 0.8 μ s GI
		1			2x HE-LTF and 0.8 μ s GI
		2			2x HE-LTF and 1.6 μ s GI
		3			4x HE-LTF and 3.2 μ s GI

Table 27-19—Interpretation of DCM, STBC, and GI+HE-LTF Size subfields (*continued*)

Subfields in the HE-SIG-A field of an HE SU PPDU or HE ER SU PPDU (see Table 27-18)			Interpretation		
DCM	STBC	GI+HE-LTF Size	DCM Applied	STBC Applied	GI+HE-LTF size
1	0	0	Yes	No	1x HE-LTF and 0.8 μs GI
		1			2x HE-LTF and 0.8 μs GI
		2			2x HE-LTF and 1.6 μs GI
		3			4x HE-LTF and 3.2 μs GI
0	1	0	No	Yes	1x HE-LTF and 0.8 μs GI
		1			2x HE-LTF and 0.8 μs GI
		2			2x HE-LTF and 1.6 μs GI
		3			4x HE-LTF and 3.2 μs GI
1	1	0–2	Reserved		
		3	No	No	4x HE-LTF and 0.8 μs GI

The HE-SIG-A field of an HE MU PPDU contains the fields listed in Table 27-20.

Table 27-20—HE-SIG-A field of an HE MU PPDU

Bit	Field	Number of bits	Description
HE-SIG-A1 subfield of HE-SIG-A field			
B0	UL/DL	1	Indicates whether the PPDU is sent UL or DL: Set to 1 if the PPDU is addressed to an AP. Set to 0 otherwise. See TXVECTOR parameter UPLINK_FLAG. NOTE—The TDLS peer can identify the TDLS frame by To DS and From DS fields in the MAC header of the frame.
B1–B3	HE-SIG-B-MCS	3	Indicates the HE-MCS of the HE-SIG-B field: Set to 0 for HE-SIG-B-MCS 0. Set to 1 for HE-SIG-B-MCS 1. Set to 2 for HE-SIG-B-MCS 2. Set to 3 for HE-SIG-B-MCS 3. Set to 4 for HE-SIG-B-MCS 4. Set to 5 for HE-SIG-B-MCS 5. Values 6 and 7 are reserved.
B4	HE-SIG-B DCM	1	Set to 1 indicates that the HE-SIG-B field is modulated with DCM for the HE-MCS. Set to 0 indicates that the HE-SIG-B field is not modulated with DCM for the HE-MCS. NOTE—DCM is applicable to only HE-SIG-B-MCS 0, 1, 3, and 4.

Table 27-20—HE-SIG-A field of an HE MU PPDU (*continued*)

Bit	Field	Number of bits	Description
B5–B10	BSS Color	6	An identifier of the BSS. Set TXVECTOR parameter BSS_COLOR.
B11–B14	Spatial Reuse	4	Indicates whether spatial reuse modes are allowed during the transmission of this PPDU: Set to a value from Table 27-22 (see 26.11.6 and 26.10). See TXVECTOR parameter SPATIAL_REUSE.
B15–B17	Bandwidth	3	Set to 0 for 20 MHz. Set to 1 for 40 MHz. Set to 2 for 80 MHz non-preamble puncturing mode. Set to 3 for 160 MHz and 80+80 MHz non-preamble puncturing mode. If the HE-SIG-B Compression field is 0: Set to 4 for preamble puncturing in 80 MHz, where in the preamble the only punctured subchannel is the secondary 20 MHz channel. Set to 5 for preamble puncturing in 80 MHz, where in the preamble the only punctured subchannel is one of the two 20 MHz subchannels in secondary 40 MHz channel. Set to 6 for preamble puncturing in 160 MHz or 80+80 MHz, where in the preamble the only punctured subchannels are the secondary 20 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel. If two of the 20 MHz subchannels in the secondary 80 MHz channel are punctured, these are either the lower two or the higher two. No more than two adjacent 20 MHz subchannels are punctured across 160 MHz. Set to 7 for preamble puncturing in 160 MHz or 80+80 MHz, where in the preamble the only punctured subchannels are zero, one or both of the 20 MHz subchannels in the secondary 40 MHz channel and zero to two of the 20 MHz subchannels in the secondary 80 MHz channel; at least one 20 MHz subchannel is punctured. If two of the 20 MHz subchannels in the secondary 80 MHz channel are punctured, these are either the lower two or the higher two. No more than two adjacent 20 MHz subchannels are punctured across 160 MHz. If the HE-SIG-B Compression field is 1, then values 4–7 are reserved.

Table 27-20—HE-SIG-A field of an HE MU PPDU (continued)

Bit	Field	Number of bits	Description
B18–B21	Number Of HE-SIG-B Symbols Or MU-MIMO Users	4	If the HE-SIG-B Compression field is 0, indicates the number of OFDM symbols in the HE-SIG-B field: Set to the number of OFDM symbols in the HE-SIG-B field minus 1 if the number of OFDM symbols in the HE-SIG-B field is less than 16. Set to 15 to indicate that the number of OFDM symbols in the HE-SIG-B field is equal to 16 if Longer Than 16 HE-SIG-B OFDM Symbols Support subfield in the HE Capabilities element transmitted by at least one recipient STA is 0. Set to 15 to indicate that the number of OFDM symbols in the HE-SIG-B field is greater than or equal to 16 if the Longer Than 16 HE-SIG-B OFDM Symbols Support subfield in the HE Capabilities element transmitted by all the recipient STAs are 1 and if the HE-SIG-B-MCS field is set to 0, 1, 2, or 3 regardless of the value of the HE-SIG-B DCM field, or the HE-SIG-B-MCS field is set to 4 and the HE-SIG-B DCM field is set to 1. The exact number of OFDM symbols in the HE-SIG-B field is calculated based on the number of User fields in the HE-SIG-B content channel, which is indicated by the Common field of the HE-SIG-B field in this case. If the HE-SIG-B Compression field is 1, indicates the number of users and is set to the number of users minus 1. If the number of users is greater than 1, then MU-MIMO is used in the HE modulated fields.
B22	HE-SIG-B Compression	1	Set to 0 if the Common field in the HE-SIG-B field is present. Set to 1 if the Common field in the HE-SIG-B field is not present.
B23–B24	GI+HE-LTF Size	2	Indicates the GI duration and HE-LTF size: Set to 0 to indicate a 4x HE-LTF and 0.8 μs GI. Set to 1 to indicate a 2x HE-LTF and 0.8 μs GI. Set to 2 to indicate a 2x HE-LTF and 1.6 μs GI. Set to 3 to indicate a 4x HE-LTF and 3.2 μs GI.
B25	Doppler	1	Set to 1 if one of the following applies: — The number of OFDM symbols in the Data field is larger than the signaled midamble periodicity plus 1 and the midamble is present. — The number of OFDM symbols in the Data field is less than or equal to the signaled midamble periodicity plus 1 (see 27.3.12.16, the midamble is not present, but the channel is fast varying. It recommends that midamble may be used for the PPDUs of the reverse link. Set to 0 otherwise.
HE-SIG-A2 subfield of HE-SIG-A field			
B0–B6	TXOP	7	Set to 127 to indicate no duration information. Set to a value less than 127 to indicate the closest minimum bound on the duration information for NAV setting and protection of the TXOP as follows: If B0 is 0, the TXOP duration indicated is B1-B6, in units of 8 μs. If B0 is 1, the TXOP duration indicated is B1-B6, in units of 128 μs, plus 512 μs. See TXVECTOR parameter TXOP_DURATION.

Table 27-20—HE-SIG-A field of an HE MU PPDU (continued)

Bit	Field	Number of bits	Description
B7	Reserved	1	Reserved and set to 1.
B8–B10	Number of HE-LTF Symbols And Midamble Periodicity	3	If the Doppler field is 0, indicates the number of HE-LTF symbols: Set to 0 for 1 HE-LTF symbol. Set to 1 for 2 HE-LTF symbols. Set to 2 for 4 HE-LTF symbols. Set to 3 for 6 HE-LTF symbols. Set to 4 for 8 HE-LTF symbols. Other values are reserved. If the Doppler field is 1, B8–B9 indicates the number of HE-LTF symbols and B10 indicates midamble periodicity. B8–B9 is encoded as follows: 0 indicates 1 HE-LTF symbol. 1 indicates 2 HE-LTF symbols. 2 indicates 4 HE-LTF symbols. 3 is reserved. B10 is set to 0 to indicate 10 symbols midamble periodicity and set to 1 to indicate 20 symbols midamble periodicity.
B11	LDPC Extra Symbol Segment	1	Indication of the presence of an LDPC extra symbol segment: Set to 1 if an LDPC extra symbol segment is present. Set to 0 otherwise.
B12	STBC	1	In an HE MU PPDU where each RU includes no more than 1 user: Set to 1 to indicate all RUs are STBC encoded in the payload. Set to 0 to indicate all RUs are not STBC encoded in the payload. STBC does not apply to the HE-SIG-B field.
B13–B14	Pre-FEC Padding Factor	2	Indicates the pre-FEC padding factor: Set to 0 to indicate a pre-FEC padding factor of 4. Set to 1 to indicate a pre-FEC padding factor of 1. Set to 2 to indicate a pre-FEC padding factor of 2. Set to 3 to indicate a pre-FEC padding factor of 3.
B15	PE Disambiguity	1	Indicates PE disambiguity as defined in 27.3.13.
B16–B19	CRC	4	CRC for bits 0–41 of the HE-SIG-A field (see 27.3.11.7.3). Bits 0–41 of the HE-SIG-A field correspond to bits 0–25 of the HE-SIG-A1 subfield followed by bits 0–15 of the HE-SIG-A2 subfield.
B20–B25	Tail	6	Used to terminate the trellis of the convolutional decoder. Set to 0.

It is optional for an HE STA to receive a preamble punctured HE MU PPDU with the Bandwidth field in the HE-SIG-A field in the range of 4 to 7. An HE STA indicates that it is capable of receiving a preamble punctured HE MU PPDU with the Bandwidth field of the HE-SIG-A field in the range of 4 to 7 using the Punctured Preamble Rx subfield in the HE PHY Capabilities Information field in the HE Capabilities element (see 9.4.2.248).

The HE-SIG-A field for an HE TB PPDU contains the fields listed in Table 27-21.

Table 27-21—HE-SIG-A field of an HE TB PPDU

Bit	Field	Number of bits	Description
HE-SIG-A1 of the HE-SIG-A field			
B0	Format	1	Differentiate an HE SU PPDU and HE ER SU PPDU from an HE TB PPDU: Set to 0 for an HE TB PPDU.
B1–B6	BSS Color	6	An identifier of the BSS. Set TXVECTOR parameter BSS_COLOR.
B7–B10	Spatial Reuse 1	4	Indicates whether specific spatial reuse modes are allowed in a subband of the PPDU during the transmission of this PPDU, and if PSR spatial reuse is allowed, indicates a value that is used to determine a limit on the transmit power of the PSRT PPDU. If the Bandwidth field indicates 20 MHz, 40 MHz, or 80 MHz, then this Spatial Reuse field applies to the first 20 MHz subband. If the Bandwidth field indicates 160/80+80 MHz, then this Spatial Reuse field applies to the first 40 MHz subband of the 160 MHz operating band. Set to a value from Table 27-23 for an HE TB PPDU (see 26.11.6 and 26.10). See the first value in the TXVECTOR parameter SPATIAL_REUSE.
B11–B14	Spatial Reuse 2	4	Indicates whether specific spatial reuse modes are allowed in a subband of the PPDU during the transmission of this PPDU, and if PSR spatial reuse is allowed, indicates a value that is used to determine a limit on the transmit power of the PSRT PPDU. If the Bandwidth field indicates 40 MHz or 80 MHz: This Spatial Reuse field applies to the second 20 MHz subband. If the STA operating channel width is 20 MHz, then this field is set to the same value as the Spatial Reuse 1 field. If the STA operating channel width is 40 MHz in the 2.4 GHz band, this field is set to the same value as the Spatial Reuse 1 field. If the Bandwidth field indicates 160/80+80 MHz, then this Spatial Reuse field applies to the second 40 MHz subband of the 160 MHz operating band. Set to a value from Table 27-23 for an HE TB PPDU (see 26.11.6 and 26.10). See the second value in the TXVECTOR parameter SPATIAL_REUSE, if present.

Table 27-21—HE-SIG-A field of an HE TB PPDU (continued)

Bit	Field	Number of bits	Description
B15–B18	Spatial Reuse 3	4	Indicates whether specific spatial reuse modes are allowed in a subband of the PPDU during the transmission of this PPDU, and if PSR spatial reuse is allowed, indicates a value that is used to determine a limit on the transmit power of the PSRT PPDU. If the Bandwidth field indicates 80 MHz: This Spatial Reuse field applies to the third 20 MHz subband. If the STA operating channel width is 20 MHz or 40 MHz, this field is set to the same value as the Spatial Reuse 1 field. If the Bandwidth field indicates 160/80+80 MHz: This Spatial Reuse field applies to the third 40 MHz subband of the 160 MHz operating band. If the STA operating channel width is 80+80 MHz, this field is set to the same value as the Spatial Reuse 1 field. Set to a value from Table 27-23 for an HE TB PPDU (see 26.11.6 and 26.10). See the third value in the TXVECTOR parameter SPATIAL_REUSE, if present.
B19–B22	Spatial Reuse 4	4	Indicates whether specific spatial reuse modes are allowed in a subband of the PPDU during the transmission of this PPDU, and if PSR spatial reuse is allowed, indicates a value that is used to determine a limit on the transmit power of the PSRT PPDU. If the Bandwidth field indicates 80 MHz: This Spatial Reuse field applies to the fourth 20 MHz subband. If the STA operating channel width is 20 MHz, then this field is set to the same value as the Spatial Reuse 1 field. If the STA operating channel width is 40 MHz, then this field is set to the same value as the Spatial Reuse 2 field. If the Bandwidth field indicates 160/80+80 MHz: This Spatial Reuse field applies to the fourth 40 MHz subband of the 160 MHz operating band. If the STA operating channel width is 80+80 MHz, then this field is set to same value as the Spatial Reuse 2 field. Set to a value from Table 27-23 for an HE TB PPDU (see 26.11.6 and 26.10). See the fourth value in the TXVECTOR parameter SPATIAL_REUSE, if present.
B23	Reserved	1	Reserved and set to 1. NOTE—Unlike other Reserved fields in the HE-SIG-A field of the HE TB PPDU, B23 does not have a corresponding bit in the Trigger frame.
B24–B25	Bandwidth	2	Set to 0 for 20 MHz. Set to 1 for 40 MHz. Set to 2 for 80 MHz. Set to 3 for 160 MHz and 80+80 MHz.

Table 27-21—HE-SIG-A field of an HE TB PPDU (continued)

Bit	Field	Number of bits	Description
HE-SIG-A2 of the HE-SIG-A field			
B0–B6	TXOP	7	Set to 127 to indicate no duration information. Set to a value less than 127 to indicate the closest minimum bound on the duration information for NAV setting and protection of the TXOP as follows: If B0 is 0, the TXOP duration indicated is B1–B6, in units of 8 μ s. If B0 is 1, the TXOP duration indicated is B1–B6, in units of 128 μ s, plus 512 μ s. See TXVECTOR parameter TXOP_DURATION.
B7–B15	Reserved	9	Reserved and set to value indicated in the UL HE-SIG-A2 Reserved subfield in the Trigger frame.
B16–B19	CRC	4	CRC of bits 0–41 of the HE-SIG-A field. See 27.3.11.7.3. Bits 0–41 of the HE-SIG-A field correspond to bits 0–25 of the HE-SIG-A1 subfield followed by bits 0–15 of the HE-SIG-A2 subfield.
B20–B25	Tail	6	Used to terminate the trellis of the convolutional decoder. Set to 0.

The four Spatial Reuse fields, 1, 2, 3, and 4, are arranged in increasing order of frequency and correspond to the following:

- For 20 MHz, one Spatial Reuse field corresponding to the entire 20 MHz (the other 3 Spatial Reuse fields indicate the same value). The Spatial Reuse fields apply to only the 20 MHz used for the transmission.
- For 40 MHz, two Spatial Reuse fields with the Spatial Reuse 3 field identical in value to the Spatial Reuse 1 field and the Spatial Reuse 4 field identical in value to the Spatial Reuse 2 field. Each pair of Spatial Reuse fields applies to only the corresponding 20 MHz used for the transmission.
- For 80 MHz, four Spatial Reuse fields, one for each 20 MHz subchannel.
 - For an OFDMA transmission of a given BW, each of the Spatial Reuse fields that corresponds to a 20 MHz subband is also applicable to the 242-tone RU that is most closely aligned in frequency (in the tone plan for that BW) with the aforementioned 20 MHz subband. The correspondence from an Spatial Reuse field to a 242-tone RU also holds for any RU within the 242-tone RU. The above also implies that a 20 MHz OBSS STA uses the Spatial Reuse field corresponding to its 20 MHz channel; a 40 MHz OBSS STA located on the lower frequency half of the 80 MHz BSS uses Spatial Reuse 1 field and Spatial Reuse 2 field values; and a 40 MHz OBSS STA located on the upper frequency half of the 80 MHz BSS uses Spatial Reuse 3 field and Spatial Reuse 4 field values.
- For 160 MHz and 80+80 MHz, four Spatial Reuse fields, one for each 40 MHz subchannel.
 - For an OFDMA transmission of a given BW, each of the Spatial Reuse fields that corresponds to a 40 MHz subband is also applicable to the 484-tone RU that is most closely aligned in frequency (in the tone-plan of that BW) with the aforementioned 40 MHz subband. The correspondence from an Spatial Reuse field to a 484-tone RU also holds for any RU within the 484-tone RU.

Table 27-22 defines the encoding for the Spatial Reuse field for an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU.

Table 27-22—Spatial Reuse field encoding for an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU

Value	Meaning
0	PSR_DISALLOW
1–12	Reserved
13	SR_RESTRICTED
14	SR_DELAYED
15	PSR_AND_NON_SRG_OBSS_PD_PROHIBITED

Table 27-23 defines the encoding for the Spatial Reuse 1, Spatial Reuse 2, Spatial Reuse 3, and Spatial Reuse 4 fields for an HE TB PPDU.

Table 27-23—Spatial Reuse field encoding for an HE TB PPDU

Value	Meaning
0	PSR_DISALLOW
1	PSR = –80 dBm
2	PSR = –74 dBm
3	PSR = –68 dBm
4	PSR = –62 dBm
5	PSR = –56 dBm
6	PSR = –50 dBm
7	PSR = –47 dBm
8	PSR = –44 dBm
9	PSR = –41 dBm
10	PSR = –38 dBm
11	PSR = –35 dBm
12	PSR = –32 dBm
13	PSR = –29 dBm
14	PSR ≥ –26 dBm
15	PSR_AND_NON_SRG_OBSS_PD_PROHIBITED

27.3.11.7.3 CRC computation

The CRC computation defined in this subclause applies to the HE-SIG-A field, the Common field of the HE-SIG-B field, and the User Block field of the HE-SIG-B field.

The CRC is calculated over bits 0 to 41 of the HE-SIG-A field and over bits 0 to L of the HE-SIG-B field ($L = x$ for each Common field where $x = N \times 8$ if the Center 26-tone RU field is present and $x = N \times 8 - 1$ otherwise; $L = 20$ for an User Block field that contains one User field; and $L = 41$ for an User Block field that contains two User fields). Bits 0 to 41 of the HE-SIG-A field correspond to bits 0–25 of the HE-SIG-A1 subfield followed by bits 0–15 of the HE-SIG-A2 subfield. Refer to Table 27-24 for N and the conditions under which the Center 26-tone RU field is present.

The value of the CRC field shall be the 1s complement of

$$crc(D) = (M(D) + I(D))D^8 \bmod G(D)$$

where

$$M(D) = \sum_{i=0}^L m_{L-i} D^i$$

$$I(D) = \sum_{i=L-7}^L D^i$$

$G(D)$ is defined in 19.3.9.4.4

$$crc(D) = c_0 D^7 + c_1 D^6 + \dots + c_6 D + c_7$$

m_L is the serial input shown in Figure 27-24

The CRC field is transmitted from c_4 to c_7 with c_7 first.

Figure 27-24 shows the operation of the CRC. First, the shift register is reset to all 1s. The bits are then passed through the XOR operation at the input. When the last bit has entered, the output is generated by shifting the bits out of the shift register, c_7 first, through an inverter.

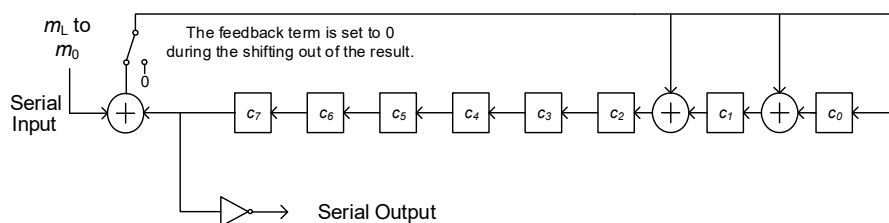


Figure 27-24—CRC calculation

As an example, if bits $\{m_0 \dots m_{41}\}$ are given by $\{1\ 1\ 0\ 1\ 1\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 1\ 0\ 0\ 1\ 1\ 0\ 1\ 0\}$, the output bits $\{B_7 \dots B_4\}$, where B_7 is output first, are $\{0\ 1\ 1\ 1\}$.

27.3.11.7.4 Encoding and modulation

For an HE SU PPDU, HE MU PPDU, and HE TB PPDU, the HE-SIG-A field is composed of two subfields, HE-SIG-A1 and HE-SIG-A2, each containing 26 data bits. The HE-SIG-A1 subfield is transmitted before the HE-SIG-A2 subfield. The data bits of the HE-SIG-A OFDM symbols shall be BCC encoded at rate $R = 1/2$, be interleaved, be mapped to a BPSK constellation, and have pilots inserted following the steps described in 17.3.5.6, 27.3.12.8, 17.3.5.8, and 17.3.5.9, respectively. The constellation mappings of the HE-SIG-A field in an HE SU PPDU, HE MU PPDU, and HE TB PPDU are shown in Figure 27-25. The first half and second half of the stream of 104 complex numbers generated by these steps (before pilot insertion) are divided into two groups of 52 complex numbers, where the first 52 complex numbers form the first OFDM symbol of the HE-SIG-A field and the second 52 complex numbers form the second OFDM symbol of the HE-SIG-A field, respectively.

If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, the time domain waveform for the HE-SIG-A field of an HE SU PPDU, HE MU PPDU, and HE TB PPDU, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , shall be as specified in Equation (27-16).

$$r_{\text{HE-SIG-A}}^{(i_{Seg} i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \cdot N_{\text{HE-SIG-A}}^{\text{Tone}} \cdot \frac{\Omega_{20\text{MHz}}}{N_{20\text{MHz}}}}} \sum_{n=0}^1 w_{T_{SYML}}(t - nT_{SYML}) \sum_{i_{BW} \in \Omega_{20\text{MHz}}} \sum_{k=-28}^{28} \eta_{\text{HE-SIG-A}, k} \left(\Upsilon_{(k - K_{\text{Shift}}(i_{BW})), \text{BW}}(D_{k, n, 20} + p_n + 2P_k) \cdot \exp(j2\pi(k - K_{\text{Shift}}(i_{BW}))\Delta_{F, \text{Pre-HE}}(t - nT_{SYML} - T_{GI, \text{Pre-HE}} - T_{CS}^{i_{TX}})) \right) \quad (27-16)$$

where

T_{SYML} is given in Table 27-12

$$K_{\text{Shift}}(i) = (N_{20\text{MHz}} - 1 - 2i) \cdot 32$$

$$D_{k, n, 20} = \begin{cases} 0, & k = 0, \pm 7, \pm 21 \\ d_{M_{20}^r(k), n}, & \text{otherwise} \end{cases}$$

$$M_{20}^r(k) = \begin{cases} k + 28, & -28 \leq k \leq -22 \\ k + 27, & -20 \leq k \leq -8 \\ k + 26, & -6 \leq k \leq -1 \\ k + 25, & 1 \leq k \leq 6 \\ k + 24, & 8 \leq k \leq 20 \\ k + 23, & 22 \leq k \leq 28 \end{cases}$$

$$\eta_{\text{HE-SIG-A}, k} = \begin{cases} \left[\frac{1}{\sqrt{2}}, 1 \right], & \text{for an HE TB PPDU} \\ 1, & \text{otherwise} \end{cases}$$

P_k and p_n are defined in 17.3.5.10

$N_{\text{HE-SIG-A}}^{\text{Tone}}$ is defined in Table 27-16

$T_{CS}^{i_{TX}}$ represents the cyclic shift for transmit chain i_{TX} with a value given in 27.3.11.2.1

If the TXVECTOR parameter BEAM_CHANGE is 0, the time domain waveform of the HE-SIG-A field shall be as given by Equation (27-17).

$$r_{\text{HE-SIG-A}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = \frac{1}{\sqrt{N_{\text{STS}} \cdot N_{\text{HE-SIG-A}}^{\text{Tone}} \cdot \frac{\Omega_{20\text{MHz}}}{N_{20\text{MHz}}}}} \sum_{n=0}^1 w_{T_{\text{SYML}}}(t - nT_{\text{SYML}}) \quad (27-17)$$

$$\sum_{i_{\text{BW}}=0}^{N_{20\text{MHz}}-1} \sum_{k=-28}^{28} \sum_{m=1}^{N_{\text{STS}}} \left(\left[Q_4^{(i_{\text{Seg}})}(k - K_{\text{Shift}}(i_{\text{BW}})) \right]_{i_{\text{TX}}, m} \left[A_{\text{HE-LTF}}^{4(k - K_{\text{Shift}}(i_{\text{BW}}))} \right]_{m, 1} (D_{k, n, 20} + p_{n+2} P_k) \right) \cdot \exp(j2\pi(k - K_{\text{Shift}}(i_{\text{BW}}))\Delta_{F, \text{Pre-HE}}(t - nT_{\text{SYML}} - T_{\text{GI, Pre-HE}} - T_{\text{CS, HE}}(m)))$$

where

$T_{\text{CS, HE}}(m)$ represents the cyclic shift for space-time stream m as defined in 27.3.11.2.2

$Q_k^{(i_{\text{Seg}})}$ is defined in 27.3.10

$A_{\text{HE-LTF}}^k$ is defined in Equation (27-55)

For an HE ER SU PPDU, the HE-SIG-A field is composed of four subfields: HE-SIG-A1, HE-SIG-A1-R, HE-SIG-A2, and HE-SIG-A2-R. Each subfield contains 26 data bits. These four subfields are transmitted sequentially from HE-SIG-A1 to HE-SIG-A2-R. The data bits of the HE-SIG-A1 and HE-SIG-A2 subfields shall be BCC encoded at rate $R = 1/2$, be interleaved, be mapped to a BPSK constellation, and have pilots inserted. The HE-SIG-A1-R subfield has the same encoded bits as the HE-SIG-A1 subfield, and the encoded bits shall be mapped to a QPSK constellation without interleaving and have pilots inserted. The constellation mappings of the HE-SIG-A field in an HE ER SU PPDU are shown in Figure 27-25. The QPSK constellation on the HE-SIG-A1-R subfield is used to differentiate between an HE ER SU PPDU and an HE MU PPDU when $m = 1$ in Equation (27-11). The HE-SIG-A2-R subfield has the same encoded bits as the HE-SIG-A2 subfield, and the encoded bits shall be mapped to a BPSK constellation without interleaving and have pilots inserted. BCC encoding, data interleaving, constellation mapping, and pilot insertion follow the steps described in 17.3.5.6, 27.3.12.8, 17.3.5.8, and 17.3.5.9, respectively.

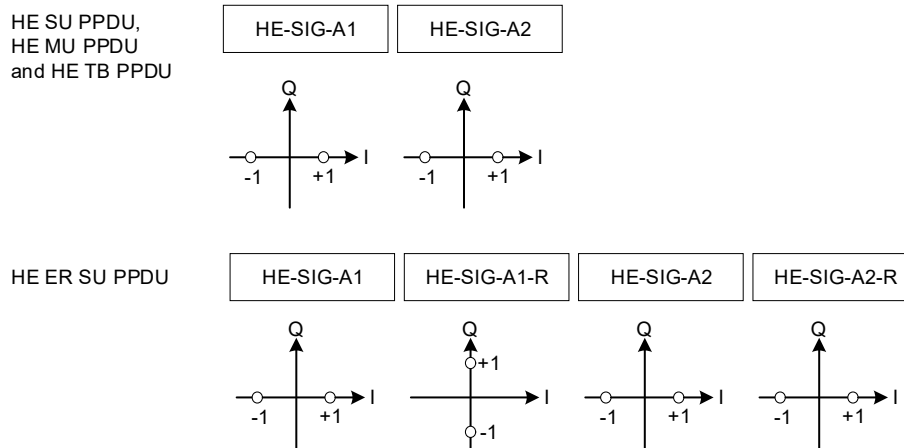


Figure 27-25—Data subcarrier constellation of HE-SIG-A symbols

If the TXVECTOR parameter BEAM_CHANGE is 1 or not present, the time domain waveform for the HE-SIG-A field in an HE ER SU PPDU shall be as specified in Equation (27-18).

$$r_{\text{HE-SIG-A}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = \frac{1}{\sqrt{N_{\text{TX}} \cdot N_{\text{HE-SIG-A}}^{\text{Tone}} \cdot \frac{|\Omega_{20\text{MHz}}|}{N_{20\text{MHz}}}}} \sum_{n=0}^3 w_{T_{\text{SYML}}}(t - nT_{\text{SYML}}) \sum_{k=-28}^{28} (R_n D_{k,n,20} + p_{n+2} P_k) \cdot \exp(j2\pi k \cdot \Delta_{F, \text{Pre-HE}}(t - nT_{\text{SYML}} - T_{\text{GI, Pre-HE}} - T_{\text{CS}}^{i_{\text{TX}}})) \quad (27-18)$$

where

R_n is a phase rotation vector defined as [1, j , 1, 1]

If the TXVECTOR parameter BEAM_CHANGE is 0, the time domain waveform for the HE-SIG-A field in an HE ER SU PPDU shall be as specified in Equation (27-19).

$$r_{\text{HE-SIG-A}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = \frac{1}{\sqrt{N_{\text{STS}} \cdot N_{\text{HE-SIG-A}}^{\text{Tone}} \cdot \frac{|\Omega_{20\text{MHz}}|}{N_{20\text{MHz}}}}} \sum_{n=0}^3 w_{T_{\text{SYML}}}(t - nT_{\text{SYML}}) \sum_{k=-28}^{28} \sum_{m=1}^{N_{\text{STS}}} \left(\left[Q_{4k}^{(i_{\text{Seg}})} \right]_{i_{\text{TX}}, m} \left[A_{\text{HE-LTF}}^{4k} \right]_{m, 1} (R_n D_{k,n,20} + p_{n+2} P_k) \cdot \exp(j2\pi k \cdot \Delta_{F, \text{Pre-HE}}(t - nT_{\text{SYML}} - T_{\text{GI, Pre-HE}} - T_{\text{CS, HE}}(m))) \right) \quad (27-19)$$

27.3.11.8 HE-SIG-B field

27.3.11.8.1 General

The HE-SIG-B field provides the necessary signaling, including the OFDMA and DL MU-MIMO resource allocation information, to allow the STAs to look up the corresponding resources to be used in the HE modulated fields of the PPDU. The integer fields of the HE-SIG-B field are transmitted in unsigned binary format, LSB first, where the LSB is in the lowest numbered bit position.

Dynamic split is defined as the split of User fields across HE-SIG-B content channels according to the Common field in each HE-SIG-B content channel and used when the HE-SIG-B Compression field in the HE-SIG-A field is set to 0.

Equitable split is defined as the split of User fields across HE-SIG-B content channels used when the HE-SIG-B Compression field in the HE-SIG-A field is set to 1.

27.3.11.8.2 HE-SIG-B content channels

The HE-SIG-B field of a 20 MHz HE MU PPDU contains one HE-SIG-B content channel. The HE-SIG-B field of an HE MU PPDU that is 40 MHz or wider contains two HE-SIG-B content channels.

The HE-SIG-B content channel format is shown in Figure 27-26. The HE-SIG-B content channel consists of a Common field, if present, followed by a User Specific field.

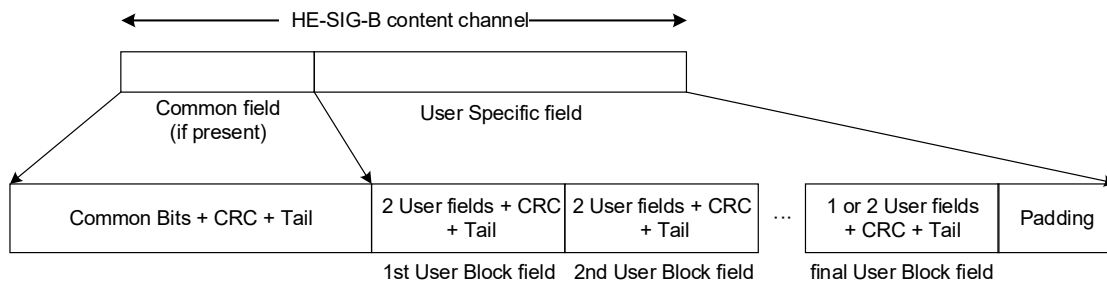


Figure 27-26—HE-SIG-B content channel format

If the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 1 (indicating full-bandwidth MU-MIMO transmission), then the Common field is not present, and the HE-SIG-B content channel consists of only the User Specific field. If the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 0, the Common field is present in the HE-SIG-B content channel.

The Common field of an HE-SIG-B content channel contains information regarding the resource unit allocation such as the RU assignment to be used in the HE modulated fields of the PPDU, the RUs allocated for MU-MIMO, and the number of users in MU-MIMO allocations. The Common field is defined in 27.3.11.8.3.

The union of the User Specific fields in the HE-SIG-B content channels contains information for all users in the PPDU on how to decode their payload. As shown in Figure 27-26, the User Specific field is organized into User Block fields that in turn contain User fields. See 27.3.11.8.4 for a description of the contents of the User Specific field.

See Annex Z for HE-SIG-B content examples.

27.3.11.8.3 Common field

This subclause is not applicable if the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 1.

The Common field format is defined in Table 27-24.

A 996-tone RU is referred to by two consecutive RU Allocation subfields per HE-SIG-B content channel, for both HE-SIG-B content channels. The two consecutive RU Allocation subfields per HE-SIG-B content channel are labeled the first RU Allocation subfield and the second RU Allocation subfield. A 484-tone RU is referred to by a single RU Allocation subfield per HE-SIG-B content channel, for both HE-SIG-B content channels. Smaller RUs are referred to by a single RU Allocation subfield in a single HE-SIG-B content channel. If a Common field is present in a 160 MHz or 80+80 MHz PPDU, a 2×996 tone RU is not permitted and cannot be indicated by the RU allocation subfield.

Table 27-24—Common field

Subfield	Number of subfields	Number of bits per subfield	Description
RU Allocation	N	8	N RU Allocation subfields are present in an HE-SIG-B content channel, where $N = 1$ if the Bandwidth field in the HE-SIG-A field is 0 or 1 (indicating a 20 MHz or 40 MHz HE MU PPDU) $N = 2$ if the Bandwidth field in the HE-SIG-A field is 2, 4, or 5 (indicating an 80 MHz HE MU PPDU) $N = 4$ if the Bandwidth field in the HE-SIG-A field is 3, 6, or 7 (indicating a 160 MHz or 80+80 MHz HE MU PPDU) Each RU Allocation subfield in an HE-SIG-B content channel corresponding to a 20 MHz frequency subchannel indicates the RU assignment, including the size of the RU(s) and their placement in the frequency domain, to be used in the HE modulated fields of the HE MU PPDU in the frequency domain and indicates information needed to compute the number of users allocated to each RU, where the subcarrier indices of the RU(s) meet the conditions in Table 27-25.
Center 26-tone RU	0 or 1	1	The Center 26-tone RU field is present if the Bandwidth field in the HE-SIG-A field indicates a bandwidth greater than 40 MHz and not present otherwise. If the Bandwidth field in the HE-SIG-A field is 2, 4, or 5 (indicating 80 MHz): Set to 1 to indicate that a user is allocated to the center 26-tone RU (see Figure 27-7) and that its User field is present in HE-SIG-B content channel 1; otherwise, set to 0. The same value is applied to both HE-SIG-B content channels. If the Bandwidth field in the HE-SIG-A field is 3, 6, or 7 (indicating 160 MHz or 80+80 MHz): For HE-SIG-B content channel 1, set to 1 to indicate that a user is allocated to the center 26-tone RU of the lower frequency 80 MHz; otherwise, set to 0. For HE-SIG-B content channel 2, set to 1 to indicate that a user is allocated to the center 26-tone RU of the higher frequency 80 MHz; otherwise, set to 0.
CRC	1	4	The CRC is calculated over bits 0 to $N \times 8$ if the Bandwidth field in the HE-SIG-A field indicates a bandwidth greater than 40 MHz, and bits 0 to $N \times 8 - 1$ otherwise. See 27.3.11.7.3.
Tail	1	6	Used to terminate the trellis of the convolutional decoder. Set to 0.

For an RU that is referred to by a first or only RU Allocation subfield in an HE-SIG-B content channel, the RU Allocation subfield encodes the number of User fields per RU contributed to the User Specific field in the same HE-SIG-B content channel as the RU Allocation subfield. This number is labeled $N_{user}(r, c)$ for RU r and HE-SIG-B content channel c as described in Table 27-26.

For an RU that is referred to by two RU Allocation subfields in an HE-SIG-B content channel (i.e., a 996-tone RU in a 160 MHz or 80+80 MHz PPDU), the second RU Allocation subfield in the HE-SIG-B content channel encodes zero additional User fields per RU contributed to the User Specific field in the same HE-SIG-B content channel as the RU Allocation subfield.

In an HE MU PPDU, an RU that is not allocated to a user can be indicated as follows:

- The Center 26-tone RU subfield in the Common field of the HE-SIG-B field is set to 0 (see Table 27-24).
- The RU Allocation subfield in the Common field of the HE-SIG-B field is set to a value between 16 and 31 or between 96 and 113 (see Table 27-26).
- Both RU Allocation subfields at the same position in each Common field of the two HE-SIG-B content channels are set to 114 (see Table 27-26).
- The STA-ID subfield in the User field of the HE-SIG-B field is set to 2046 (see 26.11.1 and 27.3.11.8.4).

The subcarriers in the HE modulated fields of a PPDU that correspond to an unallocated RU shall not be modulated.

If an RU is an unallocated RU, zero users are allocated to it. Otherwise, the number of users allocated to RU r is determined from the RU size and $N_{user}(r, c)$ as follows:

- If RU r is a 26-tone or 52-tone RU, then one user is allocated to the RU.
- If RU r is a 106-tone or 242-tone RU, then the number of users allocated to the RU is $N_{user}(r, c)$.
- If RU r is a 484-tone or larger RU, then the number of users allocated to the RU equals the number of User fields for the RU summed across both HE-SIG-B content channels, i.e., $N_{user}(r, 1) + N_{user}(r, 2)$.

NOTE 1—The exact dynamic split of User fields between the two content channels, $N_{user}(r, 1)$ and $N_{user}(r, 2)$, is not specified and might be used to reduce any disparity in the number of User fields between content channels.

NOTE 2—If the number of users per RU is greater than one, then the users in the RU are multiplexed using MU-MIMO.

For a 996-tone RU, for each HE-SIG-B content channel, the first 8-bit RU Allocation subfield referring to the RU may use values in the range 208–215 (11010 $y_2y_1y_0$ in binary representation) as in Table 27-26 with $y_2y_1y_0$ indicating the number of User fields signaled in the corresponding content channel, while the second 8-bit RU Allocation subfield referring to the RU shall be set to 115 (01110011 in binary representation).

NOTE 3—Although there are two RU Allocation subfields per HE-SIG-B content channel for the users of a 996-tone RU, each user is described by only one User field, which is located in one HE-SIG-B content channel.

As defined in Table 27-24 and Table 27-25, the Center 26-tone RU field carries the same value in both HE-SIG-B content channels. The User field that corresponds to the center 26-tone RU is carried in HE-SIG-B content channel 1.

Table 27-25—RUs associated with each RU Allocation subfield for each HE-SIG-B content channel and PPDU bandwidth

PPDU bandwidth	RU Allocation subfield and Center 26-tone RU subfield (if present)	RUs in the subcarrier range, or overlapping with the subcarrier range if the RU is larger than a 242-tone RU
20 MHz	The RU Allocation subfield in a single HE-SIG-B content channel	[−122:122]
40 MHz	The RU Allocation subfield in HE-SIG-B content channel 1	[−244:−3]
	The RU Allocation subfield in HE-SIG-B content channel 2	[3:244]

Table 27-25—RUs associated with each RU Allocation subfield for each HE-SIG-B content channel and PPDU bandwidth (*continued*)

PPDU bandwidth	RU Allocation subfield and Center 26-tone RU subfield (if present)	RUs in the subcarrier range, or overlapping with the subcarrier range if the RU is larger than a 242-tone RU
80 MHz	The first RU Allocation subfield in HE-SIG-B content channel 1	[−500:−259]
	The first RU Allocation subfield in HE-SIG-B content channel 2	[−258:−17]
	Center 26-tone RU subfield in HE-SIG-B content channel 1 and 2	[−16:−4, 4:16]
	The second RU Allocation subfield in HE-SIG-B content channel 1	[17:258]
	The second RU Allocation subfield in HE-SIG-B content channel 2	[259:500]
160 MHz or 80+80 MHz	The first RU Allocation subfield in HE-SIG-B content channel 1	[−1012:−771]
	The first RU Allocation subfield in HE-SIG-B content channel 2	[−770:−529]
	Center 26-tone RU subfield for lower frequency 80 MHz in HE-SIG-B content channel 1	[−528:−516, −508:−496]
	The second RU Allocation subfield in HE-SIG-B content channel 1	[−495:−254]
	The second RU Allocation subfield in HE-SIG-B content channel 2	[−253:−12]
	The third RU Allocation subfield in HE-SIG-B content channel 1	[12:253]
	The third RU Allocation subfield in HE-SIG-B content channel 2	[254:495]
	Center 26-tone RU subfield for higher frequency 80 MHz in HE-SIG-B content channel 2	[496:508, 516:528]
	The fourth RU Allocation subfield in HE-SIG-B content channel 1	[529:770]
	The fourth RU Allocation subfield in HE-SIG-B content channel 2	[771:1012]

The mapping from the 8-bit RU Allocation subfield to the RU assignment and the number of User fields per RU contributed to the User Specific field in the same HE-SIG-B content channel as the RU Allocation subfield are defined in Table 27-26.

If a single RU in a 40 MHz PPDU overlaps the subcarrier ranges [−244:−3] and [3:244], the corresponding RU Allocation subfields in the respective content channels shall both refer to the same RU.

If a single RU in an 80 MHz PPDU overlaps more than one of the subcarrier ranges [−500:−259], [−258:−17], [17:258] or [259:500], the corresponding RU Allocation subfields in the respective content channels shall all refer to the same RU.

Table 27-26—RU Allocation subfield

RU Allocation subfield (B7 B6 B5 B4 B3 B2 B1 B0)	#1	#2	#3	#4	#5	#6	#7	#8	#9	Number of entries
0 (00000000)	26	26	26	26	26	26	26	26	26	1
1 (00000001)	26	26	26	26	26	26	26	52		1
2 (00000010)	26	26	26	26	26	52		26	26	1
3 (00000011)	26	26	26	26	26	52		52		1
4 (00000100)	26	26	52		26	26	26	26	26	1
5 (00000101)	26	26	52		26	26	26	52		1
6 (00000110)	26	26	52		26	52		26	26	1
7 (00000111)	26	26	52		26	52		52		1
8 (00001000)	52		26	26	26	26	26	26	26	1
9 (00001001)	52		26	26	26	26	26	52		1
10 (00001010)	52		26	26	26	52		26	26	1
11 (00001011)	52		26	26	26	52		52		1
12 (00001100)	52		52		26	26	26	26	26	1
13 (00001101)	52		52		26	26	26	52		1
14 (00001110)	52		52		26	52		26	26	1
15 (00001111)	52		52		26	52		52		1
16-23 (00010y ₂ y ₁ y ₀)	52		52		—	106				8
24-31 (00011y ₂ y ₁ y ₀)	106				—	52		52		8
32-39 (00100y ₂ y ₁ y ₀)	26	26	26	26	26	106				8
40-47 (00101y ₂ y ₁ y ₀)	26	26	52		26	106				8
48-55 (00110y ₂ y ₁ y ₀)	52		26	26	26	106				8
56-63 (00111y ₂ y ₁ y ₀)	52		52		26	106				8
64-71 (01000y ₂ y ₁ y ₀)	106				26	26	26	26	26	8
72-79 (01001y ₂ y ₁ y ₀)	106				26	26	26	52		8
80-87 (01010y ₂ y ₁ y ₀)	106				26	52		26	26	8
88-95 (01011y ₂ y ₁ y ₀)	106				26	52		52		8

Table 27-26—RU Allocation subfield (*continued*)

RU Allocation subfield (B7 B6 B5 B4 B3 B2 B1 B0)	#1	#2	#3	#4	#5	#6	#7	#8	#9	Number of entries	
96-111 (0110y ₁ y ₀ z ₁ z ₀)	106				—	106				16	
112 (01110000)	52		52		—	52		52		1	
113 (01110001)	242-tone RU empty (with zero users)									1	
114 (01110010)	484-tone RU; contributes zero User fields to the User Specific field in the same HE-SIG-B content channel as this RU Allocation subfield									1	
115 (01110011)	996-tone RU; contributes zero User fields to the User Specific field in the same HE-SIG-B content channel as this RU Allocation subfield									1	
116-119 (011101x ₁ x ₀)	Reserved									4	
120-127 (01111y ₂ y ₁ y ₀)	Reserved									8	
128-191 (10y ₂ y ₁ y ₀ z ₂ z ₁ z ₀)	106				26		106				64
192-199 (11000y ₂ y ₁ y ₀)	242									8	
200-207 (11001y ₂ y ₁ y ₀)	484									8	
208-215 (11010y ₂ y ₁ y ₀)	996									8	
216-223 (11011y ₂ y ₁ y ₀)	Reserved									8	
224-255 (111x ₄ x ₃ x ₂ x ₁ x ₀)	Reserved									32	

If signaling RUs of size greater than 242 subcarriers, y₂y₁y₀ = 000–111 indicates the number of User fields in the HE-SIG-B content channel that contains the corresponding 8-bit RU Allocation subfield. Otherwise, y₂y₁y₀ = 000–111 indicates the number of users multiplexed in the 106-tone RU, 242-tone RU or the lower frequency 106-tone RU if there are two 106-tone RUs and one 26-tone RU is assigned between two 106-tone RUs. The binary vector y₂y₁y₀ indicates $N_{user}(r, c) = 2^2 \times y_2 + 2^1 \times y_1 + y_0 + 1$ users multiplexed in the RU.

z₂z₁z₀ = 000–111 indicates the number of users multiplexed in the higher frequency 106-tone RU if there are two 106-tone RUs and one 26-tone RU is assigned between two 106-tone RUs. The binary vector z₂z₁z₀ indicates $N_{user}(r, c) = 2^2 \times z_2 + 2^1 \times z_1 + z_0 + 1$ users multiplexed in the RU.

Similarly, y₁y₀ = 00–11 indicates the number of users multiplexed in the lower frequency 106-tone RU. The binary vector y₁y₀ indicates $N_{user}(r, c) = 2^1 \times y_1 + y_0 + 1$ users multiplexed in the RU.

Similarly, z₁z₀ = 00–11 indicates the number of users multiplexed in the higher frequency 106-tone RU. The binary vector z₁z₀ indicates $N_{user}(r, c) = 2^1 \times z_1 + z_0 + 1$ users multiplexed in the RU.

#1 to #9 (from left to the right) is ordered in increasing order of the absolute frequency.

x₁x₀ = 00–11, x₄x₃x₂x₁x₀ = 00000–11111.

Table 27-26—RU Allocation subfield (continued)

RU Allocation subfield (B7 B6 B5 B4 B3 B2 B1 B0)	#1	#2	#3	#4	#5	#6	#7	#8	#9	Number of entries
‘-’ means no user in that RU, i.e., $N_{user}(r, c) = 0$. For RU r that is a 106-tone or larger RU, $N_{user}(r, c)$ is indicated by the letters (such as $y_2y_1y_0$ or $z_2z_1z_0$) in the RU allocation subfield above if the letters are present in the RU allocation subfield; otherwise $N_{user}(r, c) = 0$. For RU r that is a 26-tone or 52-tone RU, $N_{user}(r, c) = 1$.										

If a single RU in a 160 MHz or 80+80 MHz PPDU overlaps more than one of the subcarrier ranges [−1012:−771], [−770:−529], [−495:−254], [−253:−12], [12:253], [254:495], [529:770] or [771:1012], the corresponding RU Allocation subfields in the respective content channels shall all refer to the same RU.

In Table 27-26, the “Number of entries” column refers to the number of RU Allocation subfield values that refer to the same RU assignment to be used in the frequency domain but differ in the number of User fields per RU. The number of User fields per RU indicated by the RU Allocation subfields and the Center 26-tone RU subfield of an HE-SIG-B content channel indicate the number of User fields in the User Specific field of the HE-SIG-B content channel.

For an MU-MIMO allocation of RU size greater than 242 subcarriers, the dynamic split of User fields between HE-SIG-B content channel 1 and HE-SIG-B content channel 2 is decided by the AP (on a per case basis) and signaled by the AP using the RU Allocation subfields in each HE-SIG-B content channel. See Annex Z for examples.

The pre-HE modulated fields (see Figure 27-23) are not transmitted in 20 MHz subchannels in which the preamble is punctured (see 27.3.7).

If the Bandwidth field in the HE-SIG-A field of an HE MU PPDU is 4, 5, 6 or 7, then one or more 20 MHz subchannels of the preamble are punctured, as defined in Table 27-20. If two adjacent 20 MHz subchannels that constitute a 40 MHz subchannel in which a 484-tone RU is located are punctured, then B7–B0 of the RU Allocation subfields corresponding to the two 20 MHz subchannels shall both be set to 113 (242-tone RU is empty) or shall both be set to 114 (see Table 27-26) to indicate that the preamble is punctured in both the 20 MHz subchannels. Each punctured 20 MHz subchannel that does not have B7–B0 of its corresponding RU Allocation subfield set to 114 shall have B7–B0 of its RU Allocation subfield set to 113.

The center 26-tone RU in a preamble punctured 80 MHz, 160 MHz, or 80+80 MHz HE MU PPDU shall not be allocated to a user if either of the two 20 MHz subchannels that the center 26-tone RU straddles have the preamble punctured.

27.3.11.8.4 User Specific field

The User Specific field of an HE-SIG-B content channel consists of zero or more User Block fields followed by padding (if present) as shown in Figure 27-26. Each non-final User Block field is made up of two User fields that contain information for two STAs that is used to decode their payloads. The final User Block field contains information for one or two users depending on the number of users in the HE-SIG-B content channel. If the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 0, then the number of User fields is indicated by the RU Allocation subfields and the Center 26-tone RU subfield in the

same HE-SIG-B content channel. If the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 1, then the number of users is indicated by the Number Of HE-SIG-B Symbols Or MU-MIMO Users field in the HE-SIG-A field.

If the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 1 (indicating full-bandwidth MU-MIMO transmission) and the Number Of HE-SIG-B Symbols Or MU-MIMO Users field in the HE-SIG-A field of an HE MU PPDU is 0 (indicating 1 user), then the User Specific field in the HE-SIG-B field consists of a single User Block field containing one User field for a non-MU-MIMO allocation as shown in Table 27-28.

The User Block field is defined in Table 27-27.

Table 27-27—User Block field

Field	Number of fields	Number of bits per field	Description
User field	N	21	<i>N</i> User fields are present, where <i>N</i> = 1 if it is the final User Block field, and if there is only one user in the final User Block field. <i>N</i> = 2 otherwise. The User field format for a non-MU-MIMO allocation is defined in Table 27-28. The User field format for a MU-MIMO allocation is defined in Table 27-29.
CRC	1	4	The CRC is calculated over bits 0 to 20 for a User Block field that contains one User field and bits 0 to 41 for a User Block field that contains two User fields. See 27.3.11.7.3.
Tail	1	6	Used to terminate the trellis of the convolutional decoder. Set to 0.

If the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 0, then for an MU-MIMO allocation of RU size greater than 242 subcarriers, the AP performs a dynamic split of the User fields between HE-SIG-B content channel 1 and HE-SIG-B content channel 2 as described in 27.3.11.8.3.

If the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 1, for bandwidths larger than 20 MHz, the AP performs an equitable split of the User fields between two HE-SIG-B content channels, i.e., User field *k* of a *K* user MU-MIMO PPDU is carried in HE-SIG-B content channel *c*, where *c* is defined in Equation (27-20).

$$c = \begin{cases} 1, & \text{if } k \in 1, \dots, \lceil K/2 \rceil \\ 2, & \text{if } (k \in \lceil K/2 \rceil + 1, \dots, K) \end{cases} \quad (27-20)$$

Since a single STA is not required to decode the data for more than one user (see 26.5.1.2), the signaling that enables a STA to decode its data is carried in only one User field.

The ordering of User fields in the User Specific field in an HE-SIG-B content channel is determined using the following three-step procedure:

- a) If the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 1, this first step is bypassed. Otherwise, the User fields in the User Specific field of an HE-SIG-B content channel are grouped into sets of User fields, where each set comprises the User fields indicated by one RU Allocation subfield or the Center 26-tone RU. These sets shall be ordered as follows:
 - 1) If the Bandwidth field in the HE-SIG-A field is 0 or 1, then there is only one set; therefore, the need for ordering in this first step does not arise.
 - 2) If the Bandwidth field in the HE-SIG-A field is 2, 4, or 5, then the set of User fields indicated by the first RU Allocation subfield is followed by the set of the User fields indicated by the second RU Allocation subfield; in turn, if the center 26-tone RU is assigned, then its User field is appended as the last User field to HE-SIG-B content channel 1 only.
 - 3) If the Bandwidth field in the HE-SIG-A field is 3, 6, or 7, then the set of User fields indicated by the first RU Allocation subfield is followed by the set of the User fields indicated by the second RU Allocation subfield; in turn these are followed by the set of the User fields indicated by the third RU Allocation subfield and then by the set of the User fields indicated by the fourth RU Allocation subfield. If the center 26-tone RU that spans subcarriers [–528:–516, –508:–496] is assigned, then its User field is appended to HE-SIG-B content channel 1. If the center 26-tone RU that spans subcarriers [496:508, 516:528] is assigned, then its User field is appended as the last User field to HE-SIG-B content channel 2.
- b) If the HE-SIG-B Compression field in the HE-SIG-A field of an HE MU PPDU is 1, then this second step is bypassed. Otherwise, each subset of User fields indicated by an RU Allocation subfield shall be ordered by increasing frequency of RU, i.e., #1 to #9 in Table 27-26.
- c) Without regard to the value of the HE-SIG-B Compression field, the ordering of User fields in the same RU shall follow the user ordering in Table 27-30.

In this way, RU Allocation subfields, Center 26-tone RU fields (if present), and the position of a user's User field in the User Specific field of an HE-SIG-B content channel indicate the user's RU assignment.

If the Bandwidth field in the HE-SIG-A field of an HE MU PPDU (see Table 27-20) is 4 or 5 (preamble is punctured), the content of HE-SIG-B content channels 1 and 2 shall be constructed as described above for an 80 MHz PPDU without preamble puncturing.

If the Bandwidth field in the HE-SIG-A field of an HE MU PPDU is 6 or 7 (preamble is punctured), the content of HE-SIG-B content channels 1 and 2 shall be constructed as described above for an 160 MHz PPDU without preamble puncturing.

An example for the mapping of the 8-bit RU Allocation subfield and the position of the User field to a user's data is illustrated in Figure 27-27. The RU Allocation subfield indicates an arrangement of one 106-tone RU followed by five 26-tone RUs and indicates that the 106-tone RU contains three User fields, i.e., the 106-tone RU supports multiplexing of three users using MU-MIMO. The 8 User fields in the User Specific field thus map to the 6 RUs, with the first three User fields indicating MU-MIMO allocations in the first 106-tone RU followed by User fields corresponding to the each of the five 26-tone RUs.

The contents of the User field differ depending on whether the field addresses a user in a non-MU-MIMO allocation in an RU or a user in an MU-MIMO allocation in an RU. Irrespective of whether the allocation is for a user in a non-MU-MIMO or an MU-MIMO allocation, the size of the User field is the same.

The User field format for a non-MU-MIMO allocation is defined in Table 27-28.

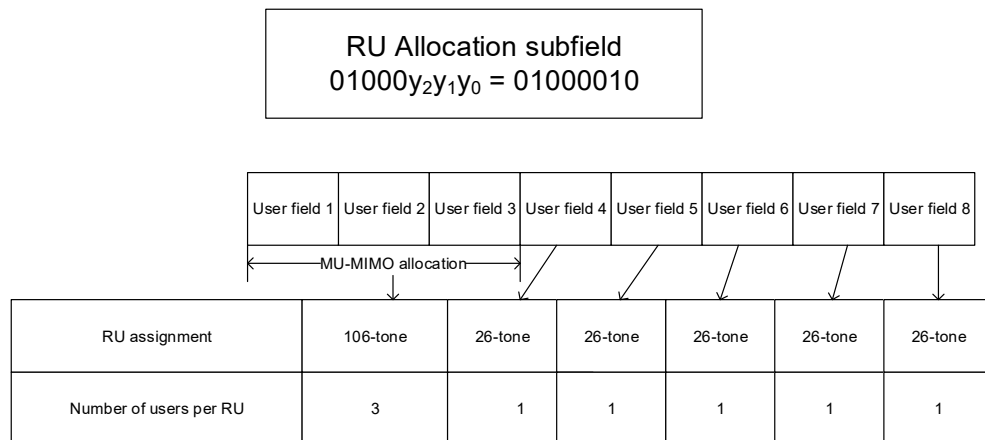


Figure 27-27—Example of the mapping of 8-bit RU Allocation subfield and the position of User field to user's assignment for one 20 MHz channel

Table 27-28—User field format for non-MU-MIMO allocation

Bit	Subfield	Number of bits	Description
B0–B10	STA-ID	11	Set to a value of the TXVECTOR parameter STA_ID (see 26.11.1).
B11–B13	NSTS	3	If the STA-ID subfield is not 2046, indicates the number of space-time streams and is set to the number of space-time streams minus 1. Set to an arbitrary value if the STA-ID subfield is 2046.
B14	Beamformed	1	If the STA-ID subfield is not 2046, used in transmit beamforming: Set to 1 if a beamforming steering matrix is applied to the waveform in a non-MU-MIMO allocation. Set to 0 otherwise. Set to an arbitrary value if the STA-ID subfield is 2046.
B15–B18	HE-MCS	4	If the STA-ID subfield is not 2046, indicates the modulation and coding scheme: Set to n for HE-MCS n , where $n = 0, 1, 2, \dots, 11$. Values 12–15 are reserved. Set to an arbitrary value if the STA-ID subfield is 2046.
B19	DCM	1	If the STA-ID subfield is not 2046, indicates whether DCM is used: Set to 1 to indicate that the payload of the corresponding user of the HE MU PPDU is modulated with DCM for the HE-MCS. Set to 0 to indicate that the payload of the corresponding user of the PPDU is not modulated with DCM for the HE-MCS. Set to an arbitrary value if the STA-ID subfield is 2046.
B20	Coding	1	If the STA-ID subfield is not 2046, indicates whether BCC or LDPC is used: Set to 0 for BCC. Set to 1 for LDPC. Set to an arbitrary value if the STA-ID subfield is 2046.

The User field format for an MU-MIMO allocation is defined in Table 27-29.

Table 27-29—User field format for MU-MIMO allocation

Bit	Subfield	Number of bits	Description
B0–B10	STA-ID	11	Set to a value indicated from TXVECTOR parameter STA_ID (see 26.11.1).
B11–B14	Spatial Configuration	4	If the STA-ID subfield is not 2046, indicates the number of spatial streams for a user in an MU-MIMO allocation (see Table 27-30). Set to an arbitrary value if the STA-ID subfield is 2046.
B15–B18	HE-MCS	4	If the STA-ID subfield is not 2046, indicates the modulation and coding scheme: Set to n for HE-MCS n , where $n = 0, 1, 2, \dots, 11$. Values 12-15 are reserved. Set to an arbitrary value if the STA-ID subfield is 2046.
B19	Reserved	1	If the STA-ID subfield is not 2046, reserved and set to 0. Set to an arbitrary value if the STA-ID subfield is 2046.
B20	Coding	1	If the STA-ID subfield is not 2046, indicates whether BCC or LDPC is used: Set to 0 for BCC. Set to 1 for LDPC. Set to an arbitrary value if the STA-ID subfield is 2046.

A User field for an MU-MIMO allocation includes a 4-bit Spatial Configuration subfield that indicates the number of spatial streams for each user and the total number of spatial streams in the MU-MIMO allocation. The subfield shown in Table 27-30 is constructed by using the entries corresponding to the value of number of users (N_{user}) multiplexed using MU-MIMO in an RU.

If the HE-SIG-B Compression field in the HE-SIG-A field is 0 and MU-MIMO is used in an RU of size less than or equal to 242 subcarriers, the number of users (N_{user}) in an MU-MIMO allocation is equal to the number of User fields per RU signaled for the RU in the associated RU Allocation subfield of the Common field in the same HE-SIG-B content channel.

If the HE-SIG-B Compression field in the HE-SIG-A field is 0 and MU-MIMO is used in RUs of size greater than 242 subcarriers, the AP performs a dynamic split of the User fields corresponding to the same MU-MIMO allocations as described in 27.3.11.8.3 into two HE-SIG-B content channels, and the number of users (N_{user}) is computed as the sum of the number of User fields indicated for the RU by the 8-bit RU Allocation subfield in each HE-SIG-B content channel.

If the HE-SIG-B Compression field in the HE-SIG-A field is 1, then the number of users, N_{user} , is signaled by the Number Of HE-SIG-B Symbols Or MU-MIMO Users field in the HE-SIG-A field, and the AP performs an equitable split of the User fields following Equation (27-20).

The User field positions within an RU are defined to be logically continuous: the last User field corresponding to an RU in HE-SIG-B content channel 1 is immediately followed by the first User field corresponding to the same RU in HE-SIG-B content channel 2.

Table 27-30—Spatial Configuration subfield encoding

N_{user}	B3...B0	N_{STS} [1]	N_{STS} [2]	N_{STS} [3]	N_{STS} [4]	N_{STS} [5]	N_{STS} [6]	N_{STS} [7]	N_{STS} [8]	Total N_{STS}	Number of entries
2	0000–0011	1–4	1							2–5	10
	0100–0110	2–4	2							4–6	
	0111–1000	3–4	3							6–7	
	1001	4	4							8	
3	0000–0011	1–4	1	1						3–6	13
	0100–0110	2–4	2	1						5–7	
	0111–1000	3–4	3	1						7–8	
	1001–1011	2–4	2	2						6–8	
	1100	3	3	2						8	
4	0000–0011	1–4	1	1	1					4–7	11
	0100–0110	2–4	2	1	1					6–8	
	0111	3	3	1	1					8	
	1000–1001	2–3	2	2	1					7–8	
	1010	2	2	2	2					8	
5	0000–0011	1–4	1	1	1	1				5–8	7
	0100–0101	2–3	2	1	1	1				7–8	
	0110	2	2	2	1	1				8	
6	0000–0010	1–3	1	1	1	1	1			6–8	4
	0011	2	2	1	1	1	1			8	
7	0000–0001	1–2	1	1	1	1	1	1		7–8	2
8	0000	1	1	1	1	1	1	1	1	8	1

For a given value of N_{users} , the four bits of the Spatial Configuration subfield are used as follows: A STA with a STA-ID that matches the 11-bit ID signaled in the User field for an MU-MIMO allocation derives the number of spatial streams allocated to it using the row corresponding to the signaled 4-bit Spatial Configuration subfield and the column corresponding to the User field position in the User Specific field. The starting stream index for the user is computed by summing the N_{STS} in the columns prior to the column indicated by the user's User field position.

The user ordering identified by the column headers $N_{STS}[s]$, $s = 1, 2, 3, \dots$ in Table 27-30 shall be the same as the user index u , $u = 0, 1, 2, \dots$ in Equation (27-38), Equation (27-59), and Equation (27-108), i.e., $u = s - 1$.

The total number of spatial streams (total N_{STS}) is computed by summing all columns for the row signaled by the Spatial Configuration field and is indicated in the “Total N_{STS} ” column of Table 27-30.

27.3.11.8.5 Encoding and modulation

The bits in the Common field of each HE-SIG-B content channel shall be BCC encoded at rate $R = 1/2$.

In the User Specific field, each User Block field of each HE-SIG-B content channel shall be BCC encoded at rate $R = 1/2$. If the number of User fields in an HE-SIG-B content channel is odd, the CRC and tail bits are added immediately after the last User field; there is only one User field in the last User Block field (see Table 27-27). Padding bits are appended immediately after the tail bits corresponding to the final User Block field in each HE-SIG-B content channel to round up to the next multiple of number of data bits per HE-SIG-B OFDM symbol. The padding bits may be set to any value. Further padding bits are appended to each HE-SIG-B content channel so that the number of OFDM symbols after encoding and modulation in different 20 MHz bands ends at the same OFDM symbol. For both the Common field and User Block field, the information bits, tail bits, and padding bits (if present) are BCC encoded at rate $R = 1/2$ using the encoder described in 17.3.5.6. If the coding rate of the HE-SIG-B-MCS is not equal to $1/2$, the convolutional encoder output bits for each field are concatenated, and then the concatenated bit streams are punctured as described in 17.3.5.6.

The coded bits are interleaved as in 27.3.12.8. The interleaved bits are mapped to constellation points from the HE-SIG-B-MCS specified in the HE-SIG-A field and have pilots inserted following the steps described in 17.3.5.8 and 17.3.5.9, respectively. Each HE-SIG-B OFDM symbol shall have 52 data tones.

The guard interval used for the HE-SIG-B field shall be 0.8 μ s.

If the HE-SIG-B Compression field in the HE-SIG-A field is 0, then the number of OFDM symbols in the HE-SIG-B field, denoted by $N_{SYM,HE-SIG-B}$, shall be signaled by the Number Of HE-SIG-B Symbols Or MU-MIMO Users field in the HE-SIG-A field of an HE MU PPDU (see 27.3.11.7.2).

If the HE-SIG-B Compression field in the HE-SIG-A field is 1, then the number of OFDM symbols in the HE-SIG-B field is calculated based on the number of User fields in the HE-SIG-B content channel, the HE-MCS of the HE-SIG-B field, whether DCM is applied to the HE-SIG-B field, and the bandwidth of HE MU PPDU, all of which are indicated by the Number Of HE-SIG-B Symbols Or MU-MIMO Users field, HE-SIG-B-MCS field, HE-SIG-B DCM field, and Bandwidth field, respectively, in the HE-SIG-A field of an HE MU PPDU (see 27.3.11.7.2).

For the HE-SIG-B content channel c ($c = 1$ or 2), denote the complex number assigned to the k^{th} data subcarrier of the n^{th} symbol by $d_{k,n,c}$. The time domain waveform for the HE-SIG-B field, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , is given by Equation (27-21).

$$r_{HE-SIG-B}^{(i_{Seg}, i_{TX})}(t) = \frac{1}{\sqrt{N_{TX} \cdot N_{HE-SIG-B}^{Tone} \cdot \frac{\Omega_{20MHz}}{N_{20MHz}}}} \sum_{n=0}^{N_{SYM,HE-SIG-B}-1} w_{T_{HE-SIG-B}}(t - nT_{SYML}) \sum_{i_{BW} \in \Omega_{20MHz}} \sum_{k=-28}^{28} \left(\Upsilon_{(k - K_{Shift}(i_{BW})), BW} \left(\Gamma_{M'_{20}(k)} D_{k,n,i_{BW}}^{i_{Seg}} + p_{n+4} P_k \right) \cdot \exp(j2\pi(k - K_{Shift}(i_{BW}))\Delta_{F, Pre-HE}(t - nT_{SYML} - T_{GI, Pre-HE} - T_{CS}^{i_{TX}})) \right) \quad (27-21)$$

where

$\Gamma_{M'_{20}(k)}$ is the phase rotation value for HE-SIG-B field PAPR reduction. If the HE-SIG-B field is modulated with HE-SIG-B-MCS 0 and DCM=1, then $\Gamma_{M'_{20}(k)} = 1$. For all other modulation schemes of the HE-SIG-B field,

$$\Gamma_{M'_{20}(k)} = \begin{cases} 1 & 0 \leq M'_{20}(k) < 26 \\ (-1)^{M'_{20}(k)} & 26 \leq M'_{20}(k) < 52 \end{cases}$$

$N_{\text{HE-SIG-B}}^{\text{Tone}}$ is given in Table 27-16

$K_{\text{Shift}}(i)$ is defined in 21.3.8.2.4

$$D_{k,n,i_{BW}}^{i_{\text{seg}}} = \begin{cases} 0, k = 0, \pm 7, \pm 21 \\ d_{M'_{20}(k), n, (i_{BW} \bmod 2) + 1}, \text{ otherwise} \end{cases}$$

$$M'_{20}(k) = \begin{cases} k + 28, -28 \leq k \leq -22 \\ k + 27, -20 \leq k \leq -8 \\ k + 26, -6 \leq k \leq -1 \\ k + 25, 1 \leq k \leq 6 \\ k + 24, 8 \leq k \leq 20 \\ k + 23, 22 \leq k \leq 28 \end{cases}$$

P_k and p_n are defined in 17.3.5.10

$N_{\text{SYM, HE-SIG-B}}$ is the number of OFDM symbols in the HE-SIG-B field

From Equation (27-21) and 27.3.11.8.2, a 20 MHz PPDU contains one HE-SIG-B content channel as shown in Figure 27-28.

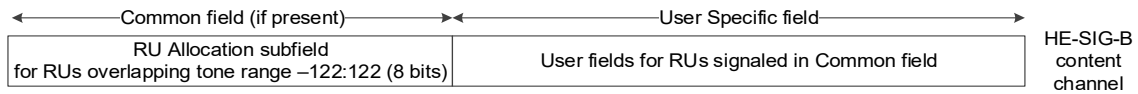


Figure 27-28—HE-SIG-B content channel for a 20 MHz PPDU

From Equation (27-21) and 27.3.11.8.2, a 40 MHz PPDU contains two HE-SIG-B content channels, each occupying a 20 MHz frequency segment as shown in Figure 27-29. HE-SIG-B content channel 1 occupies the 20 MHz frequency subchannel that is lower in frequency. HE-SIG-B content channel 2 occupies the 20 MHz frequency subchannel that is upper in frequency.

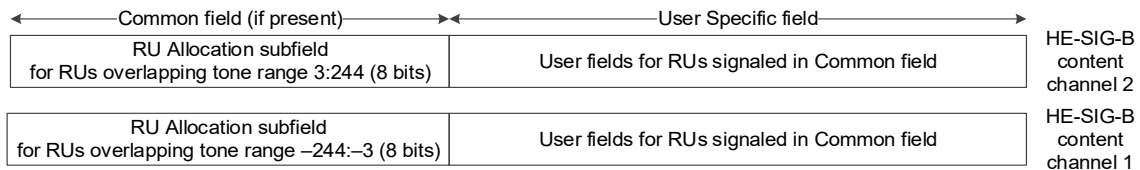


Figure 27-29—HE-SIG-B content channel for a 40 MHz PPDU

From Equation (27-21) and 27.3.11.8.2, an 80 MHz PPDU contains two HE-SIG-B content channels, each of which are duplicated once as shown in Figure 27-30. HE-SIG-B content channel 1 occupies the 20 MHz frequency subchannel that is lowest in frequency and is duplicated on the 20 MHz frequency segment that is third lowest in frequency. HE-SIG-B content channel 2 occupies the 20 MHz frequency segment that is second lowest in frequency and is duplicated on the 20 MHz frequency subchannel that is highest in frequency.

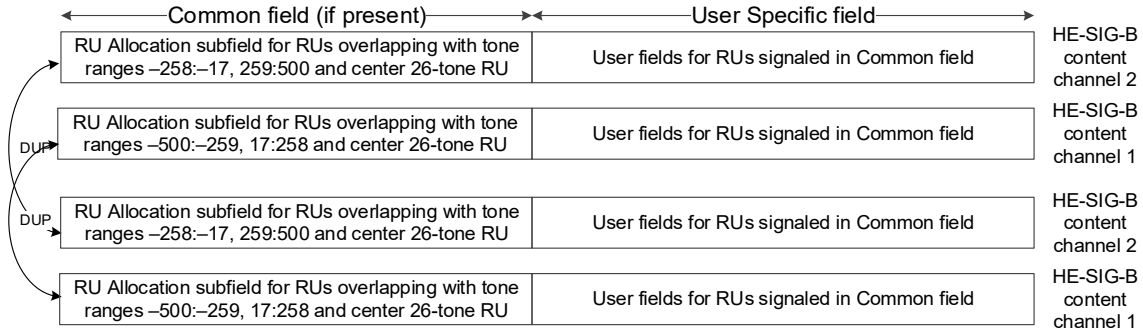


Figure 27-30—HE-SIG-B content channels and their duplication in an 80 MHz PPDU

From Equation (27-21) and 27.3.11.8.2, a 160 MHz PPDU contains two HE-SIG-B content channels, each of which are duplicated four times as shown in Figure 27-31. HE-SIG-B content channel 1 occupies the 20 MHz frequency subchannel that is lowest in frequency and is duplicated on the 20 MHz frequency subchannels that are third, fifth, and seventh lowest in frequency. HE-SIG-B content channel 2 occupies the 20 MHz frequency subchannel that is second lowest in frequency and is duplicated on the 20 MHz frequency subchannels that are fourth, sixth, and eighth lowest in frequency.

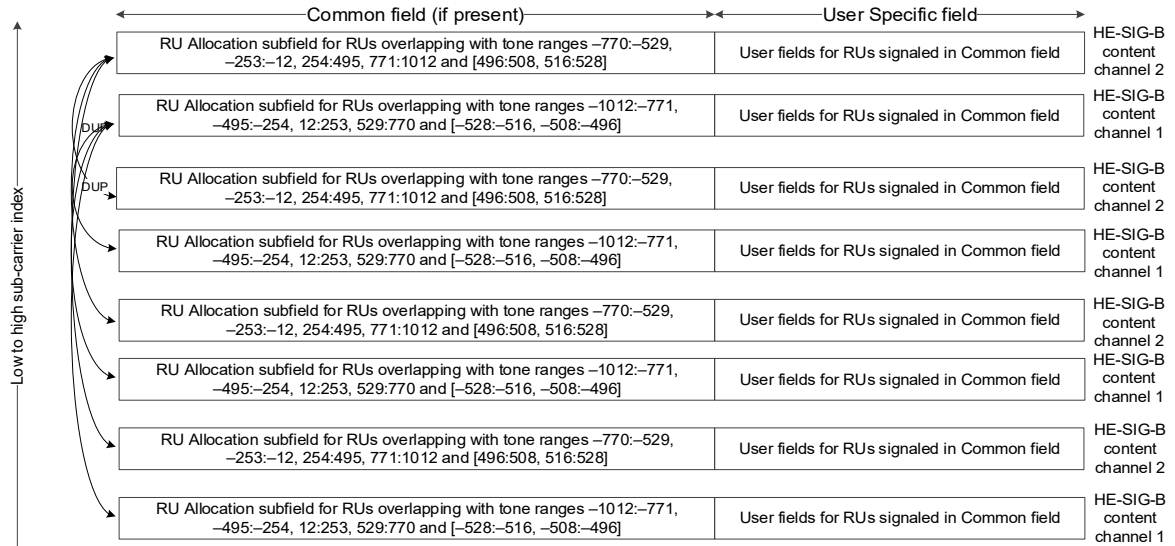


Figure 27-31—HE-SIG-B content channels and their duplication in a 160 MHz PPDU

If the Bandwidth field in the HE-SIG-A field of an HE MU PPDU (see Table 27-20) is 4 or 5 (preamble is punctured), the mapping of the HE-SIG-B content channels to 20 MHz segments shall be the same as for an 80 MHz PPDU (see Figure 27-30), with the exception that punctured 20 MHz channels shall be excluded.

If the Bandwidth field in the HE-SIG-A field of an HE MU PPDU (see Table 27-20) is 6 or 7 (preamble is punctured), the mapping of the HE-SIG-B content channels to 20 MHz segments shall be the same as for an 160 MHz PPDU (see Figure 27-31), with the exception that punctured 20 MHz channels shall be excluded.

27.3.11.9 HE-STF field

The main purpose of the HE-STF field is to improve automatic gain control estimation in a MIMO transmission. The duration of the HE-STF field for HE PPDU that are not HE TB PPDU is $T_{\text{HE-STF-NT}}$ (periodicity of 0.8 μs with 5 periods as given in Table 27-12), and the duration of the HE-STF field for an HE TB PPDU is $T_{\text{HE-STF-T}}$ (periodicity of 1.6 μs with 5 periods as given in Table 27-12).

For the HE-STF field, the M sequence is defined by Equation (27-22).

$$M = \{-1, -1, -1, 1, 1, 1, -1, 1, 1, 1, -1, 1, 1, -1, 1\} \quad (27-22)$$

The HE-STF field is constructed by mapping the M sequence(s) multiplied by $(1+j)/\sqrt{2}$ or $(-1-j)/\sqrt{2}$ to each 242-tone RU. For a transmission bandwidth greater than 40 MHz, $(1+j)/\sqrt{2}$ or $(-1-j)/\sqrt{2}$ is assigned to subcarrier indices that are inside the center 26-tone RUs.

For a 20 MHz transmission, the frequency domain sequence for HE PPDU that are not HE TB PPDU is given by Equation (27-23).

$$HES_{-112:16:112} = \{M\} \cdot (1+j)/\sqrt{2} \quad (27-23)$$

The value of the HE-STF sequence at null tone index 0 is $HES_0 = 0$

where $HES_{a:b:c}$ means coefficients of the HE-STF on every b subcarrier indices from a to c subcarrier indices and coefficients on other subcarrier indices are set to zero.

For a 40 MHz transmission, the frequency domain sequence for HE PPDU that are not HE TB PPDU is given by Equation (27-24).

$$HES_{-240:16:240} = \{M, 0, -M\} \cdot (1+j)/\sqrt{2} \quad (27-24)$$

For an 80 MHz transmission, the frequency domain sequence for HE PPDU that are not HE TB PPDU is given by Equation (27-25).

$$HES_{-496:16:496} = \{M, 1, -M, 0, -M, 1, -M\} \cdot (1+j)/\sqrt{2} \quad (27-25)$$

For a 160 MHz transmission, the frequency domain sequence for HE PPDU that are not HE TB PPDU is given by Equation (27-26).

$$HES_{-1008:16:1008} = \{M, 1, -M, 0, -M, 1, -M, 0, -M, -1, M, 0, -M, 1, -M\} \cdot (1+j)/\sqrt{2} \quad (27-26)$$

For an 80+80 MHz transmission, the lower 80 MHz segment for HE PPDU that are not HE TB PPDU shall use the HE-STF pattern for the 80 MHz defined in Equation (27-25).

For an 80+80 MHz transmission, the frequency domain sequence of the upper 80 MHz segment for HE PPDU that are not HE TB PPDU is given by Equation (27-27).

$$HES_{-496:16:496} = \{-M, -1, M, 0, -M, 1, -M\} \cdot (1+j)/\sqrt{2} \quad (27-27)$$

For a 20 MHz transmission, the frequency domain sequence for HE TB PPDU is given by Equation (27-28).

$$HES_{-120:8:120} = \{M, 0, -M\} \cdot (1+j)/\sqrt{2} \quad (27-28)$$

For an HE TB feedback NDP in 20 MHz channel width, the frequency domain sequence is given by Equation (27-29).

$$HES_{-120:8:120}^{TB\ NDP} = HES_{-120:8:120} \quad (27-29)$$

For a 40 MHz transmission, the frequency domain sequence for HE TB PPDU is given by Equation (27-30).

$$HES_{-248:8:248} = \{M, -1, -M, 0, M, -1, M\} \cdot (1+j)/\sqrt{2} \quad (27-30)$$

The value of the HE-STF sequence at edge tone indices ± 248 is $HES_{\pm 248} = 0$

For an HE TB feedback NDP in 40 MHz channel width, the frequency domain sequence is given by Equation (27-31).

$$\begin{aligned} HES_{-248:8:-8}^{TB\ NDP} &= \{M, -1, -M\} \cdot (1+j)/\sqrt{2}, \text{ if RU_TONE_SET_INDEX} \leq 18 \\ HES_{8:8:248}^{TB\ NDP} &= \{M, -1, M\} \cdot (1+j)/\sqrt{2}, \text{ if RU_TONE_SET_INDEX} > 18 \\ HES_{\pm 248}^{TB\ NDP} &= 0 \end{aligned} \quad (27-31)$$

For an 80 MHz transmission, the frequency domain sequence for HE TB PPDU is given by Equation (27-32).

$$HES_{-504:8:504} = \{M, -1, M, -1, -M, -1, M, 0, -M, 1, M, 1, -M, 1, -M\} \cdot (1+j)/\sqrt{2} \quad (27-32)$$

The value of the HE-STF sequence at edge tone indices ± 504 is $HES_{\pm 504} = 0$

For an HE TB feedback NDP in 80 MHz channel width, the frequency domain sequence is given by Equation (27-33).

$$\begin{aligned} HES_{-504:8:-264}^{TB\ NDP} &= \{M, -1, M\} \cdot (1+j)/\sqrt{2}, \text{ if RU_TONE_SET_INDEX} \leq 18 \\ HES_{-248:8:-8}^{TB\ NDP} &= \{-M, -1, M\} \cdot (1+j)/\sqrt{2}, \text{ if } 18 < \text{RU_TONE_SET_INDEX} \leq 36 \\ HES_{8:8:248}^{TB\ NDP} &= \{-M, 1, M\} \cdot (1+j)/\sqrt{2}, \text{ if } 36 < \text{RU_TONE_SET_INDEX} \leq 54 \\ HES_{264:8:504}^{TB\ NDP} &= \{-M, 1, -M\} \cdot (1+j)/\sqrt{2}, \text{ if } 54 < \text{RU_TONE_SET_INDEX} \leq 72 \\ HES_{\pm 504}^{TB\ NDP} &= 0 \end{aligned} \quad (27-33)$$

For a 160 MHz transmission, the frequency domain sequence for HE TB PPDU is given by Equation (27-34).

$$\begin{aligned} HES_{-1016:8:1016} &= \{M, -1, M, -1, -M, -1, M, 0, -M, 1, M, 1, -M, 1, -M, 0 \\ &\quad -M, 1, -M, 1, M, 1, -M, 0, -M, 1, M, 1, -M, 1, -M\} \cdot (1+j)/\sqrt{2} \end{aligned} \quad (27-34)$$

The value of the HE-STF sequence at edge tone indices ± 8 and ± 1016 is $HES_{\pm 8} = 0, HES_{\pm 1016} = 0$

For an HE TB feedback NDP in 160 MHz channel width, the frequency domain sequence is given by Equation (27-35).

$$\begin{aligned}
 HES_{-1016:8:-776}^{\text{TB NDP}} &= \{M, -1, M\} \cdot (1+j)/\sqrt{2}, \text{ if } \text{RU_TONE_SET_INDEX} \leq 18 \\
 HES_{-760:8:-520}^{\text{TB NDP}} &= \{-M, -1, M\} \cdot (1+j)/\sqrt{2}, \text{ if } 18 < \text{RU_TONE_SET_INDEX} \leq 36 \\
 HES_{-504:8:-264}^{\text{TB NDP}} &= \{-M, 1, M\} \cdot (1+j)/\sqrt{2}, \text{ if } 36 < \text{RU_TONE_SET_INDEX} \leq 54 \\
 HES_{-248:8:-8}^{\text{TB NDP}} &= \{-M, 1, -M\} \cdot (1+j)/\sqrt{2}, \text{ if } 54 < \text{RU_TONE_SET_INDEX} \leq 72 \\
 HES_{8:8:248}^{\text{TB NDP}} &= \{-M, 1, -M\} \cdot (1+j)/\sqrt{2}, \text{ if } 72 < \text{RU_TONE_SET_INDEX} \leq 90 \\
 HES_{264:8:504}^{\text{TB NDP}} &= \{M, 1, -M\} \cdot (1+j)/\sqrt{2}, \text{ if } 90 < \text{RU_TONE_SET_INDEX} \leq 108 \\
 HES_{520:8:760}^{\text{TB NDP}} &= \{-M, 1, M\} \cdot (1+j)/\sqrt{2}, \text{ if } 108 < \text{RU_TONE_SET_INDEX} \leq 126 \\
 HES_{776:8:1016}^{\text{TB NDP}} &= \{-M, 1, -M\} \cdot (1+j)/\sqrt{2}, \text{ if } 126 < \text{RU_TONE_SET_INDEX} \leq 144 \\
 HES_{\pm 8}^{\text{TB NDP}} &= HES_{\pm 1016}^{\text{TB NDP}} = 0
 \end{aligned} \tag{27-35}$$

For an 80+80 MHz transmission, the lower 80 MHz segment for HE TB PPDU shall use the HE-STF pattern for the 80 MHz defined in Equation (27-32).

For an 80+80 MHz transmission, the frequency domain sequence of the upper 80 MHz segment for HE TB PPDU is given by Equation (27-36).

$$\begin{aligned}
 HES_{-504:8:504} &= \{-M, 1, -M, 1, M, 1, -M, 0, -M, 1, M, 1, -M, 1, -M\} \cdot (1+j)/\sqrt{2} \\
 \text{The value of the HE-STF sequence at edge tone indices } \pm 504 &\text{ is } HES_{\pm 504} = 0
 \end{aligned} \tag{27-36}$$

For an HE TB feedback NDP in the lower 80 MHz segment of an 80+80 MHz channel width, the frequency domain sequence is given by Equation (27-33).

For an HE TB feedback NDP in the upper 80 MHz segment of an 80+80 MHz channel width, the frequency domain sequence is given by Equation (27-37).

$$\begin{aligned}
 HES_{-504:8:-264}^{\text{TB NDP}} &= \{-M, 1, -M\} \cdot (1+j)/\sqrt{2}, \text{ if } \text{RU_TONE_SET_INDEX} \leq 90 \\
 HES_{-248:8:-8}^{\text{TB NDP}} &= \{M, 1, -M\} \cdot (1+j)/\sqrt{2}, \text{ if } 90 < \text{RU_TONE_SET_INDEX} \leq 108 \\
 HES_{8:8:248}^{\text{TB NDP}} &= \{-M, 1, M\} \cdot (1+j)/\sqrt{2}, \text{ if } 108 < \text{RU_TONE_SET_INDEX} \leq 126 \\
 HES_{264:8:504}^{\text{TB NDP}} &= \{-M, 1, -M\} \cdot (1+j)/\sqrt{2}, \text{ if } 126 < \text{RU_TONE_SET_INDEX} \leq 144 \\
 HES_{\pm 504}^{\text{TB NDP}} &= 0
 \end{aligned} \tag{27-37}$$

If a 20 MHz operating non-AP STA sends an HE TB feedback NDP report response (see 26.5.7) on a channel width greater than 20 MHz, the HE-STF subcarrier that overlaps the DC subcarrier of the 20 MHz operating non-AP STA is not transmitted.

For an OFDMA transmission, the coefficients in Equation (27-23) to Equation (27-36) are set to zero if those values correspond to subcarrier indices that fall within RUs that have no users assigned to them.

The time domain representation of the signal for an HE PPDU that is not an HE TB PPDU on frequency segment i_{Seg} and transmit chain i_{TX} shall be as specified in Equation (27-38).

$$r_{\text{HE-STF}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = w_{T_{\text{HE-STF-NT}}}(t) \cdot \sum_{r=0}^{N_{\text{RU}}-1} \frac{\alpha_r \beta_r}{\sqrt{N_{\text{STS}, r, \text{total}}}} \sum_{k \in K_r} \eta_{\text{HE-STF}, k} \sum_{u=0}^{N_{\text{user}, r}-1} \sum_{m=1}^{N_{\text{STS}, r, u}} \left(\left[Q_k^{(i_{\text{Seg}})} \right]_{i_{\text{TX}}, M_{r,u}+m} HES_k \cdot \exp(j2\pi k \Delta_{F, \text{HE}}(t - T_{\text{CS}, \text{HE}}(M_{r,u} + m))) \right) \quad (27-38)$$

where

α_r is defined in 27.3.10

$\eta_{\text{HE-STF}, k}$ is an HE PPDU format-dependent scaling factor as defined by

$$\eta_{\text{HE-STF}, k} = \begin{cases} \sqrt{2}, & \text{for an HE ER SU PPDU} \\ 1, & \text{otherwise} \end{cases}$$

β_r is the per-RU power normalization factor as defined by

$$\beta_r = \left(\sqrt{\frac{|K_r|}{|K_r^{\text{HE-STF}}|}} \right) / \left(\sqrt{\sum_{r=0}^{N_{\text{RU}}-1} \alpha_r^2 |K_r|} \right)$$

$|K_r|$ is the cardinality of the set of subcarriers K_r as defined in 27.3.10

$K_r^{\text{HE-STF}}$ is the set of subcarriers that have nonzero values within K_r in the HE-STF field

$T_{\text{CS}, \text{HE}}(M_{r,u} + m)$

represents the cyclic shift for space-time stream $M_{r,u} + m$ as defined in 27.3.11.2.2

$Q_k^{(i_{\text{Seg}})}$ is defined in 27.3.10

$w_{T_{\text{HE-STF-NT}}}$ is the windowing function for HE-STF field in the non-HE TB PPDU

$|K_r^{\text{HE-STF}}|$ is the cardinality of the set of subcarriers $K_r^{\text{HE-STF}}$

N_{RU} , $N_{\text{STS}, r, \text{total}}$, $N_{\text{user}, r}$, and $N_{\text{STS}, r, u}$ are defined in Table 27-15

The time domain representation of the signal for an HE TB PPDU transmitted by user u in the r^{th} RU on frequency segment i_{Seg} and transmit chain i_{TX} shall be as specified in Equation (27-39).

$$r_{\text{HE-STF}, r, u}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = \frac{1}{\sqrt{|K_r^{\text{HE-STF}}| N_{\text{STS}, r, u}}} w_{T_{\text{HE-STF-T}}}(t) \sum_{k \in K_r} \sum_{m=1}^{N_{\text{STS}, r, u}} \left(\left[Q_{k, u}^{(i_{\text{Seg}})} \right]_{i_{\text{TX}}, m} HES_k \cdot \exp(j2\pi k \Delta_{F, \text{HE}}(t - T_{\text{CS}, \text{HE}}(M_{r,u} + m))) \right) \quad (27-39)$$

where

$w_{T_{\text{HE-STF-T}}}$ is the windowing function for HE-STF field in the HE TB PPDU

$Q_{k, u}^{(i_{\text{Seg}})}$ is defined in 27.3.10

HES_k is the HE-STF sequence for HE TB PPDU applied on subcarrier k . For an HE TB feedback NDP, replace HES_k with $HES_k^{\text{TB_NDP}}$ in Equation (27-39).

$M_{r,u}$ is given by the TXVECTOR parameter STARTING_STS_NUM

It is recommended that the spatial mapping matrix applied to HE-STF and beyond be chosen such that it preserves the smoothness of the physical channel, achieved by limiting the variation of each element's real and imaginary values in the spatial mapping matrix across successive tones within one RU.

27.3.11.10 HE-LTF field

The HE-LTF field provides a means for the receiver to estimate the MIMO channel between the set of constellation mapper outputs (or, if STBC is applied, the STBC encoder outputs) and the receive chains. In an HE SU PPDU and HE ER SU PPDU, the transmitter provides training for N_{STS} space-time streams (spatial mapper inputs) used for the transmission of the PSDU. In an HE MU PPDU, the transmitter provides training for $N_{STS,r,total}$ space-time streams used for the transmission of the PSDU(s) in the r^{th} RU. In an HE TB PPDU, the transmitter of user u in the r^{th} RU provides training for $N_{STS,r,u}$ space-time streams used for the transmission of the PSDU. For each subcarrier in the r^{th} RU, the MIMO channel that can be estimated is an $N_{RX} \times N_{STS,r,total}$ matrix. An HE transmission has a preamble that contains HE-LTF symbols, where the data tones of each HE-LTF symbol are multiplied by entries belonging to a matrix P_{HE-LTF} , to enable channel estimation at the receiver. When single stream pilot is used in HE-LTF, the pilot subcarriers of each HE-LTF symbol are multiplied by the entries of a matrix R_{HE-LTF} defined below to allow receivers to track phase and/or frequency offset during MIMO channel estimation using the HE-LTF. Single stream pilots shall be used in the HE-LTF field for SU, DL, and UL OFDMA, DL MU-MIMO, and partial-bandwidth UL MU-MIMO transmissions. Single stream pilots shall be used in the HE-LTF field for a full-bandwidth UL MU-MIMO transmission if single stream pilot HE-LTF mode is selected.

In an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU with a single RU (the RU having an MU-MIMO allocation or an SU allocation), the number of HE-LTF symbols, N_{HE-LTF} , is a function of the total number of space-time streams, N_{STS} , as shown in Table 21-13 in 21.3.8.3.5 with $N_{VHT-LTF}$ replaced by N_{HE-LTF} and with $N_{STS,total}$ replaced by N_{STS} . In an HE MU PPDU, N_{HE-LTF} is indicated in the HE-SIG-A field. In an HE MU PPDU with more than one RU, N_{HE-LTF} may take a value 1, 2, 4, 6, or 8 that is greater than or equal to the maximum value of the initial number of HE-LTF symbols for each RU, where the initial number of HE-LTF symbols is calculated as a function of $N_{STS,r,total}$ (where r is the index of the RU) based on Table 21-13 in 21.3.8.3.5 with $N_{VHT-LTF}$ replaced by N_{HE-LTF} .

NOTE—An AP might use the HE MU PPDU With More Than One RU Rx Max N_{HE-LTF} subfield in the PHY Capabilities Information field in the HE Capabilities element received from a non-AP STA for OFDMA scheduling and selecting an appropriate N_{HE-LTF} for an OFDMA transmission.

In an HE TB PPDU, N_{HE-LTF} is indicated in the Trigger frame that triggers the transmission of the PPDU. In a non-OFDMA HE TB PPDU, the number of HE-LTF symbols, N_{HE-LTF} , is a function of the total number of space-time streams, N_{STS} , as shown in Table 21-13 in 21.3.8.3.5 with $N_{VHT-LTF}$ replaced by N_{HE-LTF} . For an OFDMA HE TB PPDU, N_{HE-LTF} may be 1, 2, 4, 6, or 8 that is greater than or equal to the maximum value of the initial number of HE-LTF symbols for each RU r , which is calculated as a function of $N_{STS,r,total}$ separately based on Table 21-13 in 21.3.8.3.5 with $N_{VHT-LTF}$ replaced by N_{HE-LTF} .

An HE PPDU supports 3 HE-LTF types: 1x HE-LTF, 2x HE-LTF, and 4x HE-LTF. Table 27-31 defines whether a particular HE-LTF type and GI duration combination is mandatory, conditional mandatory, or optional for each HE PPDU format.

In an HE SU PPDU, HE MU PPDU, and HE ER SU PPDU, the combination of HE-LTF type and GI duration is indicated in HE-SIG-A field. In an HE TB PPDU, the combination of HE-LTF type and GI duration is indicated in the Trigger frame that triggers the transmission of the PPDU. If an HE PPDU is an HE sounding NDP, the combinations of HE-LTF types and GI durations are listed in 27.3.17. If an HE PPDU is an HE TB feedback NDP, the combinations of types and GI durations are listed in 27.3.18.

Table 27-31—HE-LTF type and GI duration combinations for various HE PPDU formats

HE-LTF type and GI duration combination	HE SU PPDU	HE MU PPDU	HE ER SU PPDU	HE TB PPDU	HE sounding NDP	HE TB feedback NDP
1x HE-LTF 0.8 μ s GI	O	N/A	O	N/A	N/A	N/A
1x HE-LTF 1.6 μ s GI	N/A	N/A	N/A	CM3	N/A	N/A
2x HE-LTF 0.8 μ s GI	M	M	M	N/A	M	N/A
2x HE-LTF 1.6 μ s GI	M	M	M	M	M	N/A
4x HE-LTF 0.8 μ s GI	CM1	CM2	O	N/A	N/A	N/A
4x HE-LTF 3.2 μ s GI	M	M	M	M	O	M
Legend M = mandatory. CM1 = Mandatory if the STA supports 4x HE-LTF 0.8 μs GI for HE ER SU PPDU. Otherwise, optional. CM2 = For an AP, mandatory for transmission if the AP supports 4x HE-LTF 0.8 μs GI for HE ER SU PPDU. For a non-AP STA, mandatory for reception if the non-AP STA supports 4x HE-LTF 0.8 μs GI for HE ER SU PPDU. Otherwise, optional. CM3 = Mandatory for full-bandwidth UL MU-MIMO if the STA supports UL MU-MIMO. Otherwise, not supported. N/A for partial-bandwidth UL MU-MIMO or UL OFDMA. O = optional. N/A = not supported by the PPDU format. If a STA does not support transmission or reception of a particular PPDU format, then the M/CM/O designation is not applicable for the transmission or reception, respectively, of that PPDU format.						

The duration of each HE-LTF symbol excluding GI is $T_{\text{HE-LTF}}$, defined in Equation (27-40).

$$T_{\text{HE-LTF}} = \begin{cases} T_{\text{HE-LTF-1X}}, & \text{if 1x HE-LTF} \\ T_{\text{HE-LTF-2X}}, & \text{if 2x HE-LTF} \\ T_{\text{HE-LTF-4X}}, & \text{if 4x HE-LTF} \end{cases} \quad (27-40)$$

where $T_{\text{HE-LTF-1X}}$, $T_{\text{HE-LTF-2X}}$, and $T_{\text{HE-LTF-4X}}$ are defined in Table 27-12.

In a 20 MHz transmission, the 1x HE-LTF sequence transmitted on subcarriers [−122: 122] is given by Equation (27-41).

$$\begin{aligned} \text{HELTF}_{-122,122} = & \{0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, \\ & 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, \\ & -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, \\ & 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, 0, 0, \\ & 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, \\ & 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, \\ & 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, \\ & -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0\} \end{aligned} \quad (27-41)$$

In a 20 MHz transmission, the 2x HE-LTF sequence transmitted on subcarriers [−122: 122] is given by Equation (27-42).

$$\begin{aligned}
 &HELTF_{-122,122} = \\
 &\{-1, 0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, \\
 &-1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, \\
 &-1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, \\
 &+1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, \\
 &+1, 0, +1, 0, -1, 0, -1, 0, +1, 0, 0, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, \\
 &+1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, \\
 &+1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, \\
 &-1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, \\
 &-1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0\}
 \end{aligned} \tag{27-42}$$

In a 20 MHz transmission, the 4x HE-LTF sequence transmitted on subcarriers [−122: 122] is given by Equation (27-43).

$$\begin{aligned}
 &HELTF_{-122,122} = \\
 &\{-1, -1, +1, -1, +1, -1, +1, +1, -1, +1, +1, +1, -1, -1, +1, -1, -1, -1, -1, +1, +1, \\
 &-1, -1, -1, -1, +1, +1, -1, +1, -1, +1, +1, +1, +1, -1, +1, -1, -1, +1, +1, -1, +1, +1, \\
 &+1, -1, -1, +1, -1, -1, -1, +1, +1, +1, +1, -1, +1, +1, -1, -1, -1, -1, +1, -1, -1, +1, +1, \\
 &-1, +1, -1, -1, -1, -1, +1, -1, +1, -1, -1, -1, -1, -1, +1, +1, -1, -1, -1, -1, -1, +1, \\
 &-1, -1, +1, +1, +1, -1, +1, +1, +1, -1, +1, -1, +1, -1, -1, -1, -1, +1, +1, +1, -1, -1, \\
 &-1, +1, -1, +1, +1, +1, 0, 0, 0, -1, +1, -1, +1, -1, +1, +1, -1, +1, +1, +1, -1, +1, -1, \\
 &-1, +1, -1, +1, -1, +1, +1, +1, -1, +1, +1, +1, -1, -1, +1, -1, -1, -1, -1, +1, +1, -1, \\
 &-1, -1, -1, -1, -1, +1, +1, -1, -1, -1, -1, -1, -1, +1, +1, -1, +1, +1, -1, -1, -1, -1, \\
 &+1, +1, -1, +1, +1, +1, +1, +1, +1, -1, +1, -1, -1, -1, -1, +1, -1, -1, +1, +1, -1, -1, \\
 &+1, -1, -1, -1, -1, +1, -1, +1, -1, -1, +1, +1, +1, -1, -1, +1, +1, +1, +1, +1, -1, +1, \\
 &+1, -1, -1, -1, +1, -1, -1, -1, +1, -1, +1, -1, +1, +1\}
 \end{aligned} \tag{27-43}$$

In a 40 MHz transmission, the 1x HE-LTF sequence transmitted on subcarriers [−244: 244] is given by Equation (27-44).

$$\begin{aligned}
 &HELTF_{-244,244} = \\
 &\{+1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, \\
 &0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, \\
 &0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, \\
 &0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, \\
 &-1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, \\
 &0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, \\
 &0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, \\
 &0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, 0, 0, 0, +1, \\
 &0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, \\
 &+1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, \\
 &0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, \\
 &0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, \\
 &0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, \\
 &0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, \\
 &0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, \\
 &+1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1\}
 \end{aligned} \tag{27-44}$$

In a 40 MHz transmission, the 2x HE-LTF sequence transmitted on subcarriers [−244: 244] is given by Equation (27-45).

$$\begin{aligned}
 &HELTF_{-244,244} = \\
 &\{+1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, \\
 &-1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, \\
 &-1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, \\
 &+1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, \\
 &+1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, +1, 0, +1, 0, \\
 &+1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, \\
 &+1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, \\
 &-1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, \\
 &+1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, \\
 &0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, \\
 &0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, \\
 &0, +1, 0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, \\
 &0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, +1, 0, +1, 0, +1, \\
 &0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, \\
 &0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, \\
 &0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, \\
 &0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, \\
 &0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1\}
 \end{aligned} \tag{27-45}$$

In a 40 MHz transmission, the 4x HE-LTF sequence transmitted on subcarriers [−244: 244] is given by Equation (27-46).

$$\begin{aligned}
 &HELTF_{-244,244} = \\
 &\{+1, -1, -1, -1, -1, +1, -1, -1, +1, +1, -1, +1, -1, +1, -1, +1, -1, -1, -1, +1, -1, +1, \\
 &+1, -1, -1, -1, -1, -1, -1, +1, -1, -1, +1, +1, -1, +1, -1, -1, -1, -1, +1, -1, +1, \\
 &+1, +1, -1, -1, +1, +1, +1, -1, -1, +1, +1, +1, +1, -1, +1, +1, -1, -1, +1, -1, +1, -1, +1, \\
 &-1, -1, +1, -1, +1, +1, +1, -1, -1, +1, +1, +1, -1, -1, -1, +1, -1, -1, +1, +1, -1, +1, \\
 &-1, -1, -1, -1, +1, -1, +1, +1, -1, -1, +1, +1, +1, +1, +1, +1, +1, +1, -1, +1, +1, \\
 &-1, -1, +1, -1, +1, +1, +1, +1, -1, +1, -1, -1, -1, +1, +1, -1, -1, -1, -1, -1, -1, \\
 &-1, +1, -1, -1, +1, +1, -1, +1, -1, +1, +1, -1, +1, -1, -1, -1, +1, +1, -1, -1, -1, \\
 &-1, -1, -1, +1, -1, -1, +1, +1, -1, +1, -1, -1, -1, -1, -1, +1, -1, +1, +1, -1, -1, \\
 &+1, +1, +1, -1, -1, -1, -1, -1, +1, +1, -1, +1, -1, +1, -1, +1, -1, +1, -1, +1, +1, \\
 &-1, -1, -1, +1, +1, -1, -1, -1, +1, +1, +1, -1, +1, +1, -1, -1, +1, -1, +1, +1, +1, \\
 &+1, -1, +1, -1, -1, -1, +1, -1, -1, -1, +1, +1, +1, -1, +1, +1, -1, -1, +1, -1, +1, \\
 &-1, -1, +1, -1, -1, +1, +1, +1, +1, +1, -1, +1, -1, -1, -1, +1, +1, -1, -1, -1, +1, \\
 &+1, -1, -1, -1, +1, -1, -1, +1, -1, +1, -1, +1, -1, +1, -1, +1, -1, -1, -1, +1, \\
 &+1, -1, +1, -1, -1, +1, +1, -1, +1, +1, -1, -1, -1, -1, -1, -1, -1, -1, -1, +1, -1, +1, \\
 &+1, -1, +1, -1, -1, -1, -1, +1, +1, +1, -1, +1, +1, -1, +1, +1, -1, -1, -1, -1, \\
 &-1, +1, -1, -1, +1, +1, -1, -1, +1, +1, -1, +1, -1, -1, -1, -1, -1, -1, -1, +1, -1, -1, \\
 &+1, +1, +1, +1, -1, +1, +1, -1, -1, +1, +1, +1, +1, +1, +1, -1, +1, -1, -1, -1, -1, \\
 &-1, -1, -1, -1\}
 \end{aligned} \tag{27-46}$$

In an 80 MHz transmission, the 1x HE-LTF sequence transmitted on subcarriers [−500: 500] is given by Equation (27-47).

$$\begin{aligned}
 &HELTF_{-500,500} = \\
 &\{-1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, \\
 &0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, \\
 &0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, \\
 &0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, \\
 &0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, \\
 &0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, \\
 &+1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, \\
 &0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, \\
 &0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, \\
 &0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, \\
 &0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, \\
 &0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, \\
 &+1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, \\
 &0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, \\
 &0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, \\
 &0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, \\
 &+1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, \\
 &-1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, \\
 &0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, \\
 &0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, \\
 &0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, \\
 &0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, \\
 &0, 0, 0, +1\}
 \end{aligned}
 \tag{27-47}$$

In an 80 MHz transmission, the 2x HE-LTF sequence transmitted on subcarriers [−500: 500] is given by Equation (27-48).

$$\begin{aligned}
 &HELTF_{-500,500} = \\
 &\{+1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, \\
 &0, +1, 0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, \\
 &0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, \\
 &0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, \\
 &0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, \\
 &0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, \\
 &0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, \\
 &0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, -1, \\
 &0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, \\
 &0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, +1, 0, +1, \\
 &0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, \\
 &0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, \\
 &0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, \\
 &0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, \\
 &0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, +1, 0, 0, 0, 0, 0, 0, 0, 0, \\
 &+1, 0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, \\
 &-1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, -1, 0, \\
 &+1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, \\
 &+1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, \\
 &+1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, \\
 &+1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, \\
 &+1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, -1, 0, \\
 &+1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, \\
 &-1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, \\
 &-1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, +1, 0, \\
 &-1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, +1, 0, \\
 &-1, 0, -1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, +1, 0, -1, 0, -1, 0, -1, 0, \\
 &-1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, \\
 &+1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, -1, 0, +1, 0, -1, 0, -1, 0, -1, 0, -1, 0, \\
 &+1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1, 0, -1, 0, +1, 0, +1\}
 \end{aligned}
 \tag{27-48}$$

In an 80 MHz transmission, the 4x HE-LTF sequence transmitted on subcarriers $[-500: 500]$ is given by Equation (27-49).

[illegible]

For an 80+80 MHz transmission using a 1x HE-LTF, the lower 80 MHz frequency segment shall use the 80 MHz 1x HE-LTF sequence, $HELTF_{-500,500} = LTF_{80\text{MHz_lower_1x}}$, and the upper 80 MHz frequency segment shall use the 80 MHz 1x HE-LTF sequence, $HELTF_{-500,500} = LTF_{80\text{MHz_upper_1x}}$.

For an 80+80 MHz transmission using a 2x HE-LTF, the lower 80 MHz frequency segment shall use the 80 MHz 2x HE-LTF sequence, $HELTF_{-500,500} = LTF_{80\text{MHz_lower_2x}}$, and the upper 80 MHz frequency segment shall use the 80 MHz 2x HE-LTF sequence, $HELTF_{-500,500} = LTF_{80\text{MHz_upper_2x}}$.

For an 80+80 MHz transmission using a 4x HE-LTF, the lower 80 MHz frequency segment shall use the 80 MHz 4x HE-LTF sequence, $HELTF_{-500,500} = LTF_{80\text{MHz_lower_4x}}$, and the upper 80 MHz frequency segment shall use the 80 MHz 4x HE-LTF sequence, $HELTF_{-500,500} = LTF_{80\text{MHz_upper_4x}}$.

For an OFDMA transmission, the values of HE-LTF sequence [defined in Equation (27-41) to Equation (27-56)] are set to zero if they are assigned to subcarriers within RUs that are not allocated to any user (see 27.3.10).

In an UL MU-MIMO transmission using the HE masked HE-LTF sequence mode, the HE-LTF sequence per frequency subblock is generated by masking the nonzero elements in the common HE-LTF sequence repeatedly with a distinct orthogonal code as defined by Equation (27-53).

$$HELTF_{k,u,m}^{\text{Mask}} = HELTF_k \cdot \left[P_{8 \times 8} \right]_{M_{r,u} + m, \left(\left(\left\lfloor \frac{k}{N_{\text{HE-LTF-Type}}} \right\rfloor - 1 \right) \bmod 8 \right) + 1} \quad (27-53)$$

where

$HELTF_k$ is the value of the common HE-LTF sequence on subcarrier k generated by one of the equations from Equation (27-41) to Equation (27-52) depending on the bandwidth and the HE-LTF type (excluding the 1x HE-LTF, which shall not be masked)

$\left[P_{8 \times 8} \right]$ is defined in Equation (21-45)

$M_{r,u} + m$ is the row index of the $\left[P_{8 \times 8} \right]$ corresponding to spatial time stream m of user u in the r^{th} RU

$$N_{\text{HE-LTF-Type}} = \begin{cases} 2, & \text{for 2x HE-LTF} \\ 1, & \text{for 4x HE-LTF} \end{cases}$$

In the HE TB feedback NDP, neither HE single stream pilot HE-LTF mode nor HE masked HE-LTF sequence mode is applied to the HE-LTF sequence. Only 4x HE-LTF shall be used in the HE TB feedback NDP. The 4x HE-LTF sequence is generated by Equation (27-54).

$$HELTF_{k,u}^{\text{TB_NDP}} = \begin{cases} HELTF_k, & \text{if } k \in K_{\text{tone_NDP}_u} \\ 0, & \text{otherwise} \end{cases} \quad (27-54)$$

where

$HELTF_k$ is the value of the common HE-LTF sequence on subcarrier k generated by one of the 4x HE-LTF equations [Equation (27-43), Equation (27-46), Equation (27-49), Equation (27-52)] according to the channel bandwidth

$K_{\text{tone_NDP}_u}$ is the set of subcarrier indices for user u and is defined in Table 27-32 according to the RU_TONE_SET_INDEX and FEEDBACK_STATUS

Table 27-32—HE-LTF subcarrier mapping for the HE TB feedback NDP

RU TONE_ SET_ INDEX	80 MHz		40 MHz		20 MHz	
	$K_{\text{tone_NDP}_u}$ if FEEDBACK_ STATUS is 1	$K_{\text{tone_NDP}_u}$ if FEEDBACK_ STATUS is 0	$K_{\text{tone_NDP}_u}$ if FEEDBACK_ STATUS is 1	$K_{\text{tone_NDP}_u}$ if FEEDBACK_ STATUS is 0	$K_{\text{tone_NDP}_u}$ if FEEDBACK_ STATUS is 1	$K_{\text{tone_NDP}_u}$ if FEEDBACK_ STATUS is 0
1	Use 20 MHz FEEDBACK_ STATUS = 1 Subcarrier Indices – 384	Use 20 MHz FEEDBACK_ STATUS = 0 Subcarrier Indices – 384	Use 20 MHz FEEDBACK_ STATUS = 1 Subcarrier Indices – 128	Use 20 MHz FEEDBACK_ STATUS = 0 Subcarrier Indices – 128	–113, –77, –41, 6, 42, 78	–112, –76, –40, 7, 43, 79
2					–111, –75, –39, 8, 44, 80	–110, –74, –38, 9, 45, 81
3					–109, –73, –37, 10, 46, 82	–108, –72, –36, 11, 47, 83
4					–107, –71, –35, 12, 48, 84	–106, –70, –34, 13, 49, 85
5					–105, –69, –33, 14, 50, 86	–104, –68, –32, 15, 51, 87
6					–103, –67, –31, 16, 52, 88	–102, –66, –30, 17, 53, 89
7					–101, –65, –29, 18, 54, 90	–100, –64, –28, 19, 55, 91
8					–99, –63, –27, 20, 56, 92	–98, –62, –26, 21, 57, 93
9					–97, –61, –25, 22, 58, 94	–96, –60, –24, 23, 59, 95
10					–95, –59, –23, 24, 60, 96	–94, –58, –22, 25, 61, 97
11					–93, –57, –21, 26, 62, 98	–92, –56, –20, 27, 63, 99
12					–91, –55, –19, 28, 64, 100	–90, –54, –18, 29, 65, 101
13					–89, –53, –17, 30, 66, 102	–88, –52, –16, 31, 67, 103
14					–87, –51, –15, 32, 68, 104	–86, –50, –14, 33, 69, 105
15					–85, –49, –13, 34, 70, 106	–84, –48, –12, 35, 71, 107
16					–83, –47, –11, 36, 72, 108	–82, –46, –10, 37, 73, 109
17					–81, –45, –9, 38, 74, 110	–80, –44, –8, 39, 75, 111
18					–79, –43, –7, 40, 76, 112	–78, –42, –6, 41, 77, 113

Table 27-32—HE-LTF subcarrier mapping for the HE TB feedback NDP (continued)

RU_TONE_SET_INDEX	80 MHz		40 MHz		20 MHz	
	$K_{\text{tone_NDP}_u}$ if FEEDBACK_STATUS is 1	$K_{\text{tone_NDP}_u}$ if FEEDBACK_STATUS is 0	$K_{\text{tone_NDP}_u}$ if FEEDBACK_STATUS is 1	$K_{\text{tone_NDP}_u}$ if FEEDBACK_STATUS is 0	$K_{\text{tone_NDP}_u}$ if FEEDBACK_STATUS is 1	$K_{\text{tone_NDP}_u}$ if FEEDBACK_STATUS is 0
19–36	Use 20 MHz FEEDBACK_STATUS = 1 Subcarrier Indices – 128	Use 20 MHz FEEDBACK_STATUS = 0 Subcarrier Indices – 128	Use 20 MHz FEEDBACK_STATUS = 1 Subcarrier Indices + 128	Use 20 MHz FEEDBACK_STATUS = 0 Subcarrier Indices + 128		
37–54	Use 20 MHz FEEDBACK_STATUS = 1 Subcarrier Indices + 128	Use 20 MHz FEEDBACK_STATUS = 0 Subcarrier Indices + 128				
55–72	Use 20 MHz FEEDBACK_STATUS = 1 Subcarrier Indices + 384	Use 20 MHz FEEDBACK_STATUS = 0 Subcarrier Indices + 384				
The RU_TONE_SET_INDEX for 80+80 MHz and 160 MHz shall use the 80 MHz RU_TONE_SET_INDEX definition for the lower and upper 80 MHz. The RU_TONE_SET_INDEX values 1–72 are mapped to the lower 80 MHz, and the RU_TONE_SET_INDEX values 73–144 are mapped to the upper 80 MHz.						

The generation of the time domain HE-LTF symbols per frequency segment in an HE SU PPDU, HE MU PPDU, and HE ER SU PPDU is shown in Figure 27-32 where $A_{\text{HE-LTF}}^k$ is given by Equation (27-55).

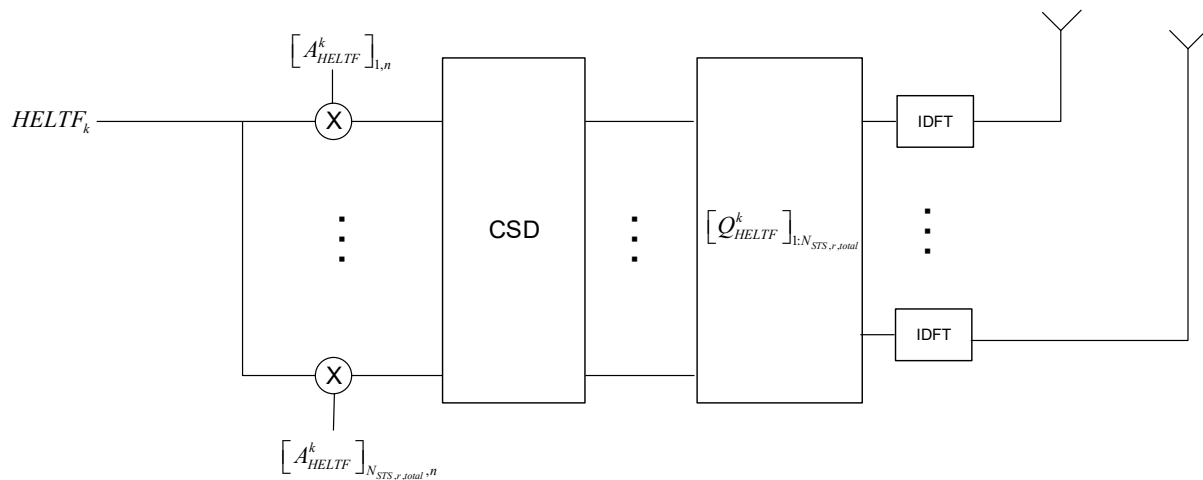


Figure 27-32—Generation of HE-LTF symbols per frequency segment in an HE SU PPDU, HE MU PPDU, and HE ER SU PPDU

The generation of the time domain HE-LTF symbols per frequency segment in an HE TB PPDU is shown in Figure 27-33.

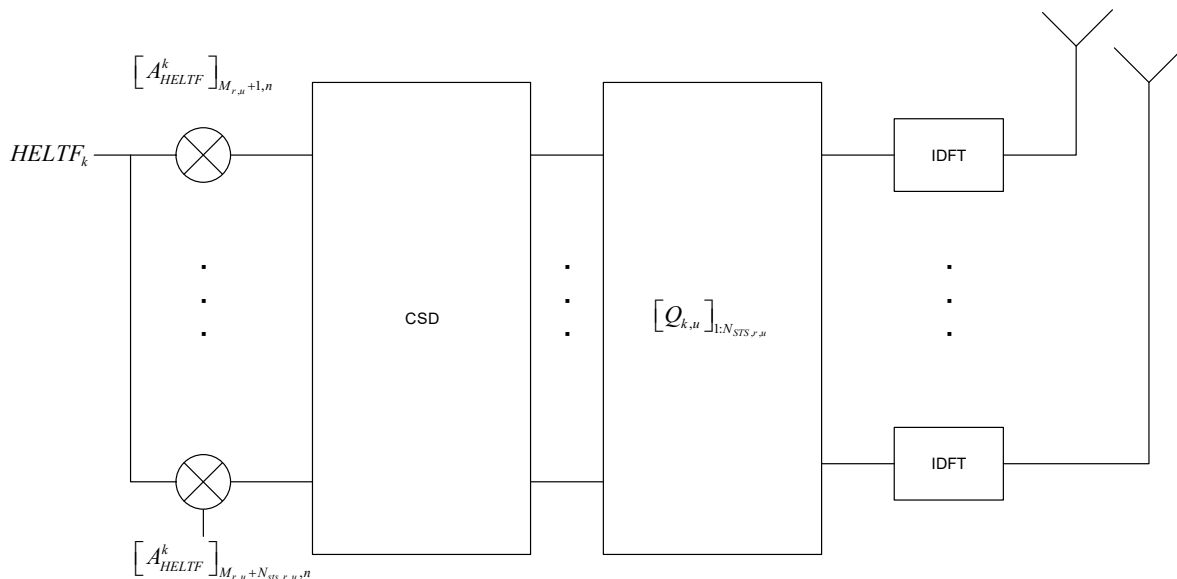


Figure 27-33—Generation of HE-LTF symbols per frequency segment in an HE TB PPDU for user u on RU r

The generation of the time domain symbol of a 1x HE-LTF is equivalent to modulating every 4 subcarriers in an OFDM symbol of 12.8 μ s excluding GI and then transmitting only the first 1/4 of the OFDM symbol in the time domain, as shown in Figure 27-34.

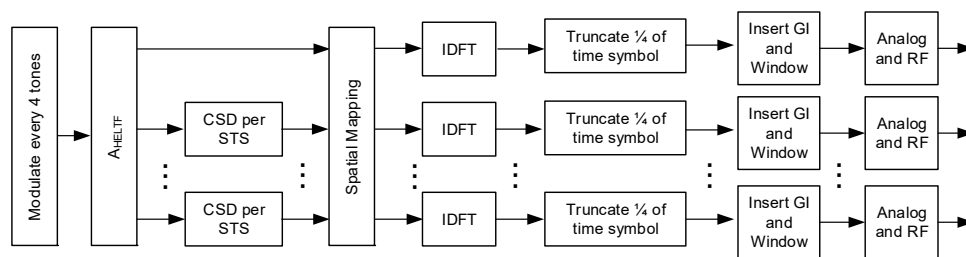


Figure 27-34—Generation of 1x HE-LTF symbols per frequency segment

The generation of the time domain symbol of a 2x HE-LTF is equivalent to modulating every other subcarrier in an OFDM symbol of 12.8 μs excluding GI and then transmitting only the first half of the OFDM symbol in time domain, as shown in Figure 27-35.

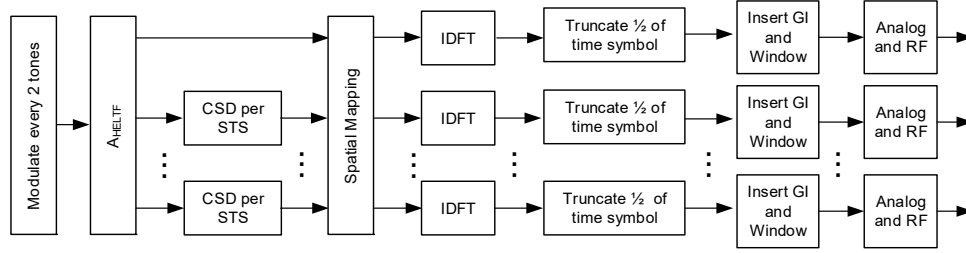


Figure 27-35—Generation of 2x HE-LTF symbols per frequency segment

$A_{\text{HE-LTF}}^k$ is given by Equation (27-55).

$$A_{\text{HE-LTF}}^k = \begin{cases} R_{\text{HE-LTF}}, & \text{if } k \in K_{\text{Pilot}} \text{ and HE single stream pilot HE-LTF mode is used} \\ P_{\text{HE-LTF}}, & \text{otherwise} \end{cases} \quad (27-55)$$

where

K_{Pilot} is the set of subcarrier indices for the pilot subcarriers as defined in 27.3.2.4

$R_{\text{HE-LTF}}$ is a $N_{\text{HE-LTF}} \times N_{\text{HE-LTF}}$ matrix whose elements are defined in Equation (27-56)

$$[R_{\text{HE-LTF}}]_{m,n} = [P_{\text{HE-LTF}}]_{1,n}, \quad 1 \leq m, n \leq N_{\text{HE-LTF}} \quad (27-56)$$

$P_{\text{HE-LTF}}$ is defined in Equation (27-57)

$$P_{\text{HE-LTF}} = \begin{cases} P_{4 \times 4}, & N_{\text{HE-LTF}} = 1, 2, 4 \\ P_{6 \times 6}, & N_{\text{HE-LTF}} = 6 \\ P_{8 \times 8}, & N_{\text{HE-LTF}} = 8 \end{cases} \quad (27-57)$$

where $P_{4 \times 4}$ is defined in Equation (19-27), $P_{6 \times 6}$ is defined in Equation (21-44), and $P_{8 \times 8}$ is defined in Equation (21-45)

If the 1x HE-LTF is used for non-OFDMA UL MU-MIMO, the HE no pilot HE-LTF mode is used.

In an HE SU PPDU, HE MU PPDU, and HE ER SU PPDU, the time domain representation of the waveform, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , shall be as described by Equation (27-58).

$$r_{\text{HE-LTF}}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = \frac{1}{\sqrt{\sum_{r=0}^{N_{\text{RU}}-1} \alpha_r^2 |K_r|}} \sum_{n=0}^{N_{\text{HE-LTF}}-1} w_{T_{\text{HE-LTF}}}(t - nT_{\text{HE-LTF}}) \sum_{r=0}^{N_{\text{RU}}-1} \frac{\alpha_r \sqrt{|K_r|}}{\sqrt{N_{\text{STS}, r, \text{total}} |K_r^{\text{HE-LTF}}|}} \left(\sum_{k \in K_r} \eta_{\text{HE-LTF}, k} \sum_{u=0}^{N_{\text{user}, r}-1} \sum_{m=1}^{N_{\text{STS}, r, u}} \left(\left[Q_k^{(i_{\text{Seg}})} \right]_{i_{\text{TX}}(M_{r,u}+m)} \left[A_{\text{HE-LTF}}^k \right]_{(M_{r,u}+m), (n+1)}^{\text{HE-LTF}''_{k,u,m}} \cdot \exp(j2\pi k \Delta_{F, \text{HE}}(t - nT_{\text{HE-LTF-SYM}} - T_{\text{GI, HE-LTF}} - T_{\text{CS, HE}}(M_{r,u}+m))) \right) \right) \quad (27-58)$$

In an HE TB PPDU, the time domain representation of the waveform of user u in the r^{th} RU, transmitted on frequency segment i_{Seg} and transmit chain i_{TX} , shall be as described by Equation (27-59).

$$r_{\text{HE-LTF}, r, u}^{(i_{\text{Seg}}, i_{\text{TX}})}(t) = \frac{1}{\sqrt{N_{\text{STS}, r, u}} |K_r^{\text{HE-LTF}}|} \sum_{n=0}^{N_{\text{HE-LTF}}-1} w_{T_{\text{HE-LTF}}} (t - nT_{\text{HE-LTF}}) \quad (27-59)$$

$$\sum_{k \in K_r} \sum_{m=1}^{N_{\text{STS}, r, u}} \left(\left[Q_{k, u}^{(i_{\text{Seg}})} \right]_{i_{\text{TX}}, m} \left[A_{\text{HE-LTF}}^k \right]_{(M_{r, u} + m), (n+1)}^{\text{HE-LTF}''_{k, u, m}} \cdot \exp(j2\pi k \Delta_{F, \text{HE}}(t - nT_{\text{HE-LTF-SYM}} - T_{\text{GI, HE-LTF}} - T_{\text{CS, HE}}(M_{r, u} + m))) \right)$$

In Equation (27-58) and Equation (27-59), the following notations are used:

$N_{\text{user}, r}$	is the number of HE MU PPDU recipients (see Table 27-15) in RU r
$\text{HE-LTF}''_{k, u, m}$	is the HE-LTF sequence applied on subcarrier k for spatial stream m of user u
$\text{HE-LTF}''_{k, u, m}$	$= \begin{cases} \text{HE-LTF}^{\text{Mask}}_{k, u, m}, & \text{if HE masked HE-LTF sequence mode is used} \\ \text{HE-LTF}^{\text{TB_NDP}}_{k, u}, & \text{for an HE TB feedback NDP} \\ \text{HE-LTF}_k, & \text{otherwise} \end{cases}$
α_r	is defined in 27.3.10
$\eta_{\text{HE-LTF}, k}$	is an HE PPDU format-dependent scaling factor, defined by
	$\eta_{\text{HE-LTF}, k} = \begin{cases} \sqrt{2}, & \text{for an HE ER SU PPDU} \\ 1, & \text{otherwise} \end{cases}$
$N_{\text{HE-LTF}}$	is the number of OFDM symbols in the HE-LTF field
$T_{\text{CS, HE}}(M_{r, u} + m)$	represents the cyclic shift for space-time stream $M_{r, u} + m$ as defined in 27.3.11.2.2
$Q_k^{(i_{\text{Seg}})}$ and $Q_{k, u}^{(i_{\text{Seg}})}$	are defined in 27.3.10
$A_{\text{HE-LTF}}^k$	is defined in Equation (27-55)
$M_{r, u}$	is given in Table 27-15 for HE SU PPDU, HE ER SU PPDU, and HE MU PPDU. For an HE TB PPDU it is given by the TXVECTOR parameter STARTING_STS_NUM.
K_r	is the set of subcarrier indices for the tones in the RU r as defined in 27.3.10
$ K_r $ and $ K_r^{\text{HE-LTF}} $	are defined after Equation (27-5)
$ K_r^{\text{HE-LTF}} $	is the cardinality of the set of modulated subcarriers within K_r for HE-LTF field, as defined in 27.3.10
other variables	are defined after Equation (27-1), Equation (27-3), Equation (27-4), Equation (27-6), Equation (27-8), Equation (27-12), and Equation (27-40)

27.3.12 Data field

27.3.12.1 General

The number of OFDM symbols in the Data field is determined by the LENGTH field in the L-SIG field [see Equation (27-11)], the preamble duration, and the settings of the GI+HE-LTF Size, Pre-FEC Padding Factor, and PE Disambiguity fields in the HE-SIG-A field (see 27.3.11.7) for an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU or in the soliciting Triggering frame for an HE TB PPDU. Data field OFDM

symbols in an HE PPDU shall use a DFT period of 12.8 μ s and subcarrier spacing of 78.125 kHz. An HE STA shall support Data field OFDM symbols in an HE PPDU with guard interval durations of 0.8 μ s, 1.6 μ s, and 3.2 μ s. HE PPDU shall have HE single stream pilot in the Data field. In UL MU-MIMO transmissions, all streams use the same pilot sequence.

If BCC encoding is used, the Data field shall consist of the SERVICE field, the PSDU, the pre-FEC PHY padding bits, the tail bits, and the post-FEC padding bits. If LDPC encoding is used, the Data field shall consist of the SERVICE field, the PSDU, the pre-FEC PHY padding bits, and the post-FEC padding bits. No tail bits are present if LDPC encoding is used.

The Data field of the HE PPDU contains data for one or more users.

27.3.12.2 Pre-FEC padding process

A two-step padding process is applied to an HE PPDU. A pre-FEC padding process including both pre-FEC MAC and pre-FEC PHY padding is applied before conducting FEC coding, and a post-FEC PHY padding process is applied on the FEC encoded bits.

Four pre-FEC padding boundaries partition the last one (in the case of non-STBC) or two (in the case of STBC) OFDM symbols of an HE PPDU into four symbol segments. The pre-FEC padding may pad toward one of the four possible boundaries. The four pre-FEC padding boundaries are represented by a pre-FEC padding factor parameter.

Figure 27-36 illustrates these four possible symbol segments in the last OFDM symbol of a non-STBC case, and the general padding process assuming the desired pre-FEC padding boundary, represented by the pre-FEC padding factor, is 1. In the case of STBC, the FEC output bits and post-FEC padding bits are modulated into the last two OFDM symbols by STBC encoding, each with the same pre-FEC padding boundary.

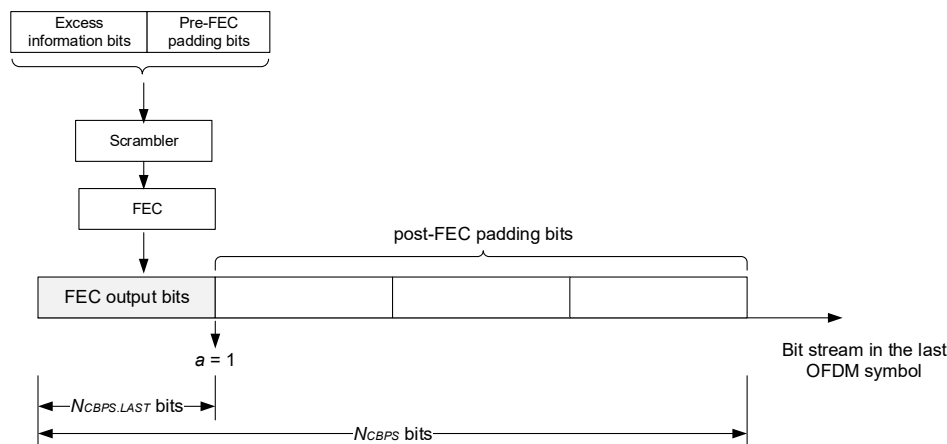


Figure 27-36—HE PPDU padding process in the last OFDM symbol (non-STBC) if $a = 1$

The pre-FEC padding process is described in this subclause, and the encoding and post-FEC padding process are described in 27.3.12.5. While this subclause describes the pre-FEC padding processing of an SU transmission, its extension to MU transmission is described in 27.3.12.5.4 and 27.3.12.5.5.

In an HE SU PPDU and HE ER SU PPDU transmission, the transmitter first computes the number of bits left in the last OFDM symbol(s) based on Equation (27-60).

$$N_{Excess} = \text{mod}(8 \cdot \text{APEP_LENGTH} + N_{Tail} + N_{service}, m_{STBC} \cdot N_{DBPS}) \quad (27-60)$$

where

m_{STBC} is 2 if STBC is used, and 1 otherwise
APEP_LENGTH is the TXVECTOR parameter APEP_LENGTH
 N_{Tail} is the number of tail bits per encoder as defined in Table 27-12
 $N_{service}$ is the number of bits in the SERVICE field as defined in Table 27-12
 N_{DBPS} is the number of data bits per symbol

Based on N_{Excess} , compute the initial number of symbol segments in the last OFDM symbol(s), initial pre-FEC padding factor value or a_{init} , as shown in Equation (27-61).

$$a_{init} = \begin{cases} 4, & \text{if } N_{Excess} = 0 \\ \min\left(\left\lceil \frac{N_{Excess}}{m_{STBC} \cdot N_{DBPS,short}} \right\rceil, 4\right), & \text{otherwise} \end{cases} \quad (27-61)$$

where

$N_{DBPS,short} = N_{CBPS,short} \cdot R$, where R is the coding rate and $N_{CBPS,short} = N_{SD,short} \cdot N_{SS} \cdot N_{BPSCS}$

The parameter $N_{SD,short}$ values for different RU sizes are as shown in Table 27-33.

Table 27-33— $N_{SD,short}$ values

RU size	$N_{SD,short}$	
	DCM = 0	DCM = 1
26-tone	6	2
52-tone	12	6
106-tone	24	12
242-tone	60	30
484-tone	120	60
996-tone	240	120
2×996-tone	492	246

Given the a_{init} values, the initial number of data bits per symbol and the initial number of coded bits per symbol in the last OFDM symbol(s) are defined in Equation (27-62).

$$N_{DBPS, last, init} = \begin{cases} a_{init} N_{DBPS, short} & \text{if } a_{init} < 4 \\ N_{DBPS} & \text{if } a_{init} = 4 \end{cases} \quad (27-62)$$

$$N_{CBPS, last, init} = \begin{cases} a_{init} N_{CBPS, short} & \text{if } a_{init} < 4 \\ N_{CBPS} & \text{if } a_{init} = 4 \end{cases}$$

For an HE SU PPDU and HE ER SU PPDU, the number of pre-FEC pad bits is calculated using Equation (27-63).

$$N_{PAD, Pre-FEC} = (N_{SYM, init} - m_{STBC}) N_{DBPS} + m_{STBC} N_{DBPS, last, init} - 8 \cdot APEP_LENGTH - N_{Tail} - N_{service} \quad (27-63)$$

where $N_{SYM, init}$ is the initial number of data OFDM symbols with BCC or LDPC encoding in an HE SU PPDU or HE ER SU PPDU as defined by Equation (27-64).

$$N_{SYM, init} = m_{STBC} \cdot \left\lceil \frac{8 \cdot APEP_LENGTH + N_{Tail} + N_{service}}{m_{STBC} N_{DBPS}} \right\rceil \quad (27-64)$$

Among the pre-FEC padding bits, the MAC delivers a PSDU that fills the available octets in the Data field of the HE PPDU (see A-MPDU padding for HE PPDUs in 26.6.2.2 and 26.6.2.3), toward the desired initial pre-FEC padding boundary, represented by a_{init} value, in the last OFDM symbol(s). The number of pre-FEC pad bits added by the MAC will always be a multiple of 8. The PHY then determines the number of remaining pad bits to add and appends them to the PSDU. The number of pre-FEC pad bits added by PHY will always be 0 to 7. The procedure is defined in Equation (27-65).

$$N_{PAD, Pre-FEC, MAC} = \left\lfloor \frac{N_{PAD, Pre-FEC}}{8} \right\rfloor \cdot 8 \quad (27-65)$$

$$N_{PAD, Pre-FEC, PHY} = N_{PAD, Pre-FEC} \bmod 8$$

27.3.12.3 SERVICE field

The SERVICE field of HE PPDU is shown in Table 27-34.

Table 27-34—SERVICE field

Bits	Field	Description
B0–B6	Scrambler Initialization	Set to 0
B7–B15	Reserved	Set to 0

27.3.12.4 Scrambler

The SERVICE field, PSDU, and pre-FEC PHY padding of the Data field shall be scrambled by the scrambler defined in 17.3.5.5 (PHY DATA scrambler and descrambler). The Clause 17 TXVECTOR parameters CH_BANDWIDTH_IN_NON_HT and DYN_BANDWIDTH_IN_NON_HT are not present; therefore, the initial state of the scrambler is set to a nonzero pseudorandom seed. A different nonzero pseudorandom seed may be used for each user in an HE MU PPDU.

27.3.12.5 Coding

The Data field shall be encoded using either the binary convolutional code (BCC) defined in 27.3.12.5.1 or the low-density parity check (LDPC) code defined in 27.3.12.5.2. The coding type is selected by the Coding field in the HE-SIG-A field in an HE SU PPDU or HE ER SU PPDU, by the Coding field in the HE-SIG-B field per user subfield(s) in an HE MU PPDU, or by the UL FEC Coding Type subfield in the User Info field in the corresponding Trigger frame in an HE TB PPDU, as defined in 27.3.11.7, 27.3.11.8, and 9.3.1.22, respectively.

When conducting BCC FEC encoding for an HE PPDU, the number of encoders is always 1.

LDPC is the only FEC coding scheme in the HE PPDU Data field for a 484-, 996-, and 2×996-tone RU. LDPC is the only FEC coding scheme in the HE PPDU Data field for HE-MCSs 10 and 11. Support for BCC coding is limited to less than or equal to four spatial streams and HE-MCSs 0 to 9 (per user in the case of MU-MIMO) and is mandatory (for both transmit and receive) for RU sizes less than or equal to a 242-tone RU. The LDPC Coding In Payload subfield in the HE Capabilities element indicates support for the transmission and reception of the LDPC encoded PPDUs. Support for LDPC coding (for both transmit and receive) is mandatory for HE STAs declaring support for at least one of HE 40/80/160/80+80 SU PPDU bandwidths, for HE STAs declaring support for more than 4 spatial streams, or for HE STAs declaring support for HE-MCSs 10 and 11, according to the LDPC Coding In Payload subfield in the HE Capabilities element as defined in 9.4.2.248. Otherwise, support of LDPC coding for either transmit or receive is optional.

27.3.12.5.1 BCC coding and puncturing

The information bits and pre-FEC padding bits of user u are encoded by a rate $R = 1/2$ convolutional encoder defined in 17.3.5.6. After encoding, the encoded data is punctured by the method defined in 17.3.5.6 (except for rate 5/6) to achieve the rate selected by the modulation and coding scheme. When rate 5/6 coding is selected, the puncturing scheme will be the same as described in 19.3.11.6.

If DCM is used with BPSK modulation in a 106-tone or 242-tone RU with $N_{SS} = 1$, then after every $2 \times N_{DBPS}$ coded bits, one padding bit is added. The padding bit may be set to any value.

For an HE SU PPDU and HE ER SU PPDU with BCC encoding,

$$N_{SYM} = N_{SYM,init} \quad (27-66)$$

and

$$a = a_{init} \quad (27-67)$$

where

- $N_{SYM,init}$ is defined in Equation (27-64)
- a_{init} is defined in Equation (27-61)
- a is the pre-FEC padding factor

The number of data bits per symbol in the last OFDM symbol(s) of an HE SU PPDU or HE ER SU PPDU is $N_{DBPS,last} = N_{DBPS,last,init}$, where $N_{DBPS,last,init}$ is defined in Equation (27-62).

The number of coded bits per symbol in the last OFDM symbol(s) of an HE SU PPDU or HE ER SU PPDU is $N_{CBPS,last} = N_{CBPS,last,init}$, where $N_{CBPS,last,init}$ is defined in Equation (27-62).

27.3.12.5.2 LDPC coding

For an HE SU PPDU or HE ER SU PPDU using LDPC coding on the Data field, the LDPC coding process described in 19.3.11.7 shall be used with the following modifications:

First, all bits in the Data field including the scrambled SERVICE, PSDU, and pre-FEC pad bits are encoded. Thus, N_{pld} for HE PPDU shall be computed using Equation (27-68) instead of Equation (19-35).

$$N_{pld} = (N_{SYM,init} - m_{STBC})N_{DBPS} + m_{STBC}N_{DBPS,last,init} \quad (27-68)$$

where $N_{SYM,init}$ is defined in Equation (27-64).

N_{avbits} shall be computed using Equation (27-69) instead of Equation (19-36).

$$N_{avbits} = (N_{SYM,init} - m_{STBC}) \cdot N_{CBPS} + m_{STBC} \cdot N_{CBPS,last,init} \quad (27-69)$$

In addition, in step d) of the LDPC encoding process in 19.3.11.7.5, if the following condition is met:

$$(N_{punc} > 0.1 \times N_{CW} \times L_{LDPC} \times (1 - R)) \text{ AND } \left(N_{shrt} < 1.2 \times N_{punc} \times \frac{R}{1 - R} \right) \text{ is true OR if } \\ N_{punc} > 0.3 \times N_{CW} \times L_{LDPC} \times (1 - R) \text{ is true,}$$

then the LDPC Extra Symbol Segment field of the HE-SIG-A field shall be set to 1, N_{avbits} shall be increased according to the Equation (27-70) instead of Equations (19-39), and N_{punc} shall be recomputed as in Equation (19-40):

$$N_{avbits} = \begin{cases} N_{avbits} + m_{STBC} \cdot (N_{CBPS} - 3N_{CBPS,short}), & \text{if } a_{init} = 3 \\ N_{avbits} + m_{STBC} \cdot N_{CBPS,short} & \text{otherwise} \end{cases} \quad (27-70)$$

and then compute the pre-FEC padding factor a and N_{SYM} using Equation (27-71).

$$\begin{cases} N_{SYM} = N_{SYM,init} + m_{STBC} \text{ and } a = 1, & \text{if } a_{init} = 4 \\ N_{SYM} = N_{SYM,init} \text{ and } a = a_{init} + 1, & \text{otherwise} \end{cases} \quad (27-71)$$

If in step d) of the LDPC encoding process in 19.3.11.7.5, the above mentioned condition is not met, then the LDPC Extra Symbol Segment field in the HE-SIG-A field shall be set to 0, and

$$\begin{aligned} N_{SYM} &= N_{SYM,init} \\ a &= a_{init} \end{aligned} \quad (27-72)$$

Using the pre-FEC padding factor a , compute N_{CBPS} of the last symbol using Equation (27-73).

$$N_{CBPS,last} = \begin{cases} a \cdot N_{CBPS,short} & \text{if } a < 4 \\ N_{CBPS} & \text{if } a = 4 \end{cases} \quad (27-73)$$

The number of data bits of the last symbol is calculated as $N_{DBPS,last} = N_{DBPS,last,init}$.

LDPC coding used in HE MU PPDU shall also follow the definitions in 19.3.11.7. Refer to 27.3.12.5.4 for a description of the LDPC encoding process for HE MU PPDU.

27.3.12.5.3 Post-FEC padding

The number of post-FEC padding bits in each of the last m_{STBC} symbols is computed by using Equation (27-74).

$$N_{PAD,Post-FEC} = N_{CBPS} - N_{CBPS,last} \quad (27-74)$$

The last m_{STBC} symbols shall consist of $N_{CBPS,last}$ bits from the FEC output followed by $N_{PAD,Post-FEC}$ post-FEC padding bits. The values of the post-FEC padding bits are not specified and are determined by the implementation.

27.3.12.5.4 Encoding process for an HE MU PPDU

For an HE MU PPDU, all the users shall use a common pre-FEC padding factor value and a common N_{SYM} value. The padding process is described as follows:

First compute initial pre-FEC padding factor value, $a_{init,u}$, for each user u using Equation (27-61) and the initial number of OFDM symbols, $N_{SYM,init,u}$, for each user u using Equation (27-64). Among all the users, derive the user index with the longest encoded packet duration, as in Equation (27-75).

$$u_{\max} = \arg \max_{u=0}^{N_{user,total}-1} (N_{SYM,init,u} - m_{STBC} + \frac{m_{STBC}a_{init,u}}{4}) \quad (27-75)$$

where

$$\arg \max f(x) := \{x | x \in [0, N_{user,total} - 1] \wedge \forall y \in [0, N_{user,total} - 1]; f(y) \leq f(x)\}$$

m_{STBC} is the common STBC setting among all the users, as described in 27.3.11.7

Then the common a_{init} and $N_{SYM,init}$ values among all the users are derived by Equation (27-76).

$$\begin{aligned} N_{SYM,init} &= N_{SYM,init,u_{\max}} \\ a_{init} &= a_{init,u_{\max}} \end{aligned} \quad (27-76)$$

Calculate each user's initial number of coded bits in its last symbol as in Equation (27-77).

$$\begin{aligned} N_{DBPS,last,init,u} &= \begin{cases} a_{init}N_{DBPS,short,u} & \text{if } a_{init} < 4 \\ N_{DBPS,u} & \text{if } a_{init} = 4 \end{cases} \\ N_{CBPS,last,init,u} &= \begin{cases} a_{init}N_{CBPS,short,u} & \text{if } a_{init} < 4 \\ N_{CBPS,u} & \text{if } a_{init} = 4 \end{cases} \end{aligned} \quad (27-77)$$

For each user with LDPC encoding, the number of pre-FEC padding bits is computed as in Equation (27-78).

$$N_{PAD,Pre-FEC,u} = (N_{SYM,init} - m_{STBC})N_{DBPS,u} + m_{STBC}N_{DBPS,last,init,u} - 8 \cdot APEP_LENGTH_u - N_{service} \quad (27-78)$$

For each user with LDPC encoding, the parameters $N_{pld,u}$ and $N_{avbits,u}$ are computed using Equation (27-79) and Equation (27-80), respectively.

$$N_{pld,u} = (N_{SYM,init} - m_{STBC})N_{DBPS,u} + m_{STBC}N_{DBPS,last,init,u} \quad (27-79)$$

$$N_{avbits,u} = (N_{SYM,init} - m_{STBC})N_{CBPS,u} + m_{STBC}N_{CBPS,last,init,u} \quad (27-80)$$

For each user with LDPC encoding, continue the LDPC encoding process in 19.3.11.7.5 starting with the parameters $N_{pld,u}$ and $N_{avbits,u}$. If there is at least one user with LDPC encoding for which step d) of the LDPC encoding process in 19.3.11.7.5 meets the following condition:

$$(N_{punc,u} > 0.1 \times N_{CW,u} \times L_{LDPC,u} \times (1 - R_u)) \text{ AND } \left(N_{shrt,u} < 1.2 \times N_{punc,u} \times \frac{R_u}{1 - R_u} \right) \text{ is true OR if } \\ N_{punc,u} > 0.3 \times N_{CW,u} \times L_{LDPC,u} \times (1 - R_u) \text{ is true,}$$

where $N_{punc,u}$, $N_{CW,u}$, $L_{LDPC,u}$ and $N_{shrt,u}$ are the LDPC encoding parameters for user u , as defined in 19.3.11.7.5, and R_u is the coding rate of user u .

Then the LDPC Extra Symbol Segment field in the HE-SIG-A field shall be set to 1, and all the users with LDPC encoding shall increment N_{avbits} and recompute N_{punc} using Equation (27-81) and Equation (27-82) once.

$$N_{avbits,u} = \begin{cases} N_{avbits,u} + m_{STBC} \cdot (N_{CBPS,u} - 3N_{CBPS,short,u}), & \text{if } a_{init} = 3 \\ N_{avbits,u} + m_{STBC} \cdot N_{CBPS,short,u} & \text{otherwise} \end{cases} \quad (27-81)$$

$$N_{punc,u} = \max(0, (N_{CW,u} \times L_{LDPC,u}) - N_{avbits,u} - N_{shrt,u}) \quad (27-82)$$

Update the common pre-FEC padding factor and N_{SYM} values for all users using Equation (27-83).

$$\begin{cases} N_{SYM} = N_{SYM,init} + m_{STBC} \text{ and } a = 1, & \text{if } a_{init} = 4 \\ N_{SYM} = N_{SYM,init} \text{ and } a = a_{init} + 1, & \text{otherwise} \end{cases} \quad (27-83)$$

If none of the users with LDPC encoding in step d) of 19.3.11.7.5 meet the condition that $(N_{punc,u} > 0.1 \times N_{CW,u} \times L_{LDPC,u} \times (1 - R_u))$ and $\left(N_{shrt,u} < 1.2 \times N_{punc,u} \times \frac{R_u}{1 - R_u} \right)$ are true or $N_{punc,u} > 0.3 \times N_{CW,u} \times L_{LDPC,u} \times (1 - R_u)$ is true, or if all the users in the HE MU PPDU are BCC encoded, then the LDPC Extra Symbol Segment field in the HE-SIG-A field shall be set to 0, and the common pre-FEC padding factor and N_{SYM} values for all users shall be updated by Equation (27-84).

$$a = a_{init}, N_{SYM} = N_{SYM,init} \quad (27-84)$$

For the users with LDPC encoding, $N_{DBPS,last,u} = N_{DBPS,last,init,u}$

For the users with BCC encoding, update the N_{DBPS} of the last symbol as

$$N_{DBPS,last,u} = \begin{cases} a \cdot N_{DBPS,short,u} & \text{if } a < 4 \\ N_{DBPS,u} & \text{if } a = 4 \end{cases} \quad (27-85)$$

For each user with either LDPC or BCC encoding, update the N_{CBPS} of the last symbol as

$$N_{CBPS, last, u} = \begin{cases} a \cdot N_{CBPS, short, u}, & \text{if } a < 4 \\ N_{CBPS, u}, & \text{if } a = 4 \end{cases}$$

For the users with BCC encoding, the number of pre-FEC padding bits is shown in Equation (27-86).

$$N_{PAD, Pre-FEC, u} = (N_{SYM} - m_{STBC})N_{DBPS, u} + m_{STBC}N_{DBPS, last, u} - 8 \cdot APEP_LENGTH_u - N_{Tail} - N_{service} \quad (27-86)$$

For each user with either LDPC or BCC encoding, the number of post-FEC padding bits in each of the last m_{STBC} symbol(s) is computed as in Equation (27-87).

$$N_{PAD, Post-FEC, u} = N_{CBPS, u} - N_{CBPS, last, u} \quad (27-87)$$

Among the pre-FEC padding bits, the MAC delivers a PSDU that fills the available octets in the Data field of the HE PPDU, toward the desired initial pre-FEC padding boundary represented by a_{init} for users encoded by LDPC and the pre-FEC padding boundary represented by a for users encoded by BCC, in the last OFDM symbol(s). The PHY then determines the number of pad bits to add and appends them to the PSDU. The number of pre-FEC pad bits added by PHY will always be 0 to 7. The procedure is defined in Equation (27-88) and Equation (27-89).

$$N_{PAD, Pre-FEC, MAC, u} = 8 \cdot \left\lfloor \frac{N_{PAD, Pre-FEC, u}}{8} \right\rfloor \quad (27-88)$$

$$N_{PAD, Pre-FEC, PHY, u} = N_{PAD, Pre-FEC, u} \bmod 8 \quad (27-89)$$

27.3.12.5.5 Encoding process for an HE TB PPDU

For an HE TB PPDU sent in response to a Trigger frame, the AP indicates the UL Length, GI And HE-LTF Type, Number Of HE-LTF Symbols And Midamble Periodicity, Pre-FEC Padding Factor, UL STBC, LDPC Extra Symbol Segment, PE Disambiguity, and Doppler fields in the Trigger frame. The common values T_{PE} and N_{SYM} are derived by non-AP STAs as shown in Equation (27-114) and Equation (27-115) (in 27.3.13). The AP should set the LDPC Extra Symbol Segment field in the Common Info field of the Trigger frame to 1 if the calculations described in the LDPC encoding process indicate the need for an LDPC extra symbol segment for any STA solicited by the AP for HE TB PPDU transmission using LDPC.

NOTE—The AP might set the LDPC Extra Symbol Segment field to 1 regardless of these calculations. The AP might select a value for the Pre-FEC Padding Factor field that differs from that derived from the calculations described in the BCC or LDPC encoding process.

For an HE TB PPDU sent in response to a frame containing a TRS Control subfield, the parameters used to derive the common values T_{PE} and N_{SYM} are described in 26.5.2.3.4.

For an HE TB PPDU with BCC encoding, follow the HE SU PPDU padding and encoding process introduced in 27.3.12.2, 27.3.12.5.1, and 27.3.12.5.3 with initial parameters as follows:

- If the TXVECTOR parameter TRIGGER_METHOD is TRIGGER_FRAME, the initial parameters are $N_{SYM, init} = N_{SYM}$ and $a_{init} = a$, where a is the TXVECTOR parameter PRE_FEC_PADDING_FACTOR and N_{SYM} is the common number of data OFDM symbols derived from the UL Length, Number Of HE-LTF Symbols And Midamble Periodicity, and Doppler subfields of the Common Info field in the Trigger frame.
- If the TXVECTOR parameter TRIGGER_METHOD is TRS, the initial parameters are $N_{SYM, init} = F_{VAL} + 1$ and $a_{init} = 4$, where F_{VAL} is the value of the UL Data Symbols subfield of the TRS Control subfield.

For an HE TB PPDU with LDPC encoding, follow the HE SU PPDU padding and encoding process introduced in 27.3.12.2, 27.3.12.5.2, and 27.3.12.5.3, with the following exceptions:

- If the TXVECTOR parameter TRIGGER_METHOD is TRIGGER_FRAME and the LDPC Extra Symbol Segment field in the Trigger frame is 1, set the initial parameters following Equation (27-90).

$$\begin{cases} a_{init} = 4 \text{ and } N_{SYM,init} = N_{SYM} - m_{STBC}, & \text{if } a = 1 \\ a_{init} = a - 1 \text{ and } N_{SYM,init} = N_{SYM}, & \text{otherwise} \end{cases} \quad (27-90)$$

where m_{STBC} is 2 if the Trigger frame indicates STBC and 1 otherwise. Then continue with the LDPC encoding process in 27.3.12.5.2, during which N_{avbits} is always incremented as in Equation (27-70) and N_{punc} is always recomputed as in Equation (19-40).

- If the TXVECTOR parameter TRIGGER_METHOD is TRIGGER_FRAME and the LDPC Extra Symbol Segment field in the Trigger frame is 0, then set the initial parameters to $N_{SYM,init} = N_{SYM}$, and $a_{init} = a$. Then continue with the LDPC encoding process as in 27.3.12.5.2, during which N_{avbits} and N_{punc} are not changed and $a = a_{init}$.
- If the TXVECTOR parameter TRIGGER_METHOD is TRS, then the parameter LDPC_EXTRA_SYMBOL is 1, and the initial parameters are set to $N_{SYM,init} = F_{VAL} + 1$ and $a_{init} = 3$, where F_{VAL} is the value of the UL Data Symbols subfield of the TRS Control subfield. Then continue with the LDPC encoding process as in 27.3.12.5.2, during which N_{avbits} is always incremented as in Equation (27-70) and N_{punc} is always recomputed as in Equation (19-40).

27.3.12.6 Stream parser

After scrambling, coding, puncturing, and post-FEC padding, the data bits are processed in groups of N_{CBPS} bits. Each of these groups is rearranged into N_{SS} blocks of N_{CBPSS} bits ($N_{SS,u}$ blocks of $N_{CBPSS,u}$ bits in the case of an HE MU transmission). This operation is referred to as *stream parsing* and is described in this subclause.

The description is given in terms of an SU transmission. For MU transmissions, the rearrangements are carried out in the same way per user.

The number of bits assigned to a single axis (real or imaginary) in a constellation point in a spatial stream is denoted by Equation (27-91).

$$s = \max(1, \frac{N_{BPSCS}}{2}) \quad (27-91)$$

The sum of these over all streams is $S = N_{SS} \cdot s$.

Consecutive blocks of s bits are assigned to different spatial streams in a round robin fashion.

For the N_{CBPS} bits of each OFDM symbol, S bits from the output of the encoder are divided among all spatial streams, s bits per stream.

NOTE—For all RU sizes the coded bits per OFDM symbol are always evenly allocated to N_{SS} spatial streams.

The following equations are an equivalent description to the above procedure. Bit i at the output of the encoder is assigned to input bit k of spatial stream i_{SS} where

$$i = (i_{SS} - 1)s + S \cdot \left\lfloor \frac{k}{s} \right\rfloor + (k \bmod s)$$

$$i_{SS} = 1, 2, \dots, N_{SS}$$

$$i = 0, 1, \dots, N_{CBPS} - 1$$

$$k = 0, 1, \dots, N_{CBPSS} - 1$$

27.3.12.7 Segment parser

The description in this subclause is for an SU transmission. For an MU transmissions, the rearrangements are carried out in the same way but per user.

For a 20 MHz, 40 MHz, 80 MHz, 160 MHz, and 80+80 MHz transmission with a 26-, 52-, 106-, 242-, 484-, or 996-tone RU, the segment parser is bypassed and the output bits are as specified in Equation (27-92).

$$y_{k,l} = x_k \quad (27-92)$$

where

x_k is bit k of a block of N_{CBPSS} bits, $k = 0$ to $N_{CBPSS} - 1$

l is the frequency subblock index and $l = 0$ for a 26-, 52-, 106-, 242-, 484-, and 996-tone RU

$y_{k,l}$ is bit k of the frequency subblock l

For a 160 MHz and 80+80 MHz transmission with a 2×996-tone RU, the output bits of each stream parser are first divided into blocks of N_{CBPSS} bits ($N_{CBPSS,u}$ bits in the case of an MU transmission). Then, each block is further divided into two frequency subblocks of $N_{CBPSS}/2$ bits as shown in Equation (27-93).

$$y_{k,l} = x_m \quad (27-93)$$

$$m = 2s \cdot \left\lfloor \frac{k}{s} \right\rfloor + l \cdot s + (k \bmod s), \quad k = 0, 1, \dots, \frac{N_{CBPSS}}{2} - 1 \quad (27-94)$$

where

x_m is bit m of a block of N_{CBPSS} bits and $m = 0, \dots, N_{CBPSS} - 1$

l is the frequency subblock index and $l = 0, 1$

$y_{k,l}$ is bit k of the frequency subblock l

s is defined in Equation (27-91)

27.3.12.8 BCC interleavers

For ease of explanation, the operation of the interleaver is described only for the SU case. For user u of an MU transmission, the interleaver operates in the same way on the output bits for the user from the stream parser by replacing N_{SS} , N_{CBPSS} , N_{CBPS} , and N_{BPSCS} with $N_{SS,u}$, $N_{CBPSS,u}$, $N_{CBPS,u}$ and $N_{BPSCS,u}$ respectively. That is, the operation of the interleaver is the same as if the transmission were an SU one, consisting of bits from only that user.

The BCC interleaver operation is specified in 21.3.10.8. The interleaver parameters, N_{COL} , N_{ROW} , and N_{ROT} , for the Data field depend on the RU size and whether or not DCM is used and are defined in the “RU size” column of Table 27-35.

Table 27-35—BCC interleaver parameters

DCM	Parameter	RU size (tones)				HE-SIG-A/ HE-SIG-B (tones)
		26	52	106	242	
Not used	N_{COL}	8	16	17	26	13
	N_{ROW}	$3 \times N_{BPSCS}$	$3 \times N_{BPSCS}$	$6 \times N_{BPSCS}$	$9 \times N_{BPSCS}$	$4 \times N_{BPSCS}$
	N_{ROT}	2	11	29	58	—
Used	N_{COL}	4	8	17	13	13
	N_{ROW}	$3 \times N_{BPSCS}$	$3 \times N_{BPSCS}$	$3 \times N_{BPSCS}$	$9 \times N_{BPSCS}$	$2 \times N_{BPSCS}$
	N_{ROT}	2	2	11	29	—

The interleaver parameters, N_{COL} and N_{ROW} , for the HE-SIG-A and HE-SIG-B fields are defined in the “HE-SIG-A/HE-SIG-B” column of Table 27-35.

NOTE—DCM is not used on the HE-SIG-A field.

27.3.12.9 Constellation mapping

The mapping between the input bits of the constellation mapper and complex constellation points for BPSK, QPSK, 16-QAM, 64-QAM, and 256-QAM is defined in 21.3.10.9.

For 1024-QAM, the mapping of the bits at the output of the stream parser or segment parser (if present) to the complex constellation points is defined in Figure 27-37, Figure 27-38, Figure 27-39, and Figure 27-40.

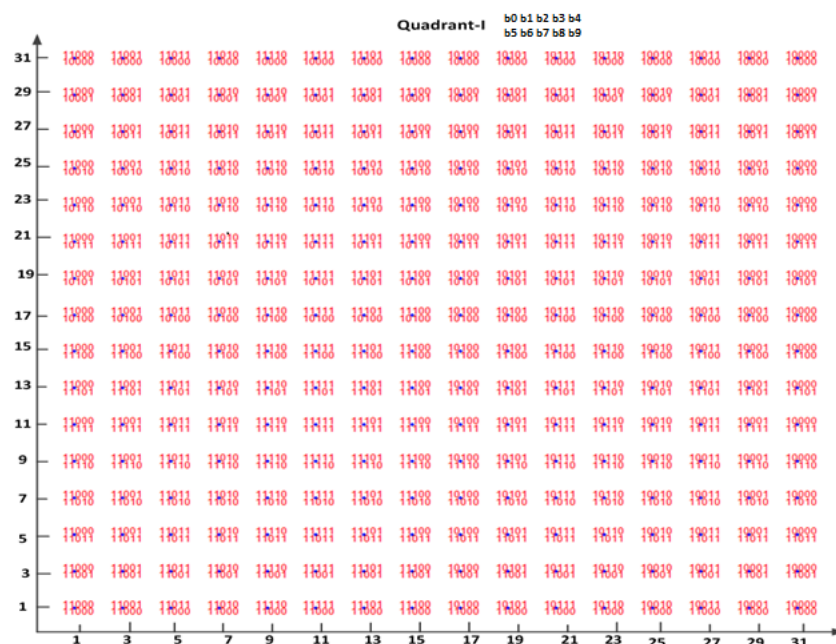


Figure 27-37—Constellation bit encoding for 1024-QAM (1st quadrant)

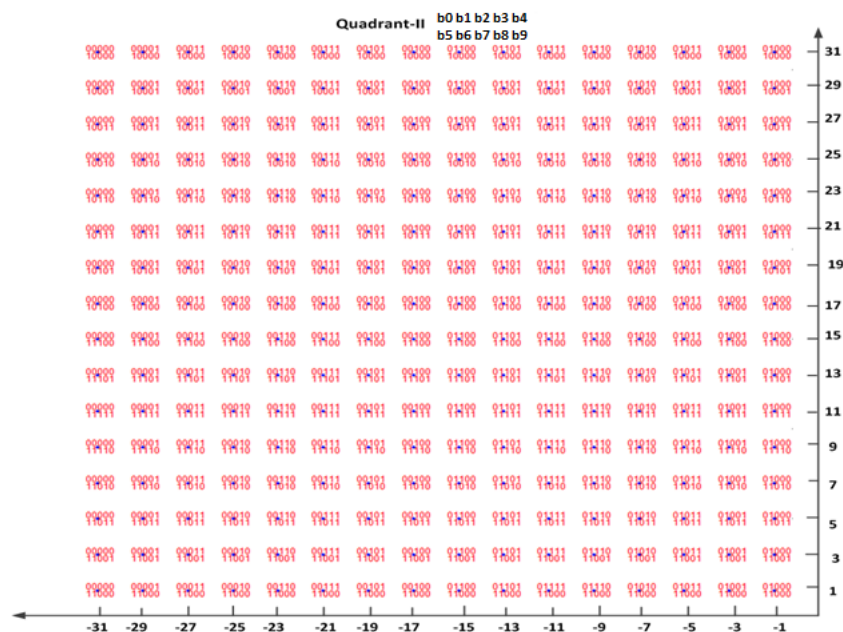


Figure 27-38—Constellation bit encoding for 1024-QAM (2nd quadrant)

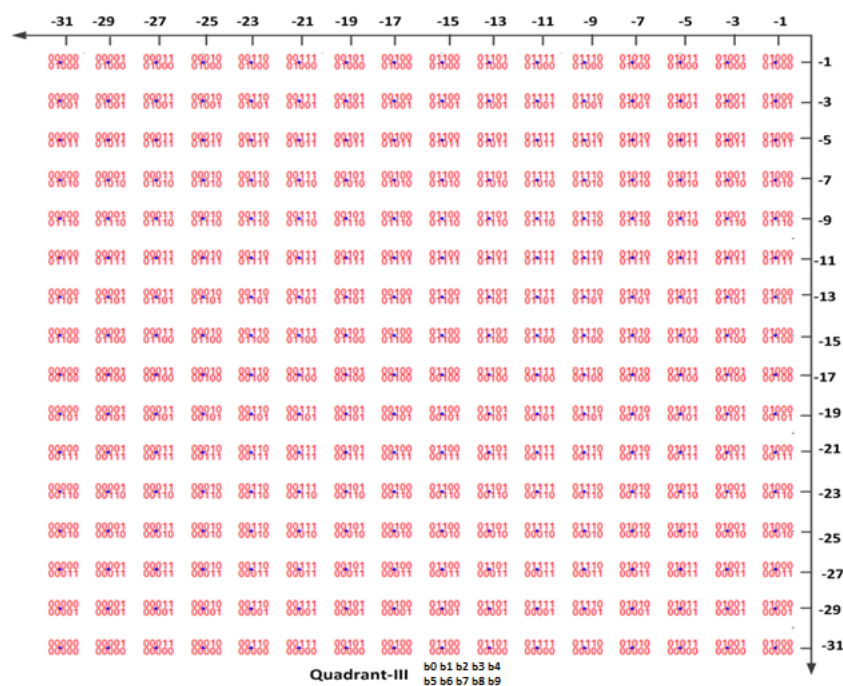


Figure 27-39—Constellation bit encoding for 1024-QAM (3rd quadrant)



Figure 27-40—Constellation bit encoding for 1024-QAM (4th quadrant)

The bit string convention in Figure 27-37, Figure 27-38, Figure 27-39, and Figure 27-40 follows the bit string convention outlined in 17.3.5.8.

The normalization factor K_{mod} for 1024 QAM is $1/\sqrt{682}$.

DCM is an optional modulation scheme for the HE-SIG-B and Data fields. DCM can be applied to an HE SU PPDU and an HE ER SU PPDU. In an HE MU PPDU or HE TB PPDU, DCM can be applied to an RU containing data for one user but cannot be applied to an RU containing data for more than one user.

DCM is applicable to only HE-MCSs 0, 1, 3, and 4. DCM is applicable to only $N_{SS} = 1$ or $N_{SS} = 2$ (in the case of single-user RU in an HE MU PPDU, $N_{SS,r,u} = 1$ or $N_{SS,r,u} = 2$). DCM is not applicable with MU-MIMO or with STBC.

If DCM is employed, bit sequences are mapped to a pair of symbols ($d_k, d_{q(k)}$) where to exploit frequency diversity for a 996-tone or smaller RU, k is in the range of $0 \leq k \leq N_{SD} - 1$ and $q(k)$ is in the range of $N_{SD} \leq q(k) \leq 2N_{SD} - 1$ and, for a 2×996-tone RU, k is in the range of $0 \leq k \leq N_{SD}/2 - 1$ and $q(k)$ is in the range of $N_{SD}/2 \leq q(k) \leq N_{SD} - 1$. To maximize the frequency diversity, the indices of a pair of DCM subcarriers ($k, q(k)$) are given by $q(k) = k + N_{SD}$ for a 996-tone or smaller RU. To reduce the implementation complexity, the indices of a pair of DCM subcarriers ($k, q(k)$) are given by $q(k) = k + N_{SD}/2$ for a 2×996-tone RU. The N_{SD} here refers to the N_{SD} with DCM = 1, which is half the value of N_{SD} with DCM = 0.

Let $N_{80seg,ru} = 1$ for a 996-tone RU or smaller and $N_{80seg,ru} = 2$ for a 2×996-tone RU.

For BPSK modulation with DCM, the input stream is broken into groups of $N_{CBPS}/N_{80seg,ru}$ or $N_{CBPS,u}/N_{80seg,ru}$ bits $(B_0, B_1, \dots, B_{N_{CBPS,u}/N_{80seg,ru}-1})$. Each bit B_k is BPSK modulated to a sample d_k^* . This generates the samples for the lower half of the data subcarriers. For the upper half of the subcarriers, the samples are generated as $d_{k+N_{SD}/N_{80seg,ru}}^* = d_k^* \times e^{j(k+N_{SD}/N_{80seg,ru})\pi}$, $k = 0, 1, \dots, N_{SD}/N_{80seg,ru} - 1$. The N_{SD} here refers to the N_{SD} with DCM = 1, which is half the value of N_{SD} with DCM = 0.

For QPSK modulation with DCM, the input stream is broken into groups of $N_{CBPS}/N_{80seg,ru}$ or $N_{CBPS,u}/N_{80seg,ru}$ bits $(B_0, B_1, \dots, B_{N_{CBPS,u}/N_{80seg,ru}-1})$. Each pair of bits (B_{2k}, B_{2k+1}) is QPSK modulated to a symbol d_k^* . This generates the constellation points for the lower half the data subcarriers in the RU. For the upper half of the data subcarriers in the RU, $d_{k+N_{SD}/N_{80seg,ru}}^* = conj(d_k^*)$, where $conj()$ represents the complex conjugate operation. The N_{SD} here refers to the N_{SD} with DCM = 1, which is half the value of N_{SD} with DCM = 0.

For 16-QAM modulation with DCM, the input stream is broken into groups of $N_{CBPS}/N_{80seg,ru}$ or $N_{CBPS,u}/N_{80seg,ru}$ bits $(B_0, B_1, \dots, B_{N_{CBPS,u}/N_{80seg,ru}-1})$. A group of 4 bits $(B_{4k}, B_{4k+1}, B_{4k+2}, B_{4k+3})$ is 16-QAM modulated to a sample d_k^* as described in 17.3.5.8. This is the sample on subcarrier k in the lower half. In the upper half, the sample $d_{k+N_{SD}/N_{80seg,ru}}^*$ on subcarrier $k + N_{SD}/N_{80seg,ru}$ is obtained by 16-QAM modulating a permutation of the bits $(B_{4k}, B_{4k+1}, B_{4k+2}, B_{4k+3})$. Specifically, $d_{k+N_{SD}/N_{80seg,ru}}^*$ is obtained by applying the 16-QAM modulation procedure in 18.3.5.8 to the bit group $(B_{4k+1}, B_{4k}, B_{4k+3}, B_{4k+2})$. The N_{SD} here refers to the N_{SD} with DCM = 1, which is half the value of N_{SD} with DCM = 0.

27.3.12.10 LDPC tone mapper

The LDPC tone mapping shall be performed on all LDPC encoded streams mapped in an RU as described in this subclause. LDPC tone mapping shall not be performed on streams that are encoded using BCC. If DCM is applied to LDPC encoded streams, D_{TM_DCM} shall be applied on both the lower half data subcarriers in an RU and the upper half data subcarriers of the RU. The LDPC tone mapping distance parameters D_{TM} and D_{TM_DCM} are constant for each RU size, and the values for different RU sizes are given in Table 27-36.

Table 27-36—LDPC tone mapping distance for each RU size

Parameter	RU size (tones)						
	26	52	106	242	484	996	2×996
D_{TM}	1	3	6	9	12	20	20
D_{TM_DCM}	1	1	3	9	9	14	14

NOTE 1—The LDPC tone mapping parameters D_{TM} and D_{TM_DCM} are applied for each frequency subblock, $l = 0$ and $l = 1$.

For an HE PPDU without DCM, the LDPC tone mapping for the LDPC encoded stream for user u in the r^{th} RU is done by permuting the stream of complex numbers generated by the constellation mappers (see 27.3.12.9) as defined by Equation (27-95).

$$d''_{t(k), i, n, l, r, u} = d'_{k, i, n, l, r, u} \quad (27-95)$$

where

$$k = \begin{cases} 0, 1, \dots, N_{SD} - 1 & \text{for a 26-, 52-, 106-, 242-, 484- and 996-tone RU} \\ 0, 1, \dots, N_{SD}/2 - 1 & \text{for a 2×996-tone RU} \end{cases}$$

$$i = 1, \dots, N_{SS, r, u}$$

$$n = 0, 1, \dots, N_{SYM} - 1$$

$$l = \begin{cases} 0 & \text{for a 26-, 52-, 106-, 242-, 484- and 996-tone RU} \\ 0, 1 & \text{for a 2×996-tone RU} \end{cases}$$

$$u = 0, \dots, N_{user, r} - 1$$

$$r = 0, \dots, N_{RU} - 1$$

N_{SD} is the number of data tones in the r^{th} RU

$$t(k) = \begin{cases} D_{TM} \left(k \bmod \frac{N_{SD}}{D_{TM}} \right) + \left\lfloor \frac{k \cdot D_{TM}}{N_{SD}} \right\rfloor, & \text{for a 26-, 52-, 106-, 242-, 484- and 996-tone RU} \\ D_{TM} \left(k \bmod \frac{N_{SD}/2}{D_{TM}} \right) + \left\lfloor \frac{k \cdot D_{TM}}{N_{SD}/2} \right\rfloor, & \text{for a 2×996-tone RU} \end{cases}$$

For an HE PPDU with DCM applied to the Data field, the LDPC tone mapping for the LDPC encoded stream corresponding to user u in the r^{th} RU is done by permuting the stream of complex numbers generated by the constellation mappers (see 27.3.12.9) as defined by Equation (27-96).

$$d''_{t(k), i, n, l, r, u} = d'_{k, i, n, l, r, u} \quad (27-96)$$

where

$$k = \begin{cases} 0, 1, \dots, 2N_{SD} - 1 & \text{for a 26-, 52-, 106-, 242-, 484- and 996-tone RU} \\ 0, 1, \dots, N_{SD} - 1 & \text{for a 2×996-tone RU} \end{cases}$$

$$i = 1, \dots, N_{SS, r, u}$$

$$n = 0, 1, \dots, N_{SYM} - 1$$

$$l = \begin{cases} 0 & \text{for a 26-, 52-, 106-, 242-, 484- and 996-tone RU} \\ 0, 1 & \text{for a 2×996-tone RU} \end{cases}$$

$$u = 0, \dots, N_{user, r} - 1$$

$$r = 0, \dots, N_{RU} - 1$$

N_{SD} is the number of data tones in the r^{th} RU if DCM is applied

$t(k)$ is as follows:

For a 26-, 52-, 106-, 242-, 484-, and 996-tone RU:

$$t(k) = \begin{cases} D_{TM_DCM} \left(k \bmod \frac{N_{SD}}{D_{TM_DCM}} \right) + \left\lfloor \frac{k \cdot D_{TM_DCM}}{N_{SD}} \right\rfloor, & \text{for } k < N_{SD} \\ D_{TM_DCM} \left((k - N_{SD}) \bmod \frac{N_{SD}}{D_{TM_DCM}} \right) + \left\lfloor \frac{(k - N_{SD}) \cdot D_{TM_DCM}}{N_{SD}} \right\rfloor + N_{SD}, & \text{for } k \geq N_{SD} \end{cases}$$

For a 2×996-tone RU:

$$t(k) = \begin{cases} D_{TM_DCM} \left(k \bmod \frac{N_{SD}/2}{D_{TM_DCM}} \right) + \left\lfloor \frac{k \cdot D_{TM_DCM}}{N_{SD}/2} \right\rfloor, & \text{for } 0 \leq k < N_{SD}/2 \\ D_{TM_DCM} \left((k - N_{SD}/2) \bmod \frac{N_{SD}/2}{D_{TM_DCM}} \right) + \left\lfloor \frac{(k - N_{SD}/2) \cdot D_{TM_DCM}}{N_{SD}/2} \right\rfloor + N_{SD}/2, & \text{for } N_{SD}/2 \leq k < N_{SD} \end{cases}$$

D_{TM_DCM}

is the LDPC tone mapping distance for the r^{th} RU if DCM is applied

NOTE 2—LDPC tone mapper for a 26-, 52-, 106-, 242-, 484-, and 996-tone RU is defined as one segment. LDPC tone mapping is performed separately for the upper and lower 80 MHz frequency subblocks of a 2×996-tone RU as indicated by the frequency subblock index l .

Since LDPC tone mapping is not performed on BCC coded streams, for BCC coded spatial streams, Equation (27-97) applies.

$$d''_{k,i,n,l,r,u} = d''_{k,i,n,l,r,u} \quad (27-97)$$

where

$k = 0, 1, \dots, N_{SD} - 1$ for a 26-, 52-, 106-, and 242-tone RU

$i = 1, \dots, N_{SS,r,u}$

$n = 0, 1, \dots, N_{SYM} - 1$

$l = 0$ for a 26-, 52-, 106-, and 242-tone RU

$u = 0, \dots, N_{user,r} - 1$

$r = 0, \dots, N_{RU} - 1$

27.3.12.11 Segment deparser

For a 26-, 52-, 106-, 242-, 484-, and 996-tone RU, the segment deparsing is not performed, and $d_{k,i,n,r,u}^{(i_{Seg})}$ is specified in Equation (27-98).

$$d_{k,i,n,r,u}^{(i_{Seg})} = d''_{k,i,n,0,r,u}, \text{ if } 0 \leq k \leq N_{SD} - 1, i_{Seg} = 0 \quad (27-98)$$

For a 2×996-tone RU in a 160 MHz HE PPDU, the two frequency subblocks at the output of the LDPC tone mapper are combined into one frequency segment as specified in Equation (27-99).

$$d_{k,i,n,r,u}^{(i_{Seg})} = \begin{cases} d''_{k,i,n,0,r,u}, & \text{if } 0 \leq k \leq \frac{N_{SD}}{2} - 1 \\ d''_{k - \frac{N_{SD}}{2}, i, n, l, r, u}, & \text{if } \frac{N_{SD}}{2} \leq k \leq N_{SD} - 1 \end{cases}, i_{Seg} = 0 \quad (27-99)$$

For a 2×996-tone RU in an 80+80 MHz HE PPDU, the segment deparsing is not performed, and $d_{k,i,n,r,u}^{(i_{Seg})}$ is specified in Equation (27-100).

$$d_{k,i,n,r,u}^{(i_{Seg})} = d_{k,i,n,i_{Seg},r,u}'' \quad \text{if } 0 \leq k \leq N_{SD} - 1, i_{Seg} = 0, 1 \quad (27-100)$$

NOTE—As per Table 21-7 (center frequency for frequency segment $i_{Seg} = 0$), $f_c^{(0)}$ is always less than $f_c^{(1)}$ in the case of an 80+80 MHz HE PPDU. Hence, $d_{k,i,n,0,r,u}''$ (frequency subblock 0) is always transmitted in the frequency segment lower in frequency, while $d_{k,i,n,1,r,u}''$ (frequency subblock 1) is always transmitted in the frequency segment higher in frequency.

27.3.12.12 Space-time block coding

For an HE PPDU, STBC is applied with only 1 spatial stream and only if DCM is not applied. Its application is indicated by the STBC field in the HE-SIG-A field. In an HE MU PPDU, STBC coding is used in all RUs or not used in any of the RUs. If any RU in an HE MU PPDU uses DL MU-MIMO, STBC shall not be used in any RU in the HE MU PPDU.

The STBC encoding process is described in 21.3.10.9.4, with $N_{SS,0} = 1$ and $N_{STS,0} = 2$.

27.3.12.13 Pilot subcarriers

For a user transmitting on the i^{th} 26-tone RU in a given PPDU BW, two pilot subcarriers shall be inserted in subcarriers $k \in K_{R26_i}$, where K_{R26_i} is given by the i^{th} pilot index set in the row of given PPDU BW of Table 27-37.

Table 27-37—Pilot indices for a 26-tone RU

PPDU BW	K_{R26_i}
20 MHz, $i = 1:9$	$\{-116, -102\}, \{-90, -76\}, \{-62, -48\}, \{-36, -22\}, \{-10, 10\}, \{22, 36\}, \{48, 62\}, \{76, 90\}, \{102, 116\}$
40 MHz, $i = 1:18$	$\{-238, -224\}, \{-212, -198\}, \{-184, -170\}, \{-158, -144\}, \{-130, -116\}, \{-104, -90\}, \{-78, -64\}, \{-50, -36\}, \{-24, -10\}, \{10, 24\}, \{36, 50\}, \{64, 78\}, \{90, 104\}, \{116, 130\}, \{144, 158\}, \{170, 184\}, \{198, 212\}, \{224, 238\}$
80 MHz, $i = 1:37$	$\{-494, -480\}, \{-468, -454\}, \{-440, -426\}, \{-414, -400\}, \{-386, -372\}, \{-360, -346\}, \{-334, -320\}, \{-306, -292\}, \{-280, -266\}, \{-252, -238\}, \{-226, -212\}, \{-198, -184\}, \{-172, -158\}, \{-144, -130\}, \{-118, -104\}, \{-92, -78\}, \{-64, -50\}, \{-38, -24\}, \{-10, 10\}, \{24, 38\}, \{50, 64\}, \{78, 92\}, \{104, 118\}, \{130, 144\}, \{158, 172\}, \{184, 198\}, \{212, 226\}, \{238, 252\}, \{266, 280\}, \{292, 306\}, \{320, 334\}, \{346, 360\}, \{372, 386\}, \{400, 414\}, \{426, 440\}, \{454, 468\}, \{480, 494\}$
160 MHz, $i = 1:74$	{pilot subcarrier indices in 80 MHz −512, pilot subcarrier indices in 80 MHz +512}

The pilot mapping P_n^k for the subcarrier k for symbol n shall be as specified in Equation (27-101).

$$P_n^{K_{R26_i}} = \{\Psi_{n \bmod 2}, \Psi_{(n+1) \bmod 2}\} \quad (27-101)$$

$$P_n^k = 0 \quad k \notin K_{R26_i}$$

where

Ψ_m is defined in Table 27-38

Table 27-38—The 2 pilot values for a 26-tone RU

Ψ_0	Ψ_1
1	−1

For a user transmitting on the i^{th} 52-tone RU in a given PPDU BW, four pilot subcarriers shall be inserted in subcarriers $k \in K_{R52_i}$, where K_{R52_i} is given by the i^{th} pilot index set in the row of given PPDU BW of Table 27-39.

Table 27-39—Pilot indices for 52-tone RU transmission

PPDU BW	K_{R52_i}
20 MHz, $i = 1:4$	$\{-116, -102, -90, -76\}, \{-62, -48, -36, -22\}, \{22, 36, 48, 62\}, \{76, 90, 102, 116\}$
40 MHz, $i = 1:8$	$\{-238, -224, -212, -198\}, \{-184, -170, -158, -144\}, \{-104, -90, -78, -64\}, \{-50, -36, -24, -10\}, \{10, 24, 36, 50\}, \{64, 78, 90, 104\}, \{144, 158, 170, 184\}, \{198, 212, 224, 238\}$
80 MHz, $i = 1:16$	$\{-494, -480, -468, -454\}, \{-440, -426, -414, -400\}, \{-360, -346, -334, -320\}, \{-306, -292, -280, -266\}, \{-252, -238, -226, -212\}, \{-198, -184, -172, -158\}, \{-118, -104, -92, -78\}, \{-64, -50, -38, -24\}, \{24, 38, 50, 64\}, \{78, 92, 104, 118\}, \{158, 172, 184, 198\}, \{212, 226, 238, 252\}, \{266, 280, 292, 306\}, \{320, 334, 346, 360\}, \{400, 414, 426, 440\}, \{454, 468, 480, 494\}$
160 MHz, $i = 1:32$	{pilot subcarrier indices in 80 MHz −512, pilot subcarrier indices in 80 MHz +512}

The pilot mapping P_n^k for the subcarrier k for symbol n shall be as specified in Equation (27-102).

$$P_n^{K_{R52_i}} = \{\Psi_{n \bmod 4}, \Psi_{(n+1) \bmod 4}, \Psi_{(n+2) \bmod 4}, \Psi_{(n+3) \bmod 4}\} \quad (27-102)$$

$$P_n^{k \notin K_{R52_i}} = 0$$

where

Ψ_m is defined in Table 27-40

Table 27-40—The 4 pilot values in a 52- and 106-tone RU

Ψ_0	Ψ_1	Ψ_2	Ψ_3
1	1	1	−1

For a user transmitting on the i^{th} 106-tone RU in a given PPDU BW, four pilot subcarriers shall be inserted at subcarriers $k \in K_{R106_i}$, where K_{R106_i} is given by the i^{th} pilot index set in the row of given PPDU BW of Table 27-41.

Table 27-41—Pilot indices for 106-tone RU transmission

PPDU BW	K_{R106_i}
20 MHz, $i = 1:2$	$\{-116, -90, -48, -22\}, \{22, 48, 90, 116\}$
40 MHz, $i = 1:4$	$\{-238, -212, -170, -144\}, \{-104, -78, -36, -10\}, \{10, 36, 78, 104\}, \{144, 170, 212, 238\}$
80 MHz, $i = 1:8$	$\{-494, -468, -426, -400\}, \{-360, -334, -292, -266\}, \{-252, -226, -184, -158\}, \{-118, -92, -50, -24\}, \{24, 50, 92, 118\}, \{158, 184, 226, 252\}, \{266, 292, 334, 360\}, \{400, 426, 468, 494\}$
160 MHz, $i = 1:16$	{pilot subcarrier indices in 80 MHz -512 , pilot subcarrier indices in 80 MHz $+512$ }

The pilot mapping P_n^k for the subcarrier k for symbol n shall be as specified in Equation (27-103).

$$P_n^{K_{R106_i}} = \{\Psi_{n \bmod 4}, \Psi_{(n+1) \bmod 4}, \Psi_{(n+2) \bmod 4}, \Psi_{(n+3) \bmod 4}\} \quad (27-103)$$

$$P_n^{k \notin K_{R106_i}} = 0$$

where

Ψ_m is defined in Table 27-40

For a user transmitting on the i^{th} 242-tone RU in a given PPDU BW, 8 pilot subcarriers shall be inserted in subcarriers $k \in K_{R242_i}$, where K_{R242_i} is given by the i^{th} pilot index set in the row of given PPDU BW of Table 27-42.

Table 27-42—Pilot indices for 242-tone RU transmission

PPDU BW	K_{R242_i}
20 MHz, $i = 1$	$\{-116, -90, -48, -22, 22, 48, 90, 116\}$
40 MHz, $i = 1:2$	$\{-238, -212, -170, -144, -104, -78, -36, -10\}, \{10, 36, 78, 104, 144, 170, 212, 238\}$
80 MHz, $i = 1:4$	$\{-494, -468, -426, -400, -360, -334, -292, -266\}, \{-252, -226, -184, -158, -118, -92, -50, -24\}, \{24, 50, 92, 118, 158, 184, 226, 252\}, \{266, 292, 334, 360, 400, 426, 468, 494\}$
160 MHz, $i = 1:8$	{pilot subcarrier indices in 80 MHz -512 , pilot subcarrier indices in 80 MHz $+512$ }

The pilot mapping P_n^k for the subcarrier k for symbol n shall be as specified in Equation (27-104).

$$P_n^{K_{R242_i}} = \{\Psi_{n \bmod 8}, \Psi_{(n+1) \bmod 8}, \Psi_{(n+2) \bmod 8}, \Psi_{(n+3) \bmod 8}, \Psi_{(n+4) \bmod 8}, \Psi_{(n+5) \bmod 8}, \Psi_{(n+6) \bmod 8}, \Psi_{(n+7) \bmod 8}\} \quad (27-104)$$

$$P_n^{k \notin K_{R242_i}} = 0$$

where

Ψ_m is defined in Table 27-43

Table 27-43—The 8 pilot values in a 242-tone RU

Ψ_0	Ψ_1	Ψ_2	Ψ_3	Ψ_4	Ψ_5	Ψ_6	Ψ_7
1	1	1	−1	−1	1	1	1

For a user transmitting on the i^{th} 484-tone RU in a given PPDU BW, 16 pilot subcarriers shall be inserted in subcarriers $k \in K_{R484_i}$, where K_{R484_i} is given by the i^{th} pilot index set in the row of given PPDU BW of Table 27-44.

Table 27-44—Pilot indices for 484-tone RU transmission

PPDU BW	K_{R484_i}
40 MHz, $i = 1$	{−238, −212, −170, −144, −104, −78, −36, −10, 10, 36, 78, 104, 144, 170, 212, 238}
80 MHz, $i = 1:2$	{−494, −468, −426, −400, −360, −334, −292, −266, −252, −226, −184, −158, −118, −92, −50, −24}, {24, 50, 92, 118, 158, 184, 226, 252, 266, 292, 334, 360, 400, 426, 468, 494}
160 MHz, $i = 1:4$	{pilot subcarrier indices in 80 MHz −512, pilot subcarrier indices in 80 MHz +512}

The pilot mapping P_n^k for the subcarrier k for symbol n shall be as specified in Equation (27-105).

$$\begin{aligned}
 P_n^{K_{R484_i}} &= \{ \Psi_{n \bmod 8}, \Psi_{(n+1) \bmod 8}, \Psi_{(n+2) \bmod 8}, \Psi_{(n+3) \bmod 8}, \\
 &\quad \Psi_{(n+4) \bmod 8}, \Psi_{(n+5) \bmod 8}, \Psi_{(n+6) \bmod 8}, \Psi_{(n+7) \bmod 8}, \\
 &\quad \Psi_{(n+8) \bmod 8}, \Psi_{(n+9) \bmod 8}, \Psi_{(n+10) \bmod 8}, \Psi_{(n+11) \bmod 8}, \\
 &\quad \Psi_{(n+12) \bmod 8}, \Psi_{(n+13) \bmod 8}, \Psi_{(n+14) \bmod 8}, \Psi_{(n+15) \bmod 8} \} \\
 P_n^{k \notin K_{R484_i}} &= 0
 \end{aligned} \tag{27-105}$$

where

Ψ_m is defined in Table 27-43

For a user transmitting on the i^{th} 996-tone RU in a given PPDU BW, 16 pilot subcarriers shall be inserted in subcarriers $k \in K_{R996_i}$, where K_{R996_i} is given by the i^{th} pilot index set in the row of given PPDU BW of Table 27-45.

Table 27-45—Pilot indices for 996-tone RU transmission

PPDU BW	K_{R996_i}
80 MHz, $i = 1$	{−468, −400, −334, −266, −226, −158, −92, −24, 24, 92, 158, 226, 266, 334, 400, 468}
160 MHz, $i = 1:2$	{for $i = 1$ pilot subcarrier indices in 80 MHz −512, for $i = 2$ pilot subcarrier indices in 80 MHz +512}

The pilot mapping P_n^k for the subcarrier k for symbol n shall be as specified in Equation (27-106).

$$\begin{aligned}
 P_n^{K_{R996_i}} &= \{ \Psi_{n \bmod 8}, \Psi_{(n+1) \bmod 8}, \Psi_{(n+2) \bmod 8}, \Psi_{(n+3) \bmod 8}, \\
 &\quad \Psi_{(n+4) \bmod 8}, \Psi_{(n+5) \bmod 8}, \Psi_{(n+6) \bmod 8}, \Psi_{(n+7) \bmod 8}, \\
 &\quad \Psi_{(n+8) \bmod 8}, \Psi_{(n+9) \bmod 8}, \Psi_{(n+10) \bmod 8}, \Psi_{(n+11) \bmod 8}, \\
 &\quad \Psi_{(n+12) \bmod 8}, \Psi_{(n+13) \bmod 8}, \Psi_{(n+14) \bmod 8}, \Psi_{(n+15) \bmod 8} \} \\
 P_n^{k \notin K_{R996_i}} &= 0
 \end{aligned} \tag{27-106}$$

where

Ψ_m is defined in Table 27-43

For a 160 MHz transmission (equivalently two 996-tone RU transmissions), the 80 MHz (equivalently 996-tone RU) pilot mapping is replicated in the two 80 MHz subchannels of the 160 MHz transmission. Specifically, 32 pilot subcarriers shall be inserted in subcarriers $k \in K_{R2 \times 996_i}$, where $K_{R2 \times 996_i}$ is given by $\{-980, -912, -846, -778, -738, -670, -604, -536, -488, -420, -354, -286, -246, -178, -112, -44, 44, 112, 178, 246, 286, 354, 420, 488, 536, 604, 670, 738, 778, 846, 912, 980\}$. The pilot mapping P_n^k for the subcarrier k for symbol n shall be as specified in Equation (27-107).

$$\begin{aligned}
 P_n^{K_{R2 \times 996_i}} &= \{ \Psi_{n \bmod 8}, \Psi_{(n+1) \bmod 8}, \Psi_{(n+2) \bmod 8}, \Psi_{(n+3) \bmod 8}, \\
 &\quad \Psi_{(n+4) \bmod 8}, \Psi_{(n+5) \bmod 8}, \Psi_{(n+6) \bmod 8}, \Psi_{(n+7) \bmod 8}, \\
 &\quad \Psi_{(n+8) \bmod 8}, \Psi_{(n+9) \bmod 8}, \Psi_{(n+10) \bmod 8}, \Psi_{(n+11) \bmod 8}, \\
 &\quad \Psi_{(n+12) \bmod 8}, \Psi_{(n+13) \bmod 8}, \Psi_{(n+14) \bmod 8}, \Psi_{(n+15) \bmod 8}, \\
 &\quad \Psi_{n \bmod 8}, \Psi_{(n+1) \bmod 8}, \Psi_{(n+2) \bmod 8}, \Psi_{(n+3) \bmod 8}, \\
 &\quad \Psi_{(n+4) \bmod 8}, \Psi_{(n+5) \bmod 8}, \Psi_{(n+6) \bmod 8}, \Psi_{(n+7) \bmod 8}, \\
 &\quad \Psi_{(n+8) \bmod 8}, \Psi_{(n+9) \bmod 8}, \Psi_{(n+10) \bmod 8}, \Psi_{(n+11) \bmod 8}, \\
 &\quad \Psi_{(n+12) \bmod 8}, \Psi_{(n+13) \bmod 8}, \Psi_{(n+14) \bmod 8}, \Psi_{(n+15) \bmod 8} \} \\
 P_n^{k \notin K_{R2 \times 996_i}} &= 0
 \end{aligned} \tag{27-107}$$

where

Ψ_m is defined in Table 27-43

For a noncontiguous 80+80 MHz transmission, each frequency segment shall follow the 80 MHz pilot subcarrier allocation and values defined for 996-tone RU in 80 MHz transmission as specified in Equation (27-106).

The pilot mapping described in this subclause shall be copied to all space-time streams before the space-time stream cyclic shifts are applied.

27.3.12.14 OFDM modulation

If midambles are not present, the time domain waveform of the Data field of an HE PPDU that is not an HE TB PPDU for transmit chain i_{TX} , where $1 \leq i_{TX} \leq N_{TX}$, and frequency segment i_{Seg} shall be as defined in Equation (27-108).

$$r_{\text{HE-Data}}^{(i_{Seg}, i_{TX})}(t) = \frac{1}{\sqrt{\sum_{r=0}^{N_{RU}-1} \alpha_r^2 |K_r|}} \sum_{n=0}^{N_{SYM}-1} w_{T_{\text{HE-Data}}}(t - nT_{SYM}) \sum_{r=0}^{N_{RU}-1} \frac{\alpha_r}{\sqrt{N_{STS, r, total}}} \sum_{k \in K_r} \sum_{u=0}^{N_{user, r}-1} \sum_{m=1}^{N_{STS, r, u}} \left(\left[\mathcal{Q}_k^{(i_{Seg})} \right]_{i_{TX}, (M_{r, u} + m)} (\tilde{D}_{k, m, n, r}^{(i_{Seg}, u)} + p_n + 2 + N_{\text{HE-SIG-A}} + N_{\text{HE-SIG-B}} P_n^k) \cdot \exp(j2\pi k \Delta_{F, \text{HE}}(t - nT_{SYM} - T_{GI, Data} - T_{CS, \text{HE}}(M_{r, u} + m))) \right) \quad (27-108)$$

where

T_{SYM}	is defined in Table 27-12
p_n	is defined in 17.3.5.10
P_n^k	can be selected from Equation (27-101) to Equation (27-107) defined in 27.3.12.13, depending on the corresponding RU size
$T_{CS, \text{HE}}(M_{r, u} + m)$	represents the cyclic shift for space-time stream $M_{r, u} + m$ as defined in 27.3.11.2.2
$T_{GI, Data}$	is the guard interval duration as defined in Table 27-12
$\tilde{D}_{k, m, n, r}^{(i_{Seg}, u)}$	is the transmitted constellation for user u in the r^{th} RU at subcarrier k , space-time stream m , and Data field OFDM symbol n and is defined by Equation (27-109)

$$\tilde{D}_{k, m, n, r}^{(i_{Seg}, u)} = \begin{cases} 0, & k \in K_{\text{Pilot}} \\ \tilde{d}_{M_r(k), m, n, r, u}^{(i_{Seg})}, & \text{otherwise} \end{cases} \quad (27-109)$$

where

K_{Pilot} is the set of pilot subcarrier indices for the Data field OFDM symbols as defined in 27.3.12.13
 $M_r(k)$ is defined in Equation (27-110)

$$M_r(k) = k - K_{r, \min} - |\{k' : K_{r, \min} \leq k' < k\} \cap K_{\text{Pilot}}| \quad (27-110)$$

where

$K_{r, \min}$ is the minimum value of the set K_r
 $|\Phi|$ is the cardinality of a set Φ

NOTE— $M_r(k)$ translates a subcarrier index ($k \in K_r$) into the index of data symbols in a transmission over RU r ($0 \leq M_r(k) \leq N_{SD}$). The subcarrier index k for the data subcarrier is first offset by the minimum value of subcarrier index $K_{r, \min}$ (for the lower edge subcarrier) in this RU and then subtracted by the number of pilot subcarriers falling in between the data subcarrier and the edge subcarrier.

In a noncontiguous 80+80 MHz transmission, each frequency segment shall follow the 80 MHz HE subcarrier mapping as specified in 27.3.10.

If midambles are not present, the time domain waveform of the Data field of an HE TB PPDU for user u in the r^{th} RU from transmit chain i_{TX} , where $1 \leq i_{TX} \leq N_{TX}$, shall be as defined in Equation (27-111).

$$r_{\text{HE-Data}, r, u}^{(i_{\text{Seg}}, i_{TX})}(t) = \frac{1}{\sqrt{|K_r|}} \sum_{n=0}^{N_{SYM}-1} w_{T_{\text{HE-Data}}}(t - nT_{SYM}) \frac{1}{\sqrt{N_{STS, r, u}}} \sum_{k \in K_r} \sum_{m=1}^{N_{STS, r, u}} \left(\left[Q_{k, u}^{(i_{\text{Seg}})} \right]_{i_{TX}, m} \left(\tilde{D}_{k, m, n, r}^{(i_{\text{Seg}}, u)} + p_{n+4} P_n^k \right) \cdot \exp(j2\pi k \Delta_{F, \text{HE}}(t - nT_{SYM} - T_{GI, \text{Data}} - T_{CS, \text{HE}}(M_{r, u} + m))) \right) \quad (27-111)$$

where

$Q_{k, u}^{(i_{\text{Seg}})}$ is defined in 27.3.10

$M_{r, u}$ is given by the TXVECTOR parameter STARTING_STS_NUM

27.3.12.15 Dual carrier modulation

DCM modulates the same information on a pair of subcarriers. DCM is an optional modulation scheme for the HE-SIG-B and Data fields. DCM is applicable to only HE-MCSs and HE-SIG-B-MCSs with indices 0, 1, 3, and 4.

The constellation mapper for DCM is defined in 27.3.12.9. The LDPC tone mapper for DCM is defined in 27.3.12.10. The BCC interleaver for DCM is defined in 27.3.12.8.

27.3.12.16 Midambles

An HE STA may include midambles in an HE PPDU transmission in fast varying channels, i.e., channels with high Doppler, to facilitate channel estimation update during the PPDU. Midambles are inserted only if $N_{STS} \leq 4$. The recipient might use the midambles to compensate the channel estimation if it is varying fast in channels with high Doppler.

If the Doppler field of the HE-SIG-A field is 1 in an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU or if the Doppler subfield in the Common Info field in the Trigger frame preceding an HE TB PPDU is 1, then midambles are present in the Data field of the HE PPDU every M_{MA} OFDM symbols, where M_{MA} is either 10 or 20 as indicated by the NSTS And Midamble Periodicity field in the HE-SIG-A field (see 27.3.11.7) or by the Number Of HE-LTF Symbols And Midamble Periodicity subfield in the Common Info field in the Trigger frame (see 9.3.1.22).

Each midamble is the same as the HE-LTF field(s) in the preamble of the same PPDU as defined in 27.3.11.10, as shown in Figure 27-41.

An HE STA shall not transmit an HE MU PPDU with midambles if there is MU-MIMO on any RU.

The scrambling and encoding process of the bits in the Data field OFDM symbols is identical for transmissions with or without midamble.

If present, the number of midamble periods, N_{MA} , in a PPDU is calculated using Equation (27-112).

$$N_{MA} = \max\left(0, \left\lceil \frac{N_{SYM} - 1}{M_{MA}} \right\rceil - 1\right) \quad (27-112)$$

where N_{SYM} is as defined in 27.3.12.5.

As shown in Figure 27-41, the first midamble is inserted immediately after the M_{MA}^{th} OFDM symbol in the Data field, and a midamble is not inserted after the last data OFDM symbol if $\text{mod}(N_{SYM}, M_{MA}) = 0$. At the end of an HE PPDU, if $\text{mod}(N_{SYM}, M_{MA}) = 1$, there is also no midamble inserted before the last OFDM symbol, as shown in Figure 27-42.

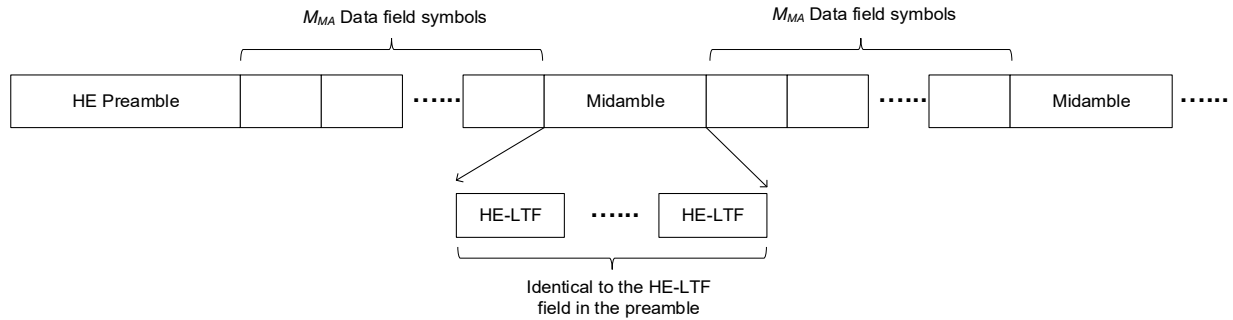


Figure 27-41—HE PPDU with midamble

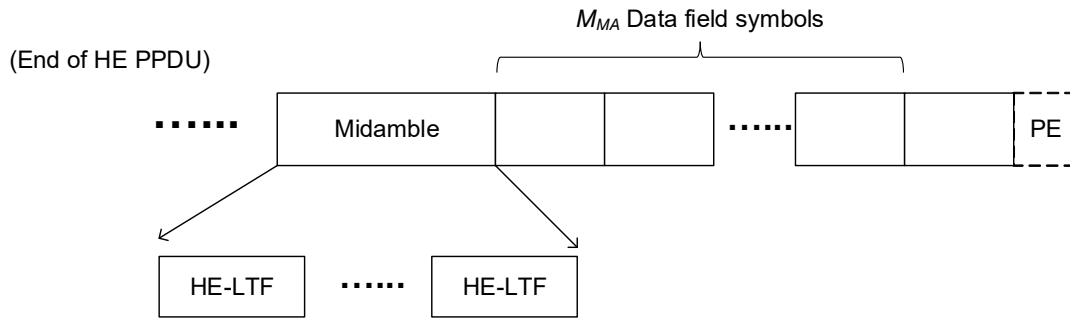


Figure 27-42—Midamble at the end of an HE PPDU

In an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU, if the Doppler field of the HE-SIG-A field is 1 and $N_{SYM} \leq M_{MA} + 1$, there is no midamble present in the current PPDU. In this case, the Doppler field setting to 1 indicates that the current channel between the transmitter and the recipient is with high channel Doppler and the transmitter recommends to the recipient that midamble may be used for the PPDUs of the reverse link.

27.3.13 Packet extension

A PE field of duration 0 μ s, 4 μ s, 8 μ s, 12 μ s, or 16 μ s is present in an HE PPDU. The PE field provides additional receive processing time at the end of the HE PPDU. The PE field, if present, shall be transmitted with the same average power as the Data field and shall not cause significant power leakage outside of the spectrum used by the Data field. Other than that, its content is arbitrary. In an OFDMA HE PPDU, the spectrum used by the Data field for the purpose of packet extension is commensurate with the locations and sizes of the occupied RUs, not the PPDU bandwidth. For example, the Data field of an OFDMA HE PPDU using a 26-tone RU would have a spectrum that is approximately 2 MHz wide.

The duration of the PE field for an HE SU PPDU, HE ER SU PPDU, or HE MU PPDU is determined by both the pre-FEC padding factor value in the last OFDM symbol(s) of the Data field and the TXVECTOR parameter NOMINAL_PACKET_PADDING.

For an HE SU or HE ER SU PPDU, the nominal T_{PE} value ($T_{PE,nominal}$) is given by Table 27-46. In this case, a in Table 27-46 is given by Equation (27-71) or Equation (27-72).

For an HE MU PPDU, the nominal T_{PE} value ($T_{PE,nominal}$) is given by Equation (27-113).

$$T_{PE,nominal} = \max_u T_{PE,nominal,u} \quad (27-113)$$

where

$T_{PE,nominal,u}$ is the nominal T_{PE} value for user u and is also given by Table 27-46

$\max_u f(u)$ is the maximum value of $f(u)$ over all values of u

In this case, a in Table 27-46 is given by Equation (27-83) or Equation (27-84).

Table 27-46—Nominal T_{PE} values

a	TXVECTOR parameter NOMINAL_PACKET_PADDING (HE SU PPDU or HE ER SU PPDU) or NOMINAL_PACKET_PADDING[u] (HE MU PPDU)		
	0 μ s	8 μ s	16 μ s
1	0 μ s	0 μ s	4 μ s
2	0 μ s	0 μ s	8 μ s
3	0 μ s	4 μ s	12 μ s
4	0 μ s	8 μ s	16 μ s

The duration of the PE field, T_{PE} , may take a value of 0, 4, 8, 12, or 16 μ s. T_{PE} for an HE SU, HE ER SU, or HE MU PPDU shall not be less than $T_{PE,nominal}$. T_{PE} for an HE SU, HE ER SU, or HE MU PPDU should be equal to $T_{PE,nominal}$ to minimize the packet extension overhead. Figure 27-43 and Figure 27-44 show examples of the PE field duration in an HE SU PPDU or HE ER SU PPDU without midambles if the TXVECTOR parameter NOMINAL_PACKET_PADDING is 8 μ s and 16 μ s, respectively, and $T_{PE} = T_{PE,nominal}$. STBC is not used in these examples.

T_{PE} for an HE sounding NDP is 4 μ s.

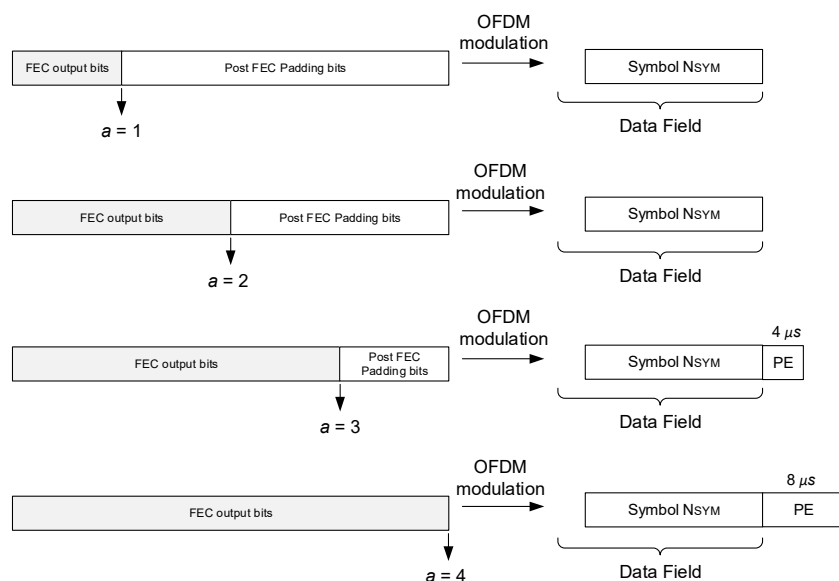


Figure 27-43—PE field duration of an HE SU PDU or HE ER SU PDU without midambles if TXVECTOR parameter $NOMINAL_PACKET_PADDING$ is $8 \mu s$ and $T_{PE} = T_{PE,nominal}$

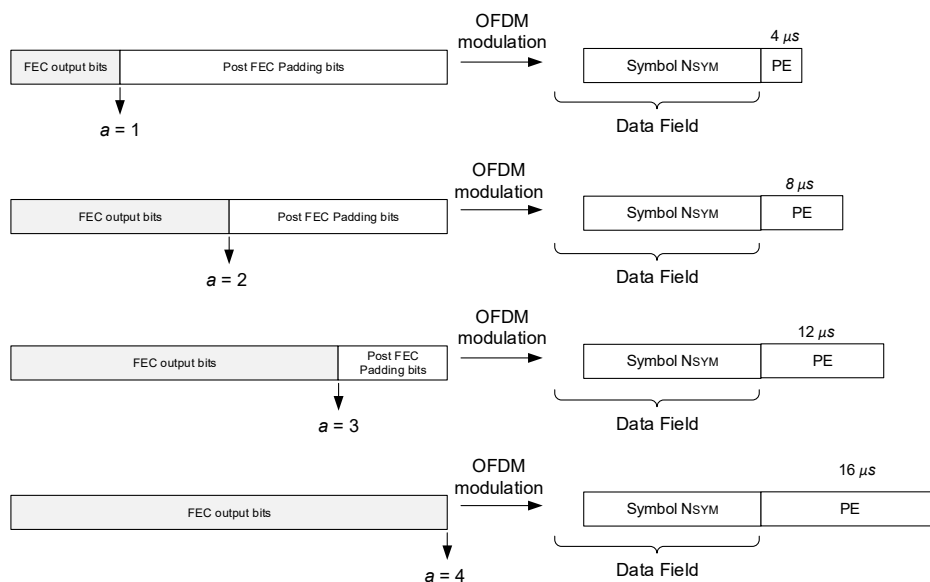


Figure 27-44—PE field duration of an HE SU PDU or HE ER SU PDU without midambles if TXVECTOR parameter $NOMINAL_PACKET_PADDING$ is $16 \mu s$ and $T_{PE} = T_{PE,nominal}$

If transmitting an HE TB PPDU for which the TXVECTOR parameter TRIGGER_METHOD is TRIGGER_FRAME, each transmitter of an HE TB PPDU shall append a PE field with a duration T_{PE} calculated using Equation (27-114), except for an HE TB feedback NDP (see 27.3.4), which has $T_{PE} = 0$.

$$T_{PE} = \left\lfloor \frac{\left(\frac{LENGTH + m + 3}{3} \times 4 - T_{HE-PREAMBLE} \right) - N_{SYM}T_{SYM} - N_{MA}N_{HE-LTF}T_{HE-LTF-SYM}}{4} \right\rfloor \times 4 \quad (27-114)$$

where

$m = 2$ for an HE TB PPDU

LENGTH is the value indicated by the UL Length subfield in the Common Info field in the Trigger frame

$T_{HE-PREAMBLE}$ is the value for an HE TB PPDU in Equation (27-121)

$T_{HE-STF-T}$, $T_{HE-LTF-SYM}$, T_{RL-SIG} , and $T_{HE-SIG-A}$ are defined in Table 27-12

N_{MA} is the number of midamble periods in the current PPDU

$$N_{SYM} = \left\lfloor \left(\frac{LENGTH + m + 3}{3} \times 4 - T_{HE-PREAMBLE} - N_{MA}N_{HE-LTF}T_{HE-LTF-SYM} \right) / T_{SYM} \right\rfloor - b_{PE-Disambiguity} \quad (27-115)$$

$b_{PE-Disambiguity}$ is the value of the TXVECTOR parameter HE_TB_PE_DISAMBIGUITY

There are multiple methods for computing N_{MA} for an HE TB PPDU that yield the same result; one example is as follows. The duration of one midamble period is defined in Equation (27-116).

$$T_{MA} = M_{MA}T_{SYM} + N_{HE-LTF}T_{HE-LTF-SYM} \quad (27-116)$$

where M_{MA} is the midamble periodicity indicated by the Number of HE-LTF Symbols And Midamble Periodicity subfield of the Common Info field in the Trigger frame. Equation (27-117) for computing N_{MA} can be used instead of Equation (27-112).

$$N_{MA} = \begin{cases} 0, & \text{if Doppler} = 0 \\ \max\left(0, \left\lfloor \frac{(LENGTH + 3 + m)}{3} \times 4 - T_{HE-PREAMBLE} - (b_{PE-Disambiguity} + 2) \cdot T_{SYM} \right\rfloor / T_{MA} \right), & \text{if Doppler} = 1 \end{cases} \quad (27-117)$$

where Doppler is indicated by the Doppler subfield of the Common Info field of the Trigger frame.

If transmitting an HE TB PPDU for which the TXVECTOR parameter TRIGGER_METHOD is TRS, each transmitter of the HE TB PPDU shall append a PE field with the duration T_{PE} equal to the value specified in the TXVECTOR parameter DEFAULT_PE_DURATION.

The PE Disambiguity field of the HE-SIG-A field for an HE SU, HE ER SU (see Table 27-18), or HE MU PPDU (see Table 27-20) shall be set to 1 if the condition in Equation (27-118) is met; otherwise, it shall be set to 0.

The PE Disambiguity subfield in the Common Info field of the Trigger frame (see Table 9-29g) shall be set to 1 if the condition in Equation (27-118) is met for the HE TB PPDU solicited by the Trigger frame. Otherwise, it shall be set to 0.

$$T_{PE} + 4 \times \left(\left\lceil \frac{\text{TXTIME} - \text{SignalExtension} - 20}{4} \right\rceil - \left(\frac{\text{TXTIME} - \text{SignalExtension} - 20}{4} \right) \right) \geq T_{SYM} \quad (27-118)$$

where

T_{PE} is the PE field duration
 T_{SYM} is the symbol duration of the Data field as defined in 27.3.9
 TXTIME is defined in 27.4.3 (in μs)
 SignalExtension is 0 μs if TXVECTOR parameter NO_SIG_EXTN is true and is aSignalExtension as defined in Table 27-54 if TXVECTOR parameter NO_SIG_EXTN is false

The receiver computes N_{SYM} , T_{PE} , and N_{MA} using Equation (27-119), Equation (27-120), and Equation (27-122), respectively.

$$N_{SYM} = \left\lceil \left(\frac{\text{L_LENGTH} + m + 3}{3} \times 4 - T_{\text{HE-PREAMBLE}} - N_{MA} N_{\text{HE-LTF}} T_{\text{HE-LTF-SYM}} \right) / T_{SYM} \right\rceil - b_{\text{PE-Disambiguity}} \quad (27-119)$$

$$T_{PE} = \left\lceil \frac{\left(\frac{\text{L_LENGTH} + m + 3}{3} \times 4 - T_{\text{HE-PREAMBLE}} \right) - N_{SYM} T_{SYM} - N_{MA} N_{\text{HE-LTF}} T_{\text{HE-LTF-SYM}}}{4} \right\rceil \times 4 \quad (27-120)$$

where

L_LENGTH is the value indicated by the LENGTH field of the L-SIG field
 $T_{\text{HE-PREAMBLE}} =$ (27-121)

$$\begin{cases} T_{\text{RL-SIG}} + T_{\text{HE-SIG-A}} + T_{\text{HE-STF-T}} + N_{\text{HE-LTF}} T_{\text{HE-LTF-SYM}}, & \text{for an HE TB PPDU} \\ T_{\text{RL-SIG}} + T_{\text{HE-SIG-A}} + T_{\text{HE-STF-NT}} + N_{\text{HE-LTF}} T_{\text{HE-LTF-SYM}}, & \text{for an HE SU PPDU} \\ T_{\text{RL-SIG}} + T_{\text{HE-SIG-A}} + N_{\text{HE-SIG-B}} T_{\text{HE-SIG-B}} + T_{\text{HE-STF-NT}} + N_{\text{HE-LTF}} T_{\text{HE-LTF-SYM}}, & \text{for an HE MU PPDU} \\ T_{\text{RL-SIG}} + T_{\text{HE-SIG-A-R}} + T_{\text{HE-STF-NT}} + N_{\text{HE-LTF}} T_{\text{HE-LTF-SYM}}, & \text{for an HE ER SU PPDU} \end{cases}$$

where

$T_{\text{RL-SIG}}$, $T_{\text{HE-STF-T}}$, $T_{\text{HE-STF-NT}}$, $T_{\text{HE-LTF-SYM}}$, $T_{\text{HE-SIG-A}}$, $T_{\text{HE-SIG-A-R}}$, and $T_{\text{HE-SIG-B}}$ are defined in Table 27-12

$N_{\text{HE-SIG-B}}$ and $N_{\text{HE-LTF}}$ are defined in Table 27-15

$b_{\text{PE-Disambiguity}}$ is the value indicated by the PE Disambiguity subfield of the HE-SIG-A field for an HE SU, HE ER SU, or HE MU PPDU or the value indicated by the PE Disambiguity subfield in the Common Info field in the Trigger frame (see Table 9-29g) for an HE TB PPDU

N_{MA} may be computed by multiple methods that yield the same result, one example of which is given in Equation (27-122)

$$N_{MA} = \begin{cases} 0, & \text{if Doppler} = 0 \\ \max\left(0, \left\lfloor \left(\frac{L_{\text{LENGTH}} + 3 + m}{3} \times 4 - T_{\text{HE-PREAMBLE}} - (b_{\text{PE-Diambiguity}} + 2) \cdot T_{\text{SYM}} \right) / T_{MA} \right\rfloor \right), & \text{if Doppler} = 1 \end{cases} \quad (27-122)$$

where

T_{MA} is defined in Equation (27-116), except that M_{MA} is the midamble periodicity indicated by the NSTS And Midamble Periodicity subfield of the HE-SIG-A field in an HE SU PPDU and HE ER SU PPDU or by the Number Of HE-LTF Symbols And Midamble Periodicity subfield of the HE-SIG-A field in an HE MU PPDU

Doppler is indicated by the Doppler field of HE-SIG-A field

27.3.14 Non-HT duplicate transmission

If the TXVECTOR parameter FORMAT is NON_HT and the TXVECTOR parameter NON_HT_MODULATION is NON_HT_DUP_OFDM, the transmitted PPDU is a non-HT duplicate. Non-HT duplicate transmission is used to transmit to non-HT STAs, HT STAs, VHT STAs, and HE STAs that may be present in a part of a 40 MHz, 80 MHz, or 160 MHz channel (see Table 21-2). The RL-SIG, HE-SIG-A, HE-SIG-B, HE-STF, HE-LTF, and PE fields are not transmitted.

The L-STF and L-LTF fields shall be transmitted in the same way as in the HE transmission. The L-SIG field shall be transmitted in the same way as in the HE transmission, with the following exceptions:

- The Rate and Length fields shall follow 17.3.4 (SIGNAL field)
- The four additional subcarriers at indices ± 27 and ± 28 are not modulated (no energy)

NOTE—The L-STF, L-LTF, and L-SIG fields are not transmitted in 20 MHz subchannels in which the preamble is punctured (see 27.3.7).

In a 40 MHz non-HT duplicate transmission, the Data field shall be as defined by Equation (19-61).

For 80 MHz and 160 MHz non-HT duplicate transmissions, the Data field shall be as defined by Equation (27-123).

$$r_{\text{non-HT}, BW}^{i_{TX}}(t) = \frac{N_{20\text{MHz}}}{\sqrt{N_{\text{Tone}}^{\text{NON_HT_DUP_OFDM_Data}} \cdot |\Omega_{20\text{MHz}}|}} \sum_{n=0}^{N_{\text{SYM}}-1} w_{T_{\text{SYM}}}(t - nT_{\text{SYM}}) \cdot \sum_{i_{BW}=0}^{N_{20\text{MHz}}-1} \left((1 - \text{INACTIVE_SUBCHANNELS}[i_{BW}]) \sum_{k=-26}^{26} \Upsilon_{(k - K_{\text{Shift}}(i_{BW})), BW}(D_{k,n} + P_{n+1}P_k) \cdot \exp(j2\pi(k - K_{\text{Shift}}(i_{BW}))\Delta_F(t - nT_{\text{SYM}} - T_{GI} - T_{CS}^{i_{TX}})) \right) \quad (27-123)$$

where

$N_{20\text{MHz}}$ and $K_{\text{Shift}}(i)$
are defined in 21.3.8.2.4

P_k and P_n are defined in 17.3.5.10

$D_{k,n}$ is defined in Equation (21-26)

$\Upsilon_{k, BW}$ is defined in Equation (21-16) and Equation (21-17)

$T_{CS}^{i_{TX}}$ represents the cyclic shift for transmit chain i_{TX} with a value given in Table 21-10

$N_{\text{NON-HT_DUP_OFDM-Data}}^{\text{Tone}}$

has the value given in Table 27-16

INACTIVE_SUBCHANNELS[x]

is bit x of the TXVECTOR parameter INACTIVE_SUBCHANNELS, if present, and is 0 otherwise.

$|\Omega_{20\text{MHz}}|$ is equal to the number of bits with value 0 in the TXVECTOR parameter INACTIVE_SUBCHANNELS, if present, and is equal to $N_{20\text{MHz}}$ otherwise

In a noncontiguous 80+80 MHz non-HT duplicate transmission, data transmission in each frequency segment shall be as defined for an 80 MHz non-HT duplicate transmission in Equation (27-123).

For each non-HT duplicate PPDU transmission that is a preamble punctured PPDU, each punctured 20 MHz subchannel is indicated as punctured by including the value of 113 (01110001 in binary representation) in the 8 bits of the TXVECTOR parameter RU_ALLOCATION corresponding to the 242-tone RU that is most closely aligned with the punctured 20 MHz subchannel. Each 20 MHz subchannel that is not punctured is indicated as such by including the value of 192 (11000000 in binary representation) in the 8 bits of the TXVECTOR parameter RU_ALLOCATION corresponding to the 242-tone RU that is most closely aligned with that 20 MHz subchannel.

27.3.15 Transmit requirements for PPDU sent in response to a triggering frame

27.3.15.1 Introduction

An AP may solicit simultaneous HE TB PPDU transmissions or simultaneous non-HT or non-HT duplicate PPDU transmissions from multiple non-AP STAs using a triggering frame. Since there are multiple transmitters, pre-correction of transmission time, frequency, sampling symbol clock, and power (in the case of an HE TB PPDU) by the non-AP STAs is necessary to mitigate synchronization and interference issues at the AP. Frequency and sampling clock pre-corrections are needed to prevent inter-carrier interference. Power pre-correction is necessary to control interference between HE TB PPDU transmissions from the non-AP STAs. An AP may solicit simultaneous HE TB PPDU transmissions from both Class A and Class B devices. A non-AP STA that supports HE TB PPDU transmission shall support power pre-correction as described in 27.3.15.2 and shall meet the pre-correction accuracy requirements described in 27.3.15.3.

27.3.15.2 Power pre-correction

A STA transmits an HE TB PPDU at the STA's maximum transmit power for the assigned HE-MCS if the UL Target Receive Power subfield of the User Info field in the Trigger frame that solicits the HE TB PPDU or the UL Target Receive Power subfield of the TRS Control field of the frame that solicits a response in an HE TB PPDU indicates that the maximum transmit power be used.

Otherwise, the STA calculates the transmit power, Tx_{pwr}^{STA} , of the HE TB PPDU for the assigned HE-MCS using Equation (27-124).

$$Tx_{pwr}^{STA} = PL_{DL} + TargetRx_{pwr} \quad (27-124)$$

where

PL_{DL} is the DL pathloss

$TargetRx_{pwr}$ is the expected receive signal power, in units of dBm, as indicated by the UL Target Receive Power subfield in the User Info field in the Trigger frame or by the UL Target Receive Power subfield in the TRS control field

The STA computes PL_{DL} using Equation (27-125).

$$PL_{DL} = Tx_{pwr}^{AP} - Rx_{pwr} \quad (27-125)$$

where

Tx_{pwr}^{AP} is the AP's transmit power, in units of dBm/20 MHz, as indicated by the AP Tx Power subfield of the Common Info field in the Trigger frame, the encoding of which is specified in 9.3.1.22, or the AP Tx Power subfield of the TRS Control field, the encoding of which is specified in 9.2.4.6a.1.

Rx_{pwr} is the receive signal power, in units of dBm/20 MHz, at the antenna connector of the STA of the triggering PPDU. Rx_{pwr} is an average of the receive signal power over the antennas on which the average PL_{DL} is being computed. If the triggering PPDU is a HT-mixed, VHT or HE PPDU, then the receive signal power is measured from the fields prior to the HT-STF, VHT-STF, or HE-STF, respectively.

NOTE 1— Tx_{pwr}^{AP} and Rx_{pwr} are in units of dBm/20 MHz, while Tx_{pwr}^{STA} and $TargetRx_{pwr}$ are in units of dBm.

A STA that applies beamforming (BF) in the UL should take the BF gain into account when calculating the transmit power needed to meet the target RSSI.

NOTE 2—An AP could account for its beamforming gain in Tx_{pwr}^{AP} or $Target_{RSSI}$ if the triggering PPDU used beamforming.

The transmit power of the HE TB PPDU is further subject to a STA's minimum and maximum transmit power limit due to hardware capability, regulatory requirements, local maximum transmit power levels (see 11.7.5), and non-IEEE-802.11 in-device coexistence requirements.

A STA includes its UL power headroom in the HE TB PPDU following the rules defined in 26.5.2.3.

27.3.15.3 Pre-correction accuracy requirements

A STA that transmits an HE TB PPDU shall support per chain $\max(P-32, -10)$ dBm as the minimum transmit power, where P is the maximum power, in dBm, that the STA can transmit at the antenna connector of that chain using HE-MCS 0 while meeting the transmit EVM and spectral mask requirements.

A STA transmitting using HE-MCS 7 or lower, at or above the minimum power but below $P_{\max, \text{MCS}7}$, shall meet the EVM requirements for HE-MCS 7 even if the HE-MCS used for the transmission is lower than HE-MCS 7, where $P_{\max, \text{MCS}7}$ is the maximum transmit power supported by the STA for HE-MCS 7 in an HE TB PPDU.

A STA that transmits an HE TB PPDU shall support the absolute and relative transmit power requirements and the RSSI measurement accuracy requirements defined in Table 27-47.

The absolute transmit power accuracy is applicable for the entire range of transmit power that the STA is intending to use for the current band of operation. The RSSI accuracy requirements shall be applied to receive signal level range from -82 dBm to -20 dBm in the 2.4 GHz band and from -82 dBm to -30 dBm in the 5 GHz and 6 GHz bands. The requirements are for nominal (room) temperature conditions. The RSSI shall be measured during the reception of the non-HE portion of the HE PPDU preamble.

Table 27-47—Transmit power and RSSI measurement accuracy

Parameter	Minimum requirement		Comments
	Class A	Class B	
Absolute transmit power accuracy	± 3 dB	± 9 dB	Accuracy of achieving a specified transmit power.
RSSI measurement accuracy	± 3 dB	± 5 dB	The difference between the RSSI and the received power. Requirements are valid from minimum receive to maximum receive input power.
Relative transmit power accuracy	N/A	± 3 dB	Accuracy of achieving a change in transmit power for consecutive HE TB PPDU. The relative transmit power accuracy is applicable only to Class B devices.

A STA compensates for carrier frequency offset (CFO) error and symbol clock error with respect to the corresponding triggering PPDU when transmitting the following types of PPDU:

- HE TB PPDU
- Non-HT or non-HT duplicate PPDU with the TXVECTOR parameter TRIGGER_RESPONDING set to true

NOTE 1—The MU-RTS Trigger frame solicits transmission of a non-HT or non-HT duplicate PPDU and not an HE TB PPDU. The non-HT or non-HT duplicate PPDU transmitted as a response to an MU-RTS Trigger frame carries a CTS frame.

After compensation, the absolute value of residual CFO error with respect to the corresponding triggering PPDU shall not exceed the following levels when measured at the 10% point of the complementary cumulative distribution function (CCDF) of CFO errors in AWGN at a received power of -60 dBm in the primary 20 MHz:

- 350 Hz for the data subcarriers of an HE TB PPDU
- 2 kHz for a non-HT PPDU or non-HT duplicate PPDU

The residual CFO error measurement on an HE TB PPDU shall be made after the HE-SIG-A field. The residual CFO error measurement on the non-HT or non-HT duplicate PPDU shall be made after the L-STF field. The symbol clock error shall be compensated by the same ppm amount as the CFO error.

A STA that transmits an HE TB PPDU, non-HT PPDU, or non-HT duplicate PPDU in response to a triggering PPDU shall ensure that the transmission start time of the HE TB PPDU, non-HT PPDU, or non-HT duplicate PPDU is within $\pm 0.4 \mu\text{s} + 16 \mu\text{s}$ from the end, at the STA's transmit antenna connector, of the last OFDM symbol of the triggering PPDU (if it contains no PE field) or of the PE field of the triggering PPDU (if the PE field is present).

NOTE 2—This end instant is before any signal extension; therefore, this is equivalent to HE TB PPDU transmission within $0.4 \mu\text{s}$ of SIFS after the end of the triggering PPDU including signal extension.

27.3.16 SU-MIMO and DL MU-MIMO beamforming

27.3.16.1 General

SU-MIMO and DL MU-MIMO beamforming are techniques used by a STA with multiple antennas (the beamformer) to steer signals using knowledge of the channel to improve throughput. With SU-MIMO beamforming, all space-time streams in the transmitted signal are intended for reception at a single STA in an RU. With DL MU-MIMO beamforming, disjoint subsets of the space-time streams are intended for reception at different STAs in an RU of size greater than or equal to 106 tones.

For SU-MIMO and DL MU-MIMO beamforming in RU r , the receive signal vector in subcarrier k (where subcarrier k is one of the subcarriers in RU r , K_r is the set of used subcarrier indices in RU r , and $k \in K_r$) at beamformee u , $\mathbf{y}_{k,u} = [y_{k,0}, y_{k,1}, \dots, y_{k,N_{RX,u}-1}]^T$, is shown in Equation (27-126), where $\mathbf{x}_k = [\mathbf{x}_{k,0}^T, \mathbf{x}_{k,1}^T, \dots, \mathbf{x}_{k,N_{user,r}-1}^T]^T$ denotes the transmit signal vector in subcarrier k for all $N_{user,r}$ beamformees, with $\mathbf{x}_{k,u} = [x_{k,0}, x_{k,1}, \dots, x_{k,N_{STS,r,u}-1}]^T$ being the transmit signal for beamformee u .

$$\mathbf{y}_{k,u} = \mathbf{H}_{k,u} \times [\mathbf{Q}_{k,0}, \mathbf{Q}_{k,1}, \dots, \mathbf{Q}_{k,N_{user,r}-1}] \times \mathbf{x}_k + \mathbf{n} \quad (27-126)$$

where

$\mathbf{H}_{k,u}$ is the channel matrix from the beamformer to beamformee u in subcarrier k with dimensions $N_{RX,u} \times N_{TX}$

$N_{RX,u}$ is the number of receive antennas at beamformee u

$\mathbf{Q}_{k,u}$ is a steering matrix for beamformee u in subcarrier k with dimensions $N_{TX} \times N_{STS,r,u}$

$N_{user,r}$ is the number of HE MU PPDU recipients (see Table 27-15) in RU r

$\mathbf{n}$ is a vector of additive noise and may include interference

The DL MU-MIMO steering matrix $\mathbf{Q}_k = [\mathbf{Q}_{k,0}, \mathbf{Q}_{k,1}, \dots, \mathbf{Q}_{k,N_{user,r}-1}]$ can be determined by the beamformer using the beamforming feedback for subcarrier k from beamformee u , where $u = 0, 1, \dots, N_{user,r}-1$. The feedback report format is described in 9.4.1.65 and 9.4.1.66. The steering matrix that is computed (or updated) using new beamforming feedback from some or all of participating beamformees might replace the existing steering matrix $\mathbf{Q}_k$ for the next DL MU-MIMO data transmission.

For SU-MIMO beamforming, the steering matrix $\mathbf{Q}_k$ can be determined from the beamforming feedback matrix $\mathbf{V}_k$ that is sent back to the beamformer by the beamformee using the compressed beamforming feedback matrix format as defined in 19.3.12.3.6. The feedback report format is described in 9.4.1.65.

27.3.16.2 Beamforming feedback matrix $\mathbf{V}$

Upon receipt of an HE sounding NDP, the beamformee computes a set of matrices for feedback to the beamformer as described in 21.3.11.2. The eligible beamformees shall remove the space-time stream CSD in Table 21-11 from the measured channel before computing a set of matrices for feedback to the beamformer.

The beamforming feedback matrix, $\mathbf{V}_{k,u}$, found by the beamformee u for subcarrier k in RU r shall be compressed in the form of angles using the method described in 19.3.12.3.6. The angles, $\phi(k,u)$ and $\psi(k,u)$, are quantized according to Table 9-74 with b_ψ defined by the Codebook Information field of the HE MIMO

Control field (see 9.4.1.64). The compressed beamforming feedback matrix as defined in 19.3.12.3.6 is the only Clause 27 beamforming feedback matrix defined.

The beamformee shall generate the beamforming feedback matrices with the number of rows (N_r) equal to the N_{STS} of the HE sounding NDP.

After receiving the angle information, $\phi(k,u)$ and $\psi(k,u)$, the beamformer reconstructs $V_{k,u}$ using Equation (19-79). For SU-MIMO beamforming, the beamformer uses the $V_{k,0}$ matrix to determine the steering matrix Q_k . For DL MU-MIMO beamforming, the beamformer may calculate a steering matrix $Q_k = [Q_{k,0}, Q_{k,1}, \dots, Q_{k,N_{user,r}-1}]$ using $V_{k,u}$ and $\Delta SNR_{k,u}$ ($0 \leq u \leq N_{user,r}-1$) in order to suppress crosstalk between participating beamformees. The method used by the beamformer to calculate the steering matrix Q_k is implementation specific.

27.3.16.3 CQI feedback

If the HE NDP Announcement frame requests CQI feedback, then upon receipt of the HE sounding NDP, the beamformee computes CQI feedback as described in 9.4.1.67. The CQI feedback, $CQI_{s,r,u}$, for beamformee u in RU r for space-time stream s shall be estimated using the method described in 9.4.1.67. The CQI values to be fed back are derived from quantized SNRs according to Table 9-91h. The beamformee shall transmit the CQI feedback for space-time stream 1, ..., N_c for each of the RU indices for which the CQI report is being requested by the beamformer. The beamformer may use the CQI feedback to determine the best range of RUs for a compressed beamforming/CQI report or for RU assignment during a subsequent MU transmissions. The actual use is implementation specific.

27.3.17 HE sounding NDP

The HE sounding NDP is a variant of the HE SU PPDU. The format of an HE sounding NDP is defined in Figure 27-45.

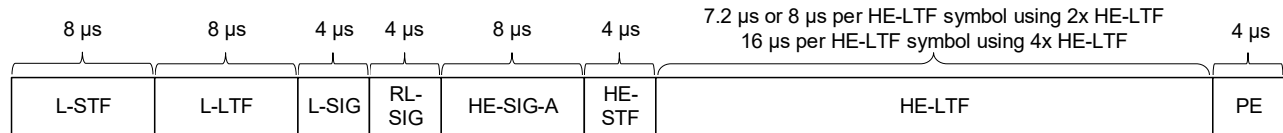


Figure 27-45—HE sounding NDP format

NOTE—The number of HE-LTF symbols in the HE sounding NDP is indicated in the NSTS And Midamble Periodicity field in the HE-SIG-A field.

The HE sounding NDP has the following properties:

- Uses the HE SU PPDU format but without the Data field
- Has a PE field that is 4 μs in duration

The HE sounding NDP overlapping the 242-tone RUs corresponding to bits with a value of 1 in the bitmap of the TXVECTOR parameter INACTIVE_SUBCHANNELS or overlapping a punctured center 26-tone RU of an HE sounding NDP is punctured. The center 26-tone RU of the HE sounding NDP is punctured if either one of the adjacent 242-tone RUs is punctured.

It is mandatory to support the 2x HE-LTF with 0.8 μ s GI and 2x HE-LTF with 1.6 μ s GI. It is optional to support the 4x HE-LTF with 3.2 μ s GI. The other combinations of HE-LTF type and GI duration are disallowed.

If the Beamformed field in the HE-SIG-A field of an HE sounding NDP is 1, then the receiver of the HE sounding NDP should not perform channel smoothing when generating the compressed beamforming feedback report.

27.3.18 HE TB feedback NDP

The HE TB feedback NDP is used to carry the NDP feedback report information as introduced in 26.5.7. The PPDU structure of an HE TB feedback NDP is shown in Figure 27-46.

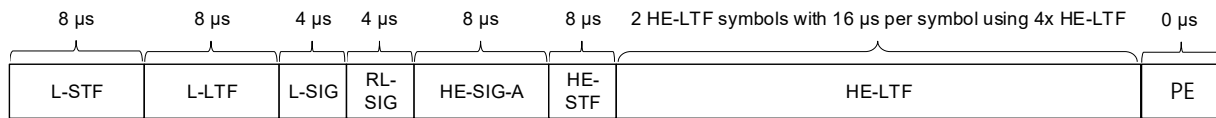


Figure 27-46—HE TB feedback NDP format

The HE TB feedback NDP has the following properties:

- Uses the HE TB PPDU format but without the Data field.
- The PE field has a duration of 0 μ s.
- Has two 4x HE-LTF symbols.
- 4x HE-LTF with 3.2 μ s GI is the only HE-LTF type and GI duration combination for the HE-LTF.
- The generation of HE-LTF symbols for the HE TB feedback NDP is defined in 27.3.11.10.
- The HE-STF and the pre-HE modulated fields are transmitted only on the 20 MHz channel where the STA is assigned.

27.3.19 Transmit specification

27.3.19.1 Transmit spectral mask

The bandwidth of the spectral mask applied to an HE SU PPDU, an HE TB PPDU, and an HE MU PPDU with the Bandwidth field of the HE-SIG-A field equal to 0, 1, 2, or 3 shall be determined by the bandwidth indicated in the Bandwidth field of the HE-SIG-A field. The bandwidth of the spectral mask applied to an HE ER SU PPDU is 20 MHz. The bandwidth of the spectral mask applied to an HE MU PPDU with the Bandwidth field of the HE-SIG-A field equal to 4 or 5 is 80 MHz. The bandwidth of the spectral mask applied to an HE MU PPDU with the Bandwidth field of the HE-SIG-A field equal to 6 or 7 is 160 MHz. All HE PPDU formats shall be compliant with the transmit spectral mask described in 27.3.19.

NOTE 1—In the presence of additional regulatory restrictions, the device has to meet both the regulatory requirements and the mask defined in this subclause.

NOTE 2—Transmit spectral mask figures in this subclause are not drawn to scale.

NOTE 3—For rules regarding transmit center frequency leakage levels, see 27.3.19.4.2. The spectral mask requirements in this subclause do not apply to the RF LO.

For a 20 MHz mask PPDU of HE format, the interim transmit spectral mask shall have a 0 dBr (dB relative to the maximum spectral density of the signal) bandwidth of 19.5 MHz, -20 dBr at 10.5 MHz frequency offset, -28 dBr at 20 MHz frequency offset, and -40 dBr at 30 MHz frequency offset and above. The interim transmit spectral mask for frequency offsets between 9.75 MHz and 10.5 MHz, 10.5 MHz and 20 MHz, and 20 MHz and 30 MHz shall be linearly interpolated in dB domain from the requirements for 9.75 MHz, 10.5 MHz, 20 MHz, and 30 MHz frequency offsets. The transmit spectrum shall not exceed the maximum of the interim transmit spectral mask and -53 dBm/MHz at any frequency offset. Figure 27-47 shows an example of the resulting overall spectral mask when the -40 dBr spectrum level is above -53 dBm/MHz.

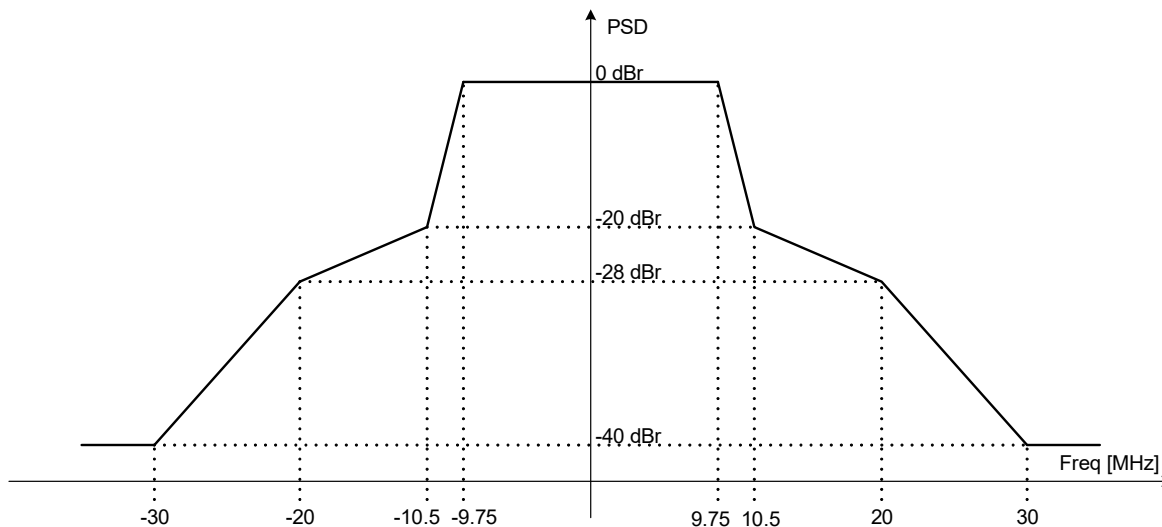


Figure 27-47—Example transmit spectral mask for a 20 MHz mask PPDU

For a 40 MHz mask PPDU of HE format, the interim transmit spectral mask shall have a 0 dBr bandwidth of 39 MHz, -20 dBr at 20.5 MHz frequency offset, -28 dBr at 40 MHz frequency offset, and -40 dBr at 60 MHz frequency offset and above. The interim transmit spectral mask for frequency offsets between 19.5 MHz and 20.5 MHz, 20.5 MHz and 40 MHz, and 40 MHz and 60 MHz shall be linearly interpolated in dB domain from the requirements for 19.5 MHz, 20.5 MHz, 40 MHz, and 60 MHz frequency offsets. The transmit spectrum shall not exceed the maximum of the interim transmit spectral mask and -56 dBm/MHz at any frequency offset greater than 19.5 MHz. Figure 27-48 shows an example of the resulting overall spectral mask when the -40 dBr spectrum level is above -56 dBm/MHz.

For an 80 MHz mask PPDU of HE format, the interim transmit spectral mask shall have a 0 dBr bandwidth of 79 MHz, -20 dBr at 40.5 MHz frequency offset, -28 dBr at 80 MHz frequency offset, and -40 dBr at 120 MHz frequency offset and above. The interim transmit spectral mask for frequency offsets between 39.5 MHz and 40.5 MHz, 40.5 MHz and 80 MHz, and 80 MHz and 120 MHz shall be linearly interpolated in dB domain from the requirements for 39.5 MHz, 40.5 MHz, 80 MHz, and 120 MHz frequency offsets. The transmit spectrum shall not exceed the maximum of the interim transmit spectrum mask and -59 dBm/MHz at any frequency offset. Figure 27-49 shows an example of the resulting overall spectral mask when the -40 dBr spectrum level is above -59 dBm/MHz.

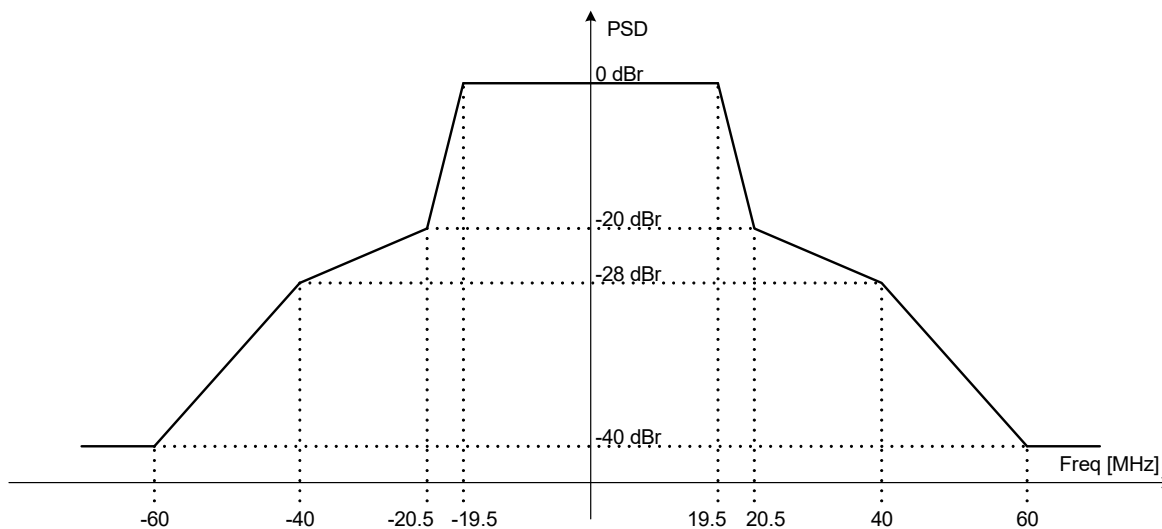


Figure 27-48—Example transmit spectral mask for a 40 MHz mask PPDU

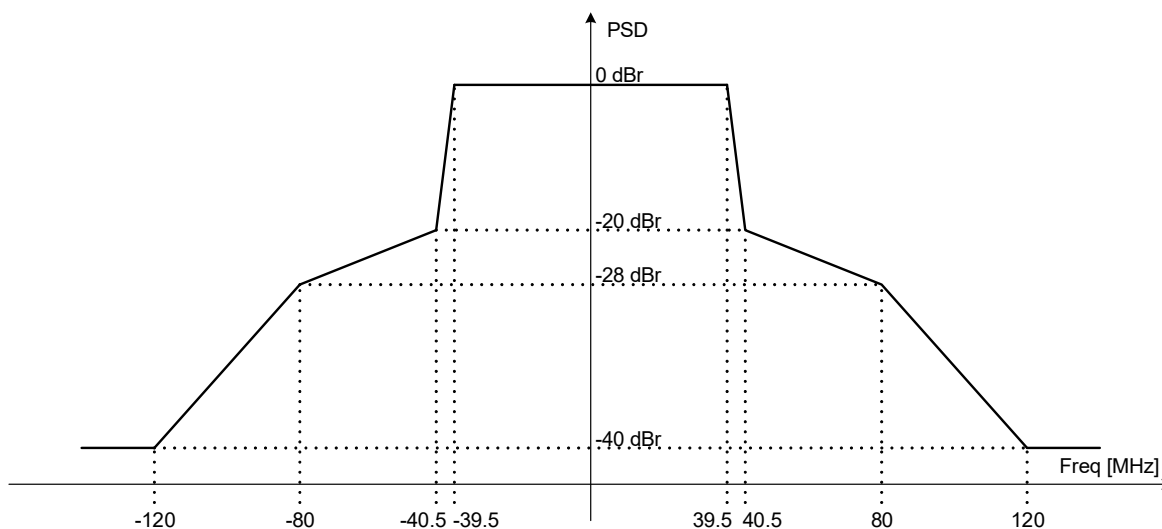


Figure 27-49—Example transmit spectral mask for an 80 MHz mask PPDU

For a 160 MHz mask PPDU of HE format, the interim transmit spectral mask shall have a 0 dBm/MHz bandwidth of 159 MHz, -20 dBm/MHz at 80.5 MHz frequency offset, -28 dBm/MHz at 160 MHz frequency offset, and -40 dBm/MHz at 240 MHz frequency offset and above. The interim transmit spectral mask for frequency offsets between 79.5 MHz and 80.5 MHz, 80.5 MHz and 160 MHz, and 160 MHz and 240 MHz shall be linearly interpolated in dB domain from the requirements for 79.5 MHz, 80.5 MHz, 160 MHz, and 240 MHz frequency offsets. The transmit spectrum shall not exceed the maximum of the interim transmit spectrum mask and -59 dBm/MHz at any frequency offset. Figure 27-50 shows an example of the resulting overall spectral mask when the -40 dBm/MHz spectrum level is above -59 dBm/MHz.

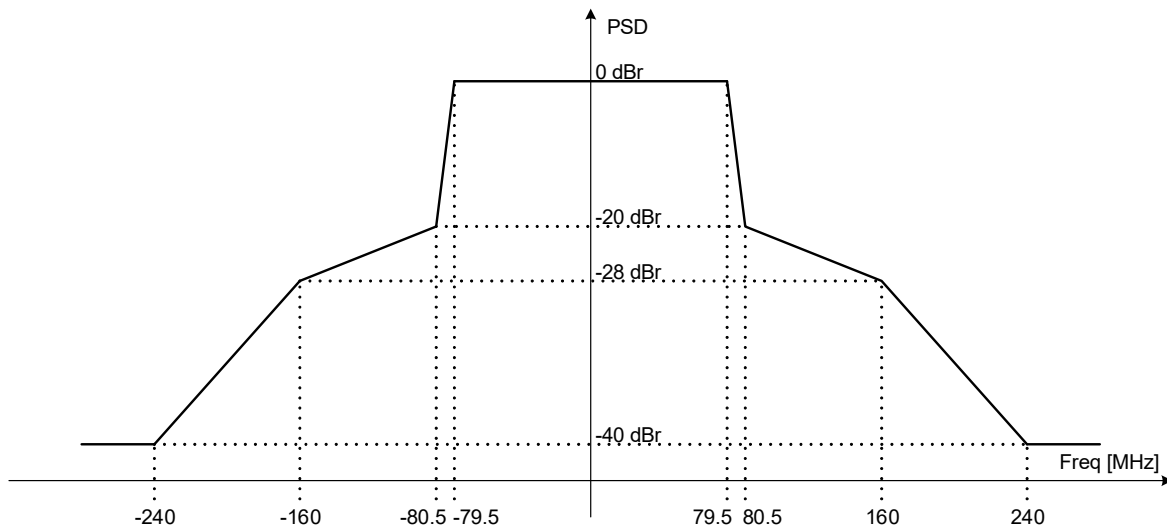


Figure 27-50—Example transmit spectral mask for a 160 MHz mask PPDU

For an 80+80 MHz mask PPDU of HE format, the overall transmit spectral mask is constructed in the following manner. First, the 80 MHz interim spectral mask is placed on each of the two 80 MHz segments. For each frequency at which either of the 80 MHz interim spectral mask has a value equal to 0 dBm/MHz, then 0 dBm/MHz shall be taken as the overall interim spectral mask value. Then, for each frequency at which both of the 80 MHz interim spectral masks have values greater than -40 dBm/MHz and less than -20 dBm/MHz, the sum of the two interim mask values (summed in linear domain) shall be taken as the overall interim spectral mask value. Next, for each frequency at which neither of the two 80 MHz interim masks has a value greater than or equal to -20 dBm/MHz, the higher value of the two interim masks shall be taken as the overall interim spectral mask value. Finally, for any frequency region where the overall interim spectral mask value has not been defined yet, linear interpolation (in dB domain) between the nearest two frequency points with the overall interim spectral mask value defined shall be used to define the overall interim spectral mask value. The transmit spectrum shall not exceed the maximum of the overall interim transmit spectrum mask and -59 dBm/MHz at any frequency offset. Figure 27-51 shows an example of a transmit spectral mask for a noncontiguous transmission using two 80 MHz channels where the center frequency of the two 80 MHz channels are separated by 160 MHz and the -40 dBm/MHz spectrum level is above -59 dBm/MHz.

Different center frequency separation between the two 80 MHz frequency segments of the spectral mask and different peak levels of each 80 MHz frequency segment of the spectral mask are possible. In such cases, a similar procedure in determining the spectral mask as in Figure 27-51 is followed.

The transmit spectral mask for noncontiguous transmissions using two nonadjacent 80 MHz channels is applicable only in regulatory domains that allow for such transmissions.

Measurements shall be made using a 100 kHz resolution bandwidth and a 7.5 kHz video bandwidth.

For preamble puncture, the signal leakage to the preamble punctured channel from the occupied subchannels shall be less than or equal to -20 dBm/MHz starting 0.5 MHz from the boundary of the preamble punctured channel. Denote the number of 20 MHz punctured channels by N .

An example transmit spectral mask for an $N \times 20$ MHz preamble punctured channel with transmission on the both the upper and lower subchannels is shown in Figure 27-52, where the X axis in the plot is centered in the middle of the punctured subbands.

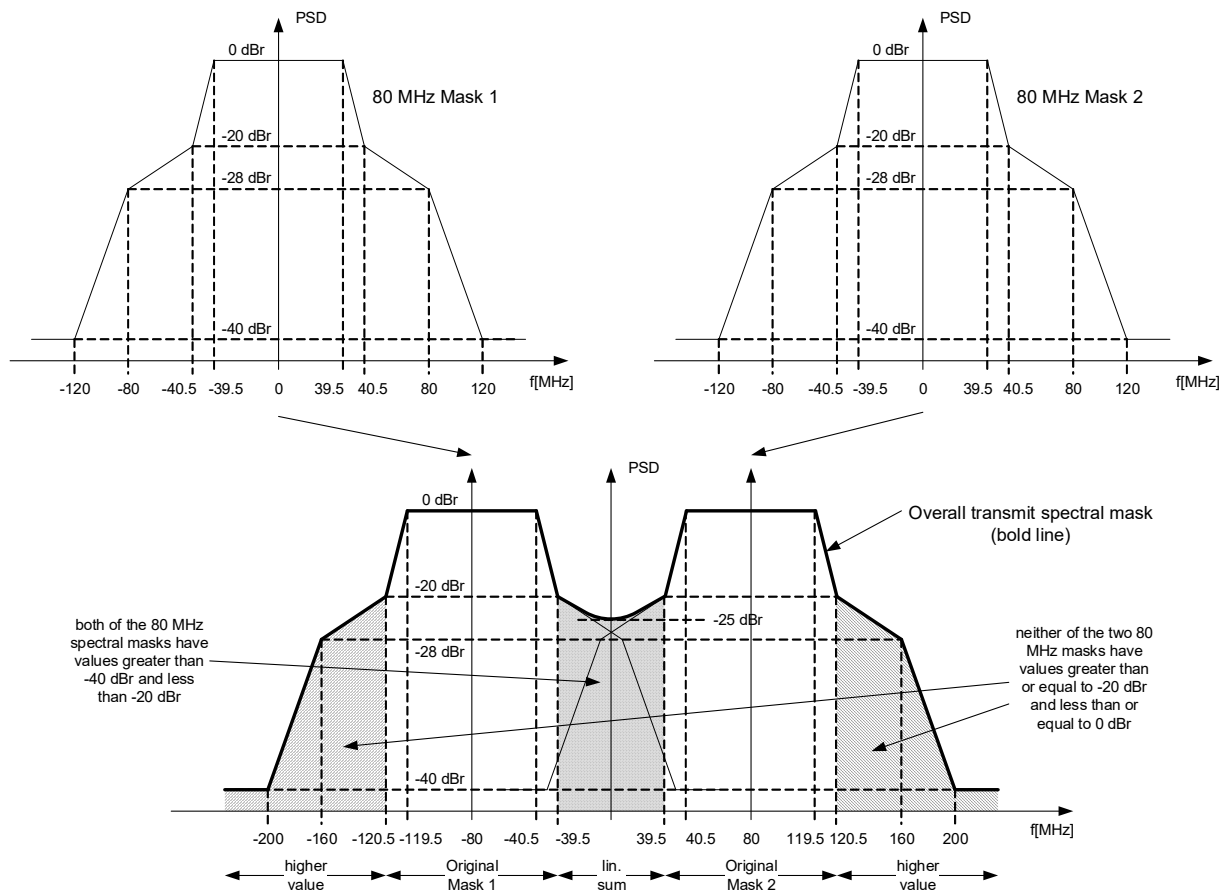


Figure 27-51—Example transmit spectral mask for an 80+80 MHz mask PPDU

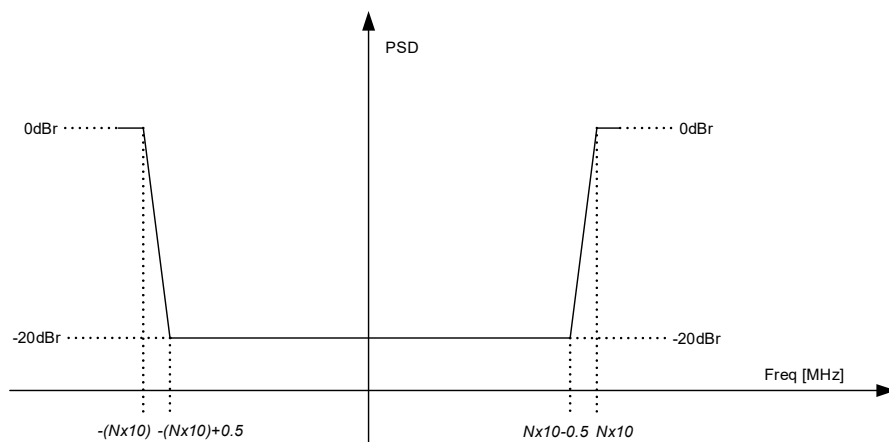


Figure 27-52—Example transmit spectral mask for the $N \times 20$ MHz preamble punctured channel with transmissions on both upper and lower subchannels

An example transmit spectral mask for an 80 MHz HE PPDU with the highest 20 MHz subchannel punctured is shown in Figure 27-53.

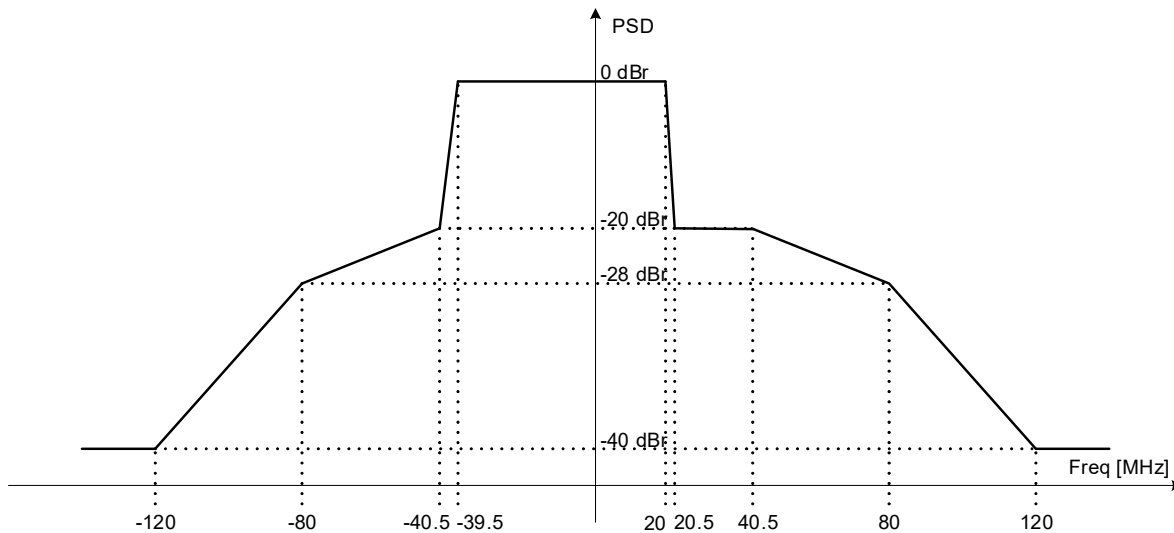


Figure 27-53—Example transmit spectral mask for an 80 MHz HE PPDU with the highest 20 MHz subchannel punctured

27.3.19.2 Spectral flatness

Spectral flatness measurements shall be conducted using BPSK modulated HE PPDU. The HE PPDU shall be demodulated using the following (or equivalent) procedure:

- Start of PPDU shall be detected.
- Transition from L-STF to L-LTF shall be detected, and fine timing shall be established.
- Coarse and fine frequency offsets shall be estimated.
- Symbols in a PPDU shall be manipulated to account for both frequency error and sampling offset drift.
- For each HE-LTF symbol, transform the symbol into subcarrier received values, estimate the phase from the pilot subcarriers, and compensate the subcarrier values according to the estimated phase.
- For each of the data OFDM symbols, transform the symbol into subcarrier received values.

The spectral flatness test shall be performed over at least 20 HE PPDU. The PPDU under test shall be at least 16 data OFDM symbols long.

Evaluate spectral flatness using the subcarrier received values or the magnitude of the channel estimation of the occupied subcarriers of the transmission HE PPDU. Nonoccupied subcarriers of the transmitted HE PPDU shall be ignored during averaging and testing. Resource unit power boosting and beamforming should not be used when measuring spectral flatness.

Let $E_{i,avg}$ denote the magnitude of the channel estimation on subcarrier i or the average constellation energy of a BPSK modulated subcarrier i in an HE data symbol. In a contiguous HE transmission having a bandwidth listed in Table 27-48, $E_{i,avg}$ of each of the subcarriers with indices listed as tested subcarrier indices shall not deviate by more than the specified maximum deviation in Table 27-48 from the average of $E_{i,avg}$ over subcarrier indices listed as averaging subcarrier indices. Averaging of $E_{i,avg}$ is done in the linear domain.

Table 27-48—Maximum transmit spectral flatness deviations

Bandwidth of transmission (MHz)	Averaging subcarrier indices (inclusive)	Tested subcarrier indices (inclusive)	Maximum deviation (dB)
20	−84 to −2 and +2 to +84	−84 to −2 and +2 to +84	± 4
		−122 to −85 and +85 to +122	+4/−6
40	−168 to −3 and +3 to +168	−168 to −3 and +3 to +168	± 4
		−244 to −169 and +169 to +244	+4/−6
80	−344 to −3 and +3 to +344	−344 to −3 and +3 to +344	± 4
		−500 to −345 and +345 to +500	+4/−6
160	−696 to −515, −509 to −166, +166 to +509, and +515 to +696	−696 to −515, −509 to −166, +166 to +509, and +515 to +696	± 4
		−1012 to −697, −165 to −12, +12 to +165, and +697 to +1012	+4/−6

In an 80+80 MHz transmission, each segment shall meet the spectral flatness requirement for an 80 MHz transmission.

For the spectral flatness test, the transmitting STA shall be configured to use a spatial mapping matrix Q_k (see 27.3.12.14) with flat frequency response. Each output port under test of the transmitting STA shall be connected through a cable to one input port of the testing instrumentation. The requirements shall apply to 20 MHz, 40 MHz, 80 MHz, and 160 MHz contiguous transmissions as well as 80+80 MHz transmissions.

27.3.19.3 Transmit center frequency and symbol clock frequency tolerance

Transmit center frequency and the symbol clock frequency for all transmit antennas and frequency segments shall be derived from the same reference oscillator. The symbol clock frequency and transmit center frequency tolerance shall be ± 20 ppm in the 5 GHz and 6 GHz bands and ± 25 ppm in the 2.4 GHz band. HE TB PPDU format is subject to additional requirements as defined in 27.3.15.

Transmit signals with TXVECTOR parameter CH_BANDWIDTH set to CBW160 or CBW80+80 may be generated using two separate RF LOs, one for each of the lower and upper 80 MHz frequency portions.

NOTE—The signal phase of the two 80 MHz frequency portions might not be correlated.

27.3.19.4 Modulation accuracy

27.3.19.4.1 Introduction to modulation accuracy tests

Transmit modulation accuracy specifications are described in 27.3.19.4.2 and 27.3.19.4.3. The test method is described in 27.3.19.4.4.

27.3.19.4.2 Transmit center frequency leakage

For an 80+80 MHz transmission where the RF LO falls outside both frequency segments, the RF LO shall meet the spectral mask requirements as defined in 27.3.19.1. Otherwise, the power measured at the location of the RF LO using resolution BW 78.125 kHz shall not exceed the maximum of -32 dB relative to the total transmit power and -20 dBm, or equivalently $\max(P - 32, -20)$, where P is the transmit power per antenna in dBm. The transmit center frequency leakage is specified per antenna.

27.3.19.4.3 Transmitter constellation error

The relative constellation RMS error in the test, calculated by first averaging over subcarriers, frequency segments, HE PPDU, and spatial streams [see Equation (27-127)] as described in 27.3.19.4.4 shall not exceed a data-rate-dependent value according to Table 27-49. The number of spatial streams under test shall be equal to the number of utilized transmitting STA antenna (output) ports and also equal to the number of utilized testing instrumentation input ports. In the test, $N_{SS} = N_{STS}$ (no STBC), and no beamforming steering matrix shall be used. Each output port of the transmitting STA shall be connected through a cable to one input port of the testing instrumentation. The requirements shall apply to 20 MHz, 40 MHz, 80 MHz, and 160 MHz contiguous transmissions as well as 80+80 MHz noncontiguous transmissions.

Table 27-49—Allowed relative constellation error versus constellation size and coding rate

Modulation		Coding rate	Relative constellation error in an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU (dB)	Relative constellation error in an HE TB PPDU when transmit power is larger than maximum power of HE-MCS 7 (dB)	Relative constellation error in an HE TB PPDU when transmit power is less than or equal to maximum power of HE-MCS 7 (dB)
Without DCM	With DCM				
N/A	BPSK	1/2	−5	−13	−27
BPSK	QPSK	1/2	−5	−13	−27
QPSK	16-QAM	1/2	−10	−13	−27
QPSK	16-QAM	3/4	−13	−13	−27
16-QAM	N/A	1/2	−16	−16	−27
16-QAM	N/A	3/4	−19	−19	−27
64-QAM	N/A	2/3	−22	−22	−27
64-QAM	N/A	3/4	−25	−25	−27
64-QAM	N/A	5/6	−27	−27	−27
256-QAM	N/A	3/4	−30	−30	−30
256-QAM	N/A	5/6	−32	−32	−32
1024-QAM	N/A	3/4	−35/−32	−35/−32	−35/−32
1024-QAM	N/A	5/6	−35/−32	−35/−32	−35/−32

NOTE—The maximum power of HE-MCS 7 can be measured by setting the UL Target RSSI subfield as defined in Table 9-29j in the Trigger frame to 127 for the RU for which the EVM test is conducted.

For 1024-QAM, the relative constellation error shall meet either one of the following requirements:

- The relative constellation error shall be less than or equal to -35 dB if amplitude drift compensation is disabled in the test equipment.
- The relative constellation error shall be less than or equal to -35 dB with amplitude drift compensation enabled in the test equipment, and the relative constellation error shall be less than or equal to -32 dB with amplitude drift compensation disabled in the test equipment.

For all other constellations, the relative constellation error shall be less than or equal to the values in Table 27-49 regardless of whether amplitude drift compensation is enabled in the test equipment.

27.3.19.4.4 Transmitter modulation accuracy (EVM) test

The transmit modulation accuracy test shall be performed by instrumentation capable of converting the transmitted signals into a stream of complex samples at a sampling rate greater than or equal to the bandwidth of the signal being transmitted, except that for a noncontiguous transmissions each frequency segment may be tested independently.

In this case, transmit modulation accuracy of each segment shall meet the required value in Table 27-49 using only the occupied data subcarriers within the corresponding segment. For HE TB PPDU transmission, two sets of EVM requirements are defined in Table 27-49 for different transmission power levels to assist AP in better managing the interference among multiple STAs responding to a Trigger frame.

LO leakage that can potentially show up at the center frequency of the HE PPDU tone plan and within ± 3 neighboring subcarriers shall be excluded from the computation of the transmitter modulation accuracy test. The potential LO leakage subcarriers for 20 MHz operating devices are the center of primary 20 MHz of the HE PPDU tone plan and ± 3 subcarriers of it. The potential LO leakage subcarriers for 40 MHz operating devices are the center of the primary 40 MHz of the PPDU tone plan and ± 3 subcarriers. The potential LO leakage subcarriers for 80 MHz operating devices are the center of the primary 80 MHz of the PPDU tone plan and ± 3 subcarriers of it. The potential LO leakage tones for 160 MHz operating devices are the center of the 160 MHz of the PPDU tone plan and ± 3 subcarriers of it. The potential LO leakage tones for 80+80 MHz operating devices exist outside the PPDU bandwidth and should not affect the transmitter modulation accuracy test. For 40 MHz capable devices that transmit 20 MHz, the potential LO leakage subcarriers exist outside the PPDU bandwidth and should not affect the transmitter modulation accuracy test. For 80 MHz capable devices that transmit 20 MHz or 40 MHz PPDU, the potential LO leakage subcarriers exist outside the PPDU bandwidth and should not affect the transmitter modulation accuracy test. For 160 or 80+80 MHz capable devices that transmit 20 MHz or 40 MHz PPDU or 80 MHz PPDU, the potential LO leakage subcarriers exist outside the PPDU bandwidth and should not affect the transmitter modulation accuracy test.

The instrument shall have sufficient accuracy in terms of I/Q branch amplitude and phase balance, DC offsets, phase noise, and analog-to-digital quantization noise. A possible embodiment of such a setup is converting the signals to a low IF frequency with a microwave synthesizer, sampling the signal with a digital oscilloscope, and decomposing it digitally into quadrature components. The sampled signal shall be processed in a manner similar to an actual receiver using the following or equivalent procedure:

- a) Start of PPDU shall be detected.
- b) Transition from L-STF to L-LTF shall be detected, and fine timing shall be established.
- c) Coarse and fine frequency offsets shall be estimated.
- d) Symbols in a PPDU shall be derotated according to estimated frequency offset. Sampling offset drift shall be also compensated.
- e) For each HE-LTF symbol, transform the symbol into subcarrier received values, estimate the phase from the pilot subcarriers, and derotate the subcarrier values according to the estimated phase.

- f) Estimate the complex channel response coefficient for each of the subcarriers and each of the transmit streams. If midambles are present in the Data field of the PPDU, the channel response coefficients shall be based upon the most recently received midamble symbols.
- g) For each of the data OFDM symbols, transform the symbol into subcarrier received values, estimate the phase from the pilot subcarriers, compensate the subcarrier values according to the estimated phase, group the results from all of the receiver chains in each subcarrier to a vector, and multiply the vector by a zero-forcing equalization matrix generated from the estimated channel.
- h) For each data-carrying subcarrier in each spatial stream of RU under test, find the closest constellation point, and compute the Euclidean distance from it. If midambles are present in the Data field of the PPDU, the midamble symbols shall not be used to compute the Euclidean distance.
- i) Compute the average across PPDU's of the RMS of all errors per PPDU as given by Equation (27-127).

$$Error_{RMS} = \quad (27-127)$$

$$\frac{1}{N_f} \sum_{i_f=1}^{N_f} \sqrt{\frac{\sum_{i_s=1}^{N_{SYM}} \sum_{i_{ss}=1}^{N_{SS}} \sum_{i_{sc}=1}^{N_{SD}} (I_e(i_f, i_s, i_{ss}, i_{sc}) - I_0(i_f, i_s, i_{ss}, i_{sc}))^2 + (Q_e(i_f, i_s, i_{ss}, i_{sc}) - Q_0(i_f, i_s, i_{ss}, i_{sc}))^2}{N_{SYM} N_{SS} N_{SD} P_0}}$$

where

$I_0(i_f, i_s, i_{ss}, i_{sc})$ $Q_0(i_f, i_s, i_{ss}, i_{sc})$
 denotes the ideal symbol point in the complex plane in data subcarrier i_{sc} of the RU under test, spatial stream i_{ss} , and OFDM symbol i_s of frame i_f

$I_e(i_f, i_s, i_{ss}, i_{sc})$ $Q_e(i_f, i_s, i_{ss}, i_{sc})$
 denotes the equalized observed symbol point in the complex plane of the data subcarrier i_{sc} of the RU under test, spatial stream i_{ss} , and OFDM symbol i_s of frame i_f

P_0 is the average power of constellation

N_f is the number of tested frames

N_{SD} is the number of data tones of the occupied RU. For an 80+80 MHz transmission, N_{SD} is the total number of data subcarriers in both 80 MHz frequency segments.

N_{SS} is the number of spatial streams of the data

N_{SYM} is the number of data OFDM symbols

The test shall be performed over at least 20 PPDU's [N_f as defined in Equation (27-127)]. If the occupied RU has 26 tones, the PPDU's under test shall be at least 32 data OFDM symbols long. For occupied RUs that have more than 26 tones, the PPDU's under test shall be at least 16 data OFDM symbols long. Random data shall be used for the symbols.

For an HE TB PPDU with an RU smaller than a 2×996-tone RU, additional transmit modulation accuracy test for the unoccupied subcarriers of the PPDU shall be performed. The transmit modulation accuracy of unoccupied subcarriers of the PPDU test shall be performed by instrumentation capable of converting the transmitted signals into a stream of complex samples at sampling rate greater than or equal to the bandwidth of the signal being transmitted, except that for noncontiguous transmissions, only the frequency segment with occupied subcarriers is tested. The transmit modulation accuracy of unoccupied subcarriers of the PPDU shall meet the relative constellation error staircase mask specified in Equation (27-131) for each modulation and coding rate using the unoccupied subcarriers within the corresponding segment.

The instrument shall have sufficient accuracy in terms of I/Q branch amplitude and phase balance, DC offsets, phase noise, and analog-to-digital quantization noise. A possible embodiment of such a setup is

converting the signals to a low IF frequency with a microwave synthesizer, sampling the signal with a digital oscilloscope and decomposing it digitally into quadrature components. The sampled signal shall be processed in a manner similar to an actual receiver, according to the following steps, or equivalent procedure:

- a) Start of PPDU shall be detected.
- b) Transition from L-STF to L-LTF shall be detected, and fine timing shall be established.
- c) Coarse and fine frequency offsets shall be estimated.
- d) Symbols in a PPDU shall be derotated according to estimated frequency offset. Sampling offset drift shall be also compensated.
- e) For each of the data OFDM symbols, transform the symbol into subcarrier received values, and estimate the power of each subcarrier.
- f) Compute the average unoccupied subcarrier error vector magnitude for each unoccupied 26-tone RU and average across PPDUs of the RMS of all errors per PPDU as given by Equation (27-128).

$$UnusedToneError_{RMS}(k) = \frac{1}{N_f} \sum_{i_f=1}^{N_f} \sqrt{\frac{\sum_{i_s=1}^{N_{SYM}} \sum_{i_{sc} \in \Omega_k} (I_u(i_f, i_s, i_{sc}))^2 + (Q_u(i_f, i_s, i_{sc}))^2}{N_{SYM} \cdot 26 \cdot P_S}} \quad (27-128)$$

where

$$I_u(i_f, i_s, i_{sc}) \quad Q_u(i_f, i_s, i_{sc})$$

denotes unequalized observed symbol point in the complex plane in subcarrier i_{sc} of the unoccupied 26-tone RU and OFDM symbol i_s of frame i_f

Ω_k is a set of subcarriers for k^{th} 26-tone RU as defined in Table 27-7, Table 27-8, and Table 27-9

P_S is the average data subcarrier power of the occupied RU under test and is given by Equation (27-129)

$$P_S = \frac{1}{N_{SYM} N_{SD}} \sum_{i_s=1}^{N_{SYM}} \sum_{i_{sc}=1}^{N_{SD}} (I_u(i_f, i_s, i_{sc}))^2 + (Q_u(i_f, i_s, i_{sc}))^2 \quad (27-129)$$

N_f is the number of tested frames

N_{SYM} is the number of data OFDM symbols

N_{SD} is the number of data subcarriers in the occupied RU

- g) For all HE-MCSs, for an occupied RU bandwidth of r in units of a 26-tone RU as defined by Equation (27-130),

$$r = \begin{cases} 1, & \text{if 26-tone RU} \\ 2, & \text{if 52-tone RU} \\ 4, & \text{if 106-tone RU} \\ 9, & \text{if 242-tone RU} \\ 18, & \text{if 484-tone RU} \\ 37, & \text{if 996-tone RU} \end{cases} \quad (27-130)$$

The average unused subcarrier error vector magnitude for each unoccupied 26-tone RU as calculated in step f) shall meet the staircase mask requirement in Equation (27-131) and Equation (27-132), where m defines the gap in the units of 26-tone RU to the occupied RU from either side with $m = \pm 1$ being the adjacent 26-tone RUs.

$$UnusedToneError(i_{RU26,start} + m) \leq \begin{cases} \max(\varepsilon - 2, -35 \text{ dB}), & \text{if } -r \leq m \leq -1 \\ \max(\varepsilon - 12, -35 \text{ dB}), & \text{if } -2r \leq m \leq -r - 1 \\ \max(\varepsilon - 22, -35 \text{ dB}), & \text{if } -3r \leq m \leq -2r - 1 \\ -35 \text{ dB}, & \text{otherwise} \end{cases} \quad (27-131)$$

The valid range for m for Equation (27-131) is as follows:

- $-i_{RU26,start} + 1 \leq m \leq -1$ for a 20 MHz, 40 MHz, 80 MHz, or 160 MHz PPDU
- $-i_{RU26,start} + 1 \leq m \leq -1$ for an 80+80 MHz PPDU and $i_{RU26,start} \leq 37$
- $-i_{RU26,start} + 38 \leq m \leq -1$ for an 80+80 MHz PPDU and $i_{RU26,start} > 37$

$$UnusedToneError(i_{RU26,end} + m) \leq \begin{cases} \max(\varepsilon - 2, -35 \text{ dB}), & \text{if } 1 \leq m \leq r \\ \max(\varepsilon - 12, -35 \text{ dB}), & \text{if } r + 1 \leq m \leq 2r \\ \max(\varepsilon - 22, -35 \text{ dB}), & \text{if } 2r + 1 \leq m \leq 3r \\ -35 \text{ dB}, & \text{otherwise} \end{cases} \quad (27-132)$$

The valid range for m for Equation (27-132) is as follows:

- $1 \leq m \leq N_{RU26} - i_{RU26,end}$ for a 20 MHz, 40 MHz, 80 MHz, or 160 MHz PPDU
- $1 \leq m \leq 37 - i_{RU26,end}$ for an 80+80 MHz PPDU and $i_{RU26,start} \leq 37$
- $1 \leq m \leq 74 - i_{RU26,end}$ for an 80+80 MHz PPDU and $i_{RU26,start} > 37$

where

$i_{RU26,start}$ is equal to i_{RU} if the occupied RU is a 26-tone RU and is defined in Table 27-50 for other RU sizes

Table 27-50— $i_{RU26,start}$ for RUs other than a 26-tone RU

i_{RU}	52-tone RU	106-tone RU	242-tone RU	484-tone RU	996-tone RU
1	1	1	1	1	1
2	3	6	10	20	38
3	6	10	20	38	
4	8	15	29	57	
5	10	20	38		
6	12	25	47		
7	15	29	57		
8	17	34	66		
9	20	38			
10	22	43			

Table 27-50— $i_{RU26,start}$ for RUs other than a 26-tone RU (continued)

i_{RU}	52-tone RU	106-tone RU	242-tone RU	484-tone RU	996-tone RU
11	25	47			
12	27	52			
13	29	57			
14	31	62			
15	34	66			
16	36	71			
17	38				
18	40				
19	43				
20	45				
21	47				
22	49				
23	52				
24	54				
25	57				
26	59				
27	62				
28	64				
29	66				
30	68				
31	71				
32	73				

$i_{RU26,end}$ is equal to $i_{RU26,start} + r - 1$

i_{RU} is the index of the occupied RU

N_{RU26} is the maximum number of 26-tone RUs for the given bandwidth of the HE TB PPDU as defined in Table 27-15

ϵ is the relative constellation error requirement for an occupied RU of an HE TB PPDU as defined in Table 27-49

NOTE—For an 80+80 MHz PPDU the unused subcarrier error is measured only in the 80 MHz segment in which the occupied RU is located. This leads to the different valid range for m in Equation (27-131) and Equation (27-132) for the 80+80 MHz PPDU.

The test shall be performed over at least 20 PPDUs [N_f as defined in Equation (27-127)]. The PPDUs under test shall be at least 16 data OFDM symbols long. The unequalized observed symbol of potential LO leakage subcarrier locations shall be treated as zero during unoccupied subcarriers transmit modulation accuracy test. Random data shall be used for the symbols.

27.3.20 Receiver specification

27.3.20.1 General

For receiver minimum input sensitivity, adjacent channel rejection, nonadjacent channel rejection, receiver maximum input level, and CCA sensitivity requirements described in this subclause, the input levels are measured at the transmit antenna connector and are referenced as the average power per receive antenna. The number of spatial streams under test shall be equal to the number of utilized transmitting STA antenna (output) ports and also equal to the number of utilized Device Under Test input ports. Each output port of the transmitting STA shall be connected through a cable to one input port of the Device Under Test.

NOTE—Additional test requirements and/or test methods may be needed to meet regulatory requirements.

The requirements on receiver minimum input sensitivity in 27.3.20.2, adjacent channel rejection in 27.3.20.3, and nonadjacent channel rejection in 27.3.20.4 apply to PPDU's that meet all the following conditions:

- STBC is not used.
- 0.8 μ s GI is used.
- If the PPDU bandwidth is 20 MHz and the HE-MCS is less than 10, then BCC is used. Otherwise, LDPC is used.
- The PPDU is an HE SU PPDU.

27.3.20.2 Receiver minimum input sensitivity

The packet error rate (PER) shall be less than 10% for a PSDU with the rate-dependent input levels listed in Table 27-51. The PSDU length shall be 2048 octets for BPSK modulation with DCM or 4096 octets for all other modulations.

Table 27-51—Receiver minimum input level sensitivity

Modulation		Rate (<i>R</i>)	Minimum sensitivity (20 MHz PPDU) (dBm)	Minimum sensitivity (40 MHz PPDU) (dBm)	Minimum sensitivity (80 MHz PPDU) (dBm)	Minimum sensitivity (160 MHz or 80+80 MHz PPDU) (dBm)
Without DCM	With DCM					
N/A	BPSK	1/2	−82	−79	−76	−73
BPSK	QPSK	1/2	−82	−79	−76	−73
QPSK	16-QAM	1/2	−79	−76	−73	−70
QPSK	16-QAM	3/4	−77	−74	−71	−68
16-QAM	N/A	1/2	−74	−71	−68	−65
16-QAM	N/A	3/4	−70	−67	−64	−61
64-QAM	N/A	2/3	−66	−63	−60	−57
64-QAM	N/A	3/4	−65	−62	−59	−56
64-QAM	N/A	5/6	−64	−61	−58	−55
256-QAM	N/A	3/4	−59	−56	−53	−50
256-QAM	N/A	5/6	−57	−54	−51	−48
1024-QAM	N/A	3/4	−54	−51	−48	−45
1024-QAM	N/A	5/6	−52	−49	−46	−43

27.3.20.3 Adjacent channel rejection

Adjacent channel rejection for W MHz (where W is 20, 40, 80, or 160) shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in Table 27-51 and raising the power of the interfering signal of W MHz bandwidth until 10% PER is caused for a PSDU length of 2048 octets for BPSK modulation with DCM or 4096 octets for all other modulations. The difference in power between the signals in the interfering channel and the desired channel is the corresponding adjacent channel rejection. The center frequency of the adjacent channel shall be placed W MHz away from the center frequency of the desired signal.

Adjacent channel rejection for 80+80 MHz channels shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in Table 27-51. Then, an interfering signal of 80 MHz bandwidth is introduced, where the center frequency of the interfering signal is placed 80 MHz away from the center frequency of the frequency segment lower in the frequency of the desired signal. The power of interfering signal is raised until 10% PER is caused for a PSDU length of 2048 octets for BPSK modulation with DCM or 4096 octets for all other modulations. Let ΔP_1 be the difference between the interfering and desired signal. Next, the interfering signal of 80 MHz bandwidth is moved to the frequency where the center frequency of the interfering signal is 80 MHz away from the center frequency of the frequency segment higher in frequency of the desired signal. The power of the interfering is raised until 10% PER is caused for a PSDU length of 2048 octets for BPSK modulation with DCM or 4096 octets for all other modulations. Let ΔP_2 be the power difference between the interfering and desired signal. The smaller value between ΔP_1 and ΔP_2 is the corresponding adjacent channel rejection.

The interfering signal in the adjacent channel shall be a signal compliant with the HE PHY, unsynchronized with the signal in the channel under test, and shall have a minimum duty cycle of 50%. The corresponding rejection shall be no less than specified in Table 27-52.

Table 27-52—Minimum required adjacent and nonadjacent channel rejection levels

Modulation		Rate (R)	Adjacent channel rejection (dB)		Nonadjacent channel rejection (dB)	
Without DCM	With DCM		20/40/80/160 MHz channel	80+80 MHz channel	20/40/80/160 MHz channel	80+80 MHz channel
N/A	BPSK	1/2	16	13	32	29
BPSK	QPSK	1/2	16	13	32	29
QPSK	16-QAM	1/2	13	10	29	26
QPSK	16-QAM	3/4	11	8	27	24
16-QAM	N/A	1/2	8	5	24	21
16-QAM	N/A	3/4	4	1	20	17
64-QAM	N/A	2/3	0	−3	16	13
64-QAM	N/A	3/4	−1	−4	15	12
64-QAM	N/A	5/6	−2	−5	14	11
256-QAM	N/A	3/4	−7	−10	9	6
256-QAM	N/A	5/6	−9	−12	7	4
1024-QAM	N/A	3/4	−12	−15	4	1
1024-QAM	N/A	5/6	−14	−17	2	−1

The measurement of adjacent channel rejection for 160 MHz operation in regulatory domain is required only if such a frequency band plan is permitted in the regulatory domain.

27.3.20.4 Nonadjacent channel rejection

Nonadjacent channel rejection for W MHz channels (where W is 20, 40, 80, or 160) shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in Table 27-51 and raising the power of the interfering signal of W MHz bandwidth until a 10% PER occurs for a PSDU length of 2048 octets for BPSK modulation with DCM or 4096 octets for all other modulations. The difference in power between the signals in the interfering channel and the desired channel is the corresponding nonadjacent channel rejection. The nonadjacent channel rejection shall be met with any nonadjacent channels located at least $2 \times W$ MHz away from the center frequency of the desired signal.

Nonadjacent channel rejection for 80+80 MHz channels shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in Table 27-51. Then, an interfering signal of 80 MHz bandwidth is introduced, where the center frequency of the interfering signal is placed at least 160 MHz away from the center frequency of the frequency segment lower in frequency of the desired signal. The center frequency of the interfering signal shall also be at least 160 MHz away from the center frequency of the frequency segment higher in frequency of the desired signal. The power of interfering signal is raised until 10% PER is caused for a PSDU length of 2048 octets for BPSK modulation with DCM or 4096 octets for all other modulations. Let ΔP_1 be the difference between the interfering and desired signal. Next, the interfering signal of 80 MHz bandwidth is moved to the frequency where the center frequency of the interfering signal is at least 160 MHz away from the center frequency of the frequency segment higher in frequency of the desired signal. The center frequency of the interfering signal shall also be at least 160 MHz away from the center frequency of the frequency segment lower in frequency of the desired signal. The power of the interfering is raised until 10% PER is caused for a PSDU length of 2048 octets for BPSK modulation with DCM or 4096 octets for all other modulations. Let ΔP_2 be the power difference between the interfering and desired signal. The smaller value between ΔP_1 and ΔP_2 is the corresponding nonadjacent channel rejection.

The interfering signal in the nonadjacent channel shall be a signal compliant with the HE PHY, unsynchronized with the signal in the channel under test, and shall have a minimum duty cycle of 50%. The corresponding rejection shall be no less than specified in Table 27-52.

The measurement of nonadjacent channel rejection for 160 MHz operation in regulatory domain is required only if such a frequency band plan is permitted in the regulatory domain.

27.3.20.5 Receiver maximum input level

The receiver shall provide a maximum PER of 10% at a PSDU length of 2048 octets for BPSK modulation with DCM or 4096 octets for all other modulations, for a maximum input level of -30 dBm in the 5 GHz and 6 GHz bands and -20 dBm in the 2.4 GHz band, measured at each antenna for any baseband HE modulation.

27.3.20.6 CCA sensitivity

27.3.20.6.1 General

The thresholds in 27.3.20.6 are compared with the signal level at each receiving antenna.

27.3.20.6.2 CCA sensitivity for operating classes requiring CCA-ED

For the operating classes requiring CCA-Energy Detect (CCA-ED), the PHY shall indicate a medium busy condition if CCA-ED detects a channel busy condition. For improved spectrum sharing, CCA-ED is required in some bands. The behavior class indicating CCA-ED is given in Table D-2. The operating classes

requiring the corresponding CCA-ED behavior class are given in E.1. The PHY of a STA that is operating within an operating class that requires CCA-ED shall operate with CCA-ED.

CCA-ED for a STA that is attempting a non-preamble puncturing transmission shall detect a channel busy condition if the received signal strength exceeds the CCA-ED threshold as given by `dot11OFDMEDThreshold` for the primary 20 MHz channel, `dot11OFDMEDThreshold` for the secondary 20 MHz channel (if present), `dot11OFDMEDThreshold + 3 dB` for the secondary 40 MHz channel (if present), and `dot11OFDMEDThreshold + 6 dB` for the secondary 80 MHz channel (if present). The CCA-ED thresholds for the operating classes requiring CCA-ED are subject to the criteria in D.2.5.

CCA-ED for a STA that is attempting a preamble puncturing transmission shall detect a channel busy condition if the received signal strength exceeds the CCA-ED threshold as given by `dot11OFDMEDThreshold` for the primary 20 MHz channel and `dot11OFDMEDThreshold` for each nonprimary 20 MHz subchannel. The CCA-ED thresholds for the operating classes requiring CCA-ED are subject to the criteria in D.2.5.

For the HE TB PPDU transmission, for each of 20 MHz sub-channels that require CCA, CCA-ED shall detect a channel busy condition if the received signal strength exceeds the CCA-ED threshold as given by `dot11OFDMEDThreshold`. The CCA-ED thresholds for the operating classes requiring CCA-ED are subject to the criteria in D.2.5.

For transmissions that carry a frame that includes a BQR Control subfield (see 9.2.4.6a), CCA-ED shall detect a channel busy condition if the received signal strength exceeds the CCA-ED threshold as given by `dot11OFDMEDThreshold` for primary 20 MHz channel and `dot11OFDMEDThreshold` for each nonprimary 20 MHz channel (if present). The CCA-ED thresholds for the operating classes requiring CCA-ED are subject to the criteria in D.2.5.

NOTE—The requirement to detect a channel busy condition as stated in 27.3.20.6.3 and 27.3.20.6.4 is a mandatory energy detect requirement on all Clause 27 receivers. Support for CCA-ED is an additional requirement that relates specifically to the sensitivities described in D.2.5.

27.3.20.6.3 CCA sensitivity for the primary 20 MHz channel

An HE STA with a W MHz operating channel width shall detect, with $> 90\%$ probability, the start of a PPDU that occupies at least the primary 20 MHz channel in an otherwise idle W MHz channel width and issue a PHY-CCA.indication with the STATUS parameter set to BUSY within a period of `aCCATime` (see 21.4.4) if one of the following conditions is met:

- The start of a non-HT PPDU as defined in 17.3.10.6 if operating in the 5 GHz or 6 GHz band and 18.4.6 if operating in the 2.4 GHz band.
- The start of an HT PPDU as defined in 19.3.19.5.
- The start of a non-HT duplicate, VHT or HE PPDU for which the power measured within the primary 20 MHz channel is at or above -82 dBm.

The channel-list parameter is present and set to {primary} if the operating channel width is greater than 20 MHz. The CCA signal shall be held busy (not issue a PHY-CCA.indication primitive with the STATUS parameter set to IDLE) for the duration of the PPDU, unless it receives a CCARESET.request primitive before the end of the PPDU, for instance, during spatial reuse operation as described in 26.10.

The receiver shall issue a PHY-CCA.indication primitive with the STATUS parameter set to BUSY for any signal that exceeds a threshold equal to 20 dB above the minimum modulation and coding rate sensitivity ($-82 + 20 = -62$ dBm) in the primary 20 MHz channel within a period of `aCCATime` after the signal arrives at the receiver's antenna(s). If the operating channel width is greater than 20 MHz, then the channel-list

parameter is present and shall be set to {primary}. Following the indication and while the threshold continues to be exceeded, the receiver shall not issue a PHY-CCA.indication primitive with the STATUS parameter set to IDLE or with a change in the channel-list parameter.

27.3.20.6.4 CCA sensitivity for signals not occupying the primary 20 MHz channel

The PHY shall issue a PHY-CCA.indication primitive with the STATUS parameter set to BUSY and with the channel-list parameter set to {secondary} if the conditions for issuing a PHY-CCA.indication primitive with the STATUS parameter set to BUSY as defined in 27.3.20.6.3 are not present and at least one of the following conditions is present in an otherwise idle 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz operating channel width:

- Any signal within the secondary 20 MHz channel at or above a threshold of -62 dBm within a period of $aCCATime$ after the signal arrives at the receiver's antenna(s).
- A 20 MHz non-HT, HT_MF, HT_GF, VHT or HE PPDU is detected in the secondary 20 MHz channel at or above $\max(-72 \text{ dBm}, OBSS_PD_{level})$ with $> 90\%$ probability within a period $aCCAMidTime$ (see 27.4.4).

Following the indication and while the threshold continues to be exceeded, the receiver shall not issue a PHY-CCA.indication primitive with the STATUS parameter set to IDLE or with the STATUS parameter set to BUSY and the channel-list parameter set to {secondary40} or {secondary80}.

The PHY shall issue a PHY-CCA.indication primitive with the STATUS parameter set to BUSY and the channel-list parameter set to {secondary40} if the conditions for issuing a PHY-CCA.indication with the STATUS parameter set to BUSY as defined in 27.3.20.6.3 and above are not present and at least one of the following conditions is present in an otherwise idle 80 MHz, 160 MHz, or 80+80 MHz operating channel width:

- Any signal within the secondary 40 MHz channel at or above a threshold of -59 dBm within a period of $aCCATime$ after the signal arrives at the receiver's antenna(s).
- A 40 MHz non-HT duplicate, HT_MF, HT_GF, VHT or HE PPDU is detected in the secondary 40 MHz channel at or above $\max(-72 \text{ dBm}, OBSS_PD_{level} + 3 \text{ dB})$ with $> 90\%$ probability within a period $aCCAMidTime$ (see 27.4.4).
- A 20 MHz non-HT, HT_MF, HT_GF, VHT or HE PPDU is detected in any 20 MHz subchannel of the secondary 40 MHz channel at or above $\max(-72 \text{ dBm}, OBSS_PD_{level})$ with $> 90\%$ probability within a period $aCCAMidTime$.

Following the indication and while the threshold continues to be exceeded, the receiver shall not issue a PHY-CCA.indication primitive with the STATUS parameter set to IDLE or with the STATUS parameter set to BUSY and the channel-list parameter set to {secondary80}.

The PHY shall issue a PHY-CCA.indication primitive with the STATUS parameter set to BUSY and the channel-list parameter set to {secondary80} if the conditions for issuing a PHY-CCA.indication primitive with the STATUS parameter set to BUSY as defined in 27.3.20.6.3 and above are not present and at least one of the following conditions is present in an otherwise idle 160 MHz or 80+80 MHz operating channel width:

- Any signal within the secondary 80 MHz channel at or above -56 dBm.
- An 80 MHz non-HT duplicate, VHT or HE PPDU is detected in the secondary 80 MHz channel at or above $\max(-69 \text{ dBm}, OBSS_PD_{level} + 6 \text{ dB})$ with $> 90\%$ probability within a period $aCCAMidTime$ (see 27.4.4).
- A 40 MHz non-HT duplicate, HT_MF, HT_GF, VHT or HE PPDU is detected in any 40 MHz subchannel of the secondary 80 MHz channel at or above $\max(-72 \text{ dBm}, OBSS_PD_{level} + 3 \text{ dB})$ with $> 90\%$ probability within a period $aCCAMidTime$.

- A 20 MHz non-HT, HT_MF, HT_GF, VHT or HE PPDU is detected in any 20 MHz subchannel of the secondary 80 MHz channel at or above $\max(-72 \text{ dBm}, OBSS_PD_{level})$ with $> 90\%$ probability within a period $aCCAMidTime$.

$OBSS_PD_{level}$ is defined in 26.10.2.4 and applied in the equations to define the detection level in this subclause if an HE STA has ignored a 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz inter-BSS PPDU following the procedure in 26.10.2.2 or 26.10.2.3. It is applied to any secondary channels within the PPDU bandwidth of the inter-BSS PPDU and during the RXTIME of the inter-BSS PPDU. Otherwise, $OBSS_PD_{level}$ is not applied in the equations to define the detection level in this subclause.

27.3.20.6.5 Per 20 MHz CCA sensitivity

If the operating channel width is greater than 20 MHz and the PHY issues a PHY-CCA.indication primitive, the PHY shall set the per20bitmap to indicate the busy/idle status of each 20 MHz subchannel. A 20 MHz subchannel is busy if at least one of the following conditions is present in an otherwise idle 40 MHz, 80 MHz, 80+80 MHz, or 160 MHz channel:

- A signal is present on the 20 MHz subchannel at or above a threshold of -62 dBm at the receiver's antenna(s). The PHY shall indicate that the 20 MHz subchannel is busy a period $aCCATime$ after the signal starts and shall continue to indicate the 20 MHz subchannel is busy while the threshold continues to be exceeded.
- The 20 MHz subchannel is in a channel on which an 80 MHz non-HT duplicate, VHT or HE PPDU at or above $\max(-69 \text{ dBm}, OBSS_PD_{level} + 6 \text{ dB})$ at the receiver's antenna(s) is present. The PHY shall indicate that the 20 MHz subchannel is busy with $> 90\%$ probability within a period $aCCAMidTime$ (see 27.4.4).
- The 20 MHz subchannel is in a channel on which a 40 MHz non-HT duplicate, HT_MF, HT_GF, VHT or HE PPDU at or above $\max(-72 \text{ dBm}, OBSS_PD_{level} + 3 \text{ dB})$ at the receiver's antenna(s) is present. The PHY shall indicate that the 20 MHz subchannel is busy with $> 90\%$ probability within a period $aCCAMidTime$ (see 27.4.4).
- A 20 MHz non-HT, HT_MF, HT_GF, VHT, or HE PPDU at or above $\max(-72 \text{ dBm}, OBSS_PD_{level})$ at the receiver's antenna(s) is present on the 20 MHz subchannel. The PHY shall indicate that the 20 MHz subchannel is busy with $> 90\%$ probability within a period $aCCAMidTime$ (see 27.4.4).

NOTE—Following the receipt of a Trigger frame with the CS Required subfield in the Common Info field set to 1, the HE PHY is required to only detect a signal at the -62 dBm threshold since the other conditions require more time than is available before the response is expected.

$OBSS_PD_{level}$ is defined in 26.10.2.4 and applied in the equations to define the detection level in this subclause if an HE STA has ignored a 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz inter-BSS PPDU following the procedure in 26.10.2.2 or 26.10.2.3. It is applied to any secondary channels within the PPDU bandwidth of the inter-BSS PPDU and during the RXTIME of the inter-BSS PPDU. Otherwise, $OBSS_PD_{level}$ is not applied in the equations to define the detection level in this subclause.

27.3.21 HE transmit procedure

There are five options for the transmit PHY procedure. The first four options, for which typical transmit procedures are shown in Figure 27-54, Figure 27-55, Figure 27-56, and Figure 27-57, are selected if the FORMAT field of the PHY-TXSTART.request(TXVECTOR) primitive is equal to HE_SU, HE_MU, HE_ER_SU, or HE_TB, respectively. These transmit procedures do not describe the operation of optional features, such as DCM.

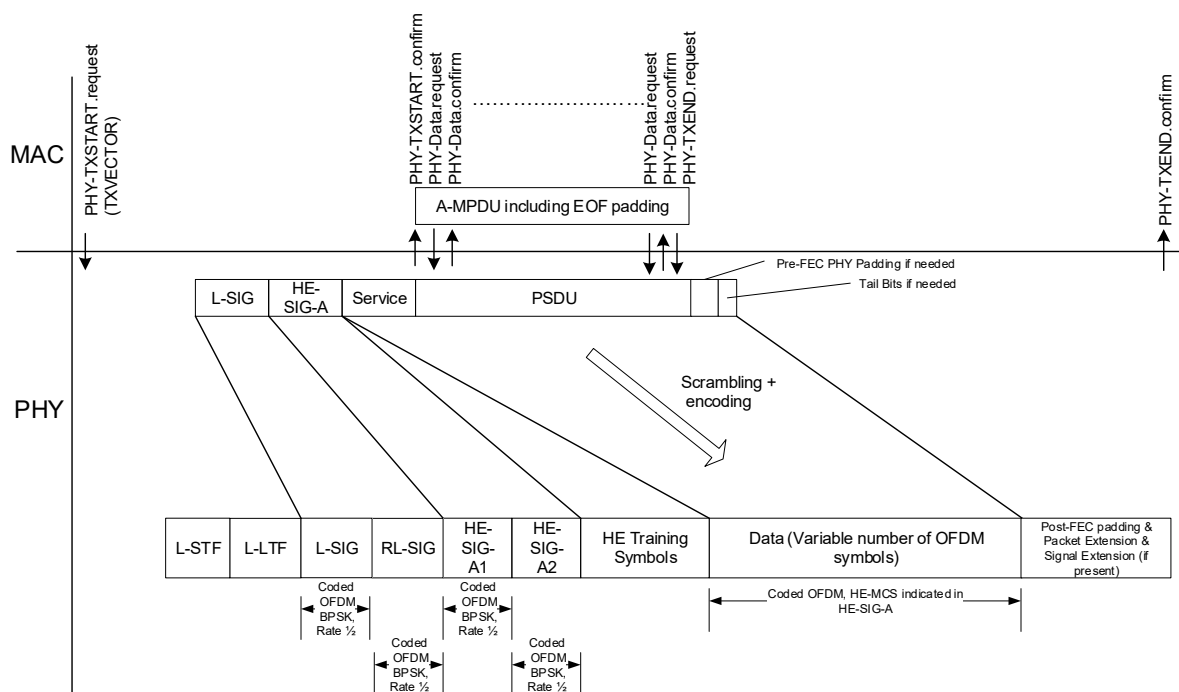


Figure 27-54—PHY transmit procedure for an HE SU PPDU

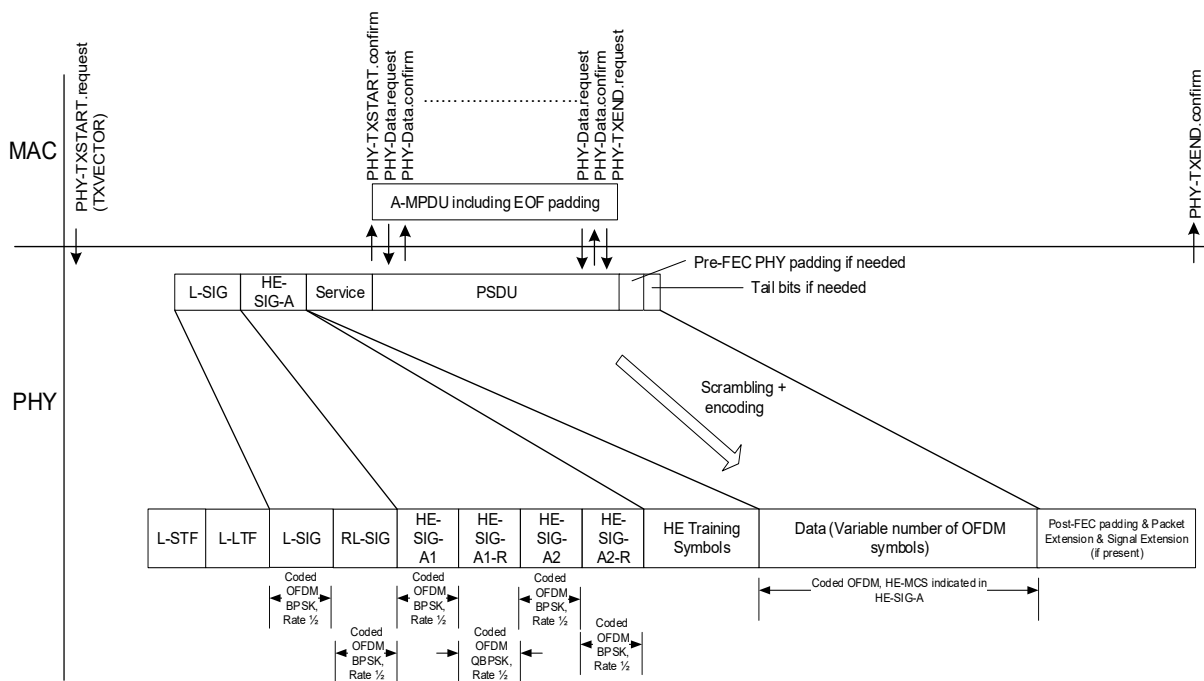


Figure 27-55—PHY transmit procedure for an HE ER SU PPDU

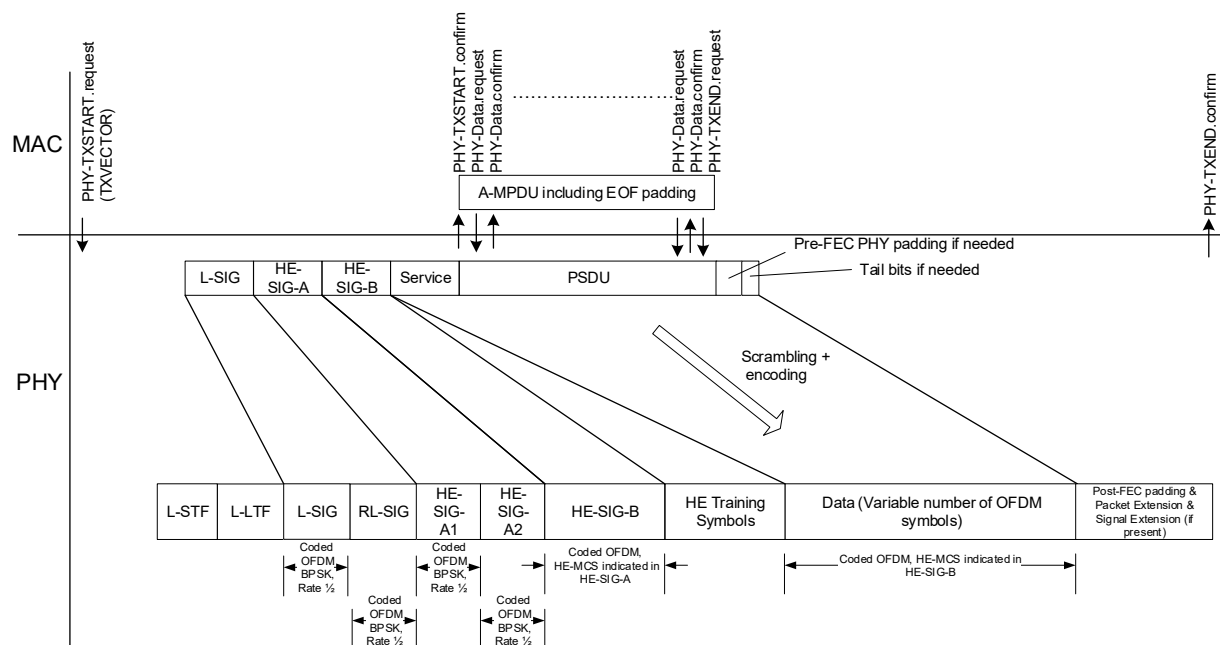


Figure 27-56—PHY transmit procedure for an HE MU PPDU

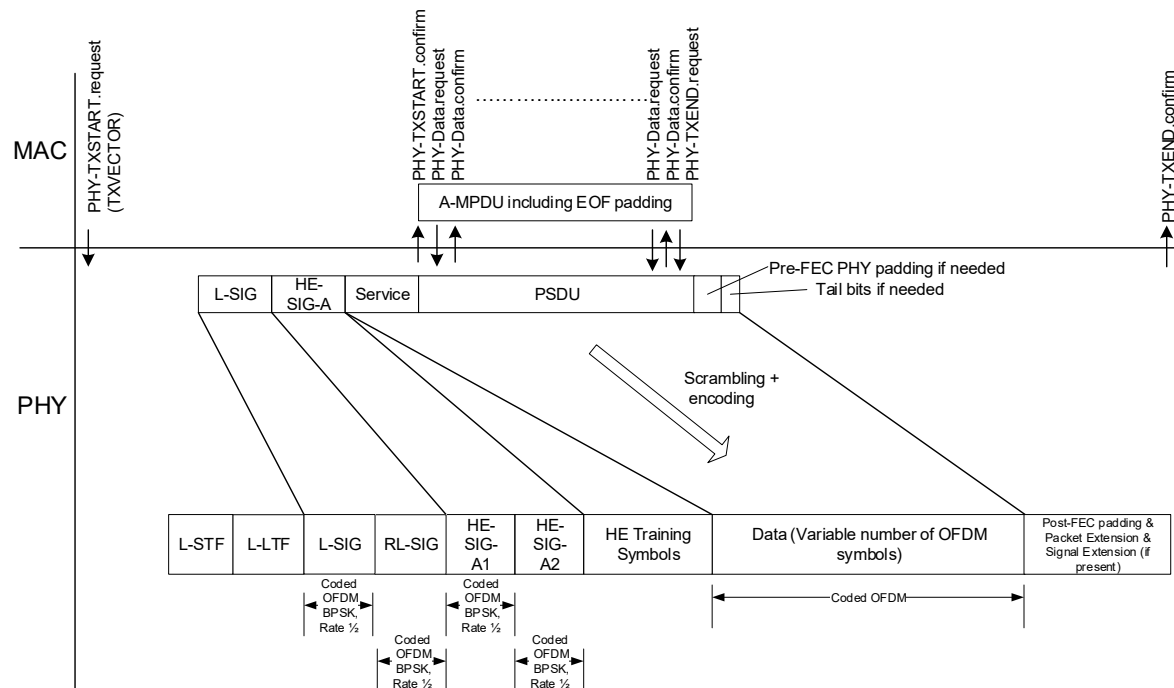


Figure 27-57—PHY transmit procedure for an HE TB PPDU

The fifth option is to follow the transmit procedure in Clause 17 or Clause 18 if the `FORMAT` parameter of the `PHYTXSTART.request(TXVECTOR)` primitive is `NON_HT` as follows:

- If the `FORMAT` parameter is `NON_HT` and the `NON_HT_MODULATION` parameter is `OFDM`, then the transmit procedure defined in Clause 17 is followed.
- If the `FORMAT` parameter is `NON_HT` and the `NON_HT_MODULATION` parameter is `NON_HT_DUP_OFDM`, then the transmit procedure defined in Clause 17 is followed, except that the signal is generated simultaneously on each of the 20 MHz channels identified by the `CH_BANDWIDTH` parameter as defined in 27.3.11 and 27.3.14.
- If the `FORMAT` parameter is `NON_HT`, the `CH_BANDWIDTH` parameter is `NON_HT_CBW20`, and the `NON_HT_MODULATION` parameter is not `OFDM`, then the transmit procedure defined in Clause 18 is followed.

NOTE 1—For an HE MU PPDU, the A-MPDU is per user in the MAC sublayer, and the HE Training Symbols and Data fields are per user in the PHY in Figure 27-56, with the number of HE Training Symbols depending on the maximum total number of space-time streams across all RUs.

NOTE 2—The transmit procedure for non-HT, HT_MF, HT_GF, and VHT format are specified in 27.2.6.

In all options, in order to transmit data, the MAC generates a `PHY-TXSTART.request` primitive, which causes the PHY entity to enter the transmit state. Further, the PHY is set to operate at the appropriate frequency through station management via the PLME, as specified in 27.4. Other transmit parameters, such as HE-MCS, coding types, and transmit power, are set via the PHY-SAP using the `PHYTXSTART.request(TXVECTOR)` primitive, as described in 27.2.2. After transmitting a PPDU that carries a Trigger frame, the MAC sublayer issues a `PHY-TRIGGER.request` with a `TRIGVECTOR` parameter that provides the PHY entity with the information needed to demodulate the expected HE TB PPDU response. The remainder of this subclause applies to the first four options.

The PHY indicates the state of the primary channel and other channels (if any) via the `PHY-CCA.indication` primitive (see 21.3.18.5 and 8.3.5.12). Transmission of the PPDU shall be initiated by the PHY after receiving the `PHY-TXSTART.request(TXVECTOR)` primitive. The `TXVECTOR` parameters for the `PHY-TXSTART.request` primitive are specified in Table 27-1.

After the PHY preamble transmission is started, the PHY entity immediately initiates data scrambling and data encoding. The encoding method for the Data field is based on the `FEC_CODING`, `CH_BANDWIDTH`, `NUM_STS`, `STBC`, `MCS`, and `NUM_USERS` parameters of the `TXVECTOR`, as described in 27.3.4.

The `SERVICE` field and PSDU are encoded as described in 27.3.5. The data shall be exchanged between the MAC and the PHY through a series of `PHY-DATA.request(DATA)` primitives issued by the MAC and `PHY-DATA.confirm` primitives issued by the PHY. PHY padding bits are appended to the PSDU to make the number of bits in the coded PSDU an integral multiple of the number of coded bits per OFDM symbol.

Transmission can be prematurely terminated by the MAC through the `PHY-TXEND.request` primitive. PSDU transmission is terminated by receiving a `PHY-TXEND.request` primitive. Each `PHY-TXEND.request` primitive is acknowledged with a `PHY-TXEND.confirm` primitive from the PHY.

A packet extension and/or a signal extension may be present in the PPDU. The `PHY-TXEND.confirm` primitive is generated at the latest of the actual ending time of the PPDU, the end of the packet extension if present, and the end of the signal extension if present.

In the PHY, the GI with GI duration indicated in `GI_TYPE` parameter of the `TXVECTOR` is inserted in every data OFDM symbol as a countermeasure against delay spread.

Once the PPDU transmission is completed, the PHY entity enters the receive state.

A typical state machine implementation for the transmission of an HE PPDU without midambles is shown in Figure 27-58. Request (.request) and confirmation (.confirm) primitives are issued once per state as shown. This state machine does not describe the operation of optional features, such as DCM.

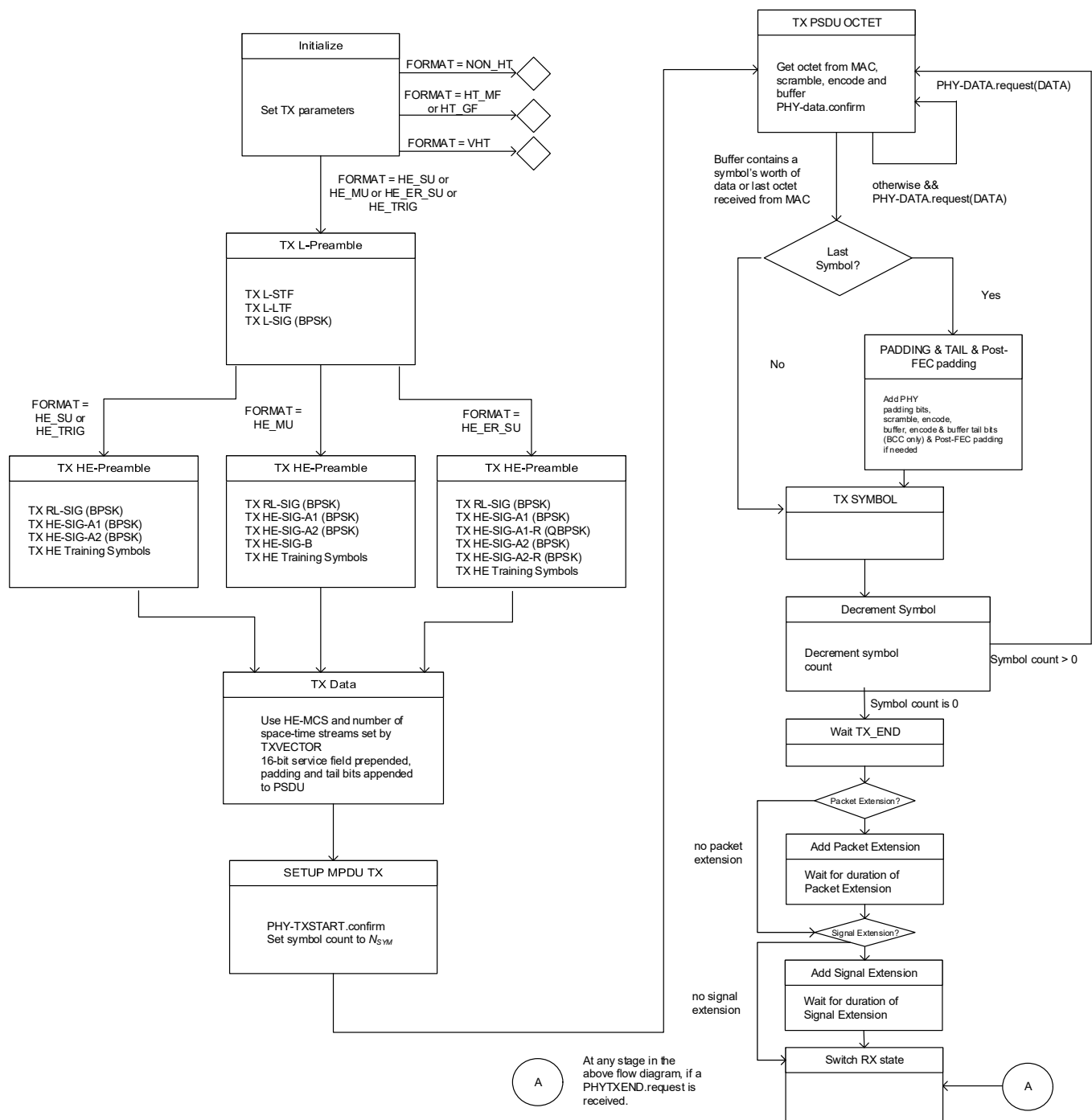


Figure 27-58—PHY transmit state machine for an HE PPDU without midambles

27.3.22 HE receive procedure

Typical PHY receive procedures are shown in Figure 27-59, Figure 27-60, Figure 27-61, and Figure 27-62, respectively.

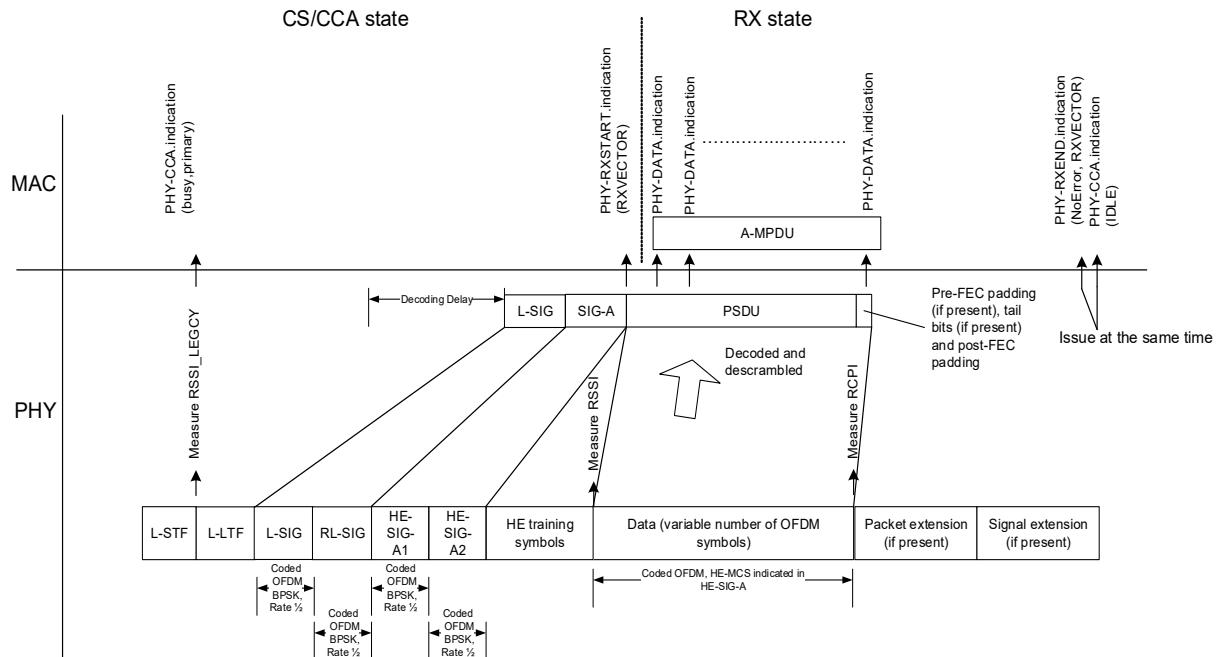


Figure 27-59—PHY receive procedure for an HE SU PPDU

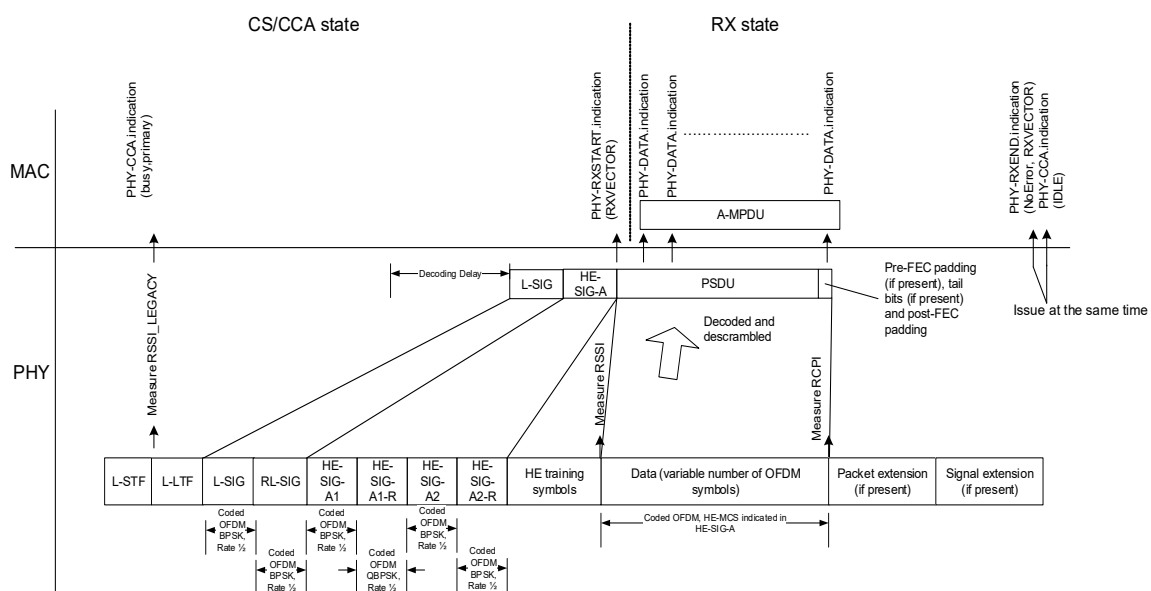


Figure 27-60—PHY receive procedure for an HE ER SU PPDU

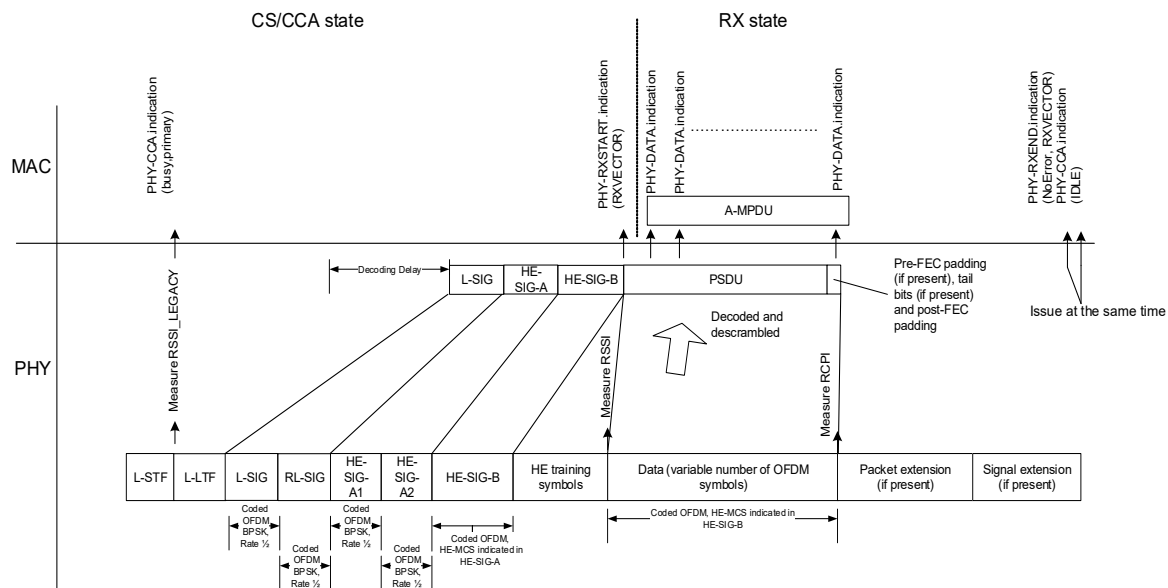


Figure 27-61—PHY receive procedure for an HE MU PPDU

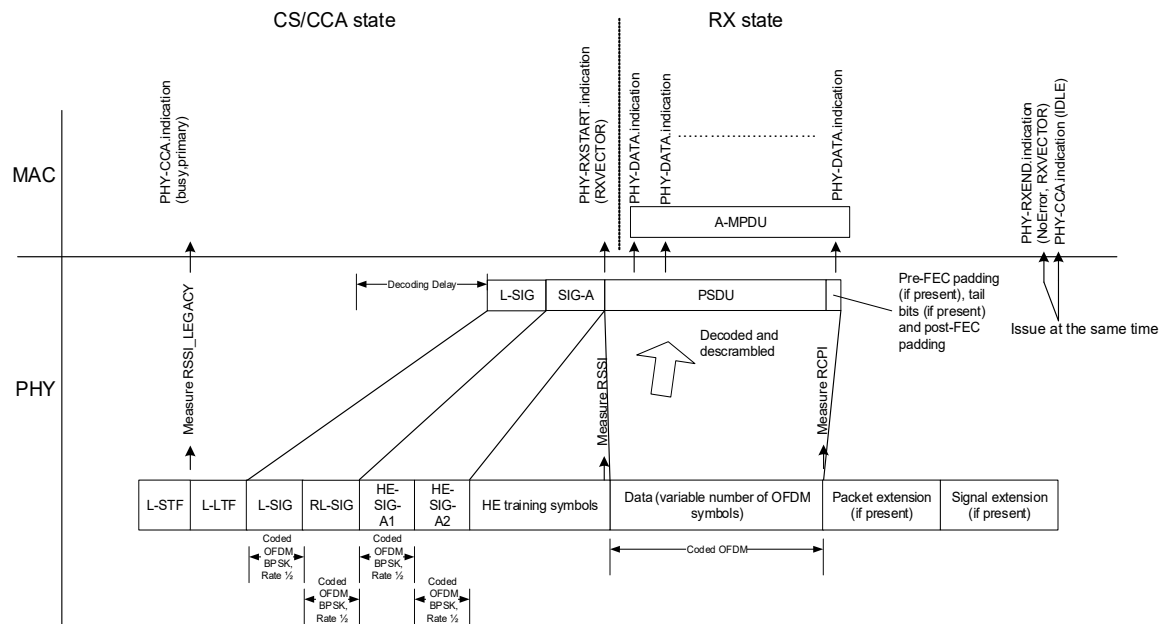


Figure 27-62—PHY receive procedure for an HE TB PPDU

A typical state machine implementation of the receive PHY is given in Figure 27-63.

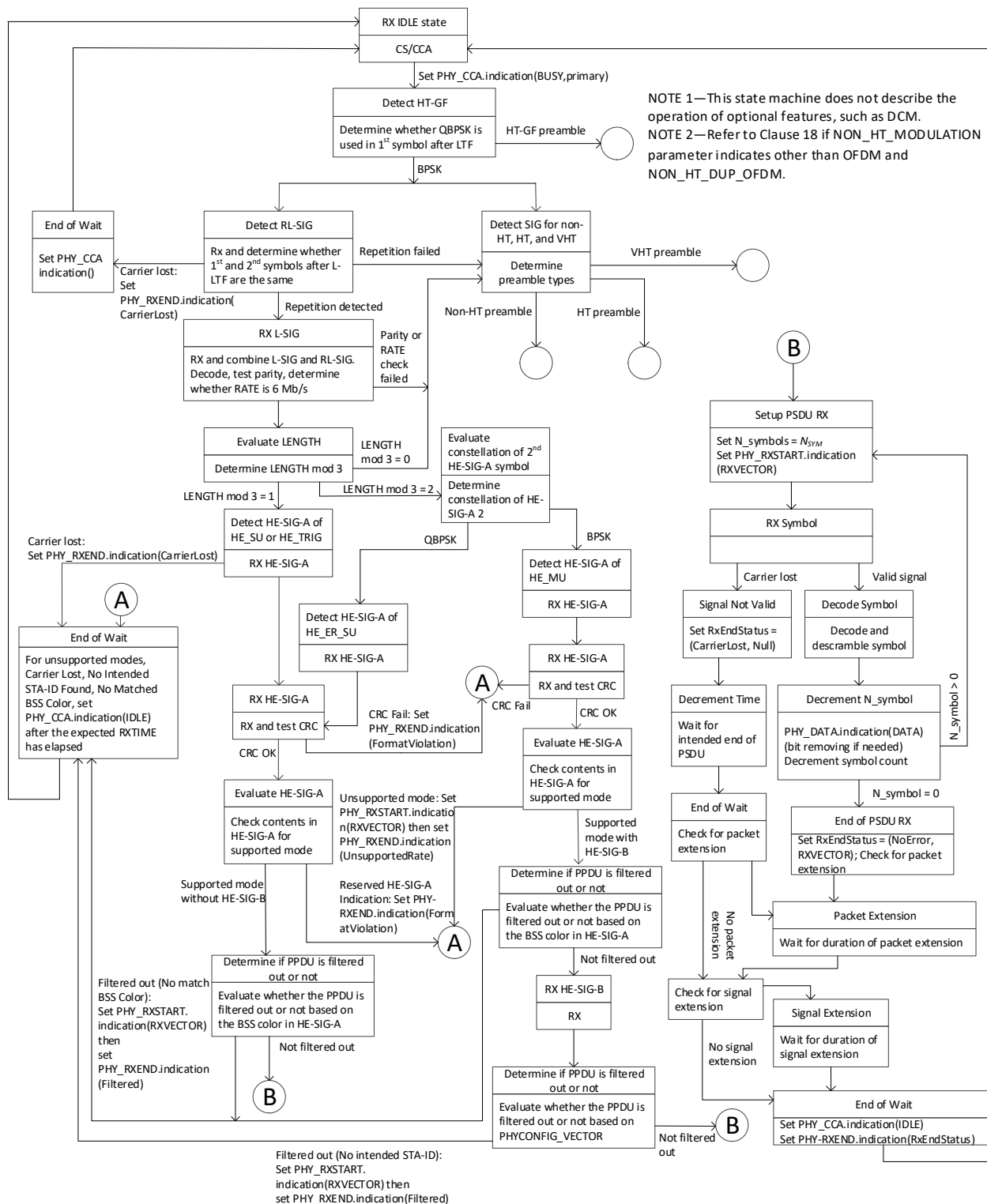


Figure 27-63—PHY receive state machine if midambles are not present

This receive procedure and state machine do not describe the operation of optional features, such as DCM. If the detected format indicates a non-HT PPDU, refer to the receive procedure and state machine in Clause 15, Clause 16, Clause 17, and Clause 18. If the detected format indicates an HT PPDU format, refer to the receive procedure and state machine in Clause 19. If the detected format indicates a VHT PPDU format, refer to the receive procedure and state machine in Clause 21. Through station management (via the PLME), the PHY is set to the appropriate frequency, as specified in 27.4. The PHY has also been configured with BSS identification information and STA identification information (i.e., BSS color value and STA-ID in the BSS) so that it can receive data intended for the STA in the specific cell. Other receive parameters, such as RSSI and indicated DATARATE, may be accessed via the PHY-SAP.

Upon receiving the transmitted PHY preamble in a greater than or equal to 20 MHz BSS, the PHY measures a receive signal strength. This activity is indicated by the PHY to the MAC via a PHY-CCA.indication primitive. A PHY-CCA.indication(BUSY, channel-list) primitive is also issued as an initial indication of reception of a signal as specified in 27.3.20.6. The channel-list parameter of the PHY-CCA.indication primitive is absent when the operating channel width is 20 MHz. The channel-list parameter is present when the operating channel width is 40 MHz, 80 MHz, 160 MHz, or 80+80 MHz.

The PHY shall not issue a PHY-RXSTART.indication primitive in response to a PPDU that does not overlap the primary channel, unless the PHY at an AP receives the HE TB PPDU solicited by the AP. For the HE TB PPDU solicited by the AP, the PHY shall issue a PHY-RXSTART.indication primitive for a PPDU received in the primary or at the secondary 20 MHz channel, the secondary 40 MHz channel, or the secondary 80 MHz channel.

The PHY includes the measured RSSI and RSSI_LEGACY value in the PHY-RXSTART.indication(RXVECTOR) primitive issued to the MAC.

After the PHY-CCA.indication(BUSY, channel-list) primitive is issued, the PHY entity shall begin receiving the training symbols and searching for the preambles for non-HT, HT, VHT, and HE PPDU, respectively. If the constellation used in the first symbol after the first long training field is QPSK, the PHY entity shall continue to detect the received signal using the receive procedure for HT-GF depicted in Clause 19. Otherwise, for detecting the HE preamble, the PHY entity shall search for L-SIG and RL-SIG fields in order to set the maximum duration of the data stream. If an RL-SIG field is detected, the PHY entity should check the parity bit and RATE fields in the L-SIG and RL-SIG fields. If either the check of the parity bit is invalid or the RATE field is not set to 6 Mb/s in non-HT, a PHY-RXSTART.indication primitive is not issued. If the check of the parity bit is valid and the RATE field indicates 6 Mb/s in non-HT but the LENGTH field value in the L-SIG field is a multiple of 3, a PHY-RXSTART.indication primitive is not issued. In both cases, the PHY should continue to detect the received signal using non-HT, HT, and VHT receive procedure in Clause 17, Clause 19, and Clause 21, respectively.

If a valid parity bit and the RATE with 6 Mb/s in non-HT are indicated in the L-SIG and RL-SIG fields and the LENGTH field value in the L-SIG and RL-SIG fields meets the condition that the remainder is 1 after LENGTH divided by 3, the PHY entity should begin receiving the sequence of HE-SIG-A, HE-STF, and HE-LTF fields for the HE SU PPDU and HE TB PPDU as shown in Figure 27-59 and Figure 27-62, respectively. After the RL-SIG field, the PHY entity shall receive two symbols of the HE-SIG-A field immediately followed by HE-STF.

The PHY entity shall check CRC of the HE-SIG-A field. If the CRC check is valid, the PHY entity shall report the TXOP and BSS color, check the Format field, and continue to receive HE-STF. The PHY entity shall report to the MAC entity the predicted duration of the TXOP in the HE-SIG-A field.

The PHY entity shall check the BSS color in the HE-SIG-A field. If the BSS color does not contain an intended value, the PHY entity shall issue a PHY-RXSTART.indication(RXVECTOR) and then issue a PHY-RXEND.indication(Filtered).

The PHY entity shall check the Format field in the HE-SIG-A field. If the Format field indicates an HE SU PPDU, the PHY entity shall receive HE-STF for 4 μ s after the HE-SIG-A field. If the Format field indicates an HE TB PPDU, the PHY entity shall receive HE-STF for 8 μ s after the HE-SIG-A field.

If the HE-SIG-A field indicates a valid CRC and a reserved HE-SIG-A indication is not indicated, for all supported modes and unsupported modes, the PHY entity shall maintain the PHY-CCA.indication(BUSY, channellist) primitive for the predicted duration of the transmitted PPDU, as defined by RXTIME in Equation (27-133), unless it receives a PHY-CCARESET.request primitive before the end of the PPDU, for instance, during spatial reuse operation as described in 26.10. If the HE-SIG-A field indicates an unsupported mode, the PHY shall issue a PHY-RXEND.indication(UnsupportedRate) primitive.

If the HE-SIG-A field indicates a reserved HE-SIG-A indication, the PHY shall issue the error condition PHY-RXEND.indication(FormatViolation) primitive and maintain the PHY-CCA.indication(BUSY, channellist) primitive for the predicted duration of the transmitted PPDU derived from the LENGTH field in the L-SIG field, as defined in Equation (27-134), unless it receives a PHY-CCARESET.request primitive before the end of the PPDU, for instance, during spatial reuse operation as described in 26.10. A reserved HE-SIG-A indication is defined as an HE-SIG-A field with Reserved bits equal to 0 or any other HE-SIG-A field bit combinations that do not correspond to modes of PHY operation defined in Clause 27.

If the HE-SIG-A field indicates an invalid CRC, the PHY shall issue the error condition PHY-RXEND.indication(FormatViolation) primitive and maintain the PHY-CCA.indication(BUSY, channellist) primitive for the predicted duration of the transmitted PPDU derived from the LENGTH field in the L-SIG field as defined in Equation (27-134).

If a valid parity bit of the L-SIG and RL-SIG fields is indicated and the LENGTH field value in the L-SIG and RL-SIG fields meets the condition that the remainder is 2 after LENGTH divided by 3, the PHY entity should detect the signal constellations in the second symbol after the RL-SIG field. If the constellation is QBPSK, the PHY entity shall continue receiving the sequence of HE-SIG-A, HE-STF, and HE-LTF fields for an HE ER SU PPDU shown in Figure 27-60. After the RL-SIG field, the PHY entity shall receive four symbols of HE-SIG-A immediately followed by HE-STF.

The PHY entity shall check the CRC of the HE-SIG-A field. If the CRC is valid, the PHY entity shall report TXOP and BSS color and continue to receive the HE-STF. The PHY entity shall report to the MAC entity the predicted duration of the TXOP in the HE-SIG-A field.

The PHY entity shall check the BSS color in the HE-SIG-A field. If the BSS color does not contain an intended value, the PHY entity shall issue a PHY-RXSTART.indication(RXVECTOR) and then issue a PHY-RXEND.indication(Filtered).

The PHY entity shall receive the HE-STF for 4 μ s after the HE-SIG-A field.

If the HE-SIG-A field indicates a valid CRC and a reserved HE-SIG-A indication is not indicated, for all supported mode and unsupported modes, the PHY entity shall maintain the PHY-CCA.indication(BUSY, channellist) primitive for the predicted duration of the transmitted PPDU, as defined by RXTIME in Equation (27-133), unless it receives a PHY-CCARESET.request primitive before the end of the PPDU, for instance, during spatial reuse operation as described in 26.10.

If the HE-SIG-A field indicates an unsupported mode, the PHY shall issue a PHY-RXEND.indication(UnsupportedRate) primitive.

If the HE-SIG-A field indicates a reserved HE-SIG-A indication, the PHY shall issue the error condition PHY-RXEND.indication(FormatViolation) primitive and maintain the PHY-CCA.indication(BUSY, channellist) primitive for the predicted duration of the transmitted PPDU derived from the LENGTH field in the L-SIG field, as defined in Equation (27-134), unless it receives a PHY-CCARESET.request primitive

before the end of the PPDU, for instance, during spatial reuse operation as described in 26.10. A reserved HE-SIG-A indication is defined as an HE-SIG-A field with Reserved bits equal to 0 or any other HE-SIG-A field bit combinations that do not correspond to modes of PHY operation defined in Clause 27.

If the HE-SIG-A field indicates an invalid CRC, the PHY shall issue the error condition PHY-RXEND.indication(FormatViolation) primitive and maintain the PHY-CCA.indication(BUSY, channellist) primitive for the predicted duration of the transmitted PPDU derived from the LENGTH field in the L-SIG field, as defined in Equation (27-134), unless it receives a PHY-CCARESET.request primitive before the end of the PPDU, for instance, during spatial reuse operation as described in 26.10.

If a STA receives an HE TB PPDU and the TRIGVECTOR parameters are not present in its PHY entity, the STA shall use Equation (27-134) to calculate the predicted duration of the HE TB PPDU.

If a valid parity bit of the L-SIG and RL-SIG fields is indicated and the LENGTH field value in the L-SIG and RL-SIG fields meets the condition that the remainder is 2 after LENGTH divided by 3, the PHY entity should detect the signal constellations in the second symbol after the RL-SIG field. If the constellation is BPSK, the PHY entity shall continue receiving the sequence of HE-SIG-A, HE-SIG-B, HE-STF, and HE-LTF fields for an HE MU PPDU shown in Figure 27-61. After the RL-SIG field, the PHY entity shall receive two symbols of the HE-SIG-A field immediately followed by the HE-SIG-B field.

The PHY entity shall check CRC of the HE-SIG-A field. If the CRC check is valid, the PHY entity shall report TXOP and BSS color and continue to receive the HE-SIG-B field. The PHY entity shall report to the MAC entity the predicted duration of the TXOP in the HE-SIG-A field.

The PHY entity shall check the BSS color in the HE-SIG-A field. If the BSS color does not contain an intended value, the PHY entity shall issue a PHY-RXSTART.indication(RXVECTOR) and then issue a PHY-RXEND.indication(Filtered).

After the HE-SIG-A field, the PHY entity shall receive the HE-SIG-B field for the number of symbols predicted from the HE-SIG-A field. If the Common field is present in the HE-SIG-B field, the PHY entity shall check the CRC of the Common field. If the CRC in the Common field is valid or the Common field is not present, the PHY entity shall search for intended STA-ID in each User Specific subfield with a valid CRC. If no CRC is valid or no intended STA-ID is detected, the PHY entity shall issue a PHY-RXSTART.indication(RXVECTOR) and then issue a PHY-RXEND.indication(Filtered). If a complete allocation of an intended STA-ID is detected in block with valid CRC, the PHY entity shall continue receiving HE-STF for 4 μ s after the HE-SIG-B field for the detected and intended STA.

If the HE-SIG-A field indicates a valid CRC and a reserved HE-SIG-A indication is not indicated, for all supported modes and unsupported modes, the PHY entity shall maintain the PHY-CCA.indication(BUSY, channellist) primitive for the predicted duration of the transmitted PPDU, as defined by RXTIME in Equation (27-133), unless it receives a PHY-CCARESET.request primitive before the end of the PPDU, for instance, during spatial reuse operation as described in 26.10.

If the HE-SIG-A field indicates an unsupported mode, the PHY shall issue a PHY-RXEND.indication(UnsupportedRate) primitive.

If the HE-SIG-A field indicates a reserved HE-SIG-A indication, the PHY shall issue the error condition PHY-RXEND.indication(FormatViolation) primitive and maintain the PHY-CCA.indication(BUSY, channellist) primitive for the predicted duration of the transmitted PPDU derived from the LENGTH field in the L-SIG field, as defined in Equation (27-134), unless it receives a PHY-CCARESET.request primitive before the end of the PPDU, for instance, during spatial reuse operation as described in 26.10. A reserved HE-SIG-A indication is defined as an HE-SIG-A field with Reserved bits equal to 0 or any other HE-SIG-A field bit combinations that do not correspond to modes of PHY operation defined in Clause 27.

If the HE-SIG-A field indicates an invalid CRC, the PHY shall issue the error condition PHY-RXEND.indication(FormatViolation) primitive and maintain the PHY-CCA.indication(BUSY, channellist) primitive for the predicted duration of the transmitted PPDU derived from the LENGTH field in the L-SIG field, as defined in Equation (27-134), unless it receives a PHY-CCARESET.request primitive before the end of the PPDU, for instance, during spatial reuse operation as described in 26.10.

If signal loss occurs during reception prior to completion of the PSDU reception, the error condition PHY-RXEND.indication(CarrierLost) shall be reported to the MAC. After waiting for the end of the PPDU as determined by Equation (27-133), the PHY shall set the PHY-CCA.indication (IDLE) primitive and return to the RX IDLE state.

(27-133)

$$RXTIME(\mu s) = 20 + T_{HE_PREAMBLE} + N_{SYM}T_{SYM} + N_{MA}N_{HE-LTF}T_{HE-LTF-SYM} + T_{PE} + SignalExtension$$

where

$T_{HE-PREAMBLE}$, N_{SYM} , T_{PE} , and N_{MA}

are defined in Equation (27-119), Equation (27-120), Equation (27-121), and Equation (27-122), respectively

$SignalExtension$

is defined in Table 27-54

$$RXTIME(\mu s) = \left\lceil \frac{LENGTH + 3}{3} \right\rceil \cdot 4 + 20 + SignalExtension \quad (27-134)$$

where

LENGTH is the LENGTH field in the L-SIG field

$SignalExtension$ is defined in Table 27-54

Except in an HE sounding NDP and HE TB feedback NDP, a Data field follows the HE-STF and HE-LTF fields. The number of symbols in the Data field and the packet extension duration are computed from Equation (27-119) and Equation (27-120), respectively.

The received PSDU bits are assembled into octets, decoded, and presented to the MAC using a series of PHY-DATA.indication(DATA) primitive exchanges. Any final bits that cannot be assembled into a complete octet are considered pad bits and discarded. After the reception of the final bit of the last PSDU octet, and possible padding and tail bits, the PHY entity shall check whether packet extension and/or signal extension is applied. If packet extension and/or signal extension is applied, the PHY entity shall wait until the packet extension and/or signal extension expires before returning to the RX IDLE state, as shown in Figure 27-63.

27.3.23 Channel numbering

27.3.23.1 General

The STA may operate in the 2.4 GHz band, 5 GHz band, or 6 GHz band. The set of valid operating channel numbers by regulatory domain is defined in Annex E. Channel allocation for each band is defined in the following subclauses:

- In 19.3.15.2 for the 2.4 GHz band
- In 19.3.15.3 for the 5 GHz band
- In 27.3.23.2 for the 6 GHz band

27.3.23.2 Channel allocation in the 6 GHz band

Channel center frequencies are defined at every integer multiple of 5 MHz above the channel starting frequency. The relationship between center frequency and channel number is given in Equation (27-135).

$$\text{Channel center frequency} = \text{Channel starting frequency} + 5 \times n_{ch} \text{ (MHz)} \quad (27-135)$$

where

$$n_{ch} = 1, \dots, 233$$

Channel starting frequency is defined as dot11ChannelStartingFactor $\times$ 500 kHz

For example, a channel center frequency of 5.955 GHz is indicated by dot11ChannelStartingFactor = 11 900 and $n_{ch} = 1$. A channel center frequency of 5.935 GHz is indicated by dot11ChannelStartingFactor = 11 850 and $n_{ch} = 2$.

27.3.24 Regulatory requirements

Wireless LANs (WLANs) implemented in accordance with this standard are subject to equipment certification and operating requirements established by regional and national regulatory administrations. The PHY specification establishes minimum technical requirements for interoperability, based upon established regulations at the time this standard was issued. These regulations are subject to revision or may be superseded. Requirements that are subject to local geographic regulations are annotated within the PHY specification. Regulatory requirements that do not affect interoperability are not addressed in this standard. Implementers are referred to the regulatory sources in Annex D for further information. Operation in countries within defined regulatory domains might be subject to additional or alternative national regulations.

27.4 HE PLME

27.4.1 PLME_SAP sublayer management primitives

Table 27-53 lists the MIB attributes that may be accessed by the PHY entities and the intralayer of higher level LMEs. These attributes are accessed via the PLME-GET, PLME-SET, PLME-RESET, and PLME-CHARACTERISTICS primitives defined in 6.5.

Table 27-53—HE PHY MIB attributes

Managed object	Default value/range	Operational semantics
dot11PHYOperationTable		
dot11PHYType	he	Static
dot11PHYTxPowerTable		
dot11NumberSupportedPowerLevelsImplemented	Implementation dependent	Static
dot11TxPowerLevel1	Implementation dependent	Static
dot11TxPowerLevel2	Implementation dependent	Static

Table 27-53—HE PHY MIB attributes (continued)

Managed object	Default value/range	Operational semantics
dot11TxPowerLevel3	Implementation dependent	Static
dot11TxPowerLevel4	Implementation dependent	Static
dot11TxPowerLevel5	Implementation dependent	Static
dot11TxPowerLevel6	Implementation dependent	Static
dot11TxPowerLevel7	Implementation dependent	Static
dot11TxPowerLevel8	Implementation dependent	Static
dot11CurrentTxPowerLevel	Implementation dependent	Static
dot11TxPowerLevelExtended	Implementation dependent	Static
dot11CurrentTxPowerLevelExtended	Implementation dependent	Static
dot11PHYOFDMTable		
dot11TwentyMHzOperationImplemented	false/Boolean	Static
dot11ChannelStartingFactor	Implementation dependent	Dynamic
dot11PHYHTTable		
dot11CurrentPrimaryChannel	Implementation dependent	Dynamic
dot11CurrentSecondaryChannel	Implementation dependent	Dynamic
dot11FortyMHzOperationImplemented	false/Boolean	Static
dot11FortyMHzOperationActivated	false/Boolean	Dynamic
dot11NumberOfSpatialStreamsImplemented	Implementation dependent	Static
dot11NumberOfSpatialStreamsActivated	Implementation dependent	Dynamic
dot11HTGreenfieldOptionImplemented	false/Boolean	Static
dot11HTGreenfieldOptionActivated	false/Boolean	Dynamic
dot11ShortGIOptionInTwentyImplemented	false/Boolean	Static
dot11ShortGIOptionInTwentyActivated	false/Boolean	Dynamic
dot11ShortGIOptionInFortyImplemented	false/Boolean	Static
dot11ShortGIOptionInFortyActivated	false/Boolean	Dynamic

Table 27-53—HE PHY MIB attributes (continued)

Managed object	Default value/range	Operational semantics
dot11LDPCCodingOptionImplemented	false/Boolean	Static
dot11LDPCCodingOptionActivated	false/Boolean	Dynamic
dot11TxSTBCOptionImplemented	false/Boolean	Static
dot11TxSTBCOptionActivated	false/Boolean	Dynamic
dot11RxSTBCOptionImplemented	false/Boolean	Static
dot11RxSTBCOptionActivated	false/Boolean	Dynamic
dot11BeamFormingOptionImplemented	false/Boolean	Static
dot11BeamFormingOptionActivated	false/Boolean	Dynamic
dot11PHYVHTTable		
dot11CurrentChannelWidth	Implementation dependent	Dynamic
dot11CurrentChannelCenterFrequencyIndex0	Implementation dependent	Dynamic
dot11CurrentChannelCenterFrequencyIndex1	Implementation dependent	Dynamic
dot11VHTChannelWidthOptionImplemented	Implementation dependent	Static
dot11EightyMHzOperationImplemented	false/Boolean	Static
dot11EightyMHzOperationActivated	false/Boolean	Dynamic
dot11VHTShortGIOptionIn80Implemented	false/Boolean	Static
dot11VHTShortGIOptionIn80Activated	false/Boolean	Dynamic
dot11VHTShortGIOptionIn160and80p80Implemented	false/Boolean	Static
dot11VHTShortGIOptionIn160and80p80Activated	false/Boolean	Dynamic
dot11VHTLDPCCodingOptionImplemented	false/Boolean	Static
dot11VHTLDPCCodingOptionActivated	false/Boolean	Dynamic
dot11VHTTxSTBCOptionImplemented	false/Boolean	Static
dot11VHTTxSTBCOptionActivated	false/Boolean	Dynamic
dot11VHTRxSTBCOptionImplemented	false/Boolean	Static
dot11VHTRxSTBCOptionActivated	false/Boolean	Dynamic
dot11VHTMaxNTxChainsImplemented	Implementation dependent	Static
dot11VHTMaxNTxChainsActivated	Implementation dependent	Dynamic

Table 27-53—HE PHY MIB attributes (continued)

Managed object	Default value/range	Operational semantics
dot11TransmitBeamformingConfigTable		
dot11ReceiveStaggerSoundingOptionImplemented	false/Boolean	Static
dot11TransmitStaggerSoundingOptionImplemented	false/Boolean	Static
dot11ReceiveNDPOptionImplemented	false/Boolean	Static
dot11TransmitNDPOptionImplemented	false/Boolean	Static
dot11ImplicitTransmitBeamformingOptionImplemented	false/Boolean	Static
dot11CalibrationOptionImplemented	Implementation dependent	Static
dot11ExplicitCSITransmitBeamformingOptionImplemented	false/Boolean	Static
dot11ExplicitNonCompressedBeamformingMatrixOptionImplemented	false/Boolean	Static
dot11ExplicitTransmitBeamformingCSIFeedbackOptionImplemented	Implementation dependent	Static
dot11ExplicitNoncompressedBeamformingFeedbackOptionImplemented	Implementation dependent	Static
dot11ExplicitCompressedBeamformingFeedbackOptionImplemented	Implementation dependent	Static
dot11NumberBeamFormingCSISupportAntenna	Implementation dependent	Static
dot11NumberNonCompressedBeamformingMatrixSupportAntenna	Implementation dependent	Static
dot11NumberCompressedBeamformingMatrixSupportAntenna	Implementation dependent	Static
dot11VHTTransmitBeamformingConfigTable		
dot11VHTSUBeamformeeOptionImplemented	false/Boolean	Static
dot11VHTSUBeamformerOptionImplemented	false/Boolean	Static
dot11VHTMUBeamformeeOptionImplemented	false/Boolean	Static
dot11VHTMUBeamformerOptionImplemented	false/Boolean	Static
dot11VHTNumberSoundingDimensions	Implementation dependent	Static
dot11VHTBeamformeeNTxSupport	Implementation dependent	Static
dot11PHYHETable		
dot11HECurrentChannelWidthSet	Implementation dependent	Dynamic
dot11HEPuncturedPreambleRxImplemented	Implementation dependent	Static

Table 27-53—HE PHY MIB attributes (continued)

Managed object	Default value/range	Operational semantics
dot11HEPuncturedSoundingOptionImplemented	Implementation dependent	Static
dot11HEDeviceClass	Implementation dependent	Static
dot11HELDPCCodingInPayloadImplemented	false/Boolean	Static
dot11HESUPPDUwith1xHELTfand0point8GIImplemented	false/Boolean	Static
dot11HESUPPDUandHEMUPPDUwith4xHELTfand0point8GIImplemented	false/Boolean	Static
dot11HEERSUPPDUwith4xHELTfand0point8GIImplemented	false/Boolean	Static
dot11HEERSUPPDUwith1xHELTfand0point8GIImplemented	false/Boolean	Static
dot11HEMidambleRxMaxNSTS	false/Boolean	Dynamic
dot11HENDPwith4xHELTfand3point2GIImplemented	false/Boolean	Static
dot11HESTBCTxLessThanOrEqualTo80Implemented	false/Boolean	Static
dot11HESTBCRxLessThanOrEqualTo80Implemented	false/Boolean	Static
dot11HESTBCTxGreaterThan80Implemented	false/Boolean	Static
dot11HESTBCRxGreaterThan80Implemented	false/Boolean	Static
dot11HEDopplerTxImplemented	false/Boolean	Static
dot11HEDopplerRxImplemented	false/Boolean	Static
dot11HEDCMIImplemented	Implementation dependent	Static
dot11HEFullBWULMUMIMOImplemented	false/Boolean	Static
dot11HEPartialBWULMUMIMOImplemented	false/Boolean	Static
dot11HEPartialBWDLMUMIMOImplemented	false/Boolean	Static
dot11HEULMUPayloadImplemented	false/Boolean	Static
dot11HEPSRbasedSRSupportImplemented	false/Boolean	Static
dot11HEPowerBoostFactorImplemented	false/Boolean	Static
dot11HEPartialBWERSUPayloadImplemented	false/Boolean	Static
dot11HETransmitBeamformingConfigTable		
dot11HESUBeamformerOptionImplemented	false/Boolean	Static
dot11HESUBeamformeeOptionImplemented	false/Boolean	Static
dot11HEMUBeamformerOptionImplemented	false/Boolean	Static
dot11HEBeamformeeSTSSupportLessThanOrEqualTo80	Implementation dependent	Static
dot11HEBeamformeeSTSSupportGreaterThan80	Implementation dependent	Static

Table 27-53—HE PHY MIB attributes (continued)

Managed object	Default value/range	Operational semantics
dot11HENumberSoundingDimensionsLessThanOrEqualTo80	Implementation dependent	Static
dot11HENumberSoundingDimensionsGreaterThan80	Implementation dependent	Static
dot11HENG16SUFeedbackSupport	false/Boolean	Static
dot11HENG16MUFeedbackSupport	false/Boolean	Static
dot11HECodebookSizePhi4Psi2SUFeedbackSupport	false/Boolean	Static
dot11HECodebookSizePhi7Psi5MUFeedbackSupport	false/Boolean	Static
dot11HETriggeredSUBeamformingFeedbackImplemented	false/Boolean	Static
dot11HETriggeredMUBeamformingFeedbackImplemented	false/Boolean	Static
dot11HETriggeredCQIFeedbackSupportImplemented	false/Boolean	Static

27.4.2 PHY MIB

HE PHY MIB attributes are defined in Annex C with specific values defined in Table 27-53. The “Operational semantics” column in Table 27-53 contains two types: static and dynamic.

- Static MIB attributes are fixed and cannot be modified for a given PHY implementation.
- Dynamic MIB attributes are interpreted according to the MAX-ACCESS field of the MIB attribute. If MAX-ACCESS is equal to read-only, the MIB attribute value may be updated by the PLME and read from the MIB attribute by management entities. If MAX-ACCESS is equal to read-write, the MIB attribute may be read and written by management entities.

27.4.3 TXTIME and PSDU_LENGTH calculation

The value of the TXTIME parameter returned by the PLME-TXTIME.confirm primitive shall be calculated for an HE PPDU using Equation (27-136).

(27-136)

$$\text{TXTIME} = 20 + T_{\text{HE-PREAMBLE}} + N_{\text{SYM}}T_{\text{SYM}} + N_{\text{MA}}N_{\text{HE-LTF}}T_{\text{HE-LTF-SYM}} + T_{\text{PE}} + \text{SignalExtension}$$

where

$T_{\text{HE-PREAMBLE}}$ is defined as in Equation (27-121)

SignalExtension takes the value of aSignalExtension as defined in Table 27-54

For an HE sounding NDP and HE TB feedback NDP, there is no Data field and $N_{\text{SYM}} = 0$.

For an HE SU PPDU and HE ER SU PPDU using BCC encoding, the total number of data OFDM symbols, N_{SYM} , is given by Equation (27-66).

For an HE SU PPDU and HE ER SU PPDU using LDPC encoding, the total number of data OFDM symbols, N_{SYM} , is given in 27.3.12.5.2.

For an HE MU PPDU (including both MU-MIMO and OFDMA), the total number of data OFDM symbols, N_{SYM} , is given in 27.3.12.5.4.

For an HE TB PPDU, the total number of data OFDM symbols, N_{SYM} , is given in 27.3.12.5.5.

T_{PE} is given in 27.3.13.

N_{MA} is the number of midambles. It is given by Equation (27-117) for an HE TB PPDU and by Equation (27-112) for an HE SU PPDU, HE ER SU PPDU, and HE MU PPDU if the TXVECTOR value DOPPLER is 1, and is 0 otherwise.

N_{HE-LTF} and T_{HE-LTF} are given in 27.3.9.

The value of the PSDU_LENGTH parameter returned in the PLME-TXTIME.confirm primitive for an HE SU PPDU, HE ER SU PPDU, and HE TB PPDU is calculated using Equation (27-137).

$$\text{PSDU_LENGTH} = \left\lfloor \frac{(N_{SYM,init} - m_{STBC})N_{DBPS} + m_{STBC}N_{DBPS,last,init} - N_{service} - N_{tail}}{8} \right\rfloor \quad (27-137)$$

where

$N_{SYM,init}$ is given by Equation (27-64) for BCC encoding, by Equation (27-64) for LDPC encoding for an HE SU PPDU and HE ER SU PPDU, and in 27.3.12.5.5 for an HE TB PPDU
 m_{STBC} is 2 when STBC is used, and 1 otherwise
 N_{DBPS} is given in 27.5
 $N_{DBPS,last,init}$ is given by Equation (27-62)

The value of the PSDU_LENGTH parameter for user u returned in the PLME-TXTIME.confirm primitive for an HE MU PPDU is calculated using Equation (27-138) and Equation (27-139) for users using BCC and LDPC, respectively.

$$\text{PSDU_LENGTH}_u = \left\lfloor \frac{(N_{SYM} - m_{STBC})N_{DBPS,u} + m_{STBC}N_{DBPS,last,u} - N_{service} - N_{tail}}{8} \right\rfloor \quad (27-138)$$

$$\text{PSDU_LENGTH}_u = \left\lfloor \frac{(N_{SYM,init} - m_{STBC})N_{DBPS,u} + m_{STBC}N_{DBPS,last,init,u} - N_{service}}{8} \right\rfloor \quad (27-139)$$

where

$N_{SYM,init}$ is given by Equation (27-76)
 $N_{DBPS,u}$ is given in Table 27-15
 $N_{DBPS,last,u}$ is given by Equation (27-85)
 $N_{DBPS,last,init,u}$ is given by Equation (27-77)

For an HE SU PPDU, HE ER SU PPDU, and HE TB PPDU, the value of the PSDU_LENGTH parameter returned in the RXVECTOR is calculated using Equation (27-140).

$$\text{PSDU_LENGTH} = \left\lfloor \frac{(N_{SYM,RX} - m_{STBC})N_{DBPS} + m_{STBC}N_{DBPS,last,RX} - N_{service} - N_{tail}}{8} \right\rfloor \quad (27-140)$$

where

$N_{SYM,RX}$ is given by Equation (27-141)

$$N_{SYM,RX} = \begin{cases} \text{if the Coding field is 1,} \\ N_{SYM} - m_{STBC}, \text{ the LDPC Extra Symbol Segment field is 1, and} \\ \text{the Pre-FEC Padding Factor field is 1 in HE-SIG-A} \\ N_{SYM}, \text{ otherwise} \end{cases} \quad (27-141)$$

where

N_{SYM} is given by Equation (27-119)

m_{STBC} is 1 if the STBC field in the HE-SIG-A field is 0 and 2 if the STBC field is 1

$N_{DBPS,last,RX}$ is given by Equation (27-142)

$$N_{DBPS,last,RX} = \begin{cases} N_{DBPS}, \text{ if } a_{RX} = 4 \\ a_{RX} \cdot N_{SD,short} \cdot N_{SS} \cdot N_{BPSCS} \cdot R, \text{ otherwise} \end{cases} \quad (27-142)$$

where

a_{RX} is given by Equation (27-143)

$N_{SD,short}$ is defined in Table 27-33

N_{SS} , N_{BPSCS} , and R are defined in Table 27-15

$$a_{RX} = \begin{cases} 4, & \text{if } a = 1, \text{ the Coding field is 1 and} \\ & \text{the LDPC Extra Symbol Segment field is 1 in HE-SIG-A} \\ a - 1, & \text{if } a > 1, \text{ the Coding field is 1 and} \\ & \text{the LDPC Extra Symbol Segment field is 1 in HE-SIG-A} \\ a, & \text{otherwise} \end{cases} \quad (27-143)$$

where

a is the pre-FEC padding factor (ranging from 1 to 4) indicated in the HE-SIG-A field

N_{DBPS} is defined in Table 27-15

$N_{service}$ and N_{tail} are defined in Table 27-12

For an HE MU PPDU, the value of the RXVECTOR parameter PSDU_LENGTH returned for user u is calculated using Equation (27-144).

$$\text{PSDU_LENGTH}_u = \left\lfloor \frac{(N_{SYM,RX,u} - m_{STBC})N_{DBPS,u} + m_{STBC}N_{DBPS,last,RX,u} - N_{service} - N_{tail,u}}{8} \right\rfloor \quad (27-144)$$

where

$N_{SYM,RX,u}$ is given by Equation (27-145)

$$N_{SYM,RX,u} = \begin{cases} N_{SYM} - m_{STBC}, & \text{if } a = 1, \text{ the Coding field is 1 in the User field of the HE-SIG-B field} \\ & \text{corresponding to user } u \\ & \text{and LDPC Extra Symbol Segment field is 1 in HE-SIG-A field} \\ N_{SYM}, & \text{otherwise} \end{cases} \quad (27-145)$$

where

N_{SYM} is given by Equation (27-119)

m_{STBC} is 1 if the STBC field in the HE-SIG-A field is 0 and 2 if the STBC field is 1

$N_{DBPS,last,RX,u}$ is given by Equation (27-146)

$$N_{DBPS,last,RX,u} = \begin{cases} N_{DBPS,u}, & \text{if } a_{RX,u} = 4 \\ a_{RX,u} \cdot N_{SD,short,u} \cdot N_{SS,u} \cdot N_{BPSCS,u} \cdot R_u, & \text{otherwise} \end{cases} \quad (27-146)$$

where

$a_{RX,u}$ is given by Equation (27-147)

$N_{SD,short,u}$ is $N_{SD,short}$ defined in Table 27-33 for user u

$N_{SS,u}$, $N_{BPSCS,u}$, and R_u are defined in Table 27-15

$$a_{RX,u} = \begin{cases} 4, & \text{if } a = 1, \text{ Coding field is 1 in User field of HE-SIG-B field} \\ & \text{corresponding to the user } u \\ & \text{and LDPC Extra Symbol Segment field is 1 in HE-SIG-A} \\ a - 1, & \text{if } a > 1, \text{ the Coding field is 1 in User field of HE-SIG-B field} \\ & \text{corresponding to the user } u \\ & \text{and LDPC Extra Symbol Segment field is 1 in HE-SIG-A field} \\ a, & \text{otherwise} \end{cases} \quad (27-147)$$

where

a is the pre-FEC padding factor (ranging from 1 to 4) indicated in the HE-SIG-A field

$N_{DBPS,u}$ is defined in Table 27-15

$N_{service}$ and N_{tail} are defined in Table 27-12

The value of the PSDU_LENGTH parameter returned in the PLME-TXTIME.confirm primitive for an HE sounding NDP is 0.

27.4.4 HE PHY

The static HE PHY characteristics, provided through the PLME-CHARACTERISTICS service primitive, shall be as shown in Table 19-25 unless otherwise listed in Table 27-54. The definitions for these characteristics are given in 6.5.

Table 27-54—HE PHY characteristics

Characteristic	Value
aTxPHYDelay	Implementation dependent
aRxPHYDelay	Implementation dependent
aSignalExtension	0 μ s if operating in the 5 GHz or 6 GHz band 6 μ s if operating in the 2.4 GHz band
aSlotTime	If operating in the 2.4 GHz band: If dot11OperatingClassesRequired is false, long = 20 μ s If dot11OperatingClassesRequired is true, long = 20 μ s plus any coverage-class-dependent aAirPropagationTime (see Table 9-95) If dot11OperatingClassesRequired is false, short = 9 μ s If dot11OperatingClassesRequired is true, short = 9 μ s plus any coverage-class-dependent aAirPropagationTime (see Table 9-95) If operating in the 5 GHz or 6 GHz band: If dot11OperatingClassesRequired is false, 9 μ s If dot11OperatingClassesRequired is true, 9 μ s plus any coverage-class-dependent aAirPropagationTime (see Table 9-95)
aSIFSTime	10 μ s if operating in the 2.4 GHz band 16 μ s if operating in the 5 GHz or 6 GHz band
aCCAMidTime	25 μ s
aPPDUMaxTime	5.484 ms
aPSDUMaxLength	6 500 631 octets (see NOTE)
aRxPHYStartDelay	32 μ s for HE SU and HE TB PPDU. 40 μ s for HE ER SU PPDU. $32 + 4 \times N_{HE-SIG-B}$ μ s for HE MU PPDU, where $N_{HE-SIG-B}$ is the number of OFDM symbols in the HE-SIG-B field.
NOTE—This is the maximum length in octets for an HE SU PDU with a bandwidth of 160 MHz or 80+80 MHz using 2×996 RU, HE-MCS 11, 8 spatial streams, 0.8 μ s GI duration, 1x HE-LTF, LDPC coding, 0 μ s duration of the PE field, pre-FEC padding factor value of 4, and limited by 398 possible Data field OFDM symbols in aPPDUMaxTime. This is the maximum PSDU length an HE PHY could support assuming no restrictions in MAC. See 10.3.2 and 9.2.4.7.1 for additional restrictions on the maximum number of octets the MAC could support.	

27.5 Parameters for HE-MCSs

27.5.1 General

The rate-dependent parameters for 26-tone RU, 52-tone RU, 106-tone RU, 242-tone RU, and non-OFDMA 20 MHz; 484-tone RU and non-OFDMA 40 MHz; 996-tone RU and non-OFDMA 80 MHz; 2×996-tone RU and non-OFDMA 160 MHz and 80+80 MHz ($N_{SS} = 1, \dots, 8$) are provided in 27.5.2 through 27.5.8. Support for HE-MCS 8, 9, 10, and 11 is optional in all cases.

HE-MCSs are defined for both SU transmission and for user u of the r^{th} RU of an MU transmission. In the case of HE-MCSs for MU transmissions, the parameters, N_{SS} , R , N_{BPSCS} , N_{CBPS} , and N_{DBPS} are replaced with $N_{SS,u}$, R_u , $N_{BPSCS,u}$, $N_{CBPS,u}$, and $N_{DBPS,u}$ respectively.

N_{DBPS} is an integer and shall be computed as follows $N_{DBPS} = \lfloor N_{CBPS}/R \rfloor$, where R is the coding rate.

27.5.2 HE-MCSs for 26-tone RU

The rate-dependent parameters for the 26-tone RU are provided in Table 27-55 through Table 27-62.

Table 27-55—HE-MCSs for 26-tone RU, $N_{SS} = 1$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	12	12	6	0.4	0.4	0.4
	0		1/2		24	24	12	0.9	0.8	0.8
1	1	QPSK	1/2	2	12	24	12	0.9	0.8	0.8
	0		1/2		24	48	24	1.8	1.7	1.5
2	N/A		3/4		24	48	36	2.6	2.5	2.3
3	1	16-QAM	1/2	4	12	48	24	1.8	1.7	1.5
	0		1/2		24	96	48	3.5	3.3	3.0
4	1		3/4		12	48	36	2.6	2.5	2.3
	0		3/4		24	96	72	5.3	5.0	4.5
5	N/A	64-QAM	2/3	6	24	144	96	7.1	6.7	6.0
6			3/4				108	7.9	7.5	6.8
7			5/6				120	8.8	8.3	7.5
8		256-QAM	3/4	8		192	144	10.6	10.0	9.0
9			5/6				160	11.8	11.1	10.0
10		1024-QAM	3/4	10		240	180	13.2	12.5	11.3
11			5/6				200	14.7	13.9	12.5

Table 27-56—HE-MCSs for 26-tone RU, $N_{SS} = 2$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	12	24	12	0.9	0.8	0.8
	0		1/2		24	48	24	1.8	1.7	1.5
1	1	QPSK	1/2	2	12	48	24	1.8	1.7	1.5
	0		1/2		24	96	48	3.5	3.3	3.0
2	N/A		3/4		24	96	72	5.3	5.0	4.5
3	1	16-QAM	1/2	4	12	96	48	3.5	3.3	3.0
	0		1/2		24	192	96	7.1	6.7	6.0
4	1		3/4		12	96	72	5.3	5.0	4.5
	0		3/4		24	192	144	10.6	10.0	9.0
5	N/A	64-QAM	2/3	6	24	288	192	14.1	13.3	12.0
6			3/4				216	15.9	15.0	13.5
7			5/6				240	17.6	16.7	15.0
8		256-QAM	3/4	8		384	288	21.2	20.0	18.0
9			5/6				320	23.5	22.2	20.0
10		1024-QAM	3/4	10		480	360	26.5	25.0	22.5
11			5/6				400	29.4	27.8	25.0

Table 27-57—HE-MCSs for 26-tone RU, $N_{SS} = 3$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	24	72	36	2.6	2.5	2.3
1		QPSK	1/2	2		144	72	5.3	5.0	4.5
2			3/4				108	7.9	7.5	6.8
3		16-QAM	1/2	4		288	144	10.6	10.0	9.0
4			3/4				216	15.9	15.0	13.5
5		64-QAM	2/3	6		432	288	21.2	20.0	18.0
6			3/4				324	23.8	22.5	20.3
7			5/6				360	26.5	25.0	22.5
8		256-QAM	3/4	8		576	432	31.8	30.0	27.0
9			5/6				480	35.3	33.3	30.0
10		1024-QAM	3/4	10		720	540	39.7	37.5	33.8
11			5/6				600	44.1	41.7	37.5

Table 27-58—HE-MCSs for 26-tone RU, $N_{SS} = 4$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	24	96	48	3.5	3.3	3.0
1		QPSK	1/2	2		192	96	7.1	6.7	6.0
2			3/4				144	10.6	10.0	9.0
3		16-QAM	1/2	4		384	192	14.1	13.3	12.0
4			3/4				288	21.2	20.0	18.0
5		64-QAM	2/3	6		576	384	28.2	26.7	24.0
6			3/4				432	31.8	30.0	27.0
7			5/6				480	35.3	33.3	30.0
8		256-QAM	3/4	8		768	576	42.4	40.0	36.0
9			5/6				640	47.1	44.4	40.0
10		1024-QAM	3/4	10		960	720	52.9	50.0	45.0
11			5/6				800	58.8	55.6	50.0

Table 27-59—HE-MCSs for 26-tone RU, $N_{SS} = 5$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	24	120	60	4.4	4.2	3.8
1		QPSK	1/2	2		240	120	8.8	8.3	7.5
2			3/4				180	13.2	12.5	11.3
3		16-QAM	1/2	4		480	240	17.6	16.7	15.0
4			3/4				360	26.5	25.0	22.5
5		64-QAM	2/3	6		720	480	35.3	33.3	30.0
6			3/4				540	39.7	37.5	33.8
7			5/6				600	44.1	41.7	37.5
8		256-QAM	3/4	8		960	720	52.9	50.0	45.0
9			5/6				800	58.8	55.6	50.0
10		1024-QAM	3/4	10		1 200	900	66.2	62.5	56.3
11			5/6				1 000	73.5	69.4	62.5

Table 27-60—HE-MCSs for 26-tone RU, $N_{SS} = 6$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	24	144	72	5.3	5.0	4.5
1		QPSK	1/2	2		288	144	10.6	10.0	9.0
2			3/4				216	15.9	15.0	13.5
3		16-QAM	1/2	4		576	288	21.2	20.0	18.0
4			3/4				432	31.8	30.0	27.0
5		64-QAM	2/3	6		864	576	42.4	40.0	36.0
6			3/4				648	47.6	45.0	40.5
7			5/6				720	52.9	50.0	45.0
8		256-QAM	3/4	8		1 152	864	63.5	60.0	54.0
9			5/6				960	70.6	66.7	60.0
10		1024-QAM	3/4	10		1 440	1 080	79.4	75.0	67.5
11			5/6				1 200	88.2	83.3	75.0

Table 27-61—HE-MCSs for 26-tone RU, $N_{SS} = 7$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	24	168	84	6.2	5.8	5.3
1		QPSK	1/2	2		336	168	12.4	11.7	10.5
2			3/4				252	18.5	17.5	15.8
3		16-QAM	1/2	4		672	336	24.7	23.3	21.0
4			3/4				504	37.1	35.0	31.5
5		64-QAM	2/3	6		1 008	672	49.4	46.7	42.0
6			3/4				756	55.6	52.5	47.3
7			5/6				840	61.8	58.3	52.5
8		256-QAM	3/4	8		1 344	1 008	74.1	70.0	63.0
9			5/6				1 120	82.4	77.8	70.0
10		1024-QAM	3/4	10		1 680	1 260	92.6	87.5	78.8
11			5/6				1 400	102.9	97.2	87.5

Table 27-62—HE-MCSs for 26-tone RU, $N_{SS} = 8$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	24	192	96	7.1	6.7	6.0
1		QPSK	1/2	2		384	192	14.1	13.3	12.0
2			3/4				288	21.2	20.0	18.0
3		16-QAM	1/2	4		768	384	28.2	26.7	24.0
4			3/4				576	42.4	40.0	36.0
5		64-QAM	2/3	6		1 152	768	56.5	53.3	48.0
6			3/4				864	63.5	60.0	54.0
7			5/6				960	70.6	66.7	60.0
8		256-QAM	3/4	8		1 536	1 152	84.7	80.0	72.0
9			5/6				1 280	94.1	88.9	80.0
10		1024-QAM	3/4	10		1 920	1 440	105.9	100.0	90.0
11			5/6				1 600	117.6	111.1	100.0

27.5.3 HE-MCSs for 52-tone RU

The rate-dependent parameters for the 52-tone RU are provided in Table 27-63 through Table 27-70.

Table 27-63—HE-MCSs for 52-tone RU, $N_{SS} = 1$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	24	24	12	0.9	0.8	0.8
	0		1/2		48	48	24	1.8	1.7	1.5
1	1	QPSK	1/2	2	24	48	24	1.8	1.7	1.5
	0		1/2		48	96	48	3.5	3.3	3.0
2	N/A		3/4		48	96	72	5.3	5.0	4.5
3	1	16-QAM	1/2	4	24	96	48	3.5	3.3	3.0
	0		1/2		48	192	96	7.1	6.7	6.0
4	1		3/4		24	96	72	5.3	5.0	4.5
	0		3/4		48	192	144	10.6	10.0	9.0
5	N/A	64-QAM	2/3	6	48	288	192	14.1	13.3	12.0
6			3/4				216	15.9	15.0	13.5
7			5/6				240	17.6	16.7	15.0
8		256-QAM	3/4	8		384	288	21.2	20.0	18.0
9			5/6				320	23.5	22.2	20.0
10		1024-QAM	3/4	10		480	360	26.5	25.0	22.5
11			5/6				400	29.4	27.8	25.0

Table 27-64—HE-MCSs for 52-tone RU, $N_{SS} = 2$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	24	48	24	1.8	1.7	1.5
	0		1/2		48	96	48	3.5	3.3	3.0
1	1	QPSK	1/2	2	24	96	48	3.5	3.3	3.0
	0		1/2		48	192	96	7.1	6.7	6.0
2	N/A		3/4		48	192	144	10.6	10.0	9.0
3	1	16-QAM	1/2	4	24	192	96	7.1	6.7	6.0
	0		1/2		48	384	192	14.1	13.3	12.0
4	1		3/4		24	192	144	10.6	10.0	9.0
	0		3/4		48	384	288	21.2	20.0	18.0
5	N/A	64-QAM	2/3	6	48	576	384	28.2	26.7	24.0
6			3/4				432	31.8	30.0	27.0
7			5/6				480	35.3	33.3	30.0
8		256-QAM	3/4	8		768	576	42.4	40.0	36.0
9			5/6				640	47.1	44.4	40.0
10		1024-QAM	3/4	10		960	720	52.9	50.0	45.0
11			5/6				800	58.8	55.6	50.0

Table 27-65—HE-MCSs for 52-tone RU, $N_{SS} = 3$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	48	144	72	5.3	5.0	4.5
1		QPSK	1/2	2		288	144	10.6	10.0	9.0
2			3/4				216	15.9	15.0	13.5
3		16-QAM	1/2	4		576	288	21.2	20.0	18.0
4			3/4				432	31.8	30.0	27.0
5		64-QAM	2/3	6		864	576	42.4	40.0	36.0
6			3/4				648	47.6	45.0	40.5
7			5/6				720	52.9	50.0	45.0
8		256-QAM	3/4	8		1 152	864	63.5	60.0	54.0
9			5/6				960	70.6	66.7	60.0
10		1024-QAM	3/4	10		1 440	1 080	79.4	75.0	67.5
11			5/6				1 200	88.2	83.3	75.0

Table 27-66—HE-MCSs for 52-tone RU, $N_{SS} = 4$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	48	192	96	7.1	6.7	6.0
1		QPSK	1/2	2		384	192	14.1	13.3	12.0
2			3/4				288	21.2	20.0	18.0
3		16-QAM	1/2	4		768	384	28.2	26.7	24.0
4			3/4				576	42.4	40.0	36.0
5		64-QAM	2/3	6		1 152	768	56.5	53.3	48.0
6			3/4				864	63.5	60.0	54.0
7			5/6				960	70.6	66.7	60.0
8		256-QAM	3/4	8		1 536	1 152	84.7	80.0	72.0
9			5/6				1 280	94.1	88.9	80.0
10		1024-QAM	3/4	10		1 920	1 440	105.9	100.0	90.0
11			5/6				1 600	117.6	111.1	100.0

Table 27-67—HE-MCSs for 52-tone RU, $N_{SS} = 5$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	48	240	120	8.8	8.3	7.5
1		QPSK	1/2	2		480	240	17.6	16.7	15.0
2			3/4				360	26.5	25.0	22.5
3		16-QAM	1/2	4		960	480	35.3	33.3	30.0
4			3/4				720	52.9	50.0	45.0
5		64-QAM	2/3	6		1 440	960	70.6	66.7	60.0
6			3/4				1080	79.4	75.0	67.5
7			5/6				1 200	88.2	83.3	75.0
8		256-QAM	3/4	8		1 920	1 440	105.9	100.0	90.0
9			5/6				1 600	117.6	111.1	100.0
10		1024-QAM	3/4	10		2 400	1 800	132.4	125.0	112.5
11			5/6				2 000	147.1	138.9	125.0

Table 27-68—HE-MCSs for 52-tone RU, $N_{SS} = 6$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	48	288	144	10.6	10.0	9.0
1		QPSK	1/2	2		576	288	21.2	20.0	18.0
2			3/4				432	31.8	30.0	27.0
3		16-QAM	1/2	4		1 152	576	42.4	40.0	36.0
4			3/4				864	63.5	60.0	54.0
5		64-QAM	2/3	6		1 728	1 152	84.7	80.0	72.0
6			3/4				1 296	95.3	90.0	81.0
7			5/6				1 440	105.9	100.0	90.0
8		256-QAM	3/4	8		2 304	1 728	127.1	120.0	108.0
9			5/6				1 920	141.2	133.3	120.0
10		1024-QAM	3/4	10		2 880	2 160	158.8	150.0	135.0
11			5/6				2 400	176.5	166.7	150.0

Table 27-69—HE-MCSs for 52-tone RU, $N_{SS} = 7$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	48	336	168	12.4	11.7	10.5
1		QPSK	1/2	2		672	336	24.7	23.3	21.0
2			3/4				504	37.1	35.0	31.5
3		16-QAM	1/2	4		1 344	672	49.4	46.7	42.0
4			3/4				1 008	74.1	70.0	63.0
5		64-QAM	2/3	6		2 016	1 344	98.8	93.3	84.0
6			3/4				1 512	111.2	105.0	94.5
7			5/6				1 680	123.5	116.7	105.0
8		256-QAM	3/4	8		2 688	2 016	148.2	140.0	126.0
9			5/6				2 240	164.7	155.6	140.0
10		1024-QAM	3/4	10		3 360	2 520	185.3	175.0	157.5
11			5/6				2 800	205.9	194.4	175.0

Table 27-70—HE-MCSs for 52-tone RU, $N_{SS} = 8$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	48	384	192	14.1	13.3	12.0
1		QPSK	1/2	2		768	384	28.2	26.7	24.0
2			3/4				576	42.4	40.0	36.0
3		16-QAM	1/2	4		1 536	768	56.5	53.3	48.0
4			3/4				1 152	84.7	80.0	72.0
5		64-QAM	2/3	6		2 304	1 536	112.9	106.7	96.0
6			3/4				1 728	127.1	120.0	108.0
7			5/6				1 920	141.2	133.3	120.0
8		256-QAM	3/4	8		3 072	2 304	169.4	160.0	144.0
9			5/6				2 560	188.2	177.8	160.0
10		1024-QAM	3/4	10		3 840	2 880	211.8	200.0	180.0
11			5/6				3 200	235.3	222.2	200.0

27.5.4 HE-MCSs for 106-tone RU

The rate-dependent parameters for the 106-tone RU are provided in Table 27-71 through Table 27-78.

Table 27-71—HE-MCSs for 106-tone RU, $N_{SS} = 1$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)			
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI	
0	1	BPSK	1/2	1	51	51	25	1.8	1.7	1.6	
	0		1/2		102	102	51	3.8	3.5	3.2	
1	1	QPSK	1/2	2	51	102	51	3.8	3.5	3.2	
	0		1/2		102	204	102	7.5	7.1	6.4	
2	N/A		3/4		102	204	153	11.3	10.6	9.6	
3	1	16-QAM	1/2	4	51	204	102	7.5	7.1	6.4	
	0		1/2		102	408	204	15.0	14.2	12.8	
4	1		3/4		51	204	153	11.3	10.6	9.6	
	0		3/4		102	408	306	22.5	21.3	19.1	
5	N/A	64-QAM	2/3	6	102	612	408	30.0	28.3	25.5	
6			3/4				459	33.8	31.9	28.7	
7			5/6				510	37.5	35.4	31.9	
8		256-QAM	3/4			8	816	612	45.0	42.5	38.3
9			5/6					680	50.0	47.2	42.5
10		1024-QAM	3/4			10	1 020	765	56.3	53.1	47.8
11			5/6					850	62.5	59.0	53.1

Table 27-72—HE-MCSs for 106-tone RU, $N_{SS} = 2$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	51	102	51	3.8	3.5	3.2
	0		1/2		102	204	102	7.5	7.1	6.4
1	1	QPSK	1/2	2	51	204	102	7.5	7.1	6.4
	0		1/2		102	408	204	15.0	14.2	12.8
2	N/A		3/4		102	408	306	22.5	21.3	19.1
3	1	16-QAM	1/2	4	51	408	204	15.0	14.2	12.8
	0		1/2		102	816	408	30.0	28.3	25.5
4	1		3/4		51	408	306	22.5	21.3	19.1
	0		3/4		102	816	612	45.0	42.5	38.3
5	N/A	64-QAM	2/3	6	102	1 224	816	60.0	56.7	51.0
6			3/4				918	67.5	63.8	57.4
7			5/6				1 020	75.0	70.8	63.8
8		256-QAM	3/4	8		1 632	1 224	90.0	85.0	76.5
9			5/6				1 360	100.0	94.4	85.0
10		1024-QAM	3/4	10		2 040	1 530	112.5	106.3	95.6
11			5/6				1 700	125.0	118.1	106.3

Table 27-73—HE-MCSs for 106-tone RU, $N_{SS} = 3$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	102	306	153	11.3	10.6	9.6
1		QPSK	1/2	2		612	306	22.5	21.3	19.1
2			3/4				459	33.8	31.9	28.7
3		16-QAM	1/2	4		1 224	612	45.0	42.5	38.3
4			3/4				918	67.5	63.8	57.4
5		64-QAM	2/3	6		1 836	1 224	90.0	85.0	76.5
6			3/4				1 377	101.3	95.6	86.1
7			5/6				1 530	112.5	106.3	95.6
8		256-QAM	3/4	8		2 448	1 836	135.0	127.5	114.8
9			5/6				2 040	150.0	141.7	127.5
10		1024-QAM	3/4	10		3 060	2 295	168.8	159.4	143.4
11			5/6				2 550	187.5	177.1	159.4

Table 27-74—HE-MCSs for 106-tone RU, $N_{SS} = 4$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	102	408	204	15.0	14.2	12.8
1		QPSK	1/2	2		816	408	30.0	28.3	25.5
2			3/4				612	45.0	42.5	38.3
3		16-QAM	1/2	4		1 632	816	60.0	56.7	51.0
4			3/4				1 224	90.0	85.0	76.5
5		64-QAM	2/3	6		2 448	1 632	120.0	113.3	102.0
6			3/4				1 836	135.0	127.5	114.8
7			5/6				2 040	150.0	141.7	127.5
8		256-QAM	3/4	8		3 264	2 448	180.0	170.0	153.0
9			5/6				2 720	200.0	188.9	170.0
10		1024-QAM	3/4	10		4 080	3 060	225.0	212.5	191.3
11			5/6				3 400	250.0	236.1	212.5

Table 27-75—HE-MCSs for 106-tone RU, $N_{SS} = 5$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	102	510	255	18.8	17.7	15.9
1		QPSK	1/2	2		1 020	510	37.5	35.4	31.9
2			3/4				765	56.3	53.1	47.8
3		16-QAM	1/2	4		2 040	1 020	75.0	70.8	63.8
4			3/4				1 530	112.5	106.3	95.6
5		64-QAM	2/3	6		3 060	2 040	150.0	141.7	127.5
6			3/4				2 295	168.8	159.4	143.4
7			5/6				2 550	187.5	177.1	159.4
8		256-QAM	3/4	8		4 080	3 060	225.0	212.5	191.3
9			5/6				3 400	250.0	236.1	212.5
10		1024-QAM	3/4	10		5 100	3 825	281.3	265.6	239.1
11			5/6				4 250	312.5	295.1	265.6

Table 27-76—HE-MCSs for 106-tone RU, $N_{SS} = 6$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	102	612	306	22.5	21.3	19.1
1		QPSK	1/2	2		1 224	612	45.0	42.5	38.3
2			3/4				918	67.5	63.8	57.4
3		16-QAM	1/2	4		2 448	1 224	90.0	85.0	76.5
4			3/4				1 836	135.0	127.5	114.8
5		64-QAM	2/3	6		3 672	2 448	180.0	170.0	153.0
6			3/4				2 754	202.5	191.3	172.1
7			5/6				3 060	225.0	212.5	191.3
8		256-QAM	3/4	8		4 896	3 672	270.0	255.0	229.5
9			5/6				4 080	300.0	283.3	255.0
10		1024-QAM	3/4	10		6 120	4 590	337.5	318.8	286.9
11			5/6				5 100	375.0	354.2	318.8

Table 27-77—HE-MCSs for 106-tone RU, $N_{SS} = 7$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	102	714	357	26.3	24.8	22.3
1		QPSK	1/2	2		1 428	714	52.5	49.6	44.6
2			3/4				1 071	78.8	74.4	66.9
3		16-QAM	1/2	4		2 856	1 428	105.0	99.2	89.3
4			3/4				2 142	157.5	148.8	133.9
5		64-QAM	2/3	6		4 284	2 856	210.0	198.3	178.5
6			3/4				3 213	236.3	223.1	200.8
7			5/6				3 570	262.5	247.9	223.1
8		256-QAM	3/4	8		5 712	4 284	315.0	297.5	267.8
9			5/6				4 760	350.0	330.6	297.5
10		1024-QAM	3/4	10		7 140	5 355	393.8	371.9	334.7
11			5/6				5 950	437.5	413.2	371.9

Table 27-78—HE-MCSs for 106-tone RU, $N_{SS} = 8$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	102	816	408	30.0	28.3	25.5
1		QPSK	1/2	2		1 632	816	60.0	56.7	51.0
2			3/4				1 224	90.0	85.0	76.5
3		16-QAM	1/2	4		3 264	1 632	120.0	113.3	102.0
4			3/4				2 448	180.0	170.0	153.0
5		64-QAM	2/3	6		4 896	3 264	240.0	226.7	204.0
6			3/4				3 672	270.0	255.0	229.5
7			5/6				4 080	300.0	283.3	255.0
8		256-QAM	3/4	8		6 528	4 896	360.0	340.0	306.0
9			5/6				5 440	400.0	377.8	340.0
10		1024-QAM	3/4	10		8 160	6 120	450.0	425.0	382.5
11			5/6				6 800	500.0	472.2	425.0

27.5.5 HE-MCSs for 242-tone RU

The rate-dependent parameters for the 242-tone RU are provided in Table 27-79 through Table 27-86.

Table 27-79—HE-MCSs for 242-tone RU, $N_{SS} = 1$

HE-MCS Index	DCM	Modulation	R	N _{BPCS}	N _{SD}	N _{CBPS}	N _{DBPS}	Data rate (Mb/s)		
								0.8 μs GI	1.6 μs GI	3.2 μs GI
0	1	BPSK	1/2	1	117	117	58	4.3	4.0	3.6
	0		1/2		234	234	117	8.6	8.1	7.3
1	1	QPSK	1/2	2	117	234	117	8.6	8.1	7.3
	0		1/2		234	468	234	17.2	16.3	14.6
2	N/A		3/4		234	468	351	25.8	24.4	21.9
3	1	16-QAM	1/2	4	117	468	234	17.2	16.3	14.6
	0		1/2		234	936	468	34.4	32.5	29.3
4	1		3/4		117	468	351	25.8	24.4	21.9
	0		3/4		234	936	702	51.6	48.8	43.9
5	N/A	64-QAM	2/3	6	234	1 404	936	68.8	65.0	58.5
6			3/4				1 053	77.4	73.1	65.8
7			5/6				1 170	86.0	81.3	73.1
8		256-QAM	3/4	8		1 872	1 404	103.2	97.5	87.8
9			5/6				1 560	114.7	108.3	97.5
10		1024-QAM	3/4	10		2 340	1 755	129.0	121.9	109.7
11			5/6				1 950	143.4	135.4	121.9

Table 27-80—HE-MCSs for 242-tone RU, $N_{SS} = 2$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	117	234	117	8.6	8.1	7.3
	0		1/2		234	468	234	17.2	16.3	14.6
1	1	QPSK	1/2	2	117	468	234	17.2	16.3	14.6
	0		1/2		234	936	468	34.4	32.5	29.3
2	N/A		3/4		234	936	702	51.6	48.8	43.9
3	1	16-QAM	1/2	4	117	936	468	34.4	32.5	29.3
	0		1/2		234	1 872	936	68.8	65.0	58.5
4	1		3/4		117	936	702	51.6	48.8	43.9
	0		3/4		234	1 872	1 404	103.2	97.5	87.8
5	N/A	64-QAM	2/3	6	234	2 808	1 872	137.6	130.0	117.0
6			3/4				2 106	154.9	146.3	131.6
7			5/6				2 340	172.1	162.5	146.3
8		256-QAM	3/4	8		3 744	2 808	206.5	195.0	175.5
9			5/6				3 120	229.4	216.7	195.0
10		1024-QAM	3/4	10		4 680	3 510	258.1	243.8	219.4
11			5/6				3 900	286.8	270.8	243.8

Table 27-81—HE-MCSs for 242-tone RU, $N_{SS} = 3$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	234	702	351	25.8	24.4	21.9
1		QPSK	1/2	2		1 404	702	51.6	48.8	43.9
2			3/4				1 053	77.4	73.1	65.8
3		16-QAM	1/2	4		2 808	1 404	103.2	97.5	87.8
4			3/4				2 106	154.9	146.3	131.6
5		64-QAM	2/3	6		4 212	2 808	206.5	195.0	175.5
6			3/4				3 159	232.3	219.4	197.4
7			5/6				3 510	258.1	243.8	219.4
8		256-QAM	3/4	8		5 616	4 212	309.7	292.5	263.3
9			5/6				4 680	344.1	325.0	292.5
10		1024-QAM	3/4	10		7 020	5 265	387.1	365.6	329.1
11			5/6				5 850	430.1	406.3	365.6

Table 27-82—HE-MCSs for 242-tone RU, $N_{SS} = 4$

HE-MCS Index	DCM	Modulation	R	N _{BPSCS}	N _{SD}	N _{CBPS}	N _{DBPS}	Data rate (Mb/s)		
								0.8 μs GI	1.6 μs GI	3.2 μs GI
0	N/A	BPSK	1/2	1	234	936	468	34.4	32.5	29.3
1		QPSK	1/2	2		1 872	936	68.8	65.0	58.5
2			3/4				1 404	103.2	97.5	87.8
3		16-QAM	1/2	4		3 744	1 872	137.6	130.0	117.0
4			3/4				2 808	206.5	195.0	175.5
5		64-QAM	2/3	6		5 616	3 744	275.3	260.0	234.0
6			3/4				4 212	309.7	292.5	263.3
7			5/6				4 680	344.1	325.0	292.5
8		256-QAM	3/4	8		7 488	5 616	412.9	390.0	351.0
9			5/6				6 240	458.8	433.3	390.0
10		1024-QAM	3/4	10		9 360	7 020	516.2	487.5	438.8
11			5/6				7 800	573.5	541.7	487.5

Table 27-83—HE-MCSs for 242-tone RU, $N_{SS} = 5$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	234	1 170	585	43.0	40.6	36.6
1		QPSK	1/2	2		2 340	1 170	86.0	81.3	73.1
2			3/4				1 755	129.0	121.9	109.7
3		16-QAM	1/2	4		4 680	2 340	172.1	162.5	146.3
4			3/4				3 510	258.1	243.8	219.4
5		64-QAM	2/3	6		7 020	4 680	344.1	325.0	292.5
6			3/4				5 265	387.1	365.6	329.1
7			5/6				5 850	430.1	406.3	365.6
8		256-QAM	3/4	8		9 360	7 020	516.2	487.5	438.8
9			5/6				7 800	573.5	541.7	487.5
10		1024-QAM	3/4	10		11 700	8 775	645.2	609.4	548.4
11			5/6				9 750	716.9	677.1	609.4

Table 27-84—HE-MCSs for 242-tone RU, $N_{SS} = 6$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	234	1 404	702	51.6	48.8	43.9
1		QPSK	1/2	2		2 808	1 404	103.2	97.5	87.8
2			3/4				2 106	154.9	146.3	131.6
3		16-QAM	1/2	4		5 616	2 808	206.5	195.0	175.5
4			3/4				4 212	309.7	292.5	263.3
5		64-QAM	2/3	6		8 424	5 616	412.9	390.0	351.0
6			3/4				6 318	464.6	438.8	394.9
7			5/6				7 020	516.2	487.5	438.8
8		256-QAM	3/4	8		11 232	8 424	619.4	585.0	526.5
9			5/6				9 360	688.2	650.0	585.0
10		1024-QAM	3/4	10		14 040	10 530	774.3	731.3	658.1
11			5/6				11 700	860.3	812.5	731.3

Table 27-85—HE-MCSs for 242-tone RU, $N_{SS} = 7$

HE-MCS Index	DCM	Modulation	R	N _{BPSCS}	N _{SD}	N _{CBPS}	N _{DBPS}	Data rate (Mb/s)		
								0.8 μs GI	1.6 μs GI	3.2 μs GI
0	N/A	BPSK	1/2	1	234	1 638	819	60.2	56.9	51.2
1		QPSK	1/2	2		3 276	1 638	120.4	113.8	102.4
2			3/4				2 457	180.7	170.6	153.6
3		16-QAM	1/2	4		6 552	3 276	240.9	227.5	204.8
4			3/4				4 914	361.3	341.3	307.1
5		64-QAM	2/3	6		9 828	6 552	481.8	455.0	409.5
6			3/4				7 371	542.0	511.9	460.7
7			5/6				8 190	602.2	568.8	511.9
8		256-QAM	3/4	8		13 104	9 828	722.6	682.5	614.3
9			5/6				10 920	802.9	758.3	682.5
10		1024-QAM	3/4	10		16 380	12 285	903.3	853.1	767.8
11			5/6				13 650	1 003.7	947.9	853.1

Table 27-86—HE-MCSs for 242-tone RU, $N_{SS} = 8$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	234	1 872	936	68.8	65.0	58.5
1		QPSK	1/2	2		3 744	1 872	137.6	130.0	117.0
2			3/4				2 808	206.5	195.0	175.5
3		16-QAM	1/2	4		7 488	3 744	275.3	260.0	234.0
4			3/4				5 616	412.9	390.0	351.0
5		64-QAM	2/3	6		11 232	7 488	550.6	520.0	468.0
6			3/4				8 424	619.4	585.0	526.5
7			5/6				9 360	688.2	650.0	585.0
8		256-QAM	3/4	8		14 976	11 232	825.9	780.0	702.0
9			5/6				12 480	917.6	866.7	780.0
10		1024-QAM	3/4	10		18 720	14 040	1 032.4	975.0	877.5
11			5/6				15 600	1 147.1	1 083.3	975.0

27.5.6 HE-MCSs for 484-tone RU

The rate-dependent parameters for the 484-tone RU are provided in Table 27-87 through Table 27-94.

Table 27-87—HE-MCSs for 484-tone RU, $N_{SS} = 1$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	234	234	117	8.6	8.1	7.3
	0		1/2		468	468	234	17.2	16.3	14.6
1	1	QPSK	1/2	2	234	468	234	17.2	16.3	14.6
	0		1/2		468	936	468	34.4	32.5	29.3
2	N/A		3/4		468	936	702	51.6	48.8	43.9
3	1	16-QAM	1/2	4	234	936	468	34.4	32.5	29.3
	0		1/2		468	1 872	936	68.8	65.0	58.5
4	1		3/4		234	936	702	51.6	48.8	43.9
	0		3/4		468	1 872	1 404	103.2	97.5	87.8
5	N/A	64-QAM	2/3	6	468	2 808	1 872	137.6	130.0	117.0
6			3/4				2 106	154.9	146.3	131.6
7			5/6				2 340	172.1	162.5	146.3
8		256-QAM	3/4	8		3 744	2 808	206.5	195.0	175.5
9			5/6				3 120	229.4	216.7	195.0
10		1024-QAM	3/4	10		4 680	3 510	258.1	243.8	219.4
11			5/6				3 900	286.8	270.8	243.8

Table 27-88—HE-MCSs for 484-tone RU, $N_{SS} = 2$

HE-MCS Index	DCM	Modulation	R	N _{BPSCS}	N _{SD}	N _{CBPS}	N _{DBPS}	Data rate (Mb/s)		
								0.8 μs GI	1.6 μs GI	3.2 μs GI
0	1	BPSK	1/2	1	234	468	234	17.2	16.3	14.6
	0		1/2		468	936	468	34.4	32.5	29.3
1	1	QPSK	1/2	2	234	936	468	34.4	32.5	29.3
	0		1/2		468	1 872	936	68.8	65.0	58.5
2	N/A		3/4		468	1 872	1 404	103.2	97.5	87.8
3	1	16-QAM	1/2	4	234	1 872	936	68.8	65.0	58.5
	0		1/2		468	3 744	1 872	137.6	130.0	117.0
4	1		3/4		234	1 872	1 404	103.2	97.5	87.8
	0		3/4		468	3 744	2 808	206.5	195.0	175.5
5	N/A	64-QAM	2/3	6	468	5 616	3 744	275.3	260.0	234.0
6			3/4				4 212	309.7	292.5	263.3
7			5/6				4 680	344.1	325.0	292.5
8		256-QAM	3/4	8		7 488	5 616	412.9	390.0	351.0
9			5/6				6 240	458.8	433.3	390.0
10		1024-QAM	3/4	10		9 360	7 020	516.2	487.5	438.8
11			5/6				7 800	573.5	541.7	487.5

Table 27-89—HE-MCSs for 484-tone RU, $N_{SS} = 3$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	468	1 404	702	51.6	48.8	43.9
1		QPSK	1/2	2		2 808	1 404	103.2	97.5	87.8
2			3/4				2 106	154.9	146.3	131.6
3		16-QAM	1/2	4		5 616	2 808	206.5	195.0	175.5
4			3/4				4 212	309.7	292.5	263.3
5		64-QAM	2/3	6		8 424	5 616	412.9	390.0	351.0
6			3/4				6 318	464.6	438.8	394.9
7			5/6				7 020	516.2	487.5	438.8
8		256-QAM	3/4	8		11 232	8 424	619.4	585.0	526.5
9			5/6				9 360	688.2	650.0	585.0
10		1024-QAM	3/4	10		14 040	10 530	774.3	731.3	658.1
11			5/6				11 700	860.3	812.5	731.3

Table 27-90—HE-MCSs for 484-tone RU, $N_{SS} = 4$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	468	1 872	936	68.8	65.0	58.5
1		QPSK	1/2	2		3 744	1 872	137.6	130.0	117.0
2			3/4				2 808	206.5	195.0	175.5
3		16-QAM	1/2	4		7 488	3 744	275.3	260.0	234.0
4			3/4				5 616	412.9	390.0	351.0
5		64-QAM	2/3	6		11 232	7 488	550.6	520.0	468.0
6			3/4				8 424	619.4	585.0	526.5
7			5/6				9 360	688.2	650.0	585.0
8		256-QAM	3/4	8		14 976	11 232	825.9	780.0	702.0
9			5/6				12 480	917.6	866.7	780.0
10		1024-QAM	3/4	10		18 720	14 040	1 032.4	975.0	877.5
11			5/6				15 600	1 147.1	1 083.3	975.0

Table 27-91—HE-MCSs for 484-tone RU, $N_{SS} = 5$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	468	2 340	1 170	86.0	81.3	73.1
1		QPSK	1/2	2		4 680	2 340	172.1	162.5	146.3
2			3/4				3 510	258.1	243.8	219.4
3		16-QAM	1/2	4		9 360	4 680	344.1	325.0	292.5
4			3/4				7 020	516.2	487.5	438.8
5		64-QAM	2/3	6		14 040	9 360	688.2	650.0	585.0
6			3/4				10 530	774.3	731.3	658.1
7			5/6				11 700	860.3	812.5	731.3
8		256-QAM	3/4	8		18 720	14 040	1 032.4	975.0	877.5
9			5/6				15 600	1 147.1	1 083.3	975.0
10		1024-QAM	3/4	10		23 400	17 550	1 290.4	1 218.8	1 096.9
11			5/6				19 500	1 433.8	1 354.2	1 218.8

Table 27-92—HE-MCSs for 484-tone RU, $N_{SS} = 6$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	468	2 808	1404	103.2	97.5	87.8
1		QPSK	1/2	2		5 616	2808	206.5	195.0	175.5
2			3/4				4212	309.7	292.5	263.3
3		16-QAM	1/2	4		11 232	5616	412.9	390.0	351.0
4			3/4				8424	619.4	585.0	526.5
5		64-QAM	2/3	6		16 848	11 232	825.9	780.0	702.0
6			3/4				12 636	929.1	877.5	789.8
7			5/6				14 040	1 032.4	975.0	877.5
8		256-QAM	3/4	8		22 464	16 848	1 238.8	1 170.0	1 053.0
9			5/6				18 720	1 376.5	1 300.0	1 170.0
10		1024-QAM	3/4	10		28 080	21 060	1 548.5	1 462.5	1 316.3
11			5/6				23 400	1 720.6	1 625.0	1 462.5

Table 27-93—HE-MCSs for 484-tone RU, $N_{SS} = 7$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	468	3 276	1 638	120.4	113.8	102.4
1		QPSK	1/2	2		6 552	3 276	240.9	227.5	204.8
2			3/4				4 914	361.3	341.3	307.1
3		16-QAM	1/2	4		13 104	6 552	481.8	455.0	409.5
4			3/4				9 828	722.6	682.5	614.3
5		64-QAM	2/3	6		19 656	13 104	963.5	910.0	819.0
6			3/4				14 742	1 084.0	1 023.8	921.4
7			5/6				16 380	1 204.4	1 137.5	1 023.8
8		256-QAM	3/4	8		26 208	19 656	1 445.3	1 365.0	1 228.5
9			5/6				21 840	1 605.9	1 516.7	1 365.0
10		1024-QAM	3/4	10		32 760	24 570	1 806.6	1 706.3	1 535.6
11			5/6				27 300	2 007.4	1 895.8	1 706.3

Table 27-94—HE-MCSs for 484-tone RU, $N_{SS} = 8$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	468	3 744	1 872	137.6	130.0	117.0
1		QPSK	1/2	2		7 488	3 744	275.3	260.0	234.0
2			3/4				5 616	412.9	390.0	351.0
3		16-QAM	1/2	4		14 976	7 488	550.6	520.0	468.0
4			3/4				11 232	825.9	780.0	702.0
5		64-QAM	2/3	6		22 464	14 976	1 101.2	1 040.0	936.0
6			3/4				16 848	1 238.8	1 170.0	1 053.0
7			5/6				18 720	1 376.5	1 300.0	1 170.0
8		256-QAM	3/4	8		29 952	22 464	1 651.8	1 560.0	1 404.0
9			5/6				24 960	1 835.3	1 733.3	1 560.0
10		1024-QAM	3/4	10		37 440	28 080	2 064.7	1 950.0	1 755.0
11			5/6				31 200	2 294.1	2 166.7	1 950.0

27.5.7 HE-MCSs for 996-tone RU

The rate-dependent parameters for the 996-tone RU are provided in Table 27-95 through Table 27-102.

Table 27-95—HE-MCSs for 996-tone RU, $N_{SS} = 1$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	490	490	245	18.0	17.0	15.3
	0		1/2		980	980	490	36.0	34.0	30.6
1	1	QPSK	1/2	2	490	980	490	36.0	34.0	30.6
	0		1/2		980	1960	980	72.1	68.1	61.3
2	N/A		3/4		980	1960	1 470	108.1	102.1	91.9
3	1	16-QAM	1/2	4	490	1960	980	72.1	68.1	61.3
	0		1/2		980	3920	1 960	144.1	136.1	122.5
4	1		3/4		490	1960	1 470	108.1	102.1	91.9
	0		3/4		980	3920	2 940	216.2	204.2	183.8
5	N/A	64-QAM	2/3	6	980	5 880	3 920	288.2	272.2	245.0
6			3/4				4 410	324.3	306.3	275.6
7			5/6				4 900	360.3	340.3	306.3
8		256-QAM	3/4	8		7 840	5 880	432.4	408.3	367.5
9			5/6				6 533	480.4	453.7	408.3
10		1024-QAM	3/4	10		9 800	7 350	540.4	510.4	459.4
11			5/6				8 166	600.4	567.1	510.4

Table 27-96—HE-MCSs for 996-tone RU, $N_{SS} = 2$

HE-MCS Index	DCM	Modulation	R	N _{BPSCS}	N _{SD}	N _{CBPS}	N _{DBPS}	Data rate (Mb/s)		
								0.8 μs GI	1.6 μs GI	3.2 μs GI
0	1	BPSK	1/2	1	490	980	490	36.0	34.0	30.6
	0		1/2		980	1 960	980	72.1	68.1	61.3
1	1	QPSK	1/2	2	490	1 960	980	72.1	68.1	61.3
	0		1/2		980	3 920	1 960	144.1	136.1	122.5
2	N/A		3/4		980	3 920	2 940	216.2	204.2	183.8
3	1	16-QAM	1/2	4	490	3 920	1 960	144.1	136.1	122.5
	0		1/2		980	7 840	3 920	288.2	272.2	245.0
4	1		3/4		490	3 920	2 940	216.2	204.2	183.8
	0		3/4		980	7 840	5 880	432.4	408.3	367.5
5	N/A	64-QAM	2/3	6	980	11 760	7 840	576.5	544.4	490.0
6			3/4				8 820	648.5	612.5	551.3
7			5/6				9 800	720.6	680.6	612.5
8		256-QAM	3/4	8		15 680	11 760	864.7	816.7	735.0
9			5/6				13 066	960.7	907.4	816.6
10		1024-QAM	3/4	10		19 600	14 700	1 080.9	1 020.8	918.8
11			5/6				16 333	1 201.0	1 134.2	1 020.8

Table 27-97—HE-MCSs for 996-tone RU, $N_{SS} = 3$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	980	2 940	1 470	108.1	102.1	91.9
1		QPSK	1/2	2		5 880	2 940	216.2	204.2	183.8
2			3/4				4 410	324.3	306.3	275.6
3		16-QAM	1/2	4		11 760	5 880	432.4	408.3	367.5
4			3/4				8 820	648.5	612.5	551.3
5		64-QAM	2/3	6		17 640	11 760	864.7	816.7	735.0
6			3/4				13 230	972.8	918.8	826.9
7			5/6				14 700	1 080.9	1 020.8	918.8
8		256-QAM	3/4	8		23 520	17 640	1 297.1	1 225.0	1 102.5
9			5/6				19 600	1 441.2	1 361.1	1 225.0
10		1024-QAM	3/4	10		29 400	22 050	1 621.3	1 531.3	1 378.1
11			5/6				24 500	1 801.5	1 701.4	1 531.3

Table 27-98—HE-MCSs for 996-tone RU, $N_{SS} = 4$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	980	3 920	1 960	144.1	136.1	122.5
1		QPSK	1/2	2		7 840	3 920	288.2	272.2	245.0
2			3/4				5 880	432.4	408.3	367.5
3		16-QAM	1/2	4		15 680	7 840	576.5	544.4	490.0
4			3/4				11 760	864.7	816.7	735.0
5		64-QAM	2/3	6		23 520	15 680	1 152.9	1 088.9	980.0
6			3/4				17 640	1 297.1	1 225.0	1 102.5
7			5/6				19 600	1 441.2	1 361.1	1 225.0
8		256-QAM	3/4	8		31 360	23 520	1 729.4	1 633.3	1 470.0
9			5/6				26 133	1 921.5	1 814.8	1 633.3
10		1024-QAM	3/4	10		39 200	29 400	2 161.8	2 041.7	1 837.5
11			5/6				32 666	2 401.9	2 268.5	2 041.6

Table 27-99—HE-MCSs for 996-tone RU, $N_{SS} = 5$

HE-MCS Index	DCM	Modulation	R	N _{BPSCS}	N _{SD}	N _{CBPS}	N _{DBPS}	Data rate (Mb/s)		
								0.8 μs GI	1.6 μs GI	3.2 μs GI
0	N/A	BPSK	1/2	1	980	4 900	2 450	180.1	170.1	153.1
1		QPSK	1/2	2		9 800	4 900	360.3	340.3	306.3
2			3/4				7 350	540.4	510.4	459.4
3		16-QAM	1/2	4		19 600	9 800	720.6	680.6	612.5
4			3/4				14 700	1 080.9	1 020.8	918.8
5		64-QAM	2/3	6		29 400	19 600	1 441.2	1 361.1	1 225.0
6			3/4				22 050	1 621.3	1 531.3	1 378.1
7			5/6				24 500	1 801.5	1 701.4	1 531.3
8		256-QAM	3/4	8		39 200	29 400	2 161.8	2 041.7	1 837.5
9			5/6				32 666	2 401.9	2 268.5	2 041.6
10		1024-QAM	3/4	10		49 000	36 750	2 702.2	2 552.1	2 296.9
11			5/6				40 833	3 002.4	2 835.6	2 552.1

Table 27-100—HE-MCSs for 996-tone RU, $N_{SS} = 6$

HE-MCS Index	DCM	Modulation	R	N _{BPSCS}	N _{SD}	N _{CBPS}	N _{DBPS}	Data rate (Mb/s)		
								0.8 μs GI	1.6 μs GI	3.2 μs GI
0	N/A	BPSK	1/2	1	980	5 880	2 940	216.2	204.2	183.8
1		QPSK	1/2	2		11 760	5 880	432.4	408.3	367.5
2			3/4				8 820	648.5	612.5	551.3
3		16-QAM	1/2	4		23 520	11 760	864.7	816.7	735.0
4			3/4				17 640	1 297.1	1 225.0	1 102.5
5		64-QAM	2/3	6		35 280	23 520	1 729.4	1 633.3	1 470.0
6			3/4				26 460	1 945.6	1 837.5	1 653.8
7			5/6				29 400	2 161.8	2 041.7	1 837.5
8		256-QAM	3/4	8		47 040	35 280	2 594.1	2 450.0	2 205.0
9			5/6				39 200	2 882.4	2 722.2	2 450.0
10		1024-QAM	3/4	10		58 800	44 100	3 242.6	3 062.5	2 756.3
11			5/6				49 000	3 602.9	3 402.8	3 062.5

Table 27-101—HE-MCSs for 996-tone RU, $N_{SS} = 7$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	980	6 860	3 430	252.2	238.2	214.4
1		QPSK	1/2	2		13 720	6 860	504.4	476.4	428.8
2			3/4				10 290	756.6	714.6	643.1
3		16-QAM	1/2	4		27 440	13 720	1 008.8	952.8	857.5
4			3/4				20 580	1 513.2	1 429.2	1 286.3
5		64-QAM	2/3	6		41 160	27 440	2 017.6	1 905.6	1 715.0
6			3/4				30 870	2 269.9	2 143.8	1 929.4
7			5/6				34 300	2 522.1	2 381.9	2 143.8
8		256-QAM	3/4	8		54 880	41 160	3 026.5	2 858.3	2 572.5
9			5/6				45 733	3 362.7	3 175.9	2 858.3
10		1024-QAM	3/4	10		68 600	51 450	3 783.1	3 572.9	3 215.6
11			5/6				57 166	4 203.4	3 969.9	3 572.9

Table 27-102—HE-MCSs for 996-tone RU, $N_{SS} = 8$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	980	7 840	3 920	288.2	272.2	245.0
1		QPSK	1/2	2		15 680	7 840	576.5	544.4	490.0
2			3/4				11 760	864.7	816.7	735.0
3		16-QAM	1/2	4		31 360	15 680	1 152.9	1 088.9	980.0
4			3/4				23 520	1 729.4	1 633.3	1 470.0
5		64-QAM	2/3	6		47 040	31 360	2 305.9	2 177.8	1 960.0
6			3/4				35 280	2 594.1	2 450.0	2 205.0
7			5/6				39 200	2 882.4	2 722.2	2 450.0
8		256-QAM	3/4	8		62 720	47 040	3 458.8	3 266.7	2 940.0
9			5/6				52 266	3 843.1	3 629.6	3 266.6
10		1024-QAM	3/4	10		78 400	58 800	4 323.5	4 083.3	3 675.0
11			5/6				65 333	4 803.9	4 537.0	4 083.3

27.5.8 HE-MCSs for 2×996-tone RU

The rate-dependent parameters for 2×996-tone RU are provided in Table 27-103 through Table 27-110.

Table 27-103—HE-MCSs for 2×996-tone RU, $N_{SS} = 1$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	980	980	490	36.0	34.0	30.6
	0		1/2		1 960	1 960	980	72.1	68.1	61.3
1	1	QPSK	1/2	2	980	1 960	980	72.1	68.1	61.3
	0		1/2		1 960	3 920	1 960	144.1	136.1	122.5
2	N/A		3/4		1 960	3 920	2 940	216.2	204.2	183.8
3	1	16-QAM	1/2	4	980	3 920	1 960	144.1	136.1	122.5
	0		1/2		1 960	7 840	3 920	288.2	272.2	245.0
4	1		3/4		980	3 920	2 940	216.2	204.2	183.8
	0		3/4		1 960	7 840	5 880	432.4	408.3	367.5
5	N/A	64-QAM	2/3	6	1 960	11 760	7 840	576.5	544.4	490.0
6			3/4				8 820	648.5	612.5	551.3
7			5/6				9 800	720.6	680.6	612.5
8		256-QAM	3/4	8		15 680	11 760	864.7	816.7	735.0
9			5/6				13 066	960.7	907.4	816.6
10		1024-QAM	3/4	10		19 600	14 700	1 080.9	1 020.8	918.8
11			5/6				16 333	1 201.0	1 134.2	1 020.8

Table 27-104—HE-MCSs for 2×996-tone RU, $N_{SS} = 2$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	1	BPSK	1/2	1	980	1 960	980	72.1	68.1	61.3
	0		1/2		1 960	3 920	1 960	144.1	136.1	122.5
1	1	QPSK	1/2	2	980	3 920	1 960	144.1	136.1	122.5
	0		1/2		1 960	7 840	3 920	288.2	272.2	245.0
2	N/A		3/4		1 960	7 840	5 880	432.4	408.3	367.5
3	1	16-QAM	1/2	4	980	7 840	3 920	288.2	272.2	245.0
	0		1/2		1 960	15 680	7 840	576.5	544.4	490.0
4	1		3/4		980	7 840	5 880	432.4	408.3	367.5
	0		3/4		1 960	15 680	11 760	864.7	816.7	735.0
5	N/A	64-QAM	2/3	6	1 960	23 520	15 680	1 152.9	1 088.9	980.0
6			3/4				17 640	1 297.1	1 225.0	1 102.5
7			5/6				19 600	1 441.2	1 361.1	1 225.0
8		256-QAM	3/4	8		31 360	23 520	1 729.4	1 633.3	1 470.0
9			5/6				26 133	1 921.5	1 814.8	1 633.3
10		1024-QAM	3/4	10		39 200	29 400	2 161.8	2 041.7	1 837.5
11			5/6				32 666	2 401.9	2 268.5	2 041.6

Table 27-105—HE-MCSs for 2×996-tone RU, $N_{SS} = 3$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	1 960	5 880	2 940	216.2	204.2	183.8
1		QPSK	1/2	2		11 760	5 880	432.4	408.3	367.5
2			3/4				8 820	648.5	612.5	551.3
3		16-QAM	1/2	4		23 520	11 760	864.7	816.7	735.0
4			3/4				17 640	1 297.1	1 225.0	1 102.5
5		64-QAM	2/3	6		35 280	23 520	1 729.4	1 633.3	1 470.0
6			3/4				26 460	1 945.6	1 837.5	1 653.8
7			5/6				29 400	2 161.8	2 041.7	1 837.5
8		256-QAM	3/4	8		47 040	35 280	2 594.1	2 450.0	2 205.0
9			5/6				39 200	2 882.4	2 722.2	2 450.0
10		1024-QAM	3/4	10		58 800	44 100	3 242.6	3 062.5	2 756.3
11			5/6				49 000	3 602.9	3 402.8	3 062.5

Table 27-106—HE-MCSs for 2×996-tone RU, $N_{SS} = 4$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	1 960	7 840	3 920	288.2	272.2	245.0
1		QPSK	1/2	2		15 680	7 840	576.5	544.4	490.0
2			3/4				11 760	864.7	816.7	735.0
3		16-QAM	1/2	4		31 360	15 680	1 152.9	1 088.9	980.0
4			3/4				23 520	1 729.4	1 633.3	1 470.0
5		64-QAM	2/3	6		47 040	31 360	2 305.9	2 177.8	1 960.0
6			3/4				35 280	2 594.1	2 450.0	2 205.0
7			5/6				39 200	2 882.4	2 722.2	2 450.0
8		256-QAM	3/4	8		62 720	47 040	3 458.8	3 266.7	2 940.0
9			5/6				52 266	3 843.1	3 629.6	3 266.6
10		1024-QAM	3/4	10		78 400	58 800	4 323.5	4 083.3	3 675.0
11			5/6				65 333	4 803.9	4 537.0	4 083.3

Table 27-107—HE-MCSs for 2×996-tone RU, $N_{SS} = 5$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	1 960	9 800	4 900	360.3	340.3	306.3
1		QPSK	1/2	2		19 600	9 800	720.6	680.6	612.5
2			3/4				14 700	1 080.9	1 020.8	918.8
3		16-QAM	1/2	4		39 200	19 600	1 441.2	1 361.1	1 225.0
4			3/4				29 400	2 161.8	2 041.7	1 837.5
5		64-QAM	2/3	6		58 800	39 200	2 882.4	2 722.2	2 450.0
6			3/4				44 100	3 242.6	3 062.5	2 756.3
7			5/6				49 000	3 602.9	3 402.8	3 062.5
8		256-QAM	3/4	8		78 400	58 800	4 323.5	4 083.3	3 675.0
9			5/6				65 333	4 803.9	4 537.0	4 083.3
10		1024-QAM	3/4	10		98 000	73 500	5 404.4	5 104.2	4 593.8
11			5/6				81 666	6 004.9	5 671.3	5 104.1

Table 27-108—HE-MCSs for 2×996-tone RU, $N_{SS} = 6$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	1 960	11 760	5 880	432.4	408.3	367.5
1		QPSK	1/2	2		23 520	11 760	864.7	816.7	735.0
2			3/4				17 640	1 297.1	1 225.0	1 102.5
3		16-QAM	1/2	4		47 040	23 520	1 729.4	1 633.3	1 470.0
4			3/4				35 280	2 594.1	2 450.0	2 205.0
5		64-QAM	2/3	6		70 560	47 040	3 458.8	3 266.7	2 940.0
6			3/4				52 920	3 891.2	3 675.0	3 307.5
7			5/6				58 800	4 323.5	4 083.3	3 675.0
8		256-QAM	3/4	8		94 080	70 560	5 188.2	4 900.0	4 410.0
9			5/6				78 400	5 764.7	5 444.4	4 900.0
10		1024-QAM	3/4	10		11 7600	88 200	6 485.3	6 125.0	5 512.5
11			5/6				98 000	7 205.9	6 805.6	6 125.0

Table 27-109—HE-MCSs for 2×996-tone RU, $N_{SS} = 7$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	1 960	13 720	6 860	504.4	476.4	428.8
1		QPSK	1/2	2		27 440	13 720	1 008.8	952.8	857.5
2			3/4				20 580	1 513.2	1 429.2	1 286.3
3		16-QAM	1/2	4		54 880	27 440	2 017.6	1 905.6	1 715.0
4			3/4				41 160	3 026.5	2 858.3	2 572.5
5		64-QAM	2/3	6		82 320	54 880	4 035.3	3 811.1	3 430.0
6			3/4				61 740	4 539.7	4 287.5	3 858.8
7			5/6				68 600	5 044.1	4 763.9	4 287.5
8		256-QAM	3/4	8		109 760	82 320	6 052.9	5 716.7	5 145.0
9			5/6				91 466	6 725.4	6 351.8	5 716.6
10		1024-QAM	3/4	10		137 200	102 900	7 566.2	7 145.8	6 431.3
11			5/6				114 333	8 406.8	7 939.8	7 145.8

Table 27-110—HE-MCSs for 2×996-tone RU, $N_{SS} = 8$

HE-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	Data rate (Mb/s)		
								0.8 μ s GI	1.6 μ s GI	3.2 μ s GI
0	N/A	BPSK	1/2	1	1 960	15 680	7 840	576.5	544.4	490.0
1		QPSK	1/2	2		31 360	15 680	1 152.9	1 088.9	980.0
2			3/4				23 520	1 729.4	1 633.3	1 470.0
3		16-QAM	1/2	4		62 720	31 360	2 305.9	2 177.8	1 960.0
4			3/4				47 040	3 458.8	3 266.7	2 940.0
5		64-QAM	2/3	6		94 080	62 720	4 611.8	4 355.6	3 920.0
6			3/4				70 560	5 188.2	4 900.0	4 410.0
7			5/6				78 400	5 764.7	5 444.4	4 900.0
8		256-QAM	3/4	8		125 440	94 080	6 917.6	6 533.3	5 880.0
9			5/6				104 533	7 686.3	7 259.2	6 533.3
10		1024-QAM	3/4	10		156 800	117 600	8 647.1	8 166.7	7 350.0
11			5/6				130 666	9 607.8	9 074.0	8 166.6

27.6 Parameters for HE-SIG-B-MCSs

The HE-SIG-B-MCSs, defined in Table 27-111, are used for the HE-SIG-B field transmission in the HE MU PPDU.

Table 27-111—HE-SIG-B-MCSs

HE-SIG-B-MCS Index	DCM	Modulation	R	N_{BPSCS}	N_{SD}	N_{CBPS}	N_{DBPS}	HE-SIG-B rate (Mb/s)
0	1	BPSK	1/2	1	26	26	13	3.3
	0		1/2		52	52	26	6.6
1	1	QPSK	1/2	2	26	52	26	6.6
	0		1/2		52	104	52	13.2
2	N/A		3/4		52	104	78	19.5
3	1	16-QAM	1/2	4	26	104	52	13.2
	0		1/2		52	208	104	26.0
4	1		3/4		26	104	78	19.5
	0		3/4		52	208	156	39.0
5	N/A	64-QAM	2/3	6	52	312	208	52.0
NOTE—The parameters N_{SD} , N_{CBPS} and N_{DBPS} are used for the HE-SIG-B field transmission in each 20 MHz subchannel.								

Annex B

(normative)

Protocol Implementation Conformance Statement (PICS) proforma

B.4 PICS proforma—IEEE Std 802.11-2020

Change the table in B.4.3 as follows (not all rows are shown):

B.4.3 IUT configuration

Item	IUT configuration	References	Status	Support
	What is the configuration of the IUT?			
* CFOFDM	Orthogonal frequency division multiplexing (OFDM) PHY	—	O.2 CFHT5G:M CFTVHT:M CFS1G:M <u>CFHE:M</u>	Yes ☐ No ☐
...				
*CFHT2G4	HT operation in the 2.4 GHz band	Clause 19	CFHT:O.6 <u>CFHE:M</u>	Yes ☐ No ☐ N/A ☐
...				
*CFVHT	Very High Throughput (VHT) features	9.4.2.157	O.2 <u>CFHE:M</u>	Yes ☐ No ☐
...				
*CFESM	Extended spectrum management	10.23.2	O CFVHT OR CFTVHT:M	Yes ☐ No ☐
<u>*CFHE</u>	<u>High-efficiency (HE) operation</u>	<u>9.4.2.248</u>	<u>O</u> <u>CFHE20:M</u> <u>CFHE80:M</u>	Yes ☐ No ☐
<u>*CFHE2G4</u>	<u>HE operation in the 2.4 GHz band</u>	<u>Clause 27</u>	<u>O.8</u>	<u>Yes ☐ No ☐</u>
<u>*CFHE5G</u>	<u>HE operation in the 5 GHz band</u>	<u>Clause 27</u>	<u>O.8</u>	<u>Yes ☐ No ☐</u>
<u>*CFHE6G</u>	<u>HE operation in the 6 GHz band</u>	<u>Clause 27</u>	<u>O.8</u>	<u>Yes ☐ No ☐</u>
<u>*CFHE20</u>	<u>HE operation as a 20 MHz-only non-AP HE STA</u>	<u>Clause 27</u>	<u>CFIndepSTA</u> <u>AND</u> <u>CFHE:O.8</u>	<u>Yes ☐ No ☐</u>
<u>*CFHE80</u>	<u>HE operation with capability of 80 MHz or wider channel width</u>	<u>Clause 27</u>	<u>CFAP AND</u> <u>CFHE</u> <u>CFIndepSTA</u> <u>AND CFHE</u>	<u>Yes ☐ No ☐</u>

B.4.3 IUT configuration *(continued)*

Item	IUT configuration	References	Status	Support
CF2G4n6G	<u>An AP that operates in the 2.4 GHz band and that is in the same co-located AP set as one or more 6 GHz APs</u>	11.53	<u>CFAP AND</u> <u>CFDSSS:O</u>	Yes ☐ No ☐ N/A ☐
CF5Gn6G	<u>An AP that operates in the 5 GHz band and that is in the same co-located AP set as one or more 6 GHz APs</u>	11.53	<u>CFAP AND</u> <u>CFOFDM:O</u>	Yes ☐ No ☐ N/A ☐

B.4.4 MAC protocol

Insert the following rows at the end of the table in B.4.4.1:

B.4.4.1 MAC protocol capabilities

Item	Protocol capability	References	Status	Support
PC46	Dynamic fragmentation	10.3, 10.4		
PC46.1	Dynamic fragmentation level 1	26.3.2.2		
PC46.1.1	Dynamic fragmentation level 1 MSDU/ MMPDU transmission	26.3.2.2	CFHE:O	Yes ☐ No ☐
PC46.1.2	Dynamic fragmentation level 1 MSDU/ MMPDU reception	26.3.2.2	CFHE:O	Yes ☐ No ☐
PC46.2	Dynamic fragmentation level 2	26.3.2.3		
PC46.2.1	Dynamic fragmentation level 2 MSDU/ MMPDU transmission	26.3.2.3	CFHE:O	Yes ☐ No ☐
PC46.2.2	Dynamic fragmentation level 2 MSDU/ MMPDU reception	26.3.2.3	CFHE:O	Yes ☐ No ☐
PC46.3	Dynamic fragmentation level 3	26.3.2.4		
PC46.3.1	Dynamic fragmentation level 3 MSDU/ MMPDU transmission	26.3.2.4	CFHE:O	Yes ☐ No ☐
PC46.3.2	Dynamic fragmentation level 3 MSDU/ MMPDU reception	26.3.2.4	CFHE:O	Yes ☐ No ☐

Insert the following rows at the end of the indicated sections of the table in B.4.4.2:

B.4.4.2 MAC frames

Item	MAC frame	References	Status	Support
	Is transmission of the following MAC frames supported?	...		
FT53	Trigger frame	Clause 9	CFHE:M	Yes ☐ No ☐ N/A ☐
	Is reception of the following MAC frames supported?	...		
FR54	Trigger frame	Clause 9	CFHE:O	Yes ☐ No ☐ N/A ☐

Insert the following row into the table in B.4.12 after QB4.4:

B.4.12 QoS base functionality

Item	Protocol capability	References	Status	Support
QB4.5	Multi-STA BlockAck	9.3.1.8.7	CFHE:M	Yes ☐ No ☐ N/A ☐

Insert the following subclauses (B.4.33, B.4.33.1, and B.4.33.2) after B.4.32:

B.4.33 High-efficiency (HE) features

B.4.33.1 HE MAC features

Item	Protocol capability	References	Status	Support
	Are the following MAC protocol features supported?			
HEM1	HE capabilities signaling			
HEM1.1	HE Capabilities element	9.4.2.248	CFHE:M	Yes ☐ No ☐ N/A ☐
HEM1.2	Signaling of STA capabilities in Probe Request, (Re)Association Request frames	9.3.3.5, 9.3.3.7, 9.3.3.9, 9.4.2.248	(CFHE AND CFIndepSTA): M	Yes ☐ No ☐ N/A ☐
HEM1.3	Signaling of HE STA capabilities and HE BSS capabilities in Beacon, Probe Response, (Re)Association Response frames	9.3.3.2, 9.3.3.6, 9.3.3.8, 9.3.3.10, 9.4.2.248	(CFHE AND CFAP):M	Yes ☐ No ☐ N/A ☐
HEM2	Signaling of HE operation	9.4.2.249	(CFHE AND CFAP):M	Yes ☐ No ☐ N/A ☐
HEM3	A-MPDU with multiple TIDs	26.6.3	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM4	HE variant HT Control			
HEM4.1	UL MU Response Scheduling	9.2.4.6a.1	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM4.2	Operating Mode	9.2.4.6a.2	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM4.3	HE Link Adaptation	9.2.4.6a.3	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM4.4	Buffer Status Report	9.2.4.6a.4	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM4.5	UL Power Headroom	9.2.4.6a.5	CFHE:M	Yes ☐ No ☐ N/A ☐
HEM4.6	BQR Control	9.2.4.6a.6	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM4.7	CAS Control	9.2.4.6a.7	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM5	Trigger frame			
HEM5.1	Basic Trigger frame	9.3.1.22.2	CFHE:M	Yes ☐ No ☐ N/A ☐
HEM5.2	BFRP Trigger frame	9.3.1.22.3	CFHE:M	Yes ☐ No ☐ N/A ☐
HEM5.3	MU-BAR Trigger frame	9.3.1.22.4	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM5.4	MU-RTS Trigger frame transmission	9.3.1.22.5	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM5.5	MU-RTS Trigger frame reception	9.3.1.22.5	CFHE:M	Yes ☐ No ☐ N/A ☐
HEM5.6	BSRP Trigger frame	9.3.1.22.6	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM5.7	GCR MU-BAR Trigger frame	9.3.1.22.7	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM5.8	BQRP Trigger frame	9.3.1.22.8	CFHE:O	Yes ☐ No ☐ N/A ☐

B.4.33.1 HE MAC features *(continued)*

Item	Protocol capability	References	Status	Support
HEM5.9	NFRP Trigger frame	9.3.1.22.9	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM6	Transmit beamforming			
*HEM6.1	SU beamformer capable if the supported maximum number of transmit spatial streams is less than 4	9.4.2.248	CFHE:O	Yes ☐ No ☐ N/A ☐
*HEM6.2	SU beamformer capable if the supported maximum number of transmit spatial streams is greater than or equal to 4	9.4.2.248	CFHE:M	Yes ☐ No ☐ N/A ☐
*HEM6.3	SU beamformee capable	9.4.2.248	CFHE:O	Yes ☐ No ☐ N/A ☐
*HEM6.4	MU beamformer capable if the supported maximum number of transmit spatial streams is less than 4	9.4.2.248	CFAP AND CFHEM6.1:O	Yes ☐ No ☐ N/A ☐
*HEM6.5	MU beamformer capable if the supported maximum number of transmit spatial streams is greater than or equal to 4	9.4.2.248	CFAP AND CFHEM6.2:M	Yes ☐ No ☐ N/A ☐
*HEM6.6	MU beamformee capable	9.4.2.248	CFIndepSTA AND VHTM7.2: O	Yes ☐ No ☐ N/A ☐
HEM6.7	Transmission of an HE sounding NDP	26.7	HEM6.1:M HEM6.2:M	Yes ☐ No ☐ N/A ☐
HEM6.8	Reception of an HE sounding NDP	26.7	HEM6.3:M	Yes ☐ No ☐ N/A ☐
*HEM6.9	Transmission of Trigger frame	26.7	HEM6.1:O HEM6.2:O	Yes ☐ No ☐ N/A ☐
HEM6.10	Reception of Trigger frame	26.7	HEM6.9:M	Yes ☐ No ☐ N/A ☐
HEM7	Sounding protocol			
HEM7.1	HE Sounding Protocol as SU beamformer	26.7	HEM6.1:M HEM6.2:M	Yes ☐ No ☐ N/A ☐
HEM7.2	HE Sounding Protocol as SU beamformee	26.7	HEM6.3:M	Yes ☐ No ☐ N/A ☐
HEM7.3	HE Sounding Protocol as MU beamformer	26.7	HEM6.4:M HEM6.5:M	Yes ☐ No ☐ N/A ☐
HEM7.4	HE Sounding Protocol as MU beamformee	26.7	HEM6.6:M	Yes ☐ No ☐ N/A ☐
HEM8	NAV update	26.2.4		
HEM8.1	Update basic NAV	26.2.4	CFHE:M	Yes ☐ No ☐ N/A ☐
HEM8.2	Update IntraBSS NAV	26.2.4	CFAP AND CFHE:O CFIndepSTA AND CFHE:M	Yes ☐ No ☐ N/A ☐
HEM9	OFDMA-based random access	26.5.4	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM10	TWT operation	26.8	CFHE:O	Yes ☐ No ☐ N/A ☐
HEM11	Quiet time period			
HEM11.1	Responding AP procedure		CFAP AND CFHE:O	
HEM11.1.1	QTP AP capability	26.17.5.1	HEM11.1:M	Yes ☐ No ☐ N/A ☐

B.4.33.1 HE MAC features *(continued)*

Item	Protocol capability	References	Status	Support
HEM11.1.2	QTP responding AP procedure	26.17.5.3	HEM11.1:M	Yes ☐ No ☐ N/A ☐
HEM11.2	Requesting STA procedure		CFSTAof AP AND CFHE:O	
HEM11.2.1	QTP non-AP STA capability	26.17.5.1	HEM11.2:M	Yes ☐ No ☐ N/A ☐
HEM11.2.2	QTP requesting STA procedure	26.17.5.2	HEM11.2:M	Yes ☐ No ☐ N/A ☐
HEM12	HE BSS operation in the 6 GHz band			
HEM12.1	Scanning in the 6 GHz band	26.17.2.3	CFHE6G:O	Yes ☐ No ☐ N/A ☐
HEM12.2	Out of band discovery	11.53	(CF2G4n6G OR CF5Gn6G):O	Yes ☐ No ☐ N/A ☐

B.4.33.2 HE PHY features

Item	Protocol capability	References	Status	Support
	Are the following PHY protocol features supported?			
HEP1	PHY operating modes			
HEP1.1	Operation according to Clause 17, Clause 19, and/or Clause 21	27.1.1	CFHE5G AND (CFAP OR CFHE80):M	Yes ☐ No ☐ N/A ☐
HEP1.2	Operation according Clause 19 in the 5 GHz band	27.1.1	CFHE5G AND CFHE20:M	Yes ☐ No ☐ N/A ☐
HEP1.3	Operation according Clause 19 in the 2.4 GHz band	27.1.1	CFHE2G4:M	Yes ☐ No ☐ N/A ☐
HEP1.4	Operation according to Clause 17 in the 6 GHz band	27.1.1	CFHE6G:M	Yes ☐ No ☐ N/A ☐
HEP2	HE PPDU format	27.1.4		
*HEP2.1	HE SU PPDU	27.1.4	CFHE:M	Yes ☐ No ☐ N/A ☐
*HEP2.2	HE ER SU PPDU	27.1.4	CFHE:M	Yes ☐ No ☐ N/A ☐
HEP2.3	HE MU PPDU	27.1.4		
*HEP2.3.1	DL OFDMA Tx	27.3.1.1	CFAP AND CFHE:M	Yes ☐ No ☐ N/A ☐
*HEP2.3.2	DL OFDMA Rx	27.3.1.1	CFSTAofAP AND CFHE:M	Yes ☐ No ☐ N/A ☐
*HEP2.3.3	DL MU-MIMO Tx	27.3.1.1	CFAP AND HEP9.1.10:M	Yes ☐ No ☐ N/A ☐
*HEP2.3.4	DL MU-MIMO Rx	27.3.1.1	CFSTAofAP AND CFHE:M	Yes ☐ No ☐ N/A ☐
*HEP2.3.5	DL MU-MIMO within OFDMA Tx	27.3.1.1	CFAP AND CFHE:O	Yes ☐ No ☐ N/A ☐
*HEP2.3.6	DL MU-MIMO within OFDMA Rx	27.3.1.1	CFSTAofAP AND CFHE:O	Yes ☐ No ☐ N/A ☐

B.4.33.2 HE PHY features *(continued)*

Item	Protocol capability	References	Status	Support
HEP2.3.7	Rx full BW with a single user using HE MU PPDU with compressed HE-SIG-B field	26.15.2	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP2.3.8	Rx full BW with a single user using HE MU PPDU with noncompressed HE-SIG-B field	26.15.2	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP2.4	HE TB PPDU	27.1.4		
HEP2.4.1	UL OFDMA Tx	27.3.1.1	CFSTAofAP AND CFHE:M	Yes ☐ No ☐ N/A ☐
HEP2.4.2	UL OFDMA Rx	27.3.1.1	CFAP AND CFHE:M	Yes ☐ No ☐ N/A ☐
*HEP2.4.3	UL MU-MIMO Tx	27.3.1.1	CFSTAofAP AND CFHE:O	Yes ☐ No ☐ N/A ☐
*HEP2.4.4	UL MU-MIMO Rx	27.3.1.1	CFAP AND CFHE:O	Yes ☐ No ☐ N/A ☐
HEP2.4.5	UL MU-MIMO within OFDMA Tx	27.3.1.1	CFSTAofAP AND CFHE:O	Yes ☐ No ☐ N/A ☐
HEP2.4.6	UL MU-MIMO within OFDMA Rx	27.3.1.1	CFAP AND CFHE:O	Yes ☐ No ☐ N/A ☐
HEP3	BSS bandwidth			
*HEP3.1	20 MHz operation	26.17	CFHE:M	Yes ☐ No ☐ N/A ☐
*HEP3.2	40 MHz operation	26.17	CFHE80 AND (CFHE5G OR CFHE6G):M CFHE2G4:O	Yes ☐ No ☐ N/A ☐
*HEP3.3	80 MHz operation	26.17	CFHE80 AND (CFHE5G OR CFHE6G):M	Yes ☐ No ☐ N/A ☐
*HEP3.4	160 MHz operation	26.17	CFHE80 AND (CFHE5G OR CFHE6G):O HEP3.5:M	Yes ☐ No ☐ N/A ☐
*HEP3.5	80+80 MHz operation	26.17	CFHE80 AND (CFHE5G OR CFHE6G):O	Yes ☐ No ☐ N/A ☐
HEP4	PHY timing information			
HEP4.1	Values in 20 MHz channel	27.3.9	CFHE:M	Yes ☐ No ☐ N/A ☐
HEP4.2	Values in 40 MHz channel	27.3.9	CFHE:M	Yes ☐ No ☐ N/A ☐
HEP4.3	Values in 80 MHz channel	27.3.9	CFHE:M	Yes ☐ No ☐ N/A ☐
HEP4.4	Values in 160 MHz channel	27.3.9	HEP3.4:M	Yes ☐ No ☐ N/A ☐
HEP4.5	Values in 80+80 MHz channel	27.3.9	HEP3.5:M	Yes ☐ No ☐ N/A ☐
HEP5	STBC	27.3.12.12	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP6	Tone allocation			
*HEP6.1	26-tone RU mapping	27.3.2.2, 27.3.2.3, and 27.3.2.4	CFHE:M	Yes ☐ No ☐ N/A ☐

B.4.33.2 HE PHY features *(continued)*

Item	Protocol capability	References	Status	Support
*HEP6.2	52-tone RU mapping	27.3.2.2, 27.3.2.3, and 27.3.2.4	CFHE:M	Yes ☐ No ☐ N/A ☐
*HEP6.3	106-tone RU mapping	27.3.2.2, 27.3.2.3, and 27.3.2.4	CFHE:M	Yes ☐ No ☐ N/A ☐
*HEP6.4	242-tone RU mapping	27.3.2.2, 27.3.2.3, and 27.3.2.4	CFHE80:M CFHE20:O	Yes ☐ No ☐ N/A ☐
*HEP6.5	484-tone RU mapping	27.3.2.2, 27.3.2.3, and 27.3.2.4	CFHE80 AND HEP3.2:M	Yes ☐ No ☐ N/A ☐
*HEP6.6	996-tone RU mapping	27.3.2.2, 27.3.2.3, and 27.3.2.4	CFHE80 AND HEP3.3:M CFHE80 AND HEP3.4:M	Yes ☐ No ☐ N/A ☐
*HEP6.7	2×996-tone RU mapping	27.3.2.2, 27.3.2.3, and 27.3.2.4	CFHE80 AND HEP3.4:M	Yes ☐ No ☐ N/A ☐
HEP7	Coding			
HEP8.1	BCC with 4 or fewer spatial streams	27.3.12.5.1	(HEP6.1 OR HEP6.2 OR HEP6.3 OR HEP6.4):M (HEP3.1 AND HEP2.1):M	Yes ☐ No ☐ N/A ☐
HEP8.2	LDPC with more than 4 spatial streams	27.3.12.5.2	CFHE80:M CFHE20:M	Yes ☐ No ☐ N/A ☐
HEP8.3	LDPC with 4 or fewer spatial streams	27.3.12.5.2	(HEP6.5 OR HEP6.6 OR HEP6.7):M ((HEP3.2 OR HEP3.3 OR HEP3.4 OR HEP3.5) AND HEP2.1):M (HEP6.1 OR HEP6.2 OR HEP6.3 OR HEP6.4):O (HEP3.1 AND HEP2.1):O CFHE20:O	Yes ☐ No ☐ N/A ☐
HEP9	Coding and modulation schemes			
HEP9.1	For 26-, 52-, 106-, 242-, 484-, and 996-tone mapping			
*HEP9.1.1	HE-MCS with Index 0-7 and $N_{SS} = 1$	27.5	CFHE:M	Yes ☐ No ☐ N/A ☐
*HEP9.1.2	HE-MCS with Index 0-8 and $N_{SS} = 1$	27.5	HEP9.1.1:O	Yes ☐ No ☐ N/A ☐

B.4.33.2 HE PHY features *(continued)*

Item	Protocol capability	References	Status	Support
HEP9.1.3	HE-MCS with Index 0-9 and $N_{SS} = 1$	27.5	HEP9.1.2:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.4	HE-MCS with Index 0-7 and $N_{SS} = 2$	27.5	CFHE:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.5	HE-MCS with Index 0-8 and $N_{SS} = 2$	27.5	HEP9.1.4:O	Yes ☐ No ☐ N/A ☐
HEP9.1.6	HE-MCS with Index 0-9 and $N_{SS} = 2$	27.5	HEP9.1.5:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.7	HE-MCS with Index 0-7 and $N_{SS} = 3$	27.5	CFHE:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.8	HE-MCS with Index 0-8 and $N_{SS} = 3$	27.5	HEP9.1.7:O	Yes ☐ No ☐ N/A ☐
HEP9.1.9	HE-MCS with Index 0-9 and $N_{SS} = 3$	27.5	HEP9.1.8:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.10	HE-MCS with Index 0-7 and $N_{SS} = 4$	27.5	CFHE:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.11	HE-MCS with Index 0-8 and $N_{SS} = 4$	27.5	HEP9.1.10:O	Yes ☐ No ☐ N/A ☐
HEP9.1.12	HE-MCS with Index 0-9 and $N_{SS} = 4$	27.5	HEP9.1.11:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.13	HE-MCS with Index 0-7 and $N_{SS} = 5$	27.5	CFHE:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.14	HE-MCS with Index 0-8 and $N_{SS} = 5$	27.5	HEP9.1.13:O	Yes ☐ No ☐ N/A ☐
HEP9.1.15	HE-MCS with Index 0-9 and $N_{SS} = 5$	27.5	HEP9.1.14:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.16	HE-MCS with Index 0-7 and $N_{SS} = 6$	27.5	CFHE:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.17	HE-MCS with Index 0-8 and $N_{SS} = 6$	27.5	HEP9.1.16:O	Yes ☐ No ☐ N/A ☐
HEP9.1.18	HE-MCS with Index 0-9 and $N_{SS} = 6$	27.5	HEP9.1.17:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.19	HE-MCS with Index 0-7 and $N_{SS} = 7$	27.5	CFHE:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.20	HE-MCS with Index 0-8 and $N_{SS} = 7$	27.5	HEP9.1.19:O	Yes ☐ No ☐ N/A ☐
HEP9.1.21	HE-MCS with Index 0-9 and $N_{SS} = 7$	27.5	HEP9.1.20:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.22	HE-MCS with Index 0-7 and $N_{SS} = 8$	27.5	CFHE:O	Yes ☐ No ☐ N/A ☐
*HEP9.1.23	HE-MCS with Index 0-8 and $N_{SS} = 8$	27.5	HEP9.1.22:O	Yes ☐ No ☐ N/A ☐
HEP9.1.24	HE-MCS with Index 0-9 and $N_{SS} = 8$	27.5	HEP9.1.23:O	Yes ☐ No ☐ N/A ☐
HEP9.2	For 242-, 484-, and 996-tone plan			
*HEP9.2.1	HE-MCS with Index 0-10 and $N_{SS} = 1$	27.5	CFHE80:O	Yes ☐ No ☐ N/A ☐
HEP9.2.2	HE-MCS with Index 0-11 and $N_{SS} = 1$	27.5	HEP9.2.1:O	Yes ☐ No ☐ N/A ☐
*HEP9.2.3	HE-MCS with Index 0-10 and $N_{SS} = 2$	27.5	CFHE80:O	Yes ☐ No ☐ N/A ☐
HEP9.2.4	HE-MCS with Index 0-11 and $N_{SS} = 2$	27.5	HEP9.2.3:O	Yes ☐ No ☐ N/A ☐
*HEP9.2.5	HE-MCS with Index 0-10 and $N_{SS} = 3$	27.5	CFHE80:O	Yes ☐ No ☐ N/A ☐
HEP9.2.6	HE-MCS with Index 0-11 and $N_{SS} = 3$	27.5	HEP9.2.5:O	Yes ☐ No ☐ N/A ☐
*HEP9.2.7	HE-MCS with Index 0-10 and $N_{SS} = 4$	27.5	CFHE80:O	Yes ☐ No ☐ N/A ☐
HEP9.2.8	HE-MCS with Index 0-11 and $N_{SS} = 4$	27.5	HEP9.2.7:O	Yes ☐ No ☐ N/A ☐
*HEP9.2.9	HE-MCS with Index 0-10 and $N_{SS} = 5$	27.5	CFHE80:O	Yes ☐ No ☐ N/A ☐
HEP9.2.10	HE-MCS with Index 0-11 and $N_{SS} = 5$	27.5	HEP9.2.9:O	Yes ☐ No ☐ N/A ☐

B.4.33.2 HE PHY features *(continued)*

Item	Protocol capability	References	Status	Support
*HEP9.2.11	HE-MCS with Index 0-10 and $N_{SS} = 6$	27.5	CFHE80:O	Yes ☐ No ☐ N/A ☐
HEP9.2.12	HE-MCS with Index 0-11 and $N_{SS} = 6$	27.5	HEP9.2.11:O	Yes ☐ No ☐ N/A ☐
*HEP9.2.13	HE-MCS with Index 0-10 and $N_{SS} = 7$	27.5	CFHE80:O	Yes ☐ No ☐ N/A ☐
HEP9.2.14	HE-MCS with Index 0-11 and $N_{SS} = 7$	27.5	HEP9.2.13:O	Yes ☐ No ☐ N/A ☐
*HEP9.2.15	HE-MCS with Index 0-10 and $N_{SS} = 8$	27.5	CFHE80:O	Yes ☐ No ☐ N/A ☐
HEP9.2.16	HE-MCS with Index 0-11 and $N_{SS} = 8$	27.5	HEP9.2.15:O	Yes ☐ No ☐ N/A ☐
HEP10	HE-LTF formats	27.3.11.10		
HEP10.1	2x HE-LTF and 0.8 μ s GI	27.3.11.10	CFHE:M	Yes ☐ No ☐ N/A ☐
HEP10.2	2x HE-LTF and 1.6 μ s GI	27.3.11.10	CFHE:M	Yes ☐ No ☐ N/A ☐
HEP10.3	4x HE-LTF and 3.2 μ s GI	27.3.11.10	CFHE:M	Yes ☐ No ☐ N/A ☐
HEP10.4	1x HE-LTF and 0.8 μ s GI	27.3.11.10	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP10.5	4x HE-LTF and 0.8 μ s GI	27.3.11.10	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP10.6	1x HE-LTF and 1.6 μ s GI	27.3.11.10	HEP2.4.3 OR HEP2.4.4:M	Yes ☐ No ☐ N/A ☐
HEP11	DCM	27.3.12.9	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP12	Preamble puncturing	27.2.2	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP13	Sounding			
HEP13.1	Punctured sounding operation	26.7.2	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP13.2	NDP with 4x HE-LTF and 3.2 μ s GI	27.3.17	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP13.3	$N_g = 16$ SU feedback	26.7.3	HEM6.3:O	Yes ☐ No ☐ N/A ☐
HEP13.4	$N_g = 16$ MU feedback	26.7.3	HEM6.6:O	Yes ☐ No ☐ N/A ☐
HEP13.5	Codebook size $(\phi, \psi) = \{4, 2\}$ SU feedback	26.7.3	HEM6.3:O	Yes ☐ No ☐ N/A ☐
HEP13.6	Codebook size $(\phi, \psi) = \{7, 5\}$ MU feedback	26.7.3	HEM6.6:O	Yes ☐ No ☐ N/A ☐
HEP13.7	Triggered SU beamforming feedback	26.7.3	CFSTAofAP AND HEM6.3:O CFAP AND (HEM6.1 OR HEM6.2):O	Yes ☐ No ☐ N/A ☐
HEP13.8	Triggered MU beamforming partial BW feedback	26.7.3	CFSTAofAP AND HEM6.6:O CFAP AND (HEM6.4 OR HEM6.5):O	Yes ☐ No ☐ N/A ☐

B.4.33.2 HE PHY features *(continued)*

Item	Protocol capability	References	Status	Support
HEP13.9	Triggered CQI feedback	26.7.3	CFSTAofAP AND HEM6.3:O CFAP AND (HEM6.1 OR HEM6.2):O	Yes ☐ No ☐ N/A ☐
HEP13.10	Non-triggered CQI feedback	26.7.3	CFSTAofAP AND HEM6.3:O CFAP AND (HEM6.1 OR HEM6.2):O	Yes ☐ No ☐ N/A ☐
HEP14	Midambles	27.3.12.16	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP15	HE-SIG-B field	27.3.11.8		
HEP15.1	HE-MCSs 0 to 5 for the HE-SIG-B field	27.3.11.8	CFHE:M	Yes ☐ No ☐ N/A ☐
HEP15.2	Greater than 16 HE-SIG-B OFDM symbols	27.3.11.8	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP16	Spatial reuse			
HEP16.1	PSR-based SR support	26.10.3	CFHE:O	Yes ☐ No ☐ N/A ☐
HEP16.2	OBSS PD-based spatial reuse operation	26.10.2	CFHE:O	Yes ☐ No ☐ N/A ☐

Annex C

(normative)

ASN.1 encoding of the MAC and PHY MIB

C.3 MIB detail

Change the dot11smt OBJECT IDENTIFIER element list in the “Major sections” part of C.3 as follows (not all lines shown):

```
dot11smt OBJECT IDENTIFIER ::= { ieee802dot11 1 }
-- ...
-- dot11GLKLinkMetricsTable ::= { dot11smt 41 }
-- dot11HEStationConfigTable ::= { dot11smt 42 }
-- dot11PPEThresholdsMappingsTable ::= { dot11smt 43 }
```

Change the dot11mac OBJECT IDENTIFIER element list in the “Major sections” part of C.3 as follows (not all lines shown):

```
dot11mac OBJECT IDENTIFIER ::= { ieee802dot11 2 }
-- MAC GROUPS
-- ...
-- dot11CDMGOperationTable ::= { dot11mac 11 }
-- dot11MUEDCatable ::= { dot11mac 15 }
```

Change the Dot11StationConfigEntry SEQUENCE list in the “dot11StationConfig TABLE” in C.3 as follows (not all lines shown):

```
Dot11StationConfigEntry ::= SEQUENCE
{
  ...
  dot11BSSMaxIdlePeriodIndicationByNonAPSTA TruthValue,
  dot11HEOptionImplemented TruthValue,
  dot11OBSSNarrowBWRUinOFDMATolerated TruthValue,
  dot11HE6GOptionImplemented TruthValue,
  dot11OCTOptionImplemented TruthValue,
  dot11QTPOptionImplemented TruthValue,
  dot11ColocatedRNRImplemented TruthValue
}
```

Insert the following element definitions into the “dot11StationConfig TABLE” in C.3 after the “dot11BSSMaxIdlePeriodIndicationByNonAPSTA” element definition:

```
dot11HEOptionImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
  "This is a capability variable.
  Its value is determined by device capabilities."
```

This attribute indicates whether the entity is HE Capable."
 ::= { dot11StationConfigEntry 181 }

dot11OBSSNarrowBWRUinOFDMATolerated OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute indicates whether the AP STA is able to tolerate 26-tone RU
UL OFDMA transmissions using HE TB PPDU from OBSS (not falsely classify
the 26-tone RU UL OFDMA transmissions as radar pulses)."
 ::= { dot11StationConfigEntry 182 }

dot11HE6GOptionImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute indicates whether the entity is capable of operating in the
6 GHz band."
 ::= { dot11StationConfigEntry 194 }

dot11OCTOptionImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the station implementation is
capable of on-channel tunneling operation. The capability is disabled
otherwise."
 ::= { dot11StationConfigEntry 187 }

dot11QTPOptionImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the QTP operation is implemented
and operational. This attribute, when false or not present, indicates that
the QTP operation is not implemented or not operational."
 ::= { dot11StationConfigEntry 198 }

dot11ColocatedRNRImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable. Its value is determined by device
capabilities.

This attribute, when true, indicates that the STA implementation is
capable of transmitting or receiving a Reduced Neighbor Report element

carrying information on APs that are in the same co-located AP set as the reporting AP in Probe Response, Beacon and FILS Discovery frames. The capability is disabled otherwise."
 DEFVAL { false }
 ::= { dot11StationConfigEntry 199 }

Insert the “dot11HEStationConfig TABLE” and “dot11PPEThresholdsMappings TABLE” into C.3 after the “dot11GLKLinkMetrics TABLE”:

```
-- *****
-- * dot11HEStationConfig TABLE
-- *****

dot11HEStationConfigTable OBJECT-TYPE
    SYNTAX SEQUENCE OF Dot11HEStationConfigEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "Station Configuration attributes. In tabular form to allow for multiple
        instances on an agent."
    ::= { dot11smt 42 }

dot11HEStationConfigEntry OBJECT-TYPE
    SYNTAX Dot11HEStationConfigEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "An entry (conceptual row) in the dot11HEStationConfig Table.

        ifIndex - Each IEEE 802.11 interface is represented by an ifEntry.
        Interface tables in this MIB module are indexed by ifIndex."
    INDEX { ifIndex }
    ::= { dot11HEStationConfigTable 1 }

Dot11HEStationConfigEntry ::=
    SEQUENCE {
        dot11TRSOptionImplemented                TruthValue,
        dot11OFDMARandomAccessOptionImplemented  TruthValue,
        dot11HEControlFieldOptionImplemented     TruthValue,
        dot11OMIOptionImplemented                 TruthValue,
        dot11HEMCSFeedbackOptionImplemented       INTEGER,
        dot11HEDynamicFragmentationLevel         INTEGER,
        dot11AMPDUwithMultipleTIDOptionImplemented TruthValue,
        dot11MPDUAskedForAckInMultiTIDAMPDU      TruthValue,
        dot11TXOPDurationRTSThreshold            Unsigned32,
        dot11PPEThresholdsRequired               TruthValue,
        dot11IntraPPDUPowerSaveOptionActivated    TruthValue,
        dot11AMSDUFragmentationOptionImplemented TruthValue,
        dot11BSSColorCollisionAPPeriod            INTEGER,
        dot11BSSColorCollisionSTAPeriod           Unsigned32,
        dot11AutonomousBSSColorCollisionReportingImplemented TruthValue,
        dot11HEPSROptionImplemented              TruthValue,
        dot11HEBSRControlImplemented             TruthValue,
        dot11HEBQRControlImplemented             TruthValue,
        dot11HECASControlImplemented             TruthValue,
        dot11PartialBSSColorImplemented          TruthValue,
        dot11ObssNbRuToleranceTime               Unsigned32,
        dot11HESubchannelSelectiveTransmissionImplemented TruthValue,
        dot11SRResponderOptionImplemented        TruthValue,
        dot11AutonomousBSSColorInUseReportingImplemented TruthValue,
        dot11ShortSSIDListImplemented            TruthValue,
        dot11SRGAPOBSSPDMinOffset               Integer32,
```

<pre> dot11SRGAPOBSSPDMaxOffset dot11SRGAPBSSColorBitmap dot11SRGAPBSSIDBitmap dot11NonSRGAPOBSSPDMaxOffset dot11HTVHTTriggerOptionImplemented dot11HEDynamicSMPowerSaveOptionImplemented dot11MUEDCAPParametersActivated dot11CoHostedBSSIDImplemented dot11UnsolicitedProbeResponseOptionActivated dot11MemberOfCollocated6GHzESSOptionActivated dot11AckEnabledAMPDUOptionImplemented dot11MinPSCProbeDelay dot11OCWmin dot11OCWmax </pre>	<pre> Integer32, OCTET STRING, OCTET STRING, Integer32, TruthValue, TruthValue, TruthValue, TruthValue, TruthValue, TruthValue, TruthValue, TruthValue, Unsigned32, Unsigned32, Unsigned32 } </pre>
---	---

dot11TRSOOptionImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.
 Its value is determined by device capabilities.

This attribute, when true, indicates that the station implementation is capable of receiving frames with a TRS Control subfield. The capability is disabled otherwise."

DEFVAL { false }

::= { dot11HEStationConfigEntry 1 }

dot11OFDMARandomAccessOptionImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.
 Its value is determined by device capabilities.

This attribute, when true, indicates that the station implementation is capable of an OFDMA random access operation. The capability is disabled otherwise."

DEFVAL { false }

::= { dot11HEStationConfigEntry 2 }

dot11HEControlFieldOptionImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.
 Its value is determined by device capabilities.

This attribute, when true, indicates that the station implementation is capable of receiving the HE variant HT Control field. The capability is disabled otherwise."

DEFVAL { false }

::= { dot11HEStationConfigEntry 3 }

dot11OMIOptionImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.
 Its value is determined by device capabilities.

This attribute, when true, indicates that the station implementation is capable of receiving frames with an OM Control subfield. The capability is disabled otherwise."

DEFVAL { false }
 ::= { dot11HEStationConfigEntry 4}

dot11HEMCSFeedbackOptionImplemented OBJECT-TYPE
 SYNTAX INTEGER {none(0), unsolicited(2), solicitedandunsolicited(3)}
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "This is a capability variable.
 Its value is determined by device capabilities.

 This attribute indicates the HE-MCS feedback capability supported by the station implementation."
 DEFVAL { 0 }
 ::= { dot11HEStationConfigEntry 5}

dot11HEDynamicFragmentationLevel OBJECT-TYPE
 SYNTAX INTEGER {hedynamicfragmentationlevel1(1),
 hedynamicfragmentationlevel2(2), hedynamicfragmentationlevel3(3)}
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "This is a capability variable.
 Its value is determined by device capabilities.

 hedynamicfragmentationlevel1 indicates support for up to one dynamic fragment that is a non-A-MPDU frame, no support for dynamic fragments within an A-MPDU that does not contain an S-MPDU.

 hedynamicfragmentationlevel2 indicates support for up to one dynamic fragment that is a non-A-MPDU frame and support for up to one dynamic fragment for each MSDU, each A-MSDU (if supported by the recipient) and one MMPDU (if present) within an A-MPDU that does not contain an S-MPDU.

 hedynamicfragmentationlevel3 indicates support for up to one dynamic fragment that is a non-A-MPDU frame and support for up to 4 dynamic fragments for each MSDU and for each A-MSDU (if supported by the recipient) within an A-MPDU and up to one dynamic fragment for one MMPDU (if present) in an A-MPDU that does not contain an S-MPDU"

 ::= { dot11HEStationConfigEntry 6}

dot11HEAMPDUwithMultipleTIDOptionImplemented OBJECT-TYPE
 SYNTAX TruthValue
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "This is a capability variable.
 Its value is determined by device capabilities.

 This attribute, when true, indicates that the station implementation is capable of receiving an A-MPDU that contains QoS Data frames with two or more different TID values. The capability is disabled otherwise."
 DEFVAL { false }
 ::= { dot11HEStationConfigEntry 7}

dot11MPDUAskedforAckInMultiTIDAMPDU OBJECT-TYPE
 SYNTAX TruthValue
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION


```
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the station implementation is
capable of receiving a multi-TID A-MPDU that can solicit either Ack or
BlockAck, or both. The capability is disabled otherwise."
DEFVAL { false }
::= { dot11HEStationConfigEntry 8 }

dot11TXOPDurationRTSThreshold OBJECT-TYPE
SYNTAX Unsigned32 (1..1023)
UNITS "32 microseconds"
MAX-ACCESS read-write
STATUS current
DESCRIPTION
    "This is a control variable.
    It is written by an external management entity or by the MAC of a non-AP
    HE STA upon receiving an HE Operation element from the HE AP with which it
    is associated.
    Changes take effect as soon as practical in the implementation.

    This attribute indicates the duration of the transmission or TXOP above
    which an RTS/CTS handshake is performed. The value 1023 means this feature
    is disabled."
DEFVAL { 1023 }
::= { dot11HEStationConfigEntry 9 }

dot11PPEThresholdsRequired OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.
    This attribute, when true, indicates that PPE thresholds exist and are
    provided in dot11PPEThresholdsTable."
DEFVAL { false }
::= { dot11HEStationConfigEntry 10 }

dot11IntraPPDUPowerSaveOptionActivated OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that the station implementation is
    capable of Intra PPDU Power Save operation. The capability is disabled
    otherwise."
DEFVAL { false }
::= { dot11HEStationConfigEntry 11 }

dot11AMSDUFragmentationOptionImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable. Its value is determined by device
    capabilities.

    This attribute, when true, indicates that the STA implementation is
    capable of receiving dynamic fragments of A-MSDUs. The capability is
    disabled otherwise."
```

```
    DEFVAL { false }
::= { dot11HEStationConfigEntry 12 }

dot11BSSColorCollisionAPPeriod OBJECT-TYPE
    SYNTAX INTEGER (-1..120)
    UNITS "seconds"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity.
        Changes take effect as soon as practical in the implementation.

        This attribute indicates the duration for which an HE AP waits before
        disabling BSS color when a color collision is detected. The value -1 means
        that this feature is disabled."
    DEFVAL { 50 }
    ::= { dot11HEStationConfigEntry 13 }

dot11BSSColorCollisionSTAPeriod OBJECT-TYPE
    SYNTAX Unsigned32 (30..120)
    UNITS "seconds"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity.
        Changes take effect as soon as practical in the implementation.

        This attribute indicates the interval between successive BSS color
        event collision reports sent by a non-AP HE STA."
    DEFVAL { 30 }
    ::= { dot11HEStationConfigEntry 14 }

dot11AutonomousBSSColorCollisionReportingImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.
        This attribute, when true, indicates that autonomously detecting and
        reporting of BSS color collision is implemented."
    ::= { dot11HEStationConfigEntry 15 }

dot11HEPSROptionImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable. Its value is determined by device
        capabilities.

        This attribute, when true, indicates that the STA implementation is
        capable of transmitting Spatial Reuse Parameters in HE PPDU. The
        capability is disabled otherwise."
    DEFVAL { false }
    ::= { dot11HEStationConfigEntry 16 }

dot11HEBSRControlImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
```

"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the station implementation is capable of receiving frames with a BSR Control subfield. The capability is disabled otherwise."
DEFVAL { false }
::= { dot11HEStationConfigEntry 17 }

dot11HEBQRControlImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the station implementation is capable of receiving frames with a BQR Control subfield. The capability is disabled otherwise."
DEFVAL { false }
::= { dot11HEStationConfigEntry 18 }

dot11HECASControlImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the station implementation is capable of receiving frames with a CAS Control subfield. The capability is disabled otherwise."
DEFVAL { false }
::= { dot11HEStationConfigEntry 19 }

dot11PartialBSSColorImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.
This attribute, when true, indicates that the partial BSS color (see 26.17.4) is implemented. The capability is disabled otherwise."
DEFVAL { false }
::= { dot11HEStationConfigEntry 20 }

dot11ObssNbRuToleranceTime OBJECT-TYPE
SYNTAX Unsigned32 (0..3600)
UNITS "seconds"
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"It is a status variable.
It is written by an external management entity. Changes take effect as soon as practical in the implementation.
This attribute indicates the minimum time that needs to pass since the reception of the last Beacon frame from an OBSS AP that did not indicate tolerance to narrow bandwidth RU in OFDMA before the STA can transmit a triggering frame that allocates a 26-tone RU, or transmit an HE TB PPDU in a 26-tone RU."
DEFVAL { 1800 }

```
::= { dot11HEStationConfigEntry 21 }

dot11HESubchannelSelectiveTransmissionImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.
        This attribute, when true, indicates that an HE subchannel selective
        transmission operation is implemented. The capability is disabled
        otherwise."
    DEFVAL { false }
::= { dot11HEStationConfigEntry 22 }

dot11SRResponderOptionImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that the PSR-based PSRT PPDU
        reception (see 26.10.3) is implemented. The capability is disabled
        otherwise."
    DEFVAL { false }
::= { dot11HEStationConfigEntry 23 }

dot11AutonomousBSSColorInUseReportingImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that autonomously
        reporting of BSS color in use is implemented."
    DEFVAL { false }
::= { dot11HEStationConfigEntry 24 }

dot11ShortSSIDListImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable. Its value is determined by device
        capabilities.

        This attribute, when true, indicates that the STA implementation is
        capable of transmitting and receiving Short SSID List element in Probe
        Request frames. The capability is disabled otherwise."
    DEFVAL { false }
::= { dot11HEStationConfigEntry 25 }

dot11SRGAPOBSSPDMinOffset OBJECT-TYPE
    SYNTAX Integer32
    UNITS "dBm"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity."
```

Changes take effect as soon as practical in the implementation.

This attribute indicates the SRG OBSS PD Min Offset for an AP."
DEFVAL { 0 }
::= { dot11HEStationConfigEntry 26 }

dot11SRGAPOBSSPDMaxOffset OBJECT-TYPE
SYNTAX Integer32
UNITS "dBm"
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "This is a control variable.
 It is written by an external management entity.
 Changes take effect as soon as practical in the implementation.

This attribute indicates the SRG OBSS PD Max Offset for an AP."
DEFVAL { 0 }
::= { dot11HEStationConfigEntry 27 }

dot11SRGAPBSSColorBitmap OBJECT-TYPE
SYNTAX OCTET STRING (SIZE(8))
MAX-ACCESS read-write
STATUS current
DESCRIPTION
 "This is a control variable.
 This variable is a 64 bit bitmap that indicates which BSS color values are
 used by members of the SRG of which the AP is a member.
 It is written by an external management entity.
 Changes take effect as soon as practical in the implementation.

This attribute indicates the SRG BSS Color Bitmap for an AP."
::= { dot11HEStationConfigEntry 28 }

dot11SRGAPBSSIDBitmap OBJECT-TYPE
SYNTAX OCTET STRING (SIZE(8))
MAX-ACCESS read-write
STATUS current
DESCRIPTION
 "This is a control variable.
 This variable is a 64 bit bitmap that indicates which Partial BSSID values
 are used by members of the SRG of which the AP is a member.
 It is written by an external management entity.
 Changes take effect as soon as practical in the implementation.

This attribute indicates the SRG BSSID Bitmap for an AP."
::= { dot11HEStationConfigEntry 29 }

dot11NonSRGAPOBSSPDMaxOffset OBJECT-TYPE
SYNTAX Integer32
UNITS "dBm"
MAX-ACCESS read-write
STATUS current
DESCRIPTION
 "This is a control variable.
 It is written by an external management entity.
 Changes take effect as soon as practical in the implementation.

This attribute indicates the Non SRG OBSS PD Max Offset for an AP."
DEFVAL { 0 }
::= { dot11HEStationConfigEntry 30 }

dot11HTVHTTriggerOptionImplemented OBJECT-TYPE
SYNTAX TruthValue

```

MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that the STA implementation is
    capable of receiving Trigger frames in HT PPDU and VHT PPDU. The
    capability is disabled, otherwise."
    DEFVAL { false }
::= { dot11HESessionConfigEntry 31 }

dot11HEDynamicSMPowerSaveOptionImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable. Its value is determined by device
    capabilities.

    This attribute, when true, indicates that the STA implementation is
    capable of enabling its multiple receive chains when it receives a
    Trigger frame. The capability is disabled otherwise."
    DEFVAL { false }
::= { dot11HESessionConfigEntry 32 }

dot11MUEDCAParametersActivated OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION
    "This is a control variable used only if the STA is an AP. It is set to
    true by an AP that advertises MU EDCA parameters so that its associated
    STAs use MU EDCA parameters."
    DEFVAL { true }
::= { dot11HESessionConfigEntry 33 }

dot11CoHostedBSSIDImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.
    This attribute, when true, indicates that the station implementation is
    capable of supporting Co-Hosted BSSID."
    DEFVAL { false }
::= { dot11HESessionConfigEntry 34 }

dot11UnsolicitedProbeResponseOptionActivated OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-write
STATUS current
DESCRIPTION
    "This is a control variable.
    It is written by an external management entity. Changes take effect as
    soon as practical in the implementation.
    This attribute, when true, indicates that the station implementation is an
    AP and is part of an ESS where all the APs operating in the 6 GHz band that
    operate in the same channel as the AP and whose transmitted frames might
    be detected by a STA receiving this frame schedules for transmission
    unsolicited Probe Response frames every 20 TUs or less (see 26.17.2.3.2).
    The capability is disabled otherwise."
::= { dot11HESessionConfigEntry 35 }

```

```
dot11MemberOfCollocated6GHzESSOptionActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity.

        This attribute, when true, indicates that the station implementation is an
        AP that operates in the 6 GHz band and is part of an ESS where each AP has
        a corresponding AP operating in the 2.4 GHz or 5 GHz band that is in the
        same co-located AP set as that AP (see 11.53). The capability is disabled
        otherwise."
 ::= { dot11HESessionConfigEntry 36 }

dot11AckEnabledAMPDUOptionImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that the station implementation is
        capable of receiving ack-enabled single-TID A-MPDU. The capability is
        disabled otherwise."
 ::= { dot11HESessionConfigEntry 37 }

dot11MinPSCProbeDelay OBJECT-TYPE
    SYNTAX Unsigned32 (5484..100000)
    UNITS "microseconds"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity.
        Changes take effect as soon as practical in the implementation.

        A STA does not send a Probe Request frame if it is scanning a preferred
        scanning channel in the 6 GHz band, unless the channel has been
        continuously idle for this duration since the start of the scan on that
        channel."
    DEFVAL { 7000 }
 ::= { dot11HESessionConfigEntry 38 }

dot11OCWmin OBJECT-TYPE
    SYNTAX Unsigned32 (0..127)
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by the MAC upon receiving an UORA Parameter Set.
        Changes take effect as soon as practical in the implementation.

        This attribute specifies the value of the minimum size of the OFDMA
        contention window (OCW) that is used by the non-AP STA for UL OFDMA-based
        random access. The value of this attribute is such that it could always be
        expressed in the form of 2^X - 1, where X is an integer."
    DEFVAL { 7 }
 ::= { dot11HESessionConfigEntry 39 }

dot11OCWmax OBJECT-TYPE
    SYNTAX Unsigned32 (0..127)
```

```

MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a control variable.
    It is written by the MAC upon receiving an UORA Parameter Set.
    Changes take effect as soon as practical in the implementation.

    This attribute specifies the value of the maximum size of the OFDMA
    contention window (OCW) that is used by the non-AP STA for UL OFDMA-based
    random access. The value of this attribute is such that it could always be
    expressed in the form of 2^X - 1, where X is an integer."
    DEFVAL { 31 }
::= { dot11HEStationConfigEntry 40 }

-- *****
-- * End of dot11HEStationConfig TABLE
-- *****

-- *****
-- * dot11PPEThresholdsMappings TABLE
-- *****
dot11PPEThresholdsMappingsTable OBJECT-TYPE
    SYNTAX SEQUENCE OF Dot11PPEThresholdsMappingsEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "A conceptual table for PPE thresholds mappings, which determines the
        nominal packet padding value as a function of the two PPE thresholds,
        PPET8 and PPET16, for an HE PPDU of a particular RU allocation size and
        NSTS value. The MIB supports the ability to share separate PPE thresholds
        for each NSTS/RU pair. The thresholds mappings table contains one entry
        for each NSTS/RU pair and contains two fields for each entry: PPET8 and
        PPET16."
    REFERENCE "IEEE Std 802.11-2020, 26.12"
    ::= { dot11smt 43}

dot11PPEThresholdsMappingsEntry OBJECT-TYPE
    SYNTAX Dot11PPEThresholdsMappingsEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "An Entry (conceptual row) in the PPE Thresholds Mappings Table.
        ifIndex - Each IEEE Std 802.11 interface is represented by an ifEntry.
        Interface tables in this MIB module are indexed by ifIndex."
    INDEX { ifIndex, dot11PPEThresholdsMappingIndex }
    ::= { dot11PPEThresholdsMappingsTable 1 }

Dot11PPEThresholdsMappingsEntry ::= SEQUENCE {
    dot11PPEThresholdsMappingIndex          Unsigned32,
    dot11PPEThresholdsMappingNSS            Unsigned32,
    dot11PPEThresholdsMappingRUIndex        Unsigned32,
    dot11PPEThresholdsMappingPPET8          INTEGER,
    dot11PPEThresholdsMappingPPET16         INTEGER,
    dot11PPEThresholdsMappingStatus         RowStatus}

dot11PPEThresholdsMappingIndex OBJECT-TYPE
    SYNTAX Unsigned32
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "The auxiliary variable used to identify instances of the columnar objects
        in the PPE Thresholds Mappings Table."
    ::= { dot11PPEThresholdsMappingsEntry 1 }
  
```



```
dot11PPEThresholdsMappingNSS OBJECT-TYPE
    SYNTAX Unsigned32
    MAX-ACCESS read-create
    STATUS current
    DESCRIPTION
        "The NSS value portion of the NSS/RU pair for which the values from this
        Thresholds mapping entry are to be used."
    ::= { dot11PPEThresholdsMappingsEntry 2 }

dot11PPEThresholdsMappingRUIndex OBJECT-TYPE
    SYNTAX Unsigned32
    MAX-ACCESS read-create
    STATUS current
    DESCRIPTION
        "The index of the RU value portion of the NSS/RU pair for which the values
        from this thresholds mapping entry are to be used. The index values map to
        an RU as follows: RU Index of 0 is 242 tones, 1 is 448 tones, 2 is 996
        tones, 3 is 2x996 tones."
    ::= { dot11PPEThresholdsMappingsEntry 3 }

dot11PPEThresholdsMappingPPET8 OBJECT-TYPE
    SYNTAX INTEGER{BPSK(0), QPSK(1), 16-QAM(2), 64-QAM(3), 256-QAM(4), 1024-
        QAM(5), NONE(7)}
    MAX-ACCESS read-create
    STATUS current
    DESCRIPTION
        "An index that determines a constellation value at or above which a
        nominal packet padding value of at least 8 microseconds is required for
        the given NSS/RU pair corresponding to the row of the entry."
    ::= { dot11PPEThresholdsMappingsEntry 4 }

dot11PPEThresholdsMappingPPET16 OBJECT-TYPE
    SYNTAX INTEGER{BPSK(0), QPSK(1), 16-QAM(2), 64-QAM(3), 256-QAM(4), 1024-
        QAM(5), NONE(7)}
    MAX-ACCESS read-create
    STATUS current
    DESCRIPTION
        "An index that determines a constellation value at or above which a
        nominal packet padding value of 16 microseconds is required for the given
        NSS/RU pair corresponding to the row of the entry."
    ::= { dot11PPEThresholdsMappingsEntry 5 }

dot11PPEThresholdsMappingStatus OBJECT-TYPE
    SYNTAX RowStatus
    MAX-ACCESS read-create
    STATUS current
    DESCRIPTION
        "The status column used for creating, modifying, and deleting instances of
        the columnar objects in the PPE thresholds mapping table."
    DEFVAL { active }
    ::= { dot11PPEThresholdsMappingsEntry 6 }

-- *****
-- * End of dot11PPEThresholdsMappings TABLE
-- *****
```

Change the dot11RTSThreshold element definition in the “dot11Operation TABLE” in C.3 as follows:

```
dot11RTSThreshold OBJECT-TYPE
    SYNTAX Unsigned32 (0..4692480 6500631)
    MAX-ACCESS read-write
    STATUS current
```

DESCRIPTION

"This is a control variable.
 It is written by an external management entity.
 Changes take effect as soon as practical in the implementation.

This attribute indicates the number of octets in a PSDU, below which an RTS/CTS handshake is not performed if dot11TXOPDurationRTSThreshold is 1023 or it is not present, except as RTS/CTS is used as a cross modulation protection mechanism as defined in 10.27 (Protection mechanisms). An RTS/CTS handshake is performed at the beginning of any frame exchange sequence where the PSDU contains an MPDU with the Type subfield equal to Data or Management and an individual address in the Address 1 field, and the length of the PSDU is greater than this threshold. Setting this attribute to be larger than the maximum PSDU size has the effect of turning off the RTS/CTS handshake for frames of Data or Management type transmitted by this STA. Setting this attribute to 0 has the effect of turning on the RTS/CTS handshake for all frames of Data or Management type transmitted by this STA."

DEFVAL { ~~4692480~~ 6500631 }
 ::= { dot11OperationEntry 2 }

Insert the "SMT MU EDCA Config TABLE" into C.3 after the "dot11CDMGOperation TABLE":

```
-- *****
-- * SMT MU EDCA Config TABLE
-- *****

dot11MUEDCATable OBJECT-TYPE
    SYNTAX SEQUENCE OF Dot11MUEDCAEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "Conceptual table for MU EDCA parameter values at a non-AP STA. This table
        contains the four entries of the MU EDCA parameters corresponding to four
        possible ACs. Index 1 corresponds to AC_BK, index 2 to AC_BE, index 3 to
        AC_VI, and index 4 to AC_VO."

    ::= { dot11mac 15 }

dot11MUEDCAEntry OBJECT-TYPE
    SYNTAX Dot11MUEDCAEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "An Entry (conceptual row) in the MU EDCA Table.

        ifIndex - Each IEEE 802.11 interface is represented by an ifEntry.
        Interface tables in this MIB module are indexed by ifIndex."
    INDEX { ifIndex, dot11MUEDCATableIndex }
    ::= { dot11MUEDCATable 1 }

Dot11MUEDCAEntry ::=
    SEQUENCE {
        dot11MUEDCATableIndex                Unsigned32,
        dot11MUEDCATableCWmin                Unsigned32,
        dot11MUEDCATableCWmax                Unsigned32,
        dot11MUEDCATableAIFSN                Unsigned32,
        dot11MUEDCATableTimer                Unsigned32
    }

dot11MUEDCATableIndex OBJECT-TYPE
    SYNTAX Unsigned32 (1..4)
    MAX-ACCESS not-accessible
```

STATUS current
DESCRIPTION
 "The auxiliary variable used to identify instances of the columnar objects in the MU EDCA Table. The value of this variable is
 1, if the value of the AC is AC_BK.
 2, if the value of the AC is AC_BE.
 3, if the value of the AC is AC_VI.
 4, if the value of the AC is AC_VO."
::= { dot11MUEDECAEntry 1 }

dot11MUEDCATableCWmin OBJECT-TYPE
SYNTAX Unsigned32 (0..255)
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "This is a control variable.
 It is written by the MAC upon receiving an MU EDCA Parameter Set.
 Changes take effect as soon as practical in the implementation.

 This attribute specifies the value of the minimum size of the window that is used by a STA for a particular AC for generating a random number for the backoff. The value of this attribute is such that it could always be expressed in the form of $2^X - 1$, where X is an integer."
::= { dot11MUEDECAEntry 2 }

dot11MUEDCATableCWmax OBJECT-TYPE
SYNTAX Unsigned32 (0..65535)
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "This is a control variable.
 It is written by the MAC upon receiving an MU EDCA Parameter Set.
 Changes take effect as soon as practical in the implementation.

 This attribute specifies the value of the maximum size of the window that is used by a STA for a particular AC for generating a random number for the backoff. The value of this attribute is such that it could always be expressed in the form of $2^X - 1$, where X is an integer."
::= { dot11MUEDECAEntry 3 }

dot11MUEDCATableAIFSN OBJECT-TYPE
SYNTAX Unsigned32 (0..15)
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "This is a control variable.
 It is written by the MAC upon receiving an MU EDCA Parameter Set element.
 Changes take effect as soon as practical in the implementation.

 This attribute specifies the number of slots, after a SIFS, that the STA, for a particular AC, senses the medium idle either before transmitting or executing a backoff, except that the value 0 indicates that EDCA is disabled for that AC."
::= { dot11MUEDECAEntry 4 }

dot11MUEDCATableTimer OBJECT-TYPE
SYNTAX Unsigned32 (1..255)
UNITS "8 TUs"
MAX-ACCESS read-only
STATUS current
DESCRIPTION
 "This is a control variable.
 It is written by an external management entity.

Changes take effect as soon as the STA sends a frame in response to a Basic Trigger frame.

This attribute specifies the duration of time during which the HE STA uses the MU EDCA parameters for the corresponding AC."

REFERENCE

"26.2.7 (EDCA operation using MU EDCA parameters)"

::= { dot11MUEDECAEntry 5 }

```
-- *****
-- *      End of SMT MU EDCA Config TABLE
-- *****
```

Change the dot11PHYType element definition in the “dot11PhyOperation TABLE” in C.3 as follows:

dot11PHYType OBJECT-TYPE

SYNTAX INTEGER {

 dsss(2),

 ofdm(4),

 hrdsss(5),

 erp(6),

 ht(7)

 dmg(8),

 vht(9),

 tvht(10),

 slg(11),

 cdmg(12),

 cmmg(13),

 he(14)}

MAX-ACCESS read-only

STATUS current

DESCRIPTION

 "This is a status variable.

 It is written by the PHY.

This is an 8-bit integer value that identifies the supported PHY type. Currently defined values and their corresponding PHY types are:

DSSS 2.4 GHz = 02, OFDM = 04, HRDSSS = 05, ERP = 06, HT = 07, DMG = 08, VHT = 09, TVHT = 10, SLG = 11, CDMG = 12, CMMG = 13, HE = 14"

::= { dot11PhyOperationEntry 1 }

Change the dot11PhyVHTEntry SEQUENCE list in the “dot11Phy VHT TABLE” in C.3 as follows:

Dot11PhyVHTEntry ::=

 SEQUENCE {

 dot11VHTChannelWidthOptionImplemented

 INTEGER,

 dot11CurrentChannelWidth

 INTEGER,

 dot11CurrentChannelCenterFrequencyIndex0

 Unsigned32,

 dot11CurrentChannelCenterFrequencyIndex1

 Unsigned32,

 dot11VHTShortGIOptionIn80Implemented

 TruthValue,

 dot11VHTShortGIOptionIn80Activated

 TruthValue,

 dot11VHTShortGIOptionIn160and80p80Implemented

 TruthValue,

 dot11VHTShortGIOptionIn160and80p80Activated

 TruthValue,

 dot11VHTLDPCCodingOptionImplemented

 TruthValue,

 dot11VHTLDPCCodingOptionActivated

 TruthValue,

 dot11VHTTxSTBCOptionImplemented

 TruthValue,

 dot11VHTTxSTBCOptionActivated

 TruthValue,

 dot11VHTRxSTBCOptionImplemented

 TruthValue,

 dot11VHTRxSTBCOptionActivated

 TruthValue,

 dot11VHTMUMaxUsersImplemented

 Unsigned32,

```

dot11VHTMUMaxNSTSPerUserImplemented      Unsigned32,
dot11VHTMUMaxNSTSTotalImplemented        Unsigned32,
dot11VHTMaxNTxChainsImplemented          Unsigned32,
dot11VHTMaxNTxChainsActivated            Unsigned32,
dot11EightyMHzOperationImplemented        TruthValue
}

```

Insert the following element definition into the “dot11 Phy VHT TABLE” in C.3 after the dot11VHTMaxNTxChainsActivated element definition:

```

dot11EightyMHzOperationImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that the 80 MHz operation is
        implemented"
    DEFVAL { false }
    ::= { dot11PhyVHTEntry 20 }

```

Insert the “dot11 Phy HE TABLE” and “dot11 HE Transmit Beamforming Config TABLE” into C.3 after the “dot11 TVHT Transmit Beamforming Config TABLE”:

```

-- *****
-- * dot11 Phy HE TABLE
-- *****
dot11PhyHETable OBJECT-TYPE
    SYNTAX SEQUENCE OF Dot11PhyHEEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "Entry of attributes for dot11PhyHETable. Implemented as a table indexed
        on ifIndex to allow for multiple instances on an Agent."
    ::= { dot11phy 31 }

dot11PhyHEEntry OBJECT-TYPE
    SYNTAX Dot11PhyHEEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "An entry in dot11PhyHEEntryTable. ifIndex - Each IEEE Std 802.11
        interface is represented by an ifEntry. Interface tables in this MIB
        module are indexed by ifIndex."
    INDEX {ifIndex}
    ::= { dot11PhyHETable 1 }

Dot11PhyHEEntry ::=
    SEQUENCE {
        dot11HECurrentChannelWidthSet      Unsigned32,
        dot11HEPuncturedPreambleRxImplemented OCTET STRING,
        dot11HEDeviceClass                  TruthValue,
        dot11HELDPCCodingInPayloadImplemented TruthValue,
        dot11HESUPPDUwith1xHELTfand0point8GIImplemented TruthValue,
        dot11HESUPPDUandHEMUPPDUwith4xHELTfand0point8GIImplemented TruthValue,
        dot11HEERSUPPDUwith4xHELTfand0point8GIImplemented TruthValue,
        dot11HEERSUPPDUwith1xHELTfand0point8GIImplemented TruthValue,

```

```

dot11HEMidambleRxMaxNSTS                               Unsigned32,
dot11HENDPwith4xHELTfand3point2GIImplemented          TruthValue,
dot11HESTBCTxLessThanOrEqualTo80Implemented            TruthValue,
dot11HESTBCRxLessThanOrEqualTo80Implemented            TruthValue,
dot11HESTBCTxGreaterThan80Implemented                  TruthValue,
dot11HESTBCRxGreaterThan80Implemented                  TruthValue,
dot11HEDopplerTxImplemented                             TruthValue,
dot11HEDopplerRxImplemented                             TruthValue,
dot11HEDCMImplemented                                  TruthValue,
dot11HEFullBWULMUMIMOImplemented                       TruthValue,
dot11HEPartialBWULMUMIMOImplemented                    TruthValue,
dot11HEPartialBWDLMUMIMOImplemented                    TruthValue,
dot11HEULMUPayloadImplemented                           TruthValue,
dot11HEPSRbasedSRSupportImplemented                    TruthValue,
dot11HEPowerBoostFactorImplemented                      TruthValue,
dot11HEPartialBWERSUPayloadImplemented                  TruthValue,
dot11HEPuncturedSoundingOptionImplemented              TruthValue
}

dot11HECurrentChannelWidthSet OBJECT-TYPE
    SYNTAX Unsigned32 (0..127)
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a status variable.

        This attribute specifies the STA's channel width set."
    REFERENCE
        "IEEE Std 802.11ax-2020, Table 9-322b (Subfields of the HE PHY
        Capabilities Information field), Support Channel Width Set subfield"
 ::= { dot11PhyHEEntry 1 }

dot11HEPuncturedPreambleRxImplemented OBJECT-TYPE
    SYNTAX OCTET STRING(SIZE(1))
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute is a bitmap that indicates the preamble pruncturing
        support, where bit 0 is set to 1 to indicate support for the reception of
        an 80 MHz preamble where the only punctured subchannel is the secondary 20
        MHz subchannel, bit 1 is set to 1 to indicate support for the reception of
        an 80 MHz preamble where the only punctured subchannel is one of the two
        20 MHz subchannels in the secondary 40 MHz channel, bit 2 is set to 1 to
        indicate support for the reception of a 160 MHz or 80+80 MHz preamble
        where the only punctured subchannels are the secondary 20 MHz channel and
        zero to two of the 20 MHz subchannels in the secondary 80 MHz channel, and
        bit 3 is set to 1 to indicate support for the reception of a 160 MHz or
        80+80 MHz preamble where the only punctured subchannels are zero, one or
        both of the 20 MHz subchannels in the secondary 40 MHz channel and zero to
        two of the 20 MHz subchannels in the secondary 80 MHz channel with at
        least one 20 MHz subchannel punctured. For bits 2 and 3, if two of the 20
        MHz subchannels in the secondary 80 MHz channel are punctured, these are
        either the lower two or the higher two. No more than two adjacent 20 MHz
        subchannels are punctured across a 160 MHz preamble."
 ::= { dot11PhyHEEntry 2 }

dot11HEDeviceClass OBJECT-TYPE
    SYNTAX INTEGER{ClassB(0), ClassA(1)}
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION

```

```

    "This is a capability variable.
    Its value is determined by device capabilities.
    This attribute specifies the device class of a non-AP STA."
 ::= { dot11PhyHEEntry 3 }

dot11HELDPCCodingInPayloadImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that the STA is capable of
        transmitting and receiving LDPC encoded packets. This capability is
        disabled otherwise."
    DEFVAL { false }
 ::= { dot11PhyHEEntry 4 }

dot11HESUPPDUwith1xHELTfand0point8GIImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that the non-AP STA is capable of
        receiving an HE SU PPDU with 1x HE-LTF and 0.8 microsecond guard interval
        duration. This capability is disabled otherwise."
    DEFVAL { false }
 ::= { dot11PhyHEEntry 5}

dot11HESUPPDUandHEMUPPDUwith4xHELTfand0point8GIImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that the non-AP STA is capable of
        receiving an HE SU PPDU and HE MU PPDU with 4x HE-LTF and 0.8 microsecond
        guard interval duration. This capability is disabled otherwise."
    DEFVAL { false }
 ::= { dot11PhyHEEntry 6}

dot11HEERSUPPDUwith4xHELTfand0point8GIImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that the non-AP STA is capable of
        receiving an HE ER SU PPDU with 4x HE-LTF and 0.8 microsecond guard
        interval duration. This capability is disabled otherwise."
    DEFVAL { false }
 ::= { dot11PhyHEEntry 7}

dot11HEERSUPPDUwith1xHELTfand0point8GIImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only

```

```
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that the non-AP STA is capable of
    receiving an HE ER SU PPDU with 1x HE-LTF and 0.8 microsecond guard
    interval duration. This capability is disabled otherwise."
    DEFVAL { false }
::= { dot11PhyHEEntry 8}

dot11HEMidambleRxMaxNSTS OBJECT-TYPE
    SYNTAX Unsigned32 (0..3)
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute specifies the maximum number of space-time streams
        supported for reception when a midamble is present in the Data field,
        equal to 0 for 1 space-time stream, equal to 1 for 2 space-time streams,
        equal to 2 for 3 space-time streams, and equal to 3 for 4 space-time
        streams."
        DEFVAL { 0 }
::= { dot11PhyHEEntry 9 }

dot11HENDPwith4xHELTFand3point2GIImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that the non-AP STA is capable of
        receiving an NDP with 4x HE-LTF and 3.2 microsecond guard interval
        duration. This capability is disabled otherwise."
        DEFVAL { false }
::= { dot11PhyHEEntry 10 }

dot11HESTBCTxLessThanOrEqualTo80Implemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that the non-AP STA is capable of
        transmitting an HE PPDU that has a bandwidth less than or equal to 80 MHz
        and is using STBC with one spatial stream. This capability is disabled
        otherwise."
        DEFVAL { false }
::= { dot11PhyHEEntry 11 }

dot11HESTBCRxLessThanOrEqualTo80Implemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.
```


This attribute, when true, indicates that the non-AP STA is capable of receiving an HE PPDU that has a bandwidth less than or equal to 80 MHz and is using STBC with one spatial stream. This capability is disabled otherwise."

```

DEFVAL { false }
 ::= { dot11PhyHEEntry 12 }

dot11HESTBCTxGreaterThan80Implemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that the non-AP STA is capable of
    transmitting an HE PPDU that has a bandwidth greater than 80 MHz and is
    using STBC with one spatial stream. This capability is disabled
    otherwise."
DEFVAL { false }
 ::= { dot11PhyHEEntry 13 }

dot11HESTBCRxGreaterThan80Implemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that the non-AP STA is capable of
    receiving an HE PPDU that has a bandwidth greater than 80 MHz and is using
    STBC with one spatial stream. This capability is disabled otherwise."
DEFVAL { false }
 ::= { dot11PhyHEEntry 14 }

dot11HEDopplerTxImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that the non-AP STA is capable of
    transmitting HE PPDU with midamble. This capability is disabled
    otherwise."
DEFVAL { false }
 ::= { dot11PhyHEEntry 15 }

dot11HEDopplerRxImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that the non-AP STA is capable of
    receiving HE PPDU with midamble. This capability is disabled otherwise."
DEFVAL { false }
 ::= { dot11PhyHEEntry 16 }

dot11HEDCMImplemented OBJECT-TYPE

```

```

SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that the non-AP STA implementation
    supports DCM. This capability is disabled otherwise."
 ::= { dot11PhyHEEntry 17 }

dot11HEFullBWULMUMIMOImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that for an AP implementation, the
    MU-MIMO reception of an HE TB PPDU on an RU that spans the entire PPD
    bandwidth is supported; and for a non-AP STA implementation, the
    transmission of an HE TB PPDU on an RU that spans the entire PPDU
    bandwidth is supported. This capability is disabled otherwise."
    DEFVAL { false }
 ::= { dot11PhyHEEntry 18 }

dot11HEPartialBWULMUMIMOImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that an AP is capable of receiving an
    RU in an HE TB PPDU where MU-MIMO is employed in the RU, the RU size is
    greater than or equal to 106 tones, and the RU does not span the entire
    PPDU bandwidth; and a non-AP STA is capable of transmitting an RU in an HE
    TB PPDU where MU-MIMO is employed in the RU, the RU size is greater than
    or equal to 106 tones, and the RU does not span the entire PPDU bandwidth.
    This capability is disabled otherwise."
    DEFVAL { false }
 ::= { dot11PhyHEEntry 19 }

dot11HEPartialBWDLMUMIMOImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that the non-AP STA is capable of
    receiving a DL MU-MIMO transmission on an RU in an HE MU PPDU where the RU
    does not span the entire PPDU bandwidth. This capability is disabled
    otherwise."
    DEFVAL { false }
 ::= { dot11PhyHEEntry 20 }

dot11HEULMUPayloadImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current

```

DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the AP is capable of receiving the payload on an RU in an HE MU PPDU where RU spans the entire PPDU bandwidth or a 106-tone RU within 20 MHz PPDU bandwidth. This capability is disabled otherwise."

DEFVAL { false }
::= { dot11PhyHEEntry 21 }

dot11HEPSRbasedSRSupportImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the non-AP STA is capable of supporting the PSR-based SR operation. This capability is disabled otherwise."

DEFVAL { false }
::= { dot11PhyHEEntry 22 }

dot11HEPowerBoostFactorImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the non-AP STA implementation supports a power boost factor for the r-th RU in the range [0.5, 2]. This capability is disabled otherwise."

DEFVAL { false }
::= { dot11PhyHEEntry 23 }

dot11HEPartialBWERSUPayloadImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the non-AP STA is capable of transmitting and receiving an HE ER SU PPDU in which the HE modulated fields are transmitted over the higher frequency 106-tone RU within primary 20 MHz channel. This capability is disabled otherwise."

DEFVAL { false }
::= { dot11PhyHEEntry 24 }

dot11HEPuncturedSoundingOptionImplemented OBJECT-TYPE
SYNTAX TruthValue
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the STA implementation is capable of operating in a mode where some 242-tone RUs are not allowed to

```

    be used within a channel of width 80 MHz or 160 MHz. The capability is
    disabled, otherwise"
  DEFVAL { false }
  ::= { dot11PhyHEEntry 25 }

-- *****
-- * End of dot11 Phy HE TABLE
-- *****

-- *****
-- * dot11 HE Transmit Beamforming Config TABLE
-- *****

dot11HETransmitBeamformingConfigTable OBJECT-TYPE
  SYNTAX SEQUENCE OF Dot11HETransmitBeamformingConfigEntry
  MAX-ACCESS not-accessible
  STATUS current
  DESCRIPTION
    "Entry of attributes for dot11HETransmitBeamformingConfigTable.
    Implemented as a table indexed on ifIndex to allow for multiple instances
    on an Agent."
  ::= { dot11phy 32 }

dot11HETransmitBeamformingConfigEntry OBJECT-TYPE
  SYNTAX Dot11HETransmitBeamformingConfigEntry
  MAX-ACCESS not-accessible
  STATUS current
  DESCRIPTION
    "An entry in dot11HETransmitBeamformingConfigTable.
    ifIndex - Each IEEE 802.11 interface is represented by an ifEntry.
    Interface tables in this MIB module are indexed by ifIndex."
  INDEX {ifIndex}
  ::= { dot11HETransmitBeamformingConfigTable 1 }

Dot11HETransmitBeamformingConfigEntry ::=
  SEQUENCE {
    dot11HESUBeamformerOptionImplemented                TruthValue,
    dot11HESUBeamformeeOptionImplemented                TruthValue,
    dot11HEMUBeamformerOptionImplemented                TruthValue,
    dot11HEBeamformeeSTSSupportLessThanOrEqualTo80     Unsigned32,
    dot11HEBeamformeeSTSSupportGreaterThan80           Unsigned32,
    dot11HENumberSoundingDimensionsLessThanOrEqualTo80 Unsigned32,
    dot11HENumberSoundingDimensionsGreaterThan80        Unsigned32,
    dot11HENG16SUFeedbackSupport                       TruthValue,
    dot11HENG16MUFeedbackSupport                       TruthValue,
    dot11HECodebookSizePhi4Psi2SUFeedbackSupport       TruthValue,
    dot11HECodebookSizePhi7Psi5MUFeedbackSupport       TruthValue,
    dot11HETriggeredSUBeamformingFeedbackImplemented   TruthValue,
    dot11HETriggeredMUBeamformingFeedbackImplemented   TruthValue,
    dot11HETriggeredCQIFeedbackSupportImplemented       TruthValue
  }

dot11HESUBeamformerOptionImplemented OBJECT-TYPE
  SYNTAX TruthValue
  MAX-ACCESS read-only
  STATUS current
  DESCRIPTION
    "This is a capability variable.
    Its value is determined by device capabilities.

    This attribute, when true, indicates that operation as an SU beamformer is
    supported. This attribute, when false, indicates that operation as an SU

```

```

    beamformer is not supported."
    DEFVAL { false }
    ::= { dot11HETransmitBeamformingConfigEntry 1 }

dot11HESUBeamformeeOptionImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that for an AP implementation
        operation as an SU beamformee is supported; for a non-AP STA
        implementation operation as an SU beamformee is supported; for an AP
        implementation operation as an SU beamformer is mandatory if the AP
        supports 4 or more spatial streams. This attribute, when false, indicates
        that for an AP implementation operation as an SU beamformee is not
        supported."
    DEFVAL { false }
    ::= { dot11HETransmitBeamformingConfigEntry 2 }

dot11HEMUBeamformerOptionImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that for an AP implementation
        operation as an MU beamformer is supported when the SU Beamformer field is
        1. This attribute, when false, indicates that for an AP implementation
        operation as an MU beamformer is not supported; for a non-AP STA
        implementation operation as an MU beamformer is not supported."
    DEFVAL { false }
    ::= { dot11HETransmitBeamformingConfigEntry 3 }

dot11HEBeamformeeSTSSupportLessThanOrEqualTo80 OBJECT-TYPE
    SYNTAX Unsigned32
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute indicates the maximum number of space-time streams that the
        non-AP STA can receive in an HE sounding NDP, the maximum value for the
        total number of space-time streams over all the users in RU r,
        NSTSr,total that can be sent in a DL MU-MIMO transmission on an RU where
        the RU might or might not span the entire PPDU bandwidth, which includes
        that non-AP STA."
    ::= { dot11HETransmitBeamformingConfigEntry 4 }

dot11HEBeamformeeSTSSupportGreaterThan80 OBJECT-TYPE
    SYNTAX Unsigned32
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute indicates the maximum number of space-time streams that the
        non-AP STA can receive in an HE sounding NDP, the maximum value for the

```

total number of space-time streams over all the users in RU r , $N_{STS,r}$, total that can be sent in a DL MU-MIMO transmission on an RU where the RU might or might not span the entire PPDU bandwidth, which includes that STA."

```
 ::= { dot11HETransmitBeamformingConfigEntry 5 }
```

dot11HENumberSoundingDimensionsLessThanOrEqualTo80 OBJECT-TYPE
 SYNTAX Unsigned32
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "This is a capability variable.
 Its value is determined by device capabilities.

 This attribute specifies, for bandwidth less than or equal to 80 MHz, the beamformer's capability to indicate the maximum value of the TXVECTOR parameter NUM_STS for an HE sounding NDP."

```
 ::= { dot11HETransmitBeamformingConfigEntry 6 }
```

dot11HENumberSoundingDimensionsGreaterThan80 OBJECT-TYPE
 SYNTAX Unsigned32
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "This is a capability variable.
 Its value is determined by device capabilities.

 This attribute specifies, for bandwidth greater than 80 MHz, the beamformer's capability to indicate the maximum value of the TXVECTOR parameter NUM_STS for an HE sounding NDP."

```
 ::= { dot11HETransmitBeamformingConfigEntry 7 }
```

dot11HENG16SUFeedbackSupport OBJECT-TYPE
 SYNTAX TruthValue
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "This is a capability variable.
 Its value is determined by device capabilities.

 This attribute, when true, indicates the HE beamformee support for a tone grouping of 16 in the HE Compressed Beamforming Report field for SU feedback. This capability is disabled otherwise."

```
 DEFVAL { false }
 ::= { dot11HETransmitBeamformingConfigEntry 8 }
```

dot11HENG16MUFeedbackSupport OBJECT-TYPE
 SYNTAX TruthValue
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "This is a capability variable.
 Its value is determined by device capabilities.

 This attribute, when true, indicates the HE beamformee support for a tone grouping of 16 in the HE Compressed Beamforming Report field for MU feedback. This capability is disabled otherwise."

```
 DEFVAL { false }
 ::= { dot11HETransmitBeamformingConfigEntry 9 }
```

dot11HECodebookSizePhi4Psi2SUFeedbackSupport OBJECT-TYPE
 SYNTAX TruthValue
 MAX-ACCESS read-only
 STATUS current

DESCRIPTION

"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates the HE beamformee support for a codebook size (psi, phy) = {4, 2} in the HE Compressed Beamforming Report field for SU feedback. This capability is disabled otherwise."

DEFVAL { false }

::= { dot11HETransmitBeamformingConfigEntry 10 }

dot11HECodebookSizePhi7Psi5MUFeedbackSupport OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates the HE beamformee support for a codebook size (psi, phy) = {7, 5} in the HE Compressed Beamforming Report field for MU feedback. This capability is disabled otherwise."

DEFVAL { false }

::= { dot11HETransmitBeamformingConfigEntry 11 }

dot11HETriggeredSUBeamformingFeedbackImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that for an AP implementation, the reception of partial- and full-bandwidth SU feedback in an HE TB sounding sequence is supported; for a non-AP STA implementation, the transmission of partial- and full-bandwidth SU feedback in an HE TB sounding sequence is supported. This capability is disabled otherwise."

DEFVAL { false }

::= { dot11HETransmitBeamformingConfigEntry 12 }

dot11HETriggeredMUBeamformingFeedbackImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that for an AP implementation, the reception of partial-bandwidth MU feedback in an HE TB sounding sequence is supported; for a non-AP STA implementation, the transmission of partial-bandwidth MU feedback in an HE TB sounding sequence is supported. This capability is disabled otherwise."

DEFVAL { false }

::= { dot11HETransmitBeamformingConfigEntry 13 }

dot11HETriggeredCQIFeedbackSupportImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.
Its value is determined by device capabilities.

```

    This attribute, when true, indicates that for an AP implementation, the
    reception of partial- and full-bandwidth CQI feedback in an HE TB sounding
    sequence is supported; for a non-AP STA implementation, the transmission
    of partial- and full-bandwidth CQI feedback in an HE TB sounding sequence
    is supported. This capability is disabled otherwise."
    DEFVAL { false }
    ::= { dot11HETransmitBeamformingConfigEntry 14 }

-- *****
-- * End of dot11 HE Transmit Beamforming Config TABLE
-- *****

```

Insert the following compliance group definitions into the “Groups - units of compliance - RSN” part of C.3 after the dot11CMMGComplianceGroup group definition:

```

dot11HEComplianceGroup OBJECT-GROUP
  OBJECTS {
    dot11HEOptionImplemented,
    dot11OBSSNarrowBWRUinOFDMATolerated,
    dot11HE6GOptionImplemented,
    dot11OCTOptionImplemented,
    dot11QTPOptionImplemented,
    dot11TRSOOptionImplemented,
    dot11OFDMARandomAccessOptionImplemented,
    dot11HEControlFieldOptionImplemented,
    dot11OMIOptionImplemented,
    dot11HEMCSFeedbackOptionImplemented,
    dot11HEDynamicFragmentationLevel,
    dot11AMPDUwithMultipleTIDOptionImplemented,
    dot11MPDUAskedforAckInMultiTIDAMPDU,
    dot11TXOPDurationRTSThreshold,
    dot11PPEThresholdsRequired,
    dot11IntraPPDUPowerSaveOptionActivated,
    dot11AMSDUFragmentationOptionImplemented,
    dot11BSSColorCollisionAPPeriod,
    dot11BSSColorCollisionSTAPeriod,
    dot11AutonomousBSSColorCollisionReportingImplemented,
    dot11HEPSROptionImplemented,
    dot11HEBSRControlImplemented,
    dot11HEBQRControlImplemented,
    dot11HECASControlImplemented,
    dot11PartialBSSColorImplemented,
    dot11ObssNbRuToleranceTime,
    dot11HESubchannelSelectiveTransmissionImplemented,
    dot11SRResponderOptionImplemented,
    dot11AutonomousBSSColorInUseReportingImplemented,
    dot11ShortSSIDListImplemented,
    dot11ColocatedRNRImplemented,
    dot11SRGAPOBSSPDMinOffset,
    dot11SRGAPOBSSPDMaxOffset,
    dot11SRGAPBSSColorBitmap,
    dot11SRGAPBSSIDBitmap,
    dot11NonSRGAPOBSSPDMaxOffset,
    dot11HTVHTTriggerOptionImplemented,
    dot11HEDynamicSMPowerSaveOptionImplemented,
    dot11MUEDCAParametersActivated,
    dot11CoHostedBSSIDImplemented,
    dot11UnsolicitedProbeResponseOptionActivated,
    dot11MemberOfColocated6GHzESSOptionActivated,
    dot11AckEnabledAMPDUOptionImplemented,
    dot11MinPSCPProbeDelay,
  }

```



```

    dot11OCWmin,
    dot11OCWmax}
STATUS current
DESCRIPTION
    "Attributes that configure the HE Group for IEEE 802.11."
::= { dot11Groups 100 }

dot11SMTbasel6 OBJECT-GROUP
OBJECTS {
    dot11HEOptionImplemented
}
STATUS current
DESCRIPTION
    "The SMTbasel6 object class provides the necessary support at the STA to
    manage the processes in the STA such that the STA may work cooperatively
    as a part of an IEEE 802.11 network."
::= { dot11Groups 101 }

dot11HETransmitBeamformingGroup OBJECT-GROUP
OBJECTS {
    dot11HESUBeamformerOptionImplemented,
    dot11HESUBeamformeeOptionImplemented,
    dot11HEMUBeamformerOptionImplemented,
    dot11HEBeamformeeSTSSupportLessThanOrEqualTo80,
    dot11HEBeamformeeSTSSupportGreaterThan80,
    dot11HENumberSoundingDimensionsLessThanOrEqualTo80,
    dot11HENumberSoundingDimensionsGreaterThan80,
    dot11HENG16SUFeedbackSupport,
    dot11HENG16MUFeedbackSupport,
    dot11HECodebookSizePhi4Psi2SUFeedbackSupport,
    dot11HECodebookSizePhi7Psi5MUFeedbackSupport,
    dot11HETriggeredSUBeamformingFeedbackImplemented,
    dot11HETriggeredMUBeamformingFeedbackImplemented,
    dot11HETriggeredCQIFeedbackSupportImplemented }
STATUS current
DESCRIPTION
    "Attributes that configure HE transmit beamforming for IEEE 802.11."
::= { dot11Groups 102 }

dot11PhyHEComplianceGroup OBJECT-GROUP
OBJECTS {
    dot11HECurrentChannelWidthSet,
    dot11HEPuncturedPreambleRxImplemented,
    dot11HEPuncturedPreambleRxActivated,
    dot11HEDeviceClass,
    dot11HELDPCCodingInPayloadImplemented,
    dot11HESUPPDUwith1xHELTfand0point8GIImplemented,
    dot11HESUPPDUandHEMUPPDUwith4xHELTfand0point8GIImplemented,
    dot11HEERSUPPDUwith4xHELTfand0point8GIImplemented,
    dot11HEERSUPPDUwith1xHELTfand0point8GIImplemented,
    dot11HEMidambleRxMaxNSTS,
    dot11HENDPwith4xHELTfand3point2GIImplemented,
    dot11HESTBCTxLessThanOrEqualTo80Implemented,
    dot11HESTBCRxLessThanOrEqualTo80Implemented,
    dot11HESTBCTxGreaterThan80Implemented,
    dot11HESTBCRxGreaterThan80Implemented,
    dot11HEDopplerTxImplemented,
    dot11HEDopplerRxImplemented,
    dot11HEDCMImplemented,
    dot11HEFullBWULMUMIMOImplemented,
    dot11HEPartialBWULMUMIMOImplemented,
    dot11HEPartialBWDLMUMIMOImplemented,
    dot11HEULMUPayloadImplemented,
    dot11HEPSRbasedSRSupportImplemented,

```

```

    dot11HEPowerBoostFactorImplemented,
    dot11HEPartialBWERSUPayloadImplemented}
STATUS current
DESCRIPTION
    "Attributes that configure the HE PHY."
::= { dot11Groups 103 }

```

Change the DESCRIPTION in the dot11SMTbase15 group definition in the “Groups - units of compliance - RSN” part in C.3 as follows (entire definition is not shown):

```

dot11SMTbase15 OBJECT-GROUP
DESCRIPTION
    "Superseded by dot11SMTbase16.
    The SMTbase15 object class provides the necessary support at the STA to
    manage the processes in the STA such that the STA may work cooperatively
    as a part of an IEEE 802.11 network."
::= { dot11Groups 109 }

```

Change the beginning of “dot11Compliance MODULE-COMPLIANCE” in the “Compliance Statements” part of C.3 as follows:

```

dot11Compliance MODULE-COMPLIANCE
STATUS current
DESCRIPTION
    "The compliance statement for SNMPv2 entities that implement the IEEE
    802.11 MIB."
MODULE -- this module
MANDATORY-GROUPS {
    dot11SMTbase15 dot11SMTbase16,
    dot11MACbase5,
    dot11CountersGroup5,
    dot11SmtAuthenticationAlgorithms,
    dot11ResourceTypeID,
    dot11PhyOperationComplianceGroup2 }

GROUP dot11PhyDSSSComplianceGroup
DESCRIPTION
    "Implementation of this group is required when object dot11PHYType is
    dsss.
    This group is mutually exclusive to the following groups:
    dot11PhyOFDMComplianceGroup3
    dot11PhyHRDSSSComplianceGroup
    dot11PhyERPComplianceGroup
    dot11PhyHTComplianceGroup
    dot11DMGComplianceGroup
    dot11PhyVHTComplianceGroup
    dot11PhyTVHTComplianceGroup
    dot11PhyS1GComplianceGroup
    dot11CDMGComplianceGroup1
    dot11CMMGComplianceGroup
    dot11PhyHEComplianceGroup"

GROUP dot11PhyOFDMComplianceGroup3
DESCRIPTION
    "Implementation of this group is required when object dot11PHYType is
    ofdm.
    This group is mutually exclusive to the following groups:
    dot11PhyDSSSComplianceGroup
    dot11PhyHRDSSSComplianceGroup

```

```
dot11PhyERPComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup"
```

GROUP dot11PhyHRDSSSComplianceGroup
DESCRIPTION
"Implementation of this group is required when object dot11PHYType is
hrdsss.
This group is mutually exclusive to the following groups:
dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyERPComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup"

GROUP dot11PhyERPComplianceGroup
DESCRIPTION
"Implementation of this group is required when object dot11PHYType is ERP.
This group is mutually exclusive to the following groups:
dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup"

GROUP dot11PhyHTComplianceGroup
DESCRIPTION
"Implementation of this group is required when object dot11PHYType has the
value of ht.
This group is mutually exclusive to the following groups:
dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup"

GROUP dot11PhyVHTComplianceGroup
DESCRIPTION
"Implementation of this group is required when object dot11PHYType has the
value of vht.

This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11DMGComplianceGroup
dot11PhyHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup"

GROUP dot11PhyTVHTComplianceGroup
DESCRIPTION

"Implementation of this group is required when object dot11PHYType has the value of tvht.

This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup"

GROUP dot11PhyS1GComplianceGroup
DESCRIPTION

"Implementation of this group is required when object dot11PHYType has the value of slg.

This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup
dot11PhyHEComplianceGroup"

GROUP dot11PhyHEComplianceGroup
DESCRIPTION

"Implementation of this group is required when object dot11PHYType has the value of HE.

This group is mutually exclusive to the following groups:

dot11PhyDSSSComplianceGroup
dot11PhyOFDMComplianceGroup3
dot11PhyHRDSSSComplianceGroup
dot11PhyERPComplianceGroup
dot11PhyHTComplianceGroup
dot11DMGComplianceGroup
dot11PhyVHTComplianceGroup
dot11PhyTVHTComplianceGroup
dot11PhyS1GComplianceGroup
dot11CDMGComplianceGroup1
dot11CMMGComplianceGroup"

Insert the following groups into “dot11Compliance MODULE-COMPLIANCE” in the “Compliance Statements” part of C.3 after “GROUP dot11VHTMACAdditions”:

```
GROUP dot11HETransmitBeamformingGroup
DESCRIPTION
    "The dot11HETransmitBeamformingGroup group is optional."

GROUP dot11HEComplianceGroup
DESCRIPTION
    "The dot11HEComplianceGroup group is optional."
```

Change “GROUP dot11DMGComplianceGroup” in “dot11Compliance MODULE-COMPLIANCE” in the “Compliance Statements” part of C.3 as follows:

```
GROUP dot11DMGComplianceGroup
DESCRIPTION
    "Implementation of this group is required when the object dot11PHYType
    is dmg.
    This group is mutually exclusive to the following groups:
    dot11PhyDSSSComplianceGroup
    dot11PhyOFDMComplianceGroup3
    dot11PhyHRDSSSComplianceGroup
    dot11PhyERPComplianceGroup
    dot11PhyHTComplianceGroup
    dot11PhyVHTComplianceGroup
    dot11PhyTVHTComplianceGroup
    dot11PhyS1GComplianceGroup
    dot11CDMGComplianceGroup1
    dot11CMMGComplianceGroup
    dot11PhyHEComplianceGroup"
```

Change OPTIONAL-GROUPS list in the “Compliance Statements” part of C.3 as follows:

```
-- OPTIONAL-GROUPS {
-- dot11SMTprivacy
-- dot11MACStatistics,
-- dot11PhyTxPowerComplianceGroup,
-- dot11PhyRegDomainsSupportGroup,
-- dot11PhyAntennasListGroup,
-- dot11PhyRateGroup,
-- dot11MultiDomainCapabilityGroup,
-- dot11RSNAAdditions,
-- dot11OperatingClassesGroup,
-- dot11Qosadditions,
-- dot11RMCompliance,
-- dot11FTComplianceGroup,
-- dot11PhyAntennaComplianceGroup2,
-- dot11HTMACAdditions3,
-- dot11PhyMCSGroup,
-- dot11TransmitBeamformingGroup,
-- dot11TVHTTransmitBeamformingGroup,
-- dot11PhyTVHTComplianceGroup,
-- dot11S1GTransmitBeamformingGroup,
-- dot11PhyS1GComplianceGroup,
-- dot11S1GComplianceGroup,
-- dot11WNMCompliance,
-- dot11BSSStatisticsGroup,
-- dot11VHTTransmitBeamformingGroup,
-- dot11PhyVHTComplianceGroup,
-- dot11VHTMACAdditions,
```

```
-- dot11TVWSComplianceGroup,  
-- dot11FILSComplianceGroup,  
-- dot11PADComplianceGroup_  
-- dot11HETransmitBeamformingGroup,  
-- dot11PhyHEComplianceGroup,  
-- dot11HEComplianceGroup}
```

Insert the following part into C.3 after the “Compliance Statements - GLK” part:

```
-- *****  
-- * Compliance Statements - HE  
-- *****  
dot11HECompliance MODULE-COMPLIANCE  
    STATUS current  
    DESCRIPTION  
        "This object class provides the objects from the IEEE 802.11  
        MIB used to operate at high efficiency."  
    MODULE -- this module  
    MANDATORY-GROUPS { dot11PhyHEComplianceGroup,  
        dot11PhyTxPowerComplianceGroup2, dot11HETransmitBeamformingGroup,  
        dot11HEComplianceGroup }  
-- OPTIONAL-GROUPS { }  
::= { dot11Compliances 22 }
```

Annex D

(normative)

Regulatory references

D.1 External regulatory references

Change the following row in Table D-1:

Table D-1—Regulatory requirement list

Geographic area	Approval standards	Documents	Approval authority
United States	Federal Communications Commission (FCC)	47 CFR [B6], Part 15, Sections 15.205, 15.209, 15.247 and 15.255; and Subpart E, Sections 15.401-15.407, and Subpart H, Sections 15.701-15.716, Section 90.210, Sections 90.371-383, Sections 90.1201-90.1217, Sections 90.1301-90.1337, Section 95.639, Sections 95.1501-1511 <u>Notice of Proposed Rulemaking, 34 FCC Rcd. 12603 (15)^a</u>	FCC

^a The FCC adopted on 12 December 2019 a Notice of Proposed Rulemaking proposing to make the 5.9 GHz band's lower 45 MHz available for unlicensed operations.

Annex E

(normative)

Country elements and operating classes

E.1 Country information and operating classes

Change Table E-4 as follows (not all rows are shown):

Table E-4—Global operating classes

Operating class	Nonglobal operating class(es)	Channel starting frequency (GHz)	Channel spacing (MHz)	Channel set	Channel center frequency index	Behavior limits set
...						
125	E-1-5, E-2-17, E-6-3	5	20	149, 153, 157, 161, 165, 169, 173, <u>177</u>		LicenseExemptBehavior, UseEirpForVHTTxPowEnv
126	E-1-25,26, E-6-6	5	40	149, 157, 165, <u>173</u>		PrimaryChannelLowerBehavior, UseEirpForVHTTxPowEnv
127	E-1-30,31	5	40	153, 161, 169, <u>177</u>		PrimaryChannelUpperBehavior, UseEirpForVHTTxPowEnv
128	E-1-128, E-2-128, E-3-128, E-6-128	5	80	—	42, 58, 106, 122, 138, 155, <u>171</u>	UseEirpForVHTTxPowEnv
129	E-1-129, E-2-129, E-3-129, E-6-129	5	160	—	50, 114, <u>163</u>	UseEirpForVHTTxPowEnv
130	E-1-130, E-2-130, E-3-130, E-6-130	5	80	—	42, 58, 106, 122, 138, 155, <u>171</u>	80+, UseEirpForVHTTxPowEnv

Table E-4—Global operating classes (continued)

Operating class	Nonglobal operating class(es)	Channel starting frequency (GHz)	Channel spacing (MHz)	Channel set	Channel center frequency index	Behavior limits set
<u>131</u>		<u>5.950</u>	<u>20</u>	<u>1, 5, 9, 13,</u> <u>17, 21, 25,</u> <u>29, 33, 37,</u> <u>41, 45, 49,</u> <u>53, 57, 61,</u> <u>65, 69, 73,</u> <u>77, 81, 85,</u> <u>89, 93, 97,</u> <u>101, 105,</u> <u>109, 113,</u> <u>117, 121,</u> <u>125, 129,</u> <u>133, 137,</u> <u>141, 145,</u> <u>149, 153,</u> <u>157, 161,</u> <u>165, 169,</u> <u>173, 177,</u> <u>181, 185,</u> <u>189, 193,</u> <u>197, 201,</u> <u>205, 209,</u> <u>213, 217,</u> <u>221, 225,</u> <u>229, 233</u>	=	
<u>132</u>		<u>5.950</u>	<u>40</u>	=	<u>3, 11, 19,</u> <u>27, 35, 43,</u> <u>51, 59, 67,</u> <u>75, 83, 91,</u> <u>99, 107,</u> <u>115, 123,</u> <u>131, 139,</u> <u>147, 155,</u> <u>163, 171,</u> <u>179, 187,</u> <u>195, 203,</u> <u>211, 219,</u> <u>227</u>	
<u>133</u>		<u>5.950</u>	<u>80</u>	=	<u>7, 23, 39,</u> <u>55, 71, 87,</u> <u>103, 119,</u> <u>135, 151,</u> <u>167, 183,</u> <u>199, 215</u>	
<u>134</u>		<u>5.950</u>	<u>160</u>	=	<u>15, 47, 79,</u> <u>111, 143,</u> <u>175, 207</u>	
<u>135</u>		<u>5.950</u>	<u>80</u>	=	<u>7, 23, 39,</u> <u>55, 71, 87,</u> <u>103, 119,</u> <u>135, 151,</u> <u>167, 183,</u> <u>199, 215</u>	<u>80+</u>

Table E-4—Global operating classes (continued)

Operating class	Nonglobal operating class(es)	Channel starting frequency (GHz)	Channel spacing (MHz)	Channel set	Channel center frequency index	Behavior limits set
<u>136</u>		<u>5.925</u>	<u>20</u>	<u>=</u>	<u>2</u>	
134 <u>137–179</u>	—	Reserved	Reserved	Reserved	Reserved	Reserved
...						

E.2 Band-specific operating requirements

Insert the following subclause (E.2.7, including Table E-12 and Table E-13) after E.2.6:

E.2.7 6 GHz band in the United States

In the United States, when operating in the 6 GHz band, the Country String field in the Country element is set to (in hexadecimal) 0x55, 0x53, 0x04.

NOTE 1—The first two octets indicate the United States. The third octet indicates that Table E-4 is in use (see Annex C).

The Regulatory Info subfield in the Control field of the 6 GHz Operation Information field of the HE Operation element is interpreted as shown in Table E-12 when operating in the 6 GHz band in the United States.

Table E-12—Regulatory Info subfield encoding in the United States

Value	Description
0	Indoor Access Point
1	Standard Power Access Point
2–7	Reserved

The Maximum Transmit Power Category subfield in the Transmit Power Information field of the Transmit Power Envelope element is interpreted as shown in Table E-13 when operating in the 6 GHz band in the United States.

Table E-13—Maximum Transmit Power Category subfield encoding in the United States

Value	Description
0	Default
1	Subordinate Device
2–3	Reserved

An AP that is an Indoor Access Point per regulatory rules shall send at least two Transmit Power Envelope elements in Beacon and Probe Response frames as follows:

- Maximum Transmit Power Category subfield = Default; Unit interpretation = Regulatory client EIRP PSD
- Maximum Transmit Power Category subfield = Subordinate Device; Unit interpretation = Regulatory client EIRP PSD

An AP that is a Standard Power Access Point per regulatory rules shall send at least one Transmit Power Envelope element in Beacon and Probe Response frames as follows:

- Maximum Transmit Power Category subfield = Default; Unit interpretation = Regulatory Client EIRP PSD

A regulatory client EIRP PSD value advertised by an AP that is a Standard Power Access Point shall be set to the highest value that meets the authorized client transmit power limits for the corresponding category obtained from the AP's AFC system and any other client PSD regulatory rules for the corresponding 20 MHz channel.

If the regulatory client EIRP PSD values advertised by an AP that is a Standard Power Access Point are insufficient to ensure that regulatory client limits on total EIRP are always met for all transmission bandwidths within the bandwidth of the AP's BSS, the AP shall also send a Transmit Power Envelope element in Beacon and Probe Response frames as follows:

- Maximum Transmit Power Category subfield = Default; Unit interpretation = Regulatory client EIRP

NOTE 2—In the case of regulatory rules where the maximum transmit power for client devices is lower than the maximum transmit power for Access Points, the regulatory client maximum transmit power advertised by the AP for client devices might be lower than the regulatory client maximum transmit power the AP is authorized to use for its own transmissions.

If a non-AP STA that is a Subordinate Device per regulatory rules receives Transmit Power Envelope elements with Maximum Transmit Power Category subfields indicating Subordinate Device, it may ignore any other received Transmit Power Envelope elements that indicate other values in the Maximum Transmit Power Category subfield.

A non-AP STA that is a Fixed Client Device per regulatory rules may ignore any received Transmit Power Envelope elements it receives from an AP that it has been identified (from interpretation of the Regulatory Info field in the HE Operation element) as a Standard Power Access Point.

NOTE 3—A non-AP STA that is a Fixed Client per regulatory rules must ensure it abides by regulatory limits it has obtained from an AFC system.

Annex G

(normative)

Frame exchange sequences

G.1 General

Change Table G.1 as follows (not all rows are shown):

Table G-1—Attributes applicable to frame exchange sequence definition

Attribute	Description
...	
<i>mu-user-respond</i>	...
<i>mu-users-respond</i>	The preceding frame or A-MPDU is part of an HE MU PPDU.
<i>mu-user-not-respond</i>	The preceding frame or A-MPDU is part of a VHT MU PPDU <u>or HE MU PPDU</u> and is addressed to a user from which no immediate response is expected. See NOTE 3 and NOTE 4.
...	

G.4 HT, VHT, and S1G sequences

Change the following text in G.4 as shown:

(* The per-user parts of a VHT MU PPDU and HE MU PPDU that do not require a response *)
other-users = {ppdu-not-requiring-response-per-user +*mu-user-not-respond*} +*mu-ppdu-end*;

Insert the following subclause (G.5) after G.4:

G.5 HE sequences

he-txop-sequence = he-nav-protected-sequence |
1 {initiator-sequence};

(* an he-nav-protected-sequence consists of setting the NAV, performing one or more initiator-sequences
and then resetting the NAV if time permits *)

he-nav-protected-sequence = he-nav-set 1 {initiator-sequence} [resync-sequence];

(* This is the sequence of frames that establish protection use MU-RTS *)

he-nav-set =
(MU-RTS Trigger 1 {CTS}) |
(Data[+HTC]+individual[+null][+QoS+normal-ack] Ack) |
Data[+HTC]+individual[+QoS+block-ack] |
Data+group[+null][+QoS] |

(1 {**Data**[+HTC]+*individual*+*QoS*+*implicit-bar*+*a-mpdu*}+*a-mpdu-end*}
BlockAck) |
(BlockAckReq (BlockAck|Ack)) |
(BlockAck|Ack);

he-dl-mu-sequence = **(BlockAck**+*delayed*[+*mu-users-respond*] **Ack** |
(BlockAckReq+*delayed*[+*mu-users-respond*] **Ack**) |
(Data[+HTC]+*individual*[+*null*][+*QoS*+*normal-ack*][+*mu-user-respond*] **Ack** |
Ack);

(* Trigger frame is sent by the AP to initiate non-AP UL transmission. A PPDU containing a Trigger frame is either a non-A-MPDU Trigger frame or an A-MPDU carrying a Trigger frame *)

he-ul-mu-sequence = **(Basic Trigger)** | **(Basic Trigger**+*a-mpdu*+*mu-user-respond*+*a-mpdu-end*)
 1 {**Data**[+HTC]+*QoS*+(*no-ack* | *block-ack*)+*a-mpdu*}
 + *a-mpdu-end* | **MU-BAR Trigger BlockAck**;

he-cascading-sequence = he-dl-mu-sequence + he-ul-mu-sequence

(* HE beamforming sequence *)

he-non-trigger-based-sounding =
(HE NDP Announcement) (HE sounding NDP)
HE Compressed Beamforming/CQI;

he-trigger-based-sounding = **(HE NDP Announcement) (HE sounding NDP)** he-feedback
{BFRP Trigger he-feedback};

he-feedback = **(HE Compressed Beamforming/CQI)** | (* S-MPDU or non-HE PPDU *)
 1 {(**HE Compressed Beamforming/CQI**)+*a-mpdu*}+*a-mpdu-end*;

he-nfrp-report = **(NFRP Trigger)** n **(HE TB feedback NDP)**

Annex T

(informative)

Overlapping BSS (OBSS) management

T.1 Introduction

Change T.1 as follows:

When two or more BSSs overlap, the available bandwidth is shared and hence reduced for each BSS. The basic access mechanism, such as DCF, is able to work across OBSSs. Similarly, if EDCA is used, the OBSS might be considered a larger network, and access to the WM is basically shared according to the EDCA access mechanism. Note that for both DCF and EDCA overlapping networks, the sharing is affected by the relative traffic; and if more than two APs are sharing, the problem of “neighbor capture” might occur. The neighbor capture effect might occur when a BSS is in the middle of two other BSSs that are hidden from each other, where it might suffer a disproportionate degradation in throughput, relative to the total traffic in all three BSSs. A particular problem arises when there is some expectation of QoS. If EDCA admission control is in use, then it can be used to regulate the QoS traffic on its own BSS, but it might not take into account the EDCA admitted traffic on an OBSS. The result is that the QoS is compromised if each BSS admits traffic up to its local maximum. Similarly a BSS using HCCA might schedule traffic in its own BSS, to “guarantee” a service, but, if not controlled, this might suppress overlapping EDCA admission control BSS. Furthermore, if two HCCA BSSs overlap and they do not coordinate their scheduled TXOPs, then a degradation of QoS might result. An HE BSS advertises BSS color information, which is a value in the range 1 to 63 that identifies the BSS. Based on the BSS color information, a receiving STA can make decisions on whether to access the medium while there is ongoing transmission on the medium or go to doze state until the end of the received PPDU or update the NAV. The features described in this annex have been introduced in order to allow a degree of management for OBSSs and for mitigation of the basic problems outlined above.

Insert the following subclause (T.6) after T.5:

T.6 BSS color and spatial reuse

The BSS color is an identifier of the BSS and is used to assist a receiving STA in identifying the BSS from which a PPDU originates so that the STA can follow the channel access rules to perform spatial reuse. The objective of spatial reuse operation is to allow the medium to be used more often between OBSSs in dense deployment scenarios by the early identification of signals from OBSSs and interference management. See 26.10.

Insert the following text (Annex Z and Annex AA) after Annex Y:

Annex Z

(informative)

HE-SIG-B content examples

Z.1 General

This annex provides a number of examples to illustrate the content of HE-SIG-B content channels.

HE-SIG-B content channels are padded to the same length and to an OFDM symbol boundary as defined in 27.3.11.8.5. For illustration simplicity, the examples do not include the complete padding bits but only pad to make the two HE-SIG-B content channels equal in length and an integer number of 4 bits. All the padding bits in the examples are set to 0.

In the examples, the binary sequence of each HE-SIG-B field is in transmission order. For the entire content of each HE-SIG-B content channel, the binary sequences are converted to hexadecimal listed in transmission order as pairs of hexadecimal digits, where the first bit transmitted is the LSB of the first transmitted octet and is the LSB of the second hexadecimal digit.

Z.2 Example 1

An example of the HE-SIG-B field with resource allocation signaling for an 80 MHz HE MU PPDU is shown in Table Z-1.

Table Z-1—Resource allocation signaling example 1

RU	484-tone RU 1	26-tone RU 19 (center 26-tone RU)	242-tone RU 3	242-tone RU 4
SS0	STA-ID 1441, HE-MCS 10, LDPC	STA-ID 1443, HE-MCS 3, BCC, 1SS, no beamforming, no DCM	STA-ID 1444, HE-MCS 4, BCC, 2SS, Tx beamforming	STA-ID 1445, HE-MCS 8, BCC
SS1				STA-ID 1446, HE-MCS 7, BCC
SS2	STA-ID 1442, HE-MCS 9, LDPC		N/A	STA-ID 1447, HE-MCS 6, BCC
SS3				STA-ID 1448, HE-MCS 5, BCC

The AP performs a dynamic split of the User fields for the two MU-MIMO STAs on 484-tone RU 1, with two User fields assigned to HE-SIG-B content channel 1 and none to HE-SIG-B content channel 2, to avoid a disparity in the number of User fields between content channels (see NOTE 1 in 27.3.11.8.3). The User field for STAs 1441, 1442, 1443, and 1444 are in HE-SIG-B content channel 1 while User field for STAs 1445, 1446, 1447, and 1448 are in HE-SIG-B content channel 2. The content of the entire HE-SIG-B field for this example is shown in Table Z-2.

Table Z-2—HE-SIG-B field content for example 1

	HE-SIG-B content channel 1		HE-SIG-B content channel 2	
Common field	10010011 00000011 1 1111 000000		01001110 11000011 1 1100 000000	
User fields	STA 1441	10000101101 0010 0101 0 1	STA 1445	10100101101 0000 0001 0 0
	STA 1442	01000101101 0010 1001 0 1	STA 1446	01100101101 0000 1110 0 0
	CRC & tail	0011 000000	CRC & tail	1101 000000
	STA 1444	00100101101 100 1 0010 0 0	STA 1447	11100101101 0000 0110 0 0
	STA 1443	11000101101 000 0 1100 0 0	STA 1448	00010101101 0000 1010 0 0
	CRC & tail	1000 000000	CRC & tail	1001 000000
	Padding	0	Padding	0
HE-SIG-B field content in binary, organized as octets (LSB first)	10010011 00000011 11111000 00010000 10110100 10010101 01000101 10100101 00101001 10000000 01001011 01100100 10001100 01011010 00011000 01000000 0000		01001110 11000011 11100000 00010100 10110100 00000100 01100101 10100001 11000110 10000001 11001011 01000001 10000001 01011010 00010100 01001000 0000	
HE-SIG-B field content in binary, organized as octets (MSB first within each octet)	11001001 11000000 00011111 00001000 00101101 10101001 10100010 10100101 10010100 00000001 11010010 00100110 00110001 01011010 00011000 00000010 0000		01110010 11000011 00000111 00101000 00101101 00100000 10100110 10000101 01100011 10000001 11010011 10000010 10000001 01011010 00101000 00010010 0000	
HE-SIG-B field content in hexadecimal, organized as octets	c9 c0 1f 08 2d a9 a2 a5 94 01 d2 26 31 5a 18 02 00		72 c3 07 28 2d 20 a6 85 63 81 d3 82 81 5a 28 12 00	

Z.3 Example 2

An example of the HE-SIG-B field with full-bandwidth MU-MIMO resource allocation for more than 1 user in an 80 MHz HE MU PPDU is shown in Table Z-3.

Table Z-3—Resource allocation signaling example 2

RU	996-tone RU 1
SS0	STA-ID 1449, HE-MCS 6, LDPC
SS1	
SS2	STA-ID 1450, HE-MCS 7, LDPC
SS3	STA-ID 1451, HE-MCS 8, LDPC

In this example, the HE-SIG-B Compression field in the HE-SIG-A field is set to 1, and the Common field is not present in either HE-SIG-B content channel. The User field for STA 1449 and STA 1450 is in HE-SIG-B content channel 1, and the User field for STA 1451 is in HE-SIG-B content channel 2. The content of the HE-SIG-B field in this example is shown in Table Z-4.

Table Z-4—HE-SIG-B field content for example 2

	HE-SIG-B content channel 1		HE-SIG-B content channel 2	
User fields	STA 1449	10010101101 1000 0110 0 1	STA 1451	11010101101 1000 0001 0 1
	STA 1450	01010101101 1000 1110 0 1	CRC & tail	0101 000000
	CRC & tail	0011 000000	Padding	00000000000000000000
HE-SIG-B field content in binary, organized as octets (LSB first)	10010101 10110000 11001010 10101101 10001110 01001100 0000		11010101 10110000 00101010 10000000 00000000 00000000 0000	
HE-SIG-B field content in binary, organized as octets (MSB first within each octet)	10101001 00001101 01010011 10110101 01110001 00110010 0000		10101011 00001101 01010100 00000001 00000000 00000000 0000	
HE-SIG-B field content in hexadecimal, organized as octets	a9 0d 53 b5 71 32 00		ab 0d 54 01 00 00 00	

Z.4 Example 3

An example of the HE-SIG-B field with signaling for a full-bandwidth single-user resource allocation in an 80 MHz HE MU PPDU is shown in Table Z-5.

Table Z-5—Resource allocation signaling example 3

RU	996-tone RU 1
SS0	STA-ID 1452, HE-MCS 8, LDPC, 2SS, Tx beamforming
SS1	

In this example, the HE-SIG-B Compression field in the HE-SIG-A field is set to 0, and the resource allocation is signaled as an OFDMA transmission with one user. The User field for STA 1452 is in HE-SIG-B content channel 1, and no User field is present in HE-SIG-B content channel 2. The content of the HE-SIG-B field is shown in Table Z-6.

Table Z-6—HE-SIG-B field content for example 3

	HE-SIG-B content channel 1		HE-SIG-B content channel 2	
Common field	00001011 11001110 0 1011 000000		11001110 11001110 0 1110 000000	
User fields	STA 1452	00110101101 100 1 0001 0 1	Padding	00000000000000000000 0000000000 00
	CRC & tail	1100 000000		
	Padding	00		
HE-SIG-B field content in binary, organized as octets (LSB first)	00001011 11001110 01011000 00000110 10110110 01000101 11000000 0000		11001110 11001110 01110000 00000000 00000000 00000000 00000000 0000	
HE-SIG-B field content in binary, organized as octets (MSB first within each octet)	11010000 01110011 00011010 01100000 01101101 10100010 00000011 0000		01110011 01110011 00001110 00000000 00000000 00000000 00000000 0000	
HE-SIG-B field content in hexadecimal, organized as octets	d0 73 1a 60 6d a2 03 00		73 73 0e 00 00 00 00 00	

Z.5 Example 4

In the following example there is an 80 MHz or wider PPDU with the lowest 484-tone RU containing two users, and there are multiple other unlisted users, which are variously split across the two HE-SIG-B content channels (e.g., because their RU sizes are up to 242 tones). For the two users in the 484-tone RU, Table Z-7 shows various dynamic split options that can be used to reduce the disparity in the number of User fields across the HE-SIG-B content channels (see NOTE 1 in 27.3.11.8.3). The example shows just the associated RU Allocation subfields (i.e., a fragment of the HE-SIG-B field), which might be sent in one of three ways.

Table Z-7—RU Allocation subfields for different dynamic splits of User fields for the example of two MU-MIMO users in the lowest 484-tone RU of an 80 MHz or wider PPDU

Dynamic split option	First RU Allocation subfield of HE-SIG-B content channel 1	Number of User fields contributed to HE SIG-B content channel 1	First RU Allocation subfield of HE SIG-B content channel 2	Number of User fields contributed to HE SIG-B content channel 2
Both User fields signaled in HE-SIG-B content channel 1	MSB first: 11001001 LSB first: 10010011	2	MSB first: 01110010 LSB first: 01001110	0
Both User fields signaled in HE-SIG-B content channel 2	MSB first: 01110010 LSB first: 01001110	0	MSB first: 11001001 LSB first: 10010011	2
One User field per HE-SIG-B content channel	MSB first: 11001000 LSB first: 00010011	1	MSB first: 11001000 LSB first: 00010011	1

Annex AA

(informative)

Multiple BSSID configuration examples

AA.1 Introduction

This annex provides examples showing the relationship between profile periodicity (indicated by the Full Set Rx Periodicity field in the Multiple BSSID Configuration element) and the DTIM interval (DTIM Period field in the Multiple BSSID-Index element) for a multiple BSSID set as described in 11.1.3.8.3. The examples provide guidance on how an AP might organize the advertisement of nontransmitted BSSID profiles in its Beacon frames if it cannot fit all the profiles in a single Beacon frame (i.e., partial list of profiles) it is advertising. By having the DTIM interval for a nontransmitted BSSID a multiple of the profile periodicity, the profile for that BSSID would always appear in its DTIM beacon. This helps an associated non-AP STA save power as it is able to receive any updates to the profile when it wakes to receive the DTIM beacon.

AA.2 Examples

In the following examples, a BSSID with BSSID-index I, DTIM count of X, and DTIM period of Y is represented as [I] X/Y. A profile appears in every p^{th} Beacon frame; where p is the value carried in the Full Set Rx Periodicity field. Each set of profiles appearing in a Beacon frame amongst the p Beacon frames is identified as set A to set P. In the following examples, a nontransmitted BSSID is considered as active if the AP corresponding to that BSSID has set up a BSS and the information of the BSSID is carried in a Beacon frame of the AP corresponding to the transmitted BSSID in the set. When an AP is advertising a partial list of profiles, the information may not be carried in every Beacon frame (see 11.1.3.8.3). For simplicity, the examples show that the BSSIDs are activated in contiguous order of BSSID-index and appear in the same order in a Beacon frame. In practice, an AP can activate BSSIDs in any order and include a certain BSSID profile in a Beacon frame based on size.

The first example illustrates the case where MaxBSSID Indicator (n) is set to 4 and there are 11 active nontransmitted BSSIDs in a multiple BSSID set. The AP is able to fit up to 3 nontransmitted BSSID profiles in each beacon, and the Full Set Rx Periodicity field is set to 4. Figure AA-1 shows the configuration for DTIM count and DTIM period of each BSSID in this set. With the DTIM period being a multiple of profile periodicity, the AP is able to include a nontransmitted BSSID in its DTIM beacon.

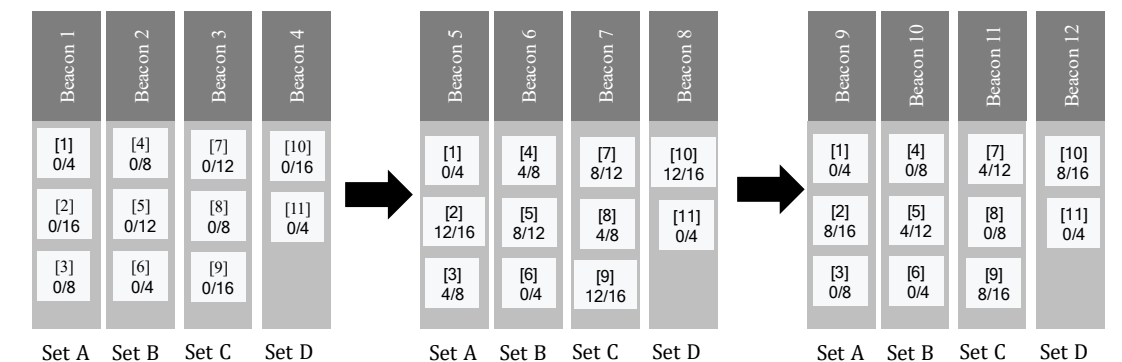


Figure AA-1—Example of partial list of profiles with a profile periodicity of 4

The next example considers the case where MaxBSSID Indicator (n) is set to 4 and there are 15 active nontransmitted BSSID in a multiple BSSID set. In this example, the AP's Beacon frame is able to fit up to 6 nontransmitted BSSID profiles; therefore, the profile periodicity for the set has a value equal to 3. Figure AA-2 shows the configuration for DTIM count and DTIM period of each BSSID in this set. A BSSID profile would be included every third Beacon frame. In addition, since the DTIM interval is a multiple of profile periodicity, it would appear in the DTIM beacon of each profile.

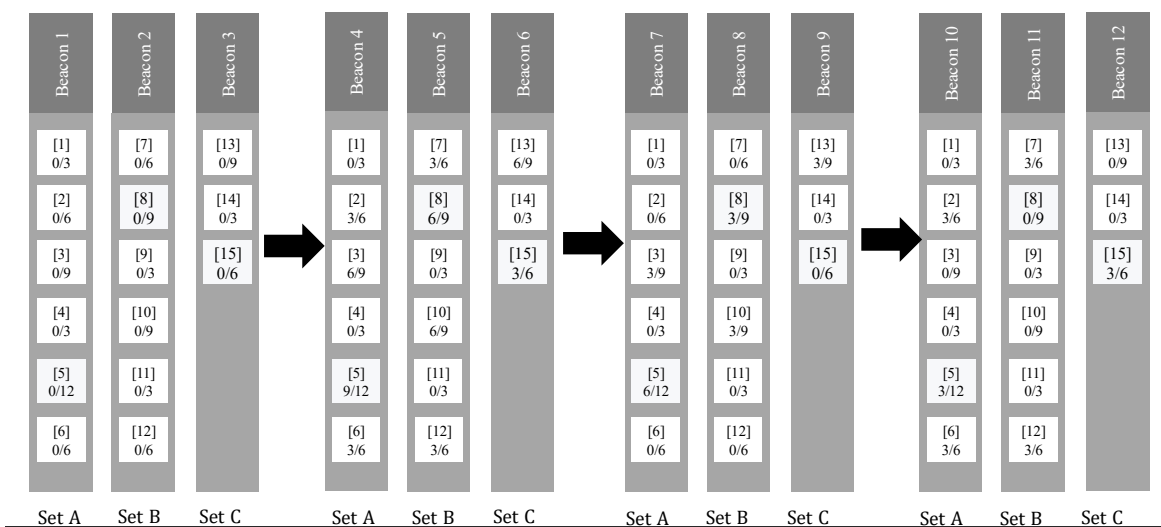


Figure AA-2—Example of partial list of profiles with a profile periodicity of 3

The next example considers the case where a BSSID is deactivated and the corresponding AP disassociates the entire BSS. In this case, there are 15 nontransmitted BSSIDs that are active, and one of the BSSID in set B (BSSID-Index = 10 with DTIM period = 9) is turned off at some point. Figure AA-3 illustrates this case. Once deactivated, this BSSID is no longer included in the Beacon frame. No other BSSIDs in set B or other sets are affected.

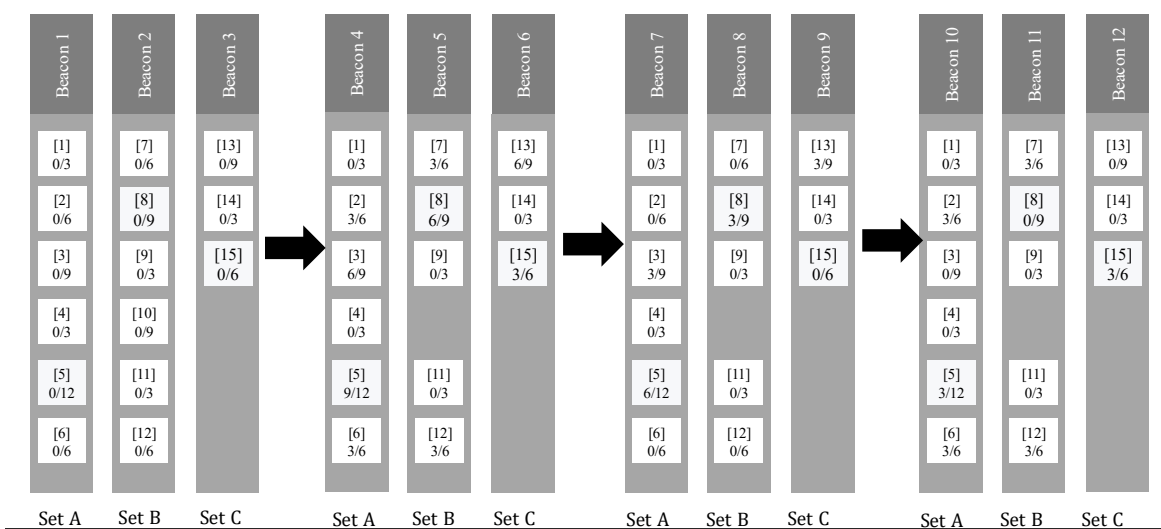


Figure AA-3—Example of a BSSID being deactivated

The next example considers the case where a BSSID that was not active earlier is activated. In this case, there are 14 nontransmitted BSSIDs that are active, and a new BSSID (BSSID-Index = 15 with DTIM period = 6) is activated at some point. The AP is able to fit it as part of set C. Figure AA-4 illustrates this case. This BSSID is included in the next set of beacons and is advertised as part of set C. When the BSSID is included in a Beacon frame for the first time, it is started with DTIM count set to 0. With the DTIM Period being a multiple of profile periodicity, the AP can ensure that the profile appears in its DTIM beacon.

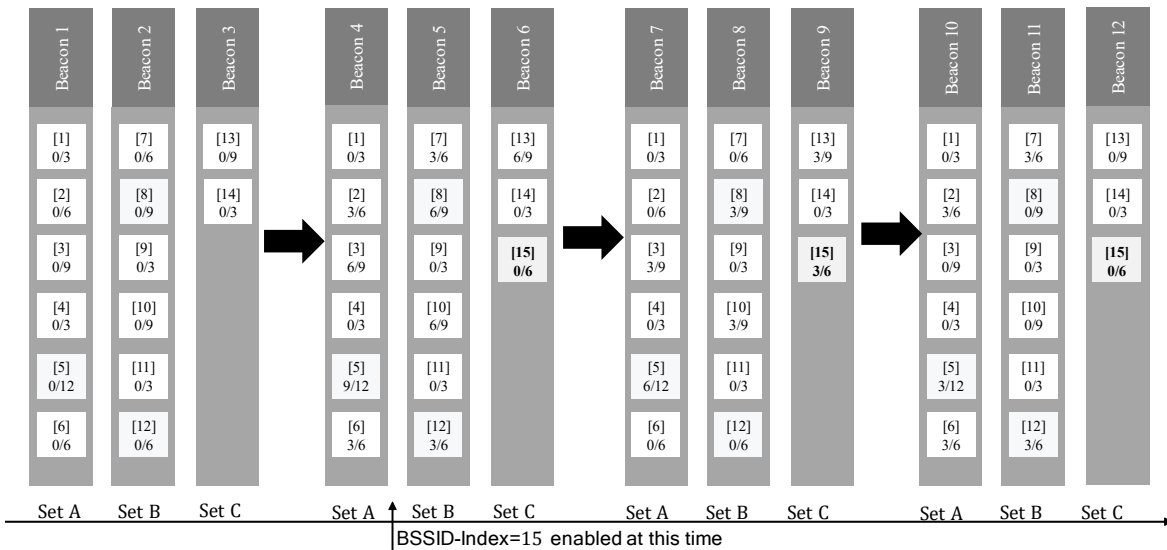


Figure AA-4—Example of a BSSID being activated

The next example considers the case where the DTIM period for a BSSID is updated. In this case, the DTIM period for a BSSID (BSSID-Index = 12) is changed from 6 to 15. This change would be indicated during a DTIM beacon for that profile. This will ensure that STAs associated with that profile are able to receive the update. After the change, when the BSSID is included in a Beacon frame for the first time, it starts with DTIM count set to 0. With the new DTIM period (of 15) being a multiple of profile periodicity (of 3), the profile will appear in its DTIM beacon. Figure AA-5 shows the configuration for DTIM count and DTIM period of each BSSID in this set.

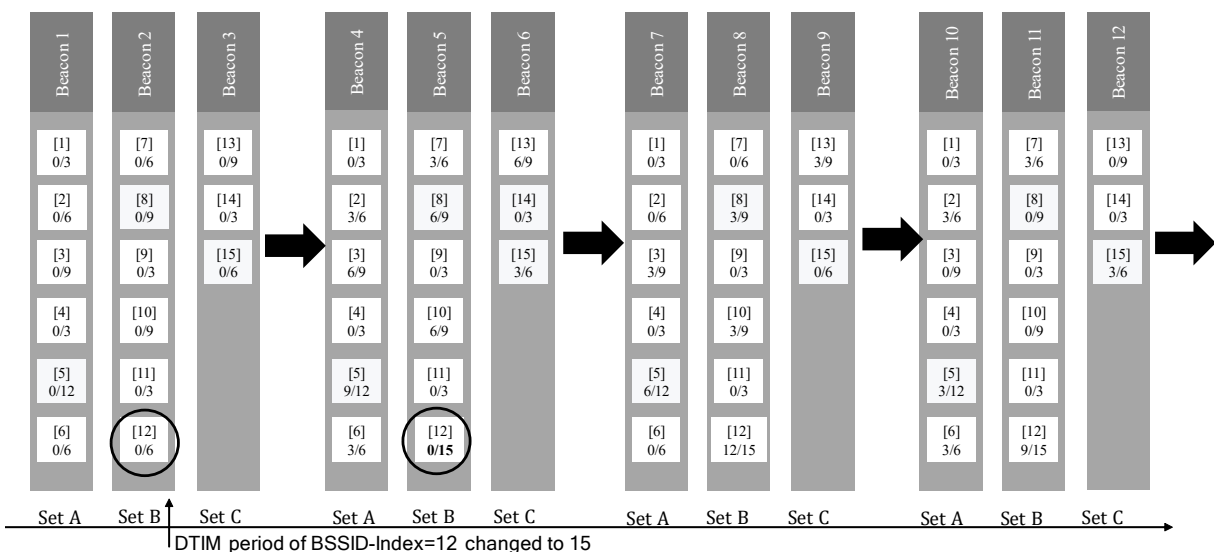


Figure AA-5—Example of DTIM period being changed for a particular BSSID

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